SCIENCE

# History of the inductive sciences, from the earliest to the present time

William Whewell

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HISTORY OF THE INDUCTIVE SCIENCES.

VOLUME I.

HISTORY OF THE INDUCTIVE SCIENCES, FROM THE EARLIEST TO THE PRESENT TIME. BY WILLIAM WHEWELL, D. D., MASTER OF TRINITY COLLEGE, CAMBRIDGE. _THE THIRD EDITION, WITH ADDITIONS._ IN TWO VOLUMES.

VOLUME I.

NEW YORK: D. APPLETON AND COMPANY, 549 & 551 BROADWAY. 1875.

TO SIR JOHN FREDERICK WILLIAM HERSCHEL, K.G.H.

MY DEAR HERSCHEL, IT is with no common pleasure that I take up my pen to dedicate these volumes to you. They are the result of trains of thought which have often been the subject of our conversation, and of which the origin goes back to the period of our early companionship at the University. And if I had ever wavered in my purpose of combining such reflections and researches into a whole, I should have derived a renewed impulse and increased animation from your delightful Discourse on a kindred subject. For I could not have read it without finding this portion of philosophy invested with a fresh charm; and though I might be well aware that I could not aspire to that large share of popularity which your work so justly gained, I should still have reflected, that something was due to the subject itself, and should have hoped that my own aim was so far similar to yours, that the present work might have a chance of exciting an interest in some of your readers. That it will interest you, I do not at all hesitate to believe.

If you were now in England I should stop here: but when a friend is removed for years to a far distant land, we seem to acquire a right to speak openly of his good qualities. I cannot, therefore, prevail upon myself to lay down my pen without alluding to the affectionate admiration of your moral and social, as well as intellectual excellencies, which springs up in the hearts of your friends, whenever you are thought of. They are much delighted to look upon the halo of deserved fame which plays round your head but still more, to recollect, {6} as one of them said, that your head is far from being the best part about you.

May your sojourn in the southern hemisphere be as happy and successful as its object is noble and worthy of you; and may your return home be speedy and prosperous, as soon as your purpose is attained.

Ever, my dear Herschel, yours,

W. WHEWELL. March 22, 1837.

P.S. So I wrote nearly ten years ago, when you were at the Cape of Good Hope, employed in your great task of making a complete standard survey of the nebulæ and double stars visible to man. Now that you are, as I trust, in a few weeks about to put the crowning stone upon your edifice by the publication of your "Observations in the Southern Hemisphere," I cannot refrain from congratulating you upon having had your life ennobled by the conception and happy execution of so great a design, and once more offering you my wishes that you may long enjoy the glory you have so well

won.

W. W. TRINITY COLLEGE, NOV. 22, 1846.

{{7}} PREFACE TO THE THIRD EDITION.

IN the Prefaces to the previous Editions of this work, several remarks were made which it is not necessary now to repeat to the same extent. That a History of the Sciences, executed as this is, has some value in the eyes of the Public, is sufficiently proved by the circulation which it has obtained. I am still able to say that I have seen no objection urged against the plan of the work, and scarcely any against the details. The attempt to throw the history of each science into EPOCHS at which some great and cardinal discovery was made, and to arrange the subordinate events of each history as belonging to the PRELUDES and the SEQUELS of such Epochs, appears to be assented to, as conveniently and fairly exhibiting the progress of scientific truth. Such a view being assumed, as it was a constant light and guide to the writer in his task, so will it also, I think, make the view of the reader far more clear and comprehensive than it could otherwise be. With regard to the manner in which this plan has been carried into effect with reference to particular writers and their researches, as I have said, I have seen scarcely any objection made. I was aware, as I stated at the outset, of the difficulty and delicacy of the office which I had undertaken; but I had various considerations to encourage me to go through it; and I had a trust, which I {8} have as yet seen nothing to disturb, that I should be able to speak impartially of the great scientific men of all ages, even of our own.

I have already said, in the Introduction, that the work aimed at being, not merely a narration of the facts in the history of Science, but a basis for the Philosophy of Science. It seemed to me that our study of the modes of discovering truth ought to be based upon a survey of the truths which have been discovered. This maxim, so stated, seems sufficiently self-evident; yet it has, even up to the present time, been very rarely acted on. Those who discourse concerning the nature of Truth and the mode of its discovery, still, commonly, make for themselves examples of truths, which for the most part are utterly frivolous and unsubstantial (as in most Treatises on Logic); or else they dig up, over and over, the narrow and special field of mathematical truth, which certainly cannot, of itself, exemplify the general mode by which man has attained to the vast body of certain truth which he now possesses.

Yet it must not be denied that the Ideas which form the basis of Mathematical Truth are concerned in the formation of Scientific Truth in general; and discussions concerning these Ideas are by no means necessarily barren of advantage. But it must be borne in mind that, besides these Ideas, there are also others, which no less lie at the root of Scientific Truth; and concerning which there have been, at various periods, discussions which have had an important bearing on the progress of Scientific Truth;--such as discussions concerning the nature and necessary attributes of Matter, of Force, of Atoms, of Mediums, of Kinds, of Organization. The controversies which have taken place concerning these have an important place in the history of Natural Science in {9} its most extended sense. Yet it appeared convenient to carry on the history of Science, so far as it depends on Observation, in

a line separate from these discussions concerning Ideas. The account of these discussions and the consequent controversies, therefore, though it be thoroughly historical, and, as appears to me, a very curious and interesting history, is reserved for the other work, the _Philosophy of the Inductive Sciences_. Such a history has, in truth, its natural place in the Philosophy of Science; for the Philosophy of Science at the present day must contain the result and summing up of all the truth which has been disentangled from error and confusion during these past controversies.

I have made a few Additions to the present Edition; partly, with a view of bringing up the history, at least of some of the Sciences, to the present time,--so far as those larger features of the History of Science are concerned, with which alone I have here to deal,--and partly also, especially in the First Volume, in order to rectify and enlarge some of the earlier portions of the history. Several works which have recently appeared suggested reconsideration of various points; and I hoped that my readers might be interested in the reflections so suggested.

I will add a few sentences from the Preface to the First Edition.

"As will easily be supposed, I have borrowed largely from other writers, both of the histories of special sciences and of philosophy in general.[1\P] I have done this without {10} scruple, since the novelty of my work was intended to consist, not in its superiority as a collection of facts, but in the point of view in which the facts were placed. I have, however, in all cases, given references to my authorities, and there are very few instances in which I have not verified the references of previous historians, and studied the original authors. According to the plan which I have pursued, the history of each science forms a whole in itself, divided into distinct but connected members, by the _Epochs_ of its successive advances. If I have satisfied the competent judges in each science by my selection of such epochs, the scheme of the work must be of permanent value, however imperfect may be the execution of any of its portions.

[Note 1\P: Among these, I may mention as works to which I have peculiar obligations, Tennemann's Geschichte der Philosophie; Degerando's Histoire Comparée des Systèmes de Philosophie; Montucla's Histoire des Mathématiques, with Delalande's continuation of it; Delambre's Astronomie Ancienne, Astronomie du Moyen Age, Astronomie Moderne, and Astronomie du Dix-huitième Siècle; Bailly's Histoire d'Astronomie Ancienne, and Histoire d'Astronomie Moderne; Voiron's Histoire d'Astronomie (published as a continuation of Bailly), Fischer's Geschichte der Physik, Gmelin's Geschichte der Chemie, Thomson's History of Chemistry, Sprengel's History of Medicine, his History of Botany, and in all branches of Natural History and Physiology, Cuvier's works; in their historical, as in all other portions, most admirable and instructive.]

"With all these grounds of hope, it is still impossible not to see that such an undertaking is, in no small degree, arduous, and its event obscure. But all who venture upon such tasks must gather trust and encouragement from reflections like those by which their great forerunner prepared himself for his endeavors;--by recollecting that they are aiming to advance the best interests and privileges of man;

and that they may expect all the best and wisest of men to join them in their aspirations and to aid them in their labors.

"'Concerning ourselves we speak not; but as touching the matter which we have in hand, this we ask;--that men deem it not to be the setting up of an Opinion, but the performing of a Work; and that they receive this as a certainty--that we are not laying the foundations of any sect or doctrine, but of the profit and dignity of mankind:--Furthermore, {11} that being well disposed to what shall advantage themselves, and putting off factions and prejudices, they take common counsel with us, to the end that being by these our aids and appliances freed and defended from wanderings and impediments, they may lend their hands also to the labors which remain to be performed:--And yet, further, that they be of good hope; neither feign and imagine to themselves this our Reform as something of infinite dimension and beyond the grasp of mortal man, when, in truth, it is, of infinite error, the end and true limit; and is by no means unmindful of the condition of mortality and humanity, not confiding that such a thing can be carried to its perfect close in the space of one single day, but assigning it as a task to a succession of generations.'--BACON--INSTAURATIO MAGNA, _Præf. ad fin._

"'If there be any man who has it at heart, not merely to take his stand on what has already been discovered, but to profit by that, and to go on to something beyond;--not to conquer an adversary by disputing, but to conquer nature by working;--not to opine probably and prettily, but to know certainly and demonstrably;--let such, as being true sons of nature (if they will consent to do so), join themselves to us; so that, leaving the porch of nature which endless multitudes have so long trod, we may at last open a way to the inner courts. And that we may mark the two ways, that old one, and our new one, by familiar names, we have been wont to call the one the _Anticipation of the Mind_, the other, the _Interpretation of Nature_.'--INST. MAG. _Præf. ad Part._ ii.

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{{39}} A HISTORY OF THE INDUCTIVE SCIENCES.

# INTRODUCTION.

"A just story of learning, containing the antiquities and originals of KNOWLEDGES, and their sects; their inventions, their diverse administrations and managings; their flourishings, their oppositions, decays, depressions, oblivions, removes; with the causes and occasions of them, and all other events concerning learning, throughout all ages of the world; I may truly affirm to be wanting.

"The use and end of which work I do not so much design for curiosity, or satisfaction of those that are the lovers of learning: but chiefly for a more serious and grave purpose; which is this, in few words--that it will make learned men more wise in the use and administration of learning." BACON, _Advancement of Learning_, book ii.

{{41}} INTRODUCTION.

IT is my purpose to write the History of some of the most important of the Physical Sciences, from the earliest to the most recent periods. I shall thus have to trace some of the most remarkable branches of human knowledge, from their first germ to their growth into a vast and varied assemblage of undisputed truths; from the acute, but fruitless, essays of the early Greek Philosophy, to the comprehensive systems, and demonstrated generalizations, which compose such sciences as the Mechanics, Astronomy, and Chemistry, of modern times.

The completeness of historical view which belongs to such a design, consists, not in accumulating all the details of the cultivation of each science, but in marking the larger features of its formation. The historian must endeavor to point out how each of the important advances was made, by which the sciences have reached their present position; and when and by whom each of the valuable truths was obtained, of which the aggregate now constitutes a costly treasure.

Such a task, if fitly executed, must have a well-founded interest for all those who look at the existing condition of human knowledge with complacency and admiration.

The present generation finds itself the heir of a vast patrimony of science; and it must needs concern us to know the steps by which these possessions were acquired, and the documents by which they are secured to us and our heirs forever. Our species, from the time of its creation, has been travelling onwards in pursuit of truth; and now that we have reached a lofty and commanding position, with the broad light of day around us, it must be grateful to look back on the line of our vast progress;--to review the journey, begun in early twilight amid primeval wilds; for a long time continued with slow advance and obscure prospects; and gradually and in later days followed along more open and lightsome paths, in a wide and fertile region. The historian of science, from early periods to the present times, may hope for favor on the score of the mere subject of his narrative, and in virtue of the curiosity which the men {42} of the present day may naturally feel respecting the events and persons of his story.

But such a survey may possess also an interest of another kind; it may be instructive as well as agreeable; it may bring before the reader the present form and extent, the future hopes and prospects of science, as well as its past progress. The eminence on which we stand may enable us to see the land of promise, as well as the wilderness through which we have passed. The examination of the steps by which our ancestors acquired our intellectual estate, may make us acquainted with our expectations as well as our possessions;--may not only remind us of what we have, but may teach us how to improve and increase our store. It will be universally expected that a History of Inductive Science should point out to us a philosophical distribution of the existing body of knowledge, and afford us some indication of the most promising mode of directing our future efforts to add to its extent and completeness.

To deduce such lessons from the past history of human knowledge, was the intention which originally gave rise to the present work. Nor is this portion of the design in any measure abandoned; but its execution, if it take place, must be attempted in a separate and future treatise, _On the Philosophy of the Inductive Sciences_. An essay of this kind may, I trust, from the progress already made in it, be laid before the public at no long interval after the present history.[1\1]

[Note 1\1: The _Philosophy of the Inductive Sciences_ was published shortly after the present work.]

Though, therefore, many of the principles and maxims of such a work will disclose themselves with more or less of distinctness in the course of the history on which we are about to enter, the systematic and complete exposition of such principles must be reserved for this other treatise. My attempts and reflections have led me to the opinion, that justice cannot be done to the subject without such a division of it.

To this future work, then, I must refer the reader who is disposed to require, at the outset, a precise explanation of the terms which occur in my title. It is not possible, without entering into this philosophy, to explain adequately how science which is INDUCTIVE differs from that which is not so; or why some portions of _knowledge_ may properly be selected from the general mass and termed SCIENCE. It will be sufficient at present to say, that the sciences of which we have {43} here to treat, are those which are commonly known as the _Physical Sciences_; and that by _Induction_

is to be understood that process of collecting general truths from the examination of particular facts, by which such sciences have been formed.

There are, however, two or three remarks, of which the application will occur so frequently, and will tend so much to give us a clearer view of some of the subjects which occur in our history, that I will state them now in a brief and general manner.

_Facts and Ideas_.[2\1]--In the first place then, I remark, that, to the formation of science, two things are requisite;--Facts and Ideas; observation of Things without, and an inward effort of Thought; or, in other words, Sense and Reason. Neither of these elements, by itself can constitute substantial general knowledge. The impressions of sense, unconnected by some rational and speculative principle, can only end in a practical acquaintance with individual objects; the operations of the rational faculties, on the other hand, if allowed to go on without a constant reference to external things, can lead only to empty abstraction and barren ingenuity. Real speculative knowledge demands the combination of the two ingredients;--right reason, and facts to reason upon. It has been well said, that true knowledge is the interpretation of nature; and therefore it requires both the interpreting mind, and nature for its subject; both the document, and the ingenuity to read it aright. Thus invention, acuteness, and connection of thought, are necessary on the one hand, for the progress of philosophical knowledge; and on the other hand, the precise and steady application of these faculties to facts well known and clearly conceived. It is easy to point out instances in which science has failed to advance, in consequence of the absence of one or other of these requisites; indeed, by far the greater part of the course of the world, the history of most times and most countries, exhibits a condition thus stationary with respect to knowledge. The facts, the impressions on the senses, on which the first successful attempts at physical knowledge proceeded, were as well known long before the time when they were thus turned to account, as at that period. The motions of the stars, and the effects of weight, were familiar to man before the rise of the Greek Astronomy and Mechanics: but the "diviner mind" was still absent; the act of thought had not been exerted, by which these facts were bound together under the form of laws and principles. And even at {44} this day, the tribes of uncivilized and half-civilized man, over the whole face of the earth, have before their eyes a vast body of facts, of exactly the same nature as those with which Europe has built the stately fabric of her physical philosophy; but, in almost every other part of the earth, the process of the intellect by which these facts become science, is unknown. The scientific faculty does not work. The scattered stones are there, but the builder's hand is wanting. And again, we have no lack of proof that mere activity of thought is equally inefficient in producing real knowledge. Almost the whole of the career of the Greek schools of philosophy; of the schoolmen of Europe in the middle ages; of the Arabian and Indian philosophers; shows us that we may have extreme ingenuity and subtlety, invention and connection, demonstration and method; and yet that out of these germs, no physical science may be developed. We may obtain, by such means, Logic and Metaphysics, and even Geometry and Algebra; but out of such materials we shall never form Mechanics and Optics, Chemistry and Physiology. How impossible the formation of these sciences is without a constant and careful reference to observation and experiment;--how rapid and prosperous their progress may be

when they draw from such sources the materials on which the mind of the philosopher employs itself;--the history of those branches of knowledge for the last three hundred years abundantly teaches us.

[Note 2\1: For the _Antithesis of Facts and Ideas_, see the _Philosophy_, book i. ch. 1, 2, 4, 5.]

Accordingly, the existence of clear Ideas applied to distinct Facts will be discernible in the History of Science, whenever any marked advance takes place. And, in tracing the progress of the various provinces of knowledge which come under our survey, it will be important for us to see that, at all such epochs, such a combination has occurred; that whenever any material step in general knowledge has been made,--whenever any philosophical discovery arrests our attention,--some man or men come before us, who have possessed, in an eminent degree, a clearness of the ideas which belong to the subject in question, and who have applied such ideas in a vigorous and distinct manner to ascertained facts and exact observations. We shall never proceed through any considerable range of our narrative, without having occasion to remind the reader of this reflection.

_Successive Steps in Science_.[3\1]--But there is another remark which we must also make. Such sciences as we have here to do with are, {45} commonly, not formed by a single act;--they are not completed by the discovery of one great principle. On the contrary, they consist in a long-continued advance; a series of changes; a repeated progress from one principle to another, different and often apparently contradictory. Now, it is important to remember that this contradiction is apparent only. The principles which constituted the triumph of the preceding stages of the science, may appear to be subverted and ejected by the later discoveries, but in fact they are (so far as they were true) taken up in the subsequent doctrines and included in them. They continue to be an essential part of the science. The earlier truths are not expelled but absorbed, not contradicted but extended; and the history of each science, which may thus appear like a succession of revolutions, is, in reality, a series of developments. In the intellectual, as in the material world,

Omnia mutantur nil interit . . . . . Nec manet ut fuerat nec formas servat easdem, Sed tamen ipsa eadem est.

All changes, naught is lost; the forms are changed, And that which has been is not what it was, Yet that which has been is.

Nothing which was done was useless or unessential, though it ceases to be conspicuous and primary.

[Note 3\1: Concerning _Successive Generalizations in Science_ see the _Philosophy_, book i. ch. 2, sect. 11.]

Thus the final form of each science contains the substance of each of its preceding modifications; and all that was at any antecedent period discovered and established, ministers to the ultimate development of its proper branch of knowledge. Such previous doctrines may require to be made precise and definite, to have their superfluous and arbitrary portions expunged, to be expressed in new language, to be

taken up into the body of science by various processes;--but they do not on such accounts cease to be true doctrines, or to form a portion of the essential constituents of our knowledge.

_Terms record Discoveries_.[4\1]--The modes in which the earlier truths of science are preserved in its later forms, are indeed various. From being asserted at first as strange discoveries, such truths come at last to be implied as almost self-evident axioms. They are recorded by some familiar maxim, or perhaps by some new word or phrase, which becomes part of the current language of the philosophical world; and thus asserts a principle, while it appears merely to indicate a transient {46} notion;--preserves as well as expresses a truth;--and, like a medal of gold, is a treasure as well as a token. We shall frequently have to notice the manner in which great discoveries thus stamp their impress upon the terms of a science; and, like great political revolutions, are recorded by the change of the current coin which has accompanied them.

[Note 4\1: Concerning _Technical Terms_, see _Philosophy_, book i. ch. 3.]

_Generalization_.--The great changes which thus take place in the history of science, the revolutions of the intellectual world, have, as a usual and leading character, this, that they are steps of _generalization_; transitions from particular truths to others of a wider extent, in which the former are included. This progress of knowledge, from individual facts to universal laws,--from particular propositions to general ones,--and from these to others still more general, with reference to which the former generalizations are particular,--is so far familiar to men's minds, that, without here entering into further explanation, its nature will be understood sufficiently to prepare the reader to recognize the exemplifications of such a process, which he will find at every step of our advance.

_Inductive Epochs; Preludes; Sequels_.--In our history, it is the _progress_ of knowledge only which we have to attend to. This is the main action of our drama; and all the events which do not bear upon this, though they may relate to the cultivation and the cultivators of philosophy, are not a necessary part of our theme. Our narrative will therefore consist mainly of successive steps of generalization, such as have just been mentioned. But among these, we shall find some of eminent and decisive importance, which have more peculiarly influenced the fortunes of physical philosophy, and to which we may consider the rest as subordinate and auxiliary. These primary movements, when the Inductive process, by which science is formed, has been exercised in a more energetic and powerful manner, may be distinguished as the _Inductive Epochs_ of scientific history; and they deserve our more express and pointed notice. They are, for the most part, marked by the great discoveries and the great philosophical names which all civilized nations have agreed in admiring. But, when we examine more clearly the history of such discoveries, we find that these epochs have not occurred suddenly and without preparation. They have been preceded by a period, which we may call their _Prelude_ during which the ideas and facts on which they turned were called into action;--were gradually evolved into clearness and connection, permanency and certainty; till at last the discovery which marks the epoch, seized and fixed forever the truth which had till then been

obscurely and {47} doubtfully discerned. And again, when this step has been made by the principal discoverers, there may generally be observed another period, which we may call the _Sequel_ of the Epoch, during which the discovery has acquired a more perfect certainty and a more complete development among the leaders of the advance; has been diffused to the wider throng of the secondary cultivators of such knowledge, and traced into its distant consequences. This is a work, always of time and labor, often of difficulty and conflict. To distribute the History of science into such Epochs, with their Preludes and Sequels, if successfully attempted, must needs make the series and connections of its occurrences more distinct and intelligible. Such periods form resting-places, where we pause till the dust of the confused march is laid, and the prospect of the path is clear.

_Inductive Charts_.[5\1]--Since the advance of science consists in collecting by induction true general laws from particular facts, and in combining several such laws into one higher generalization, in which they still retain their truth; we might form a Chart, or Table, of the progress of each science, by setting down the particular facts which have thus been combined, so as to form general truths, and by marking the further union of these general truths into others more comprehensive. The Table of the progress of any science would thus resemble the Map of a River, in which the waters from separate sources unite and make rivulets, which again meet with rivulets from other fountains, and thus go on forming by their junction trunks of a higher and higher order. The representation of the state of a science in this form, would necessarily exhibit all the principal doctrines of the science; for each general truth contains the particular truths from which it was derived, and may be followed backwards till we have these before us in their separate state. And the last and most advanced generalization would have, in such a scheme, its proper place and the evidence of its validity. Hence such an _Inductive Table_ of each science would afford a criterion of the correctness of our distribution of the inductive Epochs, by its coincidence with the views of the best judges, as to the substantial contents of the science in question. By forming, therefore, such Inductive Tables of the principal sciences of which I have here to speak, and by regulating by these tables, my views of the history of the sciences, I conceive that I have secured the distribution of my {48} history from material error; for no merely arbitrary division of the events could satisfy such conditions. But though I have constructed such charts to direct the course of the present history, I shall not insert them in the work, reserving them for the illustration of the philosophy of the subject; for to this they more properly belong, being a part of the _Logic of Induction_.

[Note 5\1: Inductive charts of the History of Astronomy and of Optics, such as are here referred to, are given in the _Philosophy_, book xi. ch. 6.]

_Stationary Periods_.--By the lines of such maps the real advance of science is depicted, and nothing else. But there are several occurrences of other kinds, too interesting and too instructive to be altogether omitted. In order to understand the conditions of the progress of knowledge, we must attend, in some measure, to the failures as well as the successes by which such attempts have been attended. When we reflect during how small a portion of the whole history of human speculations,

science has really been, in any marked degree, progressive, we must needs feel some curiosity to know what was doing in these _stationary_ periods; what field could be found which admitted of so wide a deviation, or at least so protracted a wandering. It is highly necessary to our purpose, to describe the baffled enterprises as well as the achievements of human speculation.

_Deduction_.--During a great part of such stationary periods, we shall find that the process which we have spoken of as essential to the formation of real science, the conjunction of clear Ideas with distinct Facts, was interrupted; and, in such cases, men dealt with ideas alone. They employed themselves in reasoning from principles, and they arranged, and classified, and analyzed their ideas, so as to make their reasonings satisfy the requisitions of our rational faculties. This process of drawing conclusions from our principles, by rigorous and unimpeachable trains of demonstration, is termed _Deduction_. In its due place, it is a highly important part of every science; but it has no value when the fundamental principles, on which the whole of the demonstration rests, have not first been obtained by the induction of facts, so as to supply the materials of substantial truth. Without such materials, a series of demonstrations resembles physical science only as a shadow resembles a real object. To give a real significance to our propositions, Induction must provide what Deduction cannot supply. From a pictured hook we can hang only a pictured chain.

_Distinction of common Notions and Scientific Ideas_.[6\1]--When the {49} notions with which men are conversant in the common course of practical life, which give meaning to their familiar language, and employment to their hourly thoughts, are compared with the Ideas on which exact science is founded, we find that the two classes of intellectual operations have much that is common and much that is different. Without here attempting fully to explain this relation (which, indeed, is one of the hardest problems of our philosophy), we may observe that they have this in common, that both are acquired by acts of the mind exercised in connecting external impressions, and may be employed in conducting a train of reasoning; or, speaking loosely (for we cannot here pursue the subject so as to arrive at philosophical exactness), we may say, that all notions and ideas are obtained by an _inductive_, and may be used in a _deductive_ process. But scientific Ideas and common Notions differ in this, that the former are precise and stable, the latter vague and variable; the former are possessed with clear insight, and employed in a sense rigorously limited, and always identically the same; the latter have grown up in the mind from a thousand dim and diverse suggestions, and the obscurity and incongruity which belong to their origin hang about all their applications. Scientific Ideas can often be adequately exhibited for all the purposes of reasoning, by means of Definitions and Axioms; all attempts to reason by means of Definitions from common Notions, lead to empty forms or entire confusion.

[Note 6\1: Scientific Ideas depend upon certain _Fundamental Ideas_, which are enumerated in the _Philosophy_, book i. ch. 8.]

Such common Notions are sufficient for the common practical conduct of human life: but man is not a practical creature merely; he has within him a _speculative_

tendency, a pleasure in the contemplation of ideal relations, a love of knowledge as knowledge. It is this speculative tendency which brings to light the difference of common Notions and scientific Ideas, of which we have spoken. The mind analyzes such Notions, reasons upon them, combines and connects them; for it feels assured that intellectual things ought to be able to bear such handling. Even practical knowledge, we see clearly, is not possible without the use of the reason; and the speculative reason is only the reason satisfying itself of its own consistency. The speculative faculty cannot be controlled from acting. The mind cannot but claim a right to speculate concerning all its own acts and creations; yet, when it exercises this right upon its common practical notions, we find that it runs into barren abstractions and ever-recurring cycles of subtlety. Such Notions are like waters naturally stagnant; however much we urge and agitate them, they only revolve in stationary {50} whirlpools. But the mind is capable of acquiring scientific Ideas, which are better fitted to undergo discussion and impulsion. When our speculations are duly fed from the springheads of Observation, and frequently drawn off into the region of Applied Science, we may have a living stream of consistent and progressive knowledge. That science may be both real as to its import, and logical as to its form, the examples of many existing sciences sufficiently prove.

_School Philosophy_.--So long, however, as attempts are made to form sciences, without such a verification and realization of their fundamental ideas, there is, in the natural series of speculation, no self-correcting principle. A philosophy constructed on notions obscure, vague, and unsubstantial, and held in spite of the want of correspondence between its doctrines and the actual train of physical events, may long subsist, and occupy men's minds. Such a philosophy must depend for its permanence upon the pleasure which men feel in tracing the operations of their own and other men's minds, and in reducing them to logical consistency and systematical arrangement.

In these cases the main subjects of attention are not external objects, but speculations previously delivered; the object is not to interpret nature, but man's mind. The opinions of the Masters are the facts which the Disciples endeavor to reduce to unity, or to follow into consequences. A series of speculators who pursue such a course, may properly be termed a _School_, and their philosophy a _School Philosophy_; whether their agreement in such a mode of seeking knowledge arise from personal communication and tradition, or be merely the result of a community of intellectual character and propensity. The two great periods of School Philosophy (it will be recollected that we are here directing our attention mainly to physical science) were that of the Greeks and that of the Middle Ages;--the period of the first waking of science, and that of its midday slumber.

What has been said thus briefly and imperfectly, would require great detail and much explanation, to give it its full significance and authority. But it seemed proper to state so much in this place, in order to render more intelligible and more instructive, at the first aspect, the view of the attempted or effected progress of science.

It is, perhaps, a disadvantage inevitably attending an undertaking like the present, that it must set out with statements so abstract; and must present them without their adequate development and proof. Such an Introduction, both in its character and its scale of execution, may be compared to the geographical sketch of a country, with which {51} the historian of its fortunes often begins his narration. So much of Metaphysics is as necessary to us as such a portion of Geography is to the Historian of an Empire; and what has hitherto been said, is intended as a slight outline of the Geography of that Intellectual World, of which we have here to study the History.

The name which we have given to this History--A HISTORY OF THE INDUCTIVE SCIENCES--has the fault of seeming to exclude from the rank of Inductive Sciences those which are not included in the History; as Ethnology and Glossology, Political Economy, Psychology. This exclusion I by no means wish to imply; but I could find no other way of compendiously describing my subject, which was intended to comprehend those Sciences in which, by the observation of facts and the use of reason, systems of doctrine have been established which are universally received as truths among thoughtful men; and which may therefore be studied as examples of the manner in which truth is to be discovered. Perhaps a more exact description of the work would have been, _A History of the principal Sciences hitherto established by Induction_. I may add that I do not include in the phrase "Inductive Sciences," the branches of Pure Mathematics (Geometry, Arithmetic, Algebra, and the like), because, as I have elsewhere stated (_Phil. Ind. Sc._, book ii. c. 1), these are not _Inductive_ but _Deductive_ Sciences. They do not infer true theories from observed facts, and more general from more limited laws: but they trace the conditions of all theory, the properties of space and number; and deduce results from ideas without the aid of experience. The History of these Sciences is briefly given in Chapters 13 and 14 of the Second Book of the _Philosophy_ just referred to.

I may further add that the other work to which I refer, the _Philosophy of the Inductive Sciences_, is in a great measure historical, no less than the present _History_. That work contains the history of the Sciences so far as it depends on _Ideas_; the present work contains the history so far as it depends upon _Observation_. The two works resulted simultaneously from the same examination of the principal writers on science in all ages, and may serve to supplement each other.

{{53}} BOOK I.

HISTORY OF THE GREEK SCHOOL PHILOSOPHY, WITH REFERENCE TO PHYSICAL SCIENCE.

Τίς γὰρ ἀρχὰ δέξατο ναυτιλίας; Τίς δὲ κίνδυνος κρατεροῖς ἀδάμαντος δῆσεν ἄλοις; . . . . . Ἐπεὶ δ' ἐμβόλου Κρέμασαν ἀγκύρας ὕπερθεν Χρυσέαν χείρεσσι λαβὼν φιάλαν Ἀρχὸς ἐν πρύμνᾳ πατέρ Οὐρανιδᾶν Ἐγχεικέραυνον Ζῆνα, καὶ ὠκυπόρους Κυμάτων ῥίπας, ἀνέμων τ' ἐκάλει, Ἀματά τ' εὔφρονα, καὶ Φιλίαν νόστοιο μοῖραν. PINDAR. _Pyth._ iv. 124, 349.

Whence came their voyage? them what peril held With adamantine rivets firmly bound? * * * * * * But soon as on the vessel's bow The anchor was hung up, Then took the Leader on the prow In hands a golden cup, And on great Father Jove did call, And

on the Winds and Waters all, Swept by the hurrying blast; And on the Nights, and Ocean Ways, And on the fair auspicious Days, And loved return at last.

{{55}} BOOK I.

HISTORY OF THE GREEK SCHOOL PHILOSOPHY, WITH REFERENCE TO PHYSICAL SCIENCE.

## CHAPTER I.

PRELUDE TO THE GREEK SCHOOL PHILOSOPHY.

_Sect._ 1.--_First Attempts of the Speculative Faculty in Physical Inquiries._

AT an early period of history there appeared in men a propensity to pursue speculative inquiries concerning the various parts and properties of the material world. What they saw excited them to meditate, to conjecture, and to reason: they endeavored to account for natural events, to trace their causes, to reduce them to their principles. This habit of mind, or, at least that modification of it which we have here to consider, seems to have been first unfolded among the Greeks. And during that obscure introductory interval which elapsed while the speculative tendencies of men were as yet hardly disentangled from the practical, those who were most eminent in such inquiries were distinguished by the same term of praise which is applied to sagacity in matters of action, and were called _wise_ men--σοφοὶ. But when it came to be clearly felt by such persons that their endeavors were suggested by the love of knowledge, a motive different from the motives which lead to the wisdom of active life, a name was adopted of a more appropriate, as well as of a more modest signification, and they were termed _philosophers_, or lovers of wisdom. This appellation is said[7\1] to have been first assumed by Pythagoras. Yet he, in Herodotus, instead of having this title, is called a powerful _sophist_--Ἑλλήνων οὐ τῷ ἀσθενεστάτῳ σοφιστῇ Πυθαγόρῃ;[8\1] the historian using this word, as it would seem, without intending to imply that misuse of reason which the term afterwards came to denote. The historians of literature {56} placed Pythagoras at the origin of the Italic School, one of the two main lines of succession of the early Greek philosophers: but the other, the Ionic School, which more peculiarly demands our attention, in consequence of its character and subsequent progress, is deduced from Thales, who preceded the age of _Philosophy_, and was one of the _sophi_, or "wise men of Greece."

[Note 7\1: Cic. Tusc. v. 3.]

[Note 8\1: Herod. iv. 95.]

The Ionic School was succeeded in Greece by several others; and the subjects which occupied the attention of these schools became very extensive. In fact, the first attempts were, to form systems which should explain the laws and causes of the material universe; and to these were soon added all the great questions which our moral condition and faculties suggest. The physical philosophy of these schools is especially deserving of our study, as exhibiting the character and fortunes of the most memorable attempt at universal knowledge which has ever been made. It is highly

instructive to trace the principles of this undertaking; for the course pursued was certainly one of the most natural and tempting which can be imagined; the essay was made by a nation unequalled in fine mental endowments, at the period of its greatest activity and vigor; and yet it must be allowed (for, at least so far as physical science is concerned, none will contest this), to have been entirely unsuccessful. We cannot consider otherwise than as an utter failure, an endeavor to discover the causes of things, of which the most complete results are the Aristotelian physical treatises; and which, after reaching the point which these treatises mark, left the human mind to remain stationary, at any rate on all such subjects, for nearly two thousand years.

The early philosophers of Greece entered upon the work of physical speculation in a manner which showed the vigor and confidence of the questioning spirit, as yet untamed by labors and reverses. It was for later ages to learn that man must acquire, slowly and patiently, letter by letter, the alphabet in which nature writes her answers to such inquiries. The first students wished to divine, at a single glance, the whole import of her book. They endeavored to discover the origin and principle of the universe; according to Thales, _water_ was the origin of all things, according to Anaximenes, _air_; and Heraclitus considered _fire_ as the essential principle of the universe. It has been conjectured, with great plausibility, that this tendency to give to their Philosophy the form of a Cosmogony, was owing to the influence of the poetical Cosmogonies and Theogonies which had been produced and admired at a still earlier age. Indeed, such wide and ambitious {57} doctrines as those which have been mentioned, were better suited to the dim magnificence of poetry, than to the purpose of a philosophy which was to bear the sharp scrutiny of reason. When we speak of the _principles_ of things, the term, even now, is very ambiguous and indefinite in its import, but how much more was that the case in the first attempts to use such abstractions! The term which is commonly used in this sense (ἀρχὴ), signified at first _the beginning_; and in its early philosophical applications implied some obscure mixed reference to the mechanical, chemical, organic, and historical causes of the visible state of things, besides the theological views which at this period were only just beginning to be separated from the physical. Hence we are not to be surprised if the sources from which the opinions of this period appear to be derived are rather vague suggestions and casual analogies, than any reasons which will bear examination. Aristotle conjectures, with considerable probability, that the doctrine of Thales, according to which water was the universal element, resulted from the manifest importance of moisture in the support of animal and vegetable life.[9\1] But such precarious analyses of these obscure and loose dogmas of early antiquity are of small consequence to our object.

[Note 9\1: Metaph. i. 3.]

In more limited and more definite examples of inquiry concerning the causes of natural appearances, and in the attempts made to satisfy men's curiosity in such cases, we appear to discern a more genuine prelude to the true spirit of physical inquiry. One of the most remarkable instances of this kind is to be found in the speculations which Herodotus records, relative to the cause of the floods of the Nile. "Concerning the nature of this river," says the father of history,[10\1] "I was not able

to learn any thing, either from the priests or from any one besides, though I questioned them very pressingly. For the Nile is flooded for a hundred days, beginning with the summer solstice; and after this time it diminishes, and is, during the whole winter, very small. And on this head I was not able to obtain any thing satisfactory from any one of the Egyptians, when I asked what is the power by which the Nile is in its nature the reverse of other rivers."

[Note 10\1: Herod. ii. 19.]

We may see, I think, in the historian's account, that the Grecian mind felt a craving to discover the reasons of things which other nations did not feel. The Egyptians, it appears, had no theory, and felt no want of a theory. Not so the Greeks; they had their reasons to render, though they were not such as satisfied Herodotus. "Some {58} of the Greeks," he says, "who wish to be considered great philosophers (Ἑλλήνων τινες ἐπισήμοι βουλόμενοι γενέσθαι σοφίην), have propounded three ways of accounting for these floods. Two of them," he adds, "I do not think worthy of record, except just so far as to mention them." But as these are some of the earliest Greek essays in physical philosophy, it will be worth while, even at this day, to preserve the brief notice he has given of them, and his own reasonings upon the same subject.

"One of these opinions holds that the Etesian winds [which blew from the north] are the cause of these floods, by preventing the Nile from flowing into the sea." Against this the historian reasons very simply and sensibly. "Very often when the Etesian winds do not blow, the Nile is flooded nevertheless. And moreover, if the Etesian winds were the cause, all other rivers, which have their course opposite to these winds, ought to undergo the same changes as the Nile; which the rivers of Syria and Libya so circumstanced do not."

"The next opinion is still more unscientific (ἀνεπιστημονεστέρη), and is, in truth, marvellous for its folly. This holds that the ocean flows all round the earth, and that the Nile comes out of the ocean, and by that means produces its effects." "Now," says the historian, "the man who talks about this ocean-river, goes into the region of fable, where it is not easy to demonstrate that he is wrong. I know of no such river. But I suppose that Homer and some of the earlier poets invented this fiction and introduced it into their poetry."

He then proceeds to a third account, which to a modern reasoner would appear not at all unphilosophical in itself, but which he, nevertheless, rejects in a manner no less decided than the others. "The third opinion, though much the most plausible, is still more wrong than the others; for it asserts an impossibility, namely, that the Nile proceeds from the melting of the snow. Now the Nile flows out of Libya, and through Ethiopia, which are very hot countries, and thus comes into Egypt, which is a colder region. How then can it proceed from snow?" He then offers several other reasons "to show," as he says, "to any one capable of reasoning on such subjects (ἀνδρί γε λογίζεσθαι τοιούτων πέρι οἵῳ τε ἔοντι), that the assertion cannot be true. The winds which blow from the southern regions are hot; the inhabitants are black; the swallows and kites (ἰκτῖνοι) stay in the country the whole year; the cranes fly the colds of

Scythia, and seek their warm winter-quarters there; which would not be if it snowed ever so little." He adds another reason, founded apparently upon {59} some limited empirical maxim of weather-wisdom taken from the climate of Greece. "Libya," he said, "has neither rain nor ice, and therefore no snow; _for_, in five days after a fall of snow there must be a fall of rain; so that if it snowed in those regions it must rain too." I need not observe that Herodotus was not aware of the difference between the climate of high mountains and plains in a torrid region; but it is impossible not to be struck both with the activity and the coherency of thought displayed by the Greek mind in this primitive physical inquiry.

But I must not omit the hypothesis which Herodotus himself proposes, after rejecting those which have been already given. It does not appear to me easy to catch his exact meaning, but the statement will still be curious. "If," he says, "one who has condemned opinions previously promulgated may put forward his own opinion concerning so obscure a matter, I will state why it seems to me that the Nile is flooded in summer." This opinion he propounds at first with an oracular brevity, which it is difficult to suppose that he did not intend to be impressive. "In winter the sun is carried by the seasons away from his former course, and goes to the upper parts of Libya. And _there, in short, is the whole account;_ for that region to which this divinity (the sun) is nearest, must naturally be most scant of water, and the river-sources of that country must be dried up."

But the lively and garrulous Ionian immediately relaxes from this apparent reserve. "To explain the matter more at length," he proceeds, "it is thus. The sun when he traverses the upper parts of Libya, does what he commonly does in summer;--he _draws_ the water to him (ἕλκει ἐπ' ἑωϋτὸν τὸ ὕδωρ), and having thus drawn it, he pushes it to the upper regions (of the air probably), and then the winds take it and disperse it till they dissolve in moisture. And thus the winds which blow from those countries, Libs and Notus, are the most moist of all winds. Now when the winter relaxes and the sun returns to the north, he still draws water from all the rivers, but they are increased by showers and rain torrents so that they are in flood till the summer comes; and then, the rain falling and the sun still drawing them, they become small. But the Nile, not being fed by rains, yet being drawn by the sun, is, alone of all rivers, much more scanty in the winter than in the summer. For in summer it is drawn like all other rivers, but in winter it alone has its supplies shut up. And in this way, I have been led to think the sun is the cause of the occurrence in question." We may remark that the historian here appears to {60} ascribe the inequality of the Nile at different seasons to the influence of the sun upon its springs alone, the other cause of change, the rains being here excluded; and that, on this supposition, the same relative effects would be produced whether the sun increase the sources in winter by melting the snows, or diminish them in summer by what he calls _drawing_ them upwards.

This specimen of the early efforts of the Greeks in physical speculations, appears to me to speak strongly for the opinion that their philosophy on such subjects was the native growth of the Greek mind, and owed nothing to the supposed lore of Egypt and the East; an opinion which has been adopted with regard to the Greek Philosophy in

general by the most competent judges on a full survey of the evidence.[11\1] Indeed, we have no evidence whatever that, at any period, the African or Asiatic nations (with the exception perhaps of the Indians) ever felt this importunate curiosity with regard to the definite application of the idea of cause and effect to visible phenomena; or drew so strong a line between a fabulous legend and a reason rendered; or attempted to ascend to a natural cause by classing together phenomena of the same kind. We may be well excused, therefore, for believing that they could not impart to the Greeks what they themselves did not possess; and so far as our survey goes, physical philosophy has its origin, apparently spontaneous and independent, in the active and acute intellect of Greece.

[Note 11\1: Thirlwall, _Hist. Gr._, ii. 130; and, as there quoted, Ritter, _Geschichte der Philosophie_, i. 159-173.]

_Sect._ 2.--_Primitive Mistake in Greek Physical Philosophy._

WE now proceed to examine with what success the Greeks followed the track into which they had thus struck. And here we are obliged to confess that they very soon turned aside from the right road to truth, and deviated into a vast field of error, in which they and their successors have wandered almost to the present time. It is not necessary here to inquire why those faculties which appear to be bestowed upon us for the discovery of truth, were permitted by Providence to fail so signally in answering that purpose; whether, like the powers by which we seek our happiness, they involve a responsibility on our part, and may be defeated by rejecting the guidance of a higher faculty; or whether these endowments, though they did not {61} immediately lead man to profound physical knowledge, answered some nobler and better purpose in his constitution and government. The fact undoubtedly was, that the physical philosophy of the Greeks soon became trifling and worthless; and it is proper to point out, as precisely as we can, in what the fundamental mistake consisted.

To explain this, we may in the first place return for a moment to Herodotus's account of the cause of the floods of the Nile.

The reader will probably have observed a remarkable phrase used by Herodotus, in his own explanation of these inundations. He says that the sun _draws_, or attracts, the water; a metaphorical term, obviously intended to denote some more general and abstract conception than that of the visible operation which the word primarily signifies. This abstract notion of "drawing" is, in the historian, as we see, very vague and loose; it might, with equal propriety, be explained to mean what we now understand by mechanical or by chemical attraction, or pressure, or evaporation. And in like manner, all the first attempts to comprehend the operations of nature, led to the introduction of abstract conceptions, often vague, indeed, but not, therefore, unmeaning; such as _motion_ and _velocity_, _force_ and _pressure_, _impetus_ and _momentum_ (ῥοπὴ). And the next step in philosophizing, necessarily was to endeavor to make these vague abstractions more clear and fixed, so that the logical faculty should be able to employ them securely and coherently. But there were two ways of making this attempt; the one, by examining the words only, and the thoughts which

they call up; the other, by attending to the facts and things which bring these abstract terms into use. The latter, the method of _real_ inquiry, was the way to success; but the Greeks followed the former, the _verbal_ or _notional_ course, and failed.

If Herodotus, when the notion of the sun's attracting the waters of rivers had entered into his mind, had gone on to instruct himself, by attention to facts, in what manner this notion could be made more definite, while it still remained applicable to all the knowledge which could be obtained, he would have made some progress towards a true solution of his problem. If, for instance, he had tried to ascertain whether this Attraction which the sun exerted upon the waters of rivers, depended on his influence at their fountains only, or was exerted over their whole course, and over waters which were not parts of rivers, he would have been led to reject his hypothesis; for he would have found, by observations sufficiently obvious, that the sun's Attraction, as shown in such cases, is a tendency to lessen all expanded and {62} open collections of moisture, whether flowing from a spring or not; and it would then be seen that this influence, operating on the whole surface of the Nile, must diminish it as well as other rivers, in summer, and therefore could not be the cause of its overflow. He would thus have corrected his first loose conjecture by a real study of nature, and might, in the course of his meditations, have been led to available notions of Evaporation, or other natural actions. And, in like manner, in other cases, the rude attempts at explanation, which the first exercise of the speculative faculty produced, might have been gradually concentrated and refined, so as to fall in, both with the requisitions of reason and the testimony of sense.

But this was not the direction which the Greek speculators took. On the contrary; as soon as they had introduced into their philosophy any abstract and general conceptions, they proceeded to scrutinize these by the internal light of the mind alone, without any longer looking abroad into the world of sense. They took for granted that philosophy must result from the relations of those notions which are involved in the common use of language, and they proceeded to seek their philosophical doctrines by studying such notions. They ought to have reformed and fixed their usual conceptions by Observation; they only analyzed and expanded them by Reflection: they ought to have sought by trial, among the Notions which passed through their minds, some one which admitted of exact application to Facts; they selected arbitrarily, and, consequently, erroneously, the Notions according to which Facts should be assembled and arranged: they ought to have collected clear Fundamental Ideas from the world of things by _inductive_ acts of thought; they only derived results by _Deduction_ from one or other of their familiar Conceptions.[12\1]

[Note 12\1: The course by which the Sciences were formed, and which is here referred to as that which the Greeks did _not_ follow, is described in detail in the _Philosophy_, book xi., _Of the Construction of Science_.]

When this false direction had been extensively adopted by the Greek philosophers, we may treat of it as the method of their _Schools_. Under that title we must give a further account of it. {63}

# CHAPTER II.

THE GREEK SCHOOL PHILOSOPHY.

_Sect._ 1.--_The general Foundation of the Greek School Philosophy._

THE physical philosophy of the Greek Schools was formed by looking at the material world through the medium of that common language which men employ to answer the common occasions of life; and by adopting, arbitrarily, as the grounds of comparison of facts, and of inference from them, notions more abstract and large than those with which men are practically familiar, but not less vague and obscure. Such a philosophy, however much it might be systematized, by classifying and analyzing the conceptions which it involves, could not overcome the vices of its fundamental principle. But before speaking of these defects, we must give some indications of its character.

The propensity to seek for principles in the common usages of language may be discerned at a very early period. Thus we have an example of it in a saying which is reported of Thales, the founder of Greek philosophy.[13\1] When he was asked, "What is the _greatest_ thing?" he replied, "_Place_; for all other things are _in_ the world, but the world is _in_ it." In Aristotle we have the consummation of this mode of speculation. The usual point from which he starts in his inquiries is, that we say thus or thus in common language. Thus, when he has to discuss the question, whether there be, in any part of the universe, a Void, or space in which there is nothing, he inquires first in how many senses we say that one thing is _in_ another. He enumerates many of these;[14\1] we say the part is in the whole, as the finger is _in_ the hand; again we say, the species is in the genus, as man is included _in_ animal; again, the government of Greece is _in_ the king; and various other senses are described or exemplified, but of all these _the most proper_ is when we say a thing is _in_ a vessel, and generally, _in place_. He next examines what _place_ is, and comes to this conclusion, that "if about a body there be another body including it, it is in place, and if not, not." A body _moves_ when it changes its place; but {64} he adds, that if water be in a vessel, the vessel being at rest, the parts of the water may still move, for they are included by each other; so that while the whole does not change its place, the parts may change their places in a circular order. Proceeding then to the question of a _void_, he, as usual, examines the different senses in which the term is used, and adopts, as the most proper, _place without matter_; with no useful result, as we shall soon see.

[Note 13\1: Plut. _Conv. Sept. Sap._ Diog. Laert. i. 35.]

[Note 14\1: Physic. Ausc. iv. 3.]

Again,[15\1] in a question concerning mechanical action, he says, "When a man moves a stone by pushing it with a stick, _we say_ both that the man moves the stone, and that the stick moves the stone, but the latter _more properly_."

[Note 15\1: Physic. Ausc. viii. 5.]

Again, we find the Greek philosophers applying themselves to extract their dogmas from the most general and abstract notions which they could detect; for example,--from the conception of the Universe as One or as Many things. They tried to determine how far we may, or must, combine with these conceptions that of a whole, of parts, of number, of limits, of place, of beginning or end, of full or void, of rest or motion, of cause and effect, and the like. The analysis of such conceptions with such a view, occupies, for instance, almost the whole of Aristotle's _Treatise on the Heavens_.

The Dialogue of Plato, which is entitled _Parmenides_, appears at first as if its object were to show the futility of this method of philosophizing; for the philosopher whose name it bears, is represented as arguing with an Athenian named Aristotle,[16\1] and, by a process of metaphysical analysis, reducing him at least to this conclusion, "that whether _One_ exist, or do not exist, it follows that both it and other things, with reference to themselves and to each other, all and in all respects, both are and are not, both appear and appear not." Yet the method of Plato, so far as concerns truths of that kind with which we are here concerned, was little more efficacious than that of his rival. It consists mainly, as may be seen in several of the dialogues, and especially in the _Timæus_, in the application of notions as loose as those of the Peripatetics; for example, the conceptions of the Good, the Beautiful, the Perfect; and these are rendered still more arbitrary, by assuming an acquaintance with the views of the Creator of the universe. The philosopher is thus led to maxims which agree with those {65} of the Aristotelians, that there can be no void, that things seek their own place, and the like.[17\1]

[Note 16\1: This Aristotle is not the Stagirite, who was forty-five years younger than Plato, but one of the "thirty tyrants," as they were called.]

[Note 17\1: Timæus, p. 80.]

Another mode of reasoning, very widely applied in these attempts, was the doctrine of contrarieties, in which it was assumed, that adjectives or substantives which are in common language, or in some abstract mode of conception, opposed to each other, must point at some fundamental antithesis in nature, which it is important to study. Thus Aristotle[18\1] says, that the Pythagoreans, from the contrasts which number suggests, collected ten principles,--Limited and Unlimited, Odd and Even, One and Many, Right and Left, Male and Female, Rest and Motion, Straight and Curved, Light and Darkness, Good and Evil, Square and Oblong. We shall see hereafter, that Aristotle himself deduced the doctrine of Four Elements, and other dogmas, by oppositions of the same kind.

[Note 18\1: Metaph. 1. 5.]

The physical speculator of the present day will learn without surprise, that such a mode of discussion as this, led to no truths of real or permanent value. The whole mass of the Greek philosophy, therefore, shrinks into an almost imperceptible compass, when viewed with reference to the progress of physical knowledge. Still the general character of this system, and its fortunes from the time of its founders to the overthrow of their authority, are not without their instruction, and, it may be hoped,

not without their interest. I proceed, therefore, to give some account of these doctrines in their most fully developed and permanently received form, that in which they were presented by Aristotle.

_Sect._ 2.--_The Aristotelian Physical Philosophy._

THE principal physical treatises of Aristotle are, the eight Books of "Physical Lectures," the four Books "Of the Heavens," the two Books "Of Production and Destruction:" for the Book "Of the World" is now universally acknowledged to be spurious; and the "Meteorologies," though full of physical explanations of natural phenomena, does not exhibit the doctrines and reasonings of the school in so general a form; the same may be said of the "Mechanical Problems." The treatises on the various subjects of Natural History, "On Animals," "On the Parts of Animals," "On Plants," "On Physiognomonics," "On Colors," "On Sound," contain an extraordinary {66} accumulation of facts, and manifest a wonderful power of systematizing; but are not works which expound principles, and therefore do not require to be here considered.

The Physical Lectures are possibly the work concerning which a well-known anecdote is related by Simplicius, a Greek commentator of the sixth century, as well as by Plutarch. It is said, that Alexander the Great wrote to his former tutor to this effect; "You have not done well in publishing these lectures; for how shall we, your pupils, excel other men, if you make that public to all, which we learnt from you?" To this Aristotle is said to have replied: "My Lectures are published and not published; they will be intelligible to those who heard them, and to none besides." This may very easily be a story invented and circulated among those who found the work beyond their comprehension; and it cannot be denied, that to make out the meaning and reasoning of every part, would be a task very laborious and difficult, if not impossible. But we may follow the import of a large portion of the Physical Lectures with sufficient clearness to apprehend the character and principles of the reasoning; and this is what I shall endeavor to do.

The author's introductory statement of his view of the nature of philosophy falls in very closely with what has been said, that he takes his facts and generalizations as they are implied in the structure of language. "We must in all cases proceed," he says, "from what is known to what is unknown." This will not be denied; but we can hardly follow him in his inference. He adds, "We must proceed, therefore, from universal to particular. And something of this," he pursues, "may be seen in language; for names signify things in a general and indefinite manner, as _circle_, and by defining we unfold them into particulars." He illustrates this by saying, "thus children at first call all men _father_, and all women _mother_, but afterwards distinguish."

In accordance with this view, he endeavors to settle several of the great questions concerning the universe, which had been started among subtle and speculative men, by unfolding the meaning of the words and phrases which are applied to the most general notions of things and relations. We have already noticed this method. A few examples will illustrate it further:--Whether there was or was not a _void_, or place without matter, had already been debated among rival sects of philosophers. The

antagonist arguments were briefly these:--There must be a void, because a body cannot move into a space except it is {67} empty, and therefore without a void there could be no motion:--and, on the other hand, there is no void, for the intervals between bodies are filled with air, and air is something. These opinions had even been supported by reference to experiment. On the one hand, Anaxagoras and his school had shown, that air, when confined, resisted compression, by squeezing a blown bladder, and pressing down an inverted vessel in the water; on the other hand, it was alleged that a vessel full of fine ashes held as much water as if the ashes were not there, which could only be explained by supposing void spaces among the ashes. Aristotle decides that there is no void, on such arguments as this:[19\1]--In a void there could be no difference of up and down; for as in nothing there are no differences, so there are none in a privation or negation; but a void is merely a privation or negation of matter; therefore, in a void, bodies could not move up and down, which it is in their nature to do. It is easily seen that such a mode of reasoning, elevates the familiar forms of language and the intellectual connections of terms, to a supremacy over facts; making truth depend upon whether terms are or are not privative, and whether we say that bodies fall _naturally_. In such a philosophy every new result of observation would be compelled to conform to the usual combinations of phrases, as these had become associated by the modes of apprehension previously familiar.

[Note 19\1: Physic. Ausc. iv. 7, p. 215.]

It is not intended here to intimate that the common modes of apprehension, which are the basis of common language, are limited and casual. They imply, on the contrary, universal and necessary conditions of our perceptions and conceptions; thus all things are necessarily apprehended as existing in Time and Space, and as connected by relations of Cause and Effect; and so far as the Aristotelian philosophy reasons from these assumptions, it has a real foundation, though even in this case the conclusions are often insecure. We have an example of this reasoning in the eighth Book,[20\1] where he proves that there never was a time in which change and motion did not exist; "For if all things were at rest, the first motion must have been produced by some change in some of these things; that is, there must have been a change before the first change;" and again, "How can _before_ and _after_ apply when time is not? or how can time be when motion is not? If," he adds, "time is a numeration of motion, and if time be eternal, motion must be eternal." But he sometimes {68} introduces principles of a more arbitrary character; and besides the general relations of thought, takes for granted the inventions of previous speculators; such, for instance, as the then commonly received opinions concerning the frame of the world. From the assertion that motion is eternal, proved in the manner just stated, Aristotle proceeds by a curious train of reasoning, to identify this eternal motion with the diurnal motion of the heavens. "There must," he says, "be something which is the First Mover:"[21\1] this follows from the relation of causes and effects. Again, "Motion must go on constantly, and, therefore, must be either continuous or successive. Now what is continuous is more properly said to take place _constantly_, than what is successive. Also the continuous is better; but we always suppose that which is better to take place in nature, if it be possible. The motion of the First Mover

will, therefore, be continuous, if such an eternal motion be possible." We here see the vague judgment of _better_ and _worse_ introduced, as that of _natural_ and _unnatural_ was before, into physical reasonings.

[Note 20\1: Ib. viii. 1, p. 258.]

[Note 21\1: Physic. Ausc. viii. 6. p. 258.]

I proceed with Aristotle's argument.[22\1] "We have now, therefore, to show that there may be an infinite single, continuous motion, and that this is circular." This is, in fact, proved, as may readily be conceived, from the consideration that a body may go on perpetually revolving uniformly in a circle. And thus we have a demonstration, on the principles of this philosophy, that there is and must be a First Mover, revolving eternally with a uniform circular motion.

[Note 22\1: Ib. viii. 8.]

Though this kind of philosophy may appear too trifling to deserve being dwelt upon, it is important for our purpose so far as to exemplify it, that we may afterwards advance, confident that we have done it no injustice.

I will now pass from the doctrines relating to the motions of the heavens, to those which concern the material elements of the universe. And here it may be remarked that the tendency (of which we are here tracing the development) to extract speculative opinions from the relations of words, must be very natural to man; for the very widely accepted doctrine of the Four Elements which appears to be founded on the opposition of the adjectives _hot_ and _cold_, _wet_ and _dry_, is much older than Aristotle, and was probably one of the earliest of philosophical dogmas. The great master of this philosophy, however, puts the opinion in a more systematic manner than his predecessors. {69}

"We seek," he says,[23\1] "the principles of sensible things, that is, of tangible bodies. We must take, therefore, not all the contrarieties of quality, but those only which have reference to the touch. Thus black and white, sweet and bitter, do not differ as tangible qualities, and therefore must be rejected from our consideration.

[Note 23\1: De Gen. et Corrupt. ii. 2.]

"Now the contrarieties of quality which refer to the touch are these: hot, cold; dry, wet; heavy, light; hard, soft; unctuous, meagre; rough, smooth; dense, rare." He then proceeds to reject all but the four first of these, for various reasons; heavy and light, because they are not active and passive qualities; the others, because they are combinations of the four first, which therefore he infers to be the four elementary qualities.

"[24\1] Now in four things there are six combinations of two; but the combinations of two opposites, as hot and cold, must be rejected; we have, therefore, four elementary combinations, which agree with the four apparently elementary bodies. Fire is hot and dry; air is hot and wet (for steam is air); water is cold and wet, earth is cold and dry."

[Note 24\1: Ib. iii. 8.]

It may be remarked that this disposition to assume that some common elementary quality must exist in the cases in which we habitually apply a common adjective, as it began before the reign of the Aristotelian philosophy, so also survived its influence. Not to mention other cases, it would be difficult to free Bacon's _Inquisitio in naturam calidi_, "Examination of the nature of heat," from the charge of confounding together very different classes of phenomena under the cover of the word _hot_.

The correction of these opinions concerning the elementary composition of bodies belongs to an advanced period in the history of physical knowledge, even after the revival of its progress. But there are some of the Aristotelian doctrines which particularly deserve our attention, from the prominent share they had in the very first beginnings of that revival; I mean the doctrines concerning motion.

These are still founded upon the same mode of reasoning from adjectives; but in this case, the result follows, not only from the opposition of the words, but also from the distinction of their being _absolutely_ or _relatively_ true. "Former writers," says Aristotle, "have considered heavy and light _relatively_ only, taking cases, where both things have weight, but one is lighter than the other; and they imagined that, in {70} this way, they defined what was _absolutely_ (ἁπλῶς) heavy and light." We now know that things which rise by their lightness do so only because they are pressed upwards by heavier surrounding bodies; and this assumption of absolute levity, which is evidently gratuitous, or rather merely nominal, entirely vitiated the whole of the succeeding reasoning. The inference was, that fire must be absolutely light, since it tends to take its place above the other three elements; earth absolutely heavy, since it tends to take its place below fire, air, and water. The philosopher argued also, with great acuteness, that air, which tends to take its place below fire and above water, must do so _by its nature_, and not in virtue of any combination of heavy and light elements. "For if air were composed of the parts which give fire its levity, joined with other parts which produce gravity, we might assume a quantity of air so large, that it should be lighter than a small quantity of fire, having more of the light parts." It thus follows that each of the four elements tends to its own place, fire being the highest, air the next, water the next, and earth the lowest.

The whole of this train of errors arises from fallacies which have a verbal origin;--from considering light as opposite to heavy; and from considering levity as a quality of a body, instead of regarding it as the effect of surrounding bodies.

It is worth while to notice that a difficulty which often embarrasses persons on their entrance upon physical speculations,--the difficulty of conceiving that up and down are different directions in different places,--had been completely got over by Aristotle and the Greek philosophers. They were steadily convinced of the roundness of the earth, and saw that this truth led to the conclusion that all heavy bodies tend in converging directions to the centre. And, they added, as the heavy tends to the centre, the light tends to the exterior, "for Exterior is opposite to Centre as heavy is to light."[25\1]

[Note 25\1: De Cœlo, iv. 4.]

The tendencies of bodies downwards and upwards, their weight, their fall, their floating or sinking, were thus accounted for in a manner which, however unsound, satisfied the greater part of the speculative world till the time of Galileo and Stevinus, though Archimedes in the mean time published the true theory of floating bodies, which is very different from that above stated. Other parts of the doctrines of motion were delivered by the Stagirite in the same spirit and with the same success. The motion of a body which is thrown along the {71} ground diminishes and finally ceases; the motion of a body which falls from a height goes on becoming quicker and quicker; this was accounted for on the usual principle of opposition, by saying that the former is a _violent_, the latter a _natural_ motion. And the later writers of this school expressed the characters of such motions in verse. The rule of natural motion was[26\1] Principium tepeat, medium cum fine calebit. Cool at the first, it warm and warmer glows. And of violent motion, the law was-- Principium fervet, medium calet, ultima friget. Hot at the first, then barely warm, then cold.

[Note 26\1: Alsted. Encyc. tom. i. p. 687.]

It appears to have been considered by Aristotle a difficult problem to explain why a stone thrown from the hand continues to move for some time, and then stops. If the hand was the cause of the motion, how could the stone move at all when left to itself? if not, why does it ever stop? And he answers this difficulty by saying,[27\1] "that there is a motion communicated to the air, the successive parts of which urge the stone onwards; and that each part of this medium continues to act for some while after it has been acted on, and the motion ceases when it comes to a particle which cannot act after it has ceased to be acted on." It will be readily seen that the whole of this difficulty, concerning a body which moves forward and is retarded till it stops, arises from ascribing the retardation, not to the real cause, the surrounding resistances, but to the body itself.

[Note 27\1: Phys. Ausc. viii. 10.]

One of the doctrines which was the subject of the warmest discussion between the defenders and opposers of Aristotle, at the revival of physical knowledge, was that in which he asserts,[28\1] "That body is heavier than another which in an equal bulk moves downward quicker." The opinion maintained by the **Aristotelians at the time of Galileo was, that bodies fall quicker exactly in proportion to their weight. The master himself asserts this in express terms, and reasons upon it.[29\1] Yet in another passage he appears to distinguish between weight and actual motion downwards.[30\1] "In physics, we call bodies heavy and light from their _power_ of motion; but these names are not applied to their actual operations (ἐνέργειαις) except any one thinks {72} _momentum_ (ῥοπὴ) to be a word of both applications. But heavy and light are, as it were, the _embers_ or _sparks_ of motion, and therefore proper to be treated of here."

[Note 28\1: De Cœlo, iv. 1, p. 308.]

[Note 29\1: Ib. iii. 2.]

[Note 30\1: Ib. iv. 1, p. 307.]

The distinction just alluded to, between Power or Faculty of Action, and actual Operation or Energy, is one very frequently referred to by Aristotle; and though not by any means useless, may easily be so used as to lead to mere verbal refinements instead of substantial knowledge.

The Aristotelian distinction of Causes has not any very immediate bearing upon the parts of physics of which we have here mainly spoken; but it was so extensively accepted, and so long retained, that it may be proper to notice it.[31\1] "One kind of Cause is the matter of which any thing is made, as bronze of a statue, and silver of a vial; another is the form and pattern, as the Cause of an octave is the ratio of two to one; again, there is the Cause which is the origin of the production, as the father of the child; and again, there is the End, or that for the sake of which any thing is done, as health is the cause of walking." These four kinds of Cause, the _material_, the _formal_, the _efficient_, and the _final_, were long leading points in all speculative inquiries; and our familiar forms of speech still retain traces of the influence of this division.

[Note 31\1: Phys. ii. 3.]

It is my object here to present to the reader in an intelligible shape, the principles and mode of reasoning of the Aristotelian philosophy, not its results. If this were not the case, it would be easy to excite a smile by insulating some of the passages which are most remote from modern notions. I will only mention, as specimens, two such passages, both very remarkable.

In the beginning of the book "On the Heavens," he proves[32\1] the world to be _perfect_, by reasoning of the following kind: "The bodies of which the world is composed are solids, and therefore have three dimensions: now three is the most perfect number; it is the first of numbers, for of _one_ we do not speak as a number; of _two_ we say _both_; but _three_ is the first number of which we say _all_; moreover, it has a beginning, a middle, and an end."

[Note 32\1: De Cœlo, i. 1.]

The reader will still perceive the verbal foundations of opinions thus supported.

"The simple elements must have simple motions, and thus fire and air have their natural motions upwards, and water and earth have {73} their natural motions downwards; but besides these motions, there is motion in a circle, which is unnatural to these elements, but which is a more perfect motion than the other, because a circle is a perfect line, and a straight line is not; and there must be something to which this motion is natural. From this it is evident," he adds, with obvious animation, "that there is some essence of body different from those of the four elements, more divine than those, and superior to them. If things which move in a circle move contrary to nature, it is marvellous, or rather absurd, that this, the unnatural motion, should alone be continuous and eternal; for unnatural motions decay speedily. And so, from all this, we must collect, that besides the four elements which we have here and about us, there is another removed far off, and the more excellent in proportion as it is more distant from us." This fifth element was the "_quinta essentia_," of after writers, of which we have a trace in our modern literature, in the word

_quintessence_.

_Sect._ 3.--_Technical Forms of the Greek Schools._

WE have hitherto considered only the principle of the Greek Physics; which was, as we have seen, to deduce its doctrines by an analysis of the notions which common language involves. But though the Grecian philosopher began by studying words in their common meanings, he soon found himself led to fix upon some special shades or applications of these meanings as the permanent and standard notion, which they were to express; that is, he made his language _technical_. The invention and establishment of technical terms is an important step in any philosophy, true or false; we must, therefore, say a few words on this process, as exemplified in the ancient systems.

1. _Technical Forms of the Aristotelian Philosophy._--We have already had occasion to cite some of the distinctions introduced by Aristotle, which may be considered as technical; for instance, the classification of Causes as _material_, _formal_, _efficient_, and _final_; and the opposition of Qualities as _absolute_ and _relative_. A few more of the most important examples may suffice. An analysis of objects into _Matter_ and _Form_, when metaphorically extended from visible objects to things conceived in the most general manner, became an habitual hypothesis of the Aristotelian school. Indeed this metaphor is even yet one of the most significant of those which we can employ, to suggest one of the most comprehensive and fundamental antitheses with which philosophy has to do;--the opposition of sense and reason, of {74} impressions and laws. In this application, the German philosophers have, up to the present time, rested upon this distinction a great part of the weight of their systems; as when Kant says, that Space and Time are the _Forms of Sensation_. Even in our own language, we retain a trace of the influence of this Aristotelian notion, in the word _Information_, when used for that knowledge which may be conceived as moulding the mind into a definite shape, instead of leaving it a mere mass of unimpressed susceptibility.

Another favorite Aristotelian antithesis is that of _Power_ and _Act_ (δύναμις, ἐνέργεια). This distinction is made the basis of most of the physical philosophy of the school; being, however, generally introduced with a peculiar limitation. Thus, Light is defined to be "the Act of what is lucid, as being lucid. And if," it is added, "the lucid be so in power but not in act, we have darkness." The reason of the limitation, "as being lucid," is, that a lucid body may act in other ways; thus a torch may move as well as shine, but its moving is not its act _as being a lucid_ body.

Aristotle appears to be well satisfied with this explanation, for he goes on to say, "Thus light is not Fire, nor any body whatever, or the emanation of any body (for that would be a kind of body), but it is the presence of something like Fire in the body; it is, however, impossible that two bodies should exist in the same place, so that it is not a body;" and this reasoning appears to leave him more satisfied with his doctrine, that Light is an _Energy_ or _Act_.

But we have a more distinctly technical form given to this notion. Aristotle introduced a word formed by himself to express the act which is thus opposed to

inactive power: this is the celebrated word ἐντελέχεια. Thus the noted definition of Motion in the third book of the Physics,[33\1] is that it is "the _Entelechy_, or Act, of a movable body in respect of being movable;" and the definition of the Soul is[34\1] that it is "the _Entelechy_ of a natural body which has life by reason of its power." This word has been variously translated by the followers of Aristotle, and some of them have declared it untranslatable. _Act_ and _Action_ are held to be inadequate substitutes; the _very act_, _ipse cursus actionis_, is employed by some; _primus actus_ is employed by many, but another school use _primus actus_ of a non-operating form. Budæus uses _efficacia_. Cicero[35\1] translates it "quasi quandam continuatam motionem, et perennem;" but this paraphrase, though it may {75} fall in with the description of the soul, which is the subject with which Cicero is concerned, does not appear to agree with the general applications of the term. Hermolaus Barbarus is said to have been so much oppressed with this difficulty of translation, that he consulted the evil spirit by night, entreating to be supplied with a more common and familiar substitute for this word: the mocking fiend, however, suggested only a word equally obscure, and the translator, discontented with this, invented for himself the word _perfectihabia_.

[Note 33\1: Phys. iii. 1.]

[Note 34\1: De Animâ, ii. 1.]

[Note 35\1: Tusc. i. 10.]

We need not here notice the endless apparatus of technicalities which was, in later days, introduced into the Aristotelian philosophy; but we may remark, that their long continuance and extensive use show us how powerful technical phraseology is, for the perpetuation either of truth or error. The Aristotelian terms, and the metaphysical views which they tend to preserve, are not yet extinct among us. In a very recent age of our literature it was thought a worthy employment by some of the greatest writers of the day, to attempt to expel this system of technicalities by ridicule.

"Crambe regretted extremely that _substantial forms_, a race of harmless beings, which had lasted for many years, and afforded a comfortable subsistence to many poor philosophers, should now be hunted down like so many wolves, without a possibility of retreat. He considered that it had gone much harder with them than with _essences_, which had retired from the schools into the apothecaries' shops, where some of them had been advanced to the degree of _quintessences_.**"[36\1]

[Note 36\1: Martinus Scriblerus, cap. vii.]

We must now say a few words on the technical terms which others of the Greek philosophical sects introduced.

2. _Technical Forms of the Platonists._--The other sects of the Greek philosophy, as well as the Aristotelians, invented and adopted technical terms, and thus gave fixity to their tenets and consistency to their traditionary systems; of these I will mention a few.

A technical expression of a contemporary school has acquired perhaps greater celebrity than any of the terms of Aristotle. I mean the _Ideas_ of Plato. The account which Aristotle gives of the origin of these will serve to explain their nature.[37\1] "Plato," says he, "who, in his youth, was in habits of communication first with Cratylus and the Heraclitean opinions, which represent all the objects of sense as being in a perpetual flux, so that concerning these no science nor certain {76} knowledge can exist, entertained the same opinions at a later period also. When, afterwards, Socrates treated of moral subjects, and gave no attention to physics, but, in the subjects which he did discuss, arrived at universal truths, and before any man, turned his thoughts to definitions, Plato adopted similar doctrines on this subject also; and construed them in this way, that these truths and definitions must be applicable to something else, and not to sensible things: for it was impossible, he conceived, that there should be a general common definition of any sensible object, since such were always in a state of change. The things, then, which were the subjects of universal truths he called _Ideas_; and held that objects of sense had their names according to Ideas and after them; so that things participated in that Idea which had the same name as was applied to them."

[Note 37\1: Arist. Metaph. i. 6. The same account is repeated, and the subject discussed, Metaph. xii. 4.]

In agreement with this, we find the opinions suggested in the _Parmenides_ of Plato, the dialogue which is considered by many to contain the most decided exposition of the doctrine of Ideas. In this dialogue, Parmenides is made to say to Socrates, then a young man,[38\1] "O Socrates, philosophy has not yet claimed you for her own, as, in my judgment, she will claim you, and you will not dishonor her. As yet, like a young man as you are, you look to the opinions of men. But tell me this: it appears to you, as you say, that there are certain _Kinds_ or _Ideas_ (εἰδὴ) of which things partake and receive applications according to that of which they partake: thus those things which partake of _Likeness_ are called _like_; those things which partake of _Greatness_ are called _great_; those things which partake of _Beauty_ and _Justice_ are called _beautiful_ and _just_." To this Socrates assents. And in another part of the dialogue he shows that these Ideas are not included in our common knowledge, from whence he infers that they are objects of the Divine mind.

[Note 38\1: Parmenid. p. 131.]

In the Phædo the same opinion is maintained, and is summed up in this way, by a reporter of the last conversation of Socrates,[39\1] εἶναι τι ἕκαστον τῶν εἰδῶν, καὶ τούτων τ' ἄλλα μεταλαμβάνοντα αὐτῶν τούτων τὴν ἐπωνυμίαν ἴσχειν; "that each _Kind_ has an existence, and that other things partake of these Kinds, and are called according to the Kind of which they partake."

[Note 39\1: Phædo, p. 102.]

The inference drawn from this view was, that in order to obtain true and certain knowledge, men must elevate themselves, as much as possible, to these Ideas of the qualities which they have to consider: {77} and as things were thus called after the Ideas, the Ideas had a priority and pre-eminence assigned them. The _Idea_ of Good,

Beautiful, and Wise was the "First Good," the "First Beautiful," the "First Wise." This dignity and distinction were ultimately carried to a large extent. Those Ideas were described as eternal and self-subsisting, forming an "Intelligible World," full of the models or archetypes of created things. But it is not to our purpose here to consider the Platonic Ideas in their theological bearings. In physics they were applied in the same form as in morals. The _primum calidum_, _primum frigidum_ were those Ideas of fundamental Principles by participation of which, all things were hot or cold.

This school did not much employ itself in the development of its principles as applied to physical inquiries: but we are not without examples of such speculations. Plutarch's Treatise Περὶ τοῦ Πρώτου Ψυχροῦ, "On the First Cold," may be cited as one. It is in reality a discussion of a question which has been agitated in modern times also;--whether cold be a positive quality or a mere privation. "Is there, O Favorinus," he begins, "a First Power and Essence of the Cold, as Fire is of the Hot; by a certain presence and participation of which all other things are cold: or is rather coldness a privation of heat, as darkness is of light, and rest of motion?"

3. _Technical Forms of the Pythagoreans._--The _Numbers_ of the Pythagoreans, when propounded as the explanation of physical phenomena, as they were, are still more obscure than the Ideas of the Platonists. There were, indeed, considerable resemblances in the way in which these two kinds of notions were spoken of. Plato called his Ideas _unities_, _monads_; and as, according to him, Ideas, so, according to the Pythagoreans, Numbers, were the causes of things being what they are.[40\1] But there was this difference, that things shared the nature of the Platonic Ideas "by participation," while they shared the nature of Pythagorean Numbers "by imitation." Moreover, the Pythagoreans followed their notion out into much greater development than any other school, investing particular numbers with extraordinary attributes, and applying them by very strange and forced analogies. Thus the number Four, to which they gave the name of _Tetractys_, was held to be the most perfect number, and was conceived to correspond to the human soul, in some way which appears to be very imperfectly understood by the commentators of this philosophy. {78}

[Note: 40\1: Arist. Metaph. i. 6.]

It has been observed by a distinguished modern scholar,[41\1] that the place which Pythagoras ascribed to his numbers is intelligible only by supposing that he confounded, first a numerical unit with a geometrical point, and then this with a material atom. But this criticism appears to place systems of physical philosophy under requisitions too severe. If all the essential properties and attributes of things were fully represented by the relations of number, the philosophy which supplied such an explanation of the universe, might well be excused from explaining also that existence of objects which is distinct from the existence of all their qualities and properties. The Pythagorean love of numerical speculations might have been combined with the doctrine of atoms, and the combination might have led to results well worth notice. But so far as we are aware, no such combination was attempted in the ancient schools of philosophy; and perhaps we of the present day are only just beginning to perceive, through the disclosures of chemistry and crystallography, the importance of such a line of inquiry.

[Note 41\1: Thirlwall's _Hist. Gr._ ii. 142.]

4. _Technical Forms of the Atomists and Others._--The atomic doctrine, of which we have just spoken, was one of the most definite of the physical doctrines of the ancients, and was applied with most perseverance and knowledge to the explanation of phenomena. Though, therefore, it led to no success of any consequence in ancient times, it served to transmit, through a long series of ages, a habit of really physical inquiry; and, on this account, has been thought worthy of an historical disquisition by Bacon.[42\1]

[Note 42\1: Parmenidis et Telesii et præcipue Democriti Philosophia, &c., Works, vol. ix. 317.]

The technical term, _Atom_, marks sufficiently the nature of the opinion. According to this theory, the world consists of a collection of simple particles, of one kind of matter, and of indivisible smallness (as the name indicates), and by the various configurations and motions of these particles, all kinds of matter and all material phenomena are produced.

To this, the Atomic Doctrine of Leucippus and Democritus, was opposed the _Homoiomeria_ of Anaxagoras; that is, the opinion that material things consist of particles which are homogeneous in each kind of body, but various in different kinds: thus for example, since by food the flesh and blood and bones of man increase, the author of this doctrine held that there are in food particles of flesh, and blood, {79} and bone. As the former tenet points to the corpuscular theories of modern times, so the latter may be considered as a dim glimpse of the idea of chemical analysis. The Stoics also, who were, especially at a later period, inclined to materialist views, had their technical modes of speaking on such subjects. They asserted that matter contained in itself tendencies or dispositions to certain forms, which dispositions they called λόγοι **σπερματικοὶ, _seminal proportions_, or _seminal reasons_.

Whatever of sound view, or right direction, there might be in the notions which suggested these and other technical expressions, was, in all the schools of philosophy (so far as physics was concerned) quenched and overlaid by the predominance of trifling and barren speculations; and by the love of subtilizing and commenting upon the works of earlier writers, instead of attempting to interpret the book of nature. Hence these technical terms served to give fixity and permanence to the traditional dogmas of the sect, but led to no progress of knowledge.

The advances which were made in physical science proceeded, not from these schools of philosophy (if we except, perhaps, the obligations of the science of Harmonics to the Pythagoreans), but from reasoners who followed an independent path. The sequel of the ambitious hopes, the vast schemes, the confident undertakings of the philosophers of ancient Greece, was an entire failure in the physical knowledge of which it is our business to trace the history. Yet we are not, on that account, to think slightingly of these early speculators. They were men of extraordinary acuteness, invention, and range of thought; and, above all, they had the merit of first completely unfolding the speculative faculty--of starting in that keen and vigorous chase of knowledge out of which all the subsequent culture and

improvement of man's intellectual stores have arisen. The sages of early Greece form the heroic age of science. Like the first navigators in their own mythology, they boldly ventured their untried bark in a distant and arduous voyage, urged on by the hopes of a supernatural success; and though they missed the imaginary golden prize which they sought, they unlocked the gates of distant regions, and opened the seas to the keels of the thousands of adventurers who, in succeeding times, sailed to and fro, to the indefinite increase of the mental treasures of mankind.

But inasmuch as their attempts, in one sense, and at first, failed, we must proceed to offer some account of this failure, and of its nature and causes. {80}

## CHAPTER III.

FAILURE OF THE PHYSICAL PHILOSOPHY OF THE GREEK SCHOOLS.

_Sect._ 1.--_Result of the Greek School Philosophy_.

THE methods and forms of philosophizing which we have described as employed by the Greek Schools, failed altogether in their application to physics. No discovery of general laws, no explanation of special phenomena, rewarded the acuteness and boldness of these early students of nature. Astronomy, which made considerable progress during the existence of the sects of Greek philosophers, gained perhaps something by the authority with which Plato taught the supremacy and universality of mathematical rule and order; and the truths of Harmonics, which had probably given rise to the Pythagorean passion for numbers, were cultivated with much care by that school. But after these first impulses, the sciences owed nothing to the philosophical sects; and the vast and complex accumulations and apparatus of the Stagirite do not appear to have led to any theoretical physical truths.

This assertion hardly requires proof, since in the existing body of science there are no doctrines for which we are indebted to the Aristotelian School. Real truths, when once established, remain to the end of time a part of the mental treasure of man, and may be discerned through all the additions of later days. But we can point out no physical doctrine now received, of which we trace the anticipation in Aristotle, in the way in which we see the Copernican system anticipated by Aristarchus, the resolution of the heavenly appearances into circular motions suggested by Plato, and the numerical relations of musical intervals ascribed to Pythagoras. But it may be worth while to look at this matter more closely.

Among the works of Aristotle are thirty-eight chapters of "Problems," which may serve to exemplify the progress he had really made in the reduction of phenomena to laws and causes. Of these Problems, a large proportion are physiological, and these I here pass by, as not illustrative of the state of physical knowledge. But those which are properly physical are, for the most part, questions concerning such {81} facts and difficulties as it is the peculiar business of theory to explain. Now it may be truly said, that in scarcely any one instance are the answers, which Aristotle gives to his questions, of any value. For the most part, indeed, he propounds his answer with a degree of hesitation or vacillation which of itself shows the absence of all scientific

distinctness of thought; and the opinions so offered never appear to involve any settled or general principle.

We may take, as examples of this, the problems of the simplest kind, where the principles lay nearest at hand--the mechanical ones. "Why," he asks,[43\1] "do small forces move great weights by means of a lever, when they have thus to move the lever added to the weight? Is it," he suggests, "because a greater radius moves faster?" "Why does a small wedge split great weights?[44\1] Is it because the wedge is composed of two opposite levers?" "Why,[45\1] when a man rises from a chair, does he bend his leg and his body to acute angles with his thigh? Is it because a right angle is connected with equality and rest?" "Why[46\1] can a man throw a stone further with a sling than with his hand? Is it that when he throws with his hand he moves the stone from rest, but when he uses the sling he throws it already in motion?" "Why,[47\1] if a circle be thrown on the ground, does it first describe a straight line and then a spiral, as it falls? Is it that the air first presses equally on the two sides and supports it, and afterwards presses on one side more?" "Why[48\1] is it difficult to distinguish a musical note from the octave above? Is it that proportion stands in the place of equality?" It must be allowed that these are very vague and worthless surmises; for even if we were, as some commentators have done, to interpret some of them so as to agree with sound philosophy, we should still be unable to point out, in this author's works, any clear or permanent apprehension of the general principles which such an interpretation implies.

[Note 43\1: Mech. Prob. 4.]

[Note 44\1: Ib. 18.]

[Note 45\1: Ib. 31.]

[Note 46\1: Ib. 13.]

[Note 47\1: Περὶ Ἄψυχα. 11.]

[Note 48\1: Περὶ Ἁρμον. 14.]

Thus the Aristotelian physics cannot be considered as otherwise than a complete failure. It collected no general laws from facts; and consequently, when it tried to explain facts, it had no principles which were of any avail.

The same may be said of the physical speculations of the other schools of philosophy. They arrived at no doctrines from which they could deduce, by sound reasoning, such facts as they saw; though they {82} often venture so far to trust their principles as to infer from them propositions beyond the domain of sense. Thus, the principle that each element seeks _its own place_, led to the doctrine that, the place of fire being the highest, there is, above the air, a Sphere of Fire--of which doctrine the word _Empyrean_, used by our poets, still conveys a reminiscence. The Pythagorean tenet that ten is a perfect number,[49\1] led some persons to assume that the heavenly bodies are in number ten; and as nine only were known to them, they asserted that there was an _antichthon_, or _counter-earth_, on the other side of the sun, invisible to us. Their opinions respecting numerical ratios, led to various other speculations concerning the distances and positions of the heavenly bodies: and

as they had, in other cases, found a connection between proportions of distance and musical notes, they assumed, on this suggestion, _the music of the spheres_.

[Note 49\1: Arist. Metaph. i. 5.]

Although we shall look in vain in the physical philosophy of the Greek Schools for any results more valuable than those just mentioned, we shall not be surprised to find, recollecting how much an admiration for classical antiquity has possessed the minds of men, that some writers estimate their claims much more highly than they are stated here. Among such writers we may notice Dutens, who, in 1766, published his "Origin of the Discoveries attributed to the Moderns; in which it is shown that our most celebrated Philosophers have received the greatest part of their knowledge from the Works of the Ancients." The thesis of this work is attempted to be proved, as we might expect, by very large interpretations of the general phrases used by the ancients. Thus, when Timæus, in Plato's dialogue, says of the Creator of the world,[50\1] "that he infused into it two powers, the origins of motions, both of that of the same thing and of that of different things;" Dutens[51\1] finds in this a clear indication of the projectile and attractive forces of modern science. And in some of the common declamation of the Pythagoreans and Platonists concerning the general prevalence of numerical relations in the universe, he discovers their acquaintance with the law of the inverse square of the distance by which gravitation is regulated, though he allows[52\1] that it required all the penetration of Newton and his followers to detect this law in the scanty fragments by which it is transmitted.

[Note 50\1: Tim. 96.]

[Note 51\1: 3d ed. p. 83.]

[Note 52\1: Ib. p. 88.]

Argument of this kind is palpably insufficient to cover the failure of the Greek attempts at a general physical philosophy; or rather we {83} may say, that such arguments, since they are as good as can be brought in favor of such an opinion, show more clearly how entire the failure was. I proceed now to endeavor to point out its causes.

_Sect._ 2.--_Cause of the Failure of the Greek Physical Philosophy._

THE cause of the failure of so many of the attempts of the Greeks to construct physical science is so important, that we must endeavor to bring it into view here; though the full development of such subjects belongs rather to the Philosophy of Induction. The subject must, at present, be treated very briefly.

I will first notice some errors which may naturally occur to the reader's mind, as possible causes of failure, but which, we shall be able to show, were not the real reasons in this case.

The cause of failure was _not the neglect of facts_. It is often said that the Greeks disregarded experience, and spun their philosophy out of their own thoughts alone; and this is supposed by many to be their essential error. It is, no doubt, true, that the disregard of experience is a phrase which may be so interpreted as to express almost

any defect of philosophical method; since coincidence with experience is requisite to the truth of all theory. But if we fix a more precise sense on our terms, I conceive it may be shown that the Greek philosophy did, in its opinions, recognize the necessity and paramount value of observations; did, in its origin, proceed upon observed facts; and did employ itself to no small extent in classifying and arranging phenomena. We must endeavor to illustrate these assertions, because it is important to show that these steps alone do not necessarily lead to science.

1. The acknowledgment of experience as the main ground of physical knowledge is so generally understood to be a distinguishing feature of later times, that it may excite surprise to find that Aristotle, and other ancient philosophers, not only asserted in the most pointed manner that all our knowledge must begin from experience, but also stated in language much resembling the habitual phraseology of the most modern schools of philosophizing, that particular facts must be _collected_; that from these, general principles must be obtained by _induction_; and that these principles, when of the most general kind, are _axioms_. A few passages will show this.

"The way[53\1] must be the same," says Aristotle, in speaking of the rules of reasoning, "with respect to philosophy, as it is with respect to {84} any art or science whatever; we must collect the facts, and the things to which the facts happen, in each subject, and provide as large a supply of these as possible." He then proceeds to say that "we are not to look at once at all this collected mass, but to consider small and definite portions" . . . "And thus it is the office of observation to supply principles in each subject; for instance, astronomical observation supplies the principles of astronomical science. For the phenomena being properly assumed, the astronomical demonstrations were from these discovered. And the same applies to every art and science. So that if we take the facts (τὰ ὑπάρχοντα) belonging to each subject, it is _our_ task to mark out clearly the course of the demonstrations. For if _in our natural history_ (κατὰ τὴν ἱστορίαν) we have omitted nothing of the facts and properties which belong to the subject, we shall learn what we can demonstrate and what we cannot."

[Note 53\1: Anal. Prior. i. 30.]

These facts, τὰ ὑπάρχοντα, he, at other times, includes in the term _sensation_. Thus, he says,[54\1] "It is obvious that if any sensation is wanting, there must be also some knowledge wanting which we are thus prevented from having, since we arrive at knowledge either by induction or by demonstration. Demonstration proceeds from universal propositions, Induction from particulars. But we cannot have universal theoretical propositions except from induction; and we cannot make inductions without having sensation; for sensation has to do with particulars."

[Note 54\1: Anal. Post. i. 18.]

In another place,[55\1] after stating that principles must be prior to, and better known than conclusions, he distinguishes such principles into absolutely prior, and prior relative to us: "The prior principles, relative to us, are those which are nearer to the sensation; but the principles absolutely prior are those which are more remote

from the sensation. The most general principles are the more remote, the more particular are nearer. The general principles which are necessary to knowledge are _axioms_."

[Note 55\1: Ib. i. 2.]

We may add to these passages, that in which he gives an account of the way in which Leucippus was led to the doctrine of atoms. After describing the opinions of some earlier philosophers, he says,[56\1] "Thus, proceeding in violation of sensation, and disregarding it, because, as they held, they must follow reason, some came to the conclusion that the universe was one, and infinite, and at rest. As it appeared, however, that though this ought to be by reasoning, it {85} would go near to madness to hold such opinions in practice (for no one was ever so mad as to think fire and ice to be one), Leucippus, therefore, pursued a line of reasoning which was in accordance with sensation, and which was not irreconcilable with the production and decay, the motion and multitude of things." It is obvious that the school to which Leucippus belonged (the Eclectic) must have been, at least in its origin, strongly impressed with the necessity of bringing its theories into harmony with the observed course of nature.

[Note 56\1: De Gen. et Cor. i. 8.]

2. Nor was this recognition of the fundamental value of experience a mere profession. The Greek philosophy did, in its beginning, proceed upon observation. Indeed it is obvious that the principles which it adopted were, in the first place, assumed in order to account for some classes of facts, however imperfectly they might answer their purpose. The principle of things seeking their own places, was invented in order to account for the falling and floating of bodies. Again, Aristotle says, that heat is that which brings together things of the same kind, cold is that which brings together things whether of the same or of different kinds: it is plain that in this instance he intended by his principle to explain some obvious facts, as the freezing of moist substances, and the separation of heterogeneous things by fusion; for, as he adds, if fire brings together things which are akin, it will separate those which are not akin. It would be easy to illustrate the remark further, but its truth is evident from the nature of the case; for no principles could be accepted for a moment, which were the result of an arbitrary caprice of the mind, and which were not in some measure plausible, and apparently confirmed by facts.

But the works of Aristotle show, in another way, how unjust it would be to accuse him of disregarding facts. Many large treatises of his consist almost entirely of collections of facts, as for instance, those "On Colors," "On Sounds," and the collection of Problems to which we have already referred; to say nothing of the numerous collection of facts bearing on natural history and physiology, which form a great portion of his works, and are even now treasuries of information. A moment's reflection will convince us that the physical sciences of our own times, for example. Mechanics and Hydrostatics, are founded almost entirely upon facts with which the ancients were as familiar as we are. The defect of their philosophy, therefore, wherever it may lie, consists neither in the speculative depreciation of the value of

facts, nor in the practical neglect of their use.

3. Nor again, should we hit upon the truth, if we were to say that {86} Aristotle, and other ancient philosophers, did indeed collect facts; but that they took no steps in classifying and comparing them; and that thus they failed to obtain from them any general knowledge. For, in reality, the treatises of Aristotle which we have mentioned, are as remarkable for the power of classifying and systematizing which they exhibit, as for the industry shown in the accumulation. But it is not classification of facts merely which can lead us to knowledge, except we adopt that special arrangement, which, in each case, brings into view the principles of the subject. We may easily show how unprofitable an arbitrary or random classification is, however orderly and systematic it may be.

For instance, for a long period all unusual fiery appearances in the sky were classed together as _meteors_. Comets, shooting-stars, and globes of fire, and the aurora borealis in all its forms, were thus grouped together, and classifications of considerable extent and minuteness were proposed with reference to these objects. But this classification was of a mixed and arbitrary kind. Figure, color, motion, duration, were all combined as characters, and the imagination lent its aid, transforming these striking appearances into fiery swords and spears, bears and dragons, armies and chariots. The facts so classified were, notwithstanding, worthless; and would not have been one jot the less so, had they and their classes been ten times as numerous as they were. No rule or law that would stand the test of observation was or could be thus discovered. Such classifications have, therefore, long been neglected and forgotten. Even the ancient descriptions of these objects of curiosity are unintelligible, or unworthy of trust, because the spectators had no steady conception of the usual order of such phenomena. For, however much we may fear to be misled by preconceived opinions, the caprices of imagination distort our impressions far more than the anticipations of reason. In this case men had, indeed we may say with regard to many of these meteors, they still have, no science: not for want of facts, nor even for want of classification of facts; but because the classification was one in which no real principle was contained.

4. Since, as we have said before, two things are requisite to science,--Facts and Ideas; and since, as we have seen. Facts were not wanting in the physical speculations of the ancients, we are naturally led to ask, Were they then deficient in Ideas? Was there a want among them of mental activity, and logical connection of thought? But it is so obvious that the answer to this inquiry must be in the negative, that we need not dwell upon it. No one who knows any thing of the {87} history of the ancient Greek mind, can question, that in acuteness, in ingenuity, in the power of close and distinct reasoning, they have never been surpassed. The common opinion, which considers the defect of their philosophical character to reside rather in the exclusive activity of such qualities, than in the absence of them, is at least so far just.

5. We come back again, therefore, to the question, What was the radical and fatal defect in the physical speculations of the Greek philosophical schools?

To this I answer: The defect was, that though they had in their possession Facts and Ideas, _the Ideas were not distinct and appropriate to the Facts_.

The peculiar characteristics of scientific ideas, which I have endeavored to express by speaking of them as _distinct_ and _appropriate to the facts_, must be more fully and formally set forth, when we come to the philosophy of the subject. In the mean time, the reader will probably have no difficulty in conceiving that, for each class of Facts, there is some special set of Ideas, by means of which the facts can be included in general scientific truths; and that these Ideas, which may thus be termed _appropriate_, must be possessed with entire distinctness and clearness, in order that they may be successfully applied. It was the want of Ideas having this reference to material phenomena, which rendered the ancient philosophers, with very few exceptions, helpless and unsuccessful speculators on physical subjects.

This must be illustrated by one or two examples. One of the facts which Aristotle endeavors to explain is this; that when the sun's light passes through a hole, whatever be the form of the hole, the bright image, if formed at any considerable distance from the hole, is round, instead of imitating the figure of the hole, as shadows resemble their objects in form. We shall easily perceive this appearance to be a necessary consequence of the circular figure of the sun, if we conceive light to be diffused from the luminary by means of straight rays proceeding from every point of the sun's disk and passing through every point within the boundary of the hole. By attending to the consequences of this mode of conception, it will be seen that each point of the hole will be the vertex of a double cone of rays which has the sun's disk for its base on one side and an image of the sun on the other; and the figure of the image of the hole will be determined by supposing a series of equal bright circles, images of the sun, to be placed along the boundary of an image equal to the hole itself. The figure of the image thus determined will partake of the form of the hole, and {88} of the circular form of the sun's image: but these circular images become larger and larger as they are further from the hole, while the central image of the hole remains always of the original size; and thus at a considerable distance from the hole, the trace of the hole's form is nearly obliterated, and the image is nearly a perfect circle. Instead of this distinct conception of a cone of rays which has the sun's disk for its basis, Aristotle has the following loose conjecture.[57\1] "Is it because light is emitted in a conical form; and of a cone, the base is a circle; so that on whatever the rays of the sun fall, they appear more circular?" And thus though he applies the notion of rays to this problem, he possesses this notion so _indistinctly_ that his explanation is of no value. He does not introduce into his explanation the consideration of the sun's circular figure, and is thus prevented from giving a true account of this very simple optical phenomenon.

[Note 57\1: Problem. 15, ὁσα μαθηματίκης, &c.]

6. Again, to pass to a more extensive failure: why was it that Aristotle, knowing the property of the lever, and many other mechanical truths, was unable to form them into a science of mechanics, as Archimedes afterwards did?

The reason was, that, instead of considering rest and motion directly, and distinctly, with reference to the Idea of Cause, that is Force, he wandered in search of reasons among other ideas and notions, which could not be brought into steady connection with the facts;--the ideas of properties of circles, of proportions of velocities,--the notions of "strange" and "common," of "natural" and "unnatural." Thus, in the Proem to his Mechanical Problems, after stating some of the difficulties which he has to attack, he says, "Of all such cases, the circle contains the principle of the cause. And this is what might be looked for; for it is nothing absurd, if something _wonderful_ is derived from something more wonderful still. Now the most wonderful thing is, that opposites should be combined; and the circle is constituted of such combinations of opposites. For it is constructed by a stationary point and a moving line, which are contrary to each other in nature; and hence we may the less be surprised at the resulting contrarieties. And in the first place, the circumference of the circle, though a line without breadth, has opposite qualities; for it is both _convex_ and _concave_. In the next place, it has, at the same time, opposite motions, for it moves forward and backward at the same time. For the circumference, setting out from any point, comes to the same point again, so {89} that by a continuous progression, the last point becomes the first. So that, as was before stated, it is not surprising that the circle should be the principle of all wonderful properties."

Aristotle afterwards proceeds to explain more specially how he applies the properties of the circle in this case. "The reason," he says, in his fourth Problem, "why a force, acting at a greater distance from the fulcrum, moves a weight more easily, is, that it describes a greater circle." He had already asserted that when a body at the end of a lever is put in motion, it may be considered as having two motions; one in the direction of the tangent, and one in the direction of the radius; the former motion is, he says, _according to nature_, the latter, _contrary to nature_. Now in the smaller circle, the motion, contrary to nature, is more considerable than it is in the larger circle. "Therefore," he adds, "the mover or weight at the larger arm will be transferred further by the same force than the weight moved, which is at the extremity of the shorter arm."

These loose and inappropriate notions of "natural" and "unnatural" motions, were unfit to lead to any scientific truths; and, with the habits of thought which dictated these speculations a perception of the true grounds of mechanical properties was impossible.

7. Thus, in this instance, the error of Aristotle was the neglect of the Idea _appropriate_ to the facts, namely, the Idea of Mechanical Cause, which is Force; and the substitution of vague or inapplicable notions involving only relations of space or emotions of wonder. The errors of those who failed similarly in other instances, were of the same kind. To detail or classify these would lead us too far into the philosophy of science; since we should have to enumerate the Ideas which are appropriate, and the various classes of Facts on which the different sciences are founded,--a task not to be now lightly undertaken. But it will be perceived, without further explanation, that it is necessary, in order to obtain from facts any general truth, that we should apply to them that appropriate Idea, by which permanent and definite relations are

established among them.

In such Ideas the ancients were very poor, and the stunted and deformed growth of their physical science was the result of this penury. The Ideas of Space and Time, Number and Motion, they did indeed possess distinctly; and so far as these went, their science was tolerably healthy. They also caught a glimpse of the Idea of a Medium by which the qualities of bodies, as colors and sounds, are perceived. But the idea of Substance remained barren in their hands; {90} in speculating about elements and qualities, they went the wrong way, assuming that the properties of Compounds must _resemble_ those of the Elements which determine them; and their loose notions of Contrariety never approached the form of those ideas of Polarity, which, in modern times, regulate many parts of physics and chemistry.

If this statement should seem to any one to be technical or arbitrary, we must refer, for the justification of it, to the Philosophy of Science, of which we hope hereafter to treat. But it will appear, even from what has been here said, that there are certain Ideas or Forms of mental apprehension, which may be applied to Facts in such a manner as to bring into view fundamental principles of science; while the same Facts, however arrayed or reasoned about, so long as these appropriate ideas are not employed, cannot give rise to any exact or substantial knowledge.

[2d Ed.] This account of the cause of failure in the physical speculations of the ancient Greek philosophers has been objected to as unsatisfactory. I will offer a few words in explanation of it.

The mode of accounting for the failure of the Greeks in physics is, in substance;--that the Greeks in their physical speculations fixed their attention upon the wrong aspects and relations of the phenomena; and that the aspects and relations in which phenomena are to be viewed in order to arrive at scientific truths may be arranged under certain heads, which I have termed _Ideas_; such as Space, Time, Number, Cause, Likeness. In every case, there is an Idea to which the phenomena may be referred, so as to bring into view the Laws by which they are governed; this Idea I term the _appropriate_ Idea in such case; and in order that the reference of the phenomena to the Law may be clearly seen, the Idea must be _distinctly_ possessed.

Thus the reason of Aristotle's failure in his attempts at Mechanical Science is, that he did not refer the facts to the appropriate Idea, namely Force, the Cause of Motion, but to relations of Space and the like; that is, he introduces _Geometrical_ instead of _Mechanical_ Ideas. It may be said that we learn little by being told that Aristotle's failure in this and the like cases arose from his referring to the wrong class of Ideas; or, as I have otherwise expressed it, fixing his attention upon the wrong aspects and relations of the facts; since, it may be said, this is only to state in other words that he _did_ fail. But this criticism is, I think, ill-founded. The account which I have given is not only a statement that Aristotle, and others who took a like course, did fail; but also, that they failed in one certain point out of several {91} which are enumerated. They did not fail because they neglected to observe facts; they did not fail because they omitted to class facts; they did not fail because they had not ideas to reason from; but they failed because they did not take the right ideas in each case. And so

long as they were in the wrong in this point, no industry in collecting facts, or ingenuity in classing them and reasoning about them, could lead them to solid truth.

Nor is this account of the nature of their mistake without its instruction for us; although we are not to expect to derive from the study of their failure any technical rule which shall necessarily guide us to scientific discovery. For their failure teaches us that, in the formation of science, an Error in the Ideas is as fatal to the discovery of Truth as an Error in the Facts; and may as completely impede the progress of knowledge. I have in Books II. to X. of the _Philosophy_, shown historically how large a portion of the progress of Science consists in the establishment of Appropriate Ideas as the basis of each science. Of the two main processes by which science is constructed, as stated in Book XI. of that work, namely the _Explication of Conceptions_ and the _Colligation of Facts_, the former must precede the latter. In Book XII. chap. 5, of the _Philosophy_, I have stated the maxim concerning appropriate Ideas in this form, that _the Idea and the Facts must be homogeneous_.

When I say that the failure of the Greeks in physical science arose from their not employing _appropriate_ Ideas to connect the facts, I do not use the term "appropriate" in a loose popular sense; but I employ it as a somewhat technical term, to denote _the_ appropriate Idea, out of that series of Ideas which have been made (as I have shown in the _Philosophy_) the foundation of sciences; namely, Space, Time, Number, Cause, Likeness, Substance, and the rest. It appears to me just to say that Aristotle's failure in his attempts to deal with problems of equilibrium, arose from his referring to circles, velocities, notions of natural and unnatural, and the like,--conceptions depending upon Ideas of Space, of Nature, &c.--which are not appropriate to these problems, and from his missing the Idea of Mechanical Force or Pressure, which is the appropriate Idea.

I give this, not as an account of _all_ failures in attempts at science, but only as the account of such radical and fundamental failures as this of Aristotle; who, with a knowledge of the facts, failed to connect them into a really scientific view. If I had to compare rival theories of a more complex kind, I should not necessarily say that one involved {92} an appropriate Idea and the other did not, though I might judge one to be true and the other to be false. For instance, in comparing the emissive and the undulatory theory of light, we see that both involve the same Idea;--the Idea of a Medium acting by certain mechanical properties. The question there is, What is the true view of the mechanism of the Medium?

It may be remarked, however, that the example of Aristotle's failure in physics, given in p. 87, namely, his attempted explanation of the round image of a square hole, is a specimen rather of _indistinct_ than of inappropriate ideas.

The geometrical explanation of this phenomenon, which I have there inserted, was given by Maurolycus, and before him, by Leonardo da Vinci.

We shall, in the next Book, see the influence of the appropriate general Ideas, in the formation of various sciences. It need only be observed, before we proceed, that, in order to do full justice to the physical knowledge of the Greek Schools of philosophy, it is not necessary to study their course after the time of their founders.

Their fortunes, in respect of such acquisitions as we are now considering, were not progressive. The later chiefs of the Schools followed the earlier masters; and though they varied much, they added little. The Romans adopted the philosophy of their Greek subjects; but they were always, and, indeed, acknowledged themselves to be, inferior to their teachers. They were as arbitrary and loose in their ideas as the Greeks, without possessing their invention, acuteness, and spirit of system.

In addition to the vagueness which was combined with the more elevated trains of philosophical speculation among the Greeks, the Romans introduced into their treatises a kind of declamatory rhetoric, which arose probably from their forensic and political habits, and which still further obscured the waning gleams of truth. Yet we may also trace in the Roman philosophers to whom this charge mostly applies (Lucretius, Pliny, Seneca), the national vigor and ambition. There is something Roman in the public spirit and anticipation of universal empire which they display, as citizens of the intellectual republic. Though they speak sadly or slightingly of the achievements of their own generation, they betray a more abiding and vivid belief in the dignity and destined advance of human knowledge as a whole, than is obvious among the Greeks.

We must, however, turn back, in order to describe steps of more definite value to the progress of science than those which we have hitherto noticed.

{{93}} BOOK II.

HISTORY OF THE PHYSICAL SCIENCES IN ANCIENT GREECE.

Ναρθηκοπλήρωτον δὲ θηρῶμαι πυρὸς Πηγὴν κλοπαίαν, ἣ διδάσκαλος τέχνης Πάσης βροτοῖς πεφῆνε καὶ μέγας πόρος. Prom. Vinct. 109.

I brought to earth the spark of heavenly fire, Concealed at first, and small, but spreading soon Among the sons of men, and burning on, Teacher of art and use, and fount of power.

{{95}} INTRODUCTION.

IN order to the acquisition of any such exact and real knowledge of nature as that which we properly call Physical Science, it is requisite, as has already been said, that men should possess Ideas both distinct and appropriate, and should apply them to ascertained Facts. They are thus led to propositions of a general character, which are obtained by Induction, as will elsewhere be more fully explained. We proceed now to trace the formation of Sciences among the Greeks by such processes. The provinces of knowledge which thus demand our attention are, Astronomy, Mechanics and Hydrostatics, Optics and Harmonics; of which I must relate, first, the earliest stages, and next, the subsequent progress.

Of these portions of human knowledge, Astronomy is, beyond doubt or comparison, much the most ancient and the most remarkable; and probably existed, in somewhat of a scientific form, in Chaldea and Egypt, and other countries, before the period of the intellectual activity of the Greeks. But I will give a brief account of some of the other Sciences before I proceed to Astronomy, for two reasons; first, because the origin of Astronomy is lost in the obscurity of a remote antiquity; and therefore we

cannot exemplify the conditions of the first rise of science so well in that subject as we can in others which assumed their scientific form at known periods; and next, in order that I may not have to interrupt, after I have once begun it, the history of the only progressive Science which the ancient world produced.

It has been objected to the arrangement here employed that it is not symmetrical; and that Astronomy, as being one of the Physical Sciences, ought to have occupied a chapter in this Second Book, instead of having a whole Book to itself (Book III). I do not pretend that the arrangement is symmetrical, and have employed it only on the ground of convenience. The importance and extent of the history of Astronomy are such that this science could not, with a view to our purposes, be made co-ordinate with Mechanics or Optics. {96}

## CHAPTER I.

EARLIEST STAGES OF MECHANICS AND HYDROSTATICS.

_Sect._ 1.--_Mechanics._

ASTRONOMY is a science so ancient that we can hardly ascend to a period when it did not exist; Mechanics, on the other hand, is a science which did not begin to be till after the time of Aristotle; for Archimedes must be looked upon as the author of the first sound knowledge on this subject. What is still more curious, and shows remarkably how little the continued progress of science follows inevitably from the nature of man, this department of knowledge, after the right road had been fairly entered upon, remained absolutely stationary for nearly two thousand years; no single step was made, in addition to the propositions established by Archimedes, till the time of Galileo and Stevinus. This extraordinary halt will be a subject of attention hereafter; at present we must consider the original advance.

The great step made by Archimedes in Mechanics was the establishing, upon true grounds, the general proposition concerning a straight lever, loaded with two heavy bodies, and resting upon a fulcrum. The proposition is, that two bodies so circumstanced will balance each other, when the distance of the smaller body from the fulcrum is greater than the distance of the other, in exactly the same proportion in which the weight of the body is less.

This proposition is proved by Archimedes in a work which is still extant, and the proof holds its place in our treatises to this day, as the simplest which can be given. The demonstration is made to rest on assumptions which amount in effect to such Definitions and Axioms as these: That those bodies are of equal weight which balance each other at equal arms of a straight lever; and that in every heavy body there is a definite point called a _Centre of Gravity_, in which point we may suppose the weight of the body collected.

The principle, which is really the foundation of the validity of the demonstration thus given, and which is the condition of all experimental knowledge on the subject, is this: that when two equal weights are supported on a lever, they act on the fulcrum of the lever with the {97} same effect as if they were both together supported

immediately at that point. Or more generally, we may state the principle to be this: that the pressure by which a heavy body is supported continues the same, however we alter the form or position of the body, so long as the magnitude and material continue the same.

The experimental truth of this principle is a matter of obvious and universal experience. The weight of a basket of stones is not altered by shaking the stones into new positions. We cannot make the direct burden of a stone less by altering its position in our hands; and if we try the effect on a balance or a machine of any kind, we shall see still more clearly and exactly that the altered position of one weight, or the altered arrangement of several, produces no change in their effect, so long as their point of support remains unchanged.

This general fact is obvious, when we possess in our minds the ideas which are requisite to apprehend it clearly. But when we are so prepared, the truth appears to be manifest, even independent of experience, and is seen to be a rule to which experience must conform. What, then, is the leading idea which thus enables us to reason effectively upon mechanical subjects? By attention to the course of such reasonings, we perceive that it is the idea of _Pressure_; Pressure being conceived as a measurable effect of heavy bodies at rest, distinguishable from all other effects, such as motion, change of figure, and the like. It is not here necessary to attempt to trace the history of this idea in our minds; but it is certain that such an idea may be distinctly formed, and that upon it the whole science of statics may be built. _Pressure_, _load_, _weight_, are names by which this idea is denoted when the effect tends directly downwards; but we may have pressure without motion, or _dead pull_, in other cases, as at the critical instant when two nicely-matched wrestlers are balanced by the exertion of the utmost strength of each.

Pressure in any direction may thus exist without any motion whatever. But the causes which produce such pressure are capable of producing motion, and are generally seen producing motion, as in the above instance of the wrestlers, or in a pair of scales employed in weighing; and thus men come to consider pressure as the exception, and motion as the rule: or perhaps they image to themselves the motion which _might_ or _would_ take place; for instance, the motion which the arms of a lever _would_ have if they _did_ move. They turn away from the case really before them, which is that of bodies at rest, and balancing each other, and pass to another case, which is arbitrarily {98} assumed to represent the first. Now this arbitrary and capricious evasion of the question we consider as opposed to the introduction of the distinct and proper idea of Pressure, by means of which the true principles of this subject can be apprehended.

We have already seen that Aristotle was in the number of those who thus evaded the difficulties of the problem of the lever, and consequently lost the reward of success. He failed, as has before been stated, in consequence of his seeking his principles in notions, either vague and loose, as the distinction of natural and unnatural motions, or else inappropriate, as the circle which the weight _would_ describe, the velocity which it _would_ have if it moved; circumstances which are not part of the fact under consideration. The influence of such modes of speculation was

the main hindrance to the prosecution of the true Archimedean form of the science of Mechanics.

The mechanical doctrine of Equilibrium, is _Statics_. It is to be distinguished from the mechanical doctrine of Motion, which is termed _Dynamics_, and which was not successfully treated till the time of Galileo.

_Sect._ 2.--_Hydrostatics._

ARCHIMEDES not only laid the foundations of the Statics of solid bodies, but also solved the principal problem of _Hydrostatics_, or the Statics of Fluids; namely, the conditions of the floating of bodies. This is the more remarkable, since not only did the principles which Archimedes established on this subject remain unpursued till the revival of science in modern times, but, when they were again put forward, the main proposition was so far from obvious that it was termed, and is to this day called, the _hydrostatic paradox_. The true doctrine of Hydrostatics, however, assuming the Idea of Pressure, which it involves, in common with the Mechanics of solid bodies, requires also a distinct Idea of a Fluid, as a body of which the parts are perfectly movable among each other by the slightest partial pressure, and in which all pressure exerted on one part is transferred to all other parts. From this idea of Fluidity, necessarily follows that multiplication of pressure which constitutes the hydrostatic paradox; and the notion being seen to be verified in nature, the consequences were also realized as facts. This notion of Fluidity is expressed in the postulate which stands at the head of Archimedes' "Treatise on Floating Bodies." And from this principle are deduced the solutions, not only of the simple problems of the science, but of some problems of considerable complexity. {99}

The difficulty of holding fast this Idea of Fluidity so as to trace its consequences with infallible strictness of demonstration, may be judged of from the circumstance that, even at the present day, men of great talents, not unfamiliar with the subject, sometimes admit into their reasonings an oversight or fallacy with regard to this very point. The importance of the Idea when clearly apprehended and securely held, may be judged of from this, that the whole science of Hydrostatics in its most modern form is only the development of the Idea. And what kind of attempts at science would be made by persons destitute of this Idea, we may see in the speculations of Aristotle concerning light and heavy bodies, which we have already quoted; where, by considering light and heavy as opposite qualities, residing in things themselves, and by an inability to apprehend the effect of surrounding fluids in supporting bodies, the subject was made a mass of false or frivolous assertions, which the utmost ingenuity could not reconcile with facts, and could still less deduce from the asserted doctrines any new practical truths.

In the case of Statics and Hydrostatics, the most important condition of their advance was undoubtedly the distinct apprehension of these two _appropriate Ideas_--_Statical Pressure_, and _Hydrostatical Pressure_ as included in the idea of Fluidity. For the Ideas being once clearly possessed, the experimental laws which they served to express (that the whole pressure of a body downwards was always the same; and that water, and the like, were fluids according to the above idea of

fluidity), were so obvious, that there was no doubt nor difficulty about them. These two ideas lie at the root of all mechanical science; and the firm possession of them is, to this day, the first requisite for a student of the subject. After being clearly awakened in the mind of Archimedes, these ideas slept for many centuries, till they were again called up in Galileo, and more remarkably in Stevinus. This time, they were not destined again to slumber; and the results of their activity have been the formation of two Sciences, which are as certain and severe in their demonstrations as geometry itself and as copious and interesting in their conclusions; but which, besides this recommendation, possess one of a different order,--that they exhibit the exact impress of the laws of the physical world, and unfold a portion of the rules according to which the phenomena of nature take place, and must take place, till nature herself shall alter. {100}

## CHAPTER II.

EARLIEST STAGES OF OPTICS.

THE progress made by the ancients in Optics was nearly proportional to that which they made in Statics. As they discovered the true grounds of the doctrine of Equilibrium, without obtaining any sound principles concerning Motion, so they discovered the law of the Reflection of light, but had none but the most indistinct notions concerning Refraction.

The extent of the principles which they really possessed is easily stated. They knew that vision is performed by _rays_ which proceed in straight lines, and that these rays are _reflected_ by certain surfaces (mirrors) in such manner that the angles which they make with the surface on each side are equal. They drew various conclusions from these premises by the aid of geometry; as, for instance, the convergence of rays which fall on a concave speculum.

It may be observed that the _Idea_ which is here introduced, is that of visual _rays_, or lines along which vision is produced and light carried. This idea once clearly apprehended, it was not difficult to show that these lines are straight lines, both in the case of light and of sight. In the beginning of Euclid's "Treatise on Optics," some of the arguments are mentioned by which this was established. We are told in the Proem, "In explaining what concerns the sight, he adduced certain arguments from which he inferred that all light is carried in straight lines. The greatest proof of this is shadows, and the bright spots which are produced by light coming through windows and cracks, and which could not be, except the rays of the sun were carried in straight lines. So in fires, the shadows are greater than the bodies if the fire be small, but less than the bodies if the fire be greater." A clear comprehension of the principle would lead to the perception of innumerable proofs of its truth on every side.

The Law of Equality of Angles of Incidence and Reflection was not quite so easy to verify; but the exact resemblance of the object and its image in a plane mirror, (as the surface of still water, for instance), which is a consequence of this law, would afford convincing evidence of its truth in that case, and would be confirmed by the

examination of other cases. {101}

With these true principles was mixed much error and indistinctness, even in the best writers. Euclid, and the Platonists, maintained that vision is exercised by rays proceeding _from_ the eye, not _to_ it; so that when we see objects, we learn their form as a blind man would do, by feeling it out with his staff. This mistake, however, though Montucla speaks severely of it, was neither very discreditable nor very injurious; for the mathematical conclusions on each supposition are necessarily the same. Another curious and false assumption is, that those visual rays are not close together, but separated by intervals, like the fingers when the hand is spread. The motive for this invention was the wish to account for the fact, that in looking for a small object, as a needle, we often cannot see it when it is under our nose; which it was conceived would be impossible if the visual rays reached to all points of the surface before us.

These errors would not have prevented the progress of the science. But the Aristotelian physics, as usual, contained speculations more essentially faulty. Aristotle's views led him to try to describe the kind of causation by which vision is produced, instead of the laws by which it is exercised; and the attempt consisted, as in other subjects, of indistinct principles, and ill-combined facts. According to him, vision must be produced by a Medium,--by something _between_ the object and the eye,--for if we press the object on the eye, we do not see it; this Medium is Light, or "the transparent in action;" darkness occurs when the transparency is potential, not actual; color is not the "absolute visible," but something which is _on_ the absolute visible; color has the power of setting the transparent in action; it is not, however, all colors that are seen by means of light, but only the proper color of each object; for some things, as the heads, and scales, and eyes of fish, are seen in the dark; but they are not seen with their proper color.**[1\2]

[Note 1\2: De Anim. ii. **7.]

In all this there is no steady adherence either to one notion, or to one class of facts. The distinction of Power and Act is introduced to modify the Idea of Transparency, according to the formula of the school; then Color is made to be something unknown in addition to Visibility; and the distinction of "proper" and "improper" colors is assumed, as sufficient to account for a phenomenon. Such classifications have in them nothing of which the mind can take steady hold; nor is it difficult to see that they do not come under those {102} conditions of successful physical speculation, which we have laid down.

It is proper to notice more distinctly the nature of the Geometrical Propositions contained in Euclid's work. The _Optica_ contains Propositions concerning Vision and Shadows, derived from the principle that the rays of light are rectilinear: for instance, the Proposition that the shadow is greater than the object, if the illuminating body be less and _vice versa_. The _Catoptrica_ contains Propositions concerning the effects of Reflection, derived from the principle that the Angles of Incidence and Reflection are equal: as, that in a convex mirror the object appears convex, and smaller than the object. We see here an example of the promptitude of the Greeks in deduction. When

they had once obtained a knowledge of a principle, they followed it to its mathematical consequences with great acuteness. The subject of concave mirrors is pursued further in Ptolemy's _Optics_.

The Greek writers also cultivated the subject of _Perspective_ speculatively, in mathematical treatises, as well as practically, in pictures. The whole of this theory is a consequence of the principle that vision takes place in straight lines drawn from the object to the eye.

"The ancients were in some measure acquainted with the Refraction as well as the Reflection of Light," as I have shown in Book IX. Chap. 2 [2d Ed.] of the _Philosophy_. The current knowledge on this subject must have been very slight and confused; for it does not appear to have enabled them to account for one of the simplest results of Refraction, the magnifying effect of convex transparent bodies. I have noticed in the passage just referred to, Seneca's crude notions on this subject; and in like manner Ptolemy in his _Optics_ asserts that an object placed in water must always appear larger then when taken out. Aristotle uses the term ἀνακλάσις (_Meteorol_. iii. 2), but apparently in a very vague manner. It is not evident that he distinguished Refraction from Reflection. His Commentators however do distinguish these as διακλάσις and ἀνακλάσις. See Olympiodorus in Schneider's _Eclogæ Physicæ_, vol. i. p. 397. And Refraction had been the subject of special attention among the Greek Mathematicians. Archimedes had noticed (as we learn from the same writer) that in certain cases, a ring which cannot be seen over the edge of the empty vessel in which it is placed, becomes visible when the vessel is filled with water. The same fact is stated in the _Optics_ of Euclid. We do not find this fact explained in that work as we now have it; but in Ptolemy's _Optics_ the fact is explained by a flexure of the visual ray: it is {103} noticed that this flexure is different at different angles from the perpendicular, and there is an elaborate collection of measures of the flexure at different angles, made by means of an instrument devised for the purpose. There is also a collection of similar measures of the refraction when the ray passes from air to glass, and when it passes from glass to water. This part of Ptolemy's work is, I think, the oldest extant example of a collection of experimental measures in any other subject than astronomy; and in astronomy our measures are the result of _observation_ rather than of _experiment_. As Delambre says (_Astron. Anc._ vol. ii. p. 427), "On y voit des expériences de physique bien faites, ce qui est sans exemple chez les anciens."

Ptolemy's Optical work was known only by Roger Bacon's references to it (_Opus Majus_, p. 286, &c.) till 1816; but copies of Latin translations of it were known to exist in the Royal Library at Paris, and in the Bodleian at Oxford. Delambre has given an account of the contents of the Paris copy in his _Astron. Anc._ ii. 414, and in the _Connoissance des Temps_ for 1816; and Prof. Rigaud's account of the Oxford copy is given in the article _Optics_, in the _Encyclopædia Britannica_. Ptolemy shows great sagacity in applying the notion of Refraction to the explanation of the displacement of astronomical objects which is produced by the atmosphere,--_Astronomical Refraction_, as it is commonly called. He represents the visual ray as refracted in passing from the _ether_, which is above the air, into the air; the air being bounded

by a spherical surface which has for its centre "the centre of all the elements, the centre of the earth;" and the refraction being a flexure towards the line drawn perpendicular to this surface. He thus constructs, says Delambre, the same figure on which Cassini afterwards founded the whole of his theory; and gives a theory more complete than that of any astronomer previous to him. Tycho, for instance, believed that astronomical refraction was caused only by the _vapors_ of the atmosphere, and did not exist above the altitude of 45°.

Cleomedes, about the time of Augustus, had guessed at Refraction, as an explanation of an eclipse in which the sun and moon are both seen at the same time. "Is it not possible," he says, "that the ray which proceeds from the eye and traverses moist and cloudy air may bend downwards to the sun, even when he is below the horizon?" And Sextus Empiricus, a century later, says, "The air being dense, by the refraction of the visual ray, a constellation may be seen above the horizon when it is yet below the horizon." But from what follows, it {104} appears doubtful whether he clearly distinguished Refraction and Reflection.

In order that we may not attach too much value to the vague expressions of Cleomedes and Sextus Empiricus, we may remark that Cleomedes conceives such an eclipse as he describes not to be possible, though he offers an explanation of it if it be: (the fact must really occur whenever the moon is seen in the horizon in the middle of an eclipse:) and that Sextus Empiricus gives his suggestion of the effect of refraction as an argument why the Chaldean astrology cannot be true, since the constellation which appears to be rising at the moment of a birth is not the one which is truly rising. The Chaldeans might have answered, says Delambre, that the star begins to shed its influence, not when it is really in the horizon, but when its light is seen. (_Ast. Anc._ vol. i. p. 231, and vol. ii. p. 548.)

It has been said that Vitellio, or Vitello, whom we shall hereafter have to speak of in the history of Optics, took his Tables of Refractions from Ptolemy. This is contrary to what Delambre states. He says that Vitello may be accused of plagiarism from Alhazen, and that Alhazen did not borrow his Tables from Ptolemy. Roger Bacon had said (_Opus Majus_, p. 288), "Ptolemæus in libro de Opticis, id est, de Aspectibus, seu in Perspectivâ suâ, qui prius quam Alhazen dedit hanc sententiam, quam a Ptolemæo acceptam Alhazen exposuit." This refers only to the opinion that visual rays proceed from the eye. But this also is erroneous; for Alhazen maintains the contrary: "Visio fit radiis a visibili extrinsecus ad visum manantibus." (_Opt._ Lib. i. cap. 5.) Vitello says of his Table of Refractions, "Acceptis instrumentaliter, prout potuimus propinquius, angulis omnium refractionum . . . invenimus quod semper iidem sunt anguli refractionum: . . . secundum hoc fecimus has tabulas." "Having measured, by means of instruments, as exactly as we could, the whole range of the angles of refraction, we found that the refraction is always the same for the same angle; and hence we have constructed these Tables." {105}

## CHAPTER III.

EARLIEST STAGES OF HARMONICS.

AMONG the ancients, the science of Music was an application of Arithmetic, as Optics and Mechanics were of Geometry. The story which is told concerning the origin of their arithmetical music, is the following, as it stands in the Arithmetical Treatise of Nicomachus.

Pythagoras, walking one day, meditating on the means of measuring musical notes, happened to pass near a blacksmith's shop, and had his attention arrested by hearing the hammers, as they struck the anvil, produce the sounds which had a musical relation to each other. On listening further, he found that the intervals were a Fourth, a Fifth, and an Octave; and on weighing the hammers, it appeared that the one which gave the Octave was _one-half_ the heaviest, the one which gave the Fifth was _two-thirds_, and the one which gave the Fourth was _three-quarters_. He returned home, reflected upon this phenomenon, made trials, and finally discovered, that if he stretched musical strings of equal lengths, by weights which have the proportion of one-half, two-thirds, and three-fourths, they produced intervals which were an Octave, a Fifth, and a Fourth. This observation gave an arithmetical measure of the principal Musical Intervals, and made Music an arithmetical subject of speculation.

This story, if not entirely a philosophical fable, is undoubtedly inaccurate; for the musical intervals thus spoken of would not be produced by striking with hammers of the weights there stated. But it is true that the notes of strings have a definite relation to the forces which stretch them; and this truth is still the groundwork of the theory of musical concords and discords.

Nicomachus says that Pythagoras found the weights to be, as I have mentioned, in the proportion of 12, 6, 8, 9; and the intervals, an Octave, corresponding to the proportion 12 to 6, or 2 to 1; a Fifth, corresponding to the proportion 12 to 8, or 3 to 2; and a Fourth, corresponding to the proportion 12 to 9, or 4 to 3. There is no doubt that this statement of the ancient writer is inexact as to the physical fact, for the rate of vibration of a string, on which its note depends, is, {106} other things being equal, not as the weight, but as the square root of the weight. But he is right as to the essential point, that those ratios of 2 to 1, 3 to 2, and 4 to 3, are the characteristic ratios of the Octave, Fifth, and Fourth. In order to produce these intervals, the appended weights must be, not as 12, 9, 8, and 6, but as 12, 6¾, 5⅓, and 3.

The numerical relations of the other intervals of the musical scale, as well as of the Octave, Fifth, and Fourth, were discovered by the Greeks. Thus they found that the proportion in a Major Third was 5 to 4; in a Minor Third, 6 to 5; in a Major Tone, 9 to 8; in a Semitone or _Diesis_, 16 to 15. They even went so far as to determine the _Comma_, in which the interval of two notes is so small that they are in the proportion of 81 to 80. This is the interval between two notes, each of which may be called the Seventeenth above the key-note;--the one note being obtained by ascending a Fifth four times over; the other being obtained by ascending through two Octaves and a Major Third. The want of exact coincidence between these two notes is an inherent arithmetical imperfection in the musical scale, of which the consequences are very extensive.

The numerical properties of the musical scale were worked out to a very great extent by the Greeks, and many of their Treatises on this subject remain to us. The principal ones are the seven authors published by Meibomius.[2\2] These arithmetical elements of Music are to the present day important and fundamental portions of the Science of Harmonics.

[Note 2\2: _Antiquæ Musicæ Scriptores septem_, 1652.]

It may at first appear that the truth, or even the possibility of this history, by referring the discovery to accident, disproves our doctrine, that this, like all other fundamental discoveries, required a distinct and well-pondered Idea as its condition. In this, however, as in all cases of supposed accidental discoveries in science, it will be found, that it was exactly the possession of such an Idea which made the accident possible.

Pythagoras, assuming the truth of the tradition, must have had an exact and ready apprehension of those relations of musical sounds, which are called respectively an Octave, a Fifth, and a Fourth. If he had not been able to conceive distinctly this relation, and to apprehend it when heard, the sounds of the anvil would have struck his ears to no more purpose than they did those of the smiths themselves. He {107} must have had, too, a ready familiarity with numerical ratios; and, moreover (that in which, probably, his superiority most consisted), a disposition to connect one notion with the other--the musical relation with the arithmetical, if it were found possible. When the connection was once suggested, it was easy to devise experiments by which it might be confirmed.

"The philosophers of the Pythagorean School,[3\2] and in particular, Lasus of Hermione, and Hippasus of Metapontum, made many such experiments upon strings; varying both their lengths and the weights which stretched them; and also upon vessels filled with water, in a greater or less degree." And thus was established that connection of the Idea with the Fact, which this Science, like all others, requires.

[Note 3\2: Montucla, iii. 10.]

I shall quit the Physical Sciences of Ancient Greece, with the above brief statement of the discovery of the fundamental principles which they involved; not only because such initial steps must always be the most important in the progress of science, but because, in reality, the Greeks made no advances beyond these. There took place among them no additional inductive processes, by which new facts were brought under the dominion of principles, or by which principles were presented in a more comprehensive shape than before. Their advance terminated in a single stride. Archimedes had stirred the intellectual world, but had not put it in progressive motion: the science of Mechanics stopped where he left it. And though, in some objects, as in Harmonics, much was written, the works thus produced consisted of deductions from the fundamental principles, by means of arithmetical calculations; occasionally modified, indeed, by reference to the pleasures which music, as an art, affords, but not enriched by any new scientific truths.

[3d Ed.] We should, however, quit the philosophy of the ancient Greeks without a due sense of the obligations which Physical Science in all succeeding ages owes to

the acute and penetrating spirit in which their inquiries in that region of human knowledge were conducted, and to the large and lofty aspirations which were displayed, even in their failure, if we did not bear in mind both the multifarious and comprehensive character of their attempts, and some of the causes which limited their progress in positive science. They speculated and {108} theorized under a lively persuasion that a Science of every part of nature was possible, and was a fit object for the exercise of man's best faculties; and they were speedily led to the conviction that such a science must clothe its conclusions in the language of mathematics. This conviction is eminently conspicuous in the writings of Plato. In the _Republic_, in the _Epinomis_, and above all in the _Timæus_, this conviction makes him return, again and again, to a discussion of the laws which had been established or conjectured in his time, respecting Harmonics and Optics, such as we have seen, and still more, respecting Astronomy, such as we shall see in the next Book. Probably no succeeding step in the discovery of the Laws of Nature was of so much importance as the full adoption of this pervading conviction, that there must be Mathematical Laws of Nature, and that it is the business of Philosophy to discover these Laws. This conviction continues, through all the succeeding ages of the history of science, to be the animating and supporting principle of scientific investigation and discovery. And, especially in Astronomy, many of the erroneous guesses which the Greeks made, contain, if not the germ, at least the vivifying life-blood, of great truths, reserved for future ages.

Moreover, the Greeks not only sought such theories of special parts of nature, but a general Theory of the Universe. An essay at such a theory is the _Timæus_ of Plato; too wide and too ambitious an attempt to succeed at that time; or, indeed, on the scale on which he unfolds it, even in our time; but a vigorous and instructive example of the claim which man's Intellect feels that it may make to understand the universal frame of things, and to render a reason for all that is presented to it by the outward senses.

Further; we see in Plato, that one of the grounds of the failure in this attempt, was the assumption that the _reason why_ every thing is what it is and as it is, must be that so it is _best_, according to some view of better or worse attainable by man. Socrates, in his dying conversation, as given in the _Phædo_, declares this to have been what he sought in the philosophy of his time; and tells his friends that he turned away from the speculations of Anaxagoras because they did not give him such reasons for the constitution of the world; and Plato's _Timæus_ is, in reality, an attempt to supply this deficiency, and to present a Theory of the Universe, in which every thing is accounted for by such reasons. Though this is a failure, it is a noble as well as an instructive failure.

{{109}} BOOK III.

HISTORY OF GREEK ASTRONOMY.

Τόδε δὲ μηδείς ποτε φοβηθῇ τῶν Ἑλλήνων, ὡς οὐ χρὴ περὶ τὰ θεῖα ποτὲ πραγματεύεσθαι θνητοὺς ὄντας· πᾶν δε τούτου διανοηθῆναι τοὐναντίον, ὡς οὔτε ἄφρον ἔστι ποτὲ τὸ θεῖον, οὔτε ἀγνοεῖ που τὴν ἀνθρωπίνην φυσιν· ἀλλ' οἶδεν ὅτι,

διδάσκοντος αὐτοῦ, ξυνακολουθήσει καὶ μαθήσεται τὰ διδάσκομενα.--PLATO, _Epinomis_, p. 988.

Nor should any Greek have any misgiving of this kind; that it is not fitting for us to inquire narrowly into the operations of Superior Powers, such as those by which the motions of the heavenly bodies are produced: but, on the contrary, men should consider that the Divine Powers never act without purpose, and that they know the nature of man: they know that by their guidance and aid, man may follow and comprehend the lessons which are vouchsafed him on such subjects.

{{111}} INTRODUCTION.

THE earliest and fundamental conceptions of men respecting the objects with which Astronomy is concerned, are formed by familiar processes of thought, without appearing to have in them any thing technical or scientific. Days, Years, Months, the Sky, the Constellations, are notions which the most uncultured and incurious minds possess. Yet these are elements of the Science of Astronomy. The reasons why, in this case alone, of all the provinces of human knowledge, men were able, at an early and unenlightened period, to construct a science out of the obvious facts of observation, with the help of the common furniture of their minds, will be more apparent in the course of the philosophy of science: but I may here barely mention two of these reasons. They are, first, that the familiar act of thought, exercised for the common purposes of life, by which we give to an assemblage of our impressions such a unity as is implied in the above notions and terms, a Month, a Year, the Sky, and the like, is, in reality, an _inductive act_, and shares the nature of the processes by which all sciences are formed; and, in the next place, that the ideas appropriate to the induction in this case, are those which, even in the least cultivated minds, are very clear and definite; namely, the ideas of Space and Figure, Time and Number, Motion and Recurrence. Hence, from their first origin, the modifications of those ideas assume a scientific form.

We must now trace in detail the peculiar course which, in consequence of these causes, the knowledge of man respecting the heavenly bodies took, from the earliest period of his history. {112}

## CHAPTER I.

EARLIEST STAGES OF ASTRONOMY.

_Sect._ 1.--_Formation of the Notion of a Year._

THE notion of a _Day_ is early and obviously impressed upon man in almost any condition in which we can imagine him. The recurrence of light and darkness, of comparative warmth and cold, of noise and silence, of the activity and repose of animals;--the rising, mounting, descending, and setting of the sun;--the varying colors of the clouds, generally, notwithstanding their variety, marked by a daily progression of appearances;--the calls of the desire of food and of sleep in man himself, either exactly adjusted to the period of this change, or at least readily capable of being accommodated to it;--the recurrence of these circumstances at intervals, equal, so far

as our obvious judgment of the passage of time can decide; and these intervals so short that the repetition is noticed with no effort of attention or memory;--this assemblage of suggestions makes the notion of a Day necessarily occur to man, if we suppose him to have the conception of Time, and of Recurrence. He naturally marks by a term such a portion of time, and such a cycle of recurrence; he calls each portion of time, in which this series of appearances and occurrences come round, a _Day_; and such a group of particulars are considered as appearing or happening _in_ the same day.

_A Year_ is a notion formed in the same manner; implying in the same way the notion of recurring facts; and also the faculty of arranging facts in time, and of appreciating their recurrence. But the notion of a Year, though undoubtedly very obvious, is, on many accounts, less so than that of a Day. The repetition of similar circumstances, at equal intervals, is less manifest in this case, and the intervals being much longer, some exertion of memory becomes requisite in order that the recurrence may be perceived. A child might easily be persuaded that successive years were of unequal length; or, if the summer were cold, and the spring and autumn warm, might be made to believe, if all who spoke in its hearing agreed to support the delusion, that one year was two. It would be impossible to practise such a deception with regard to the day, without the use of some artifice beyond mere words. {113}

Still, the recurrence of the appearances which suggest the notion of a Year is so obvious, that we can hardly conceive man without it. But though, in all climes and times, there would be a recurrence, and at the same interval in all, the recurring appearances would be extremely different in different countries; and the contrasts and resemblances of the seasons would be widely varied. In some places the winter utterly alters the face of the country, converting grassy hills, deep leafy woods of various hues of green, and running waters, into snowy and icy wastes, and bare snow-laden branches; while in others, the field retains its herbage, and the tree its leaves, all the year; and the rains and the sunshine alone, or various agricultural employments quite different from ours, mark the passing seasons. Yet in all parts of the world the yearly cycle of changes has been singled out from all others, and designated by a peculiar name. The inhabitant of the equatorial regions has the sun vertically over him at the end of every period of six months, and similar trains of celestial phenomena fill up each of these intervals, yet we do not find years of six months among such nations. The Arabs alone,[1\3] who practise neither agriculture nor navigation, have a year depending upon the moon only; and borrow the word from other languages, when they speak of the solar year.

[Note 1\3: Ideler, _Berl. Trans._ 1813, p. 51.]

In general, nations have marked this portion of time by some word which has a reference to the returning circle of seasons and employments. Thus the Latin _annus_ signified a ring, as we see in the derivative _annulus_: the Greek term ἐνιαυτὸς implies something which _returns into itself_: and the word as it exists in Teutonic languages, of which our word _year_ is an example, is said to have its origin in the word _yra_ which means a ring in Swedish, and is perhaps connected with the Latin

_gyrus_.

_Sect._ 2.--_Fixation of the Civil Year._

THE year, considered as a recurring cycle of seasons and of general appearances, must attract the notice of man as soon as his attention and memory suffice to bind together the parts of a succession of the length of several years. But to make the same term imply a certain fixed number of days, we must know how many days the cycle of the seasons occupies; a knowledge which requires faculties and artifices beyond what we have already mentioned. For instance, men cannot reckon as far as any number at all approaching the number of days in the year, without possessing a system of numeral terms, and methods {114} of practical numeration on which such a system of terms is always founded.[2\3] The South American Indians, the Koussa Caffres and Hottentots, and the natives of New Holland, all of whom are said to be unable to reckon further than the fingers of their hands and feet,[3\3] cannot, as we do, include in their notion of a year the fact of its consisting of 365 days. This fact is not likely to be known to any nation except those which have advanced far beyond that which may be considered as the earliest scientific process which we can trace in the history of the human race, the formation of a method of designating the successive numbers to an indefinite extent, by means of names, framed according to the decimal, quinary, or vigenary scale.

[Note 2\3: _Arithmetic_ in _Encyc. Metrop._ (by Dr. Peacock), Art. 8.]

[Note 3\3: Ibid. Art. 32.]

But even if we suppose men to have the habit of recording the passage of each day, and of counting the score thus recorded, it would be by no means easy for them to determine the exact number of days in which the cycle of the seasons recurs; for the indefiniteness of the appearances which mark the same season of the year, and the changes to which they are subject as the seasons are early or late, would leave much uncertainty respecting the duration of the year. They would not obtain any accuracy on this head, till they had attended for a considerable time to the motions and places of the sun; circumstances which require more precision of notice than the general facts of the degrees of heat and light. The motions of the sun, the succession of the places of his rising and setting at different times of the year, the greatest heights which he reaches, the proportion of the length of day and night, would all exhibit several cycles. The turning back of the sun, when he had reached the greatest distance to the south or to the north, as shown either by his rising or by his height at noon, would perhaps be the most observable of such circumstances. Accordingly the τροπαὶ ἠελίοιο, the turnings of the sun, are used repeatedly by Hesiod as a mark from which he reckons the seasons of various employments. "Fifty days," he says, "after the turning of the sun, is a seasonable time for beginning a voyage."[4\3]

[Note 4\3: Ἤματα πεντήκοντα μετὰ τροπὰς ἠελίοιο Ἐς τέλος ἐλθόντος θέρεος.--_Op. et Dies_, 661.]

The phenomena would be different in different climates, but the recurrence would be common to all. Any one of these kinds of phenomena, noted with moderate care for a year, would show what was the number of days of which a year consisted; and if

several years {115} were included in the interval through which the scrutiny extended, the knowledge of the length of the year so acquired would be proportionally more exact.

Besides those notices of the sun which offered exact indications of the seasons, other more indefinite natural occurrences were used; as the arrival of the swallow (χελιδών) and the kite (ἰκτίν), The birds, in Aristophanes' play of that name, mention it as one of their offices to mark the seasons; Hesiod similarly notices the cry of the crane as an indication of the departure of winter.[5\3]

[Note 5\3: Ideler, i. 240.]

Among the Greeks the seasons were at first only summer and winter (θέρος and χειμών), the latter including all the rainy and cold portion of the year. The winter was then subdivided into the χειμών and ἔαρ (winter proper and spring), and the summer, less definitely, into θέρος and ὀπώρα (summer and autumn). Tacitus says that the Germans knew neither the blessings nor the name of autumn, "Autumni perinde nomen ac bona ignorantur." Yet _harvest_, _herbst_, is certainly an old German word.[6\3]

[Note 6\3: Ib. i. 243.]

In the same period in which the sun goes through his cycle of positions, the stars also go through a cycle of appearances belonging to them; and these appearances were perhaps employed at as early a period as those of the sun, in determining the exact length of the year. Many of the groups of fixed stars are readily recognized, as exhibiting always the same configuration; and particular bright stars are singled out as objects of attention. These are observed, at particular seasons, to appear in the west after sunset; but it is noted that when they do this, they are found nearer and nearer to the sun every successive evening, and at last disappear in his light. It is observed also, that at a certain interval after this, they rise visibly before the dawn of day renders the stars invisible; and after they are seen to do this, they rise every day at a longer interval before the sun. The risings and settings of the stars under these circumstances, or under others which are easily recognized, were, in countries where the sky is usually clear, employed at an early period to mark the seasons of the year. Eschylus[7\3] makes Prometheus mention this among the benefits of which {116} he, the teacher of arts to the earliest race of men, was the communicator.

[Note 7\3: Οὔκ ἦν γαρ αὐτοῖς οὔτε χείματος τέκμαρ, Οὔτ' ἀνθεμώδους ἦρος, οὔδε καρπίμου Θέρους βέβαιον· ἀλλ' ἄτερ γνώμης τὸ πᾶν Ἔπρασσον, ἔστε δή σφιν ἀνατολὰς ἐγὼ Ἄστρων ἔδειξα, τάς τε δυσκρίτους δύσεις.--_Prom. V._ 454.]

Thus, for instance, the rising[8\3] of the Pleiades in the evening was a mark of the approach of winter. The rising of the waters of the Nile in Egypt coincided with the heliacal rising of Sirius, which star the Egyptians called Sothis. Even without any artificial measure of time or position, it was not difficult to carry observations of this kind to such a degree of accuracy as to learn from them the number of days which compose the year; and to fix the precise season from the appearance of the stars.

[Note 8\3: Ideler (Chronol. i. 242) says that _this_ rising of the Pleiades took place at a time of the year which corresponds to our 11th May, and the setting to the 20th October; but this does not agree with the forty days of their being "concealed," which, from the context, must mean, I conceive, the interval between their setting and rising. Pliny, however, says, "Vergiliarum exortu æstas incipit, occasu hiems; _semestri_ spatio intra se messes vindemiasque et omnium maturitatem complexæ." (H. N. xviii. 69.)

The autumn of the Greeks, ὀπώρα, was earlier than our autumn, for Homer calls Sirius ἀστὴρ ὀπωρινός, which rose at the end of July.]

A knowledge concerning the stars appears to have been first cultivated with the last-mentioned view, and makes its first appearance in literature with this for its object. Thus Hesiod directs the husbandman when to reap by the rising, and when to plough by the setting of the Pleiades.[9\3] In like manner Sirius,[10\3] Arcturus,[11\3] the Hyades and Orion,[12\3] are noticed. {117}

[Note 9\3: Πληίαδων Ἀτλαγενέων ἐπιτελλομενάων. Ἄρχεσθ' ἀμητοῦ· ἀρότοιο δὲ, δυσομενάων. Αἵ δή τοι νύκτας τε καὶ ἤματα τεσσεράκοντα Κεκρύφαται, αὖτις δὲ περιπλομένου ἐνιαυτοῦ Φαίνονται. _Op. et Dies_, l. 381.]

[Note 10\3: Ib. l. 413.]

[Note 11\3: Εὖτ' ἂν δ' ἐξήκοντα μετὰ τροπὰς ἠελίοιο Χειμέρι', ἐκτελέσῃ Ζεὺς ἤματα, δή ῥα τότ' ἀστὴρ Ἀρκτοῦρος, προλιπὼν ἱερὸν ῥόον Ὠκεανοῖο Πρῶτον παμφαίνων ἐπιτέλλεται ἀκροκνέφαιος. _Op. et Dies_, l. 562.

Εὖτ' ἂν δ' Ὠρίων καὶ Σείριος ἐς μέσον ἔλθῃ Οὐρανὸν, Ἀρκτοῦρον δ' ἐσὶδῃ ῥοδοδάκτυλος ἠὼς. Ib. 607.]

[Note 12\3: . . . . . αὐτὰρ ἐπὴν δὴ Πληϊάδες Ὑάδες τε τὸ τε σθένος Ὠρίωνος Δύνωσιν. Ib. 612.

These methods were employed to a late period, because the Greek months, being lunar, did not correspond to the seasons. Tables of such motions were called παραπήγματα.--Ideler, _Hist. Untersuchungen_, p. 209.]

By such means it was determined that the year consisted, at least, nearly, of 365 days. The Egyptians, as we learn from Herodotus,[13\3] claimed the honor of this discovery. The priests informed him, he says, "that the Egyptians were the first men who discovered the year, dividing it into twelve equal parts; and this they asserted that they discovered from the stars." Each of these parts or months consisted of 30 days, and they added 5 days more at the end of the year, "and thus the circle of the seasons come round." It seems, also, that the Jews, at an early period, had a similar reckoning of time, for the Deluge which continued 150 days (Gen. vii. 24), is stated to have lasted from the 17th day of the second month (Gen. vii. 11) to the 17th day of the seventh month (Gen. viii. 4), that is, 5 months of 30 days.

[Note 13\3: Ib. ii. 4.]

A year thus settled as a period of a certain number of days is called a _Civil Year_. It is one of the earliest discoverable institutions of States possessing any germ of

civilization; and one of the earliest portions of human systematic knowledge is the discovery of the length of the civil year, so that it should agree with the natural year, or year of the seasons.

_Sect._ 3.--_Correction of the Civil Year._ (_Julian Calendar._)

IN reality, by such a mode of reckoning as we have described, the circle of the seasons would not come round exactly. The real length of the year is very nearly 365 days and a quarter. If a year of 365 days were used, in four years the year would begin a day too soon, when considered with reference to the sun and stars; and in 60 years it would begin 15 days too soon: a quantity perceptible to the loosest degree of attention. The civil year would be found not to coincide with the year of the seasons; the beginning of the former would take place at different periods of the latter; it would _wander_ into various seasons, instead of remaining fixed to the same season; the term _year_, and any number of years, would become ambiguous: some correction, at least some comparison, would be requisite.

We do not know by whom the insufficiency of the year of 365 days was first discovered;[14\3] we find this knowledge diffused among all civilized nations, and various artifices used in making the correction. The method which we employ, and which consists in reckoning an {118} additional day at the end of February every fourth or _leap_ year, is an example of the principle of _intercalation_, by which the correction was most commonly made. Methods of intercalation for the same purpose were found to exist in the new world. The Mexicans added 13 days at the end of every 52 years. The method of the Greeks was more complex (by means of the _octaëteris_ or cycle of 8 years); but it had the additional object of accommodating itself to the motions of the moon, and therefore must be treated of hereafter. The Egyptians, on the other hand, knowingly permitted their civil year to _wander_, at least so far as their religious observances were concerned. "They do not wish," says Geminus,[15\3] "the same sacrifices of the gods to be made perpetually at the same time of the year, but that they should go through all the seasons, so that the same feast may happen in summer and winter, in spring and autumn." The period in which any festival would thus pass through all the seasons of the year is 1461 years; for 1460 years of 365¼ days are equal to 1461 years of 365 days. This period of 1461 years is called the _Sothic_ Period, from Sothis, the name of the Dog-star, by which their _fixed_ year was determined; and for the same reason it is called the _Canicular_ Period.[16\3]

[Note 14\3: Syncellus (_Chronographia_, p. 123) says that according to the legend, it was King Aseth who first added the 5 additional days to 360, for the year, in the eighteenth century, B. C.]

[Note 15\3: _Uranol._ p. 33.]

[Note 16\3: Censorinus _de Die Natali_, c. 18.]

Other nations did not regulate their civil year by intercalation at short intervals, but rectified it by a _reform_ when this became necessary. The Persians are said to have added a month of 30 days every 120 years. The Roman calendar, at first very rude in its structure, was reformed by Numa, and was directed to be kept in order by the perpetual interposition of the augurs. This, however, was, from various causes,

not properly done; and the consequence was, that the reckoning fell into utter disorder, in which state it was found by Julius Cæsar, when he became dictator. By the advice of Sosigenes, he adopted the mode of intercalation of one day in 4 years, which we still retain; and in order to correct the derangement which had already been produced, he added 90 days to a year of the usual length, which thus became what was called _the year of confusion_. The _Julian Calendar_, thus reformed, came into use, January 1, B. C. 45.

_Sect._ 4.--_Attempts at the Fixation of the Month._

THE circle of changes through which the moon passes in about thirty days, is marked, in the earliest stages of language, by a word which implies the space of time which one such circle occupies; just {119} as the circle of changes of the seasons is designated by the word _year_. The lunar changes are, indeed, more obvious to the sense, and strike a more careless person, than the annual; the moon, when the sun is absent, is almost the sole natural object which attracts our notice; and we look at her with a far more tranquil and agreeable attention than we bestow on any other celestial object. Her changes of form and place are definite and striking to all eyes; they are uninterrupted, and the duration of their cycle is so short as to require no effort of memory to embrace it. Hence it appears to be more easy, and in earlier stages of civilization more common, to count time by _moons_ than by years.

The words by which this period of time is designated in various languages, seem to refer us to the early history of language. Our word _month_ is connected with the word _moon_, and a similar connection is noticeable in the other branches of the Teutonic. The Greek word μὴν in like manner is related to μήνη, which though not the common word for the moon, is found in Homer with that signification. The Latin word _mensis_ is probably connected with the same group.[17\3]

[Note 17\3: Cicero derives this word from the verb _to measure_: "quia _mensa_ spatia conficiunt, _menses_ nominantur;" and other etymologists, with similar views, connect the above-mentioned words with the Hebrew _manah_, to measure (with which the Arabic word _almanach_ is connected). Such a derivation would have some analogy with that of _annus_, &c., noticed above: but if we are to attempt to ascend to the earliest condition of language, we must conceive it probable that men would have a name for a most conspicuous visible object, _the moon_, before they would have a verb denoting the very abstract and general notion, _to measure_.]

The month is not any exact number of days, being more than 29, and less than 30. The latter number was first tried, for men more readily select numbers possessing some distinction of regularity. It existed for a long period in many countries. A very few months of 30 days, however, would suffice to derange the agreement between the days of the months and the moon's appearance. A little further trial would show that months of 29 and 30 days alternately, would preserve, for a considerable period, this agreement.

The Greeks adopted this calendar, and, in consequence, considered the days of their month as representing the changes of the moon: the last day of the month was called ἕνη καὶ νέα, "the old and new" as belonging to both the waning and the

reappearing moon:[18\3] and their {120} festivals and sacrifices, as determined by the calendar, were conceived to be necessarily connected with the same periods of the cycles of the sun and moon. "The laws and the oracles," says Geminus, "which directed that they should in sacrifices observe three things, months, days, years, were so understood." With this persuasion, a correct system of intercalation became a religious duty.

[Note 18\3: Aratus says of the moon, in a passage quoted by Geminus, p. 33: Αἴει δ' ἄλλοθεν ἄλλα παρακλίνουσα μετωπὰ Εἴρῃ, ὁποσταίη μήνος περιτέλλεται ἠὼς As still her shifting visage changing turns, By her we count the monthly round of morns.]

The above rule of alternate months of 29 and 30 days, supposes the length of the months 29 days and a half, which is not exactly the length of a lunar month. Accordingly the Months and the Moon were soon at variance. Aristophanes, in "The Clouds," makes the Moon complain of the disorder when the calendar was deranged.

Οὐκ ἄγειν τὰς ἡμέρας Οὐδὲν ὀρθῶς, ἀλλ' ἀνω τε καὶ κάτω κυδοιδοπᾶν Ὥστ' ἀπειλεῖν φησὶν αὐτῇ τοὺς θεοὺς ἑκάστοτε Ἠνίκ' ἂν ψευσθῶσι δείπνου κἀπίωσιν οἴκαδε Τῆς ἑορτῆς μὴ τυχόντες κατὰ λόγον τῶν ἡμερῶν. _Nubes_, 615-19.

CHORUS OF CLOUDS.

The Moon by us to you her greeting sends, But bids us say that she's an ill-used moon, And takes it much amiss that you should still Shuffle her days, and turn them topsy-turvy: And that the gods (who know their feast-days well) By your false count are sent home supperless, And scold and storm at her for your neglect.[19\3]

[Note 19\3: This passage is supposed by the commentators to be intended as a satire upon those who had introduced the cycle of Meton (spoken of in Sect. 5), which had been done at Athens a few years before "The Clouds" was acted.]

The correction of this inaccuracy, however, was not pursued separately, but was combined with another object, the securing a correspondence between the lunar and solar years, the main purpose of all early cycles.

_Sect._ 5.--_Invention of Lunisolar Years._

THERE are 12 complete lunations in a year; which according to the above rule (of 29½ days to a lunation) would make 354 days, leaving 12¼ days of difference between such a lunar year and a solar year. It is said that, at an early period, this was attempted to be corrected by interpolating a month of 30 days every alternate year; and Herodotus[20\3] relates a conversation of Solon, implying a still ruder mode of {121} intercalation. This can hardly be considered as an improvement in the Greek calendar already described.

[Note 20\3: B. i. c. 15.]

The first cycle which produced any near correspondence of the reckoning of the moon and the sun, was the _Octaëteris_, or period of 8 years: 8 years of 354 days, together with 3 months of 30 days each, making up (in 99 lunations) 2922 days; which is exactly the amount of 8 years of 365¼ days each. Hence this period would answer its purpose, so far as the above lengths of the lunar and solar cycles are

exact; and it might assume various forms, according to the manner in which the three intercalary months were distributed. The customary method was to add a thirteenth month at the end of the third, fifth, and eighth year of the cycle. This period is ascribed to various persons and times; probably different persons proposed different forms of it. Dodwell places its introduction in the 59th Olympiad, or in the 6th century, B. C.: but Ideler thinks the astronomical knowledge of the Greeks of that age was too limited to allow of such a discovery.

This cycle, however, was imperfect. The duration of 99 lunations is something more than 2922 days; it is more nearly 2923½; hence in 16 years there was a deficiency of 3 days, with regard to the motions of the moon. This cycle of 16 years (_Heccædecaëteris_), with 3 interpolated days at the end, was used, it is said, to bring the calculation right with regard to the moon; but in this way the origin of the year was displaced with regard to the sun. After 10 revolutions of this cycle, or 160 years, the interpolated days would amount to 30, and hence the end of the lunar year would be a month in advance of the end of the solar. By terminating the lunar year at the end of the preceding month, the two years would again be brought into agreement: and we have thus a cycle of 160 years.[21\3]

[Note 21\3: Geminus. Ideler.]

This cycle of 160 years, however, was calculated from the cycle of 16 years; and it was probably never used in civil reckoning; which the others, or at least that of 8 years, appear to have been.

The cycles of 16 and 160 years were corrections of the cycle of 8 years; and were readily suggested, when the length of the solar and lunar periods became known with accuracy. But a much more exact cycle, independent of these, was discovered and introduced by Meton,[22\3] 432 years B. C. This cycle consisted of 19 years, and is so correct and convenient, that it is in use among ourselves to this day. The time occupied by 19 years, and by 235 lunations, is very nearly the same; {122} (the former time is less than 6940 days by 9½ hours, the latter, by 7½ hours). Hence, if the 19 years be divided into 235 months, so as to agree with the changes of the moon, at the end of that period the same succession may begin again with great exactness.

[Note 22\3: Ideler, _Hist. Unters._ p. 208.]

In order that 235 months, of 30 and 29 days, may make up 6940 days, we must have 125 of the former, which were called _full_ months, and 110 of the latter, which were termed _hollow_. An artifice was used in order to distribute 110 hollow months among 6940 days. It will be found that there is a hollow month for each 63 days nearly. Hence if we reckon 30 days to every month, but at every 63d day leap over a day in the reckoning, we shall, in the 19 years, omit 110 days; and this accordingly was done. Thus the 3d day of the 3d month, the 6th day of the 5th month, the 9th day of the 7th, must be omitted, so as to make these months "hollow." Of the 19 years, seven must consist of 13 months; and it does not appear to be known according to what order these seven years were selected. Some say they were the 3d, 6th, 8th, 11th, 14th, 17th, and 19th; others, the 3d, 5th, 8th, 11th, 13th, 16th, and 19th.

The near coincidence of the solar and lunar periods in this cycle of 19 years, was undoubtedly a considerable discovery at the time when it was first accomplished. It is not easy to trace the way in which such a discovery was made at that time; for we do not even know the manner in which men then recorded the agreement or difference between the calendar day and the celestial phenomenon which ought to correspond to it. It is most probable that the length of the month was obtained with some exactness by the observation of eclipses, at considerable intervals of time from each other; for eclipses are very noticeable phenomena, and must have been very soon observed to occur only at new and full moon.[23\3]

[Note 23\3: Thucyd. vii. 50. Ἡ σελήνη ἐκλείπει· ἐτύγχανε γὰρ _πανσέληνος_ οὖσα. iv. 52, Τοῦ ἡλίου ἐκλιπές τι ἐγένετο _περὶ νουμηνίαν_. ii. 28. Νουμηνίᾳ κατὰ _σελήνην_ (ὥσπερ καὶ μόνον δοκεῖ εἶναι γίγνεσθαι δυνατὸν) ὁ ἡλίος ἐξέλιπε μετὰ μεσημβρίαν καὶ πάλιν ἀν ἐπληρώθη, γενόμενος μηνοειδὴς καὶ ἀστέρων τινῶν ἐκφανέντων.]

The exact length of a certain number of months being thus known, the discovery of a cycle which should regulate the calendar with sufficient accuracy would be a business of arithmetical skill, and would depend, in part, on the existing knowledge of arithmetical methods; but in making the discovery, a natural arithmetical sagacity was probably more efficacious than method. It is very possible that the _Cycle of Meton_ is correct more nearly than its author was aware, and {123} nearly than he could ascertain from any evidence and calculation known to him. It is so exact that it is still used in calculating the new moon for the time of Easter; and the _Golden Number_, which is spoken of in stating such rules, is the number of this Cycle corresponding to the current year.[24\3]

[Note 24\3: The same cycle of 19 years has been used by the Chinese for a very great length of time; their civil year consisting, like that of the Greeks, of months of 29 and 30 days. The Siamese also have this period. (_Astron._ Lib. U. K.)]

Meton's Cycle was corrected a hundred years later (330 B. C.), by Calippus, who discovered the error of it by observing an eclipse of the moon six years before the death of Alexander.[25\3] In this corrected period, four cycles of 19 years were taken, and a day left out at the end of the 76 years, in order to make allowance for the hours by which, as already observed, 6940 days are greater than 19 years, and than 235 lunations: and this _Calippic period_ is used in Ptolemy's Almagest, in stating observations of eclipses.

[Note 25\3: Delamb. _A. A._ p. 17.]

The Metonic and Calippic periods undoubtedly imply a very considerable degree of accuracy in the knowledge which the astronomers, to whom they are due, had of the length of the month; and the first is a very happy invention for bringing the solar and lunar calendars into agreement.

The Roman Calendar, from which our own is derived, appears to have been a much less skilful contrivance than the Greek; though scholars are not agreed on the subject of its construction, we can hardly doubt that months, in this as in other cases, were intended originally to have a reference to the moon. In whatever manner the solar

and lunar motions were intended to be reconciled, the attempt seems altogether to have failed, and to have been soon abandoned. The Roman months, both before and after the Julian correction, were portions of the year, having no reference to full and new moons; and we, having adopted this division of the year, have thus, in our common calendar, the traces of one of the early attempts of mankind to seize the law of the succession of celestial phenomena, in a case where the attempt was a complete failure.

Considered as a part of the progress of our astronomical knowledge, improvements in the calendar do not offer many points to our observation, but they exhibit a few very important steps. Calendars which, belonging apparently to unscientific ages and nations, possess a great degree of accordance with the true motions of the sun and moon (like {124} the solar calendar of the Mexicans, and the lunar calendar of the Greeks), contain the only record now extant of discoveries which must have required a great deal of observation, of thought, and probably of time. The later improvements in calendars, which take place when astronomical observation has been attentively pursued, are of little consequence to the history of science; for they are generally founded on astronomical determinations, and are posterior in time, and inferior in accuracy, to the knowledge on which they depend. But cycles of correction, which are both short and close to exactness, like that of Meton, may perhaps be the original form of the knowledge which they imply; and certainly require both accurate facts and sagacious arithmetical reasonings. The discovery of such a cycle must always have the appearance of a happy guess, like other discoveries of laws of nature. Beyond this point, the interest of the study of calendars, as bearing on our subject, ceases: they may be considered as belonging rather to Art than to Science; rather as an application of a part of our knowledge to the uses of life, than a means or an evidence of its extension.

_Sect._ 6.--_The Constellations._

SOME tendency to consider the stars as formed into groups, is inevitable when men begin to attend to them; but how men were led to the fanciful system of names of Stars and of Constellations, which we find to have prevailed in early times, it is very difficult to determine. Single stars, and very close groups, as the Pleiades, were named in the time of Homer and Hesiod, and at a still earlier period, as we find in the book of Job.[26\3]

[Note 26\3: Job xxxviii. 31. "Canst thou bind the sweet influences of Chima (the Pleiades), or loose the bands of Kesil (Orion)? Canst thou bring forth Mazzaroth (Sirius) in his season? or canst thou guide Ash (or Aisch) (Arcturus) with his sons?"

And ix. 9. "Which maketh Arcturus, Orion, and Pleiades, and the chambers of the south."

Dupuis, vi. 545, thinks that Aisch was αἴξ, the goat and kids. See Hyde, _Ulughbeigh_.]

Two remarkable circumstances with respect to the Constellations are, first, that they appear in most cases to be arbitrary combinations; the artificial figures which are made to include the stars, not having any resemblance to their obvious

configurations; and second, that these figures, in different countries, are so far similar, as to imply some communication. The arbitrary nature of these figures shows that they {125} were rather the work of the imaginative and mythological tendencies of man, than of mere convenience and love of arrangement. "The constellations," says an astronomer of our own time,[27\3] "seem to have been almost purposely named and delineated to cause as much confusion and inconvenience as possible. Innumerable snakes twine through long and contorted areas of the heavens, where no memory can follow them: bears, lions, and fishes, large and small, northern and southern, confuse all nomenclature. A better system of constellations might have been a material help as an artificial memory." When men indicate the stars by figures, borrowed from obvious resemblances, they are led to combinations quite different from the received constellations. Thus the common people in our own country find a wain or wagon, or a plough, in a portion of the great bear.[28\3]

[Note 27\3: Sir J. Herschel.]

[Note 28\3: So also the Greeks, Homer, _Il._ XVIII. 487. Ἄρκτον ἣν καὶ ἅμαξαν ἐπίκλησιν καλέουσιν. The Northern Bear which oft the Wain they call. Ἄρκτος was the traditional name; ἅμαξα, that suggested by the form.]

The similarity of the constellations recognized in different countries is very remarkable. The Chaldean, the Egyptian, and the Grecian skies have a resemblance which cannot be overlooked. Some have conceived that this resemblance may be traced also in the Indian and Arabic constellations, at least in those of the zodiac.[29\3] But while the figures are the same, the names and traditions connected with them are different, according to the histories and localities of each country;[30\3] the river among the stars which the Greeks called the Eridanus, the Egyptians asserted to be the Nile. Some conceive that the Signs of the _Zodiac_, or path along which the sun and moon pass, had its divisions marked by signs which had a reference to the course of the seasons, to the motion of the sun, or the employments of the husbandman. If we take the position of the heavens, which, from the knowledge we now possess, we are sure they must have had 15,000 years ago, the significance of the signs of the zodiac, in which the sun was, as referred to the Egyptian year, becomes very marked,[31\3] and has led some to suppose that the zodiac was invented at such a period. Others have rejected this as an improbably great antiquity, and have thought it more likely that the constellation assigned to each season was that which, at that season, rose at the beginning of the night: {126} thus the balance (which is conceived to designate the equality of days and nights) was placed among the stars which rose in the evening when the spring began: this would fix the origin of these signs 2500 years before our era.

[Note 29\3: Dupuis, vi. 548. The Indian zodiac contains, in the place of our Capricorn, a ram _and_ a fish, which proves the resemblance without chance of mistake. Bailly, i. p. 157.]

[Note 30\3: Dupuis, vi. 549.]

[Note 31\3: Laplace, _Hist. Astron._ p. 8.]

It is clear, as has already been said, that Fancy, and probably Superstition, had a share in forming the collection of constellations. It is certain that, at an early period, superstitious notions were associated with the stars.[32\3] Astrology is of very high antiquity in the East. The stars were supposed to influence the character and destiny of man, and to be in some way connected with superior natures and powers.

[Note 32\3: Dupuis, vi. 546.]

We may, I conceive, look upon the formation of the constellations, and the notions thus connected with them, as a very early attempt to find a meaning in the relations of the stars; and as an utter failure. The first effort to associate the appearances and motions of the skies by conceptions implying unity and connection, was made in a wrong direction, as may very easily be supposed. Instead of considering the appearances only with reference to space, time, number, in a manner purely rational, a number of other elements, imagination, tradition, hope, fear, awe of the supernatural, belief in destiny, were called into action. Man, still young, as a philosopher at least, had yet to learn what notions his successful guesses on these subjects must involve, and what they must exclude. At that period, nothing could be more natural or excusable than this ignorance; but it is curious to see how long and how obstinately the belief lingered (if indeed it be yet extinct) that the motions of the stars, and the dispositions and fortunes of men, may come under some common conceptions and laws, by which a connection between the one and the other may be established.

We cannot, therefore, agree with those who consider Astrology in the early ages as "only a degraded Astronomy, the abuse of a more ancient science."[33\3] It was the first step to astronomy by leading to habits and means of grouping phenomena; and, after a while, by showing that pictorial and mythological relations among the stars had no very obvious value. From that time, the inductive process went on steadily in the true road, under the guidance of ideas of space, time, and number.

[Note 33\3: Ib. vi. 546.]

_Sect._ 7.--_The Planets._

WHILE men were becoming familiar with the fixed stars, the planets must have attracted their notice. Venus, from her brightness, and {127} from her accompanying the sun at no great distance, and thus appearing as the morning and evening star, was very conspicuous. Pythagoras is said to have maintained that the evening and morning star are the same body, which certainly must have been one of the earliest discoveries on this subject; and indeed we can hardly conceive men noticing the stars for a year or two without coming to this conclusion.

Jupiter and Mars, sometimes still brighter than Venus, were also very noticeable. Saturn and Mercury were less so, but in fine climates they and their motion would soon be detected by persons observant of the heavens. To reduce to any rule the movements of these luminaries must have taken time and thought; probably before this was done, certainly very early, these heavenly bodies were brought more peculiarly under those views which we have noticed as leading to astrology.

At a time beyond the reach of certain history, the planets, along with the sun and moon, had been arranged in a certain recognized order by the Egyptians or some other ancient nation. Probably this arrangement had been made according to the slowness of their motions among the stars; for though the motion of each is very variable, the gradation of their velocities is, on the whole, very manifest; and the different rate of travelling of the different planets, and probably other circumstances of difference, led, in the ready fancy of early times, to the attribution of a peculiar character to each luminary. Thus Saturn was held to be of a cold and gelid nature; Jupiter, who, from his more rapid motion, was supposed to be lower in place, was temperate; Mars, fiery, and the like.[34\3]

[Note 34\3: Achilles Tatius (_Uranol._ pp. 135, 136), gives the Grecian and Egyptian names of the planets. Egyptian. Greek. Saturn Νεμεσέως Κρόνου ἀστὴρ φαίνων Jupiter Ὸσίριδος Δῖος φαέθων Mars Ἡρακλεοῦς Ἀρέος πυρόεις Venus Ἀφροδίτης ἑώσφορος Mercury Ἀπόλλωνος Ἑρμοῦ στίλβων]

It is not necessary to dwell on the details of these speculations, but we may notice a very remarkable evidence of their antiquity and generality in the structure of one of the most familiar of our measures of time, the _Week_. This distribution of time according to periods of seven days, comes down to us, as we learn from the Jewish scriptures, from the beginning of man's existence on the earth. The same usage is found over all the East; it existed among the Arabians, Assyrians, {128} Egyptians.[35\3] The same week is found in India among the Bramins; it has there, also, its days marked by those of the heavenly bodies; and it has been ascertained that the same day has, in that country, the name corresponding with its designation in other nations.

[Note 35\3: Laplace, _Hist. Astron._ p. 16.]

The notion which led to the usual designations of the days of the week is not easily unravelled. The days each correspond to one of the heavenly bodies, which were, in the earliest systems of the world, conceived to be the following, enumerating them in the order of their remoteness from the earth:[36\3] Saturn, Jupiter, Mars, the Sun, Venus, Mercury, the Moon. At a later period, the received systems placed the seven luminaries in _the seven spheres_. The knowledge which was implied in this view, and the time when it was obtained, we must consider hereafter. The order in which the names are assigned to the days of the week (beginning with Saturday) is, Saturn, the Sun, the Moon, Mars, Mercury, Jupiter, Venus; and various accounts are given of the manner in which one of these orders is obtained from the other; all the methods proceeding upon certain arbitrary arithmetical processes, connected in some way with astrological views. It is perhaps not worth our while here to examine further the steps of this process; it would be difficult to determine with certainty why the former order of the planets was adopted, and how and why the latter was deduced from it. But there is something very remarkable in the universality of the notions, apparently so fantastic, which have produced this result; and we may probably consider the Week, with Laplace,[37\3] as "the most ancient monument of astronomical knowledge." This period has gone on without interruption or irregularity from the earliest recorded times to our own days, traversing the extent of ages and the

revolutions of empires; the names of the ancient deities which were associated with the stars have been replaced by those of the objects of the worship of our Teutonic ancestors, according to their views of the correspondence of the two mythologies; and the Quakers, in rejecting these names of days, have cast aside the most ancient existing relic of astrological as well as idolatrous superstition.

[Note 36\3: _Philol. Mus._ No. 1.]

[Note 37\3: _Hist. Ast._ p. 17.]

_Sect._ 8.--_The Circles of the Sphere._

THE inventions hitherto noticed, though undoubtedly they were steps in astronomical knowledge, can hardly be considered as purely abstract and scientific speculations; for the exact reckoning of time is one of {129} the wants, even of the least civilized nations. But the distribution of the places and motions of the heavenly bodies by means of a celestial sphere with imaginary lines drawn upon it, is a step in _speculative_ astronomy, and was occasioned and rendered important by the scientific propensities of man.

It is not easy to say with whom this notion originated. Some parts of it are obvious. The appearance of the sky naturally suggests the idea of a concave Sphere, with the stars fixed on its surface. Their motions during any one night, it would be readily seen, might be represented by supposing this Sphere to turn round a Pole or Axis; for there is a conspicuous star in the heavens which apparently stands still (the Pole-star); all the others travel round this in circles, and keep the same positions with respect to each other. This stationary star is every night the same, and in the same place; the other stars also have the same relative position; but their general position at the same time of night varies gradually from night to night, so as to go through its cycle of appearances once a year. All this would obviously agree with the supposition that the sky is a concave sphere or dome, that the stars have fixed places on this sphere, and that it revolves perpetually and uniformly about the Pole or fixed point.

But this supposition does not at all explain the way in which the appearances of different nights succeed each other. This, however, may be explained, it appears, by supposing the _sun_ also _to move among the stars_ on the surface of the concave sphere. The sun by his brightness makes the stars invisible which are on his side of the heavens: this we can easily believe; for the moon, when bright, also puts out all but the largest stars; and we see the stars appearing in the evening, each in its place, according to their degree of splendor, as fast as the declining light of day allows them to become visible. And as the sun brings day, and his absence night, if he move through the circuit of the stars in a year, we shall have, in the course of that time, every part of the starry sphere in succession presented to us as our nocturnal sky.

This notion, _that the sun moves round among the stars in a year_, is the basis of astronomy, and a considerable part of the science is only the development and particularization of this general conception. It is not easy to ascertain either the exact method by which the path of the sun among the stars was determined, or the author and date of the discovery. That there is some difficulty in tracing the course of the sun among the stars will be clearly seen, when it is considered that no {130} star can

ever be seen at the same time with the sun. If the whole circuit of the sky be divided into twelve parts or _signs_, it is estimated by Autolycus, the oldest writer on these subjects whose works remain to us,[38\3] that the stars which occupy one of these parts are absorbed by the solar rays, so that they cannot be seen. Hence the stars which are seen nearest to the place of the setting and the rising sun in the evening and in the morning, are distant from him by the half of a sign: the evening stars being to the west, and the morning stars to the east of him. If the observer had previously obtained a knowledge of the places of all the principal stars, he might in this way determine the position of the sun each night, and thus trace his path in a year.

[Note 38\3: Delamb. _A. A._ p. xiii.]

In this, or some such way, the sun's path was determined by the early astronomers of Egypt. Thales, who is mentioned as the father of Greek astronomy, probably learnt among the Egyptians the results of such speculations, and introduced them into his own country. His knowledge, indeed, must have been a great deal more advanced than that which we are now describing, if it be true, as is asserted, that he predicted an eclipse. But his having done so is not very consistent with what we are told of the steps which his successors had still to make.

The Circle of the Signs, in which the sun moves among the stars, is obliquely situated with regard to the circles in which the stars move about the poles. Pliny[39\3] states that Anaximander,[40\3] a scholar of Thales, was the first person who pointed out this obliquity, and thus, as he says, "opened the gate of nature." Certainly, the person who first had a clear view of the nature of the sun's path in the celestial sphere, made that step which led to all the rest; but it is difficult to conceive that the Egyptians and Chaldeans had not already advanced so far.

[Note 39\3: Lib. ii. c. (viii.)]

[Note 40\3: Plutarch, _De Plac. Phil._ lib. ii. cap. xii. says Pythagoras was the author of this discovery.]

The diurnal motion of the celestial sphere, and the motion of the moon in the circle of the signs, gave rise to a mathematical science, _the Doctrine of the Sphere_, which was one of the earliest branches of applied mathematics. A number of technical conceptions and terms were soon introduced. The _Sphere_ of the heavens was conceived to be complete, though we see but a part of it; it was supposed to turn about the visible _pole_ and another pole opposite to this, and these poles were connected by an imaginary _Axis_. The circle which divided the sphere exactly midway between these poles was called the _Equator_ (ἰσημέρινος). {131} The two circles parallel to this which bounded the sun's path among the stars were called _Tropics_ (τροπικαί), because the sun _turns_ back again towards the equator when he reaches them. The stars which never set are bounded by a circle called the _Arctic Circle_ (ἄρκτικος, from ἄρκτος, the Bear, the constellation to which some of the principal stars within that circle belong.) A circle about the opposite pole is called _Antarctic_, and the stars which are within it can never rise to us.[41\3] The sun's path or circle of the signs is called the _Zodiac_, or circle of animals; the points where this circle meets the equator are the _Equinoctial Points_, the days and nights being

equal when the sun is in them; the _Solstitial Points_ are those where the sun's path touches the tropics; his motion to the south or to the north ceases when he is there, and he appears in that respect to stand still. The _Colures_ (κόλουροι, mutilated) are circles which pass through the poles and through the equinoctial and solstitial points; they have their name because they are only visible in part, a portion of them being below the horizon.

[Note 41\3: The Arctic and Antarctic Circles of modern astronomers are different from these.]

The _Horizon_ (ὁρίζων) is commonly understood as the boundary of the visible earth and heaven. In the doctrine of the sphere, this boundary is _a great circle_, that is, a circle of which the plane passes through the centre of the sphere; and, therefore, an entire hemisphere is always above the horizon. The term occurs for the first time in the work of Euclid, called _Phænomena_ (Φαινόμενα). We possess two treatises written by Autolycus[42\3] (who lived about 300 B. C.) which trace _deductively_ the results of the doctrine of the sphere. Supposing its diurnal motion to be uniform, in a work entitled Περὶ Κινουμένης Σφαῖρας, "On the Moving Sphere," he demonstrates various properties of the diurnal risings, settings, and motions of the stars. In another work, Περὶ Ἐπιτολῶν καὶ Δύσεων, "On Risings and Settings,"[43\3] _tacitly_ assuming the sun's motion in his circle to be uniform, he proves certain propositions, with regard to those risings and settings of the stars, which take place at the same time when the sun rises and sets,[44\3] or _vice versâ_;[45\3] and also their _apparent_ risings and settings when they cease to be visible after sunset, or begin to be visible after sunrise.[46\3] {132} Several of the propositions contained in the former of these treatises are still necessary to be understood, as fundamental parts of astronomy.

[Note 42\3: Delambre, _Astron. Ancienne_, p. 19.]

[Note 43\3: Delambre, _Astron. Anc._ p. 25.]

[Note 44\3: _Cosmical_ rising and setting.]

[Note 45\3: _Acronycal_ rising and setting; (ἀκρονυκίος, happening at the extremity of the night.)]

[Note 46\3: _Heliacal_ rising and setting.]

The work of Euclid, just mentioned, is of the same kind. Delambre[47\3] finds in it evidence that Euclid was merely a book-astronomer, who had never observed the heavens.

[Note 47\3: _Ast. Anc._ p. 53.]

We may here remark the first instance of that which we shall find abundantly illustrated in every part of the history of science; that man is _prone_ to become a deductive reasoner;--that as soon as he obtains principles which can be traced to details by logical consequence, he sets about forming a body of science, by making a system of such reasonings. Geometry has always been a favorite mode of exercising this propensity: and that science, along with Trigonometry, Plane and Spherical, to

which the early problems of astronomy gave rise, have, up to the present day, been a constant field for the exercise of mathematical ingenuity; a few simple astronomical truths being assumed as the basis of the reasoning.

_Sect._ 9.--_The Globular Form of the Earth._

THE establishment of the globular form of the earth is an important step in astronomy, for it is the first of those convictions, directly opposed to the apparent evidence of the senses, which astronomy irresistibly proves. To make men believe that _up_ and _down_ are different directions in different places; that the sea, which seems so level, is, in fact, convex; that the earth, which appears to rest on a solid foundation, is, in fact, not supported at all; are great triumphs both of the power of discovering and the power of convincing. We may readily allow this, when we recollect how recently the doctrine of the _antipodes_, or the existence of inhabitants of the earth, who stand on the opposite side of it, with their feet turned towards ours, was considered both monstrous and heretical.

Yet the different positions of the horizon at different places, necessarily led the student of spherical astronomy towards this notion of the earth as a round body. Anaximander[48\3] is said by some to have held the earth to be globular, and to be detached or suspended; he is also stated to have constructed a sphere, on which were shown the extent of land and water. As, however, we do not know the arguments upon which he maintained the earth's globular form, we cannot judge of the {133} value of his opinion; it may have been no better founded than a different opinion ascribed to him by Laertius, that the earth had the shape of a pillar. Probably, the authors of the doctrine of the globular form of the earth were led to it, as we have said, by observing the different height of the pole at different places. They would find that the space which they passed over from north to south on the earth, was proportional to the change of place of the horizon in the celestial sphere; and as the horizon is, at every place, in the direction of the earth's apparently level surface, this observation would naturally suggest to them the opinion that the earth is placed within the celestial sphere, as a small globe in the middle of a much larger one.

[Note 48\3: See Brucker, _Hist. Phil._ vol. i. p. 486.]

We find this doctrine so distinctly insisted on by Aristotle, that we may almost look on him as the establisher of it.[49\3] "As to the figure of the earth, it must necessarily be spherical." This he proves, first by the tendency of things, in all places, downwards. He then adds,[50\3] "And, moreover, from the phenomena according to the sense: for if it were not so, the eclipses of the moon would not have such sections as they have. For in the configurations in the course of a month, the deficient part takes all different shapes; it is straight, and concave, and convex; but in eclipses it always has the line of division convex; wherefore, since the moon is eclipsed in consequence of the interposition of the earth, the periphery of the earth must be the cause of this by having a spherical form. And again, from the appearances of the stars, it is clear, not only that the earth is round, but that its size is not very large: for when we make a small removal to the south or the north, the circle of the horizon becomes palpably different, so that the stars overhead undergo a great change, and

are not the same to those that travel to the north and to the south. For some stars are seen in Egypt or at Cyprus, but are not seen in the countries to the north of these; and the stars that in the north are visible while they make a complete circuit, there undergo a setting. So that from this it is manifest, not only that the form of the earth is round, but also that it is a part of not a very large sphere: for otherwise the difference would not be so obvious to persons making so small a change of place. Wherefore we may judge that those persons _who connect the region in the neighborhood of the pillars of Hercules with that towards India, and who assert that in this way the sea is_ ONE, do not assert things very improbable. They confirm this conjecture moreover by the {134} elephants, which are said to be of the same species (γένος) towards each extreme; as if this circumstance was a consequence of the conjunction of the extremes. The mathematicians, who try to calculate the measure of the circumference, make it amount to 400,000 stadia; whence we collect that the earth is not only spherical, but is not large compared with the magnitude of the other stars."

[Note 49\3: Arist. _de Cœlo_, lib. ii. cap. xiv. ed. Casaub. p. 290.]

[Note 50\3: p. 291 C.]

When this notion was once suggested, it was defended and confirmed by such arguments as we find in later writers: for instance,[51\3] that the tendency of all things was to fall to the place of heavy bodies, and that this place being the centre of the earth, the whole earth had no such tendency; that the inequalities on the surface were so small as not materially to affect the shape of so vast a mass; that drops of water naturally form themselves into figures with a convex surface; that the end of the ocean would fall if it were not rounded off; that we see ships, when they go out to sea, disappearing downwards, which shows the surface to be convex. These are the arguments still employed in impressing the doctrines of astronomy upon the student of our own days; and thus we find that, even at the early period of which we are now speaking, truths had begun to accumulate which form a part of our present treasures.

[Note 51\3: Pliny, _Nat. Hist._ ii. LXV.]

_Sect._ 10.--_The Phases of the Moon._

WHEN men had formed a steady notion of the Moon as a solid body, revolving about the earth, they had only further to conceive it spherical, and to suppose the sun to be beyond the region of the moon, and they would find that they had obtained an explanation of the varying forms which the bright part of the moon assumes in the course of a month. For the convex side of the crescent-moon, and her full edge when she is gibbous, are always turned towards the sun. And this explanation, once suggested, would be confirmed, the more it was examined. For instance, if there be near us a spherical stone, on which the sun is shining, and if we place ourselves so that this stone and the moon are seen in the same direction (the moon appearing just over the top of the stone), we shall find that the visible part of the stone, which is then illuminated by the sun, is exactly similar in form to the moon, at whatever period of her changes she may be. The stone and the moon being in the same position with respect to us, and both being enlightened by the sun, the bright parts are the same in

figure; {135} the only difference is, that the dark part of the moon is usually not visible at all.

This doctrine is ascribed to Anaximander. Aristotle was fully aware of it.[52\3] It could not well escape the Chaldeans and Egyptians, if they speculated at all about the causes of the appearances in the heavens.

[Note 52\3: Probl. Cap. XV. Art. 7.]

_Sect._ 11.--_Eclipses._

ECLIPSES of the sun and moon were from the earliest tunes regarded with a peculiar interest. The notions of superhuman influences and relations, which, as we have seen, were associated with the luminaries of the sky, made men look with alarm at any sudden and striking change in those objects; and as the constant and steady course of the celestial revolutions was contemplated with a feeling of admiration and awe, any marked interruption and deviation in this course, was regarded with surprise and terror. This appears to be the case with all nations at an early stage of their civilization.

This impression would cause Eclipses to be noted and remembered; and accordingly we find that the records of Eclipses are the earliest astronomical information which we possess. When men had discovered some of the laws of succession of other astronomical phenomena, for instance, of the usual appearances of the moon and sun, it might then occur to them that these unusual appearances also might probably be governed by some rule.

The search after this rule was successful at an early period. The Chaldeans were able to predict Eclipses of the Moon. This they did, probably, by means of their Cycle of 223 months, or about 18 years; for at the end of this time, the eclipses of the moon begin to return, at the same intervals and in the same order as at the beginning.[53\3] Probably this was the first instance of the prediction of peculiar astronomical phenomena. The Chinese have, indeed, a legend, in which it is related that a solar eclipse happened in the reign of Tchongkang, above 2000 years before Christ, and that the emperor was so much irritated against two great officers of state, who had neglected to predict this eclipse, that he put them to death. But this cannot be accepted as a real event: for, during the next ten centuries, we find no single observation or fact connected with astronomy in the Chinese {136} histories; and their astronomy has never advanced beyond a very rude and imperfect condition.

[Note 53\3: The eclipses of the sun are more difficult to calculate; since they depend upon the place of the spectator on the earth.]

We can only conjecture the mode in which the Chaldeans discovered their Period of 18 years; and we may make very different suppositions with regard to the degree of science by which they were led to it. We may suppose, with Delambre,[54\3] that they carefully recorded the eclipses which happened, and then, by the inspection of their registers, discovered that those of the moon recurred after a certain period. Or we may suppose, with other authors, that they sedulously determined the motions of the moon, and having obtained these with considerable accuracy, sought and found a

period which should include cycles of these motions. This latter mode of proceeding would imply a considerable degree of knowledge.

[Note 54\3: _A. A._ p. 212.]

It appears probable rather that such a period was discovered by noticing the _recurrence_ of eclipses, than by studying the moon's _motions_. After 6585⅓ days, or 223 lunations, the same eclipses nearly will recur. It is not contested that the Chaldeans were acquainted with this period, which they called _Saros_; or that they calculated eclipses by means of it.

_Sect._ 12.--_Sequel to the Early Stages of Astronomy._

EVERY stage of science has its train of practical applications and systematic inferences, arising both from the demands of convenience and curiosity, and from the pleasure which, as we have already said, ingenuous and active-minded men feel in exercising the process of deduction. The earliest condition of astronomy, in which it can be looked upon as a science, exhibits several examples of such applications and inferences, of which we may mention a few.

_Prediction of Eclipses._--The Cycles which served to keep in order the Calendar of the early nations of antiquity, in some instances enabled them also, as has just been stated, to predict Eclipses; and this application of knowledge necessarily excited great notice. Cleomedes, in the time of Augustus, says, "We never see an eclipse happen which has not been predicted by those who made use of the Tables." (ὑπὸ τῶν κανονικῶν.)

_Terrestrial Zones._--The globular form of the earth being assented to, the doctrine of the sphere was applied to the earth as well as the heavens; and the earth's surface was divided by various imaginary {137} circles; among the rest, the equator, the tropics, and circles, at the same distance from the poles as the tropics are from the equator. One of the curious consequences of this division was the _assumption_ that there must be some marked difference in the stripes or _zones_ into which the earth's surface was thus divided. In going to the south, Europeans found countries hotter and hotter, in going to the north, colder and colder; and it was supposed that the space between the tropical circles must be uninhabitable from heat, and that within the polar circles, again, uninhabitable from cold. This fancy was, as we now know, entirely unfounded. But the principle of the globular form of the earth, when dealt with by means of spherical geometry, led to many true and important propositions concerning the lengths of days and nights at different places. These propositions still form a part of our Elementary Astronomy.

_Gnomonic._--Another important result of the doctrine of the sphere was _Gnomonic_ or _Dialling_. Anaximenes is said by Pliny to have first taught this art in Greece; and both he and Anaximander are reported to have erected the first dial at Lacedemon. Many of the ancient dials remain to us; some of these are of complex forms, and must have required great ingenuity and considerable geometrical knowledge in their construction.

_Measure of the Sun's Distance._--The explanation of the phases of the moon led to no result so remarkable as the attempt of Aristarchus of Samos to obtain from this doctrine a measure of the Distance of the Sun as compared with that of the Moon. If the moon was a perfectly smooth sphere, when she was exactly midway between the new and full in position (that is, a quadrant from the sun), she would be somewhat more than a half moon; and the place when she was _dichotomized_, that is, was an exact semicircle, the bright part being bounded by a straight line, would depend upon the sun's distance from the earth. Aristarchus endeavored to fix the exact place of this Dichotomy; but the irregularity of the edge which bounds the bright part of the moon, and the difficulty of measuring with accuracy, by means then in use, either the precise time when the boundary was most nearly a straight line, or the exact distance of the moon from the sun at that time, rendered his conclusion false and valueless. He collected that the sun is at 18 times the distance of the moon from us; we now know that he is at 400 times the moon's distance.

It would be easy to dwell longer on subjects of this kind; but we have already perhaps entered too much in detail. We have been {138} tempted to do this by the interest which the mathematical spirit of the Greeks gave to the earliest astronomical discoveries, when these were the subjects of their reasonings; but we must now proceed to contemplate them engaged in a worthier employment, namely, in adding to these discoveries.

## CHAPTER II.

PRELUDE TO THE INDUCTIVE EPOCH OF HIPPARCHUS.

WITHOUT pretending that we have exhausted the consequences of the elementary discoveries which we have enumerated, we now proceed to consider the nature and circumstances of the next great discovery which makes an Epoch in the history of Astronomy; and this we shall find to be the Theory of Epicycles and Eccentrics. Before, however, we relate the establishment of this theory, we must, according to the general plan we have marked out, notice some of the conjectures and attempts by which it was preceded, and the growing acquaintance with facts, which made the want of such an explanation felt.

In the steps previously made in astronomical knowledge, no ingenuity had been required to devise the view which was adopted. The motions of the stars and sun were most naturally and almost irresistibly conceived as the results of motion in a revolving sphere; the indications of position which we obtain from different places on the earth's surface, when clearly combined, obviously imply a globular shape. In these cases, the first conjectures, the supposition of the simplest form, of the most uniform motion, required no after-correction. But this manifest simplicity, this easy and obvious explanation, did not apply to the movement of all the heavenly bodies. The Planets, the "wandering stars," could not be so easily understood; the motion of each, as Cicero says, "undergoing very remarkable changes in its course, going before and behind, quicker and slower, appearing in the evening, but gradually lost there, and emerging again in the morning."[55\3] A continued attention to these stars

would, however, {139} detect a kind of intricate regularity in their motions, which might naturally be described as "a dance." The Chaldeans are stated by Diodorus[56\3] to have observed assiduously the risings and settings of the planets, from the top of the temple of Belus. By doing this, they would find the times in which the forward and backward movements of Saturn, Jupiter, and Mars recur; and also the time in which they come round to the same part of the heavens.[57\3] Venus and Mercury never recede far from the sun, and the intervals which elapse while either of them leaves its greatest distance from the sun and returns again to the greatest distance on the same side, would easily be observed.

[Note 55\3: Cic. _de Nat. D._ lib. ii. p. 450. "Ea quæ Saturni stella dicitur, φαίνωνque a Græcis nominatur, quæ a terra abest plurimum, xxx fere annis cursum suum conficit; in quo cursu multa mirabiliter efficiens, tum antecedendo, tum retardando, tum vespertinis temporibus delitescendo, tum matutinis se rursum aperiendo, nihil immutat sempiternis sæculorum ætatibus, quin eadem iisdem temporibus efficiat." And so of the other planets.]

[Note 56\3: _A. A._ i. p. 4.]

[Note 57\3: Plin. _H. N._ ii. p. 204.]

Probably the manner in which the motions of the planets were originally reduced to rule was something like the following:--In about 30 of our years, Saturn goes 29 times through his _Anomaly_, that is, the succession of varied motions by which he sometimes goes forwards and sometimes backwards among the stars. During this time, he goes once round the heavens, and returns nearly to the same place. This is the cycle of his apparent motions.

Perhaps the eastern nations contented themselves with thus referring these motions to cycles of time, so as to determine their recurrence. Something of this kind was done at an early period, as we have seen.

But the Greeks soon attempted to frame to themselves a sensible image of the mechanism by which these complex motions were produced; nor did they find this difficult. Venus, for instance, who, upon the whole, moves from west to east among the stars, is seen, at certain intervals, to return or move _retrograde_ a short way back from east to west, then to become for a short time _stationary_, then to turn again and resume her _direct_ motion westward, and so on. Now this can be explained by supposing that she is placed in the rim of a wheel, which is turned edgeways to us, and of which the centre turns round in the heavens from west to east, while the wheel, carrying the planet in its motion, moves round its own centre. In this way the motion of the wheel about its centre, would, in some situations, counterbalance the general motion of the centre, and make the planet retrograde, while, on the whole, the westerly motion would prevail. Just as if we suppose that a person, holding a lamp in his hand in the dark, and at a {140} distance, so that the lamp alone is visible, should run on turning himself round; we should see the light sometimes stationary, sometimes retrograde, but on the whole progressive.

A mechanism of this kind was imagined for each of the planets, and the wheels of which we have spoken were in the end called _Epicycles_.

The application of such mechanism to the planets appears to have arisen in Greece about the time of Aristotle. In the works of Plato we find a strong taste for this kind of mechanical speculation. In the tenth book of the "Polity," we have the apologue of Alcinus the Pamphylian, who, being supposed to be killed in battle, revived when he was placed on the funeral pyre, and related what he had seen during his trance. Among other revelations, he beheld the machinery by which all the celestial bodies revolve. The axis of these revolutions is the adamantine distaff which Destiny holds between her knees; on this are fixed, by means of different sockets, flat rings, by which the planets are carried. The order and magnitude of these spindles are minutely detailed. Also, in the "Epilogue to the Laws" (_Epinomis_), he again describes the various movements of the sky, so as to show a distinct acquaintance with the general character of the planetary motions; and, after speaking of the Egyptians and Syrians as the original cultivators of such knowledge, he adds some very remarkable exhortations to his countrymen to prosecute the subject. "Whatever we Greeks," he says, "receive from the barbarians, we improve and perfect; there is good hope and promise, therefore, that Greeks will carry this knowledge far beyond that which was introduced from abroad." To this task, however, he looks with a due appreciation of the qualities and preparation which it requires. "An astronomer must be," he says, "the wisest of men; his mind must be duly disciplined in youth; especially is mathematical study necessary; both an acquaintance with the doctrine of number, and also with that other branch of mathematics, which, closely connected as it is with the science of the _heavens_, we very absurdly call _geometry_, the measurement of the _earth_."[58\3]

[Note 58\3: _Epinomis_, pp. 988, 990.]

Those anticipations were very remarkably verified in the subsequent career of the Greek Astronomy.

The theory, once suggested, probably made rapid progress. Simplicius[59\3] relates, that Eudoxus of Cnidus introduced the hypothesis of revolving circles or spheres. Calippus of Cyzicus, having visited {141} Polemarchus, an intimate friend of Eudoxus, they went together to Athens, and communicated to Aristotle the invention of Eudoxus, and with his help improved and corrected it.

[Note 59\3: Lib. ii. _de Cœlo_. Bullialdus, p. 18.]

Probably at first this hypothesis was applied only to account for the general phenomena of the progressions, retrogradations, and stations of the planet; but it was soon found that the motions of the sun and moon, and the circular motions of the planets, which the hypothesis supposed, had other _anomalies_ or irregularities, which made a further extension of the hypothesis necessary.

The defect of uniformity in these motions of the sun and moon, though less apparent than in the planets, is easily detected, as soon as men endeavor to obtain any accuracy in their observations. We have already stated (Chap. I.) that the Chaldeans were in possession of a period of about eighteen years, which they used in the calculation of eclipses, and which might have been discovered by close observation of the moon's motions; although it was probably rather hit upon by noting

the recurrence of eclipses. The moon moves in a manner which is not reducible to regularity without considerable care and time. If we trace her path among the stars, we find that, like the path of the sun, it is oblique to the equator, but it does not, like that of the sun, pass over the same stars in successive revolutions. Thus its _latitude_, or distance from the equator, has a cycle different from its revolution among the stars; and its _Nodes_, or the points where it cuts the equator, are perpetually changing their position. In addition to this, the moon's motion in her own path is not uniform; in the course of each lunation, she moves alternately slower and quicker, passing gradually through the intermediate degrees of velocity; and goes through the cycle of these changes in something less than a month; this is called a revolution of _Anomaly_. When the moon has gone through a complete number of revolutions of Anomaly, and has, in the same time, returned to the same position with regard to the sun, and also with regard to her Nodes, her motions with respect to the sun will thenceforth be the same as at the first, and all the circumstances on which lunar eclipses depend being the same, the eclipses will occur in the same order. In 6585⅓ days there are 239 revolutions of anomaly, 241 revolutions with regard to one of the Nodes, and, as we have said, 223 lunations or revolutions with regard to the sun. Hence this Period will bring about a succession of the same lunar eclipses.

If the Chaldeans observed the moon's motion among the stars with any considerable accuracy, so as to detect this period by that means, {142} they could hardly avoid discovering the anomaly or unequal motion of the moon; for in every revolution, her daily progression in the heavens varies from about twenty-two to twenty-six times her own diameter. But there is not, in their knowledge of this Period, any evidence that they had measured the amount of this variation; and Delambre[60\3] is probably right in attributing all such observations to the Greeks.

[Note 60\3: _Astronomie Ancienne_, i. 212.]

The sun's motion would also be seen to be irregular as soon as men had any exact mode of determining the lengths of the four seasons, by means of the passage of the sun through the equinoctial and solstitial points. For spring, summer, autumn, and winter, which would each consist of an equal number of days if the motions were uniform, are, in fact, found to be unequal in length.

It was not very difficult to see that the mechanism of epicycles might be applied so as to explain irregularities of this kind. A wheel travelling round the earth, while it revolved upon its centre, might produce the effect of making the sun or moon fixed in its rim go sometimes faster and sometimes slower in appearance, just in the same way as the same suppositions would account for a planet going sometimes forwards and sometimes backwards: the epicycles of the sun and moon would, for this purpose, be less than those of the planets. Accordingly, it is probable that, at the time of Plato and Aristotle, philosophers were already endeavoring to apply the hypothesis to these cases, though it does not appear that any one fully succeeded before Hipparchus.

The problem which was thus present to the minds of astronomers, and which Plato is said to have proposed to them in a distinct form, was, "To reconcile the celestial phenomena by the combination of equable circular motions." That the circular

motions should be equable as well as circular, was a condition, which, if it had been merely tried at first, as the most simple and definite conjecture, would have deserved praise. But this condition, which is, in reality, inconsistent with nature, was, in the sequel, adhered to with a pertinacity which introduced endless complexity into the system. The history of this assumption is one of the most marked instances of that love of simplicity and symmetry which is the source of all general truths, though it so often produces and perpetuates error. At present we can easily see how fancifully the notion of simplicity and perfection was interpreted, in the arguments by which the opinion was defended, that the {143} real motions of the heavenly bodies must be circular and uniform. The Pythagoreans, as well as the Platonists, maintained this dogma. According to Geminus, "They supposed the motions of the sun, and the moon, and the five planets, to be circular and equable: for they would not allow of such disorder among divine and eternal things, as that they should sometimes move quicker, and sometimes slower, and sometimes stand still; for no one would tolerate such anomaly in the movements, even of a man, who was decent and orderly. The occasions of life, however, are often reasons for men going quicker or slower, but in the incorruptible nature of the stars, it is not possible that any cause can be alleged of quickness and slowness. Whereupon they propounded this question, how the phenomena might be represented by equable and circular motions."

These conjectures and assumptions led naturally to the establishment of the various parts of the Theory of Epicycles. It is probable that this theory was adopted with respect to the Planets at or before the time of Plato. And Aristotle gives us an account of the system thus devised.[61\3] "Eudoxus," he says, "attributed four spheres to each Planet: the first revolved with the fixed stars (and this produced the diurnal motion); the second gave the planet a motion along the ecliptic (the mean motion in longitude); the third had its axis perpendicular[62\3] to the ecliptic (and this gave the inequality of each planetary motion, really arising from its special motion about the sun); the fourth produced the oblique motion transverse to this (the motion in latitude)." He is also said to have attributed a motion in latitude and a corresponding sphere to the Sun as well as to the Moon, of which it is difficult to understand the meaning, if Aristotle has reported rightly of the theory; for it would be absurd to ascribe to Eudoxus a knowledge of the motions by which the sun deviates from the ecliptic. Calippus conceived that two additional spheres must be given to the sun and to the moon, in order to explain the phenomena: probably he was aware of the inequalities of the motions of these luminaries. He also proposed an additional sphere for each planet, to account, we may suppose, for the results of the eccentricity of the orbits.

[Note 61\3: Metaph. xi. 8.]

[Note 62\3: Aristotle says "has its poles in the ecliptic," but this must be a mistake of his. He professes merely to receive these opinions from the mathematical astronomers, "ἐκ τῆς οἰκειοτάτης φιλοσοφίας τῶν μαθηματικῶν."]

The hypothesis, in this form, does not appear to have been reduced to measure, and was, moreover, unnecessarily complex. The resolution {144} of the oblique motion of the moon into two separate motions, by Eudoxus, was not the simplest way

of conceiving it; and Calippus imagined the connection of these spheres in some way which made it necessary nearly to double their number; in this manner his system had no less than 55 spheres.

Such was the progress which the _Idea_ of the hypothesis of epicycles had made in men's minds, previously to the establishment of the theory by Hipparchus. There had also been a preparation for this step, on the other side, by the collection of _Facts_. We know that observations of the Eclipses of the Moon were made by the Chaldeans 367 B. C. at Babylon, and were known to the Greeks; for Hipparchus and Ptolemy founded their Theory of the Moon on these observations. Perhaps we cannot consider, as equally certain, the story that, at the time of Alexander's conquest, the Chaldeans possessed a series of observations, which went back 1903 years, and which Aristotle caused Callisthenes to bring to him in Greece. All the Greek observations which are of any value, begin with the school of Alexandria. Aristyllus and Timocharis appear, by the citations of Hipparchus, to have observed the Places of Stars and Planets, and the Times of the Solstices, at various periods from B. C. 295 to B. C. 269. Without their observations, indeed, it would not have been easy for Hipparchus to establish either the Theory of the Sun or the Precession of the Equinoxes.

In order that observations at distant intervals may be compared with each other, they must be referred to some common era. The Chaldeans dated by the era of Nabonassar, which commenced 749 B. C. The Greek observations were referred to the Calippic periods of 76 years, of which the first began 331 B. C. These are the dates used by Hipparchus and Ptolemy. {145}

## CHAPTER III.

INDUCTIVE EPOCH OF HIPPARCHUS.

_Sect._ 1.--_Establishment of the Theory of Epicycles and Eccentrics._

ALTHOUGH, as we have already seen, at the time of Plato, the Idea of Epicycles had been suggested, and the problem of its general application proposed, and solutions of this problem offered by his followers; we still consider Hipparchus as the real discoverer and founder of that theory; inasmuch as he not only guessed that it _might_, but showed that it _must_, account for the phenomena, both as to their nature and as to their quantity. The assertion that "he only discovers who proves," is just; not only because, until a theory is proved to be the true one, it has no pre-eminence over the numerous other guesses among which it circulates, and above which the proof alone elevates it; but also because he who takes hold of the theory so as to apply calculation to it, possesses it with a distinctness of conception which makes it peculiarly his.

In order to establish the Theory of Epicycles, it was necessary to assign the magnitudes, distances, and positions of the circles or spheres in which the heavenly bodies were moved, in such a manner as to account for their apparently irregular motions. We may best understand what was the problem to be solved, by calling to

mind what we now know to be the real motions of the heavens. The true motion of the earth round the sun, and therefore the apparent annual motion of the sun, is performed, not in a circle of which the earth is the centre, but in an ellipse or oval, the earth being nearer to one end than to the other; and the motion is most rapid when the sun is at the nearer end of this oval. But instead of an oval, we may suppose the sun to move uniformly in a circle, the earth being now, not in the centre, but nearer to one side; for on this supposition, the sun will appear to move most quickly when he is nearest to the earth, or in his _Perigee_, as that point is called. Such an orbit is called an _Eccentric_, and the distance of the earth from the centre of the circle is called the _Eccentricity_. It may easily be shown by geometrical reasoning, that the inequality of apparent motion so produced, is exactly the same in {146} detail, as the inequality which follows from the hypothesis of a small _Epicycle_, turning uniformly on its axis, and carrying the sun in its circumference, while the centre of this epicycle moves uniformly in a circle of which the earth is the centre. This identity of the results of the hypothesis of the Eccentric and the Epicycle is proved by Ptolemy in the third book of the "Almagest."

_The Sun's Eccentric._--When Hipparchus had clearly conceived these hypotheses, as _possible_ ways of accounting for the sun's motion, the task which he had to perform, in order to show that they deserved to be adopted, was to assign a place to the _Perigee_, a magnitude to the _Eccentricity_, and an _Epoch_ at which the sun was at the perigee; and to show that, in this way, he had produced a true representation of the motions of the sun. This, accordingly, he did; and having thus determined, with considerable exactness, both the law of the solar irregularities, and the numbers on which their amount depends, he was able to assign the motions and places of the sun for any moment of future time with corresponding exactness; he was able, in short, to construct _Solar Tables_, by means of which the sun's place with respect to the stars could be correctly found at any time. These tables (as they are given by Ptolemy)[63\3] give the _Anomaly_, or inequality of the sun's motion; and this they exhibit by means of the _Prosthapheresis_, the quantity of which, at any distance of the sun from the _Apogee_, it is requisite to add to or subtract from the arc, which he would have described if his motion had been equable.

[Note 63\3: Syntax. 1. iii.]

The reader might perhaps expect that the calculations which thus exhibited the motions of the sun for an indefinite future period must depend upon a considerable number of observations made at all seasons of the year. That, however, was not the case; and the genius of the discoverer appeared, as such genius usually does appear, in his perceiving how small a number of facts, rightly considered, were sufficient to form a foundation for the theory. The number of days contained in two seasons of the year sufficed for this purpose to Hipparchus. "Having ascertained," says Ptolemy, "that the time from the vernal equinox to the summer tropic is 94½ days, and the time from the summer tropic to the autumnal equinox 92½ days, from these phenomena alone he demonstrates that the straight line joining the centre of the sun's eccentric path with the centre of the zodiac (the spectator's eye) is nearly the 24th part of the radius of the eccentric path; and that {147} its _apogee_ precedes

the summer solstice by 24½ degrees nearly, the zodiac containing 360."

The exactness of the Solar Tables, or _Canon_, which was founded on these data, was manifested, not only by the coincidence of the sun's calculated place with such observations as the Greek astronomers of this period were able to make (which were indeed very rude), but by its enabling them to calculate solar and lunar eclipses; phenomena which are a very precise and severe trial of the accuracy of such tables, inasmuch as a very minute change in the apparent place of the sun or moon would completely alter the obvious features of the eclipse. Though the tables of this period were by no means perfect, they bore with tolerable credit this trying and perpetually recurring test; and thus proved the soundness of the theory on which the tables were calculated.

_The Moon's Eccentric._--The moon's motions have many irregularities; but when the hypothesis of an Eccentric or an Epicycle had sufficed in the case of the sun, it was natural to try to explain, in the same way, the motions of the moon; and it was shown by Hipparchus that such hypotheses would account for the more obvious anomalies. It is not very easy to describe the several ways in which these hypotheses were applied, for it is, in truth, very difficult to explain in words even the mere facts of the moon's motion. If she were to leave a visible bright line behind her in the heavens wherever she moved, the path thus exhibited would be of an extremely complex nature; the circle of each revolution slipping away from the preceding, and the traces of successive revolutions forming a sort of band of net-work running round the middle of the sky.[64\3] In each revolution, the motion in longitude is affected by an anomaly of the same nature as the sun's anomaly already spoken of; but besides this, the path of the moon deviates from the ecliptic to the north and to the south of the ecliptic, and thus she has a motion in latitude. This motion in latitude would be sufficiently known if we knew the period of its _restoration_, that is, the time which the moon occupies in moving from any latitude till she is restored to the same latitude; as, for instance, from the ecliptic on one side of the heavens to the ecliptic on the same side of the heavens again. But it is found that the period of the restoration of the latitude is not the same as the period of the restoration of the longitude, that is, as the period of the moon's revolution among the {148} stars; and thus the moon describes a different path among the stars in every successive revolution, and her path, as well as her velocity, is constantly variable.

[Note 64\3: The reader will find an attempt to make the nature of this path generally intelligible in the _Companion to the British Almanac_ for 1814.]

Hipparchus, however, reduced the motions of the moon to rule and to Tables, as he did those of the sun, and in the same manner. He determined, with much greater accuracy than any preceding astronomer, the mean or average equable motions of the moon in longitude and in latitude; and he then represented the anomaly of the motion in longitude by means of an eccentric, in the same manner as he had done for the sun.

But here there occurred still an additional change, besides those of which we have spoken. The Apogee of the Sun was always in the same place in the heavens; or at

least so nearly so, that Ptolemy could detect no error in the place assigned to it by Hipparchus 250 years before. But the Apogee of the Moon was found to have a motion among the stars. It had been observed before the time of Hipparchus, that in 6585⅓ days, there are 241 revolutions of the moon with regard to the stars, but only 239 revolutions with regard to the anomaly. This difference could be suitably represented by supposing the eccentric, in which the moon moves, to have itself an angular motion, perpetually carrying its apogee in the same direction in which the moon travels; but this supposition being made, it was necessary to determine, not only the eccentricity of the orbit, and place of the apogee at a certain time, but also the rate of motion of the apogee itself, in order to form tables of the moon.

This task, as we have said, Hipparchus executed; and in this instance, as in the problem of the reduction of the sun's motion to tables, the data which he found it necessary to employ were very few. He deduced all his conclusions from six eclipses of the moon.[65\3] Three of these, the records of which were brought from Babylon, where a register of such occurrences was kept, happened in the 366th and 367th years from the era of Nabonassar, and enabled Hipparchus to determine the eccentricity and apogee of the moon's orbit at that time. The three others were observed at Alexandria, in the 547th year of Nabonassar, which gave him another position of the orbit at an interval of 180 years; and he thus became acquainted with the motion of the orbit itself, as well as its form.[66\3] {149}

[Note 65\3: Ptol. _Syn._ iv. 10.]

[Note 66\3: Ptolemy uses the hypothesis of an epicycle for the moon's first inequality; but Hipparchus employs an eccentric.]

The moon's motions are really affected by several other inequalities, of very considerable amount, besides those which were thus considered by Hipparchus; but the lunar paths, constructed on the above data, possessed a considerable degree of correctness, and especially when applied, as they were principally, to the calculation of eclipses; for the greatest of the additional irregularities which we have mentioned disappear at new and full moon, which are the only times when eclipses take place.

The numerical explanation of the motions of the sun and moon, by means of the Hypothesis of Eccentrics, and the consequent construction of tables, was one of the great achievements of Hipparchus. The general explanation of the motions of the planets, by means of the hypothesis of epicycles, was in circulation previously, as we have seen. But the special motions of the planets, in their epicycles, are, in reality, affected by anomalies of the same kind as those which render it necessary to introduce eccentrics in the cases of the sun and moon.

Hipparchus determined, with great exactness, the _Mean Motions_ of the Planets; but he was not able, from want of data, to explain the planetary _Irregularities_ by means of Eccentrics. The whole mass of good observations of the planets which he received from preceding ages, did not contain so many, says Ptolemy, as those which he has transmitted to us of his own. "Hence[67\3] it was," he adds, "that while he labored, in the most assiduous manner to represent the motions of the sun and moon by means of equable circular motions; with respect to the planets, so far as his works

show, he did not even make the attempt, but merely put the extant observations in order, added to them himself more than the whole of what he received from preceding ages, and showed the insufficiency of the hypothesis current among astronomers to explain the phenomena." It appears that preceding mathematicians had already pretended to construct "a Perpetual Canon," that is, Tables which should give the places of the planets at any future time; but these being constructed without regard to the eccentricity of the orbits, must have been very erroneous.

[Note 67\3: _Synt._ ix. 2.]

Ptolemy declares, with great reason, that Hipparchus showed his usual love of truth, and his right sense of the responsibility of his task, in leaving this part of it to future ages. The Theories of the Sun and Moon, which we have already described, constitute him a great astronomical discoverer, and justify the reputation he has always {150} possessed. There is, indeed, no philosopher who is so uniformly spoken of in terms of admiration. Ptolemy, to whom we owe our principal knowledge of him, perpetually couples with his name epithets of praise: he is not only an excellent and careful observer, but "a[68\3] most truth-loving and labor-loving person," one who had shown extraordinary sagacity and remarkable desire of truth in every part of science. Pliny, after mentioning him and Thales, breaks out into one of his passages of declamatory vehemence: "Great men! elevated above the common standard of human nature, by discovering the laws which celestial occurrences obey, and by freeing the wretched mind of man from the fears which eclipses inspired--Hail to you and to your genius, interpreters of heaven, worthy recipients of the laws of the universe, authors of principles which connect gods and men!" Modern writers have spoken of Hipparchus with the same admiration; and even the exact but severe historian of astronomy, Delambre, who bestows his praise so sparingly, and his sarcasm so generally;--who says[69\3] that it is unfortunate for the memory of Aristarchus that his work has come to us entire, and who cannot refer[70\3] to the statement of an eclipse rightly predicted by Halicon of Cyzicus without adding, that if the story be true, Halicon was more lucky than prudent;--loses all his bitterness when he comes to Hipparchus.[71\3] "In Hipparchus," says he, "we find one of the most extraordinary men of antiquity; the _very greatest_, in the sciences which require a combination of observation with geometry." Delambre adds, apparently in the wish to reconcile this eulogium with the depreciating manner in which he habitually speaks of all astronomers whose observations are inexact, "a long period and the continued efforts of many industrious men are requisite to produce good instruments, but energy and assiduity depend on the man himself."

[Note 68\3: _Synt._ ix. 2.]

[Note 69\3: _Astronomie Ancienne_, i. 75.]

[Note 70\3: Ib. i. 17.]

[Note 71\3: Ib. i. 186.]

Hipparchus was the author of other great discoveries and improvements in astronomy, besides the establishment of the Doctrine of Eccentrics and Epicycles; but this, being the greatest advance in the _theory_ of the celestial motions which was

made by the ancients, must be the leading subject of our attention in the present work; our object being to discover in what the progress of real theoretical knowledge consists, and under what circumstances it has gone on. {151}

_Sect._ 2.--_Estimate of the Value of the Theory of Eccentrics and Epicycles._

IT may be useful here to explain the value of the theoretical step which Hipparchus thus made; and the more so, as there are, perhaps, opinions in popular circulation, which might lead men to think lightly of the merit of introducing or establishing the Doctrine of Epicycles. For, in the first place, this doctrine is now acknowledged to be false; and some of the greatest men in the more modern history of astronomy owe the brightest part of their fame to their having been instrumental in overturning this hypothesis. And, moreover, in the next place, the theory is not only false, but extremely perplexed and entangled, so that it is usually looked upon as a mass of arbitrary and absurd complication. Most persons are familiar with passages in which it is thus spoken of.[72\3]

. . . . . He his fabric of the heavens Hath left to their disputes, perhaps to move His laughter at their quaint opinions wide; Hereafter, when they come to model heaven And calculate the stars, how will they wield The mighty frame! how build, unbuild, contrive, To save appearances! how gird the sphere With centric and eccentric scribbled o'er, Cycle in epicycle, orb in orb!

And every one will recollect the celebrated saying of Alphonso X., king of Castile,[73\3] when this complex system was explained to him; that "if God had consulted him at the creation, the universe should have been on a better and simpler plan." In addition to this, the system is represented as involving an extravagant conception of the nature of the orbs which it introduces; that they are crystalline spheres, and that the vast spaces which intervene between the celestial luminaries are a solid mass, formed by the fitting together of many masses perpetually in motion; an imagination which is presumed to be incredible and monstrous.

[Note 72\3: _Paradise Lost_, viii.]

[Note 73\3: A. D. 1252.]

We must endeavor to correct or remove these prejudices, not only in order that we may do justice to the Hipparchian, or, as it is usually called, Ptolemaic system of astronomy, and to its founder; but for another reason, much more important to the purpose of this work; {152} namely, that we may see how theories may be highly estimable, though they contain false representations of the real state of things, and may be extremely useful, though they involve unnecessary complexity. In the advance of knowledge, the value of the true part of a theory may much outweigh the accompanying error, and the use of a rule may be little impaired by its want of simplicity. The first steps of our progress do not lose their importance because they are not the last; and the outset of the journey may require no less vigor and activity than its close.

That which is true in the Hipparchian theory, and which no succeeding discoveries have deprived of its value, is the _Resolution_ of the apparent motions of the heavenly

bodies into an assemblage of circular motions. The test of the truth and reality of this Resolution is, that it leads to the construction of theoretical Tables of the motions of the luminaries, by which their places are given at any time, agreeing nearly with their places as actually observed. The assumption that these circular motions, thus introduced, are all exactly uniform, is the fundamental principle of the whole process. This assumption is, it may be said, false; and we have seen how fantastic some of the arguments were, which were originally urged in its favor. But _some_ assumption is necessary, in order that the motions, at different points of a revolution, may be somehow connected, that is, in order that we may have any theory of the motions; and no assumption more simple than the one now mentioned can be selected. The merit of the theory is this;--that obtaining the amount of the eccentricity, the place of the apogee, and, it may be, other elements, from _few_ observations, it deduces from these, results agreeing with _all_ observations, however numerous and distant. To express an inequality by means of an epicycle, implies, not only that there is an inequality, but further,--that the inequality is at its greatest value at a certain known place,--diminishes in proceeding from that place by a known law,--continues its diminution for a known portion of the revolution of the luminary,--then increases again; and so on: that is, the introduction of the epicycle represents the inequality of motion, as completely as it can be represented with respect to its _quantity_.

We may further illustrate this, by remarking that such a Resolution of the unequal motions of the heavenly bodies into equable circular motions, is, in fact, equivalent to the most recent and improved processes by which modern astronomers deal with such motions. Their universal method is to resolve all unequal motions into a series of {153} _terms_, or expressions of partial motions; and these terms involve _sines_ and _cosines_, that is, certain technical modes of measuring circular motion, the circular motion having some constant relation to the time. And thus the problem of the resolution of the celestial motions into equable circular ones, which was propounded above two thousand years ago in the school of Plato, is still the great object of the study of modern astronomers, whether observers or calculators.

That Hipparchus should have succeeded in the first great steps of this resolution for the sun and moon, and should have seen its applicability in other cases, is a circumstance which gives him one of the most distinguished places in the roll of great astronomers. As to the charges or the sneers against the complexity of his system, to which we have referred, it is easy to see that they are of no force. As a system of _calculation_, his is not only good, but, as we have just said, in many cases no better has yet been discovered. If, when the actual motions of the heavens are calculated in the best possible way, the process is complex and difficult, and if we are discontented at this, nature, and not the astronomer, must be the object of our displeasure. This plea of the astronomers must be allowed to be reasonable. "We must not be repelled," says Ptolemy,[74\3] "by the complexity of the hypotheses, but explain the phenomena as well as we can. If the hypotheses satisfy each apparent inequality separately, the combination of them will represent the truth; and why should it appear wonderful to any that such a complexity should exist in the heavens, when we know nothing of their nature which entitles us to suppose that any inconsistency will result?"

[Note 74\3: _Synt._ xiii. 2.]

But it may be said, we now know that the motions are more simple than they were thus represented, and that the Theory of Epicycles was false, as a conception of the real construction of the heavens. And to this we may reply, that it does not appear that the best astronomers of antiquity conceived the cycles and epicycles to have a material existence. Though the dogmatic philosophers, as the Aristotelians, appear to have taught that the celestial spheres were real solid bodies, they are spoken of by Ptolemy as imaginary;[75\3] and it is clear, from his proof of the identity of the results of the hypothesis of an eccentric and an epicycle, that they are intended to pass for no more than geometrical conceptions, in which view they are true representations of the apparent motions. {154}

[Note 75\3: Ibid. iii. 3.]

It is true, that the real motions of the heavenly bodies are simpler than the apparent motions; and that we, who are in the habit of representing to our minds their real arrangement, become impatient of the seeming confusion and disorder of the ancient hypotheses. But this real arrangement never could have been detected by philosophers, if the apparent motions had not been strictly examined and successfully analyzed. How far the connection between the facts and the true theory is from being obvious or easily traced, any one may satisfy himself by endeavoring, from a general conception of the moon's real motions, to discover the rules which regulate the occurrences of eclipses; or even to explain to a learner, of what nature the apparent motions of the moon among the stars will be.

The unquestionable evidence of the merit and value of the Theory of Epicycles is to be found in this circumstance;--that it served to embody all the most exact knowledge then extant, to direct astronomers to the proper methods of making it more exact and complete, to point out new objects of attention and research; and that, after doing this at first, it was also able to take in, and preserve, all the new results of the active and persevering labors of a long series of Greek, Latin, Arabian, and modern European astronomers, till a new theory arose which could discharge this office. It may, perhaps, surprise some readers to be told, that the author of this next _great_ step in astronomical theory, Copernicus, adopted the theory of epicycles; that is, he employed that which we have spoken of as its really valuable characteristic. "We[76\3] must confess," he says, "that the celestial motions are circular, or compounded of several circles, since their inequalities observe a fixed law and recur in value at certain intervals, which could not be, except that they were circular; for a circle alone can make that which has been, recur again."

[Note 76\3: Copernicus. _De Rev._ 1. i. c. 4.]

In this sense, therefore, the Hipparchian theory was a real and indestructible truth, which was not rejected, and replaced by different truths, but was adopted and incorporated into every succeeding astronomical theory; and which can never cease to be one of the most important and fundamental parts of our astronomical knowledge.

A moment's reflection will show that, in the events just spoken of, the introduction and establishment of the Theory of Epicycles, those characteristics were strictly exemplified, which we have asserted to be the conditions of every real advance in progressive science; namely, {155} the application of distinct and appropriate Ideas to a real series of Facts. The distinctness of the geometrical conceptions which enabled Hipparchus to assign the Orbits of the Sun and Moon, requires no illustration; and we have just explained how these ideas combined into a connected whole the various motions and places of those luminaries. To make this step in astronomy, required diligence and care, exerted in collecting observations, and mathematical clearness and steadiness of view, exercised in seeing and showing that the theory was a successful analysis of them.

_Sect._ 3.--_Discovery of the Precession of the Equinoxes._

THE same qualities which we trace in the researches of Hipparchus already examined,--diligence in collecting observations, and clearness of idea in representing them,--appear also in other discoveries of his, which we must not pass unnoticed. The Precession of the Equinoxes, in particular, is one of the most important of these discoveries.

The circumstance here brought into notice was a Change of Longitude of the Fixed Stars. The longitudes of the heavenly bodies, being measured from the point where the sun's annual path cuts the equator, will change if that path changes. Whether this happens, however, is not very easy to decide; for the sun's path among the stars is made out, not by merely looking at the heavens, but by a series of inferences from other observable facts. Hipparchus used for this purpose eclipses of the moon; for these, being exactly opposite to the sun, afford data in marking out his path. By comparing the eclipses of his own time with those observed at an earlier period by Timocharis, he found that the bright star, Spica Virginis, was six degrees behind the equinoctial point in his own time, and had been eight degrees behind the same point at an earlier epoch. The suspicion was thus suggested, that the longitudes of all the stars increase perpetually; but Hipparchus had too truly philosophical a spirit to take this for granted. He examined the places of Regulus, and those of other stars, as he had done those of Spica; and he found, in all these instances, a change of place which could be explained by a certain alteration of position in the circles to which the stars are referred, which alteration is described as the Precession of the Equinoxes.

The distinctness with which Hipparchus conceived this change of relation of the heavens, is manifested by the question which, as we are told by Ptolemy, he examined and decided;--that this motion of the {156} heavens takes place about the poles of the ecliptic, and not about those of the equator. The care with which he collected this motion from the stars themselves, may be judged of from this, that having made his first observations for this purpose on Spica and Regulus, zodiacal stars, his first suspicion was that the stars of the zodiac alone changed their longitude, which suspicion he disproved by the examination of other stars. By his processes, the idea of the nature of the motion, and the evidence of its existence, the two conditions of a discovery, were fully brought into view. The scale of the facts which Hipparchus was thus able to reduce to law, may be in some measure judged of by recollecting that the

precession, from his time to ours, has only carried the stars through one sign of the zodiac; and that, to complete one revolution of the sky by the motion thus discovered, would require a period of 25,000 years. Thus this discovery connected the various aspects of the heavens at the most remote periods of human history; and, accordingly, the novel and ingenious views which Newton published in his chronology, are founded on this single astronomical fact, the Precession of the Equinoxes.

The two discoveries which have been described, the mode of constructing Solar and Lunar Tables, and the Precession, were advances of the greatest importance in astronomy, not only in themselves, but in the new objects and undertakings which they suggested to astronomers. The one discovery detected a constant law and order in the midst of perpetual change and apparent disorder; the other disclosed mutation and movement perpetually operating where every thing had been supposed fixed and stationary. Such discoveries were well adapted to call up many questionings in the minds of speculative men; for, after this, nothing could be supposed constant till it had been ascertained to be so by close examination; and no apparent complexity or confusion could justify the philosopher in turning away in despair from the task of simplification. To answer the inquiries thus suggested, new methods of observing the facts were requisite, more exact and uniform than those hitherto employed. Moreover, the discoveries which were made, and others which could not fail to follow in their train, led to many consequences, required to be reasoned upon, systematized, completed, enlarged. In short, the _Epoch of Induction_ led, as we have stated that such epochs must always lead, to a _Period of Development_, _of Verification_, _Application_, _and Extension_. {157}

## CHAPTER IV.

SEQUEL TO THE INDUCTIVE EPOCH OF HIPPARCHUS.

_Sect._ 1.--_Researches which verified the Theory._

THE discovery of the leading Laws of the Solar and Lunar Motions, and the detection of the Precession, may be considered as the great positive steps in the Hipparchian astronomy;--the parent discoveries, from which many minor improvements proceeded. The task of pursuing the collateral and consequent researches which now offered themselves,--of bringing the other parts of astronomy up to the level of its most improved portions,--was prosecuted by a succession of zealous observers and calculators, first, in the school of Alexandria, and afterwards in other parts of the world. We must notice the various labors of this series of astronomers; but we shall do so very briefly; for the ulterior development of doctrines once established is not so important an object of contemplation for our present purpose, as the first conception and proof of those fundamental truths on which systematic doctrines are founded. Yet Periods of Verification, as well as Epochs of Induction, deserve to be attended to; and they can nowhere be studied with so much advantage as in the history of astronomy.

In truth, however, Hipparchus did not leave to his successors the task of pursuing into detail those views of the heavens to which his discoveries led him. He examined with scrupulous care almost every part of the subject. We must briefly mention some of the principal points which were thus settled by him.

The verification of the laws of the changes which he assigned to the skies, implied that the condition of the heavens was constant, except so far as it was affected by those changes. Thus, the doctrine that the changes of position of the stars were rightly represented by the precession of the equinoxes, supposed that the stars were fixed with regard to each other; and the doctrine that the unequal number of days, in certain subdivisions of months and years, was adequately explained by the theory of epicycles, assumed that years and days were always of constant lengths. But Hipparchus was not content with assuming these bases of his theory, he endeavored to prove them. {158}

1. _Fixity of the Stars._--The question necessarily arose after the discovery of the precession, even if such a question had never suggested itself before, whether the stars which were called _fixed_, and to which the motions of the other luminaries are referred, do really retain constantly the same relative position. In order to determine this fundamental question, Hipparchus undertook to construct a _Map_ of the heavens; for though the result of his survey was expressed in words, we may give this name to his Catalogue of the positions of the most conspicuous stars. These positions are described by means of _alineations_; that is, three or more such stars are selected as can be touched by an apparent straight line drawn in the heavens. Thus Hipparchus observed that the southern claw of Cancer, the bright star in the same constellation which precedes the head of the Hydra, and the bright star Procyon, were nearly in the same line. Ptolemy quotes this and many other of the configurations which Hipparchus had noted, in order to show that the positions of the stars had not changed in the intermediate time; a truth which the catalogue of Hipparchus thus gave astronomers the means of ascertaining. It contained 1080 stars.

The construction of this catalogue of the stars by Hipparchus is an event of great celebrity in the history of astronomy. Pliny,[77\3] who speaks of it with admiration as a wonderful and superhuman task ("ausus rem etiam Deo improbam, annumerare posteris stellas"), asserts the undertaking to have been suggested by a remarkable astronomical event, the appearance of a new star; "novam stellam et alium in ævo suo genitam deprehendit; ejusque motu, qua die fulsit, ad dubitationem est adductus anne hoc sæpius fieret, moverenturque et eæ quas putamus affixas." There is nothing inherently improbable in this tradition, but we may observe, with Delambre,[78\3] that we are not informed whether this new star remained in the sky, or soon disappeared again. Ptolemy makes no mention of the star or the story; and his catalogue contains no _bright_ star which is not found in the "Catasterisms" of Eratosthenes. These Catasterisms were an enumeration of 475 of the principal stars, according to the constellations in which they are, and were published about sixty years before Hipparchus.

[Note 77\3: _Nat. Hist._ lib. ii. (xxvi.)]

[Note 78\3: _A. A._ i. 290.]

2. _Constant Length of Years._--Hipparchus also attempted to ascertain whether successive years are all of the same length; and though, with his scrupulous love of accuracy,[79\3] he does not appear to have {159} thought himself justified in asserting that the years were always exactly equal, he showed, both by observations of the time when the sun passed the equinoxes, and by eclipses, that the difference of successive years, if there were any difference, must be extremely slight. The observations of succeeding astronomers, and especially of Ptolemy, confirmed this opinion, and proved, with certainty, that there is no progressive increase or diminution in the duration of the year.

[Note 79\3: Ptolem. _Synt._ iii. 2.]

3. _Constant Length of Days. Equation of Time._--The equality of days was more difficult to ascertain than that of years; for the year is measured, as on a natural scale, by the number of days which it contains; but the day can be subdivided into hours only by artificial means; and the mechanical skill of the ancients did not enable them to attain any considerable accuracy in the measure of such portions of time; though clepsydras and similar instruments were used by astronomers. The equality of days could only be proved, therefore, by the consequences of such a supposition; and in this manner it appears to have been assumed, as the fact really is, that the apparent revolution of the stars is accurately uniform, never becoming either quicker or slower. It followed, as a consequence of this, that the solar days (or rather the _nycthemers_, compounded of a night and a day) would be unequal, in consequence of the sun's unequal motion, thus giving rise to what we now call the _Equation of Time_,--the interval by which the time, as marked on a dial, is before or after the time, as indicated by the accurate timepieces which modern skill can produce. This inequality was fully taken account of by the ancient astronomers; and they thus in fact assumed the equality of the sidereal days.

_Sect._ 2.--_Researches which did not verify the Theory._

SOME of the researches of Hipparchus and his followers fell upon the weak parts of his theory; and if the observations had been sufficiently exact, must have led to its being corrected or rejected.

Among these we may notice the researches which were made concerning the _Parallax_ of the heavenly bodies, that is, their apparent displacement by the alteration of position of the observer from one part of the earth's surface to the other. This subject is treated of at length by Ptolemy; and there can be no doubt that it was well examined by Hipparchus, who invented a _parallactic instrument_ for that purpose. The idea of parallax, as a geometrical possibility, was indeed too obvious to be overlooked by geometers at any time; and when the doctrine of the sphere was established, it must have appeared strange {160} to the student, that every place on the earth's surface might alike be considered as the centre of the celestial motions. But if this was true with respect to the motions of the fixed stars, was it also true with regard to those of the sun and moon? The displacement of the sun by parallax is so small, that the best observers among the ancients could never be sure of its

existence; but with respect to the moon, the case is different. She may be displaced by this cause to the amount of twice her own breadth, a quantity easily noticed by the rudest process of instrumental observation. The law of the displacement thus produced is easily obtained by theory, the globular form of the earth being supposed known; but the amount of the displacement depends upon the distance of the moon from the earth, and requires at least one good observation to determine it. Ptolemy has given a table of the effects of parallax, calculated according to the apparent altitude of the moon, assuming certain supposed distances; these distances, however, do not follow the real law of the moon's distances, in consequence of their being founded upon the Hypothesis of the Eccentric and Epicycle.

In fact this Hypothesis, though a very close representation of the truth, so far as the _positions_ of the luminaries are concerned, fails altogether when we apply it to their _distances_. The radius of the epicycle, or the eccentricity of the eccentric, are determined so as to satisfy the observations of the apparent _motions_ of the bodies; but, inasmuch as the hypothetical motions are different altogether from the real motions, the Hypothesis does not, at the same time, satisfy the observations of the _distances_ of the bodies, if we are able to make any such observations.

Parallax is one method by which the distances of the moon, at different times, may be compared; her Apparent Diameters afford another method. Neither of these modes, however, is easily capable of such accuracy as to overturn at once the Hypothesis of epicycles; and, accordingly, the Hypothesis continued to be entertained in spite of such measures; the measures being, indeed, in some degree falsified in consequence of the reigning opinion. In fact, however, the imperfection of the methods of measuring parallax and magnitude, which were in use at this period, was such, their results could not lead to any degree of conviction deserving to be set in opposition to a theory which was so satisfactory with regard to the more certain observations, namely, those of the motions.

The Eccentricity, or the Radius of the Epicycle, which would satisfy {161} the inequality of the _motions_ of the moon, would, in fact, double the inequality of the _distances_. The Eccentricity of the moon's orbit is determined by Ptolemy as 1/12 of the radius of the orbit; but its real amount is only half as great; this difference is a necessary consequence of the supposition of uniform circular motions, on which the Epicyclic Hypothesis proceeds.

We see, therefore, that this part of the Hipparchian theory carries in itself the germ of its own destruction. As soon as the art of celestial measurement was so far perfected, that astronomers could be sure of the apparent diameter of the moon within 1/30 or 1/40 of the whole, the inconsistency of the theory with itself would become manifest. We shall see, hereafter, the way in which this inconsistency operated; in reality a very long period elapsed before the methods of observing were sufficiently good to bring it clearly into view.

_Sect._ 3._--Methods of Observation of the Greek Astronomers._

WE must now say a word concerning the Methods above spoken of. Since one of the most important tasks of verification is to ascertain with accuracy the magnitude

of the quantities which enter, as elements, into the theory which occupies men during the period; the improvement of instruments, and the methods of observing and experimenting, are principal features in such periods. We shall, therefore, mention some of the facts which bear upon this point.

The estimation of distances among the stars by the eye, is an extremely inexact process. In some of the ancient observations, however, this appears to have been the method employed; and stars are described as being _a cubit_ or _two cubits_ from other stars. We may form some notion of the scale of this kind of measurement, from what Cleomedes remarks,[80\3] that the sun appears to be about a foot broad; an opinion which he confutes at length.

[Note 80\3: Del. _A. A._ i. 222.]

A method of determining the positions of the stars, susceptible of a little more exactness than the former, is the use of _alineations_, already noticed in speaking of Hipparchus's catalogue. Thus, a straight line passing through two stars of the Great Bear passes also through the pole-star; this is, indeed, even now a method usually employed to enable us readily to fix on the pole-star; and the two stars β and α of Ursa Major, are hence often called "the pointers." {162}

But nothing like accurate measurements of any portions of the sky were obtained, till astronomers adopted the method of making visual coincidences of the objects with the instruments, either by means of _shadows_ or of _sights_.

Probably the oldest and most obvious measurements of the positions of the heavenly bodies were those in which the elevation of the sun was determined by comparing the length of the shadow of an upright staff or _gnomon_, with the length of the staff itself. It appears,[81\3] from a memoir of Gautil, first printed in the _Connaissance des Temps_ for 1809, that, at the lower town of Loyang, now called Hon-anfou, Tchon-kong found the length of the shadow of the gnomon, at the summer solstice, equal to one foot and a half, the gnomon itself being eight feet in length. This was about 1100 B. C. The Greeks, at an early period, used the same method. Strabo says[82\3] that "Byzantium and Marseilles are on the same parallel of latitude, because the shadows at those places have the same proportion to the gnomon, according to the statement of Hipparchus, who follows Pytheas."

[Note 81\3: Lib. U. K. _Hist. Ast._ p. 5.]

[Note 82\3: Del. _A. A._ i. 257.]

But the relations of position which astronomy considers, are, for the most part, angular distances; and these are most simply expressed by the intercepted portion of a circumference described about the angular point. The use of the gnomon might lead to the determination of the angle by the graphical methods of geometry; but the numerical expression of the circumference required some progress in trigonometry; for instance, a table of the tangents of angles.

Instruments were soon invented for measuring angles, by means of circles, which had a border or _limb_, divided into equal parts. The whole circumference was divided into 360 _degrees_: perhaps because the circles, first so divided, were those

which represented the sun's annual path; one such degree would be the sun's daily advance, more nearly than any other convenient aliquot part which could be taken. The position of the sun was determined by means of the shadow of one part of the instrument upon the other. The most ancient instrument of this kind appears to be the _Hemisphere of Berosus_. A hollow hemisphere was placed with its rim horizontal, and a style was erected in such a manner that the extremity of the style was exactly at the centre of the sphere. The shadow of this extremity, on the concave surface, had the same position with regard to the lowest point of the sphere which the sun had with regard to the highest point of the heavens. {163} But this instrument was in fact used rather for dividing the day into portions of time than for determining position.

Eratosthenes[83\3] observed the amount of the obliquity of the sun's path to the equator: we are not informed what instruments he used for this purpose; but he is said to have obtained, from the munificence of Ptolemy Euergetes, two _Armils_, or instruments composed of circles, which were placed in the portico at Alexandria, and long used for observations. If a circular rim or hoop were placed so as to coincide with the plane of the equator, the inner concave edge would be enlightened by the sun's rays which came under the front edge, when the sun was south of the equator, and by the rays which came over the front edge, when the sun was north of the equator: the moment of the transition would be the time of the equinox. Such an instrument appears to be referred to by Hipparchus, as quoted by Ptolemy.[84\3] "The circle of copper, which stands at Alexandria in what is called the Square Porch, appears to mark, as the day of the equinox, that on which the concave surface begins to be enlightened from the other side." Such an instrument was called an _equinoctial armil_.

[Note 83\3: Delambre, _A. A._ i. 86.]

[Note 84\3: Ptol. _Synt._ iii. 2.]

A _solstitial armil_ is described by Ptolemy, consisting of two circular rims, one sliding round within the other, and the inner one furnished with two pegs standing out from its surface at right angles, and diametrically opposite to each other. These circles being fixed in the plane of the meridian, and the inner one turned, till, at noon, the shadow of the peg in front falls upon the peg behind, the position of the sun at noon would be determined by the degrees on the outer circle.

In calculation, the degree was conceived to be divided into 60 _minutes_, the minute into 60 _seconds_, and so on. But in practice it was impossible to divide the limb of the instrument into parts so small. The armils of Alexandria were divided into no parts smaller than sixths of degrees, or divisions of 10 minutes.

The angles, observed by means of these divisions, were expressed as a fraction of the circumference. Thus Eratosthenes stated the interval between the tropics to be 11/83 of the circumference.[85\3]

[Note 85\3: Delambre, _A. A._ i. 87. It is probable that his observation gave him 47⅔ degrees. The fraction 47⅔/360 = 143/1080 = 11 • 13/1080 = 11/(83+1/13), which is very nearly 11/83.]

It was soon remarked that the whole circumference of the circle {164} was not wanted for such observations. Ptolemy[86\3] says that he found it more convenient to observe altitudes by means of a square flat piece of stone or wood, with a _quadrant_ of a circle described on one of its flat faces, about a centre near one of the angles. A peg was placed at the centre, and one of the extreme radii of the quadrant being perpendicular to the horizon, the elevation of the sun above the horizon was determined by observing the point of the arc of the quadrant on which the shadow of the peg fell.

[Note 86\3: _Synt._ i. 1.]

As the necessity of accuracy in the observations was more and more felt, various adjustments of such instruments were practised. The instruments were placed in the meridian by means of a _meridian line_ drawn by astronomical methods on the floor on which they stood. The plane of the instrument was made vertical by means of a plumb-line: the bounding radius, from which angles were measured, was also adjusted by the _plumb-line_.[87\3]

[Note 87\3: The curvature of the plane of the circle, by warping, was noticed. Ptol. iii. 2. p. 155, observes that his equatorial circle was illuminated on the hollow side twice in the same day. (He did not know that this might arise from refraction.)]

In this manner, the places of the sun and of the moon could be observed by means of the shadows which they cast. In order to observe the stars,[88\3] the observer looked along the face of the circle of the armil, so as to see its two edges apparently brought together, and the star apparently touching them.[89\3]

[Note 88\3: Delamb. _A. A._ i. 185.]

[Note 89\3: Ptol. _Synt._ i. 1. Ὥσπερ κεκολλήμενος ἀμφοτέραις αὐτῶν ταῖς ἐπιφανείαις ὁ ἀστὴρ ἐν τῷ δι' αὐτῶν ἐπιπέδῳ διοπτεύηται.]

It was afterwards found important to ascertain the position of the sun with regard to the ecliptic: and, for this purpose, an instrument, called an _astrolabe_, was invented, of which we have a description in Ptolemy.[90\3] This also consisted of circular rims, movable within one another, or about poles; and contained circles which were to be brought into the position of the ecliptic, and of a plane passing through the sun and the poles of the ecliptic. The position of the moon with regard to the ecliptic, and its position in longitude with regard to the sun or a star, were thus determined.

[Note 90\3: _Synt._ v. 1.]

The astrolabe continued long in use, but not so long as the quadrant described by Ptolemy; this, in a larger form, is the _mural quadrant_, which has been used up to the most recent times.

It may be considered surprising,[91\3] that Hipparchus, after having {165} observed, for some time, right ascensions and declinations, quitted equatorial armils for the astrolabe, which immediately refers the stars to the ecliptic. He probably did this because, after the discovery of precession, he found the latitudes of the stars

constant, and wanted to ascertain their motion in longitude.

[Note 91\3: Del. _A. A._ 181.]

To the above instruments, may be added the _dioptra_, and the _parallactic instrument_ of Hipparchus and Ptolemy. In the latter, the distance of a star from the zenith was observed by looking through two sights fixed in a rule, this being annexed to another rule, which was kept in a vertical position by a plumb-line; and the angle between the two rules was measured.

The following example of an observation, taken from Ptolemy, may serve to show the form in which the results of the instruments, just described, were usually stated.[92\3]

[Note 92\3: Del. _A. A._ ii. 248.]

"In the 2d year of Antoninus, the 9th day of Pharmouthi, the sun being near setting, the last division of Taurus being on the meridian (that is, 5½ equinoctial hours after noon), the moon was in 3 degrees of Pisces, by her distance from the sun (which was 92 degrees, 8 minutes); and half an hour after, the sun being set, and the quarter of Gemini on the meridian, Regulus appeared, by the other circle of the astrolabe, 57½ degrees more forwards than the moon in longitude." From these data the longitude of Regulus is calculated.

From what has been said respecting the observations of the Alexandrian astronomers, it will have been seen that their instrumental observations could not be depended on for any close accuracy. This defect, after the general reception of the Hipparchian theory, operated very unfavorably on the progress of the science. If they could have traced the moon's place distinctly from day to day, they must soon have discovered all the inequalities which were known to Tycho Brahe; and if they could have measured her parallax or her diameter with any considerable accuracy, they must have obtained a confutation of the epicycloidal form of her orbit. By the badness of their observations, and the imperfect agreement of these with calculation, they not only were prevented making such steps, but were led to receive the theory with a servile assent and an indistinct apprehension, instead of that rational conviction and intuitive clearness which would have given a progressive impulse to their knowledge. {166}

_Sect._ 4.--_Period from Hipparchus to Ptolemy._

WE have now to speak of the cultivators of astronomy from the time of Hipparchus to that of Ptolemy, the next great name which occurs in the history of this science; though even he holds place only among those who verified, developed, and extended the theory of Hipparchus. The astronomers who lived in the intermediate time, indeed, did little, even in this way; though it might have been supposed that their studies were carried on under considerable advantages, inasmuch as they all enjoyed the liberal patronage of the kings of Egypt.[93\3] The "divine school of Alexandria," as it is called by Synesius, in the fourth century, appears to have produced few persons capable of carrying forwards, or even of verifying, the labors of its great astronomical teacher. The mathematicians of the school wrote much, and apparently they observed

sometimes; but their observations are of little value; and their books are expositions of the theory and its geometrical consequences, without any attempt to compare it with observation. For instance, it does not appear that any one verified the remarkable discovery of the precession, till the time of Ptolemy, 250 years after; nor does the statement of this motion of the heavens appear in the treatises of the intermediate writers; nor does Ptolemy quote a single observation of any person made in this long interval of time; while his references to those of Hipparchus are perpetual; and to those of Aristyllus and Timocharis, and of others, as Conon, who preceded Hipparchus, are not unfrequent.

[Note 93\3: Delamb. _A. A._ ii. 240.]

This Alexandrian period, so inactive and barren in the history of science, was prosperous, civilized, and literary; and many of the works which belong to it are come down to us, though those of Hipparchus are lost. We have the "Uranologion" of Geminus,[94\3] a systematic treatise on Astronomy, expounding correctly the Hipparchian Theories and their consequences, and containing a good account of the use of the various Cycles, which ended in the adoption of the Calippic Period. We have likewise "The Circular Theory of the Celestial Bodies" of Cleomedes,[95\3] of which the principal part is a development of the doctrine of the sphere, including the consequences of the globular form of the earth. We have also another work on "Spherics" by Theodosius of Bithynia,[96\3] which contains some of the most important propositions of the subject, and has been used as a book of {167} instruction even in modern times. Another writer on the same subject is Menelaus, who lived somewhat later, and whose Three Books on Spherics still remain.

[Note 94\3: B. C. 70.]

[Note 95\3: B. C. 60.]

[Note 96\3: B. C. 50.]

One of the most important kinds of deduction from a geometrical theory, such as that of the doctrine of the sphere, or that of epicycles, is the calculation of its numerical results in particular cases. With regard to the latter theory, this was done in the construction of Solar and Lunar Tables, as we have already seen; and this process required the formation of a _Trigonometry_, or system of rules for calculating the relations between the sides and angles of triangles. Such a science had been formed by Hipparchus, who appears to be the author of every great step in ancient astronomy.[97\3] He wrote a work in twelve books, "On the Construction of the Tables of Chords of Arcs;" such a table being the means by which the Greeks solved their triangles. The Doctrine of the Sphere required, in like manner, a _Spherical Trigonometry_, in order to enable mathematicians to calculate its results; and this branch of science also appears to have been formed by Hipparchus,[98\3] who gives results that imply the possession of such a method. Hypsicles, who was a contemporary of Ptolemy, also made some attempts at the solution of such problems: but it is extraordinary that the writers whom we have mentioned as coming after Hipparchus, namely, Theodosius, Cleomedes, and Menelaus, do not even mention the calculation of triangles,[99\3] either plain or spherical; though the latter

writer[100\3] is said to have written on "the Table of Chords," a work which is now lost.

[Note 97\3: Delamb. _A. A._ ii. 37.]

[Note 98\3: _A. A._ i. 117.]

[Note 99\3: _A. A._ i. 249.]

[Note 100\3: _A. A._ ii. 37.]

We shall see, hereafter, how prevalent a disposition in literary ages is that which induces authors to become commentators. This tendency showed itself at an early period in the school of Alexandria. Aratus,[101\3] who lived 270 B. C. at the court of Antigonus, king of Macedonia, described the celestial constellations in two poems, entitled "Phænomena," and "Prognostics." These poems were little more than a versification of the treatise of Eudoxus on the acronycal and heliacal risings and settings of the stars. The work was the subject of a comment by Hipparchus, who perhaps found this the easiest way of giving connection and circulation to his knowledge. Three Latin translations of this poem gave the Romans the means of becoming acquainted with it: the first is by Cicero, of which we have numerous fragments {168} extant;[102\3] Germanicus Cæsar, one of the sons-in-law of Augustus, also translated the poem, and this translation remains almost entire. Finally, we have a complete translation by Avienus.[103\3] The "Astronomica" of Manilius, the "Poeticon Astronomicon" of Hyginus, both belonging to the time of Augustus, are, like the work of Aratus, poems which combine mythological ornament with elementary astronomical exposition; but have no value in the history of science. We may pass nearly the same judgment upon the explanations and declamations of Cicero, Seneca, and Pliny, for they do not apprise us of any additions to astronomical knowledge; and they do not always indicate a very clear apprehension of the doctrines which the writers adopt.

[Note 101\3: _A. A._ i. 74.]

[Note 102\3: Two copies of this translation, illustrated by drawings of different ages, one set Roman, and the other Saxon, according to Mr. Ottley, are described in the _Archæologia_, vol. xviii.]

[Note 103\3: Montucla, i. 221.]

Perhaps the most remarkable feature in the two last-named writers, is the declamatory expression of their admiration for the discoverers of physical knowledge; and in one of them, Seneca, the persuasion of a boundless progress in science to which man was destined. Though this belief was no more than a vague and arbitrary conjecture, it suggested other conjectures in detail, some of which, having been verified, have attracted much notice. For instance, in speaking of comets,[104\3] Seneca says, "The time will come when those things which are now hidden shall be brought to light by time and persevering diligence. Our posterity will wonder that we should be ignorant of what is so obvious." "The motions of the planets," he adds, "complex and seemingly confused, have been reduced to rule; and some one will come hereafter, who will reveal to us the paths of comets." Such convictions and

conjectures are not to be admired for their wisdom; for Seneca was led rather by enthusiasm, than by any solid reasons, to entertain this opinion; nor, again, are they to be considered as merely lucky guesses, implying no merit; they are remarkable as showing how the persuasion of the universality of law, and the belief of the probability of its discovery by man, grow up in men's minds, when speculative knowledge becomes a prominent object of attention.

[Note 104\3: Seneca, _Qu. N._ vii. 25.]

An important practical application of astronomical knowledge was made by Julius Cæsar, in his correction of the calendar, which we have already noticed; and this was strictly due to the Alexandrian School: Sosigenes, an astronomer belonging to that school, came from Egypt to Rome for the purpose. {169}

_Sect._ 5.--_Measures of the Earth._

THERE were, as we have said, few attempts made, at the period of which we are speaking, to improve the accuracy of any of the determinations of the early Alexandrian astronomers. One question naturally excited much attention at all times, the _magnitude_ of the earth, its figure being universally acknowledged to be a globe. The Chaldeans, at an earlier period, had asserted that a man, walking without stopping, might go round the circuit of the earth in a year; but this might be a mere fancy, or a mere guess. The attempt of Eratosthenes to decide this question went upon principles entirely correct. Syene was situated on the tropic; for there, on the day of the solstice, at noon, objects cast no shadow; and a well was enlightened to the bottom by the sun's rays. At Alexandria, on the same day, the sun was, at noon, distant from the zenith by a fiftieth part of the circumference. Those two cities were north and south from each other: and the distance had been determined, by the royal overseers of the roads, to be 5000 stadia. This gave a circumference of 250,000 stadia to the earth, and a radius of about 40,000. Aristotle[105\3] says that the mathematicians make the circumference 400,000 stadia. Hipparchus conceived that the measure of Eratosthenes ought to be increased by about one-tenth.[106\3] Posidonius, the friend of Cicero, made another attempt of the same kind. At Rhodes, the star Canopus but just appeared above the horizon; at Alexandria, the same star rose to an altitude of 1/48th of the circumference; the direct distance on the meridian was 5000 stadia, which gave 240,000 for the whole circuit. We cannot look upon these measures as very precise; the stadium employed is not certainly known; and no peculiar care appears to have been bestowed on the measure of the direct distance.

[Note 105\3: _De Cœlo_, ii. ad fin.]

[Note 106\3: Plin. ii. (cviii.)]

When the Arabians, in the ninth century, came to be the principal cultivators of astronomy, they repeated this observation in a manner more suited to its real importance and capacity of exactness. Under the Caliph Almamon,[107\3] the vast plain of Singiar, in Mesopotamia, was the scene of this undertaking. The Arabian astronomers there divided themselves into two bands, one under the direction of Chalid ben Abdolmalic, and the other having at its head Alis ben Isa. These two parties proceeded, the one north, the other south, determining the distance by the

actual application of their measuring-rods to the ground, {170} till each was found, by astronomical observation, to be a degree from the place at which they started. It then appeared that these terrestrial degrees were respectively 56 miles, and 56 miles and two-thirds, the mile being 4000 cubits. In order to remove all doubt concerning the scale of this measure, we are informed that the cubit is that called the black cubit, which consists of 27 inches, each inch being the thickness of six grains of barley.

[Note 107\3: Montu. 357.]

_Sect._ 6.--_Ptolemy's Discovery of Evection._

BY referring, in this place, to the last-mentioned measure of the earth, we include the labors of the Arabian as well as the Alexandrian astronomers, in the period of mere detail, which forms the sequel to the great astronomical revolution of the Hipparchian epoch. And this period of verification is rightly extended to those later times; not merely because astronomers were then still employed in determining the magnitude of the earth, and the amount of other elements of the theory,--for these are some of their employments to the present day,--but because no great intervening discovery marks a new epoch, and begins a new period;--because no great revolution in the theory added to the objects of investigation, or presented them in a new point of view. This being the case, it will be more instructive for our purpose to consider the general character and broad intellectual features of this period, than to offer a useless catalogue of obscure and worthless writers, and of opinions either borrowed or unsound. But before we do this, there is one writer whom we cannot leave undistinguished in the crowd; since his name is more celebrated even than that of Hipparchus; his works contain ninety-nine hundredths of what we know of the Greek astronomy; and though he was not the author of a new theory, he made some very remarkable steps in the verification, correction, and extension of the theory which he received. I speak of Ptolemy, whose work, "The Mathematical Construction" (of the heavens), contains a complete exposition of the state of astronomy in his time, the reigns of Adrian and Antonine. This book is familiarly known to us by a term which contains the record of our having received our first knowledge of it from the Arabic writers. The "_Megiste_ Syntaxis," or Great Construction, gave rise, among them, to the title _Al Magisti_, or _Almagest_, by which the work is commonly described. As a mathematical exposition of the Theory of Epicycles and Eccentrics, of the observations and calculations which were employed in {171} order to apply this theory to the sun, moon, and planets, and of the other calculations which are requisite, in order to deduce the consequences of this theory, the work is a splendid and lasting monument of diligence, skill, and judgment. Indeed, all the other astronomical works of the ancients hardly add any thing whatever to the information we obtain from the Almagest; and the knowledge which the student possesses of the ancient astronomy must depend mainly upon his acquaintance with Ptolemy. Among other merits, Ptolemy has that of giving us a very copious account of the manner in which Hipparchus established the main points of his theories; an account the more agreeable, in consequence of the admiration and enthusiasm with which this author everywhere speaks of the great master of the astronomical school.

In our present survey of the writings of Ptolemy, we are concerned less with his exposition of what had been done before him, than with his own original labors. In most of the branches of the subject, he gave additional exactness to what Hipparchus had done; but our main business, at present, is with those parts of the Almagest which contain new steps in the application of the Hipparchian hypothesis. There are two such cases, both very remarkable,--that of the moon's _Evection_, and that of the _Planetary Motions_.

The law of the moon's anomaly, that is, of the leading and obvious inequality of her motion, could be represented, as we have seen, either by an eccentric or an epicycle; and the amount of this inequality had been collected by observations of eclipses. But though the hypothesis of an epicycle, for instance, would bring the moon to her proper place, so far as eclipses could show it, that is, at new and full moon, this hypothesis did not rightly represent her motions at other points of her course. This appeared, when Ptolemy set about measuring her distances from the sun at different times. "These," he[108\3] says, "sometimes agreed, and sometimes disagreed." But by further attention to the facts, a rule was detected in these differences. "As my knowledge became more complete and more connected, so as to show the order of this new inequality, I perceived that this difference was small, or nothing, at new and full moon; and that at both the _dichotomies_ (when the moon is half illuminated) it was small, or nothing, if the moon was at the apogee or perigee of the epicycle, and was greatest when she was in the middle of the interval, and therefore when the first {172} inequality was greatest also." He then adds some further remarks on the circumstances according to which the moon's place, as affected by this new inequality, is before or behind the place, as given by the epicyclical hypothesis.

[Note 108\3: _Synth._ v. 2.]

Such is the announcement of the celebrated discovery of the moon's second inequality, afterwards called (by Bullialdus) the _Evection_. Ptolemy soon proceeded to represent this inequality by a combination of circular motions, uniting, for this purpose, the hypothesis of an epicycle, already employed to explain the first inequality, with the hypothesis of an eccentric, in the circumference of which the centre of the epicycle was supposed to move. The mode of combining these was somewhat complex; more complex we may, perhaps, say, than was absolutely requisite;[109\3] the apogee of the eccentric moved backwards, or contrary to the order of the signs, and the centre of the epicycle moved forwards nearly twice as fast upon the circumference of the eccentric, so as to reach a place nearly, but not exactly, the same, as if it had moved in a concentric instead of an eccentric path. Thus the centre of the epicycle went twice round the eccentric in the course of one month: and in this manner it satisfied the condition that it should vanish at new and full moon, and be greatest when the moon was in the quarters of her monthly course.[110\3]

[Note 109\3: If Ptolemy had used the hypothesis of an eccentric instead of an epicycle for the first inequality of the moon, an epicycle would have represented the second inequality more simply than his method did.]

[Note 110\3: I will insert here the explanation which my German translator, the late distinguished astronomer Littrow, has given of this point. The Rule of this Inequality, the Evection, may be most simply expressed thus. If _a_ denote the excess of the Moon's Longitude over the Sun's, and _b_ the Anomaly of the Moon reckoned from her Perigee, the Evection is equal to 1°. 3. sin (2_a_ - _b_). At New and Full Moon, _a_ is 0 or 180°, and thus the Evection is - 1°.3.sin _b_. At both quarters, or dichotomies, _a_ is 90° or 270°, and consequently the Evection is + 1°.3 . sin _b_. The Moon's Elliptical Equation of the centre is at all points of her orbit equal to 6°.3.sin _b_. The Greek Astronomers before Ptolemy observed the moon only at the time of eclipses; and hence they necessarily found for the sum of these two greatest inequalities of the moon's motion the quantity 6°.3. sin _b_ - 1°.3.sin _b_, or 5°.sin _b_: and as they took this for the moon's equation of the centre, which depends upon the eccentricity of the moon's orbit, we obtain from this too small equation of the centre, an eccentricity also smaller than the truth. Ptolemy, who first observed the moon in her quarters, found for the sum of those Inequalities at those points the quantity 6°.3.sin _b_ + 1°.3.sin _b_, or 7°.6.sin _b_; and thus made the eccentricity of the moon as much too great at the quarters as the observers of eclipses had made it too small. He hence concluded that the eccentricity of the Moon's orbit is variable, which is not the case.]

The discovery of the Evection, and the reduction of it to the {173} epicyclical theory, was, for several reasons, an important step in astronomy; some of these reasons may be stated.

1. It obviously suggested, or confirmed, the suspicion that the motions of the heavenly bodies might be subject to _many_ inequalities:--that when one set of anomalies had been discovered and reduced to rule, another set might come into view;--that the discovery of a rule was a step to the discovery of deviations from the rule, which would require to be expressed in other rules;--that in the application of theory to observation, we find, not only the _stated phenomena_, for which the theory does account, but also _residual phenomena_, which remain unaccounted for, and stand out beyond the calculation;--that thus nature is not simple and regular, by conforming to the simplicity and regularity of our hypotheses, but leads us forwards to apparent complexity, and to an accumulation of rules and relations. A fact like the Evection, explained by an Hypothesis like Ptolemy's, tended altogether to discourage any disposition to guess at the laws of nature from mere ideal views, or from a few phenomena.

2. The discovery of Evection had an importance which did not come into view till long afterwards, in being the first of a numerous series of inequalities of the moon, which results from the _Disturbing Force_ of the sun. These inequalities were successfully discovered; and led finally to the establishment of the law of universal gravitation. The moon's first inequality arises from a different cause;--from the same cause as the inequality of the sun's motion;--from the motion in an ellipse, so far as the central attraction is undisturbed by any other. This first inequality is called the Elliptic Inequality, or, more usually, the _Equation of the Centre_.[111\3] All the planets have such inequalities, but the Evection is peculiar to the moon. The

discovery of other inequalities of the moon's motion, the Variation and Annual Equation, made an immediate sequel in the order of the subject to {174} the discoveries of Ptolemy, although separated by a long interval of time; for these discoveries were only made by Tycho Brahe in the sixteenth century. The imperfection of astronomical instruments was the great cause of this long delay.

[Note 111\3: The Equation of the Centre is the difference between the place of the Planet in its elliptical orbit, and that place which a Planet would have, which revolved uniformly round the Sun as a centre in a circular orbit in the same time. An imaginary Planet moving in the manner last described, is called the _mean_ Planet, while the actual Planet which moves in the ellipse is called the _true_ Planet. The Longitude of the mean Planet at a given time is easily found, because its motion is uniform. By adding to it the Equation of the Centre, we find the Longitude of the true Planet, and thus, its place in its orbit.--_Littrow's Note_.

I may add that the word _Equation_, used in such cases, denotes in general a quantity which must be added to or subtracted from a mean quantity, to make it _equal_ to the true quantity; or rather, a quantity which must be added to or subtracted from a variably increasing quantity to make it increase _equably_.]

3. The Epicyclical Hypothesis was found capable of accommodating itself to such new discoveries. These new inequalities could be represented by new combinations of eccentrics and epicycles: all the real and imaginary discoveries by astronomers, up to Copernicus, were actually embodied in these hypotheses; Copernicus, as we have said, did not reject such hypotheses; the lunar inequalities which Tycho detected might have been similarly exhibited; and even Newton[112\3] represents the motion of the moon's apogee by means of an epicycle. As a mode of expressing the law of the irregularity, and of calculating its results in particular cases, the epicyclical theory was capable of continuing to render great service to astronomy, however extensive the progress of the science might be. It was, in fact, as we have already said, the modern process of representing the motion by means of a series of circular functions.

[Note 112\3: _Principia_, lib. iii. prop. xxxv.]

4. But though the doctrine of eccentrics and epicycles was thus admissible as an Hypothesis, and convenient as a means of expressing the laws of the heavenly motions, the successive occasions on which it was called into use, gave no countenance to it as a Theory; that is, as a true view of the nature of these motions, and their causes. By the steps of the progress of this Hypothesis, it became more and more complex, instead of becoming more simple, which, as we shall see, was the course of the true Theory. The notions concerning the position and connection of the heavenly bodies, which were suggested by one set of phenomena, were not confirmed by the indications of another set of phenomena; for instance, those relations of the epicycles which were adopted to account for the Motions of the heavenly bodies, were not found to fall in with the consequences of their apparent Diameters and Parallaxes. In reality, as we have said, if the relative distances of the sun and moon at different times could have been accurately determined, the Theory of Epicycles must have been forthwith overturned. The insecurity of such measurements alone

maintained the theory to later times.[113\3] {175}

[Note 113\3: The alteration of the apparent diameter of the moon is so great that it cannot escape us, even with very moderate instruments. This apparent diameter contains, when the moon is nearest the earth, 2010 seconds; when she is furthest off 1762 seconds; that is, 248 seconds, or 4 minutes 8 seconds, less than in the former case. [The two quantities are in the proportion of 8 to 7, nearly.]--_Littrow's Note_.]

_Sect._ 7.--_Conclusion of the History of Greek Astronomy._

I MIGHT now proceed to give an account of Ptolemy's other great step, the determination of the Planetary Orbits; but as this, though in itself very curious, would not illustrate any point beyond those already noticed, I shall refer to it very briefly. The planets all move in ellipses about the sun, as the moon moves about the earth; and as the sun apparently moves about the earth. They will therefore each have an Elliptic Inequality or Equation of the centre, for the same reason that the sun and moon have such inequalities. And this inequality may be represented, in the cases of the planets, just as in the other two, by means of an eccentric; the epicycle, it will be recollected, had already been used in order to represent the more obvious changes of the planetary motions. To determine the amount of the Eccentricities and the places of the Apogees of the planetary orbits, was the task which Ptolemy undertook; Hipparchus, as we have seen, having been destitute of the observations which such a process required. The determination of the Eccentricities in these cases involved some peculiarities which might not at first sight occur to the reader. The **elcliptical motion of the planets takes place about the sun; but Ptolemy considered their movements as altogether independent of the sun, and referred them to the earth alone; and thus the apparent eccentricities which he had to account for, were the compound result of the Eccentricity of the earth's orbit, and of the proper eccentricity of the orbit of the Planet. He explained this result by the received mechanism of an eccentric _Deferent_, carrying an Epicycle; but the motion in the Deferent is uniform, not about the centre of the circle, but about another point, the _Equant_. Without going further into detail, it may be sufficient to state that, by a combination of Eccentrics and Epicycles, he did account for the leading features of these motions; and by using his own observations, compared with more ancient ones (for instance, those of Timocharis for Venus), he was able to determine the Dimensions and Positions of the orbits.[114\3] {176}

[Note 114\3: Ptolemy determined the Radius and the Periodic Time of his two circles for each Planet in the following manner: For the _inferior_ Planets, that is, Mercury and Venus, he took the Radius of the Deferent equal to the Radius of the Earth's orbit, and the Radius of the Epicycle equal to that of the Planet's orbit. For these Planets, according to his assumption, the Periodic Time of the Planet in its Epicycle was to the Periodic Time of the Epicyclical Centre on the Deferent, as the _synodical_ Revolution of the Planet to the _tropical_ Revolution of the Earth above the Sun. For the three _superior_ Planets, Mars, Jupiter, and Saturn, the Radius of the Deferent was equal to the Radius of the Planet's orbit, and the Radius of the Epicycle was equal to the Radius of the Earth's orbit; the Periodic Time on the Planet in its Epicycle was to the Periodic Time of the Epicyclical Centre on the Deferent, as

the _synodical_ Revolution of the Planet to the _tropical_ Revolution of the same Planet.

Ptolemy might obviously have made the geometrical motions of all the Planets correspond with the observations by one of these two modes of construction; but he appears to have adopted this double form of the theory, in order that in the inferior, as well as in the superior Planets, he might give the smaller of the two Radii to the Epicycle: that is, in order that he might make the smaller circle move round the larger, not _vice versâ_.--_Littrow's Notes._]

I shall here close my account of the astronomical progress of the Greek School. My purpose is only to illustrate the principles on which the progress of science depends, and therefore I have not at all pretended to touch upon every part of the subject. Some portion of the ancient theories, as, for instance, the mode of accounting for the motions of the moon and planets in latitude, are sufficiently analogous to what has been explained, not to require any more especial notice. Other parts of Greek astronomical knowledge, as, for instance, their acquaintance with refraction, did not assume any clear or definite form, and can only be considered as the prelude to modern discoveries on the same subject. And before we can with propriety pass on to these, there is a long and remarkable, though unproductive interval, of which some account must be given.

_Sect._ 8.--_Arabian Astronomy._

THE interval to which I have just alluded may be considered as extending from Ptolemy to Copernicus; we have no advance in Greek astronomy after the former; no signs of a revival of the power of discovery till the latter. During this interval of 1350 years,[115\3] the principal cultivators of astronomy were the Arabians, who adopted this science from the Greeks whom they conquered, and from whom the conquerors of western Europe again received back their treasure, when the love of science and the capacity for it had been awakened in their minds. In the intervening time, the precious deposit had undergone little change. The Arab astronomer had been the scrupulous but unprofitable servant, who kept his talent without apparent danger of loss, but also without prospect of increase. There is little in {177} Arabic literature which bears upon the _progress_ of astronomy; but as the little that there is must be considered as a sequel to the Greek science, I shall notice one or two points before I treat of the stationary period in general.

[Note 115\3: Ptolemy died about A. D. 150. Copernicus was living A. D. 1500.]

When the sceptre of western Asia had passed into the hands of the Abasside caliphs,[116\3] Bagdad, "the city of peace," rose to splendor and refinement, and became the metropolis of science under the successors of Almansor the Victorious, as Alexandria had been under the successors of Alexander the Great. Astronomy attracted peculiarly the favor of the powerful as well as the learned; and almost all the culture which was bestowed upon the science, appears to have had its source in the patronage, often also in the personal studies, of Saracen princes. Under such encouragement, much was done, in those scientific labors which money and rank can command. Translations of Greek works were made, large instruments were erected,

observers were maintained; and accordingly as observation showed the defects and imperfection of the extant tables of the celestial motions, new ones were constructed. Thus under Almansor, the Grecian works of science were collected from all quarters, and many of them translated into Arabic.[117\3] The translation of the "Megiste Syntaxis" of Ptolemy, which thus became the Almagest, is ascribed to Isaac ben Homain in this reign.

[Note 116\3: Gibbon, x. 31.]

[Note 117\3: Id. x. 36.]

The greatest of the Arabian Astronomers comes half a century later. This is Albategnius, as he is commonly called; or more exactly, Mohammed ben Geber Albatani, the last appellation indicating that he was born at Batan, a city of Mesopotamia.[118\3] He was a Syrian prince, whose residence was at Aracte or Racha in Mesopotamia: a part of his observations were made at Antioch. His work still remains to us in Latin. "After having read," he says, "the Syntaxis of Ptolemy, and learnt the methods of calculation employed by the Greeks, his observations led him to conceive that some improvements might be made in their results. He found it necessary to add to Ptolemy's observations as Ptolemy had added to those of Abrachis" (Hipparchus). He then published Tables of the motions of the sun, moon, and planets, which long maintained a high reputation.

[Note 118\3: Del. _Astronomie du Moyen Age_, 4.]

These, however, did not prevent the publication of others. Under the Caliph Hakem (about A. D. 1000) Ebon Iounis published Tables of the Sun, Moon, and Planets, which were hence called the _Hakemite Tables_. Not long after, Arzachel of Toledo published the _Toletan_ {178} Tables. In the 13th century, Nasir Eddin published Tables of the Stars, dedicated to Ilchan, a Tartar prince, and hence termed the _Ilchanic_ Tables. Two centuries later, Ulugh Beigh, the grandson of Tamerlane, and prince of the countries beyond the Oxus, was a zealous practical astronomer; and his Tables, which were published in Europe by Hyde in 1665, are referred to as important authority by modern astronomers. The series of Astronomical Tables which we have thus noticed, in which, however, many are omitted, leads us to the _Alphonsine_ Tables, which were put forth in 1488, and in succeeding years, under the auspices of Alphonso, king of Castile; and thus brings us to the verge of modern astronomy.

For all these Tables, the Ptolemaic hypotheses were employed; and, for the most part, without alteration. The Arabs sometimes felt the extreme complexity and difficulty of the doctrine which they studied; but their minds did not possess that kind of invention and energy by which the philosophers of Europe, at a later period, won their way into a simpler and better system.

Thus Alpetragius states, in the outset of his "Planetarum Theorica," that he was at first astonished and stupefied with this complexity, but that afterwards "God was pleased to open to him the occult secret in the theory of his orbs, and to make known to him the truth of their essence and the rectitude of the quality of their motion." His system consists, according to Delambre,[119\3] in attributing to the planets a spiral

motion from east to west, an idea already refuted by Ptolemy. Geber of Seville criticises Ptolemy very severely,[120\3] but without introducing any essential alteration into his system. The Arabian observations are in many cases valuable; both because they were made with more skill and with better instruments than those of the Greeks; and also because they illustrate the permanence or variability of important elements, such as the obliquity of the ecliptic and the inclination of the moon's orbit.

[Note 119\3: Delambre, _M. A._ p. 7.]

[Note 120\3: _M. A._ p. 180, &c.]

We must, however, notice one or two peculiar Arabian doctrines. The most important of these is the discovery of the Motion of the Son's Apogee by Albategnius. He found the Apogee to be in longitude 82 degrees; Ptolemy had placed it in longitude 65 degrees. The difference of 17 degrees was beyond all limit of probable error of calculation, though the process is not capable of great precision; and the inference of the Motion of the Apogee was so obvious, that we cannot {179} agree with Delambre, in doubting or extenuating the claim of Albategnius to this discovery, on the ground of his not having expressly stated it.

In detecting this motion, the Arabian astronomers reasoned rightly from facts well observed: they were not always so fortunate. Arzachel, in the 11th century, found the apogee of the sun to be less advanced than Albategnius had found it, by some degrees; he inferred that it had receded in the intermediate time; but we now know, from an acquaintance with its real rate of moving, that the true inference would have been, that Albategnius, whose method was less trustworthy than that of Arzachel, had made an error to the amount of the difference thus arising. A curious, but utterly false hypothesis was founded on observations thus erroneously appreciated; namely, the _Trepidation of the fixed stars_. Arzachel conceived that a uniform Precession of the equinoctial points would not account for the apparent changes of position of the stars, and that for this purpose, it was necessary to conceive two circles of about eight degrees radius described round the equinoctial points of the immovable sphere, and to suppose the first points of Aries and Libra to describe the circumference of these circles in about 800 years. This would produce, at one time a progression, and at another a regression, of the apparent equinoxes, and would moreover change the latitude of the stars. Such a motion is entirely visionary; but the doctrine made a sect among astronomers, and was adopted in the first edition of the Alphonsine Tables, though afterwards rejected.

An important exception to the general unprogressive character of Arabian science has been pointed out recently by M. Sedillot.[121\3] It appears that Mohammed-Aboul Wefa-al-Bouzdjani, an Arabian astronomer of the tenth century, who resided at Cairo, and observed at Bagdad in 975, discovered a third inequality of the moon, in addition to the two expounded by Ptolemy, the Equation of the Centre, and the Evection. This third inequality, the _Variation_, is usually supposed to have been discovered by Tycho Brahe, six centuries later. It is an inequality of the moon's motion, in virtue of which she moves quickest when she is at new or full, and slowest

at the first and third quarter; in consequence of this, from the first quarter to the full, she is behind her mean place; at the full, she does not differ from her mean place; from the full to the third quarter, she is before her true {180} place; and so on; and the greatest effect of the inequality is in the _octants_, or points half-way between the four quarters. In an Almagest of Aboul Wefa, a part of which exists in the Royal Library at Paris, after describing the two inequalities of the moon, he has a Section ix., "Of the Third Anomaly of the moon called _Muhazal_ or _Prosneusis_." He there says, that taking cases when the moon was in apogee or perigee, and when, consequently, the effect of the two first inequalities vanishes, he found, _by observation of the moon_, when she was nearly _in trine_ and _in sextile_ with the sun, that she was a degree and a quarter from her calculated place. "And hence," he adds, "I perceived that this anomaly exists independently of the two first: and this can only take place by a declination of the diameter of the epicycle with respect to the centre of the zodiac."

[Note 121\3: Sedillot, Nouvelles Rech. sur l'Hist. de l'Astron. chez les Arabes. _Nouveau Journal Asiatique_. 1836.]

We may remark that we have here this inequality of the moon made out in a really philosophical manner; a residual quantity in the moon's longitude being detected by observation, and the cases in which it occurs selected and grouped by an inductive effort of the mind. The advance is not great; for Aboul Wefa appears only to have detected the existence, and not to have fixed the law or the exact quantity of the inequality; but still it places the scientific capacity of the Arabs in a more favorable point of view than any circumstance with which we were previously acquainted.

But this discovery of Aboul Wefa appears to have excited no notice among his contemporaries and followers: at least it had been long quite forgotten when Tycho Brahe rediscovered the same lunar inequality. We can hardly help looking upon this circumstance as an evidence of a servility of intellect belonging to the Arabian period. The learned Arabians were so little in the habit of considering science as progressive, and looking with pride and confidence at examples of its progress, that they had not the courage to believe in a discovery which they themselves had made, and were dragged back by the chain of authority, even when they had advanced beyond their Greek masters.

As the Arabians took the whole of their theory (with such slight exceptions as we have been noticing) from the Greeks, they took from them also the mathematical processes by which the consequences of the theory were obtained. Arithmetic and Trigonometry, two main branches of these processes, received considerable improvements at their hands. In the former, especially, they rendered a service to the world which it is difficult to estimate too highly, in abolishing the {181} cumbrous Sexagesimal Arithmetic of the Greeks, and introducing the notation by means of the digits 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, which we now employ.[122\3] These numerals appear to be of Indian origin, as is acknowledged by the Arabs themselves; and thus form no exception to the sterility of the Arabian genius as to great scientific inventions. Another improvement, of a subordinate kind, but of great utility, was Arabian, being made by Albategnius. He introduced into calculation the _sine_, or half-chord of the

double arc, instead of the chord of the arc itself, which had been employed by the Greek astronomers. There have been various conjectures concerning the origin of the word _sine_; the most probable appears to be that _sinus_ is the Latin translation of the Arabic word _gib_, which signifies a fold, the two halves of the chord being conceived to be folded together.

[Note 122\3: Mont. i. 376.]

The great obligation which Science owes to the Arabians, is to have preserved it during a period of darkness and desolation, so that Europe might receive it back again when the evil days were past. We shall see hereafter how differently the European intellect dealt with this hereditary treasure when once recovered.

Before quitting the subject, we may observe that Astronomy brought back, from her sojourn among the Arabs, a few terms which may still be perceived in her phraseology. Such are the _zenith_, and the opposite imaginary point, the _nadir_;--the circles of the sphere termed _almacantars_ and _azimuth_ circles. The _alidad_ of an instrument is its index, which possesses an angular motion. Some of the stars still retain their Arabic names; _Aldebran_, _Rigel_, _Fomalhaut_; many others were known by such appellations a little while ago. Perhaps the word _almanac_ is the most familiar vestige of the Arabian period of astronomy.

It is foreign to my purpose to note any efforts of the intellectual faculties among other nations, which may have taken place independently of the great system of progressive European culture, from which all our existing science is derived. Otherwise I might speak of the astronomy of some of the Orientals, for example, the Chinese, who are said, by Montucla (i. 465), to have discovered the first equation of the moon, and the proper motion of the fixed stars (the Precession), in the third century of our era. The Greeks had made these discoveries 500 years earlier.

{{183}} BOOK IV.

HISTORY OF PHYSICAL SCIENCE IN THE MIDDLE AGES; OR, VIEW OF THE STATIONARY PERIOD OF INDUCTIVE SCIENCE.

In vain, in vain! the all-composing hour Resistless falls . . . . . As one by one, at dread Medea's strain, The sickening stars fade off th' ethereal plain; As Argus' eyes, by Hermes' wand opprest, Closed one by one to everlasting rest; Thus at her felt approach and secret might, Art after art goes out, and all is night. See skulking Truth to her old cavern fled, Mountains of casuistry heaped on her head; Philosophy, that reached the heavens before, Shrinks to her hidden cause, and is no more. Physic of Metaphysic begs defence, And Metaphysic calls for aid to Sense: See Mystery to Mathematics fly! In vain! they gaze, turn giddy, rave, and die.

_Dunciad_, B. iv.

{{185}} INTRODUCTION.

WE have now to consider more especially a long and barren period, which intervened between the scientific activity of ancient Greece and that of modern Europe; and which we may, therefore, call the Stationary Period of Science. It would

be to no purpose to enumerate the various forms in which, during these times, men reproduced the discoveries of the inventive ages; or to trace in them the small successes of Art, void of any principle of genuine Philosophy. Our object requires rather that we should point out the general and distinguishing features of the intellect and habits of those times. We must endeavor to delineate the character of the Stationary Period, and, as far as possible, to analyze its defects and errors; and thus obtain some knowledge of the causes of its barrenness and darkness.

We have already stated, that real scientific progress requires distinct general Ideas, applied to many special and certain Facts. In the period of which we now have to speak, men's Ideas were obscured; their disposition to bring their general views into accordance with Facts was enfeebled. They were thus led to employ themselves unprofitably, among indistinct and unreal notions. And the evil of these tendencies was further inflamed by moral peculiarities in the character of those times;--by an abjectness of thought on the one hand, which could not help looking towards some intellectual superior, and by an impatience of dissent on the other. To this must be added an enthusiastic temper, which, when introduced into speculation, tends to subject the mind's operations to ideas altogether distorted and delusive.

These characteristics of the stationary period, its obscurity of thought, its servility, its intolerant disposition, and its enthusiastic temper, will be treated of in the four following chapters, on the Indistinctness of Ideas, the Commentatorial Spirit, the Dogmatism, and the Mysticism of the Middle Ages. {186}

## CHAPTER I.

ON THE INDISTINCTNESS OF IDEAS OF THE MIDDLE AGES.

THAT firm and entire possession of certain clear and distinct general ideas which is necessary to sound science, was the character of the minds of those among the ancients who created the several sciences which arose among them. It was indispensable that such inventors should have a luminous and steadfast apprehension of certain general relations, such as those of space and number, order and cause; and should be able to apply these notions with perfect readiness and precision to special facts and cases. It is necessary that such scientific notions should be more definite and precise than those which common language conveys; and in this state of unusual clearness, they must be so familiar to the philosopher, that they are the language in which he thinks. The discoverer is thus led to doctrines which other men adopt and follow out, in proportion as they seize the fundamental ideas, and become acquainted with the leading facts. Thus Hipparchus, conceiving clearly the motions and combinations of motion which enter into his theory, saw that the relative lengths of the seasons were sufficient data for determining the form of the sun's orbit; thus Archimedes, possessing a steady notion of mechanical pressure, was able, not only to deduce the properties of the lever and of the centre of gravity, but also to see the truth of those principles respecting the distribution of pressure in fluids, on which the science of hydrostatics depends.

With the progress of such distinct ideas, the inductive sciences rise and flourish; with the decay and loss of such distinct ideas, these sciences become stationary, languid, and retrograde. When men merely repeat the terms of science, without attaching to them any clear conceptions;--when their apprehensions become vague and dim;--when they assent to scientific doctrines as a matter of tradition, rather than of conviction, on trust rather than on sight;--when science is considered as a collection of opinions, rather than a record of laws by which the universe is really governed;--it must inevitably happen, that men will lose their hold on the knowledge which the great discoverers who preceded them have brought to light. They are not able to push forwards the truths on which they lay so {187} feeble and irresolute a hand; probably they cannot even prevent their sliding back towards the obscurity from which they had been drawn, or from being lost altogether. Such indistinctness and vacillation of thought appear to have prevailed in the stationary period, and to be, in fact, intimately connected with its stationary character. I shall point out some indications of the intellectual peculiarity of which I speak.

1. _Collections of Opinions._--The fact, that mere Collections of the opinions of physical philosophers came to hold a prominent place in literature, already indicated a tendency to an indistinct and wandering apprehension of such opinions. I speak of such works as Plutarch's five Books "on the Opinions of Philosophers," or the physical opinions which Diogenes Laërtius gives in his "Lives of the Philosophers." At an earlier period still, books of this kind appear; as for instance, a large portion of Pliny's Natural History, a work which has very appropriately been called the Encyclopædia of Antiquity; even Aristotle himself is much in the habit of enumerating the opinions of those who had preceded him. To present such statements as an important part of physical philosophy, shows an erroneous and loose apprehension of its nature. For the only proof of which its doctrines admit, is the possibility of applying the general theory to each particular case; the authority of great men, which in moral and practical matters may or must have its weight, is here of no force; and the technical precision of ideas which the terms of a sound physical theory usually demand, renders a mere statement of the doctrines very imperfectly intelligible to readers familiar with common notions only. To dwell upon such collections of opinions, therefore, both implies, and produces, in writers and readers, an obscure and inadequate apprehension of the full meaning of the doctrines thus collected; supposing there be among them any which really possess such a clearness, solidity, and reality, as to make them important in the history of science. Such diversities of opinion convey no truth; such a multiplicity of statements of what has been _said_, in no degree teaches us what _is_; such accumulations of indistinct notions, however vast and varied, do not make up one distinct idea. On the contrary, the habit of dwelling upon the verbal expressions of the views of other persons, and of being content with such an apprehension of doctrines as a transient notice can give us, is fatal to firm and clear thought: it indicates wavering and feeble conceptions, which are inconsistent with speculation. {188}

We may, therefore, consider the prevalence of Collections of the kind just referred to, as indicating a deficiency of philosophical talent in the ages now under review. As evidence of the same character, we may add the long train of publishers of Abstracts,

Epitomes, Bibliographical Notices, and similar writers. All such writers are worthless for all purposes of _science_, and their labors may be considered as dead works; they have in them no principle of philosophical vitality; they draw their origin and nutriment from the death of true physical knowledge; and resemble the swarms of insects that are born from the perishing carcass of some noble animal.

2. _Indistinctness of Ideas in Mechanics._--But the indistinctness of thought which is so fatal a feature in the intellect of the stationary period, may be traced more directly in the works, even of the best authors, of those times. We find that they did not retain steadily the ideas on which the scientific success of the previous period had depended. For instance, it is a remarkable circumstance in the history of the science of Mechanics, that it did not make any advance from the time of Archimedes to that of Stevinus and Galileo. Archimedes had established the doctrine of the lever; several persons tried, in the intermediate time, to prove the property of the inclined plane, and none of them succeeded. But let us look to the attempts; for example, that of Pappus, in the eighth Book of his Mathematical Collections, and we may see the reason of the failure. His Problem shows, in the very terms in which it is propounded, the want of a clear apprehension of the subject. "Having given the power which will draw a given weight along the horizontal plane, to find the additional power which will draw the same weight along a given inclined plane." This is proposed without previously defining how Powers, producing such effects, are to be measured; and as if the speed with which the body were drawn, and the nature of the surface of the plane, were of no consequence. The proper elementary Problem is, To find the force which will _support_ a body on a smooth inclined plane; and no doubt the solution of Pappus has more reference to this problem than to his own. His reasoning is, however, totally at variance with mechanical ideas on any view of the problem. He supposes the weight to be formed into a sphere; and this sphere being placed in contact with the inclined plane, he assumes that the effect will be the same as if the weight were supported on a horizontal lever, the fulcrum being the point of contact of the sphere with the plane, and the power acting at the circumference of the sphere. Such an assumption implies an entire {189} absence of those distinct ideas of force and mechanical pressure, on which our perception of the identity or difference of different modes of action must depend;--of those ideas by the help of which Archimedes had been able to demonstrate the properties of the lever, and Stevinus afterwards discovered the true solution of the problem of the inclined plane. The motive to Pappus's assumption was probably no more than this;--he perceived that the additional power, which he thus obtained, vanished when the plane became horizontal, and increased as the inclination became greater. Thus his views were vague; he had no clear conception of mechanical action, and he tried a geometrical conjecture. This is not the way to real knowledge.

Pappus (who lived about A. D. 400) was one of the best mathematicians of the Alexandrian school; and, on subjects where his ideas were so indistinct, it is not likely that any much clearer were to be found in the minds of his contemporaries. Accordingly, on all subjects of speculative mechanics, there appears to have been an entire confusion and obscurity of thought till modern times. Men's minds were busy in endeavoring to systematize the distinctions and subtleties of the Aristotelian

school, concerning Motion and Power; and, being thus employed among doctrines in which there was involved no definite meaning capable of real exemplification, they, of course, could not acquire sound physical knowledge. We have already seen that the physical opinions of Aristotle, even as they came from him, had no proper scientific precision. His followers, in their endeavors to perfect and develop his statements, never attempted to introduce clearer ideas than those of their master; and as they never referred, in any steady manner, to facts, the vagueness of their notions was not corrected by any collision with observation. The physical doctrines which they extracted from Aristotle were, in the course of time, built up into a regular system; and though these doctrines could not be followed into a practical application without introducing distinctions and changes, such as deprived the terms of all steady signification, the dogmas continued to be repeated, till the world was persuaded that they were self-evident; and when, at a later period, experimental philosophers, such as Galileo and Boyle, ventured to contradict these current maxims, their new principles sounded in men's ears as strange as they now sound familiar. Thus Boyle promulgated his opinions on the mechanics of fluids, as "Hydrostatical _Paradoxes_, proved and illustrated by experiments." And the opinions which he there opposes, are those which the Aristotelian philosophers habitually propounded as certain {190} and indisputable; such, for instance, as that "in fluids the upper parts do not gravitate on the lower;" that "a lighter fluid will not gravitate on a heavier;" that "levity is a positive quality of bodies as well as gravity." So long as these assertions were left uncontested and untried, men heard and repeated them, without perceiving the incongruities which they involved: and thus they long evaded refutation, amid the vague notions and undoubting habits of the stationary period. But when the controversies of Galileo's time had made men think with more acuteness and steadiness, it was discovered that many of these doctrines were inconsistent with themselves, as well as with experiment. We have an example of the confusion of thought to which the Aristotelians were liable, in their doctrine concerning falling bodies. "Heavy bodies," said they, "must fall quicker than light ones; for weight is the cause of their fall, and the weight of the greater bodies is greater." They did not perceive that, if they considered the weight of the body as a power acting to produce motion, they must consider the body itself as offering a resistance to motion; and that the effect must depend on the proportion of the power to the resistance; in short, they had no clear idea of _accelerating force_. This defect runs through all their mechanical speculations, and renders them entirely valueless.

We may exemplify the same confusion of thought on mechanical subjects in writers of a less technical character. Thus, if men had any distinct idea of mechanical action, they could not have accepted for a moment the fable of the Echineis or Remora, a little fish which was said to be able to stop a large ship merely by sticking to it.[1\4] Lucan refers to this legend in a poetical manner, and notices this creature only in bringing together a collection of monstrosities; but Pliny relates the tale gravely, and moralizes upon it after his manner. "What," he cries,[2\4] "is more violent than the sea and the winds? what a greater work of art than a ship? Yet one little fish (the Echineis) can hold back all these when they all strain the same way. The winds may {191} blow, the waves may rage; but this small creature controls their fury, and

stops a vessel, when chains and anchors would not hold it: and this it does, not by hard labor, but merely by adhering to it. Alas, for human vanity! when the turreted ships which man has built, that he may fight from castle-walls, at sea as well as at land, are held captive and motionless by a fish a foot and a half long! Such a fish is said to have stopped the admiral's ship at the battle of Actium, and compelled Antony to go into another. And in our own memory, one of these animals held fast the ship of Caius, the emperor, when he was sailing from Astura to Antium. The stopping of this ship, when all the rest of the fleet went on, caused surprise; but this did not last long, for some of the men jumped into the water to look for the fish, and found it sticking to the rudder; they showed it to Caius, who was indignant that this animal should interpose its prohibition to his progress, when impelled by four hundred rowers. It was like a slug; and had no power, after it was taken into the ship."

[Note 1\4: Lucan is describing one of the poetical compounds produced in incantations. Huc quicquid fœtu genuit Natura sinistro Miscetur: non spuma canum quibus unda timori est, Viscera non lyncis, non duræ nodus hyænæ Defuit, et cervi pasti serpente medullæ; In mediis _Echineis_ aquis, oculique draconum. Etc. _Pharsalia_, **vi. 670.]

[Note 2\4: Plin. _Hist. N._ xxxii. 5.]

A very little advance in the power of thinking clearly on the force which it exerted in pulling, would have enabled the Romans to see that the ship and its rowers must pull the adhering fish by the hold the oars had upon the water; and that, except the fish had a hold equally strong on some external body, it could not resist this force.

3. _Indistinctness of Ideas shown in Architecture._--Perhaps it may serve to illustrate still further the extent to which, under the Roman empire, men's notions of mechanical relations became faint, wavered, and disappeared, if we observe the change which took place in architecture. All architecture, to possess genuine beauty, must be mechanically consistent. The decorative members must represent a structure which has in it a principle of support and stability. Thus the Grecian colonnade was a straight horizontal beam, resting on vertical props; and the pediment imitated a frame like a roof, where oppositely inclined beams support each other. These forms of building were, therefore, proper models of art, because they implied supporting forces. But to be content with colonnades and pediments, which, though they imitated the forms of the Grecian ones, were destitute of their mechanical truth, belonged to the decline of art; and showed that men had lost the idea of force, and retained only that of shape. Yet this was what the architects of the Roman empire did. Under their hands, the pediment was severed at its vertex, and divided into separate halves, so that it was no longer a mechanical possibility. The entablature no longer lay straight from pillar to pillar, but, projecting over each {192} column, turned back to the wall, and adhered to it in the intervening space. The splendid remains of Palmyra, Balbec, Petra, exhibit endless examples of this kind of perverse inventiveness; and show us, very instructively, how the decay of art and of science alike accompany this indistinctness of ideas which we are now endeavoring to illustrate.

4. _Indistinctness of Ideas in Astronomy._--Returning to the sciences, it may be supposed, at first sight, that, with regard to astronomy, we have not the same ground for charging the stationary period with indistinctness of ideas on that subject, since they were able to acquire and verify, and, in some measure, to apply, the doctrines previously established. And, undoubtedly, it must be confessed that men's notions of the relations of space and number are never very indistinct. It appears to be impossible for these chains of elementary perception ever to be much entangled. The later Greeks, the Arabians, and the earliest modern astronomers, must have conceived the hypotheses of the Ptolemaic system with tolerable completeness. And yet, we may assert, that during the stationary period, men did not possess the notions, even of space and number, in that vivid and vigorous manner which enables them to discover new truths. If they had perceived distinctly that the astronomical theorist had merely to do with _relative_ motions, they must have been led to see the possibility, at least, of the Copernican system; as the Greeks, at an earlier period, had already perceived it. We find no trace of this. Indeed, the mode in which the Arabian mathematicians present the solutions of their problems, does not indicate that clear apprehension of the relations of space, and that delight in the contemplation of them, which the Greek geometrical speculations imply. The Arabs are in the habit of giving conclusions without demonstrations, precepts without the investigations by which they are obtained; as if their main object were practical rather than speculative,--the calculation of results rather than the exposition of theory. Delambre[3\4] has been obliged to exercise great ingenuity, in order to discover the method by which Ibn Iounis proved his solution of certain difficult problems.

[Note 3\4: Delamb. _M. A._ p. 125-8.]

5. _Indistinctness of Ideas shown by Skeptics._--The same unsteadiness of ideas which prevents men from obtaining clear views, and steady and just convictions, on special subjects, may lead them to despair of or deny the possibility of acquiring certainty at all, and may thus make them skeptics with regard to all knowledge. Such skeptics {193} are themselves men of indistinct views, for they could not otherwise avoid assenting to the demonstrated truths of science; and, so far as they may be taken as specimens of their contemporaries, they prove that indistinct ideas prevail in the age in which they appear. In the stationary period, moreover, the indefinite speculations and unprofitable subtleties of the schools might further impel a man of bold and acute mind to this universal skepticism, because they offered nothing which could fix or satisfy him. And thus the skeptical spirit may deserve our notice as indicative of the defects of a system of doctrine too feeble in demonstration to control such resistance.

The most remarkable of these philosophical skeptics is Sextus Empiricus; so called, from his belonging to that medical sect which was termed the _empirical_, in contradistinction to the _rational_ and _methodical_ sects. His works contain a series of treatises, directed against all the divisions of the science of his time. He has chapters against the Geometers, against the Arithmeticians, against the Astrologers, against the Musicians, as well as against Grammarians, Rhetoricians, and Logicians; and, in short, as a modern writer has said, his skepticism is employed as a sort of

frame-work which embraces an encyclopedical view of human knowledge. It must be stated, however, that his objections are rather to the metaphysical grounds, than to the details of the sciences; he rather denies the possibility of speculative truth in general, than the experimental truths which had been then obtained. Thus his objections to geometry and arithmetic are founded on abstract cavils concerning the nature of points, letters, unities, &c. And when he comes to speak against astrology, he says, "I am not going to consider that perfect science which rests upon geometry and arithmetic; for I have already shown the weakness of those sciences: nor that faculty of prediction (of the motions of the heavens) which belongs to the pupils of Eudoxus, and Hipparchus, and the rest, which some call Astronomy; for that is an observation of phenomena, like agriculture or navigation: but against the Art of Prediction from the time of birth, which the Chaldeans exercise." Sextus, therefore, though a skeptic by profession, was not insensible to the difference between experimental knowledge and mystical dogmas, though even the former had nothing which excited his admiration.

The skepticism which denies the evidence of the truths of which the best established physical sciences consist, must necessarily involve a very indistinct apprehension of those truths; for such truths, properly exhibited, contain their own evidence, and are the best antidote {194} to this skepticism. But an incredulity or contempt towards the asserted truths of physical science may arise also from the attention being mainly directed to the certainty and importance of religious truths. A veneration for revealed religion may thus assume the aspect of a skepticism with regard to natural knowledge. Such appears to be the case with Algazel or Algezeli, who is adduced by Degerando[4\4] as an example of an Arabian skeptic. He was a celebrated teacher at Bagdad in the eleventh century, and he declared himself the enemy, not only of the mixed Peripatetic and Platonic philosophy of the time, but of Aristotle himself. His work entitled _The Destructions of the Philosophers_, is known to us by the refutation of it which Averrhoes published, under the title of _Destruction of Algazel's Destructions of the Philosophers_. It appears that he contested the fundamental principles both of the Platonic and of the Aristotelian schools, and denied the possibility of a known connection between cause and effect; thus making a prelude, says Degerando, to the celebrated argumentation of Hume.

[Note 4\4: Degerando, _Hist. Comp. de Systèmes_, iv. 224.]

[2d Ed.] Since the publication of my first edition, an account of Algazel or Algazzali and his works has been published under the title of _Essai sur les Ecoles Philosophiques chez les Arabes, et notamment sur la Doctrine d'Algazzali_, par August Schmölders. Paris. 1842. From this book it appears that Degerando's account of Algazzali is correct, when he says[5\4] that "his skepticism seems to have essentially for its object to destroy all systems of merely rational theology, in order to open an indefinite career, not only to faith guided by revelation, but also to the free exaltation of a mystical enthusiasm." It is remarked by Dr. Schmölders, following M. de Hammer-Purgstall, that the title of the work referred to in the text ought rather to be _Mutual Refutation of the Philosophers_: and that its object is to show that Philosophy consists of a mass of systems, each of which overturns the others. The

work of Algazzali which Dr. Schmölders has published, _On the Errors of Sects, &c._, contains a kind of autobiographical account of the way in which the author was led to his views. He does not reject the truths of science, but he condemns the mental habits which are caused by laying too much stress upon science. Religious men, he says, are, by such a course, led to reject all science, even what relates to eclipses of the moon and sun; and men of science are led to hate religion.[6\4] {195}

[Note 5\4: _Hist. Comp._ iv. p. 227.]

[Note 6\4: _Essai_, p. 33.]

6. _Neglect of Physical Reasoning in Christendom._--If the Arabians, who, during the ages of which we are speaking, were the most eminent cultivators of science, entertained only such comparatively feeble and servile notions of its doctrines, it will easily be supposed, that in the Christendom of that period, where physical knowledge was comparatively neglected, there was still less distinctness and vividness in the prevalent ideas on such subjects. Indeed, during a considerable period of the history of the Christian Church, and by many of its principal authorities, the study of natural philosophy was not only disregarded but discommended. The great practical doctrines which were presented to men's minds, and the serious tasks, of the regulation of the will and affections, which religion impressed upon them, made inquiries of mere curiosity seem to be a reprehensible misapplication of human powers; and many of the fathers of the Church revived, in a still more peremptory form, the opinion of Socrates, that the only valuable philosophy is that which teaches us our moral duties and religious hopes.[7\4] Thus Eusebius says,[8\4] "It is not through ignorance of the things admired by them, but through contempt of their useless labor, that we think little of these matters, turning our souls to the exercise of better things." When the thoughts were thus intentionally averted from those ideas which natural philosophy involves, the ideas inevitably became very indistinct in their minds; and they could not conceive that any other persons could find, on such subjects, grounds of clear conviction and certainty. They held the whole of their philosophy to be, as Lactantius[9\4] asserts it to be, "empty and false." "To search," says he, "for the causes of natural things; to inquire whether the sun be as large as he seems, whether the moon is convex or concave, whether the stars are fixed in the sky or float freely in the air; of what size and of what material are the heavens; whether they be at rest or in motion; what is the magnitude of the earth; on what foundations it is suspended and balanced;--to dispute and conjecture on such matters, is just as if we chose to discuss what we think of a city in a remote country, of which we never heard but the name." It is impossible to express more forcibly that absence of any definite notions on physical subjects which led to this tone of thought.

[Note 7\4: Brucker, iii. 317.]

[Note 8\4: _Præp. Ev._ xv. 61.]

[Note 9\4: _Inst._ 1. iii. init.]

7. _Question of Antipodes._--With such habits of thought, we are not to be surprised if the relations resulting from the best established theories were apprehended in an imperfect and incongruous manner. {196} We have some

remarkable examples of this; and a very notable one is the celebrated question of the existence of _Antipodes_, or persons inhabiting the opposite side of the globe of the earth, and consequently having the soles of their feet directly opposed to ours. The doctrine of the globular form of the earth results, as we have seen, by a geometrical necessity, from a clear conception of the various points of knowledge which we obtain, bearing upon that subject. This doctrine was held distinctly by the Greeks; it was adopted by all astronomers, Arabian and European, who followed them; and was, in fact, an inevitable part of every system of astronomy which gave a consistent and intelligible representation of phenomena. But those who did not call before their minds any distinct representation at all, and who referred the whole question to other relations than those of space, might still deny this doctrine; and they did so. The existence of inhabitants on the opposite side of the terraqueous globe, was a fact of which experience alone could teach the truth or falsehood; but the religious relations, which extend alike to all mankind, were supposed to give the Christian philosopher grounds for deciding against the possibility of such a race of men. Lactantius,[10\4] in the fourth century, argues this matter in a way very illustrative of that impatience of such speculations, and consequent confusion of thought, which we have mentioned. "Is it possible," he says, "that men can be so absurd as to believe that the crops and trees on the other side of the earth hang downwards, and that men there have their feet higher than their heads? If you ask of them how they defend these monstrosities--how things do not fall away from the earth on that side--they reply, that the nature of things is such that heavy bodies tend towards the centre, like the spokes of a wheel, while light bodies, as clouds, smoke, fire, tend from the centre towards the heavens on all sides. Now I am really at a loss what to say of those who, when they have once gone wrong, steadily persevere in their folly, and defend one absurd opinion by another." It is obvious that so long as the writer refused to admit into his thoughts the fundamental conception of their theory, he must needs be at a loss what to say to their arguments without being on that account in any degree convinced of their doctrines.

[Note 10\4: _Inst._ 1. iii. 23.]

In the sixth century, indeed, in the reign of Justinian, we find a writer (Cosmas Indicopleustes[11\4]) who does not rest in this obscurity of {197} representation; but in this case, the distinctness of the pictures only serves to show his want of any clear conception as to what suppositions would explain the phenomena. He describes the earth as an oblong floor, surrounded by upright walls, and covered by a vault, below which the heavenly bodies perform their revolutions, going round a certain high mountain, which occupies the northern parts of the earth, and makes night by intercepting the light of the sun. In Augustin[12\4] (who flourished A. D. 400) the opinion is treated on other grounds; and without denying the globular form of the earth, it is asserted that there are no inhabitants on the opposite side, because no such race is recorded by Scripture among the descendants of Adam.[13\4] Considerations of the same kind operated in the well-known instance of Virgil, Bishop of Salzburg, in the eighth century. When he was reported to Boniface, Archbishop of Mentz, as holding the existence of Antipodes, the prelate was shocked at the assumption, as it seemed to him, of a world of human beings, out of the reach of the

conditions of salvation; and application was made to Pope Zachary for a censure of the holder of this dangerous doctrine. It does not, however, appear that this led to any severity; and the story of the deposition of Virgil from his bishopric, which is circulated by Kepler and by more modern writers, is undoubtedly altogether false. The same scruples continued to prevail among Christian writers to a later period; and Tostatus[14\4] notes the opinion of the rotundity of the earth as an "unsafe" doctrine, only a few years before Columbus visited the other hemisphere.

[Note 11\4: Montfaucon, _Collectio Nova Patrum_, t. ii. p. 113. Cosmas Indicopleustes. Christianorum Opiniones de Mundo, sive Topographia Christiana.]

[Note 12\4: _Civ. D._ xvi. 9.]

[Note 13\4: It appears, however, that scriptural arguments were found on the other side. St. Jerome says (_Comm. in Ezech._ i. 6), speaking of the two cherubims with four faces, seen by the prophet, and the interpretation of the vision: "Alii vero qui philosophorum stultam sequuntur sapientiam, duo hemispheria in duobus templi cherubim, nos et antipodes, quasi supinos et cadentes homines suspicantur."]

[Note 14\4: Montfauc. _Patr._ t. ii.]

8. _Intellectual Condition of the Religious Orders._--It must be recollected, however, that though these were the views and tenets of many religious writers, and though they may be taken as indications of the prevalent and characteristic temper of the times of which we speak, they never were universal. Such a confusion of thought affects the minds of many persons, even in the most enlightened times; and in what we call the Dark Ages, though clear views on such subjects might be more rare, those who gave their minds to science, entertained the true opinion of the figure of the earth. Thus Boëthius[15\4] (in the sixth century) urges the smallness of the globe of the earth, {198} compared with the heavens, as a reason to repress our love of glory. This work, it will be recollected, was translated into the Anglo-Saxon by our own Alfred. It was also commented on by Bede, who, in what he says on this passage, assents to the doctrine, and shows an acquaintance with Ptolemy and his commentators, both Arabian and Greek. Gerbert, in the tenth century, went from France to Spain to study astronomy with the Arabians, and soon surpassed his masters. He is reported to have fabricated clocks, and an astrolabe of peculiar construction. Gerbert afterwards (in the last year of the first thousand from the birth of Christ) became pope, by the name of Sylvester II. Among other cultivators of the sciences, some of whom, from their proficiency, must have possessed with considerable clearness and steadiness the elementary ideas on which it depends, we may here mention, after Montucla,[16\4] Adelbold, whose work On the Sphere was addressed to Pope Sylvester, and whose geometrical reasonings are, according to Montucla,[17\4] vague and chimerical; Hermann Contractus, a monk of St Gall, who, in 1050, published astronomical works; William of Hirsaugen, who followed his example in 1080; Robert of Lorraine, who was made Bishop of Hereford by William the Conqueror, in consequence of his astronomical knowledge. In the next century, Adelhard Goth, an Englishman, travelled among the Arabs for purposes of study, as Gerbert had done in the preceding age; and on his return, translated the Elements of

Euclid, which he had brought from Spain or Egypt. Robert Grostête, Bishop of Lincoln, was the author of an Epitome on the Sphere; Roger Bacon, in his youth the contemporary of Robert, and of his brother Adam Marsh, praises very highly their knowledge in mathematics.

[Note 15\4: Boëthius, _Cons._ ii. pr. 7.]

[Note 16\4: Mont. i. 502.]

[Note 17\4: Ib. i. 503.]

"And here," says the French historian of mathematics, whom I have followed in the preceding relation, "it is impossible not to reflect that all those men who, if they did not augment the treasure of the sciences, at least served to transmit it, were monks, or had been such originally. Convents were, during these stormy ages, the asylum of sciences and letters. Without these religious men, who, in the silence of their monasteries, occupied themselves in transcribing, in studying, and in imitating the works of the ancients, well or ill, those works would have perished; perhaps not one of them would have come down to us. The thread which connects us with the Greeks and Romans would have been snapt asunder; the precious productions of {199} ancient literature would no more exist for us, than the works, if any there were, published before the catastrophe that annihilated that highly scientific nation, which, according to Bailly, existed in remote ages in the centre of Tartary, or at the roots of Caucasus. In the sciences we should have had all to create; and at the moment when the human mind should have emerged from its stupor and shaken off its slumbers, we should have been no more advanced than the Greeks were after the taking of Troy." He adds, that this consideration inspires feelings towards the religious orders very different from those which, when he wrote, were prevalent among his countrymen.

Except so far as their religious opinions interfered, it was natural that men who lived a life of quiet and study, and were necessarily in a great measure removed from the absorbing and blinding interests with which practical life occupies the thoughts, should cultivate science more successfully than others, precisely because their ideas on speculative subjects had time and opportunity to become clear and steady. The studies which were cultivated under the name of the Seven Liberal Arts, necessarily tended to favor this effect. The _Trivium_,[18\4] indeed, which consisted of Grammar, Logic, and Rhetoric, had no direct bearing upon those ideas with which physical science is concerned; but the _Quadrivium_, Music, Arithmetic, Geometry, Astronomy, could not be pursued with any attention, without a corresponding improvement of the mind for the purposes of sound knowledge.[19\4]

[Note 18\4: Bruck. iii. 597.]

[Note 19\4: Roger Bacon, in his _Specula Mathematica_, cap. i., says "Harum scientiarum porta et clavis est mathematica, quam sancti a principio mundi invenerunt, etc. Cujus negligentia _jam per triginta vel quadraginta annos_ destruxit totum studium Latinorum." I do not know on what occasion this neglect took place.]

9. _Popular Opinions._--That, even in the best intellects, something was wanting to fit them for scientific progress and discovery, is obvious from the fact that science

was so long absolutely stationary. And I have endeavored to show that one part of this deficiency was the want of the requisite clearness and vigor of the fundamental scientific ideas. If these were wanting, even in the most powerful and most cultivated minds, we may easily conceive that still greater confusion and obscurity prevailed in the common class of mankind. They actually adopted the belief, however crude and inconsistent, that the form of the earth and heavens really is what at any place it appears to be; that the earth is flat, and the waters of the sky sustained above a material floor, through which in showers they descend. Yet the true doctrines of {200} astronomy appear to have had some popular circulation. For instance, a French poem of the time of Edward the Second, called _Ymage du Monde_, contains a metrical account of the earth and heavens, according to the Ptolemaic views; and in a manuscript of this poem, preserved in the library of the University of Cambridge, there are representations, in accordance with the text, of a spherical earth, with men standing upright upon it on every side; and by way of illustrating the tendency of all things to the centre, perforations of the earth, entirely through its mass, are described and depicted; and figures are exhibited dropping balls down each of these holes, so as to meet in the interior. And, as bearing upon the perplexity which attends the motions of _up_ and _down_, when applied to the globular earth, and the change of the direction of gravity which would occur in passing the centre, the readers of Dante will recollect the extraordinary manner in which the poet and his guide emerge from the bottom of the abyss; and the explanation which Virgil imparts to him of what he there sees. After they have crept through the aperture in which Lucifer is placed, the poet says, "Io levai gli occhi e credetti vedere Lucifero com' io l' avea lasciato, E vidile le gambe in su tenere." . . . . . "Questi come è fitto Si sottasopra!" . . . . . "Quando mi volsi, tu passast' il punto Al qual si traggon d' ogni parte i pesi." _Inferno_, xxxiv.

. . . . . "I raised mine eyes, Believing that I Lucifer should see Where he was lately left, but saw him now With legs held upward." . . . . . "How standeth he in posture thus reversed?" . . . . . "Thou wast on the other side so long as I Descended; when I turned, thou didst o'erpass That point to which from every part is dragged All heavy substance." CARY.

This is more philosophical than Milton's representation, in a more scientific age, of Uriel sliding to the earth on a sunbeam, and sliding back again, when the sun had sunk below the horizon. . . . . "Uriel to his charge Returned on that bright beam whose point now raised, Bore him slope downward to the sun, now fallen Beneath the Azores." _Par. Lost_, B. iv. {201}

The philosophical notions of up and down are too much at variance with the obvious suggestions of our senses, to be held steadily and justly by minds undisciplined in science. Perhaps it was some misunderstood statement of the curved surface of the ocean, which gave rise to the tradition of there being a part of the sea directly over the earth, from which at times an object has been known to fall or an anchor to be let down. Even such whimsical fancies are not without instruction, and may serve to show the reader what that vagueness and obscurity of ideas is, of which I have been endeavoring to trace the prevalence in the dark ages.

We now proceed to another of the features which appears to me to mark, in a very prominent manner, the character of the stationary period.

## CHAPTER II.

THE COMMENTATORIAL SPIRIT OF THE MIDDLE AGES.

WE have already noticed, that, after the first great achievements of the founders of sound speculation, in the different departments of human knowledge, had attracted the interest and admiration which those who became acquainted with them could not but give to them, there appeared a disposition among men to lean on the authority of some of these teachers;--to study the opinions of others as the only mode of forming their own;--to read nature through books;--to attend to what had been already thought and said, rather than to what really is and happens. This tendency of men's minds requires our particular consideration. Its manifestations were very important, and highly characteristic of the stationary period; it gave, in a great degree, a peculiar bias and direction to the intellectual activity of many centuries; and the kind of labor with which speculative men were occupied in consequence of this bias, took the place of that examination of realities which must be their employment, in order that real knowledge may make any decided progress.

In some subjects, indeed, as, for instance, in the domains of morals, poetry, and the arts, whose aim is the production of beauty, this opposition between the study of former opinion and present reality, may not be so distinct; inasmuch as it may be said by some, that, in these subjects, opinions are realities; that the thoughts and feelings which {202} prevail in men's minds are the material upon which we must work, the particulars from which we are to generalize, the instruments which we are to use; and that, therefore, to reject the study of antiquity, or even its authority, would be to show ourselves ignorant of the extent and mutual bearing of the elements with which we have to deal;--would be to cut asunder that which we ought to unite into a vital whole. Yet even in the provinces of history and poetry, the poverty and servility of men's minds during the middle ages, are shown by indications so strong as to be truly remarkable; for instance, in the efforts of the antiquarians of almost every European country to assimilate the early history of their own state to the poet's account of the foundation of Rome, by bringing from the sack of Troy, Brutus to England, Bavo to Flanders, and so on. But however this may be, our business at present is, to trace the varying spirit of the _physical_ philosophy of different ages; trusting that, hereafter, this prefatory study will enable us to throw some light upon the other parts of philosophy. And in physics the case undoubtedly was, that the labor of observation, which is one of the two great elements of the progress of knowledge, was in a great measure superseded by the collection, the analysis, the explanation, of previous authors and opinions; experimenters were replaced by commentators; criticism took the place of induction; and instead of great discoverers we had learned men.

1. _Natural Bias to Authority._--It is very evident that, in such a bias of men's studies, there is something very natural; however strained and technical this erudition may have been, the propensities on which it depends are very general, and

are easily seen. Deference to the authority of thoughtful and sagacious men, a disposition which men in general neither reject nor think they ought to reject in practical matters, naturally clings to them, even in speculation. It is a satisfaction to us to suppose that there are, or have been, minds of transcendent powers, of wide and wise views, superior to the common errors and blindness of our nature. The pleasure of admiration, and the repose of confidence, are inducements to such a belief. There are also other reasons why we willingly believe that there are in philosophy great teachers, so profound and sagacious, that, in order to arrive at truth, we have only to learn their thoughts, to understand their writings. There is a peculiar interest which men feel in dealing with the thoughts of their fellow-men, rather than with brute matter. Matter feels and excites no sympathies: in seeking for mere laws of nature, there is nothing of mental intercourse with the great spirits of the past, as there is in {203} studying Aristotle or Plato. Moreover, a large portion of this employment is of a kind the most agreeable to most speculative minds; it consists in tracing the consequences of assumed principles: it is deductive like geometry: and the principles of the teachers being known, and being undisputed, the deduction and application of their results is an obvious, self-satisfying, and inexhaustible exercise of ingenuity.

These causes, and probably others, make criticism and commentation flourish, when invention begins to fail, oppressed and bewildered by the acquisitions it has already made; and when the vigor and hope of men's minds are enfeebled by civil and political changes. Accordingly,[20\4] the Alexandrian school was eminently characterized by a spirit of erudition, of literary criticism, of interpretation, of imitation. These practices, which reigned first in their full vigor in "the Museum," are likely to be, at all times, the leading propensities of similar academical institutions.

[Note 20\4: Degerando, _Hist. des Syst. de Philos._ iii. p. 134.]

How natural it is to select a great writer as a paramount authority, and to ascribe to him extraordinary profundity and sagacity, we may see, in the manner in which the Greeks looked upon Homer; and the fancy which detected in his poems traces of the origin of all arts and sciences, has, as we know, found favor even in modern times. To pass over earlier instances of this feeling, we may observe, that Strabo begins his Geography by saying that he agrees with Hipparchus, who had declared Homer to be the first author of our geographical knowledge; and he does not confine the application of this assertion to the various and curious topographical information which the Iliad and Odyssey contain, concerning the countries surrounding the Mediterranean; but in phrases which, to most persons, might appear the mere play of a poetical fancy, or a casual selection of circumstances, he finds unquestionable evidence of a correct knowledge of general geographical truths. Thus,[21\4] when Homer speaks of the sun "rising from the soft and deep-flowing ocean," of his "splendid blaze plunging in the ocean;" of the northern constellation "Alone unwashen by the ocean wave;" and of Jupiter, "who goes to the ocean to feast with the blameless Ethiopians;" Strabo is satisfied from these passages that Homer knew the dry land to be surrounded with water: and he reasons in like manner with respect to other points of geography. {204}

[Note 21\4: Strabo, i. p. 5.]

2. _Character of Commentators._--The spirit of commentation, as has already been suggested, turns to questions of taste, of metaphysics, of morals, with far more avidity than to physics. Accordingly, critics and grammarians were peculiarly the growth of this school; and, though the commentators sometimes chose works of mathematical or physical science for their subject (as Proclus, who commented on Euclid's Geometry, and Simplicius, on Aristotle's Physics), these commentaries were, in fact, rather metaphysical than mathematical. It does not appear that the commentators have, in any instance, illustrated the author by bringing his assertions of facts to the test of experiment. Thus, when Simplicius comments on the passage concerning a vacuum, which we formerly adduced, he notices the argument which went upon the assertion, that a vessel full of ashes would contain as much water as an empty vessel; and he mentions various opinions of different authors, but no trial of the fact. Eudemus had said, that the ashes contained something hot, as quicklime does, and that by means of this, a part of the water was evaporated; others supposed the water to be condensed, and so on.[22\4]

[Note 22\4: Simplicius, p. 170.]

The Commentator's professed object is to explain, to enforce, to illustrate doctrines assumed as true. He endeavors to adapt the work on which he employs himself to the state of information and of opinion in his own time; to elucidate obscurities and technicalities; to supply steps omitted in the reasoning; but he does not seek to obtain additional truths or new generalizations. He undertakes only to give what is virtually contained in his author; to develop, but not to create. He is a cultivator of the thoughts of others: his labor is not spent on a field of his own; he ploughs but to enrich the granary of another man. Thus he does not work as a freeman, but as one in a servile condition; or rather, his is a menial, and not a productive service: his office is to adorn the appearance of his master, not to increase his wealth.

Yet though the Commentator's employment is thus subordinate and dependent, he is easily led to attribute to it the greatest importance and dignity. To elucidate good books is, indeed, a useful task; and when those who undertake this work execute it well, it would be most unreasonable to find fault with them for not doing more. But the critic, long and earnestly employed on one author, may easily underrate the relative value of other kinds of mental exertion. He may {205} ascribe too large dimensions to that which occupies the whole of his own field of vision. Thus he may come to consider such study as the highest aim, and best evidence of human genius. To understand Aristotle, or Plato, may appear to him to comprise all that is possible of profundity and acuteness. And when he has travelled over a portion of their domain, and satisfied himself that of this he too is master, he may look with complacency at the circuit he has made, and speak of it as a labor of vast effort and difficulty. We may quote, as an expression of this temper, the language of Sir Henry Savile, in concluding a course of lectures on Euclid, delivered at Oxford.[23\4] "By the grace of God, gentlemen hearers, I have performed my promise; I have redeemed my pledge. I have explained, according to my ability, the definitions, postulates, axioms,

and _first eight propositions_ of the Elements of Euclid. Here, sinking under the weight of years, I lay down my art and my instruments."

[Note 23\4: Exolvi per Dei gratiam, Domini auditores, promissum; liberavi fldem meam; explicavi pro meo modulo, definitiones, petitiones, communes sententias, et _octo priores propositiones_ Elementorum Euclidis. Hic, annis fessus, cyclos artemque repono.]

We here speak of the peculiar province of the Commentator; for undoubtedly, in many instances, a commentary on a received author has been made the vehicle of conveying systems and doctrines entirely different from those of the author himself; as, for instance, when the New Platonists wrote, taking Plato for their text. The labors of learned men in the stationary period, which came under this description, belong to another class.

3. _Greek Commentators on Aristotle._--The commentators or disciples of the great philosophers did not assume at once their servile character. At first their object was to supply and correct, as well as to explain their teacher. Thus among the earlier commentators of Aristotle, Theophrastus invented five moods of syllogism in the first figure, in addition to the four invented by Aristotle, and stated with additional accuracy the rules of hypothetical syllogisms. He also not only collected much information concerning animals, and natural events, which Aristotle had omitted, but often differed with his master; as, for instance, concerning the saltness of the sea: this, which the Stagirite attributed to the effect of the evaporation produced by the sun's rays, was ascribed by Theophrastus to beds of salt at the bottom. Porphyry,[24\4] who flourished in the third century, wrote a book on the _Predicables_, which was found to be so suitable a complement {206} to the _Predicaments_ or Categories of Aristotle, that it was usually prefixed to that treatise; and the two have been used as an elementary work together, up to modern times. The Predicables are the five steps which the gradations of generality and particularity introduce;--_genus_, _species_, _difference_, _individual_, _accident_:--the Categories are the ten heads under which assertions or predications may be arranged:--_substance_, _quantity_, _relation_, _quality_, _place_, _time_, _position_, _habit_, _action_, _passion_.

[Note 24\4: Buhle, Arist. i. 284.]

At a later period, the Aristotelian commentators became more servile, and followed the author step by step, explaining, according to their views, his expressions and doctrines; often, indeed, with extreme prolixity, expanding his clauses into sentences, and his sentences into paragraphs. Alexander Aphrodisiensis, who lived at the end of the second century, is of this class; "sometimes useful," as one of the recent editors of Aristotle says;[25\4] "but by the prolixity of his interpretation, by his perverse itch for himself discussing the argument expounded by Aristotle, for defending his opinions, and for refuting or reconciling those of others, he rather obscures than enlightens." At various times, also, some of the commentators, and especially those of the Alexandrian school, endeavored to reconcile, or combined without reconciling, opposing doctrines of the great philosophers of the earlier times. Simplicius, for

instance, and, indeed, a great number of the Alexandrian Philosophers,[26\4] as Alexander, Ammonius, and others, employed themselves in the futile task of reconciling the doctrines of the Pythagoreans, of the Eleatics, of Plato, and of the Stoics, with those of Aristotle. Boethius[27\4] entertained the design of translating into Latin the whole of Aristotle's and Plato's works, and of showing their agreement; a gigantic plan, which he never executed. Others employed themselves in disentangling the confusion which such attempts produced, as John the Grammarian, surnamed Philoponus, "the Labor-loving;" who, towards the end of the seventh century, maintained that Aristotle was entirely misunderstood by Porphyry and Proclus,[28\4] who had pretended to incorporate his doctrines into those of the New Platonic school, or even to reconcile him with Plato himself on the subject of _ideas_. Others, again, wrote Epitomes, Compounds, Abstracts; and endeavored to throw the works of the philosopher into some simpler and more obviously regular form, as John of Damascus, in {207} the middle of the eighth century, who made abstracts of some of Aristotle's works, and introduced the study of the author into theological education. These two writers lived under the patronage of the Arabs; the former was favored by Amrou, the conqueror of Egypt; the latter was at first secretary to the Caliph, but afterwards withdrew to a monastery.[29\4]

[Note 25\4: Ib. i. 288.]

[Note 26\4: Ib. i. 311.]

[Note 27\4: Degerando, _Hist. des Syst._ iv. 100.]

[Note 28\4: Ib. iv. 155.]

[Note 29\4: Deg. iv. 150.]

At this period the Arabians became the fosterers and patrons of philosophy, rather than the Greeks. Justinian had, by an edict, closed the school of Athens, the last of the schools of heathen philosophy. Leo, the Isaurian, who was a zealous Iconoclast, abolished also the schools where general knowledge had been taught, in combination with Christianity,[30\4] yet the line of the Aristotelian commentators was continued, though feebly, to the later ages of the Greek empire. Anna Comnena[31\4] mentions a Eustratus who employed himself upon the dialectic and moral treatises, and whom she does not hesitate to elevate above the Stoics and Platonists, for his talent in philosophical discussions. Nicephorus Blemmydes wrote logical and physical epitomes for the use of John Ducas; George Pachymerus composed an epitome of the philosophy of Aristotle, and a compend of his logic; Theodore Metochytes, who was famous in his time alike for his eloquence and his learning, has left a paraphrase of the books of Aristotle on Physics, on the Soul, the Heavens,[32\4] &c. Fabricius states that this writer has a chapter, the object of which is to prove, that all philosophers, and Aristotle and Plato in particular, have disdained the authority of their predecessors. He could hardly help remarking in how different a spirit philosophy had been pursued since their time.

[Note 30\4: Ib. iv. 163.]

[Note 31\4: Ib. 167.]

[Note 32\4: Ib. 168.]

4. _Greek Commentators of Plato and others._--I have spoken principally of the commentators of Aristotle, for he was the great subject of the commentators proper; and though the name of his rival, Plato, was graced by a list of attendants, hardly less numerous, these, the Neoplatonists, as they are called, had introduced new elements into the doctrines of their nominal master, to such an extent that they must be placed in a different class. We may observe here, however, how, in this school as in the Peripatetic, the race of commentators multiplied itself. Porphyry, who commented on Aristotle, was commented on by Ammonius; Plotinus's Enneads were commented on by Proclus and Dexippus. Psellus[33\4] the elder was a paraphrast of {208} Aristotle; Psellus the younger, in the eleventh century, attempted to restore the New Platonic school. The former of these two writers had for his pupils two men, the emperor Leo, surnamed the Philosopher, and Photius the patriarch, who exerted themselves to restore the study of literature at Constantinople. We still possess the Collection of Extracts of Photius, which, like that of Stobæus and others, shows the tendency of the age to compilations, abstracts, and epitomes,--the extinction of philosophical vitality.

[Note 33\4: Deg. iv. 169.]

5. _Arabian Commentators of Aristotle._--The reader might perhaps have expected, that when the philosophy of the Greeks was carried among a new race of intellects, of a different national character and condition, the train of this servile tradition would have been broken; that some new thoughts would have started forth; that some new direction, some new impulse, would have been given to the search for truth. It might have been anticipated that we should have had schools among the Arabians which should rival the Peripatetic, Academic, and Stoic among the Greeks;--that they would preoccupy the ground on which Copernicus and Galileo, Lavoisier and Linnæus, won their fame;--that they would make the next great steps in the progressive sciences. Nothing of this, however, happened. The Arabians cannot claim, in science or philosophy, any really great names; they produced no men and no discoveries which have materially influenced the course and destinies of human knowledge; they tamely adopted the intellectual servitude of the nation which they conquered by their arms; they joined themselves at once to the string of slaves who were dragging the car of Aristotle and Plotinus. Nor, perhaps, on a little further reflection, shall we be surprised at this want of vigor and productive power, in this period of apparent national youth. The Arabians had not been duly prepared rightly to enjoy and use the treasures of which they became possessed. They had, like most uncivilized nations, been passionately fond of their indigenous poetry; their imagination had been awakened, but their rational powers and speculative tendencies were still torpid. They received the Greek philosophy without having passed through those gradations of ardent curiosity and keen research, of obscurity brightening into clearness, of doubt succeeded by the joy of discovery, by which the Greek mind had been enlarged and exercised. Nor had the Arabians ever enjoyed, as the Greeks had, the individual consciousness, the independent volition, the intellectual freedom, arising from the freedom of political institutions. They had not felt the contagious mental activity of a small city,--the elation arising from the general {209} sympathy in speculative

pursuits diffused through an intelligent and acute audience; in short, they had not had a national education such as fitted the Greeks to be disciples of Plato and Hipparchus. Hence, their new literary wealth rather encumbered and enslaved, than enriched and strengthened them: in their want of taste for intellectual freedom, they were glad to give themselves up to the guidance of Aristotle and other dogmatists. Their military habits had accustomed them to look to a leader; their reverence for the book of their law had prepared them to accept a philosophical Koran also. Thus the Arabians, though they never translated the Greek poetry, translated, and merely translated, the Greek philosophy; they followed the Greek philosophers without deviation, or, at least, without any philosophical deviations. They became for the most part Aristotelians;--studied not only Aristotle, but the commentators of Aristotle; and themselves swelled the vast and unprofitable herd.

The philosophical works of Aristotle had, in some measure, made their way in the East, before the growth of the Saracen power. In the sixth century, a Syrian, Uranus,[34\4] encouraged by the love of philosophy manifested by Cosroes, had translated some of the writings of the Stagirite; about the same time, Sergius had given some translations in Syriac. In the seventh century, Jacob of Edessa translated into this language the Dialectics, and added Notes to the work. Such labors became numerous; and the first Arabic translations of Aristotle were formed upon these Persian or Syriac texts. In this succession of transfusions, some mistakes must inevitably have been introduced.

[Note 34\4: Deg. iv. 196.]

The Arabian interpreters of Aristotle, like a large portion of the Alexandrian ones, gave to the philosopher a tinge of opinions borrowed from another source, of which I shall have to speak under the head of _Mysticism_. But they are, for the most part, sufficiently strong examples of the peculiar spirit of commentation, to make it fitting to notice them here. At the head of them stands[35\4] Alkindi, who appears to have lived at the court of Almamon, and who wrote commentaries on the Organon of Aristotle. But Alfarabi was the glory of the school of Bagdad; his knowledge included mathematics, astronomy, medicine, and philosophy. Born in an elevated rank, and possessed of a rich patrimony, he led an austere life, and devoted himself altogether to study and meditation. He employed himself particularly in unfolding the import of Aristotle's treatise On the Soul.[36\4] Avicenna (Ebn Sina) {210} was at once the Hippocrates and the Aristotle of the Arabians; and certainly the most extraordinary man that the nation produced. In the course of an unfortunate and stormy life, occupied by politics and by pleasures, he produced works which were long revered as a sort of code of science. In particular, his writings on medicine, though they contain little besides a compilation of Hippocrates and Galen, took the place of both, even in the universities of Europe; and were studied as models at Paris and Montpelier, till the end of the seventeenth century, at which period they fell into an almost complete oblivion. Avicenna is conceived, by some modern writers,[37\4] to have shown some power of original thinking in his representations of the Aristotelian Logic and Metaphysics. Averroes (Ebn Roshd) of Cordova, was the most illustrious of the Spanish Aristotelians, and became the guide of the schoolmen,[38\4] being placed by

them on a level with Aristotle himself, or above him. He translated Aristotle from the first Syriac version, not being able to read the Greek text. He aspired to, and retained for centuries, the title of the _Commentator_; and he deserves this title by the servility with which he maintains that Aristotle[39\4] carried the sciences to the highest possible degree, measured their whole extent, and fixed their ultimate and permanent boundaries; although his works are conceived to exhibit a trace of the New Platonism. Some of his writings are directed against an Arabian skeptic, of the name of Algazel, whom we have already noticed.

[Note 35\4: Ib. iv. 187.]

[Note 36\4: Ib. iv. 205.]

[Note 37\4: Deg. iv. 206.]

[Note 38\4: Ib. iv. 247. Averroes died A. D. 1206.]

[Note 39\4: Ib. iv. 248.]

When the schoolmen had adopted the supremacy of Aristotle to the extent in which Averroes maintained it, their philosophy went further than a system of mere commentation, and became a system of dogmatism; we must, therefore, in another chapter, say a few words more of the Aristotelians in this point of view, before we proceed to the revival of science; but we must previously consider some other features in the character of the Stationary Period. {211}

## CHAPTER III.

OF THE MYSTICISM OF THE MIDDLE AGES.

IT has been already several times hinted, that a new and peculiar element was introduced into the Greek philosophy which occupied the attention of the Alexandrian school; and that this element tinged a large portion of the speculations of succeeding ages. We may speak of this peculiar element as _Mysticism_; for, from the notion usually conveyed by this term, the reader will easily apprehend the general character of the tendency now spoken of; and especially when he sees its effect pointed out in various subjects. Thus, instead of referring the events of the external world to space and time, to sensible connection and causation, men attempted to reduce such occurrences under spiritual and supersensual relations and dependencies; they referred them to superior intelligences, to theological conditions, to past and future events in the moral world, to states of mind and feelings, to the creatures of an imaginary mythology or demonology. And thus their physical Science became Magic, their Astronomy became Astrology, the study of the Composition of bodies became Alchemy, Mathematics became the contemplation of the Spiritual Relations of number and figure, and Philosophy became Theosophy.

The examination of this feature in the history of the human mind is important for us, in consequence of its influence upon the employments and the thoughts of the times now under our notice. This tendency materially affected both men's speculations and their labours in the pursuit of knowledge. By its direct operation, it

gave rise to the newer Platonic philosophy among the Greeks, and to corresponding doctrines among the Arabians; and by calling into a prominent place astrology, alchemy, and magic, it long occupied most of the real observers of the material world. In this manner it delayed and impeded the progress of true science; for we shall see reason to believe that human knowledge lost more by the perversion of men's minds and the misdirection of their efforts, than it gained by any increase of zeal arising from the peculiar hopes and objects of the mystics.

It is not our purpose to attempt any general view of the progress and fortunes of the various forms of Mystical Philosophy; but only to exhibit some of its characters, in so far as they illustrate those {212} tendencies of thought which accompanied the retrogradation of inductive science. And of these, the leading feature which demands our notice is that already alluded to; namely, the practice of referring things and events, not to clear and distinct relations, obviously applicable to such cases;--not to general rules capable of direct verification; but to notions vague, distant, and vast, which we cannot bring into contact with facts, because they belong to a different region from the facts; as when we connect natural events with moral or historical causes, or seek spiritual meanings in the properties of number and figure. Thus the character of Mysticism is, that it refers particulars, not to generalizations homogeneous and immediate, but to such as are heterogeneous and remote; to which we must add, that the process of this reference is not a calm act of the intellect, but is accompanied with a glow of enthusiastic feeling.

1. _Neoplatonic Theosophy._--The _Newer Platonism_ is the first example of this Mystical Philosophy which I shall consider. The main points which here require our notice are, the doctrine of an Intellectual World resulting from the act of the Divine Mind, as the only reality; and the aspiration after the union of the human soul with this Divine Mind, as the object of human existence. The "Ideas" of Plato were Forms of our knowledge; but among the Neoplatonists they became really existing, indeed the only really existing, Objects; and the inaccessible scheme of the universe which these ideas constitute, was offered as the great subject of philosophical contemplation. The desire of the human mind to approach towards its Creator and Preserver, and to obtain a spiritual access to Him, leads to an employment of the thoughts which is well worth the notice of the religious philosopher; but such an effort, even when founded on revelation and well regulated, is not a means of advance in physics; and when it is the mere result of natural enthusiasm, it may easily obtain such a place in men's minds as to unfit them for the successful prosecution of natural philosophy. The temper, therefore, which introduces such supernatural communion into the general course of its speculations, may be properly treated as mystical, and as one of the causes of the decline of science in the Stationary Period. The Neoplatonic philosophy requires our notice as one of the most remarkable forms of this Mysticism.

Though Ammonius Saccas, who flourished at the end of the second century, is looked upon as the beginner of the Neoplatonists, his disciple Plotinus is, in reality, the great founder of the school, both by his {213} works, which still remain to us, and by the enthusiasm which his character and manners inspired among his

followers. He lived a life of meditation, gentleness, and self-denial, and died in the second year of the reign of Claudius (A. D. 270). His disciple, Porphyry, has given us a Life of him, from which we may see how well his habitual manners were suited to make his doctrines impressive. "Plotinus, the philosopher of our time," Porphyry thus begins his biography, "appeared like a person ashamed that he was in the body. In consequence of this disposition, he could not bear to talk concerning his family, or his parents, or his country. He would not allow himself to be represented by a painter or statuary; and once, when Aurelius entreated him to permit a likeness of him to be taken, he said, 'Is it not enough for us to carry this image in which nature has enclosed us, but we must also try to leave a more durable image of this image, as if it were so great a sight?' And he retained the same temper to the last. When he was dying, he said, 'I am trying to bring the divinity which is in us to the divinity which is in the universe.'" He was looked upon by his successors with extraordinary admiration and reverence; and his disciple Porphyry collected from his lips, or from fragmental notes, the six _Enneads_ of his doctrines (that is, parts each consisting of _nine_ Books), which he arranged and annotated.

We have no difficulty in finding in this remarkable work examples of mystical speculation. The Intelligible World of realities or essences corresponds to the world of sense[40\4] in the classes of things which it includes. To the Intelligible World, man's mind ascends, by a triple road which Plotinus figuratively calls that of the Musician, the Lover, the Philosopher.[41\4] The activity of the human soul is identified by analogy with the motion of the heavens. "This activity is about a middle point, and thus it is circular; but a middle point is not the same in body and in the soul: in that, the middle point is local; in this, it is that on which the rest depends. There is, however, an analogy; for as in one case, so in the other, there must be a middle point, and as the sphere revolves about its centre, the soul revolves about God through its affections."

[Note 40\4: vi. Ennead, iii. 1.]

[Note 41\4: ii. E. ii. 2.]

The conclusion of the work is,[42\4] as might be supposed, upon the approach to, union with, and fruition of God. The author refers again to the analogy between the movements of the soul and those of the heavens. "We move round him like a choral dance; even when we {214} look from him we revolve about him: we do not always look at him, but when we do, we have satisfaction and rest, and the harmony which belongs to that divine movement. In this movement, the mind beholds the fountain of life, the fountain of mind, the origin of being, the cause of good, the root of the soul."[43\4] "There will be a time when this vision shall be continual; the mind being no more interrupted, nor suffering any perturbation from the body. Yet that which beholds is not that which is disturbed; and when this vision becomes dim, it does not obscure the knowledge which resides in demonstration, and faith, and reasoning; but the vision itself is not reason, but greater than reason, and before reason."[44\4]

[Note 42\4: vi. Enn. ix. 8.]

[Note 43\4: vi. Enn. ix. 9.]

[Note 44\4: vi. Enn. ix. 10.]

The fifth book of the third Ennead has for its subject the Dæmon which belongs to each man. It is entitled "Concerning Love;" and the doctrine appears to be, that the Love, or common source of the passions which is in each man's mind, is "the Dæmon which they say accompanies each man."[45\4] These dæmons were, however (at least by later writers), invested with a visible aspect and with a personal character, including a resemblance of human passions and motives. It is curious thus to see an untenable and visionary generalization falling back into the domain of the senses and the fancy, after a vain attempt to support itself in the region of the reason. This imagination soon produced pretensions to the power of making these dæmons or genii visible; and the Treatise on the Mysteries of the Egyptians, which is attributed to Iamblichus, gives an account of the secret ceremonies, the mysterious words, the sacrifices and expiations, by which this was to be done.

[Note 45\4: Ficinus, _Comm._ in v. Enn. iii.]

It is unnecessary for us to dwell on the progress of this school; to point out the growth of the Theurgy which thus arose; or to describe the attempts to claim a high antiquity for this system, and to make Orpheus, the poet, the first promulgator of its doctrines. The system, like all mystical systems, assumed the character rather of religion than of a theory. The opinions of its disciples materially influenced their lives. It gave the world the spectacle of an austere morality, a devotional exaltation, combined with the grossest superstitions of Paganism. The successors of Iamblichus appeared rather to hold a priesthood, than the chair of a philosophical school.[46\4] They were persecuted by Constantine and Constantius, as opponents of Christianity. Sopater, a {215} Syrian philosopher of this school, was beheaded by the former emperor on a charge that he had bound the winds by the power of magic.[47\4] But Julian, who shortly after succeeded to the purple, embraced with ardor the opinions of Iamblichus. Proclus (who died A. D. 487) was one of the greatest of the teachers of this school;[48\4] and was, both in his life and doctrines, a worthy successor of Plotinus, Porphyry, and Iamblichus. We possess a biography, or rather a panegyric of him, by his disciple Marinus, in which he is exhibited as a representation of the ideal perfection of the philosophic character, according to the views of the Neoplatonists. His virtues are arranged as physical, moral, purificatory, theoretic, and theurgic. Even in his boyhood, Apollo and Minerva visited him in his dreams: he studied oratory at Alexandria, but it was at Athens that Plutarch and Lysianus initiated him in the mysteries of the New Platonists. He received a kind of consecration at the hands of the daughter of Plutarch, the celebrated Asclepigenia, who introduced him to the traditions of the Chaldeans, and the practices of theurgy; he was also admitted to the mysteries of Eleusis. He became celebrated for his knowledge and eloquence; but especially for his skill in the supernatural arts which were connected with the doctrines of his sect. He appears before us rather as a hierophant than a philosopher. A large portion of his life was spent in evocations, purifications, fastings, prayers, hymns, intercourse with apparitions, and with the gods, and in the celebration of the festivals of Paganism, especially those which were held in honor of the Mother of the Gods. His religious admiration extended to all forms of mythology. The philosopher,

said he, is not the priest of a single religion, but of all the religions of the world. Accordingly, he composed hymns in honor of all the divinities of Greece, Rome, Egypt, Arabia;--Christianity alone was excluded from his favor.

[Note 46\4: Deg. iii. 407]

[Note 47\4: Gibbon, iii. 352.]

[Note 48\4: Deg. iii. 419.]

The reader will find an interesting view of the _School of Alexandria_, in M. Barthelemy Saint-Hilaire's _Rapport_ on the _Mémoires_ sent to the Academy of Moral and Political Sciences at Paris, in consequence of its having, in 1841, proposed this as the subject of a prize, which was awarded in 1844. M. Saint-Hilaire has prefixed to this _Rapport_ a dissertation on the Mysticism of that school. He, however, uses the term _Mysticism_ in a wider sense than my purpose, which regarded mainly the bearing of the doctrines of this school upon the progress of the Inductive Sciences, has led me to do. Although he finds much to {216} admire in the Alexandrian philosophy, he declares that they were incapable of treating scientific questions. The extent to which this is true is well illustrated by the extract which he gives from Plotinus, on the question, "Why objects appear smaller in proportion as they are more distant." Plotinus denies that the reason of this is that the angles of vision become smaller. His reason for this denial is curious enough. If it were so, he says, how could the heaven appear smaller than it is, since it occupies the whole of the visual angle?

2. _Mystical Arithmetic._--It is unnecessary further to exemplify, from Proclus, the general mystical character of the school and time to which he belonged; but we may notice more specially one of the forms of this mysticism, which very frequently offers itself to our notice, especially in him; and which we may call _Mystical Arithmetic_. Like all the kinds of Mysticism, this consists in the attempt to connect our conceptions of external objects by general and inappropriate notions of goodness, perfection, and relation to the divine essence and government; instead of referring such conceptions to those appropriate ideas, which, by due attention, become perfectly distinct, and capable of being positively applied and verified. The subject which is thus dealt with, in the doctrines of which we now speak, is Number; a notion which tempts men into these visionary speculations more naturally than any other. For number is really applicable to moral notions--to emotions and feelings, and to their objects--as well as to the things of the material world. Moreover, by the discovery of the principle of musical concords, it had been found, probably most unexpectedly, that numerical relations were closely connected with sounds which could hardly be distinguished from the expression of thought and feeling; and a suspicion might easily arise, that the universe, both of matter and of thought, might contain many general and abstract truths of some analogous kind. The relations of number have so wide a bearing, that the ramifications of such a suspicion could not easily be exhausted, supposing men willing to follow them into darkness and vagueness; which it is precisely the mystical tendency to do. Accordingly, this kind of speculation appeared very early, and showed itself first among the Pythagoreans, as

we might have expected, from the attention which they gave to the theory of harmony: and this, as well as some other of the doctrines of the Pythagorean philosophy, was adopted by the later Platonists, and, indeed, by Plato himself, whose speculations concerning number have decidedly a mystical character. The mere mathematical relations of numbers,--as odd and even, perfect and imperfect, {217} abundant and defective,--were, by a willing submission to an enthusiastic bias, connected with the notions of good and beauty, which were suggested by the terms expressing their relations; and principles resulting from such a connection were woven into a wide and complex system. It is not necessary to dwell long on this subject; the mere titles of the works which treated of it show its nature. Archytas[49\4] is said to have written a treatise on the number _ten_: Telaugé, the daughter of Pythagoras, wrote on the number _four_. This number, indeed, which was known by the name of the _Tetractys_, was very celebrated in the school of Pythagoras. It is mentioned in the "Golden Verses," which are ascribed to him: the pupil is conjured to be virtuous, Ναὶ μὰ τὸν ἀμετέρᾳ ψυχᾷ παραδόντα τετρακτὺν Παγὰν ἀεννάου φύσεως . . . . By him who stampt _The Four_ upon the mind,-- _The Four_, the fount of nature's endless stream.

[Note 49\4: Mont. ii. 123.]

In Plato's works, we have evidence of a similar belief in religious relations of Number; and in the new Platonists, this doctrine was established as a system. Proclus, of whom we have been speaking, founds his philosophy, in a great measure, on the relation of Unity and Multiple; from this, he is led to represent the causality of the Divine Mind by three Triads of abstractions; and in the development of one part of this system, the number seven is introduced.[50\4] "The intelligible and intellectual gods produce all things triadically; for the monads in these latter are divided according to number; and what the monad was in the former, the number is in these latter. And the intellectual gods produce all things hebdomically; for they evolve the intelligible, and at the same time intellectual triads, into intellectual hebdomads, and expand their contracted powers into intellectual variety." Seven is what is called by arithmeticians a _prime_ number, that is, it cannot be produced by the multiplication of other numbers. In the language of the New Platonists, the number seven is said to be a virgin, and without a mother, and it is therefore sacred to Minerva. The number six is a perfect number, and is consecrated to Venus.

[Note 50\4: Procl. v. 3, Taylor's translation.]

The relations of space were dealt with in like manner, the Geometrical properties being associated with such physical and metaphysical notions as vague thought and lively feeling could anyhow connect with them. We may consider, as an example of this,[51\4] Plato's opinion {218} concerning the particles of the four elements. He gave to each kind of particle one of the five regular solids, about which the geometrical speculations of himself and his pupils had been employed. The particles of fire were pyramids, because they are sharp, and tend upwards; those of earth are cubes, because they are stable, and fill space; the particles of air are octahedral, as most nearly resembling those of fire; those of water are the icositetrahedron, as most nearly spherical. The dodecahedron is the figure of the element of the heavens, and

shows its influence in other things, as in the twelve signs of the zodiac. In such examples we see how loosely space and number are combined or confounded by these mystical visionaries.

[Note 51\4: Stanley, _Hist. Phil._]

These numerical dreams of ancient philosophers have been imitated by modern writers; for instance, by Peter Bungo and Kircher, who have written De Mysteriis Numerorum. Bungo treats of the mystical properties of each of the numbers in order, at great length. And such speculations have influenced astronomical theories. In the first edition of the Alphonsine Tables,[52\4] the precession was represented by making the first point of Aries move, in a period of 7000 years, through a circle of which the radius was 18 degrees, while the circle moved round the ecliptic in 49,000 years; and these numbers, 7000 and 49,000, were chosen probably by Jewish calculators, or with reference to Jewish Sabbatarian notions.

[Note 52\4: Montucla, i. 511.]

3. _Astrology._--Of all the forms which mysticism assumed, none was cultivated more assiduously than astrology. Although this art prevailed most universally and powerfully during the stationary period, its existence, even as a detailed technical system, goes back to a very early age. It probably had its origin in the East; it is universally ascribed to the Babylonians and Chaldeans; the name _Chaldean_ was, at Rome, synonymous with _mathematicus_, or astrologer; and we read repeatedly that this class of persons were expelled from Italy by a decree of the senate, both during the times of the republic and of the empire.[53\4] The recurrence of this act of legislation shows that it was not effectual: "It is a class of men," says Tacitus, "which, in our city, will always be prohibited, and will always exist." In Greece, it does not appear that the state showed any hostility to the professors of this art. They undertook, it would seem, then, as at a later period, to determine the course of a man's character and life from the configuration of the stars at the moment of his birth. We do not possess any of the {219} speculations of the early astrologers; and we cannot therefore be certain that the notions which operated in men's minds when the art had its birth, agreed with the views on which it was afterwards defended, when it became a matter of controversy. But it appears probable, that, though it was at later periods supported by physical analogies, it was originally suggested by mythological belief. The Greeks spoke of the _influences_ or _effluxes_ (ἀπόῤῥοιας) which proceeded from the stars; but the Chaldeans had probably thought rather of the powers which they exercised as _deities_. In whatever manner the sun, moon, and planets came to be identified with gods and goddesses, it is clear that the characters ascribed to these gods and goddesses regulate the virtues and powers of the stars which bear their names. This association, so manifestly visionary, was retained, amplified, and pursued, in an enthusiastic spirit, instead of being rejected for more distinct and substantial connections; and a pretended science was thus formed, which bears the obvious stamp of mysticism.

[Note 53\4: Tacit. _Ann._ ii. 32. xii. 52. _Hist._ I. 22, II. 62.]

That common sense of mankind which teaches them that theoretical opinions are to be calmly tried by their consequences and their accordance with facts, appears to have counteracted the prevalence of astrology in the better times of the human mind. Eudoxus, as we are informed by Cicero,[54\4] rejected the pretensions of the Chaldeans; and Cicero himself reasons against them with arguments as sensible and intelligent as could be adduced by a writer of the present day; such as the different fortunes and characters of persons born at the same time; and the failure of the predictions, in the case of Pompey, Crassus, Cæsar, to whom the astrologers had foretold glorious old age and peaceful death. He also employs an argument which the reader would perhaps not expect from him,--the very great remoteness of the planets as compared with the distance of the moon. "What contagion can reach us," he asks, "from a distance almost infinite?"

[Note 54\4: Cic. _de Div._ ii. 42.]

Pliny argues on the same side, and with some of the same arguments.[55\4] "Homer," he says, "tells us that Hector and Polydamus were born the same night;--men of such different fortune. And every hour, in every part of the world, are born lords and slaves, kings and beggars."

[Note 55\4: _Hist. Nat._ vii. 49.]

The impression made by these arguments is marked in an anecdote told concerning Publius Nigidius Figulus, a Roman of the time of Julius Cæsar, whom Lucan mentions as a celebrated astrologer. It is {220} said, that when an opponent of the art urged as an objection the different fates of persons born in two successive instants, Nigidius bade him make two contiguous marks on a potter's wheel, which was revolving rapidly near them. On stopping the wheel, the two marks were found to be really far removed from each other; and Nigidius is said to have received the name of Figulus (the potter), in remembrance of this story, His argument, says St. Augustine, who gives us the narrative, was as fragile as the ware which the wheel manufactured.

As the darkening times of the Roman empire advanced, even the stronger minds seem to have lost the clear energy which was requisite to throw off this delusion. Seneca appears to take the influence of the planets for granted; and even Tacitus[56\4] seems to hesitate. "For my own part," says he, "I doubt; but certainly the majority of mankind cannot be weaned from the opinion, that, at the birth of each man, his future destiny is fixed; though some things may fall out differently from the predictions, by the ignorance of those who profess the art; and that thus the art is unjustly blamed, confirmed as it is by noted examples in all ages." The occasion which gives rise to these reflections of the historian is the mention of Thrasyllus, the favorite astrologer of the Emperor Tiberius, whose skill is exemplified in the following narrative. Those who were brought to Tiberius on any important matter, were admitted to an interview in an apartment situated on a lofty cliff in the island of Capreæ. They reached this place by a narrow path, accompanied by a single freedman of great bodily strength; and on their return, if the emperor had conceived any doubts of their trustworthiness, a single blow buried the secret and its victim in

the ocean below. After Thrasyllus had, in this retreat, stated the results of his art as they concerned the emperor, Tiberius asked him whether he had calculated how long he himself had to live. The astrologer examined the aspect of the stars, and while he did this, as the narrative states, showed hesitation, alarm, increasing terror, and at last declared that, "the present hour was for him critical, perhaps fatal." Tiberius embraced him, and told him "he was right in supposing he had been in danger, but that he should escape it;" and made him thenceforth his confidential counsellor.

[Note 56\4: _Ann._ vi. 22.]

The belief in the power of astrological prediction which thus obtained dominion over the minds of men of literary cultivation and practical energy, naturally had a more complete sway among the speculative {221} but unstable minds of the later philosophical schools of Alexandria, Athens, and Rome. We have a treatise on astrology by Proclus, which will serve to exemplify the mystical principle in this form. It appears as a commentary on a work on the same subject called "Tetrabiblos," ascribed to Ptolemy; though we may reasonably doubt whether the author of the "Megale Syntaxis" was also the writer of the astrological work. A few notices of the commentary of Proclus will suffice.[57\4] The science is defended by urging how powerful we know the physical effects of the heavenly bodies to be. "The sun regulates all things on earth;--the birth of animals, the growth of fruits, the flowing of waters, the change of health, according to the seasons: he produces heat, moisture, dryness, cold, according to his approach to our zenith. The moon, which is the nearest of all bodies to the earth, gives out much _influence_; and all things, animate and inanimate, sympathize with her: rivers increase and diminish according to her light; the advance of the sea, and its recess, are regulated by her rising and setting; and along with her, fruits and animals wax and wane, either wholly or in part." It is easy to see that by pursuing this train of associations (some real and some imaginary) very vaguely and very enthusiastically, the connections which astrology supposes would receive a kind of countenance. Proclus then proceeds to state[58\4] the doctrines of the science. "The sun," he says, "is productive of heat and dryness; this power is moderate in its nature, but is more perceived than that of the other luminaries, from his magnitude, and from the change of seasons. The nature of the moon is for the most part moist; for being the nearest to the earth, she receives the vapors which rise from moist bodies, and thus she causes bodies to soften and rot. But by the illumination she receives from the sun, she partakes in a moderate degree of heat. Saturn is cold and dry, being most distant both from the heating power of the sun, and the moist vapors of the earth. His cold, however, is most prevalent, his dryness is more moderate. Both he and the rest receive additional powers from the configurations which they make with respect to the sun and moon." In the same manner it is remarked that Mars is dry and caustic, from his fiery nature, which, indeed, his color shows. Jupiter is well compounded of warm and moist, as is Venus. Mercury is variable in his character. From these notions were derived others concerning the beneficial or hurtful effect of these stars. Heat and {222} moisture are generative and creative elements; hence the ancients, says Proclus, deemed Jupiter, and Venus, and the Moon to have a good power; Saturn and Mercury, on the other hand, had an evil nature.

[Note 57\4: I. 2.]

[Note 58\4: I. 4.]

Other distinctions of the character of the stars are enumerated, equally visionary, and suggested by the most fanciful connections. Some are masculine, and some feminine: the Moon and Venus are of the latter kind. This appears to be merely a mythological or etymological association. Some are diurnal, some nocturnal: the Moon and Venus are of the latter kind, the Sun and Jupiter of the former; Saturn and Mars are both.

The fixed stars, also, and especially those of the zodiac, had especial influences and subjects assigned to them. In particular, each sign was supposed to preside over a particular part of the body; thus Aries had the head assigned to it, Taurus the neck, and so on.

The most important part of the sky in the astrologer's consideration, was that sign of the zodiac which rose at the moment of the child's birth; this was, properly speaking, the _horoscope_, the _ascendant_, or the _first house_; the whole circuit of the heavens being divided into twelve _houses_, in which life and death, marriage and children, riches and honors, friends and enemies, were distributed.

We need not attempt to trace the progress of this science. It prevailed extensively among the Arabians, as we might expect from the character of that nation. Albumasar, of Balkh in Khorasan, who flourished in the ninth century, who was one of their greatest astronomers, was also a great astrologer; and his work on the latter subject, "De Magnis Conjunctionibus, Annorum Revolutionibus ac eorum Perfectionibus," was long celebrated in Europe. Aboazen Haly (the writer of a treatise "De Judiciis Astrorum"), who lived in Spain in the thirteenth century, was one of the classical authors on this subject.

It will easily be supposed that when this _apotelesmatic_ or _judicial_ astrology obtained firm possession of men's minds, it would be pursued into innumerable subtle distinctions and extravagant conceits; and the more so, as experience could offer little or no check to such exercises of fancy and subtlety. For the correction of rules of astrological divination by comparison with known events, though pretended to by many professors of the art, was far too vague and fallible a guidance to be of any real advantage. Even in what has been called Natural Astrology, the dependence of the weather on the heavenly bodies, it is easy to see what a vast accumulation of well-observed facts is requisite to establish {223} any true rule; and it is well known how long, in spite of facts, false and groundless rules (as the dependence of the weather on the moon) may keep their hold on men's minds. When the facts are such loose and many-sided things as human characters, passions, and happiness, it was hardly to be expected that even the most powerful minds should be able to find a footing sufficiently firm, to enable them to resist the impression of a theory constructed of sweeping and bold assertions, and filled out into a complete system of details. Accordingly, the connection of the stars with human persons and actions was, for a long period, undisputed. The vague, obscure, and heterogeneous character of such a connection, and its unfitness for any really scientific reasoning, could, of

course, never be got rid of; and the bewildering feeling of earnestness and solemnity, with which the connection of the heavens with man was contemplated, never died away. In other respects, however, the astrologers fell into a servile commentatorial spirit; and employed themselves in annotating and illustrating the works of their predecessors to a considerable extent, before the revival of true science.

It may be mentioned, that astrology has long been, and probably is, an art held in great esteem and admiration among other eastern nations besides the Mohammedans; for instance, the Jews, the Indians, the Siamese, and the Chinese. The prevalence of vague, visionary, and barren notions among these nations, cannot surprise us; for with regard to them we have no evidence, as with regard to Europeans we have, that they are capable, on subjects of physical speculation, of originating sound and rational general principles. The Arts may have had their birth in all parts of the globe; but it is only Europe, at particular favored periods of its history, which has ever produced Sciences.

We are, however, now speaking of a long period, during which this productive energy was interrupted and suspended. During this period Europe descended, in intellectual character, to the level at which the other parts of the world have always stood. Her Science was then a mixture of Art and Mysticism; we have considered several forms of this Mysticism, but there are two others which must not pass unnoticed, Alchemy and Magic.

We may observe, before we proceed, that the deep and settled influence which Astrology had obtained among them, appears perhaps most strongly in the circumstance, that the most vigorous and clear-sighted minds which were concerned in the revival of science, did not, for a long period, shake off the persuasion that there was, in this art, some element of truth. Roger Bacon, Cardan, Kepler, Tycho Brahe, {224} Francis Bacon, are examples of this. These, or most of them, rejected all the more obvious and extravagant absurdities with which the subject had been loaded; but still conceived that some real and valuable truth remained when all these were removed. Thus Campanella,[59\4] whom we shall have to speak of as one of the first opponents of Aristotle, wrote an "Astrology purified from all the Superstitions of the Jews and Arabians, and treated physiologically."

[Note 59\4: Bacon, _De Aug._ iii. 4.]

4. _Alchemy._--Like other kinds of Mysticism, Alchemy seems to have grown out of the notions of moral, personal, and mythological qualities, which men associated with terms, of which the primary application was to physical properties. This is the form in which the subject is presented to us in the earliest writings which we possess on the subject of chemistry;--those of Geber[60\4] of Seville, who is supposed to have lived in the eighth or ninth century. The very titles of Geber's works show the notions on which this pretended science proceeds. They are, "Of the Search of Perfection;" "Of the Sum of Perfection, or of the Perfect Magistery;" "Of the Invention of Verity, or Perfection." The basis of this phraseology is the distinction of metals into more or less _perfect_; gold being the most perfect, as being the most valuable, most beautiful, most pure, most durable; silver the next; and so on. The "Search of Perfection" was,

therefore, the attempt to convert other metals into gold; and doctrines were adopted which represented the metals as all compounded of the same elements, so that this was theoretically possible. But the mystical trains of association were pursued much further than this; gold and silver were held to be the most noble of metals; gold was their King, and silver their Queen. Mythological associations were called in aid of these fancies, as had been done in astrology. Gold was Sol, silver was Luna, the moon; copper, iron, tin, lead, were assigned to Venus, Mars, Jupiter, Saturn. The processes of mixture and heat were spoken of as personal actions and relations, struggles and victories. Some elements were conquerors, some conquered; there existed preparations which possessed the power of changing the whole of a body into a substance of another kind: these were called _magisteries_.[61\4] When gold and quicksilver are combined, the king and the queen are married, to produce children of their own kind. It will easily be conceived, that when chemical operations were described in phraseology of this sort, the enthusiasm of the {225} fancy would be added to that of the hopes, and observation would not be permitted to correct the delusion, or to suggest sounder and more rational views.

[Note 60\4: Thomson's _Hist. of Chem._ i. 117.]

[Note 61\4: Boyle, Thomson's _Hist. Ch._ i. 25. Carolus Musitanus.]

The exaggeration of the vague notion of perfection and power in the object of the alchemist's search, was carried further still. The same preparation which possessed the faculty of turning baser metals into gold, was imagined to be also a universal medicine, to have the gift of curing or preventing diseases, prolonging life, producing bodily strength and beauty: the _philosophers' stone_ was finally invested with every desirable efficacy which the fancy of the "philosophers" could devise.

It has been usual to say that Alchemy was the mother of Chemistry; and that men would never have made the experiments on which the real science is founded, if they had not been animated by the hopes and the energy which the delusive art inspired. To judge whether this is truly said, we must be able to estimate the degree of interest which men feel in purely speculative truth, and in the real and substantial improvement of art to which it leads. Since the fall of Alchemy, and the progress of real Chemistry, these motives have been powerful enough to engage in the study of the science, a body far larger than the Alchemists ever were, and no less zealous. There is no apparent reason why the result should not have been the same, if the progress of true science had begun sooner. Astronomy was long cultivated without the bribe of Astrology. But, perhaps, we may justly say this;--that, in the stationary period, men's minds were so far enfeebled and degraded, that pure speculative truth had not its full effect upon them; and the mystical pursuits in which some dim and disfigured images of truth were sought with avidity, were among the provisions by which the human soul, even when sunk below its best condition, is perpetually directed to something above the mere objects of sense and appetite;--a contrivance of compensation, as it were, in the intellectual and spiritual constitution of man.

5. _Magic._--Magical Arts, so far as they were believed in by those who professed to practise them, and so far as they have a bearing in science, stand on the same

footing as astrology; and, indeed, a close alliance has generally been maintained between the two pursuits. Incapacity and indisposition to perceive natural and philosophical causation, an enthusiastic imagination, and such a faith as can devise and maintain supernatural and spiritual connexions, are the elements of this, as of other forms of Mysticism. And thus, that temper which led men to aim at the magician's supposed authority over the elements, {226} is an additional exemplification of those habits of thought which prevented the progress of real science, and the acquisition of that command over nature which is founded on science, during the interval now before us.

But there is another aspect under which the opinions connected with this pursuit may serve to illustrate the mental character of the Stationary Period.

The tendency, during the middle ages, to attribute the character of Magician to almost all persons eminent for great speculative or practical knowledge, is a feature of those times, which shows how extensive and complete was the inability to apprehend the nature of real science. In cultivated and enlightened periods, such as those of ancient Greece, or modern Europe, knowledge is wished for and admired, even by those who least possess it: but in dark and degraded periods, superior knowledge is a butt for hatred and fear. In the one case, men's eyes are open; their thoughts are clear; and, however high the philosopher may be raised above the multitude, they can catch glimpses of the intervening path, and see that it is free to all, and that elevation is the reward of energy and labor. In the other case, the crowd are not only ignorant, but spiritless; they have lost the pleasure in knowledge, the appetite for it, and the feeling of dignity which it gives: there is no sympathy which connects them with the learned man: they see him above them, but know not how he is raised or supported: he becomes an object of aversion and envy, of vague suspicion and terror; and these emotions are embodied and confirmed by association with the fancies and dogmas of superstition. To consider superior knowledge as Magic, and Magic as a detestable and criminal employment, was the form which these feelings of dislike assumed; and at one period in the history of Europe, almost every one who had gained any eminent literary fame, was spoken of as a magician. Naudæus, a learned Frenchman, in the seventeenth century, wrote "An Apology for all the Wise Men who have been unjustly reported Magicians, from the Creation to the present Age." The list of persons whom he thus thinks it necessary to protect, are of various classes and ages. Alkindi, Geber, Artephius, Thebit, Raymund Lully, Arnold de Villâ Novâ, Peter of Apono, and Paracelsus, had incurred the black suspicion as physicians or alchemists. Thomas Aquinas, Roger Bacon, Michael Scott, Picus of Mirandula, and Trithemius, had not escaped it, though ministers of religion. Even dignitaries, such as Robert Grosteste, Bishop of Lincoln, Albertus Magnus, Bishop of Ratisbon, {227} Popes Sylvester the Second, and Gregory the Seventh, had been involved in the wide calumny. In the same way in which the vulgar confounded the eminent learning and knowledge which had appeared in recent times, with skill in dark and supernatural arts, they converted into wizards all the best-known names in the rolls of fame; as Aristotle, Solomon, Joseph, Pythagoras; and, finally, the poet Virgil was a powerful and skilful necromancer, and this fancy was exemplified by many strange stories of his achievements and practices.

The various results of the tendency of the human mind to mysticism, which we have here noticed, form prominent features in the intellectual character of the world, for a long course of centuries. The theosophy and theurgy of the Neoplatonists, the mystical arithmetic of the Pythagoreans and their successors, the predictions of the astrologers, the pretences of alchemy and magic, represent, not unfairly, the general character and disposition of men's thoughts, with reference to philosophy and science. That there were stronger minds, which threw off in a greater or less degree this train of delusive and unsubstantial ideas, is true; as, on the other hand, Mysticism, among the vulgar or the foolish, often went to an extent of extravagance and superstition, of which I have not attempted to convey any conception. The lesson which the preceding survey teaches us is, that during the Stationary Period, Mysticism, in its various forms, was a leading character, both of the common mind, and of the speculations of the most intelligent and profound reasoners; and that this Mysticism was the opposite of that habit of thought which we have stated Science to require; namely, clear Ideas, distinctly employed to connect well-ascertained Facts; inasmuch as the Ideas in which it dealt were vague and unstable, and the temper in which they were contemplated was an urgent and aspiring enthusiasm, which could not submit to a calm conference with experience upon even terms. The fervor of thought in some degree supplied the place of reason in producing belief; but opinions so obtained had no enduring value; they did not exhibit a permanent record of old truths, nor a firm foundation for new. Experience collected her stores in vain, or ceased to collect them, when she had only to pour them into the flimsy folds of the lap of Mysticism; who was, in truth, so much absorbed in looking for the treasures which were to fall from the skies, that she heeded little how scantily she obtained, or how loosely she held, such riches as might be found near her. {228}

## CHAPTER IV.

OF THE DOGMATISM OF THE STATIONARY PERIOD.

IN speaking of the character of the age of commentators, we noticed principally the ingenious servility which it displays;--the acuteness with which it finds ground for speculation in the expression of other men's thoughts;--the want of all vigor and fertility in acquiring any real and new truths. Such was the character of the reasoners of the stationary period from the first; but, at a later day, this character, from various causes, was modified by new features. The servility which had yielded itself to the yoke, insisted upon forcing it on the necks of others: the subtlety which found all the truth it needed in certain accredited writings, resolved that no one should find there, or in any other region, any other truths; speculative men became tyrants without ceasing to be slaves; to their character of Commentators they added that of Dogmatists.

1. _Origin of the Scholastic Philosophy._--The causes of this change have been very happily analyzed and described by several modern writers.[62\4] The general nature of the process may be briefly stated to have been the following.

[Note 62\4: Dr. Hampden, in the Life of Thomas Aquinas, in the _Encyc. Metrop._ Degerando, _Hist. Comparée_, vol. iv. Also Tennemann, _Hist. of Phil._ vol. viii. Introduction.]

The tendencies of the later times of the Roman empire to a commenting literature, and a second-hand philosophy, have already been noticed. The loss of the dignity of political freedom, the want of the cheerfulness of advancing prosperity, and the substitution of the less philosophical structure of the Latin language for the delicate intellectual mechanism of the Greek, fixed and augmented the prevalent feebleness and barrenness of intellect. Men forgot, or feared, to consult nature, to seek for new truths, to do what the great discoverers of other times had done; they were content to consult libraries, to study and defend old opinions, to talk of what great geniuses had said. They sought their philosophy in accredited treatises, and dared not question such doctrines as they there found.

The character of the philosophy to which they were thus led, was determined by this want of courage and originality. There are various {229} antagonist principles of opinion, which seem alike to have their root in the intellectual constitution of man, and which are maintained and developed by opposing sects, when the intellect is in vigorous action. Such principles are, for instance,--the claims of Authority and of Reason to our assent;--the source of our knowledge in Experience or in Ideas;--the superiority of a Mystical or of a Skeptical turn of thought. Such oppositions of doctrine were found in writers of the greatest fame; and two of those, who most occupied the attention of students, Plato and Aristotle, were, on several points of this nature, very diverse from each other in their tendency. The attempt to reconcile these philosophers by Boëthius and others, we have already noticed; and the attempt was so far successful, that it left on men's minds the belief in the possibility of a great philosophical system which should be based on both these writers, and have a claim to the assent of all sober speculators.

But, in the mean time, the Christian Religion had become the leading subject of men's thoughts; and divines had put forward its claims to be, not merely the guide of men's lives, and the means of reconciling them to their heavenly Master, but also to be a Philosophy in the widest sense in which the term had been used;--a consistent speculative view of man's condition and nature, and of the world in which he is placed.

These claims had been acknowledged; and, unfortunately, from the intellectual condition of the times, with no due apprehension of the necessary ministry of Observation, and Reason dealing with observation, by which alone such a system can be embodied. It was held without any regulating principle, that the philosophy which had been bequeathed to the world by the great geniuses of heathen antiquity, and the Philosophy which was deduced from, and implied by, the Revelations made by God to man, must be identical; and, therefore, that Theology is the only true philosophy. Indeed, the Neoplatonists had already arrived, by other roads, at the same conviction. John Scot Erigena, in the reign of Alfred, and consequently before the existence of the Scholastic Philosophy, properly so called, had reasserted this doctrine.[63\4] Anselm, in the eleventh century, again brought it forward;[64\4] and Bernard de Chartres, in

the thirteenth.[65\4]

[Note 63\4: Deg. iv. 351.]

[Note 64\4: Ib. iv. 388.]

[Note 65\4: Ib. iv. 418.]

This view was confirmed by the opinion which prevailed, concerning the nature of philosophical truth; a view supported by the theory {230} of Plato, the practice of Aristotle, and the general propensities of the human mind: I mean the opinion that all science may be obtained by the use of reasoning alone;--that by analysing and combining the notions which common language brings before us, we may learn all that we can know. Thus Logic came to include the whole of Science; and accordingly this Abelard expressly maintained.[66\4] I have already explained, in some measure, the fallacy of this belief, which consists, as has been well said,[67\4] "in mistaking the universality of the theory of language for the generalization of facts." But on all accounts this opinion is readily accepted; and it led at once to the conclusion, that the Theological Philosophy which we have described, is complete as well as true.

[Note 66\4: Deg. iv. 407.]

[Note 67\4: _Enc. Met._ 807.]

Thus a Universal Science was established, with the authority of a Religious Creed. Its universality rested on erroneous views of the relation of words and truths; its pretensions as a science were admitted by the servile temper of men's intellects; and its religious authority was assigned it, by making all truth part of religion. And as Religion claimed assent within her own jurisdiction under the most solemn and imperative sanctions, Philosophy shared in her imperial power, and dissent from their doctrines was no longer blameless or allowable. Error became wicked, dissent became heresy; to reject the received human doctrines, was nearly the same as to doubt the Divine declarations. The _Scholastic Philosophy_ claimed the assent of all believers.

The external form, the details, and the text of this philosophy, were taken, in a great measure, from Aristotle; though, in the spirit, the general notions, and the style of interpretation, Plato and the Platonists had no inconsiderable share. Various causes contributed to the elevation of Aristotle to this distinction. His Logic had early been adopted as an instrument of theological disputation; and his spirit of systematization, of subtle distinction, and of analysis of words, as well as his disposition to argumentation, afforded the most natural and grateful employment to the commentating propensities. Those principles which we before noted as the leading points of his physical philosophy, were selected and adopted; and these, presented in a most technical form, and applied in a systematic manner, constitute a large portion of the philosophy of which we now speak, so far as it pretends to deal with physics.

2. _Scholastic Dogmas._--But before the complete ascendancy of Aristotle was thus established, when something of an intellectual waking {231} took place after the darkness and sleep of the ninth and tenth centuries, the Platonic doctrines seem to

have had, at first, a strong attraction for men's minds, as better falling in with the mystical speculations and contemplative piety which belonged to the times. John Scot Erigena[68\4] may be looked upon as the reviver of the New Platonism in the tenth century. Towards the end of the eleventh, Peter Damien,[69\4] in Italy, reproduced, involved in a theological discussion, some Neoplatonic ideas. Godefroy[70\4] also, censor of St. Victor, has left a treatise, entitled _Microcosmus_; this is founded on a mystical analogy, often afterwards again brought forward, between Man and the Universe. "Philosophers and theologians," says the writer, "agree in considering man as a little world; and as the world is composed of four elements, man is endowed with four faculties, the senses, the imagination, reason, and understanding." Bernard of Chartres,[71\4] in his _Megascosmus_ and _Microcosmus_, took up the same notions. Hugo, abbot of St. Victor, made a contemplative life the main point and crown of his philosophy; and is said to have been the first of the scholastic writers who made psychology his special study.[72\4] He says the faculties of the mind are "the senses, the imagination, the reason, the memory, the understanding, and the intelligence."

[Note 68\4: Deg. iv. 35.]

[Note 69\4: Ib. iv. 367.]

[Note 70\4: Ib. iv. 413.]

[Note 71\4: Ib. iv. 419.]

[Note 72\4: Ib. iv. 415.]

Physics does not originally and properly form any prominent part of the Scholastic Philosophy, which consists mainly of a series of questions and determinations upon the various points of a certain technical divinity. Of this kind is the _Book of Sentences_ of Peter the Lombard (bishop of Paris), who is, on that account, usually called "Magister Sententiarum;" a work which was published in the twelfth century, and was long the text and standard of such discussions. The questions are decided by the authority of Scripture and of the Fathers of the Church, and are divided into four Books, of which the first contains questions concerning God and the doctrine of the Trinity in particular; the second is concerning the Creation; the third, concerning Christ and the Christian Religion; and the fourth treats of Religious and Moral Duties. In the second book, as in many of the writers of this time, the nature of Angels is considered in detail, and the Orders of their Hierarchy, of which there were held to be nine. The physical discussions enter only as bearing upon the scriptural history of the creation, and cannot be taken as a specimen of the work; but I may observe, that in speaking of the division of the waters above the {232} firmament, he gives one opinion, that of Bede, that the former waters are the solid crystalline heavens in which the stars are fixed,[73\4] "for crystal, which is so hard and transparent, is made of water." But he mentions also the opinion of St. Augustine, that the waters above the heavens are in a state of vapor, (_vaporaliter_) and in minute drops; "if, then, water can, as we see in clouds, be so minutely divided that it may be thus supported as vapor on air, which is naturally lighter than water; why may we not believe that it floats above that lighter celestial element in still minuter drops and still lighter vapors? But in whatever manner the waters are there, we do not doubt that

they are there."

[Note 73\4: Lib. ii. Distinct. xiv. _De opere secundæ diei_.]

The celebrated _Summa Theologicæ_ of Thomas Aquinas is a work of the same kind; and anything which has a physical bearing forms an equally small part of it. Thus, of the 512 Questions of the _Summa_, there is only one (Part I., Quest. 115), "on Corporeal Action," or on any part of the material world; though there are several concerning the celestial Hierarchies, as "on the Act of Angels," "on the Speaking of Angels," "on the Subordination of Angels," "on Guardian Angels," and the like. This, of course, would not be remarkable in a treatise on Theology, except this Theology were intended to constitute the whole of Philosophy.

We may observe, that in this work, though Plato, Avecibron, and many other heathen as well as Christian philosophers, are adduced as authority, Aristotle is referred to in a peculiar manner as "the philosopher." This is noticed by John of Salisbury, as attracting attention in his time (he died A.D. 1182). "The various Masters of Dialectic," says he,[74\4] "shine each with his peculiar merit; but all are proud to worship the footsteps of Aristotle; so much so, indeed, that the name of _philosopher_, which belongs to them all, has been pre-eminently appropriated to him. He is called the philosopher _autonomatice_, that is, by excellence."

[Note 74\4: _Metalogicus_, lib. ii. cap. 16.]

The Question concerning Corporeal Action, in Aquinas, is divided into six Articles; and the conclusion delivered upon the first is,[75\4] that "Body being compounded of power and act, is active as well as passive." Against this it is urged, that quantity is an attribute of body, and that quantity prevents action; that this appears in fact, since a larger body is more difficult to move. The author replies, that {233} "quantity does not prevent corporeal form from action altogether, but prevents it from being a universal agent, inasmuch as the form is individualized, which, in matter subject to quantity, it is. Moreover, the illustration deduced from the ponderousness of bodies is not to the purpose; first, because the addition of quantity is not the cause of gravity, as is proved in the fourth book, De Cœlo and De Mundo" (we see that he quotes familiarly the physical treatises of Aristotle); "second, because it is false that ponderousness makes motion slower; on the contrary, in proportion as any thing is heavier, the more does it move with its proper motion; thirdly, because action does not take place by local motion, as Democritus asserted; but by this, that something is drawn from power into act."

[Note 75\4: _**Summa_, P. i. Q. 115. Art. 1.]

It does not belong to our purpose to consider either the theological or the metaphysical doctrines which form so large a portion of the treatises of the schoolmen. Perhaps it may hereafter appear, that some light is thrown on some of the questions which have occupied metaphysicians in all ages, by that examination of the history of the Progressive Sciences in which we are now engaged; but till we are able to analyze the leading controversies of this kind, it would be of little service to speak of them in detail. It may be noticed, however, that many of the most prominent of them refer to the great question, "What is the relation between actual things and

general terms?" Perhaps in modern times, the actual things would be more commonly taken as the point to start from; and men would begin by considering how classes and universals are obtained from individuals. But the schoolmen, founding their speculations on the received modes of considering such subjects, to which both Aristotle and Plato had contributed, travelled in the opposite direction, and endeavored to discover how individuals were deduced from genera and species;--what was "the Principle of Individuation." This was variously stated by different reasoners. Thus Bonaventura[76\4] solves the difficulty by the aid of the Aristotelian distinction of Matter and Form. The individual derives from the Form the property of _being something_, and from the Matter the property of being that _particular thing_. Duns Scotus,[77\4] the great adversary of Thomas Aquinas in theology, placed the principle of Individuation in "a certain determining positive entity," which his school called _Hæcceity_ or _thisness_. "Thus an individual man is Peter, because his _humanity_ is combined with_ Petreity_." The force {234} of abstract terms is a curious question, and some remarkable experiments in their use had been made by the Latin Aristotelians before this time. In the same way in which we talk of the _quantity_ and _quality_ of a thing, they spoke of its _quiddity_.[78\4]

[Note 76\4: Deg. iv. 573.]

[Note 77\4: Ib. iv. 523.]

[Note 78\4: Deg. iv. 494.]

We may consider the reign of mere disputation as fully established at the time of which we are now speaking; and the only kind of philosophy henceforth studied was one in which no sound physical science had or could have a place. The wavering abstractions, indistinct generalizations, and loose classifications of common language, which we have already noted as the fountain of the physics of the Greek Schools of philosophy, were also the only source from which the Schoolmen of the middle ages drew their views, or rather their arguments: and though these notional and verbal relations were invested with a most complex and pedantic technicality, they did not, on that account, become at all more precise as notions, or more likely to lead to a single real truth. Instead of acquiring distinct ideas, they multiplied abstract terms; instead of real generalizations, they had recourse to verbal distinctions. The whole course of their employments tended to make them, not only ignorant of physical truth, but incapable of conceiving its nature.

Having thus taken upon themselves the task of raising and discussing questions by means of abstract terms, verbal distinctions, and logical rules alone, there was no tendency in their activity to come to an end, as there was no progress. The same questions, the same answers, the same difficulties, the same solutions, the same verbal subtleties,--sought for, admired, cavilled at, abandoned, reproduced, and again admired,--might recur without limit. John of Salisbury[79\4] observes of the Parisian teachers, that, after several years' absence, he found them not a step advanced, and still employed in urging and parrying the same arguments; and this, as Mr. Hallam remarks,[80\4] "was equally applicable to the period of centuries." The same knots were tied and {235} untied; the same clouds were formed and dissipated. The poet's

censure of "the Sons of Aristotle," is just as happily expressed: They stand Locked up together hand in hand Every one leads as he is led, The same bare path they tread, And dance like Fairies a fantastic round, But neither change their motion nor their ground.

[Note 79\4: He studied logic at Paris, at St. Geneviève, and then left them. "Duodecennium mihi elapsum est diversis studiis occupatum. Jucundum itaque visum est veteres quos reliqueram, et quos adhuc Dialectica detinebat in monte, (Sanctæ Genovefæ) revisere socios, conferre cum eis super ambiguitatibus pristinis; ut nostrûm invicem collatione mutuâ commetiremur profectum. Inventi sunt, qui fuerant, et ubi; neque enim ad palmam visi sunt processisse ad quæstiones pristinis dirimendas, neque propositiunculam unam adjecerant. Quibus urgebant stimulis eisdem et ipsi urgebantur," &c. _Metalogicus_, lib. ii. cap. 10.]

[Note 80\4: _Middle Ages_, iii. 537.]

It will therefore be unnecessary to go into any detail respecting the history of the School Philosophy of the thirteenth, fourteenth, and fifteenth centuries. We may suppose it to have been, during the intermediate time, such as it was at first and at last. An occasion to consider its later days will be brought before us by the course of our subject. But, even during the most entire ascendency of the scholastic doctrines, the elements of change were at work. While the doctors and the philosophers received all the ostensible homage of men, a doctrine and a philosophy of another kind were gradually forming: the practical instincts of man, their impatience of tyranny, the progress of the useful arts, the promises of alchemy, were all disposing men to reject the authority and deny the pretensions of the received philosophical creed. Two antagonist forms of opinion were in existence, which for some time went on detached, and almost independent of each other; but, finally, these came into conflict, at the time of Galileo; and the war speedily extended to every part of civilized Europe.

3. _Scholastic Physics._--It is difficult to give briefly any appropriate examples of the nature of the Aristotelian physics which are to be found in the works of this time. As the gravity of bodies was one of the first subjects of dispute when the struggle of the rival methods began, we may notice the mode in which it was treated.[81\4] "Zabarella maintains that the proximate cause of the motion of elements is the _form_, in the Aristotelian sense of the term: but to this sentence we," says Keckerman, "cannot agree; for in all other things the _form_ is the proximate cause, not of the _act_, but of the power or faculty from which the act flows. Thus in man, the rational soul is not the cause of the act of laughing, but of the risible faculty or power." Keckerman's system was at one time a work of considerable authority: it was published in 1614. By comparing and systematizing what he finds in Aristotle, he is led to state his results in the form of definitions {236} and theorems. Thus, "gravity is a motive quality, arising from cold, density, and bulk, by which the elements are carried downwards." "Water is the lower, intermediate element, cold and moist." The first theorem concerning water is, "The moistness of the water is controlled by its coldness, so that it is less than the moistness of the air; though, according to the sense of the vulgar, water appears to moisten more than air." It is obvious that the

two properties of fluids, to have their parts easily moved, and to wet other bodies, are here confounded. I may, as a concluding specimen of this kind, mention those propositions or maxims concerning fluids, which were so firmly established, that, when Boyle propounded the true mechanical principles of fluid action, he was obliged to state his opinions as "hydrostatical _paradoxes_." These were,--that fluids do not gravitate _in proprio loco_; that is, that water has no gravity in or on water, since it is in its own place;--that air has no gravity on water, since it is above water, which is its proper place;--that earth in water tends to descend, since its place is below water;--that the water rises in a pump or siphon, because nature abhors a vacuum;--that some bodies have a positive levity in others, as oil in water; and the like.

[Note 81\4: Keckerman, p. 1428.]

4. _Authority of Aristotle among the Schoolmen._--The authority of Aristotle, and the practice of making him the text and basis of the system, especially as it regarded physics, prevailed during the period of which we speak. This authority was not, however, without its fluctuations. Launoy has traced one part of its history in a book _On the various Fortune of Aristotle in the University of Paris_. The most material turns of this fortune depend on the bearing which the works of Aristotle were supposed to have upon theology. Several of Aristotle's works, and more especially his metaphysical writings, had been translated into Latin, and were explained in the schools of the University of Paris, as early as the beginning of the thirteenth century.[82\4] At a council held at Paris in 1209, they were prohibited, as having given occasion to the heresy of Almeric (or Amauri), and because "they might give occasion to other heresies not yet invented." The Logic of Aristotle recovered its credit some years after this, and was publicly taught in the University of Paris in the year 1215; but the Natural Philosophy and Metaphysics were prohibited by a decree of Gregory the Ninth, in 1231. The Emperor Frederic the Second employed a number of learned men to translate into Latin, from the Greek and {237} Arabic, certain books of Aristotle, and of other ancient sages; and we have a letter of Peter de Vineis, in which they are recommended to the attention of the University of Bologna: probably the same recommendation was addressed to other Universities. Both Albertus Magnus and Thomas Aquinas wrote commentaries on Aristotle's works; and as this was done soon after the decree of Gregory the Ninth, Launoy is much perplexed to reconcile the fact with the orthodoxy of the two doctors. Campanella, who was one of the first to cast off the authority of Aristotle, says, "We are by no means to think that St. Thomas _aristotelized_; he only expounded Aristotle, that he might correct his errors; and I should conceive he did this with the license of the Pope." This statement, however, by no means gives a just view of the nature of Albertus's and Aquinas's commentaries. Both have followed their author with profound deference.[83\4] For instance, Aquinas[84\4] attempts to defend Aristotle's assertion, that if there were no resistance, a body would move through a space in no time; and the same defence is given by Scotus.

[Note 82\4: Mosheim, iii. 157.]

[Note 83\4: Deg. N. 475.]

[Note 84\4: F. Piccolomini, ii. 835.]

We may imagine the extent of authority and admiration which Aristotle would attain, when thus countenanced, both by the powerful and the learned. In universities, no degree could be taken without a knowledge of the philosopher. In 1452, Cardinal Totaril established this rule in the University of Paris.[85\4] When Ramus, in 1543, published an attack upon Aristotle, it was repelled by the power of the court, and the severity of the law. Francis the First published an edict, in which he states that he had appointed certain judges, who had been of opinion,[86\4] "que le dit Ramus avoit été téméraire, arrogant et impudent; et que parcequ'en son livre des animadversions il reprenait Aristotle, estait évidemment connue et manifeste son ignorance." The books are then declared to be suppressed. It was often a complaint of pious men, that theology was corrupted by the influence of Aristotle and his commentators. Petrarch says,[87\4] that one of the Italian learned men conversing with him, after expressing much contempt for the apostles and fathers, exclaimed, "Utinam tu Averroen pati posses, ut videres quanto ille tuis his nugatoribus major sit!"

[Note 85\4: Launoy, pp. 108, 128.]

[Note 86\4: Launoy, p. 132.]

[Note 87\4: Hallam, _M. A._ iii. 536.]

When the revival of letters began to take place, and a number of men of ardent and elegant minds, susceptible to the impressions of beauty of style and dignity of thought, were brought into contact with Greek literature, Plato had naturally greater charms for them. A {238} powerful school of Platonists (not Neoplatonists) was formed in Italy, including some of the principal scholars and men of genius of the time; as Picus of Mirandula in the middle, Marsilius Ficinus at the end, of the fifteenth century. At one time, it appeared as if the ascendency of Aristotle was about to be overturned; but, in physics at least, his authority passed unshaken through this trial. It was not by disputation that Aristotle could be overthrown; and the Platonists were not persons whose doctrines led them to use the only decisive method in such cases, the observation and unfettered interpretation of facts.

The history of their controversies, therefore, does not belong to our design. For like reasons we do not here speak of other authors, who opposed the scholastic philosophy on general theoretical grounds of various kinds. Such examples of insurrection against the dogmatism which we have been reviewing, are extremely interesting events in the history of the philosophy of science. But, in the present work, we are to confine ourselves to the history of science itself; in the hope that we may thus be able, hereafter, to throw a steadier light upon that philosophy by which the succession of stationary and progressive periods, which we are here tracing, may be in some measure explained. We are now to close our account of the stationary period, and to enter upon the great subject of the progress of physical science in modern times.

5. _Subjects omitted. Civil Law, Medicine._--My object has been to make my way, as rapidly as possible, to this period of progress; and in doing this, I have had to pass

over a long and barren track, where almost all traces of the right road disappear. In exploring this region, it is not without some difficulty that he who is travelling with objects such as mine, continues a steady progress in the proper direction; for many curious and attractive subjects of research come in his way: he crosses the track of many a controversy, which in its time divided the world of speculators, and of which the results may be traced, even now, in the conduct of moral, or political, or metaphysical discussions; or in the common associations of thought, and forms of language. The wars of the Nominalists and Realists; the disputes concerning the foundations of morals, and the motives of human actions; the controversies concerning predestination, free will, grace, and the many other points of metaphysical divinity; the influence of theology and metaphysics upon each other, and upon other subjects of human curiosity; the effects of opinion upon politics, and of political condition upon opinion; the influence of literature and philosophy {239} upon each other, and upon society; and many other subjects;--might be well worth examination, if our hope of success did not reside in pursuing, steadily and directly, those inquiries in which we can look for a definite and certain reply. We must even neglect two of the leading studies of those times, which occupied much of men's time and thoughts, and had a very great influence on society; the one dealing with Notions, the other with Things; the one employed about moral rules, the other about material causes, but both for practical ends; I mean, the study of the _Civil Law_, and of _Medicine_. The second of these studies will hereafter come before us, as one of the principal occasions which led to the cultivation of chemistry; but, in itself, its progress is of too complex and indefinite a nature to be advantageously compared with that of the more exact sciences. The Roman Law is held, by its admirers, to be a system of deductive science, as exact as the mathematical sciences themselves; and it may, therefore, be useful to consider it, if we should, in the sequel, have to examine how far there can exist an analogy between moral and physical science. But, after a few more words on the middle ages, we must return to our task of tracing the progress of the latter.

## CHAPTER V.

PROGRESS OF THE ARTS IN THE MIDDLE AGES.

ART AND SCIENCE.--I shall, before I resume the history of science, say a few words on the subject described in the title of this chapter, both because I might otherwise be accused of doing injustice to the period now treated of; and also, because we shall by this means bring under our notice some circumstances which were important as being the harbingers of the revival of progressive knowledge.

The accusation of injustice towards the state of science in the middle ages, if we were to terminate our survey of them with what has hitherto been said, might be urged from obvious topics. How do we recognize, it might be asked, in a picture of mere confusion and mysticism of thought, of servility and dogmatism of character, the powers and acquirements to which we owe so many of the most important inventions which we now enjoy? Parchment and paper, printing and engraving, improved glass and steel, gunpowder, clocks, telescopes, {240} the mariner's

compass, the reformed calendar, the decimal notation, algebra, trigonometry, chemistry, counterpoint, an invention equivalent to a new creation of music;--these are all possessions which we inherit from that which has been so disparagingly termed the Stationary Period. Above all, let us look at the monuments of architecture of this period;--the admiration and the despair of modern architects, not only for their beauty, but for the skill disclosed in their construction. With all these evidences before us, how can we avoid allowing that the masters of the middle ages not only made some small progress in Astronomy, which has, grudgingly as it would seem, been admitted in a former Book; but also that they were no small proficients in other sciences, in Optics, in Harmonics, in Physics, and, above all, in Mechanics?

If, it may be added, we are allowed, in the present day, to refer to the perfection of our arts as evidence of the advanced state of our physical philosophy;--if our steam-engines, our gas-illumination, our buildings, our navigation, our manufactures, are cited as triumphs of science;--shall not prior inventions, made under far heavier disadvantages,--shall not greater works, produced in an earlier state of knowledge, also be admitted as witnesses that the middle ages had their share, and that not a small or doubtful one, of science?

To these questions I answer, by distinguishing between Art, and Science in that sense of general Inductive Systematic Truth, which it bears in this work. To separate and compare, with precision, these two processes, belongs to the Philosophy of Induction; and the attempt must be reserved for another place: but the leading differences are sufficiently obvious. Art is practical, Science is speculative: the former is seen in doing; the latter rests in the contemplation of what is known. The Art of the builder appears in his edifice, though he may never have meditated on the abstract propositions on which its stability and strength depends. The Science of the mathematical mechanician consists in his seeing that, under certain conditions, bodies must sustain each other's pressure, though he may never have applied his knowledge in a single case.

Now the remark which I have to make is this:--in all cases the Arts are prior to the related Sciences. Art is the parent, not the progeny, of Science; the realization of principles in practice forms part of the prelude, as well as of the sequel, of theoretical discovery. And thus the inventions of the middle ages, which have been above enumerated, though at the present day they may be portions of our sciences, are no evidence that the sciences then existed; but only that {241} those powers of practical observation and practical skill were at work, which prepare the way for theoretical views and scientific discoveries.

It may be urged, that the great works of art do virtually take for granted principles of science; and that, therefore, it is unreasonable to deny science to great artists. It may be said, that the grand structures of Cologne, or Amiens, or Canterbury, could not have been erected without a profound knowledge of mechanical principles.

To this we reply, that _such_ knowledge is manifestly not of the nature of that which we call _science_. If the beautiful and skilful structures of the middle ages prove that mechanics then existed as a science, mechanics must have existed as a

science also among the builders of the Cyclopean walls of Greece and Italy, or of our own Stonehenge; for the masses which are there piled on each other, could not be raised without considerable mechanical skill. But we may go much further. The actions of every man who raises and balances weights, or walks along a pole, take for granted the laws of equilibrium; and even animals constantly avail themselves of such principles. Are these, then, acquainted with mechanics as a science? Again, if actions which are performed by taking advantage of mechanical properties prove a knowledge of the science of mechanics, they must also be allowed to prove a knowledge of the science of geometry, when they proceed on geometrical properties. But the most familiar actions of men and animals proceed upon geometrical truths. The Epicureans held, as Proclus informs us, that even asses knew that two sides of a triangle are greater than the third. And animals may truly be said to have a practical knowledge of this truth; but they have not, therefore, a science of geometry. And in like manner among men, if we consider the matter strictly, a practical assumption of a principle does not imply a speculative knowledge of it.

We may, in another way also, show how inadmissible are the works of the Master Artists of the middle ages into the series of events which mark the advance of Science. The following maxim is applicable to a history, such as we are here endeavoring to write. We are employed in tracing the progress of such general principles as constitute each of the sciences which we are reviewing; and no facts or subordinate truths belong to our scheme, except so far as they tend to or are included in these higher principles; nor are they important to us, any further than as they prove such principles. Now with regard to processes of art like those which we have referred to, namely, the inventions of the middle ages, let us ask, _what_ principle each of them {242} illustrates? What chemical doctrine rests for its support on the phenomena of gunpowder, or glass, or steel? What new harmonical truth was illustrated in the Gregorian chant? What mechanical principle unknown to Archimedes was displayed in the printing-press? The practical value and use, the ingenuity and skill of these inventions is not questioned; but what is their place in the history of speculative knowledge? Even in those cases in which they enter into such a history, how minute a figure do they make! how great is the contrast between their practical and theoretical importance! They may in their operation have changed the face of the world; but in the history of the principles of the sciences to which they belong, they may be omitted without being missed.

As to that part of the objection which was stated by asking, why, if the arts of our age prove its scientific eminence, the arts of the middle ages should not be received as proof of theirs; we must reply to it, by giving up some of the pretensions which are often put forwards on behalf of the science of our times. The perfection of the mechanical and other arts among us proves the advanced condition of our sciences, only in so far as these arts have been perfected by the application of some great scientific truth, with a clear insight into its nature. The greatest improvement of the steam-engine was due to the steady apprehension of an atmological doctrine by Watt; but what distinct theoretical principle is illustrated by the beautiful manufactures of porcelain, or steel, or glass? A chemical view of these compounds, which would explain the conditions of success and failure in their manufacture, would be of great

value in art; and it would also be a novelty in chemical theory; so little is the present condition of those processes a triumph of science, shedding intellectual glory on our age. And the same might be said of many, or of most, of the processes of the arts as now practised.

2. _Arabian Science._--Having, I trust, established the view I have stated, respecting the relation of Art and Science, we shall be able very rapidly to dispose of a number of subjects which otherwise might seem to require a detailed notice. Though this distinction has been recognized by others, it has hardly been rigorously adhered to, in consequence of the indistinct notion of _science_ which has commonly prevailed. Thus Gibbon, in speaking of the knowledge of the period now under our notice, says,[88\4] "Much useful experience had been acquired in {243} the practice of arts and manufactures; but the _science_ of chemistry owes its origin and improvement to the industry of the Saracens. They," he adds, "first invented and named the alembic for the purposes of distillation, analyzed the substances of the three kingdoms of nature, tried the distinction and affinities of alkalies and acids, and converted the poisonous minerals into soft and salutary medicines." The formation and realization of the notions of _analysis_ and of _affinity_, were important steps in chemical science, which, as I shall hereafter endeavor to show, it remained for the chemists of Europe to make at a much later period. If the Arabians had done this, they might with justice have been called the authors of the science of chemistry; but no doctrines can be adduced from their works which give them any title to this eminent distinction. Their claims are dissipated at once by the application of the maxim above stated. _What_ analysis of theirs tended to establish any received principle of chemistry? _What_ true doctrine concerning the differences and affinities of acids and alkalies did they teach? We need not wonder if Gibbon, whose views of the boundaries of scientific chemistry were probably very wide and indistinct, could include the arts of the Arabians within its domain; but they cannot pass the frontier of science if philosophically defined, and steadily guarded.

[Note 88\4: _Decline and Fall_, vol. x. p. 43.]

The judgment which we are thus led to form respecting the chemical knowledge of the middle ages, and of the Arabians in particular, may serve to measure the condition of science in other departments; for chemistry has justly been considered one of their strongest points. In botany, anatomy, zoology, optics, acoustics, we have still the same observations to make, that the steps in science which, in the order of progress, next followed what the Greeks had done, were left for the Europeans of the sixteenth and seventeenth centuries. The merits and advances of the Arabian philosophers in astronomy and pure mathematics, we have already described.

3. _Experimental Philosophy of the Arabians._--The estimate to which we have thus been led, of the scientific merits of the learned men of the middle ages, is much less exalted than that which has been formed by many writers; and, among the rest, by some of our own time. But I am persuaded that any attempt to answer the questions just asked, will expose the untenable nature of the higher claims which have been advanced in favor of the Arabians. We can deliver no just decision, except we will consent to use the terms of science in a strict and precise sense: and if we do this, we

shall find little, either in the {244} particular discoveries or general processes of the Arabians, which is important in the history of the Inductive Sciences.[89\4]

[Note 89\4: If I might take the liberty of criticising an author who has given a very interesting view of the period in question (_Mahometanism Unveiled_, by the Rev. Charles Forster, 1829), I would remark, that in his work this caution is perhaps too little observed. Thus, he says, in speaking of Alhazen (vol. ii. p. 270), "the theory of the telescope may be found in the work of this astronomer;" and of another, "the uses of magnifying glasses and telescopes, and the principle of their construction, are explained in the Great Work of (Roger) Bacon, with a truth and clearness which have commanded universal admiration." Such phrases would be much too strong, even if used respecting the optical doctrines of Kepler, which were yet incomparably more true and clear than those of Bacon. To employ such language, in such cases, is to deprive such terms as _theory_ and _principle_ of all meaning.]

The credit due to the Arabians for improvements in the general methods of philosophizing, is a more difficult question; and cannot be discussed at length by us, till we examine the history of such methods in the abstract, which, in the present work, it is not our intention to do. But we may observe, that we cannot agree with those who rank their merits high in this respect. We have already seen, that their minds were completely devoured by the worst habits of the stationary period,--Mysticism and Commentation. They followed their Greek leaders, for the most part, with abject servility, and with only that kind of acuteness and independent speculation which the Commentator's vocation implies. And in their choice of the standard subjects of their studies, they fixed upon those works, the Physical Books of Aristotle, which have never promoted the progress of science, except in so far as they incited men to refute them; an effect which they never produced on the Arabians. That the Arabian astronomers made some advances beyond the Greeks, we have already stated: the two great instances are, the discovery of the Motion of the Sun's Apogee by Albategnius, and the discovery (recently brought to light) of the existence of the Moon's Second Inequality, by Aboul Wefa. But we cannot but observe in how different a manner they treated these discoveries, from that with which Hipparchus or Ptolemy would have done. The Variation of the Moon, in particular, instead of being incorporated into the system by means of an Epicycle, as Ptolemy had done with the Evection, was allowed, almost immediately, so far as we can judge, to fall into neglect and oblivion: so little were the learned Arabians prepared to take their lessons from observation as well as from books. That in many subjects they made experiments, may easily be allowed: there never was a period of the earth's history, and least of all a period of commerce {245} and manufactures, luxury and art, medicine and engineering, in which there were not going on innumerable processes, which may be termed Experiments; and, in addition to these, the Arabians adopted the pursuit of alchemy, and the love of exotic plants and animals. But so far from their being, as has been maintained,[90\4] a people whose "experimental intellect" fitted them to form sciences which the "abstract intellect" of the Greeks failed in producing, it rather appears, that several of the sciences which the Greeks had founded, were never even comprehended by the Arabians. I do not know any evidence that these pupils ever attained to understand the real principles of Mechanics, Hydrostatics, and

Harmonics, which their masters had established. At any rate, when these sciences again became progressive, Europe had to start where Europe had stopped. There is no Arabian name which any one has thought of interposing between Archimedes the ancient, and Stevinus and Galileo the moderns.

[Note 90\4: _Mahometanism Unveiled_, ii. 271.]

4. _Roger Bacon._--There is one writer of the middle ages, on whom much stress has been laid, and who was certainly a most remarkable person. Roger Bacon's works are not only so far beyond his age in the knowledge which they contain, but so different from the temper of the times, in his assertion of the supremacy of experiment, and in his contemplation of the future progress of knowledge, that it is difficult to conceive how such a character could then exist. That he received much of his knowledge from Arabic writers, there can be no doubt; for they were in his time the repositories of all traditionary knowledge. But that he derived from them his disposition to shake off the authority of Aristotle, to maintain the importance of experiment, and to look upon knowledge as in its infancy, I cannot believe, because I have not myself hit upon, nor seen quoted by others, any passages in which Arabian writers express such a disposition. On the other hand, we do find in European writers, in the authors of Greece and Rome, the solid sense, the bold and hopeful spirit, which suggest such tendencies. We have already seen that Aristotle asserts, as distinctly as words can express, that all knowledge must depend on observation, and that science must be collected from facts by induction. We have seen, too, that the Roman writers, and Seneca in particular, speak with an enthusiastic confidence of the progress which science must make in the course of ages. When Roger Bacon holds similar language in the thirteenth century, the resemblance is probably rather a sympathy of character, than a matter of direct derivation; but I know of nothing {246} which proves even so much as this sympathy in the case of Arabian philosophers.

A good deal has been said of late of the coincidences between his views, and those of his great namesake in later times, Francis Bacon.[91\4] The resemblances consist mainly in such points as I have just noticed; and we cannot but acknowledge, that many of the expressions of the Franciscan Friar remind us of the large thoughts and lofty phrases of the Philosophical Chancellor. How far the one can be considered as having anticipated the method of the other, we shall examine more advantageously, when we come to consider what the character and effect of Francis Bacon's works really are.[92\4]

[Note 91\4: Hallam's _Middle Ages_, iii. 549. Forster's _Mahom. U._ ii. 313.]

[Note 92\4: In the _Philosophy of the Inductive Sciences_, I have given an account at considerable length of Roger Bacon's mode of treating Arts and Sciences; and have also compared more fully his philosophy with that of Francis Bacon; and I have given a view of the bearing of this latter upon the progress of Science in modern times. See _Phil. Ind. Sc._ book xii. chaps. 7 and 11. See also the Appendix to this volume.]

5. _Architecture of the Middle Ages._--But though we are thus compelled to disallow several of the claims which have been put forwards in support of the

scientific character of the middle ages, there are two points in which we may, I conceive, really trace the progress of scientific ideas among them; and which, therefore, may be considered as the prelude to the period of discovery. I mean their practical architecture, and their architectural treatises.

In a previous chapter of this book, we have endeavored to explain how the indistinctness of ideas, which attended the decline of the Roman empire, appears in the forms of their architecture;--in the disregard, which the decorative construction exhibits, of the necessary mechanical conditions of support. The original scheme of Greek ornamental architecture had been horizontal masses resting on vertical columns: when the arch was introduced by the Romans, it was concealed, or kept in a state of subordination: and the lateral support which it required was supplied latently, marked by some artifice. But the struggle between the _mechanical_ and the _decorative construction_[93\4] ended in the complete disorganization of the classical style. The {247} inconsistencies and extravagances of which we have noticed the occurrence, were results and indications of the fall of good architecture. The elements of the ancient system had lost all principle of connection and regard to rule. Building became not only a mere art, but an art exercised by masters without skill, and without feeling for real beauty.

[Note 93\4: See Mr. Willis's admirable _Remarks on the Architecture of the Middle Ages_, chap. ii.

Since the publication of my first edition, Mr. Willis has shown that much of the "mason-craft" of the middle ages consisted in the geometrical methods by which the artists wrought out of the blocks the complex forms of their decorative system.

To the general indistinctness of speculative notions on mechanical subjects prevalent in the middle ages, there may have been some exceptions, and especially so long as there were readers of Archimedes. Boëthius had translated the mechanical works of Archimedes into Latin, as we learn from the enumeration of his works by his friend Cassiodorus (_Variar._ lib i. cap. 45), "_Mechanicum_ etiam Archimedem latialem siculis reddidisti." But _Mechanicus_ was used in those times rather for one skilled in the art of constructing wonderful machines than in the speculative theory of them. The letter from which the quotation is taken is sent by King Theodoric to Boëthius, to urge him to send the king a water-clock.]

When, after this deep decline, architecture rose again, as it did in the twelfth and succeeding centuries, in the exquisitely beautiful and skilful forms of the Gothic style, what was the nature of the change which had taken place, so far as it bears upon the progress of science? It was this:--the idea of true mechanical relations in an edifice had been revived in men's minds, as far as was requisite for the purposes of art and beauty: and this, though a very different thing from the possession of the idea as an element of speculative science, was the proper preparation for that acquisition. The notion of support and stability again became conspicuous in the decorative construction, and universal in the forms of building. The eye which, looking for beauty in definite and significant relations of parts, is never satisfied except the weights appear to be duly supported,[94\4] was again gratified. Architecture threw

off its barbarous characters: a new decorative construction was matured, not thwarting and controlling, but assisting and harmonizing with the mechanical construction. All the ornamental parts were made to enter into the apparent construction. Every member, almost every moulding, became a sustainer of weight; and by the multiplicity of props assisting each other, and the consequent subdivision of weight, the eye was satisfied of the stability of the structure, notwithstanding the curiously-slender forms of the separate parts. The arch and the vault, no longer trammelled by an incompatible system of decoration, but favoured by more tractable forms, were only limited by the skill of the builders. Everything showed that, practically at least, men possessed and applied, with steadiness and pleasure, the idea of mechanical pressure and support.

[Note 94\4: Willis, pp. 15-21. I have throughout this description of the formation of the Gothic style availed myself of Mr. Willis's well-chosen expressions.]

The possession of this idea, as a principle of art, led, in the course of time, to its speculative development as the foundation of a science; {248} and thus Architecture prepared the way for Mechanics. But this advance required several centuries. The interval between the admirable cathedrals of Salisbury, Amiens, Cologne, and the mechanical treatises of Stevinus, is not less than three hundred years. During this time, men were advancing towards science; but in the mean time, and perhaps from the very beginning of the time, art had begun to decline. The buildings of the fifteenth century, erected when the principles of mechanical support were just on the verge of being enunciated in general terms, exhibit those principles with a far less impressive simplicity and elegance than those of the thirteenth. We may hereafter inquire whether we find any other examples to countenance the belief, that the formation of Science is commonly accompanied by the decline of Art.

The leading principle of the style of the Gothic edifices was, not merely that the weights were supported, but that they were seen to be so; and that not only the mechanical relations of the larger masses, but of the smaller members also, were displayed. Hence we cannot admit, as an origin or anticipation of the Gothic, a style in which this principle is not manifested. I do not see, in any of the representations of the early Arabic buildings, that distribution of weights to supports, and that mechanical consistency of parts, which would elevate them above the character of barbarous architecture. Their masses are broken into innumerable members, without subordination or meaning, in a manner suggested apparently by caprice and the love of the marvellous. "In the construction of their mosques, it was a favorite artifice of the Arabs to sustain immense and ponderous masses of stone by the support of pillars so slender, that the incumbent weight seemed, as it were, suspended in the air by an invisible hand."[95\4] This pleasure in the contemplation of apparent impossibilities is a very general disposition among mankind; but it appears to belong to the infancy, rather than the maturity of intellect. On the other hand, the pleasure in the contemplation of what is clear, the craving for a thorough insight into the reasons of things, which marks the European mind, is the temper which leads to science.

[Note 95\4: _Mahometanism Unveiled_, ii. 255.]

6. _Treatises on Architecture._--No one who has attended to the architecture which prevailed in England, France, and Germany, from the twelfth to the fifteenth century, so far as to comprehend its beauty, harmony, consistency, and uniformity, even in the minutest parts and most obscure relations, can look upon it otherwise than as a {249} remarkably connected and definite artificial system. Nor can we doubt that it was exercised by a class of artists who formed themselves by laborious study and practice, and by communication with each other. There must have been bodies of masters and of scholars, discipline, traditions, precepts of art. How these associated artists diffused themselves over Europe, and whether history enables us to trace them in a distinct form, I shall not here discuss. But the existence of a course of instruction, and of a body of rules of practice, is proved beyond dispute by the great series of European cathedrals and churches, so nearly identical in their general arrangements, and in their particular details. The question then occurs, have these rules and this system of instruction anywhere been committed to writing? Can we, by such evidence, trace the progress of the scientific idea, of which we see the working in these buildings?

We are not to be surprised, if, during the most flourishing and vigorous period of the art of the middle ages, we find none of its precepts in books. Art has, in all ages and countries, been taught and transmitted by practice and verbal tradition, not by writing. It is only in our own times, that the thought occurs as familiar, of committing to books all that we wish to preserve and convey. And, even in our own times, most of the Arts are learned far more by practice, and by intercourse with practitioners, than by reading. Such is the case, not only with Manufactures and Handicrafts, but with the Fine Arts, with Engineering, and even yet, with that art, Building, of which we are now speaking.

We are not, therefore, to wonder, if we have no treatises on Architecture belonging to the great period of the Gothic masters;--or if it appears to have required some other incitement and some other help, besides their own possession of their practical skill, to lead them to shape into a literary form the precepts of the art which they knew so well how to exercise:--or if, when they did write on such subjects, they seem, instead of delivering their own sound practical principles, to satisfy themselves with pursuing some of the frivolous notions and speculations which were then current in the world of letters.

Such appears to be the case. The earliest treatises on Architecture come before us under the form which the commentatorial spirit of the middle ages inspired. They are Translations of Vitruvius, with Annotations. In some of these, particularly that of Cesare Cesariano, published at Como, in 1521, we see, in a very curious manner, how the habit of assuming that, in every department of literature, the ancients {250} must needs be their masters, led these writers to subordinate the members of their own architecture to the precepts of the Roman author. We have Gothic shafts, mouldings, and arrangements, given as parallelisms to others, which profess to represent the Roman style, but which are, in fact, examples of that mixed manner which is called the style of the _Cinque cento_ by the Italians, of the _Renaissance_ by the French, and which is commonly included in our _Elizabethan_. But in the early

architectural works, besides the superstitions and mistaken erudition which thus choked the growth of real architectural doctrines, another of the peculiar elements of the middle ages comes into view;--its mysticism. The dimensions and positions of the various parts of edifices and of their members, are determined by drawing triangles, squares, circles, and other figures, in such a manner as to bound them; and to these geometrical figures were assigned many abstruse significations. The plan and the front of the Cathedral at Milan are thus represented in Cesariano's work, bounded and subdivided by various equilateral triangles; and it is easy to see, in the earnestness with which he points out these relations, the evidence of a fanciful and mystical turn of thought.[96\4]

[Note 96\4: The plan which he has given, fol. 14, he has entitled "Ichnographia Fundamenti sacræ Ædis baricephalæ, Germanico more, à Trigono ac Pariquadrato perstructa, uti etiam ea quæ nunc Milani videtur."

The work of Cesariano was translated into German by Gualter Rivius, and published at Nuremberg, in 1548, under the title of _Vitruvius Teutsch_, with copies of the Italian diagrams. A few years ago, in an article in the _Wiener Jahrbücher_ (Oct.-Dec., 1821), the reviewer maintained, on the authority of the diagrams in Rivius's book, that Gothic architecture had its origin in Germany and not in England.]

We thus find erudition and mysticism take the place of much of that development of the architectural principles of the middle ages which would be so interesting to us. Still, however, these works are by no means without their value. Indeed many of the arts appear to flourish not at all the worse, for being treated in a manner somewhat mystical; and it may easily be, that the relations of geometrical figures, for which fantastical reasons are given, may really involve principles of beauty or stability. But independently of this, we find, in the best works of the architects of all ages (including engineers), evidence that the true idea of mechanical pressure exists among them more distinctly than among men in general, although it may not be developed in a scientific form. This is true up to our own time, and the arts which such persons cultivate could not be successfully {251} exercised if it were not so. Hence the writings of architects and engineers during the middle ages do really form a prelude to the works on scientific mechanics. Vitruvius, in his _Architecture_, and Julius Frontinus, who, under Vespasian, wrote _On Aqueducts_, of which he was superintendent, have transmitted to us the principal part of what we know respecting the practical mechanics and hydraulics of the Romans. In modern times the series is resumed. The early writers on architecture are also writers on engineering, and often on hydrostatics: for example, Leonardo da Vinci wrote on the equilibrium of water. And thus we are led up to Stevinus of Bruges, who was engineer to Prince Maurice of Nassau, and inspector of the dykes in Holland; and in whose work, on the processes of his art, is contained the first clear modern statement of the scientific principles of hydrostatics.

Having thus explained both the obstacles and the prospects which the middle ages offered to the progress of science, I now proceed to the history of the progress, when that progress was once again resumed.

{{253}} BOOK V.

HISTORY OF FORMAL ASTRONOMY AFTER THE STATIONARY PERIOD.

. . . Cyclopum educta caminis Mœnia conspicio, atque adverso fornice portas. . . . . His demum exactis, perfecto munere Divæ, Devenere locos lætos et amœna vireta Fortunatorum nemorum sedesque beatas. Largior hic campos æther et lumine vestit Purpureo: solemque suum, sua sidera norunt. VIRGIL, _Æn._ vi. 630.

They leave at length the nether gloom, and stand Before the portals of a better land: To happier plains they come, and fairer groves, The seats of those whom heaven, benignant, loves; A brighter day, a bluer ether, spreads Its lucid depths above their favored heads; And, purged from mists that veil our earthly skies, Shine suns and stars unseen by mortal eyes.

{{255}} INTRODUCTION.

_Of Formal and Physical Astronomy._

WE have thus rapidly traced the causes of the almost complete blank which the history of physical science offers, from the decline of the Roman empire, for a thousand years. Along with the breaking up of the ancient forms of society, were broken up the ancient energy of thinking, the clearness of idea, and steadiness of intellectual action. This mental declension produced a servile admiration for the genius of the better periods, and thus, the spirit of Commentation: Christianity established the claim of truth to govern the world; and this principle, misinterpreted and combined with the ignorance and servility of the times, gave rise to the Dogmatic System: and the love of speculation, finding no secure and permitted path on solid ground, went off into the regions of Mysticism.

The causes which produced the inertness and blindness of the stationary period of human knowledge, began at last to yield to the influence of the principles which tended to progression. The indistinctness of thought, which was the original feature in the decline of sound knowledge, was in a measure remedied by the steady cultivation of Pure Mathematics and Astronomy, and by the progress of inventions in the Arts, which call out and fix the distinctness of our conceptions of the relations of natural phenomena. As men's minds became clear, they became less servile: the perception of the nature of truth drew men away from controversies about mere opinion; when they saw distinctly the relations of _things_, they ceased to give their whole attention to what had been _said_ concerning them; and thus, as science rose into view, the spirit of commentation lost its way. And when men came to feel what it was to think for themselves on subjects of science, they soon rebelled against the right of others to impose opinions upon them. When they threw off their blind admiration for the ancients, they were disposed to cast away also their passive obedience to the ancient system of doctrines. When they were no longer inspired by the spirit of commentation, they were no longer submissive to the dogmatism of the schools. When they began to feel that they could {256} discover truths, they felt also a persuasion of a right and a growing will so to do.

Thus the revived clearness of ideas, which made its appearance at the revival of letters, brought on a struggle with the authority, intellectual and civil, of the established schools of philosophy. This clearness of idea showed itself, in the first instance, in Astronomy, and was embodied in the system of Copernicus; but the contest did not come to a crisis till a century later, in the time of Galileo and other disciples of the new doctrine. It is our present business to trace the principles of this series of events in the history of philosophy.

I do not profess to write a history of Astronomy, any further than is necessary in order to exhibit the principles on which the progression of science proceeds; and, therefore, I neglect subordinate persons and occurrences, in order to bring into view the leading features of great changes. Now in the introduction of the Copernican system into general acceptation, two leading views operated upon men's minds; the consideration of the system as exhibiting the apparent motions of the universe, and the consideration of this system with reference to its causes;--the _formal_ and the _physical_ aspect of the Theory;--the relations of Space and Time, and the relations of Force and Matter. These two divisions of the subject were at first not clearly separated; the second was long mixed, in a manner very dim and obscure, with the first, without appearing as a distinct subject of attention; but at last it was extricated and treated in a manner suitable to its nature. The views of Copernicus rested mainly on the formal condition of the universe, the relations of space and time; but Kepler, Galileo, and others, were led, by controversies and other causes, to give a gradually increasing attention to the physical relations of the heavenly bodies; an impulse was given to the study of Mechanics (the Doctrine of Motion), which became very soon an important and extensive science; and in no long period, the discoveries of Kepler, suggested by a vague but intense belief in the physical connection of the parts of the universe, led to the decisive and sublime generalizations of Newton.

The distinction of _formal_ and _physical_ Astronomy thus becomes necessary, in order to treat clearly of the discussions which the propounding of the Copernican theory occasioned. But it may be observed that, besides this great change, Astronomy made very great advances in the same path which we have already been tracing, namely, the determination of the quantities and laws of the celestial motions, in so far as they were exhibited by the ancient theories, or {257} might be represented by obvious modifications of those theories. I speak of new Inequalities, new Phenomena, such as Copernicus, Galileo, and Tycho Brahe discovered. As, however, these were very soon referred to the Copernican rather than the Ptolemaic hypothesis, they may be considered as developments rather of the new than of the old Theory; and I shall, therefore, treat of them, agreeably to the plan of the former part, as the sequel of the Copernican Induction.

## CHAPTER I.

PRELUDE TO THE INDUCTIVE EPOCH OF COPERNICUS.

THE Doctrine of Copernicus, that the Sun is the true centre of the celestial motions, depends primarily upon the consideration that such a supposition explains

very simply and completely all the obvious appearances of the heavens. In order to see that it does this, nothing more is requisite than a distinct conception of the nature of Relative Motion, and a knowledge of the principal Astronomical Phenomena. There was, therefore, no reason why such a doctrine might not be _discovered_, that is, suggested as a theory plausible at first sight, long before the time of Copernicus; or rather, it was impossible that this guess, among others, should not be propounded as a solution of the appearances of the heavens. We are not, therefore, to be surprised if we find, in the earliest times of Astronomy, and at various succeeding periods, such a system spoken of by astronomers, and maintained by some as true, though rejected by the majority, and by the principal writers.

When we look back at such a difference of opinion, having in our minds, as we unavoidably have, the clear and irresistible considerations by which the Copernican Doctrine is established _for us_, it is difficult for us not to attribute superior sagacity and candor to those who held that side of the question, and to imagine those who clung to the Ptolemaic Hypothesis to have been blind and prejudiced; incapable of seeing the beauty of simplicity and symmetry, or indisposed to resign established errors, and to accept novel and comprehensive truths. Yet in judging thus, we are probably ourselves influenced by prejudices arising from the knowledge and received opinions of our own times. For is it, in reality, clear that, before the time of Copernicus, the {258} _Heliocentric_ Theory (that which places the centre of the celestial motions in the Sun) had a claim to assent so decidedly superior to the Geocentric Theory, which places the Earth in the centre? What is the basis of the heliocentric theory?--That the _relative_ motions are _the same_, on that and on the other supposition. So far, therefore, the two hypotheses are exactly on the same footing. But, it is urged, on the heliocentric side we have the advantage of simplicity:--true; but we have, on the other side, the testimony of our senses; that is, the geocentric doctrine (which asserts that the Earth rests and the heavenly bodies move) is the obvious and spontaneous interpretation of the appearances. Both these arguments, _simplicity_ on the one side, and _obviousness_ on the other, are vague, and we may venture to say, both indecisive. We cannot establish any strong preponderance of probability in favor of the former doctrine, without going much further into the arguments of the question.

Nor, when we speak of the superior _simplicity_ of the Copernican theory, must we forget, that though this theory has undoubtedly, in this respect, a great advantage over the Ptolemaic, yet that the Copernican system itself is very complex, when it undertakes to account, as the Ptolemaic did, for the _Inequalities_ of the Motions of the sun, moon, and planets; and, that in the hands of Copernicus, it retained a large share of the eccentrics and epicycles of its predecessor, and, in some parts, with increased machinery. The heliocentric theory, without these appendages, would not approach the Ptolemaic, in the accurate explanation of facts; and as those who had placed the sun in the centre had never, till the time of Copernicus, shown how the inequalities were to be explained on that supposition, we may assert that after the promulgation of the theory of eccentrics and epicycles on the geocentric hypothesis, there was no _published_ heliocentric theory which could bear a comparison with that hypothesis.

It is true, that all the contrivances of epicycles, and the like, by which the geocentric hypothesis was made to represent the phenomena, were susceptible of an easy adaptation to a heliocentric method, _when a good mathematician had once proposed to himself the problem_: and this was precisely what Copernicus undertook and executed. But, till the appearance of his work, the heliocentric system had never come before the world except as a hasty and imperfect hypothesis; which bore a favorable comparison with the phenomena, so long as their general features only were known; but which had been completely thrown into the shade by the labor and intelligence bestowed upon {259} the Hipparchian or Ptolemaic theories by a long series of great astronomers of all civilized countries.

But, though the astronomers who, before Copernicus, held the heliocentric opinion, cannot, on any good grounds, be considered as much more enlightened than their opponents, it is curious to trace the early and repeated manifestations of this view of the universe. The distinct assertion of the heliocentric theory among the Greeks is an evidence of the clearness of their thoughts, and the vigour of their minds; and it is a proof of the feebleness and servility of intellect in the stationary period, that, till the period of Copernicus, no one was found to try the fortune of this hypothesis, modified according to the improved astronomical knowledge of the time.

The most ancient of the Greek philosophers to whom the ancients ascribe the heliocentric doctrine, is Pythagoras; but Diogenes Laertius makes Philolaus, one of the followers of Pythagoras, the first author of this doctrine. We learn from Archimedes, that it was held by his contemporary, Aristarchus. "Aristarchus of Samos," says he,[1\5] "makes this supposition,--that the fixed stars and the sun remain at rest, and that the earth revolves round the sun in a circle." Plutarch[2\5] asserts that this, which was only a hypothesis in the hands of Aristarchus, was _proved_ by Seleucus; but we may venture to say that, at that time, no such proof was possible. Aristotle had recognized the existence of this doctrine by arguing against it. "All things," says he,[3\5] "tend to the centre of the earth, and rest there, and therefore the whole mass of the earth cannot rest except there." Ptolemy had in like manner argued against the diurnal motion of the earth: such a revolution would, he urged, disperse into surrounding space all the loose parts of the earth. Yet he allowed that such a supposition would facilitate the explanation of some phenomena. Cicero appears to make Mercury and Venus revolve about the sun, as does Martianus Capella at a later period; and Seneca says[4\5] it is a worthy subject of contemplation, whether the earth be at rest or in motion: but at this period, as we may see from Seneca himself, that habit of intellect which was requisite for the solution of such a question, had been succeeded by indistinct views, and rhetorical forms of speech. If there were any good mathematicians and good observers at this period, they were employed in cultivating and verifying the Hipparchian theory.

[Note 1\5: Archim. _Arenarius._]

[Note 2\5: _Quest. Plat._ Delamb. _A. A._ vi.]

[Note 3\5: Quoted by Copernic. i. 7.]

[Note 4\5: _Quest. Nat._ vii. 2.]

Next to the Greeks, the Indians appear to have possessed that {260} original vigor and clearness of thought, from which true science springs. It is remarkable that the Indians, also, had their heliocentric theorists. Aryabatta[5\5] (A. D. 1322), and other astronomers of that country, are said to have advocated the doctrine of the earth's revolution on its axis; which opinion, however, was rejected by subsequent philosophers among the Hindoos.

[Note 5\5: Lib. U. K. _Hist. Ast._ p. 11.]

Some writers have thought that the heliocentric doctrine was _derived_ by Pythagoras and other European philosophers, from some of the oriental nations. This opinion, however, will appear to have little weight, if we consider that the heliocentric hypothesis, in the only shape in which the ancients knew it, was too obvious to require much teaching; that it did not and could not, so far as we know, receive any additional strength from any thing which the oriental nations could teach; and that each astronomer was induced to adopt or reject it, not by any information which a master could give him, but by his love of geometrical simplicity on the one hand, or the prejudices of sense on the other. Real science, depending on a clear view of the relation of phenomena to general theoretical ideas, cannot be communicated in the way of secret and exclusive traditions, like the mysteries of certain arts and crafts. If the philosopher do not _see_ that the theory is true, he is little the better for having heard or read the words which assert its truth.

It is impossible, therefore, for us to assent to those views which would discover in the heliocentric doctrines of the ancients, traces of a more profound astronomy than any which they have transmitted to us. Those doctrines were merely the plausible conjectures of men with sound geometrical notions; but they were never extended so as to embrace the details of the existing astronomical knowledge; and perhaps we may say, that the analysis of the phenomena into the arrangements of the Ptolemaic system, was so much more obvious than any other, that it must necessarily come first, in order to form an introduction to the Copernican.

The true foundation of the heliocentric theory for the ancients was, as we have intimated, its perfect geometrical consistency with the general features of the phenomena, and its simplicity. But it was unlikely that the human mind would be content to consider the subject under this strict and limited aspect alone. In its eagerness for wide speculative views, it naturally looked out for other and vaguer principles of connection and relation. Thus, as it had been urged in {261} favor of the geocentric doctrine, that the heaviest body must be in the centre, it was maintained, as a leading recommendation of the opposite opinion, that it placed the Fire, the noblest element, in the Centre of the Universe. The authority of mythological ideas was called in on both sides to support these views. Numa, as Plutarch[6\5] informs us, built a circular temple over the ever-burning Fire of Vesta; typifying, not the earth, but the Universe, which, according to the Pythagoreans, has the Fire seated at its Centre. The same writer, in another of his works, makes one of his interlocutors say, "Only, my friend, do not bring me before a court of law on a charge of impiety; as Cleanthes said, that Aristarchus the Samian ought to be tried for impiety, because he removed the Hearth of the Universe." This, however, seems to have been intended as

a pleasantry.

[Note 6\5: _De Facie in Orbe Lunæ_, 6.]

The prevalent physical views, and the opinions concerning the causes of the motions of the parts of the universe, were scarcely more definite than the ancient opinions concerning the relations of the four elements, till Galileo had founded the true Doctrine of Motion. Though, therefore, arguments on this part of the subject were the most important part of the controversy after Copernicus, the force of such arguments was at his time almost balanced. Even if more had been known on such subjects, the arguments would not have been conclusive: for instance, the vast mass of the heavens, which is commonly urged as a reason why the heavens do not move round the earth, would not make such a motion impossible; and, on the other hand, the motions of bodies at the earth's surface, which were alleged as inconsistent with its motion, did not really disprove such an opinion. But according to the state of the science of motion before Copernicus, all reasonings from such principles were utterly vague and obscure.

We must not omit to mention a modern who preceded Copernicus, in the assertion at least of the heliocentric doctrine. This was Nicholas of Cusa (a village near Treves), a cardinal and bishop, who, in the first half of the fifteenth century, was very eminent as a divine and mathematician; and who in a work, _De Doctâ Ignorantiâ_, propounded the doctrine of the motion of the earth; more, however, as a paradox than as a reality. We cannot consider this as any distinct anticipation of a profound and consistent view of the truth.

We shall now examine further the promulgation of the Heliocentric System by Copernicus, and its consequences. {262}

## CHAPTER II.

INDUCTION OF COPERNICUS.--THE HELIOCENTRIC THEORY ASSERTED ON FORMAL GROUNDS.

IT will be recollected that the _formal_ are opposed to the _physical_ grounds of a theory; the former term indicating that it gives a satisfactory account of the relations of the phenomena in Space and Time, that is, of the Motions themselves; while the latter expression implies further that we include in our explanation the Causes of the motions, the laws of Force and Matter. The strongest of the considerations by which Copernicus was led to invent and adopt his system of the universe were of the former kind. He was dissatisfied, he says, in his Preface addressed to the Pope, with the want of symmetry in the Eccentric Theory, as it prevailed in his days; and weary of the uncertainty of the mathematical traditions. He then sought through all the works of philosophers, whether any had held opinions concerning the motions of the world, different from those received in the established mathematical schools. He found, in ancient authors, accounts of Philolaus and others, who had asserted the motion of the earth. "Then," he adds, "I, too, began to meditate concerning the motion of the earth; and though it appeared an absurd opinion, yet since I knew that, in previous times,

others had been allowed the privilege of feigning what circles they chose, in order to explain the phenomena, I conceived that I also might take the liberty of trying whether, on the supposition of the earth's motion, it was possible to find better explanations than the ancient ones, of the revolutions of the celestial orbs.

"Having then assumed the motions of the earth, which are hereafter explained, by laborious and long observation I at length found, that if the motions of the other planets be compared with the revolution of the earth, not only their phenomena follow from the suppositions, but also that the several orbs, and the whole system, are so connected in order and magnitude, that no one part can be transposed without disturbing the rest, and introducing confusion into the whole universe."

Thus the satisfactory explanation of the apparent motions of the planets, and the simplicity and symmetry of the system, were the {263} grounds on which Copernicus adopted his theory; as the craving for these qualities was the feeling which led him to seek for a new theory. It is manifest that in this, as in other cases of discovery, a clear and steady possession of abstract Ideas, and an aptitude in comprehending real Facts under these general conceptions, must have been leading characters in the discoverer's mind. He must have had a good geometrical head, and great astronomical knowledge. He must have seen, with peculiar distinctness, the consequences which flowed from his suppositions as to the relations of space and time,--the apparent motions which resulted from the assumed real ones; and he must also have known well all the irregularities of the apparent motions for which he had to account. We find indications of these qualities in his expressions. A steady and calm contemplation of the theory is what he asks for, as the main requisite to its reception. If you suppose the earth to revolve and the heaven to be at rest, you will find, he says, "_si serio animadvertas_," if you think steadily, that the apparent diurnal motion will follow. And after alleging his reasons for his system, he says,[7\5] "We are, therefore, not ashamed to confess, that the whole of the space within the orbit of the moon, along with the centre of the earth, moves round the sun in a year among the other planets; the magnitude of the world being so great, that the distance of the earth from the sun has no apparent magnitude when compared with the sphere of the fixed stars." "All which things, though they be difficult and almost inconceivable, and against the opinion of the majority, yet, in the sequel, by God's favor, we will make clearer than the sun, at least to those who are not ignorant of mathematics."

[Note 7\5: Nicolai Copernici Torinensis _de Revolutionibus Orbium Cœlestium Libri VI_. Norimbergæ, M.D.XLIII. p. 9.]

It will easily be understood, that since the ancient geocentric hypothesis ascribed to the planets those motions which were apparent only, and which really arose from the motion of the earth round the sun in the new hypothesis, the latter scheme must much simplify the planetary theory. Kepler[8\5] enumerates eleven motions of the Ptolemaic system, which are at once exterminated and rendered unnecessary by the new system. Still, as the real motions, both of the earth and the planets, are unequable, it was requisite to have some mode of representing their inequalities; and, accordingly, the ancient theory of eccentrics and epicycles was retained, so far as

was requisite for this purpose. The planets revolved round the sun by means of a Deferent, and a {264} great and small Epicycle; or else by means of an Eccentric and Epicycle, modified from Ptolemy's, for reasons which we shall shortly mention. This mode of representing the motions of the planets continued in use, until it was expelled by the discoveries of Kepler.

[Note 8\5: _Myst. Cosm._ cap. 1.]

Besides the daily rotation of the earth on its axis, and its annual circuit about the sun, Copernicus attributed to the axis a "motion of declination," by which, during the whole annual revolution, the pole was constantly directed towards the same part of the heavens. This constancy in the absolute direction of the axis, or its moving parallel to itself, may be more correctly viewed as not indicating any separate motion. The axis continues in the same direction, because there is nothing to make it change its direction; just as a straw, lying on the surface of a cup of water, continues to point nearly in the same direction when the cup is carried round a room. And this was noticed by Copernicus's adherent, Rothman,[9\5] a few years after the publication of the work _De Revolutionibus_. "There is no occasion," he says, in a letter to Tycho Brahe, "for the triple motion of the earth: the annual and diurnal motions suffice." This error of Copernicus, if it be looked upon as an error, arose from his referring the position of the axis to a limited space, which he conceived to be carried round the sun along with the earth, instead of referring it to fixed or absolute space. When, in a Planetarium (a machine in which the motions of the planets are imitated), the earth is carried round the sun by being fastened to a material radius, it is requisite to give a motion to the axis by _additional_ machinery, in order to enable it to _preserve_ its parallelism. A similar confusion of geometrical conception, produced by a double reference to absolute space and to the centre of revolution, often leads persons to dispute whether the moon, which revolves about the earth, always turning to it the same face, revolves about her axis or not.

[Note 9\5: Tycho. Epist. i. p. 184, A. D. 1590.]

It is also to be noticed that the precession of the equinoxes made it necessary to suppose the axis of the earth to be not _exactly_ parallel to itself, but to deviate from that position by a slight annual difference. Copernicus erroneously supposes the precession to be unequable; and his method of explaining this change, which is simpler than that of the ancients, becomes more simple still, when applied to the true state of the facts.

The tendencies of our speculative nature, which carry us onwards in {265} pursuit of symmetry and rule, and which thus produced the theory of Copernicus, as they produce all theories, perpetually show their vigor by overshooting their mark. They obtain something by aiming at much more. They detect the order and connection which exist, by imagining relations of order and connection which have no existence. Real discoveries are thus mixed with baseless assumptions; profound sagacity is combined with fanciful conjecture; not rarely, or in peculiar instances, but commonly, and in most cases; probably in all, if we could read the thoughts of the discoverers as we read the books of Kepler. To try wrong guesses is apparently the only way to hit

upon right ones. The character of the true philosopher is, not that he never conjectures hazardously, but that his conjectures are clearly conceived and brought into rigid contact with facts. He sees and compares distinctly the ideas and the things,--the relations of his notions to each other and to phenomena. Under these conditions it is not only excusable, but necessary for him, to snatch at every semblance of general rule;--to try all promising forms of simplicity and symmetry.

Copernicus is not exempt from giving us, in his work, an example of this character of the inventive spirit. The axiom that the celestial motions must be _circular_ and _uniform_, appeared to him to have strong claims to acceptation; and his theory of the inequalities of the planetary motions is fashioned upon it. His great desire was to apply it more rigidly than Ptolemy had done. The time did not come for rejecting this axiom, till the observations of Tycho Brahe and the calculations of Kepler had been made.

I shall not attempt to explain, in detail, Copernicus's system of the planetary inequalities. He retained epicycles and eccentrics, altering their centres of motion; that is, he retained what was _true_ in the old system, _translating_ it into his own. The peculiarities of his method consisted in making such a combination of epicycles as to supply the place of the _equant_,[10\5] and to make all the motions equable about the centres of motion. This device was admired for a time, till Kepler's elliptic theory expelled it, with all other forms of the theory of epicycles: but we must observe that Copernicus was aware of some of the discrepancies which belonged to that theory as it had, up to that time, been propounded. In the case of Mercury's orbit, which is more eccentric than that of the other planets, he makes suppositions which are complex indeed, but which show his perception of the imperfection of {266} the common theory; and he proposes a new theory of the moon, for the very reason which did at last overturn the doctrine of epicycles, namely, that the ratio of their distances from the earth at different times was inconsistent with the circular hypothesis.[11\5]

[Note 10\5: See B. iii. Chap. **iv. Sect. 7.]

[Note 11\5: _De Rev._ iv. c. 2.]

It is obvious, that, along with his mathematical clearness of view, and his astronomical knowledge, Copernicus must have had great intellectual boldness and vigor, to conceive and fully develop a theory so different as his was from all received doctrines. His pupil and expositor, Rheticus, says to Schener, "I beg you to have this opinion concerning that learned man, my Preceptor; that he was an ardent admirer and follower of Ptolemy; but when he was compelled by phenomena and demonstration, he thought he did well to aim at the same mark at which Ptolemy had aimed, though with a bow and shafts of a very different material from his. We must recollect what Ptolemy says, Δεῖ δ' ἐλευθέρον εἶναι τῇ γνώμῃ τὸν μέλλοντα φιλοσοφεῖν. 'He who is to follow philosophy must be a freeman in mind.'" Rheticus then goes on to defend his master from the charge of disrespect to the ancients: "That temper," he says, "is alien from the disposition of every good man, and most especially from the spirit of philosophy, and from no one more utterly than from my

Preceptor. He was very far from rashly rejecting the opinions of ancient philosophers, except for weighty reasons and irresistible facts, through any love of novelty. His years, his gravity of character, his excellent learning, his magnanimity and nobleness of spirit, are very far from having any liability to such a temper, which belongs either to youth, or to ardent and light minds, or to those τῶν μέγα φρονούντων ἐπὶ θεωρίᾳ μικρῇ, 'who think much of themselves and know little,' as Aristotle says." Undoubtedly this deference for the great men of the past, joined with the talent of seizing the spirit of their methods when the letter of their theories is no longer tenable, _is_ the true mental constitution of discoverers.

Besides the intellectual energy which was requisite in order to construct a system of doctrines so novel as those of Copernicus, some courage was necessary to the publication of such opinions; certain, as they were, to be met, to a great extent, by rejection and dispute, and perhaps by charges of heresy and mischievous tendency. This last danger, however, must not be judged so great as we might infer from the angry controversies and acts of authority which occurred in {267} Galileo's time. The Dogmatism of the stationary period, which identified the cause of philosophical and religious truth, had not yet distinctly felt itself attacked by the advance of physical knowledge; and therefore had not begun to look with alarm on such movements. Still, the claims of Scripture and of ecclesiastical authority were asserted as paramount on all subjects; and it was obvious that many persons would be disquieted or offended with the new interpretation of many scriptural expressions, which the true theory would make necessary. This evil Copernicus appears to have foreseen; and this and other causes long withheld him from publication. He was himself an ecclesiastic; and, by the patronage of his maternal uncle, was prebendary of the church of St. John at Thorn, and a canon of the church of Frauenburg, in the diocese of Ermeland.[12\5] He had been a student at Bologna, and had taught mathematics at Rome in the year 1500; and he afterwards pursued his studies and observations at his residence near the mouth of the Vistula.[13\5] His discovery of his system must have occurred before 1507, for in 1543 he informs Pope Paulus the Third, in his dedication, that he had kept his book by him for four times the nine years recommended by Horace, and then only published it at the earnest entreaty of his friend Cardinal Schomberg, whose letter is prefixed to the work. "Though I know," he says, "that the thoughts of a philosopher do not depend on the judgment of the many, his study being to seek out truth in all things as far as that is permitted by God to human reason: yet when I considered," he adds, "how absurd my doctrine would appear, I long hesitated whether I should publish my book, or whether it were not better to follow the example of the Pythagoreans and others, who delivered their doctrines only by tradition and to friends." It will be observed that he speaks here of the opposition of the established school of Astronomers, not of Divines. The latter, indeed, he appears to consider as a less formidable danger. "If perchance," he says at the end of his preface, "there be ματαιολόγοι, vain babblers, who knowing nothing of mathematics, yet assume the right of judging on account of some place of Scripture perversely wrested to their purpose, and who blame and attack my undertaking; I heed them not, and look upon their judgments as rash and contemptible." He then goes on to show that the globular figure of the earth (which was, of course, at that time, an

undisputed point among astronomers), had been opposed on similar grounds by Lactantius, who, {268} though a writer of credit in other respects, had spoken very childishly in that matter. In another epistle prefixed to the work (by Andreas Osiander), the reader is reminded that the hypotheses of astronomers are not necessarily asserted to be true, by those who propose them, but only to be a way of _representing_ facts. We may observe that, in the time of Copernicus, when the motion of the earth had not been connected with the physical laws of matter and motion, it could not be considered so distinctly real as it necessarily was held to be in after times.

[Note 12\5: Rheticus, _Nar._ p. 94.]

[Note 13\5: Riccioli.]

The delay of the publication of Copernicus's work brought it to the end of his life; he died in the year 1543, in which it was published. It was entitled _De Revolutionibus Orbium Cœlestium Libri VI_. He received the only copy he ever saw on the day of his death, and never opened it: he had then, says Gassendi, his biographer, other cares. His system was, however, to a certain extent, promulgated, and his fame diffused before that time. Cardinal Schomberg, in his letter of 1536, which has been already mentioned, says, "Some years ago, when I heard tidings of your merit by the constant report of all persons, my affection for you was augmented, and I congratulated the men of our time, among whom you flourish in so much honor. For I had understood that you were not only acquainted with the discoveries of ancient mathematicians, but also had formed a new system of the world, in which you teach that the Earth moves, the Sun occupies the lowest, and consequently, the middle place, the sphere of the fixed stars remains immovable and fixed, and the Moon, along with the elements included in her sphere, placed between the orbits (_cœlum_) of Mars and Venus, travels round the sun in a yearly revolution."[14\5] The writer goes on to say that he has heard that Copernicus has written a book (_Commentarios_), in which this system is applied to the construction of Tables of the Planetary Motions (_erraticarum stellarum_). He then proceeds to entreat him earnestly to publish his lucubrations. {269}

[Note 14\5: This passage has so important a place in the history, that I will give it in the original:--"Intellexeram te non modo veterum mathematicorum inventa egregie callere sed etiam novam mundi rationem constituisse: Qua doceas terram moveri: solem imum mundi, atque medium locum obtinere: cœlum octavum immotum atque fixum perpetuo manere: Lunam se una cum inclusis suæ spheræ elementis, inter Martis et Veneris cœlum sitam, anniversario cursu circum solem convertere. Atque de hac tota astronomiæ ratione commentarios a te confectos esse, ac erraticarum stellarum motus calculis subductos tabulis te contulisse, maxima omnium cum admiratione. Quamobrem vir doctissime, nisi tibi molestus sum, te etiam atque etiam oro vehementer ut hoc tuum inventum studiosis communices, et tuas de mundi sphæra lucubrationes, una cum Tabulis et si quid habes præterea quod ad eandem rem pertineat primo quoque tempore ad me mittas."]

This letter is dated 1536, and implies that the work of Copernicus was then written, and known to persons who studied astronomy. Delambre says that Achilles Gassarus of Lindau, in a letter dated 1540, sends to his friend George Vogelin of Constance, the book _De Revolutionibus_. But Mr. De Morgan[15\5] has pointed out that the printed work which Gassarus sent to Vogelin was the _Narratio_ by Rheticus of Feldkirch, a eulogium of Copernicus and his system prefixed to the second edition of the _De Revolutionibus_, which appeared in 1566. In this Narration, Rheticus speaks of the work of Copernicus as a Palingenesia, or New Birth of astronomy. Rheticus, it appears, had gone to Copernicus for the purpose of getting knowledge about triangles and trigonometrical tables, and had had his attention called to the heliocentric theory, of which he became an ardent admirer. He speaks of his "Preceptor" with strong admiration, as we have seen. "He appears to me," says he, "more to resemble Ptolemy than any other astronomers." This, it must be recollected, was selecting the highest known subject of comparison.

[Note 15\5: _Ast. Mod._ i. p. 138. I owe this and many other corrections to the personal kindness of Mr. De Morgan.]

## CHAPTER III.

SEQUEL TO COPERNICUS.--THE RECEPTION AND DEVELOPMENT OF THE COPERNICAN THEORY.

_Sect._ 1.--_First Reception of the Copernican Theory._

THE theories of Copernicus made their way among astronomers, in the manner in which true astronomical theories always obtain the assent of competent judges. They led to the construction of Tables of the motion of the sun, moon, and planets, as the theories of Hipparchus and Ptolemy had done; and the verification of the doctrines was to be looked for, from the agreement of these Tables with observation, through a sufficient course of time. The work _De Revolutionibus_ contains such Tables. In 1551 Reinhold improved and republished Tables founded on the principles of Copernicus. "We owe," he says in his preface, "great obligations to Copernicus, both for his laborious {270} observations, and for restoring the doctrine of the Motions. But though his geometry is perfect, the good old man appears to have been, at times, careless in his numerical calculations. I have, therefore, recalculated the whole, from a comparison of his observations with those of Ptolemy and others, following nothing but the general plan of Copernicus's demonstrations." These "Prutenic Tables" were republished in 1571 and 1585, and continued in repute for some time; till superseded by the Rudolphine Tables of Kepler in 1627. The name _Prutenic_, or Prussian, was employed by the author as a mark of gratitude to his benefactor Albert, Markgrave of Brandenbourg. The discoveries of Copernicus had inspired neighboring nations with the ambition of claiming a place in the literary community of Europe. In something of the same spirit, Rheticus wrote an _Encomium Borussiæ_, which was published along with his _Narratio_.

The Tables founded upon the Copernican system were, at first, much more generally adopted than the heliocentric doctrine on which they were founded. Thus

Magini published at Venice, in 1587, _New Theories of the Celestial Orbits, agreeing with the Observations of Nicholas Copernicus_. But in the preface, after praising Copernicus, he says, "Since, however, he, either for the sake of showing his talents, or induced by his own reasons, has revived the opinion of Nicetas, Aristarchus, and others, concerning the motion of the earth, and has disturbed the established constitution of the world, which was a reason why many rejected, or received with dislike, his hypothesis, I have thought it worth while, that, rejecting the suppositions of Copernicus, I should accommodate other causes to his observations, and to the Prutenic Tables."

This doctrine, however, was, as we have shown, received with favor by many persons, even before its general publication. The doctrine of the motion of the earth was first publicly maintained at Rome by Widmanstadt,[16\5] who professed to have received it from Copernicus, and explained the System before the Pope and the Cardinals, but did not teach it to the public.

[Note 16\5: See Venturi, _Essai sur les Ouvrages Physico-Mathématiques de Leonard da Vinci, avec des Fragmens tirés de ses Manuscrits apportés d'Italie_. Paris, 1797; and, as there quoted, _Marini Archiatri Pontificii_, tom. ii. p. 251.]

Leonardo da Vinci, who was an eminent mathematician, as well as painter, about 1510, explained how a body, by describing a kind of spiral, might descend towards a revolving globe, so that its apparent motion relative to a point in the surface of the globe, might be in a {271} straight line leading to the centre. He thus showed that he had entertained in his thoughts the hypothesis of the earth's rotation, and was employed in removing the difficulties which accompanied this supposition, by means of the consideration of the composition of motions.

In like manner we find the question stirred by other eminent men. Thus John Muller of Konigsberg, a celebrated astronomer who died in 1476, better known by the name of Regiomontanus, wrote a dissertation on the subject "Whether the earth be in motion or at rest," in which he decides _ex professo_[17\5] against the motion. Yet such discussions must have made generally known the arguments for the heliocentric theory.

[Note 17\5: Schoneri _Opera_, part ii. p. 129.]

We have already seen the enthusiasm with which Rheticus, who was Copernicus's pupil in the latter years of his life, speaks of him. "Thus," says he, "God has given to my excellent preceptor a reign without end; which may He vouchsafe to guide, govern, and increase, to the restoration of astronomical truth. Amen."

Of the immediate converts of the Copernican system, who adopted it before the controversy on the subject had attracted attention, I shall only add **Mæstlin, and his pupil, Kepler. **Mæstlin published in 1588 an _Epitome Astronomiæ_, in which the immobility of the earth is asserted; but in 1596 he edited Kepler's _Mysterium Cosmographicum_, and the _Narratio_ of Rheticus: and in an epistle of his own, which he inserts, he defends the Copernican system by those physical reasonings which we shall shortly have to mention, as the usual arguments in this dispute. Kepler himself, in the outset of the work just named, says, "When I was at Tübingen, attending to

Michael Mæstlin, being disturbed by the manifold inconveniences of the usual opinion concerning the world, I was so delighted with Copernicus, of whom he made great mention in his lectures, that I not only defended his opinions in our disputations of the candidates, but wrote a thesis concerning the First Motion which is produced by the revolution of the earth." This must have been in 1590.

The differences of opinion respecting the Copernican system, of which we thus see traces, led to a controversy of some length and extent. This controversy turned principally upon physical considerations, which were much more distinctly dealt with by Kepler, and others of the followers of Copernicus, than they had been by the {272} discoverer himself. I shall, therefore, give a separate consideration to this part of the subject. It may be proper, however, in the first place, to make a few observations on the progress of the doctrine, independently of these physical speculations.

_Sect._ 2.--_Diffusion of the Copernican Theory._

THE diffusion of the Copernican opinions in the world did not take place rapidly at first. Indeed, it was necessarily some time before the progress of observation and of theoretical mechanics gave the heliocentric doctrine that superiority in argument, which now makes us wonder that men should have hesitated when it was presented to them. Yet there were some speculators of this kind, who were attracted at once by the enlarged views of the universe which it opened to them. Among these was the unfortunate Giordano Bruno of Nola, who was burnt as a heretic at Rome in 1600. The heresies which led to his unhappy fate were, however, not his astronomical opinions, but a work which he published in England, and dedicated to Sir Philip Sydney, under the title of _Spaccio della Bestia Trionfante_, and which is understood to contain a bitter satire of religion and the papal government. Montucla conceives that, by his rashness in visiting Italy after putting forth such a work, he compelled the government to act against him. Bruno embraced the Copernican opinions at an early period, and connected with them the belief in innumerable worlds besides that which we inhabit; as also certain metaphysical or theological doctrines which he called the Nolan philosophy. In 1591 he published _De innumerabilibus, immenso, et infigurabili, seu de Universo et Mundis_, in which he maintains that each star is a sun, about which revolve planets like our earth; but this opinion is mixed up with a large mass of baseless verbal speculations.

Giordano Bruno is a disciple of Copernicus on whom we may look with peculiar interest, since he probably had a considerable share in introducing the new opinions into England;[18\5] although other persons, as Recorde, Field, Dee, had adopted it nearly thirty years earlier; and Thomas Digges ten years before, much more expressly. Bruno visited this country in the reign of Queen Elizabeth, and speaks of her and of her councillors in terms of praise, which appear to show that {273} his book was intended for English readers; though he describes the mob which was usually to be met with in the streets of London with expressions of great disgust: "Una plebe la quale in essere irrespettevole, incivile, rozza, rustica, selvatica, et male allevata, non cede ad altra che pascer possa la terra nel suo seno."[19\5] The work to which I refer is _La Cena de le Cenere_, and narrates what took place at a supper

held on the evening of Ash Wednesday (about 1583, see p. 145 of the book), at the house of Sir Fulk Greville, in order to give "Il Nolano" an opportunity of defending his peculiar opinions. His principal antagonists are two "Dottori d' Oxonia," whom Bruno calls Nundinio and Torquato. The subject is not treated in any very masterly manner on either side; but the author makes himself have greatly the advantage not only in argument, but in temper and courtesy: and in support of his representations of "pedantesca, ostinatissima ignoranza et presunzione, mista con una rustica incivilità, che farebbe prevaricar la pazienza di Giobbe," in his opponents, he refers to a public disputation which he had held at Oxford with these doctors of theology, in presence of Prince Alasco, and many of the English nobility.[20\5]

[Note 18\5: See Burton's _Anat. Mel._ Pref. "Some prodigious tenet or paradox of the earth's motion," &c. "Bruno," &c.]

[Note 19\5: _Opere di Giordano Bruno_, vol. i. p. 146.]

[Note 20\5: Ib. vol. i. p. 179.]

Among the evidences of the difficulties which still lay in the way of the reception of the Copernican system, we may notice Bacon, who, as is well known, never gave a full assent to it. It is to be observed, however, that he does not reject the opinion of the earth's motion in so peremptory and dogmatical a manner as he is sometimes accused of doing: thus in the _Thema Cœli_ he says, "The earth, then, being supposed to be at rest (for that now appears to us the _more true_ opinion)." And in his tract _On the Cause of the Tides_, he says, "If the tide of the sea be the extreme and diminished limit of the diurnal motion of the heavens, it will follow that the earth is immovable; or at least that it moves with a much slower motion than the water." In the _Descriptio Globi Intellectualis_ he gives his reasons for not accepting the heliocentric theory. "In the system of Copernicus there are many and grave difficulties: for the threefold motion with which he encumbers the earth is a serious inconvenience; and the separation of the sun from the planets, with which he has so many affections in common, is likewise a harsh step; and the introduction of so many immovable bodies into nature, as when he makes the sun and the stars immovable, the bodies which are peculiarly lucid and radiant; and his making the moon adhere to the earth in a sort of epicycle; and some {274} other things which he assumes, are proceedings which mark a man who thinks nothing of introducing fictions of any kind into nature, provided his calculations turn out well." We have already explained that, in attributing _three_ motions to the earth, Copernicus had presented his system encumbered with a complexity not really belonging to it. But it will be seen shortly, that Bacon's fundamental objection to this system was his wish for a system which could be supported by sound physical considerations; and it must be allowed, that at the period of which we are speaking, this had not yet been done in favor of the Copernican hypothesis. We may add, however, that it is not quite clear that Bacon was in full possession of the details of the astronomical systems which that of Copernicus was intended to supersede; and that thus he, perhaps, did not see how much less harsh were these fictions, as he called them, than those which were the inevitable alternatives. Perhaps he might even be liable to a little of that indistinctness, with respect to strictly geometrical conceptions, which we have

remarked in Aristotle. We can hardly otherwise account for his not seeing any use in resolving the apparently irregular motion of a planet into separate regular motions. Yet he speaks slightingly of this important step.[21\5] "The motion of planets, which is constantly talked of as the motion of regression, or renitency, from west to east, and which is ascribed to the planets as a proper motion, is not true; but only arises from appearance, from the greater advance of the starry heavens towards the west, by which the planets are left behind to the east." Undoubtedly those who spoke of such a motion of _regression_ were aware of this; but they saw how the motion was simplified by this way of conceiving it, which Bacon seems not to have seen. Though, therefore, we may admire Bacon for the steadfastness with which he looked forward to physical astronomy as the great and proper object of philosophical interest, we cannot give him credit for seeing the full value and meaning of what had been done, up to his time, in Formal Astronomy.

[Note 21\5: _Thema Cœli_, p. 246.]

Bacon's contemporary, Gilbert, whom he frequently praises as a philosopher, was much more disposed to adopt the Copernican opinions, though even he does not appear to have made up his mind to assent to the whole of the system. In his work. _De Magnete_ (printed 1600), he gives the principal arguments in favor of the Copernican system, and decides that the earth revolves on its axis.[22\5] He connects {275} this opinion with his magnetic doctrines; and especially endeavors by that means to account for the precession of the equinoxes. But he does not seem to have been equally confident of its annual motion. In a posthumous work, published in 1661 (_De Mundo Nostra Sublunari Philosophia Nova_) he appears to hesitate between the systems of Tycho and Copernicus.[23\5] Indeed, it is probable that at this period many persons were in a state of doubt on such subjects. Milton, at a period somewhat later, appears to have been still undecided. In the opening of the eighth book of the _Paradise Lost_, he makes Adam state the difficulties of the Ptolemaic hypothesis, to which the archangel Raphael opposes the usual answers; but afterwards suggests to his pupil the newer system: . . . . What if seventh to these The planet earth, so steadfast though she seem, Insensibly three different motions move? _Par. Lost_, b. viii.

[Note 22\5: Lib. vi. cap. 3, 4.]

[Note 23\5: Lib. ii. cap. 20.]

Milton's leaning, however, seems to have been for the new system; we can hardly believe that he would otherwise have conceived so distinctly, and described with such obvious pleasure, the motion of the earth: Or she from west her silent course advance With inoffensive pace, that spinning sleeps On her soft axle, while she paces even, And bears thee soft with the smooth air along. _Par. Lost_, b. viii.

Perhaps the works of the celebrated Bishop Wilkins tended more than any others to the diffusion of the Copernican system in England, since even their extravagances drew a stronger attention to them. In 1638, when he was only twenty-four years old, he published a book entitled _The Discovery of a New World; or a Discourse tending to prove that it is probable there may be another habitable World in the Moon; with a

Discourse concerning_ the possibility of a passage thither. The latter part of his subject was, of course, an obvious mark for the sneers and witticisms of critics. Two years afterwards, in 1640, appeared his _Discourse concerning a new Planet; tending to prove that it is probable our Earth is one of the Planets_: in which he urged the reasons in favor of the heliocentric system; and explained away the opposite arguments, especially those drawn from the {276} supposed declarations of Scripture. Probably a good deal was done for the establishment of those opinions by Thomas Salusbury, who was a warm admirer of Galileo, and published, in 1661, a translation of several of his works bearing upon this subject. The mathematicians of this country, in the seventeenth century, as Napier and Briggs, Horrox and Crabtree, Oughtred and Seth Ward, Wallis and Wren, were probably all decided Copernicans. Kepler dedicates one of his works to Napier, and Ward invented an approximate method of solving Kepler's problem, still known as "the simple elliptical hypothesis." Horrox wrote, and wrote well, in defence of the Copernican opinion, in his _Keplerian Astronomy defended and promoted_, composed (in Latin) probably about 1635, but not published till 1673, the author having died at the age of twenty-two, and his papers having been lost. But Salusbury's work was calculated for another circle of readers. "The book," he says in the introductory address, "being, for subject and design, intended chiefly for gentlemen, I have been as careless of using a studied pedantry in my style, as careful in contriving a pleasant and beautiful impression." In order, however, to judge of the advantage under which the Copernican system now came forward, we must consider the additional evidence for it which was brought to light by Galileo's astronomical discoveries.

_Sect._ 3.--_The Heliocentric Theory confirmed by Facts.--Galileo's Astronomical Discoveries._

THE long interval which elapsed between the last great discoveries made by the ancients and the first made by the moderns, had afforded ample time for the development of all the important consequences of the ancient doctrines. But when the human mind had been thoroughly roused again into activity, this was no longer the course of events. Discoveries crowded on each other; one wide field of speculation was only just opened, when a richer promise tempted the laborers away into another quarter. Hence the history of this period contains the beginnings of many sciences, but exhibits none fully worked out into a complete or final form. Thus the science of Statics, soon after its revival, was eclipsed and overlaid by that of Dynamics; and the Copernican system, considered merely with reference to the views of its author, was absorbed in the commanding interest of Physical Astronomy.

Still, advances were made which had an important bearing on the {277} heliocentric theory, in other ways than by throwing light upon its physical principles. I speak of the new views of the heavens which the Telescope gave; the visible inequalities of the moon's surface; the moon-like phases of the planet Venus; the discovery of the Satellites of Jupiter, and of the Ring of Saturn. These discoveries excited at the time the strongest interest; both from the novelty and beauty of the objects they presented to the sense; from the way in which they seemed to gratify man's curiosity with regard to the remote parts of the universe; and also from that of

which we have here to speak, their bearing upon the conflict of the old and the new philosophy, the heliocentric and geocentric theories. It may be true, as Lagrange and Montucla say, that the laws which Galileo discovered in Mechanics implied a profounder genius than the novelties he detected in the sky: but the latter naturally attracted the greater share of the attention of the world, and were matter of keener discussion.

It is not to our purpose to speak here of the details and of the occasion of the invention of the Telescope; it is well known that Galileo constructed his about 1609, and proceeded immediately to apply it to the heavens. The discovery of the Satellites of Jupiter was almost immediately the reward of his activity; and these were announced in his _Nuncius Sidereus_, published at Venice in 1610. The title of this work will best convey an idea of the claim it made to public notice: "The _Sidereal Messenger_, announcing great and very wonderful spectacles, and offering them to the consideration of every one, but especially of philosophers and astronomers; which have been observed by _Galileo Galilei_, &c. &c., by the assistance of a perspective glass lately invented by him; namely, in the face of the moon, in innumerable fixed stars in the milky-way, in nebulous stars, but especially in four planets which revolve round Jupiter at different intervals and periods with a wonderful celerity; which, hitherto not known to any one, the author has recently been the first to detect, and has decreed to call the _Medicean stars_."

The interest this discovery excited was intense: and men were at this period so little habituated to accommodate their convictions on matters of science to newly observed facts, that several of the "paper-philosophers," as Galileo termed them, appear to have thought they could get rid of these new objects by writing books against them. The effect which the discovery had upon the reception of the Copernican system was immediately very considerable. It showed that the real universe was very different from that which ancient philosophers had imagined, {278} and suggested at once the thought that it contained mechanism more various and more vast than had yet been conjectured. And when the system of the planet Jupiter thus offered to the bodily eye a model or image of the solar system according to the views of Copernicus, it supported the belief of such an arrangement of the planets, by an analogy all but irresistible. It thus, as a writer[24\5] of our own times has said, "gave the _holding turn_ to the opinions of mankind respecting the Copernican system." We may trace this effect in Bacon, even though he does not assent to the motion of the earth. "We affirm," he says,[25\5] "the _sun-following arrangement_ (solisequium) of Venus and Mercury; since it has been found by Galileo that Jupiter also has attendants."

[Note 24\5: Sir J. Herschel.]

[Note 25\5: _Thema Cœli_, ix. p. 253.]

The _Nuncius Sidereus_ contained other discoveries which had the same tendency in other ways. The examination of the moon showed, or at least seemed to show, that she was a solid body, with a surface extremely rugged and irregular. This, though perhaps not bearing directly upon the question of the heliocentric theory, was yet a

blow to the Aristotelians, who had, in their philosophy, made the moon a body of a kind altogether different from this, and had given an abundant quantity of reasons for the visible marks on her surface, all proceeding on these preconceived views. Others of his discoveries produced the same effect; for instance, the new stars invisible to the naked eye, and those extraordinary appearances called Nebulæ.

But before the end of the year, Galileo had new information to communicate, bearing more decidedly on the Copernican controversy. This intelligence was indeed decisive with regard to the motion of Venus about the sun; for he found that that planet, in the course of her revolution, assumes the same succession of phases which the moon exhibits in the course of a month. This he expressed by a Latin verse: Cynthiæ figuras æmulatur mater amorum: The Queen of Love like Cynthia shapes her forms: transposing the letters of this line in the published account, according to the practice of the age; which thus showed the ancient love for combining verbal puzzles with scientific discoveries, while it betrayed the newer feeling, of jealousy respecting the priority of discovery of physical facts.

It had always been a formidable objection to the Copernican theory that this appearance of the planets had not been observed. The author {279} of that theory had endeavored to account for this, by supposing that the rays of the sun passed freely through the body of the planet; and Galileo takes occasion to praise him for not being deterred from adopting the system which, on the whole, appeared to agree best with the phenomena, by meeting with some appearances which it did not enable him to explain.[26\5] Yet while the fate of the theory was yet undecided, this could not but be looked upon as a weak point in its defences.

[Note 26\5: Drinkwater-Bethune, _Life of Galileo_, p. 35.]

The objection, in another form also, was embarrassing alike to the Ptolemaic and Copernican systems. Why, it was asked, did not Venus appear four times as large when nearest to the earth, as when furthest from it? The author of the Epistle prefixed to Copernicus's work had taken refuge in this argument from the danger of being supposed to believe in the reality of the system; and Bruno had attempted to answer it by saying, that luminous bodies were not governed by the same laws of perspective as opake ones. But a more satisfactory answer now readily offered itself. Venus does not appear four times as large when she is four times as near, because her _bright part_ is _not_ four times as large, though her visible diameter is; and as she is too small for us to see her shape with the naked eye, we judge of her size only by the quantity of light.

The other great discoveries made in the heavens by means of telescopes, as that of Saturn's ring and his satellites, the spots in the sun, and others, belong to the further progress of astronomy. But we may here observe, that this doctrine of the motion of Mercury and Venus about the sun was further confirmed by Kepler's observation of the transit of the former planet over the sun in 1631. Our countryman Horrox was the first person who, in 1639, had the satisfaction of seeing a transit of Venus.

These events are a remarkable instance of the way in which a discovery in art (for at this period, the making of telescopes must be mainly so considered) may influence

the progress of science. We shall soon have to notice a still more remarkable example of the way in which two sciences (Astronomy and Mechanics) may influence and promote the progress of each other. {280}

_Sect._ 4.--_The Copernican System opposed on Theological Grounds._

THE doctrine of the Earth's motion round the Sun, when it was asserted and promulgated by Copernicus, soon after 1500, excited no visible alarm among the theologians of his own time. Indeed, it was received with favor by the most intelligent ecclesiastics; and lectures in support of the heliocentric doctrine were delivered in the ecclesiastical colleges. But the assertion and confirmation of this doctrine by Galileo, about a century later, excited a storm of controversy, and was visited with severe condemnation. Galileo's own behavior appears to have provoked the interference of the ecclesiastical authorities; but there must have been a great change in the temper of the times to make it possible for his adversaries to bring down the sentence of the Inquisition upon opinions which had been so long current without giving any serious offence.

[2d Ed.] [It appears to me that the different degree of toleration accorded to the heliocentric theory in the time of Copernicus and of Galileo, must be ascribed in a great measure to the controversies and alarms which had in the mean time arisen out of the Reformation in religion, and which had rendered the Romish Church more jealous of innovations in received opinions than it had previously been. It appears too that the discussion of such novel doctrines was, at that time at least, less freely tolerated in Italy than in other countries. In 1597, Kepler writes to Galileo thus: "Confide Galilæe et progredere. Si bene conjecto, pauci de præcipuis Europæ Mathematicis a nobis secedere volent; tanta vis est veritatis. Si tibi Italia minus est idonea ad publicationem et si aliqua habitures es impedimenta, forsan Germania nobis hanc libertatem concedet."--Venturi, _Mem. di Galileo_, vol. i. p. 19.

I would not however be understood to assert the condemnation of new doctrines in science to be either a general or a characteristic practice of the Romish Church. Certainly the intelligent and cultivated minds of Italy, and many of the most eminent of her ecclesiastics among them, have always been the foremost in promoting and welcoming the progress of science: and, as I have stated, there were found among the Italian ecclesiastics of Galileo's time many of the earliest and most enlightened adherents of the Copernican system. The condemnation of the doctrine of the earth's motion, is, so far as I am aware, the only instance in which the Papal authority has pronounced a decree upon a point of science. And the most candid of the {281} adherents of the Romish Church condemn the assumption of authority in such matters, which in this one instance, at least, was made by the ecclesiastical tribunals. The author of the _Ages of Faith_ (book viii. p. 248) says, "A congregation, it is to be lamented, declared the new system to be opposed to Scripture, and therefore heretical." In more recent times, as I have elsewhere remarked,[27\5] the Church of Authority and the Church of Private Judgment have each its peculiar temptations and dangers, when there appears to be a discrepance between Scripture and Philosophy.

[Note 27\5: _Phil. Ind. Sci._ book x. chap. 4.]

But though we may acquit the popes and cardinals in Galileo's time of stupidity and perverseness in rejecting manifest scientific truths, I do not see how we can acquit them of dissimulation and duplicity. Those persons appear to me to defend in a very strange manner the conduct of the ecclesiastical authorities of that period, who boast of the liberality with which Copernican professors were placed by them in important offices, at the very time when the motion of the earth had been declared by the same authorities contrary to Scripture. Such merits cannot make us approve of their conduct in demanding from Galileo a public recantation of the system which they thus favored in other ways, and which they had repeatedly told Galileo he might hold as much as he pleased. Nor can any one, reading the plain language of the Sentence passed upon Galileo, and of the Abjuration forced from him, find any value in the plea which has been urged, that the opinion was denominated a _heresy_ only in a wide, improper, and technical sense.

But if we are thus unable to excuse the conduct of Galileo's judges, I do not see how we can give our unconditional admiration to the philosopher himself. Perhaps the conventional decorum which, as we have seen, was required in treating of the Copernican system, may excuse or explain the furtive mode of insinuating his doctrines which he often employs, and which some of his historians admire as subtle irony, while others blame it as insincerity. But I do not see with what propriety Galileo can be looked upon as a "Martyr of Science." Undoubtedly he was very desirous of promoting what he conceived to be the cause of philosophical truth; but it would seem that, while he was restless and eager in urging his opinions, he was always ready to make such submissions as the spiritual tribunals required. He would really have acted as a martyr, if he had uttered {282} his "E pur si muove," in the place of his abjuration, not after it. But even in this case he would have been a martyr to a cause of which the merit was of a mingled scientific character; for his own special and favorite share in the reasonings by which the Copernican system was supported, was the argument drawn from the flux and reflux of the sea, which argument is altogether false. He considered this as supplying a mechanical ground of belief, without which the mere astronomical reasons were quite insufficient; but in this case he was deserted by the mechanical sagacity which appeared in his other speculations.]

The heliocentric doctrine had for a century been making its way into the minds of thoughtful men, on the general ground of its simplicity and symmetry. Galileo appears to have thought that now, when these original recommendations of the system had been reinforced by his own discoveries and reasonings, it ought to be universally acknowledged as a truth and a reality. And when arguments against the fixity of the sun and the motion of the earth were adduced from the expressions of Scripture, he could not be satisfied without maintaining his favorite opinion to be conformable to Scripture as well as to Philosophy; and he was very eager in his attempts to obtain from authority a declaration to this effect. The ecclesiastical authorities were naturally averse to express themselves in favor of a novel opinion, startling to the common mind, and contrary to the most obvious meaning of the words of the Bible; and when they were compelled to pronounce, they decided against Galileo and his doctrines. He was accused before the Inquisition in 1615; but at that

period the result was that he was merely recommended to confine himself to the mathematical reasonings upon the system, and to abstain from meddling with the Scripture. Galileo's zeal for his opinions soon led him again to bring the question under the notice of the Pope, and the result was a declaration of the Inquisition that the doctrine of the earth's motion appeared to be contrary to the Sacred Scripture. Galileo was prohibited from defending and teaching this doctrine in any manner, and promised obedience to this injunction. But in 1632 he published his **"_Dialogo delli due Massimi Sistemi del Mondo, Tolemaico e Copernicano_:" and in this he defended the heliocentric system by all the strongest arguments which its admirers used. Not only so, but he introduced into this _Dialogue_ a character under the name of Simplicius, in whose mouth was put the defence of all the ancient dogmas, and who was represented as defeated at all points in the discussion; {283} and he prefixed to the _Dialogue_ a Notice, _To the Discreet Reader_, in which, in a vein of transparent irony, he assigned his reasons for the publication. "Some years ago," he says, "a wholesome edict was promulgated at Rome, which, in order to check the perilous scandals of the present age, imposed silence upon the Pythagorean opinion of the motion of the earth. There were not wanting," he adds, "persons who rashly asserted that this decree was the result, not of a judicious inquiry, but of a passion ill-informed; and complaints were heard that counsellors, utterly unacquainted with astronomical observations, ought not to be allowed, with their undue prohibitions, to clip the wings of speculative intellects. At the hearing of rash lamentations like these, my zeal could not keep silence." And he then goes on to say that he wishes, by the publication of his _Dialogue_ to show that the subject had been fully examined at Rome. The result of this was that Galileo was condemned for his infraction of the injunction laid upon him in 1616; his _Dialogue_ was prohibited; he himself was commanded to abjure on his knees the doctrine which he had taught; and this abjuration he performed.

This celebrated event must be looked upon rather as a question of decorum than a struggle in which the interests of truth and free inquiry were deeply concerned. The general acceptance of the Copernican System was no longer a matter of doubt. Several persons in the highest positions, including the Pope himself, looked upon the doctrine with favorable eyes; and had shown their interest in Galileo and his discoveries. They had tried to prevent his involving himself in trouble by discussing the question on scriptural grounds. It is probable that his knowledge of those favorable dispositions towards himself and his opinions led him to suppose that the slightest color of professed submission to the Church in his belief, would enable his arguments in favor of the system to pass unvisited: the notice which I have quoted, in which the irony is quite transparent and the sarcasm glaringly obvious, was deemed too flimsy a veil for the purpose of decency, and indeed must have aggravated the offence. But it is not to be supposed that the inquisitors believed Galileo's abjuration to be sincere, or even that they wished it to be so. It is stated that when Galileo had made his renunciation of the earth's motion, he rose from his knees, and stamping on the earth with his foot, said, _E pur si muove_--"And yet it _does_ move." This is sometimes represented as the heroic soliloquy of a mind cherishing its conviction of the truth in spite of persecution; I think we may more naturally conceive it uttered as

a playful {284} epigram in the ear of a cardinal's secretary, with a full knowledge that it would be immediately repeated to his master.

[2d Ed.] [Throughout the course of the proceedings against him, Galileo was treated with great courtesy and indulgence. He was condemned to a formal imprisonment and a very light discipline. "Te damnamus ad formalem carcerem hujus S. Officii ad tempus arbitrio nostro limitandum; et titulo pœnitentiæ salutaris præcipimus ut tribus annis futuris recites **semel in hebdomadâ septem psalmos penitentiales." But this confinement was reduced to his being placed under some slight restrictions, first at the house of Nicolini, the ambassador of his own sovereign, and afterwards at the country seat of Archbishop Piccolomini, one of his own warmest friends.

It has sometimes been asserted or insinuated that Galileo was subjected to bodily torture. An argument has been drawn from the expressions used in his sentence: "Cum vero nobis videretur non esse a te integram veritatem pronunciatam circa tuam intentionem; judicavimus necesse esse venire ad rigorosum examen tui, in quo respondisti catholicè." It has been argued by M. Libri (_Hist. des Sciences Mathématiques en Italie_, vol. IV. p. 259), and M. Quinet (_L'Ultramontanisme_, IV. Leçon, p. 104), that the _rigorosum examen_ necessarily implies bodily torture, notwithstanding that no such thing is mentioned by Galileo and his contemporaries, and notwithstanding the consideration with which he was treated in all other respects: but M. Biot more justly remarks (_Biogr. Univ._ Art. _Galileo_), that such a procedure is incredible.

To the opinion of M. Biot, we may add that of Delambre, who rejects the notion of Galileo's having been put to the torture, as inconsistent with the general conduct of the authorities towards him, and as irreconcilable with the accounts of the trial given by Galileo himself, and by a servant of his, who never quitted him for an instant. He adds also, that it is inconsistent with the words of his sentence, "ne tuus iste gravis et perniciosus error ac transgressio remaneat _omnino impunitus_;" for the error would have been already very far from impunity, if Galileo had been previously subjected to the rack. He adds, very reasonably, "il ne faut noircir personne sans preuve, pas même l'Inquisition;"--we must not calumniate even the Inquisition.]

The ecclesiastical authorities having once declared the doctrine of the earth's motion to be contrary to Scripture and heretical, long adhered in form to this declaration, and did not allow the Copernican system to be taught in any other way than as an "hypothesis." The {285} Padua edition of Galileo's works, published in 1744, contains the _Dialogue_ which now, the editors say, "Esce finalmente a pubblico libero uso colle debite licenze," is now at last freely published with the requisite license; but they add, "quanto alla Quistione principale del moto della terra, anche noi ci conformiamo alla ritrazione et protesta dell' Autore, dichiarando nella piu solenne forma, che non può, nè dee ammetersi se non come pura Ipotesi Mathematice, che serve a spiegare piu agevolamento certi fenomeni;" "neither can nor ought to be admitted except as a convenient hypothesis." And in the edition of Newton's _Principia_, published in 1760, by Le Sueur and Jacquier, of the Order of Minims, the editors prefix to the Third Book their _Declaratio_, that though Newton

assumes the hypothesis of the motion of the earth, and therefore they had used similar language, they were, in doing this, assuming a character which did not belong to them. "Hinc alienam coacti sumus gerere personam." They add, "Cæterum latis a summis Pontificibus contra telluris motum Decretis, nos obsequi profitemur."

By thus making decrees against a doctrine which in the course of time was established as an indisputable scientific truth, the See of Rome was guilty of an unwise and unfortunate stretch of ecclesiastical authority. But though we do not hesitate to pronounce such a judgment on this case, we may add that there is a question of no small real difficulty, which the progress of science often brings into notice, as it did then. The Revelation on which our religion is founded, seems to declare, or to take for granted, opinions on points on which Science also gives her decision; and we then come to this dilemma,--that doctrines, established by a scientific use of reason, may seem to contradict the declarations of Revelation, according to our view of its meaning;--and yet, that we cannot, in consistency with our religious views, make reason a judge of the truth of revealed doctrines. In the case of Astronomy, on which Galileo was called in question, the general sense of cultivated and sober-minded men has long ago drawn that distinction between religious and physical tenets, which is necessary to resolve this dilemma. On this point, it is reasonably held, that the phrases which are employed in Scripture respecting astronomical facts, are not to be made use of to guide our scientific opinions; they may be supposed to answer their end if they fall in with common notions, and are thus effectually subservient to the moral and religions import of Revelation. But the establishment of this distinction was not accomplished without long and distressing controversies. Nor, if we wish to {286} include all cases in which the same dilemma may again come into play is it easy to lay down an adequate canon for the purpose. For we can hardly foresee, beforehand, what part of the past history of the universe may eventually be found to come within the domain of science; or what bearing the tenets, which science establishes, may have upon our view of the providential and revealed government of the world. But without attempting here to generalize on this subject, there are two reflections which may be worth our notice: they are supported by what took place in reference to Astronomy on the occasion of which we are speaking; and may, at other periods, be applicable to other sciences.

In the first place, the meaning which any generation puts upon the phrases of Scripture, depends, more than is at first sight supposed upon the received philosophy of the time. Hence, while men imagine that they are contending for Revelation, they are, in fact, contending for their own interpretation of Revelation, unconsciously adapted to what they believe to be rationally probable. And the new interpretation, which the new philosophy requires, and which appears to the older school to be a fatal violence done to the authority of religion, is accepted by their successors without the dangerous results which were apprehended. When the language of Scripture, invested with its new meaning, has become familiar to men, it is found that the ideas which it calls up, are quite as reconcilable as the former ones were with the soundest religious views. And the world then looks back with surprise at the error of those who thought that the essence of Revelation was involved in their own arbitrary version of some collateral circumstance. At the present day we can hardly conceive

how reasonable men should have imagined that religious reflections on the stability of the earth, and the beauty and use of the luminaries which revolve round it, would be interfered with by its being acknowledged that this rest and motion are apparent only.

In the next place, we may observe that those who thus adhere tenaciously to the traditionary or arbitrary mode of understanding Scriptural expressions of physical events, are always strongly condemned by succeeding generations. They are looked upon with contempt by the world at large, who cannot enter into the obsolete difficulties with which they encumbered themselves; and with pity by the more considerate and serious, who know how much sagacity and rightmindedness are requisite for the conduct of philosophers and religious men on such occasions; but who know also how weak and vain is the attempt {287} to get rid of the difficulty by merely denouncing the new tenets as inconsistent with religious belief, and by visiting the promulgators of them with severity such as the state of opinions and institutions may allow. The prosecutors of Galileo are still up to the scorn and aversion of mankind: although, as we have seen, they did not act till it seemed that their position compelled them to do so, and then proceeded with all the gentleness and moderation which were compatible with judicial forms.

_Sect._ 5.--_The Heliocentric Theory confirmed on Physical considerations.--_(_Prelude to Kepler's Astronomical Discoveries._)

BY physical views, I mean, as I have already said, those which depend on the causes of the motions of matter, as, for instance, the consideration of the nature and laws of the force by which bodies fall downwards. Such considerations were necessarily and immediately brought under notice by the examination of the Copernican theory; but the loose and inaccurate notions which prevailed respecting the nature and laws of force, prevented, for some time, all distinct reasoning on this subject, and gave truth little advantage over error. The formation of a new Science, the Science of Motion and its Causes, was requisite, before the heliocentric system could have justice done it with regard to this part of the subject.

This discussion was at first carried on, as was to be expected, in terms of the received, that is, the Aristotelian doctrines. Thus, Copernicus says that terrestrial things appear to be at rest when they have a motion according to nature, that is, a circular motion; and ascend or descend when they have, in addition to this, a rectilinear motion by which they **endeavor to get into their own place. But his disciples soon began to question the Aristotelian dogmas, and to seek for sounder views by the use of their own reason. "The great argument against this system," says Mæstlin, "is that heavy bodies are said to move to the centre of the universe, and light bodies from the centre. But I would ask, where do we get this experience of heavy and light bodies? and how is our knowledge on these subjects extended so far that we can reason with certainty concerning the centre of the whole universe? Is not the only residence and home of all the things which are heavy and light to us, the earth and the air which surrounds it? and what is the earth and the ambient air, with respect to the immensity of the universe? It is a point, a punctule, or something, if there be any thing, still less. As our light and heavy bodies tend to {288} the centre

of our earth, it is credible that the sun, the moon, and the other lights, have a similar affection, by which they remain round as we see them; but none of these centres is necessarily the centre of the universe."

The most obvious and important physical difficulty attendant upon the supposition of the motion of the earth was thus stated: If the earth move, how is it that a stone, dropped from the top of a high tower, falls exactly at the foot of the tower? since the tower being carried from west to east by the diurnal revolution of the earth, the stone must be left behind to the west of the place from which it was let fall. The proper answer to this was, that the motion which the falling body received from its tendency downwards was _compounded_ with the motion which, before it fell, it had in virtue of the earth's rotation: but this answer could not be clearly made or apprehended, till Galileo and his pupils had established the laws of such Compositions of motion arising from different forces. Rothman, Kepler, and other defenders of the Copernican system, gave their reply somewhat at a venture, when they asserted that the motion of the earth was communicated to bodies at its surface. Still, the facts which indicate and establish this truth are obvious, when the subject is steadily considered; and the Copernicans soon found that they had the superiority of argument on this point as well as others. The attacks upon the Copernican system by Durret, Morin, Riccioli, and the defence of it by Galileo, Lansberg, Gassendi,[28\5] left on all candid reasoners a clear impression in favour of the system. Morin attempted to stop the motion of the earth, which he called breaking its wings; his _Alæ Terræ Fractæ_ was published in 1643, and answered by Gassendi. And Riccioli, as late as 1653, in his _Almagestum Novum_, enumerated fifty-seven Copernican arguments, and pretended to refute them all: but such reasonings now made no converts; and by this time the mechanical objections to the motion of the earth were generally seen to be baseless, as we shall relate when we come to speak of the progress of Mechanics as a distinct science. In the mean time, the beauty and simplicity of the heliocentric theory were perpetually winning the admiration even of those who, from one cause or other, refused their assent to it. Thus Riccioli, the last of its considerable opponents, allows its superiority in these respects; and acknowledges (in 1653) that the Copernican belief appears rather to increase than diminish under the condemnation of the decrees of the Cardinals. He applies to it the lines of Horace:[29\5] {289} Per damna per cædes, ab ipso Sumit opes animumque ferro. Untamed its pride, unchecked its course, From foes and wounds it gathers force.

[Note 28\5: Del. _A. M._ vol. i. p. 594.]

[Note 29\5: _Almag. Nov._ p. 102.]

We have spoken of the influence of the motion of the earth on the motions of bodies at its surface; but the notion of a physical connection among the parts of the universe was taken up by Kepler in another point of view, which would probably have been considered as highly fantastical, if the result had not been, that it led to by far the most magnificent and most certain train of truths which the whole expanse of human knowledge can show. I speak of the persuasion of the existence of numerical and geometrical laws connecting the distances, times, and forces of the bodies which

revolve about the central sun. That steady and intense conviction of this governing principle, which made its development and verification the leading employment of Kepler's most active and busy life, cannot be considered otherwise than as an example of profound sagacity. That it was connected, though dimly and obscurely, with the notion of a central agency or influence of some sort, emanating from the sun, cannot be doubted. Kepler, in his first essay of this kind, the _Mysterium Cosmographicum_, says, "The motion of the earth, which Copernicus had proved by _mathematical_ reasons, I wanted to prove by _physical_, or, if you prefer it, metaphysical." In the twentieth chapter of that work, he endeavors to make out some relation between the distances of the Planets from the Sun and their velocities. The inveterate yet vague notions of forces which preside in this attempt, may be judged of by such passages as the following:--"We must suppose one of two things; either that the moving spirits, in proportion as they are more removed from the sun, are more feeble; or that there is one moving spirit in the centre of all the orbits, namely, in the sun, which urges each body the more vehemently in proportion as it is nearer; but in more distant spaces languishes in consequence of the remoteness and attenuation of its virtue."

We must not forget, in reading such passages, that they were written under a belief that force was requisite to keep up, as well as to change the motion of each planet; and that a body, moving in a circle, would _stop_ when the force of the central point ceased, instead of moving off in a tangent to the circle, as we now know it would do. The force which Kepler supposes is a tangential force, in the direction of the body's motion, and nearly perpendicular to the radius; the {290} force which modern philosophy has established, is in the direction of the radius, and nearly perpendicular to the body's path. Kepler was right no further than in his suspicion of a connection between the cause of motion and the distance from the centre; not only was his knowledge imperfect in all particulars, but his most general conception of the mode of action of a cause of motion was erroneous.

With these general convictions and these physical notions in his mind, Kepler endeavored to detect numerical and geometrical relations among the parts of the solar system. After extraordinary labor, perseverance, and ingenuity, he was eminently successful in discovering such relations; but the glory and merit of interpreting them according to their physical meaning, was reserved for his greater successor, Newton.

## CHAPTER IV.

INDUCTIVE EPOCH OF KEPLER.

_Sect._ 1.--_Intellectual Character of Kepler._

SEVERAL persons,[30\5] especially in recent times, who have taken a view of the discoveries of Kepler, appear to have been surprised and somewhat discontented that conjectures, apparently so fanciful and arbitrary as his, should have led to important discoveries. They seem to have been alarmed at the _Moral_ that their readers might draw, from the tale of a Quest of Knowledge, in which the Hero, though fantastical

and self-willed, and violating in his conduct, as they conceived, all right rule and sound philosophy, is rewarded with the most signal triumphs. Perhaps one or two reflections may in some measure reconcile us to this result. {291}

[Note 30\5: Laplace, _Précis de l'Hist. d'Ast._ p. 94. "Il est affligeant pour l'esprit humain de voir ce grand homme, même dans ses derniers ouvrages, se complaire avec délices dans ses chimériques spéculations, et les regarder comme l'âme et la vie de l'astronomie."

_Hist. of Ast._, L. U. K., p. 53. "This success [of Kepler] may well inspire with dismay those who are accustomed to consider experiment and rigorous induction as the only means to interrogate nature with success."

_Life of Kepler_, L. U. K., p. 14, "Bad philosophy." P. 15, "Kepler's miraculous good fortune in seizing truths across the wildest and most absurd theories." P. 54, "The danger of attempting to follow his method in the pursuit of truth."]

In the first place, we may observe that the leading thought which suggested and animated all Kepler's attempts was true, and we may add, sagacious and philosophical; namely, that there must be _some_ numerical or geometrical relations among the times, distances, and velocities of the revolving bodies of the solar system. This settled and constant conviction of an important truth regulated all the conjectures, apparently so capricious and fanciful, which he made and examined, respecting particular relations in the system.

In the next place, we may venture to say, that advances in knowledge are not commonly made without the previous exercise of some boldness and license in guessing. The discovery of new truths requires, undoubtedly, minds careful and scrupulous in examining what is suggested; but it requires, no less, such as are quick and fertile in suggesting. What is Invention, except the talent of rapidly calling before us many possibilities, and selecting the appropriate one? It is true, that when we have rejected all the inadmissible suppositions, they are quickly forgotten by most persons; and few think it necessary to dwell on these discarded hypotheses, and on the process by which they were condemned, as Kepler has done. But all who discover truths must have reasoned upon many errors, to obtain each truth; every accepted doctrine must have been one selected out of many candidates. In making many conjectures, which on trial proved erroneous, Kepler was no more fanciful or unphilosophical than other discoverers have been. Discovery is not a "cautious" or "rigorous" process, in the sense of abstaining from such suppositions. But there are great differences in different cases, in the facility with which guesses are proved to be errors, and in the degree of attention with which the error and the proof are afterwards dwelt on. Kepler certainly was remarkable for the labor which he gave to such self-refutations, and for the candor and copiousness with which he narrated them; his works are in this way extremely curious and amusing; and are a very instructive exhibition of the mental process of discovery. But in this respect, I venture to believe, they exhibit to us the usual process (somewhat caricatured) of inventive minds: they rather exemplify the _rule_ of genius than (as has generally been hitherto taught) the _exception_. We may add, that if many of Kepler's guesses now appear

fanciful and absurd, because time and observation have refuted them, others, which were at the time equally gratuitous, have been confirmed by succeeding discoveries in a manner which makes them appear marvellously sagacious; as, for instance, his assertion of the rotation of {292} the sun on his axis, before the invention of the telescope, and his opinion that the obliquity of the ecliptic was decreasing, but would, after a long-continued diminution, stop, and then increase again.[31\5] Nothing can be more just, as well as more poetically happy, than Kepler's picture of the philosopher's pursuit of scientific truth, conveyed by means of an allusion to Virgil's shepherd and shepherdess: Malo me Galatea petit, lasciva puella Et fugit ad salices et se cupit ante videri. Coy yet inviting, Galatea loves To sport in sight, then plunge into the groves; The challenge given, she darts along the green, Will not be caught, yet would not run unseen.

[Note 31\5: Bailly, _A. M._ iii. 175.]

We may notice as another peculiarity of Kepler's reasonings, the length and laboriousness of the processes by which he discovered the errors of his first guesses. One of the most important talents requisite for a discoverer, is the ingenuity and skill which devises means for rapidly testing false suppositions as they offer themselves. This talent Kepler did not possess: he was not even a good arithmetical calculator, often making mistakes, some of which he detected and laments, while others escaped him to the last. But his defects in this respect were compensated by his courage and perseverance in undertaking and executing such tasks; and, what was still more admirable, he never allowed the labor he had spent upon any conjecture to produce any reluctance in abandoning the hypothesis, as soon as he had evidence of its inaccuracy. The only way in which he rewarded himself for his trouble, was by describing to the world, in his lively manner, his schemes, exertions, and feelings.

The _mystical_ parts of Kepler's opinions, as his belief in astrology, his persuasion that the earth was an animal, and many of the loose moral and spiritual as well as sensible analyses by which he represented to himself the powers which he supposed to prevail in the universe, do not appear to have interfered with his discovery, but rather to have stimulated his invention, and animated his exertions. Indeed, where there are clear scientific ideas on one subject in the mind, it does not appear that mysticism on others is at all unfavorable to the successful prosecution of research.

I conceive, then, that we may consider Kepler's character as containing the general features of the character of a scientific discoverer, {293} though some of the features are exaggerated, and some too feebly marked. His spirit of invention was undoubtedly very fertile and ready, and this and his perseverance served to remedy his deficiency in mathematical artifice and method. But the peculiar physiognomy is given to his intellectual aspect by his dwelling in a most prominent manner on those erroneous trains of thought which other persons conceal from the world, and often themselves forget, because they find means of stopping them at the outset. In the beginning of his book (_Argumenta Capitum_) he says, "if Christopher Columbus, if Magellan, if the Portuguese, when they narrate their wanderings, are not only excused, but if we do not wish these passages omitted, and should lose much pleasure if they were, let no one blame me for doing the same." Kepler's talents were

a kindly and fertile soil, which he cultivated with abundant toil and vigor; but with great scantiness of agricultural skill and implements. Weeds and the grain throve and flourished side by side almost undistinguished; and he gave a peculiar appearance to his harvest, by gathering and preserving the one class of plants with as much care and diligence as the other.

_Sect._ 2.--_Kepler's Discovery of his Third Law._

I SHALL now give some account of Kepler's speculations and discoveries. The first discovery which he attempted, the relation among the successive distances of the planets from the sun, was a failure; his doctrine being without any solid foundation, although propounded by him with great triumph, in a work which he called _Mysterium Cosmographicum_, and which was published in 1596. The account which he gives of the train of his thoughts on this subject, namely, the various suppositions assumed, examined, and rejected, is curious and instructive, for the reasons just stated; but we shall not dwell upon these essays, since they led only to an opinion now entirely abandoned. The doctrine which professed to give the true relation of the orbits of the different planets, was thus delivered:[32\5] "The orbit of the earth is a circle: round the sphere to which this circle belongs, describe a dodecahedron; the sphere including this will give the orbit of Mars. Round Mars describe a tetrahedron; the circle including this will be the orbit of Jupiter. Describe a cube round Jupiter's orbit; the circle including this will be the orbit of Saturn. Now inscribe in the Earth's orbit an icosahedron; the circle inscribed in it will be the orbit of Venus. {294} Inscribe an octahedron in the orbit of Venus; the circle inscribed in it will be Mercury's orbit. This is the reason of the number of the planets." The five kinds of polyhedral bodies here mentioned are the only "Regular Solids."

[Note 32\5: L. U. K. Kepler, 6.]

But though this part of the _Mysterium Cosmographicum_ was a failure, the same researches continued to occupy Kepler's mind; and twenty-two years later led him to one of the important rules known to us as "Kepler's Laws;" namely, to the rule connecting the mean distances of the planets from the sun with the times of their revolutions. This rule is expressed in mathematical terms, by saying that the squares of the periodic times are in the same proportion as the cubes of the distances; and was of great importance to Newton in leading him to the law of the sun's attractive force. We may properly consider this discovery as the sequel of the train of thought already noticed. In the beginning of the _Mysterium_, Kepler had said, "In the year 1595, I brooded with the whole energy of my mind on the subject of the Copernican system. There were three things in particular of which I pertinaciously sought the causes why they are not other than they are; the number, the size, and the motion of the orbits." We have seen the nature of his attempt to account for the two first of these points. He had also made some essays to connect the motions of the planets with their distances, but with his success in this respect he was not himself completely satisfied. But in the fifth book of the _Harmonice Mundi_, published in 1619, he says, "What I prophesied two-and-twenty years ago as soon as I had discovered the Five Solids among the Heavenly Bodies; what I firmly believed before I had seen the _Harmonics_ of Ptolemy; what I promised my friends in the title of this

book (_On the most perfect Harmony of the Celestial Motions_) which I named before I was sure of my discovery; what sixteen years ago I regarded as a thing to be sought; that for which I joined Tycho Brahe, for which I settled in Prague, for which I have devoted the best part of my life to astronomical contemplations; at length I have brought to light, and have recognized its truth beyond my most sanguine expectations."

The rule thus referred to is stated in the third Chapter of this fifth Book. "It is," he says, "a most certain and exact thing that the proportion which exists between the periodic times of any two planets is precisely the sesquiplicate of the proportion of their mean distances; that is, of the radii of the orbits. Thus, the period of the earth is one year, that of Saturn thirty years; if any one trisect the proportion, that {295} is, take the cube root of it, and double the proportion so found, that is, square it, he will find the exact proportion of the distances of the Earth and of Saturn from the sun. For the cube root of 1 is 1, and the square of this is 1; and the cube root of 30 is greater than 3, and therefore the square of it is greater than 9. And Saturn at his mean distance from the sun is at a little more than 9 times the mean distance of the Earth."

When we now look back at the time and exertions which the establishment of this law cost Kepler, we are tempted to imagine that he was strangely blind in not seeing it sooner. His object, we might reason, was to discover a law connecting the distances and the periodic times. What law of connection could be more simple and obvious, we might say, than that one of these quantities should vary as some _power_ of the other, or as some _root_; or as some combination of the two, which in a more general view, may still be called a _power_? And if the problem had been viewed in this way, the question must have occurred, to _what_ power of the periodic times are the distances proportional? And the answer must have been, the trial being made, that they are proportional to the square of the cube root. This _ex-post-facto_ obviousness of discoveries is a delusion to which we are liable with regard to many of the most important principles. In the case of Kepler, we may observe, that the process of connecting two classes of quantities by comparing their _powers_, is obvious only to those who are familiar with general algebraical views; and that in Kepler's time, algebra had not taken the place of geometry, as the most usual vehicle of mathematical reasoning. It may be added, also, that Kepler always sought his _formal_ laws by means of _physical_ reasonings; and these, though vague or erroneous, determined the nature of the mathematical connection which he assumed. Thus in the _Mysterium_ he had been led by his notions of moving virtue of the sun to this conjecture, among others--that, in the planets, the increase of the periods will be double of the difference of the distances; which supposition he found to give him an approach to the actual proportion of the distances, but one not sufficiently close to satisfy him.

The greater part of the fifth Book of the _Harmonics of the Universe_ consists in attempts to explain various relations among the distances, times, and eccentricities of the planets, by means of the ratios which belong to certain concords and discords. This portion of the work is so complex and laborious, that probably few modern readers have had courage to go through it. Delambre acknowledged that his patience

{296} often failed him during the task;[33\5] and subscribes to the judgment of Bailly: "After this sublime effort, Kepler replunges himself in the relations of music to the motions, the distance, and the eccentricities of the planets. In all these harmonic ratios there is not one true relation; in a crowd of ideas there is not one truth: he becomes a man after being a spirit of light." Certainly these speculations are of no value, but we may look on them with toleration, when we recollect that Newton has sought for analogies between the spaces occupied by the prismatic colors and the notes of the gamut.[34\5] The numerical relations of Concords are so peculiar that we can easily suppose them to have other bearings than those which first offer themselves.

[Note 33\5: _A. M._ a. 358.]

[Note 34\5: _Optics_, b. ii. p. iv. Obs. 5.]

It does not belong to my present purpose to speak at length of the speculations concerning the forces producing the celestial motions by which Kepler was led to this celebrated law, or of those which he deduced from it, and which are found in the _Epitome Astronomiæ Copernicanæ_, published in 1622. In that work also (p. 554), he extended this law, though in a loose manner, to the satellites of Jupiter. These _physical_ speculations were only a vague and distant prelude to Newton's discoveries; and the law, as a _formal_ rule, was complete in itself. We must now attend to the history of the other two laws with which Kepler's name is associated.

_Sect._ 3.--_Kepler's Discovery of his First and Second Laws.--Elliptical Theory of the Planets._

THE propositions designated as Kepler's First and Second Laws are these: That the orbits of the planets are elliptical; and, That the areas described, or _swept_, by lines drawn from the sun to the planet, are proportional to the times employed in the motion.

The occasion of the discovery of these laws was the attempt to reconcile the theory of Mars to the theory of eccentrics and epicycles; the event of it was the complete overthrow of that theory, and the establishment, in its stead, of the Elliptical Theory of the planets. Astronomy was now ripe for such a change. As soon as Copernicus had taught men that the orbits of the planets were to be referred to the sun, it obviously became a question, what was the true form of these orbits, and the rule of motion of each planet in its own orbit. Copernicus represented the motions in longitude by means of {297} eccentrics and epicycles, as we have already said; and the motions in latitude by certain _librations_, or alternate elevations and depressions of epicycles. If a mathematician had obtained a collection of true positions of a planet, the form of the orbit and the motion of the star would have been determined with reference to the sun as well as to the earth; but this was not possible, for though the _geocentric_ position, or the direction in which the planet was seen, could be observed, its distance from the earth was not known. Hence, when Kepler attempted to determine the orbit of a planet, he combined the observed geocentric places with successive modifications of the theory of epicycles, till at last he was led, by one step after another, to change the epicyclical into the elliptical theory. We may observe,

moreover, that at every step he endeavored to support his new suppositions by what he called, in his fanciful phraseology, "sending into the field a reserve of new physical reasonings on the rout and dispersion of the veterans;"[35\5] that is, by connecting his astronomical hypotheses with new imaginations, when the old ones became untenable. We find, indeed, that this is the spirit in which the pursuit of knowledge is generally carried on with success; those men arrive at truth who eagerly endeavor to connect remote points of their knowledge, not those who stop cautiously at each point till something compels them to go beyond it.

[Note 35\5: I will insert this passage, as a specimen of Kepler's fanciful mode of narrating the defeats which he received in the war which he carried on with Mars. "Dum in hunc modum de Martis motibus triumpho, eique ut planè devicto tabularum carceres et equationum compedes necto, diversis nuntiatur locis, futilem victoriam ut bellam totâ mole recrudescere. Nam domi quidam hostis ut captivus contemptus, rupit omnia equationum vincula, carceresque tabularum effregit. Foris speculatores profligerunt meas causarum physicarum arcessitas copias earumque jugum excusserunt resumtà libertate. Jamque parum abfuit quia hostis fugitivus sese cum rebellibus suis conjungeret meque in desperationem adigeret: nisi raptim, nova rationum physicarum subsidia, fusis et palantibus veteribus, submisissem, et qua se captivus proripuisset, omni diligentia, edoctus vestigiis ipsius nullâ morâ interpositâ inhæsisserem."]

Kepler joined Tycho Brahe at Prague in 1600, and found him and Longomontanus busily employed in correcting the theory of Mars; and he also then entered upon that train of researches which he published in 1609 in his extraordinary work _On the Motions of Mars_. In this work, as in others, he gives an account, not only of his success, but of his failures, explaining, at length, the various suppositions which he had made, the notions by which he had been led to invent or to entertain them, the processes by which he had proved their {298} falsehood, and the alternations of hope and sorrow, of vexation and triumph, through which he had gone. It will not be necessary for us to cite many passages of these kinds, curious and amusing as they are.

One of the most important truths contained in the motions of Man is the discovery that the plane of the orbit of the planet should be considered with reference to the sun itself, instead of referring it to any of the other centres of motion which the eccentric hypothesis introduced: and that, when so considered, it had none of the librations which Ptolemy and Copernicus had attributed to it. The fourteenth chapter of the second part asserts, "Plana eccentricorum esse ἀτάλαντα;" that the planes are _unlibrating_; retaining always the same inclination to the ecliptic, and the same _line of nodes_. With this step Kepler appears to have been justly delighted. "Copernicus," he says, "not knowing the value of what he possessed (his system), undertook to represent Ptolemy, rather than nature, to which, however, he had approached more nearly than any other person. For being rejoiced that the quantity of the latitude of each planet was increased by the approach of the earth to the planet, according to his theory, he did not venture to reject the rest of Ptolemy's increase of latitude, but in order to express it, devised librations of the planes of the eccentric, depending not

upon its own eccentric, but (most improbably) upon the orbit of the earth, which has nothing to do with it. I always fought against this impertinent tying together of two orbits, even before I saw the observations of Tycho; and I therefore rejoice much that in this, as in others of my preconceived opinions, the observations were found to be on my side." Kepler established his point by a fair and laborious calculation of the results of observations of Mars made by himself and Tycho Brahe; and had a right to exult when the result of these calculations confirmed his views of the symmetry and simplicity of nature.

We may judge of the difficulty of casting off the theory of eccentrics and epicycles, by recollecting that Copernicus did not do it at all, and that Kepler only did it after repeated struggles; the history of which occupies thirty-nine Chapters of his book. At the end of them he says, "This prolix disputation was necessary, in order to prepare the way to the natural form of the equations, of which I am now to treat.[36\5] My first error was, that the path of a planet is a perfect circle;--an opinion which was a more mischievous thief of my time, {299} in proportion as it was supported by the authority of all philosophers, and apparently agreeable to metaphysics." But before he attempts to correct this erroneous part of his hypothesis, he sets about discovering the law according to which the different parts of the orbit are described in the case of the earth, in which case the eccentricity is so small that the effect of the oval form is insensible. The result of this inquiry was[37\5] the Rule, that the time of describing any arc of the orbit is proportional to the area intercepted between the curve and two lines drawn from the sun to the extremities of the arc. It is to be observed that this rule, at first, though it had the recommendation of being selected after the unavoidable abandonment of many, which were suggested by the notions of those times, was far from being adopted upon any very rigid or cautious grounds. A rule had been proved at the apsides of the orbit, by calculation from observations, and had then been extended by conjecture to other parts of the orbit; and the rule of the areas was only an approximate and inaccurate mode of representing this rule, employed for the purpose of brevity and convenience, in consequence of the difficulty of applying, geometrically, that which Kepler now conceived to be the true rule, and which required him to find the sum of the lines drawn from the sun to _every_ point of the orbit. When he proceeded to apply this rule to Mars, in whose orbit the oval form is much more marked, additional difficulties came in his way; and here again the true supposition, that the _oval_ is of that special kind called _ellipse_, was adopted at first only in order to simplify calculation,[38\5] and the deviation from exactness in the result was attributed to the inaccuracy of those approximate processes. The supposition of the oval had already been forced upon Purbach in the case of Mercury, and upon Reinhold in the case of the Moon. The centre of the epicycle was made to describe an egg-shaped figure in the former case, and a lenticular figure in the latter.[39\5]

[Note 36\5: _De Stellâ Martis_, iii. 40.]

[Note 37\5: _De Stellâ Martis_, p. 194.]

[Note 38\5: Ib. iv. c. 47.]

[Note 39\5: L. U. K. Kepler, p. 30.]

It may serve to show the kind of labor by which Kepler was led to his result, if we here enumerate, as he does in his forty-seventh Chapter,[40\5] six hypotheses, on which he calculated the longitude of Mars, in order to see which best agreed with observation.

[Note 40\5: _De Stellâ Martis_, p. 228.]

1. The simple eccentricity.

2. The bisection of the eccentricity, and the duplication of the superior part of the equation. {300}

3. The bisection of the eccentricity, and a stationary point of equations, after the manner of Ptolemy.

4. The vicarious hypothesis by a free section of the eccentricity made to agree as nearly as possible with the truth.

5. The physical hypothesis on the supposition of a perfect circle.

6. The physical hypothesis on the supposition of a perfect ellipse.

By the physical hypothesis, he meant the doctrine that the time of a planet's describing any part of its orbit is proportional to the distance of the planet from the sun, for which supposition, as we have said, he conceived that he had assigned physical reasons.

The two last hypotheses came the nearest to the truth, and differed from it only by about eight minutes, the one in excess and the other in defect. And, after being much perplexed by this remaining error, it at last occurred to him[41\5] that he might take another ellipsis, exactly intermediate between the former one and the circle, and that this must give the path and the motion of the planet. Making this assumption, and taking the areas to represent the times, he now saw[42\5] that both the longitude and the distances of Mars would agree with observation to the requisite degree of accuracy. The rectification of the former hypothesis, when thus stated, may, perhaps, appear obvious. And Kepler informs us that he had nearly been anticipated in this step (c. 55). "David Fabricius, to whom I had communicated my hypothesis of cap. 45, was able, by his observations, to show that it erred in making the distances too short at mean longitudes; of which he informed me by letter while I was laboring, by repeated efforts, to discover the true hypothesis. So nearly did he get the start of me in detecting the truth." But this was less easy than it might seem. When Kepler's first hypothesis was enveloped in the complex construction requisite in order to apply it to each point of the orbit, it was far more difficult to see where the error lay, and Kepler hit upon it only by noticing the coincidences of certain numbers, which, as he says, raised him as if from sleep, and gave him a new light. We may observe, also, that he was perplexed to reconcile this new view, according to which the planet described an exact ellipse, with his former opinion, which represented the motion by means of libration in an epicycle. "This," he says, "was my greatest trouble, that, though I considered and reflected till I was almost mad, I could not find why the planet to

which, with so much probability, and with such an exact {301} accordance of the distances, libration in the diameter of the epicycle was attributed, should, according to the indication of the equations, go in an elliptical path. What an absurdity on my part! as if libration in the diameter might not be a way to the ellipse!"

[Note 41\5: _De Stellâ Martis_, c. 58.]

[Note 42\5: Ibid. p. 235.]

Another scruple respecting this theory arose from the impossibility of solving, by any geometrical construction, the problem to which Kepler was thus led, namely, "To divide the area of a semicircle in a given ratio, by a line drawn from any point of the diameter." This is still termed "Kepler's Problem," and is, in fact, incapable of exact geometrical solution. As, however, the calculation can be performed, and, indeed, was performed by Kepler himself, with a sufficient degree of accuracy to show that the elliptical hypothesis is true, the insolubility of this problem is a mere mathematical difficulty in the deductive process, to which Kepler's induction gave rise.

Of Kepler's physical reasonings we shall speak more at length on another occasion. His numerous and fanciful hypotheses had discharged their office, when they had suggested to him his many lines of laborious calculation, and encouraged him under the exertions and disappointments to which these led. The result of this work was the formal laws of the motion of Mars, established by a clear induction, since they represented, with sufficient accuracy, the best observations. And we may allow that Kepler was entitled to the praise which he claims in the motto on his first leaf. Ramus had said that if any one would construct an astronomy without hypothesis, he would be ready to resign to him his professorship in the University of Paris. Kepler quotes this passage, and adds, "it is well, Ramus, that you have run from this pledge, by quitting life and your professorship;[43\5] if you held it still, I should, with justice, claim it." This was not saying too much, since he had entirely overturned the hypothesis of eccentrics and epicycles, and had obtained a theory which was a mere representation of the motions and distances as they were observed. {302}

[Note 43\5: Ramus perished in the Massacre of St. Bartholomew.]

## CHAPTER V.

SEQUEL TO THE EPOCH OF KEPLER. RECEPTION, VERIFICATION, AND EXTENSION OF THE ELLIPTICAL THEORY.

_Sect._ 1.--_Application of the Elliptical Theory to the Planets._

THE extension of Kepler's discoveries concerning the orbit of Mars to the other planets, obviously offered itself as a strong probability, and was confirmed by trial. This was made in the first place upon the orbit of Mercury; which planet, in consequence of the largeness of its eccentricity, exhibits more clearly than the others the circumstances of the elliptical motion. These and various other supplementary portions of the views to which Kepler's discoveries had led, appeared in the latter part of his _Epitome Astronomiæ Copernicanæ_, published in 1622.

The real verification of the new doctrine concerning the orbits and motions of the heavenly bodies was, of course, to be found in the construction of tables of those motions, and in the continued comparison of such tables with observation. Kepler's discoveries had been founded, as we have seen, principally on Tycho's observations. Longomontanus (so called as being a native of Langberg in Denmark), published in 1621, in his _Astronomia Danica_, tables founded upon the theories as well as the observations of his countryman. Kepler[44\5] in 1627 published his tables of the planets, which he called _Rudolphine Tables_, the result and application of his own theory. In 1633, Lansberg, a Belgian, published also _Tabulæ Perpetuæ_, a work which was ushered into the world with considerable pomp and pretension, and in which the author cavils very keenly at Kepler and Brahe. We may judge of the impression made upon the astronomical world in general by these rival works, from the account which our countryman Jeremy Horrox has given of their effect on him. He had been seduced by the magnificent promises of Lansberg, and the praises of his admirers, which are prefixed to the work, and was persuaded that the common opinion which preferred Tycho and Kepler to him was a prejudice. In 1636, however, he became acquainted with Crabtree, another young {303} astronomer, who lived in the same part of Lancashire. By him Horrox was warned that Lansberg was not to be depended on; that his hypotheses were vicious, and his observations falsified or forced into agreement with his theories. He then read the works and adopted the opinions of Kepler; and after some hesitation which he felt at the thought of attacking the object of his former idolatry, he wrote a dissertation on the points of difference between them. It appears that, at one time, he intended to offer himself as the umpire who was to adjudge the prize of excellence among the three rival theories of Longomontanus, Kepler, and Lansberg; and, in allusion to the story of ancient mythology, his work was to have been called _Paris Astronomicus_; we easily see that he would have given the golden apple to the Keplerian goddess. Succeeding observations confirmed his judgment: and the _Rudolphine Tables_, thus published seventy-six years after the Prutenic, which were founded on the doctrines of Copernicus, were for a long time those universally used.

[Note 44\5: Rheticus, _Narratio_, p. 98.]

_Sect._ 2.--_Application of the Elliptical Theory to the Moon._

THE reduction of the Moon's motions to rule was a harder task than the formation of planetary tables, if accuracy was required; for the Moon's motion is affected by an incredible number of different and complex inequalities, which, till their law is detected, appear to defy all theory. Still, however, progress was made in this work. The most important advances were due to Tycho Brahe. In addition to the first and second inequalities of the moon (the _Equation of the Centre_, known very early, and the _Evection_, which Ptolemy had discovered), Tycho proved that there was another inequality, which he termed the _Variation_,[45\5] which depended on the moon's position with respect to the sun, and which at its maximum was forty minutes and a half, about a quarter of the evection. He also perceived, though not very distinctly, the necessity of another correction of the moon's place depending on the sun's longitude, which has since been termed the _Annual Equation_.

[Note 45\5: We have seen (chap. iii.), that Aboul-Wefa, in the tenth century, had already noticed this inequality; but his discovery had been entirely forgotten long before the time of Tycho, and has only recently been brought again into notice.]

These steps concerned the Longitude of the Moon; Tycho also made important advances in the knowledge of the Latitude. The Inclination of the Orbit had hitherto been assumed to be the same at all {304} times; and the motion of the Node had been supposed uniform. He found that the inclination increased and diminished by twenty minutes, according to the position of the line of nodes; and that the nodes, though they regress upon the whole, sometimes go forwards and sometimes go backwards.

Tycho's discoveries concerning the moon are given in his _Progymnasmata_, which was published in 1603, two years after the author's death. He represents the Moon's motion in longitude by means of certain combinations of epicycles and eccentrics. But after Kepler had shown that such devices are to be banished from the planetary system, it was impossible not to think of extending the elliptical theory to the moon. Horrox succeeded in doing this; and in 1638 sent this essay to his friend Crabtree. It was published in 1673, with the numerical elements requisite for its application added by Flamsteed. Flamsteed had also (in 1671-2) compared this theory with observation, and found that it agreed far more nearly than the _Philolaic Tables_ of Bullialdus, or the _Carolinian Tables_ of Street (_Epilogus ad Tabulas_). Moreover Horrox, by making the centre of the ellipse revolve in an epicycle, gave an explanation of the evection, as well as of the equation of the centre.[46\5]

[Note 46\5: Horrox (_Horrockes_ as he himself spelt his name) gave a first sketch of his theory in letters to his friend Crabtree in 1638: in which the variation of the eccentricity is not alluded to. But in Crabtree's letter to Gascoigne in 1642, he gives Horrox's rule concerning it; and Flamsteed in his _Epilogue_ to the Tables, published by Wallis along with Horrox's works in 1673, gave an explanation of the theory which made it amount very nearly to a revolution of the centre of the ellipse in an epicycle. Halley afterwards made a slight alteration; but hardly, I think, enough to justify Newton's assertion (_Princip._ Lib. iii. Prop. 35, Schol.), "Halleius centrum ellipseos in epicyclo locavit." See Baily's _Flamsteed_, p. 683.]

Modern astronomers, by calculating the effects of the perturbing forces of the solar system, and comparing their calculations with observation, have added many new corrections or equations to those known at the time of Horrox; and since the Motions of the heavenly bodies were even then affected by these variations as yet undetected, it is clear that the Tables of that time must have shown some errors when compared with observation. These errors much perplexed astronomers, and naturally gave rise to the question whether the motions of the heavenly bodies really were exactly regular, or whether they were not affected by accidents as little reducible to rule as wind and weather. Kepler had held the opinion of the _casualty_ of such errors; but Horrox, far more philosophically, argues against this opinion, though he {305} allows that he is much embarrassed by the deviations. His arguments show a singularly clear and strong apprehension of the features of the case, and their real import. He says,[47\5] "these errors of the tables are alternately in excess and defect;

how could this constant compensation happen if they were casual? Moreover, the alternation from excess to defect is most rapid in the Moon, most slow in Jupiter and Saturn, in which planets the error continues sometimes for years. If the errors were casual, why should they not last as long in the Moon as in Saturn? But if we suppose the tables to be right in the mean motions, but wrong in the equations, these facts are just what must happen; since Saturn's inequalities are of long period, while those of the Moon are numerous, and rapidly changing." It would be impossible, at the present moment, to reason better on this subject; and the doctrine, that all the apparent irregularities of the celestial motions are really regular, was one of great consequence to establish at this period of the science.

[Note 47\5: _Astron. Kepler._ Proleg. p. 17.]

_Sect._ 3.--_Causes of the further Progress of Astronomy._

WE are now arrived at the time when theory and observation sprang forwards with emulous energy. The physical theories of Kepler, and the reasonings of other defenders of the Copernican theory, led inevitably, after some vagueness and perplexity, to a sound science of Mechanics; and this science in time gave a new face to Astronomy. But in the mean time, while mechanical mathematicians were generalizing from the astronomy already established, astronomers were accumulating new facts, which pointed the way to new theories and new generalizations. Copernicus, while he had established the permanent length of the year, had confirmed the motion of the sun's apogee, and had shown that the eccentricity of the earth's orbit, and the obliquity of the ecliptic, were gradually, though slowly, diminishing. Tycho had accumulated a store of excellent observations. These, as well as the laws of the motions of the moon and planets already explained, were materials on which the Mechanics of the Universe was afterwards to employ its most matured powers. In the mean time, the telescope had opened other new subjects of notice and speculation; not only confirming the Copernican doctrine by the phases of Venus, and the analogical examples of Jupiter and Saturn, which with their Satellites {306} appeared like models of the Solar System; but disclosing unexpected objects, as the Ring of Saturn, and the Spots of the Sun. The art of observing made rapid advances, both by the use of the telescope, and by the sounder notions of the construction of instruments which Tycho introduced. Copernicus had laughed at Rheticus, when he was disturbed about single minutes; and declared that if he could be sure to ten minutes of space, he should be as much delighted as Pythagoras was when he discovered the property of the right-angled triangle. But Kepler founded the revolution which he introduced on a quantity less than this. "Since," he says,[48\5] "the Divine Goodness has given us in Tycho an observer so exact that this error of eight minutes is impossible, we must be thankful to God for this, and turn it to account. And these eight minutes, which we must not neglect, will, of themselves, enable us to reconstruct the whole of astronomy." In addition to other improvements, the art of numerical calculation made an inestimable advance by means of Napier's invention of Logarithms; and the progress of other parts of pure mathematics was proportional to the calls which astronomy and physics made upon them.

[Note 48\5: _De Stellâ Martis_, c. 19.]

The exactness which observation had attained enabled astronomers both to verify and improve the existing theories, and to study the yet unsystematized facts. The science was, therefore, forced along by a strong impulse on all sides, and its career assumed a new character. Up to this point, the history of European Astronomy was only the sequel of the history of Greek Astronomy; for the heliocentric system, as we have seen, had had a place among the guesses, at least, of the inventive and acute intellects of the Greek philosophers. But the discovery of Kepler's Laws, accompanied, as from the first they were, with a conviction that the relations thus brought to light were the effects and exponents of physical causes, led rapidly and irresistibly to the Mechanical Science of the skies, and collaterally, to the Mechanical Science of the other parts of Nature: Sound, and Light, and Heat; and Magnetism, and Electricity, and Chemistry. The history of these Sciences, thus treated, forms the sequel of the present work, and will be the subject of the succeeding volumes. And since, as I have said, our main object in this work is to deduce, from the history of science, the philosophy of scientific discovery, it may be regarded as fortunate for our purpose that the history, after this point, so far changes its aspect as to offer new materials for such speculations. The details of {307} a history of astronomy, such as the history of astronomy since Newton has been, though interesting to the special lovers of that science, would be too technical, and the features of the narrative too monotonous and unimpressive, to interest the general reader, or to suggest a comprehensive philosophy of science. But when we pass from the Ideas of Space and Time to the Ideas of Force and Matter, of Mediums by which action and sensation are produced, and of the Intimate Constitution of material bodies, we have new fields of inquiry opened to us. And when we find that in these fields, as well as in astronomy, there are large and striking trains of unquestioned discovery to be narrated, we may gird ourselves afresh to the task of writing, and I hope, of reading, the remaining part of the History of the Inductive Sciences, in the trust that it will in some measure help us to answer the important questions, What is Truth? and, How is it to be discovered?

{{309}} BOOK VI.

# _THE MECHANICAL SCIENCES._

HISTORY OF MECHANICS, INCLUDING FLUID MECHANICS.

ΚΡΑΤΟΣ ΒΙΑ ΤΕ, σφῷν μὲν ἐντολὴ Διὸς Ἔχει Τέλος δὴ, κ' οὐδὲν ἐμποδῶν ἔτι ÆSCHYLUS. _Prom. Vinct._ 13.

You, FORCE and POWER, have done your destined task: And naught impedes the work of other hands.

{{311}} INTRODUCTION.

WE enter now upon a new region of the human mind. In passing from Astronomy to Mechanics we make a transition from the _formal_ to the _physical_ sciences;--from time and space to force and matter;--from _phenomena_ to _causes_. Hitherto we have been concerned only with the paths and orbits, the periods and cycles, the

angles and distances, of the objects to which our sciences applied, namely, the heavenly bodies. How these motions are produced;--by what agencies, impulses, powers, they are determined to be what they are;--of what nature are the objects themselves;--are speculations which we have hitherto not dwelt upon. The history of such speculations now comes before us; but, in the first place, we must consider the history of speculations concerning motion in general, terrestrial as well as celestial. We must first attend to Mechanics, and afterwards return to Physical Astronomy.

In the same way in which the development of Pure Mathematics, which began with the Greeks, was a necessary condition of the progress of Formal Astronomy, the creation of the science of Mechanics now became necessary to the formation and progress of Physical Astronomy. Geometry and Mechanics were studied for their own sakes; but they also supplied ideas, language, and reasoning to other sciences. If the Greeks had not cultivated Conic Sections, Kepler could not have superseded Ptolemy; if the Greeks had cultivated Dynamics,[1\6] Kepler might have anticipated Newton. {312}

[Note 1\6: _Dynamics_ is the science which treats of the Motions of Bodies; _Statics_ is the science which treats of the Pressure of Bodies which are in equilibrium, and therefore at rest.]

# CHAPTER I.

PRELUDE TO THE EPOCH OF GALILEO.

_Sect._ 1.--_Prelude to the Science of Statics._

SOME steps in the science of Motion, or rather in the science of Equilibrium, had been made by the ancients, as we have seen. Archimedes established satisfactorily the doctrine of the Lever, some important properties of the Centre of Gravity, and the fundamental proposition of Hydrostatics. But this beginning led to no permanent progress. Whether the distinction between the principles of the doctrine of Equilibrium and of Motion was clearly seen by Archimedes, we do not know; but it never was caught hold of by any of the other writers of antiquity, or by those of the Stationary Period. What was still worse, the point which Archimedes had won was not steadily maintained.

We have given some examples of the general ignorance of the Greek philosophers on such subjects, in noticing the strange manner in which Aristotle refers to mathematical properties, in order to account for the equilibrium of a lever, and the attitude of a man rising from a chair. And we have seen, in speaking of the indistinct ideas of the Stationary Period, that the attempts which were made to extend the statical doctrine of Archimedes, failed, in such a manner as to show that his followers had not clearly apprehended the idea on which his reasoning altogether depended. The clouds which he had, for a moment, cloven in his advance, closed after him, and the former dimness and confusion settled again on the land.

This dimness and confusion, with respect to all subjects of mechanical reasoning, prevailed still, at the period we now have to consider; namely, the period of the first

promulgation of the Copernican opinions. This is so important a point that I must illustrate it further.

Certain general notions of the connection of cause and effect in motion, exist in the human mind at all periods of its development, and are implied in the formation of language and in the most familiar employments of men's thoughts. But these do not constitute a _science_ of {313} Mechanics, any more than the notions of _square_ and _round_ make a Geometry, or the notions of _months_ and _years_ make an Astronomy. The unfolding these Notions into distinct Ideas, on which can be founded principles and reasonings, is further requisite, in order to produce a science; and, with respect to the doctrines of Motion, this was long in coming to pass; men's thoughts remained long entangled in their primitive and unscientific confusion.

We may mention one or two features of this confusion, such as we find in authors belonging to the period now under review.

We have already, in speaking of the Greek School Philosophy, noticed the attempt to explain some of the differences among Motions, by classifying them into Natural Motions and Violent Motions; and we have spoken of the assertion that heavy bodies fall quicker in proportion to their greater weight. These doctrines were still retained: yet the views which they implied were essentially erroneous and unsound; for they did not refer distinctly to a measurable Force as the cause of all motion or change of motion; and they confounded the causes which _produce_ and those which _preserve_, motion. Hence such principles did not lead immediately to any advance of knowledge, though efforts were made to apply them, in the cases both of terrestrial Mechanics and of the motions of the heavenly bodies.

The effect of the Inclined Plane was one of the first, as it was one of the most important, propositions, on which modern writers employed themselves. It was found that a body, when supported on a sloping surface, might be sustained or raised by a force or exertion which would not have been able to sustain or raise it without such support. And hence, _The Inclined Plane_ was placed in the list of Mechanical Powers, or simple machines by which the efficacy of forces is increased: the question was, in what proportion this increase of efficiency takes place. It is easily seen that the force requisite to sustain a body is smaller, as the slope on which it rests is smaller; Cardan (whose work, _De Proportionibus Numerorum, Motuum, Ponderum,_ &c., was published in 1545) asserts that the force is double when the angle of inclination is double, and so on for other proportions; this is probably a guess, and is an erroneous one. Guido Ubaldi, of Marchmont, published at Pesaro, in 1577, a work which he called _Mechanicorum Liber_, in which he endeavors to prove that an acute wedge will produce a greater mechanical effect than an obtuse one, without determining in what proportion. There is, he observes, "a certain repugnance" between the direction in which the side of the wedge tends to {314} move the obstacle, and the direction in which it really does move. Thus the Wedge and the Inclined Plane are connected in principle. He also refers the Screw to the Inclined Plane and the Wedge, in a manner which shows a just apprehension of the question. Benedetti (1585) treats the Wedge in a different manner; not exact, but still showing some powers of thought on mechanical subjects. Michael Varro, whose _Tractatus de

Motu_ was published at Geneva in 1584, deduces the wedge from the composition of hypothetical motions, in a way which may appear to some persons an anticipation of the doctrine of the Composition of Forces.

There is another work on subjects of this kind, of which several editions were published in the sixteenth century, and which treats this matter in nearly the same way as Varro, and in favour of which a claim has been made[2\6] (I think an unfounded one), as if it contained the true principle of this problem. The work is "Jordanus Nemorarius _De Ponderositate_." The date and history of this author were probably even then unknown; for in 1599, Benedetti, correcting some of the errors of Tartalea, says they are taken "a Jordano quodam antiquo." The book was probably a kind of school-book, and much used; for an edition printed at Frankfort, in 1533, is stated to be _Cum gratia et privilegio Imperiali, Petro Apiano mathematico Ingolstadiano ad xxx annos concesso_. But this edition does not contain the Inclined Plane. Though those who compiled the work assert in words something like the inverse proportion of Weights and their Velocities, they had not learnt at that time how to apply this maxim to the Inclined Plane; nor were they ever able to render a sound reason for it. In the edition of Venice, 1565, however, such an application is attempted. The reasonings are founded on the Aristotelian assumption, "that bodies descend more quickly in proportion as they are heavier." To this principle are added some others; as, that "a body is heavier in proportion as it descends more directly to the centre," and that, in proportion as a body descends more obliquely, the intercepted part of the direct descent is smaller. By means of these principles, the "descending force" of bodies, on inclined planes, was compared, by a process, which, so far as it forms a line of proof at all, is a somewhat curious example of confused and vicious reasoning. When two bodies are supported on two inclined planes, and are connected by a string passing over the junction of the planes, so that when one descends the other ascends, {315} they must move through equal spaces on the planes; but on the plane which is more oblique (that is, more nearly horizontal), the vertical descent will be smaller in the same proportion in which the plane is longer. Hence, by the Aristotelian principle, the weight of the body on the longer plane is less; and, to produce an equality of effect, the body must be greater in the same proportion. We may observe that the Aristotelian principle is not only false, but is here misapplied; for its genuine meaning is, that when bodies _fall freely_ by gravity, they move quicker in proportion as they are heavier; but the rule is here applied to the motions which bodies _would_ have, if they were moved by a force extraneous to their gravity. The proposition was supposed by the Aristotelians to be true of _actual_ velocities; it is applied by Jordanus to _virtual_ velocities, without his being aware what he was doing. This confusion being made, the result is got at by taking for granted that bodies _thus_ proved to be equally _heavy_, have equal powers of descent on the inclined planes; whereas, in the previous part of the reasoning, the weight was supposed to be proportional to the descent in the vertical direction. It is obvious, in all this, that though the author had adopted the false Aristotelian principle, he had not settled in his own mind whether the motions of which it spoke were actual or virtual motions;--motions in the direction of the inclined plane, or of the intercepted parts of the vertical, corresponding to these; nor whether the

"descending force" of a body was something different from its weight. We cannot doubt that, if he had been required to point out, with any exactness, the cases to which his reasoning applied, he would have been unable to do so; not possessing any of those clear fundamental Ideas of Pressure and Force, on which alone any real knowledge on such subjects must depend. The whole of Jordanus's reasoning is an example of the confusion of thought of his period, and of nothing more. It no more supplied the want of some man of genius, who should give the subject a real scientific foundation, than Aristotle's knowledge of the proportion of the weights on the lever superseded the necessity of Archimedes's proof of it.

[Note 2\6: Mr. Drinkwater's _Life of Galileo_, in the Lib. Usef. Kn. p. 83.]

We are not, therefore, to wonder that, though this pretended theorem was copied by other writers, as by Tartalea, in his _Quesiti et Inventioni Diversi_, published in 1554, no progress was made in the real solution of any one mechanical problem by means of it. Guido Ubaldi, who, in 1577, writes in such a manner as to show that he had taken a good hold of his subject for his time, refers to Pappus's solution of the problem of the Inclined Plane, but makes no mention of that of {316} Jordanus and Tartalea.[3\6] No progress was likely to occur, till the mathematicians had distinctly recovered the genuine Idea of Pressure, as a Force producing equilibrium, which Archimedes had possessed, and which was soon to reappear in Stevinus.

[Note 3\6: Ubaldi mentions and blames Jordanus's way of treating the Lever. (See his Preface.)]

The properties of the Lever had always continued known to mathematicians, although, in the dark period, the superiority of the proof given by Archimedes had not been recognized. We are not to be surprised, if reasonings like those of Jordanus were applied to demonstrate the theories of the Lever with apparent success. Writers on Mechanics were, as we have seen, so vacillating in their mode of dealing with words and propositions, that their maxims could be made to prove any thing which was already known to be true.

We proceed to speak of the beginning of the real progress of Mechanics in modern times.

_Sect._ 2.--_Revival of the Scientific Idea of Pressure.--Stevinus.--Equilibrium of Oblique Forces._

THE doctrine of the Centre of Gravity was the part of the mechanical speculations of Archimedes which was most diligently prosecuted after his time. Pappus and others, among the ancients, had solved some new problems on this subject, and Commandinus, in 1565, published _De Centro Gravitatis Solidorum_. Such treatises contained, for the most part, only mathematical consequences of the doctrines of Archimedes; but the mathematicians also retained a steady conviction of the mechanical property of the Centre of Gravity, namely, that all the weight of the body might be collected there, without any change in the mechanical results; a conviction which is closely connected with our fundamental conceptions of mechanical action. Such a principle, also, will enable us to determine the result of many simple mechanical arrangements; for instance, if a mathematician of those days had been

asked whether a solid ball could be made of such a form, that, when placed on a horizontal plane, it should go on rolling forwards without limit merely by the effect of its own weight, he would probably have answered, that it could not; for that the centre of gravity of the ball would seek the lowest position it could find, and that, when it had found this, the ball could have no tendency to roll any further. And, in making this assertion, the supposed reasoner would not be {317} anticipating any wider proof of the impossibility of a _perpetual motion_ drawn from principles subsequently discovered, but would be referring the question to certain fundamental convictions, which, whether put into Axioms or not, inevitably accompany our mechanical conceptions.

In the same way, Stevinus of Bruges, in 1586, when he published his _Beghinselen der Waaghconst_ (Principles of Equilibrium), had been asked why a loop of chain, hung over a triangular beam, could not, as he asserted it could not, go on moving round and round perpetually, by the action of its own weight, he would probably have answered, that the weight of the chain, if it produced motion at all, must have a tendency to bring it into some certain position, and that when the chain had reached this position, it would have no tendency to go any further; and thus he would have reduced the impossibility of such a perpetual motion, to the conception of gravity, as a force tending to produce equilibrium; a principle perfectly sound and correct.

Upon this principle thus applied, Stevinus did establish the fundamental property of the Inclined Plane. He supposed a loop of string, loaded with fourteen equal balls at equal distances, to hang over a triangular support which was composed of two inclined planes with a horizontal base, and whose sides, being unequal in the proportion of two to one, supported four and two balls respectively. He showed that this loop must hang at rest, because any motion would only bring it into the same condition in which it was at first; and that the festoon of eight balls which hung down below the triangle might be removed without disturbing the equilibrium; so that four balls on the longer plane would balance two balls on the shorter plane; or in other words, the weights would be as the lengths of the planes intercepted by the horizontal line.

Stevinus showed his firm possession of the truth contained in this principle, by deducing from it the properties of forces acting in oblique directions under all kinds of conditions; in short, he showed his entire ability to found upon it a complete doctrine of equilibrium; and upon his foundations, and without any additional support, the mathematical doctrines of Statics might have been carried to the highest pitch of perfection they have yet reached. The formation of the science was finished; the mathematical development and exposition of it were alone open to extension and change.

[2d Ed.] ["Simon Stevin of Bruges," as he usually designates himself in the title-page of his work, has lately become an object of general interest in his own country, and it has been resolved to erect a {318} statue in honor of him in one of the public places of his native city. He was born in 1548, as I learn from M. Quetelet's notice of him, and died in 1620. Montucla says that he died in 1633; misled apparently by the preface to Albert Girard's edition of Stevin's works, which was

published in 1634, and which speaks of a death which took place in the preceding year; but on examination it will be seen that this refers to Girard, not to Stevin.

I ought to have mentioned, in consideration of the importance of the proposition, that Stevin distinctly states the _triangle of forces_; namely, that three forces which act upon a point are in equilibrium when they are parallel and proportional to the three sides of any plane triangle. This includes the principle of the _Composition of Statical Forces_. Stevin also applies his principle of equilibrium to cordage, pulleys, funicular polygons, and especially to the bits of bridles; a branch of mechanics which he calls _Chalinothlipsis_.

He has also the merit of having seen very clearly, the distinction of statical and dynamical problems. He remarks that the question, "What force will _support_ a loaded wagon on an inclined plane? is a statical question, depending on simple conditions; but that the question, What force will _move_ the wagon? requires additional considerations to be introduced.

In Chapter iv. of this Book, I have noticed Stevin's share in the rediscovery of the _Laws of the Equilibrium of Fluids_. He distinctly explains the _hydrostatic paradox_, of which the discovery is generally ascribed to Pascal.

Earlier than Stevinus, Leonardo da Vinci must have a place among the discoverers of the Conditions of Equilibrium of Oblique Forces. He published no work on this subject; but extracts from his manuscripts have been published by Venturi, in his _Essai sur les Ouvrages Physico-Mathematiques de Leonard da Vinci, avec des Fragmens tirés de ses Manuscrits apportés d'Italie_, Paris, 1797: and by Libri, in his _Hist. des Sc. Math. en Italie_, 1839. I have also myself examined these manuscripts in the Royal Library at Paris.

It appears that, as early as 1499, Leonardo gave a perfectly correct statement of the proportion of the forces exerted by a cord which acts obliquely and supports a weight on a lever. He distinguishes between the real lever, and the _potential levers_, that is, the perpendiculars drawn from the centre upon the directions of the forces. This is quite sound and satisfactory. These views must in all probability have been sufficiently promulgated in Italy to influence the speculations of Galileo; {319} whose reasonings respecting the lever much resemble those of Leonardo.--Da Vinci also anticipated Galileo in _asserting_ that the time of descent of a body down an inclined plane is to the time of descent down its vertical length in the proportion of the length of the plane to the height. But this cannot, I think, have been more than a guess: there is no vestige of a proof given.]

The contemporaneous progress of the other branch of mechanics, the Doctrine of Motion, interfered with this independent advance of Statics; and to that we must now turn. We may observe, however, that true propositions respecting the composition of forces appear to have rapidly diffused themselves. The _Tractatus de Motu_ of Michael Varro of Geneva, already noticed, printed in 1584, had asserted, that the forces which balance each other, acting on the sides of a right-angled triangular wedge, are in the proportion of the sides of the triangle; and although this assertion does not appear to have been derived from a distinct idea of pressure, the author had

hence rightly deduced the properties of the wedge and the screw. And shortly after this time, Galileo also established the same results on different principles. In his Treatise _Delle Scienze Mecaniche_ (1592), he refers the Inclined Plane to the Lever, in a sound and nearly satisfactory manner; imagining a lever so placed, that the motion of a body at the extremity of one of its arms should be in the same direction as it is upon the plane. A slight modification makes this an unexceptionable proof.

_Sect._ 3.--_Prelude to the Science of Dynamics.--Attempts at the First Law of Motion._

WE have already seen, that Aristotle divided Motions into Natural and Violent. Cardan endeavored to improve this division by making three classes: _Voluntary_ Motion, which is circular and uniform, and which is intended to include the celestial motions; _Natural_ Motion, which is stronger towards the end, as the motion of a falling body,--this is in a straight line, because it is motion to an end, and nature seeks her ends by the shortest road; and thirdly, _Violent_ Motion, including in this term all kinds different from the former two. Cardan was aware that such Violent Motion might be produced by a very small force; thus he asserts, that a spherical body resting on a horizontal plane may be put in motion by any force which is sufficient to cleave the air; for which, however, he erroneously assigns as a reason, {320} the smallness of the point of contact.[4\6] But the most common mistake of this period was, that of supposing that as force is requisite to move a body, so a perpetual supply of force is requisite to keep it in motion. The whole of what Kepler called his "physical" reasoning, depended upon this assumption. He endeavored to discover the forces by which the motions of the planets about the sun might be produced; but, in all cases, he considered the velocity of the planet as produced by, and exhibiting the effect of, a force which acted in the direction of the motion. Kepler's essays, which are in this respect so feeble and unmeaning, have sometimes been considered as disclosing some distant anticipation of Newton's discovery of the existence and law of central forces. There is, however, in reality, no other connection between these speculations than that which arises from the use of the term _force_ by the two writers in two utterly different meanings. Kepler's Forces were certain imaginary qualities which appeared in the actual motion which the bodies had; Newton's Forces were causes which appeared by the change of motion: Kepler's Forces urged the bodies forwards; Newton's deflected the bodies from such a progress. If Kepler's Forces were destroyed, the body would instantly stop; if Newton's were annihilated, the body would go on uniformly in a straight line. Kepler compares the action of his Forces to the way in which a body might be driven round, by being placed among the sails of a windmill; Newton's Forces would be represented by a rope pulling the body to the centre. Newton's Force is merely mutual attraction; Kepler's is something quite different from this; for though he perpetually illustrates his views by the example of a magnet, he warns us that the sun differs from the magnet in this respect, that its force is not attractive, but directive.[5\6] Kepler's essays may with considerable reason be asserted to be an anticipation of the Vortices of Descartes; but they can with no propriety whatever be said to anticipate Newton's Dynamical Theory.

[Note 4\6: In speaking of the force which would draw a body up an inclined plane he observes, that "per communem animi sententiam," when the plane becomes horizontal, the requisite force is nothing.]

[Note 5\6: _Epitome Astron. Copern._ p. 176.]

The confusion of thought which prevented mathematicians from seeing the difference between producing and preserving motion, was, indeed, fatal to all attempts at progress on this subject. We have already noticed the perplexity in which Aristotle involved himself, by his endeavors to find a reason for the continued motion of a stone {321} after the moving power had ceased to act; and that he had ascribed it to the effect of the air or other medium in which the stone moves. Tartalea, whose _Nuova Scienza_ is dated 1550, though a good _pure_ mathematician, is still quite in the dark on mechanical matters. One of his propositions, in the work just mentioned, is (B. i. Prop. 3), "The more a heavy body recedes from the beginning, or approaches the end of violent motion, the slower and more inertly it goes;" which he applies to the horizontal motion of projectiles. In like manner most other writers about this period conceived that a cannon-ball goes forwards till it loses all its projectile motion, and then falls downwards. Benedetti, who has already been mentioned, must be considered as one of the first enlightened opponents of this and other Aristotelian errors or puzzles. In his _Speculationum Liber_ (Venice, 1585), he opposes Aristotle's mechanical opinions, with great expressions of respect, but in a very sweeping manner. His chapter xxiv. is headed, "Whether this eminent man was right in his opinion concerning violent and natural motion." And after stating the Aristotelian opinion just mentioned, that the body is impelled by the air, he says that the air must impede rather than impel the body, and that[6\6] "the motion of the body, separated from the mover, arises by a certain natural impression from the impetuosity (_ex impetuositate_) received from the mover." He adds, that in natural motions this _impetuosity_ continually increases by the continued action of the cause,--namely, the propension of going to the place assigned it by nature; and that thus the velocity increases as the body moves from the beginning of its path. This statement shows a clearness of conception with regard to the cause of accelerated motion, which Galileo himself was long in acquiring.

[Note 6\6: P. 184.]

Though Benedetti was thus on the way to the First Law of Motion,--that all motion is uniform and rectilinear, except so far as it is affected by extraneous forces;--this Law was not likely to be either generally conceived, or satisfactorily proved, till the other Laws of Motion, by which the action of Forces is regulated, had come into view. Hence, though a partial apprehension of this principle had preceded the discovery of the Laws of Motion, we must place the establishment of the principle in the period when those Laws were detected and established, the period of Galileo and his followers. {322}

## CHAPTER II.

INDUCTIVE EPOCH OF GALILEO.--DISCOVERY OF THE LAWS OF MOTION IN SIMPLE CASES.

_Sect._ 1.--_Establishment of the First Law of Motion._

AFTER mathematicians had begun to doubt or reject the authority of Aristotle, they were still some time in coming to the conclusion, that the distinction of Natural and Violent Motions was altogether untenable;--that the velocity of a body in motion increased or diminished in consequence of the action of extrinsic causes, not of any property of the motion itself;--and that the apparently universal fact, of bodies growing slower and slower, as if by their own disposition, till they finally stopped, from which Motions had been called Violent, arose from the action of external obstacles not immediately obvious, as the friction and the resistance of the air when a ball runs on the ground, and the action of gravity, when it is thrown upwards. But the truth to which they were at last led, was, that such causes would account for _all_ the diminution of velocity which bodies experience when apparently left to themselves and that without such causes, the motion of all bodies would go on forever, in a straight line and with a uniform velocity.

Who first announced this Law in a general form, it may be difficult to point out; its exact or approximate truth was necessarily taken for granted in all complete investigations on the subject of the laws of motion of falling bodies, and of bodies projected so as to describe curves. In Galileo's first attempt to solve the problem of falling bodies, he did not carry his analysis back to the notion of force, and therefore this law does not appear. In 1604 he had an erroneous opinion on this subject and we do not know when he was led to the true doctrine which he published in his _Discorso_, in 1638. In his third Dialogue he gives the instance of water in a vessel, for the purpose of showing that circular motion has a tendency to continue. And in his first Dialogue on the Copernican System[7\6] (published in 1630), he asserts {323} Circular Motion alone to be naturally uniform, and retains the distinction between Natural and Violent Motion. In the _Dialogues on Mechanics_, however, published in 1638, but written apparently at an earlier period, in treating of Projectiles,[8\6] he asserts the true Law. "Mobile super planum horizontale projectum mente concipio omni secluso impedimento; jam constat ex his quæ fusius alibi dicta sunt, illius motum equabilem et perpetuum super ipso plano futurum esse, si planum in infinitum extendatur." "Conceive a movable body upon a horizontal plane, and suppose all obstacles to motion to be removed; it is then manifest, from what has been said more at large in another place, that the body's motion will be uniform and perpetual upon the plane, if the plane be indefinitely extended." His pupil Borelli, in 1667 (in the treatise _De Vi Percussionis_), states the proposition generally, that "Velocity is, by its nature, uniform, and perpetual;" and this opinion appears to have been, at that time, generally diffused, as we find evidence in Wallis and others. It is commonly said that Descartes was the first to state this generally. His _Principia_ were published in 1644; but his proofs of this First Law of Motion are rather of a theological than of a mechanical kind. His reason for this Law is,[9\6] "the immutability and simplicity of the operation by which God preserves motion in matter. For he only preserves it precisely as it is in that moment in which he preserves it, taking no account of that

which may have been previously." Reasoning of this abstract and _à priori_ kind, though it may be urged in favor of true opinions after they have been inductively established, is almost equally capable of being called in on the side of error, as we have seen in the case of Aristotle's philosophy. We ought not, however, to forget that the reference to these abstract and _à priori_ principles is an indication of the absolute universality and necessity which we look for in complete Sciences, and a result of those faculties by which such Science is rendered possible, and suitable to man's intellectual nature.

[Note 7\6: Dial. **i. p. 40.]

[Note 8\6: **p. 141.]

[Note 9\6: _Princip._ p. 34.]

The induction by which the First Law of Motion is established, consists, as induction consists in all cases, in conceiving clearly the Law, and in perceiving the subordination of Facts to it. But the Law speaks of bodies not acted upon by any external force,--a case which never occurs in fact; and the difficulty of the step consisted in bringing all the common cases in which motion is gradually extinguished, under the notion of the action of a retarding force. In order to do this, {324} Hooke and others showed that, by diminishing the obvious resistances, the retardation also became less; and men were gradually led to a distinct appreciation of the Resistance, Friction, &c., which, in all terrestrial motions, prevent the Law from being evident; and thus they at last established by experiment a Law which cannot be experimentally exemplified. The natural uniformity of motion was proved by examining all kinds of cases in which motion was not uniform. Men culled the abstract Rule out of the concrete Experiment; although the Rule was, in every case, mixed with other Rules, and each Rule could be collected from the Experiment only by supposing the others known. The perfect simplicity which we necessarily seek for in a law of nature, enables us to disentangle the complexity which this combination appears at first sight to occasion.

The First Law of Motion asserts that the motion of a body, when left to itself will not only be uniform, but rectilinear also. This latter part of the law is indeed obvious of itself as soon as we conceive a body detached from all special reference to external points and objects. Yet, as we have seen, Galileo asserted that the naturally uniform motion of bodies was that which takes place in a circle. Benedetti, however, in 1585, had entertained sound notions on this subject. In commenting on Aristotle's question, why we obtain an advantage in throwing by using a sling, he says,[10\6] that the body, when whirled round, tends to go on in a straight line. In Galileo's second Dialogue, he makes one of his interlocutors (Simplicio), when appealed to on this subject, after thinking intently for a little while, give the same opinion; and the principle is, from this time, taken for granted by the authors who treat of the motion of projectiles. Descartes, as might be supposed, gives the same reason for this as for the other part of the law, namely, the immutability of the Deity.

[Note 10\6: "Corpus vellet recta iter peragere." _**Speculationum Liber_, p. 160.]

_Sect._ 2.--_Formation and Application of the Notion of Accelerating Force.--Laws of Falling Bodies._

WE have seen how rude and vague were the attempts of Aristotle and his followers to obtain a philosophy of bodies falling downwards or thrown in any direction. If the First Law of Motion had been clearly known, it would then, perhaps, have been seen that the way to understand and analyze the motion of any body, is to consider the {325} Causes of _change_ of motion which at each instant operate upon it; and thus men would have been led to the notion of Accelerating Forces, that is, Forces which act upon bodies already in motion, and accelerate, retard, or deflect their motions. It was, however, only after many attempts that they reached this point. They began by considering the _whole motion_ with reference to certain ill-defined abstract Notions, instead of considering, with a clear apprehension of the conditions of Causation, the _successive parts_ of which the motion consists. Thus, they spoke of the tendency of bodies to the Centre, or to their Own Place;--of Projecting Force, of Impetus, of Retraction;--with little or no profit to knowledge. The indistinctness of their notions may, perhaps, be judged of from their speculations concerning projectiles. Santbach,[11\6] in 1561, imagined that a body thrown with great velocity, as, for instance, a ball from a cannon, went in a straight line till all its velocity was exhausted, and then fell directly downwards. He has written a treatise on gunnery, founded on this absurd assumption. To this succeeded another doctrine, which, though not much more philosophical than the former, agreed much better with the phenomena. Nicolo Tartalea (_Nuova Scienza_, Venice, 1550; _Quesiti et Inventioni Diversi_, 1554) and **Gualter Rivius (_Architectura_, &c., Basil, 1582) represented the path of a cannon-ball as consisting, first of a straight line in the direction of the original projection, then of an arc of a circle in which it went on till its motion became vertical downwards, and then of a vertical line in which it continued to fall. The latter of these writers, however, was aware that the path must, from the first, be a curve; and treated it as a straight line, only because the curvature is very slight. Even Santbach's figure represents the path of the ball as partially descending before its final fall, but then it descends by _steps_, not in a curve. Santbach, therefore, did not conceive the _Composition_ of the effect of gravity with the existing motion, but supposed them to act alternately; Rivius, however, understood this Composition, and saw that gravity must act as a deflecting force at every point of the path. Galileo, in his second Dialogue,[12\6] makes Simplicius come to the same conclusion. "Since," he says, "there is nothing to support the body, when it quits that which projects it, it cannot be but that its proper gravity must operate," and it must immediately begin to decline downwards. {326}

[Note 11\6: _Problematum Astronomicorum et Geometricorum Sectiones_ vii. &c. &c. Auctore Daniele Santbach, Noviomago. Basileæ, 1561.]

[Note 12\6: P. 147.]

The Force of Gravity which thus produces deflection and curvature in the path of a body thrown _obliquely_, constantly increases the velocity of a body when it falls _vertically_ downwards. The universality of this increase was obvious, both from reasoning and in fact; the law of it could only be discovered by closer consideration;

and the full analysis of the problem required a distinct measure of the quantity of Accelerating Force. Galileo, who first solved this problem, began by viewing it as a question of fact, but conjectured the solution by taking for granted that the rule must be the simplest possible. "Bodies," he says,[13\6] "will fall in the most simple way, because Natural Motions are always the most simple. When a stone falls, if we consider the matter attentively, we shall find that there is no addition, no increase, of the velocity more simple than that which is always added in the same manner," that is, when equal additions take place in equal times; "which we shall easily understand if we attend to the close connection of motion and time." From this Law, thus assumed, he deduced that the spaces described from the beginning of the motion must be as the squares of the times; and, again, assuming that the laws of descent for balls rolling down inclined planes, must be the same as for bodies falling freely, he verified this conclusion by experiment.

[Note 13\6: _Dial. Sc._ iv. p. 91.]

It will, perhaps, occur to the reader that this argument, from the simplicity of the assumed law, is somewhat insecure. It is not always easy for us to discern what that greatest simplicity is, which nature adopts in her laws. Accordingly, Galileo was led wrong by this way of viewing the subject before he was led right. He at first supposed, that the Velocity which the body had acquired at any point must be proportional to the _Space_ described from the point where the motion began. This false law is as simple in its enunciation as the true law, that the Velocity is proportional to the _Time_: it had been asserted as the true law by M. Varro (_De Motu Tractatus_, Genevæ, 1584), and by Baliani, a gentleman of Genoa, who published it in 1638. It was, however, soon rejected by Galileo, though it was afterwards taken up and defended by Casræus, one of Galileo's opponents. It so happens, indeed, that the false law is not only at variance with fact, but with itself: it involves a mathematical self-contradiction. This circumstance, however, was accidental: it would be easy to state laws of the increase of velocity which should be simple, and yet false in fact, though quite possible in their own nature. {327}

The Law of Velocity was hitherto, as we have seen, treated as a law of phenomena, without reference to the Causes of the law. "The cause of the acceleration of the motions of falling bodies is not," Galileo observes, "a necessary part of the investigation. Opinions are different. Some refer it to the approach to the centre; others say that there is a certain extension of the centrical medium, which, closing behind the body, pushes it forwards. For the present, it is enough for us to demonstrate certain properties of Accelerated Motion, the acceleration being according to the very simple Law, that the Velocity is proportional to the Time. And if we find that the properties of such motion are verified by the motions of bodies descending freely, we may suppose that the assumption agrees with the laws of bodies falling freely by the action of gravity."[14\6]

[Note 14\6: Gal. _Op._ iii. 91, 92.]

It was, however, an easy step to conceive this acceleration as caused by the continual action of Gravity. This account had already been given by Benedetti, as we

have seen. When it was once adopted, Gravity was considered as a _constant_ or _uniform_ force; on this point, indeed, the adherents of the law of Galileo and of that of Casræus were agreed; but the question was, what _is_ a Uniform Force? The answer which Galileo was led to give was obviously this;--_that_ is a Uniform Force which generates equal velocities in equal successive times; and this principle leads at once to the doctrine, that Forces are to be compared by comparing the Velocities generated by them in equal times.

Though, however, this was a consequence of the rule by which Gravity is represented as a Uniform Force, the subject presents some difficulty at first sight. It is not immediately obvious that we may thus measure forces by the Velocity _added_ in a given time, without taking into account the velocity they have already. If we communicate velocity to a body by the hand or by a spring, the effect we produce in a second of time is lessened, when the body has already a velocity which withdraws it from the pressure of the agent. But it appears that this is not so in the case of gravity; the velocity added in one second is the same, whatever downward motion the body already possesses. A body falling from rest acquires a velocity, in one second, of thirty-two feet; and if a cannon-ball were shot downwards with a velocity of 1000 feet a second, it would equally, at the end of one second, have received an accession of 32 feet to its velocity.

This conception of Gravity as a Uniform Force,--as constantly and {328} equally _increasing_ the velocity of a descending body,--will become clear by a little attention; but it undoubtedly presents difficulty at first. Accordingly, we find that Descartes did not accept it. "It is certain," he says, "that a stone is not equally disposed to receive a new motion or increase of velocity when it is already moving very quickly, and when it is moving slowly."

Descartes showed, by other expressions, that he had not caught hold of the true notion of accelerating force. Thus, he says in a letter to Mersenne, "I am astonished at what you tell me, of having found, by experiment, that bodies thrown up in the air take neither more nor less time to rise than to fall again; and you will excuse me if I say that I look upon the experiment as a very difficult one to make accurately." Yet it is clear from the Notion of a Constant Force that (omitting the resistance of the air) this equality must take place; for the Force which will gradually destroy the whole velocity in a certain time in ascending, will, in the same time, generate again the same velocity by the same gradations inverted; and therefore the same space will be passed over in the same time in the descent and in the ascent.

Another difficulty arose from a necessary consequence of the Laws of Falling Bodies thus established;--the proposition, namely, that in acquiring its motion, a body passes through every intermediate degree of velocity, from the smallest conceivable, up to that which it at last acquires. When a body falls from rest, it begins to fall with _no_ velocity; the velocity increases with the time; and in one-thousandth part of a second, the body has only acquired one-thousandth part of the velocity which it has at the end of one second.

This is certain, and manifest on consideration; yet there was at first much difficulty raised on the subject of this assertion; and disputes took place concerning the velocity with which a body _begins_ to fall. On this subject also Descartes did not form clear notions. He writes to a correspondent, "I have been revising my notes on Galileo, in which I have not said expressly that falling bodies do not pass through every degree of slowness, but I said that this cannot be known without knowing what Weight is, which comes to the same thing; as to your example, I grant that it proves that every degree of velocity is infinitely divisible, but not that a falling body actually passes through all these divisions."

The Principles of the Motion of Falling Bodies being thus established by Galileo, the Deduction of the principal mathematical consequences was, as is usual, effected with great rapidity, and is to be found {329} in his works, and in those of his scholars and successors. The motion of bodies falling freely was, however, in such treatises, generally combined with the motion of bodies Falling along Inclined Planes; a part of the theory of which we have still to speak.

The Notion of Accelerating Force and of its operation, once formed, was naturally applied in other cases than that of bodies falling freely. The different velocities with which heavy and light bodies fall were explained by the different resistance of the air, which diminishes the accelerating force;[15\6] and it was boldly asserted, that in a vacuum a lock of wool and a piece of lead would fall equally quickly. It was also maintained[16\6] that any falling body, however large and heavy, would always have its velocity in some degree diminished by the air in which it falls, and would at last be reduced to a state of uniform motion, as soon as the resistance upwards became equal to the accelerating force downwards. Though the law of progress of a body to this limiting velocity was not made out till the _Principia_ of Newton appeared, the views on which Galileo made this assertion are perfectly sound, and show that he had clearly conceived the nature and operation of accelerating and retarding force.

[Note 15\6: Galileo, iii. 43.]

[Note 16\6: iii. 54.]

When Uniform Accelerating Forces had once been mastered, there remained only mathematical difficulties in the treatment of Variable Forces. A Variable Force was measured by the _Limit_ of the increment of the Velocity, compared with the increment of the Time; just as a Variable Velocity was measured by the Limit of the increment of the Space compared with that of the Time.

With this introduction of the Notion of Limits, we are, of course, led to the Higher Geometry, either in its geometrical or its analytical form. The general laws of bodies falling by the action of any Variable Forces were given by Newton in the Seventh Section of the _Principia_. The subject is there, according to Newton's preference of geometrical methods, treated by means of the Quadrature of Curves; the Doctrine of Limits being exhibited in a peculiar manner in the First Section of the work, in order to prepare the way for such applications of it. Leibnitz, the Bernouillis, Euler, and since their time, many other mathematicians, have treated such questions by means of the analytical method of limits, the Differential Calculus. The Rectilinear Motion of

bodies acted upon by variable forces is, of course, a simpler problem than their Curvilinear Motion, to which we have now to proceed. But it {330} may be remarked that Newton, having established the laws of Curvilinear Motion independently, has, in a great part of his Seventh Section, deduced the simpler case of the Rectilinear Motion from the move complex problem, by reasonings of great ingenuity and beauty.

_Sect._ 3.--_Establishment of the Second Law of Motion.--Curvilinear Motions._

A SLIGHT degree of distinctness in men's mechanical notions enabled them to perceive, as we have already explained, that a body which traces a curved line must be urged by some force, by which it is constantly made to deviate from that rectilinear path, which it would pursue if acted upon by no force. Thus, when a body is made to describe a circle, as when a stone is whirled round in a sling, we find that the string does exert such a force on the stone; for the string is stretched by the effort, and if it be too slender, it may thus be broken. This _centrifugal force_ of bodies moving in circles was noticed even by the ancients. The effect of force to produce curvilinear motion also appears in the paths described by projectiles. We have already seen that though Tartalea did not perceive this correctly, Rivius, about the same time, did.

To see that a transverse force would produce a curve, was one step; to determine what the curve is, was another step, which involved the discovery of the Second Law of Motion. This step was made by Galileo. In his _Dialogues on Motion_, he asserts that a body projected horizontally will retain a uniform motion in the horizontal direction, and will have, compounded with this, a uniformly accelerated motion downwards, that is, the motion of a body falling vertically from rest; and will thus describe the curve called a parabola.

The Second Law of Motion consists of this assertion in a general form;--namely, that in all cases the motion which the force will produce is compounded with the motion which the body previously has. This was not obvious; for Cardan had maintained,[17\6] that "if a body is moved by two motions at once, it will come to the place resulting from their composition slower than by either of them." The proof of the truth of the law to Galileo's mind was, so far as we collect from the Dialogue itself, the simplicity of the supposition, and his clear perception of the causes which, in some cases, produced an obvious deviation in practice {331} from this theoretical result. For it may be observed, that the curvilinear paths ascribed to military projectiles by Rivius and Tartalea, and by other writers who followed them, as Digges and Norton in our own country, though utterly different from the theoretical form, the parabola, do, in fact, approach nearer the true paths of a cannon or musket ball than a parabola would do; and this approximation more especially exists in that which at first sight appears most absurd in the old theory; namely, the assertion that the ball, which ascends in a sloping direction, finally descends vertically. In consequence of the resistance of the air, this is really the path of a projectile; and when the velocity is very great, as in military projectiles, the deviation from the parabolic form is very manifest. This cause of discrepancy between the theory, which does not take resistance into the account, and the fact, Galileo perceived; and accordingly he says,[18\6] that the velocities of the projectiles, in such cases, may be considered as

excessive and supernatural. With the due allowance to such causes, he maintained that his theory was verified, and might be applied in practice. Such practical applications of the doctrine of projectiles no doubt had a share in establishing the truth of Galileo's views. We must not forget, however, that the full establishment of this second law of motion was the result of the theoretical and experimental discussions concerning the motion of the earth: its fortunes were involved in those of the Copernican system; and it shared the triumph of that doctrine. This triumph was already decisive, indeed, in the time of Galileo, but not complete till the time of Newton.

[Note 17\6: _Op._ vol. iv. p. 490.]

[Note 18\6: _Op._ vol. iii. p. 147.]

_Sect._ 4.--_Generalization of the Laws of Equilibrium.--Principle of Virtual Velocities._

IT was known, even as early as Aristotle, that the two weights which balance each other on the lever, if they move at all, move with velocities which are in the inverse proportions of the weights. The peculiar resources of the Greek language, which could state this relation of inverse proportionality in a single word (ἀντιπέπονθεν), fixed it in men's minds, and prompted them to generalize from this property. Such attempts were at first made with indistinct ideas, and on conjecture only, and had, therefore, no scientific value. This is the judgment which we must pass on the book of Jordanus Nemorarius, which {332} we have already mentioned. Its reasonings are professedly on Aristotelian principles, and exhibit the common Aristotelian absence of all distinct mechanical ideas. But in Varro, whose _Tractatus de Motu_ appeared in 1584, we find the principle, in a general form, not satisfactorily proved, indeed, but much more distinctly conceived. This is his first theorem: "Duarum virium connexarum quarum (si moveantur) motus erunt ipsis ἀντιπεπονθῶς proportionales, neutra alteram movebit, sed equilibrium facient." The proof offered of this is, that the resistance to a force is as the motion produced; and, as we have seen, the theorem is rightly applied in the example of the wedge. From this time it appears to have been usual to prove the properties of machines by means of this principle. This is done, for instance, in _Les Raisons des Forces Mouvantes_, the production of Solomon de Caus, engineer to the Elector Palatine, published at Antwerp in 1616; in which the effect of Toothed-Wheels and of the Screw is determined in this manner, but the Inclined Plane is not treated of. The same is the case in Bishop Wilkins's _Mathematical Magic_, in 1648.

When the true doctrine of the Inclined Plane had been established, the laws of equilibrium for all the simple machines or Mechanical Powers, as they had usually been enumerated in books on Mechanics, were brought into view; for it was easy to see that the _Wedge_ and the _Screw_ involved the same principle as the _Inclined Plane_, and the _Pulley_ could obviously be reduced to the _Lever_. It was, also, not difficult for a person with clear mechanical ideas to perceive how any other combination of bodies, on which pressure and traction are exerted, may be reduced to these simple machines, so as to disclose the relation of the forces. Hence by the

discovery of Stevinus, all problems of equilibrium were essentially solved.

The conjectural generalization of the property of the lever, which we have just mentioned, enabled mathematicians to express the solution of all these problems by means of one proposition. This was done by saying, that in raising a weight by any machine, we _lose_ in Time what we _gain_ in Force; the weight raised moves as much _slower_ than the power, as it is _larger_ than the power. This was explained with great clearness by Galileo, in the preface to his _Treatise on Mechanical Science_, published in 1592.

The motions, however, which we here suppose the parts of the machine to have, are not motions which the forces produce; for at present we are dealing with the case in which the forces balance each other, and therefore produce no motion. But we ascribe to the {333} Weights and Powers hypothetical motions, arising from some other cause; and then, by the construction of the machine, the velocities of the Weights and Powers must have certain definite ratios. These velocities, being thus hypothetically supposed and not actually produced, are called _Virtual_ Velocities. And the general law of equilibrium is, that in any machine, the Weights which balance each other, are reciprocally to each other as their Virtual Velocities. This is called the _Principle of Virtual Velocities_.

This Principle (which was afterwards still further generalized) is, by some of the admirers of Galileo, dwelt upon as one of his great services to Mechanics. But if we examine it more nearly, we shall see that it has not much importance in our history. It is a generalization, but a generalization established rather by enumeration of cases, than by any induction proceeding upon one distinct Idea, like those generalizations of Facts by which Laws are primarily established. It rather serves verbally to conjoin Laws previously known, than to exhibit a connection in them: it is rather a help for the memory than a proof for the reason.

The Principle of Virtual Velocities is so far from implying any clear possession of mechanical ideas, that any one who knows the property of the Lever, whether he is capable of seeing the reason for it or not, can see that the greater weight moves slower in the exact proportion of its greater magnitude. Accordingly, Aristotle, whose entire want of sound mechanical views we have shown, has yet noticed this truth. When Galileo treats of it, instead of offering any reasons which could independently establish this principle, he gives his readers a number of analogies and illustrations, many of them very loose ones. Thus the raising a great weight by a small force, he illustrates by supposing the weight broken into many small parts, and conceiving those parts raised one by one. By other persons, the analogy, already intimated, of gain and loss is referred to as an argument for the principle in question. Such images may please the fancy, but they cannot be accepted as mechanical reasons.

Since Galileo neither first enunciated this rule, nor ever proved it as an independent principle of Mechanics, we cannot consider the discovery of it as one of his mechanical achievements. Still less can we compare his reference to this principle with Stevinus's proof of the Inclined Plane; which, as we have seen, was rigorously inferred from the sound axiom, that a body cannot put itself in motion. If we were to

assent to the really self-evident axioms of Stevinus, only in virtue {334} of the unproved verbal generalization of Galileo, we should be in great danger of allowing ourselves to be referred successively from one truth to another, without any reasonable hope of ever arriving at any thing ultimate and fundamental.

But though this Principle of Virtual Velocity cannot be looked upon as a great discovery of Galileo, it is a highly useful rule; and the various forms under which he and his successors urged it, tended much to dissipate the vague wonder with which the effects of machines had been looked upon; and thus to diffuse sounder and clearer notions on such subjects.

The Principle of Virtual Velocities also affected the progress of mechanical science in another way: it suggested some of the analogies by the aid of which the Third Law of Motion was made out; leading to the adoption of the notion of _Momentum_ as the arithmetical product of weight and velocity. Since on a machine on which a weight of two pounds at one part balances three pounds at another part, the former weight would move through three inches while the latter would move through two inches; we see (since three multiplied into two is equal to two multiplied into three) that the _Product_ of the weight and the velocity is the same for the two balancing weights; and if we call this Product _Momentum_, the Law of Equilibrium is, that when two weights balance on a machine, the Momentum of the two would be the same, if they were put in motion.

The Notion of Momentum was here employed in connection with Virtual Velocities; but it also came under consideration in treating of Actual Velocities, as we shall soon see.

_Sect._ 5.--_Attempts at the Third Law of Motion.--Notion of Momentum._

IN the questions we have hitherto had to consider respecting Motion, no regard is had to the Size of the body moved, but only to the Velocity and Direction of the motion. We must now trace the progress of knowledge respecting the mode in which the Mass of the body influences the effect of Force. This is a more difficult and complex branch of the subject; but it is one which requires to be noticed, as obviously as the former. Questions belonging to this department of Mechanics, as well as to the others, occur in Aristotle's Mechanical Problems. "Why," says he, "is it, that neither very small nor very large bodies go far when we throw them; but, in order that this may {335} happen, the thing thrown must have a certain proportion to the agent which throws it? Is it that what is thrown or pushed must react[19\6] against that which pushes it; and that a body so large as not to yield at all, or so small as to yield entirely, and not to react, produces no throw or push?" The same confusion of ideas prevailed after his time; and mechanical questions were in vain discussed by means of general and abstract terms, employed with no distinct and steady meaning; such as _impetus_, _power_, _momentum_, _virtue_, _energy_, and the like. From some of these speculations we may judge how thorough the confusion in men's heads had become. Cardan perplexes himself with the difficulty, already mentioned, of the comparison of the forces of bodies at rest and in motion. If the Force of a body depends on its velocity, as it appears to do, how is it that a body at rest has any Force

at all, and how can it resist the slightest effort, or exert any pressure? He flatters himself that he solves the question, by asserting that bodies at rest have an occult motion. "Corpus movetur occulto motu quiescendo."--Another puzzle, with which he appears to distress himself rather more wantonly, is this: "If one man can draw half of a certain weight, and another man also one half; when the two act together, these proportions should be compounded; so that they ought to be able to draw one half of one half, or one quarter only." The talent which ingenious men had for getting into such perplexities, was certainly at one time very great. Arriaga,[20\6] who wrote in 1639, is troubled to discover how several flat weights, lying one upon another on a board, should produce a greater pressure than the lowest one alone produces, since that alone touches the board. Among other solutions, he suggests that the board affects the upper weight, which it does not touch, by determining its _ubication_, or _whereness_.

[Note 19\6: ἀντερείδειν.]

[Note 20\6: Rod. de Arriaga, _Cursus Philosophicus_. Paris, 1639.]

Aristotle's doctrine, that a body ten times as heavy as another, will fall ten times as fast, is another instance of the confusion of Statical and Dynamical Forces: the Force of the greater body, while _at rest_, is ten times as great as that of the other; but the Force as measured by the _velocity_ produced, is equal in the two cases. The two bodies would fall downwards with the same rapidity, except so far as they are affected by accidental causes. The merit of proving this by experiment, and thus refuting the Aristotelian dogma, is usually ascribed to Galileo, who made his experiment from the famous leaning tower of Pisa, about 1590. But others about the same time had not {336} overlooked so obvious a fact--F. Piccolomini, in his _Liber Scientiæ de Natura_, published at Padua, in 1597, says, "On the subject of the motion of heavy and light bodies, Aristotle has put forth various opinions, which are contrary to sense and experience, and has delivered rules concerning the proportion of quickness and slowness, which are palpably false. For a stone twice as great does _not_ move twice as fast." And Stevinus, in the Appendix to his Statics, describes his having made the experiment, and speaks with great correctness of the apparent deviations from the rule, arising from the resistance of the air. Indeed, the result followed by very obvious reasoning; for ten bricks, in contact with each other, side by side, would obviously fall in the same time as one; and these might be conceived to form a body ten times as large as one of them. Accordingly, Benedetti, in 1585, reasons in this manner with regard to bodies of different size, though he retains Aristotle's error as to the different velocity of bodies of different density.

The next step in this subject is more clearly due to Galileo; he discovered the true proportion which the Accelerating Force of a body falling down an inclined plane bears to the Accelerating Force of the same body falling freely. This was at first a happy conjecture; it was then confirmed by experiments, and, finally, after some hesitation, it was referred to its true principle, the Third Law of Motion, with proper elementary simplicity. The Principle here spoken of is this:--that for the same body, the Dynamical effect of force is as the Statical effect; that is, the Velocity which any force generates in a given time when it puts the body in motion, is proportional to the

Pressure which the same force produces in a body at rest. The Principle, so stated, appears very simple and obvious; yet this was not the form in which it suggested itself either to Galileo or to other persons who sought to prove it. Galileo, in his _Dialogues on Motion_, assumes, as his fundamental proposition on this subject, one much less evident than that we have quoted, but one in which that is involved. His Postulate is,[21\6] that when the same body falls down different planes of the same height, the velocities acquired are equal. He confirms and illustrates this by a very ingenious experiment on a pendulum, showing that the weight swings to the same height whatever path it be compelled to follow. Torricelli, in his treatise published 1644, says that he had heard that Galileo had, towards the end of his life, proved his {337} assumption, but that, not having seen the proof, he will give his own. In this he refers us to the right principle, but appears not distinctly to conceive the proof, since he estimates _momentum_ indiscriminately by the statical Pressure of a body, and by its Velocity when in motion; as if these two quantities were self-evidently equal. Huyghens, in 1673, expresses himself dissatisfied with the proof by which Galileo's assumption was supported in the later editions of his works. His own proof rests on this principle;--that if a body fall down one inclined plane, and proceed up another with the velocity thus acquired, it cannot, under any circumstances, ascend to a higher position than that from which it fell. This principle coincides very nearly with Galileo's experimental illustration. In truth, however, Galileo's principle, which Huyghens thus slights, may be looked upon as a satisfactory statement of the true law namely, that, in the same body, the velocity produced is as the pressure which produces it. "We are agreed," he says,[22\6] "that, in a movable body, the _impetus_, _energy_, _momentum_, or _propension to motion_, is as great as is the _force_ or _least resistance_ which suffices to _support_ it." The various terms here used, both for dynamical and statical Force, show that Galileo's ideas were not confused by the ambiguity of any one term, as appears to have happened to some mathematicians. The principle thus announced, is, as we shall see, one of great extent and value; and we read with interest the circumstances of its discovery, which are thus narrated.[23\6] When Viviani was studying with Galileo, he expressed his dissatisfaction at the want of any clear reason for Galileo's postulate respecting the equality of velocities acquired down inclined planes of the same heights; the consequence of which was, that Galileo, as he lay, the same night, sleepless through indisposition, discovered the proof which he had long sought in vain, and introduced it in the subsequent editions. It is easy to see, by looking at the proof, that the discoverer had had to struggle, not for intermediate steps of reasoning between remote notions, as in a problem of geometry, but for a clear possession of ideas which were near each other, and which he had not yet been able to bring into contact, because he had not yet a sufficiently firm grasp of them. Such terms as Momentum and Force had been sources of confusion from the time of Aristotle; and it required considerable steadiness of thought to compare the forces of bodies at rest and in motion under the obscurity and vacillation thus produced. {338}

[Note 21\6: _Opere_, iii. 96.]

[Note 22\6: Galileo, _Op._ iii. 104.]

[Note 23\6: Drinkwater, _Life of Galileo_, p. 59.]

The term _Momentum_ had been introduced to express the force of bodies in motion, before it was known what that effect was. Galileo, in his _Discorso intorno alle Cose che stanno in su l' Acqua_, says, that "Momentum is the force, efficacy, or virtue, with which the motion moves and the body moved resists, depending not upon weight only, but upon the velocity, inclination, and any other cause of such virtue." When he arrived at more precision in his views, he determined, as we have seen, that, in the same body, the Momentum is _proportional_ to the Velocity; and, hence it was easily seen that in different bodies it was proportional to the Velocity and Mass jointly. The principle thus enunciated is capable of very extensive application, and, among other consequences, leads to a determination of the results of the mutual Percussion of Bodies. But though Galileo, like others of his predecessors and contemporaries, had speculated concerning the problem of Percussion, he did not arrive at any satisfactory conclusion; and the problem remained for the mathematicians of the next generation to solve.

We may here notice Descartes and his Laws of Motion, the publication of which is sometimes spoken of as an important event in the history of Mechanics. This is saying far too much. The _Principia_ of Descartes did little for physical science. His assertion of the Laws of Motion, in their most general shape, was perhaps an improvement in form; but his Third Law is false in substance. Descartes claimed several of the discoveries of Galileo and others of his contemporaries; but we cannot assent to such claims, when we find that, as we shall see, he did not understand, or would not apply, the Laws of Motion when he had them before him. If we were to compare Descartes with Galileo, we might say, that of the mechanical truths which were easily attainable in the beginning of the seventeenth century, Galileo took hold of as many, and Descartes of as few, as was well possible for a man of genius.

[2d Ed.] [The following remarks of M. Libri appear to be just. After giving an account of the doctrines put forth on the subject of Astronomy, Mechanics, and other branches of science, by Leonardo da Vinci, Fracastoro, Maurolycus, Commandinus, Benedetti, he adds (_Hist. des Sciences Mathématiques en Italie_, t. iii. p. 131): "This short analysis is sufficient to show that, at the period at which we are arrived, Aristotle no longer reigned unquestioned in the Italian Schools. If we had to write the history of philosophy, we should prove by a multitude of facts that it was the Italians who overthrew the ancient idol of philosophers. Men go on incessantly repeating that the {339} struggle was begun by Descartes, and they proclaim him the legislator of modern philosophers. But when we examine the philosophical writings of Fracastoro, of Benedetti, of Cardan, and above all, those of Galileo; when we see on all sides energetic protests raised against the peripatetic doctrines; we ask, what there remained for the inventor of vortices to do, in overturning the natural philosophy of Aristotle? In addition to this, the memorable labors of the School of Cosenza, of Telesius, of Giordano Bruno, of Campanella; the writings of Patricius, who was, besides, a good geometer; of Nizolius, whom Leibnitz esteemed so highly, and of the other metaphysicians of the same epoch,--prove that the ancient philosophy had already lost its empire on that side the Alps, when Descartes threw himself upon the

enemy now put to the rout. The yoke was cast off in Italy, and all Europe had only to follow the example, without its being necessary to give a new impulse to real science."

In England, we are accustomed to hear Francis Bacon, rather than Descartes, spoken of as the first great antagonist of the Aristotelian schools, and the legislator of modern philosophy. But it is true, both of one and the other, that the overthrow of the ancient system had been effectively begun before their time by the practical discoverers here mentioned, and others who, by experiment and reasoning, established truths inconsistent with the received Aristotelian doctrines. Gilbert in England, Kepler in Germany, as well as Benedetti and Galileo in Italy, gave a powerful impulse to the cause of real knowledge, before the influence of Bacon and Descartes had produced any general effect. What Bacon really did was this;--that by the august image which he presented of a future Philosophy, the rival of the Aristotelian, and far more powerful and extensive, he drew to it the affections and hopes of all men of comprehensive and vigorous minds, as well as of those who attended to special trains of discovery. He announced a New Method, not merely a correction of special current errors; he thus converted the Insurrection into a Revolution, and established a new philosophical Dynasty. Descartes had, in some degree, the same purpose; and, in addition to this, he not only proclaimed himself the author of a New Method, but professed to give a complete system of the results of the Method. His physical philosophy was put forth as complete and demonstrative, and thus involved the vices of the ancient dogmatism. Telesius and Campanella had also grand notions of an entire reform in the method of philosophizing, as I have noticed in the _Philosophy of the Inductive Sciences_, Book xii.] {340}

## CHAPTER III.

SEQUEL TO THE EPOCH OF GALILEO.--PERIOD OF VERIFICATION AND DEDUCTION.

THE evidence on which Galileo rested the truth of the Laws of Motion which he asserted, was, as we have seen, the simplicity of the laws themselves, and the agreement of their consequences with facts; proper allowances being made for disturbing causes. His successors took up and continued the task of making repeated comparisons of the theory with practice, till no doubt remained of the exactness of the fundamental doctrines: they also employed themselves in simplifying, as much as possible, the mode of stating these doctrines, and in tracing their consequences in various problems by the aid of mathematical reasoning. These employments led to the publication of various Treatises on Falling Bodies, Inclined Planes, Pendulums, Projectiles, Spouting Fluids, which occupied a great part of the seventeenth century.

The authors of these treatises may be considered as the School of Galileo. Several of them were, indeed, his pupils or personal friends. Castelli was his disciple and astronomical assistant at Florence, and afterwards his correspondent. Torricelli was at first a pupil of Castelli, but became the inmate and amanuensis of Galileo in 1641, and succeeded him in his situation at the court of Florence on his death, which took

place a few months afterwards. Viviani formed one of his family during the three last years of his life; and surviving him and his contemporaries (for Viviani lived even into the eighteenth century), has a manifest pleasure and pride in calling himself the last of the disciples of Galileo. Gassendi, an eminent French mathematician and professor, visited him in 1628; and it shows us the extent of his reputation when we find Milton referring thus to his travels in Italy:[24\6] "There it was that I found and visited the famous Galileo, grown old, a prisoner in the Inquisition, for thinking in astronomy otherwise than the Franciscan and Dominican licensers thought."

[Note 24\6: _Speech for the Liberty of Unlicensed Printing._]

Besides the above writers, we may mention, as persons who pursued and illustrated Galileo's doctrines, Borelli, who was professor at Florence and Pisa; Mersenne, the correspondent of Descartes, who was {341} professor at Paris; Wallis, who was appointed Savilian professor at Oxford in 1649, his predecessor being ejected by the parliamentary commissioners. It is not necessary for us to trace the progress of purely mathematical inventions, which constitute a great part of the works of these authors; but a few circumstances may be mentioned.

The question of the proof of the Second Law of Motion was, from the first, identified with the controversy respecting the truth of the Copernican System; for this law supplied the true answer to the most formidable of the objections against the motion of the earth; namely, that if the earth were moving, bodies which were dropt from an elevated object would be left behind by the place from which they fell. This argument was reproduced in various forms by the opponents of the new doctrine; and the answers to the argument, though they belong to the history of Astronomy, and form part of the Sequel to the Epoch of Copernicus, belong more peculiarly to the history of Mechanics, and are events in the sequel to the Discoveries of Galileo. So far, indeed, as the mechanical controversy was concerned, the advocates of the Second Law of Motion appealed, very triumphantly, to experiment. Gassendi made many experiments on this subject publicly, of which an account is given in his _Epistolæ tres de Motu Impresso a Motore Translato_[25\6] It appeared in these experiments, that bodies let fall downwards, or cast upwards, forwards, or backwards, from a ship, or chariot, or man, whether at rest, or in any degree of motion, had always the same motion relatively to the _motor_. In the application of this principle to the system of the world, indeed, Gassendi and other philosophers of his time were greatly hampered; for the deference which religious scruples required, did not allow them to say that the earth really moved, but only that the physical reasons against its motion were invalid. This restriction enabled Riccioli and other writers on the geocentric side to involve the subject in metaphysical difficulties; but the conviction of men was not permanently shaken by these, and the Second Law of Motion was soon assumed as unquestioned.

[Note 25\6: Mont. ii. 199.]

The Laws of the Motion of Falling Bodies, as assigned by Galileo, were confirmed by the reasonings of Gassendi and Fermat, and the experiments of Riccioli and Grimaldi; and the effect of resistance was pointed out by **Mersenne and Dechales.

The parabolic motion of Projectiles was more especially illustrated by experiments on the jet which spouts from an orifice in a vessel full of fluid. This mode of experimenting {342} is well adapted to attract notice, since the curve described, which is transient and invisible in the case of a single projectile, becomes permanent and visible when we have a continuous stream. The doctrine of the motions of fluids has always been zealously cultivated by the Italians. Castelli's treatise, _Della Misura dell' Acque Corrente_ (1638), is the first work on this subject, and Montucla with justice calls him "the creator of a new branch of hydraulics;"[26\6] although he mistakenly supposed the velocity of efflux to be as the depth of the orifice from the surface. **Mersenne and Torricelli also pursued this subject, and after them, many others.

[Note 26\6: Mont. ii. 201.]

Galileo's belief in the near approximation of the curve described by a cannon-ball or musket-ball to the theoretical parabola, was somewhat too obsequiously adopted by succeeding practical writers on artillery. They underrated, as he had done, the effect of the resistance of the air, which is in effect so great as entirely to change the form and properties of the curve. Notwithstanding this, the parabolic theory was employed, as in Anderson's _Art of Gunnery_ (1674); and Blondel, in his _Art de jeter les Bombes_ (1688), not only calculated Tables on this supposition, but attempted to answer the objections which had been made respecting the form of the curve described. It was not till a later period (1740), when Robins made a series of careful and sagacious experiments on artillery, and when some of the most eminent mathematicians calculated the curve, taking into account the resistance, that the Theory of Projectiles could be said to be verified in fact.

The Third Law of Motion was still in some confusion when Galileo died, as we have seen. The next great step made in the school of Galileo was the determination of the Laws of the motions of bodies in their Direct Impact, so far as this impact affects the motion of translation. The difficulties of the problem of Percussion arose, in part, from the heterogeneous nature of Pressure (of a body at rest), and Momentum (of a body in motion); and, in part, from mixing together the effects of percussion on the parts of a body, as, for instance, cutting, bruising, and breaking, with its effect in moving the whole.

The former difficulty had been seen with some clearness by Galileo himself. In a posthumous addition to his _Mechanical Dialogues_, he says, "There are two kinds of resistance in a movable body, one internal, as when we say it is more difficult to lift a weight of a thousand pounds than a weight of a hundred; another respecting space, as {343} when we say that it requires more force to throw a stone one hundred paces than fifty."[27\6] Reasoning upon this difference, he comes to the conclusion that "the Momentum of percussion is infinite, since there is no resistance, however great, which is not overcome by a force of percussion, however small."[28\6] He further explains this by observing that the resistance to percussion must occupy some portion of time, although this portion may be insensible. This correct mode of removing the apparent incongruity of continuous and instantaneous force, was a material step in the solution of the problem.

[Note 27\6: _Op._ iii. 210.]

[Note 28\6: iii. 211.]

The Laws of the mutual Impact of bodies were erroneously given by Descartes in his _Principia_; and appear to have been first correctly stated by Wren, Wallis, and Huyghens, who about the same time (1669) sent papers to the Royal Society of London on the subject. In these solutions, we perceive that men were gradually coming to apprehend the Third Law of Motion in its most general sense; namely, that the Momentum (which is proportional to the Mass of the body and its Velocity jointly) may be taken for the measure of the effect; so that this Momentum is as much diminished in the striking body by the resistance it experiences, as it is increased in the body struck by the Impact. This was sometimes expressed by saying that "the Quantity of Motion remains unaltered," _Quantity of Motion_ being used as synonymous with _Momentum_. Newton expressed it by saying that "Action and Reaction are equal and opposite," which is still one of the most familiar modes of expressing the Third Law of Motion.

In this mode of stating the Law, we see an example of a propensity which has prevailed very generally among mathematicians; namely, a disposition to present the fundamental laws of rest and of motion as if they were equally manifest, and, indeed, identical. The close analogy and connection which exists between the principles of equilibrium and of motion, often led men to confound the evidence of the two; and this confusion introduced an ambiguity in the use of words, as we have seen in the case of Momentum, Force, and others. The same may be said of _Action_ and _Reaction_, which have both a statical and a dynamical signification. And by this means, the most general statements of the laws of motion are made to coincide with the most general statical propositions. For instance, Newton deduced from his principles the conclusion, that by the mutual action of bodies, the motion of their centre of gravity cannot be affected. Marriotte, in his _Traité de la_ {344} _Percussion_ (1684), had asserted this proposition for the case of direct impact. But by the reasoners of Newton's time, the dynamical proposition, that the motion of the centre of gravity is not altered by the actual free motion and impact of bodies, was associated with the statical proposition, that when bodies are in equilibrium, the centre of gravity cannot be made to ascend or descend by the _virtual_ motions of the bodies. This latter is a proposition which was assumed as self-evident by Torricelli; but which may more philosophically be proved from elementary statical principles.

This disposition to identify the elementary laws of equilibrium and of motion, led men to think too slightingly of the ancient solid and sufficient foundation of Statics, the doctrine of the lever. When the progress of thought had opened men's minds to a more general view of the subject, it was considered as a blemish in the science to found it on the properties of one particular machine. Descartes says in his Letters, that "it is ridiculous to prove the pulley by means of the lever." And Varignon was led by similar reflections to the project of his _Nouvelle Mécanique_, in which the whole of statics should be founded on the composition of forces. This project was published in 1687; but the work did not appear till 1725, after the death of the author. Though the attempt to reduce the equilibrium of all machines to the composition of forces, is

philosophical and meritorious, the attempt to reduce the composition of Pressures to the composition of _Motions_, with which Varignon's work is occupied, was a retrograde step in the subject, so far as the progress of distinct mechanical ideas was concerned.

Thus, at the period at which we have now arrived, the Principles of Elementary Mechanics were generally known and accepted; and there was in the minds of mathematicians a prevalent tendency to reduce them to the most simple and comprehensive form of which they admitted. The execution of this simplification and extension, which we term the generalization of the laws, is so important an event, that though it forms part of the natural sequel of Galileo, we shall treat of it in a separate chapter. But we must first bring up the history of the mechanics of fluids to the corresponding point. {345}

## CHAPTER IV.

DISCOVERY OF THE MECHANICAL PRINCIPLES OF FLUIDS.

_Sect._ 1.--_Rediscovery of the Laws of Equilibrium of Fluids._

WE have already said, that the true laws of the equilibrium of fluids were discovered by Archimedes, and rediscovered by Galileo and Stevinus; the intermediate time having been occupied by a vagueness and confusion of thought on physical subjects, which made it impossible for men to retain such clear views as Archimedes had disclosed. Stevinus must be considered as the earliest of the authors of this rediscovery; for his work (_Principles of Statik and Hydrostatik_) was published in Dutch about 1585; and in this, his views are perfectly distinct and correct. He restates the doctrines of Archimedes, and shows that, as a consequence of them, it follows that the pressure of a fluid on the bottom of a vessel may be much greater than the weight of the fluid itself: this he proves, by imagining some of the upper portions of the vessel to be filled with fixed solid bodies, which take the place of the fluid, and yet do not alter the pressure on the base. He also shows what will be the pressure on any portion of a base in an oblique position; and hence, by certain mathematical artifices which make an approach to the Infinitesimal Calculus, he finds the whole pressure on the base in such cases. This mode of treating the subject would take in a large portion of our elementary Hydrostatics as the science now stands. Galileo saw the properties of fluids no less clearly, and explained them very distinctly, in 1612, in his _Discourse on Floating Bodies_. It had been maintained by the Aristotelians, that _form_ was the cause of bodies floating; and collaterally, that ice was _condensed_ water; apparently from a confusion of thought between _rigidity_ and _density_. Galileo asserted, on the contrary, that ice is _rarefied_ water, as appears by its floating: and in support of this, he proved, by various experiments, that the floating of bodies does not depend on their form. The happy genius of Galileo is the more remarkable in this case, as the controversy was a good deal perplexed by the mixture of phenomena of another kind, due to what is usually called _capillary_ or _molecular attraction_. Thus it is a fact, that a _ball_ {346} of ebony sinks in water, while a _flat slip_ of the same material lies on the surface; and it required

considerable sagacity to separate such cases from the general rule. Galileo's opinions were attacked by various writers, as Nozzolini, Vincenzio di Grazia, Ludovico delle Colombe; and defended by his pupil Castelli, who published a reply in 1615. These opinions were generally adopted and diffused; but somewhat later, Pascal pursued the subject more systematically, and wrote his _Treatise of the Equilibrium of Fluids_ in 1653; in which he shows that a fluid, inclosed in a vessel, necessarily presses equally in all directions, by imagining two _pistons_ or sliding plugs, applied at different parts, the surface of one being centuple that of the other: it is clear, as he observes, that the force of one man acting at the first piston, will balance the force of one hundred men acting at the other. "And thus," says he, "it appears that a vessel full of water is a new Principle of Mechanics, and a new Machine which will multiply force to any degree we choose." Pascal also referred the equilibrium of fluids to the "principle of virtual velocities," which regulates the equilibrium of other machines. This, indeed, Galileo had done before him. It followed from this doctrine, that the pressure which is exercised by the lower parts of a fluid arises from the weight of the upper parts.

In all this there was nothing which was not easily assented to; but the extension of these doctrines to the air required an additional effort of mechanical conception. The pressure of the air on all sides of us, and its weight above us, were two truths which had never yet been apprehended with any kind of clearness. Seneca, indeed,[29\6] talks of the "gravity of the air," and of its power of diffusing itself when condensed, as the causes of wind; but we can hardly consider such propriety of phraseology in him as more than a chance; for we see the value of his philosophy by what he immediately adds: "Do you think that we have forces by which we move ourselves, and that the air is left without any power of moving? when even water has a motion of its own, as we see in the growth of plants." We can hardly attach much value to such a recognition of the gravity and elasticity of the air.

[Note 29\6: _Quæst. Nat._ v. 5.]

Yet the effects of these causes were so numerous and obvious, that the Aristotelians had been obliged to invent a principle to account for them; namely, "Nature's Horror of a Vacuum." To this principle were referred many familiar phenomena, as suction, breathing, the {347} action of a pair of bellows, its drawing water if immersed in water, its refusing to open when the rent is stopped up. The action of a cupping instrument, in which the air is rarefied by fire; the fact that water is supported when a full inverted bottle is placed in a basin; or when a full tube, open below and closed above, is similarly placed; the running out of the water, in this instance, when the top is opened; the action of a siphon, of a syringe, of a pump; the adhesion of two polished plates, and other facts, were all explained by the _fuga vacui_. Indeed, we must contend that the principle was a very good one, inasmuch as it brought together all these facts which are really of the same kind, and referred them to a common cause. But when urged as an ultimate principle, it was not only _unphilosophical_, but _imperfect_ and _wrong_. It was _unphilosophical_, because it introduced the notion of an emotion, Horror, as an account of physical facts; it was _imperfect_, because it was at best only a law of phenomena, not pointing out any

physical cause; and it was _wrong_, because it gave an unlimited extent to the effect. Accordingly, it led to mistakes. Thus Mersenne, in 1644, speaks of a siphon which shall go over a mountain, being ignorant then that the effect of such an instrument was limited to a height of thirty-four feet. A few years later, however, he had detected this mistake; and in his third volume, published in 1647, he puts his siphon in his _emendanda_, and speaks correctly of the weight of air as supporting the mercury in the tube of Torricelli. It was, indeed, by finding this horror of a vacuum to have a limit at the height of thirty-four feet, that the true principle was suggested. It was discovered that when attempts were made to raise water higher than this. Nature tolerated a vacuum above the water which rose. In 1643, Torricelli tried to produce this vacuum at a smaller height, by using, instead of water, the heavier fluid, quicksilver; an attempt which shows that the true explanation, the balance of the weight of the water by another pressure, had already suggested itself. Indeed, this appears from other evidence. Galileo had already taught that the air has weight; and Baliani, writing to him in 1630, says,[30\6] "If we were in a vacuum, the weight of the air above our heads would be felt." Descartes also appears to have some share in this discovery; for, in a letter of the date of 1631, he explains the suspension of mercury in a tube, closed at top, by the pressure of the column of air reaching to the clouds. {348}

[Note 30\6: Drinkwater's _Galileo_, p. 90.]

Still men's minds wanted confirmation in this view; and they found such confirmation, when, in 1647, Pascal showed practically, that if we alter the length of the superincumbent column of air by going to a high place, we alter the weight which it will support. This celebrated experiment was made by Pascal himself on a church-steeple in Paris, the column of mercury in the Torricellian tube being used to compare the weights of the air; but he wrote to his brother-in-law, who lived near the high mountain of Puy de Dôme in Auvergne, to request him to make the experiment there, where the result would be more decisive. "You see," he says, "that if it happens that the height of the mercury at the top of the hill be less than at the bottom (which I have many reasons to believe, though all those who have thought about it are of a different opinion), it will follow that the weight and pressure of the air are the sole cause of this suspension, and not the horror of a vacuum: since it is very certain that there is more air to weigh on it at the bottom than at the top; while we cannot say that nature abhors a vacuum at the foot of a mountain more than on its summit."--M. Perrier, Pascal's correspondent, made the observation as he had desired, and found a difference of three inches of mercury, "which," he says, "ravished us with admiration and astonishment."

When the least obvious case of the operation of the pressure and weight of fluids had thus been made out, there were no further difficulties in the progress of the theory of Hydrostatics. When mathematicians began to consider more general cases than those of the action of gravity, there arose differences in the way of stating the appropriate principles: but none of these differences imply any different conception of the fundamental nature of fluid equilibrium.

_Sect._ 2.--_Discovery of the Laws of Motion of Fluids._

THE art of conducting water in pipes, and of directing its motion for various purposes, is very old. When treated systematically, it has been termed _Hydraulics_: but _Hydrodynamics_ is the general name of the science of the laws of the motions of fluids, under those or other circumstances. The Art is as old as the commencement of civilization: the Science does not ascend higher than the time of Newton, though attempts on such subjects were made by Galileo and his scholars.

When a fluid spouts from an orifice in a vessel, Castelli saw that the velocity of efflux depends on the depth of the orifice below the {349} surface: but he erroneously judged the velocity to be exactly proportional to the depth. Torricelli found that the fluid, under the inevitable causes of defect which occur in the experiment, would spout nearly to the height of the surface: he therefore inferred, that the full velocity is that which a body would acquire in falling through the depth; and that it is consequently proportional to the square root of the depth.--This, however, he stated only as a result of experience, or law of phenomena, at the end of his treatise, _De Motu Naturaliter Accelerato_, printed in 1643.

Newton treated the subject theoretically in the _Principia_ (1687); but we must allow, as Lagrange says, that this is the least satisfactory passage of that great work. Newton, having made his experiments in another manner than Torricelli, namely, by measuring the quantity of the efflux instead of its velocity, found a result inconsistent with that of Torricelli. The velocity inferred from the quantity discharged, was only that due to _half_ the depth of the fluid.

In the first edition of the _Principia_,[31\6] Newton gave a train of reasoning by which he theoretically demonstrated his own result, going upon the principle, that the momentum of the issuing fluid is equal to the momentum which the column vertically over the orifice would generate by its gravity. But Torricelli's experiments, which had given the velocity due to the whole depth, were confirmed on repetition: how was this discrepancy to be explained?

[Note 31\6: B. ii. Prop. xxxvii.]

Newton explained the discrepancy by observing the contraction which the jet, or vein of water, undergoes, just after it leaves the orifice, and which he called the _vena contracta_. At the orifice, the velocity is that due to half the height; at the _vena contracta_ it is that due to the whole height. The former velocity regulates the quantity of the discharge; the latter, the path of the jet.

This explanation was an important step in the subject; but it made Newton's original proof appear very defective, to say the least. In the second edition of the _Principia_ (1714), Newton attacked the problem in a manner altogether different from his former investigation. He there assumed, that when a round vessel, containing fluid, has a hole in its bottom, the descending fluid may be conceived to be a conoidal mass, which has its base at the surface of the fluid, and its narrow end at the orifice. This portion of the fluid he calls the _cataract_; and supposes that while this part descends, the surrounding {350} parts remain immovable, as if they were frozen; in this way he finds a result agreeing with Torricelli's experiments on the velocity of the efflux.

We must allow that the assumptions by which this result is obtained are somewhat arbitrary; and those which Newton introduces in attempting to connect the problem of issuing fluids with that of the resistance to a body moving in a fluid, are no less so. But even up to the present time, mathematicians have not been able to reduce problems concerning the motions of fluids to mathematical principles and calculations, without introducing some steps of this arbitrary kind. And one of the uses of experiments on this subject is, to suggest those hypotheses which may enable us, in the manner most consonant with the true state of things, to reduce the motions of fluids to those general laws of mechanics, to which we know they must be subject.

Hence the science of the Motion of Fluids, unlike all the other primary departments of Mechanics, is a subject on which we still need experiments, to point out the fundamental principles. Many such experiments have been made, with a view either to compare the results of deduction and observation, or, when this comparison failed, to obtain purely empirical rules. In this way the resistance of fluids, and the motion of water in pipes, canals, and rivers, has been treated. Italy has possessed, from early times, a large body of such writers. The earlier works of this kind have been collected in sixteen quarto volumes. Lecchi and Michelotti about 1765, Bidone more recently, have pursued these inquiries. Bossut, Buat, Hachette, in France, have labored at the same task, as have Coulomb and Prony, Girard and Poncelet. Eytelwein's German treatise (_Hydraulik_) contains an account of what others and himself have done. Many of these trains of experiments, both in France and Italy, were made at the expense of governments, and on a very magnificent scale. In England less was done in this way during the last century, than in most other countries. The _Philosophical Transactions_, for instance, scarcely contain a single paper on this subject founded on experimental investigations.[32\6] Dr. Thomas Young, who was at the head of his countrymen in so many branches of science, was one of the first to call back attention to this: and Mr. Rennie and others have recently made valuable experiments. In many of the questions now spoken of, the accordance which engineers are able to obtain, between their calculated and observed results, {351} is very great: but these calculations are performed by means of empirical formulæ, which do not connect the facts with their causes, and still leave a wide space to be traversed, in order to complete the science.

[Note 32\6: Rennie, _Report to Brit. Assoc._]

In the mean time, all the other portions of Mechanics were reduced to general laws, and analytical processes; and means were found of including Hydrodynamics, notwithstanding the difficulties which attend its special problems, in this common improvement of form. This progress we must relate.

[2d Ed.] [The hydrodynamical problems referred to above are, the laws of a fluid issuing from a vessel, the laws of the motion of water in pipes, canals, and rivers, and the laws of the resistance of fluids. To these may be added, as an hydrodynamical problem important in theory, in experiment, and in the comparison of the two, the laws of waves. Newton gave, in the _Principia_, an explanation of the waves of water (Lib. ii. Prop. 44), which appears to proceed upon an erroneous view of the nature of the motion of the fluid: but in his solution of the problem of sound, appeared, for the

first time, a correct view of the propagation of an undulation in a fluid. The history of this subject, as bearing upon the theory of sound, is given in Book viii.: but I may here remark, that the laws of the motion of waves have been pursued experimentally by various persons, as Bremontier (_Recherches sur le Mouvement des Ondes_, 1809), Emy (_Du Mouvement des Ondes_, 1831), the Webers (_Wellenlehre_, 1825); and by Mr. Scott Russell (_Reports of the British Association_, 1844). The analytical theory has been carried on by Poisson, Cauchy, and, among ourselves, by Prof. Kelland (_Edin. Trans._) and Mr. Airy (in the article _Tides_, in the _Encyclopædia Metropolitana_). And though theory and experiment have not yet been brought into complete accordance, great progress has been made in that work, and the remaining chasm between the two is manifestly due only to the incompleteness of both.]

Perhaps the most remarkable case of fluid motion recently discussed, is one which Mr. Scott Russell has presented experimentally; and which, though novel, is easily seen to follow from known principles; namely, the _Great Solitary Wave_. A wave may be produced, which shall move along a canal unaccompanied by any other wave: and the simplicity of this case makes the mathematical conditions and consequences more simple than they are in most other problems of Hydrodynamics. {352}

## CHAPTER V.

GENERALIZATION OF THE PRINCIPLES OF MECHANICS.

_Sect._ 1.--_Generalization of the Second Law of Motion.--Central Forces._

THE Second Law of Motion being proved for constant Forces which act in parallel lines, and the Third Law for the Direct Action of bodies, it still required great mathematical talent, and some inductive power, to see clearly the laws which govern the motion of any number of bodies, acted upon by each other, and by any forces, anyhow varying in magnitude and direction. This was the task of the generalization of the laws of motion.

Galileo had convinced himself that the velocity of projection, and that which gravity alone would produce, are "both maintained, without being altered, perturbed, or impeded in their mixture." It is to be observed, however, that the truth of this result depends upon a particular circumstance, namely, that gravity, at all points, acts in lines, which, as to sense, are parallel. When we have to consider cases in which this is not true, as when the force tends to the centre of a circle, the law of composition cannot be applied in the same way; and, in this case, mathematicians were met by some peculiar difficulties.

One of these difficulties arises from the apparent inconsistency of the statical and dynamical measures of force. When a body moves in a circle, the force which urges the body to the centre is only a _tendency_ to motion; for the body does not, in fact, approach to the centre; and this mere tendency to motion is combined with an actual motion, which takes place in the circumference. We appear to have to compare two things which are heterogeneous. Descartes had noticed this difficulty, but without giving any satisfactory solution of it.[33\6] If we combine the actual motion to or from

the centre with the traverse motion about the centre, we obtain a result which is false on mechanical principles. Galileo endeavored in this way to find the curve described by a body which falls towards the earth's centre, and is, at the same time, carried {353} round by the motion of the earth; and obtained an erroneous result. Kepler and Fermat attempted the same problem, and obtained solutions different from that of Galileo, but not more correct.

[Note 33\6: _Princip._ P. iii. 59.]

Even Newton, at an early period of his speculations, had an erroneous opinion respecting this curve, which he imagined to be a kind of spiral. Hooke animadverted upon this opinion when it was laid before the Royal Society of London in 1679, and stated, more truly, that, supposing no resistance, it would be "an eccentric ellipsoid," that is, a figure resembling an ellipse. But though he had made out the approximate form of the curve, in some unexplained way, we have no reason to believe that he possessed any means of determining the mathematical properties of the curve described in such a case. The perpetual composition of a central force with the previous motion of the body, could not be successfully treated without the consideration of the Doctrine of Limits, or something equivalent to that doctrine. The first example which we have of the right solution of such a problem occurs, so far as I know, in the Theorems of Huyghens concerning Circular Motion, which were published, without demonstration, at the end of his _Horologium Oscillatorium_, in 1673. It was there asserted that when equal bodies describe circles, if the times are equal, the centrifugal forces will be as the diameters of the circles; if the velocities are equal, the forces will be reciprocally as the diameters, and so on. In order to arrive at these propositions, Huyghens must, virtually at least, have applied the Second Law of Motion to the limiting elements of the curve, according to the way in which Newton, a few years later, gave the demonstration of the theorems of Huyghens in the _Principia_.

The growing persuasion that the motions of the heavenly bodies about the sun might be explained by the action of central forces, gave a peculiar interest to these mechanical speculations, at the period now under review. Indeed, it is not easy to state separately, as our present object requires us to do, the progress of Mechanics, and the progress of Astronomy. Yet the distinction which we have to make is, in its nature, sufficiently marked. It is, in fact, no less marked than the distinction between speaking logically and speaking truly. The framers of the science of motion were employed in establishing those notions, names, and rules, in conformity to which _all_ mechanical _truth must_ be expressed; but _what was the truth_ with regard to the mechanism of the universe remained to be determined by other means. Physical Astronomy, at the period of which we speak, eclipsed and overlaid {354} theoretical Mechanics, as, a little previously, Dynamics had eclipsed and superseded Statics.

The laws of variable force and of curvilinear motion were not much pursued, till the invention of Fluxions and of the Differential Calculus again turned men's minds to these subjects, as easy and interesting exercises of the powers of these new methods. Newton's _Principia_, of which the first two Books are purely dynamical, is the great exception to this assertion; inasmuch as it contains correct solutions of a great

variety of the most general problems of the science; and indeed is, even yet, one of the most complete treatises which we possess upon the subject.

We have seen that Kepler, in his attempts to explain the curvilinear motions of the planets by means of a central force, failed, in consequence of his belief that a continued transverse action of the central body was requisite to keep up a continual motion. Galileo had founded his theory of projectiles on the principle that such an action was not necessary; yet Borelli, a pupil of Galileo, when, in 1666, he published his theory of the Medicean Stars (the satellites of Jupiter), did not keep quite clear of the same errors which had vitiated Kepler's reasonings. In the same way, though Descartes is sometimes spoken of as the first promulgator of the First Law of Motion, yet his theory of Vortices must have been mainly suggested by a want of an entire confidence in that law. When he represented the planets and satellites as owing their motions to oceans of fluid diffused through the celestial spaces, and constantly whirling round the central bodies, he must have felt afraid of trusting the planets to the operation of the laws of motion in free space. Sounder physical philosophers, however, began to perceive the real nature of the question. As early as 1666, we read, in the Journals of the Royal Society, that "there was read a paper of Mr. Hooke's explicating the inflexion of a direct motion into a curve by a supervening attractive principle;" and before the publication of the _Principia_ in 1687, Huyghens, as we have seen, in Holland, and, in our own country, Wren, Halley, and Hooke, had made some progress in the true mechanics of circular motion,[34\6] and had distinctly contemplated the problem of the motion of a body in an ellipse by a central force, though they could not solve it. Halley went to Cambridge in 1684,[35\6] for the express purpose of consulting Newton upon the subject of the production of the elliptical motion of the planets by means of a central {355} force, and, on the 10th of December,[36\6] announced to the Royal Society that he had seen Mr. Newton's book, _De Motu Corporum_. The feeling that mathematicians were on the brink of discoveries such as are contained in this work was so strong, that Dr. Halley was requested to remind Mr. Newton of his promise of entering them in the Register of the Society, "for securing the invention to himself till such time as he can be at leisure to publish it." The manuscript, with the title _Philosophiæ Naturalis Principia Mathematica_, was presented to the society (to which it was dedicated) on the 28th of April, 1686. Dr. Vincent, who presented it, spoke of the novelty and dignity of the subject; and the president (Sir J. Hoskins) added, with great truth, "that the method was so much the more to be prized as it was both invented and perfected at the same time."

[Note 34\6: Newt. _Princip._ Schol. to Prop. iv.]

[Note 35\6: Sir D. Brewster's _Life of Newton_, p. 154.]

[Note 36\6: Id. p. 184.]

The reader will recollect that we are here speaking of the _Principia_ as a Mechanical Treatise only; we shall afterwards have to consider it as containing the greatest discoveries of Physical Astronomy. As a work on Dynamics, its merit is, that it exhibits a wonderful store of refined and beautiful mathematical artifices, applied

to solve all the most general problems which the subject offered. The _Principia_ can hardly be said to contain any new inductive discovery respecting the principles of mechanics; for though Newton's _Axioms or Laws of Motion_ which stand at the beginning of the book, are a much clearer and more general statement of the grounds of Mechanics than had yet appeared, they do not involve any doctrines which had not been previously stated or taken for granted by other mathematicians.

The work, however, besides its unrivalled mathematical skill, employed in tracing out, deductively, the consequences of the laws of motion, and its great cosmical discoveries, which we shall hereafter treat of, had great philosophical value in the history of Dynamics, as exhibiting a clear conception of the new character and functions of that science. In his Preface, Newton says, "Rational Mechanics must be the science of the Motions which result from any Forces, and of the Forces which are required for any Motions, accurately propounded and demonstrated. For many things induce me to suspect, that all natural phenomena may depend upon some Forces by which the particles of bodies are either drawn towards each other, and cohere, or repel and recede from each other: and these Forces being hitherto unknown, philosophers have pursued their researches in vain. And I hope {356} that the principles expounded in this work will afford some light, either to this mode of philosophizing, or to some mode which is more true."

Before we pursue this subject further, we must trace the remainder of the history of the Third Law.

_Sect._ 2.--_Generalization of the Third Law of Motion.--Centre of Oscillation.--Huyghens._

THE Third Law of Motion, whether expressed according to Newton's formula (by the equality of Action and Reaction), or in any other of the ways employed about the same time, easily gave the solution of mechanical problems in all cases of _direct_ action; that is, when each body acted directly on others. But there still remained the problems in which the action is _indirect_;--when bodies, in motion, act on each other by the intervention of levers, or in any other way. If a rigid rod, passing through two weights, be made to swing about its upper point, so as to form a pendulum, each weight will act and react on the other by means of the rod, considered as a lever turning about the point of suspension. What, in this case, will be the effect of this action and reaction? In what time will the pendulum oscillate by the force of gravity? Where is the point at which a single weight must be placed to oscillate in the same time? in other words, where is the _Centre of Oscillation_?

Such was the problem--an example only of the general problem of indirect action--which mathematicians had to solve. That it was by no means easy to see in what manner the law of the communication of motion was to be extended from simpler cases to those where rotatory motion was produced, is shown by this;--that Newton, in attempting to solve the mechanical problem of the Precession of the Equinoxes, fell into a serious error on this very subject. He assumed that, when a part has to communicate rotatory movement to the whole (as the protuberant portion of the terrestrial spheroid, attracted by the sun and moon, communicates a small

movement to the whole mass of the earth), the quantity of the _motion_, "motus," will not be altered by being communicated. This principle is true, if, by _motion_, we understand what is called _moment of inertia_, a quantity in which both the velocity of each particle and its distance from the axis of rotation are taken into account: but Newton, in his calculations of its amount, considered the velocity only; thus making _motion_, in this case, identical with the _momentum_ which he introduces in treating of the simpler case {357} of the third law of motion, when the action is direct. This error was retained even in the later editions of the _Principia_.[37\6]

[Note 37\6: B. iii. Lemma iii. to Prop, xxxix.]

The question of the centre of oscillation had been proposed by Mersenne somewhat earlier,[38\6] in 1646. And though the problem was out of the reach of any principles at that time known and understood, some of the mathematicians of the day had rightly solved some cases of it, by proceeding as if the question had been to find the _Centre of Percussion_. The Centre of Percussion is the point about which the momenta of all the parts of a body balance each other, when it is in motion about any axis, and is stopped by striking against an obstacle placed at that centre. Roberval found this point in some easy cases; Descartes also attempted the problem; their rival labors led to an angry controversy: and Descartes was, as in his physical speculations he often was, very presumptuous, though not more than half right.

[Note 38\6: Mont. ii. 423.]

Huyghens was hardly advanced beyond boyhood when Mersenne first proposed this problem; and, as he says,[39\6] could see no principle which even offered an opening to the solution, and had thus been repelled at the threshold. When, however, he published his _Horologium Oscillatorium_ in 1673, the fourth part of that work was on the Centre of Oscillation or Agitation; and the principle which he then assumed, though not so simple and self-evident as those to which such problems were afterwards referred, was perfectly correct and general, and led to exact solutions in all cases. The reader has already seen repeatedly in the course of this history, complex and derivative principles presenting themselves to men's minds, before simple and elementary ones. The "hypothesis" assumed by Huyghens was this; "that if any weights are put in motion by the force of gravity, they _cannot_ move so that the centre of gravity of them all shall rise _higher_ than the place from which it descended." This being assumed, it is easy to show that the centre of gravity will, under all circumstances, rise _as high_ as its original position; and this consideration leads to a determination of the oscillation of a compound pendulum. We may observe, in the principle thus selected, a conviction that, in all mechanical action, the centre of gravity may be taken as the representative of the whole system. This conviction, as we have seen, may be traced in the axioms of Archimedes and Stevinus; and Huyghens, when he proceeds upon it, undertakes to show,[40\6] that he assumes only this, that a heavy body cannot, of itself, move upwards. {358}

[Note 39\6: _Hor. Osc._ Pref.]

[Note 40\6: _Hor. Osc._ p. 121.]

Clear as Huyghen's principle appeared to himself, it was, after some time, attacked by the Abbé Catelan, a zealous Cartesian. Catelan also put forth principles which he conceived were evident, and deduced from them conclusions contradictory to those of Huyghens. His principles, now that we know them to be false, appear to us very gratuitous. They are these; "that in a compound pendulum, the sum of the velocities of the component weights is equal to the sum of the velocities which they would have acquired if they had been detached pendulums;" and "that the time of the vibration of a compound pendulum is an arithmetic mean between the times of the vibrations of the weights, moving as detached pendulums." Huyghens easily showed that these suppositions would make the centre of gravity ascend to a greater height than that from which it fell; and after some time, James Bernoulli stept into the arena, and ranged himself on the side of Huyghens. As the discussion thus proceeded, it began to be seen that the question really was, in what manner the Third Law of Motion was to be extended to cases of indirect action; whether by distributing the action and reaction according to statical principles, or in some other way. "I propose it to the consideration of mathematicians," says Bernoulli in 1686, "what law of the communication of velocity is observed by bodies in motion, which are sustained at one extremity by a fixed fulcrum, and at the other by a body also moving, but more slowly. Is the excess of velocity which must be communicated from the one body to the other to be distributed in the same proportion in which a load supported on the lever would be distributed?" He adds, that if this question be answered in the affirmative, Huyghens will be found to be in error; but this is a mistake. The principle, that the action and reaction of bodies thus moving are to be distributed according to the rules of the lever, is true; but Bernoulli mistook, in estimating this action and reaction by the _velocity_ acquired at any moment; instead of taking, as he should have done, the _increment_ of velocity which gravity tended to impress in the next instant. This was shown by the Marquis de l'Hôpital; who adds, with justice, "I conceive that I have thus fully answered the call of Bernoulli, when he says, I propose it to the consideration of mathematicians, &c."

We may, from this time, consider as known, but not as fully established, the principle that "When bodies in motion affect each other, the action and reaction are distributed according to the laws of Statics;" although there were still found occasional difficulties in the {359} generalization and application of the role. James Bernoulli, in 1703, gave "a General Demonstration of the Centre of Oscillation, drawn from the nature of the Lever." In this demonstration[41\6] he takes as a fundamental principle, that bodies in motion, connected by levers, balance, when the products of their momenta and the lengths of the levers are equal in opposite directions. For the proof of this proposition, he refers to Marriotte, who had asserted it of weights acting by percussion,[42\6] and in order to prove it, had balanced the effect of a weight on a lever by the effect of a jet of water, and had confirmed it by other experiments.[43\6] Moreover, says Bernoulli, there is no one who denies it. Still, this kind of proof was hardly satisfactory or elementary enough. John Bernoulli took up the subject after the death of his brother James, which happened in 1705. The former published in 1714 his _Meditatio de Naturâ Centri Oscillationis_. In this memoir, he assumes, as his brother had done, that the effects of forces on a lever in motion are distributed

according to the common rules of the lever.[44\6] The principal generalization which he introduced was, that he considered gravity as a force soliciting to motion, which might have different intensities in different bodies. At the same time, Brook Taylor in England solved the problem, upon the same principles as Bernoulli; and the question of priority on this subject was one point in the angry intercourse which, about this time, became common between the English mathematicians and those of the Continent. Hermann also, in his _Phoronomia_, published in 1716, gave a proof which, as he informs us, he had devised before he saw John Bernoulli's. This proof is founded on the statical equivalence of the "_solicitations of gravity_" and the "_vicarious solicitations_" which correspond to the actual motion of each part; or, as it has been expressed by more modern writers, the equilibrium of the _impressed_ and _effective forces_.

[Note 41\6: _Op._ ii. 930.]

[Note 42\6: _Choq. des Corps_, p. 296.]

[Note 43\6: Ib. Prop. xi.]

[Note 44\6: P. 172.]

It was shown by John Bernoulli and Hermann, and was indeed easily proved, that the proposition assumed by Huyghens as the foundation of his solution, was, in fact, a consequence of the elementary principles which belong to this branch of mechanics. But this assumption of Huyghens was an example of a more general proposition, which by some mathematicians at this time had been put forward as an original and elementary law; and as a principle which ought to supersede the usual measure of the forces of bodies in motion; this principle they called "_the Conservation of Vis Viva_." The attempt to {360} make this change was the commencement of one of the most obstinate and curious of the controversies which form part of the history of mechanical science. The celebrated Leibnitz was the author of the new opinion. In 1686, he published, in the Leipsic Acts, "A short Demonstration of a memorable Error of Descartes and others, concerning the natural law by which they think that God always preserves the same quantity of motion; in which they pervert mechanics." The principle that the same quantity of motion, and therefore of moving force, is always preserved in the world, follows from the equality of action and reaction; though Descartes had, after his fashion, given a theological reason for it; Leibnitz allowed that the quantity of moving force remains always the same, but denied that this force is measured by the quantity of motion or momentum. He maintained that the same force is requisite to raise a weight of one pound through four feet, and a weight of four pounds through one foot, though the momenta in this case are as one to two. This was answered by the Abbé de Conti; who truly observed, that allowing the effects in the two cases to be equal, this did not prove the forces to be equal; since the effect, in the first case, was produced in a double time, and therefore it was quite consistent to suppose the force only half as great. Leibnitz, however, persisted in his innovation; and in 1695 laid down the distinction between _vires mortuæ_, or pressures, and _vires vivæ_, the name he gave to his own measure of force. He kept up a correspondence with John Bernoulli, whom he converted to his peculiar opinions

on this subject; or rather, as Bernoulli says,[45\6] made him think for himself, which ended in his proving directly that which Leibnitz had defended by indirect reasons. Among other arguments, he had pretended to show (what is certainly not true), that if the common measure of forces be adhered to, a perpetual motion would be possible. It is easy to collect many cases which admit of being very simply and conveniently reasoned upon by means of the _vis viva_, that is, by taking the force to be proportional to the _square_ of the velocity, and not to the velocity itself. Thus, in order to give the arrow _twice_ the velocity, the bow must be _four_ times as strong; and in all cases in which no account is taken of the time of producing the effect, we may conveniently use similar methods.

[Note 45\6: _Op._ iii. 40.]

But it was not till a later period that the question excited any general notice. The Academy of Sciences of Paris in 1724 proposed {361} as a subject for their prize dissertation the laws of the impact of bodies. Bernoulli, as a competitor, wrote a treatise, upon Leibnitzian principles, which, though not honored with the prize, was printed by the Academy with commendation.[46\6] The opinions which he here defended and illustrated were adopted by several mathematicians; the controversy extended from the mathematical to the literary world, at that time more attentive than usual to mathematical disputes, in consequence of the great struggle then going on between the Cartesian and the Newtonian system. It was, however, obvious that by this time the interest of the question, so far as the progress of Dynamics was concerned, was at an end; for the combatants all agreed as to the results in each particular case. The Laws of Motion were now established; and the question was, by means of what definitions and abstractions could they be best expressed;--a metaphysical, not a physical discussion, and therefore one in which "the paper philosophers," as Galileo called them, could bear a part. In the first volume of the _Transactions of the Academy of St. Petersburg_, published in 1728, there are three Leibnitzian memoirs by Hermann, Bullfinger, and Wolff. In England, Clarke was an angry assailant of the German opinion, which S'Gravesande maintained. In France, Mairan attacked the _vis viva_ in 1728; "with strong and victorious reasons," as the Marquise du Chatelet declared, in the first edition of her _Treatise on Fire_.[47\6] But shortly after this praise was published, the Chateau de Cirey, where the Marquise usually lived, became a school of Leibnitzian opinions, and the resort of the principal partisans of the _vis viva_. "Soon," observes Mairan, "their language was changed; the _vis viva_ was enthroned by the side of the _monads_." The Marquise tried to retract or explain away her praises; she urged arguments on the other side. Still the question was not decided; even her friend Voltaire was not converted. In 1741 he read a memoir _On the Measure and Nature of Moving Forces_, in which he maintained the old opinion. Finally, D'Alembert in 1743 declared it to be, as it truly was, a mere question of words; and by the turn which Dynamics then took, it ceased to be of any possible interest or importance to mathematicians.

[Note 46\6: _Discours sur les Loix de la Communication du Mouvement_.]

[Note 47\6: Mont. iii. 640.]

The representation of the laws of motion and of the reasonings depending on them, in the most general form, by means of analytical language, cannot be said to have been fully achieved till the time of D'Alembert; but as we have already seen, the discovery of these laws {362} had taken place somewhat earlier; and that law which is more particularly expressed in D'Alembert's Principle (_the equality of the action gained and lost_) was, it has been seen, rather led to by the general current of the reasoning of mathematicians about the end of the seventeenth century than discovered by any one. Huyghens, Marriotte, the two Bernoulli's, L'Hôpital, Taylor, and Hermann, have each of them their name in the history of this advance; but we cannot ascribe to any of them any great real inductive sagacity shown in what they thus contributed, except to Huyghens, who first seized the principle in such a form as to find the centre of oscillation by means of it. Indeed, in the steps taken by the others, language itself had almost made the generalization for them at the time when they wrote; and it required no small degree of acuteness and care to distinguish the old cases, in which the law had already been applied, from the new cases, in which they had to apply it.

## CHAPTER VI.

SEQUEL TO THE GENERALIZATION OF THE PRINCIPLES OF MECHANICS.--PERIOD OF MATHEMATICAL DEDUCTION.--ANALYTICAL MECHANICS.

WE have now finished the history of the discovery of Mechanical Principles, strictly so called. The three Laws of Motion, generalized in the manner we have described, contain the materials of the whole structure of Mechanics; and in the remaining progress of the science, we are led to no new truth which was not implicitly involved in those previously known. It may be thought, therefore, that the narrative of this progress is of comparatively small interest. Nor do we maintain that the application and development of principles is a matter of so much importance to the philosophy of science, as the advance towards and to them. Still, there are many circumstances in the latter stages of the progress of the science of Mechanics, which well deserve notice, and make a rapid survey of that part of its history indispensable to our purpose.

The Laws of Motion are expressed in terms of Space and Number; the development of the consequences of these laws must, therefore, be performed by means of the reasonings of mathematics; and the science {363} of Mechanics may assume the various aspects which belong to the different modes of dealing with mathematical quantities. Mechanics, like pure mathematics, may be geometrical or may be analytical; that is, it may treat space either by a direct consideration of its properties, or by a symbolical representation of them: Mechanics, like pure mathematics, may proceed from special cases, to problems and methods of extreme generality;--may summon to its aid the curious and refined relations of symmetry, by which general and complex conditions are simplified;--may become more powerful by the discovery of more powerful analytical artifices;--may even have the generality of its principles further expanded, inasmuch as symbols are a more general language than words. We

shall very briefly notice a series of modifications of this kind.

1. _Geometrical Mechanics. Newton, &c._--The first great systematical Treatise on Mechanics, in the most general sense, is the two first Books of the _Principia_ of Newton. In this work, the method employed is predominantly geometrical: not only space is not represented symbolically, or by reference to number; but numbers, as, for instance, those which measure time and force, are represented by spaces; and the laws of their changes are indicated by the properties of curve lines. It is well known that Newton employed, by preference, methods of this kind in the exposition of his theorems, even where he had made the discovery of them by analytical calculations. The intuitions of space appeared to him, as they have appeared to many of his followers, to be a more clear and satisfactory road to knowledge, than the operations of symbolical language. Hermann, whose _Phoronomia_ was the next great work on this subject, pursued a like course; employing curves, which he calls "the scale of velocities," "of forces," &c. Methods nearly similar were employed by the two first Bernoullis, and other mathematicians of that period; and were, indeed, so long familiar, that the influence of them may still be traced in some of the terms which are used on such subjects; as, for instance, when we talk of "reducing a problem to quadratures," that is, to the finding the area of the curves employed in these methods.

2. _Analytical Mechanics. Euler._--As analysis was more cultivated, it gained a predominancy over geometry; being found to be a far more powerful instrument for obtaining results; and possessing a beauty and an evidence, which, though different from those of geometry, had great attractions for minds to which they became familiar. The person who did most to give to analysis the generality and {364} symmetry which are now its pride, was also the person who made Mechanics analytical; I mean Euler. He began his execution of this task in various memoirs which appeared in the _Transactions of the Academy of Sciences at St. Petersburg_, commencing with its earliest volumes; and in 1736, he published there his _Mechanics, or the Science of Motion analytically expounded; in the way of a Supplement to the Transactions of the Imperial Academy of Sciences_. In the preface to this work, he says, that though the solutions of problems by Newton and Hermann were quite satisfactory, yet he found that he had a difficulty in applying them to new problems, differing little from theirs; and that, therefore, he thought it would be useful to extract an analysis out of their synthesis.

3. _Mechanical Problems._--In reality, however, Euler has done much more than merely give analytical methods, which may be applied to mechanical problems: he has himself applied such methods to an immense number of cases. His transcendent mathematical powers, his long and studious life, and the interest with which he pursued the subject, led him to solve an almost inconceivable number and variety of mechanical problems. Such problems suggested themselves to him on all occasions. One of his memoirs begins, by stating that, happening to think of the line of Virgil, Anchora de prorà jacitur stant litore puppes; The anchor drops, the rushing keel is staid; he could not help inquiring what would be the nature of the ship's motion under the circumstances here described. And in the last few days of his life, after his mortal

illness had begun, having seen in the newspapers some statements respecting balloons, he proceeded to calculate their motions; and performed a difficult integration, in which this undertaking engaged him. His Memoirs occupy a very large portion of the _Petropolitan Transactions_ during his life, from 1728 to 1783; and he declared that he should leave papers which might enrich the publications of the Academy of Petersburg for twenty years after his death;--a promise which has been more than fulfilled; for, up to 1818, the volumes usually contain several Memoirs of his. He and his contemporaries may be said to have exhausted the subject; for there are few mechanical problems which have been since treated, which they have not in some manner touched upon.

I do not dwell upon the details of such problems; for the next great step in Analytical Mechanics, the publication of D'Alembert's {365} Principle in 1743, in a great degree superseded their interest. The Transactions of the Academies of Paris and Berlin, as well as St. Petersburg, are filled, up to this time, with various questions of this kind. They require, for the most part, the determination of the motions of several bodies, with or without weight, which pull or push each other by means of threads, or levers, to which they are fastened, or along which they can slide; and which, having a certain impulse given them at first, are then left to themselves, or are compelled to move in given lines and surfaces. The postulate of Huyghens, respecting the motion of the centre of gravity, was generally one of the principles of the solution; but other principles were always needed in addition to this; and it required the exercise of ingenuity and skill to detect the most suitable in each case. Such problems were, for some time, a sort of trial of strength among mathematicians: the principle of D'Alembert put an end to this kind of challenges, by supplying a direct and general method of resolving, or at least of throwing into equations, any imaginable problem. The mechanical difficulties were in this way reduced to difficulties of pure mathematics.

4. _D'Alembert's Principle._--D'Alembert's Principle is only the expression, in the most general form, of the principle upon which John Bernoulli, Hermann, and others, had solved the problem of the centre of oscillation. It was thus stated, "The motion _impressed_ on each particle of any system by the forces which act upon it, may be resolved into two, the _effective_ motion, and the motion gained or _lost_: the effective motions will be the real motions of the parts, and the motions gained and lost will be such as would keep the system at rest." The distinction of _statics_, the doctrine of equilibrium, and _dynamics_, the doctrine of motion, was, as we have seen, fundamental; and the difference of difficulty and complexity in the two subjects was well understood, and generally recognized by mathematicians. D'Alembert's principle reduces every dynamical question to a statical one; and hence, by means of the conditions which connect the possible motions of the system, we can determine what the actual motions must be. The difficulty of determining the laws of equilibrium, in the application of this principle in complex cases is, however, often as great as if we apply more simple and direct considerations.

5. _Motion in Resisting Media. Ballistics._--We shall notice more particularly the history of some of the problems of mechanics. Though John Bernoulli always spoke

with admiration of Newton's _Principia_, and of its author, he appears to have been well disposed to point out {366} real or imagined blemishes in the work. Against the validity of Newton's determination of the path described by a body projected in any part of the solar system, Bernoulli urges a cavil which it is difficult to conceive that a mathematician, such as he was, could seriously believe to be well founded. On Newton's determination of the path of a body in a resisting medium, his criticism is more just. He pointed out a material error in this solution: this correction came to Newton's knowledge in London, in October, 1712, when the impression of the second edition of the Principia was just drawing to a close, under the care of Cotes at Cambridge; and Newton immediately cancelled the leaf and corrected the error.[48\6]

[Note 48\6: MS. Correspondence in Trin. Coll. Library.]

This problem of the motion of a body in a resisting medium, led to another collision between the English and the German mathematicians. The proposition to which we have referred, gave only an indirect view of the nature of the curve described by a projectile in the air; and it is probable that Newton, when he wrote the _Principia_, did not see his way to any direct and complete solution of this problem. At a later period, in 1718, when the quarrel had waxed hot between the admirers of Newton and Leibnitz, Keill, who had come forward as a champion on the English side, proposed this problem to the foreigners as a challenge. Keill probably imagined that what Newton had not discovered, no one of his time would be able to discover. But the sedulous cultivation of analysis by the Germans had given them mathematical powers beyond the expectations of the English; who, whatever might be their talents, had made little advance in the effective use of general methods; and for a long period seemed to be fascinated to the spot, in their admiration of Newton's excellence. Bernoulli speedily solved the problem; and reasonably enough, according to the law of honor of such challenges, called upon the challenger to produce his solution. Keill was unable to do this; and after some attempts at procrastination, was driven to very paltry evasions. Bernoulli then published his solution, with very just expressions of scorn towards his antagonist. And this may, perhaps, be considered as the first material addition which was made to the _Principia_ by subsequent writers.

6. _Constellation of Mathematicians._--We pass with admiration along the great series of mathematicians, by whom the science of theoretical mechanics has been cultivated, from the time of Newton to our own. There is no group of men of science whose fame is {367} higher or brighter. The great discoveries of Copernicus, Galileo, Newton, had fixed all eyes on those portions of human knowledge on which their successors employed their labors. The certainty belonging to this line of speculation seemed to elevate mathematicians above the students of other subjects; and the beauty of mathematical relations, and the subtlety of intellect which may be shown in dealing with them, were fitted to win unbounded applause. The successors of Newton and the Bernoullis, as Euler, Clairaut, D'Alembert, Lagrange, Laplace, not to introduce living names, have been some of the most remarkable men of talent which the world has seen. That their talent is, for the most part, of a different kind from that by which the laws of nature were discovered, I shall have occasion to explain

elsewhere; for the present, I must endeavor to arrange the principal achievements of those whom I have mentioned.

The series of persons is connected by social relations. Euler was the pupil of the first generation of Bernoullis, and the intimate friend of the second generation; and all these extraordinary men, as well as Hermann, were of the city of Basil, in that age a spot fertile of great mathematicians to an unparalleled degree. In 1740, Clairaut and Maupertuis visited John Bernoulli, at that time the Nestor of mathematicians, who died, full of age and honors, in 1748. Euler, several of the Bernoullis, Maupertuis, Lagrange, among other mathematicians of smaller note, were called into the north by Catharine of Russia and Frederic of Prussia, to inspire and instruct academies which the brilliant fame then attached to science, had induced those monarchs to establish. The prizes proposed by these societies, and by the French Academy of Sciences, gave occasion to many of the most valuable mathematical works of the century.

7. _The Problem of Three Bodies._--In 1747, Clairaut and D'Alembert sent, on the same day, to this body, their solutions of the celebrated "Problem of Three Bodies," which, from that time, became the great object of attention of mathematicians;--the bow in which each tried his strength, and endeavored to shoot further than his predecessors.

This problem was, in fact, the astronomical question of the effect produced by the attraction of the sun, in disturbing the motions of the moon about the earth; or by the attraction of one planet, disturbing the motion of another planet about the sun; but being expressed generally, as referring to one body which disturbs any two others, it became a mechanical problem, and the history of it belongs to the present subject. {368}

One consequence of the synthetical form adopted by Newton in the _Principia_, was, that his successors had the problem of the solar system to begin entirely anew. Those who would not do this, made no progress, as was long the case with the English. Clairaut says, that he tried for a long time to make some use of Newton's labors; but that, at last, he resolved to take up the subject in an independent manner. This, accordingly, he did, using analysis throughout, and following methods not much different from those still employed. We do not now speak of the comparison of this theory with observation, except to remark, that both by the agreements and by the discrepancies of this comparison, Clairaut and other writers were perpetually driven on to carry forwards the calculation to a greater and greater degree of accuracy.

One of the most important of the cases in which this happened, was that of the movement of the Apogee of the Moon; and in this case, a mode of approximating to the truth, which had been depended on as nearly exact, was, after having caused great perplexity, found by Clairaut and Euler to give only half the truth. This same Problem of Three Bodies was the occasion of a memoir of Clairaut, which gained the prize of the Academy of St. Petersburg in 1751; and, finally, of his _Théorie de la Lune_, published in 1765. D'Alembert labored at the same time on the same problem; and the value of their methods, and the merit of the inventors, unhappily became a

subject of controversy between those two great mathematicians. Euler also, in 1753, published a _Theory of the Moon_, which was, perhaps, more useful than either of the others, since it was afterwards the basis of Mayer's method, and of his Tables. It is difficult to give the general reader any distinct notion of these solutions. We may observe, that the quantities which determine the moon's position, are to be determined by means of certain algebraical equations, which express the mechanical conditions of the motion. The operation, by which the result is to be obtained, involves the process of integration; which, in this instance, cannot be performed in an immediate and definite manner; since the quantities thus to be operated on depend upon the moon's position, and thus require us to know the very thing which we have to determine by the operation. The result must be got at, therefore, by successive approximations: we must first find a quantity near the truth; and then, by the help of this, one nearer still; and so on; and, in this manner, the moon's place will be given by a converging series of terms. The form of these terms depends upon the relations of position between the sun {369} and moon, their apogees, the moon's nodes, and other quantities; and by the variety of combinations of which these admit, the terms become very numerous and complex. The magnitude of the terms depends also upon various circumstances; as the relative force of the sun and earth, the relative times of the solar and lunar revolutions, the eccentricities and inclinations of the two orbits. These are combined so as to give terms of different orders of magnitudes; and it depends upon the skill and perseverance of the mathematician how far he will continue this series of terms. For there is no limit to their number: and though the methods of which we have spoken do theoretically enable us to calculate as many terms as we please, the labor and the complexity of the operations are so serious that common calculators are stopped by them. None but very great mathematicians have been able to walk safely any considerable distance into this avenue,--so rapidly does it darken as we proceed. And even the possibility of doing what has been done, depends upon what we may call accidental circumstances; the smallness of the inclinations and eccentricities of the system, and the like. "If nature had not favored us in this way," Lagrange used to say, "there would have been an end of the geometers in this problem." The expected return of the comet of 1682 in 1759, gave a new interest to the problem, and Clairaut proceeded to calculate the case which was thus suggested. When this was treated by the methods which had succeeded for the moon, it offered no prospect of success, in consequence of the absence of the favorable circumstances just referred to, and, accordingly, Clairaut, after obtaining the six equations to which he reduces the solution,[49\6] adds, "Integrate them who can" (Intègre maintenant qui pourra). New methods of approximation were devised for this case.

[Note 49\6: _Journal des Sçavans_, Aug. 1759.]

The problem of three bodies was not prosecuted in consequence of its analytical beauty, or its intrinsic attraction; but its great difficulties were thus resolutely combated from necessity; because in no other way could the theory of universal gravitation be known to be true or made to be useful. The construction of _Tables of the Moon_, an object which offered a large pecuniary reward, as well as mathematical glory, to the successful adventurer, was the main purpose of these labors.

The _Theory of the Planets_ presented the Problem of Three Bodies in a new form, and involved in peculiar difficulties; for the {370} approximations which succeed in the Lunar theory fail here. Artifices somewhat modified are required to overcome the difficulties of this case.

Euler had investigated, in particular, the motions of Jupiter and Saturn, in which there was a secular acceleration and retardation, known by observation, but not easily explicable by theory. Euler's memoirs, which gained the prize of the French Academy, in 1748 and 1752, contained much beautiful analysis; and Lagrange published also a theory of Jupiter and Saturn, in which he obtained results different from those of Euler. Laplace, in 1787, showed that this inequality arose from the circumstance that two of Saturn's years are very nearly equal to five of Jupiter's.

The problems relating to Jupiter's _Satellites_, were found to be even more complex than those which refer to the planets: for it was necessary to consider each satellite as disturbed by the other three at once; and thus there occurred the Problem of _Five_ Bodies. This problem was resolved by Lagrange.[50\6]

[Note 50\6: Bailly, _Ast. Mod._ iii. 178.]

Again, the newly-discovered _small Planets_, Juno, Ceres, Vesta, Pallas, whose orbits almost coincide with each other, and are more inclined and more eccentric than those of the ancient planets, give rise, by their perturbations, to new forms of the problem, and require new artifices.

In the course of these researches respecting Jupiter, Lagrange and Laplace were led to consider particularly the _secular Inequalities_ of the solar system; that is, those inequalities in which the duration of the cycle of change embraces very many revolutions of the bodies themselves. Euler in 1749 and 1755, and Lagrange[51\6] in 1766, had introduced the method of the _Variation of the Elements_ of the orbit; which consists in tracing the effect of the perturbing forces, not as directly altering the place of the planet, but as producing a change from one instant to another, in the dimensions and position of the Elliptical orbit which the planet describes.[52\6] Taking this view, he {371} determines the secular changes of each of the _elements_ or determining quantities of the orbit. In 1773, Laplace also attacked this subject of secular changes, and obtained expressions for them. On this occasion, he proved the celebrated proposition that, "the mean motions of the planets are invariable:" that is, that there is, in the revolutions of the system, no progressive change which is not finally stopped and reversed; no increase, which is not, after some period, changed into decrease; no retardation which is not at last succeeded by acceleration; although, in some cases, millions of years may elapse before the system reaches the turning-point. Thomas Simpson noticed the same consequence of the laws of universal attraction. In 1774 and 1776, Lagrange[53\6] still labored at the secular equations; extending his researches to the nodes and inclinations; and showed that the invariability of the mean motions of the planets, which Laplace had proved, neglecting the fourth powers of the eccentricities and inclinations of the orbits,[54\6] was true, however far the approximation was carried, so long as the squares of the disturbing masses were neglected. He afterwards improved his methods;[55\6] and,

in 1783, he endeavored to extend the calculation of the changes of the elements to the periodical equations, as well as the secular.

[Note 51\6: Gautier, _Prob. de Trois Corps_, p. 155.]

[Note 52\6: In the first edition of this History, I had ascribed to Lagrange the invention of the Method of Variation of Elements in the theory of Perturbations. But justice to Euler requires that we should assign this distinction to him; at least, next to Newton, whose mode of representing the paths of bodies by means of a _Revolving Orbit_, in the Ninth Section of the _Principia_, may be considered as an anticipation of the method of variation of elements. In the fifth volume of the _Mécanique Céleste_, livre xv. p. 305, is an abstract of Euler's paper of 1749; where Laplace adds, "C'est le premier essai de la méthode de la variation des constantes arbitraires." And in page 310 is an abstract of the paper of 1756: and speaking of the method, Laplace says, "It consists in regarding the elements of the elliptical motion as variable in virtue of the perturbing forces. Those elements are, 1, the axis major; 2, the epoch of the body being at the apse; 3, the eccentricity; 4, the movement of the apse; 5, the inclination; 6, the longitude of the node;" and he then proceeds to show how Euler did this. It is possible that Lagrange knew nothing of Euler's paper. See _Méc. Cél._ vol. v. p. 312. But Euler's conception and treatment of the method are complete, so that he must be looked upon as the author of it.]

[Note 53\6: Gautier, p. 104.]

[Note 54\6: Ib. p. 184.]

[Note 55\6: Ib. p. 196.]

8. _Mécanique Céleste_, _&c._--Laplace also resumed the consideration of the secular changes; and, finally, undertook his vast work, the _Mécanique Céleste_, which he intended to contain a complete view of the existing state of this splendid department of science. We may see, in the exultation which the author obviously feels at the thought of erecting this monument of his age, the enthusiasm which had been excited by the splendid course of mathematical successes of which I have given a sketch. The two first volumes of this great work appeared in 1799. The third and fourth volumes were published in 1802 and 1805 respectively. Since its publication, little has been added to the solution of the great problems of which it treats. In 1808, Laplace presented to the French Bureau des Longitudes, a Supplement to the _Mécanique Céleste_; the object of which was to improve still further {372} the mode of obtaining the secular variations of the elements. Poisson and Lagrange proved the invariability of the major axes of the orbits, as far as the second order of the perturbing forces. Various other authors have since labored at this subject. Burckhardt, in 1808, extended the perturbing function as far as the sixth order of the eccentricities. Gauss, Hansen, and Bessel, Ivory, MM. Lubbock, Plana, Pontécoulant, and Airy, have, at different periods up to the present time, either extended or illustrated some particular part of the theory, or applied it to special cases; as in the instance of Professor Airy's calculation of an inequality of Venus and the earth, of which the period is 240 years. The approximation of the Moon's motions has been pushed to an almost incredible extent by M. Damoiseau, and, finally, Plana has once

more attempted to present, in a single work (three thick quarto volumes), all that has hitherto been executed with regard to the theory of the Moon.

I give only the leading points of the progress of analytical dynamics. Hence I have not spoken in detail of the theory of the Satellites of Jupiter, a subject on which Lagrange gained a prize for a Memoir, in 1766, and in which Laplace discovered some most curious properties in 1784. Still less have I referred to the purely speculative question of _Tautochronous Curves_ in a resisting medium, though it was a subject of the labors of Bernoulli, Euler, Fontaine, D'Alembert, Lagrange, and Laplace. The reader will rightly suppose that many other curious investigations are passed over in utter silence.

[2d Ed.] [Although the analytical calculations of the great mathematicians of the last century had determined, in a demonstrative manner, a vast series of inequalities to which the motions of the sun, moon, and planets were subject in virtue of their mutual attraction, there were still unsatisfactory points in the solutions thus given of the great mechanical problems suggested by the System of the Universe. One of these points was the want of any evident mechanical significance in the successive members of these series. Lindenau relates that Lagrange, near the end of his life, expressed his sorrow that the methods of approximation employed in Physical Astronomy rested on arbitrary processes, and not on any insight into the results of mechanical action. But something was subsequently done to remove the ground of this complaint. In 1818, Gauss pointed out that secular equations may be conceived to result from the disturbing body being distributed along its orbit so as to form a ring, and thus made the result conceivable more distinctly than as a mere result of calculation. And it appears {373} to me that Professor Airy's treatise entitled _Gravitation_, published at Cambridge in 1834, is of great value in supplying similar modes of conception with regard to the mechanical origin of many of the principal inequalities of the solar system.

Bessel in 1824, and Hansen in 1828, published works which are considered as belonging, along with those of Gauss, to a new era in physical astronomy.[56\6] Gauss's _Theoria Motuum Corporum Celestium_, which had Lalande's medal assigned to it by the French Institute, had already (1810) resolved all problems concerning the determination of the place of a planet or comet in its orbit in function of the elements. The value of Hansen's labors respecting the Perturbations of the Planets was recognized by the Astronomical Society of London, which awarded to them its gold medal.

[Note 56\6: _Abhand. der Akad. d. Wissensch. zu Berlin_. 1824; and _Disquisitiones circa Theoriam Perturbationum_. See Jahn. _Gesch. der Astron._ p. 84.]

The investigations of M. Damoiseau, and of MM. Plana and Carlini, on the Problem of the Lunar Theory, followed nearly the same course as those of their predecessors. In these, as in the _Mécanique Céleste_ and in preceding works on the same subject, the Moon's co-ordinates (time, radius vector, and latitude) were expressed in function of her true longitude. The integrations were effected in series, and then by reversion of the series, the longitude was expressed in function of the time; and then in the

same manner the other two co-ordinates. But Sir John Lubbock and M. Pontécoulant have made the _mean_ longitude of the moon, that is, the time, the independent variable, and have expressed the moon's co-ordinates in terms of sines and cosines of angles increasing proportionally to the time. And this method has been adopted by M. Poisson (_Mem. Inst._ xiii. 1835, p. 212). M. Damoiseau, like Laplace and Clairaut, had deduced the successive coefficients of the lunar inequalities by numerical equations. But M. Plana expresses explicitly each coefficient in general terms of the letters expressing the constants of the problem, arranging them according to the order of the quantities, and substituting numbers at the end of the operation only. By attending to this arrangement, MM. Lubbock and Pontécoulant have verified or corrected a large portion of the terms contained in the investigations of MM. Damoiseau and Plana. Sir John Lubbock has calculated the polar co-ordinates of the Moon directly; M. Poisson, on the other hand, has obtained the variable elliptical elements; M. Pontécoulant conceives that the method of variation or arbitrary {374} constants may most conveniently be reserved for secular inequalities and inequalities of long periods.

MM. Lubbock and Pontécoulant have made the mode of treating the Lunar Theory and the Planetary Theory agree with each other, instead of following two different paths in the calculation of the two problems, which had previously been done.

Prof. Hansen, also, in his _Fundamenta Nova Investigationis Orbitæ veræ quam Luna perlustrat_ (_Gothæ_, 1838), gives a general method, including the Lunar Theory and the Planetary Theory as two special cases. To this is annexed a solution of the _Problem of Four Bodies_.

I am here speaking of the Lunar and Planetary Theories as Mechanical Problems only. Connected with this subject, I will not omit to notice a very general and beautiful method of solving problems respecting the motion of systems **of mutually attracting bodies, given by Sir W. R. Hamilton, in the _Philosophical Transactions_ for 1834-5 ("On a General Method in Dynamics"). His method consists in investigating the _Principal Function_ of the co-ordinates of the bodies: this function being one, by the differentiation of which, the co-ordinates of the bodies of the system may be found. Moreover, an approximate value of this function being obtained, the same formulæ supply a means of successive approximation without limit.]

9. _Precession. Motion of Rigid Bodies._--The series of investigations of which I have spoken, extensive and complex as it is, treats the moving bodies as points only, and takes no account of any peculiarity of their form or motion of their parts. The investigation of the motion of a body of any magnitude and form, is another branch of analytical mechanics, which well deserves notice. Like the former branch, it mainly owed its cultivation to the problems suggested by the solar system. Newton, as we have seen, endeavored to calculate the effect of the attraction of the sun and moon in producing the _precession of the equinoxes_; but in doing this he made some mistakes. In 1747, D'Alembert solved this problem by the aid of his "Principle;" and it was not difficult for him to show, as he did in his _Opuscules_, in 1761, that the same method enabled him to determine the motion of a body of any figure acted upon by any forces. But, as the reader will have observed in the course of this narrative, the

great mathematicians of this period were always nearly abreast of each other in their advances.--Euler,[57\6] in the mean time, had published, in 1751, a solution of the {375} problem of the precession; and in 1752, a memoir which he entitled _Discovery of a New Principle of Mechanics_, and which contains a solution of the general problem of the alteration of rotary motion by forces. D'Alembert noticed with disapprobation the assumption of priority which this title implied, though allowing the merit of the memoir. Various improvements were made in these solutions; but the final form was given them by Euler; and they were applied to a great variety of problems in his _Theory of the Motion of Solid and Rigid Bodies_, which was written[58\6] about 1760, and published in 1765. The formulæ in this work were much simplified by the use of a discovery of Segner, that every body has three axes which were called Principal Axes, about which alone (in general) it would permanently revolve. The equations which Euler and other writers had obtained, were attacked as erroneous by Landen in the Philosophical Transactions for 1785; but I think it is impossible to consider this criticism otherwise than as an example of the inability of the English mathematicians of that period to take a steady hold of the analytical generalizations to which the great Continental authors had been led. Perhaps one of the most remarkable calculations of the motion of a rigid body is that which Lagrange performed with regard to the _Moon's Libration_; and by which he showed that the Nodes of the Moon's Equator and those of her Orbit must always coincide.

[Note 57\6: _Ac. Berl._ 1745, 1750.]

[Note 58\6: See the preface to the book.]

10. _Vibrating Strings._--Other mechanical questions, unconnected with astronomy, were also pursued with great zeal and success. Among these was the problem of a vibrating string, stretched between two fixed points. There is not much complexity in the mechanical conceptions which belong to this case, but considerable difficulty in reducing them to analysis. Taylor, in his _Method of Increments_, published in 1716, had annexed to his work a solution of this problem; obtained on suppositions, limited indeed, but apparently conformable to the most common circumstances of practice. John Bernoulli, in 1728, had also treated the same problem. But it assumed an interest altogether new, when, in 1747, D'Alembert published his views on the subject; in which he maintained that, instead of one kind of curve only, there were an infinite number of different curves, which answered the conditions of the question. The problem, thus put forward by one great mathematician, was, as usual, taken up by the others, whose names the reader is now so familiar with in such an association. In {376} 1748, Euler not only assented to the generalization of D'Alembert, but held that it was not necessary that the curves so introduced should be defined by any algebraical condition whatever. From this extreme indeterminateness D'Alembert dissented; while Daniel Bernoulli, trusting more to physical and less to analytical reasonings, maintained that both these generalizations were inapplicable in fact, and that the solution was really restricted, as had at first been supposed, to the form of the trochoid, and to other forms derivable from that. He introduced, in such problems, the "Law of Coexistent

Vibrations," which is of eminent use in enabling us to conceive the results of complex mechanical conditions, and the real import of many analytical expressions. In the mean time, the wonderful analytical genius of Lagrange had applied itself to this problem. He had formed the Academy of Turin, in conjunction with his friends Saluces and Cigna; and the first memoir in their Transactions was one by him on this subject: in this and in subsequent writings he has established, to the satisfaction of the mathematical world, that the functions introduced in such cases are not necessarily continuous, but are arbitrary to the same degree that the motion is so practically; though capable of expression by a series of circular functions. This controversy, concerning the degree of lawlessness with which the conditions of the solution may be assumed, is of consequence, not only with respect to vibrating strings, but also with respect to many problems, belonging to a branch of Mechanics which we now have to mention, the Doctrine of Fluids.

11. _Equilibrium of Fluids. Figure of the Earth. Tides._--The application of the general doctrines of Mechanics to fluids was a natural and inevitable step, when the principles of the science had been generalized. It was easily seen that a fluid is, for this purpose, nothing more than a body of which the parts are movable amongst each other with entire facility; and that the mathematician must trace the consequences of this condition upon his equations. This accordingly was done, by the founders of mechanics, both for the cases of the equilibrium and of motion. Newton's attempt to solve the problem of the _figure of the earth_, supposing it fluid, is the first example of such an investigation: and this solution rested upon principles which we have already explained, applied with the skill and sagacity which distinguished all that Newton did.

We have already seen how the generality of the principle, that fluids press equally in all directions, was established. In applying it to calculation, Newton took for his fundamental principle, the equal {377} weight of columns of the fluid reaching to the centre; Huyghens took, as his basis, the **perpendicularity of the resulting force at each point to the surface of the fluid; Bouguer conceived that both principles were necessary; and Clairaut showed that the equilibrium of _all_ canals is requisite. He also was the first mathematician who deduced from this principle the Equations of Partial Differentials by which these laws are expressed; a step which, as Lagrange says,[59\6] changed the face of Hydrostatics, and made it a new science. Euler simplified the mode of obtaining the Equations of Equilibrium for any forces whatever; and put them in the form which is now generally adopted in our treatises.

[Note 59\6: _Méc. Analyt._ ii. p. 180.]

The explanation of the _Tides_, in the way in which Newton attempted it in the third book of the _Principia_, is another example of a hydrostatical investigation: for he considered only the form that the ocean would have if it were at rest. The memoirs of Maclaurin, Daniel Bernoulli, and Euler, on the question of the Tides, which shared among them the prize of the Academy of Sciences in 1740, went upon the same views.

The _Treatise of the Figure of the Earth_, by Clairaut, in 1743, extended Newton's solution of the same problem, by supposing a solid nucleus covered with a fluid of different density. No peculiar novelty has been introduced into this subject, except a method employed by Laplace for determining the attractions of spheroids of small eccentricity, which is, as Professor Airy has said,[60\6] "a calculus the most singular in its nature, and the most powerful in its effects, of any which has yet appeared."

[Note 60\6: _Enc. Met._ Fig. of Earth, p. 192.]

12. _Capillary Action._--There is only one other problem of the statics of fluids on which it is necessary to say a word,--the doctrine of Capillary Attraction. Daniel Bernoulli,[61\6] in 1738, states that he passes over the subject, because he could not reduce the facts to general laws: but Clairaut was more successful, and Laplace and Poisson have since given great analytical completeness to his theory. At present our business is, not so much with the sufficiency of the theory to explain phenomena, as with the mechanical problem of which this is an example, which is one of a very remarkable and important character; namely, to determine the effect of attractions which are exercised by all the particles of bodies, on the hypothesis that the {378} attraction of each particle, though sensible when it acts upon another particle at an extremely small distance from it, becomes insensible and vanishes the moment this distance assumes a perceptible magnitude. It may easily be imagined that the analysis by which results are obtained under conditions so general and so peculiar, is curious and abstract; the problem has been resolved in some very extensive cases.

[Note 61\6: _Hydrodyn._ Pref. p. 5.]

13. _Motion of Fluids._--The only branch of mathematical mechanics which remains to be considered, is that which is, we may venture to say, hitherto incomparably the most incomplete of all,--Hydrodynamics. It may easily be imagined that the mere hypothesis of absolute relative mobility in the parts, combined with the laws of motion and nothing more, are conditions too vague and general to lead to definite conclusions. Yet such are the conditions of the problems which relate to the motion of fluids. Accordingly, the mode of solving them has been, to introduce certain other hypotheses, often acknowledged to be false, and almost always in some measure arbitrary, which may assist in determining and obtaining the solution. The Velocity of a fluid issuing from an orifice in a vessel, and the Resistance which a solid body suffers in moving in a fluid, have been the two main problems on which mathematicians have employed themselves. We have already spoken of the manner in which Newton attacked both these, and endeavored to connect them. The subject became a branch of Analytical Mechanics by the labors of D. Bernoulli, whose _Hydrodynamica_ was published in 1738. This work rests upon the Huyghenian principle of which we have already spoken in the history of the centre of oscillation; namely, the equality of the _actual descent_ of the particles and the _potential ascent_; or, in other words, the conservation of _vis viva_. This was the first analytical treatise; and the analysis is declared by Lagrange to be as elegant in its steps as it is simple in its results. Maclaurin also treated the subject; but is accused of reasoning in such a way as to show that he had determined upon his result beforehand; and the method of John Bernoulli, who likewise wrote upon it, has been strongly objected to

by D'Alembert. D'Alembert himself applied the principle which bears his name to this subject; publishing a _Treatise on the Equilibrium and Motion of Fluids_ in 1744, and on the _Resistance of Fluids_ in 1753. His _Réflexions sur la Cause Générale des Vents_, printed in 1747, are also a celebrated work, belonging to this part of mathematics. Euler, in this as in other cases, was one of those who most contributed to give analytical elegance to the subject. In addition to the questions which {379} have been mentioned, he and Lagrange treated the problems of the small vibrations of fluids, both inelastic and elastic;--a subject which leads, like the question of vibrating strings, to some subtle and abstruse considerations concerning the significations of the integrals of partial differential equations. Laplace also took up the subject of waves propagated along the surface of water; and deduced a very celebrated theory of the tides, in which he considered the ocean to be, not in equilibrium, as preceding writers had supposed, but agitated by a constant series of undulations, produced by the solar and lunar forces. The difficulty of such an investigation may be judged of from this, that Laplace, in order to carry it on, is obliged to assume a mechanical proposition, unproved, and only conjectured to be true; namely,[62\6] that, "in a system of bodies acted upon by forces which are periodical, the state of the system is periodical like the forces." Even with this assumption, various other arbitrary processes are requisite; and it appears still very doubtful whether Laplace's theory is either a better mechanical solution of the problem, or a nearer approximation to the laws of the phenomena, than that obtained by D. Bernoulli, following the views of Newton.

[Note 62\6: _Méc. Cél._ t. ii. p. 218.]

In most cases, the solutions of problems of hydrodynamics are not satisfactorily confirmed by the results of observation. Poisson and Cauchy have prosecuted the subject of waves, and have deduced very curious conclusions by a very recondite and profound analysis. The assumptions of the mathematician here do not represent the conditions of nature; the rules of theory, therefore, are not a good standard to which we may refer the aberrations of particular cases; and the laws which we obtain from experiment are very imperfectly illustrated by _à priori_ calculation. The case of this department of knowledge, Hydrodynamics, is very peculiar; we have reached the highest point of the science,--the laws of extreme simplicity and generality from which the phenomena flow; we cannot doubt that the ultimate principles which we have obtained are the true ones, and those which really apply to the facts; and yet we are far from being able to apply the principles to explain or find out the facts. In order to do this, we want, in addition to what we have, true and useful principles, intermediate between the highest and the lowest;--between the extreme and almost barren generality of the laws of motion, and the endless varieties and inextricable complexity of fluid motions in special cases. {380} The reason of this peculiarity in the science of Hydrodynamics appears to be, that its general principles were not discovered with reference to the science itself, but by extension from the sister science of the Mechanics of Solids; they were not obtained by ascending gradually from particulars, to truths more and more general, respecting the motions of fluids; but were caught at once, by a perception that the parts of fluids are included in that range of generality which we are entitled to give to the supreme laws of motions of

solids. Thus, Solid Dynamics and Fluid Dynamics resemble two edifices which have their highest apartment in common, and though we can explore every part of the former building, we have not yet succeeded in traversing the staircase of the latter, either from the top or from the bottom. If we had lived in a world in which there were no solid bodies, we should probably not have yet discovered the laws of motion; if we had lived in a world in which there were no fluids, we should have no idea how insufficient a complete possession of the general laws of motion may be, to give us a true knowledge of particular results.

14. _Various General Mechanical Principles._--The generalized laws of motion, the points to which I have endeavored to conduct my history, include in them all other laws by which the motions of bodies can be regulated; and among such, several laws which had been discovered before the highest point of generalization was reached, and which thus served as stepping-stones to the ultimate principles. Such were, as we have seen, the Principles of the Conservation of _vis viva_, the Principle of the Conservation of the Motion of the Centre of Gravity, and the like. These principles may, of course, be deduced from our elementary laws, and were finally established by mathematicians on that footing. There are other principles which may be similarly demonstrated; among the rest, I may mention the Principle of _the Conservation of areas_, which extends to any number of bodies a law analogous to that which Kepler had observed, and Newton demonstrated, respecting the areas described by each planet round the sun. I may mention also, the Principle of the _Immobility of the plane of maximum areas_, a plane which is not disturbed by any mutual action of the parts of any system. The former of these principles was published about the same time by Euler, D. Bernoulli, and Darcy, under different forms, in 1746 and 1747; the latter by Laplace.

To these may be added a law, very celebrated in its time, and the occasion of an angry controversy, _the Principle of least action_. {381} Maupertuis conceived that he could establish _à priori_, by theological arguments, that all mechanical changes must take place in the world so as to occasion the least possible quantity of _action_. In asserting this, it was proposed to measure the Action by the product of Velocity and Space; and this measure being adopted, the mathematicians, though they did not generally assent to Maupertuis' reasonings, found that his principle expressed a remarkable and useful truth, which might be established on known mechanical grounds.

15. _Analytical Generality. Connection of Statics and Dynamics._--Before I quit this subject, it is important to remark the peculiar character which the science of Mechanics has now assumed, in consequence of the extreme analytical generality which has been given it. Symbols, and operations upon symbols, include the whole of the reasoner's task; and though the relations of space are the leading subjects in the science, the great analytical treatises upon it do not contain a single diagram. The _Mécanique Analytique_ of Lagrange, of which the first edition appeared in 1788, is by far the most consummate example of this analytical generality. "The plan of this work," says the author, "is entirely new. I have proposed to myself to reduce the whole theory of this science, and the art of resolving the problems which it includes,

to general formulæ, of which the simple development gives all the equations necessary for the solution of the problem."--"The reader will find no figures in the work. The methods which I deliver do not require either constructions, or geometrical or mechanical reasonings; but only algebraical operations, subject to a regular and uniform rule of proceeding." Thus this writer makes Mechanics a branch of Analysis; instead of making, as had previously been done, Analysis an implement of Mechanics.[63\6] The transcendent generalizing genius of Lagrange, and his matchless analytical skill and elegance, have made this undertaking as successful as it is striking.

[Note 63\6: Lagrange himself terms Mechanics, "An Analytical Geometry of four dimensions." Besides the _three co-ordinates_ which determine the place of a body in _space_, the _time_ enters as a _fourth co-ordinate_. [Note by Littrow.]]

The mathematical reader is aware that the language of mathematical symbols is, in its nature, more general than the language of words: and that in this way truths, translated into symbols, often suggest their own generalizations. Something of this kind has happened in Mechanics. The same Formula expresses the general condition of Statics and that of Dynamics. The tendency to generalization which is thus introduced by analysis, makes mathematicians unwilling to {382} acknowledge a plurality of Mechanical principles; and in the most recent analytical treatises on the subject, all the doctrines are deduced from the single Law of Inertia. Indeed, if we identify Forces with the Velocities which produce them, and allow the Composition of Forces to be applicable to force _so understood_, it is easy to see that we can reduce the Laws of Motion to the Principles of Statics; and this conjunction, though it may not be considered as philosophically just, is verbally correct. If we thus multiply or extend the meanings of the term Force, we make our elementary principles simpler and fewer than before; and those persons, therefore, who are willing to assent to such a use of words, can thus obtain an additional generalisation of dynamical principles; and this, as I have stated, has been adopted in several recent treatises. I shall not further discuss here how far this is a real advance in science.

Having thus rapidly gone through the history of Force and Attraction in the abstract, we return to the attempt to interpret the phenomena of the universe by the aid of these abstractions thus established.

But before we do so, we may make one remark on the history of this part of science. In consequence of the vast career into which the Doctrine of Motion has been drawn by the splendid problems proposed to it by Astronomy, the origin and starting-point of Mechanics, namely Machines, had almost been lost out of sight. _Machines_ had become the smallest part of _Mechanics_, as _Land-measuring_ had become the smallest part of _Geometry_. Yet the application of Mathematics to the doctrine of Machines has led, at all periods of the Science, and especially in our own time, to curious and valuable results. Some of these will be noticed in the _Additions_ to this volume.

{{383}} BOOK VII.

THE MECHANICAL SCIENCES. (CONTINUED.)

HISTORY OF PHYSICAL ASTRONOMY.

DESCEND from heaven, Urania, by that name If rightly thou art called, whose voice divine Following, above the Olympian hill I soar, Above the flight of Pegasean wing. The meaning, not the name, I call, for thou Nor of the muses nine, nor on the top Of old Olympus dwell'st: but heavenly-born, Before the hills appeared, or fountain flowed, Thou with Eternal Wisdom didst converse, Wisdom, thy sister.

_Paradise Lost_, B. vii.

{{385}} CHAPTER I.

PRELUDE TO THE INDUCTIVE EPOCH OF NEWTON.

WE have now to contemplate the last and most splendid period of the progress of Astronomy;--the grand completion of the history of the most ancient and prosperous province of human knowledge;--the steps which elevated this science to an unrivalled eminence above other sciences;--the first great example of a wide and complex assemblage of phenomena indubitably traced to their single simple cause;--in short, the first example of the formation of a perfect Inductive Science.

In this, as in other considerable advances in real science, the complete disclosure of the new truths by the principal discoverer, was preceded by movements and glimpses, by trials, seekings, and guesses on the part of others; by indications, in short, that men's minds were already carried by their intellectual impulses in the direction in which the truth lay, and were beginning to detect its nature. In a case so important and interesting as this, it is more peculiarly proper to give some view of this Prelude to the Epoch of the full discovery.

(_Francis Bacon._) That Astronomy should become Physical Astronomy,--that the motions of the heavenly bodies should be traced to their causes, as well as reduced to rule,--was felt by all persons of active and philosophical minds as a pressing and irresistible need, at the time of which we speak. We have already seen how much this feeling had to do in impelling Kepler to the train of laborious research by which he made his discoveries. Perhaps it may be interesting to point out how strongly this persuasion of the necessity of giving a physical character to astronomy, had taken possession of the mind of Bacon, who, looking at the progress of knowledge with a more comprehensive spirit, and from a higher point of view than Kepler, could have none of his astronomical prejudices, since on that subject he was of a different school, and of far inferior knowledge. In his "Description of the Intellectual Globe," Bacon says that while Astronomy had, up to that time, had it for her business to inquire into the rules of the heavenly motions, and Philosophy into their causes, they had both so far worked without due appreciation of their respective tasks; Philosophy neglecting facts, and Astronomy claiming assent to her {386} mathematical hypotheses, which ought to be considered as mere steps of calculation. "Since, therefore," he continues,[1\7] "each science has hitherto been a slight and ill-constructed thing, we must assuredly take a firmer stand; our ground being, that these two subjects, which on account of the narrowness of men's views and the traditions of professors have been so long dissevered, are, in fact, one and the same thing, and compose one body of science." It must be allowed that, however erroneous might be the points of

Bacon's positive astronomical creed, these general views of the nature and position of the science are most sound and philosophical.

[Note 1\7: Vol. ix. 221.]

(_Kepler_) In his attempts to suggest a right physical view of the starry heavens and their relation to the earth, Bacon failed, along with all the writers of his time. It has already been stated that the main cause of this failure was the want of a knowledge of the true theory of motion;--the non-existence of the science of Dynamics. At the time of Bacon and Kepler, it was only just beginning to be possible to reduce the heavenly motions to the laws of earthly motion, because the latter were only just then divulged. Accordingly, we have seen that the whole of Kepler's physical speculations proceed upon an ignorance of the first law of motion, and assume it to be the main problem of the physical astronomer to assign the cause which _keeps up_ the motions of the planets. Kepler's doctrine is, that a certain Force or Virtue resides in the sun, by which all bodies within his influence are carried round him. He illustrates[2\7] the nature of this Virtue in various ways, comparing it to Light, and to the Magnetic Power, which it resembles in the circumstances of operating at a distance, and also in exercising a feebler influence as the distance becomes greater. But it was obvious that these comparisons were very imperfect; for they do not explain how the sun produces in a body at a distance a motion _athwart_ the line of emanation; and though Kepler introduced an assumed rotation of the sun on his axis as the cause of this effect, that such a cause could produce the result could not be established by any analogy of terrestrial motions. But another image to which he referred, suggested a much more substantial and conceivable kind of mechanical action by which the celestial motions might be produced, namely, a current of fluid matter circulating round the sun, and carrying the planet with it, like a boat in a stream. In the Table of Contents of the work on the planet Mars, the purport of the chapter to which I have alluded is {387} stated as follows: "A physical speculation, in which it is demonstrated that the vehicle of that Virtue which urges the planets, circulates through the spaces of the universe after the manner of a river or whirlpool (_vortex_), moving quicker than the planets." I think it will be found, by any one who reads Kepler's phrases concerning the _moving force,--the magnetic nature,--the immaterial virtue_ of the sun, that they convey no distinct conception, except so far as they are interpreted by the expressions just quoted. A vortex of fluid constantly whirling round the sun, kept in this whirling motion by the rotation of the sun himself, and carrying the planets round the sun by its revolution, as a whirlpool carries straws, could be readily understood; and though it appears to have been held by Kepler that this current and vortex was immaterial, he ascribes to it the power of overcoming the inertia of bodies, and of putting them and keeping them in motion, the only material properties with which he had any thing to do. Kepler's physical reasonings, therefore, amount, in fact, to the doctrine of Vortices round the central bodies, and are occasionally so stated by himself; though by asserting these vortices to be "an immaterial species," and by the fickleness and variety of his phraseology on the subject, he leaves this theory in some confusion;--a proceeding, indeed, which both his want of sound mechanical conceptions, and his busy and inventive fancy, might have led us to expect. Nor, we may venture to say, was it easy for any one at

Kepler's time to devise a more plausible theory than the theory of vortices might have been made. It was only with the formation and progress of the science of Mechanics that this theory became untenable.

[Note 2\7: _De Stellâ Martis_, P. 3. c. xxxiv.]

(_Descartes_) But if Kepler might be excused, or indeed admired, for propounding the theory of Vortices at his time, the case was different when the laws of motion had been fully developed, and when those who knew the state of mechanical science ought to have learned to consider the motions of the stars as a mechanical problem, subject to the same conditions as other mechanical problems, and capable of the same exactness of solution. And there was an especial inconsistency in the circumstance of the Theory of Vortices being put forwards by Descartes, who pretended, or was asserted by his admirers, to have been one of the discoverers of the true Laws of Motion. It certainly shows both great conceit and great shallowness, that he should have proclaimed with much pomp this crude invention of the ante-mechanical period, at the time when the best mathematicians of Europe, as Borelli in Italy, Hooke and Wallis in England, Huyghens in Holland, {388} were patiently laboring to bring the mechanical problem of the universe into its most distinct form, in order that it might be solved at last and forever.

I do not mean to assert that Descartes borrowed his doctrines from Kepler, or from any of his predecessors, for the theory was sufficiently obvious; and especially if we suppose the inventor to seek his suggestions rather in the casual examples offered to the sense than in the exact laws of motion. Nor would it be reasonable to rob this philosopher of that credit, of the plausible deduction of a vast system from apparently simple principles, which, at the time, was so much admired; and which undoubtedly was the great cause of the many converts to his views. At the same time we may venture to say that a system of doctrine thus deduced from assumed principles by a long chain of reasoning, and not verified and confirmed at every step by detailed and exact facts, has hardly a chance of containing any truth. Descartes said that he should think it little to show how the world _is_ constructed, if he could not also show that it _must_ of necessity have been so constructed. The more modest philosophy which has survived the boastings of his school is content to receive all its knowledge of facts from experience, and never dreams of interposing its peremptory _must be_ when nature is ready to tell us what _is_. The _à priori_ philosopher has, however, always a strong feeling in his favor among men. The deductive form of his speculations gives them something of the charm and the apparent certainty of pure mathematics; and while he avoids that laborious recurrence to experiments, and measures, and multiplied observations, which is irksome and distasteful to those who are impatient to grow wise at once, every fact of which the theory appears to give an explanation, seems to be an unasked and almost an infallible witness in its favor.

My business with Descartes here is only with his physical Theory of Vortices; which, great as was its glory at one time, is now utterly extinguished. It was propounded in his _Principia Philosophiæ_, in 1644. In order to arrive at this theory, he begins, as might be expected of him, from reasonings sufficiently general. He lays it down as a maxim, in the first sentence of his book, that a person who seeks for

truth must, once in his life, doubt of all that he most believes. Conceiving himself thus to have stripped himself of all his belief on all subjects, in order to resume that part of it which merits to be retained, he begins with his celebrated assertion, "I think, therefore I am;" which appears to him a certain and immovable principle, by means of {389} which he may proceed to something more. Accordingly, to this he soon adds the idea, and hence the certain existence, of God and his perfections. He then asserts it to be also manifest, that a vacuum in any part of the universe is impossible; the whole must be filled with matter, and the matter must be divided into equal angular parts, this being the most simple, and therefore the most natural supposition.[3\7] This matter being in motion, the parts are necessarily ground into a spherical form; and the corners thus rubbed off (like filings or sawdust) form a second and more subtle matter.[4\7] There is, besides, a third kind of matter, of parts more coarse and less fitted for motion. The first matter makes luminous bodies, as the sun, and the fixed stars; the second is the transparent substance of the skies; the third is the material of opake bodies, as the earth, planets, and comets. We may suppose, also,[5\7] that the motions of these parts take the form of revolving circular currents,[6\7] or _vortices_. By this means, the first matter will be collected to the centre of each vortex, while the second, or subtle matter, surrounds it, and, by its centrifugal effort, constitutes light. The planets are carried round the sun by the motion of his vortex,[7\7] each planet being at such a distance from the sun as to be in a part of the vortex suitable to its solidity and mobility. The motions are prevented from being exactly circular and regular by various causes; for instance, a vortex may be pressed into an oval shape by contiguous vortices. The satellites are, in like manner, carried round their primary planets by subordinate vortices; while the comets have sometimes the liberty of gliding out of one vortex into the one next contiguous, and thus travelling in a sinuous course, from system to system, through the universe. It is not necessary for us to speak here of the entire deficiency of this system in mechanical consistency, and in a correspondency to observation in details and measures. Its general reception and temporary sway, in some instances even among intelligent men and good mathematicians, are the most remarkable facts connected with it. These may be ascribed, in part, to the circumstance that philosophers were now ready and eager for a physical astronomy commensurate with the existing state of knowledge; they may have been owing also, in some measure, to the character and position of Descartes. He was a man of high claims in every department of speculation, and, in pure mathematics, a genuine inventor of great eminence;--a man of family and a soldier;--an inoffensive philosopher, attacked and persecuted {390} for his opinions with great bigotry and fury by a Dutch divine, Voet;--the favorite and teacher of two distinguished princesses, and, it is said, the lover of one of them. This was Elizabeth, the daughter of the Elector Frederick, and consequently grand-daughter of our James the First. His other royal disciple, the celebrated Christiana of Sweden, showed her zeal for his instructions by appointing the hour of five in the morning for their interviews. This, in the climate of Sweden, and in the winter, was too severe a trial for the constitution of the philosopher, born in the sunny valley of the Loire; and, after a short residence at Stockholm, he died of an inflammation of the chest in 1650. He always kept up an active correspondence with his friend Mersenne, who was called, by some of the Parisians, "the Resident of

Descartes at Paris;" and who informed him of all that was done in the world of science. It is said that he at first sent to Mersenne an account of a system of the universe which he had devised, which went on the assumption of a vacuum; Mersenne informed him that the _vacuum_ was no longer the fashion at Paris; upon which he proceeded to remodel his system, and to re-establish it on the principle of a _plenum_. Undoubtedly he tried to avoid promulgating opinions which might bring him into trouble. He, on all occasions, endeavored to explain away the doctrine of the motion of the earth, so as to evade the scruples to which the decrees of the pope had given rise; and, in stating the theory of vortices, he says,[8\7] "There is no doubt that the world was created at first with all its perfection; nevertheless, it is well to consider how it might have arisen from certain principles, although we know that it did not." Indeed, in the whole of his philosophy, he appears to deserve the character of being both rash and cowardly, "_pusillanimus simul et audax_," far more than Aristotle, to whose physical speculations Bacon applies this description.[9\7]

[Note 3\7: _Prin._ p. 58.]

[Note 4\7: Ib. p. 59.]

[Note 5\7: Ib. p. 56.]

[Note 6\7: Ib. p. 61.]

[Note 7\7: Ib. c. 140, p. 114.]

[Note 8\7: _Prin._ p. 56.]

[Note 9\7: Bacon, _Descriptio Globi Intellectualis_.]

Whatever the causes might be, his system was well received and rapidly adopted. Gassendi, indeed, says that he found nobody who had the courage to read the _Principia_ through;[10\7] but the system was soon embraced by the younger professors, who were eager to dispute in its favor. It is said[11\7] that the University of Paris was on the point of publishing an edict against these new doctrines, and was only prevented from doing so by a pasquinade which is worth mentioning. It was composed by the poet Boileau (about 1684), and professed to be a Request in favor of Aristotle, and an Edict issued from Mount {391} Parnassus in consequence. It is obvious that, at this time, the cause of Cartesianism was looked upon as the cause of free inquiry and modern discovery, in opposition to that of bigotry, prejudice, and ignorance. Probably the poet was far from being a very severe or profound critic of the truth of such claims. "This petition of the Masters of Arts, Professors and Regents of the University of Paris, humbly showeth, that it is of public notoriety that the sublime and incomparable Aristotle was, without contest, the first founder of the four elements, fire, air, earth, and water; that he did, by special grace, accord unto them a simplicity which belongeth not to them of natural right;" and so on. "Nevertheless, since, a certain time past, two individuals, named Reason and Experience, have leagued themselves together to dispute his claim to the rank which of justice pertains to him, and have tried to erect themselves a throne on the ruins of his authority; and, in order the better to gain their ends, have excited certain factious spirits, who, under the names of Cartesians and Gassendists, have begun to shake off the yoke of their

master, Aristotle; and, contemning his authority, with unexampled temerity, would dispute the right which he had acquired of making true pass for false and false for true;"--In fact, this production does not exhibit any of the peculiar tenets of Descartes, although, probably, the positive points of his doctrines obtained a footing in the University of Paris, under the cover of this assault on his adversaries. The Physics of Rohault, a zealous disciple of Descartes, was published at Paris about 1670,[12\7] and was, for a time, the standard book for students of this subject, both in France and in England. I do not here speak of the later defenders of the Cartesian system, for, in their hands, it was much modified by the struggle which it had to maintain against the Newtonian system.

[Note 10\7: Del. _A. M._ ii. 193.]

[Note 11\7: _Enc. Brit._ art. _Cartesianism._]

[Note 12\7: And a second edition in 1672.]

We are concerned with Descartes and his school only as they form part of the picture of the intellectual condition of Europe just before the publication of Newton's discoveries. Beyond this, the Cartesian speculations are without value. When, indeed, Descartes' countrymen could no longer refuse their assent and admiration to the Newtonian theory, it came to be the fashion among them to say that Descartes had been the necessary precursor of Newton; and to adopt a favorite saying of Leibnitz, that the Cartesian philosophy was the antechamber of Truth. Yet this comparison is far from being happy: it appeared rather as if these suitors had mistaken the door; for those {392} who first came into the presence of Truth herself, were those who never entered this imagined antechamber, and those who were in the antechamber first, were the last in penetrating further. In partly the same spirit, Playfair has noted it as a service which Newton perhaps owed to Descartes, that "he had exhausted one of the most tempting forms of error." We shall see soon that this temptation had no attraction for those who looked at the problem in its true light, as the Italian and English philosophers already did. Voltaire has observed, far more truly, that Newton's edifice rested on no stone of Descartes' foundations. He illustrates this by relating that Newton only once read the work of Descartes, and, in doing so, wrote the word "_error_," repeatedly, on the first seven or eight pages; after which he read no more. This volume, Voltaire adds, was for some time in the possession of Newton's nephew.[13\7]

[Note 13\7: _Cartesianism_, Enc. Phil.]

(_Gassendi._) Even in his own country, the system of Descartes was by no means universally adopted. We have seen that though Gassendi was coupled with Descartes as one of the leaders of the new philosophy, he was far from admiring his work. Gassendi's own views of the causes of the motions of the heavenly bodies are not very clear, nor even very clearly referrible to the laws of mechanics; although he was one of those who had most share in showing that those laws apply to astronomical motions. In a chapter, headed[14\7] "Quæ sit motrix siderum causa," he reviews several opinions; but the one which he seems to adopt, is that which ascribes the motion of the celestial globes to certain fibres, of which the action is similar to that of

the muscles of animals. It does not appear, therefore, that he had distinctly apprehended, either the continuation of the movements of the planets by the First Law of Motion, or their deflection by the Second Law;--the two main steps on the road to the discovery of the true forces by which they are made to describe their orbits.

[Note 14\7: Gassendi, _Opera_, vol. i. p. 639.]

(_Leibnitz, &c._) Nor does it appear that in Germany mathematicians had attained this point of view. Leibnitz, as we have seen, did not assent to the opinions of Descartes, as containing the complete truth; and yet his own views of the physics of the universe do not seem to have any great advantage over these. In 1671 he published _A new physical hypothesis, by which the causes of most phenomena are deduced from a certain single universal motion supposed in our globe;--not to be despised either by the Tychonians or the Copernicans_. He supposes {393} the particles of the earth to have separate motions, which produce collisions, and thus propagate[15\7] an "agitation of the ether," radiating in all directions; and,[16\7] "by the rotation of the sun on its axis, concurring with its rectilinear action on the earth, arises the motion of the earth about the sun." The other motions of the solar system are, as we might expect, accounted for in a similar manner; but it appears difficult to invest such an hypothesis with any mechanical consistency.

[Note 15\7: Art. 5.]

[Note 16\7: Ib. 8.]

John Bernoulli maintained to the last the Cartesian hypothesis, though with several modifications of his own, and even pretended to apply mathematical calculation to his principles. This, however, belongs to a later period of our history; to the reception, not to the prelude, of the Newtonian theory.

(_Borelli._) In Italy, Holland, and England, mathematicians appear to have looked much more steadily at the problem of the celestial motions, by the light which the discovery of the real laws of motion threw upon it. In Borelli's _Theories of the Medicean Planets_, printed at Florence in 1666, we have already a conception of the nature of central action, in which true notions begin to appear. The attraction of a body upon another which revolves about it is spoken of and likened to magnetic action; not converting the attracting force into a transverse force, according to the erroneous views of Kepler, but taking it as a tendency of the bodies to meet. "It is manifest," says he,[17\7] "that every planet and satellite revolves round some principal globe of the universe as a fountain of virtue, which so draws and holds them that they cannot by any means be separated from it, but are compelled to follow it wherever it goes, in constant and continuous revolutions." And, further on, he describes[18\7] the nature of the action, as a matter of conjecture indeed, but with remarkable correctness.[19\7] "We shall account for these motions by supposing, that which can hardly be denied, that the planets have a certain natural appetite for uniting themselves with the globe round which they revolve, and that they really tend, with all their efforts, to approach to such globe; the planets, for instance, to the sun, the Medicean Stars to Jupiter. It is certain, also, that circular motion gives a body a tendency to recede from the centre of such revolution, as we find in a wheel,

or a stone whirled in a sling. Let us suppose, then, the planet to endeavor to approach the sun; since, in the mean time, it requires, by the circular motion, a force to recede from the same central body, it comes to pass, that when {394} those two opposite forces are equal, each compensates the other, and the planet cannot go nearer to the sun nor further from him than a certain determinate space, and thus appears balanced and floating about him."

[Note 17\7: Cap. 2.]

[Note 18\7: Ib. 11.]

[Note 19\7: P. 47.]

This is a very remarkable passage; but it will be observed, at the same time, that the author has no distinct conception of the manner in which the change of direction of the planet's motion is regulated from one instant to another; still less do his views lead to any mode of calculating the distance from the central body at which the planet would be thus balanced, or the space through which it might approach to the centre and recede from it. There is a great interval from Borelli's guesses, even to Huyghens' theorems and a much greater to the beginning of Newton's discoveries.

(_England._) It is peculiarly interesting to us to trace the gradual approach towards these discoveries which took place in the minds of English mathematicians and this we can do with tolerable distinctness. Gilbert, in his work, _De Magnete_, printed in 1600, has only some vague notions that the magnetic virtue of the earth in some way determines the direction of the earth's axis, the rate of its diurnal rotation, and that of the revolution of the moon about it.[20\7] He died in 1603, and, in his posthumous work, already mentioned (_De Mundo nostro Sublunari Philosophia nova_, 1651), we have already a more distinct statement of the attraction of one body by another.[21\7] "The force which emanates from the moon reaches to the earth, and, in like manner, the magnetic virtue of the earth pervades the region of the moon: both correspond and conspire by the joint action of both, according to a proportion and conformity of motions; but the earth has more effect, in consequence of its superior mass; the earth attracts and repels the moon, and the moon, within certain limits, the earth; not so as to make the bodies come together, as magnetic bodies do, but so that they may go on in a continuous course." Though this phraseology is capable of representing a good deal of the truth, it does not appear to have been connected, in the author's mind, with any very definite notions of mechanical action in detail. We may probably say the same of Milton's language: What if the sun Be centre to the world; and other stars, By his attractive virtue and their own Incited, dance about him various rounds? _Par. Lost_, B. viii. {395}

[Note 20\7: Lib. vi. cap. 6, 7.]

[Note 21\7: Ib. ii. c. 19.]

Boyle, about the same period, seems to have inclined to the Cartesian hypothesis. Thus, in order to show the advantage of the natural theology which contemplates organic contrivances, over that which refers to astronomy, he remarks: "It may be said, that in bodies inanimate,[22\7] the contrivance is very rarely so exquisite but

that the various motions and occurrences of their parts may, without much improbability, be suspected capable, after many essays, to cast one another into several of those circumvolutions called by Epicurus συστροφὰς and by Descartes, _vortices_; which being once made, may continue a long time after the manner explained by the latter." Neither Milton nor Boyle, however, can be supposed to have had an exact knowledge of the laws of mechanics; and therefore they do not fully represent the views of their mathematical contemporaries. But there arose about this time a group of philosophers, who began to knock at the door where Truth was to be found, although it was left for Newton to force it open. These were the founders of the Royal Society, Wilkins, Wallis, Seth Ward, Wren, Hooke, and others. The time of the beginning of the speculations and association of these men corresponds to the time of the civil wars between the king and parliament in England and it does not appear a fanciful account of their scientific zeal and activity, to say, that while they shared the common mental ferment of the times, they sought in the calm and peaceful pursuit of knowledge a contrast to the vexatious and angry struggles which at that time disturbed the repose of society. It was well if these dissensions produced any good to science to balance the obvious evils which flowed from them. Gascoigne, the inventor of the micrometer, a friend of Horrox, was killed in the battle of Marston Moor. Milburne, another friend of Horrox, who like him detected the errors of Lansberg's astronomical tables, left papers on this subject, which were lost by the coming of the Scotch army into England in 1639; in the civil war which ensued, the anatomical collections of Harvey were plundered and destroyed. Most of these persons of whom I have lately had to speak, were involved in the changes of fortune of the Commonwealth, some on one side, and some on the other. Wilkins was made Warden of Wadham by the committee of parliament appointed for reforming the University of Oxford; and was, in 1659, made Master of Trinity College, Cambridge, by Richard Cromwell, but ejected thence the year following, upon the restoration of the {396} royal sway. Seth Ward, who was a Fellow of Sidney College, Cambridge, was deprived of his Fellowship by the parliamentary committee; but at a later period (1649) he took the engagement to be faithful to the Commonwealth, and became Savilian Professor of Astronomy at Oxford. Wallis held a Fellowship of Queen's College, Cambridge, but vacated it by marriage. He was afterwards much employed by the royal party in deciphering secret writings, in which art he had peculiar skill. Yet he was appointed by the parliamentary commissioners Savilian Professor of Geometry at Oxford, in which situation he was continued by Charles II. after his restoration. Christopher Wren was somewhat later, and escaped these changes. He was chosen Fellow of All-Souls in 1652, and succeeded Ward as Savilian Professor of Astronomy. These men, along with Boyle and several others, formed themselves into a club, which they called the Philosophical, or the Invisible College; and met, from about the year 1645, sometimes in London, and sometimes in Oxford, according to the changes of fortune and residence of the members. Hooke went to Christ Church, Oxford, in 1663, where he was patronized by Boyle, Ward, and Wallis; and when the Philosophical College resumed its meetings in London, after the Restoration, as the Royal Society, Hooke was made "curator of experiments." Halley was of the next generation, and comes after Newton; he studied at Queen's College, Oxford, in 1673; but was at first a man of some fortune, and not engaged in any official situation. His

talents and zeal, however, made him an active and effective ally in the promotion of science.

[Note 22\7: Shaw's Boyle's _Works_, ii. 160.]

The connection of the persons of whom we have been speaking has a bearing on our subject, for it led, historically speaking, to the publication of Newton's discoveries in physical astronomy. Rightly to propose a problem is no inconsiderable step to its solution; and it was undoubtedly a great advance towards the true theory of the universe to consider the motion of the planets round the sun as a mechanical question, to be solved by a reference to the laws of motion, and by the use of mathematics. So far the English philosophers appear to have gone, before the time of Newton. Hooke, indeed, when the doctrine of gravitation was published, asserted that he had discovered it previously to Newton; and though this pretension could not be maintained, he certainly had perceived that the thing to be done was, to determine the effect of a central force in producing curvilinear motion; which effect, as we have already seen, he illustrated by experiment as early as 1666. Hooke had also spoken more clearly on this subject {397} in _An Attempt to prove the Motion of the Earth from Observations_, published in 1674. In this, he distinctly states that the planets would move in straight lines, if they were not deflected by central forces; and that the central attractive power increases in approaching the centre in certain degrees, dependent on the distance. "Now what these degrees are," he adds, "I have not yet experimentally verified;" but he ventures to promise to any one who succeeds in this undertaking, a discovery of the cause of the heavenly motions. He asserted, in conversation, to Halley and Wren, that he had solved this problem, but his solution was never produced. The proposition that the attractive force of the sun varies inversely as the square of the distance from the centre, had already been divined, if not fully established. If the orbits of the planets were circles, this proportion of the forces might be deduced in the same manner as the propositions concerning circular motion, which Huyghens published in 1673; yet it does not appear that Huyghens made this application of his principles. Newton, however, had already made this step some years before this time. Accordingly, he says in a letter to Halley, on Hooke's claim to this discovery,[23\7] "When Huygenius put out his _Horologium Oscillatorium_, a copy being presented to me, in my letter of thanks I gave those rules in the end thereof a particular commendation for their usefulness in computing the forces of the moon from the earth, and the earth from the sun." He says, moreover, "I am almost confident by circumstances, that Sir Christopher Wren knew the duplicate proportion when I gave him a visit; and then Mr. Hooke, by his book _Cometa_, will prove the last of us three that knew it." Hooke's _Cometa_ was published in 1678. These inferences were all connected with Kepler's law, that the times are in the sesquiplicate ratio of the major axes of the orbits. But Halley had also been led to the duplicate proportion by another train of reasoning, namely, by considering the force of the sun as an emanation, which must become more feeble in proportion to the increased spherical surface over which it is diffused, and therefore in the inverse proportion of the square of the distances.[24\7] In this view of the matter, however, the difficulty was to determine what would be the motion of a body acted on by such a force, when the orbit is not circular but oblong. The investigation

of this case was a problem which, we can {398} easily conceive, must have appeared of very formidable complexity while it was unsolved, and the first of its kind. Accordingly Halley, as his biographer says, "finding himself unable to make it out in any geometrical way, first applied to Mr. Hooke and Sir Christopher Wren, and meeting with no assistance from either of them, he went to Cambridge in August (1684), to Mr. Newton, who supplied him fully with what he had so ardently sought."

[Note 23\7: _Biog. Brit._, art. _Hooke._]

[Note 24\7: Bullialdus, in 1645, had asserted that the force by which the sun "prehendit et harpagat," takes hold of and grapples the planets, must be as the inverse square of the distance.]

A paper of Halley's in the _Philosophical Transactions_ for January, 1686, professedly inserted as a preparation for Newton's work, contains some arguments against the Cartesian hypothesis of gravity, which seem to imply that Cartesian opinions had some footing among English philosophers; and we are told by Whiston, Newton's successor in his professorship at Cambridge, that Cartesianism formed a part of the studies of that place. Indeed, Rohault's _Physics_ was used as a classbook at that University long after the time of which we are speaking; but the peculiar Cartesian doctrines which it contained were soon superseded by others.

With regard, then, to this part of the discovery, that the force of the sun follows the inverse duplicate proportion of the distances, we see that several other persons were on the verge of it at the same time with Newton; though he alone possessed that combination of distinctness of thought and power of mathematical invention, which enabled him to force his way across the barrier. But another, and so far as we know, an earlier train of thought, led by a different path to the same result; and it was the convergence of these two lines of reasoning that brought the conclusion to men's minds with irresistible force. I speak now of the identification of the force which retains the moon in her orbit with the force of gravity by which bodies fall at the earth's surface. In this comparison Newton had, so far as I am aware, no forerunner. We are now, therefore, arrived at the point at which the history of Newton's great discovery properly begins. {399}

## CHAPTER II.

THE INDUCTIVE EPOCH OF NEWTON.--DISCOVERY OF THE UNIVERSAL GRAVITATION OF MATTER, ACCORDING TO THE LAW OF THE INVERSE SQUARE OF THE DISTANCE.

IN order that we may the more clearly consider the bearing of this, the greatest scientific discovery ever made, we shall resolve it into the partial propositions of which it consists. Of these we may enumerate five. The doctrine of universal gravitation asserts,

1. That the force by which the _different_ planets are attracted to the sun is in the inverse proportion of the squares of their distances;

2. That the force by which the _same_ planet is attracted to the sun, in different parts of its orbit, is also in the inverse proportion of the squares of the distances;

3. That the _earth_ also exerts such a force on the _moon_, and that this force is identical with the force of _gravity_;

4. That bodies thus act on _other_ bodies, besides those which revolve round them; thus, that the sun exerts such a force on the moon and satellites, and that the planets exert such forces on _one another_;

5. That this force, thus exerted by the general masses of the sun, earth, and planets, arises from the attraction of _each particle_ of these masses; which attraction follows the above law, and belongs to all matter alike.

The history of the establishment of these five truths will be given in order.

1. _Sun's Force on Different Planets._--With regard to the first of the above five propositions, that the different planets are attracted to the sun by a force which is inversely as the square of the distance, Newton had so far been anticipated, that several persons had discovered it to be true, or nearly true; that is, they had discovered that if the orbits of the planets were circles, the proportions of the central force to the inverse square of the distance would follow from Kepler's third law, of the sesquiplicate proportion of the periodic times. As we have seen, Huyghens' theorems would have proved this, if they had been so applied; Wren knew it; Hooke not only knew it, but claimed a prior knowledge to Newton; and Halley had satisfied himself that it was at {400} least nearly true, before he visited Newton. Hooke was reported to Newton at Cambridge, as having applied to the Royal Society to do him justice with regard to his claims; but when Halley wrote and informed Newton (in a letter dated June 29, 1686), that Hooke's conduct "had been represented in worse colors than it ought," Newton inserted in his book a notice of these his predecessors, in order, as he said, "to compose the dispute."[25\7] This notice appears in a Scholium to the fourth Proposition of the _Principia_, which states the general law of revolutions in circles. "The case of the sixth corollary," Newton there says, "obtains in the celestial bodies, as has been separately inferred by our countrymen, Wren, Hooke, and Halley;" he soon after names Huyghens, "who, in his excellent treatise _De Horologio Oscillatorio_, compares the force of gravity with the centrifugal forces of revolving bodies."

[Note 25\7: _Biog. Brit._ folio, art. _Hooke._]

The two steps requisite for this discovery were, to propose the motions of the planets as simply a mechanical problem, and to apply mathematical reasoning so as to solve this problem, with reference to Kepler's third law considered as a fact. The former step was a consequence of the mechanical discoveries of Galileo and his school; the result of the firm and clear place which these gradually obtained in men's mind, and of the utter abolition of all the notions of solid spheres by Kepler. The mathematical step required no small mathematical powers; as appears, when we consider that this was the first example of such a problem, and that the method of limits, under all its forms, was at this time in its infancy, or rather, at its birth. Accordingly, even this step, though much the easiest in the path of deduction, no one

before Newton completely executed.

2. _Force in different Points of an Orbit._--The inference of the law of the force from Kepler's two laws concerning the elliptical motion, was a problem quite different from the preceding, and much more difficult; but the dispute with respect to priority in the two propositions was intermingled. Borelli, in 1666, had, as we have seen, endeavored to reconcile the general form of the orbit with the notion of a central attractive force, by taking centrifugal force into the account; and Hooke, in 1679, had asserted that the result of the law of the inverse square in the force of the earth would be an ellipse,[26\7] or a curve like an ellipse.[27\7] But it does not appear that this was any thing more than {401} a conjecture. Halley says[28\7] that "Hooke, in 1683, told him he had demonstrated all the laws of the celestial motions by the reciprocally duplicate proportion of the force of gravity; but that, being offered forty shillings by Sir Christopher Wren to produce such a demonstration, his answer was, that he had it, but would conceal it for some time, that others, trying and failing, might know how to value it when he should make it public." Halley, however, truly observes, that after the publication of the demonstration in the _Principia_, this reason no longer held; and adds, "I have plainly told him, that unless he produce another differing demonstration, and let the world judge of it, neither I nor any one else can believe it."

[Note 26\7: Newton's Letter, _Biog. Brit._, Hooke, p. 2660.]

[Note 27\7: Birch's _Hist. R. S._, Wallis's Life.]

[Note 28\7: _Enc. Brit._, Hooke, p. 2660.]

Newton allows that Hooke's assertions in 1679 gave occasion to his investigation on this point of the theory. His demonstration is contained in the second and third Sections of the _Principia_. He first treats of the general law of central forces in any curve; and then, on account, as he states, of the application to the motion of the heavenly bodies, he treats of the case of force varying inversely as the square of the distance, in a more diffuse manner.

In this, as in the former portion of his discovery, the two steps were, the proposing the heavenly motions as a mechanical problem, and the solving this problem. Borelli and Hooke had certainly made the former step, with considerable distinctness; but the mathematical solution required no common inventive power.

Newton seems to have been much ruffled by Hooke's speaking slightly of the value of this second step; and is moved in return to deny Hooke's pretensions with some asperity, and to assert his own. He says, in a letter to Halley, "Borelli did something in it, and wrote modestly; he (Hooke) has done nothing; and yet written in such a way as if he knew, and had sufficiently hinted all but what remained to be determined by the drudgery of calculations and observations; excusing himself from that labor by reason of his other business; whereas he should rather have excused himself by reason of his inability; for it is very plain, by his words, he knew not how to go about it. Now is not this very fine? Mathematicians that find out, settle, and do all the business, must content themselves with being nothing but dry calculators and drudges; and another that does nothing but pretend and grasp at all things, must

carry away all the inventions, as well of those that were to follow him as of those that {402} went before." This was written, however, under the influence of some degree of mistake; and in a subsequent letter, Newton says, "Now I understand he was in some respects misrepresented to me, I wish I had spared the postscript to my last," in which is the passage just quoted. We see, by the melting away of rival claims, the undivided honor which belongs to Newton, as the real discoverer of the proposition now under notice. We may add, that in the sequel of the third Section of the _Principia_, he has traced its consequences, and solved various problems flowing from it with his usual fertility and beauty of mathematical resource; and has there shown the necessary connection of Kepler's third law with his first and second.

3. _Moon's Gravity to the Earth._--Though others had considered cosmical forces as governed by the general laws of motion, it does not appear that they had identified such forces with the force of terrestrial gravity. This step in Newton's discoveries has generally been the most spoken of by superficial thinkers; and a false kind of interest has been attached to it, from the story of its being suggested by the fall of an apple. The popular mind is caught by the character of an eventful narrative which the anecdote gives to this occurrence; and by the antithesis which makes a profound theory appear the result of a trivial accident. How inappropriate is such a view of the matter we shall soon see. The narrative of the progress of Newton's thoughts, is given by Pemberton (who had it from Newton himself) in his preface to his _View of Newton's Philosophy_, and by Voltaire, who had it from Mrs. Conduit, Newton's niece.[29\7] "The first thoughts," we are told, "which gave rise to his _Principia_, he had when he retired from Cambridge, in 1666, on account of the plague (he was then twenty-four years of age). As he sat alone in a garden, he fell into a speculation on the power of gravity; that as this power is not found sensibly diminished at the remotest distance from the centre of the earth to which we can rise, neither at the tops of the loftiest buildings, nor even on the summits of the highest mountains, it appeared to him reasonable to conclude that this power must extend much further than was usually thought: Why not as high as the moon? said he to himself; and if so, her motion must be influenced by it; perhaps she is retained in her orbit thereby."

[Note 29\7: _Elémens de Phil. de Newton_, 3me partie, chap. iii.]

The thought of cosmical gravitation was thus distinctly brought into being; and Newton's superiority here was, that he conceived the {403} celestial motions as distinctly as the motions which took place close to him;--considered them as of the same kind, and applied the same rules to each, without hesitation or obscurity. But so far, this thought was merely a guess: its occurrence showed the activity of the thinker; but to give it any value, it required much more than a "why not?"--a "perhaps." Accordingly, Newton's "why not?" was immediately succeeded by his "if so, what then?" His reasoning was, that if gravity reach to the moon, it is probably of the same kind as the central force of the sun, and follows the same rule with respect to the distance. What is this rule? We have already seen that, by calculating from Kepler's laws, and supposing the orbits to be circles, the rule of the force appears to be the inverse duplicate proportion of the distance; and this, which had been current as a conjecture among the previous generation of mathematicians, Newton had

already proved by indisputable reasonings, and was thus prepared to proceed in his train of inquiry. If, then, he went on, pursuing his train of thought, the earth's gravity extend to the moon, diminishing according to the inverse square of the distance, will it, at the moon's orbit, be of the proper magnitude for retaining her in her path? Here again came in calculation, and a calculation of extreme interest; for how important and how critical was the decision which depended on the resulting numbers? According to Newton's calculations, made at this time, the moon by her motion in her orbit, was deflected from the tangent every minute through a space of thirteen feet. But by noticing the space through which bodies would fall in one minute at the earth's surface, and supposing this to be diminished in the ratio of the inverse square, it appeared that gravity would, at the moon's orbit, draw a body through more than fifteen feet. The difference seems small, the approximation encouraging, the theory plausible; a man in love with his own fancies would readily have discovered or invented some probable cause of this difference. But Newton acquiesced in it as a disproof of his conjecture, and "laid aside at that time any further thoughts of this matter;**" thus resigning a favorite hypothesis, with a candor and openness to conviction not inferior to Kepler, though his notion had been taken up on far stronger and sounder grounds than Kepler dealt in; and without even, so far as we know, Kepler's regrets and struggles. Nor was this levity or indifference; the idea, though thus laid aside, was not finally condemned and abandoned. When Hooke, in 1679, contradicted Newton on the subject of the curve described by a falling body, and asserted it to be an ellipse, Newton {404} was led to investigate the subject, and was then again conducted, by another road, to the same law of the inverse square of the distance. This naturally turned his thoughts to his former speculations. Was there really no way of explaining the discrepancy which this law gave, when he attempted to reduce the moon's motion to the action of gravity? A scientific operation then recently completed, gave the explanation at once. He had been mistaken in the magnitude of the earth, and consequently in the distance of the moon, which is determined by measurements of which the earth's radius is the base. He had taken the common estimate, current among geographers and seamen, that sixty English miles are contained in one degree of latitude. But Picard, in 1670, had measured the length of a certain portion of the meridian in France, with far greater accuracy than had yet been attained and this measure enabled Newton to repeat his calculations with these amended data. We may imagine the strong curiosity which he must have felt as to the result of these calculations. His former conjecture was now found to agree with the phenomena to a remarkable degree of precision. This conclusion, thus coming after long doubts and delays, and falling in with the other results of mechanical calculation for the solar system, gave a stamp from that moment to his opinions, and through him to those of the whole philosophical world.

[2d Ed.] [Dr. Robison (_Mechanical Philosophy_, p. 288) says that Newton having become a member of the Royal Society, there learned the accurate measurement of the earth by Picard, differing very much from the estimation by which he had made his calculations in 1666. And M. Biot, in his Life of Newton, published in the _Biographie Universelle_, says, "According to conjecture, about the month of June, 1682, Newton being in London at a meeting of the Royal Society, mention was made

of the new measure of a degree of the earth's surface, recently executed in France by Picard; and great praise was given to the care which had been employed in making this measure exact."

I had adopted this conjecture as a fact in my first edition; but it has been pointed out by Prof. Rigaud (_Historical Essay on the First Publication of the Principia_, 1838), that Picard's measurement was probably well known to the Fellows of the Royal Society as early as 1675, there being an account of the results of it given in the _Philosophical Transactions_ for that year. Newton appears to have discovered the method of determining that a body might describe an ellipse when acted upon by a force residing in the focus, and varying {405} inversely as the square of the distance, in 1679, upon occasion of his correspondence with Hooke. In 1684, at Halley's request, he returned to the subject, and in February, 1685, there was inserted in the Register of the Royal Society a paper of Newton's (_Isaaci Newtoni Propositiones de Motu_) which contained some of the principal Propositions of the first two Books of the _Principia_. This paper, however, does not contain the Proposition "Lunam gravitare in terram," nor any of the other propositions of the third Book. The _Principia_ was printed in 1686 and 7, apparently at the expense of Halley. On the 6th of April, 1687, the third Book was presented to the Royal Society.]

It does not appear, I think, that before Newton, philosophers in general had supposed that terrestrial gravity was the very force by which the moon's motions are produced. Men had, as we have seen, taken up the conception of such forces, and had probably called them gravity: but this was done only to explain, by analogy, what _kind_ of forces they were, just as at other times they compared them with magnetism; and it did not imply that terrestrial gravity was a force which acted in the celestial spaces. After Newton had discovered that this was so, the application of the term "gravity" did undoubtedly convey such a suggestion; but we should err if we inferred from this coincidence of expression that the notion was commonly entertained before him. Thus Huyghens appears to use language which may be mistaken, when he says,[30\7] that Borelli was of opinion that the primary planets were urged by "gravity" towards the sun, and the satellites towards the primaries. The notion of terrestrial gravity, as being actually a cosmical force, is foreign to all Borelli's speculations.[31\7] But Horrox, as early as 1635, appears to have entertained the true view on this subject, although vitiated by Keplerian errors concerning the connection between the rotation of the central body and its effect on the body which revolves about it. Thus he says,[32\7] that the emanation of the earth carries a projected stone along with the motion of the earth, just in the same way as it carries the moon in her orbit; and that this force is greater on the stone than on the moon, because the distance is less.

[Note 30\7: _**Cosmotheoros_, l. 2. p. 720.]

[Note 31\7: I have found no instance in which the word is so used by him.]

[Note 32\7: _Astronomia Kepleriana defensa et promota_, cap. 2. See further on this subject in the _Additions_ to this volume.]

The Proposition in which Newton has stated the discovery of which we are now speaking, is the fourth of his third Book: "That the moon gravitates to the earth, and by the force of gravity is perpetually {406} deflected from a rectilinear motion, and retained in her orbit." The proof consists in the numerical calculation, of which he only gives the elements, and points out the method; but we may observe, that no small degree of knowledge of the way in which astronomers had obtained these elements, and judgment in selecting among them, were necessary: thus, the mean distance of the moon had been made as little as fifty-six and a half semidiameters of the earth by Tycho, and as much as sixty-two and a half by Kircher: Newton gives good reasons for adopting sixty-one.

The term "gravity," and the expression "to gravitate," which, as we have just seen, Newton uses of the moon, were to receive a still wider application in consequence of his discoveries; but in order to make this extension clearer, we consider it as a separate step.

4. _Mutual Attraction of all the Celestial Bodies._--If the preceding parts of the discovery of gravitation were comparatively easy to conjecture, and difficult to prove, this was much more the case with the part of which we have now to speak, the attraction of other bodies, besides the central ones, upon the planets and satellites. If the mathematical calculation of the unmixed effect of a central force required transcendent talents, how much must the difficulty be increased, when other influences prevented those first results from being accurately verified, while the deviations from accuracy were far more complex than the original action! If it had not been that these deviations, though surprisingly numerous and complicated in their nature, were very small in their quantity, it would have been impossible for the intellect of man to deal with the subject; as it was, the struggle with its difficulties is even now a matter of wonder.

The conjecture that there is some mutual action of the planets, had been put forth by Hooke in his _Attempt to prove the Motion of the Earth_ (1674). It followed, he said, from his doctrine, that not only the sun and moon act upon the course and motion of the earth, but that Mercury, Venus, Mars, Jupiter, and Saturn, have also, by their attractive power, a considerable influence upon the motion of the earth, and the earth in like manner powerfully affects the motions of those bodies. And Borelli, in attempting to form "theories" of the satellites of Jupiter, had seen, though dimly and confusedly, the probability that the sun would disturb the motions of these bodies. Thus he says (cap. 14), "How can we believe that the Medicean globes are not, like other planets, impelled with a greater velocity when they approach the sun: and thus they are acted upon by two moving forces, one of {407} which produces their proper revolution about Jupiter, the other regulates their motion round the sun." And in another place (cap. 20), he attempts to show an effect of this principle upon the inclination of the orbit; though, as might be expected, without any real result.

The case which most obviously suggests the notion that the sun exerts a power to disturb the motions of secondary planets about primary ones, might seem to be our own moon; for the great inequalities which had hitherto been discovered, had all, except the first, or elliptical anomaly, a reference to the position of the sun.

Nevertheless, I do not know that any one had attempted thus to explain the curiously irregular course of the earth's attendant. To calculate, from the disturbing agency, the amount of the irregularities, was a problem which could not, at any former period, have been dreamt of as likely to be at any time within the verge of human power.

Newton both made the step of inferring that there were such forces, and, to a very great extent, calculated the effects of them. The inference is made on mechanical principles, in the sixth Theorem of the third Book of the _Principia_;--that the moon is attracted by the sun, as the earth is;--that the satellites of Jupiter and Saturn are attracted as the primaries are; in the same manner, and with the same forces. If this were not so, it is shown that these attendant bodies could not accompany the principal ones in the regular manner in which they do. All those bodies at equal distances from the sun would be equally attracted.

But the complexity which must occur in tracing the results of this principle will easily be seen. The satellite and the primary, though nearly at the same distance, and in the same direction, from the sun, are not exactly so. Moreover the difference of the distances and of the directions is perpetually changing; and if the motion of the satellite be elliptical, the cycle of change is long and intricate: on this account alone the effects of the sun's action will inevitably follow cycles as long and as perplexed as those of the positions. But on another account they will be still more complicated; for in the continued action of a force, the effect which takes place at first, modifies and alters the effect afterwards. The result at any moment is the sum of the results in preceding instants: and since the terms, in this series of instantaneous effects, follow very complex rules, the sums of such series will be, it might be expected, utterly incapable of being reduced to any manageable degree of simplicity.

It certainly does not appear that any one but Newton could make {408} any impression on this problem, or course of problems. No one for sixty years after the publication of the _Principia_, and, with Newton's methods, no one up to the present day, had added any thing of any value to his deductions. We know that he calculated all the principal lunar inequalities; in many of the cases, he has given us his processes; in others, only his results. But who has presented, in his beautiful geometry, or deduced from his simple principles, any of the inequalities which he left untouched? The ponderous instrument of synthesis, so effective in his hands, has never since been grasped by one who could use it for such purposes; and we gaze at it with admiring curiosity, as on some gigantic implement of war, which stands idle among the memorials of ancient days, and makes us wonder what manner of man he was who could wield as a weapon what we can hardly lift as a burden.

It is not necessary to point out in detail the sagacity and skill which mark this part of the _Principia_. The mode in which the author obtains the effect of a disturbing force in producing a motion of the apse of an elliptical orbit (the ninth Section of the first Book), has always been admired for its ingenuity and elegance. The general statement of the nature of the principal inequalities produced by the sun in the motion of a satellite, given in the sixty-sixth Proposition, is, even yet, one of the best explanations of such action; and the calculations of the quantity of the effects in the

third Book, for instance, the _variation_ of the moon, the _motion of the nodes_ and its inequalities, the _change of inclination_ of the orbit,--are full of beautiful and efficacious artifices. But Newton's inventive faculty was exercised to an extent greater than these published investigations show. In several cases he has suppressed the demonstration of his method, and given us the result only; either from haste or from mere weariness, which might well overtake one who, while he was struggling with facts and numbers, with difficulties of conception and practice, was aiming also at that geometrical elegance of exposition, which he considered as alone fit for the public eye. Thus, in stating the effect of the eccentricity of the moon's orbit upon the motion of the apogee, he says,[33\7] "The computations, as too intricate and embarrassed with approximations, I do not choose to introduce."

[Note 33\7: Schol. to Prop. 35, first edit.]

The computations of the theoretical motion of the moon being thus difficult, and its irregularities numerous and complex, we may ask {409} whether Newton's reasoning was sufficient to establish this part of his theory; namely, that her actual motions arise from her gravitation to the sun. And to this we may reply, that it was sufficient for that purpose,--since it showed that, from Newton's hypothesis, inequalities must result, following the laws which the moon's inequalities were known to follow;--since the amount of the inequalities given by the theory agreed nearly with the rules which astronomers had collected from observation;--and since, by the very intricacy of the calculation, it was rendered probable, that the first results might be somewhat inaccurate, and thus might give rise to the still remaining differences between the calculations and the facts. A _Progression of the Apogee_; a _Regression of the Nodes_; and, besides the Elliptical, or first Inequality, an inequality, following the law of the _Evection_, or second inequality discovered by Ptolemy; another, following the law of the _Variation_ discovered by Tycho;--were pointed out in the first edition of the _Principia_, as the consequences of the theory. Moreover, the quantities of these inequalities were calculated and compared with observation with the utmost confidence, and the agreement in most instances was striking. The Variation agreed with Halley's recent observations within a minute of a degree.[34\7] The Mean Motion of the Nodes in a year agreed within less than one-hundredth of the whole.[35\7] The Equation of the Motion of the Nodes also agreed well.[36\7] The Inclination of the Plane of the Orbit to the ecliptic, and its changes, according to the different situations of the nodes, likewise agreed.[37\7] The Evection has been already noticed as encumbered with peculiar difficulties: here the accordance was less close. The Difference of the daily progress of the Apogee in syzygy, and its daily Regress in Quadratures, is, Newton says, "4¼ minutes by the Tables, 6⅔ by our calculation." He boldly adds, "I suspect this difference to be due to the fault of the Tables." In the second edition (1711) he added the calculation of several other inequalities, as the _Annual Equation_, also discovered by Tycho; and he compared them with more recent observations made by Flamsteed at Greenwich; but even in what has already been stated, it must be allowed that there is a wonderful accordance of theory with phenomena, both being very complex in the rules which they educe.

[Note 34\7: B. iii. Prop. 29.]

[Note 35\7: Prop. 32.]

[Note 36\7: Prop. 33.]

[Note 37\7: Prop. 35.]

The same theory which gave these Inequalities in the motion of the Moon produced by the disturbing force of the sun, gave also {410} corresponding Inequalities in the motions of the Satellites of other planets, arising from the same cause; and likewise pointed out the necessary existence of irregularities in the motions of the Planets arising from their mutual attraction. Newton gave propositions by which the Irregularities of the motion of Jupiter's moons might be deduced from those of our own;[38\7] and it was shown that the motions of their nodes would be slow by theory, as Flamsteed had found it to be by observation.[39\7] But Newton did not attempt to calculate the effect of the mutual action of the planets, though he observes, that in the case of Jupiter and Saturn this effect is too considerable to be neglected;[40\7] and he notices in the second edition,[41\7] that it follows from the theory of gravity, that the aphelia of Mercury, Venus, the Earth, and Mars, slightly progress.

[Note 38\7: B. i. Prop. 66.]

[Note 39\7: B. iii. Prop. 23.]

[Note 40\7: B. iii. Prop. 13.]

[Note 41\7: Scholium to Prop. 14. B. iii.]

In one celebrated instance, indeed, the deviation of the theory of the _Principia_ from observation was wider, and more difficult to explain; and as this deviation for a time resisted the analysis of Euler and Clairaut, as it had resisted the synthesis of Newton, it at one period staggered the faith of mathematicians in the exactness of the law of the inverse square of the distance. I speak of the Motion of the Moon's Apogee, a problem which has already been referred to; and in which Newton's method, and all the methods which could be devised for some time afterwards, gave only half the observed motion; a circumstance which arose, as was discovered by Clairaut in 1750, from the insufficiency of the method of approximation. Newton does not attempt to conceal this discrepancy. After calculating what the motion of apse would be, upon the assumption of a disturbing force of the same amount as that which the sun exerts on the moon, he simply says,[42\7] "the apse of the moon moves about twice as fast."

[Note 42\7: B. i. Prop. 44, second edit. There is reason to believe, however, that Newton had, in his unpublished calculations, rectified this discrepancy.]

The difficulty of doing what Newton did in this branch of the subject, and the powers it must have required, may be judged of from what has already been stated;--that no one, with his methods, has yet been able to add any thing to his labors: few have undertaken to illustrate what he has written, and no great number have understood it throughout. The extreme complication of the forces, and of the conditions under which they act, makes the subject by far the most thorny walk of mathematics. It is necessary to resolve the action {411} into many elements, such as

can be separated; to invent artifices for dealing with each of these; and then to recompound the laws thus obtained into one common conception. The moon's motion cannot be conceived without comprehending a scheme more complex than the Ptolemaic epicycles and eccentrics in their worst form; and the component parts of the system are not, in this instance, mere geometrical ideas, requiring only a distinct apprehension of relations of space in order to hold them securely; they are the foundations of mechanical notions, and require to be grasped so that we can apply to them sound mechanical reasonings. Newton's successors, in the next generation, abandoned the hope of imitating him in this intense mental effort; they gave the subject over to the operation of algebraical reasoning, in which symbols think for us, without our dwelling constantly upon their meaning, and obtain for us the consequences which result from the relations of space and the laws of force, however complicated be the conditions under which they are combined. Even Newton's countrymen, though they were long before they applied themselves to the method thus opposed to his, did not produce any thing which showed that they had mastered, or could retrace, the Newtonian investigations.

Thus the Problem of Three Bodies,[43\7] treated geometrically, belongs exclusively to Newton; and the proofs of the mutual action of the sun, planets, and satellites, which depend upon such reasoning, could not be discovered by any one but him.

[Note 43\7: See the history of the _Problem of Three Bodies_, _ante_, in Book vi. Chap. vi. Sect. 7.]

But we have not yet done with his achievements on this subject; for some of the most remarkable and beautiful of the reasonings which he connected with this problem, belong to the next step of his generalization.

5. _Mutual Attraction of all Particles of Matter._--That all the parts of the universe are drawn and held together by love, or harmony, or some affection to which, among other names, that of attraction may have been given, is an assertion which may very possibly have been made at various times, by speculators writing at random, and taking their chance of meaning and truth. The authors of such casual dogmas have generally nothing accurate or substantial, either in their conception of the general proposition, or in their reference to examples of it; and, therefore, their doctrines are no concern of ours at present. But among those who were really the first to think of the mutual {412} attraction of matter, we cannot help noticing Francis Bacon; for his notions were so far from being chargeable with the looseness and indistinctness to which we have alluded, that he proposed an experiment[44\7] which was to decide whether the facts were so or not;--whether the gravity of bodies to the earth arose from an attraction of the parts of matter towards each other, or was a tendency towards the centre of the earth. And this experiment is, even to this day, one of the best which can be devised, in order to exhibit the universal gravitation of matter: it consists in the comparison of the rate of going of a clock in a deep mine, and on a high place. Huyghens, in his book _De Causâ Gravitatis_, published in 1690, showed that the earth would have an oblate form, in consequence of the action of the centrifugal force; but his reasoning does not suppose gravity to arise from the mutual attraction of the parts of the earth. The apparent influence of the moon upon the tides

had long been remarked; but no one had made any progress in truly explaining the mechanism of this influence; and all the analogies to which reference had been made, on this and similar subjects, as magnetic and other attractions, were rather delusive than illustrative, since they represented the attraction as something peculiar in particular bodies, depending upon the nature of each body.

[Note 44\7: _Nov. Org._ Lib. ii. Aph. 36.]

That all such forces, cosmical and terrestrial, were the same single force, and that this was nothing more than the insensible attraction which subsists between one stone and another, was a conception equally bold and grand; and would have been an incomprehensible thought, if the views which we have already explained had not prepared the mind for it. But the preceding steps having disclosed, between all the bodies of the universe, forces of the same kind as those which produce the weight of bodies at the earth, and, therefore, such as exist in every particle of terrestrial matter; it became an obvious question, whether such forces did not also belong to all particles of planetary matter, and whether this was not, in fact, the whole account of the forces of the solar system. But, supposing this conjecture to be thus suggested, how formidable, on first appearance at least, was the undertaking of verifying it! For if this be so, every finite mass of matter exerts forces which are the result of the infinitely numerous forces of its particles, these forces acting in different directions. It does not appear, at first sight, that the law by which the force is related to the distance, will be the same for the particles as it is for the masses; and, in reality, it {413} is not so, except in special cases. And, again, in the instance of any effect produced by the force of a body, how are we to know whether the force resides in the whole mass as a unit, or in the separate particles? We may reason, as Newton does,[45\7] that the rule which proves gravity to belong universally to the planets, proves it also to belong to their parts; but the mind will not be satisfied with this extension of the rule, except we can find decisive instances, and calculate the effects of both suppositions, under the appropriate conditions. Accordingly, Newton had to solve a new series of problems suggested by this inquiry; and this he did.

[Note 45\7: _Princip._ B. iii. Prop. 7.]

These solutions are no less remarkable for the mathematical power which they exhibit, than the other parts of the _Principia_. The propositions in which it is shown that the law of the inverse square for the particles gives the same law for spherical masses, have that kind of beauty which might well have justified their being published for their mathematical elegance alone, even if they had not applied to any real case. Great ingenuity is also employed in other instances, as in the case of spheroids of small eccentricity. And when the amount of the mechanical action of masses of various forms has thus been assigned, the sagacity shown in tracing the results of such action in the solar system is truly admirable; not only the general nature of the effect being pointed out, but its quantity calculated. I speak in particular of the reasonings concerning the Figure of the Earth, the Tides, the Precession of the Equinoxes, the Regression of the Nodes of a ring such as Saturn's; and of some effects which, at that time, had not been ascertained even as facts of observation; for instance, the difference of gravity in different latitudes, and the

Nutation of the earth's axis. It is true, that in most of these cases, Newton's process could be considered only as a rude approximation. In one (the Precession) he committed an error, and in all, his means of calculation were insufficient. Indeed these are much more difficult investigations than the Problem of Three Bodies, in which three points act on each other by explicit laws. Up to this day, the resources of modern analysis have been employed upon some of them with very partial success; and the facts, in all of them, required to be accurately ascertained and measured, a process which is not completed even now. Nevertheless the form and nature of the conclusions which Newton did obtain, were such as to inspire a strong confidence in the competency of his theory to explain {414} all such phenomena as have been spoken of. We shall afterwards have to speak of the labors, undertaken in order to examine the phenomena more exactly, to which the theory gave occasion.

Thus, then, the theory of the universal mutual gravitation of all the particles of matter, according to the law of the inverse square of the distances, was conceived, its consequences calculated, and its results shown to agree with phenomena. It was found that this theory took up all the facts of astronomy as far as they had hitherto been ascertained; while it pointed out an interminable vista of new facts, too minute or too complex for observation alone to disentangle, but capable of being detected when theory had pointed out their laws, and of being used as criteria or confirmations of the truth of the doctrine. For the same reasoning which explained the evection, variation, and annual equation of the moon, showed that there must be many other inequalities besides these; since these resulted from approximate methods of calculation, in which small quantities were neglected. And it was known that, in fact, the inequalities hitherto detected by astronomers did not give the place of the moon with satisfactory accuracy; so that there was room, among these hitherto untractable irregularities, for the additional results of the theory. To work out this comparison was the employment of the succeeding century; but Newton began it. Thus, at the end of the proposition in which he asserts,[46\7] that "all the lunar motions and their irregularities follow from the principles here stated," he makes the observation which we have just made; and gives, as examples, the different motions of the apogee and nodes, the difference of the change of the eccentricity, and the difference of the moon's variation, according to the different distances of the sun. "But this inequality," he says, "in astronomical calculations, is usually referred to the prosthaphæresis of the moon, and confounded with it."

[Note 46\7: B. iii. Prop. 22.]

_Reflections on the Discovery._--Such, then, is the great Newtonian Induction of Universal Gravitation, and such its history. It is indisputably and incomparably the greatest scientific discovery ever made, whether we look at the advance which it involved, the extent of the truth disclosed, or the fundamental and satisfactory nature of this truth. As to the first point, we may observe that any one of the five steps into which we have separated the doctrine, would, of itself, have been considered as an important advance;--would have conferred distinction on the persons who made it, and the time to which it belonged. All {415} the five steps made at once, formed not a leap, but a flight,--not an improvement merely, but a metamorphosis,--not an epoch,

but a termination. Astronomy passed at once from its boyhood to mature manhood. Again, with regard to the extent of the truth, we obtain as wide a generalization as our physical knowledge admits, when we learn that every particle of matter, in all times, places, and circumstances, attracts every other particle in the universe by one common law of action. And by saying that the truth was of a fundamental and satisfactory nature, I mean that it assigned, not a rule merely, but a cause, for the heavenly motions; and that kind of cause which most eminently and peculiarly we distinctly and thoroughly conceive, namely, mechanical force. Kepler's laws were merely _formal_ rules, governing the celestial motions according to the relations of space, time, and number; Newton's was a _**causal_ law, referring these motions to mechanical reasons. It is no doubt conceivable that future discoveries may both extend and further explain Newton's doctrines;--may make gravitation a case of some wider law, and may disclose something of the mode in which it operates; questions with which Newton himself struggled. But, in the mean time, few persons will dispute, that both in generality and profundity, both in width and depth, Newton's theory is altogether without a rival or neighbor.[47\7]

[Note 47\7: The value and nature of this step have long been generally acknowledged wherever science is cultivated. Yet it would appear that there is, in one part of Europe, a school of philosophers who contest the merit of this part of Newton's discoveries. "Kepler," says a celebrated German metaphysician,* "discovered the laws of free motion; a discovery of immortal glory. It has since been the fashion to say that Newton first found out the proof of these rules. It has seldom happened that the glory of the first discoverer has been more unjustly transferred to another person." It may appear strange that any one in the present day should hold such language; but if we examine the reasons which this author gives, they will be found, I think, to amount to this: that his mind is in the condition in which Kepler's was; and that the whole range of mechanical ideas and modes of conception which made the transition from Kepler and Newton possible, are extraneous to the domain of his philosophy. Even this author, however, if I understand him rightly, recognizes Newton as the author of the doctrine of Perturbations.

I have given a further account of these views, in a Memoir _On Hegel's Criticism of Newton's Principia_. Cambridge Transactions, 1849.

* Hegel, _Encyclopædia_, § 270.]

The requisite conditions of such a discovery in the mind of its author were, in this as in other cases, the idea, and its comparison with facts;--the conception of the law, and the moulding this conception in such a form as to correspond with known realities. The idea of mechanical {416} force as the cause of the celestial motions, had, as we have seen, been for some time growing up in men's minds; had gone on becoming more distinct and more general; and had, in some persons, approached the form in which it was entertained by Newton. Still, in the mere conception of universal gravitation, Newton must have gone far beyond his predecessors and contemporaries, both in generality and distinctness; and in the inventiveness and sagacity with which he traced the consequences of this conception, he was, as we have shown, without a rival, and almost without a second. As to the facts which he

had to include in his law, they had been accumulating from the very birth of astronomy; but those which he had more peculiarly to take hold of were the facts of the planetary motions as given by Kepler, and those of the moon's motions as given by Tycho Brahe and Jeremy Horrox.

We find here occasion to make a remark which is important in its bearing on the nature of progressive science. What Newton thus used and referred to as _facts_, were the _laws_ which his predecessors had established. What Kepler and Horrox had put forth as "theories," were now established truths, fit to be used in the construction of other theories. It is in this manner that one theory is built upon another;--that we rise from particulars to generals, and from one generalization to another;--that we have, in short, successive steps of induction. As Newton's laws assumed Kepler's, Kepler's laws assumed as facts the results of the planetary theory of Ptolemy; and thus the theories of each generation in the scientific world are (when thoroughly verified and established,**) the facts of the next generation. Newton's theory is the circle of generalization which includes all the others;--the highest point of the inductive ascent;--the catastrophe of the philosophic drama to which Plato had prologized;--the point to which men's minds had been journeying for two thousand years.

_Character of Newton._--It is not easy to anatomize the constitution and the operations of the mind which makes such an advance in knowledge. Yet we may observe that there must exist in it, in an eminent degree, the elements which compose the mathematical talent. It must possess distinctness of intuition, tenacity and facility in tracing logical connection, fertility of invention, and a strong tendency to generalization. It is easy to discover indications of these characteristics in Newton. The distinctness of his intuitions of space, and we may add of force also, was seen in the amusements of his youth; in his constructing clocks and mills, carts and dials, as well as the facility with which he {417} mastered geometry. This fondness for handicraft employments, and for making models and machines, appears to be a common prelude of excellence in physical science;[48\7] probably on this very account, that it arises from the distinctness of intuitive power with which the child conceives the shapes and the working of such material combinations. Newton's inventive power appears in the number and variety of the mathematical artifices and combinations which he devised, and of which his books are full. If we conceive the operation of the inventive faculty in the only way in which it appears possible to conceive it;--that while some hidden source supplies a rapid stream of possible suggestions, the mind is on the watch to seize and detain any one of these which will suit the case in hand, allowing the rest to pass by and be forgotten;--we shall see what extraordinary fertility of mind is implied by so many successful efforts; what an innumerable host of thoughts must have been produced, to supply so many that deserved to be selected. And since the selection is performed by tracing the consequences of each suggestion, so as to compare them with the requisite conditions, we see also what rapidity and certainty in drawing conclusions the mind must possess as a talent, and what watchfulness and patience as a habit.

[Note 48\7: As in Galileo, Hooke, Huyghens, and others.]

The hidden fountain of our unbidden thoughts is for us a mystery; and we have, in our consciousness, no standard by which we can measure our own talents; but our acts and habits are something of which we are conscious; and we can understand, therefore, how it was that Newton could not admit that there was any difference between himself and other men, except in his possession of such habits as we have mentioned, perseverance and vigilance. When he was asked how he made his discoveries, he answered, "by always thinking about them;" and at another time he declared that if he had done any thing, it was due to nothing but industry and patient thought: "I keep the subject of my inquiry constantly before me, and wait till the first dawning opens gradually, by little and little, into a full and clear light." No better account can be given of the nature of the mental _effort_ which gives to the philosopher the full benefit of his powers; but the natural _powers_ of men's minds are not on that account the less different. There are many who might wait through ages of darkness without being visited by any dawn.

The habit to which Newton thus, in some sense, owed his {418} discoveries, this constant attention to the rising thought, and development of its results in every direction, necessarily engaged and absorbed his spirit, and made him inattentive and almost insensible to external impressions and common impulses. The stories which are told of his extreme absence of mind, probably refer to the two years during which he was composing his _Principia_, and thus following out a train of reasoning the most fertile, the most complex, and the most important, which any philosopher had ever had to deal with. The magnificent and striking questions which, during this period, he must have had daily rising before him; the perpetual succession of difficult problems of which the solution was necessary to his great object; may well have entirely occupied and possessed him. "He existed only to calculate and to think."[49\7] Often, lost in meditation, he knew not what he did, and his mind appeared to have quite forgotten its connection with the body. His servant reported that, on rising in a morning, he frequently sat a large portion of the day, half-dressed, on the side of his bed and that his meals waited on the table for hours before he came to take them. Even with his transcendent powers, to do what he did was almost irreconcilable with the common conditions of human life; and required the utmost devotion of thought, energy of effort, and steadiness of will--the strongest character, as well as the highest endowments, which belong to man.

[Note 49\7: Biot.]

Newton has been so universally considered as the greatest example of a natural philosopher, that his moral qualities, as well as his intellect, have been referred to as models of the philosophical character; and those who love to think that great talents are naturally associated with virtue, have always dwelt with pleasure upon the views given of Newton by his contemporaries; for they have uniformly represented him as candid and humble, mild and good. We may take as an example of the impressions prevalent about him in his own time, the expressions of Thomson, in the Poem on his Death.[50\7] {419} Say ye who best can tell, ye happy few, Who saw him in the softest lights of life, All unwithheld, indulging to his friends The vast unborrowed treasures of his mind, Oh, speak the wondrous man! how mild, how calm How greatly

humble, how divinely good, How firm established on eternal truth! Fervent in doing well, with every nerve Still pressing on, forgetful of the past, And panting for perfection; far above Those little cares and visionary joys That so perplex the fond impassioned heart Of ever-cheated, ever-trusting man.

[Note 50\7: In the same strain we find the general voice of the time. For instance, one of Loggan's "Views of Cambridge" is dedicated "Isaaco Newtono . . Mathematico, Physico, Chymico consummatissimo; nec minus suavitate morum et candore animi . . . spectabili."

In opposition to the general current of such testimony, we have the complaints of Flamsteed, who ascribes to Newton angry language and harsh conduct in the matter of the publication of the Greenwich Observations, and of Whiston. Yet even Flamsteed speaks well of his general disposition. Whiston was himself so weak and prejudiced that his testimony is worth very little.]

[2d Ed.] [In the first edition of the _Principia_, published in 1687, Newton showed that the nature of all the then known inequalities of the moon, and in some cases, their quantities, might be deduced from the principles which he laid down but the determination of the amount and law of most of the inequalities was deferred to a more favorable opportunity, when he might be furnished with better astronomical observations. Such observations as he needed for this purpose had been made by Flamsteed, and for these he applied, representing how much value their use would add to the observations. "If," he says, in 1694, "you publish them without such a theory to recommend them, they will only be thrown into the heap of the observations of former astronomers, till somebody shall arise that by perfecting the theory of the moon shall discover your observations to be exacter than the rest; but when that shall be, God knows: I fear, not in your lifetime, if I should die before it is done. For I find this theory so very intricate, and the theory of gravity so necessary to it, that I am satisfied it will never be perfected but by somebody who understands the theory of gravity as well, or better than I do." He obtained from Flamsteed the lunar observations for which he applied, and by using these he framed the Theory of the Moon which is given as his in David Gregory's _Astronomiæ Elementa_.[51\7] He also obtained from Flamsteed the diameters of the planets as observed at various times, and the greatest elongation of Jupiter's Satellites, both of which, Flamsteed says, he made use of in his _Principia_.

[Note 51\7: In the Preface to a _Treatise on Dynamics_, Part i., published in 1836, I have endeavored to show that Newton's modes of determining several of the lunar inequalities admitted of an accuracy not very inferior to the modern analytical methods.]

Newton, in his letters to Flamsteed in 1694 and 5, acknowledges this service.[52\7]**] {420}

[Note 52\7: The quarrel on the subject of the publication of Flamsteed's Observations took place at a later period. Flamsteed wished to have his Observations printed complete and entire. Halley, who, under the authority of Newton and others, had the management of the printing, made many alterations and omissions, which

Flamsteed considered as deforming and spoiling the work. The advantages of publishing a _complete_ series of observations, now generally understood, were not then known to astronomers in general, though well known to Flamsteed, and earnestly insisted upon in his remonstrances. The result was that Flamsteed published his Observations at his own expense, and finally obtained permission to destroy the copies printed by Halley, which he did. In 1726, after Flamsteed's death, his widow applied to the Vice-Chancellor of Oxford, requesting that the volume printed by Halley might be removed out of the Bodleian Library, where it exists, as being "nothing more than an erroneous abridgment of Mr. Flamsteed's works," and unfit to see the light.]

## CHAPTER III.

SEQUEL TO THE EPOCH OF NEWTON.--RECEPTION OF THE NEWTONIAN THEORY.

_Sect._ 1.--_General Remarks._

THE doctrine of universal gravitation, like other great steps in science, required a certain time to make its way into men's minds; and had to be confirmed, illustrated, and completed, by the labors of succeeding philosophers. As the discovery itself was great beyond former example, the features of the natural sequel to the discovery were also on a gigantic scale; and many vast and laborious trains of research, each of which might, in itself, be considered as forming a wide science, and several of which have occupied many profound and zealous inquirers from that time to our own day, come before us as parts only of the verification of Newton's Theory. Almost every thing that has been done, and is doing, in astronomy, falls inevitably under this description; and it is only when the astronomer travels to the very limits of his vast field of labor, that he falls in with phenomena which do not acknowledge the jurisdiction of the Newtonian legislation. We must give some account of the events of this part of the history of astronomy; but our narrative must necessarily be extremely brief and imperfect; for the subject is most large and copious, and our limits are fixed and narrow. We have here to do with the history of discoveries, only so far as it illustrates their philosophy. And though the {421} astronomical discoveries of the last century are by no means poor, even in interest of this kind, the generalizations which they involve are far less important for our object, in consequence of being included in a previous generalization. Newton shines out so brightly, that all who follow seem faint and dim. It is not precisely the case which the poet describes-- As in a theatre the eyes of men, After some well-graced actor leaves the stage, Are idly bent on him that enters next, Thinking his prattle to be tedious: but our eyes are at least less intently bent on the astronomers who succeeded, and we attend to their communications with less curiosity, because we know the end, if not the course of their story; we know that their speeches have all closed with Newton's sublime declaration, asserted in some new form.

Still, however, the account of the verification and extension of any great discovery is a highly important part of its history. In this instance it is most important; both

from the weight and dignity of the theory concerned, and the ingenuity and extent of the methods employed: and, of course, so long as the Newtonian theory still required verification, the question of the truth or falsehood of such a grand system of doctrines could not but excite the most intense curiosity. In what I have said, I am very far from wishing to depreciate the value of the achievements of modern astronomers, but it is essential to my purpose to mark the subordination of narrower to wider truths--the different character and import of the labors of those who come before and after the promulgation of a master-truth. With this warning I now proceed to my narrative.

_Sect._ 2.--_Reception of the Newtonian Theory in England._

THERE appears to be a popular persuasion that great discoveries are usually received with a prejudiced and contentious opposition, and the authors of them neglected or persecuted. The reverse of this was certainly the case in England with regard to the discoveries of Newton. As we have already seen, even before they were published, they were proclaimed by Halley to be something of transcendent value; and from the moment of their appearance, they rapidly made their way from one class of thinkers to another, nearly as fast as the nature of men's intellectual capacity allows. Halley, Wren, and all the leading {422} members of the Royal Society, appear to have embraced the system immediately and zealously. Men whose pursuits had lain rather in literature than in science, and who had not the knowledge and habits of mind which the strict study of the system required, adopted, on the credit of their mathematical friends, the highest estimation of the _Principia_, and a strong regard for its author, as Evelyn, Locke, and Pepys. Only five years after the publication, the principles of the work were referred to from the pulpit, as so incontestably proved that they might be made the basis of a theological argument. This was done by Dr. Bentley, when he preached the Boyle's Lectures in London, in 1692. Newton himself, from the time when his work appeared, is never mentioned except in terms of profound admiration; as, for instance, when he is called by Dr. Bentley, in his sermon,[53\7] "That very excellent and divine theorist, Mr. Isaac Newton." It appears to have been soon suggested, that the Government ought to provide in some way for a person who was so great an honor to the nation. Some delay took place with regard to this; but, in 1695, his friend Mr. Montague, afterwards Earl of Halifax, at that time Chancellor of the Exchequer, made him Warden of the Mint; and in 1699, he succeeded to the higher office of Master of the Mint, a situation worth £1200 or £1500 a year, which he filled to the end of his life. In 1703, he became President of the Royal Society, and was annually re-elected to this office during the remaining twenty-five years of his life. In 1705, he was knighted in the Master's Lodge, at Trinity College, by Queen Anne, then on a visit to the University of Cambridge. After the accession of George the First, Newton's conversation was frequently sought by the Princess, afterwards Queen Caroline, who had a taste for speculative studies, and was often heard to declare in public, that she thought herself fortunate in living at a time which enabled her to enjoy the society of so great a genius. His fame, and the respect paid him, went on increasing to the end of his life; and when, in 1727, full of years and glory, his earthly career was ended, his death was mourned as a national calamity, with the forms usually confined to royalty. His body lay in state in the Jerusalem chamber; his pall was borne by the first nobles of the land and his earthly

remains were deposited in the centre of Westminster Abbey, in the midst of the memorials of the greatest and wisest men whom England has produced.

[Note 53\7: Serm. vii. 221.]

It cannot be superfluous to say a word or two on the reception of {423} his philosophy in the universities of England. These are often represented as places where bigotry and ignorance resist, as long as it is possible to resist, the invasion of new truths. We cannot doubt that such opinions have prevailed extensively, when we find an intelligent and generally temperate writer, like the late Professor Playfair of Edinburgh, so far possessed by them, as to be incapable of seeing, or interpreting, in any other way, any facts respecting Oxford and Cambridge. Yet, notwithstanding these opinions, it will be found that, in the English universities, new views, whether in science or in other subjects, have been introduced as soon as they were clearly established;--that they have been diffused from the few to the many more rapidly there than elsewhere occurs;--and that from these points, the light of newly-discovered truths has most usually spread over the land. In most instances undoubtedly there has been something of a struggle, on such occasions, between the old and the new opinions. Few men's minds can at once shake off a familiar and consistent system of doctrines, and adopt a novel and strange set of principles as soon as presented; but all can see that one change produces many, and that change, in itself, is a source of inconvenience and danger. In the case of the admission of the Newtonian opinions into Cambridge and Oxford, however, there are no traces even of a struggle. Cartesianism had never struck its roots deep in this country; that is, the peculiar hypotheses of Descartes. The Cartesian books, such, for instance, as that of Rohault, were indeed in use; and with good reason, for they contained by far the best treatises on most of the physical sciences, such as Mechanics, Hydrostatics, Optics, and Formal Astronomy, which could then be found. But I do not conceive that the Vortices were ever dwelt upon as a matter of importance in our academic teaching. At any rate, if they were brought among us, they were soon dissipated. Newton's College, and his University, exulted in his fame, and did their utmost to honor and aid him. He was exempted by the king from the obligation of taking orders, under which the fellows of Trinity College in general are; by his college he was relieved from all offices which might interfere, however slightly, with his studious employments, though he resided within the walls of the society thirty-five years, almost without the interruption of a month.[54\7] By the University he was elected their representative in parliament in 1688, {424} and again in 1701; and though he was rejected in the dissolution of 1705, those who opposed him acknowledged him[55\7] to be "the glory of the University and nation," but considered the question as a political one, and Newton as sent "to tempt them from their duty, by the great and just veneration they had for him." Instruments and other memorials, valued because they belonged to him, are still preserved in his college, along with the tradition of the chambers which he occupied.

[Note 54\7: His name is nowhere found on the college-books, as appointed to any of the offices which usually pass down the list of resident fellows in rotation. This might be owing in part, however, to his being Lucasian Professor. The constancy of

his residence in college appears from the _exit_ and _redit_ book of that time, which is still preserved.]

[Note 55\7: A pamphlet by Styan Thurlby.]

The most active and powerful minds at Cambridge became at once disciples and followers of Newton. Samuel Clarke, afterwards his friend, defended in the public schools a thesis taken from his philosophy, as early as 1694; and in 1697 published an edition of Rohault's _Physics_, with notes, in which Newton is frequently referred to with expressions of profound respect, though the leading doctrines of the _Principia_ are not introduced till a later edition, in 1703. In 1699, Bentley, whom we have already mentioned as a Newtonian, became Master of Trinity College; and in the same year, Whiston, another of Newton's disciples, was appointed his deputy as professor of mathematics. Whiston delivered the Newtonian doctrines, both from the professor's chair, and in works written for the use of the University; yet it is remarkable that a taunt respecting the late introduction of the Newtonian system into the Cambridge course of education, has been founded on some peevish expressions which he uses in his Memoirs, written at a period when, having incurred expulsion from his professorship and the University, he was naturally querulous and jaundiced in his views. In 1709-10, Dr. Laughton, who was tutor in Clare Hall, procured himself to be appointed moderator of the University disputations, in order to promote the diffusion of the new mathematical doctrines. By this time the first edition of the _Principia_ was become rare, and fetched a great price. Bentley urged Newton to publish a new one; and Cotes, by far the first, at that time, of the mathematicians of Cambridge, undertook to superintend the printing, and the edition was accordingly published in 1713.

[2d Ed.] [I perceive that my accomplished German translator, Littrow, has incautiously copied the insinuations of some modern writers to the effect that Clarke's reference to Newton, in his Edition of Rohault's _Physics_, was a mode of introducing Newtonian doctrines covertly, when it was not allowed him to introduce such novelties {425} openly. I am quite sure that any one who looks into this matter will see that this supposition of any unwillingness at Cambridge to receive Newton's doctrine is quite absurd, and can prove nothing but the intense prejudices of those who maintain such an opinion. Newton received and held his professorship amid the unexampled admiration of all contemporary members of the University. Whiston, who is sometimes brought as an evidence against Cambridge on this point, says, "I with immense pains set myself with the utmost zeal to the study of Sir Isaac Newton's wonderful discoveries in his _Philosophiæ Naturalis Principia Mathematica_, one or two of which _lectures I had heard him read in the public schools_, though I understood them not at the time." As to Rohault's _Physics_, it really did contain the best mechanical philosophy of the time;--the doctrines which were held by Descartes in common with Galileo, and with all the sound mathematicians who succeeded them. Nor does it look like any great antipathy to novelty in the University of Cambridge, that this book, which was quite as novel in its doctrines as Newton's _Principia_, and which had only been published at Paris in 1671, had obtained a firm hold on the University in less than twenty years. Nor is there any attempt made in Clarke's notes

to conceal the novelty of Newton's discoveries, but on the contrary, admiration is claimed for them as new.

The promptitude with which the Mathematicians of the University of Cambridge adopted the best parts of the mechanical philosophy of Descartes, and the greater philosophy of Newton, in the seventeenth century, has been paralleled in our own times, in the promptitude with which they have adopted and followed into their consequences the Mathematical Theory of Heat of Fourier and Laplace, and the Undulatory Theory of Light of Young and Fresnel.

In Newton's College, we possess, besides the memorials of him mentioned above (which include two locks of his silver-white hair), a paper in his own handwriting, describing the preparatory reading which was necessary in order that our College students might be able to read the _Principia_. I have printed this paper in the Preface to my Edition of the First Three Sections of the _Principia_ in the original Latin (1846).

Bentley, who had expressed his admiration for Newton in his Boyle's Lectures in 1692, was made Master of the College in 1699, as I have stated; and partly, no doubt, in consequence of the Newtonian sermons which he had preached. In his administration of the College, he zealously stimulated and assisted the exertions of Cotes, Whiston, and other disciples of Newton. Smith, Bentley's successor as Master of {426} the College, erected a statue of Newton in the College Chapel (a noble work of Roubiliac), with the inscription, _Qui genus humanum ingenio superavit._]

At Oxford, David Gregory and Halley, both zealous and distinguished disciples of Newton, obtained the Savilian professorships of astronomy and geometry in 1691 and 1703.

David Gregory's _Astronomiæ Physicæ et Geometricæ Elementa_ issued from the Oxford Press in 1702. The author, in the first sentence of the Preface, states his object to be to explain the mechanics of the universe (Physica Cœlestis), which Isaac Newton, the Prince of Geometers, has carried to a point of elevation which all look up to with admiration. And this design is executed by a full exposition of the Newtonian doctrines and their results. Keill, a pupil of Gregory, followed his tutor to Oxford, and taught the Newtonian philosophy there in 1700, being then Deputy Sedleian Professor. He illustrated his lectures by experiments, and published an Introduction to the _Principia_ which is not out of use even yet.

In Scotland, the Newtonian philosophy was accepted with great alacrity, as appears by the instances of David Gregory and Keill. David Gregory was professor at Edinburgh before he removed to Oxford, and was succeeded there by his brother James. The latter had, as early as 1690, printed a thesis, containing in twenty-two propositions, a compend of Newton's _Principia_.[56\7] Probably these were intended as theses for academical disputations; as Laughton at Cambridge introduced the Newtonian philosophy into these exercises. The formula at Cambridge, in use till very recently in these disputations, was "_Rectè statuit Newtonus de Motu Lunæ_;" or the like.

[Note 56\7: See Hutton's _Math. Dict._, art. _James Gregory_. If it fell in with my plan to notice derivative works, I might speak of Maclaurin's admirable _Account of Sir Isaac Newton's Discoveries_, published in 1748. This is still one of the best books on the subject. The late Professor Rigaud's _Historical Essay on the First Publication of Sir Isaac Newton's "Principia"_ (Oxf. 1838) contains a careful and candid view of the circumstances of that event.]

The general diffusion of these opinions in England took place, not only by means of books, but through the labors of various experimental lecturers, like Desaguliers, who removed from Oxford to London in 1713; when he informs us,[57\7] that "he found the Newtonian philosophy generally received among persons of all ranks and professions, and even among the ladies by the help of experiments." {427}

[Note 57\7: Desag. _Pref._]

We might easily trace in our literature indications of the gradual progress of the Newtonian doctrines. For instance, in the earlier editions of Pope's _Dunciad_, this couplet occurred, in the description of the effects of the reign of Dulness: Philosophy, that reached the heavens before, Shrinks to her hidden cause, and is no more. "And this," says his editor, Warburton, "was intended as a censure on the Newtonian philosophy. For the poet had been misled by the prejudices of foreigners, as if that philosophy had recurred to the occult qualities of Aristotle. This was the idea he received of it from a man educated much abroad, who had read every thing, but every thing superficially.[58\7] When I hinted to him how he had been imposed upon, he changed the lines with great pleasure into a compliment (as they now stand) on that divine genius, and a satire on that very folly by which he himself had been misled." In 1743 it was printed, Philosophy, that leaned on heaven before, Shrinks to her second cause, and is no more. The Newtonians repelled the charge of dealing in occult causes;[59\7] and, referring gravity to the will of the Deity, as the First Cause, assumed a superiority over those whose philosophy rested in second causes.

[Note 58\7: I presume Bolingbroke is here meant.]

[Note 59\7: See Cotes's Pref. to the _Principia_.]

To the cordial reception of the Newtonian theory by the English astronomers, there is only one conspicuous exception; which is, however, one of some note, being no other than Flamsteed, the Astronomer Royal, a most laborious and exact observer. Flamsteed at first listened with complacency to the promises of improvements in the Lunar Tables, which the new doctrines held forth, and was willing to assist Newton, and to receive assistance from him. But after a time, he lost his respect for Newton's theory, and ceased to take any interest in it. He then declared to one of his correspondents,[60\7] "I have determined to lay these crotchets of Sir Isaac Newton's wholly aside." We need not, however, find any difficulty in this, if we recollect that Flamsteed, though a good observer, was no philosopher;--never understood by a Theory any thing more than a Formula which should predict results;--and was incapable of comprehending the object of Newton's theory, which was to assign causes as well as rules, and to satisfy the conditions of Mechanics as well as of Geometry. {428}

[Note 60\7: Baily's _Account of Flamsteed, &c._, p. 309.]

[2d Ed.] [I do not see any reason to retract what was thus said; but it ought perhaps to be distinctly said that on these very accounts Flamsteed's rejection of Newton's rules did not imply a denial of the doctrine of gravitation. In the letter above quoted, Flamsteed says that he has been employed upon the Moon, and that "the heavens reject that equation of Sir I. Newton which Gregory and Newton called his sixth: I had then [when he wrote before] compared but 72 of my observations with the tables, now I have examined above 100 more. I find them all firm in the same, and the seventh [equation] too." And thereupon he comes to the determination above stated.

At an earlier period Flamsteed, as I have said, had received Newton's suggestions with great deference, and had regulated his own observations and theories with reference to them. The calculation of the lunar inequalities upon the theory of gravitation was found by Newton and his successors to be a more difficult and laborious task than he had anticipated, and was not performed without several trials and errors. One of the equations was at first published (in Gregory's _Astronomiæ Elementa_) with a wrong sign. And when Newton had done all, Flamsteed found that the rules were far from coming up to the degree of accuracy which had been claimed for them, that they could give the moon's place true to 2 or 3 minutes. It was not till considerably later that this amount of exactness was attained.

The late Mr. Baily, to whom astronomy and astronomical literature are so deeply indebted, in his _Supplement to the Account of Flamsteed_, has examined with great care and great candor the assertion that Flamsteed did not understand Newton's Theory. He remarks, very justly, that what Newton himself at first presented as his Theory, might more properly be called Rules for computing lunar tables, than a physical Theory in the modern acceptation of the term. He shows, too, that Flamsteed had read the _Principia_ with attention.[61\7] Nor do I doubt that many considerable mathematicians gave the same imperfect assent to Newton's doctrine which Flamsteed did. But when we find that others, as Halley, David Gregory, and Cotes, at once not only saw in the doctrine a source of true formulæ, but also a magnificent physical discovery, we are obliged, I think, to make Flamsteed, in this respect, an exception to the first class of astronomers of his own time.

[Note 61\7: _Supp._ p. 691.]

Mr. Baily's suggestion that the annual equations for the corrections of the lunar apogee and node were collected from Flamsteed's tables {429} and observations independently of their suggestion by Newton as the results of Theory (_Supp._ p. 692, Note, and p. 698), appears to me not to be adequately supported by the evidence given.]

_Sect._ 3.--_Reception of the Newtonian Theory abroad._

THE reception of the Newtonian theory on the Continent, was much more tardy and unwilling than in its native island. Even those whose mathematical attainments most fitted them to appreciate its proofs, were prevented by some peculiarity of view from adopting it as a system; as Leibnitz, Bernoulli, Huyghens; who all clung to one

modification or other of the system of vortices. In France, the Cartesian system had obtained a wide and popular reception, having been recommended by Fontenelle with the graces of his style; and its empire was so firm and well established in that country, that it resisted for a long time the pressure of Newtonian arguments. Indeed, the Newtonian opinions had scarcely any disciples in France, till Voltaire asserted their claims, on his return from England in 1728: until then, as he himself says, there were not twenty Newtonians out of England.

The hold which the Philosophy of Descartes had upon the minds of his countrymen is, perhaps, not surprising. He really had the merit, a great one in the history of science, of having completely overturned the Aristotelian system, and introduced the philosophy of matter and motion. In all branches of mixed mathematics, as we have already said, his followers were the best guides who had yet appeared. His hypothesis of vortices, as an explanation of the celestial motions, had an apparent advantage over the Newtonian doctrine, in this respect;--that it referred effects to the most intelligible, or at least most familiar kinds of mechanical causation, namely, pressure and impulse. And above all, the system was acceptable to most minds, in consequence of being, as was pretended, deduced from a few simple principles by necessary consequences; and of being also directly connected with metaphysical and theological speculations. We may add, that it was modified by its mathematical adherents in such a way as to remove most of the objections to it. A vortex revolving about a centre could be constructed, or at least it was supposed that it could be constructed, so as to produce a tendency of bodies to the centre. In all cases, therefore, where a central force acted, a vortex was supposed; but in reasoning to the results of this hypothesis, it was {430} easy to leave out of sight all other effects of the vortex, and to consider only the central force; and when this was done, the Cartesian mathematician could apply to his problems a mechanical principle of some degree of consistency. This reflection will, in some degree, account for what at first seems so strange;--the fact that the language of the French mathematicians is Cartesian, for almost half a century after the publication of the _Principia_ of Newton.

There was, however, a controversy between the two opinions going on all this time, and every day showed the insurmountable difficulties under which the Cartesians labored. Newton, in the _Principia_, had inserted a series of propositions, the object of which was to prove, that the machinery of vortices could not be accommodated to one part of the celestial phenomena, without contradicting another part. A more obvious difficulty was the case of gravity of the earth; if this force arose, as Descartes asserted, from the rotation of the earth's vortex about its axis, it ought to tend directly to the axis, and not to the centre. The asserters of vortices often tried their skill in remedying this vice in the hypothesis, but never with much success. Huyghens supposed the ethereal matter of the vortices to revolve about the centre in all directions; Perrault made the strata of the vortex increase in velocity of rotation as they recede from the centre; Saurin maintained that the circumambient resistance which comprises the vortex will produce a pressure passing through the centre. The elliptic form of the orbits of the planets was another difficulty. Descartes had supposed the vortices themselves to be oval but others, as John Bernoulli, contrived ways of having elliptical motion in a circular vortex.

The mathematical prize-questions proposed by the French Academy, naturally brought the two sets of opinions into conflict. The Cartesian memoir of John Bernoulli, to which we have just referred, was the one which gained the prize in 1730. It not unfrequently happened that the Academy, as if desirous to show its impartiality, divided the prize between the Cartesians and Newtonians. Thus in 1734, the question being, the cause of the inclination of the orbits of the planets, the prize was shared between John Bernoulli, whose Memoir was founded on the system of vortices, and his son Daniel, who was a Newtonian. The last act of homage of this kind to the Cartesian system was performed in 1740, when the prize on the question of the Tides was distributed between Daniel Bernoulli, Euler, Maclaurin, and Cavallieri; the last of whom had tried to patch up and amend the Cartesian hypothesis on this subject. {431}

Thus the Newtonian system was not adopted in France till the Cartesian generation had died off; Fontenelle, who was secretary to the Academy of Sciences, and who lived till 1756, died a Cartesian. There were exceptions; for instance, Delisle, an astronomer who was selected by Peter the Great of Russia, to found the Academy of St Petersburg; who visited England in 1724, and to whom Newton then gave his picture, and Halley his Tables. But in general, during the interval, that country and this had a national difference of creed on physical subjects. Voltaire, who visited England in 1727, notices this difference in his lively manner. "A Frenchman who arrives in London, finds a great alteration in philosophy, as in other things. He left the world full [**a _plenum_], he finds it empty. At Paris you see the universe composed of vortices of subtle matter, in London we see nothing of the kind. With you it is the pressure of the moon which causes the tides of the sea, in England it is the sea which gravitates towards the moon; so that when you think the moon ought to give us high water, these gentlemen believe that you ought to have low water; which unfortunately we cannot test by experience; for in order to do that, we should have examined the Moon and the Tides at the moment of the creation. You will observe also that the sun, which in France has nothing to do with the business, here comes in for a quarter of it. Among you Cartesians, all is done by an impulsion which one does not well understand; with the Newtonians, it is done by an attraction of which we know the cause no better. At Paris you fancy the earth shaped like a melon, at London it is flattened on the two sides."

It was Voltaire himself as we have said, who was mainly instrumental in giving the Newtonian doctrines currency in France. He was at first refused permission to print his _Elements of the Newtonian Philosophy_, by the Chancellor, D'Aguesseaux, who was a Cartesian; but after the appearance of this work in 1738, and of other writings by him on the same subject, the Cartesian edifice, already without real support or consistency, crumbled to pieces and disappeared. The first Memoir in the _Transactions of the French Academy_ in which the doctrine of central force is applied to the solar system, is one by the Chevalier de Louville in 1720, _On the Construction and Theory of Tables of the Sun_. In this, however, the mode of explaining the motions of the planets by means of an original impulse and an attractive force is attributed to Kepler, not to Newton. The first Memoir which refers to the universal gravitation of matter is by Maupertuis, in {432} 1736. But Newton

was not unknown or despised in France till this time. In 1699 he was admitted one of the very small number of foreign associates of the French Academy of Sciences. Even Fontenelle, who, as we have said, never adopted his opinions, spoke of him in a worthy manner, in the _Eloge_ which he composed on the occasion of his death. At a much earlier period too, Fontenelle did homage to his fame. The following passage refers, I presume, to Newton. In the _History_ of the Academy for 1708, which is written by the secretary, he says,[62\7] in referring to the difficulty which the comets occasion in the Cartesian hypothesis: "We might relieve ourselves at once from all the embarrassment which arises from the directions of these motions, by suppressing, as has been done _by one of the greatest geniuses of the age_, all this immense fluid matter, which we commonly suppose between the planets, and conceiving them suspended in a perfect void."

[Note 62\7: _Hist. Ac. Sc._ 1708. p. 103.]

Comets, as the above passage implies, were a kind of artillery which the Cartesian _plenum_ could not resist. When it appeared that the paths of such wanderers traversed the vortices in all directions, it was impossible to maintain that these imaginary currents governed the movements of bodies immersed in them and the mechanism ceased to have any real efficacy. Both these phenomena of comets, and many others, became objects of a stronger and more general interest, in consequence of the controversy between the rival parties; and thus the prevalence of the Cartesian system did not seriously impede the progress of sound knowledge. In some cases, no doubt, it made men unwilling to receive the truth, as in the instance of the deviation of the comets from the zodiacal motion; and again, when Römer discovered that light was not instantaneously propagated. But it encouraged observation and calculation, and thus forwarded the verification and extension of the Newtonian system; of which process we must now consider some of the incidents. {433}

## CHAPTER IV.

SEQUEL TO THE EPOCH OF NEWTON, CONTINUED.--VERIFICATION AND COMPLETION OF THE NEWTONIAN THEORY.

_Sect._ 1.--_Division of the Subject._

THE verification of the Law of Universal Gravitation as the governing principle of all cosmical phenomena, led, as we have already stated, to a number of different lines of research, all long and difficult. Of these we may treat successively, the motions of the Moon, of the Sun, of the Planets, of the Satellites, of Comets; we may also consider separately the Secular Inequalities, which at first sight appear to follow a different law from the other changes; we may then speak of the results of the principle as they affect this Earth, in its Figure, in the amount of Gravity at different places, and in the phenomena of the Tides. Each of these subjects has lent its aid to confirm the general law: but in each the confirmation has had its peculiar difficulties, and has its separate history. Our sketch of this history must be very rapid, for our aim is only to show what is the kind and course of the confirmation which such a theory demands and receives.

For the same reason we pass over many events of this period which are highly important in the history of astronomy. They have lost much of their interest for us, and even for common readers, because they are of a class with which we are already familiar, truths included in more general truths to which our eyes now most readily turn. Thus, the discovery of new satellites and planets is but a repetition of what was done by Galileo: the determination of their nodes and apses, the reduction of their motions to the law of the ellipse, is but a fresh exemplification of the discoveries of Kepler. Otherwise, the formation of Tables of the satellites of Jupiter and Saturn, the discovery of the eccentricities of the orbits, and of the motions of the nodes and apses, by Cassini, Halley, and others, would rank with the great achievements in astronomy. Newton's peculiar advance in the _Tables_ of the celestial motions is the introduction of Perturbations. To these motions, so affected, we now proceed. {434}

_Sect._ 2.--_Application of the Newtonian Theory to the Moon._

THE Motions of the Moon may be first spoken of, as the most obvious and the most important of the applications of the Newtonian Theory. The verification of such a theory consists, as we have seen in previous cases, in the construction of Tables derived from the theory, and the comparison of these with observation. The advancement of astronomy would alone have been a sufficient motive for this labor; but there were other reasons which urged it on with a stronger impulse. A perfect Lunar Theory, if the theory could be perfected, promised to supply a method of finding the Longitude of any place on the earth's surface; and thus the verification of a theory which professed to be complete in its foundations, was identified with an object of immediate practical use to navigators and geographers, and of vast acknowledged value. A good method for the near discovery of the longitude had been estimated by nations and princes at large sums of money. The Dutch were willing to tempt Galileo to this task by the offer of a chain of gold: Philip the Third of Spain had promised a reward for this object still earlier;[63\7] the parliament of England, in 1714, proposed a recompense of 20,000_l._ sterling; the Regent Duke of Orléans, two years afterwards, offered 100,000 francs for the same purpose. These prizes, added to the love of truth and of fame, kept this object constantly before the eyes of mathematicians, during the first half of the last century.

[Note 63\7: Del. _A. M._ i. 39, 66.]

If the Tables could be so constructed as to represent the moon's real place in the heavens with extreme precision, as it would be seen from a _standard_ observatory, the observation of her apparent place, as seen from any other point of the earth's surface, would enable the observer to find his longitude from the standard point. The motions of the moon had hitherto so ill agreed with the best Tables, that this method failed altogether. Newton had discovered the ground of this want of agreement. He had shown that the same force which produces the Evection, Variation, and Annual Equation, must produce also a long series of other Inequalities, of various magnitudes and cycles, which perpetually drag the moon before or behind the place where she would be sought by an astronomer who knew only of those principal and notorious inequalities. But to calculate and apply the new inequalities, was no slight undertaking. {435}

In the first edition of the _Principia_ in 1687, Newton had not given any calculations of new inequalities affecting the longitude of the moon. But in David Gregory's _Elements of Physical and Geometrical Astronomy_, published in 1702, is inserted[64\7] "Newton's Lunar Theory as applied by him to Practice;" in which the great discoverer has given the results of his calculations of eight of the lunar Equations, their quantities, epochs, and periods. These calculations were for a long period the basis of new Tables of the Moon, which were published by various persons;[65\7] as by Delisle in 1715 or 1716, Grammatici at Ingoldstadt in 1726, Wright in 1732, Angelo Capelli at Venice in 1733, Dunthorne at Cambridge in 1739.

[Note 64\7: P. 332.]

[Note 65\7: Lalande, 1457.]

Flamsteed had given Tables of the Moon upon Horrox's theory in 1681, and wished to improve them; and though, as we have seen, he would not, or could not, accept Newton's doctrines in their whole extent, Newton communicated his theory to the observer in the shape in which he could understand it and use it:[66\7] and Flamsteed employed these directions in constructing new Lunar Tables, which he called his _Theory_.[67\7] These Tables were not published till long after his death, by Le Monnier at Paris in 1746. They are said, by Lalande,[68\7] not to differ much from Halley's. Halley's Tables of the Moon were printed in 1719 or 1720, but not published till after his death in 1749. They had been founded on Flamsteed's observations and his own; and when, in 1720, Halley succeeded Flamsteed in the post of Astronomer Royal at Greenwich, and conceived that he had the means of much improving what he had done before, he began by printing what he had already executed.[69\7]

[Note 66\7: Baily. _Account of Flamsteed_, p. 72.]

[Note 67\7: P. 211.]

[Note 68\7: Lal. 1459.]

[Note 69\7: Mr. Baily* says that Mayer's _Nouvelles Tables de la Lune_ in **1753, published upwards of fifty years after Gregory's _Astronomy_, may be considered as the first lunar tables formed _solely_ on Newton's principles. Though Wright in 1732 published _New and Correct Tables of the Lunar Motions according to the Newtonian Theory_, Newton's rules were in them only partially adopted. In 1735 Leadbetter published his _Uranoscopia_, in which those rules were more fully followed. But these _Newtonian Tables_ did not supersede Flamsteed's Horroxian Tables, till both were supplanted by those of Mayer.

* _Supp._ p. 702.]

But Halley had long proposed a method, different from that of Newton, but marked by great ingenuity, for amending the Lunar Tables. He proposed to do this by the use of a cycle, which we have mentioned as one of the earliest discoveries in astronomy;--the Period of 223 lunations, or eighteen years and eleven days, the Chaldean {436} Saros. This period was anciently used for predicting the eclipses of the sun and moon; for those eclipses which happen during this period, are repeated again in the same order, and with nearly the same circumstances, after the expiration

of one such period and the commencement of a second. The reason of this is, that at the end of such a cycle, the moon is in nearly the same position with respect to the sun, her nodes, and her apogee, as she was at first; and is only a few degrees distant from the same part of the heavens. But on the strength of this consideration, Halley conjectured that all the irregularities of the moon's motion, however complex they may be, would recur after such an interval; and that, therefore, if the requisite corrections were determined by observation for one such period, we might by means of them give accuracy to the Tables for all succeeding periods. This idea occurred to him before he was acquainted with Newton's views.[70\7] After the lunar theory of the _Principia_ had appeared, he could not help seeing that the idea was confirmed; for the inequalities of the moon's motion, which arise from the attraction of the sun, will depend on her positions with regard to the sun, the apogee, and the node; and therefore, however numerous, will recur when these positions recur.

[Note 70\7: _Phil. Trans._ 1731, p. 188.]

Halley announced, in 1691,[71\7] his intention of following this idea into practice; in a paper in which he corrected the text of three passages in Pliny, in which this period is mentioned, and from which it is sometimes called the Plinian period. In 1710, in the preface to a new edition to Street's _Caroline Tables_, he stated that he had already confirmed it to a considerable extent.[72\7] And even after Newton's theory had been applied, he still resolved to use his cycle as a means of obtaining further accuracy. On succeeding to the Observatory at Greenwich in 1720, he was further delayed by finding that the instruments had belonged to Flamsteed, and were removed by his executors. "And this," he says,[73\7] "was the more grievous to me, on account of my advanced age, being then in my sixty-fourth year: which put me past all hopes of ever living to see a complete period of eighteen years' observation. But, thanks to God, he has been pleased hitherto (in 1731) to afford me sufficient health and strength to execute my office, in all its parts, with my own hands and eyes, without any assistance or interruption, during one whole period of the moon's {437} apogee, which period is performed in somewhat less than nine years." He found the agreement very remarkable, and conceived hopes of attaining the great object, of finding the Longitude with the requisite degree of exactness; nor did he give up his labors on this subject till he had completed his Plinian period in 1739.

[Note 71\7: Ib. p. 536.]

[Note 72\7: Ib. 1731, p. 187.]

[Note 73\7: Ib. p. 193.]

The accuracy with which Halley conceived himself able to predict the moon's place[74\7] was within two minutes of space, or one fifteenth of the breadth of the moon herself. The accuracy required for obtaining the national reward was considerably greater. Le Monnier pursued the idea of Halley.[75\7] But before Halley's method had been completed, it was superseded by the more direct prosecution of Newton's views.

[Note 74\7: _Phil. Trans._ 1731, p. 195.]

[Note 75\7: Bailly, _A. M._ c. 131.]

We have already remarked, in the history of analytical mechanics, that in the Lunar Theory, considered as one of the cases of the Problem of Three Bodies, no advance was made beyond what Newton had done, till mathematicians threw aside the Newtonian artifices, and applied the newly developed generalizations of the analytical method. The first great apparent deficiency in the agreement of the law of universal gravitation with astronomical observation, was removed by Clairaut's improved approximation to the theoretical Motion of the Moon's Apogee, in 1750; yet not till it had caused so much disquietude, that Clairaut himself had suggested a modification of the law of attraction; and it was only in tracing the consequences of this suggestion, that he found the Newtonian law of the inverse square to be that which, when rightly developed, agreed with the facts. Euler solved the problem by the aid of his analysis in 1745,[76\7] and published Tables of the Moon in 1746. His tables were not very accurate at first;[77\7] but he, D'Alembert, and Clairaut, continued to labor at this object, and the two latter published Tables of the Moon in 1754.[78\7] Finally, Tobias Mayer, an astronomer of Göttingen, having compared Euler's tables with observations, corrected them so successfully, that in 1753 he published Tables of the Moon, which really did possess the accuracy which Halley only flattered himself that he had attained. Mayer's success in his first Tables encouraged him to make them still more perfect. He applied himself to the mechanical theory of the moon's orbit; corrected all the coefficients of the series by a great number of observations; and in 1755, sent his new Tables to London as worthy to claim the prize offered for the discovery of longitude. He died soon after {438} (in 1762), at the early age of thirty-nine, worn out by his incessant labors; and his widow sent to London a copy of his Tables with additional corrections. These Tables were committed to Bradley, then Astronomer Royal, in order to be compared with observation. Bradley labored at this task with unremitting zeal and industry, having himself long entertained hopes that the Lunar Method of finding the Longitude might be brought into general use. He and his assistant, Gael Morris, introduced corrections into Mayer's Tables of 1755. In his report of 1756, he says,[79\7] that he did not find any difference so great as a minute and a quarter; and in 1760, he adds, that this deviation had been further diminished by his corrections. It is not foreign to our purpose to observe the great labor which this verification required. Not less than 1220 observations, and long calculations founded upon each, were employed. The accuracy which Mayer's Tables possessed was considered to entitle them to a part of the parliamentary reward; they were printed in 1770, and his widow received 3000_l._ from the English nation. At the same time, Euler, whose Tables had been the origin and foundation of Mayer's, also had a recompense of the same amount.

[Note 76\7: Lal. 1460.]

[Note 77\7: Bradley's Correspondence.]

[Note 78\7: Lal. 1460.]

[Note 79\7: Bradley's _Mem._ p. xcviii.]

This public national acknowledgment of the practical accuracy of these Tables is, it will be observed, also a solemn recognition of the truth of the Newtonian theory, as far as truth can be judged of by men acting under the highest official responsibility, and aided by the most complete command of the resources of the skill and talents of others. The finding the Longitude is thus the seal of the moon's gravitation to the sun and earth; and with this occurrence, therefore, our main concern with the history of the Lunar Theory ends. Various improvements have been since introduced into this research; but on these we, with so many other subjects before us, must forbear to enter.

_Sect._ 3.--_Application of the Newtonian Theory to the Planets, Satellites, and Earth._

THE theories of the Planets and Satellites, as affected by the law of universal gravitation, and therefore by perturbations, were naturally subjects of interest, after the promulgation of that law. Some of the effects of the mutual attraction of the planets had, indeed, already attracted notice. The inequality produced by the mutual attraction of Jupiter and Saturn cannot be overlooked by a good observer. In the {439} preface to the second edition of the _Principia_, Cotes remarks,[80\7] that the perturbation of Jupiter and Saturn is not unknown to astronomers. In Halley's Tables it was noticed[81\7] that there are very great deviations from regularity in these two planets, and these deviations are ascribed to the perturbing force of the planets on each other; but the correction of these by a suitable equation is left to succeeding astronomers.

[Note 80\7: Preface to _Principia_, p. xxi.]

[Note 81\7: End of Planetary Tables.]

The motion of the planes and apsides of the planetary orbits was one of the first results of their mutual perturbation which was observed. In 1706, La Hire and Maraldi compared Jupiter with the Rudolphine Tables, and those of Bullialdus: it appeared that his aphelion had advanced, and that his nodes had regressed. In 1728, J. Cassini found that Saturn's aphelion had in like manner travelled forwards. In 1720, when Louville refused to allow in his solar tables the motion of the aphelion of the earth, Fontenelle observed that this was a misplaced scrupulousness, since the aphelion of Mercury certainly advances. Yet this reluctance to admit change and irregularity was not yet overcome. When astronomers had found an approximate and apparent constancy and regularity, they were willing to believe it absolute and exact. In the satellites of Jupiter, for instance, they were unwilling to admit even the eccentricity of the orbits; and still more, the variation of the nodes, inclinations, and apsides. But all the fixedness of these was successively disproved. Fontenelle in 1732, on the occasion of Maraldi's discovery of the change of inclination of the fourth satellite, expresses a suspicion that all the elements might prove liable to change. "We see," says he, "the constancy of the inclination already shaken in the three first satellites, and the eccentricity in the fourth. The immobility of the nodes holds out so far, but there are strong indications that it will share the same fate."

The motions of the nodes and apsides of the satellites are a necessary part of the Newtonian theory; and even the Cartesian astronomers now required only data, in order to introduce these changes into their Tables.

The complete reformation of the Tables of the Sun, Planets, and Satellites, which followed as a natural consequence from the revolution which Newton had introduced, was rendered possible by the labors of the great constellation of mathematicians of whom we have spoken in the last book, Clairaut, Euler, D'Alembert, and their successors; and {440} it was carried into effect in the course of the last century. Thus Lalande applied Clairaut's theory to Mars, as did Mayer; and the inequalities in this case, says Bailly[82\7] in 1785, may amount to two minutes, and therefore must not be neglected. Lalande determined the inequalities of Venus, as did Father Walmesley, an English mathematician; these were found to reach only to thirty seconds.

[Note 82\7: _Ast. Mod._ iii. 170.]

The Planetary Tables[83\7] which were in highest repute, up to the end of the last century, were those of Lalande. In these, the perturbations of Jupiter and Saturn were introduced, their magnitude being such that they cannot be dispensed with; but the Tables of Mercury, Venus, and Mars, had no perturbations. Hence these latter Tables might be considered as accurate enough to enable the observer to find the object, but not to test the theory of perturbations. But when the calculation of the mutual disturbances of the planets was applied, it was always found that it enabled mathematicians to bring the theoretical places to coincide more exactly with those observed. In improving, as much as possible, this coincidence, it is necessary to determine the mass of each planet; for upon that, according to the law of universal gravitation, its disturbing power depends. Thus, in 1813, Lindenau published Tables of Mercury, and concluded, from them, that a considerable increase of the supposed mass of Venus was necessary to reconcile theory with observation.[84\7] He had published Tables of Venus in 1810, and of Mars in 1811. And, in proving Bouvard's Tables of Jupiter and Saturn, values were obtained of the masses of those planets. The form in which the question of the truth of the doctrine of universal gravitation now offers itself to the minds of astronomers, is this:--that it is taken for granted that it will account for the motions of the heavenly bodies, and the question is, with what supposed masses it will give the _best_ account.[85\7] The continually increasing accuracy of the table shows the truth of the fundamental assumption.

[Note 83\7: Airy, _Report on Ast. to Brit. Ass._ 1832.]

[Note 84\7: Airy, _Report on Ast. to Brit. **Ass._ 1832.]

[Note 85\7: Among the most important corrections of the supposed masses of the planets, we may notice that of Jupiter, by Professor Airy. This determination of Jupiter's mass was founded, not on the effect as seen in perturbations, but on a much more direct datum, the time of revolution of his fourth satellite. It appeared, from this calculation, that Jupiter's mass required to be increased by about 1⁄80th. This result agrees with that which has been derived by German astronomers from the perturbations which the attractions of Jupiter produce in the four new planets, and

has been generally adopted as an improvement of the elements of our system.]

The question of perturbation is exemplified in the satellites also. {441} Thus the satellites of Jupiter are not only disturbed by the sun, as the moon is, but also by each other, as the planets are. This mutual action gives rise to some very curious relations among their motions; which, like most of the other leading inequalities, were forced upon the notice of astronomers by observation before they were obtained by mathematical calculation. In Bradley's remarks upon his own Tables of Jupiter's Satellites, published among Halley's Tables, he observes that the places of the three interior satellites are affected by errors which recur in a cycle of 437 days, answering to the time in which they return to the same relative position with regard to each other, and to the axis of Jupiter's shadow. Wargentin, who had noticed the same circumstance without knowledge of what Bradley had done, applied it, with all diligence, to the purpose of improving the tables of the satellites in 1746. But, at a later period, Laplace established, by mathematical reasoning, the very curious theorem on which this cycle depends, which he calls the _libration of Jupiter's satellites_; and Delambre was then able to publish Tables of Jupiter's Satellites more accurate than those of Wargentin, which he did in 1789.[86\7]

[Note 86\7: Voiron, _Hist. Ast._ p. 322.]

The progress of physical astronomy from the time of Euler and Clairaut, has consisted of a series of calculations and comparisons of the most abstruse and recondite kind. The formation of Tables of the Planets and Satellites from the theory, required the solution of problems much more complex than the original case of the Problem of Three Bodies. The real motions of the planets and their orbits are rendered still further intricate by this, that all the lines and points to which we can refer them, are themselves in motion. The task of carrying order and law into this mass of apparent confusion, has required a long series of men of transcendent intellectual powers; and a perseverance and delicacy of observation, such as we have not the smallest example of in any other subject. It is impossible here to give any detailed account of these labors; but we may mention one instance of the complex considerations which enter into them. The nodes of Jupiter's fourth satellite do not go backwards,[87\7] as the Newtonian theory seems to require; they advance upon Jupiter's orbit. But then, it is to be recollected that the theory requires the nodes to retrograde upon the orbit of the perturbing body, which is here the third satellite; and Lalande showed that, by the necessary relations of space, the latter motion may be retrograde though the former is direct. {442}

[Note 87\7: Bailly, iii. 175.]

Attempts have been made, from the time of the solution of the Problem of three bodies to the present, to give the greatest possible accuracy to the Tables of the Sun, by considering the effect of the various perturbations to which the earth is subject. Thus, in 1756, Euler calculated the effect of the attractions of the planets on the earth (the prize-question of the French Academy of Sciences), and Clairaut soon after. Lacaille, making use of these results, and of his own numerous observations, published Tables of the Sun. In 1786, Delambre[88\7] undertook to verify and

improve these tables, by comparing them with 314 observations made by Maskelyne, at Greenwich, in 1775 and 1784, and in some of the intermediate years. He corrected most of the elements; but he could not remove the uncertainty which occurred respecting the amount of the inequality produced by the reaction of the moon. He admitted also, in pursuance of Clairaut's theory, a second term of this inequality depending on the moon's latitude; but irresolutely, and half disposed to reject it on the authority of the observations. Succeeding researches of mathematicians have shown, that this term is not admissible as a result of mechanical principles. Delambre's Tables, thus improved, were exact to seven or eight seconds;[89\7] which was thought, and truly, a very close coincidence for the time. But astronomers were far from resting content with this. In 1806, the French Board of Longitude published Delambre's improved Solar Tables; and in the _Connaissance des Tems_ for 1816, Burckhardt gave the results of a comparison of Delambre's Tables with a great number of Maskelyne's observations;--far greater than the number on which they were founded.[90\7] It appeared that the epoch, the perigee, and the eccentricity, required sensible alterations, and that the mass of Venus ought to be reduced about one-ninth, and that of the Moon to be sensibly diminished. In 1827, Professor Airy[91\7] compared Delambre's tables with 2000 Greenwich observations, made with the new transit-instrument at Cambridge, and deduced from this comparison the correction of the elements. These in general agreed closely with Burckhardt's, excepting that a diminution of Mars appeared necessary. Some discordances, however, led Professor Airy to suspect the existence of an inequality which had escaped the sagacity of Laplace and Burckhardt. And, a few weeks after this suspicion had been expressed, the same mathematician announced to the Royal Society that he had {443} detected, in the planetary theory such an inequality, hitherto unnoticed, arising from the mutual attraction of Venus and the Earth. Its whole effect on the earth's longitude, would be to increase or diminish it by nearly three seconds of space, and its period is about 240 years. "This term," he adds, "accounts completely for the difference of the secular motions given by the comparison of the epochs of 1783 and 1821, and by that of the epochs of 1801 and 1821."

[Note 88\7: Voiron, _Hist._ p. 315.]

[Note 89\7: Montucla, iv. 42.]

[Note 90\7: Airy, _Report_, p. 150.]

[Note 91\7: _Phil. Trans._ 1828.]

Many excellent Tables of the motions of the sun, moon, and planets, were published in the latter part of the last century; but the Bureau des Longitudes which was established in France in 1795, endeavored to give new or improved tables of most of these motions. Thus were produced Delambre's Tables of the Sun, Burg's Tables of the Moon, Bouvard's Tables of Jupiter, Saturn, and Uranus. The agreement between these and observation is, in general, truly marvellous.

We may notice here a difference in the mode of referring to observation when a theory is first established, and when it is afterwards to be confirmed and corrected. It

was remarked as a merit in the method of Hipparchus, and an evidence of the mathematical coherence of his theory, that in order to determine the place of the sun's apogee, and the eccentricity of his orbit, he required to know nothing besides the lengths of winter and spring. But if the fewness of the requisite data is a beauty in the first fixation of a theory, the multitude of observations to which it applies is its excellence when it is established; and in correcting Tables, mathematicians take far more data than would be requisite to determine the elements. For the theory ought to account for _all_ the facts: and since it will not do this with mathematical rigor (for observation is not perfect), the elements are determined, not so as to satisfy any selected observations, but so as to make the whole mass of error as small as possible. And thus, in the adaptation of theory to observation, even in its most advanced state, there is room for sagacity and skill, prudence and judgment.

In this manner, by selecting the best mean elements of the motions of the heavenly bodies, the observed motions deviate from this mean in the way the theory points out, and constantly return to it. To this general rule, of the constant return to a mean, there are, however, some apparent exceptions, of which we shall now speak. {444}

_Sect._ 4.--_Application of the Newtonian Theory to Secular Inequalities._

SECULAR Inequalities in the motions of the heavenly bodies occur in consequence of changes in the elements of the solar system, which go on progressively from age to age. The example of such changes which was first studied by astronomers, was the Acceleration of the Moon's Mean Motion, discovered by Halley. The observed fact was, that the moon now moves in a very small degree quicker than she did in the earlier ages of the world. When this was ascertained, the various hypotheses which appeared likely to account for the fact were reduced to calculation. The resistance of the medium in which the heavenly bodies move was the most obvious of these hypotheses. Another, which was for some time dwelt upon by Laplace, was the successive transmission of gravity, that is, the hypothesis that the gravity of the earth takes a certain finite time to reach the moon. But none of these suppositions gave satisfactory conclusions; and the strength of Euler, D'Alembert, Lagrange, and Laplace, was for a time foiled by this difficulty. At length, in 1787, Laplace announced to the Academy that he had discovered the true cause of this acceleration, and that it arose from the action of the sun upon the moon, combined with the secular variation of the eccentricity of the earth's orbit. It was found that the effects of this combination would exactly account for the changes which had hitherto so perplexed mathematicians. A very remarkable result of this investigation was, that "this Secular Inequality of the motion of the moon is periodical, but it requires millions of years to re-establish itself;" so that after an almost inconceivable time, the acceleration will become a retardation. Laplace some time after (in 1797), announced other discoveries, relative to the secular motions of the apogee and the nodes of the moon's orbit. Laplace collected these researches in his "Theory of the Moon," which he published in the third volume of the _Mécanique Céleste_ in 1802.

A similar case occurred with regard to an acceleration of Jupiter's mean motion, and a retardation of Saturn's, which had been observed by Cassini, Maraldi, and Horrox. After several imperfect attempts by other mathematicians, Laplace, in 1787,

found that there resulted from the mutual attraction of these two planets a great Inequality, of which the period is 929 years and a half, and which has accelerated Jupiter and retarded Saturn ever since the restoration of astronomy. {445}

Thus the secular inequalities of the celestial motions, like all the others, confirm the law of universal gravitation. They are called "secular," because ages are requisite to unfold their existence, and because they are not obviously periodical. They might, in some measure, be considered as extensions of the Newtonian theory, for though Newton's law accounts for such facts, he did not, so far as we know, foresee such a result of it. But on the other hand, they are exactly of the same nature as those which he did foresee and calculate. And when we call them _secular_ in opposition to _periodical_, it is not that there is any real difference, for they, too, have their cycle; but it is that we have assumed our _mean_ motion without allowing for these long inequalities. And thus, as Laplace observes on this very occasion,[92\7] the lot of this great discovery of gravitation is no less than this, that every apparent exception becomes a proof, every difficulty a new occasion of a triumph. And such, as he truly adds, is the character of a true theory,--of a real representation of nature.

[Note 92\7: _Syst. du Monde_, 8vo, ii. 37.]

It is impossible for us here to enumerate even the principal objects which have thus filled the triumphal march of the Newtonian theory from its outset up to the present time. But among these secular changes, we may mention the Diminution of the Obliquity of the Ecliptic, which has been going on from the earliest times to the present. This change has been explained by theory, and shown to have, like all the other changes of the system, a limit, after which the diminution will be converted into an increase.

We may mention here some subjects of a kind somewhat different from those just spoken of. The true theoretical quantity of the Precession of the Equinoxes, which had been erroneously calculated by Newton, was shown by D'Alembert to agree with observation. The constant coincidence of the Nodes of the Moon's Equator with those of her Orbit, was proved to result from mechanical principles by Lagrange. The curious circumstance that the Time of the Moon's rotation on her axis is equal to the Time of her revolution about the earth, was shown to be consistent with the results of the laws of motion by Laplace. Laplace also, as we have seen, explained certain remarkable relations which constantly connect the longitudes of the three first satellites of Jupiter; Bailly and Lagrange analyzed and explained the curious librations of the nodes and inclinations of their orbits; and Laplace traced the effect of Jupiter's oblate figure on their motions, {446} which masks the other causes of inequality, by determining the direction of the motions of the _perijove_ and node of each satellite.

_Sect._ 5.--_Application of the Newtonian Theory to the New Planets._

WE are now so accustomed to consider the Newtonian theory as true, that we can hardly imagine to ourselves the possibility that those planets which were not discovered when the theory was founded, should contradict its doctrines. We can scarcely conceive it possible that Uranus or Ceres should have been found to violate

Kepler's laws, or to move without suffering perturbations from Jupiter and Saturn. Yet if we can suppose men to have had any doubt of the exact and universal truth of the doctrine of universal gravitation, at the period of these discoveries, they must have scrutinized the motions of these new bodies with an interest far more lively than that with which we now look for the predicted return of a comet. The solid establishment of the Newtonian theory is thus shown by the manner in which we take it for granted not only in our reasonings, but in our feelings. But though this is so, a short notice of the process by which the new planets were brought within the domain of the theory may properly find a place here.

William Herschel, a man of great energy and ingenuity, who had made material improvements in reflecting telescopes, observing at Bath on the 13th of March, 1781, discovered, in the constellation Gemini, a star larger and less luminous than the fixed stars. On the application of a more powerful telescope, it was seen magnified, and two days afterwards he perceived that it had changed its place. The attention of the astronomical world was directed to this new object, and the best astronomers in every part of Europe employed themselves in following it along the sky.[93\7]

[Note 93\7: Voiron, _Hist. Ast._ p. 12.]

The admission of an eighth planet into the long-established list, was a notion so foreign to men's thoughts at that time, that other suppositions were first tried. The orbit of the new body was at first calculated as if it had been a comet running in a parabolic path. But in a few days the star deviated from the course thus assigned it: and it was in vain that in order to represent the observations, the perihelion distance of the parabola was increased from fourteen to eighteen times the earth's distance from the sun. Saron, of the Academy of Sciences of Paris, is said[94\7] to have been the first person who perceived that the {447} places were better represented by a circle than by a parabola: and Lexell, a celebrated mathematician of Petersburg, found that a motion in a circular orbit, with a radius double of that of Saturn, would satisfy all the observations. This made its period about eighty-two years.

[Note 94\7: Ibid.]

Lalande soon discovered that the circular motion was subject to a sensible inequality: the orbit was, in fact, an ellipse, like those of the other planets. To determine the equation of the centre of a body which revolves so slowly, would, according to the ancient methods, have required many years; but Laplace contrived methods by which the elliptical elements were determined from four observations, within little more than a year from its first discovery by Herschel. These calculations were soon followed by tables of the new planet, published by Nouet.

In order to obtain additional accuracy, it now became necessary to take account of the perturbations. The French Academy of Sciences proposed, in 1789, the construction of new Tables of this Planet as its prize-question. It is a curious illustration of the constantly accumulating evidence of the theory, that the calculation of the perturbations of the planet enabled astronomers to discover that it had been observed as a star in three different positions in former times; namely, by Flamsteed in 1690, by Mayer in 1756, and by Le Monnier in 1769. Delambre, aided by this

discovery and by the theory of Laplace, calculated Tables of the planet, which, being compared with observation for three years, never deviated from it more than seven seconds. The Academy awarded its prize to these Tables, they were adopted by the astronomers of Europe, and the planet of Herschel now conforms to the laws of attraction, along with those ancient members of the known system from which the theory was inferred.

The history of the discovery of the other new planets, Ceres, Pallas, Juno, and Vesta, is nearly similar to that just related, except that their planetary character was more readily believed. The first of these was discovered on the first day of this century by Piazzi, the astronomer at Palermo; but he had only begun to suspect its nature, and had not completed his third observation, when his labors were suspended by a dangerous illness; and on his recovery the star was invisible, being lost in the rays of the sun.

He declared it to be a planet with an elliptical orbit; but the path which it followed, on emerging from the neighborhood of the sun, was not that which Piazzi had traced out for it. Its extreme smallness made it difficult to rediscover; and the whole of the year 1801 was {448} employed in searching the sky for it in vain. At last, after many trials, Von Zach and Olbers again found it, the one on the last day of 1801, the other on the first day of 1802. Gauss and Burckhardt immediately used the new observations in determining the elements of the orbit; and the former invented a new method for the purpose. Ceres now moves in a path of which the course and inequalities are known, and can no more escape the scrutiny of astronomers.

The second year of the nineteenth century also produced its planet. This was discovered by Dr. Olbers, a physician of Bremen, while he was searching for Ceres among the stars of the constellation Virgo. He found a star which had a perceptible motion even in the space of two hours. It was soon announced as a new planet, and received from its discoverer the name of Pallas. As in the case of Ceres, Burckhardt and Gauss employed themselves in calculating its orbit. But some peculiar difficulties here occurred. Its eccentricity is greater than that of any of the old planets, and the inclination of its orbit to the ecliptic is not less than thirty-five degrees. These circumstances both made its perturbations large, and rendered them difficult to calculate. Burckhardt employed the known processes of analysis, but they were found insufficient: and the Imperial Institute (as the French Academy was termed during the reign of Napoleon) proposed the Perturbations of Pallas as a prize-question.

To these discoveries succeeded others of the same kind. The German astronomers agreed to examine the whole of the zone in which Ceres and Pallas move; in the hope of finding other planets, fragments, as Olbers conceived they might possibly be, of one original mass. In the course of this research, Mr. Harding of Lilienthal, on the first of September, 1804, found a new star, which he soon was led to consider as a planet. Gauss and Burckhardt also calculated the elements of this orbit, and the planet was named Juno.

After this discovery, Olbers sought the sky for additional fragments of his planet with extraordinary perseverance. He conceived that one of two opposite

constellations, the Virgin or the Whale, was the place where its separation must have taken place; and where, therefore, all the orbits of all the portions must pass. He resolved to survey, three times a year, all the small stars in these two regions. This undertaking, so curious in its nature, was successful. The 29th of March, 1807, he discovered Vesta, which was soon found to be a planet. And to show the manner in which Olbers pursued his labors, we may state that he afterwards published a notification that he had examined the {449} same parts of the heavens with such regularity, that he was certain no new planet had passed that way between 1808 and 1816. Gauss and Burckhardt computed the orbit of Vesta; and when Gauss compared one of his orbits with twenty-two observations of M. Bouvard, he found the errors below seventeen seconds of space in right ascension, and still less in declination.

The elements of all these orbits have been successively improved, and this has been done entirely by the German mathematicians.[95\7] These perturbations are calculated, and the places for some time before and after opposition are now given in the Berlin Ephemeris. "I have lately observed," says Professor Airy, "and compared with the Berlin Ephemeris, the right ascensions of Juno and Vesta, and I find that they are rather more accurate than those of Venus:" so complete is the confirmation of the theory by these new bodies; so exact are the methods of tracing the theory to its consequences.

[Note 95\7: Airy, _Rep._ 157.]

We may observe that all these new-discovered bodies have received names taken from the ancient mythology. In the case of the first of these, astronomers were originally divided; the discoverer himself named it the _Georgium Sidus_, in honor of his patron, George the Third; Lalande and others called it _Herschel_. Nothing can be more just than this mode of perpetuating the fame of the author of a discovery; but it was felt to be ungraceful to violate the homogeneity of the ancient system of names. Astronomers tried to find for the hitherto neglected denizen of the skies, an appropriate place among the deities to whose assembly he was at last admitted; and _Uranus_, the father of Saturn, was fixed upon as best suiting the order of the course.

The mythological nomenclature of planets appeared, from this time, to be generally agreed to. Piazzi termed his _Ceres Ferdinandea_. The first term, which contains a happy allusion to Sicily, the country of the discovery in modern, and of the goddess in ancient, times, has been accepted; the attempt to pay a compliment to royalty out of the products of science, in this as in most other cases, has been set aside. Pallas, Juno, and Vesta, were named, without any peculiar propriety of selection, according to the choice of their discoverers.

_Sect._ 6.--_Application of the Newtonian Theory to Comets._

A FEW words must be said upon another class of bodies, which at first seemed as lawless as the clouds and winds; and which astronomy {450} has reduced to a regularity as complete as that of the sun;--upon _Comets_. No part of the Newtonian discoveries excited a more intense interest than this. These anomalous visitants were anciently gazed at with wonder and alarm; and might still, as in former times, be accused of "perplexing nations," though with very different fears and questionings.

The conjecture that they, too, obeyed the law of universal gravitation, was to be verified by showing that they described a curve such as that force would produce. Hevelius, who was a most diligent observer of these objects, had, without reference to gravitation, satisfied himself that they moved in parabolas.[96\7] To determine the elements of the parabola from observations, even Newton called[97\7] "problema longe difficillimum." Newton determined the orbit of the comet of 1680 by certain graphical methods. His methods supposed the orbit to be a parabola, and satisfactorily represented the motion in the visible part of the comet's path. But this method did not apply to the possible return of the wandering star. Halley has the glory of having first detected a periodical comet, in the case of that which has since borne his name. But this great discovery was not made without labor. In 1705, Halley[98\7] explained how the parabolic orbit of a planet may be determined from three observations; and, joining example to precept, himself calculated the positions and orbits of twenty-four comets. He found, as the reward of this industry, that the comets of 1607 and of 1531 had the same orbit as that of 1682. And here the intervals are also nearly the same, namely, about seventy-five years. Are the three comets then identical? In looking back into the history of such appearances, he found comets recorded in 1456, in 1380, and in 1305; the intervals are still the same, seventy-five or seventy-six years. It was impossible now to doubt that they were the periods of a revolving body; that the comet was a planet; its orbit a long ellipse, not a parabola.[99\7]

[Note 96\7: Bailly, ii. 246.]

[Note 97\7: _Principia_, ed. 1. p. 494.]

[Note 98\7: Bailly, ii. 646.]

[Note 99\7: The importance of Halley's labors on Comets has always been acknowledged. In speaking of Halley's _Synopsis Astronomicæ Cometicæ_, Delambre says (_Ast._ xviii. _Siècle_, p. 130), "Voilà bien, depuis Kepler, ce qu'on a fait de plus grand, de plus beau, de plus neuf en astronomie." Halley, in predicting the comet of 1758, says, if it returns, "Hoc primum ab homine Anglo iuventum fuisse non inficiabitur æqua posteritas."]

But if this were so, the Comet must reappear in 1758 or 1759. Halley predicted that it would do so; and the fulfilment of this prediction was naturally looked forwards to, as an additional stamp of the truths of the theory of gravitation. {451}

But in all this, the Comet had been supposed to be affected only by the attraction of the sun. The planets must disturb its motion as they disturb each other. How would this disturbance affect the time and circumstances of its reappearance? Halley had proposed, but not attempted to solve, this question.

The effect of perturbations upon a comet defeats all known methods of approximation, and requires immense labor. "Clairaut," says Bailly,[100\7] "undertook this: with courage enough to dare the adventure, he had talent enough to obtain a memorable victory;" the difficulties, the labors, grew upon him as he advanced, but he fought his way through them, assisted by Lalande, and by a female calculator, Madame Lepaute. He predicted that the comet would reach its perihelion

April 13, 1759, but claimed the license of a month for the inevitable inaccuracy of a calculation which, in addition to all other sources of error, was made in haste, that it might appear as a prediction. The comet justified his calculations and his caution together; for it arrived at its perihelion on the 13th of March.

[Note 100\7: Bailly, _A. M._ iii. 190.]

Two other Comets, of much shorter period, have been detected of late years; Encke's, which revolves round the sun in three years and one-third, and Biela's which describes an ellipse, not extremely eccentric, in six years and three-quarters. These bodies, apparently thin and vaporous masses, like other comets, have, since their orbits were calculated, punctually conformed to the law of gravitation. If it were still doubtful whether the more conspicuous comets do so, these bodies would tend to prove the fact, by showing it to be true in an intermediate case.

[2d Ed.] [A third Comet of short period was discovered by Faye, at the Observatory of Paris, Nov. 22, 1843. It is included between the orbits of Mars and Saturn, and its period is seven years and three-tenths.

This is commonly called _Faye's Comet_, as the two mentioned in the text are called _Encke's_ and _Biela's_. In the former edition I had expressed my assent to the rule proposed by M. Arago, that the latter ought to be called _Gambart's Comet_, in honor of the astronomer who first proved it to revolve round the Sun. But astronomers in general have used the former name, considering that the discovery and observation of the object are more distinct and conspicuous merits than a calculation founded upon the observations of others. And in reality {452} Biela had great merit in the discovery of his Comet's periodicity, having set about his search of it from an anticipation of its return founded upon former observations.

Also a Comet was discovered by De Vico at Rome on Aug. 22, 1844, which was found to describe an elliptical orbit having its aphelion near the orbit of Jupiter, which is consequently one of those of short period. And on Feb. 26, 1846, M. Brorsen of Kiel discovered a telescopic Comet whose orbit is found to be elliptical.]

We may add to the history of Comets, that of Lexell's, which, in 1770, appeared to be revolving in a period of about five years, and whose motion was predicted accordingly. The prediction was disappointed; but the failure was sufficiently explained by the comet's having passed close to Jupiter, by which occurrence its orbit was utterly deranged.

It results from the theory of universal gravitation, that Comets are collections of extremely attenuated matter. Lexell's is supposed to have passed twice (in 1767 and 1779) through the system of Jupiter's Satellites, without disturbing their motions, though suffering itself so great a disturbance as to have its orbit entirely altered. The same result is still more decidedly proved by the last appearance of Biela's Comet. It appeared double, but the two bodies did not perceptibly affect each other's motions, as I am informed by Professor Challis of Cambridge, who observed both of them from Jan. 23 to Mar. 25, 1846. This proves the quantity of matter in each body to have been exceedingly small.

Thus, no verification of the Newtonian theory, which was possible in the motions of the stars, has yet been wanting. The return of Halley's Comet again in 1835, and the extreme exactitude with which it conformed to its predicted course, is a testimony of truth, which must appear striking even to the most incurious respecting such matters.[101\7]

[Note 101\7: M. de Humboldt (_Kosmos_, p. 116) speaks of _nine_ returns of Halley's Comet, the comet observed in China in 1378 being identified with this. But whether we take 1378 or 1380 for the appearance in that century, if we begin with that, we have only _seven_ appearances, namely, in 1378 or 1380, in 1456, in 1531, in 1607, in 1682, in 1759, and in 1835.]

_Sect._ 7.--_Application of the Newtonian Theory to the Figure of the Earth._

THE Heavens had thus been consulted respecting the Newtonian doctrine, and the answer given, over and over again, in a thousand {453} different forms, had been, that it was true; nor had the most persevering cross-examination been able to establish any thing of contradiction or prevarication. The same question was also to be put to the Earth and the Ocean, and we must briefly notice the result.

According to the Newtonian principles, the form of the earth must be a globe somewhat flattened at the poles. This conclusion, or at least the amount of the flattening, depends not only upon the existence and law of attraction, but upon its belonging to each particle of the mass separately; and thus the experimental confirmation of the form asserted from calculation, would be a verification of the theory in its widest sense. The application of such a test was the more necessary to the interests of science, inasmuch as the French astronomers had collected from their measures, and had connected with their Cartesian system, the opinion that the earth was not _oblate_ but _oblong_. Dominic Cassini had measured seven degrees of latitude from Amiens to Perpignan, in 1701, and found them to decrease in going from south to north. The prolongation of this measure to Dunkirk confirmed the same result. But if the Newtonian doctrine was true, the contrary ought to be the case, and the degrees ought to increase in proceeding towards the pole.

The only answer which the Newtonians could at this time make to the difficulty thus presented, was, that an arc so short as that thus measured, was not to be depended upon for the determination of such a question; inasmuch as the inevitable errors of observation might exceed the differences which were the object of research. It would, undoubtedly, have become the English to have given a more complete answer, by executing measurements under circumstances not liable to this uncertainty. The glory of doing this, however, they for a long time abandoned to other nations. The French undertook the task with great spirit.[102\7] In 1733, in one of the meetings of the French Academy, when this question was discussed, De la Condamine, an ardent and eager man, proposed to settle this question by sending members of the Academy to measure a degree of the meridian near the equator, in order to compare it with the French degrees, and offered himself for the expedition. Maupertuis, in like manner, urged the necessity of another expedition to measure a degree in the neighborhood of the pole. The government received the applications

favorably, and these remarkable scientific missions were sent out at the national expense. {454}

[Note 102\7: Bailly, iii. 11.]

As soon as the result of these measurements was known, there was no longer any doubt as to the fact of the earth's oblateness, and the question only turned upon its quantity. Even before the return of the academicians, the Cassinis and Lacaille had measured the French arc, and found errors which subverted the former result, making the earth oblate to the amount of 1/168th of its diameter. The expeditions to Peru and to Lapland had to struggle with difficulties in the execution of their design, which make their narratives resemble some romantic history of irregular warfare, rather than the monotonous records of mere measurements. The equatorial degree employed the observers not less than eight years. When they did return, and the results were compared, their discrepancy, as to quantity, was considerable. The comparison of the Peruvian and French arcs gave an ellipticity of nearly 1/314th, that of the Peruvian and Swedish arcs gave 1/213th for its value.

Newton had deduced from his theory, by reasonings of singular ingenuity, an ellipticity of 1/230th; but this result had been obtained by supposing the earth homogeneous. If the earth be, as we should most readily conjecture it to be, more dense in its interior than at its exterior, its ellipticity will be less than that of a homogeneous spheroid revolving in the same time. It does not appear that Newton was aware of this; but Clairaut, in 1743, in his _Figure of the Earth_, proved this and many other important results of the attraction of the particles. Especially he established that, in proportion as the fraction expressing the Ellipticity becomes smaller, that expressing the Excess of the polar over the equatorial gravity becomes larger; and he thus connected the measures of the ellipticity obtained by means of Degrees, with those obtained by means of Pendulums in different latitudes.

The altered rate of a Pendulum when carried towards the equator, had been long ago observed by Richer and Halley, and had been quoted by Newton as confirmatory of his theory. Pendulums were swung by the academicians who measured the degrees, and confirmed the general character of the results.

But having reached this point of the verification of the Newtonian theory, any additional step becomes more difficult. Many excellent measures, both of Degrees and of Pendulums, have been made since those just mentioned. The results of the Arcs[103\7] is an Ellipticity of 1/298th;--of the Pendulums, an Ellipticity of about 1/285th. This difference {455} is considerable, if compared with the quantities themselves; but does not throw a shadow of doubt on the truth of the theory. Indeed, the observations of each kind exhibit irregularities which we may easily account for, by ascribing them to the unknown distribution of the denser portions of the earth; but which preclude the extreme of accuracy and certainty in our result.

[Note 103\7: Airy, _Fig. Earth_, p. 230.]

But the near agreement of the determination, from Degrees and from Pendulums, is not the only coincidence by which the doctrine is confirmed. We can trace the effect of the earth's Oblateness in certain minute apparent motions of the stars; for

the attraction of the sun and moon on the protuberant matter of the spheroid produces the Precession of the equinoxes, and a Nutation of the earth's axis. The Precession had been known from the time of Hipparchus, and the existence of Nutation was foreseen by Newton; but the quantity is so small, that it required consummate skill and great labor in Bradley to detect it by astronomical observation. Being, however, so detected, its amount, as well as that of the Precession, gives us the means of determining the amount of Terrestrial Ellipticity, by which the effect is produced. But it is found, upon calculation, that we cannot obtain this determination without assuming some law of density in the homogeneous strata of which we suppose the earth to consist[104\7] The density will certainly increase in proceeding towards the centre, and there is a simple and probable law of this increase, which will give 1⁄300th for the Ellipticity, from the amount of two lunar Inequalities (one in latitude and one in longitude), which are produced by the earth's oblateness. Nearly the same result follows from the quantity of Nutation. Thus every thing tends to convince us that the ellipticity cannot deviate much from this fraction.

[Note 104\7: Airy, _Fig. Earth_, p. 235.]

[2d Ed.] [I ought not to omit another class of phenomena in which the effects of the Earth's Oblateness, acting according to the law of universal gravitation, have manifested themselves;--I speak of the Moon's Motion, as affected by the Earth's Ellipticity. In this case, as in most others, observation anticipated theory. Mason had inferred from lunar observations a certain Inequality in Longitude, depending upon the distance of the Moon's Node from the Equinox. Doubts were entertained by astronomers whether this inequality really existed; but Laplace showed that such an inequality would arise from the oblate form of the earth; and that its magnitude might serve to {456} determine the amount of the oblateness. Laplace showed, at the same time, that along with this Inequality in Longitude there must be an Inequality in Latitude; and this assertion Burg confirmed by the discussion of observations. The two Inequalities, as shown in the observations, agree in assigning to the earth's form an Ellipticity of 1⁄305th.]

_Sect._ 8.--_Confirmation of the Newtonian Theory by Experiments on Attraction._

THE attraction of all the parts of the earth to one another was thus proved by experiments, in which the whole mass of the earth is concerned. But attempts have also been made to measure the attraction of smaller portions; as mountains, or artificial masses. This is an experiment of great difficulty; for the attraction of such masses must be compared with that of the earth, of which it is a scarcely perceptible fraction; and, moreover, in the case of mountains, the effect of the mountain will be modified or disguised by unknown or unappreciable circumstances. In many of the measurements of degrees, indications of the attraction of mountains had been perceived; but at the suggestion of Maskelyne, the experiment was carefully made, in 1774, upon the mountain Schehallien, in Scotland, the mountain being mineralogically surveyed by Playfair. The result obtained was, that the attraction of the mountain drew the plumb-line about six seconds from the vertical; and it was deduced from this, by Hutton's calculations, that the density of the earth was about once and four-fifths that of Schehallien, or four and a half times that of water.

Cavendish, who had suggested many of the artifices in this calculation, himself made the experiment in the other form, by using leaden balls, about nine inches diameter. This observation was conducted with an extreme degree of ingenuity and delicacy, which could alone make it valuable; and the result agreed very nearly with that of the Schehallien experiment, giving for the density of the earth about five and one-third times that of water. Nearly the same result was obtained by Carlini, in 1824, from observations of the pendulum, made at a point of the Alps (the Hospice, on Mount Cenis) at a considerable elevation above the average surface of the earth. {457}

_Sect._ 9.--_Application of the Newtonian Theory to the Tides._

WE come, finally, to that result, in which most remains to be done for the verification of the general law of attraction--the subject of the Tides. Yet, even here, the verification is striking, as far as observations have been carried. Newton's theory explained, with singular felicity, all the prominent circumstances of the tides then known;--the difference of spring and neap tides; the effect of the moon's and sun's declination and parallax; even the difference of morning and evening tides, and the anomalous tides of particular places. About, and after, this time, attempts were made both by the Royal Society of England, and by the French Academy, to collect numerous observations but these were not followed up with sufficient perseverance. Perhaps, indeed, the theory had not been at that time sufficiently developed but the admirable prize-essays of Euler, Bernoulli, and D'Alembert, in 1740, removed, in a great measure, this deficiency. These dissertations supplied the means of bringing this subject to the same test to which all the other consequences of gravitation had been subjected;--namely, the calculation of tables, and the continued and orderly comparison of these with observation. Laplace has attempted this verification in another way, by calculating the results of the theory (which he has done with an extraordinary command of analysis), and then by comparing these, in supposed critical cases, with the Brest observations. This method has confirmed the theory as far as it could do so; but such a process cannot supersede the necessity of applying the proper criterion of truth in such cases, the construction and verification of Tables. Bernoulli's theory, on the other hand, has been used for the construction of Tide-tables; but these have not been properly compared with experiment; and when the comparison has been made, having been executed for purposes of gain rather than of science, it has not been published, and cannot be quoted as a verification of the theory.

Thus we have, as yet, no sufficient comparison of fact with theory, for Laplace's is far from a complete comparison. In this, as in other parts of physical astronomy, our theory ought not only to agree with observations selected and grouped in a particular manner, but with the whole course of observation, and with every part of the phenomena. In this, as in other cases, the true theory should be verified by its giving us the best Tables; but Tide-tables were never, I believe, {458} calculated upon Laplace's theory, and thus it was never fairly brought to the test.

It is, perhaps, remarkable, considering all the experience which astronomy had furnished, that men should have expected to reach the completion of this branch of

science by improving the mathematical theory, without, at the same time, ascertaining the laws of the facts. In all other departments of astronomy, as, for instance, in the cases of the moon and the planets, the leading features of the phenomena had been made out empirically, before the theory explained them. The course which analogy would have recommended for the cultivation of our knowledge of the tides, would have been, to ascertain, by an analysis of long series of observations, the effect of changes in the time of transit, parallax, and declination of the moon, and thus to obtain the laws of phenomena and then proceed to investigate the laws of causation.

Though this was not the course followed by mathematical theorists, it was really pursued by those who practically calculated Tide-tables; and the application of knowledge to the useful purposes of life being thus separated from the promotion of the theory, was naturally treated as a gainful property, and preserved by secrecy. Art, in this instance, having cast off her legitimate subordination to Science, or rather, being deprived of the guidance which it was the duty of Science to afford, resumed her ancient practices of exclusiveness and mystery. Liverpool, London, and other places, had their Tide-tables, constructed by undivulged methods, which methods, in some instances at least, were handed down from father to son for several generations as a family possession; and the publication of new Tables, accompanied by a statement of the mode of calculation, was resented as an infringement of the rights of property.

The mode in which these secret methods were invented, was that which we have pointed out;--the analysis of a considerable series of observations. Probably the best example of this was afforded by the Liverpool Tide-tables. These were deduced by a clergyman named Holden, from observations made at that port by a harbor-master of the name of Hutchinson; who was led, by a love of such pursuits, to observe the tides carefully for above twenty years, day and night. Holden's Tables, founded on four years of these observations, were remarkably accurate.

At length men of science began to perceive that such calculations were part of their business; and that they were called upon, as the {459} guardians of the established theory of the universe, to compare it in the greatest possible detail with the facts. Mr. Lubbock was the first mathematician who undertook the extensive labors which such a conviction suggested. Finding that regular tide-observations had been made at the London Docks from 1795, he took nineteen years of these (purposely selecting the length of a cycle of the motions of the lunar orbit), and caused them (in 1831) to be analyzed by Mr. Dessiou, an expert calculator. He thus obtained[105\7] Tables for the effect of the Moon's Declination, Parallax, and hour of Transit, on the tides; and was enabled to produce Tide-tables founded upon the data thus obtained. Some mistakes in these as first published (mistakes unimportant as to the theoretical value of the work), served to show the jealousy of the practical tide-table calculators, by the acrimony with which the oversights were dwelt upon; but in a very few years, the tables thus produced by an open and scientific process were more exact than those which resulted from any of the secrets; and thus practice was brought into its proper subordination to theory.

[Note 105\7: _Phil. Trans._ 1831. _British Almanac_, 1832.]

The theory with which Mr. Lubbock was led to compare his results, was the Equilibrium-theory of Daniel Bernoulli; and it was found that this theory, with certain modifications of its elements, represented the facts to a remarkable degree of precision. Mr. Lubbock pointed out this agreement especially in the semi-mensual inequality of the times of high water. The like agreement was afterwards (in 1833) shown by Mr. Whewell[106\7] to obtain still more accurately at Liverpool, both for the Times and Heights; for by this time, nineteen years of Hutchinson's Liverpool Observations had also been discussed by Mr. Lubbock. The other inequalities of the Times and Heights (depending upon the Declination and Parallax of the Moon and Sun,) were variously compared with the Equilibrium-theory by Mr. Lubbock and Mr. Whewell; and the general result was, that the facts agreed with the condition of equilibrium at a certain anterior time, but that this anterior time was different for different phenomena. In like manner it appeared to follow from these researches, that in order to explain the facts, the mass of the moon must be supposed different in the calculation at different places. A result in effect the same was obtained by M. Daussy,[107\7] an active French Hydrographer; for he found that observations at various stations could not be reconciled with the formulæ of Laplace's _Mécanique_ {460} _Céleste_ (in which the ratio of the heights of spring-tides and neap-tides was computed on an assumed mass of the moon) without an alteration of level which was, in fact, equivalent to an alteration of the moon's mass. Thus all things appeared to tend to show that the Equilibrium-theory would give the _formulæ_ for the inequalities of the tides, but that the _magnitudes_ which enter into these formulæ must be sought from observation.

[Note 106\7: _Phil. Trans._ 1834.]

[Note 107\7: _Connaissance des Tems_, 1838.]

Whether this result is consistent with theory, is a question not so much of Physical Astronomy as of Hydrodynamics, and has not yet been solved. A Theory of the Tides which should include in its conditions the phenomena of Derivative Tides, and of their combinations, will probably require all the resources of the mathematical mechanician.

As a contribution of empirical materials to the treatment of this hydrodynamical problem, it may be allowable to mention here Mr. Whewell's attempts to trace the progress of the tide into all the seas of the globe, by drawing on maps of the ocean what he calls _Cotidal Lines_;--lines marking the contemporaneous position of the various points of the great wave which carries high water from shore to shore.[108\7] This is necessarily a task of labor and difficulty, since it requires us to know the time of high water on the same day in every part of the world; but in proportion as it is completed, it supplies steps between our general view of the movements of the ocean and the phenomena of particular ports.

[Note 108\7: Essay towards a First Approximation to a Map of Cotidal Lines. _Phil. Trans._ 1833, 1836.]

Looking at this subject by the light which the example of the history of astronomy affords, we may venture to repeat, that it will never have justice done it till it is treated as other parts of astronomy are treated; that is, till Tables of all the phenomena which can be observed, are calculated by means of the best knowledge which we at present possess, and till these tables are constantly improved by a comparison of the predicted with the observed fact. A set of Tide-observations and Tide-ephemerides of this kind, would soon give to this subject that precision which marks the other parts of astronomy; and would leave an assemblage of unexplained _residual phenomena_, in which a careful research might find the materials of other truths as yet unsuspected.

[2d Ed.] [That there would be, in the tidal movements of the ocean, inequalities of the heights and times of high and low water {461} _corresponding_ to those which the equilibrium theory gives, could be considered only as a conjecture, till the comparison with observation was made. It was, however, a natural conjecture; since the waters of the ocean are at every moment _tending_ to acquire the form assumed in the equilibrium theory: and it may be considered likely that the causes which prevent their assuming this form produce an effect nearly constant for each place. Whatever be thought of this reasoning, the conjecture is confirmed by observation with curious exactness. The laws of a great number of the tidal phenomena--namely, of the Semi-mensual Inequality of the Heights, of the Semi-mensual Inequality of the Times, of the Diurnal Inequality, of the effect of the Moon's Declination, of the effect of the Moon's Parallax--are represented very closely by formulæ derived from the equilibrium theory. The hydrodynamical mode of treating the subject has not added any thing to the knowledge of the laws of the phenomena to which the other view had conducted us.

We may add, that Laplace's assumption, that in the moving fluid the motions must have a _periodicity_ corresponding to that of the forces, is also a conjecture. And though this conjecture may, in some cases of the problem, be verified, by substituting the resulting expressions in the equations of motion, this cannot be done in the actual case, where the revolving motion of the ocean is prevented by the intrusion of tracts of land running nearly from pole to pole.

Yet in Mr. Airy's Treatise _On Tides and Waves_ (in the _Encyclopædia Metropolitana_) much has been done to bring the hydrodynamical theory of oceanic tides into agreement with observation. In this admirable work, Mr. Airy has, by peculiar artifices, solved problems which come so near the actual cases that they may represent them. He has, in this way, deduced the laws of the semi-diurnal and the diurnal tide, and the other features of the tides which the equilibrium theory in some degree imitates; but he has also, taking into account the effect of friction, shown that the actual tide may be represented as the tide of an earlier epoch;--that the relative mass of the moon and sun, as inferred from the tides, would depend upon the depth of the ocean (Art. 455);--with many other results remarkably explaining the observed phenomena. He has also shown that the relation of the cotidal lines to the tide waves really propagated is, in complex cases, very obscure, because different waves of different magnitudes, travelling in different directions, may coexist, and the cotidal

line is the compound result of all these. {462}

With reference to the _Maps of Cotidal Lines_, mentioned in the text, I may add, that we are as yet destitute of observations which should supply the means of drawing such lines on a large scale in the Pacific Ocean. Admiral Lütke has however supplied us with some valuable materials and remarks on this subject in his _Notice sur les Marées Périodiques dans le grand Océan Boréal et dans la Mer Glaciale_; and has drawn them, apparently on sufficient data, in the White Sea.]

## CHAPTER V.

DISCOVERIES ADDED TO THE NEWTONIAN THEORY.

_Sect._ 1.--_Tables of Astronomical Refraction._

WE have travelled over an immense field of astronomical and mathematical labor in the last few pages, and have yet, at the end of every step, still found ourselves under the jurisdiction of the Newtonian laws. We are reminded of the universal monarchies, where a man could not escape from the empire without quitting the world. We have now to notice some other discoveries, in which this reference to the law of universal gravitation is less immediate and obvious; I mean the astronomical discoveries respecting Light.

The general truths to which the establishment of the true laws of Atmospheric Refraction led astronomers, were the law of Deflection of the rays of light, which applies to all refractions, and the real structure and size of the Atmosphere, so far as it became known. The great discoveries of Römer and Bradley, namely, the Velocity of Light, the Aberration of Light, and the Nutation of the earth's axis, gave a new distinctness to the conceptions of the propagation of light in the minds of philosophers, and confirmed the doctrines of Copernicus, Kepler, and Newton, respecting the motions which belong to the earth.

The true laws of Atmospheric Refraction were slowly discovered. Tycho attributed the apparent displacement of the heavenly bodies to the low and gross part of the atmosphere only, and hence made it cease at a point half-way to the zenith; but Kepler rightly extended it to the zenith itself. Dominic Cassini endeavored to discover the law of this correction by observation, and gave his result in the form {463} which, as we have said, sound science prescribes, a Table to be habitually used for all observations. But great difficulties at this time embarrassed this investigation, for the parallaxes of the sun and of the planets were unknown, and very diverse values had been assigned them by different astronomers. To remove some of these difficulties, Richer, in 1762, went to observe at the equator; and on his return, Cassini was able to confirm and amend his former estimations of parallax and refraction. But there were still difficulties. According to La Hire, though the phenomena of twilight give an altitude of 34,000 toises to the atmosphere,[109\7] those of refraction make it only 2000. John Cassini undertook to support and improve the calculations of his father Dominic, and took the true supposition, that the light follows a curvilinear path through the air. The Royal Society of London had already ascertained experimentally

the refractive power of air.[110\7] Newton calculated a Table of Refractions, which was published under Halley's name in the _**Philosophical Transactions_ for 1721, without any indication of the method by which it was constructed. But M. Biot has recently shown,[111\7] by means of the published correspondence of Flamsteed, that Newton had solved the problem in a manner nearly corresponding to the most improved methods of modern analysis.

[Note 109\7: Bailly, ii. 612.]

[Note 110\7: Ibid. ii. 607.]

[Note 111\7: Biot, _Acad. Sc. Compte Rendu_, Sept. 5, 1836.]

Dominic Cassini and Picard proved,[112\7] Le Monnier in 1738 confirmed more fully, the fact that the variations of the Thermometer affect the Refraction. Mayer, taking into account both these changes, and the changes indicated by the Barometer, formed a theory, which Lacaille, with immense labor, applied to the construction of a Table of Refractions from observation. But Bradley's Table (published in 1763 by Maskelyne) was more commonly adopted in England; and his formula, originally obtained empirically, has been shown by Young to result from the most probable suppositions we can make respecting the atmosphere. Bessel's Refraction Tables are now considered the best of those which have appeared.

[Note 112\7: Bailly, iii. 92.]

_Sect._ 2.--_Discovery of the Velocity of Light.--Römer._

THE astronomical history of Refraction is not marked by any great discoveries, and was, for the most part, a work of labor only. The progress of the other portions of our knowledge respecting light is {464} more striking. In 1676, a great number of observations of eclipses of Jupiter's satellites were accumulated, and could be compared with Cassini's Tables. Römer, a Danish astronomer, whom Picard had brought to Paris, perceived that these eclipses happened constantly later than the calculated time at one season of the year, and earlier at another season;--a difference for which astronomy could offer no account. The error was the same for all the satellites; if it had depended on a defect in the Tables of Jupiter, it might have affected all, but the effect would have had a reference to the velocities of the satellites. The cause, then, was something extraneous to Jupiter. Römer had the happy thought of comparing the error with the earth's distance from Jupiter, and it was found that the eclipses happened later in proportion as Jupiter was further off.[113\7] Thus we see the eclipse later, as it is more remote; and thus light, the messenger which brings us intelligence of the occurrence, travels over its course in a measurable time. By this evidence, light appeared to take about eleven minutes in describing the diameter of the earth's orbit.

[Note 113\7: Bailly, ii. 17.]

This discovery, like so many others, once made, appears easy and inevitable; yet Dominic Cassini had entertained the idea for a moment,[114\7] and had rejected it; and Fontenelle had congratulated himself publicly on having narrowly escaped this seductive error. The objections to the admission of the truth arose principally from

the inaccuracy of observation, and from the persuasion that the motions of the satellites were circular and uniform. Their irregularities disguised the fact in question. As these irregularities became clearly known, Römer's discovery was finally established, and the "Equation of Light" took its place in the Tables.

[Note 114\7: Ib. ii. 419.]

_Sect._ 3.--_Discovery of Aberration.--Bradley._

IMPROVEMENTS in instruments, and in the art of observing, were requisite for making the next great step in tracing the effect of the laws of light. It appears clear, on consideration, that since light and the spectator on the earth are both in motion, the apparent direction of an object will be determined by the composition of these motions. But yet the effect of this composition of motions was (as is usual in such cases) traced as a fact in observation, before it was clearly seen as a consequence of reasoning. This fact, the Aberration of Light, the greatest astronomical discovery of the eighteenth century, belongs to Bradley, {465} who was then Professor of Astronomy at Oxford, and afterwards Astronomer Royal at Greenwich. Molyneux and Bradley, in 1725, began a series of observations for the purpose of ascertaining, by observations near the zenith, the existence of an annual parallax of the fixed stars, which Hooke had hoped to detect, and Flamsteed thought he had discovered. Bradley[115\7] soon found that the star observed by him had a minute apparent motion different from that which the annual parallax would produce. He thought of a nutation of the earth's axis as a mode of accounting for this; but found, by comparison of a star on the other side of the pole, that this explanation would not apply. Bradley and Molyneux then considered for a moment an annual alteration of figure in the earth's atmosphere, such as might affect the refractions, but this hypothesis was soon rejected.[116\7] In 1727, Bradley resumed his observations, with a new instrument, at Wanstead, and obtained empirical rules for the changes of declination of different stars. At last, accident turned his thoughts to the direction in which he was to find the cause of the variations which he had discovered. Being in a boat on the Thames, he observed that the vane on the top of the mast gave a different apparent direction to the wind, as the boat sailed one way or the other. Here was an image of his case: the boat represented the earth moving in different directions at different seasons, and the wind represented the light of a star. He had now to trace the consequences of this idea; he found that it led to the empirical rules, which he had already discovered, and, in 1729, he gave his discovery to the Royal Society. His paper is a very happy narrative of his labors and his thoughts. His theory was so sound that no astronomer ever contested it; and his observations were so accurate, that the quantity which he assigned as the greatest amount of the change (one nineteenth of a degree) has hardly been corrected by more recent astronomers. It must be noticed, however, that he considered the effects in declination only; the effects in right ascension required a different mode of observation, and a consummate goodness in the machinery of clocks, which at that time was hardly attained.

[Note 115\7: Rigaud's Bradley.]

[Note 116\7: Rigaud, p. xxiii.]

_Sect._ 4.--_Discovery of Nutation._

WHEN Bradley went to Greenwich as Astronomer Royal, he continued with perseverance observations of the same kind as those by which he had detected Aberration. The result of these was another {466} discovery; namely, that very Nutation which he had formerly rejected. This may appear strange, but it is easily explained. The aberration is an annual change, and is detected by observing a star at different seasons of the year: the Nutation is a change of which the cycle is eighteen years; and which, therefore, though it does not much change the place of a star in one year, is discoverable in the alterations of several successive years. A very few years' observations showed Bradley the effect of this change;[117\7] and long before the half cycle of nine years had elapsed, he had connected it in his mind with the true cause, the motion of the moon's nodes. Machin was then Secretary to the Royal Society,[118\7] and was "employed in considering the theory of gravity, and its consequences with regard to the celestial motions:" to him Bradley communicated his conjectures; from him he soon received a Table containing the results of his calculations; and the law was found to be the same in the Table and in observation, though the quantities were somewhat different. It appeared by both, that the earth's pole, besides the motion which the precession of the equinoxes gives it, moves, in eighteen years, through a small circle;--or rather, as was afterwards found by Bradley, an ellipse, of which the axes are nineteen and fourteen seconds.[119\7]

[Note 117\7: Rigaud, lxiv.]

[Note 118\7: Ib. 25.]

[Note 119\7: Ib. lxvi.]

For the rigorous establishment of the mechanical theory of that effect of the moon's attraction from which the phenomena of Nutation flow, Bradley rightly and prudently invited the assistance of the great mathematicians of his time. D'Alembert, Thomas Simpson, Euler, and others, answered this call, and the result was, as we have already said in the last chapter (Sect. 7), that this investigation added another to the recondite and profound evidences of the doctrine of universal gravitation.

It has been said[120\7] that Bradley's discoveries "assure him the most distinguished place among astronomers after Hipparchus and Kepler." If his discoveries had been made before Newton's, there could have been no hesitation as to placing him on a level with those great men. The existence of such suggestions as the Newtonian theory offered on all astronomical subjects, may perhaps dim, in our eyes, the brilliance of Bradley's achievements; but this circumstance cannot place any other person above the author of such discoveries, and therefore we may consider Delambre's adjudication of precedence as well warranted, and deserving to be permanent. {467}

[Note 120\7: Delambre, _Ast. du_ 18 _Sièc._ p. 420. Rigaud, xxxvii.]

_Sect._ 5.--_Discovery of the Laws of Double Stars.--The two Herschels._

NO truth, then, can be more certainly established, than that the law of gravitation prevails to the very boundaries of the solar system. But does it hold good further? Do

the fixed stars also obey this universal sway? The idea, the question, is an obvious one--but where are we to find the means of submitting it to the test of observation?

If the Stars were each insulated from the rest, as our Sun appears to be from them, we should have been quite unable to answer this inquiry. But among the stars, there are some which are called _Double Stars_, and which consist of two stars, so near to each other that the telescope alone can separate them. The elder Herschel diligently observed and measured the relative positions of the two stars in such pairs; and as has so often happened in astronomical history, pursuing one object he fell in with another. Supposing such pairs to be really unconnected, he wished to learn, from their phenomena, something respecting the annual parallax of the earth's orbit. But in the course of twenty years' observations he made the discovery (in 1803) that some of these couples were turning round each other with various angular velocities. These revolutions were for the most part so slow that he was obliged to leave their complete determination as an inheritance to the next generation. His son was not careless of the bequest, and after having added an enormous mass of observations to those of his father, he applied himself to determine the laws of these revolutions. A problem so obvious and so tempting was attacked also by others, as Savary and Encke, in 1830 and 1832, with the resources of analysis. But a problem in which the data are so minute and inevitably imperfect, required the mathematician to employ much judgment, as well as skill in using and combining these data; and Sir John Herschel, by employing positions only of the line joining the pair of stars (which can be observed with comparative exactness), to the exclusion of their distances (which cannot be measured with much correctness), and by inventing a method which depended upon the whole body of observations, and not upon selected ones only, for the determination of the motion, has made his investigations by far the most satisfactory of those which have appeared. The result is, that it has been rendered very probable, that in several of the double stars the two stars describe ellipses about each other; and therefore that here also, at an {468} immeasurable distance from our system, the law of attraction according to the inverse square of the distance, prevails. And, according to the practice of astronomers when a law has been established, Tables have been calculated for the future motions; and we have Ephemerides of the revolutions of suns round each other, in a region so remote, that the whole circle of our earth's orbit, if placed there, would be imperceptible by our strongest telescopes. The permanent comparison of the observed with the predicted motions, continued for more than one revolution, is the severe and decisive test of the truth of the theory; and the result of this test astronomers are now awaiting.

[2d Ed.] [In calculating the orbits of revolving systems of double stars, there is a peculiar difficulty, arising from the plane of the orbit being in a position unknown, but probably oblique, to the visual ray. Hence it comes to pass that even if the orbit be an ellipse described about the focus by the laws of planetary motion, it will appear otherwise; and the true orbit will have to be deduced from the apparent one.

With regard to a difficulty which has been mentioned, that the two stars, if they are governed by gravity, will not revolve the one about the other, but both about their common centre of gravity;--this circumstance adds little difficulty to the problem.

Newton has shown (_Princip._ lib. i. Prop. 61) in the _problem of two bodies_, the relation between the relative orbits and the orbit about the common centre of gravity.

_How many of the apparently double stars have orbitual motions?_ Sir John Herschel in 1833 gave, in his _Astronomy_ (Art. 606), a list of nine stars, with periods extending from 43 years (η Coronæ) to 1200 years (γ Leonis), which he presented as the chief results then obtained in this department. In his work on Double Stars, the fruit of his labors in both hemispheres, which the astronomical world are looking for with eager expectation, he will, I believe, have a few more to add to these.

_Is it well established that such double stars attract each other according to the law of the inverse square of the distance?_ The answer to this question must be determined by ascertaining whether the above cases are regulated by the laws of elliptical motion. This is a matter which it must require a long course of careful observation to determine in such a number of cases as to prove the universality of the rule. Perhaps the minds of astronomers are still in suspense upon the subject. When Sir John Herschel's work shall appear, it will probably {469} be found that with regard to some of these stars, and γ Virginis in particular, the conformity of the observations with the laws of elliptical motion amounts to a degree of exactness which must give astronomers a strong conviction of the truth of the law. For since Sir W. Herschel's first measures in 1781, the arc described by one star about the other is above 305 degrees; and during this period the angular annual motion has been very various, passing through all gradations from about 20 minutes to 80 degrees. Yet in the whole of this change, the two curves constructed, the one from the observations, the other from the elliptical elements, for the purpose of comparison, having a total ordinate of 305 parts, do not, in any part of their course, deviate from each other so much as _two_ such parts.]

The verification of Newton's discoveries was sufficient employment for the last century; the first step in the extension of them belongs to this century. We cannot at present foresee the magnitude of this task, but every one must feel that the law of gravitation, before verified in all the particles of our own system, and now probably extended to the all but infinite distance of the fixed stars, presses upon our minds with a strong claim to be accepted as a universal law of the whole material creation.

Thus, in this and the preceding chapter, I have given a brief sketch of the history of the verification and extension of Newton's great discovery. By the mass of labor and of skill which this head of our subject includes, we may judge of the magnitude of the advance in our knowledge which that discovery made. A wonderful amount of talent and industry have been requisite for this purpose; but with these, external means have co-operated. Wealth, authority, mechanical skill, the division of labor, the power of associations and of governments, have been largely and worthily applied in bringing astronomy to its present high and flourishing condition. We must consider briefly what has thus been done. {470}

## CHAPTER VI.

## THE INSTRUMENTS AND AIDS OF ASTRONOMY DURING THE NEWTONIAN PERIOD.

### _Sect._ 1.--_Instruments._

SOME instruments or other were employed at all periods of astronomical observation. But it was only when observation had attained a considerable degree of delicacy, that the exact construction of instruments became an object of serious care. Gradually, as the possibility and the value of increased exactness became manifest, it was seen that every thing which could improve the astronomer's instruments was of high importance to him. And hence in some cases a vast increase of size and of expense was introduced; in other cases new combinations, or the result of improvements in other sciences, were brought into play. Extensive knowledge, intense thought, and great ingenuity, were requisite in the astronomical instrument maker. Instead of ranking with artisans, he became a man of science, sharing the honor and dignity of the astronomer himself.

1. _Measure of Angles._--Tycho Brahe was the first astronomer who acted upon a due appreciation of the importance of good instruments. The collection of such at Uraniburg was by far the finest which had ever existed. He endeavored to give steadiness to the frame, and accuracy to the divisions of his instruments. His Mural Quadrant was well adapted for this purpose; its radius was five cubits: it is clear, that as we enlarge the instrument we are enabled to measure smaller arcs. On this principle many large _gnomons_ were erected. Cassini's celebrated one in the church of St. Petronius at Bologna, was eighty-three feet (French) high. But this mode of obtaining accuracy was soon abandoned for better methods. Three great improvements were introduced about the same time. The application of the Micrometer to the telescope, by Huyghens, Malvasia, and Auzout; the application of the Telescope to the astronomical quadrant; and the fixation of the centre of its field by a Cross of fine wires placed in the focus by Gascoigne, and afterwards by Picard. We may judge how great was the improvement which these contrivances introduced into the art of {471} observing, by finding that Hevelius refused to adopt them because they would make all the old observations of no value. He had spent a laborious and active life in the exercise of the old methods, and could not bear to think that all the treasures which he had accumulated had lost their worth by the discovery of a new mine of richer ore.

[2d Ed.] [Littrow, in his _Die Wunder des Himmels_, Ed. 2, pp. 684, 685, says that Gascoigne invented and used the telescope with wires in the common focus of the lenses in 1640. He refers to _Phil. Trans._ xxx. 603. Picard reinvented this arrangement in 1667. I have already spoken of Gascoigne as the inventor of the micrometer.

Römer (already mentioned, p. 464) brought into use the Transit Instrument, and the employment of complete Circles, instead of the Quadrants used till then; and by these means gave to practical astronomy a new form, of which the full value was not discovered till long afterwards.**]

The apparent place of the object in the instrument being so precisely determined by the new methods, the exact Division of the arc into degrees and their subdivisions became a matter of great consequence. A series of artists, principally English, have acquired distinguished places in the lists of scientific fame by their performances in this way; and from that period, particular instruments have possessed historical interest and individual reputation. Graham was one of the first of these artists. He executed a great Mural Arc for Halley at Greenwich; for Bradley he constructed the Sector which detected aberration. He also made the Sector which the French academicians carried to Lapland; and probably the goodness of this instrument, compared with the imperfection of those which were sent to Peru, was one main cause of the great difference of duration in the two series of observations. Bird, somewhat later[121\7] (about 1750), divided several Quadrants for public observatories. His method of dividing was considered so perfect, that the knowledge of it was purchased by the English government, and published in 1767. Ramsden was equally celebrated. The error of one of his best Quadrants (that at Padua) is said to be never greater than two seconds. But at a later period, Ramsden constructed Mural Circles only, holding this to be a kind of instrument far superior to the quadrant. He made one of five feet diameter, in 1788, for M. Piazzi at Palermo; and one of eight feet for the observatory of Dublin. Troughton, a worthy successor of the {472} artists we have mentioned, has invented a method of dividing the circle still superior to the former ones; indeed, one which is theoretically perfect, and practically capable of consummate accuracy. In this way, circles have been constructed for Greenwich, Armagh, Cambridge, and many other places; and probably this method, carefully applied, offers to the astronomer as much exactness as his other implements allow him to receive; but the slightest casualty happening to such an instrument, after it has been constructed, or any doubt whether the method of graduation has been rightly applied, makes it unfit for the jealous scrupulosity of modern astronomy.

[Note 121\7: Mont. iv. 337.]

The English artists sought to attain accurate measurements by continued bisection and other aliquot subdivision of the limb of their circle; but Mayer proposed to obtain this end otherwise, by _repeating_ the measure on different parts of the circumference till the error of the division becomes unimportant, instead of attempting to divide an instrument without error. This invention of the Repeating Circle was zealously adopted by the French, and the relative superiority of the rival methods is still a matter of difference of opinion.

[2d Ed.] [In the series of these great astronomical mechanists, we must also reckon George Reichenbach. He was born Aug. 24, 1772, at Durlach; became Lieutenant of Artillery in the Bavarian service in 1794; (Salinenrath) Commissioner of Salt-works in 1811; and in 1820, First Commissioner of Water-works and Roads. He became, with Fraunhofer, the ornament of the mechanical and optical Institute erected in 1805 at Benedictbeuern by Utzschneider; and his astronomical instruments, meridian circles, transit instruments, equatorials, heliometers, make an epoch in Observing Astronomy. His contrivances in the Salt-works at Berchtesgaden and Reichenhall, in the Arms Manufactory at Amberg, and in the works for boring cannon at Vienna, are

enduring monuments of his rare mechanical talent. He died May 21, 1826, at Munich.]

2. _Clocks._--The improvements in the measures of space require corresponding improvements in the measure of time. The beginning of any thing which we can call accuracy, in this subject, was the application of the Pendulum to clocks, by Huyghens, in 1656. That the successive oscillations of a pendulum occupy equal times, had been noticed by Galileo; but in order to take advantage of this property, the pendulum must be connected with machinery by which its motion is kept from languishing, and by which the number of its swings is recorded. By inventing such machinery, Huyghens at once obtained {473} a measure of time more accurate than the sun itself. Hence astronomers were soon led to obtain the right ascension of a star, not directly, by measuring a Distance in the heavens, but indirectly, by observing the Moment of its Transit. This observation is now made with a degree of accuracy which might, at first sight, appear beyond the limits of human sense, being noted to a _tenth of a second of time_: but we may explain this, by remarking that though the number of the second at which the transit happens is given by the clock, and is reckoned according to the course of time, the subdivision of the second of time into smaller fractions is performed by the eye,--by seeing the space described by the heavenly body in a whole second, and hence estimating a smaller time, according to the space which its description occupies.

But in order to make clocks so accurate as to justify this degree of precision, their construction was improved by various persons in succession. Picard soon found that Huyghens' clocks were affected in their going by temperature, for heat caused expansion of the metallic pendulum. This cause of error was remedied by combining different metals, as iron and copper, which expand in a different degree, in such a way that their effects compensate each other. Graham afterwards used quicksilver for the same purpose. The _Escapement_ too (which connects the force which impels the clock with the pendulum which regulates it), and other parts of the machinery, had the most refined mechanical skill and ingenuity of the best artists constantly bestowed upon then. The astronomer of the present day, constantly testing the going of such a clock by the motions of the fixed stars, has a scale of time as stable and as minutely exact as the scales on which he measures distance.

The construction of good Watches, that is, portable or marine clocks, was important on another account, namely, because they might be used in determining the longitude of places. Hence the improvement of this little machine became an object of national interest, and was included in the reward of 20,000_l._, which we have already noticed as offered by the English parliament for the discovery of the longitude. Harrison,[122\7] originally a carpenter, turned his mind to this subject with success. After thirty years of labor, in which he was encouraged by many eminent persons, he produced, in 1758, a time-keeper, which was sent on a voyage to Jamaica for trial. After 161 days, the error {474} of the watch was only one minute five seconds, and the artist received from the nation 5000_l._ At a later period,[123\7] at the age of seventy-five years, after a life devoted to this object, having still further satisfied the commissioners, he received, in 1765, 10,000_l._, at the same time that

Euler and the heirs of Mayer received each 3000_l._ for the lunar tables which they had constructed.

[Note 122\7: Mont. iv. 554.]

[Note 123\7: Mont. iv. 560.]

The two methods of finding the longitude, by Chronometers and by Lunar Observations, have solved the problem for all practical purposes; but the latter could not have been employed at sea without the aid of that invaluable instrument, the Sextant, in which the distance of two objects is observed, by bringing one to coincide apparently with the reflected image of the other. This instrument was invented by Hadley, in 1731. Though the problem of finding the longitude be, in fact, one of geography rather than astronomy, it is an application of astronomical science which has so materially affected the progress of our knowledge, that it deserves the notice we have bestowed upon it.

3. _Telescopes._--We have spoken of the application of the telescope to astronomical measurements, but not of the improvement of the telescope itself. If we endeavor to augment the optical power of this instrument, we run, according to the path we take, into various inconveniences;--distortion, confusion, want of light, or colored images. Distortion and confusion are produced, if we increase the magnifying power, retaining the length and the aperture of the object-glass. If we diminish the aperture we suffer from loss of light. What remains then is to increase the focal length. This was done to an extraordinary extent, in telescopes constructed in the beginning of the last century. Huyghens, in his first attempts, made them 22 feet long;[124\7] afterwards, Campani, by order of Louis the Fourteenth, made them of 86, 100, and 136 feet. Huyghens, by new exertions, made a telescope 210 feet long. Auzout and Hartsoecker are said to have gone much further, and to have succeeded in making an object-glass of 600 feet focus. But even such telescopes as those of Campani are almost unmanageable: in that of Huyghens, the object-glass was placed on a pole, and the observer was placed at the focus with an eye-glass.

[Note 124\7: Bailly, ii. 253.]

The most serious objection to the increase of the aperture of object-glasses, was the coloration of the image produced, in consequence of the unequal refrangibility of differently colored rays. Newton, who discovered the principle of this defect in lenses, had maintained that {475} the evil was irremediable, and that a compound lens could no more refract without producing color, than a single lens could. Euler and Klingenstierna doubted the exactness of Newton's proposition; and, in 1755, Dollond disproved it by experiment. This discovery pointed out a method of making object-glasses which should give no color;--which should be _achromatic_. For this purpose Dollond fabricated various kinds of glass (flint and crown glass); and Clairaut and D'Alembert calculated formulæ. Dollond and his son[125\7] succeeded in constructing telescopes of three feet long (with a triple object-glass) which produced an effect as great as those of forty-five feet on the ancient principles. At first it was conceived that these discoveries opened the way to a vast extension of the astronomer's power of vision; but it was found that the most material improvement

was the compendious size of the new instruments; for, in increasing the dimensions, the optician was stopped by the impossibility of obtaining lenses of flint-glass of very large dimensions. And this branch of art remained long stationary; but, after a time, its epoch of advance again arrived. In the present century, Fraunhofer, at Munich, with the help of Guinand and the pecuniary support of Utzschneider, succeeded in forming lenses of flint-glass of a magnitude till then unheard of. Achromatic object-glasses, of a foot in diameter, and twenty feet focal length, are now no longer impossible; although in such attempts the artist cannot reckon on certain success.

[Note 125\7: Bailly, iii. 118.]

[2d Ed.] [Joseph Fraunhofer was born March 6, 1787, at Straubing in Bavaria, the son of a poor glazier. He was in his earlier years employed in his father's trade, so that he was not able to attend school, and remained ignorant of writing and arithmetic till his fourteenth year. At a later period he was assisted by Utzschneider, and tried rapidly to recover his lost ground. In the year 1806 he entered the establishment of Utzschneider as an optician. In this establishment (transferred from Benedictbeuern to Munich in 1819) he soon came to be the greatest Optician of Germany. His excellent telescopes and microscopes are known throughout Europe. His greatest telescope, that in the Observatory at Dorpat, has an object-glass of 9 inches diameter, and a focal length of 13⅓ feet. His written productions are to be found in the _Memoirs_ of the Bavarian Academy, in Gilbert's _Annalen der Physik_, and in Schumacher's _Astronomische Nachrichten_. He died the 7th of June, 1826.] {476}

Such telescopes might be expected to add something to our knowledge of the heavens, if they had not been anticipated by reflectors of an equal or greater scale. James Gregory had invented, and Newton had more efficaciously introduced, reflecting telescopes. But these were not used with any peculiar effect, till the elder Herschel made them his especial study. His skill and perseverance in grinding specula, and in contriving the best apparatus for their use, were rewarded by a number of curious and striking discoveries, among which, as we have already related, was the discovery of a new planet beyond Saturn. In 1789, Herschel surpassed all his former attempts, by bringing into action a reflecting telescope of forty feet length, with a speculum of four feet in diameter. The first application of this magnificent instrument showed a new satellite (the sixth) of Saturn. He and his son have, with reflectors of twenty feet, made a complete survey of the heavens, so far as they are visible in this country; and the latter is now in a distant region completing this survey, by adding to it the other hemisphere.

In speaking of the improvements of telescopes we ought to notice, that they have been pursued in the eye-glasses as well as in the object-glasses. Instead of the single lens, Huyghens substituted an eye-piece of two lenses, which, though introduced for another purpose, attained the object of destroying color.[126\7] Ramsden's eye-piece is one fit to be used with a micrometer, and others of more complex construction have been used for various purposes.

[Note 126\7: Coddington's _Optics_, ii. 21.]

_Sect._ 2.--_Observatories._

ASTRONOMY, which is thus benefited by the erection of large and stable instruments, requires also the establishment of permanent Observatories, supplied with funds for their support, and for that of the observers. Such observatories have existed at all periods of the history of the science; but from the commencement of the period which we are now reviewing, they multiplied to such an extent that we cannot even enumerate them. Yet we must undoubtedly look upon such establishments, and the labors of which they have been the scene, as important and essential parts of the history of the progress of astronomy. Some of the most distinguished of the observatories of modern times we may mention. The first of these were that of Tycho Brahe {477} at Uraniburg, and that of the Landgrave of Hesse Cassel, at Cassel, where Rothman and Byrgius observed. But by far the most important observations, at least since those of Tycho, which were the basis of the discoveries of Kepler and Newton, have been made at Paris and Greenwich. The Observatory of Paris was built in 1667. It was there that the first Cassini made many of his discoveries; three of his descendants have since labored in the same place, and two others of his family, the Maraldis;[127\7] besides many other eminent astronomers, as Picard, La Hire, Lefêvre, Fouchy, Legentil, Chappe, Méchain, Bouvard. Greenwich Observatory was built a few years later (1675); and ever since its erection, the observations there made have been the foundation of the greatest improvements which astronomy, for the time, received. Flamsteed, Halley, Bradley, Bliss, Maskelyne, Pond, have occupied the place in succession: on the retirement of the last-named astronomer in 1835, Professor Airy was removed thither from the Cambridge Observatory. In every state, and in almost every principality in Europe, Observatories have been established; but these have often fallen speedily into inaction, or have contributed little to the progress of astronomy, because their observations have not been published. From the same causes, the numerous private observatories which exist throughout Europe have added little to our knowledge, except where the attention of the astronomer has been directed to some definite points; as, for instance, the magnificent labors of the Herschels, or the skilful observations made by Mr. Pond with the Westbury circle, which first pointed out the error of graduation of the Greenwich quadrants. The Observations, now regularly published,[128\7] are those of Greenwich, begun by Maskelyne, and continued quarterly by Mr. Pond; those of Königsberg, published by Bessel since 1814; of Vienna, by Littrow since 1820; of Speier, by Schwerd since 1826; those of Cambridge, commenced by Airy in 1828; of Armagh, by Robinson in 1829. Besides these, a number of useful observations have been published in journals and occasional forms; as, for instance, those of Zach, made by Seeberg, near Gotha, since 1788; and others have been employed in forming catalogues, of which we shall speak shortly.

[Note 127\7: Mont. iv. 346.]

[Note 128\7: Airy, _Rep._ p. 128.]

[2d Ed.] [I have left the statement of published Observations in the text as it stood originally. I believe that at present (1847) the twelve places contained in the following list publish their Observations quite regularly, or nearly so;--Greenwich,

Oxford, Cambridge, Vienna, {478} Berlin, Dorpat, Munich, Geneva, Paris, Königsberg, Madras, the Cape of Good Hope.

Littrow, in his translation, adds to the publications noticed in the text as containing astronomical Observations, Zach's _Monatliche Correspondenz_, Lindenau and Bohnenberger's _Zeitschrift für Astronomie_, Bode's _Astronomisches Jahrbuch_, Schumacher's _Astronomische Nachrichten_.]

Nor has the establishment of observatories been confined to Europe. In 1786, M. de Beauchamp, at the expense of Louis the Sixteenth, erected an observatory at Bagdad, "built to restore the Chaldean and Arabian observations," as the inscription stated; but, probably, the restoration once effected, the main intention had been fulfilled, and little perseverance in observing was thought necessary. In 1828, the British government completed the building of an observatory at the Cape of Good Hope, which Lacaille had already made an astronomical station by his observations there at an earlier period (1750); and an observatory formed in New South Wales by Sir T. M. Brisbane in 1822, and presented by him to the government, is also in activity. The East India Company has founded observatories at Madras, Bombay, and St. Helena; and observations made at the former of these places, and at St. Helena, have been published.

The bearing of the work done at such observatories upon the past progress of astronomy, has already been seen in the preceding narrative. Their bearing upon the present condition of the science will be the subject of a few remarks hereafter.

_Sect._ 3.--_Scientific Societies._

THE influence of Scientific Societies, or Academical Bodies, has also been very powerful in the subject before us. In all branches of knowledge, the use of such associations of studious and inquiring men is great; the clearness and coherence of a speculator's ideas, and their agreement with facts (the two main conditions of scientific truth), are severally but beneficially tested by collision with other minds. In astronomy, moreover, the vast extent of the subject makes requisite the division of labor and the support of sympathy. The Royal Societies of London and of Paris were founded nearly at the same time as the metropolitan Observatories of the two countries. We have seen what constellations of philosophers, and what activity of research, existed at those periods; these philosophers appear in the lists, their discoveries {479} in the publications, of the above-mentioned eminent Societies. As the progress of physical science, and principally of astronomy, attracted more and more admiration, Academies were created in other countries. That of Berlin was founded by Leibnitz in 1710; that of St Petersburg was established by Peter the Great in 1725; and both these have produced highly valuable Memoirs. In more modern times these associations have multiplied almost beyond the power of estimation. They have been formed according to divisions, both of locality and of subject, conformable to the present extent of science, and the vast population of its cultivators. It would be useless to attempt to give a view either of their number or of the enormous mass of scientific literature which their Transactions present. But we may notice, as especially connected with our present subject, the Astronomical Society of London,

founded in 1820, which gave a strong impulse to the pursuit of the science in England.

_Sect._ 4.--_Patrons of Astronomy._

The advantages which letters and philosophy derive from the patronage of the great have sometimes been questioned; that love of knowledge, it has been thought, cannot be genuine which requires such stimulation, nor those speculations free and true which are thus forced into being. In the sciences of observation and calculation, however, in which disputed questions can be experimentally decided, and in which opinions are not disturbed by men's practical principles and interests, there is nothing necessarily operating to poison or neutralize the resources which wealth and power supply to the investigation of truth.

Astronomy has, in all ages, flourished under the favor of the rich and powerful; in the period of which we speak, this was eminently the case. Louis the Fourteenth gave to the astronomy of France a distinction which, without him, it could not have attained. No step perhaps tended more to this than his bringing the celebrated Dominic Cassini to Paris. This Italian astronomer (for he was born at Permaldo, in the county of Nice, and was professor at Bologna), was already in possession of a brilliant reputation, when the French ambassador, in the name of his sovereign, applied to Pope Clement the Ninth, and to the senate of Bologna, that he should be allowed to remove to Paris. The request was granted only so far as an absence of six years; but at the end of that time, the benefits and honors which {480} the king had conferred upon him, fixed him in France. The impulse which his arrival (in 1669) and his residence gave to astronomy, showed the wisdom of the measure. In the same spirit, the French government drew to Paris Römer from Denmark, Huyghens from Holland, and gave a pension to Hevelius, and a large sum when his observatory at Dantzic had been destroyed by fire in 1679.

When the sovereigns of Prussia and Russia were exerting themselves to encourage the sciences in their countries, they followed the same course which had been so successful in France. Thus, as we have said, the Czar Peter took Delisle to Petersburg in 1725; the celebrated Frederick the Great drew to Berlin, Voltaire and Maupertuis, Euler and Lagrange; and the Empress Catharine obtained in the same way Euler, two of the Bernoulli's, and other mathematicians. In none of these instances, however, did it happen that "the generous plant did still its stock renew," as we have seen was the case at Paris, with the Cassinis, and their kinsmen the Maraldis.

[2d Ed.] [I may notice among instances of the patronage of Astronomy, the reward at present offered by the King of Denmark for the discovery of a Comet.]

It is not necessary to mention here the more recent cases in which sovereigns or statesmen have attempted to patronize individual astronomers.

_Sect._ 5.--_Astronomical Expeditions._

BESIDES the pensions thus bestowed upon resident mathematicians and astronomers, the governments of Europe have wisely and usefully employed considerable sums upon expeditions and travels undertaken by men of science for

some appropriate object. Thus Picard, in 1671, was sent to Uraniburg, the scene of Tycho's observations, to determine its latitude and its longitude. He found that "the City of the Skies" had utterly disappeared from the earth; and even its foundations were retraced with difficulty. With the same object, that of accurately connecting the labors of the places which had been at different periods the metropolis of astronomy, Chazelles was sent, in 1693, to Alexandria. We have already mentioned Richer's astronomical expedition to Cayenne in 1672. Varin and Deshayes[129\7] were sent a few years later into the same regions for similar purposes. Halley's expedition to St. {481} Helena in 1677, with the view of observing the southern stars, was at his own expense; but at a later period (in 1698), he was appointed to the command of a small vessel by King William the Third, in order that he might make his magnetical observations in all parts of the world. Lacaille was maintained by the French government four years at the Cape of Good Hope (1750-4), for the purpose of observing the stars of the southern hemisphere. The two transits of Venus in 1761 and 1769, occasioned expeditions to be sent to Kamtschatka and Tobolsk by the Russians; to the Isle of France, and to Coromandel, by the French;[130\7] to the isles of St. Helena and Otaheite by the English; to Lapland and to Drontheim, by the Swedes and Danes. I shall not here refer to the measures of degrees executed by various nations, still less the innumerable surveys by land and sea; but I may just notice the successive English expeditions of Captains Basil Hall, Sabine, and Foster, for the purpose of determining the length of the seconds' pendulum in different latitudes; and the voyages of M. Biot and others, sent by the French government for the same purpose. Much has been done in this way, but not more than the progress of astronomy absolutely required; and only a small portion of that which the completion of the subject calls for.

[Note 129\7: Bailly, ii. 374.]

[Note 130\7: Bailly, iii. 107.]

_Sect._ 6.--_Present State of Astronomy._

ASTRONOMY, in its present condition, is not only much the most advanced of the sciences, but is also in far more favorable circumstances than any other science for making any future advance, as soon as this is possible. The general methods and conditions by which such an advantage is to be obtained for the various sciences, we shall endeavor hereafter to throw some light upon; but in the mean time, we may notice here some of the circumstances in which this peculiar felicity of the present state of astronomy may be traced.

The science is cultivated by a number of votaries, with an assiduity and labor, and with an expenditure of private and public resources, to which no other subject approaches; and the mode of its cultivation in all public and most private observatories, has this character--that it forms, at the same time, a constant process of verification of existing discoveries, and a strict search for any new discoverable laws. The observations made are immediately referred to the best tables, and {482} corrected by the best formulæ which are known; and if the result of such a reduction leaves any thing unaccounted for, the astronomer is forthwith curious and anxious to

trace this deviation from the expected numbers to its rule and its origin; and till the first, at least, of these things is performed, he is dissatisfied and unquiet. The reference of observations to the state of the heavens as known by previous researches, implies a great amount of calculation. The exact places of the stars at some standard period are recorded in _Catalogues_; their movements, according to the laws hitherto detected, are arranged in _Tables_; and if these tables are applied to predict the numbers which observation on each day ought to give, they form _Ephemerides_. Thus the catalogues of fixed stars of Flamsteed, of Piazzi, of Maskelyne, of the Astronomical Society, are the basis of all observation. To these are applied the Corrections for Refraction of Bradley or Bessel, and those for Aberration, for Nutation, for Precession, of the best modern astronomers. The observations so corrected enable the observer to satisfy himself of the delicacy and fidelity of his measures of time and space; his Clocks and his Arcs. But this being done, different stars so observed can be compared with each other, and the astronomer can then endeavor further to correct his fundamental Elements;--his Catalogue, or his Tables of Corrections. In these Tables, though previous discovery has ascertained the law, yet the exact quantity, the _constant_ or _coefficient_ of the formula, can be exactly fixed only by numerous observations and comparisons. This is a labor which is still going on, and in which there are differences of opinion on almost every point; but the amount of these differences is the strongest evidence of the certainty and exactness of those doctrines in which all agree. Thus Lindenau makes the coefficient of Nutation rather less than nine seconds, which other astronomers give as about nine seconds and three-tenths. The Tables of Refraction are still the subject of much discussion, and of many attempts at improvement. And after or amid these discussions, arise questions whether there be not other corrections of which the law has not yet been assigned. The most remarkable example of such questions is the controversy concerning the existence of an Annual Parallax of the fixed stars, which Brinkley asserted, and which Pond denied. Such a dispute between two of the best modern observers, only proves that the quantity in question, if it really exist, is of the same order as the hitherto unsurmounted errors of instruments and corrections.

[2d Ed.] [The belief in an appreciable parallax of some of the fixed {483} stars appears to gain ground among astronomers. The parallax of 61 _Cygni_, as determined by Bessel, is 0"·34; about one-third of a second, or 1⁄10000 of a degree. That of _α Centauri_, as determined by Maclear, is 0"·9, or 1⁄4000 of a degree.]

But besides the fixed stars and their corrections, the astronomer has the motions of the planets for his field of action. The established theories have given us tables of these, from which their daily places are calculated and given in our Ephemerides, as the _Berliner Jahrbuch_ of Encke, or the _Nautical Almanac_, published by the government of this country, the _Connaissance des Tems_ which appears at Paris, or the _Effemeridi di Milano_. The comparison of the observed with the tabular place, gives us the means of correcting the coefficients of the tables; and thus of obtaining greater exactness in the constants of the solar system. But these constants depend upon the mass and form of the bodies of which the system is composed; and in this province, as well as in sidereal astronomy, different determinations, obtained by different paths, may be compared; and doubts may be raised and may be solved. In

this way, the perturbations produced by Jupiter on different planets gave rise to a doubt whether his attraction be really proportional to his mass, as the law of universal gravitation asserts. The doubt has been solved by Nicolai and Encke in Germany, and by Airy in England. The mass of Jupiter, as shown by the perturbations of Juno, of Vesta, and of Encke's Comet, and by the motion of his outermost Satellite, is found to agree, though different from the mass previously received on the authority of Laplace. Thus also Burckhardt, Littrow, and Airy, have corrected the elements of the Solar Tables. In other cases, the astronomer finds that no change of the coefficients will bring the Tables and the observations to a coincidence;--that a new term in the formula is wanting. He obtains, as far as he can, the law of this unknown term; if possible, he traces it to some known or probable cause. Thus Mr. Airy, in his examination of the Solar Tables, not only found that a diminution of the received mass of Mars was necessary, but perceived discordances which led him to suspect the existence of a new inequality. Such an inequality was at length found to result theoretically from the attraction of Venus. Encke, in his examination of his comet, found a diminution of the periodic time in the successive revolutions; from which he inferred the existence of a resisting medium. Uranus still deviates from his tabular place, and the cause remains yet to be discovered. (But see the _Additions_ to this volume.) {484}

Thus it is impossible that an assertion, false to any amount which the existing state of observation can easily detect, should have any abiding prevalence in astronomy. Such errors may long keep their ground in any science which is contained mainly in didactic works, and studied in the closet, but not acted upon elsewhere;--which is reasoned upon much, but brought to the test of experiment rarely or never. Here, on the contrary, an error, if it arise, makes its way into the Tables, into the Ephemeris, into the observer's nightly List, or his sheet of Reductions; the evidence of sense flies in its face in a thousand observatories; the discrepancy is traced to its source, and soon disappears forever.

In this favored branch of knowledge, the most recondite and delicate discoveries can no more suffer doubt or contradiction, than the most palpable facts of sense which the face of nature offers to our notice. The last great discovery in astronomy--the motion of the stars arising from Aberration--is as obvious to the vast population of astronomical observers in all parts of the world, as the motion of the stars about the pole is to the casual night wanderer. And this immunity from the danger of any large error in the received doctrines, is a firm platform on which the astronomer can stand and exert himself to reach perpetually further and further into the region of the unknown.

The same scrupulous care and diligence in recording all that has hitherto been ascertained, has been extended to those departments of astronomy in which we have as yet no general principles which serve to bind together our acquired treasures. These records may be considered as constituting a _Descriptive Astronomy_; such are, for instance, Catalogues of Stars, and Maps of the Heavens, Maps of the Moon, representations of the appearance of the Sun and Planets as seen through powerful telescopes, pictures of Nebulæ, of Comets, and the like. Thus, besides the Catalogue

of Fundamental Stars which may be considered as standard points of reference for all observations of the Sun, Moon, and Planets, there exist many large catalogues of smaller stars. Flamsteed's _Historia Celestis_, which much surpassed any previous catalogue, contained above 3000 stars. But in 1801, the French _Histoire Céleste_ appeared, comprising observations of 50,000 stars. Catalogues or charts of other special portions of the sky have been published more recently; and in 1825, the Berlin Academy proposed to the astronomers of Europe to carry on this work by portioning out the heavens among them.

[2d Ed.] [Before Flamsteed, the best Catalogue of the Stars was {485} Tycho Brahe's, containing the places of about 1000 stars, determined very roughly with the naked eye. On the occasion of a project of finding the longitude, which was offered to Charles II., in 1674, Flamsteed represented that the method was quite useless, in consequence, among other things, of the inaccuracy of Tycho's places of the stars. Flamsteed's letters being shown King Charles, he was startled at the assertion of the fixed stars' places being false in the Catalogue, and said, with some vehemence, "He must have them anew observed, examined, and corrected for the use of his seamen." This was the immediate occasion of building Greenwich Observatory, and placing Flamsteed there as an observer. Flamsteed's _Historia Celestis_ contained above 3000 stars, observed with telescopic sights. It has recently been republished with important improvements by Mr. Baily. See Baily's _Flamsteed_, p. 38.

The French _Histoire Céleste_ was published in 1801 by Lalande, containing 50,000 stars, simply as observed by himself and other French astronomers. The reduction of the observations contained in this Catalogue to the mean places at the beginning of the year 1800 may be effected by means of Tables published by Schumacher for that purpose in 1825.

In 1807, Piazzi's Catalogue of 6748 stars, founded on Maskelyne's Catalogue of 1700, was published; afterwards extended to 7646 stars in 1814. This is considered as the greatest work undertaken by any modern astronomer; the observations being well made, reduced, and compared with those of former astronomers. Piazzi's Catalogue is the standard and accurate Catalogue, as the _Histoire Céleste_ is the standard approximate Catalogue for small stars. But the new planets were discovered mostly by a comparison of the heavens with Bode's (Berlin) Catalogue.

I may mention other Catalogues of Stars which have recently been published. Pond's Catalogue contains 1112 Northern stars; Johnson's, 606; Wrottesley's, 1318 (in Right Ascension only); Airy's First Cambridge Catalogue, 726; his Greenwich Catalogue, 1439. Pearson's has 520 zodiacal stars; Groombridge's, 4243 circumpolar stars as far as 50 degrees of North Polar distance; Santini's, a zone 18 degrees North of the equator. Besides these, Mr. Taylor has published, by order of the Madras government, a Catalogue of 11,000 stars observed by him at Madras; and Rumker, who observed in the Observatory established by Sir Thomas Brisbane at Paramatta (in Australia), has commenced a Catalogue which is to contain 12,000. Mr. Baily {486} published two Standard Catalogues; that of the Royal Astronomical Society, containing 2881 stars; and that of the British Association, containing 8377 stars. I omit other Catalogues, as those of Argelander, &c., and Catalogues of Southern

Stars.

Of the Berlin Maps, fourteen hours in Right Ascension have been published; and their value may be judged of by this circumstance, that it was in a great measure by comparing the heavens with these Maps that the new planet Astræa was discovered. The Zone observations made at Königsberg, by the late illustrious astronomer Bessel, deserve to be mentioned, as embracing a vast number of stars.

The common mode of _designating the Stars_ is founded upon the ancient constellations as given by Ptolemy; to which Bayer, of Augsburg, in his _Uranometria_, added the artifice of designating the brightest stars in each constellation by the Greek letters, α, β, γ, &c., applied in order of brightness, and when these were exhausted, the Latin letters. Flamsteed used numbers. As the number of observed stars increased, various methods were employed for designating them; and the confusion which has been thus introduced, both with regard to the boundaries of the constellations and the nomenclature of the stars in each, has been much complained of lately. Some attempts have been made to remedy this variety and disorder. Mr. Argelander has recently recorded stars, according to their magnitudes as seen by the naked eye, in a _Neue Uranometrie_.

Among representations of the Moon I may mention Hevelius's _Selenographia_, a work of former times, and Beer and Madler's Map of the Moon, recently published.]

I have already said something of the observations of the two Herschels on _Double Stars_, which have led to a knowledge of the law of the revolution of such systems. But besides these, the same illustrious astronomers have accumulated enormous treasures of observations of _Nebulæ_; the materials, it may be, hereafter, of some vast new generalization with respect to the history of the system of the universe.

[2d Ed.] [A few measures of Double Stars are to be found in previous astronomical records. But the epoch of the creation of this part of the science of astronomy must be placed at the beginning of the present century, when Sir William Herschel (in 1802) published in the _Phil. Trans._ a Catalogue of 500 new Nebulæ of various classes, and in the _Phil. Trans._ 1803, a paper "On the changes in the relative situation of the Double Stars in 25 years." In succeeding papers he pursued the subject. In one in 1814 he noticed the breaking up of the {487} Milky Way in different places, apparently from some principle of Attraction; and in this, and in one in 1817, he published those remarkable views on the distribution of the stars in our own cluster as forming a large stratum, and on the connection of stars and nebulæ (the stars appearing sometimes to be accompanied by nebulæ, sometimes to have absorbed a part of the nebula, and sometimes to have been formed from nebulæ), which have been accepted and propounded by others as the _Nebular Theory_. Sir William Herschel's last paper was a Catalogue of 145 new Double Stars communicated to the Astronomical Society in 1822. In 1827 M. Struve, of Dorpat (in Russia), published his _Catalogus Novus_, containing the places of 3112 double stars. While this was going on, Sir John Herschel and Sir James South published (in the _Phil. Trans._ 1824) accurate measures of 380 Double and Triple Stars, to which Sir J. South afterwards added 458. Mr. Dunlop published measures of 253 Southern Double

Stars. Other Observations have been published by Capt. Smyth, Mr. Dawes, &c. The great work of Struve, _Mensuræ Micrometricæ_, &c., contains 3134 such objects, including most of Sir W. Herschel's Double Stars. Sir J. Herschel in 1826, 7, and 8 presented to the Astronomical Society about 1000 measures of Double Stars; and in 1830, good measures of 1236, made with his 20-feet reflector. His paper in vol. v. of the _Ast. Soc. Mem._, besides measures of 364 such stars, exhibits all the most striking results, as to the motion of Double Stars, which have yet been obtained. In 1835 he carried his 20-feet reflector to the Cape of Good Hope for the purpose of completing the survey of Double Stars and Nebulæ in the southern hemisphere with the same instruments which had explored the northern skies. He returned from the Cape in 1838, and is now (1846) about to give the world the results of his labors. Besides the stars just mentioned, his work will contain from 1500 to 2000 additional double stars; making a gross number of above 8000; in which of course are included a number of objects of no great scientific interest, but in which also are contained the materials of the most important discoveries which remain to be made by astronomers. The publication of Sir John Herschel's great work upon Double Stars and Nebulæ is looked for with eager interest by astronomers.

Of the observations of Nebulæ we may say what has just been said of the observations of Double Stars;--that they probably contain the materials of important future discoveries. It is impossible not to regard these phenomena with reference to the _Nebular Hypothesis_, which has been propounded by Laplace, and much more strongly {488} insisted upon by other persons;--namely, the hypothesis that systems of revolving planets, of which the Solar System is an example, arise from the gradual contraction and separation of vast masses of nebulous matter. Yet it does not appear that any changes have been observed in nebulæ which tend to confirm this hypothesis; and the most powerful telescope in the world, recently erected by the Earl of Rosse, has given results which militate against the hypothesis; inasmuch as it has shown that what appeared a diffused nebulous mass is, by a greater power of vision, resolved, in all cases yet examined, into separate stars.

When astronomical phenomena are viewed with reference to the Nebular Hypothesis, they do not belong so properly to Astronomy, in the view here taken of it, as to Cosmogony. If such speculations should acquire any scientific value, we shall have to arrange them among those which I have called _Palætiological_ Sciences; namely, those Sciences which contemplate the universe, the earth, and its inhabitants, with reference to their historical changes and the causes of those changes.]

{{489}} ADDITIONS TO THE THIRD EDITION.

## INTRODUCTION.

THERE is a difficulty in writing for popular readers a History of the Inductive Sciences, arising from this;--that the sympathy of such readers goes most readily and naturally along the course which leads to false science and to failure. Men, in the outset of their attempts at knowledge, are prone to rush from a few hasty

observations of facts to some wide and comprehensive principles; and then, to frame a system on these principles. This is the opposite of the method by which the Sciences have really and historically been conducted; namely, the method of a gradual and cautious ascent from observation to principles of limited generality, and from them to others more general. This latter, the true Scientific Method, is _Induction_, and has led to the _Inductive Sciences_. The other, the spontaneous and delusive course, has been termed by Francis Bacon, who first clearly pointed out the distinction, and warned men of the error, _Anticipation_. The hopelessness of this course is the great lesson of his philosophy; but by this course proceeded all the earlier attempts of the Greek philosophers to obtain a knowledge of the Universe.

Laborious observation, narrow and modest inference, caution, slow and gradual advance, limited knowledge, are all unwelcome efforts and restraints to the mind of man, when his speculative spirit is once roused: yet these are the necessary conditions of all advance in the Inductive Sciences. Hence, as I have said, it is difficult to win the sympathy of popular readers to the true history of these sciences. The career of bold systems and fanciful pretences of knowledge is more entertaining and striking. Not only so, but the bold guesses and fanciful reasonings of men unchecked by doubt or fear of failure are often presented as the dictates of _Common Sense_;--as the plain, unsophisticated, unforced reason of man, acting according to no artificial rules, but following its own natural course. Such Common Sense, while it {490} complacently plumes itself on its clear-sightedness in rejecting arbitrary systems of others, is no less arbitrary in its own arguments, and often no less fanciful in its inventions, than those whom it condemns.

We cannot take a better representative of the Common Sense of the ancient Greeks than Socrates: and we find that his Common Sense, judging with such admirable sagacity and acuteness respecting moral and practical matters, offered, when he applied it to physical questions, examples of the unconscious assumptions and fanciful reasonings which, as we have said, Common Sense on such subjects commonly involves.

Socrates, Xenophon tells us (_Memorabilia_, iv. 7), recommended his friends not to study astronomy, so as to pursue it into scientific details. This was practical advice: but he proceeded further to speak of the palpable mistakes made by those who had carried such studies farthest. Anaxagoras, for instance, he said, held that the Sun was a Fire:--he did not consider that men can look at a fire, but they cannot look at the Sun; they become dark by the Sun shining upon them, but not so by the fire. He did not consider that no plants can grow well except they have sunshine, but if they are exposed to the fire they are spoiled. Again, when he said that the Sun was a stone red-hot, he did not consider that a stone heated by the fire is not luminous, and soon cools, but the Sun is always luminous and always hot.

We may easily conceive how a disciple of Anaxagoras would reply to these arguments. He would say, for example, as we should probably say at present, that if there were a mass of matter so large and so hot as Anaxagoras supposed the Sun to be, its light might be as great and its heat as permanent as the heat and light of the Sun are, as yet, known to be. In this case the arguments of Socrates are at any rate

no better than the doctrine of Anaxagoras.

{{491}} BOOK I.

THE GREEK SCHOOL PHILOSOPHY.

## CHAPTER II.

THE GREEK SCHOOLS.

_The Platonic Doctrine of Ideas._

IN speaking of the Foundation of the Greek School Philosophy, I have referred to the dialogue entitled _Parmenides_, commonly ascribed to Plato. And the doctrines ascribed to Parmenides, in that and in other works of ancient authors, are certainly remarkable examples of the tendency which prevailed among the Greeks to rush at once to the highest generalizations of which the human mind is capable. The distinctive dogma of the Eleatic School, of which Parmenides was one of the most illustrious teachers, was that _All Things are One_. This indeed was rather a doctrine of metaphysical theology than of physical science. It tended to, or agreed with, the doctrine that All things are God:--the doctrine commonly called _Pantheism_. But the tenet of the Platonists which was commonly put in opposition to this, that we must seek _The One in the Many_, had a bearing upon physical science; at least, if we interpret it, as it is generally interpreted, that we must seek the one Law which pervades a multiplicity of Phenomena. We may however take the liberty of remarking, that to speak of a Rule which is exemplified in many cases, as being "the One in the Many" (a way of speaking by which we put out of sight the consideration what very different kinds of things _the One_ and _the Many_ are), is a mode of expression which makes a very simple matter look very mysterious; and is another example of the tendency which urges speculative men to aim at metaphysical generality rather than scientific truth.

The Dialogue _Parmenides_ is, as I have said, commonly referred to Plato. Yet it is entirely different in substance, manner, and tendency {492} from the most characteristic of the Platonic Dialogues. In these, Socrates is represented as finally successful in refuting or routing his adversaries, however confident their tone and however popular their assertions. They are angered or humbled; he retains his good temper and his air of superiority, and when they are exhausted, he sums up in his own way.

In the _Parmenides_, on the contrary, everything is the reverse of this. Parmenides and Zeno exchange good-humoured smiles at Socrates's criticism, when the bystanders expect them to grow angry. They listen to Socrates while he propounds Plato's doctrine of Ideas; and reply to him with solid arguments which he does not answer, and which have never yet been answered. Parmenides, in a patronising way, lets him off; and having done this, being much entreated, he pronounces a discourse concerning the One and the Many; which, obscure as it may seem to us, was obviously intended to be irrefutable: and during the whole of this part of the Dialogue, the friend of Socrates appears only as a passive respondent, saying _Yes_ or

_No_ as the assertions of Parmenides require him to do; just in the same way in which the opponents of Socrates are represented in other Dialogues.

These circumstances, to which other historical difficulties might be added, seem to show plainly that the _Parmenides_ must be regarded as an Eleatic, not as a Platonic Dialogue;--as composed to confute, not to assert, the Platonic doctrine of Ideas.

The Platonic doctrine of Ideas has an important bearing upon the philosophy of Science, and was suggested in a great measure by the progress which the Greeks had really made in Geometry, Astronomy, and other Sciences, as I shall elsewhere endeavor to show. This doctrine has been recommended in our own time,[1\A] as containing "a mighty substance of imperishable truth." It cannot fail to be interesting to see in what manner the doctrine is presented by those who thus judge of it. The following is the statement of its leading features which they give us.

[Note 1\A: A. Butler's _Lectures_, Second Series, Lect. viii. p. 132.]

Man's soul is made to contain not merely a consistent scheme of its own notions, but a direct apprehension of _real and eternal laws beyond it_. These real and eternal laws are things _intelligible_, and not things sensible. The laws, impressed upon creation by its Creator, and apprehended by man, are something equally distinct from the Creator {493} and from man; and the whole mass of them may be termed the World of Things purely Intelligible.

Further; there are qualities in the Supreme and Ultimate Cause of all, which are manifested in his creation; and not merely manifested, but in a manner--after being brought out of his super-essential nature into the stage of being which is below him, but next to him--are then, by the causative act of creation, deposited in things, differencing them one from the other, so that the things participate of them (μετέχουσι), communicate with them (κοινωνοῦσι).

The Intelligence of man, excited to reflection by the impressions of these objects, thus (though themselves transitory) participant of a divine quality, may rise to higher conceptions of the perfections thus faintly exhibited; and inasmuch as the perfections are unquestionably _real_ existences, and known to be such in the very act of contemplation, this may be regarded as a distinct intellectual apprehension of them;--a union of the Reason with the Ideas in that sphere of being which is common to both.

Finally, the Reason, in proportion as it learns to contemplate the Perfect and Eternal, desires the enjoyment of such contemplations in a more consummate degree, and cannot be fully satisfied except in the actual fruition of the Perfect itself.

These propositions taken together constitute the THEORY OF IDEAS. When we have to treat of the Philosophy of Science, it may be worth our while to resume the consideration of this subject.

In this part of the History, the _Timæus_ of Plato is referred to as an example of the loose notions of the Greek philosophers in their physical reasonings. And undoubtedly this Dialogue does remarkably exemplify the boldness of the early Greek attempts at generalization on such subjects. Yet in this and in other parts the writings

of Plato contain speculations which may be regarded as containing germs of true physical science; inasmuch as they assume that the phenomena of the world are governed by mathematical laws;--by relations of space and number;--and endeavor, too boldly, no doubt, but not vaguely or loosely, to assign those laws. The Platonic writings offer, in this way, so much that forms a Prelude to the Astronomy and other Physical Sciences of the Greeks, that they will deserve our notice, as supplying materials for the next two Books of the History, in which these subjects are treated of. {494}

## CHAPTER III.

FAILURE OF THE GREEK PHYSICAL PHILOSOPHY.

_Francis Bacon's Remarks._

THOUGH we do not accept, as authority, even the judgments of Francis Bacon, and shall have to estimate the strong and the weak parts of his, no less than of other philosophies, we shall find his remarks on the Greek philosophers very instructive. Thus he says of Aristotle, (_Nov. Org._ 1. Aph. lxiii.):

"He is an example of the kind of philosophy in which much is made out of little; so that the basis of experience is too narrow. He corrupted Natural Philosophy by his Logic, and made the world out of his Categories. He disposed of the distinction of _dense_ and _rare_, by which bodies occupy more or less dimensions or space, by the frigid distinction of _act_ and _power_. He assigned to each kind of body a single proper motion, so that if they have any other motion they must receive it from some extraneous source; and imposed many other arbitrary rules upon Nature; being everywhere more careful how one may give a ready answer, and make a positive assertion, than how he may apprehend the variety of nature.

"And this appears most evidently by the comparison of his philosophy with the other philosophies which had any vogue in Greece. For the _Homoiomeria_[2\A] of Anaxagoras, the _Atoms_ of Leucippus and Democritus, the Heaven and Earth of Parmenides, the Love and Hate of Empedocles, the Fire of Heraclitus, had some trace of the thoughts of a natural philosopher; some savor of experience, and nature, and bodily things; while the Physics of Aristotle, in general, sound only of Logical Terms.

[Note 2\A: For these technical forms of the Greeks, see Sec. 3 of this chapter.]

"Nor let any one be moved by this--that in his books _Of Animals_, and in his _Problems_, and in others of his tracts, there is often a quoting of experiments. For he had made up his mind beforehand; and did not consult experience in order to make right propositions and axioms, but when he had settled his system to his will, he twisted experience {495} round, and made her bend to his system: so that in this way he is even more wrong than his modern followers, the Schoolmen, who have deserted experience altogether."

We may note also what Bacon says of the term _Sophist_. (Aph. lxxi.) "The wisdom of the Greeks was professorial, and prone to run into disputations: which kind is very adverse to the discovery of Truth. And the name of _Sophists_, which was cast in the

way of contempt, by those who wished to be reckoned philosophers, upon the old professors of rhetoric, Gorgias, Protagoras, Hippias, Polus, does, in fact, fit the whole race of them, Plato,[3\A] Aristotle, Zeno, Epicurus, Theophrastus; and their successors, Chrysippus, Carneades, and the rest."

[Note 3\A: It is curious that the attempt to show that Plato's opponents were not commonly illusive and immoral reasoners, has been represented as an attempt to obliterate the distinction of "Sophist" and "Philosopher."--See A. Butler's _Lectures_, i. 357. Note.]

That these two classes of teachers, as moralists, were not different in their kind, has been urged by Mr. Grote in a very striking and amusing manner. But Bacon speaks of them here as physical philosophers; in which character he holds that all of them were _sophists_, that is, illusory reasoners.

_Aristotle's Account of the Rainbow._

To exemplify the state of physical knowledge among the Greeks, we may notice briefly Aristotle's account of the _Rainbow_; a phenomenon so striking and definite, and so completely explained by the optical science of later times. We shall see that not only the explanations there offered were of no value, but that even the observation of facts, so common and so palpable, was inexact. In his _Meteorologica_ (lib. iii. c. 2) he says, "The Rainbow is never more than a semicircle. And at sunset and sunrise, the circle is least, but the arch is greatest; when the sun is high, the circle is larger, but the arch is less." This is erroneous, for the diameter of the circle of which the arch of the rainbow forms a part, is always the same, namely 82°. "After the autumnal equinox," he adds, "it appears at every hour of the day; but in the summer season, it does not appear about noon." It is curious that he did not see the reason of this. The centre of the circle of which the rainbow is part, is always opposite to the sun. And therefore if the sun be more than 41° above the horizon, the centre of the rainbow will be so much below the horizon, that the place of the rainbow will {496} be entirely below the horizon. In the latitude of Athens, which is 38°, the equator is 52° above the horizon, and the rainbow can be visible only when the sun is 11° lower than it is at the equinoctial noon. These remarks, however, show a certain amount of careful observation; and so do those which Aristotle makes respecting the colors. "Two rainbows at most appear: and of these, each has three colors; but those in the outer bow are duller; and their order opposite to those in the inner. For in the inner bow the first and largest arch is red; but in the outer bow the smallest arch is red, the nearest to the inner; and the others in order. The colors are red, green, and purple, such as painters cannot imitate." It is curious to observe how often modern painters disregard even the order of the colors, which they could imitate, if they attended to it.

It may serve to show the loose speculation which we oppose to science, if we give Aristotle's attempt to explain the phenomenon of the Rainbow. It is produced, he says (c. iv.), by Reflexion (ἀνάκλασις) from a cloud opposite to the sun, when the cloud forms into drops. And as a reason for the red color, he says that a bright object seen through darkness appears red, as the flame through the smoke of a fire of green

wood. This notion hardly deserves notice; and yet it was taken up again by Göthe in our own time, in his speculations concerning colors.

{{497}} BOOK II.

THE PHYSICAL SCIENCES IN ANCIENT GREECE.

_Plato's Timæus and Republic._

ALTHOUGH a great portion of the physical speculations of the Greek philosophers was fanciful, and consisted of doctrines which were rejected in the subsequent progress of the Inductive Sciences; still many of these speculations must be considered as forming a Prelude to more exact knowledge afterwards attained; and thus, as really belonging to the Progress of knowledge. These speculations express, as we have already said, the conviction that the phenomena of nature are governed by laws of space and number; and commonly, the mathematical laws which are thus asserted have some foundation in the facts of nature. This is more especially the case in the speculations of Plato. It has been justly stated by Professor Thompson (A. Butler's _Lectures_, Third Series, Lect. i. Note 11), that it is Plato's merit to have discovered that the laws of the physical universe are resolvable into numerical relations, and therefore capable of being represented by mathematical formulæ. Of this truth, it is there said, Aristotle does not betray the slightest consciousness.

The _Timæus_ of Plato contains a scheme of mathematical and physical doctrines concerning the universe, which make it far more analogous than any work of Aristotle to Treatises which, in modern times, have borne the titles of _Principia_, _System of the World_, and the like. And fortunately the work has recently been well and carefully studied, with attention, not only to the language, but to the doctrines and their bearing upon our real knowledge. Stallbaum has published an edition of the Dialogue, and has compared the opinions of Plato with those of Aristotle on the like subjects. Professor Archer Butler of Dublin has devoted to it several of his striking and eloquent Lectures; and these have been furnished with valuable annotations by Professor Thompson of Cambridge; and M. The. Henri Martin, then Professor at Rennes, published in 1841 two volumes of _Etudes sur le Timée de Platon_, in {498} which the bearings of the work on Science are very fully discussed. The Dialogue treats not only concerning the numerical laws of harmonical sounds, of visual appearances, and of the motions of planets and stars, but also concerning heat, as well as light; and concerning water, ice, gold, gems, iron, rust, and other natural objects;--concerning odors, tastes, hearing, sight, light, colors, and the powers of sense in general:--concerning the parts and organs of the body, as the bones, the marrow, the brain, the flesh, muscles, tendons, ligaments, nerves; the skin, the hair, the nails; the veins and arteries; respiration; generation; and in short every obvious point of physiology.

But the opinions delivered in the _Timæus_ upon these latter subjects have little to do with the progress of real knowledge. The doctrines, on the other hand, which depend upon geometrical and arithmetical relations, are portions or preludes of the sciences which, in the fulness of time, assumed a mathematical form for the expression of truth.

Among these may be mentioned the arithmetical relations of harmonical sounds, to which I have referred in the History. These occur in various parts of Plato's writings. In the _Timæus_, in which the numbers are most fully given, the meaning of the numbers is, at first sight, least obvious. The numbers are given as representing the proportion of the parts of the Soul (_Tim._ pp. 35, 36), which does not immediately refer us to the relations of Sounds. But in a subsequent part of the Dialogue (47, D), we are told that music is a privilege of the hearing given on account of Harmony; and that Harmony has Cycles corresponding to the movements of the Soul; (referring plainly to those already asserted.) And the numbers which are thus given by Plato as elements of harmony, are in a great measure the same as those which express the musical relations of the tones of the musical scale at this day in use, as M. Henri Martin shows (_Et. sur le Timée_, note xxiii.) The intervals C to D, C to F, C to G, C to C, are expressed by the fractions 9/8, 4/3, 3/2, 2/1, and are now called a Tone, a Fourth, a Fifth, an Octave. They were expressed by the same fractions among the Greeks, and were called _Tone_, _Diatessaron_, _Diapente_, _Diapason_. The Major and Minor Third, and the Major and Minor Sixth, were however wanting, it is conceived, in the musical scale of Plato.

The _Timæus_ contains also a kind of theory of vision by reflexion from a plane, and in a concave mirror; although the theory is in this case less mathematical and less precise than that of Euclid, referred to in chap. ii. of this Book.

One of the most remarkable speculations in the _Timæus_ is that in {499} which the Regular Solids are assigned as the forms of the Elements of which the Universe is composed. This curious branch of mathematics, Solid Geometry, had been pursued with great zeal by Plato and his friends, and with remarkable success. The five Regular Solids, the Tetrahedron or regular Triangular Pyramid, the Cube, the Octahedron, the Dodecahedron, and the Icosahedron, had been discovered; and the remarkable theorem, that of regular solids there can be just so many, these and no others, was known. And in the _Timæus_ it is asserted that the particles of the various elements have the forms of these solids. Fire has the Pyramid; Earth has the Cube; Water the Octahedron; Air the Icosahedron; and the Dodecahedron is the plan of the Universe itself. It was natural that when Plato had learnt that other mathematical properties had a bearing upon the constitution of the Universe, he should suppose that the singular property of space, which the existence of this limited and varied class of solids implied, should have some corresponding property in the Universe, which exists in space.

We find afterwards, in Kepler and others, a recurrence to this assumption; and we may say perhaps that Crystallography shows us that there are properties of bodies, of the most intimate kind, which involve such spatial relations as are exhibited in the Regular Solids. If the distinctions of Crystalline System in bodies were hereafter to be found to depend upon the chemical elements which predominate in their composition, the admirers of Plato might point to his doctrine, of the different form of the particles of the different elements of the Universe, as a remote Prelude to such a discovery.

But the mathematical doctrines concerning the parts and elements of the Universe are put forwards by Plato, not so much as assertions concerning physical facts, of

which the truth or falsehood is to be determined by a reference to nature herself. They are rather propounded as examples of a truth of a higher kind than any reference to observation can give or can test, and as revelations of principles such as must have prevailed in the mind of the Creator of the Universe; or else as contemplations by which the mind of man is to be raised above the region of sense, and brought nearer to the Divine Mind. In the _Timæus_ these doctrines appear rather in the former of the two lights; as an exposition of the necessary scheme of creation, so far as its leading features are concerned. In the seventh Book of the _Polity_, the same doctrines are regarded more as a mental discipline; as the necessary study of the true philosopher. But in both places these mathematical {500} propositions are represented as Realities more real than the Phenomena;--as a Natural Philosophy of a higher kind than the study of Nature itself can teach. This is no doubt an erroneous assumption: yet even in this there is a germ of truth; namely, that the mathematical laws, which prevail in the universe, involve mathematical truths which being demonstrative, are of a higher and more cogent kind than mere experimental truths.

Notions, such as these of Plato, respecting a truth at which science is to aim, which is of an exact and demonstrative kind, and is imperfectly manifested in the phenomena of nature, may help or may mislead inquirers; they may be the impulse and the occasion to great discoveries; or they may lead to the assertion of false and the loss of true doctrines. Plato considers the phenomena which nature offers to the senses as mere suggestions and rude sketches of the objects which the philosophic mind is to contemplate. The heavenly bodies and all the splendors of the sky, though the most beautiful of visible objects, being only visible objects, are far inferior to the true objects of which they are the representatives. They are merely diagrams which may assist in the study of the higher truth as we might study geometry by the aid of diagrams constructed by some consummate artist. Even then, the true object about which we reason is the conception which we have in the mind.

We have, I conceive, an instance of the error as well as of the truth, to which such views may lead, in the speculations of Plato concerning Harmony, contained in that part of his writings (the seventh Book of the _Republic_), in which these views are especially urged. He there, by way of illustrating the superiority of philosophical truth over such exactness as the senses can attest, speaks slightingly of those who take immense pains in measuring musical notes and intervals by the ear, as the astronomers measure the heavenly motions by the eye. "They screw their pegs and pinch their strings, and dispute whether two notes are the same or not." Now, in truth, the ear is the final and supreme judge whether two notes are the same or not. But there is a case in which notes which are nominally the same, are different really and to the ear; and it is probably to disputes on this subject, which we know did prevail among the Greek musicians, that Plato here refers. We may ascend from a note A_{1} to a note C_{3} by two octaves and a third. We may also ascend from the same note A_{1} to C_{3} by fifths four times repeated. But the two notes C_{3} thus arrived at are not the same: they differ by a small interval, which the Greeks called a {501} Comma, of which the notes are in the ratio of 80 to 81. That the ear really detects this defect of the musical coincidence of the two notes under the proper

conditions, is a proof of the coincidence of our musical perceptions with the mathematical relations of the notes; and is therefore an experimental confirmation of the mathematical principles of harmony. But it seems to be represented by Plato, that to look out for such confirmation of mathematical principles, implies a disposition to lean on the senses, which he regards as very unphilosophical.

_Hero of Alexandria._

THE other branches of mathematical science which I have spoken of in the History as cultivated by the Greeks, namely Mechanics and Hydrostatics, are not treated expressly by Plato; though we know from Aristotle and others that some of the propositions of those sciences were known about his time. Machines moved not only by weights and springs, but by water and air, were constructed at an early period. Ctesibius, who lived probably about B. C. 250, under the Ptolemies, is said to have invented a clepsydra or water-clock, and an hydraulic organ; and to have been the first to discover the elastic power of air, and to apply it as a moving power. Of his pupil Hero, the name is to this day familiar, through the little pneumatic instrument called _Hero's Fountain_. He also described pumps and hydraulic machines of various kinds; and an instrument which has been spoken of by some modern writers as a _steam-engine_, but which was merely a toy made to whirl round by the steam emitted from holes in its arms. Concerning mechanism, besides descriptions of _Automatons_, Hero composed two works: the one entitled _Mechanics_, or _Mechanical Introductions_; the other _Barulcos_, the _Weight-lifter_. In these works the elementary contrivances by which weights may be lifted or drawn were spoken of as the _Five Mechanical Powers_, the same enumeration of such machines as prevails to this day; namely, the Lever, the Wheel and Axle, the Pulley, the Wedge, and the Screw. In his Mechanics, it appears that Hero reduced all these machines to one single machine, namely to the lever. In the _Barulcos_, Hero proposed and solved the problem which it was the glory of Archimedes to have solved: To move any object (however large) by any power (however small). This, as may easily be conceived by any one acquainted with the elements of Mechanics, is done by means of a combination of the mechanical powers, and especially by means of a train of toothed-wheels and axles. {502}

The remaining writings of Hero of Alexandria have been the subject of a special, careful, and learned examination by M. Th. H. Martin (Paris, 1854), in which the works of this writer, Hero the Ancient, as he is sometimes called, are distinguished from those of another writer of the same name of later date.

Hero of Alexandria wrote also, as it appears, a treatise on _Pneumatics_, in which he described machines, either useful or amusing, moved by the force of air and vapor.

He also wrote a work called _Catoptrics_, which contained proofs of properties of the rays of reflected light.

And a treatise _On the Dioptra_; which subject however must be carefully distinguished from the subject entitled _Dioptrics_ by the moderns. This latter subject treats of the properties of refracted light; a subject on which the ancients had little exact knowledge till a later period; as I have shown in the History. The _Dioptra_, as

understood by Hero, was an instrument for taking angles so as to measure the position and hence to determine the distance of inaccessible objects; as is done by the _Theodolite_ in our times.

M. Martin is of opinion that Hero of Alexandria lived at a later period than is generally supposed; namely, after B. C. 81.

{{503}} BOOK III.

THE GREEK ASTRONOMY.

## INTRODUCTION.

THE mathematical opinions of Plato respecting the philosophy of nature, and especially respecting what we commonly call "the heavenly bodies," the Sun, Moon, and Planets, were founded upon the view which I have already described: namely, that it is the business of philosophy to aim at a truth higher than observation can teach; and to solve problems which the phenomena of the universe only suggest. And though the students of nature in more recent times have learnt that this is too presumptuous a notion of human knowledge, yet the very boldness and hopefulness which it involved impelled men in the pursuit of truth, with more vigor than a more timorous temper could have done; and the belief that there must be, in nature, mathematical laws more exact than experience could discover, stimulated men often to discover true laws, though often also to invent false laws. Plato's writings, supplying examples of both these processes, belong to the Prelude of true Astronomy, as well as to the errors of false philosophy. We may find specimens of both kinds in those parts of his Dialogues to which we have referred in the preceding Book of our History.

To Plato's merits in preparing the way for the Theory of Epicycles, I have already referred in Chapter ii. of this Book. I conceive that he had a great share in that which is an important step in every discovery, the proposing distinctly the problem to be solved; which was, in this case, as he states it, To account for the apparent movements of the planets by a combination of two circular motions for each:--the motion of identity, and the motion of difference. (_Tim._ 39, A.) In the tenth Book of the _Republic_, quoted in our text, the spindle which Destiny or Necessity holds between her knees, and on which are rings, by means of which the planets revolve round it as an axis, is a step towards the conception of the problem, as the construction of a machine.

It will not be thought surprising that Plato expected that {504} Astronomy, when further advanced, would be able to render an account of many things for which she has not accounted even to this day. Thus, in the passage in the seventh Book of the _Republic_, he says that the philosopher requires a reason for the proportion of the day to the month, and the month to the year, deeper and more substantial than mere observation can give. Yet Astronomy has not yet shown us any reason why the proportion of the times of the earth's rotation on its axis, the moon's revolution round the earth, and the earth's revolution round the sun, might not have been made by the

Creator quite different from what they are. But in thus asking Mathematical Astronomy for reasons which she cannot give, Plato was only doing what a great astronomical discoverer, Kepler, did at a later period. One of the questions which Kepler especially wished to have answered was, why there are five planets, and why at such particular distances from the sun? And it is still more curious that he thought he had found the reason of these things, in the relations of those Five Regular Solids which, as we have seen, Plato was desirous of introducing into the philosophy of the universe. We have Kepler's account of this, his imaginary discovery, in the _Mysterium Cosmographicum_, published in 1596, as stated in our History, Book v. Chap. iv. Sect. 2.

Kepler regards the law which thus determines the number and magnitude of the planetary orbits by means of the five regular solids as a discovery no less remarkable and certain than the Three Laws which give his name its imperishable place in the history of astronomy.

We are not on this account to think that there is no steady criterion of the difference between imaginary and real discoveries in science. As discovery becomes possible by the liberty of guessing, it becomes real by allowing observation constantly and authoritatively to determine the value of guesses. Kepler added to Plato's boldness of fancy his own patient and candid habit of testing his fancies by a rigorous and laborious comparison with the phenomena; and thus his discoveries led to those of Newton. {505}

## CHAPTER I.

EARLIEST STAGES OF ASTRONOMY.

_The Globular Form of the Earth._

THERE are parts of Plato's writings which have been adduced as bearing upon the subsequent progress of science; and especially upon the globular form of the earth, and the other views which led to the discovery of America. In the _Timæus_ we read of a great continent lying in the Ocean west of the Pillars of Hercules, which Plato calls _Atlantis_. He makes the personage in his Dialogue who speaks of this put it forward as an Egyptian tradition. M. H. Martin, who has discussed what has been written respecting the Atlantis of Plato, and has given therein a dissertation rich in erudition and of the most lively interest, conceives that Plato's notions on this subject arose from his combining his conviction of the spherical form of the earth, with interpretations of Homer, and perhaps with traditions which were current in Egypt (_Etudes sur le Timée_, Note xiii. § ix.). He does not consider that the belief in Plato's Atlantis had any share in the discoveries of Columbus.

It may perhaps surprise modern readers who have a difficulty in getting rid of the persuasion that there is a natural direction _upwards_ and a natural direction _downwards_, to learn that both Plato and Aristotle, and of course other philosophers also, had completely overcome this difficulty. They were quite ready to allow and to conceive that _down_ meant nothing but towards some centre, and _up_, the opposite

direction. (Aristotle has, besides, an ingenious notion that while heavy bodies, as earth and water, tend to the centre, and light bodies, as fire, tend from the centre, the fifth element, of which the heavenly bodies are composed, tends to move _round_ the centre.)

Plato explains this in the most decided manner in the _Timæus_ (62, C). "It is quite erroneous to suppose that there are two opposite regions in the universe, one above and the other below; and that heavy things naturally tend to the latter place. The heavens are spherical, and every thing tends to the centre; and thus _above_ and _below_ have no real meaning. If there be a solid globe in the middle, {506} and if a person walk round it, he will become the antipodes to himself, and the direction which is _up_ at one time will be _down_ at another."

The notion of _antipodes_, the inhabitants of the part of the globe of the earth opposite to ourselves, was very familiar. Thus in Cicero's _Academic Questions_ (ii. 39) one of the speakers says, "Etiam dicitis esse e regione nobis, e contraria parte terræ, qui adversis vestigiis stant contra nostra vestigia, quos Antipodas vocatis." See also _Tusc. Disp._ i. 28 and v. 24.

_The Heliocentric System among the Ancients._

As the more clear-sighted of the ancients had overcome the natural prejudice of believing that there is an absolute _up_ and _down_, so had they also overcome the natural prejudice of believing that the earth is at rest. Cicero says (_Acad. Quest._ ii. 39), "Hicetas of Syracuse, as Theophrastus tells us, thinks that the heavens, the sun, the moon, the stars, do not move; and that nothing does move but the earth. The earth revolves about her axis with immense velocity; and thus the same effect is produced as if the earth were at rest and the heavens moved; and this, he says, Plato teaches in the _Timæus_, though somewhat obscurely." Of course the assertion that the moon and planets do not move, was meant of the diurnal motion only. The passage referred to in the _Timæus_ seems to be this (40, C)--"As to the Earth, which is our nurse, and which _clings to_ the axis which stretches through the universe, God made her the producer and preserver of day and night." The word εἰλλομένην, which I have translated _clings to_, some translate _revolves_; and an extensive controversy has prevailed, both in ancient and modern times (beginning with Aristotle), whether Plato did or did not believe in the rotation of the earth on her axis. (See M. Cousin's Note on the _Timæus_, and M. Henri Martin's Dissertation, Note xxxvii., in his _Etudes sur le Timée_.) The result of this discussion seems to be that, in the _Timæus_, the Earth is supposed to be at rest. It is however related by Plutarch (_Platonic Questions_, viii. 1), that Plato in his old age repented of having given to the Earth the place in the centre of the universe which did not belong to it.

In describing the Prelude to the Epoch of Copernicus (Book v. Chap. i.), I have spoken of Philolaus, one of the followers of Pythagoras, who lived at the time of Socrates, as having held the doctrine that the earth revolves about the sun. This has been a current {507} opinion;--so current, indeed, that the Abbé Bouillaud, or Bullialdus, as we more commonly call him, gave the title of _Philolaus_ to the defence of Copernicus which he published in 1639; and Chiaramonti, an Aristotelian,

published his answer under the title of _Antiphilolaus_. In 1645 Bullialdus published his _Astronomia Philolaica_, which was another exposition of the heliocentric doctrine.

Yet notwithstanding this general belief, it appears to be tolerably certain that Philolaus did not hold the doctrine of the earth's motion round the sun. (M. H. Martin, _Etudes sur le Timée_, 1841, Note xxxvii. Sect. i.; and Bœckh, _De vera Indole Astronomiæ Philolaicæ_, 1810.) In the system of Philolaus, the earth revolved about _the central fire_; but this central fire was not the sun. The Sun, along with the moon and planets, revolved in circles external to the earth. The Earth had the _Antichthon_ or _Counter-Earth_ between it and the centre; and revolving round this centre in one day, the Antichthon, being always between it and the centre, was, during a portion of the revolution, interposed between the Earth and the Sun, and thus made night; while the Sun, by his proper motion, produced the changes of the year.

When men were willing to suppose the earth to be in motion, in order to account for the recurrence of day and night, it is curious that they did not see that the revolution of a spherical earth about an axis passing through its centre was a scheme both simple and quite satisfactory. Yet the illumination of a globular earth by a distant sun, and the circumstances and phenomena thence resulting, appear to have been conceived in a very confused manner by many persons. Thus Tacitus (_Agric._ xii.), after stating that he has heard that in the northern part of the island of Britain, the night disappears in the height of summer, says, as his account of this phenomenon, that "the extreme parts of the earth are low and level, and do not throw their shadow upwards; so that the shade of night falls below the sky and the stars." But, as a little consideration will show, it is the globular form of the earth, and not the level character of the country, which produces this effect.

It is not in any degree probable that Pythagoras taught that the Earth revolves round the Sun, or that it rotates on its own axis. Nor did Plato hold either of these motions of the Earth. They got so far as to believe in the Spherical Form of the Earth; and this was apparently such an effort that the human mind made a pause before going any further. "It required," says M. H. Martin, "a great struggle for {508} men to free themselves from the prejudices of the senses, and to interpret their testimony in such a manner as to conceive the sphericity of the earth. It is natural that they should have stopped at this point, before putting the earth in motion in space."

Some of the expressions which have been understood, as describing a system in which the Sun is the _centre of motion_, do really imply merely the Sun is the _middle term_ of the series of heavenly bodies which revolve round the earth: the series being Moon, Mercury, Venus, Sun, Mars, Jupiter, Saturn. This is the case, for instance, in a passage of Cicero's _Vision of Scipio_, which has been supposed to imply, (as I have stated in the History,) that Mercury and Venus revolve about the Sun.

But though the doctrine of the diurnal rotation and annual revolution of the earth is not the doctrine of Pythagoras, or of Philolaus, or of Plato, it was nevertheless held by some of the philosophers of antiquity. The testimony of Archimedes that this doctrine was held by his contemporary Aristarchus of Samos, is unquestionable and

there is no reason to doubt Plutarch's assertion that Seleucus further enforced it.

It is curious that Copernicus appears not to have known anything of the opinions of Aristarchus and Seleucus, which were really anticipations of his doctrine; and to have derived his notion from passages which, as I have been showing, contain no such doctrine. He says, in his Dedication to Pope Paul III., "I found in Cicero that Nicetas [or Hicetas] held that the earth was in motion: and in Plutarch I found that some others had been of that opinion: and his words I will transcribe that any one may read them: 'Philosophers in general hold that the earth is at rest. But Philolaus the Pythagorean teaches that it moves round the central fire in an oblique circle, in the same direction as the Sun and the Moon. Heraclides of Pontus and Ecphantus the Pythagorean give the earth a motion, but not a motion of translation; they make it revolve like a wheel about its own centre from west to east.'" This last opinion was a correct assertion of the diurnal motion.

_The Eclipse of Thales._

"THE Eclipse of Thales" is so remarkable a point in the history of astronomy, and has been the subject of so much discussion among astronomers, that it ought to be more especially noticed. The original {509} record is in the first Book of Herodotus's History (chap. lxxiv.) He says that there was a war between the Lydians and the Medes; and after various turns of fortune, "in the sixth year a conflict took place; and on the battle being joined, it happened that the day suddenly became night. And this change, Thales of Miletus had predicted to them, definitely naming this year, in which the event really took place. The Lydians and the Medes, when they saw day turned into night, ceased from fighting; and both sides were desirous of peace." Probably this prediction was founded upon the Chaldean period of eighteen years, of which I have spoken in Section 11. It is probable, as I have already said, that this period was discovered by noticing the recurrence of eclipses. It is to be observed that Thales predicted only the year of the eclipse, not the day or the month. In fact, the exact prediction of the circumstances of an eclipse of the sun is a very difficult problem; much more difficult, it may be remarked, than the prediction of the circumstance of an eclipse of the moon.

Now that the Theory of the Moon is brought so far towards completeness, astronomers are able to calculate backwards the eclipses of the sun which have taken place in former times; and the question has been much discussed in what year this Eclipse of Thales really occurred. The Memoir of Mr. Airy, the Astronomer Royal, on this subject, in the _Phil. Trans._ for 1853, gives an account of the modern examinations of this subject. Mr. Airy starts from the assumption that the eclipse must have been one decidedly total; the difference between such a one and an eclipse only _nearly_ total being very marked. A total eclipse alone was likely to produce so strong an effect on the minds of the combatants. Mr. Airy concludes from his calculations that the eclipse predicted by Thales took place B. C. 585.

Ancient eclipses of the Moon and Sun, if they can be identified, are of great value for modern astronomy; for in the long interval of between two and three thousand years which separates them from our time, those of the _inequalities_, that is,

accelerations or retardations of the Moon's motion, which go on increasing constantly,[4\A] accumulate to a large amount; so that the actual time and circumstances of the eclipse give astronomers the means of determining what the rate of these accelerations or retardations has been. Accordingly Mr. Airy has discussed, as even more important than the eclipse of Thales, an eclipse which Diodorus relates to have happened during an expedition of {510} Agathocles, the ruler of Sicily, and which is hence known as the Eclipse of Agathocles. He determines it to have occurred B. C. 310.

[Note 4\A: Or at least for very long periods.]

M. H. Martin, in Note xxxvii. to his _Etudes sur le Timée_, discusses among other astronomical matters, the Eclipse of Thales. He does not appear to render a very cordial belief to the historical fact of Thales having delivered the prediction before the event. He says that even if Thales did make such a prediction of an eclipse of the sun, as he might do, by means of the Chaldean period of 18 years, or 223 lunations, he would have to take the chance of its being visible in Greece, about which he could only guess:--that no author asserts that Thales, or his successors Anaximander and Anaxagoras, ever tried their luck in the same way again:--that "en revanche" we are told that Anaximander predicted an earthquake, and Anaxagoras the fall of aërolites, which are plainly fabulous stories, though as well attested as the Eclipse of Thales. He adds that according to Aristotle, Thales and Anaximenes were so far from having sound notions of cosmography, that they did not even believe in the roundness of the earth.

{{511}} BOOK IV.

PHYSICAL SCIENCE IN THE MIDDLE AGES.

GENERAL REMARKS.

IN the twelfth Book of the _Philosophy_, in which I have given a Review of Opinions on the Nature of Knowledge and the method of seeking it, I have given some account of several of the most important persons belonging to the ages now under consideration. I have there (vol. ii. b. xii. p. 146) spoken of the manner in which remarks made by Aristotle came to be accepted as fundamental maxims in the schools of the middle ages, and of the manner in which they were discussed by the greatest of the schoolmen, as Thomas Aquinas, Albertus Magnus, and the like. I have spoken also (p. 149) of a certain kind of recognition of the derivation of our knowledge from experience; as shown in Richard of St. Victor, in the twelfth century. I have considered (p. 152) the plea of the admirers of those ages, that religious authority was not claimed for physical science.

I have noticed that the rise of Experimental Philosophy exhibited two features (chap. vii. p. 155), the Insurrection against Authority, and the Appeal to Experience: and as exemplifying these features, I have spoken of Raymond Lully and of Roger Bacon. I have further noticed the opposition to the prevailing Aristotelian dogmatism manifested (chap. viii.) by Nicolas of Cus, Marsilius Ficinus, Francis Patricius, Picus of Mirandula, Cornelius Agrippa, Theophrastus Paracelsus, Robert Fludd. I have gone on to notice the Theoretical Reformers of Science (chap. ix.), Bernardinus Telesius,

Thomas Campanella, Andreas Cæsalpinus, Peter Ramus; and the Protestant Reformers, as Melancthon. After these come the Practical Reformers of Science, who have their place in the subsequent history of Inductive Philosophy; Leonardo da Vinci, and the Heralds of the dawning light of real science, whom Francis Bacon welcomes, as Heralds are accosted in Homer: Χαίρετε Κήρυκες Διὸς ἄγγελοι ἠδὲ καὶ ἀνδρῶν. Hail, Heralds, messengers of Gods and men! {512}

I have, in the part of the _Philosophy_ referred to, discussed the merits and defects of Francis Bacon's _Method_, and I shall have occasion, in the next Book, to speak of his mode of dealing with the positive science of his time. There is room for much more reflexion on these subjects, but the references now made may suffice at present.

## CHAPTER V.

PROGRESS IN THE MIDDLE AGES.

_Thomas Aquinas._

AQUINAS wrote (besides the _Summa_ mentioned in the text) a Commentary on the Physics of Aristotle: _Commentaria in Aristotelis Libros Physicorum_, Venice, 1492. This work is of course of no scientific value; and the commentary consists of empty permutations of abstract terms, similar to those which constitute the main substance of the text in Aristotle's physical speculations. There is, however, an attempt to give a more technical form to the propositions and their demonstrations. As specimens of these, I may mention that in Book vi. c. 2, we have a demonstration that when bodies move, the time and the magnitude (that is, the space described), are divided similarly; with many like propositions. And in Book viii. we have such propositions as this (c. 10): "Demonstration that a finite mover (_movens_) cannot move anything in an infinite time." This is illustrated by a diagram in which two hands are represented as engaged in moving a whole sphere, and one hand in moving a hemisphere.

This mode of representing force, in diagrams illustrative of mechanical reasonings, by human hands pushing, pulling, and the like, is still employed in elementary books. Probably this is the first example of such a mode of representation.

_Roger Bacon._

THIS writer, a contemporary of Thomas Aquinas, exhibits to us a kind of knowledge, speculation, and opinion, so different from that of any known person near his time, that he deserves especial notice here; {513} and I shall transfer to this place the account which I have given of him in the _Philosophy_. I do this the more willingly because I regard the existence of such a work as the _Opus Majus_ at that period as a problem which has never yet been solved. Also I may add, that the scheme of the Contents of this work which I have given, deserves, as I conceive, more notice than it has yet received.

"Roger Bacon was born in 1214, near Ilchester, in Somersetshire, of an old family. In his youth he was a student at Oxford, and made extraordinary progress in all

branches of learning. He then went to the University of Paris, as was at that time the custom of learned Englishmen, and there received the degree of Doctor of Theology. At the persuasion of Robert Grostête, bishop of Lincoln, he entered the brotherhood of Franciscans in Oxford, and gave himself up to study with extraordinary fervor. He was termed by his brother monks _Doctor Mirabilis_. We know from his own works, as well as from the traditions concerning him, that he possessed an intimate acquaintance with all the science of his time which could be acquired from books; and that he had made many remarkable advances by means of his own experimental labors. He was acquainted with Arabic, as well as with the other languages common in his time. In the title of his works, we find the whole range of science and philosophy, Mathematics and Mechanics, Optics, Astronomy, Geography, Chronology, Chemistry, Magic, Music, Medicine, Grammar, Logics, Metaphysics, Ethics, and Theology; and judging from those which are published, these works are full of sound and exact knowledge. He is, with good reason, supposed to have discovered, or to have had some knowledge of, several of the most remarkable inventions which were made generally known soon afterwards; as gunpowder, lenses, burning specula, telescopes, clocks, the correction of the calendar, and the explanation of the rainbow.

"Thus possessing, in the acquirements and habits of his own mind, abundant examples of the nature of knowledge and of the process of invention, Roger Bacon felt also a deep interest in the growth and progress of science, a spirit of inquiry respecting the causes which produced or prevented its advance, and a fervent hope and trust in its future destinies; and these feelings impelled him to speculate worthily and wisely respecting a Reform of the Method of Philosophizing. The manuscripts of his works have existed for nearly six hundred years in many of the libraries of Europe, and especially in those of England; and for a long period the very imperfect portions of them which were {514} generally known, left the character and attainments of the author shrouded in a kind of mysterious obscurity. About a century ago, however, his _Opus Majus_ was published[5\A] by Dr. S. Jebb, principally from a manuscript in the library of Trinity College, Dublin; and this contained most or all of the separate works which were previously known to the public, along with others still more peculiar and characteristic. We are thus able to judge of Roger Bacon's knowledge and of his views, and they are in every way well worthy our attention.

[Note 5\A: _Fratris Rogeri Bacon Ordinis Minorum_ Opus Majus _ad Clementem Quartum, Pontificem Romanum, ex MS. Codice Dubliniensi cum aliis quibusdam collato nunc primum edidit_ S. Jebb, M.D. Londini, 1733.]

"The _Opus Majus_ is addressed to Pope Clement the Fourth, whom Bacon had known when he was legate in England as Cardinal-bishop of Sabina, and who admired the talents of the monk, and pitied him for the persecutions to which he was exposed. On his elevation to the papal chair, this account of Bacon's labours and views was sent, at the earnest request of the pontiff. Besides the _Opus Majus_, he wrote two others, the _Opus Minus_ and _Opus Tertium_; which were also sent to the pope, as the author says,[6\A] 'on account of the danger of roads, and the possible loss of the work.' These works still exist unpublished, in the Cottonian and other libraries.

[Note 6\A: _Opus Majus_, Præf.]

"The _Opus Majus_ is a work equally wonderful with regard to its general scheme, and to the special treatises with which the outlines of the plan are filled up. The professed object of the work is to urge the necessity of a reform in the mode of philosophizing, to set forth the reasons why knowledge had not made a greater progress, to draw back attention to the sources of knowledge which had been unwisely neglected, to discover other sources which were yet almost untouched, and to animate men in the undertaking, by a prospect of the vast advantages which it offered. In the developement of this plan, all the leading portions of science are expounded in the most complete shape which they had at that time assumed; and improvements of a very wide and striking kind are proposed in some of the principal of these departments. Even if the work had had no leading purpose, it would have been highly valuable as a treasure of the most solid knowledge and soundest speculations of the time; even if it had contained no such details, it would have been a work most remarkable for its general views and scope. It may be considered as, at the same time, the _Encyclopedia_ and the _Novum Organon_ of the thirteenth century. {515}

"Since this work is thus so important in the history of Inductive Philosophy I shall give, in a Note, a view[7\A] of its divisions and contents. But I must now endeavor to point out more especially the way in which the various principles, which the reform of scientific method involved, are here brought into view.

[Note 7\A: Contents of Roger Bacon's _Opus Majus_: Part I. On the four causes of human ignorance:--Authority, Custom, Popular Opinion, and the Pride of supposed Knowledge. Part II. On the source of perfect wisdom in the Sacred Scripture. Part III. On the Usefulness of Grammar. Part IV. On the Usefulness of Mathematics. (1.) The Necessity of Mathematics in Human Things (published separately as the _Specula Mathematica_). (2.) The Necessity of Mathematics in Divine Things.--1°. This study has occupied holy men: 2°. Geography: 3°. Chronology: 4°. Cycles; the Golden Number, &c.: 5°. Natural Phenomena, as the Rainbow: 6°. Arithmetic: 7°. Music. (3.) The Necessity of Mathematics in Ecclesiastical Things. 1°. The Certification of Faith: 2°. The Correction of the Calendar. (4.) The Necessity of Mathematics in the State.--1°. Of Climates: 2°. Hydrography: 3°. Geography: 4°. Astrology. Part V. On Perspective (published separately as _Perspectiva_). (1.) The organs of vision. (2.) Vision in straight lines. (3.) Vision reflected and refracted. (4.) De multiplicatione specierum (on the propagation of the impressions of light, heat, &c.) Part VI. On Experimental Science.]

"One of the first points to be noticed for this purpose, is the resistance to authority; and at the stage of philosophical history with which we here have to do, this means resistance to the authority of Aristotle, as adopted and interpreted by the Doctors of the Schools. Bacon's work[8\A] is divided into Six Parts; and of these Parts, the First is, Of the four universal Causes of all Human Ignorance. The causes thus enumerated[9\A] are:--the force of unworthy authority;--traditionary habit;--the imperfection of the undisciplined senses;--and the disposition to conceal our ignorance and to make an ostentatious show of our knowledge. These influences involve every man, occupy every condition. They prevent our obtaining the most

useful and large and fair doctrines of wisdom, the secrets of all sciences and arts. He then proceeds to argue, from the testimony of philosophers themselves, that the authority of antiquity, and especially of Aristotle, is not infallible. 'We find[10\A] their books full of doubts, obscurities, and perplexities. They {516} scarce agree with each other in one empty question or one worthless sophism, or one operation of science, as one man agrees with another in the practical operations of medicine, surgery, and the like arts of secular men. Indeed,' he adds,[11\A] 'not only the philosophers, but the saints have fallen into errors which they have afterwards retracted,' and this he instances in Augustin, Jerome, and others. He gives an admirable sketch of the progress of philosophy from the Ionic School to Aristotle; of whom he speaks with great applause. 'Yet,' he adds, 'those who came after him corrected him in some things, and added many things to his works, and shall go on adding to the end of the world.' Aristotle, he adds, is now called peculiarly[12\A] the Philosopher, 'yet there was a time when his philosophy was silent and unregarded, either on account of the rarity of copies of his works, or their difficulty, or from envy; till after the time of Mahomet, when Avicenna and Averroes, and others, recalled this philosophy into the full light of exposition. And although the Logic and some other works were translated by Boethius from the Greek, yet the philosophy of Aristotle first received a quick increase among the Latins at the time of Michael Scot; who, in the year of our Lord 1230, appeared, bringing with him portions of the books of Aristotle on Natural Philosophy and Mathematics. And yet a small part only of the works of this author is translated, and a still smaller part is in the hands of common students.' He adds further[13\A] (in the Third Part of the _Opus Majus_, which is a Dissertation on Language) that the translations which are current of these writings, are very bad and imperfect. With these views, he is moved to express himself somewhat impatiently[14\A] respecting these works: 'If I had,' he says, 'power over the works of Aristotle, I would have them all burnt; for it is only a loss of time to study in them, and a course of error, and a multiplication of ignorance beyond expression.' 'The common herd of students,' he says, 'with their heads, have no principle by which they can be excited to any worthy employment; and hence they mope and make asses of themselves over their bad translations, and lose their time, and trouble, and money.' {517}

[Note 8\A: _Op. Maj._ p. 1.]

[Note 9\A: Ib. p. 2.]

[Note 10\A: Ib. p. 10.]

[Note 11\A: _Op. Maj._ p. 36.]

[Note 12\A: _Autonomaticè_.]

[Note 13\A: _Op. Maj._ p. 46.]

[Note 14\A: See _Pref._ to Jebb's edition. The passages there quoted, however, are not extracts from the _Opus Majus_, but (apparently) from the _Opus Minus_ (_MS. Cott._ Tib. c . 5). "Si haberem potestatem supra libros Aristotelis, ego facerem omnes cremari; quia non est nisi temporis amissio studere in illis, et causa erroris, et multiplicatio ignorantiæ ultra id quod valeat explicari. . . . Vulgus studentum cum

capitibus suis non habet unde excitetur ad aliquid dignum, et ideo languet et _asininat_ circa male translata, et tempus et studium amittit in omnibus et expensas."]

"The remedies which he recommends for these evils, are, in the first place, the study of that only perfect wisdom which is to be found in the Sacred Scripture;[15\A] in the next place, the study of mathematics and the use of experiment.[16\A] By the aid of these methods, Bacon anticipates the most splendid progress for human knowledge. He takes up the strain of hope and confidence which we have noticed as so peculiar in the Roman writers; and quotes some of the passages of Seneca which we adduced in illustration of this:--that the attempts in science were at first rude and imperfect, and were afterwards improved;--that the day will come, when what is still unknown shall be brought to light by the progress of time and the labors of a longer period;--that one age does not suffice for inquiries so wide and various;--that the people of future times shall know many things unknown to us;--and that the time shall arrive when posterity will wonder that we overlooked what was so obvious. Bacon himself adds anticipations more peculiarly in the spirit of his own time. 'We have seen,' he says, at the end of the work, 'how Aristotle, by the ways which wisdom teaches, could give to Alexander the empire of the world. And this the Church ought to take into consideration against the infidels and rebels, that there may be a sparing of Christian blood, and especially on account of the troubles that shall come to pass in the days of Antichrist; which by the grace of God it would be easy to obviate, if prelates and princes would encourage study, and join in searching out the secrets of nature and art.'

[Note 15\A: Part ii.]

[Note 16\A: Parts iv. v. and vi.]

"It may not be improper to observe here that this belief in the appointed progress of knowledge, is not combined with any overweening belief in the unbounded and independent power of the human intellect. On the contrary, one of the lessons which Bacon draws from the state and prospects of knowledge, is the duty of faith and humility. 'To him,' he says,[17\A] 'who denies the truth of the faith because he is unable to understand it, I will propose in reply the course of nature, and as we have seen it in examples.' And after giving some instances, he adds, 'These, and the like, ought to move men and to excite them to the reception of divine truths. For if, in the vilest objects of creation, truths are found, before which the inward pride of man must bow, and believe though it cannot understand, how much more should man humble his mind before the glorious truths of God!' He had before said:[18\A] 'Man is incapable of perfect wisdom in this life; it is hard for {518} him to ascend towards perfection, easy to glide downwards to falsehoods and vanities: let him then not boast of his wisdom, or extol his knowledge. What he knows is little and worthless, in respect of that which he believes without knowing; and still less, in respect of that which he is ignorant of. He is mad who thinks highly of his wisdom; he most mad, who exhibits it as something to be wondered at.' He adds, as another reason for humility, that he has proved by trial, he could teach in one year, to a poor boy, the marrow of all that the most diligent person could acquire in forty years' laborious and expensive

study.

[Note 17\A: _Op. Maj._ p. 476.]

[Note 18\A: Ib. p. 15.]

"To proceed somewhat more in detail with regard to Roger Bacon's views of a Reform in Scientific Inquiry, we may observe that by making Mathematics and Experiment the two great points of his recommendation, he directed his improvement to the two essential parts of all knowledge, Ideas and Facts, and thus took the course which the most enlightened philosophy would have suggested. He did not urge the prosecution of experiment, to the comparative neglect of the existing mathematical sciences and conceptions; a fault which there is some ground for ascribing to his great namesake and successor Francis Bacon: still less did he content himself with a mere protest against the authority of the schools, and a vague demand for change, which was almost all that was done by those who put themselves forward as reformers in the intermediate time. Roger Bacon holds his way steadily between the two poles of human knowledge; which, as we have seen, it is far from easy to do. 'There are two modes of knowing,' says he;[19\A] 'by argument, and by experiment. Argument concludes a question; but it does not make us feel certain, or acquiesce in the contemplation of truth, except the truth be also found to be so by experience.' It is not easy to express more decidedly the clearly-seen union of exact conceptions with certain facts, which, as we have explained, constitutes real knowledge.

[Note 19\A: _Op. Maj._ p. 445; see also p. 448. "Scientiæ aliæ sciunt sua principia invenire per experimenta, sed conclusiones per argumenta facta ex principiis inventis. Si vero debeant habere experientiam conclusionum suarum particularem et completam, tunc oportet quod habeant per adjutorium istius scientiæ nobilis (experimentalis)."]

"One large division of the _Opus Majus_ is 'On the Usefulness of Mathematics,' which is shown by a copious enumeration of existing branches of knowledge, as Chronology, Geography, the Calendar and (in a separate Part) Optics. There is a chapter,[20\A] in which it is proved {519} by reason, that all science requires mathematics. And the arguments which are used to establish this doctrine, show a most just appreciation of the office of mathematics in science. They are such as follows:--That other sciences use examples taken from mathematics as the most evident:--That mathematical knowledge is, as it were, innate to us, on which point he refers to the well-known dialogue of Plato, as quoted by Cicero:--That this science, being the easiest, offers the best introduction to the more difficult:--That in mathematics, things as known to us are identical with things as known to nature:--That we can here entirely avoid doubt and error, and obtain certainty and truth:--That mathematics is prior to other sciences in nature, because it takes cognizance of quantity, which is apprehended by intuition (_intuitu intellectus_). 'Moreover,' he adds,[21\A] 'there have been found famous men, as Robert, bishop of Lincoln, and Brother Adam Marshman (de Marisco), and many others, who by the power of mathematics have been able to explain the causes of things; as may be seen in the writings of these men, for instance, concerning the Rainbow and Comets, and

the generation of heat, and climates, and the celestial bodies.'

[Note 20\A: Ib. p. 60.]

[Note 21\A: _Op. Maj._ p. 64.]

"But undoubtedly the most remarkable portion of the _Opus Majus_ is the Sixth and last Part, which is entitled 'De Scientia experimentali.' It is indeed an extraordinary circumstance to find a writer of the thirteenth century, not only recognizing experiment as one source of knowledge, but urging its claims as something far more important than men had yet been aware of, exemplifying its value by striking and just examples, and speaking of its authority with a dignity of diction which sounds like a foremurmur of the Baconian sentences uttered nearly four hundred years later. Yet this is the character of what we here find.[22\A] 'Experimental science, the sole mistress of speculative sciences, has three great Prerogatives among other parts of knowledge: First she tests by experiment the noblest conclusions of all other sciences: Next she discovers respecting the notions which other sciences deal with, magnificent truths to which these sciences of themselves can by no means attain: her Third dignity is, that she by her own power and without respect of other sciences, investigates the secrets of nature.' {520}

[Note 22\A: "Veritates magnificas in terminis aliarum scientiarum in quas per nullam viam possunt illæ scientiæ, hæc sola scientiarum domina speculativarum, potest dare."--_Op. Maj._ p. 465.]

"The examples which Bacon gives of these 'Prerogatives' are very curious, exhibiting, among some error and credulity, sound and clear views. His leading example of the First Prerogative is the Rainbow, of which the cause, as given by Aristotle, is tested by reference to experiment with a skill which is, even to us now, truly admirable. The examples of the Second Prerogative are three--_first_, the art of making an artificial sphere which shall move with the heavens by natural influences, which Bacon trusts may be done, though astronomy herself cannot do it--'et tunc,' he says, 'thesaurum unius regis valeret hoc instrumentum;'--_secondly_, the art of prolonging life, which experiment may teach, though medicine has no means of securing it except by regimen;[23\A]--_thirdly_, the art of making gold finer than fine gold, which goes beyond the power of alchemy. The Third Prerogative of experimental science, arts independent of the received sciences, is exemplified in many curious examples, many of them whimsical traditions. Thus it is said that the character of a people may be altered by altering the air.[24\A] Alexander, it seems, applied to Aristotle to know whether he should exterminate certain nations which he had discovered, as being irreclaimably barbarous; to which the philosopher replied, 'If you can alter their air, permit them to live; if not, put them to death.' In this part, we find the suggestion that the fire-works made by children, of saltpetre, might lead to the invention of a formidable military weapon.

[Note 23\A: One of the ingredients of a preparation here mentioned, is the flesh of a dragon, which, it appears, is used as food by the Ethiopians. The mode of preparing this food cannot fail to amuse the reader. "Where there are good flying dragons, by the art which they possess, they draw them out of their dens, and have bridles and

saddles in readiness, and they ride upon them, and make them bound about in the air in a violent manner, that the hardness and toughness of the flesh may be reduced, as boars are hunted and bulls are baited before they are killed for eating."--_Op. Maj._ p. 470.]

[Note 24\A: _Op. Maj._ p. 472.]

"It could not be expected that Roger Bacon, at a time when experimental science hardly existed, could give any _precepts_ for the discovery of truth by experiment. But nothing can be a better _example_ of the method of such investigation, than his inquiry concerning the cause of the Rainbow. Neither Aristotle, nor Avicenna, nor Seneca, he says, have given us any clear knowledge of this matter, but experimental science can do so. Let the experimenter (_experimentator_) consider the cases in which he finds the same colors, as the hexagonal crystals from Ireland and India; by looking into these he will see colors like those of the rainbow. Many think that this arises from some {521} special virtue of these stones and their hexagonal figure; let therefore the experimenter go on, and he will find the same in other transparent stones, in dark ones as well as in light-colored. He will find the same effect also in other forms than the hexagon, if they be furrowed in the surface, as the Irish crystals are. Let him consider too, that he sees the same colors in the drops which are dashed from oars in the sunshine;--and in the spray thrown by a mill wheel;--and in the dew drops which lie on the grass in a meadow on a summer morning;--and if a man takes water in his mouth and projects it on one side into a sunbeam;--and if in an oil lamp hanging in the air, the rays fall in certain positions upon the surface of the oil;--and in many other ways, are colors produced. We have here a collection of instances, which are almost all examples of the same kind as the phenomenon under consideration; and by the help of a principle collected by induction from these facts, the colors of the rainbow were afterwards really explained.

"With regard to the form and other circumstances of the bow he is still more precise. He bids us measure the height of the bow and of the sun, to show that the centre of the bow is exactly opposite to the sun. He explains the circular form of the bow,--its being independent of the form of the cloud, its moving when we move, its flying when we follow,--by its consisting of the reflections from a vast number of minute drops. He does not, indeed, trace the course of the rays through the drop, or account for the precise magnitude which the bow assumes; but he approaches to the verge of this part of the explanation; and must be considered as having given a most happy example of experimental inquiry into nature, at a time when such examples were exceedingly scanty. In this respect, he was more fortunate than Francis Bacon, as we shall hereafter see.

"We know but little of the biography of Roger Bacon, but we have every reason to believe that his influence upon his age was not great. He was suspected of magic, and is said to have been put into close confinement in consequence of this charge. In his work he speaks of Astrology, as a science well worth cultivating. 'But,' says he, 'Theologians and Decretists, not being learned in such matters, and seeing that evil as well as good may be done, neglect and abhor such things, and reckon them among Magic Arts.' We have already seen, that at the very time when Bacon was thus raising

his voice against the habit of blindly following authority, and seeking for all science in Aristotle, Thomas Aquinas was employed in fashioning Aristotle's tenets into that fixed form in which they became the great impediment to the {522} progress of knowledge. It would seem, indeed, that something of a struggle between the progressive and stationary powers of the human mind was going on at this time. Bacon himself says,[25\A] 'Never was there so great an appearance of wisdom, nor so much exercise of study in so many Faculties, in so many regions, as for this last forty years. Doctors are dispersed everywhere, in every castle, in every burgh, and especially by the students of two Orders, (he means the Franciscans and Dominicans, who were almost the only religious orders that distinguished themselves by an application to study,[26\A]) which has not happened except for about forty years. And yet there was never so much ignorance, so much error.' And in the part of his work which refers to Mathematics, he says of that study,[27\A] that it is the door and the key of the sciences; and that the neglect of it for thirty or forty years has entirely ruined the studies of the Latins. According to these statements, some change, disastrous to the fortunes of science, must have taken place about 1230, soon after the foundation of the Dominican and Franciscan Orders.[28\A] Nor can we doubt that the adoption of the Aristotelian philosophy by these two Orders, in the form in which the Angelical Doctor had systematized it, was one of the events which most tended to defer, for three centuries, the reform which Roger Bacon urged as a matter of crying necessity in his own time."

[Note 25\A: Quoted by Jebb, Pref. to _Op. Maj._]

[Note 26\A: Mosheim, _Hist._ iii. 161.]

[Note 27\A: _Op. Maj._ p. 57.]

[Note 28\A: Mosheim, iii. 161.]

It is worthy of remark that in the _Opus Majus_ of Roger Bacon, as afterwards in the _Novum Organon_ of Francis Bacon, we have certain features of experimental research pointed out conspicuously as _Prærogativæ_: although in the former, this term is employed to designate the superiority of experimental science in general to the science of the schools; in the latter work, the term is applied to certain classes of experiments as superior to others.

{{523}} BOOK V.

FORMAL ASTRONOMY.

## CHAPTER I.

PRELUDE TO COPERNICUS.

_Nicolas of Cus._

I WILL quote the passage, in the writings of this author, which bears upon the subject in question. I translate it from the edition of his book _De Docta Ignorantia_, from his works published at Basil in 1565. He praises _Learned Ignorance_--that is, Acknowledged Ignorance--as the source of knowledge. His ground for asserting the

motions of the earth is, that there is no such thing as perfect rest, or an exact centre, or a perfect circle, nor perfect uniformity of motion. "Neque verus circulus dabilis est, quinetiam verior dari possit, neque unquam uno tempore sicut alio æqualiter præcisè, aut movetur, aut circulum veri similem, æqualem describit, etiamsi nobis hoc non appareat. Et ubicumque quis fuerit, se in centro esse credit." (Lib. i. cap. xi. p. 39.) He adds, "The Ancients did not attain to this knowledge, because they were wanting in Learned Ignorance. Now it is manifest to us that the Earth is truly in motion, although this do not appear to us; since we do not apprehend motion except by comparison with something fixed. For if any one were in a boat in the middle of a river, ignorant that the water was flowing, and not seeing the banks, how could he apprehend that the boat was moving? And thus since every one, whether he be in the Earth, or in the Sun, or in any other star, thinks that he is in an immovable centre, and that everything else is moving; he would assign different poles for himself, others as being in the Sun, others in the Earth, and others in the Moon, and so of the rest. Whence the machine of the world is as if it had its centre everywhere and its circumference nowhere." This train of thought {524} might be a preparation for the reception of the Copernican system; but it is very different from the doctrine that the Sun is the centre of the Planetary Motions.

## CHAPTER II.

THE COPERNICAN THEORY.

_The Moon's Rotation._

I HAVE said, in page 264, that a confusion of mind produced by the double reference of motion to absolute space, and to a centre of revolution, often leads persons to dispute whether the Moon, while she revolves about the Earth, always turning to it the same face, revolves about her axis or not.

This dispute has been revived very lately, and has been conducted in a manner which shows that popular readers and writers have made little progress in the clearness of their notions during the last two or three centuries; and that they have accepted the Newtonian doctrines in words with a very dim apprehension of their real import.

If the Moon were carried round the Earth by a rigid arm revolving about the Earth as a centre, being rigidly fastened to this arm, as a mimic Moon might be, in a machine constructed to represent her motions, this contrivance, while it made her revolve round the Earth would make her also turn the same face to the Earth: and if we were to make such a machine the standard example of rotation, the Moon might be said not to rotate on her axis.

But we are speedily led to endless confusion by taking this case as the standard of rotation. For the selection of the centre of rotation in a system which includes several bodies is arbitrary. The Moon turns all her faces successively to the Sun, and therefore with regard to the Sun, she does rotate on her axis; and yet she revolves round the Sun as truly as she revolves round the Earth. And the only really simple

and consistent mode of speaking of rotation, is to refer the motion not to any relative centre, but to absolute space.

This is the argument merely on the ground of simplicity and consistency. But we find physical reasons, as well as mathematical, for referring the motion to absolute space. If a cup of water be carried round a centre so as to describe a circle, a straw floating on the surface {525} of the water, if it point to the centre of the circle at first, does not continue to do so, but remains parallel to itself during the whole revolution. Now there is no cause to make the water (and therefore the straw) rotate on its axis; and therefore it is not a clear or convenient way of speaking, to say that the water in this case does revolve on its axis. But if the water in this case do not revolve on its axis, a body in the case of the Moon does revolve on its axis.

The difficulty, as I have said in the text, is of the same nature as that which the Copernicans at first found in the parallel motion of the Earth's axis. In order to make the axis of the Earth's rotation remain parallel to itself while the Earth revolves about the Sun, in a mechanical representation, some machinery is needed _in addition_ to the machinery which produces the revolution round the centre (the Sun): but the simplest way of regarding the parallel motion is, to conceive that the axis has no motion except that which carries it round the central Sun. And it was seen, when the science of Mechanics was established, that no force was needed in nature to produce this parallelism of the Earth's axis. It was therefore the only scientific course, to conceive this parallelism as not being a rotation: and in like manner we are to conceive the parallelism of a revolving body as not being a rotation.

_M. Foucault's Proofs of the Earth's Motion._

IT was hardly to be expected that we should discover, in our own day, a new physical proof of the earth's motion, yet so it has been. The experiments of M. Foucault have enabled us to see the Rotation of the Earth on its axis, as taking place, we may say, before our eyes. These experiments are, in fact, a result of what has been said in speaking of the Moon's rotation: namely, That the mechanical causes of motion operate with reference to absolute, not relative, space; so that where there is no cause operating to change a motion, it will retain its direction in _absolute_ space; and may on that account seem to change, if regarded relatively in a _limited_ space.

In M. Foucault's first experiment, the motion employed was that of a pendulum. If a pendulum oscillate quite freely, there is no cause acting to change the vertical plane of oscillation _absolutely_; for the forces which produce the oscillation are _in_ the vertical plane. But if the vertical plane remain the same _absolutely_, at a spot on the surface of the revolving Earth, it will change _relatively_ to the spectator. He will see the pendulum oscillate in a vertical plane which gradually {526} turns away from its first position. Now this is what really happens; and thus the revolution of the Earth in absolute space is experimentally proved.

In subsequent experiments, M. Foucault has used the rotation of a body to prove the same thing. For when a body rotates freely, acted upon by no power, there is nothing to change the position of the axis of rotation in absolute space. But if the position of the axis remain the same in absolute space, it will, in virtue of its relative

motion, change as seen by a spectator at any spot on the rotating Earth. By taking a heavy disk or globe and making it rotate on its axis rapidly, the force of absolute permanence (as compared with the inevitable casual disturbances arising from the machinery which supports the revolving disk) becomes considerable and hence the relative motion can, in this way also, be made visible.

Mr. De Morgan has said (_Comp. to Brit. Alm._ 1836, p. 18) that astronomy does not supply any argument for the earth's motion which is absolutely and demonstrably conclusive, till we come to the Aberration of Light. But we may now venture to say that the experiments of M. Foucault prove the diurnal motion of the Earth in the most conclusive manner, by palpable and broad effects, if we accept the doctrines of the Science of Mechanics: while Aberration proves the annual motion, if we suppose that we can observe the places of the fixed stars to the accuracy of a few seconds; and if we accept, in addition to the doctrines of Mechanics, the doctrine of the motion of light with a certain great velocity.

## CHAPTER III.

SEQUEL TO COPERNICUS.

_English Copernicans._

PROFESSOR DE MORGAN has made numerous and interesting contributions to the history of the progress and reception of the Copernican System. These are given mainly in the _Companion to the British Almanac_; especially in his papers entitled "Old Arguments against the Motion of the Earth" (1836); "English Mathematical and Astronomical Writers" (1837); "On the Difficulty of Correct {527} Description of Books" (1853); "The Progress of the Doctrine of the Earth's Motion between the Times of Copernicus and Galileo" (1855). In these papers he insists very rightly upon the distinction between the _mathematical_ and the _physical_ aspect of the doctrines of Copernicus: a distinction corresponding very nearly with the distinction which we have drawn between Formal and Physical Astronomy; and in accordance with which we have given the history of the Heliocentric Doctrine as a Formal Theory in Book v., and as a Physical Theory in Book vii.

Another interesting part of Mr. De Morgan's researches are the notices which he has given of the early assertors of the heliocentric doctrine in England. These make their appearance as soon as it was well possible they should exist. The work of Copernicus was published, as we have said, in 1543. In September 1556, John Field published an Ephemeris for 1557, "juxta Copernici et Reinholdi Canones," in the preface to which he avows his conviction of the truth of the Copernican hypothesis. Robert Recorde, the author of various works on Arithmetic, published among others, "The Pathway to Knowledge" in 1551. In this book, the author discusses the question of the "quietnes of the earth," and professes to leave it undecided: but Mr. De Morgan (_Comp. A._ 1837, p. 33) conceives that it appears from what is said, that he was really a Copernican, but did not think the world ripe for any such doctrine.

Mr. Joseph Hunter also has brought to notice[29\A] the claims of Field, whom he designates as the _Proto-Copernican_ of England. He quotes the Address to the Reader prefixed to his first _Ephemeris_, and dated May 31, 1556, in which he says that, since abler men decline the task, "I have therefore published this Ephemeris of the year 1557, following in it as my authorities, N. Copernicus and Erasmus Reinhold, whose writings are established and founded on true, certain, and authentic demonstrations." I conceive that this passage, however, only shows that Field had adopted the Copernican scheme as a basis for the calculation of Ephemerides; which, as Mr. De Morgan has remarked, is a very different thing from accepting it as a physical truth. Field, in this same address, makes mention of the errors "illius turbæ quæ Alphonsi utitur hypothesi;" but the word _hypothesis_ is still indecisive.

[Note 29\A: _Ast. Soc. Notices_, vol. iii. p. 3 (1833).]

As evidence that Field was regarded in his own day as a man who {528} had rendered good service to science, Mr. Hunter notices that, in 1558, the Heralds granted to him the right of using, with his arms, the crest or additional device of a red right arm issuing from the clouds, and presenting a golden armillary sphere.

Recorde's claims depend upon a passage in a Dialogue between _Master_ and _Scholar_, in which the Master expounds the doctrine of Copernicus, and the authorities against it; to which the Scholar answers, taking the common view: "Nay, sir, in good faith I desire not to hear such vaine phantasies, so far against common reason, and repugnant to all the learned multitude of wryters, and therefore let it passe for ever and a day longer." The Master, more sagely, warns him against a hasty judgment, and says, "Another time I will so declare his supposition, that you shall not only wonder to hear it, but also peradventure be as earnest then to credit it, as you now are now to condemne it." I conceive that this passage proves Mr. De Morgan's assertion, that Recorde was a Copernican, and very likely the first in England.

In 1555, also, Leonard Digges published his "Prognostication Everlasting;" but this is, as Mr. De Morgan says (_Comp. A._ 1837, p. 40) a meteorological work. It was republished in 1592 by his son Thomas Digges with additions; and as these have been the occasion of some confusion among those who have written on the history of astronomy, I am glad to be able, through the kindness of Professor Walker of Oxford, to give a distinct account of the editions of the work.

In the Bodleian Library, besides the editions of 1555 and 1592 of the "Prognostication Everlasting," there is an edition of 1564. It is still decidedly Ptolemaic, and contains a Diagram representing a number of concentric circles, which are marked, in order, as-- "The Earth, Moone, Venus, Mercury, Sunne, Mars, Jupiter, Saturne, The Starrie Firmament, The Crystalline Heavens, The First Mover, The Abode of God and the Elect. Here the Learned do approve." {529}

The third edition, of 1592, contains an Addition, by the son, of twenty pages. He there speaks of having found, apparently among his father's papers, "A description or modile of the world and situation of Spheres Cœlestiall and elementare according to the doctrine of Ptolemie, whereunto all universities (led thereunto chiefly by the authoritie of Aristotle) do consent." He adds: "But in this our age, one rare witte

(seeing the continuall errors that from time to time more and more continually have been discovered, besides the infinite absurdities in their Theoricks, which they have been forced to admit that would not confesse any Mobilitie in the ball of the Earth) hath by long studye, paynfull practise, and rare invention, delivered a new Theorick or Model of the world, shewing that the Earth resteth not in the Center of the whole world or globe of elements, which encircled and enclosed in the Moone's orbe, and together with the whole globe of mortalitye is carried yearely round about the Sunne, which like a king in the middest of all, raygneth and giveth lawes of motion to all the rest, sphærically dispersing his glorious beames of light through all this sacred cœlestiall Temple. And the Earth itselfe to be one of the Planets, having his peculiar and strange courses, turning every 24 hours rounde upon his owne centre, whereby the Sunne and great globe of fixed Starres seem to sway about and turne, albeit indeed they remaine fixed--So many ways is the sense of mortal man abused."

This Addition is headed: "A Perfit Description of the Cœlestiall Orbes, according to the most ancient doctrine of the Pythagoreans: lately revived by Copernicus, and by Geometrical Demonstrations approved." Mr. De Morgan, not having seen this edition, and knowing the title-page only as far as the word "Pythagoreans," says "their _astrological_ doctrines we presume, not their reputed _Copernican_ ones." But it now appears that in this, as in other cases, the authority of the Pythagoreans was claimed for the Copernican system. Antony a Wood quotes the latter part of the title thus: "Cui subnectitur _orbium_ Copernicarum accurata descriptio;" which is inaccurate. Weidler, still more inaccurately, cites it, "Cui subnectitur _operum_ Copernici accurata descriptio." Lalande goes still further, attempting, it would seem, to recover the English title-page from the Latin: we find in the _Bibl. Astron._ the following: "1592 . . Leonard Digges, Accurate Description of the Copernican System to the Astronomical perpetual Prognostication."

Thomas Digges appears, by others also of his writings, to have been {530} a clear and decided Copernican. In his "Alæ sive Scalæ Mathematicæ," 1573, he bestows high praise upon Copernicus and upon his system: and appears to have been a believer in the real motion of the Earth, and not merely an admirer of the system of Copernicus as an explanatory hypothesis.

_Giordano Bruno._

The complete title of the work referred to is:

"Jordani Bruni Nolani De Monade Numero et Figura liber consequens Quinque De Minimo Magno et Mensura, item De Innumerabilibus, Immenso et Infigurabili; seu De Universo et Mundis libri octo. (Francofurti, 1591.)"

That the Reader may judge of the value of Bruno's speculations, I give the following quotations:

Lib. iv. c. 11 (Index). "Tellurem totam habitabilem esse _intus_ et extra, et innumerabilia animantium complecti tum nobis sensibilium tum _occultorum_ genera."

C. 13. "Ut Mundorum Synodi in Universo et particulares Mundi in Synodis ordinentur,' &c.

He says (Lib. v. c. 1, p. 461): "Besides the stars and the great worlds there are smaller living creatures carried through the etherial space, in the form of a small sphere which has the aspect of a bright fire, and is by the vulgar regarded as a fiery beam. They are below the clouds, and I saw one which seemed to touch the roofs of the houses. Now this sphere, or beam as they call it, was really a living creature (_animal_), which I once saw moving in a straight path, and grazing as it were the roofs of the city of Nola, as if it were going to impinge on Mount Cicada; which however it went over."

There are two recent editions of the works of Giordano Bruno; by Adolf Wagner, Leipsick, 1830, in two volumes; and by Gfrörer, Berlin, 1833. Of the latter I do not know that more than one volume (vol. ii.) has appeared.

_Did Francis Bacon reject the Copernican System?_

MR. DE MORGAN has very properly remarked (_Comp. B. A._ 1855, p. 11) that the notice of the heliocentric question in the _Novum Organon_ must be considered one of the most important passages in his works upon this point, as being probably the latest written and best {531} matured. It occurs in Lib. ii. Aphorism xxxvi., in which he is speaking of _Prerogative Instances_, of which he gives twenty-seven species. In the passage now referred to, he is speaking of a kind of Prerogative Instances, better known to ordinary readers than most of the kinds by name, the _Instantia Crucis_: though probably the metaphor from which this name is derived is commonly wrongly apprehended. Bacon's meaning is _Guide-Post Instances_: and the _Crux_ which he alludes to is not a Cross, but a Guide-Post at Cross-roads. And among the cases to which such Instances may be applied, he mentions the diurnal motion of the heavens from east to west, and the special motion of the particular heavenly bodies from west to east. And he suggests what he conceives may be an _Instantia Crucis_ in each case. If, he says, we find any motion from east to west in the bodies which surround the earth, slow in the ocean, quicker in the air, quicker still in comets, gradually quicker in planets according to their greater distance from the earth: _then_ we may suppose that there is a cosmical diurnal motion, and the motion of the earth must be denied.

With regard to the special motions of the heavenly bodies, he first remarks that each body not coming quite so far westwards as before, after one revolution of the heavens, and going to the north or the south, does not imply any special motion; since it may be accounted for by a modification of the diurnal motion in each, which produces a defect of the return, and a spiral path; and he says that if we look at the matter as common people[30\A] and disregard the devices of astronomers, the motion is really so to the senses; and that he has made an imitation of it by means of wires. The _instantia crucis_ which he here suggests is, to see if we can find in any credible history an account of any comet which did not share in the diurnal revolution of the skies.

[Note 30\A: Et certissimum est si paulisper pro plebeiis nos geramus (missis astronomorum et scholæ commentis, quibus illud in more est, ut sensui in multis

immerito vim faciant et obscuriora malint) talem esse motum istum ad sensum qualem diximus.]

On his assertion that the motion of each separate planet is, to sense, a spiral, we may remark that it is certainly true; but that the business of science, here, as elsewhere, consists in _resolving_ the complex phenomenon into simple phenomena; the complex spiral motion into simple circular motions.

With regard to the diurnal motion of the earth, it would seem as if Bacon himself had a leaning to believe it when he wrote this passage; for neither is he himself, nor are any of the Anticopernicans, {532} accustomed to assert that the immensely rapid motion of the sphere of the Fixed Stars graduates by a slower and slower motion of Planets, Comets, Air, and Ocean, into the immobility of the Earth. So that the conditions are not satisfied on which he hypothetically says, "tum abnegandus est motus terræ."

With regard to the proper motions of the planets, this passage seems to me to confirm what I have already said of him; that he does not appear to have seen the full value and meaning of what had been done, up to his time, in Formal Astronomy.

We may however fully agree with Mr. De Morgan; that the whole of what he has said on this subject, when put together, does not justify Hume's assertion that he rejected the Copernican system "with the most positive disdain."

Mr. De Morgan, in order to balance the Copernican argument derived from the immense velocity of the stars in their diurnal velocity on the other supposition, has reminded us that those who reject this great velocity as improbable, accept without scruple the greater velocity of light. It is curious that Bacon also has made this comparison, though using it for a different purpose; namely, to show that the transmission of the visual impression may be instantaneous. In Aphorism xlvi. of Book ii. of the _Novum Organon_ he is speaking of what he calls _Instantiæ curriculi_, or _Instantiæ ad aquam_, which we may call _Instances by the clock_: and he says that the great velocity of the diurnal sphere makes the marvellous velocity of the rays of light more credible.

"Immensa illa velocitas in ipso corpore, quæ cernitur in motu diurno (quæ etiam viros graves ita obstupefecit ut _mallent credere motum terræ_), facit motum illum ejaculationis ab ipsis [stellis] (licet celeritate ut diximus admirabilem) magis credibilem." This passage shows an inclination towards the opinion of the earth's being at rest, but not a very strong conviction.

_Kepler persecuted._

WE have seen (p. 280) that Kepler writes to Galileo in 1597--"Be trustful and go forwards. If Italy is not a convenient place for the publication of your views, and if you are likely to meet with any obstacles, perhaps Germany will grant us the necessary liberty." Kepler however had soon afterwards occasion to learn that in Germany also, the cultivators of science were exposed to persecution. It is true that {533} in his case the persecution went mainly on the broad ground of his being a Protestant, and extended to great numbers of persons at that time. The circumstances of this and

other portions of Kepler's life have been brought to light only recently through an examination of public documents in the Archives of Würtemberg and unpublished letters of Kepler. (Johann Keppler's Leben und Wirken, nach neuerlich aufgefundenen Manuscripten bearbeitet von J. L. C. Freiherrn v. Breitschwart, K. Würtemberg. Staats-Rath. Stuttgart, 1831.)

Schiller, in his _History of the Thirty Years' War_, says that when Ferdinand of Austria succeeded to the Archduchy of Stiria, and found a great number of Protestants among his subjects, he suppressed their public worship without cruelty and almost without noise. But it appears now that the Protestants were treated with great severity. Kepler held a professorship in Stiria, and had married, in 1507, Barbara Müller, who had landed property in that province. On the 11th of June, 1598, he writes to his friend Mæstlin that the arrival of the Prince out of Italy is looked forwards to with terror. In December he writes that the Protestants had irritated the Catholics by attacks from the pulpit and by caricatures; that hereupon the Prince, at the prayer of the Estates, had declared the Letter of License granted by his father to be forfeited, and had ordered all the Evangelical Teachers to leave the country on pain of death. They went to the frontiers of Hungary and Croatia; but after a month, Kepler was allowed to return, on condition of keeping quiet. His discoveries appear to have operated in his favor. But the next year he found his situation in Stiria intolerable, and longed to return to his native country of Würtemberg, and to find some position there. This he did not obtain. He wrote a circular letter to his Brother Protestants, to give them consolation and courage; and this was held to be a violation of the conditions on which his residence was tolerated. Fortunately, at this time he was invited to join Tycho Brahe, who had also been driven from his native country, and was living at Prague. The two astronomers worked together under the patronage of the Emperor Rudolph II.; and when Tycho died in 1601, Kepler became the Imperial _Mathematicus_.

We are not to imagine that even among Protestants, astronomical notions were out of the sphere of religious considerations. When Kepler was established in Stiria, his first official business was the calculation of the Calendar for the Evangelical Community. They protested against the new Calendar, as manifestly calculated for the furtherance of an impious papistry: and, say they, "We hold the Pope for a {534} horrible roaring Lion. If we take his Calendar, we must needs go into the church when he rings us in." Kepler however did not fail to see, and to say, that the Papal Reformation of the Calendar was a vast improvement.

Kepler, as court-astronomer, was of course required to provide such observations of the heavens as were requisite for the calculations of the Astrologers. That he considered Astrology to be valuable only as the nurse of Astronomy, he did not hesitate to reveal. He wrote a work with a title of which the following is the best translation which I can give: "_Tertius interveniens_, or: A Warning to certain _Theologi_, _Medici_, _Philosophi_, that while they reasonably reject star-gazing superstition, they do not throw away the kernel with the shell.[31\A] 1610." In this he says, "You over-clever Philosophers blame this Daughter of Astronomy more than is reasonable. Do you not know that she must maintain her mother with her charms?

How many men would be able to make Astronomy their business, if men did not cherish the hope to read the Future in the skies?"

[Note 31\A: The German passage involves a curious image, borrowed, I suppose, from some odd story: "dass sie mit billiger Verwerfung des sternguckerischen Aberglaubens das Kind nicht mit dem Bade ausschütten." "That they do not throw away the child along with the dirty water of his bath."]

_Were the Papal Edicts against the Copernican System repealed?_

ADMIRAL SMYTH, in his _Cycle of Celestial Objects_, vol. i. p. 65, says--"At length, in 1818, the voice of truth was so prevailing that Pius VII. repealed the edicts against the Copernican system, and thus, in the emphatic words of Cardinal Toriozzi, 'wiped off this scandal from the Church.'"

A like story is referred to by Sir Francis Palgrave, in his entertaining and instructive fiction, _The Merchant and the Friar_.

Having made inquiry of persons most likely to be well informed on this subject, I have not been able to learn that there is any further foundation for these statements than this: In 1818, on the revisal of the _Index Expurgatorius_, Galileo's writings were, after some opposition, expunged from that Catalogue.

Monsignor Marino Marini, an eminent Roman Prelate, had addressed to the _Romana Accademia di Archeologia_, certain historico-critical Memoirs, which he published in 1850, with the title _Galileo e l'Inquisizione_. In these, he confirms the conclusion which, I think, almost {535} all persons who have studied the facts have arrived at;[32\A] that Galileo trifled with authority to which he professed to submit, and was punished for obstinate contumacy, not for heresy. M. Marini renders full justice to Galileo's ability, and does not at all hesitate to regard his scientific attainments as among the glories of Italy. He quotes, what Galileo himself quoted, an expression of Cardinal Baronius, that "the intention of the Holy Spirit was to teach how to go to heaven, not how heaven goes."[33\A] He shows that Galileo pleaded (p. 62) that he had not held the Copernican opinion after it had been intimated to him (by Bellarmine in 1616), that he was not to hold it; and that his breach of promise in this respect was the cause of the proceedings against him.

[Note 32\A: M. Marini (p. 29) mentions Leibnitz, Guizot, Spittler, Eichhorn, Raumer, Ranke, among the "storici eterodossi" who have at last done justice to the Roman Church.]

[Note 33\A: Come si vada al Cielo, e non come vada il Cielo.]

Those who admire Galileo and regard him as a martyr because, after escaping punishment by saying "It _does not_ move," he forthwith said "And yet it _does_ move," will perhaps be interested to know that the former answer was suggested to him by friends anxious for his safety. Niccolini writes to Bali Cioli (April 9, 1633) that Galileo continued to be so persuaded of the truth of his opinions that "he was resolved (some moments before his sentence) to defend them stoutly; but I (continues Niccolini) exhorted him to make an end of this; not to mind defending them; and to submit himself to that which he sees that they may desire him to believe or to hold

about this matter of the motion of the earth. He was extremely afflicted." But the Inquisition was satisfied with his answers, and required no more.[34\A]

[Note 34\A: Marini, p. 61.]

{{536}} BOOK VI.

MECHANICS.

## CHAPTER III.

PRINCIPLES AND PROBLEMS.

_Significance of Analytical Mechanics._

IN the text, page 372, I have stated that Lagrange, near the end of his life, expressed his sorrow that the methods of approximation employed in Physical Astronomy rested on arbitrary processes, and not on any insight into the results of mechanical action. From the recent biography of Gauss, the greatest physical mathematician of modern times, we learn that he congratulated himself on having escaped this error. He remarked[35\A] that many of the most celebrated mathematicians, Euler very often, Lagrange sometimes, had trusted too much to the symbolical calculation of their problems, and would not have been able to give an account of the meaning of each successive step of their investigation. He said that he himself, on the other hand, could assert that at every step which he took, he always had the aim and purpose of his operations before his eyes without ever turning aside from the way. The same, he remarked, might be said of Newton.

[Note 35\A: Gauss, _Zum Gedächtniss, von W. Sartorius v. Waltershausen_, p. 80.]

_Engineering Mechanics._

The principles of the science of Mechanics were discovered by observations made upon bodies within the reach of men; as we have seen in speaking of the discoveries of Stevinus, Galileo, and others, up to the time of Newton. And when there arose the controversy about _vis viva_ (Chap. v. Sect. 2 of this Book);--namely, whether the "living force" of a body is measured by the product of the weight into the {537} velocity, or of the weight into the square of the velocity;--still the examples taken were cases of action in machines and the like terrestrial objects. But Newton's discoveries identified celestial with terrestrial mechanics; and from that time the mechanical problems of the heavens became more important and attractive to mathematicians than the problems about earthly machines. And thus the generalizations of the problems, principles, and methods of the mathematical science of Mechanics from this period are principally those which have reference to the motions of the heavenly bodies: such as the Problem of Three Bodies, the Principles of the Conservation of Areas, and of the Immovable Plane, the Method of Variation of Parameters, and the like (Chap. vi. Sect. 7 and 14). And the same is the case in the more recent progress of that subject, in the hands of Gauss, Bessel, Hansen, and others.

But yet the science of Mechanics as applied to terrestrial machines--_Industrial Mechanics_, as it has been termed--has made some steps which it may be worth while to notice, even in a general history of science. For the most part, all the most general laws of mechanical action being already finally established, in the way which we have had to narrate, the determination of the results and conditions of any combination of materials and movements becomes really a mathematical deduction from known principles. But such deductions may be made much more easy and much more luminous by the establishment of general terms and general propositions suited to their special conditions. Among these I may mention a new abstract term, introduced because a general mechanical principle can be expressed by means of it, which has lately been much employed by the mathematical engineers of France, MM. Poncelet, Navier, Morin, &c. The abstract term is _Travail_, which has been translated _Laboring Force_; and the principle which gives it its value, and makes it useful in the solution of problems, is this;--that the _work done_ (in overcoming resistance or producing any other effect) is equal to the _Laboring Force_, by whatever contrivances the force be applied. This is not a new principle, being in fact mathematically equivalent to the conservation of Vis Viva; but it has been employed by the mathematicians of whom I have spoken with a fertility and simplicity which make it the mark of a new school of _The Mechanics of Engineering_.

The Laboring Force expended and the work done have been described by various terms, as _Theoretical Effect_ and _Practical Effect_, and the like. The usual term among English engineers for the work {538} which an Engine usually does, is _Duty_; but as this word naturally signifies what the engine _ought_ to do, rather than what it does, we should at least distinguish between the Theoretical and the Actual Duty.

The difference between the Theoretical and Actual Duty of a Machine arises from this: that a portion of the Laboring Force is absorbed in producing effects, that is, in doing work which is not reckoned as Duty: for instance, overcoming the resistance and waste of the machine itself. And so long as this resistance and waste are not rightly estimated, no correspondence can be established between the theoretical and the practical Duty. Though much had been written previously upon the theory of the steam-engine, the correspondence between the Force expended and the Work done was not clearly made out till Comte De Pambour published his _Treatise on Locomotive Engines_ in 1835, and his _Theory of the Steam-Engine_ in 1839.

_Strength of Materials._

Among the subjects which have specially engaged the attention of those who have applied the science of Mechanics to practical matters, is the strength of materials: for example, the strength of a horizontal beam to resist being broken by a weight pressing upon it. This was one of the problems which Galileo took up. He was led to his study of it by a visit which he made to the arsenal and dockyards of Venice, and the conclusions which he drew were published in his _Dialogues_, in 1633. In his mode of regarding the problem, he considers the section at which the beam breaks as the short arm of a bent lever which resists fracture, and the part of the beam which is broken off as the longer arm of the lever, the lever turning about the fracture as a

hinge. So far this is true; and from this principle he obtained results which are also true as, that the strength of a rectangular beam is proportional to the breadth multiplied into the square of the depth:--that a hollow beam is stronger than a solid beam of the same mass; and the like.

But he erred in this, that he supposed the hinge about which the breaking beam turns, to be exactly at the unrent surface, that surface resisting all change, and the beam being rent all the way across. Whereas the fact is, that the unrent surface yields to compression, while the opposite surface is rent; and the hinge about which the breaking beam turns is at an intermediate point, where the extension {539} and rupture end, and the compression and crushing begin: a point which has been called _the neutral axis_. This was pointed out by Mariotte; and the notion, once suggested, was so manifestly true that it was adopted by mathematicians in general. James Bernoulli,[36\A] in 1705, investigated the strength of beams on this view; and several eminent mathematicians pursued the subject; as Varignon, Parent, and Bulfinger; and at a later period, Dr. Robison in our own country.

[Note 36\A: _Opera_, ii. p. 976.]

But along with the fracture of beams, the mathematicians considered also another subject, the flexure of beams, which they undergo before they break, in virtue of their elasticity. What is the _elastic curve_?--the curve into which an elastic line forms itself under the pressure of a weight--is a problem which had been proposed by Galileo, and was fully solved, as a mathematical problem, by Euler and others.

But beams in practice are not mere lines: they are solids. And their resistance to flexure, and the amount of it, depends upon the resistance of their internal parts to extension and compression, and is different for different substances. To measure these differences, Dr. Thomas Young introduced the notion of the _Modulus of Elasticity_:[37\A] meaning thereby a column of the substance of the same diameter, such as would by its weight produce a compression equal to the whole length of the beam, the rate of compression being supposed to continue the same throughout. Thus if a rod of any kind, 100 inches long, were compressed 1 inch by a weight 1000 pounds, the weight of its modulus of elasticity would be 100,000 pounds. This notion assumes Hooke's law that the extension of a substance is as its tension; and extends this law to compression also.

[Note 37\A: Lecture xiii. The height of the modulus is the same for the same substance, whatever its breadth and thickness may be; for atmospheric air it is about five miles, and for steel nearly 1500 miles.]

There is this great advantage in introducing the definition of the Modulus of Elasticity,--that it applies equally to the flexure of a substance and to the minute vibrations which propagate sound, and the like. And the notion was applied so as to lead to curious and important results with regard to the power of beams to resist flexure, not only when loaded transversely, but when pressed in the direction of their length, and in any oblique direction.

But in the fracture of beams, the resistance to extension and to compression are not practically equal; and it was necessary to determine {540} the difference of these

two forces by experiments. Several persons pursued researches on this subject; especially Mr. Barlow, of the Royal Military Academy,[38\A] who investigated the subject with great labor and skill, so far as wood is concerned. But the difference between the resistance to tension and to compression requires more special study in the case of iron; and has been especially attended to in recent times, in consequence of the vast increase in the number of iron structures, and in particular, railways. It appears that wrought iron yields to compressive somewhat more easily than to tensile force, while cast iron yields far more easily to tensile than to compressive strains. In all cases the power of a beam to resist fracture resides mainly in the upper and the under side, for there the tenacity of the material acts at the greatest leverage round the hinge of fracture. Hence the practice was introduced of making iron beams with a broad _flange_ at the upper and another flange at the under side, connected by a vertical plate or _web_, of which the office was to keep the two flanges asunder. Mr. Hodgkinson made many valuable experiments, on a large scale, to determine the forms and properties of such beams.

[Note 38\A: _An Essay on the Strength and Shape of Timber_. 3d edition, 1826.]

But though engineers were, by such experiments and reasonings, enabled to calculate the strength of a given iron beam, and the dimensions of a beam which should bear a given load, it would hardly have occurred to the boldest speculator, a few years ago, to predict that there might be constructed beams nearly 500 feet long, resting merely on their two extremities, of which it could be known beforehand, that they would sustain, without bending or yielding in any perceptible degree, the weight of a railroad train, and the jar of its unchecked motion. Yet of such beams, constructed beforehand with the most perfect confidence, crowned with the most complete success, is composed the great tubular bridge which that consummate engineer, Mr. Robert Stephenson, has thrown across the Menai Strait, joining Wales with the island of Anglesey. The upper and under surfaces of this quadrangular tube are the flanges of the beam, and the two sides are the webs which connect them. In planning this wonderful structure, the point which required especial care was to make the upper surface strong enough to resist the compressive force which it has to sustain; and this was done by constructing the upper part of the beam of a series of cells, made of iron plate. The application of the arch, of the dome, and of groined vaulting, to the widest space over which they have ever been thrown, {541} are achievements which have, in the ages in which they occurred, been received with great admiration and applause; but in those cases the principle of the structure had been tried and verified for ages upon a smaller scale. Here not only was the space thus spanned wider than any ever spanned before, but the principle of such a beam with a cellular structure of its parts, was invented for this very purpose, experimentally verified with care, and applied with the most exact calculation of its results.

_Roofs--Arches--Vaults._

The calculations of the mechanical conditions of structures consisting of several beams, as for instance, the frames of roofs, depends upon elementary principles of mechanics; and was a subject of investigation at an early period of the science. Such

frames may be regarded as assemblages of levers. The parts of which they consist are rigid beams which sustain and convey force, and _Ties_ which resist such force by their tension. The former parts must be made rigid in the way just spoken of with regard to iron beams; but ties may be rods merely. The wide structures of many of the roofs of railway stations, compared with the massive wooden roofs of ancient buildings, may show us how boldly and how successfully this distinction has been carried out in modern times. The investigation of the conditions and strength of structures consisting of wooden beams has been cultivated by Mathematicians and Engineers, and is often entitled _Carpentry_ in our Mechanical Treatises. In our own time, Dr. Robison and Dr. Thomas Young have been two of the most eminent mathematicians who have written upon this subject.

The properties of the simple machines have been known, as we have narrated, from the time of the Ancient Greeks. But it is plain that such machines are prevented from producing their full effect by various causes. Among the rest, the rubbing of one part of the machine upon another produces an obstacle to the effectiveness of a machine: for instance, the rubbing of the axle of a wheel in the hole in which it rests, the rubbing of a screw against the sides of its hollow screw; the rubbing of a wedge against the sides of its notch; the rubbing of a cord against its pulley. In all these cases, the effect of the machine to produce motion is diminished by the friction. And this _Friction_ may be measured and its effects calculated; and thus we have a new branch of mechanics, which has been much cultivated. {542}

Among the effects of friction, we may notice the standing of a stone arch. For each of the vaulting stones of an arch is a truncated wedge; and though a collection of such stones might be so proportioned in their weights as to balance exactly, yet this balance would be a tottering equilibrium, which the slightest shock would throw down, and which would not practically subsist. But the friction of the vaulting stones against one another prevents this instability from being a practical inconvenience; and makes an equilibrated arch to be an arch strong for practical purposes. The _Theory of Arches_ is a portion of Mechanics which has been much cultivated, and which has led to conclusions of practical use, as well as of theoretical beauty.

I have already spoken of the invention of the Arch, the Dome, and Groined Vaulting, as marked steps in building. In all these cases the invention was devised by practical builders; and mechanical theory, though it can afterwards justify these structures, did not originally suggest them. They are not part of the result, nor even of the application of theory, but only of its exemplification. The authors of all these inventions are unknown; and the inventions themselves may be regarded as a part of the Prelude of the science of mechanics, because they indicate that the ideas of mechanical pressure and support, in various forms, are acquiring clearness and fixity.

In this point of view, I spoke (Book iv. chap. v. sect. 5) of the Architecture of the Middle Ages as indicating a progress of thought which led men towards the formation of Statics as a science.

As particular instances of the operation of such ideas, we have the _Flying Buttresses_ which support stone vaults; and especially, as already noted, the various

contrivances by which stone vaults are made to intersect one another, so as to cover a complex pillared space below with _Groined Vaulting_. This invention, executed as it was by the builders of the twelfth and succeeding centuries, is the most remarkable advance in the mechanics of building, after the invention of the _Arch_ itself.

It is curious that it has been the fortune of our times, among its many inventions, to have produced one in this department, of which we may say that it is the most remarkable step in the mechanics of arches which has been made since the introduction of pointed groined vaults. I speak of what are called _Skew Arches_, in which the courses of stone or brick of which the bridge is built run obliquely to the walls of the bridge. Such bridges have become very common in the works of railroads; for they save space and material, and the {543} invention once made, the cost of the ingenuity is nothing. Of course, the mechanical principles involved in such structures are obvious to the mathematician, when the problem has been practically solved. And in this case, as in the previous cardinal inventions in structure, though the event has taken place within a few years, no single person, so far as I am aware, can be named as the inventor.[39\A]

[Note 39\A: Since this was written, I have been referred to Rees's _Cyclopædia_, Article _Oblique Arches_, where this invention is correctly explained, and is claimed for an engineer named Chapman. It is there said, that the first arch of this kind was erected in 1787 at Naas, near Kildare in Ireland.]

{{544}} BOOK VII.

PHYSICAL ASTRONOMY.

## CHAPTER I.

PRELUDE TO NEWTON.

_The Ancients._

EXPRESSIONS in ancient writers which may be interpreted as indicating a notion of gravitation in the Newtonian sense, no doubt occur. But such a notion, we may be sure, must have been in the highest degree obscure, wavering, and partial. I have mentioned (Book i. Chap. 3) an author who has fancied that he traces in the works of the ancients the origin of most of the vaunted discoveries of the moderns. But to ascribe much importance to such expressions would be to give a false representation of the real progress of science. Yet some of Newton's followers put forward these passages as well deserving notice; and Newton himself appears to have had some pleasure in citing such expressions; probably with the feeling that they relieved him of some of the odium which, he seems to have apprehended, hung over new discoveries. The Preface to the _Principia_, begins by quoting[40\A] the authority of the ancients, as well as the moderns, in favor of applying the science of Mechanics to Natural Philosophy. In the Preface to David Gregory's _Astronomiæ Physicæ et Geometricæ Elementa_, published in 1702, is a large array of names of ancient authors, and of quotations, to prove the early and wide diffusion of the doctrine of the gravity of the Heavenly Bodies. And it appears to be now made out, that this

collection of ancient authorities {545} was supplied to Gregory by Newton himself. The late Professor Rigaud, in his _Historical Essay on the First Publication of Sir Isaac Newton's Principia_, says (pp. 80 and 101) that having been allowed to examine Gregory's papers, he found that the quotations given by him in his Preface are copied or abridged from notes which Newton had supplied to him in his own handwriting. Some of the most noticeable of the quotations are those taken from Plutarch's Dialogue _on the Face which appears in the Moon's Disk_: it is there said, for example, by one of the speakers, that the Moon is perhaps prevented from falling to the earth by the rapidity of her revolution round it; as a stone whirled in a sling keeps it stretched. Lucretius also is quoted, as teaching that all bodies would descend with an equal celerity in a vacuum: Omnia quapropter debent per inane quietum Æque ponderibus non æquis concita ferri. Lib. ii. v. 238.

[Note 40\A: Cum veteres _Mechanicam_ (uti author est _Pappus_), in rerum Naturalium investigatione maximi fecerint, et recentiores, missis formis substantialibus et qualitatibus occultis, Phenonmena Naturæ ad leges mathematicas revocare aggressa sunt; visum est in hoc Tractatu _Mathesin_ excolere quatenus ea ad _Philosophiam_ spectat.]

It is asserted in Gregory's Preface that Pythagoras was not unacquainted with the important law of gravity, the inverse squares of the distances from the centre. For, it is argued, the seven strings of Apollo's lyre mean the seven planets; and the proportions of the notes of strings are reciprocally as the inverse squares of the weights which stretch them.

I have attempted, throughout this work, to trace the progress of the discovery of the great truths which constitute real science, in a more precise manner than that which these interpretations of ancient authors exemplify.

_Jeremiah Horrox._

In describing the Prelude to the Epoch of Newton, I have spoken (p. 395) of a group of philosophers in England who began, in the first half of the seventeenth century, to knock at the door where Truth was to be found, although it was left for Newton to force it open; and I have there noticed the influence of the civil wars on the progress of philosophical studies. To the persons thus tending towards the true physical theory of the solar system, I ought to have added Jeremy Horrox, whom I have mentioned in a former part (Book v. chap. 5) as one of the earliest admirers of Kepler's discoveries. He died at the early age of twenty-two, having been the first person who ever saw Venus pass across the disk of the Sun according to astronomical prediction, which took place in 1639. His _Venus in sole visa_, {546} in which this is described, did not appear till 1661, when it was published by Hevelius of Dantzic. Some of his papers were destroyed by the soldiers in the English civil wars; and his remaining works were finally published by Wallis, in 1673. The passage to which I here specially wish to refer is contained in a letter to his astronomical ally, William Crabtree, dated 1638. He appears to have been asked by his friend to suggest some cause for the motion of the aphelion of a planet; and in reply, he uses an experimental illustration which was afterwards employed by Hooke in 1666. A ball at the end of a

string is made to swing so that it describes an oval. This contrivance Hooke employed to show the way in which an orbit results from the combination of a projectile motion with a central force. But the oval does not keep its axis constantly in the same position. The apsides, as Horrox remarked, move in the same direction as the pendulum, though much slower. And it is true, that this experiment does illustrate, in a general way, the cause of the motion of the aphelia of the Planetary Orbits; although the form of the orbit is different in the experiment and in the solar system; being an ellipse with the centre of force in the centre of the ellipse, in the former case, and an ellipse with the centre of force in the focus, in the latter case. These two forms of orbits correspond to a central force varying directly as the distance, and a central force varying inversely as the square of the distance; as Newton proved in the _Principia_. But the illustration appears to show that Horrox pretty clearly saw how an orbit arose from a central force. So far, and no further, Newton's contemporaries could get; and then he had to help them onwards by showing what was the law of the force, and what larger truths were now attainable.

_Newton's Discovery of Gravitation._

[Page 402.] As I have already remarked, men have a willingness to believe that great discoveries are governed by casual coincidences, and accompanied by sudden revolutions of feeling. Newton had entertained the thought of the moon being retained in her orbit by gravitation as early as 1665 or 1666. He resumed the subject and worked the thought out into a system in 1684 and 5. What induced him to return to the question? What led to his success on this last occasion? With what feelings was the success attended? It is easy to make an imaginary connection of facts. "His optical discoveries had recommended him to the Royal Society, and he was now a member. He {547} _there_ learned the accurate measurement of the Earth by Picard, differing very much from the estimation by which he had made his calculation in 1666; and he thought his conjecture now more likely to be just."[41\A] M. Biot gives his assent to this guess.[42\A] The English translation of M. Biot's biography[43\A] converts the guess into an assertion. But, says Professor Rigaud,[44\A] Picard's measurement of the Earth was well known to the Fellows of the Royal Society as early as 1675, there being an account of the results of it given in the _Philosophical Transactions_ for that year. Moreover, Norwood, in his _Seaman's Practice_, dated 1636, had given a much more exact measure than Newton employed in 1666. But Norwood, says Voltaire, had been buried in oblivion by the civil wars. No, again says the exact and truth-loving Professor Rigaud, Norwood was in communication with the Royal Society in 1667 and 1668. So these guesses at the accident which made the apple of 1665 germinate in 1684, are to be carefully distinguished from history.

[Note 41\A: Robison's _Mechanical Philosophy_, vol. iii. p. 94. (Art. 195.)]

[Note 42\A: _Biographie Universelle_.]

[Note 43\A: _Library of Useful Knowledge_.]

[Note 44\A: _Historical Essay on the First Publication of the Principia_ (1838).]

But with what feelings did Newton attain to his success? Here again we have, I fear, nothing better than conjecture. "He went home, took out his old papers, and

resumed his calculations. As they drew near to a close, he was so much agitated that he was obliged to desire a friend to finish them. His former conjecture was now found to agree with the phænomena with the utmost precision."[45\A] This conjectural story has been called "a tradition;" but he who relates it does not call it so. Every one must decide, says Professor Rigaud, from his view of Newton's character, how far he thinks it consistent with this statement. Is it likely that Newton, so calm and so indifferent to fame as he generally showed himself, should be thus agitated on such an occasion? "No," says Sir David Brewster; "it is not supported by what we know of Newton's character."[46\A] To this we may assent; and this conjectural incident we must therefore, I conceive, separate from history. I had incautiously admitted it into the text of the first Edition.

[Note 45\A: Robison, ibid.]

[Note 46\A: _Life of Newton_, vol. i. p. 292.]

Newton appears to have discovered the method of demonstrating that a body might describe an ellipse when acted upon by a force residing in the focus, and varying inversely as the square of the distance, in 1669, upon occasion of his correspondence with Hooke. In 1684, {548} at Halley's request, he returned to the subject; and in February, 1685 there was inserted in the Register of the Royal Society a paper of Newton's (_Isaaci Newtoni Propositiones de Motu_), which contained some of the principal propositions of the first two Books of the _Principia_. This paper, however, does not contain the proposition "Lunam gravitare in Terram," nor any of the propositions of the Third Book.

## CHAPTER III.

THE PRINCIPIA.

_Sect._ 2.--_Reception of the Principia._

LORD BROUGHAM has very recently (_Analytical View of Sir Isaac Newton's Principia_, 1855) shown a strong disposition still to maintain, what he says has frequently been alleged, that the reception of the work was not, even in this country, "such as might have been expected." He says, in explanation of the facts which I have adduced, showing the high estimation in which Newton was held immediately after the publication of the _Principia_, that Newton's previous fame was great by former discoveries. This is true; but the effect of this was precisely what was most honorable to Newton's countrymen, that they received with immediate acclamations this new and greater discovery. Lord Brougham adds, "after its appearance the _Principia_ was more admired than studied;" which is probably true of the _Principia_ still, and of all great works of like novelty and difficulty at all times. But, says Lord Brougham, "there is no getting over the inference on this head which arises from the dates of the two first editions. There elapsed an interval of no less than twenty-seven years between them; and although Cotes [in his Preface] speaks of the copies having become scarce and in very great demand when the second edition appeared in 1713, yet had this urgent demand been of many years' continuance, the reprinting could

never have been so long delayed." But Lord Brougham might have learnt from Sir David Brewster's _Life of Newton_ (vol. i. p. 312), which he extols so emphatically, that already in 1691 (only four years after the publication), a copy of the _Principia_ could hardly be procured, and that even at that {549} time an improved edition was in contemplation; that Newton had been pressed by his friends to undertake it, and had refused.

When Bentley had induced Newton to consent that a new edition should be printed, he announces his success with obvious exultation to Cotes, who was to superintend the work. And in the mean time the _Astronomy_ of David Gregory, published in 1702, showed in every page how familiar the Newtonian doctrines were to English philosophers, and tended to make them more so, as the sermons of Bentley himself had done in 1692.

Newton's Cambridge contemporaries were among those who took a part in bringing the _Principia_ before the world. The manuscript draft of it was conveyed to the Royal Society (April 28, 1686) by Dr. Vincent, Fellow of Clare Hall, who was the tutor of Whiston, Newton's deputy in his professorship; and he, in presenting the work, spoke of the novelty and dignity of the subject. There exists in the library of the University of Cambridge a manuscript containing the early Propositions of the _Principia_ as far as Prop. xxxiii. (which is a part of Section vii., about Falling Bodies). This appears to have been a transcript of Newton's Lectures, delivered as Lucasian Professor: it is dated October, 1684.

_Is Gravitation proportional to Quantity of Matter?_

It was a portion of Newton's assertion in his great discovery, that all the bodies of the universe attract each other with forces which are _as the quantity of matter_ in each: that is, for instance, the sun attracts the satellites of any planet just as much as he attracts the planet itself, in proportion to the quantity of matter in each; and the planets attract one another just as much as they attract the sun, according to the quantity of matter.

To prove this part of the law _exactly_ is a matter which requires careful experiments; and though proved experimentally by Newton, has been considered in our time worthy of re-examination by the great astronomer Bessel. There was some ground for doubt; for the mass of Jupiter, as deduced from the perturbations of Saturn, was only 1/1070 of the mass of the sun; the mass of the same planet as deduced from the perturbations of Juno and Pallas was 1/1045 of that of the Sun. If this difference were to be confirmed by accurate observations and calculations, it would follow that the attractive power exercised by Jupiter upon the minor planets was greater than that exercised upon {550} Saturn. And in the same way, if the attraction of the Earth had any _specific_ relation to different kinds of matter, the time of oscillation of a pendulum of equal length composed wholly or in part of the two substances would be different. If, for instance, it were more intense for magnetized iron than for stone, the iron pendulum would oscillate more quickly. Bessel showed[47\A] that it was possible to assume hypothetically a constitution of the sun, planets, and their appendages, such that the attraction of the Sun on the

Planets and Satellites should be proportional to the quantity of matter in each; but that the attraction of the Planets on one another would not be on the same scale.

[Note 47\A: _Berlin Mem._ 1824.]

Newton had made experiments (described in the _Principia_, Book iii., Prop. vi.) by which it was shown that there could be no considerable or palpable amount of such specific difference among terrestrial bodies, but his experiments could not be regarded as exact enough for the requirements of modern science. Bessel instituted a laborious series of experiments (presented to the Berlin Academy in 1832) which completely disproved the conjecture of such a difference; every substance examined having given exactly the same coefficient of gravitating intensity as compared with inertia. Among the substances examined were metallic and stony masses of meteoric origin, which might be supposed, if any bodies could, to come from other parts of the solar system.

## CHAPTER IV.

VERIFICATION AND COMPLETION OF THE NEWTONIAN THEORY.

_Tables of the Moon and Planets._

THE Newtonian discovery of Universal Gravitation, so remarkable in other respects, is also remarkable as exemplifying the immense extent to which the verification of a great truth may be carried, the amount of human labor which may be requisite to do it justice, and the striking extension of human knowledge to which it may lead. I have said that it is remarked as a beauty in the first fixation of a theory that its measures or elements are established by means of a few {551} data; but that its excellence when established is in the number of observations which it explains. The multiplicity of observations which are explained by astronomy, and which are made because astronomy explains them, is immense, as I have noted in the text. And the multitude of observations thus made is employed for the purpose of correcting the first adopted elements of the theory. I have mentioned some of the examples of this process: I might mention many others in order to continue the history of this part of Astronomy up to the present time. But I will notice only those which seem to me the most remarkable.

In 1812, Burckhardt's _Tables de la Lune_ were published by the French Bureau des Longitudes. A comparison of these and Burg's with a considerable number of observations, gave 9-100ths of a second as the mean error of the former in the Moon's longitude, while the mean error of Burg's was 18-100ths. The preference was therefore accorded to Burckhardt's.

Yet the Lunar Tables were still as much as thirty seconds wrong in single observations. This circumstance, and Laplace's expressed wish, induced the French Academy to offer a prize for a complete and purely theoretical determination of the Lunar path, instead of determinations resting, as hitherto, partly upon theory and partly upon observations. In 1820, two prize essays appeared, the one by Damoiseau, the other by Plana and Carlini. And some years afterwards (in 1824, and again in

1828), Damoiseau published _Tables de la Lune formées sur la seule Théorie d'Attraction_. These agree very closely with observation. That we may form some notion of the complexity of the problem, I may state that the longitude of the Moon is in these Tables affected by no fewer than forty-seven _equations_; and the other quantities which determine her place are subject to inequalities not much less in number.

Still I had to state in the second Edition, published in 1847, that there remained an unexplained discordance between theory and observation in the motions of the Moon; an inequality of long period as it seemed, which the theory did not give.

A careful examination of a long series of the best observations of the Moon, compared throughout with the theory in its most perfect form, would afford the means both of correcting the numerical elements of the theory, and of detecting the nature, and perhaps the law, of any still remaining discrepancies. Such a work, however, required vast labor, as well as great skill and profound mathematical knowledge. {552} Mr. Airy undertook the task; employing for that purpose, the Observations of the Moon made at Greenwich from 1750 to 1830. Above 8000 observed places of the Moon were compared with theory by the computation of the same number of places, each separately and independently calculated from Plana's Formulæ. A body of calculators (sometimes sixteen), at the expense of the British Government, was employed for about eight years in this work. When we take this in conjunction with the labor which the observations themselves imply, it may serve to show on what a scale the verification of the Newtonian theory has been conducted. The first results of this labor were published in two quarto volumes; the final deductions as to correction of elements, &c., were given in the Memoirs of the Astronomical Society in 1848.[48\A]

[Note 48\A: The total expense of computers, to the end of reading the proof-sheets, was 4300_l._

Mr. Airy's estimate of days' works [made before beginning], for the heavy part of calculations only, was thirty-six years of one computer. This was somewhat exceeded, but not very greatly, in that part.]

Even while the calculations were going on, it became apparent that there were some differences between the observed places of the Moon, and the theory so far as it had then been developed. M. Hansen, an eminent German mathematician who had devised new and powerful methods for the mathematical determination of the results of the law of gravitation, was thus led to explore still further the motions of the Moon in pursuance of this law. The result was that he found there must exist two lunar inequalities, hitherto not known; the one of 273, and the other of 239 years, the coefficients of which are respectively 27 and 23 seconds. Both these originate in the attraction of Venus; one of them being connected with the long inequality in the Solar Tables, of which Mr. Airy had already proved the existence, as stated in Chap. vi. Sect. 6 of this Book.

These inequalities fell in with the discrepancies between the actual observations and the previously calculated Tables, which Mr. Airy had discovered. And again,

shortly afterwards, M. Hansen found that there resulted from the theory two other new equations of the Moon; one in latitude and one in longitude, agreeing with two which were found by Mr. Airy in deducing from the observations the correction of the elements of the Lunar Tables. And again, a little later, there was detected by these mathematicians a theoretical correction for the {553} motion of the Node of the Moon's orbit, coinciding exactly with one which had been found to appear in the observations.

Nothing can more strikingly exhibit the confirmation which increased scrutiny brings to light between the Newtonian theory on the one hand, and the celestial motions on the other. We have here a very large mass of the best observations which have ever been made, systematically examined, with immense labor, and with the set purpose of correcting at once all the elements of the Lunar Tables. The corrections of the elements thus deduced imply of course some error in the theory as previously developed. But at the same time, and with the like determination thoroughly to explore the subject, the theory is again pressed to yield its most complete results, by the invention of new and powerful mathematical methods; and the event is, that residual errors of the old Tables, several in number, following the most diverse laws, occurring in several detached parts, agree with the residual results of the Theory thus newly extracted from it. And thus every additional exactness of scrutiny into the celestial motions on the one hand and the Newtonian theory on the other, has ended, sooner or later, in showing the exactness of their coincidence.

The comparison of the theory with observation in the case of the motions of the Planets, the motion of each being disturbed by the attraction of all the others, is a subject in some respects still more complicated and laborious. This work also was undertaken by the same indefatigable astronomer; and here also his materials belonged to the same period as before; being the admirable observations made at Greenwich from 1750 to 1830, during the time that Bradley, Maskelyne, and Pond were the Astronomers Royal.[49\A] These Planetary observations were deduced, and the observed places were compared with the tabular places: with Lindenau's Tables of Mercury, Venus, and Mars; and with Bouvard's Tables of Jupiter, Saturn, and Uranus; and thus, while the received theory and its elements were confirmed, the means of testing any improvement which may hereafter be proposed, either in the form of the theoretical results or in the constant elements which they involved, was placed within the reach of the {554} astronomers of all future time. The work appeared in 1845; the expense of the compilations and the publication being defrayed by the British Government.

[Note 49\A: The observations of stars made by Bradley, who preceded Maskelyne at Greenwich, had already been discussed by Bessel, a great German astronomer; and the results published in 1818, with a title that well showed the estimation in which he held those materials: _Fundamenta Astronomiæ pro anno_ 1775, _deducta ex Ohservationibus viri incomparabilis James Bradley in specula Astronomica Grenovicensi per annos_ 1750-1762 _institutis_.]

_The Discovery of Neptune._

The theory of gravitation was destined to receive a confirmation more striking than any which could arise from any explanation, however perfect, given by the motions of a known planet; namely, in revealing the existence of an unknown planet, disclosed to astronomers by the attraction which it exerted upon a known one. The story of the discovery of Neptune by the calculations of Mr. Adams and M. Le Verrier was partly told in the former edition of this History. I had there stated (vol. ii. p. 306) that "a deviation of observation from the theory occurs at the very extremity of the solar system, and that its existence appears to be beyond doubt. Uranus does not conform to the Tables calculated for him on the theory of gravitation. In 1821, Bouvard said in the Preface to the Tables of this Planet, "the formation of these Tables offers to us this alternative, that we cannot satisfy modern observations to the requisite degree of precision without making our Tables deviate from the ancient observations." But when we have done this, there is still a discordance between the Tables and the more modern observations, and this discordance goes on increasing. At present the Tables make the Planet come upon the meridian about eight seconds later than he really does. This discrepancy has turned the thoughts of astronomers to the effects which would result from a planet external to Uranus. It appears that the observed motion would be explained by applying a planet at twice the distance of Uranus from the Sun to exercise a disturbing force, and it is found that the present longitude of this disturbing body must be about 325 degrees.

I added, "M. Le Verrier (_Comptes Rendus_, Jan. 1, 1846) and, as I am informed by the Astronomer Royal, Mr. Adams, of St. John's College, Cambridge, have both arrived independently at this result."

To this Edition I added a Postscript, dated, Nov. 7, 1846, in which I said:

"The planet exterior to Uranus, of which the existence was inferred by M. Le Verrier and Mr. Adams from the motions of Uranus (vol. ii. Note (L.)), has since been discovered. This confirmation of calculations founded upon the doctrine of universal gravitation, may be looked upon as the most remarkable event of the kind since the return of Halley's comet in 1757 and in some respects, as a more striking event {555} even than that; inasmuch as the new planet had never been seen at all, and was discovered by mathematicians entirely by their feeling of its influence, which they perceived through the organ of mathematical calculation.

"There can be no doubt that to M. Le Verrier belongs the glory of having first published a prediction of the place and appearance of the new planet, and of having thus occasioned its discovery by astronomical observers. M. Le Verrier's first prediction was published in the _Comptes Rendus de l'Acad. des Sciences_, for _June_ 1, 1846 (not _Jan._ 1, as erroneously printed in my Note). A subsequent paper on the subject was read Aug. 31. The planet was seen by M. Galle, at the Observatory of Berlin, on September 23, on which day he had received an express application from M. Le Verrier, recommending him to endeavor to recognize the stranger by its having a visible disk. Professor Challis, at the Observatory of Cambridge, was looking out for the new planet from July 29, and saw it on August 4, and again on August 12, but without recognizing it, in consequence of his plan of not comparing his observations till he had accumulated a greater number of them. On Sept. 29, having read for the

first time M. Le Verrier's second paper, he altered his plan, and paid attention to the physical appearance rather than the position of the star. On that very evening, not having then heard of M. Galle's discovery, he singled out the star by its seeming to have a disk.

"M. Le Verrier's mode of discussing the circumstances of Uranus's motion, and inferring the new planet from these circumstances, is in the highest degree sagacious and masterly. Justice to him cannot require that the contemporaneous, though unpublished, labors of Mr. Adams, of St John's College, Cambridge, should not also be recorded. Mr. Adams made his first calculations to account for the anomalies in the motion of Uranus, on the hypothesis of a more distant planet, in 1843. At first he had not taken into account the earlier Greenwich observations; but these were supplied to him by the Astronomer Royal, in 1844. In September, 1845, Mr. Adams communicated to Professor Challis values of the elements of the supposed disturbing body; namely, its mean distance, mean longitude at a given epoch, longitude of perihelion, eccentricity of orbit, and mass. In the next month, he communicated to the Astronomer Royal values of the same elements, somewhat corrected. The note (L.), vol. ii., of the present work (2d Ed.), in which the names of MM. Le Verrier and Adams are mentioned in conjunction, was in the press in August, 1846, a {556} month before the planet was seen. As I have stated in the text, Mr. Adams and M. Le Verrier assigned to the unseen planet nearly the same position; they also assigned to it nearly the same mass; namely, 2½ times the mass of Uranus. And hence, supposing the density to be not greater than that of Uranus, it followed that the visible diameter would be about 3", an apparent magnitude not much smaller than Uranus himself.

"M. Le Verrier has mentioned for the new planet the name _Neptunus_; and probably, deference to his authority as its discoverer, will obtain general currency for this name."

Mr. Airy has given a very complete history of the circumstances attending the discovery of Neptune, in the Memoirs of the Astronomical Society (read November 13, 1846). In this he shows that the probability of some disturbing body beyond Uranus had suggested itself to M. A. Bouvard and Mr. Hussey as early as 1834. Mr. Airy himself then thought that the time was not ripe for making out the nature of any external action on the planets. But Mr. Adams soon afterwards proceeded to work at the problem. As early as 1841 (as he himself informs me) he conjectured the existence of a planet exterior to Uranus, and recorded in a memorandum his design of examining its effect; but deferred the calculations till he had completed his preparations for the University examination which he was to undergo in January, 1843, in order to receive the Degree of Bachelor of Arts. He was the Senior Wrangler of that occasion, and soon afterwards proceeded to carry his design into effect; applying to the Astronomer Royal for recorded observations which might aid him in his task. On one of the last days of October, 1845, Mr. Adams went to the Observatory at Greenwich; and finding the Astronomer Royal abroad, he left there a paper containing the elements of the extra-Uranian Planet: the longitude was in this paper stated as 323½ degrees. It was, as we have seen, in June, 1846, that M. Le Verrier's Memoir appeared, in which he assigned to the disturbing body a longitude

of 325 degrees. The coincidence was striking. "I cannot sufficiently express," says Mr. Airy, "the feeling of delight and satisfaction which I received from the Memoir of M. Le Verrier." This feeling communicated itself to others. Sir John Herschel said in September, 1846, at a meeting of the British Association at Southampton, "We see it (the probable new planet) as Columbus saw America from the shores of Spain. Its movements have been felt, trembling along the far-reaching line of our analysis, with a certainty hardly inferior to that of ocular demonstration." {557}

In truth, at the moment when this was uttered, the new Planet had already been seen by Professor Challis; for, as we have said, he had seen it in the early part of August. He had included it in the net which he had cast among the stars for this very purpose; but employing a slow and cautious process, he had deferred for a time that examination of his capture which would have enabled him to detect the object sought. As soon as he received M. Le Verrier's paper of August 31 on September 29, he was so much impressed with the sagacity and clearness of the limitations of the field of observation there laid down, that he instantly changed his plan of observation, and noted the planet, as an object having a visible disk, on the evening of the same day.

In this manner the theory of gravitation predicted and produced the discovery. Thus to predict unknown facts found afterwards to be true, is, as I have said, a confirmation of a theory which in impressiveness and value goes beyond any explanation of known facts. It is a confirmation which has only occurred a few times in the history of science; and in the case only of the most refined and complete theories, such as those of Astronomy and Optics. The mathematical skill which was requisite in order to arrive at such a discovery, may in some measure be judged of by the account which we have had to give of the previous mathematical progress of the theory of gravitation. It there appeared that the lives of many of the most acute, clear-sighted, and laborious of mankind, had been employed for generations in solving the problem. Given the planetary bodies, to find their mutual perturbations: but here we have the inverse problem--Given the perturbations, to find the planets.[50\A]

[Note 50\A: This may be called the _inverse_ problem with reference to the older and more familiar problem; but we may remark that the usual phraseology of the Problem of Central Forces differs from this analogy. In Newton's _Principia_, the earlier Sections, in which the motion is given to find the force, are spoken of as containing the Direct Problem of Central Forces: the Eighth Section of the First Book, where the Force is given to find the orbit, is spoken of as containing the _Inverse_ Problem of Central Forces.]

_The Minor Planets._

The discovery of the Minor Planets which revolve between the orbits of Mars and Jupiter was not a consequence or confirmation of the Newtonian theory. That theory gives no reason for the distance of {558} the Planets from the Sun; nor does any theory yet devised give such reason. But an empirical formula proposed by the Astronomer Bode of Berlin, gives a law of these distances (_Bode's Law_), which, to make it coherent, requires a planet between Mars and Jupiter. With such an addition,

the distance of Mercury, Venus, Earth, Mars, the Missing Planet, Jupiter, Saturn, and Uranus, are nearly as the numbers 4, 7, 10, 16, 28, 52, 100, 196, in which the excesses of each number above the preceding are the series 3, 3, 6, 12, 24, 48, 96. On the strength of this law the Germans wrote _on the long-expected Planet_, and formed themselves into associations for the discovery of it.

Not only did this law stimulate the inquiries for the Missing Planet, and thus lead to the discovery of the Minor Planets, but it had also a share in the discovery of Neptune. According to the law, a planet beyond Uranus may be expected to be at the distance represented by 388. Mr. Adams and M. Le Verrier both of them began by assuming a distance of nearly this magnitude for the Planet which they sought; that is, a distance more than 38 times the earth's distance. It was found afterwards that the distance of Neptune is only 30 times that of the earth; yet the assumption was of essential use in obtaining the result and Mr. Airy remarks that the history of the discovery shows the importance of using any received theory as far as it will go, even if the theory can claim no higher merit than that of being plausible.[51\A]

[Note 51\A: Account of the Discovery of Neptune, &c., _Mem. Ast. Soc._, vol. xvi. p. 414.]

The discovery of Minor Planets in a certain region of the interval between Mars and Jupiter has gone on to such an extent, that their number makes them assume in a peculiar manner the character of representatives of a Missing Planet. At first, as I have said in the text, it was supposed that all these portions must pass through or near a common node; this opinion being founded on the very bold doctrine, that the portions must at one time have been united in one Planet, and must then have separated. At this node, as I have stated, Olbers lay in wait for them, as for a hostile army at a defile. Ceres, Pallas, and Juno had been discovered in this way in the period from 1801 to 1804; and Vesta was caught in 1807. For a time the chase for new planets in this region seemed to have exhausted the stock. But after thirty-eight years, to the astonishment of astronomers, they began to be again detected in extraordinary numbers. In 1845, M. Hencke of {559} Driessen discovered a fifth of these planets, which was termed Astræa. In various quarters the chase was resumed with great ardor. In 1847 were found Hebe, Iris, and Flora; in 1848, Metis; in 1849, Hygæa; in 1850, Parthenope, Victoria, and Egeria; in 1861, Irene and Eunomia; in 1852, Psyche, Thetis, Melpomene, Fortuna, Massilia, Lutetia, Calliope. To these we have now (at the close of 1856) to add _nineteen_ others; making up the whole number of these Minor Planets at present known to _forty-two_.

As their enumeration will show, the ancient practice has been continued of giving to the Planets mythological names. And for a time, till the numbers became too great, each of the Minor Planets was designated in astronomical books by some symbol appropriate to the character of the mythological person; as from ancient times Mars has been denoted by a mark indicating a spear, and Venus by one representing a looking-glass. Thus, when a Minor Planet was discovered at London in 1851, the year in which the peace of the world was, in a manner, celebrated by the Great Exhibition of the Products of All Nations, held at that metropolis, the name _Irene_ was given to the new star, as a memorial of the auspicious time of its discovery. And it was agreed,

for awhile, that its symbol should be a dove with an olive-branch. But the vast multitude of the Minor Planets, as discovery went on, made any mode of designation, except a numerical one, practically inconvenient. They are now denoted by a small circle inclosing a figure in the order of their discovery. Thus, _Ceres_ is (1), _Irene_ is (14), and _Isis_ is (42).

The rapidity with which these discoveries were made was owing in part to the formation of star-maps, in which all known fixed stars being represented, the existence of a new and movable star might be recognized by comparison of the sky with the map. These maps were first constructed by astronomers of different countries at the suggestion of the Academy of Berlin; but they have since been greatly extended, and now include much smaller stars than were originally laid down.

I will mention the number of planets discovered in each year. After the start was once made, by Hencke's discovery of Astræa in 1845, the same astronomer discovered Hebe in 1847; and in the same year Mr. Hind, of London, discovered two others, Iris and Flora. The years 1848 and 1849 each supplied one; the year 1850, three; 1851, two; 1852 was marked by the extraordinary discovery of _eight_ new members of the planetary system. The year 1853 supplied four; 1854, six; 1855, four; and 1856 has already given us five. {560}

These discoveries have been distributed among the observatories of Europe. The bright sky of Naples has revealed seven new planets to the telescope of Signer Gasparis. Marseilles has given us one; Germany, four, discovered by M. Luther at Bilk; Paris has furnished seven; and Mr. Hind, in Mr. Bishop's private observatory in London, notwithstanding our turbid skies, has discovered no less than ten planets; and there also Mr. Marth discovered (29) Amphitrite. Mr. Graham, at the private observatory of Mr. Cooper, in Ireland, discovered (9) Metis.

America has supplied its planet, namely (31) Euphrosyne, discovered by Mr. Ferguson at Washington and the most recent of these discoveries is that by Mr. Pogson, of Oxford, who has found the forty-second of these Minor Planets, which has been named Isis.[52\A]

[Note 52\A: I take this list from a Memoir of M. Bruhns, Berlin, 1856.]

I may add that it appears to follow from the best calculations that the total mass of all these bodies is very small. Herschel reckoned the diameters of Ceres at 35, and of Pallas at 26 miles. It has since been calculated[53\A] that some of them are smaller still; Victoria having a diameter of 9 miles, Lutetia of 8, and Atalanta of little more than 4. It follows from this that the whole mass would probably be less than the sixth part of our moon. Hence their perturbing effects on each other or on other planets are null; but they are not the less disturbed by the action of the other planets, and especially of Jupiter.

[Note 53\A: Bruhns, as above.]

_Anomalies in the Action of Gravitation._

The complete and exact manner in which the doctrine of gravitation explains the motions of the Comets as well as of the Planets, has made astronomers very bold in

proposing hypotheses to account for any deviations from the motion which the theory requires. Thus Encke's Comet is found to have its motion accelerated by about one-eighth of a day in every revolution. This result was conceived to be established by former observations, and is confirmed by the facts of the appearance of 1852.[54\A] The hypothesis which is proposed in order to explain this result is, that the Comet moves in a resisting medium, which makes it fall inwards from its path, towards the Sun, and thus, by narrowing its orbit, diminishes its periodic time. On the other hand, M. Le Verrier has found that Mercury's mean motion has gone on diminishing; {561} as if the planet were, in the progress of his revolutions, receding further from the Sun. This is explained, if we suppose that there is, in the region of Mercury, a resisting medium which moves round the Sun in the same direction as the Planets move. Evidence of a kind of nebulous disk surrounding the Sun, and extending beyond the orbits of Mercury and Venus, appears to be afforded us by the phenomenon called the _Zodiacal Light_; and as the Sun itself rotates on its axis, it is most probable that this kind of atmosphere rotates also.[55\A] On the other hand, M. Le Verrier conceives that the Comets which now revolve within the ordinary planetary limits have not always done so, but have been caught and detained by the Planets among which they move. In this way the action of Jupiter has brought the Comets of Faye and Vico into their present limited orbits, as it drew the Comet of Lexell out of its known orbit, when the Comet passed over the Planet in 1779, since which time it has not been seen.

[Note 54\A: _Berlin Memoirs_, 1854.]

[Note 55\A: M. Le Verrier, _Annales de l'Obs. de Paris_, vol. i. p. 89.]

Among the examples of the boldness with which astronomers assume the doctrine of gravitation even beyond the limits of the solar system to be so entirely established, that hypotheses may and must be assumed to explain any apparent irregularity of motion, we may reckon the mode of accounting for certain supposed irregularities in the proper motion of Sirius, which has been proposed by Bessel, and which M. Peters thinks is proved to be true by his recent researches (_Astr. Nach._ xxxi. p. 219, and xxxii. p. 1). The hypothesis is, that Sirius has a companion star, dark, and therefore invisible to us; and that the two, revolving round their common centre as the system moves on, the motion of Sirius is seen to be sometimes quicker and sometimes slower.

_The Earth's Density._

"Cavendish's experiment," as it is commonly called--the measure of the attractions of manageable masses by the torsion balance, in order to determine the density of the Earth--has been repeated recently by Professor Reich at Freiberg, and by Mr. Baily in England, with great attention to the means of attaining accuracy. Professor Reich's result for the density of the Earth is 5·44; Mr. Baily's is 5·92. Cavendish's result was 5·48; according to recent revisions[56\A] it is 5·52. {562}

[Note 56\A: The calculation has been revised by M. Edward Schmidt. Humboldt's _Kosmos_, ii. p. 425.]

But the statical effect of the attraction of manageable masses, or even of mountains, is very small. The effect of a small change in gravity may be accumulated by being constantly repeated in the oscillations of a pendulum, and thus may become perceptible. Mr. Airy attempted to determine the density of the Earth by a method depending on this view. A pendulum oscillating at the surface was to be compared with an equal pendulum at a great depth below the surface. The difference of their rates would disclose the different force of gravity at the two positions; and hence, the density of the Earth. In 1826 and 1828, Mr. Airy attempted this experiment at the copper mine of Dolcoath in Cornwall, but failed from various causes. But in 1854, he resumed it at the Harton coal mine in Durham, the depth of which is 1260 feet; having in this new trial, the advantage of transmitting the time from one station to the other by the instantaneous effect of galvanism, instead of by portable watches. The result was a density of 6·56; which is much larger than the preceding results, but, as Mr. Airy holds, is entitled to compete with the others on at least equal terms.

_Tides._

I should be wanting in the expression of gratitude to those who have practically assisted me in Researches on the Tides, if I did not mention the grand series of Tide Observations made on the coast of Europe and America in June, 1835, through the authority of the Board of Admiralty, and the interposition of the late Duke of Wellington, at that time Foreign Secretary. Tide observations were made for a fortnight at all the Coast-guard stations of Great Britain and Ireland in June, 1834; and these were repeated in June, 1835, with corresponding observations on all the coasts of Europe, from the North Cape of Norway to the Straits of Gibraltar; and from the mouth of the St. Lawrence to the mouth of the Mississippi. The results of these observations, which were very complete so far as the coast tides were concerned, were given in the _Philosophical Transactions_ for 1836.

Additional accuracy respecting the Tides of the North American coast may be expected from the survey now going on under the direction of Superintendent A. Bache. The Tides of the English Channel have been further investigated, and the phenomena presented under a new point of view by Admiral Beechey. {563}

The Tides of the Coast of Ireland have been examined with great care by Mr. Airy. Numerous and careful observations were made with a view, in the first instance, of determining what was to be regarded as "the Level of the Sea;" but the results were discussed so as to bring into view the laws and progress, on the Irish coast, of the various inequalities of the Tides mentioned in Chap. iv. Sect. 9 of this Book.

I may notice as one of the curious results of the Tide Observations of 1836, that it appeared to me, from a comparison of the Observations, that there must be a point in the German Ocean, about midway between Lowestoft on the English coast, and the Brill on the Dutch coast, where the tide would vanish: and this was ascertained to be the case by observation; the observations being made by Captain Hewett, then employed in a survey of that sea.

_Cotidal Lines_ supply, as I conceive, a good and simple method of representing the progress and connection of _littoral_ tides. But to draw cotidal lines across

oceans, is a very precarious mode of representing the facts, except we had much more knowledge on the subject than we at present possess. In the _Phil. Trans._ for 1848, I have resumed the subject of the Tides of the Pacific; and I have there expressed my opinion, that while the littoral tides are produced by progressive waves, the oceanic tides are more of the nature of stationary undulations.

But many points of this kind might be decided, and our knowledge on this subject might be brought to a condition of completeness, if a ship or ships were sent expressly to follow the phenomena of the Tides from point to point, as the observations themselves might suggest a course. Till this is done, our knowledge cannot be completed. Detached and casual observations, made _aliud agendo_, can never carry us much beyond the point where we at present are.

_Double Stars._

Sir John Herschel's work, referred to in the History (2d Ed.) as then about to appear, was published in 1847.[57\A] In this work, besides a vast amount of valuable observations and reasonings on other subjects {564} (as Nebulæ, the Magnitude of Stars, and the like), the orbits of several double stars are computed by the aid of the new observations. But Sir John Herschel's conviction on the point in question, the operation of the Newtonian law of gravitation in the region of the stars, is expressed perhaps more clearly in another work which he published in 1849.[58\A] He there speaks of Double Stars, and especially of _gamma Virginis_, the one which has been most assiduously watched, and has offered phenomena of the greatest interest.[59\A] He then finds that the two components of this star revolve round each other in a period of 182 years; and says that the elements of the calculated orbit represent the whole series of recorded observations, comprising an angular movement of nearly nine-tenths of a complete circuit, both in angle and distance, with a degree of exactness fully equal to that of observation itself. "No doubt can therefore," he adds, "remain as to the **prevalence in this remote system of the Newtonian Law of Gravitation."

[Note 57\A: _Results of Astronomical Observations made during the years_ 1834, 5, 6, 7, 8, _at the Cape of Good Hope, being the completion of a Telescopic Survey of the whole Surface of the visible Heavens commenced in_ 1825.]

[Note 58\A: _Outlines of Astronomy_.]

[Note 59\A: _Out._ 844.]

Yet M. Yvon de Villarceau has endeavored to show[60\A] that this conclusion, however probable, is not yet proved. He holds, even for the Double Stars, which have been most observed, the observations are only equivalent to seven or eight really distinct data, and that seven data are not sufficient to determine that an ellipse is described according to the Newtonian law. Without going into the details of this reasoning, I may remark, that the more rapid relative angular motion of the components of a Double Star when they are more near each other, proves, as is allowed on all hands, that they revolve under the influence of a mutual attractive force, obeying the Keplerian Law of Areas. But that, whether this force follows the law of the inverse square or some other law, can hardly have been rigorously proved

as yet, we may easily conceive, when we recollect the manner in which that law was proved for the Solar System. It was by means of an error of _eight minutes_, observed by Tycho, that Kepler was enabled, as he justly boasted, to reform the scheme of the Solar System,--to show, that is, that the planetary orbits are ellipses with the sun in the focus. Now, the observations of Double Stars cannot pretend to such accuracy as this; and therefore the Keplerian theorem cannot, as yet, have been fully demonstrated from those observations. But when we know {565} that Double Stars are held together by a central force, to prove that this force follows a different law from the only law which has hitherto been found to obtain in the universe, and which obtains between all the known masses of the universe, would require very clear and distinct evidence, of which astronomers have as yet seen no trace.

[Note 60\A: _Connaissance des Temps_, for 1852; published in 1849.]

## CHAPTER VI.

_Sect._ 1. _Instruments._--2. _Clocks._

IN page 473, I have described the manner in which astronomers are able to observe the transit of a star, and other astronomical phenomena, to the exactness of a tenth of a second of time. The mode of observation there described implies that the observer at the moment of observation compares the impressions of the eye and of the ear. Now it is found that the habit which the observer must form of doing this operates differently in different observers, so that one observer notes the same fact as happening a fraction of a second earlier or later than another observer does; and this in every case. Thus, using the term _equation_, as we use it in Astronomy, to express a correction by which we get regularity from irregularity, there is a _personal equation_ belonging to this mode of observation, showing that it is liable to error. Can this error be got rid of?

It is at any rate much diminished by a method of observation recently introduced into observatories, and first practised in America. The essential feature of this mode of observation consists in combining the impression of sight with that of touch, instead of with that of hearing. The observer at the moment of observation presses with his finger so as to make a mark on a machine which by its motion measures time with great accuracy and on a large scale; and thus small intervals of time are made visible.

A universal, though not a necessary, part of this machinery, as hitherto adopted, is, that a galvanic circuit has been employed in conveying the impression from the finger to the part where time is measured and marked. The facility with which galvanic wires can {566} thus lead the impression by any path to any distance, and increase its force in any degree, has led to this combination, and almost identification, of observation by touch with its record by galvanism.

The method having been first used by Mr. Bond at Cambridge, in North America, has been adopted elsewhere, and especially at Greenwich, where it is used for all the instruments; and consequently a collection of galvanic batteries is thus as necessary

a part of the apparatus of the establishment as its graduated circles and arcs.

END OF VOL. I.

HISTORY OF THE INDUCTIVE SCIENCES.

VOLUME II.

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{{21}} BOOK VIII.

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. . . . . Go, demand Of mighty Nature, if 'twas ever meant That we should pry far off and be unraised, That we should pore, and dwindle as we pore, Viewing all objects unremittingly In disconnexion dead and spiritless; And still dividing, and dividing still, Break down all grandeur, still unsatisfied With the perverse attempt, while littleness May yet become more little; waging thus An impious warfare 'gainst the very life Of our own souls. WORDSWORTH, _Excursion_.

. . . . . Ἐσσυμένη δὲ Ἠερίην ἀψῖδα διερῥοίζησε πεδίλῳ Εἰς δόμον ἌΡΜΟΝΙΗΣ παμμητόρος, ὁππόθι νύμφη Ἵκελον οἶκον ἐναίε τύπῳ τετράζυγι κόσμου Αὐτοπαγῆ NONNUS. _Dionysiac_. xli. 275.

Along the skiey arch the goddess trode, And sought Harmonia's august abode; The universal plan, the mystic Four, Defines the figure of the palace-floor. Solid and square the ancient fabric stands, Raised by the labors of unnumbered hands.

{{23}} BOOK VIII.

## INTRODUCTION.

_The Secondary Mechanical Sciences._

IN the sciences of Mechanics and Physical Astronomy, Motion and Force are the direct and primary objects of our attention. But there is another class of sciences in which we endeavor to reduce phenomena, not evidently mechanical, to a known dependence upon mechanical properties and laws. In the cases to which I refer, the facts do not present themselves to the senses as modifications of position and motion, but as _secondary qualities_, which are found to be in some way derived from those primary attributes. Also, in these cases the phenomena are reduced to their mechanical laws and causes in a secondary manner; namely, by treating them as the operation of a _medium_ interposed between the object and the organ of sense. These, then, we may call _Secondary Mechanical Sciences_. The sciences of this kind which require our notice are those which treat of the sensible qualities, Sound, Light, and Heat; that is. Acoustics, Optics, and Thermotics.

It will be recollected that our object is not by any means to give a full statement of all the additions which have been successively made to our knowledge on the subjects under review, or a complete list of the persons by whom such additions have been made; but to present a view of the progress of each of those branches of knowledge _as a theoretical science_;--to point out the Epochs of the discovery of those general principles which reduce many facts to one theory; and to note all that is most characteristic and instructive in the circumstances and persons which bear

upon such Epochs. A history of any science, written with such objects, will not need to be long; but it will fail in its purpose altogether, if it do not distinctly exhibit some well-marked and prominent features. {24}

We begin our account of the Secondary Mechanical Sciences with Acoustics, because the progress towards right theoretical views, was, in fact, made much earlier in the science of Sound, than in those of Light and of Heat; and also, because a clear comprehension of the theory to which we are led in this case, is the best preparation for the difficulties (by no means inconsiderable) of the reasonings of theorists on the other subjects.

## CHAPTER I.

PRELUDE TO THE SOLUTION OF PROBLEMS IN ACOUSTICS.

IN some measure the true theory of sound was guessed by very early speculators on the subject; though undoubtedly conceived in a very vague and wavering manner. That sound is caused by some motion of the sounding body, and conveyed by some motion of the air to the ear, is an opinion which we trace to the earliest times of physical philosophy. We may take Aristotle as the best expounder of this stage of opinion. In his Treatise _On Sound and Hearing_, he says, "Sound takes place when bodies strike the air, not by the air having a _form_ impressed upon it (σχηματίζομενον), as some think, but by its being moved in a corresponding manner; (probably he means in a manner corresponding to the impulse;) the air being contracted, and expanded, and overtaken, and again struck by the impulses of the breath and of the strings. For when the breath falls upon and strikes the air which is next it, the air is carried forwards with an impetus, and that which is contiguous to the first is carried onwards; so that the same voice spreads every way as far as the motion of the air takes place."

As is the case with all such specimens of ancient physics, different persons would find in such a statement very different measures of truth and distinctness. The admirers of antiquity might easily, by pressing the language closely, and using the light of modern discovery, detect in this passage an exact account of the production and propagation of sound: while others might maintain that in Aristotle's own mind, there were only vague notions, and verbal generalizations. This {25} latter opinion is very emphatically expressed by Bacon.[1\8] "The collision or thrusting of air, which they will have to be the cause of sound, neither denotes the _form_ nor the latent process of sound; but is a term of ignorance and of superficial contemplation." Nor can it be justly denied, that an exact and distinct apprehension of the kind of motion of the air by which sound is diffused, was beyond the reach of the ancient philosophers, and made its way into the world long afterwards. It was by no means easy to reconcile the nature of such motion with obvious phenomena. For the process is not evident as motion; since, as Bacon also observes,[2\8] it does not visibly agitate the flame of a candle, or a feather, or any light floating substance, by which the slightest motions of the air are betrayed. Still, the persuasion that sound is some motion of the air, continued to keep hold of men's minds, and acquired additional

distinctness. The illustration employed by Vitruvius, in the following passage, is even now one of the best we can offer.[3\8] "Voice is breath, flowing, and made sensible to the hearing by striking the air. It moves in infinite circumferences of circles, as when, by throwing a stone into still water, you produce innumerable circles of waves, increasing from the centre and spreading outwards, till the boundary of the space, or some obstacle, prevents their outlines from going further. In the same manner the voice makes its motion in circles. But in water the circle moves breadthways upon a level plain; the voice proceeds in breadth, and also successively ascends in height."

[Note 1\8: _Hist. Son. et Aud._ vol. ix. p. 68.]

[Note 2\8: _Ibid._]

[Note 3\8: _De Arch._ v. 3.]

Both the comparison, and the notice of the difference of the two cases, prove the architect to have had very clear notions on the subject; which he further shows by comparing the resonance of the walls of a building to the disturbance of the outline of the waves of water when they meet with a boundary, and are thrown back. "Therefore, as in the outlines of waves in water, so in the voice, if no obstacle interrupt the foremost, it does not disturb the second and the following ones, so that all come to the ears of persons, whether high up or low down, without resonance. But when they strike against obstacles, the foremost, being thrown back, disturb the lines of those which follow." Similar analogies were employed by the ancients in order to explain the occurrence of Echoes. Aristotle says,[4\8] "An Echo takes place, when the air, being as one body in consequence of the vessel which bounds it, and being prevented from being thrust forwards, is reflected {26} back like a ball." Nothing material was added to such views till modern times.

[Note 4\8: _De Animâ_, ii. 8.]

Thus the first conjectures of those who philosophized concerning sound, led them to an opinion concerning its causes and laws, which only required to be distinctly understood, and traced to mechanical principles, in order to form a genuine science of Acoustics. It was, no doubt, a work which required a long time and sagacious reasoners, to supply what was thus wanting; but still, in consequence of this peculiar circumstance in the early condition of the prevalent doctrine concerning sound, the history of Acoustics assumes a peculiar form. Instead of containing, like the history of Astronomy or of Optics, a series of generalizations, each including and rising above preceding generalizations; in this case, the highest generalization is in view from the first; and the object of the philosopher is to determine its precise meaning and circumstances in each example. Instead of having a series of inductive Truths, successively dawning on men's minds, we have a series of Explanations, in which certain experimental facts and laws are reconciled, as to their mechanical principles and their measures, with the general doctrine already in our possession. Instead of having to travel gradually towards a great discovery, like Universal Gravitation, or Luminiferous Undulations, we take our stand upon acknowledged truths, the production and propagation of sound by the motion of bodies and of air; and we connect these with other truths, the laws of motion and the known properties of

bodies, as, for instance, their elasticity. Instead of _Epochs of Discovery_, we have _Solutions of Problems_; and to these we must now proceed.

We must, however, in the first place, notice that these Problems include other subjects than the mere production and propagation of sound generally. For such questions as these obviously occur:--what are the laws and cause of the differences of sounds;--of acute and grave, loud and low, continued and instantaneous;--and, again, of the differences of articulate sounds, and of the quality of different voices and different instruments? The first of these questions, in particular, the real nature of the difference of acute and grave sounds, could not help attracting attention; since the difference of notes in this respect was the foundation of one of the most remarkable mathematical sciences of antiquity. Accordingly, we find attempts to explain this difference in the ancient writers on music. In Ptolemy's _Harmonics_, the third Chapter of the first Book is entitled, "How the {27} acuteness and graveness of notes is produced;" and in this, after noting generally the difference of sounds, and the causes of difference (which he states to be the force of the striking body, the physical constitution of the body struck, and other causes), he comes to the conclusion, that "the things which produce acuteness in sounds, are a greater density and a smaller size; the things which produce graveness, are a greater rarity and a bulkier form." He afterwards explains this so as to include a considerable portion of truth. Thus he says, "That in strings, and in pipes, other things remaining the same, those which are stopped at the smaller distance from the bridge give the most acute note; and in pipes, those notes which come through holes nearest to the mouth-hole are most acute." He even attempts a further generalization, and says that the greater acuteness arises, in fact, from the body being more tense; and that thus "hardness may counteract the effect of greater density, as we see that brass produces a more acute sound than lead." But this author's notions of tension, since they were applied so generally as to include both the tension of a string, and the tension of a piece of solid brass, must necessarily have been very vague. And he seems to have been destitute of any knowledge of the precise nature of the motion or impulse by which sound is produced; and, of course, still more ignorant of the mechanical principles by which these motions are explained. The notion of _vibrations_ of the parts of sounding bodies, does not appear to have been dwelt upon as an essential circumstance; though in some cases, as in sounding strings, the fact is very obvious. And the notion of vibrations of the air does not at all appear in ancient writers, except so far as it may be conceived to be implied in the comparison of aërial and watery waves, which we have quoted from Vitruvius. It is however, very unlikely that, even in the case of water, the motions of the particles were distinctly conceived, for such conception is far from obvious.

The attempts to apprehend distinctly, and to explain mechanically, the phenomena of sound, gave rise to a series of Problems, of which we most now give a brief history. The questions which more peculiarly constitute the Science of Acoustics, are the questions concerning those motions or affections of the air by which it is the medium of hearing. But the motions of sounding bodies have both so much connexion with those of the medium, and so much resemblance to them, that we shall include in our survey researches on that subject also. {28}

# CHAPTER II.

PROBLEM OF THE VIBRATIONS OF STRINGS.

THAT the continuation of sound depends on a continued minute and rapid motion, a shaking or trembling, of the parts of the sounding body, was soon seen. Thus Bacon says,[5\8] "The duration of the sound of a bell or a string when struck, which appears to be prolonged and gradually extinguished, does not proceed from the first percussion; but the trepidation of the body struck perpetually generates a new sound. For if that trepidation be prevented, and the bell or string be stopped, the sound soon dies: as in _spinets_, as soon as the _spine_ is let fall so as to touch the string, the sound ceases." In the case of a stretched string, it is not difficult to perceive that the motion is a motion back and forwards across the straight line which the string occupies when at rest. The further examination of the quantitative circumstances of this oscillatory motion was an obvious problem; and especially after oscillations, though of another kind (those of a pendulous body), had attracted attention, as they had done in the school of Galileo. Mersenne, one of the promulgators of Galileo's philosophy in France, is the first author in whom I find an examination of the details of this case (_Harmonicorum Liber_, Paris, 1636). He asserts,[6\8] that the differences and concords of acute and grave sounds depend on the rapidity of vibrations, and their ratio; and he proves this doctrine by a series of experimental comparisons. Thus he finds[7\8] that the note of a string is as its length, by taking a string first twice, and then four times as long as the original string, other things remaining the same. This, indeed, was known to the ancients, and was the basis of that numerical indication of the notes which the proposition expresses. Mersenne further proceeds to show the effect of thickness and tension. He finds (Prop. 7) that a string must be four times as thick as another, to give the octave below; he finds, also (Prop. 8), that the tension must be about four times as great in order to produce the octave above. From these proportions various others are deduced, and the _law of the_ {29} _phenomena_ of this kind may be considered as determined. Mersenne also undertook to _measure_ the phenomena numerically, that is to determine the number of vibrations of the string in each of such cases; which at first might appear difficult, since it is obviously impossible to count with the eye the passages of a sounding string backwards and forwards. But Mersenne rightly assumed, that the number of vibrations is the same so long as the tone is the same, and that the ratios of the numbers of vibrations of different strings may be determined from the numerical relations of their notes. He had, therefore, only to determine the number of vibrations of one certain string, or one known note, to know those of all others. He took a musical string of three-quarters of a foot long, stretched with a weight of six pounds and five eighths, which he found gave him by its vibrations a certain standard note in his organ: he found that a string of the same material and tension, fifteen feet, that is, twenty times as long, made ten recurrences in a second; and he inferred that the number of vibrations of the shorter string must also be twenty times as great; and thus such a string must make in one second of time two hundred vibrations.

[Note 5\8: _Hist. Son. et Aud._ vol. ix. p. 71.]

[Note 6\8: L. i. Prop. 15.]

[Note 7\8: L. ii. Prop. 6.]

This determination of Mersenne does not appear to have attracted due notice; but some time afterwards attempts were made to ascertain the connexion between the sound and its elementary pulsations in a more direct manner. Hooke, in 1681, produced sounds by the striking of the teeth of brass wheels,[8\8] and Stancari, in 1706, by whirling round a large wheel in air, showed, before the Academy of Bologna, how the number of vibrations in a given note might be known. Sauveur, who, though deaf for the first seven years of his life, was one of the greatest promoters of the science of sound, and gave it its name of _Acoustics_, endeavored also, about the same time, to determine the number of vibrations of a standard note, or, as he called it, Fixed Sound. He employed two methods, both ingenious and both indirect. The first was the method of _beats_. Two organ-pipes, which form a discord, are often heard to produce a kind of _howl_, or _wavy_ noise, the sound swelling and declining at small intervals of time. This was readily and rightly ascribed to the coincidences of the pulsations of sound of the two notes after certain cycles. Thus, if the number of vibrations of the notes were as fifteen to sixteen in the same time, every fifteenth vibration of the one would coincide with every {30} sixteenth vibration of the other, while all the intermediate vibrations of the two tones would, in various degrees, disagree with each other; and thus every such cycle, of fifteen and sixteen vibrations, might be heard as a separate beat of sound. Now, Sauveur wished to take a case in which these beats were so slow as to be counted,[9\8] and in which the ratio of the vibrations of the notes was known from a knowledge of their musical relations. Thus if the two notes form an interval of a semitone, their ratio will be that above supposed, fifteen to sixteen; and if the beats be found to be six in a second, we know that, in that time, the graver note makes ninety and the acuter ninety-six vibrations. In this manner Sauveur found that an open organ-pipe, five feet long, gave one hundred vibrations in a second.

[Note 8\8: _Life_, p. xxiii.]

[Note 9\8: _Ac. Sc. Hist._ 1700, p. 131.]

Sauveur's other method is more recondite, and approaches to a mechanical view of the question.[10\8] He proceeded on this basis; a string, horizontally stretched, cannot be drawn into a mathematical straight line, but always hangs in a very flat curve, or _festoon_. Hence Sauveur assumed that its transverse vibrations may be conceived to be identical with the lateral swingings of such a festoon. Observing that the string C, in the middle of a harpsichord, hangs in such a festoon to the amount of 1⁄323rd of an inch, he calculates, by the laws of pendulums, the time of oscillation, and finds it 1⁄122nd of a second. Thus this C, his _fixed note_, makes one hundred and twenty-two vibrations in a second. It is curious that this process, seemingly so arbitrary, is capable of being justified on mechanical principles; though we can hardly give the author credit for the views which this justification implies. It is, therefore, easy to understand that it agreed with other experiments, in the laws which it gave for the dependence of the tone on the length and tension.

[Note 10\8: _Ac. Sc. Hist._ 1713.]

The problem of satisfactorily explaining this dependence, on mechanical principles, naturally pressed upon the attention of mathematicians when the law of the phenomena was thus completely determined by Mersenne and Sauveur. It was desirable to show that both the circumstances and the measure of the phenomena were such as known mechanical causes and laws would explain. But this problem, as might be expected, was not attacked till mechanical principles, and the modes of applying them, had become tolerably familiar.

As the vibrations of a string are produced by its tension, it appeared to be necessary, in the first place, to determine the law of the tension {31} which is called into action by the motion of the string; for it is manifest that, when the string is drawn aside from the straight line into which it is stretched, there arises an additional tension, which aids in drawing it back to the straight line as soon as it is let go. Hooke (_On Spring_, 1678) determined the law of this additional tension, which he expressed in his noted formula, "Ut tensio sic vis," the Force is as the Tension; or rather, to express his meaning more clearly, the Force of tension is as the Extension, or, in a string, as the increase of length. But, in reality, this principle, which is important in many acoustical problems, is, in the one now before us, unimportant; the force which urges the string towards the straight line, depends, with such small extensions as we have now to consider, not on the extension, but on the curvature; and the power of treating the mathematical difficulty of curvature, and its mechanical consequences, was what was requisite for the solution of this problem.

The problem, in its proper aspect, was first attacked and mastered by Brook Taylor, an English mathematician of the school of Newton, by whom the solution was published in 1715, in his _Methodus Incrementorum_. Taylor's solution was indeed imperfect, for it only pointed out a form and a mode of vibration, with which the string _might_ move consistently with the laws of mechanics; not the mode in which it _must_ move, supposing its form to be any whatever. It showed that the curve might be of the nature of that which is called _the companion to the cycloid_; and, on the supposition of the curve of the string being of this form, the calculation confirmed the previously established laws by which the tone, or the time of vibration, had been discovered to depend on the length, tension, and bulk of the string. The mathematical incompleteness of Taylor's reasoning must not prevent us from looking upon his solution of the problem as the most important step in the progress of this part of the subject: for the difficulty of applying mechanical principles to the question being once overcome, the extension and correction of the application was sure to be undertaken by succeeding mathematicians; and, accordingly, this soon happened. We may add, moreover, that the subsequent and more general solutions require to be considered with reference to Taylor's, in order to apprehend distinctly their import; and further, that it was almost evident to a mathematician, even before the general solution had appeared, that the dependence of the time of vibration on the length and tension, would be the same in the general case as in the {32} Taylorian curve; so that, for the ends of physical philosophy, the solution was not very incomplete.

John Bernoulli, a few years afterwards,[11\8] solved the problem of vibrating chords on nearly the same principles and suppositions as Taylor; but a little later (in

1747), the next generation of great mathematicians, D'Alembert, Euler, and Daniel Bernoulli, applied the increased powers of analysis to give generality to the mode of treating this question; and especially the calculus of partial differentials, invented for this purpose. But at this epoch, the discussion, so far as it bore on physics, belonged rather to the history of another problem, which comes under our notice hereafter, that of the composition of vibrations; we shall, therefore, defer the further history of the problem of vibrating strings, till we have to consider it in connexion with new experimental facts.

[Note 11\8: _Op._ iii. p. 207.]

## CHAPTER III.

PROBLEM OF THE PROPAGATION OF SOUND.

WE have seen that the ancient philosophers, for the most part, held that sound was transmitted, as well as produced, by some motion of the air, without defining what kind of motion this was; that some writers, however, applied to it a very happy similitude, the expansive motion of the circular waves produced by throwing a stone into still water; but that notwithstanding, some rejected this mode of conception, as, for instance, Bacon, who ascribed the transmission of sound to certain "spiritual species."

Though it was an obvious thought to ascribe the motion of sound to some motion of air; to conceive what kind of motion could and did produce this effect, must have been a matter of grave perplexity at the time of which we are speaking; and is far from easy to most persons even now. We may judge of the difficulty of forming this conception, when we recollect that John Bernoulli the younger[12\8] declared, that he could not understand Newton's proposition on this subject. The difficulty consists in this; that the movement of the parts of air, in which sound consists, travels along, but that the parts {33} of air themselves do not so travel. Accordingly Otto Guericke,[13\8] the inventor of the air-pump, asks, "How can sound be conveyed by the motion of the air? when we find that it is better conveyed through air that is still, than when there is a wind." We may observe, however, that he was partly misled by finding, as he thought, that a bell could be heard in the vacuum of his air-pump; a result which arose, probably, from some imperfection in his apparatus.

[Note 12\8:_ Prize Dis. on Light_, 1736.]

[Note 13\8: _De Vac. Spat._ p. 138.]

Attempts were made to determine, by experiment, the circumstances of the motion of sound; and especially its velocity. Gassendi[14\8] was one of the first who did this. He employed fire-arms for the purpose, and thus found the velocity to be 1473 Paris feet in a second. Roberval found a velocity so small (560 feet) that it threw uncertainty upon the rest, and affected Newton's reasonings subsequently.[15\8] Cassini, Huyghens, Picard, Römer, found a velocity of 1172 Paris feet, which is more accurate than the former. Gassendi had been surprised to find that the velocity with which sounds travel, is the same whether they are loud or gentle.

[Note 14\8: Fischer, _Gesch. d. Physik_. vol. i. 171.]

[Note 15\8: Newt. _Prin._ B. ii. P. 50, Schol.]

The explanation of this constant velocity of sound, and of its amount, was one of the problems of which a solution was given in the Great Charter of modern science, Newton's _Principia_ (1687). There, for the first time, were explained the real nature of the motions and mutual action of the parts of the air through which sound is transmitted. It was shown[16\8] that a body vibrating in an elastic medium, will propagate _pulses_ through the medium; that is, the parts of the medium will move forwards and backwards, and this motion will affect successively those parts which are at a greater and greater distance from the origin of motion. The parts, in going forwards, produce condensation; in returning to their first places, they allow extension; and the play of the elasticities developed by these expansions and contractions, supplies the forces which continue to propagate the motion.

[Note 16\8: Newt. _Prin._ B. ii. P. 43.]

The idea of such a motion as this, is, as we have said, far from easy to apprehend distinctly: but a distinct apprehension of it is a step essential to the physical part of the sciences now under notice; for it is by means of such _pulses_, or _undulations_, that not only sound, but light, and probably heat, are propagated. We constantly meet with evidence of the difficulty which men have in conceiving this undulatory motion, and in separating it from a local motion of the medium as a {34} mass. For instance, it is not easy at first to conceive the waters of a great river flowing constantly _down_ towards the sea, while waves are rolling _up_ the very same part of the stream; and while the great elevation, which makes the tide, is travelling from the sea perhaps with a velocity of fifty miles an hour. The motion of such a wave, or elevation, is distinct from any stream, and is of the nature of undulations in general. The parts of the fluid stir for a short time and for a small distance, so as to accumulate themselves on a neighboring part, and then retire to their former place; and this movement affects the parts in the order of their places. Perhaps if the reader looks at a field of standing corn when gusts of wind are sweeping over it in visible waves, he will have his conception of this matter aided; for he will see that here, where each ear of grain is anchored by its stalk, there can be no permanent local motion of the substance, but only a successive stooping and rising of the separate straws, producing hollows and waves, closer and laxer strips of the crowded ears.

Newton had, moreover, to consider the mechanical consequences which such condensations and rarefactions of the elastic medium, air, would produce in the parts of the fluid itself. Employing known laws of the elasticity of air, he showed, in a very remarkable proposition,[17\8] the law according to which the particles of air might vibrate. We may observe, that in this solution, as in that of the vibrating string already mentioned, a rule was exhibited according to which the particles _might_ oscillate, but not the law to which they _must_ conform. It was proved that, by taking the motion of each particle to be perfectly similar to that of a pendulum, the forces, developed by contraction and expansion, were precisely such as the motion required; but it was not shown that no other type of oscillation would give rise to the same

accordance of force and motion. Newton's reasoning also gave a determination of the speed of propagation of the pulses: it appeared that sound ought to travel with the velocity which a body would acquire by falling freely through half _the height of a homogeneous atmosphere_; "the height of a homogeneous atmosphere" being the height which the air must have, in order to produce, at the earth's surface, the actual atmospheric pressure, supposing no diminution of density to take place in ascending. This height is about 29,000 feet; and hence it followed that the velocity was 968 feet. This velocity is really considerably less than that of sound; but at the time of which {35} we speak, no accurate measure had been established; and Newton persuaded himself, by experiments made in the cloister of Trinity College, his residence, that his calculation was not far from the fact. When, afterwards, more exact experiments showed the velocity to be 1142 English feet, Newton attempted to explain the difference by various considerations, none of which were adequate to the purpose;--as, the dimensions of the solid particles of which the fluid air consists;--or the vapors which are mixed with it. Other writers offered other suggestions; but the true solution of the difficulty was reserved for a period considerably subsequent.

[Note 17\8: _Princ._ B. ii. P. 48.]

Newton's calculation of the motion of sound, though logically incomplete, was the great step in the solution of the problem; for mathematicians could not but presume that his result was not restricted to the hypothesis on which he had obtained it; and the extension of the solution required only mere ordinary talents. The logical defect of his solution was assailed, as might have been expected. Cranmer (professor at Geneva), in 1741, conceived that he was destroying the conclusiveness of Newton's reasoning, by showing that it applied equally to other modes of oscillation. This, indeed, contradicted the enunciation of the 48th Prop. of the Second Book of the _Principia_; but it confirmed and extended all the general results of the demonstration; for it left even the velocity of sound unaltered, and thus showed that the velocity did not depend mechanically on the type of the oscillation. But the satisfactory establishment of this physical generalization was to be supplied from the vast generalizations of analysis, which mathematicians were now becoming able to deal with. Accordingly this task was performed by the great master of analytical generalization, Lagrange, in 1759, when, at the age of twenty-three, he and two friends published the first volume of the _Turin Memoirs_. Euler, as his manner was, at once perceived the merit of the new solution, and pursued the subject on the views thus suggested. Various analytical improvements and extensions were introduced into the solution by the two great mathematicians; but none of these at all altered the formula by which the velocity of sound was expressed; and the discrepancy between calculation and observation, about one-sixth of the whole, which had perplexed Newton, remained still unaccounted for.

The merit of satisfactorily explaining this discrepancy belongs to Laplace. He was the first to remark[18\8] that the common law of the {36} changes of elasticity in the air, as dependent on its compression, cannot be applied to those rapid vibrations in which sound consists, since the sudden compression produces a degree of heat which additionally increases the elasticity. The ratio of this increase depended on the

experiments by which the relation of heat and air is established. Laplace, in 1816, published[19\8] the theorem on which the correction depends. On applying it, the calculated velocity of sound agreed very closely with the best antecedent experiments, and was confirmed by more exact ones instituted for that purpose.

[Note 18\8: _Méc. Cél._ t. v. l. xii. p. 96.]

[Note 19\8: _Ann. Phys. et Chim._ t. iii. p. 288.]

This step completes the solution of the problem of the propagation of sound, as a mathematical induction, obtained from, and verified by, facts. Most of the discussions concerning points of analysis to which the investigations on this subject gave rise, as, for instance, the admissibility of _discontinuous functions_ into the solutions of partial differential equations, belong to the history of pure mathematics. Those which really concern the physical theory of sound may be referred to the problem of the motion of air in tubes, to which we shall soon have to proceed; but we must first speak of another form which the problem of vibrating strings assumed.

It deserves to be noticed that the ultimate result of the study of the undulations of fluids seems to show that the comparison of the motion of air in the diffusion of sound with the motion of circular waves from a centre in water, which is mentioned at the beginning of this chapter, though pertinent in a certain way, is not exact. It appears by Mr. Scott's recent investigations concerning waves,[20\8] that the circular waves are oscillating waves of the Second order, and are _gregarious_. The sound-wave seems rather to resemble the great solitary Wave of Translation of the First order, of which we have already spoken in Book vi. chapter vi.

[Note 20\8: _Brit. Ass. Reports for_ 1844, p. 361.]

## CHAPTER IV.

PROBLEM OF DIFFERENT SOUNDS OF THE SAME STRING.

IT had been observed at an early period of acoustical knowledge, that one string might give several sounds. Mersenne and others {37} had noticed[21\8] that when a string vibrates, one which is in unison with it vibrates without being touched. He was also aware that this was true if the second string was an octave or a twelfth below the first. This was observed as a new fact in England in 1674, and communicated to the Royal Society by Wallis.[22\8] But the later observers ascertained further, that the longer string divides itself into two, or into three equal parts, separated by _nodes_, or points of rest; this they proved by hanging bits of paper on different parts of the string. The discovery so modified was again made by Sauveur[23\8] about 1700. The sounds thus produced in one string by the vibration of another, have been termed _Sympathetic Sounds_. Similar sounds are often produced by performers on stringed instruments, by touching the string at one of its aliquot divisions, and are then called the _Acute harmonics_. Such facts were not difficult to explain on Taylor's view of the mechanical condition of the string; but the difficulty was increased when it was noticed that a sounding body could produce these different notes _at the same time_. Mersenne had remarked this, and the fact was more distinctly observed and pursued

by Sauveur. The notes thus produced in addition to the genuine note of the string, have been called _Secondary Notes_; those usually heard are, the Octave, the Twelfth, and the Seventeenth above the note itself. To supply a mode of conceiving distinctly, and explaining mechanically, vibrations which should allow of such an effect, was therefore a requisite step in acoustics.

[Note 21\8: _Harm._ lib. iv. Prop. 28 (1636).]

[Note 22\8: _Ph. Tr._ 1677, April.]

[Note 23\8: _A. P._ 1701.]

This task was performed by Daniel Bernoulli in a memoir published in 1755.[24\8] He there stated and proved the Principle of _the coexistence of small vibrations_. It was already established, that a string might vibrate either in a single _swelling_ (if we use this word to express the curve between two nodes which Bernoulli calls a _ventre_), or in two or three or any number of equal swellings with immoveable nodes between. Daniel Bernoulli showed further, that these nodes might be combined, each taking place as if it were the only one. This appears sufficient to explain the coexistence of the harmonic sounds just noticed. D'Alembert, indeed, in the article _Fundamental_ in the French _Encyclopédie_, and Lagrange in his _Dissertation on Sound_ in the _Turin Memoirs_,[25\8] offer several objections to this explanation; and it cannot be denied that the subject has its difficulties; but {38} still these do not deprive Bernoulli of the merit of having pointed out the principle of Coexistent Vibrations, or divest that principle of its value in physical science.

[Note 24\8: _Berlin Mem._ 1753, p. 147.]

[Note 25\8: T. i. pp. 64, 103.]

Daniel Bernoulli's Memoir, of which we speak, was published at a period when the clouds which involve the general analytical treatment of the problem of vibrating strings, were thickening about Euler and D'Alembert, and darkening into a controversial hue; and as Bernoulli ventured to interpose his view, as a solution of these difficulties, which, in a mathematical sense, it is not, we can hardly be surprised that he met with a rebuff. The further prosecution of the different modes of vibration of the same body need not be here considered.

The sounds which are called _Grave Harmonics_, have no analogy with the Acute Harmonics above-mentioned; nor do they belong to this section; for in the case of Grave Harmonics, we have one sound from the co-operation of two strings, instead of several sounds from one string. These harmonics are, in fact, connected with beats, of which we have already spoken; the beats becoming so close as to produce a note of definite musical quality. The discovery of the Grave Harmonics is usually ascribed to Tartini, who mentions them in 1754; but they are first noticed[26\8] in the work of Sorge _On tuning Organs_, 1744. He there expresses this discovery in a query. "Whence comes it, that if we tune a fifth (2 : 3), a _third_ sound is faintly heard, the octave below the lower of the two notes? Nature shows that with 2 : 3, she still requires the unity, to perfect the order 1, 2, 3." The truth is, that these numbers express the frequency of the vibrations, and thus there will be coincidences of the

notes 2 and 3, which are of the frequency 1, and consequently give the octave below the sound 2. This is the explanation given by Lagrange,[27\8] and is indeed obvious.

[Note 26\8: Chladni. _Acoust._ p. 254.]

[Note 27\8: _Mem. Tur._ i. p. 104.]

## CHAPTER V.

PROBLEM OF THE SOUNDS OF PIPES.

IT was taken for granted by those who reasoned on sounds, that the sounds of flutes, organ-pipes, and wind-instruments in general, {39} consisted in vibrations of some kind; but to determine the nature and laws of these vibrations, and to reconcile them with mechanical principles, was far from easy. The leading facts which had been noticed were, that the note of a pipe was proportional to its length, and that a flute and similar instruments might be made to produce some of the acute harmonics, as well as the genuine note. It had further been noticed,[28\8] that pipes closed at the end, instead of giving the series of harmonics 1, ½, ⅓, ¼, &c., would give only those notes which answer to the odd numbers 1, ⅓, ⅕, &c. In this problem also, Newton[29\8] made the first step to the solution. At the end of the propositions respecting the velocity of sound, of which we have spoken, he noticed that it appeared by taking Mersenne's or Sauveur's determination of the number of vibrations corresponding to a given note, that the pulse of air runs over twice the length of the pipe in the time of each vibration. He does not follow out this observation, but it obviously points to the theory, that the sound of a pipe consists of pulses which travel back and forwards along its length, and are kept in motion by the breath of the player. This supposition would account for the observed dependence of the note on the length of the pipe. The subject does not appear to have been again taken up in a theoretical way till about 1760; when Lagrange in the second volume of the _Turin Memoirs_, and D. Bernoulli in the _Memoirs of the French Academy_ for 1762, published important essays, in which some of the leading facts were satisfactorily explained, and which may therefore be considered as the principal solutions of the problem.

[Note 28\8: D. Bernoulli, _Berlin. Mem._ 1753, p. 150.]

[Note 29\8: _Princip._ Schol. Prop. 50.]

In these solutions there was necessarily something hypothetical. In the case of vibrating strings, as we have seen, the Form of the vibrating curve was guessed at only, but the existence and position of the Nodes could be rendered visible to the eye. In the vibrations of air, we cannot see either the places of nodes, or the mode of vibration; but several of the results are independent of these circumstances. Thus both of the solutions explain the fact, that a tube closed at one end is in unison with an open tube of double the length; and, by supposing nodes to occur, they account for the existence of the odd series of harmonics alone, 1, 3, 5, in closed tubes, while the whole series, 1, 2, 3, 4, 5, &c., occurs in open ones. Both views of the nature of the vibration appear to be nearly the same; though Lagrange's is expressed with an

analytical generality which renders it obscure, and Bernoulli has perhaps {40} laid down an hypothesis more special than was necessary. Lagrange[30\8] considers the vibration of open flutes as "the oscillations of a fibre of air," under the condition that its elasticity at the two ends is, during the whole oscillation, the same as that of the surrounding atmosphere. Bernoulli supposes[31\8] the whole inertia of the air in the flute to be collected into one particle, and this to be moved by the whole elasticity arising from this displacement. It may be observed that both these modes of treating the matter come very near to what we have stated as Newton's theory; for though Bernoulli supposes all the air in the flute to be moved at once, and not successively, as by Newton's pulse, in either case the whole elasticity moves the whole air in the tube, and requires more time to do this according to its quantity. Since that time, the subject has received further mathematical developement from Euler,[32\8] Lambert,[33\8] and Poisson;[34\8] but no new explanation of facts has arisen. Attempts have however been made to ascertain experimentally the places of the nodes. Bernoulli himself had shown that this place was affected by the amount of the opening, and Lambert[35\8] had examined other cases with the same view. Savart traced the node in various musical pipes under different conditions; and very recently Mr. Hopkins, of Cambridge, has pursued the same experimental inquiry.[36\8] It appears from these researches, that the early assumptions of mathematicians with regard to the position of the nodes, are not exactly verified by the facts. When the air in a pipe is made to vibrate so as to have several nodes which divide it into equal parts, it had been supposed by acoustical writers that the part adjacent to the open end was half of the other parts; the outermost node, however, is found experimentally to be _displaced_ from the position thus assigned to it, by a quantity depending on several collateral circumstances.

[Note 30\8: _Mém. Turin_, vol. ii. p. 154.]

[Note 31\8: _Mém. Berlin_, 1753, p. 446.]

[Note 32\8: _Nov. Act. Petrop._ tom. xvi.]

[Note 33\8: _Acad. Berlin_, 1775.]

[Note 34\5: _Journ. Ec. Polyt._ cap. 14.]

[Note 35\8: _Acad. Berlin_, 1775.]

[Note 36\8: _Camb. Trans._ vol. v. p. 234.]

Since our purpose was to consider this problem only so far as it has tended towards its mathematical solution, we have avoided saying anything of the dependence of the mode of vibration on the cause by which the sound is produced; and consequently, the researches on the effects of reeds, embouchures, and the like, by Chladni, Savart, Willis, and others, do not belong to our subject. It is easily seen that the complex effect of the elasticity and other properties of the reed and of the air together, is a problem of which we can hardly {41} hope to give a complete solution till our knowledge has advanced much beyond its present condition.

Indeed, in the science of Acoustics there is a vast body of facts to which we might apply what has just been said; but for the sake of pointing out some of them, we shall

consider them as the subjects of one extensive and yet unsolved problem.

## CHAPTER VI.

PROBLEM OF DIFFERENT MODES OF VIBRATION OF BODIES IN GENERAL.

NOT only the objects of which we have spoken hitherto, strings and pipes, but almost all bodies are capable of vibration. Bells, gongs, tuning-forks, are examples of solid bodies; drums and tambourines, of membranes; if we run a wet finger along the edge of a glass goblet, we throw the fluid which it contains into a regular vibration; and the various character which sounds possess according to the room in which they are uttered, shows that large masses of air have peculiar modes of vibration. Vibrations are generally accompanied by sound, and they may, therefore, be considered as acoustical phenomena, especially as the sound is one of the most decisive facts in indicating the mode of vibration. Moreover, every body of this kind can vibrate in many different ways, the vibrating segments being divided by Nodal Lines and Surfaces of various form and number. The mode of vibration, selected by the body in each case, is determined by the way in which it is held, the way in which it is set in vibration, and the like circumstances.

The general problem of such vibrations includes the discovery and classification of the phenomena; the detection of their formal laws; and, finally, the explanation of these on mechanical principles. We must speak very briefly of what has been done in these ways. The facts which indicate Nodal Lines had been remarked by Galileo, on the sounding board of a musical instrument; and Hooke had proposed to observe the vibrations of a bell by strewing flour upon it. But it was Chladni, a German philosopher, who enriched acoustics with the discovery of the vast variety of symmetrical figures of Nodal Lines, which are exhibited on plates of regular forms, when {42} made to sound. His first investigations on this subject, _Entdeckungen über die Theorie des Klangs_, were published 1787; and in 1802 and 1817 he added other discoveries. In these works he not only related a vast number of new and curious facts, but in some measure reduced some of them to order and law. For instance, he has traced all the vibrations of square plates to a resemblance with those forms of vibration in which Nodal Lines are parallel to one side of the square, and those in which they are parallel to another side; and he has established a notation for the modes of vibration founded on this classification. Thus, 5-2 denotes a form in which there are five nodal lines parallel to one side, and two to another; or a form which can be traced to a disfigurement of such a standard type. Savart pursued this subject still further; and traced, by actual observation, the forms of the Nodal Surfaces which divide solid bodies, and masses of air, when in a state of vibration.

The dependence of such vibrations upon their physical cause, namely, the elasticity of the substance, we can conceive in a general way; but the mathematical theory of such cases is, as might be supposed, very difficult, even if we confine ourselves to the obvious question of the mechanical possibility of these different modes of vibration, and leave out of consideration their dependence upon the mode of excitation. The transverse vibrations of elastic rods, plates, and rings, had been considered by Euler

in 1779; but his calculation concerning plates had foretold only a small part of the curious phenomena observed by Chladni;[37\8] and the several notes which, according to his calculation, the same ring ought to give, were not in agreement with experiment.[38\8] Indeed, researches of this kind, as conducted by Euler, and other authors,[39\8] rather were, and were intended for, examples of analytical skill, than explanations of physical facts. James Bernoulli, after the publication of Chladni's experiments in 1787, attempted to solve the problem for plates, by treating a plate as a collection of fibres; but, as Chladni observes, the justice of this mode of conception is disproved, by the disagreement of the results with experiment.

[Note 37\8: Fischer, vi. 587.]

[Note 38\8: Ib. vi. 596.]

[Note 39\8: See Chladni, p. 474.]

The Institute of France, which had approved of Chladni's labours, proposed, in 1809, the problem now before us as a prize-question:[40\8]--"To give the mathematical theory of the vibrations of elastic {43} surfaces, and to compare it with experiment." Only one memoir was sent in as a candidate for the prize; and this was not crowned, though honorable mention was made of it.[41\8] The formulæ of James Bernoulli were, according to M. Poisson's statement, defective, in consequence of his not taking into account the normal force which acts at the exterior boundary of the plate.[42\8] The author of the anonymous memoir corrected this error, and calculated the note corresponding to various figures of the nodal lines; and he found an agreement with experiment sufficient to justify his theory. He had not, however, proved his fundamental equation, which M. Poisson demonstrated in a Memoir, read in 1814.[43\8] At a more recent period also, MM. Poisson and Cauchy (as well as a lady, Mlle. Sophie Germain) have applied to this problem the artifices of the most improved analysis. M. Poisson[44\8] determined the relation of the notes given by the longitudinal and the transverse vibrations of a rod; and solved the problem of vibrating circular plates when the nodal lines are concentric circles. In both these cases, the numerical agreement of his results with experience, seemed to confirm the justice of his fundamental views.[45\8] He proceeds upon the hypothesis, that elastic bodies are composed of separate particles held together by the attractive forces which they exert upon each other, and distended by the repulsive force of heat. M. Cauchy[46\8] has also calculated the transverse, longitudinal, and rotatory vibrations of elastic rods, and has obtained results agreeing closely with experiment through a considerable list of comparisons. The combined authority of two profound analysts, as MM. Poisson and Cauchy are, leads us to believe that, for the simpler cases of the vibrations of elastic bodies, Mathematics has executed her task; but most of the more complex cases remain as yet unsubdued.

[Note 40\8: See Chladni, p. 357.]

[Note 41\8: Poisson's _Mém. in Ac. Sc._ 1812, p. 169.]

[Note 42\8: Ib. p. 220.]

[Note 43\8: Ib. 1812, p. 2.]

[Note 44\8: Ib. t. viii. 1829.]

[Note 45\8: _An. Chim._ tom. xxxvi. 1827, p. 90.]

[Note 46\8: _Exercices de Mathématique_, iii. and iv.]

The two brothers, Ernest and William Weber, made many curious observations on undulations, which are contained in their _Wellenlehre_, (Doctrine of Waves,) published at Leipsig in 1825. They were led to suppose, (as Young had suggested at an earlier period,) that Chladni's figures of nodal lines in plates were to be accounted for by the superposition of undulations.[47\8] Mr. Wheatstone[48\8] has undertaken to account for Chladni's figures of vibrating _square_ plates by this {44} superposition of two or more simple and obviously allowable modes of nodal division, which have the same time of vibration. He assumes, for this purpose, certain "primary figures," containing only _parallel_ nodal lines; and by combining these, first in twos, and then in fours, he obtains most of Chladni's observed figures, and accounts for their transitions and deviations from regularity.

[Note 47\8: _Wellenlehre_, p. 474.]

[Note 48\8: _Phil. Trans._ 1833, p. 593.]

The principle of the superposition of vibrations is so solidly established as a mechanical truth, that we may consider an acoustical problem as satisfactorily disposed of when it is reduced to that principle, as well as when it is solved by analytical mechanics: but at the same time we may recollect, that the right application and limitation of this law involves no small difficulty; and in this case, as in all advances in physical science, we cannot but wish to have the new ground which has been gained, gone over by some other person in some other manner; and thus secured to us as a permanent possession.

_Savart's Laws._--In what has preceded, the vibrations of bodies have been referred to certain general classes, the separation of which was suggested by observation; for example, the _transverse_, _longitudinal_, and _rotatory_,[49\8] vibrations of rods. The transverse vibrations, in which the rod goes backwards and forwards across the line of its length, were the only ones noticed by the earlier acousticians: the others were principally brought into notice by Chladni. As we have already seen in the preceding pages, this classification serves to express important laws; as, for instance, a law obtained by M. Poisson which gives the relation of the notes produced by the transverse and longitudinal vibrations of a rod. But this distinction was employed by M. Felix Savart to express laws of a more general kind; and then, as often happens in the progress of science, by pursuing these laws to a higher point of generality, the distinction again seemed to vanish. A very few words will explain these steps.

[Note 49\8: Vibrations tournantes.]

It was long ago known that vibrations may be communicated by contact. The distinction of transverse and longitudinal vibrations being established, Savart found that if one rod touched another perpendicularly, the longitudinal vibrations of the first occasion transverse vibrations in the second, and _vice versâ_. This is the more

remarkable, since the two sets of vibrations are not equal in rapidity, and therefore cannot sympathize in any obvious manner.[50\8] Savart found himself {45} able to generalize this proposition, and to assert that in any combination of rods, strings, and laminæ, at right angles to each other, the longitudinal and transverse vibrations affect respectively the rods in the one and other direction,[51\8] so that when the horizontal rods, for example, vibrate in the one way, the vertical rods vibrate in the other.

[Note 50\8: _An. Chim._ 1819, tom. xiv. p. 138.]

[Note 51\8: _An. Chim._ p. 152.]

This law was thus expressed in terms of that classification of vibrations of which we have spoken. Yet we easily see that we may express it in a more general manner, without referring to that classification, by saying, that vibrations are communicated so as always to be parallel to their original direction. And by following it out in this shape by means of experiment, M. Savart was led, a short time afterwards, to deny that there is any essential distinction in these different kinds of vibration. "We are thus led," he says[52\8] in 1822, "to consider _normal_ [transverse] vibrations as only one circumstance in a more general motion common to all bodies, analogous to _tangential_ [longitudinal and rotatory] vibrations; that is, as produced by small _molecular oscillations_, and differently modified according to the direction which it affects, relatively to the dimensions of the vibrating body."

[Note 52\8: Ib. t. xxv. p. 33.]

These "inductions," as he properly calls them, are supported by a great mass of ingenious experiments; and may be considered as well established, when they are limited to molecular oscillations, employing this phrase in the sense in which it is understood in the above statement; and also when they are confined to bodies in which the play of elasticity is not interrupted by parts more rigid than the rest, as the sound-post of a violin.[53\8] And before I quit the subject, I may notice a consequence which M. Savart has deduced from his views, and which, at first sight, appears to overturn most of the earlier doctrines respecting vibrating bodies. It was formerly held that tense strings and elastic rods could vibrate only in a determinate series of modes of division, with no intermediate steps. But M. Savart maintains,[54\8] on the contrary, that they produce sounds which are gradually transformed into one another, by indefinite intermediate degrees. The reader may naturally ask, what is the solution of this apparent {46} contradiction between the earliest and the latest discoveries in acoustics. And the answer must be, that these intermediate modes of vibration are complex in their nature, and difficult to produce; and that those which were formerly believed to be the only possible vibrating conditions, are so eminent above all the rest by their features, their simplicity, and their facility, that we may still, for common purposes, consider them as a class apart; although for the sake of reaching a general theorem, we may associate them with the general mass of cases of molecular vibrations. And thus we have no exception here, as we can have none in any case, to our maxim, that what formed part of the early discoveries of science, forms part of its latest systems.

[Note 53\8: For the suggestion of the necessity of this limitation I am indebted to Mr. Willis.]

[Note 54\8: _An. Chim._ 1826, t. xxxii. p. 384.]

We have thus surveyed the progress of the science of sound up to recent times, with respect both to the discovery of laws of phenomena, and the reduction of these to their mechanical causes. The former branch of the science has necessarily been inductively pursued; and therefore has been more peculiarly the subject of our attention. And this consideration will explain why we have not dwelt more upon the deductive labors of the great analysts who have treated of this problem.

To those who are acquainted with the high and deserved fame which the labors of D'Alembert, Euler, Lagrange, and others, upon this subject, enjoy among mathematicians, it may seem as if we had not given them their due prominence in our sketch. But it is to be recollected here, as we have already observed in the case of hydrodynamics, that even when the general principles are uncontested, mere mathematical deductions from them do not belong to the history of physical science, except when they point out laws which are intermediate between the general principle and the individual facts, and which observation may confirm.

The business of constructing any science may be figured as the task of forming a road on which our reason can travel through a certain province of the external world. We have to throw a bridge which may lead from the chambers of our own thoughts, from our speculative principles, to the distant shore of material facts. But in all cases the abyss is too wide to be crossed, except we can find some intermediate points on which the piers of our structure may rest. Mere facts, without connexion or law, are only the rude stones hewn from the opposite bank, of which our arches may, at some time, be built. But mere hypothetical mathematical calculations are only plans of projected structures; and those plans which exhibit only one vast {47} and single arch, or which suppose no support but that which our own position supplies, will assuredly never become realities. We must have a firm basis of intermediate generalizations in order to frame a continuous and stable edifice.

In the subject before us, we have no want of such points of intermediate support, although they are in many instances irregularly distributed and obscurely seen. The number of observed laws and relations of the phenomena of sound, is already very great; and though the time may be distant, there seems to be no reason to despair of one day uniting them by clear ideas of mechanical causation, and thus of making acoustics a perfect secondary mechanical science.

The historical sketch just given includes only such parts of acoustics as have been in some degree reduced to general laws and physical causes; and thus excludes much that is usually treated of under that head. Moreover, many of the numerical calculations connected with sound belong to its agreeable effect upon the ear; as the properties of the various systems of _Temperament_. These are parts of Theoretical Music, not of Acoustics; of the Philosophy of the Fine Arts, not of Physical Science; and may be referred to in a future portion of this work, so far as they bear upon our object.

The science of Acoustics may, however, properly consider other differences of sound than those of acute and grave,--for instance, the _articulate_ differences, or those by which the various letters are formed. Some progress has been made in reducing this part of the subject to general rules; for though Kempelen's "talking machine" was only a work of art, Mr. Willis's machine,[55\8] which exhibits the relation among the vowels, gives us a law such as forms a step in science. We may, however, consider this instrument as a _phthongometer_, or measure of vowel quality; and in that point of view we shall have to refer to it again when we come to speak of such measures.

[Note 55\8: On the Vowel Sounds, and on Reed Organ-pipes. _Camb. Trans._ iii. 237.]

{{49}} BOOK IX.

# _SECONDARY MECHANICAL SCIENCES._

# (CONTINUED)

# HISTORY OF OPTICS,

FORMAL AND PHYSICAL.

Ω Διὸς ὑψιμέλαθρον ἔχων κράτος αἰὲν ἀτειρὲς Ἄστρων, Ἠελίου τε, Σεληναίης τε μέρισμα Πανδαμάτωρ, πυρίπνου, πᾶσιν ζωοῖσιν ἔναυσμα **Ὑψιφάνης ἼΑΙΘΗΡ, κόσμου στοιχεῖον, **ἄριστον· Ἀγλαὸν ὦ βλάστημα, σελασφόρον, ἀστεροφεγγὲς Κικλήσκων λίτομαι σε, κεκραμένον **εὔδιον εἶναι. ORPHEUS. HYMN.

O thou who fillest the palaces of Jove; Who flowest round moon, and sun, and stars above; Pervading, bright, life-giving element, Supernal ETHER, fair and excellent; Fountain of hope and joy, of light and day, We own at length thy tranquil, steady sway.

{{51}} INTRODUCTION.

_Formal and Physical Optics._

THE history of the science of Optics, written at length, would be very voluminous; but we shall not need to make our history so; since our main object is to illustrate the nature of science and the conditions of its progress. In this way Optics is peculiarly instructive; the more so, as its history has followed a course in some respects different from both the sciences previously reviewed. Astronomy, as we have seen, advanced with a steady and continuous movement from one generation to another, from the earliest time, till her career was crowned by the great unforeseen discovery of Newton; Acoustics had her extreme generalization in view from the first, and her history consists in the correct application of it to successive problems; Optics advanced through a scale of generalizations as remarkable as those of Astronomy; but for a long period she was almost stationary; and, at last, was rapidly impelled through all those stages by the energy of two or three discoverers. The highest point

of generality which Optics has reached is little different from that which Acoustics occupied at once; but in the older and earlier science we still want that palpable and pointed confirmation of the general principle, which the undulatory theory receives from optical phenomena. Astronomy has amassed her vast fortune by long-continued industry and labor; Optics has obtained hers in a few years by sagacious and happy speculations; Acoustics, having early acquired a competence, has since been employed rather in improving and adorning than in extending her estate.

The successive inductions by which Optics made her advances, might, of course, be treated in the same manner as those of Astronomy, each having its prelude and its sequel. But most of the discoveries in Optics are of a smaller character, and have less employed the minds of men, than those of Astronomy; and it will not be necessary to exhibit them in this detailed manner, till we come to the great generalization by which the theory was established. I shall, therefore, now pass rapidly in review the earlier optical discoveries, without any such division of the series. {52}

Optics, like Astronomy, has for its object of inquiry, first, the laws of phenomena, and next, their causes; and we may hence divide this science, like the other, into _Formal Optics_ and _Physical Optics_. The distinction is clear and substantive, but it is not easy to adhere to it in our narrative; for, after the theory had begun to make its rapid advance, many of the laws of phenomena were studied and discovered in immediate reference to the theoretical cause, and do not occupy a separate place in the history of science, as in Astronomy they do. We may add, that the reason why Formal Astronomy was almost complete before Physical Astronomy began to exist, was, that it was necessary to construct the science of Mechanics in the mean time, in order to be able to go on; whereas, in Optics, mathematicians were able to calculate the results of the undulatory theory as soon as it had suggested itself from the earlier facts, and while the great mass of facts were only becoming known.

We shall, then, in the first _nine_ chapters of the History of Optics treat of the Formal Science, that is, the discovery of the laws of phenomena. The classes of phenomena which will thus pass under oar notice are numerous; namely, reflection, refraction, chromatic dispersion, achromatization, double refraction, polarization, dipolarization, the colors of thin plates, the colors of thick plates, and the fringes and bands which accompany shadows. All these cases had been studied, and, in most of them, the laws had been in a great measure discovered, before the physical theory of the subject gave to our knowledge a simpler and more solid form.

{{53}} FORMAL OPTICS.

## CHAPTER I.

PRIMARY INDUCTION OF OPTICS.--RAYS OF LIGHT AND LAWS OF REFLECTION.

IN speaking of the Ancient History of Physics, we have already noticed that the optical philosophers of antiquity had satisfied themselves that vision is performed in straight lines;--that they had fixed their attention upon those straight lines, or _rays_, as the proper object of the science;--they had ascertained that rays reflected from a

bright surface make the _angle of reflection_ equal to the _angle of incidence_;--and they had drawn several consequences from these principles.

We may add to the consequences already mentioned, the art of _perspective_, which is merely a corollary from the doctrine of rectilinear visual rays; for if we suppose objects to be referred by such rays to a plane interposed between them and the eye, all the rules of perspective follow directly. The ancients practised this art, as we see in the pictures which remain to us and we learn from Vitruvius,[1\9] that they also wrote upon it. Agatharchus, who had been instructed by Eschylus in the art of making decorations for the theatre, was the first author on this subject, and Anaxagoras, who was a pupil of Agatharchus, also wrote an _Actinographia_, or doctrine of drawing by rays: but none of these treatises are come down to us. The moderns re-invented the art in the flourishing times of the art of painting, that is, about the end of the fifteenth century; and, belonging to that period also, we have treatises[2\9] upon it.

[Note 1\9: _De Arch._ ix. Mont. i. 707.]

[Note 2\9: Gauricus, 1504.]

But these are only deductive applications of the most elementary optical doctrines; we must proceed to the inductions by which further discoveries were made. {54}

## CHAPTER II.

DISCOVERY OF THE LAW OF REFRACTION.

WE have seen in the former part of this history that the Greeks had formed a tolerably clear conception of the refraction as well as the reflection of the rays of light; and that Ptolemy had measured the amount of refraction of glass and water at various angles. If we give the names of the _angle of incidence_ and the _angle of refraction_ respectively to the angles which a ray of light makes with the line perpendicular to surface of glass or water (or any other medium) within and without the medium, Ptolemy had observed that the angle of refraction is always less than the angle of incidence. He had supposed it to be less in a given proportion, but this opinion is false; and was afterwards rightly denied by the Arabian mathematician Alhazen. The optical views which occur in the work of Alhazen are far sounder than those of his predecessors; and the book may be regarded as the most considerable monument which we have of the scientific genius of the Arabians; for it appears, for the most part, not to be borrowed from Greek authorities. The author not only asserts (lib. vii.), that refraction takes place towards the perpendicular, and refers to experiment for the truth of this: and that the quantities of the refraction differ according to the magnitudes of the angles which the directions of the incidental rays (_primæ lineæ_) make with the perpendiculars to the surface; but he also says distinctly and decidedly that the angles of refraction do not follow the proportion of the angles of incidence.

[2nd Ed.] [There appears to be good ground to assent to the assertion of Alhazen's originality, made by his editor Risner, who says, "Euclideum hic vel Ptolemaicum nihil

fere est." Besides the doctrine of reflection and refraction of light, the Arabian author gives a description of the eye. He distinguishes three fluids, _humor aqueus_, _crystallinus_, _vitreus_, and four coats of the eye, _tunica adherens_, _cornea_, _uvea_, _tunica reti similis_. He distinguishes also three kinds of vision: "Visibile percipitur aut solo visu, aut visu et syllogismo, aut visu et anticipatâ notione." He has several propositions relating to what we sometimes call the Philosophy of Vision: for instance this: "E visibili sæpius viso remanet in anima generalis notio," &c.] {55}

The assertion, that the angles of refraction are not proportional to the angles of incidence, was an important remark; and if it had been steadily kept in mind, the next thing to be done with regard to refraction was to go on experimenting and conjecturing till the true law of refraction was discovered; and in the mean time to apply the principle as far as it was known. Alhazen, though he gives directions for making experimental measures of refraction, does not give any Table of the results of such experiments, as Ptolemy had done. Vitello, a Pole, who in the 13th century published an extensive work upon Optics, does give such a table; and asserts it to be deduced from experiment, as I have already said (vol. i.). But this assertion is still liable to doubt in consequence of the table containing impossible observations.

[2nd Ed.] [As I have already stated, Vitello asserts that his Tables were derived from his own observations. Their near agreement with those of Ptolemy does not make this improbable: for where the observations were only made to half a degree, there was not much room for observers to differ. It is not unlikely that the observations of refraction out of air into water and glass, and out of water into glass, were actually made; while the impossible values which accompany them, of the refraction out of water and glass into air, and out of glass into water, were calculated, and calculated from an erroneous rule.]

The principle that a ray refracted in glass or water is turned towards the perpendicular, without knowing the exact law of refraction, enabled mathematicians to trace the effects of transparent bodies in various cases. Thus in Roger Bacon's works we find a tolerably distinct explanation of the effect of a convex glass; and in the work of Vitello the effect of refraction at the two surfaces of a glass globe is clearly traceable.

Notwithstanding Alhazen's assertion of the contrary, the opinion was still current among mathematicians that the angle of refraction was proportional to the angle of incidence. But when Kepler's attention was drawn to the subject, he saw that this was plainly inconsistent with the observations of Vitello for large angles; and he convinced himself by his own experiments that the true law was something different from the one commonly supposed. The discovery of this true law excited in him an eager curiosity; and this point had the more interest for him in consequence of the introduction of a correction for atmospheric refraction into astronomical calculations, which had been made by Tycho, and of the invention of the telescope. In {56} his _Supplement to Vitello_, published in 1604, Kepler attempts to reduce to a rule the measured quantities of refraction. The reader who recollects what we have already narrated, the manner in which Kepler attempted to reduce to law the astronomical observations of Tycho,--devising an almost endless variety of possible formulæ,

tracing their consequences with undaunted industry, and relating, with a vivacious garrulity, his disappointments and his hopes,--will not be surprised to find that he proceeded in the same manner with regard to the Tables of Observed Refractions. He tried a variety of constructions by triangles, conic sections, &c., without being able to satisfy himself; and he at last[3\9] is obliged to content himself with an approximate rule, which makes the refraction partly proportional to the angle of incidence, and partly, to the secant of that angle. In this way he satisfies the observed refractions within a difference of less than half a degree each way. When we consider how simple the law of refraction is, (that the ratio of the sines of the angles of incidence and refraction is constant for the same medium,) it appears strange that a person attempting to discover it, and drawing triangles for the purpose, should fail; but this lot of missing what afterwards seems to have been obvious, is a common one in the pursuit of truth.

[Note 3\9: L. U. K. _Life of Kepler_, p. 115.]

The person who did discover the Law of the Sines, was Willebrord Snell, about 1621; but the law was first published by Descartes, who had seen Snell's papers.[4\9] Descartes does not acknowledge this law to have been first detected by another; and after his manner, instead of establishing its reality by reference to experiment, he pretends to prove _à priori_ that it must be true,[5\9] comparing, for this purpose, the particles of light to balls striking a substance which _accelerates_ them.

[Note 4\9: Huyghens, _Dioptrica_, p. 2.]

[Note 5\9: _Diopt._ p. 53.]

[2nd Ed.] [Huyghens says of Snell's papers, "Quæ et nos vidimus aliquando, et Cartesium quoque vidisse accepimus, et hinc fortasse mensuram illam quæ in sinibus consistit elicuerit." Isaac Vossius, _De Lucis Naturâ et Proprietate_, says that he also had seen this law in Snell's unpublished optical Treatise. The same writer says, "Quod itaque (Cartesius) habet, refractionum momenta non exigenda esse ad angulos sed ad lineas, id tuo Snellio, acceptum ferre debuisset, cujus nomen _more solito_ dissimulavit." "Cartesius got his law from Snell, and _in his usual way_, concealed it." {57}

Huyghens' assertion, that Snell did not _attend to_ the proportion of the sines, is very captious; and becomes absurdly so, when it is made to mean that Snell did not _know_ the law of the sines. It is not denied that Snell knew the true law, or that the true law is the law of the sines. Snell does not use the trigonometrical term _sine_, but he expresses the law in a geometrical form more simply. Even if he _had_ attended to the law of the sines, he might reasonably have preferred his own way of stating it.

James Gregory also independently discovered the true law of refraction; and, in publishing it, states that he had learnt that it had already been published by Descartes.]

But though Descartes does not, in this instance, produce any good claims to the character of an inductive philosopher, he showed considerable skill in tracing the

consequences of the principle when once adopted. In particular we must consider him as the genuine author of the explanation of the rainbow. It is true that Fleischer[6\9] and Kepler had previously ascribed this phenomenon to the rays of sunlight which, falling on drops of rain, are refracted into each drop, reflected at its inner surface, and refracted out again: Antonio de Dominis had found that a glass globe of water, when placed in a particular position with respect to the eye, exhibited bright colors; and had hence explained the circular form of the bow, which, indeed, Aristotle had done before.[7\9] But none of these writers had shown why there was a narrow bright circle of a definite diameter; for the drops which send rays to the eye after two refractions and a reflection, occupy a much wider space in the heavens. Descartes assigned the reason for this in the most satisfactory manner,[8\9] by showing that the rays which, after two refractions and a reflection, come to the eye at an angle of about forty-one degrees with their original direction, are far more dense than those in any other position. He showed, in the same manner, that the existence and position of the _secondary bow_ resulted from the same laws. This is the complete and adequate account of the state of things, so far as the brightness of the bows only is concerned; the explanation of the colors belongs to the next article of our survey.

[Note 6\9: Mont. i. 701.]

[Note 7\9: _Meteorol._ iii. 3.]

[Note 8\9: _Meteorum_, cap. viii. p. 196.]

The explanation of the rainbow and of its magnitude, afforded by Snell's law of sines, was perhaps one of the leading points in the verification of the law. The principle, being once established, was applied, by the aid of mathematical reasoning, to atmospheric refractions, {58} optical instruments, _diacaustic_ curves, (that is, the curves of intense light produced by refraction,) and to various other cases; and was, of course, tested and confirmed by such applications. It was, however, impossible to pursue these applications far, without a due knowledge of the laws by which, in such cases, colors are produced. To these we now proceed.

[2nd Ed.] [I have omitted many interesting parts of the history of Optics about this period, because I was concerned with the _inductive_ discovery of laws, rather than with mathematical _deductions_ from such laws when established, or _applications_ of them in the form of instruments. I might otherwise have noticed the discovery of Spectacle Glasses, of the Telescope, of the Microscope, of the Camera Obscura, and the mathematical explanation of these and other phenomena, as given by Kepler and others. I might also have noticed the progress of knowledge respecting the Eye and Vision. We have seen that Alhazen described the structure of the eye. The operation of the parts was gradually made out. Baptista Porta compares the eye to his _Camera Obscura_ (_Magia Naturalis_, 1579). Scheiner, in his _Oculus_, published 1652, completed the Theory of the Eye. And Kepler discussed some of the questions even now often agitated; as the causes and conditions of our seeing objects single with two eyes, and erect with inverted images.]

# CHAPTER III.

DISCOVERY OF THE LAW OF DISPERSION BY REFRACTION.

EARLY attempts were made to account for the colors of the rainbow, and various other phenomena in which colors are seen to arise from transient and unsubstantial combinations of media. Thus Aristotle explains the colors of the rainbow by supposing[9\9] that it is light seen through a dark medium: "Now," says he, "the bright seen through the dark appears red, as, for instance, the fire of green wood seen through the smoke, and the sun through mist. Also[10\9] the weaker is the light, or the visual power, and the nearer the color approaches to the black; becoming first red, then green, then purple. But[11\9] the {59} vision is strongest in the outer circle, because the periphery is greater;--thus we shall have a gradation from red, through green, to purple, in passing from the outer to the inner circle." This account would hardly have deserved much notice, if it had not been for a strange attempt to revive it, or something very like it, in modern times. The same doctrine is found in the work of De Dominis.[12\9] According to him, light is white: but if we mix with the light something dark, the colors arise,--first red, then green, then blue or violet. He applies this to explain the colors of the rainbow,[13\9] by means of the consideration that, of the rays which come to the eye from the globes of water, some go through a larger thickness of the globe than others, whence he obtains the gradation of colors just described.

[Note 9\9: _Meteor._ iii. 3, p. 373.]

[Note 10\9: Ib. p. 374.]

[Note 11\9: Ib. p. 375.]

[Note 12\9: Cap. iii. p. 9. See also Göthe, _Farbenl._ vol. ii. p. 251.]

[Note 13\9: Göthe, p. 263.]

Descartes came far nearer the true philosophy of the iridal colors. He found that a similar series of colors was produced by refraction of light bounded by shade, through a prism;[14\9] and he rightly inferred that neither the curvature of the surface of the drops of water, nor the reflection, nor the repetition of refraction, were necessary to the generation of such colors. In further examining the course of the rays, he approaches very near to the true conception of the case; and we are led to believe that he might have anticipated Newton in his discovery of the unequal refrangibility of different colors, if it had been possible for him to reason any otherwise than in the terms and notions of his preconceived hypotheses. The conclusion which he draws is,[15\9] that "the particles of the subtile matter which transmit the action of light, endeavor to rotate with so great a force and impetus, that they cannot move in a straight line (whence comes refraction): and that those particles which endeavor to revolve much more strongly produce a red color, those which endeavor to move only a little more strongly produce yellow." Here we have a clear perception that colors and unequal refraction are connected, though the cause of refraction is expressed by a gratuitous hypothesis. And we may add, that he applies this notion rightly, so far as he explains himself,[16\9] to account for the colors of the

rainbow.

[Note 14\9: _Meteor._ Sect. viii. p. 190.]

[Note 15\9: Sect. vii. p. 192.]

[Note 16\9: _Meteor._ Sect. ix.]

It appears to me that Newton and others have done Descartes injustice, in ascribing to De Dominis the true theory of the rainbow. There are two main points of this theory, namely, the showing that a _bright_ circular band, of a certain definite diameter, arises from the {60} great intensity of the light returned at a certain angle; and the referring the different _colors_ to the _different quantity of the refraction_; and both these steps appear indubitably to be the discoveries of Descartes. And he informs us that these discoveries were not made without some exertion of thought. "At first," he says,[17\9] "I doubted whether the iridal colors were produced in the same way as those in the prism; but, at last, taking my pen, and carefully calculating the course of the rays which fell on each part of the drop, I found that many more come at an angle of forty-one degrees, than either at a greater or a less angle. So that there is a bright bow terminated by a shade; and hence the colors are the same as those produced through a prism."

[Note 17\9: Sect. ix. p. 193.]

The subject was left nearly in the same state, in the work of Grimaldi, _Physico-Mathesis, de Lumine, Coloribus et Iride_, published at Bologna in 1665. There is in this work a constant reference to numerous experiments, and a systematic exposition of the science in an improved state. The author's calculations concerning the rainbow are put in the same form as those of Descartes; but he is further from seizing the true principle on which its coloration depends. He rightly groups together a number of experiments in which colors arise from refraction;[18\9] and explains them by saying that the color is brighter where the light is denser: and the light is denser on the side from which the refraction turns the ray, because the increments of refraction are greater in the rays that are more inclined.[19\9] This way of treating the question might be made to give a sort of explanation of most of the facts, but is much more erroneous than a developement of Descartes's view would have been.

[Note 18\9: Prop. 35, p. 254.]

[Note 19\9: Ib. p. 256.]

At length, in 1672, Newton gave[20\9] the true explanation of the facts; namely, that light consists of rays of different colors and different refrangibility. This now appears to us so obvious a mode of interpreting the phenomena, that we can hardly understand how they can be conceived in any other manner; but yet the impression which this discovery made, both upon Newton and upon his contemporaries, shows how remote it was from the then accepted opinions. There appears to have been a general persuasion that the coloration was produced, not by any peculiarity in the law of refraction itself but by some collateral circumstance,--some dispersion or variation of density of the light, in addition to the refraction. Newton's discovery consisted in {61} teaching distinctly that the law of refraction was to be applied, not

to the beam of light in general, but to the colors in particular.

[Note 20\9: _Phil. Trans._ t. vii. p. 3075.]

When Newton produced a bright spot on the wall of his chamber, by admitting the sun's light through a small hole in his window-shutter, and making it pass through a prism, he expected the image to be round; which, of course, it would have been, if the colors had been produced by an equal dispersion in all directions; but to his surprise he saw the image, or _spectrum_, five times as long as it was broad. He found that no consideration of the different thickness of the glass, the possible unevenness of its surface, or the different angles of rays proceeding from the two sides of the sun, could be the cause of this shape. He found, also, that the rays did not go from the prism to the image in curves; he was then convinced that the different colors were refracted separately, and at different angles; and he confirmed this opinion by transmitting and refracting the rays of each color separately.

The experiments are so easy and common, and Newton's interpretation of them so simple and evident, that we might have expected it to receive general assent; indeed, as we have shown, Descartes had already been led very near the same point. In fact, Newton's opinions were not long in obtaining general acceptance; but they met with enough of cavil and misapprehension to annoy extremely the discoverer, whose clear views and quiet temper made him impatient alike of stupidity and of contentiousness.

We need not dwell long on the early objections which were made to Newton's doctrine. A Jesuit, of the name of Ignatius Pardies, professor at Clermont, at first attempted to account for the elongation of the image by the difference of the angles made by the rays from the two edges of the sun, which would produce a difference in the amount of refraction of the two borders; but when Newton pointed out the calculations which showed the insufficiency of this explanation, he withdrew his opposition. Another more pertinacious opponent appeared in Francis Linus, a physician of Liege; who maintained, that having tried the experiment, he found the sun's image, when the sky was clear, to be round and not oblong; and he ascribed the elongation noticed by Newton, to the effect of clouds. Newton for some time refused to reply to this contradiction of his assertions, though obstinately persisted in; and his answer was at last sent, just about the time of Linus's death, in 1675. But Gascoigne, a friend of Linus, still maintained that he and others had seen what the Dutch physician had described; and Newton, who was pleased with the candor of {62} Gascoigne's letter, suggested that the Dutch experimenters might have taken one of the images reflected from the surfaces of the prism, of which there are several, instead of the proper refracted one. By the aid of this hint, Lucas of Liege repeated Newton's experiments, and obtained Newton's result, except that he never could obtain a spectrum whose length was more than three and a half times its breadth. Newton, on his side, persisted in asserting that the image would be five times as long as it was broad, if the experiment were properly made. It is curious that he should have been so confident of this, as to conceive himself certain that such would be the result in all cases. We now know that the dispersion, and consequently the length, of the spectrum, is very different for different kinds of glass, and it is very probable that the Dutch prism was really less dispersive than the English one.[21\9] The erroneous

assumption which Newton made in this instance, he held by to the last; and was thus prevented from making the discovery of which we have next to speak.

[Note 21\9: Brewster's _Newton_, p. 50.]

Newton was attacked by persons of more importance than those we have yet mentioned; namely, Hooke and Huyghens. These philosophers, however, did not object so much to the laws of refraction of different colors, as to some expressions used by Newton, which, they conceived, conveyed false notions respecting the composition and nature of light. Newton had asserted that all the different colors are of distinct kinds, and that, by their composition, they form white light. This is true of colors as far as their analysis and composition by refraction are concerned; but Hooke maintained that all natural colors are produced by various combinations of two primary ones, red and violet;[22\9] and Huyghens held a similar doctrine, taking, however, yellow and blue for his basis. Newton answers, that such compositions as they speak of are not compositions of simple colors in his sense of the expressions. These writers also had both of them adopted an opinion that light consisted in vibrations; and objected to Newton that his language was erroneous, as involving the hypothesis that light was a body. Newton appears to have had a horror of the word _hypothesis_, and protests against its being supposed that his "theory" rests on such a foundation.

[Note 22\9: Brewster's _Newton_, p. 54. _Phil. Trans._ viii. 5084, 6086.]

The doctrine of the unequal refrangibility of different rays is clearly exemplified in the effects of lenses, which produce images more or {63} less bordered with color, in consequence of this property. The improvement of telescopes was, in Newton's time, the great practical motive for aiming at the improvement of theoretical optics. Newton's theory showed why telescopes were imperfect, namely, in consequence of the different refraction of different colors, which produces a _chromatic_ aberration: and the theory was confirmed by the circumstances of such imperfections. The false opinion of which we have already spoken, that the dispersion must be the same when the refraction is the same, led him to believe that the imperfection was insurmountable,--that _achromatic_ refraction could not be obtained: and this view made him turn his attention to the construction of reflecting instead of refracting telescopes. But the rectification of Newton's error was a further confirmation of the general truth of his principles in other respects; and since that time, the soundness of the Newtonian law of refraction has hardly been questioned among physical philosophers.

It has, however, in modern times, been very vehemently controverted in a quarter from which we might not readily have expected a detailed discussion on such a subject. The celebrated Göthe has written a work on _The Doctrine of Colors_, (_Farbenlehre_; Tübingen, 1810,) one main purpose of which is, to represent Newton's opinions, and the work in which they are formally published, (his _Opticks_,) as utterly false and mistaken, and capable of being assented to only by the most blind and obstinate prejudice. Those who are acquainted with the extent to which such an opinion, promulgated by Göthe, was likely to be widely adopted in

Germany, will not be surprised that similar language is used by other writers of that nation. Thus Schelling[23\9] says: "Newton's _Opticks_ is the greatest proof of the possibility of a whole structure of fallacies, which, in all its parts, is founded upon observation and experiment." Göthe, however, does not concede even so much to Newton's work. He goes over a large portion of it, page by page, quarrelling with the experiments, diagrams, reasoning, and language, without intermission; and holds that it is not reconcileable with the most simple facts. He declares,[24\9] that the first time he looked through a prism, he saw the white walls of the room still look white, "and though alone, I pronounced, as by an instinct, that the Newtonian doctrine is false." We need not here point out how inconsistent with the Newtonian doctrine it was, to expect, as Göthe expected, that the wall should be all over colored various colors. {64}

[Note 23\9: _Vorlesungen_, p. 270.]

[Note 24\9: _Farbenlehre_, vol. ii. p. 678.]

Göthe not only adopted and strenuously maintained the opinion that the Newtonian theory was false, but he framed a system of his own to explain the phenomena of color. As a matter of curiosity, it may be worth our while to state the nature of this system; although undoubtedly it forms no part of the _progress_ of physical science. Göthe's views are, in fact, little different from those of Aristotle and Antonio de Dominis, though more completely and systematically developed. According to him, colors arise when we see through a dim medium ("ein trübes mittel"). Light in itself is colorless; but if it be seen through a somewhat dim medium, it appears yellow; if the dimness of the medium increases, or if its depth be augmented, we see the light gradually assume a yellow-red color, which finally is heightened to a ruby-red. On the other hand, if darkness is seen through a dim medium which is illuminated by a light falling on it, a blue color is seen, which becomes clearer and paler, the more the dimness of the medium increases, and darker and fuller, as the medium becomes more transparent; and when we come to "the smallest degree of the purest dimness," we see the most perfect violet.[25\9] In addition to this "doctrine of the dim medium," we have a second principle asserted concerning refraction. In a vast variety of cases, images are accompanied by "accessory images," as when we see bright objects in a looking-glass.[26\9] Now, when an image is displaced by refraction, the displacement is not complete, clear and sharp, but incomplete, so that there is an accessory image along with the principal one.[27\9] From these principles, the colors produced by refraction in the image of a bright object on a dark ground, are at once derivable. The accessory image is semitransparent;[28\9] and hence that border of it which is pushed forwards, is drawn from the dark over the bright, and there the yellow appears; on the other hand, where the clear border laps over the dark ground, the blue is seen;[29\9] and hence we easily see that the image must appear red and yellow at one end, and blue and violet at the other.

[Note 25\9: _Farbenlehre_, § 150, p. 151.]

[Note 26\9: Ib. § 223.]

[Note 27\9: Ib. § 227.]

[Note 28\9: Ib. § 238.]

[Note 29\9: Ib. § 239.]

We need not explain this system further, or attempt to show how vague and loose, as well as baseless, are the notions and modes of conception which it introduces. Perhaps it is not difficult to point out the peculiarities in Göthe's intellectual character which led to his singularly unphilosophical views on this subject. One important {65} circumstance is, that he appears, like many persons in whom the poetical imagination is very active, to have been destitute of the talent and the habit of geometrical thought. In all probability, he never apprehended clearly and steadily those relations of position on which the Newtonian doctrine depends. Another cause of his inability to accept the doctrine probably was, that he had conceived the "composition" of colors in some way altogether different from that which Newton understands by composition. What Göthe expected to see, we cannot clearly collect; but we know, from his own statement, that his intention of experimenting with a prism arose from his speculations on the roles of coloring in pictures; and we can easily see that any notion of the composition of colors which such researches would suggest, would require to be laid aside, before he could understand Newton's theory of the composition of light.

Other objections to Newton's theory, of a kind very different, have been recently made by that eminent master of optical science, Sir David Brewster. He contests Newton's opinion, that the colored rays into which light is separated by refraction are altogether simple and homogeneous, and incapable of being further analysed and modified. For he finds that by passing such rays through colored media (as blue glass for instance), they are not only absorbed and transmitted in very various degrees, but that some of them have their color altered; which effect he conceives as a further analysis of the rays, one component color being absorbed and the other transmitted.[30\9] And on this subject we can only say, as we have before said, that Newton has incontestably and completely established his doctrine, so far as analysis and decomposition _by refraction_ are concerned; but that with regard to any other analysis, which absorbing media or other agents may produce, we have no right from his experiments to assert, that the colors of the spectrum are incapable of _such_ decomposition. The whole subject of the colors of objects, both opake and transparent, is still in obscurity. Newton's conjectures concerning the causes of the colors of natural bodies, appear to help us little; and his opinions on that subject are to be separated altogether from the important step which he made in optical science, by the establishment of the true doctrine of refractive dispersion.

[Note 30\9: This latter fact has, however, been denied by other experimenters.]

[2nd Ed.] [After a careful re-consideration of Sir D. Brewster's asserted analysis of the solar light into three colors by means of {66} absorbing media, I cannot consider that he has established his point as an exception to Newton's doctrine. In the first place, the analysis of light into _three_ colors appears to be quite arbitrary, granting all his experimental facts. I do not see why, using other media, he might not just as

well have obtained other elementary colors. In the next place, this cannot be called an _analysis_ in the same sense as Newton's analysis, except the relation between the two is shown. Is it meant that Newton's experiments prove nothing? Or is Newton's conclusion allowed to be true of light which has not been analysed by absorption? And where are we to find such light, since the atmosphere absorbs? But, I must add, in the third place, that with a very sincere admiration of Sir D. Brewster's skill as an experimenter, I think his experiment requires, not only limitation, but confirmation by other experimenters. Mr. Airy repeated the experiments with about thirty different absorbing substances, and could not satisfy himself that in any case they changed the color of a ray of given refractive power. These experiments were described by him at a meeting of the Cambridge Philosophical Society.]

We now proceed to the corrections which the next generation introduced into the details of this doctrine.

## CHAPTER IV.

DISCOVERY OF ACHROMATISM.

THE discovery that the laws of refractive dispersion of different substances were such as to allow of combinations which neutralised the dispersion without neutralizing the refraction, is one which has hitherto been of more value to art than to science. The property has no definite bearing, which has yet been satisfactorily explained, upon the _theory_ of light; but it is of the greatest importance in its application to the construction of telescopes; and it excited the more notice, in consequence of the prejudices and difficulties which for a time retarded the discovery.

Newton conceived that he had proved by experiment,[31\9] that light {67} is white after refraction, when the emergent rays are parallel to the incident, and in no other case. If this were so, the production of colorless images by refracting media would be impossible; and such, in deference to Newton's great authority, was for some time the general persuasion. Euler[32\9] observed, that a combination of lenses which does not color the image must be possible, since we have an example of such a combination in the human eye; and he investigated mathematically the conditions requisite for such a result. Klingenstierna,[33\9] a Swedish mathematician, also showed that Newton's rule could not be universally true. Finally, John Dollond,[34\9] in 1757, repeated Newton's experiment, and obtained an opposite result. He found that when an object was seen through two prisms, one of glass and one of water, of such angles that it did not appear displaced by refraction, it was colored. Hence it followed that, without being colored, the rays might be made to undergo refraction; and that thus, substituting lenses for prisms, a combination might be formed, which should produce an image without coloring it, and make the construction of an _achromatic_ telescope possible.

[Note 31\9: _Opticks_, B. i. p. ii. Prop. 3.]

[Note 32\9: _Ac. Berlin._ 1747.]

[Note 33\9: _Swedish Trans._ 1754.]

[Note 34\9: _Phil. Trans._ 1758.]

Euler at first hesitated to confide in Dollond's experiments; but he was assured of their correctness by Clairaut, who had throughout paid great attention to the subject; and those two great mathematicians, as well as D'Alembert, proceeded to investigate mathematical formulæ which might be useful in the application of the discovery. The remainder of the deductions, which were founded upon the laws of dispersion of various refractive substances, belongs rather to the history of art than of science. Dollond used at first, for his achromatic object-glass, a lens of crown-glass, and one of flint-glass. He afterwards employed two lenses of the former substance, including between them one of the latter, adjusting the curvatures of his lenses in such a way as to correct the imperfections arising from the spherical form of the glasses, as well as the fault of color. Afterwards, Blair used fluid media along with glass lenses, in order to produce improved object-glasses. This has more recently been done in another form by Mr. Barlow. The inductive laws of refraction being established, their results have been deduced by various mathematicians, as Sir J. Herschel and Professor Airy among ourselves, who have simplified and extended the investigation of the formulæ which determine the best combination of lenses in the object-glasses and eye-glasses of {68} telescopes, both with reference to spherical and to _chromatic_ aberrations.

According to Dollond's discovery, the colored spectra produced by prisms of two substances, as flint-glass and crown-glass, would be of the same length when the refraction was different. But a question then occurred: When the whole distance from the red to the violet in one spectrum was the same as the whole distance in the other, were the intermediate colors, yellow, green, &c., in corresponding places in the two? This point also could not be determined any otherwise than by experiment. It appeared that such a correspondence did not exist; and, therefore, when the extreme colors were corrected by combinations of the different media, there still remained an uncorrected residue of color arising from the rest of the spectrum. This defect was a consequence of the property, that the spectra belonging to different media were not divided in the _same ratio_ by the same colors, and was hence termed the _irrationality_ of the spectrum. By using three prisms, or three lenses, three colors may be made to coincide instead of two, and the effects of this irrationality greatly diminished.

For the reasons already mentioned, we do not pursue this subject further,[35\9] but turn to those optical facts which finally led to a great and comprehensive theory.

[Note 35\9: The discovery of the _fixed lines_ in the spectrum, by Wollaston and Fraunhofer, has more recently supplied the means of determining, with extreme accuracy, the corresponding portions of the spectrum in different refracting substances.]

[2nd Ed.] [Mr. Chester More Hall, of More Hall, in Essex, is said to have been led by the study of the human eye, which he conceived to be achromatic, to construct achromatic telescopes as early as 1729. Mr. Hall, however, kept his invention a

secret. David Gregory, in his _Catoptrics_ (1713), had suggested that it would perhaps be an improvement of telescopes, if, in imitation of the human eye, the object-glass were composed of different media. _Encyc. Brit._ art. _Optics_.

It is said that Clairaut first discovered the irrationality of the colored spaces in the spectrum. In consequence of this irrationality, it follows that when two refracting media are so combined as to correct each other's extreme dispersion, (the separation of the red and violet rays,) this first step of correction still leaves a residue of {69} coloration arising from the unequal dispersion of the intermediate rays (the green, &c.). These _outstanding_ colors, as they were termed by Professor Robison, form the residual, or _secondary_ spectrum.

Dr. Blair, by very ingenious devices, succeeded in producing an object-glass, corrected by a fluid lens, in which this aberration of color was completely corrected, and which performed wonderfully well.

The dispersion produced by a prism may be corrected by another prism of the _same substance_ and of a different angle. In this case also there is an irrationality in the colored spaces, which prevents the correction of color from being complete; and hence, a new residuary spectrum, which has been called the _tertiary_ spectrum, by Sir David Brewster, who first noticed it.

I have omitted, in the notice of discoveries respecting the spectrum, many remarkable trains of experimental research, and especially the investigations respecting the power of various media to absorb the light of different parts of the spectrum, prosecuted by Sir David Brewster with extraordinary skill and sagacity. The observations are referred to in chapter iii. Sir John Herschel, Prof. Miller, Mr. Daniel, Dr. Faraday, and Mr. Talbot, have also contributed to this part of our knowledge.]

## CHAPTER V.

DISCOVERY OF THE LAWS OF DOUBLE REFRACTION.

THE laws of refraction which we have hitherto described, were simple and uniform, and had a symmetrical reference to the surface of the refracting medium. It appeared strange to men, when their attention was drawn to a class of phenomena in which this symmetry was wanting, and in which a refraction took place which was not even in the plane of incidence. The subject was not unworthy the notice and admiration it attracted; for the prosecution of it ended in the discovery of the general laws of light. The phenomena of which I now speak, are those exhibited by various kinds of crystalline bodies; but observed for a long time in one kind only, namely, the rhombohedral calc-spar; or, as it was usually termed, from the country which supplied the largest and clearest crystals, _Iceland spar_. These {70} rhombohedral crystals are usually very smooth and transparent, and often of considerable size; and it was observed, on looking through them, that all objects appeared double. The phenomena, even as early as 1669, had been considered so curious, that Erasmus Bartholin published a work upon them at Copenhagen,[36\9] (_Experimenta Crystalli

Islandici_, Hafniæ, 1669.) He analysed the phenomena into their laws, so far as to discover that one of the two images was produced by refraction after the usual rule, and the other by an unusual refraction. This latter refraction Bartholin found to vary in different positions; to be regulated by a line parallel to the sides of the rhombohedron; and to be greatest in the direction of a line bisecting two of the angles of the rhombic face of the crystal.

[Note 36\9: Priestley's _Optics_, p. 550.]

These rules were exact as far as they went; and when we consider how geometrically complex the law is, which really regulates the unusual or extraordinary refraction;--that Newton altogether mistook it, and that it was not verified till the experiments of Haüy and Wollaston in our own time;--we might expect that it would not be soon or easily detected. But Huyghens possessed a key to the secret, in the theory, which he had devised, of the propagation of light by undulations, and which he conceived with perfect distinctness and correctness, so far as its application to these phenomena is concerned. Hence he was enabled to lay down the law of the phenomena (the only part of his discovery which we have here to consider), with a precision and success which excited deserved admiration, when the subject, at a much later period, regained its due share of attention. His Treatise was written[37\9] in 1678, but not published till 1690.

[Note 37\9: See his _Traité de la Lumière_. Preface.]

The laws of the _ordinary_ and the _extraordinary_ refraction in Iceland spar are related to each other; they are, in fact, similar constructions, made, in the one case, by means of an imaginary sphere, in the other, by means of a spheroid; the spheroid being of such oblateness as to suit the rhombohedral form of the crystal, and the axis of the spheroid being the axis of symmetry of the crystal. Huyghens followed this general conception into particular positions and conditions; and thus obtained rules, which he compared with observation, for cutting the crystal and transmitting the rays in various manners. "I have examined in detail," says he,[38] "the properties of the {71} extraordinary refraction of this crystal, to see if each phenomenon which is deduced from theory, would agree with what is really observed. And this being so, it is no slight proof of the truth of our suppositions and principles; but what I am going to add here confirms them still more wonderfully; that is, the different modes of cutting this crystal, in which the surfaces produced give rise to refractions exactly such as they ought to be, and as I had foreseen them, according to the preceding theory."

[Note 38\9: See Maseres's _Tracts on Optics_, p. 250; or Huyghens, _Tr. sur la Lum._ ch. v. Art. 43.]

Statements of this kind, coming from a philosopher like Huyghens, were entitled to great confidence; Newton, however, appears not to have noticed, or to have disregarded them. In his _Opticks_, he gives a rule for the extraordinary refraction of Iceland spar which is altogether erroneous, without assigning any reason for rejecting the law published by Huyghens; and, so far as appears, without having made any experiments of his own. The Huyghenian doctrine of double refraction fell,

along with his theory of undulations, into temporary neglect, of which we shall have hereafter to speak. But in 1788, Haüy showed that Huyghens's rule agreed much better than Newton's with the phenomena: and in 1802, Wollaston, applying a method of his own for measuring refraction, came to the same result. "He made," says Young,[39\9] "a number of accurate experiments with an apparatus singularly well calculated to examine the phenomena, but could find no general principle to connect them, until the work of Huyghens was pointed out to him." In 1808, the subject of double refraction was proposed as a prize-question by the French Institute; and Malus, whose Memoir obtained the prize, says, "I began by observing and measuring a long series of phenomena on natural and artificial faces of Iceland spar. Then, testing by means of these observations the different laws proposed up to the present time by physical writers, I was struck with the admirable agreement of the law of Huyghens with the phenomena, and I was soon convinced that it is really the law of nature." Pursuing the consequences of the law, he found that it satisfied phenomena which Huyghens himself had not observed. From this time, then, the truth of the Huyghenian law was universally allowed, and soon afterwards, the theory by which it had been suggested was generally received.

[Note 39\9: _Quart. Rev._ 1809, Nov. p. 338.]

The property of double refraction had been first studied only in Iceland spar, in which it is very obvious. The same property belongs, {72} though less conspicuously, to many other kinds of crystals. Huyghens had noticed the same fact in rock-crystal;[40\9] and Malus found it to belong to a large list of bodies besides; for instance, arragonite, sulphate of lime, of baryta, of strontia, of iron; carbonate of lead; zircon, corundum, cymophane, emerald, euclase, felspar, mesotype, peridote, sulphur, and mellite. Attempts were made, with imperfect success, to reduce all these to the law which had been established for Iceland spar. In the first instance, Malus took for granted that the extraordinary refraction depended always upon an oblate spheroid; but M. Biot[41\9] pointed out a distinction between two classes of crystals in which this spheroid was oblong and oblate respectively, and these he called _attractive_ and _repulsive_ crystals. With this correction, the law could be extended to a considerable number of cases; but it was afterwards proved by Sir D. Brewster's discoveries, that even in this form, it belonged only to substances of which the crystallization has relation to a single axis of symmetry, as the rhombohedron, or the square pyramid. In other cases, as the rhombic prism, in which the form, considered with reference to its crystalline symmetry, is _biaxal_, the law is much more complicated. In that case, the sphere and the spheroid, which are used in the construction for uniaxal crystals, transform themselves into the two successful convolutions of a single continuous curve surface; neither of the two rays follows the law of ordinary refraction; and the formula which determines their position is very complex. It is, however, capable of being tested by measures of the refractions of crystals cut in a peculiar manner for the purpose, and this was done by MM. Fresnel and Arago. But this complex law of double refraction was only discovered through the aid of the theory of a luminiferous ether, and therefore we must now return to the other facts which led to such a theory.

[Note 40\9: _ Traité de la Lumière_, ch. v. Art. 20]

[Note 41\9: Biot, _Traité de Phys._ iii. 330.]

## CHAPTER VI.

DISCOVERY OF THE LAWS OF POLARIZATION.

IF the Extraordinary Refraction of Iceland spar had appeared strange, another phenomenon was soon noticed in the same {73} substance, which appeared stranger still, and which in the sequel was found to be no less important. I speak of the facts which were afterwards described under the term _Polarization_. Huyghens was the discoverer of this class of facts. At the end of the treatise which we have already quoted, he says,[42\9] "Before I quit the subject of this crystal, I will add one other marvellous phenomenon, which I have discovered since writing the above; for though hitherto I have not been able to find out its cause, I will not, on that account, omit pointing it out, that I may give occasion to others to examine it." He then states the phenomena; which are, that when two rhombohedrons of Iceland spar are in parallel positions, a ray doubly refracted by the first, is not further divided when it falls on the second: the ordinarily refracted ray is ordinarily refracted _only_, and the extraordinary ray is only extraordinarily refracted by the second crystal, neither ray being doubly refracted. The same is still the case, if the two crystals have their _principal planes_ parallel, though they themselves are not parallel. But if the principal plane of the second crystal be perpendicular to that of the first, the reverse of what has been described takes place; the ordinarily refracted ray of the first crystal suffers, at the second, extraordinary refraction _only_, and the extraordinary ray of the first suffers ordinary refraction only at the second. Thus, in each of these positions, the double refraction of each ray at the second crystal is reduced to single refraction, though in a different manner in the two cases. But in any other position of the crystals, each ray, produced by the first, is doubly refracted by the second, so as to produce four rays.

[Note 42\9: _Tr. Opt._ p. 252.]

A step in the right conception of these phenomena was made by Newton, in the second edition of his _Opticks_ (1717). He represented them as resulting from this;--that the rays of light have "sides," and that they undergo the ordinary or extraordinary refraction, according as these sides are parallel to the principal plane of the crystal, or at right angles to it (Query 26). In this way, it is clear, that those rays which, in the first crystal, had been selected for extraordinary refraction, because their sides were perpendicular to the principal plane, would all suffer extraordinary refraction at the second crystal for the same reason, if its principal plane were parallel to that of the first; and would all suffer ordinary refraction, if the principal plane of the second crystal were perpendicular to that of the first, and {74} consequently parallel to the sides of the refracted ray. This view of the subject includes some of the leading features of the case, but still leaves several considerable difficulties.

No material advance was made in the subject till it was taken up by Malus,[43\9] along with the other circumstances of double refraction, about a hundred years afterwards. He verified what had been observed by Huyghens and Newton, on the subject of the variations which light thus exhibits; but he discovered that this modification, in virtue of which light undergoes the ordinary, or the extraordinary, refraction, according to the position of the plane of the crystal, may be impressed upon it many other ways. One part of this discovery was made accidentally.[44\9] In 1808, Malus happened to be observing the light of the setting sun, reflected from the windows of the Luxembourg, through a rhombohedron of Iceland spar; and he observed that in turning round the crystal, the two images varied in their intensity. Neither of the images completely vanished, because the light from the windows was not properly modified, or, to use the term which Malus soon adopted, was not completely _polarized_. The complete polarization of light by reflection from glass, or any other transparent substance, was found to take place at a certain definite angle, different for each substance. It was found also that in all crystals in which double refraction occurred, the separation of the refracted rays was accompanied by polarization; the two rays, the ordinary and the extraordinary, being always polarized _oppositely_, that is, in planes at right angles to each other. The term _poles_, used by Malus, conveyed nearly the same notion as the term _sides_ which had been employed by Newton, with the additional conception of a property which appeared or disappeared according as the _poles_ of the particles were or were not in a certain direction; a property thus resembling the _polarity_ of magnetic bodies. When a spot of polarized light is looked at through a transparent crystal of Iceland spar, each of the two images produced by the double refraction varies in brightness as the crystal is turned round. If, for the sake of example, we suppose the crystal to be turned round in the direction of the points of the compass, N, E, S, W, and if one image be brightest when the crystal marks N and S, it will disappear when the crystal marks E and W: and on the contrary, the second image will vanish when the crystal marks N and S, {75} and will be brightest when the crystal marks E and W. The first of these images is polarized _in the plane_ NS passing through the ray, and the second _in the plane_ EW, perpendicular to the other. And these rays are _oppositely_ polarized. It was further found that whether the ray were polarized by reflection from glass, or from water, or by double refraction, the modification of light so produced, or the nature of the polarization, was identical in all these cases;--that the alternatives of ordinary and extraordinary refraction and non-refraction, were the same, by whatever crystal they were tested, or in whatever manner the polarization had been impressed upon the light; in short, that the property, when once acquired, was independent of everything except the sides or _poles_ of the ray; and thus, in 1811, the term "polarization" was introduced.[45\9]

[Note 43\9: Malus, _Th. de la Doub. Réf._ p. 296.]

[Note 44\9: Arago, art. _Polarization_, Supp. _Enc. Brit._]

[Note 45\9: _Mém. Inst._ 1810.]

This being the state of the subject, it became an obvious question, by what other means, and according to what laws, this property was communicated. It was found

that some crystals, instead of giving, by double refraction, two images oppositely polarized, give a single polarized image. This property was discovered in the agate by Sir D. Brewster, and in tourmaline by M. Biot and Dr. Seebeck. The latter mineral became, in consequence, a very convenient part of the apparatus used in such observations. Various peculiarities bearing upon this subject, were detected by different experimenters. It was in a short time discovered, that light might be polarized by refraction, as well as by reflection, at the surface of uncrystallized bodies, as glass; the plane of polarization being perpendicular to the plane of refraction; further, that when a portion of a ray of light was polarized by reflection, a corresponding portion was polarized by transmission, the planes of the two polarizations being at right angles to each other. It was found also that the polarization which was incomplete with a single plate, either by reflection or refraction, might be made more and more complete by increasing the number of plates.

Among an accumulation of phenomena like this, it is our business to inquire what general laws were discovered. To make such discoveries without possessing the general theory of the facts, required no ordinary sagacity and good fortune. Yet several laws were detected at this stage of the subject. Malus, in 1811, obtained the important generalization that, whenever we obtain, by any means, a polarized ray of light, we produce also another ray, polarized in a contrary {76} direction; thus when reflection gives a polarized ray, the companion-ray is refracted polarized oppositely, along with a quantity of unpolarized light. And we must particularly notice _Sir D. Brewster's rule_ for the _polarizing angle_ of different bodies.

Malus[46\9] had said that the angle of reflection from transparent bodies which most completely polarizes the reflected ray, does not follow any discoverable rule with regard to the order of refractive or dispersive powers of the substances. Yet the rule was in reality very simple. In 1815, Sir D. Brewster stated[47\9] as the law, which in all cases determines this angle, that "the index of refraction is the tangent of the angle of polarization." It follows from this, that the polarization takes place when the reflected and refracted rays are at right angles to each other. This simple and elegant rule has been fully confirmed by all subsequent observations, as by those of MM. Biot and Seebeck; and must be considered one of the happiest and most important discoveries of the laws of phenomena in Optics.

[Note 46\9: _Mém. Inst._ 1810.]

[Note 47\9: _Phil. Trans._ 1815.]

The rule for polarization by one reflection being thus discovered, tentative formulæ were proposed by Sir D. Brewster and M. Biot, for the cases in which several reflections or refractions take place. Fresnel also in 1817 and 1818, traced the effect of reflection in modifying the direction of polarization, which Malus had done inaccurately in 1810. But the complexity of the subject made all such attempts extremely precarious, till the theory of the phenomena was understood, a period which now comes under notice. The laws which we have spoken of were important materials for the establishment of the theory; but in the mean time, its progress at

first had been more forwarded by some other classes of facts, of a different kind, and of a longer standing notoriety, to which we must now turn our attention.

## CHAPTER VII.

DISCOVERY OF THE LAWS OF THE COLOURS OF THIN PLATES.

THE facts which we have now to consider are remarkable, inasmuch as the colours are produced merely by the smallness of dimensions of the bodies employed. The light is not analysed by any peculiar {77} property of the substances, but dissected by the minuteness of their parts. On this account, these phenomena give very important indications of the real structure of light; and at an early period, suggested views which are, in a great measure, just.

Hooke appears to be the first person who made any progress in discovering the laws of the colors of thin plates. In his _Micrographia_, printed by the Royal Society in 1664, he describes, in a detailed and systematic manner, several phenomena of this kind, which he calls "fantastical colors." He examined them in _Muscovy glass_ or mica, a transparent mineral which is capable of being split into the exceedingly thin films which are requisite for such colors; he noticed them also in the fissures of the same substance, in bubbles blown of water, rosin, gum, glass; in the films on the surface of tempered steel; between two plane pieces of glass; and in other cases. He perceived also,[48\9] that the production of each color required a plate of determinate thickness, and he employed this circumstance as one of the grounds of his theory of light.

[Note 48\9: _Micrographia_, p. 53.]

Newton took up the subject where Hooke had left it; and followed it out with his accustomed skill and clearness, in his _Discourse on Light and Colors_, communicated to the Royal Society in 1675. He determined, what Hooke had not ascertained, the thickness of the film which was requisite for the production of each color; and in this way explained, in a complete and admirable manner, the colored rings which occur when two lenses are pressed together, and the _scale of color_ which the rings follow; a step of the more consequence, as the same scale occurs in many other optical phenomena.

It is not our business here to state the hypothesis with regard to the properties of light which Newton founded on these facts;--the "fits of easy transmission and reflection." We shall see hereafter that his attempted induction was imperfect; and his endeavor to account, by means of the laws of thin plates, for the colors of natural bodies, is altogether unsatisfactory. But notwithstanding these failures in the speculations on this subject, he did make in it some very important steps; for he clearly ascertained that when the thickness of the plate was about 1⁄178000th of an inch, or three times, five times, seven times that magnitude, there was a bright color produced; but blackness, when the thickness was exactly intermediate between those magnitudes. He found, also, that the thicknesses which gave red and {78} violet[49\9] were as fourteen to nine; and the intermediate colors of course

corresponded to intermediate thicknesses, and therefore, in his apparatus, consisting of two lenses pressed together, appeared as rings of intermediate sizes. His mode of confirming the rule, by throwing upon this apparatus differently colored homogeneous light, is striking and elegant. "It was very pleasant," he says, "to see the rings gradually swell and contract as the color of the light was changed."

[Note 49\9: _Opticks_, p. 184.]

It is not necessary to enter further into the detail of these phenomena, or to notice the rings seen by transmission, and other circumstances. The important step made by Newton in this matter was, the showing that the rays of light, in these experiments, as they pass onwards go periodically through certain cycles of modification, each period occupying nearly the small fraction of an inch mentioned above; and this interval being different for different colors. Although Newton did not correctly disentangle the conditions under which this periodical character is manifestly disclosed, the discovery that, under some circumstances, such a periodical character does exist, was likely to influence, and did influence, materially and beneficially, the subsequent progress of Optics towards a connected theory.

We must now trace this progress; but before we proceed to this task, we will briefly notice a number of optical phenomena which had been collected, and which waited for the touch of sound theory to introduce among them that rule and order which mere observation had sought for in vain.

## CHAPTER VIII.

ATTEMPTS TO DISCOVER THE LAWS OF OTHER PHENOMENA.

THE phenomena which result from optical combinations, even of a comparatively simple nature, are extremely complex. The theory which is now known accounts for these results with the most curious exactness, and points out the laws which pervade the apparent confusion; but without this key to the appearances, it was scarcely possible that any rule or order should be detected. The undertaking was of {79} the same kind as it would have been, to discover all the inequalities of the moon's motion without the aid of the doctrine of gravity. We will enumerate some of the phenomena which thus employed and perplexed the cultivators of optics.

The fringes of shadows were one of the most curious and noted of such classes of facts. These were first remarked by Grimaldi[50\9] (1665), and referred by him to a property of light which he called _Diffraction_. When shadows are made in a dark room, by light admitted through a very small hole, these appearances are very conspicuous and beautiful. Hooke, in 1672, communicated similar observations to the Royal Society, as "a new property of light not mentioned by any optical writer before;" by which we see that he had not heard of Grimaldi's experiments. Newton, in his _Opticks_, treats of the same phenomena, which he ascribes to the _inflexion_ of the rays of light. He asks (Qu. 3), "Are not the rays of light, in passing by the edges and sides of bodies, bent several times backward and forward with a motion like that of an eel? And do not the three fringes of colored light in shadows arise from three

such bendings?" It is remarkable that Newton should not have noticed, that it is impossible, in this way, to account for the facts, or even to express their laws; since the light which produces the fringes must, on this theory, be propagated, even after it leaves the neighborhood of the opake body, in curves, and not in straight lines. Accordingly, all who have taken up Newton's notion of inflexion, have inevitably failed in giving anything like an intelligible and coherent character to these phenomena. This is, for example, the case with Mr. (now Lord) Brougham's attempts in the _Philosophical Transactions_ for 1796. The same may be said of other experimenters, as Mairan[51\9] and Du Four,[52\9] who attempted to explain the facts by supposing an atmosphere about the opake body. Several authors, as Maraldi,[53\9] and Comparetti,[54\9] repeated or varied these experiments in different ways.

[Note 50\9: _Physico-Mathesis, de Lumine, Coloribus et Iride._ Bologna, 1665.]

[Note 51\9: _Ac. Par._ 1738.]

[Note 52\9: _Mémoires Présentés_, vol. v.]

[Note 53\9: _Ac. Par._ 1723.]

[Note 54\9: _Observationes Opticæ de Luce Inflexâ et Coloribus._ Padua, 1787.]

Newton had noticed certain rings of color produced by a glass speculum, which he called "colors of thick plates," and which he attempted to connect with the colors of thin plates. His reasoning is by no means satisfactory; but it was of use, by pointing out this as a case in which his "fits" (the small periods, or cycles in the rays of light, of {80} which we have spoken) continued to occur for a considerable length of the ray. But other persons, attempting to repeat his experiments, confounded with them extraneous phenomena of other kinds; as the Duc de Chaulnes, who spread muslin before his mirror,[55\9] and Dr. Herschel, who scattered hair-powder before his.[56\9] The colors produced by the muslin were those belonging to shadows of _gratings_, afterwards examined more successfully by Fraunhofer, when in possession of the theory. We may mention here also the colors which appear on finely-striated surfaces, and on mother-of-pearl, feathers, and similar substances. These had been examined by various persons (as Boyle, Mazeas, Lord Brougham), but could still, at this period, be only looked upon as insulated and lawless facts.

[Note 55\9: _Ac. Par._ 1755.]

[Note 56\9: _Phil. Trans._ 1807.]

## CHAPTER IX.

DISCOVERY OF THE LAWS OF PHENOMENA OF DIPOLARIZED LIGHT.

BESIDES the above-mentioned perplexing cases of colors produced by common light, cases of _periodical colors produced by polarized light_ began to be discovered, and soon became numerous. In August, 1811, M. Arago communicated to the Institute of France an account of colors seen by passing polarized light through mica, and _analysing_[57\9] it with a prism of Iceland spar. It is remarkable that the light

which produced the colors in this case was the light polarized by the sky, a cause of polarization not previously known. The effect which the mica thus produced was termed _depolarization_;--not a very happy term, since the effect is not the destruction of the polarization, but the combination of a new polarizing influence with the former. The word _dipolarization_, which has since been proposed, is a much more appropriate expression. Several other curious phenomena of the same kind were observed in quartz, and in flint-glass. M. Arago was not able to reduce these phenomena to laws, but he had a full conviction of their value, and ventures to class them with the great steps in {81} this part of optics. "To Bartholin we owe the knowledge of double refraction; to Huyghens, that of the accompanying polarization; to Malus, polarization by reflection; to Arago, depolarization." Sir D. Brewster was at the same time engaged in a similar train of research; and made discoveries of the same nature, which, though not published till some time after those of Arago, were obtained without a knowledge of what had been done by him. Sir D. Brewster's _Treatise on New Philosophical Instruments_, published in 1813, contains many curious experiments on the "depolarizing" properties of minerals. Both these observers noticed the changes of color which are produced by changes in the position of the ray, and the alternations of color in the two oppositely polarized images; and Sir D. Brewster discovered that, in topaz, the phenomena had a certain reference to lines which he called the _neutral_ and _depolarizing_ axes. M. Biot had endeavored to reduce the phenomena to a law; and had succeeded so far, that he found that in the plates of sulphate of lime, the place of the tint, estimated in Newton's _scale_ (see _ante_, chap. vii.), was as the square of the sine of the inclination. But the laws of these phenomena became much more obvious when they were observed by Sir D. Brewster with a larger field of view.[58\9] He found that the colors of topaz, under the circumstances now described, exhibited themselves in the form of elliptical rings, crossed by a black bar, "the most brilliant class of phenomena," as he justly says, "in the whole range of optics." In 1814, also, Wollaston observed the circular rings with a black cross, produced by similar means in calc-spar; and M. Biot, in 1815, made the same observation. The rings in several of these cases were carefully measured by M. Biot and Sir D. Brewster, and a great mass of similar phenomena was discovered. These were added to by various persons, as M. Seebeck, and Sir John Herschel.

[Note 57\9: The prism of Iceland spar produces the colors by separating the transmitted rays according to the laws of double refraction. Hence it is said to _analyse_ the light.]

[Note 58\9: _Phil. Trans._ 1814.]

Sir D. Brewster, in 1818, discovered a general relation between the crystalline form and the optical properties, which gave an incalculable impulse and a new clearness to these researches. He found that there was a correspondence between the degree of symmetry of the optical phenomena and the crystalline form; those crystals which are uniaxal in the crystallographical sense, are also uniaxal in their optical properties, and give circular rings; those which are of other forms are, generally speaking, biaxal; they give oval and knotted _isochromatic_ lines, with two _poles_. He also discovered a rule for the tint at each point {82} in such cases; and

thus explained, so far as an empirical law of phenomena went, the curious and various forms of the colored curves. This law, when simplified by M. Biot,[59\9] made the tint proportional to the product of the distances of the point from the two poles. In the following year, Sir J. Herschel confirmed this law by showing, from actual measurement, that the curve of the isochromatic lines in these cases was the curve termed the _lemniscata_, which has, for each point, the product of the distances from two fixed poles equal to a constant quantity.[60\9] He also reduced to rule some other apparent anomalies in phenomena of the same class.

[Note 59\9: _Mém. Inst._ 1818, p. 192.]

[Note 60\9: _Phil. Trans._ 1819.]

M. Biot, too, gave a rule for the directions of the planes of polarization of the two rays produced by double refraction in biaxal crystals, a circumstance which has a close bearing upon the phenomena of dipolarization. His rule was, that the one plane of polarization bisects the dihedral angle formed by the two planes which pass through the optic axes, and that the other is perpendicular to such a plane. When, however, Fresnel had discovered from the theory the true laws of double refraction, it appeared that the above rule is inaccurate, although in a degree which observation could hardly detect without the aid of theory.[61\9]

[Note 61\9: Fresnel, _Mém. Inst._ 1827, p. 162.]

There were still other classes of optical phenomena which attracted notice; especially those which are exhibited by plates of quartz cut perpendicular to the axis. M. Arago had observed, in 1811, that this substance produced a _twist_ of the plane of polarization to the right or left hand, the amount of this twist being different for different colors; a result which was afterwards traced to a modification of light different both from common and from polarized light, and subsequently known as _circular polarization_. Sir J. Herschel had the good fortune and sagacity to discover that this peculiar kind of polarization in quartz was connected with an equally peculiar modification of crystallization, the _plagihedral_ faces which are seen, on some crystals, obliquely disposed, and, as it were, following each other round the crystal from left to right, or from right to left. Sir J. Herschel found that the _right-handed_ or _left-handed_ character of the circular polarization corresponded, in all cases, to that of the crystal.

In 1815, M. Biot, in his researches on the subject of circular polarization, was led to the unexpected and curious discovery, that this {83} property which seemed to require for its very conception a crystalline structure in the body, belonged nevertheless to several fluids, and in different directions for different fluids. Oil of turpentine, and an essential oil of laurel, gave the plane of polarization a rotation to the left hand; oil of citron, syrup of sugar, and a solution of camphor, gave a rotation to the right hand. Soon after, the like discovery was made independently by Dr. Seebeck, of Berlin.

It will easily be supposed that all those brilliant phenomena could not be observed, and the laws of many of the phenomena discovered, without attempts on the part of philosophers to combine them all under the dominion of some wide and profound

theory. Endeavors to ascend from such knowledge as we have spoken of, to the general theory of light, were, in fact, made at every stage of the subject, and with a success which at last won almost all suffrages. We are now arrived at the point at which we are called upon to trace the history of this theory; to pass from the laws of phenomena to their causes;--from Formal to Physical Optics. The undulatory theory of light, the only discovery which can stand by the side of the theory of universal gravitation, as a doctrine belonging to the same order, for its generality, its fertility, and its certainty, may properly be treated of with that ceremony which we have hitherto bestowed only on the great advances of astronomy; and I shall therefore now proceed to speak of the Prelude to this epoch, the Epoch itself, and its Sequel, according to the form of the preceding Book which treats of astronomy.

[2nd Ed.] [I ought to have stated, in the beginning of this chapter, that Malus discovered the depolarization of _white light_ in 1811. He found that a pencil of light which, being polarized, refused to be reflected by a surface properly placed, recovered its power of being reflected after being transmitted through certain crystals and other transparent bodies. Malus intended to pursue this subject, when his researches were terminated by his death, Feb. 7, 1812. M. Arago, about the same time, announced his important discovery of the depolarization of _colors_ by crystals.

I may add, to what is above said of M. Biot's discoveries respecting the circular polarizing power of fluids, that he pursued his researches so as to bring into view some most curious relations among the elements of bodies. It appeared that certain substances, as sugar of canes, had a right-handed effect, and certain other substances, as gum, a left-handed effect; and that the molecular value of this effect was not altered by dilution. It appeared also that a certain element of the {84} substance of fruits, which had been supposed to be gum, and which is changed into sugar by the operation of acids, is not gum, and has a very energetic right-handed effect. This substance M. Biot called _dextrine_, and he has since traced its effects into many highly curious and important results.**]



## CHAPTER X.

PRELUDE TO THE EPOCH OF YOUNG AND FRESNEL.

BY _Physical_ Optics we mean, as has already been stated, the theories which explain optical phenomena on mechanical principles. No such explanation could be given till true mechanical principles had been obtained; and, accordingly, we must date the commencement of the essays towards physical optics from Descartes, the founder of the modern mechanical philosophy. His hypothesis concerning light is, that it consists of small particles emitted by the luminous body. He compares these particles to balls, and endeavors to explain, by means of this comparison, the laws of reflection and refraction.[62\9] In order to account for the production of colors by refraction, he ascribes to these balls an alternating rotatory motion.[63\9] This form of the _emission theory_, was, like most of the physical speculations of its author, hasty and gratuitous; but was extensively accepted, like the rest of the Cartesian

doctrines, in consequence of the love which men have for sweeping and simple dogmas, and deductive reasonings from them. In a short time, however, the rival optical _theory of undulations_ made its appearance. Hooke in his _Micrographia_ (1664) propounds it, upon occasion of his observations, already noticed, (chap. **vii.,) on the colors of thin plates. He there asserts[64\9] light to consist in a "quick, short, vibrating motion," and that it is propagated in a homogeneous medium, in such a way that "every pulse or vibration of the luminous body will generate a sphere, which will continually increase and grow bigger, just after the same manner (though indefinitely swifter) as the waves or rings on the surface of water do swell into bigger and bigger circles about a point in it."[65\9] He applies this to the explanation of refraction, {86} by supposing that the rays in a denser medium move more easily, and hence that the pulses become oblique; a far less satisfactory and consistent hypothesis than that of Huyghens, of which we shall next have to speak. But Hooke has the merit of having also combined with his theory, though somewhat obscurely, the _Principle of Interferences_, in the application which he makes of it to the colors of thin plates. Thus[66\9] he supposes the light to be reflected at the first surface of such plates; and he adds, "after two refractions and one reflection (from the second surface) there is propagated a kind of fainter ray," which comes behind the other reflected pulse; "so that hereby (the surfaces AB and EF being so near together that the eye cannot discriminate them from one), this compound or duplicated pulse does produce on the retina the sensation of a yellow." The reason for the production of this particular color, in the case of which he here speaks, depends on his views concerning the kind of pulses appropriate to each color; and, for the same reason, when the thickness is different, he finds that the result will be a red or a green. This is a very remarkable anticipation of the explanation ultimately given of these colors; and we may observe that if Hooke could have measured the thickness of his thin plates, he could hardly have avoided making considerable progress in the doctrine of interferences.

[Note 62\9: _Diopt._ c. ii. 4.]

[Note 63\9: _Meteor._ c. viii. 6.]

[Note 64\9: _Micrographia_, p. 56.]

[Note 65\9: _Micrographia_, p. 57.]

[Note 66\9: _Micrographia_, p. 66.]

But the person who is generally, and with justice, looked upon as the great author of the undulatory theory, at the period now under notice, is Huyghens, whose _Traité de la Lumière_, containing a developement of his theory, was written in 1678, though not published till 1690. In this work he maintained, as Hooke had done, that light consists in undulations, and expands itself spherically, nearly in the same manner as sound does; and he referred to the observations of Römer on Jupiter's satellites, both to prove that this difference takes place successively, and to show its exceeding swiftness. In order to trace the effect of an undulation, Huyghens considers that every point of a wave diffuses its motion in all directions; and hence he draws the conclusion, so long looked upon as the turning-point of the combat between the rival theories, that the light will not be _diffused_ beyond the rectilinear space, when it

passes through an aperture; "for," says he,[67\9] "although the _partial_ waves, produced by the particles comprised in the aperture, do diffuse themselves beyond the rectilinear space, these waves do not _concur_ anywhere except in front of the {87} aperture." He rightly considers this observation as of the most essential value. "This," he says, "was not known by those who began to consider the waves of light, among whom are Mr. Hooke in his _Micrography_, and Father Pardies; who, in a treatise of which he showed me a part, and which he did not live to finish, had undertaken to prove, by these waves, the effects of reflection and refraction. But the principal foundation, which consists in the remark I have just made, was wanting in his demonstrations."

[Note 67\9: _Tracts on Optics_, p. 209.]

By the help of this view, Huyghens gave a perfectly satisfactory and correct explanation of the laws of reflection and refraction; and he also applied the same theory, as we have seen, to the double refraction of Iceland spar with great sagacity and success. He conceived that in this crystal, besides the spherical waves, there might be others of a spheroidal form, the axis of the spheroid being symmetrically disposed with regard to the faces of the rhombohedron, for to these faces the optical phenomena are symmetrically related. He found[68\9] that the position of the refracted ray, determined by such spheroidal undulations, would give an oblique refraction, which would coincide in its laws with the refraction observed in Iceland spar; and, as we have stated, this coincidence was long after fully confirmed by other observers.

[Note 68\9: _Tracts on Optics_, 237.]

Since Huyghens, at this early period, expounded the undulatory theory with so much distinctness, and applied it with so much skill, it may be asked why we do not hold him up as the great Author of the induction of undulations of light;--the person who marks the epoch of the theory? To this we reply, that though Huyghens discovered strong presumptions in favor of the undulatory theory, it was not _established_ till a later era, when the fringes of shadows, rightly understood, made the waves visible, and when the hypothesis which had been assumed to account for double refraction, was found to contain also an explanation of polarization. It is _then_ that this theory of light assumes its commanding form; and the persons who gave it this form, we must make the great names of our narrative; without, however, denying the genius and merit of Huyghens, who is, undoubtedly, the leading character in the prelude to the discovery.

The undulatory theory, from this time to our own, was unfortunate in its career. It was by no means destitute of defenders, but these were not experimenters; and none of them thought of applying it to {88} Grimaldi's experiments on fringes, of which we have spoken a little while ago. And the great authority of the period, Newton, adopted the opposite hypothesis, that of emission, and gave it a currency among his followers which kept down the sounder theory for above a century.

Newton's first disposition appears to have been by no means averse to the assumption of an ether as the vehicle of luminiferous undulations. When Hooke

brought against his prismatic analysis of light some objections, founded on his own hypothetical notions, Newton, in his reply, said,[69\9] "The hypothesis has a much greater affinity with his own hypothesis than he seems to be aware of; the vibrations of the ether being as useful and necessary in this as in his." This was in 1672; and we might produce, from Newton's writing, passages of the same kind, of a much later date. Indeed it would seem that, to the last, Newton considered the assumption of an ether as highly probable, and its vibrations important parts of the phenomena of light; but he also introduced into his system the hypothesis of emission, and having followed this hypothesis into mathematical detail, while he has left all that concerns the ether in the form of queries and conjectures, the emission theory has naturally been treated as the leading part of his optical doctrines.

[Note 69\9: _Phil. Trans._ vii. 5087.]

The principal propositions of the _Principia_ which bear upon the question of optical theory are those of the fourteenth Section of the first Book,[70\9] in which the law of the sines in refraction is proved on the hypothesis that the particles of bodies act on light only at very small distances; and the proposition of the eighth Section of the second Book;[71\9] in which it is pretended to be demonstrated that the motion propagated in a fluid must diverge when it has passed through an aperture. The former proposition shows that the law of refraction, an optical truth which mainly affected the choice of a theory, (for about reflection there is no difficulty on any mechanical hypothesis,) follows from the theory of emission: the latter proposition was intended to prove the inadmissibility of the rival hypothesis, that of undulations. As to the former point,--the hypothetical explanation of refraction, on the assumptions there made,--the conclusion is quite satisfactory; but the reasoning in the latter case, (respecting the propagation of undulations,) is certainly inconclusive and vague; and something better might the more reasonably have been expected, since Huyghens had at least {89} endeavored to prove the opposite proposition. But supposing we leave these properties, the rectilinear course, the reflection, and the refraction of light, as problems in which neither theory has a decided advantage, what is the next material point? The colors of thin plates. Now, how does Newton's theory explain these? By a new and special supposition;--that of _fits of easy transmission and reflection_: a supposition which, though it truly expresses these facts, is not borne out by any other phenomena. But, passing over this, when we come to the peculiar laws of polarization in Iceland spar, how does Newton's meet this? Again by a special and new supposition;--that the rays of light have _sides_. Thus we find no fresh evidence in favor of the emission hypothesis springing out of the fresh demands made upon it. It may be urged, in reply, that the same is true of the undulatory theory; and it must be allowed that, at the time of which we now speak, its superiority in this respect was not manifested; though Hooke, as we have seen, had caught a glimpse of the explanation, which this theory supplies, of the colors of thin plates.

[Note 70\9: _Principia_, Prop. 94, _et seq._]

[Note 71\9: Ib. Prop. 42.]

At a later period, Newton certainly seems to have been strongly disinclined to believe light to consist in undulations merely. "Are not," he says, in Question twenty-eight of the _Opticks_, "all hypotheses erroneous, in which light is supposed to consist in pression or motion propagated through a fluid medium?" The arguments which most weighed with him to produce this conviction, appear to have been the one already mentioned,--that, on the undulatory hypothesis, undulations passing through an aperture would be diffused; and again,--his conviction, that the properties of light, developed in various optical phenomena, "depend not upon new modifications, but upon the original and unchangeable properties of the rays." (Question twenty-seven.)

But yet, even in this state of his views, he was very far from abandoning the machinery of vibrations altogether. He is disposed to use such machinery to produce his "fits of easy transmission." In his seventeenth Query, he says,[72\9] "when a ray of light falls upon the surface of any pellucid body, and is there refracted or reflected; may not waves of vibrations or tremors be thereby excited in the refracting or reflecting medium at the point of incidence? . . . . and do not these vibrations overtake the rays of light, and by overtaking them successively, do they not put them into the fits of easy reflection and easy {90} transmission described above?" Several of the other queries imply the same persuasion, of the necessity for the assumption of an ether and its vibrations. And it might have been asked, whether any good reason could be given for the hypothesis of an ether as a _part_ of the mechanism of light, which would not be equally valid in favor of this being the _whole_ of the mechanism, especially if it could be shown that nothing more was wanted to produce the results.

[Note 72\9: _Opticks_, p. 322.]

The emission theory was, however, embraced in the most strenuous manner by the disciples of Newton. That propositions existed in the _Principia_ which proceeded on this hypothesis, was, with many of these persons, ground enough for adopting the doctrine; and it had also the advantage of being more ready of conception, for though the propagation of a wave is not very difficult to conceive, at least by a mathematician, the motion of a particle is still easier.

On the other hand, the undulation theory was maintained by no less a person than Euler; and the war between the two opinions was carried on with great earnestness. The arguments on one side and on the other soon became trite and familiar, for no person explained any new class of facts by either theory. Thus it was urged by Euler against the system of emission,[73\9]--that the perpetual emanation of light from the sun must have diminished the mass;--that the stream of matter thus constantly flowing must affect the motions of the planets and comets; that the rays must disturb each other;--that the passage of light through transparent bodies is, on this system, inconceivable: all such arguments were answered by representations of the exceeding minuteness and velocity of the matter of light. On the other hand, there was urged against the theory of waves, the favorite Newtonian argument, that on this theory the light passing through an aperture ought to be diffused, as sound is. It is curious that Euler does not make to this argument the reply which Huyghens had made before. The fact really was, that he was not aware of the true ground of the difference of the result in the cases of sound and light; namely, that any ordinary

aperture bears an immense ratio to the length of an undulation of light, but does not bear a very great ratio to the length of an undulation of sound. The demonstrable consequence of this difference is, that light darts through such an orifice in straight rays, while sound is diffused in all directions. Euler, not perceiving this difference, rested his answer mainly upon a circumstance by no means {91} unimportant, that the partitions usually employed are not impermeable to sound, as opake bodies are to light. He observes that the sound does not all come through the aperture; for we hear, though the aperture be stopped. These were the main original points of attack and defence, and they continued nearly the same for the whole of the last century; the same difficulties were over and over again proposed, and the same solutions given, much in the manner of the disputations of the schoolmen of the middle ages.

[Note 73\9: Fischer, iv. 449.]

The struggle being thus apparently balanced, the scale was naturally turned by the general ascendancy of the Newtonian doctrines: and the emission theory was the one most generally adopted. It was still more firmly established, in consequence of the turn generally taken by the scientific activity of the latter half of the eighteenth century: for while nothing was added to our knowledge of optical laws, the chemical effects of light were studied to a considerable extent by various inquirers;[74\9] and the opinions at which these persons arrived, they found that they could express most readily, in consistency with the reigning chemical views, by assuming the materiality of light. It is, however, clear, that no reasonings of the inevitably vague and doubtful character which belong to these portions of chemistry, ought to be allowed to interfere with the steady and regular progress of induction and generalization, founded on relations of space and number, by which procedure the mechanical sciences are formed. We reject, therefore, all these chemical speculations, as belonging to other subjects; and consider the history of optical theory as a blank, till we arrive at some very different events, of which we have now to speak. {92}

[Note 74\9: As Scheele, Selle, Lavoisier, De Luc, Richter, Leonhardi, Gren, Girtanner, Link, Hagen, Voigt, De la Metherie, Scherer, Dizé, Brugnatelli. See Fischer, vii. p. 20.]

## CHAPTER XI.

EPOCH OF YOUNG AND FRESNEL.

_Sect._ 1.--_Introduction._

THE man whose name must occupy the most distinguished place in the history of Physical Optics, in consequence of what he did in reviving and establishing the undulatory theory of light, is Dr. Thomas Young. He was born in 1773, at Milverton in Somersetshire, of Quaker parents; and after distinguishing himself during youth by the variety and accuracy of his attainments, he settled in London as a physician in 1801; but continued to give much of his attention to general science. His optical theory, for a long time, made few proselytes; and several years afterwards, Auguste Fresnel, an eminent French mathematician, an engineer officer, took up similar

views, proved their truth, and traced their consequences, by a series of labors almost independent of those of Dr. Young. It was not till the theory was thus re-echoed from another land, that it was able to take any strong hold on the attention of the countrymen of its earlier promulgator.

The theory of undulations, like that of universal gravitation, may be divided into several successive steps of generalization. In both cases, all these steps were made by the same persons; but there is this difference:--all the parts of the law of universal gravitation were worked out in one burst of inspiration by its author, and published at one time;--in the doctrine of light, on the other hand, the different steps of the advance were made and published at separate times, with intervals between. We see the theory in a narrower form, and in detached portions, before the widest generalizations and principles of unity are reached; we see the authors struggling with the difficulties before we see them successful. They appear to us as men like ourselves, liable to perplexity and failure, instead of coming before us, as Newton does in the history of Physical Astronomy, as the irresistible and almost supernatural hero of a philosophical romance. {93}

The main subdivisions of the great advance in physical optics, of which we have now to give an account, are the following:--

1. The explanation of the _periodical colors_ of thin plates, thick plates, fringed shadows, striated surfaces, and other phenomena of the same kind, by means of the doctrine of the _interference_ of undulations.

2. The explanation of the phenomena of _double refraction_ by the propagation of undulations in a medium of which the optical _elasticity_ is different in different directions.

3. The conception of _polarization_ as the result of the vibrations being _transverse_; and the consequent explanation of the production of polarization, and the necessary connexion between polarization and double refraction, on mechanical principles.

4. The explanation of the phenomena of _dipolarization_, by means of the interference of the _resolved parts_ of the vibrations after double refraction.

The history of each of these discoveries will be given separately to a certain extent; by which means the force of proof arising from their combination will be more apparent.

_Sect._ 2.--_Explanation of the Periodical Colors of Thin Plates and Shadows by the Undulatory Theory._

THE explanation of periodical colors by the principle of interference of vibrations, was the first step which Young made in his confirmation of the undulatory theory. In a paper on Sound and Light, dated Emmanuel College, Cambridge, 9th July, 1799, and read before the Royal Society in January following, he appears to incline strongly to the Huyghenian theory; not however offering any new facts or calculations in its favor, but pointing out the great difficulties of the Newtonian hypothesis. But in a paper read before the Royal Society, November 12, 1801, he says, "A further

consideration of the colors of thin plates has converted that prepossession which I before entertained for the undulatory theory of light, into a very strong conviction of its truth and efficiency; a conviction which has since been most strikingly confirmed by an analysis of the colors of _striated surfaces_." He here states the general principle of interferences in the form of a proposition. (Prop. viii.) "When two undulations from different origins coincide either perfectly or very nearly in direction, their joint effect is a combination of the motions belonging to them." He explains, by the help of this proposition, the colors which were observed in Coventry's {94} micrometers, in which instrument lines were drawn on glass at a distance of 1⁄500th of an inch. The interference of the undulations of the rays reflected from the two sides of these fine lines, produced periodical colors. In the same manner, he accounts for the colors of thin plates, by the interference of the light partially reflected from the two surfaces of the plates. We have already seen that Hooke had long before suggested the same explanation; and Young says at the end of his paper, "It was not till I had satisfied myself respecting all these phenomena, that I found in Hooke's _Micrographia_ a passage which might have led me earlier to a similar opinion." He also quotes from Newton many passages which assume the existence of an ether; of which, as we have already seen, Newton suggests the necessity in these very phenomena, though he would apply it in combination with the emission of material light. In July, 1802, Young explained, on the same principle, some facts in indistinct vision, and other similar appearances. And in 1803,[75\9] he speaks more positively still. "In making," he says, "some experiments on the fringes of colors accompanying shadows, I have found so simple and so demonstrative a proof of the general law of interference of two portions of light, which I have already endeavored to establish, that I think it right to lay before the Royal Society a short statement of the facts which appear to me to be thus decisive." The two papers just mentioned certainly ought to have convinced all scientific men of the truth of the doctrine thus urged; for the number and exactness of the explanations is very remarkable. They include the colored fringes which are seen with the shadows of fibres; the colors produced by a dew between two pieces of glass, which, according to the theory, should appear when the thickness of the plate is _six_ times that of thin plates, and which do so; the changes resulting from the employment of other fluids than water; the effect of inclining the plates; also the fringes and bands which accompany shadows, the phenomena observed by Grimaldi, Newton, Maraldi, and others, and hitherto never at all reduced to rule. Young observes, very justly, "whatever may be thought of the theory, we have got a simple and general law" of the phenomena. He moreover calculated the length of an undulation from the measurements of fringes of shadows, as he had done before from the colors of thin plates; and found a very close accordance of the results of the various cases with one another. {95}

[Note 75\9: _Phil. Trans._ Memoir, read Nov. 24.]

There is one difficulty, and one inaccuracy, in Young's views at this period, which it may be proper to note. The difficulty was, that he found it necessary to suppose that light, when reflected at a rarer medium, is retarded by half an undulation. This assumption, though often urged at a later period as an argument against the theory, was fully justified as the mechanical principles of the subject were unfolded; and the

necessity of it was clear to Young from the first. On the strength of this, says he, "I ventured to predict, that if the reflections were of the same kind, made at the surfaces of a thin plate, of a density intermediate between the densities of the mediums surrounding it, the central spot would be white; and I have now the pleasure of stating, that I have fully verified this prediction by interposing a drop of oil of sassafras between a prism of flint-glass and a lens of crown-glass."

The inaccuracy of his calculations consisted in his considering the external fringe of shadows to be produced by the interference of a ray _reflected_ from the _edge_ of the object, with a ray which passes clear of it; instead of supposing _all the parts_ of the wave of light to corroborate or interfere with one another. The mathematical treatment of the question on the latter hypothesis was by no means easy. Young was a mathematician of considerable power in the solution of the problems which came before him: though his methods possessed none of the analytical elegance which, in his time, had become general in France. But it does not appear that he ever solved the problem of undulations as applied to fringes, with its true conditions. He did, however, rectify his conceptions of the nature of the interference; and we may add, that the numerical error of the consequences of the defective hypothesis is not such as to prevent their confirming the undulatory theory.[76\9]

[Note 76\9: I may mention, in addition to the applications which Young made of the principle of interferences, his _Eriometer_, an instrument invented for the purpose of measuring the thickness of the fibres of wood; and the explanation of the supernumerary bands of the rainbow. These explanations involve calculations founded on the length of an undulation of light, and were confirmed by experiment, as far as experiment went.]

But though this theory was thus so powerfully recommended by experiment and calculation, it met with little favor in the scientific world. Perhaps this will be in some measure accounted for, when we come, in the next chapter, to speak of the mode of its reception by {96} the supposed judges of science and letters. Its author went on laboring at the completion and application of the theory in other parts of the subject; but his extraordinary success in unravelling the complex phenomena of which we have been speaking, appears to have excited none of the notice and admiration which properly belonged to it, till Fresnel's Memoir _On Diffraction_ was delivered to the Institute, in October, 1815.

MM. Arago and Poinsot were commissioned to make a report upon this Memoir; and the former of these philosophers threw himself upon the subject with a zeal and intelligence which peculiarly belonged to him. He verified the laws announced by Fresnel: "laws," he says, "which appear to be destined to make an epoch in science." He then cast a rapid glance at the history of the subject, and recognized, at once, the place which Young occupied in it. Grimaldi, Newton, Maraldi, he states, had observed the facts, and tried in vain to reduce them to rule or cause. "Such[77\9] was the state of our knowledge on this difficult question, when Dr. Thomas Young made the very remarkable experiment which is described in the _Philosophical Transactions_ for 1803;" namely, that to obliterate all the bands within the shadow, we need only stop the ray which is going to graze, or has grazed, one border of the object. To this,

Arago added the important observation, that the same obliteration takes place, if we stop the ray, with a transparent plate; except the plate be very thin, in which case the bands are displaced, and not extinguished. "Fresnel," says he, "guessed the effect which a thin plate would produce, when I had told him of the effect of a thick glass." Fresnel himself declares[78\9] that he was not, at the time, aware of Young's previous labors. After stating nearly the same reasonings concerning fringes which Young had put forward in 1801, he adds, "it is therefore the meeting, the actual crossing of the rays, which produces the fringes. This consequence, which is only, so to speak, the translation of the phenomena, seems to me entirely opposed to the hypothesis of emission, and confirms the system which makes light consist in the vibrations of a peculiar fluid." And thus the Principle of Interferences, and the theory of undulations, so far as that principle depends upon the theory, was a second time established by Fresnel in France, fourteen years after it had been discovered, fully proved, and repeatedly published by Young in England. {97}

[Note 77\9: _An. Chim._ 1815, Febr.]

[Note 78\9: Ib. tom. xvii. p. 402.]

In this Memoir of Fresnel's, he takes very nearly the same course as Young had done; considering the interference of the direct light with that reflected at the edge, as the cause of the external fringes; and he observes, that in this reflection it is necessary to suppose half an undulation lost: but a few years later, he considered the propagation of undulations in a more true and general manner, and obtained the solution of this difficulty of the half-undulation. His more complete Memoir on _Diffraction_ was delivered to the Institute of France, July 29, 1818; and had the prize awarded it in 1819:[79\9] but by the delays which at that period occurred in the publication of the _Parisian Academical Transactions_, it was not published[80\9] till 1826, when the theory was no longer generally doubtful or unknown in the scientific world. In this Memoir, Fresnel observes, that we must consider the effect of _every portion_ of a wave of light upon a distant point, and must, on this principle, find the illumination produced by any number of such waves together. Hence, in general, the process of integration is requisite; and though the integrals which here offer themselves are of a new and difficult kind, he succeeded in making the calculation for the cases in which he experimented. His _Table of the Correspondences of Theory and Observation_,[81\9] is very remarkable for the closeness of the agreement; the errors being generally less than one hundredth of the whole, in the distances of the black bands. He justly adds, "A more striking agreement could not be expected between experiment and theory. If we compare the smallness of the differences with the extent of the breadths measured; and if we remark the great variations which _a_ and _b_ (the distance of the object from the luminous point and from the screen) have received in the different observations, we shall find it difficult not to regard the integral which has led us to these results as the faithful expression of the law of the phenomena."

[Note 79\9: _Ann. Chim._ May, 1819.]

[Note 80\9: _Mém. Inst._ for 1821-2.]

[Note 81\9: _Mém. Inst._ p. 420-424.]

A mathematical theory, applied, with this success, to a variety of cases of very different kinds, could not now fail to take strong hold of the attention of mathematicians; and accordingly, from this time, the undulatory doctrine of diffraction has been generally assented to, and the mathematical difficulties which it involves, have been duly studied and struggled with.

Among the remarkable applications of the undulatory doctrine to diffraction, we may notice those of Joseph Fraunhofer, a {98} mathematical optician of Munich. He made a great number of experiments on the shadows produced by small holes, and groups of small holes, very near each other. These were published[82\9] in his _New Modifications of Light_, in 1823. The greater part of this Memoir is employed in tracing the laws of phenomena of the extremely complex and splendid appearances which he obtained; but at the conclusion he observes, "It is remarkable that the laws of the reciprocal influence and of the diffraction of the rays, can be deduced from the principles of the undulatory theory: knowing the conditions, we may, by means of an extremely simple equation, determine the extent of a luminous wave for each of the different colors; and in every case, the calculation corresponds with observation." This mention of "an extremely simple equation," appears to imply that he employed only Young's and Fresnel's earlier mode of calculating interferences, by considering two portions of light, and not the method of integration. Both from the late period at which they were published, and from the absence of mathematical details, Fraunhofer's labors had not any strong influence on the establishment of the undulatory theory; although they are excellent verifications of it, both from the goodness of the observations, and the complexity and beauty of the phenomena.

[Note 82\9: In Schumacher's _Astronomische Abhandlungen_, in French; earlier in German.]

We have now to consider the progress of the undulatory theory in another of its departments, according to the division already stated.

_Sect._ 3.--_Explanation of Double Refraction by the Undulatory Theory._

WE have traced the history of the undulatory theory applied to diffraction, into the period when Young came to have Fresnel for his fellow-laborer. But in the mean time, Young had considered the theory in its reference to other phenomena, and especially to those of _double refraction_.

In this case, indeed, Huyghens's explanation of the facts of Iceland spar, by means of spheroidal undulations, was so complete, and had been so fully confirmed by the measurements of Haüy and Wollaston, that little remained to be done, except to connect the Huyghenian hypothesis with the mechanical views belonging to the theory, and to extend his law to other cases. The former part of this task Young executed, by remarking that we may conceive the _elasticity_ of the {99} crystal, on which the velocity of propagation of the luminiferous undulation depends, to be different, in the direction of the crystallographic axis, and in the direction of the planes at right angles to this axis; and from such a difference, he deduces the existence of spheroidal undulations. This suggestion appeared in the _Quarterly

Review_ for November, 1809, in a critique upon an attempt of Laplace to account for the same phenomena. Laplace had proposed to reduce the double refraction of such crystals as Iceland spar, to his favorite machinery of forces which are sensible at small distances only. The peculiar forces which produce the effect in this case, he conceives to emanate from the crystallographic axis: so that the velocity of light within the crystal will depend only on the situation of the ray with respect to this axis. But the establishment of this condition is, as Young observes, the main difficulty of the problem. How are we to conceive refracting forces, independent of the surface of the refracting medium, and regulated only by a certain internal line? Moreover, the law of force which Laplace was obliged to assume, namely, that it varied as the square of the sine of the angle which the ray made with the axis, could hardly be reconciled with mechanical principles. In the critique just mentioned, Young appears to feel that the undulatory theory, and perhaps he himself, had not received justice at the hands of men of science; he complains that a person so eminent in the world of science as Laplace then was, should employ his influence in propagating error, and should disregard the extraordinary confirmations which the Huyghenian theory had recently received.

The extension of this view, of the different elasticity of crystals in different directions, to other than uniaxal crystals, was a more complex and difficult problem. The general notion was perhaps obvious, after what Young had done; but its application and verification involved mathematical calculations of great generality, and required also very exact experiments. In fact, this application was not made till Fresnel, a pupil of the Polytechnic School, brought the resources of the modern analysis to bear upon the problem;--till the phenomena of dipolarized light presented the properties of biaxal crystals in a vast variety of forms;--and till the theory received its grand impulse by the combination of the explanation of polarization with the explanation of double refraction. To the history of this last-mentioned great step we now proceed. {100}

_Sect._ 4.--_Explanation of Polarization by the Undulatory Theory._

EVEN while the only phenomena of _polarization_ which were known were those which affect the two images in Iceland spar, the difficulty which these facts seemed at first to throw in the way of the undulatory theory was felt and acknowledged by Young. Malus's discovery of polarization by reflection increased the difficulty, and this Young did not attempt to conceal. In his review of the papers containing this discovery[83\9] he says, "The discovery related in these papers appears to us to be by far the most important and interesting which has been made in France concerning the properties of light, at least since the time of Huyghens; and it is so much the more deserving of notice, as it greatly influences the general balance of evidence in the comparison of the undulatory and projectile theories of the nature of light." He then proceeds to point out the main features in this comparison, claiming justly a great advantage for the theory of undulations on the two points we have been considering, the phenomena of diffraction and of double refraction. And he adds, with reference to the embarrassment introduced by polarization, that we are not to expect the course of scientific discovery to run smooth and uninterrupted; but that we are to lay our

account with partial obscurity and seeming contradiction, which we may hope that time and enlarged research will dissipate. And thus he steadfastly held, with no blind prejudice, but with unshaken confidence, his great philosophical trust, the fortunes of the undulatory theory. It is here, after the difficulties of polarization had come into view, and before their solution had been discovered, that we may place the darkest time of the history of the theory; and at this period Young was alone in the field.

[Note 83\9: _Quart. Rev._ May, 1810.]

It does not appear that the light dawned upon him for some years. In the mean time, Young found that his theory would explain dipolarized colors; and he had the satisfaction to see Fresnel re-discover, and M. Arago adopt, his views on diffraction. He became engaged in friendly intercourse with the latter philosopher, who visited him in England in 1816. On January the 12th, 1817, in writing to this gentleman, among other remarks on the subject of optics, he says, "I have also been reflecting on the possibility of giving an imperfect explanation of the affection of light which constitutes polarization, {101} without departing from the genuine doctrine of undulation." He then proceeds to suggest the possibility of "a _transverse_ vibration, propagated in the direction of the radius, the motions of the particles being in a certain constant direction with respect to that radius; and this," he adds, "is _polarization_." From his further explanation of his views, it appears that he conceived the motions of the particles to be oblique to the direction of the ray, and not perpendicular, as the theory was afterwards framed; but still, here was the essential condition for the explanation of the facts of polarization,--the transverse nature of the vibrations. This idea at once made it possible to conceive how the rays of light could have _sides_; for the direction in which the vibration was transverse to the ray, might be marked by peculiar properties. And after the idea was once started, it was comparatively easy for men like Young and Fresnel to pursue and modify it till it assumed its true and distinct form.

We may judge of the difficulty of taking firmly hold of the conception of transverse vibrations of the ether, as those which constitute light, by observing how long the great philosophers of whom we are speaking lingered within reach of it, before they ventured to grasp it. Fresnel says, in 1821, "When M. Arago and I had remarked (in 1816) that two rays polarized at right angles always give the same quantity of light by their union, I thought this might be explained by supposing the vibrations to be transverse, and to be at right angles when the rays are polarized at right angles. But this supposition was so contrary to the received ideas on the nature of the vibrations of elastic fluids," that Fresnel hesitated to adopt it till he could reconcile it better to his mechanical notions. "Mr. Young, more bold in his conjectures, and less confiding in the views of geometers, published it before me, though perhaps he thought it after me." And M. Arago was afterwards wont to relate[84\9] that when he and Fresnel had obtained their joint experimental results of the non-interference of oppositely-polarized pencils, and when Fresnel pointed out that transverse vibrations were the only possible translation of this fact into the undulatory theory, he himself protested that he had not courage to publish such a conception; and accordingly, the second part of the Memoir was published in Fresnel's name alone. What renders this

more remarkable is, that it occurred when M. Arago had in his possession the very letter of Young, in which he proposed the same suggestion. {102}

[Note 84\9: I take the liberty of stating this from personal knowledge.]

Young's first published statement of the doctrine of transverse vibrations was given in the explanation of the phenomena of dipolarization, of which we shall have to speak in the next Section. But the primary and immense value of this conception, as a step in the progress of the undulatory theory, was the connexion which it established between polarization and double refraction; for it held forth a promise of accounting for polarization, if any conditions could be found which might determine what was the direction of the transverse vibrations. The analysis of these conditions is, in a great measure, the work of Fresnel; a task performed with profound philosophical sagacity and great mathematical skill.

Since the double refraction of uniaxal crystals could be explained by undulations of the form of a spheroid, it was perhaps not difficult to conjecture that the undulations of biaxal crystals would be accounted for by undulations of the form of an ellipsoid, which differs from the spheroid in having its three axes unequal, instead of two only; and consequently has that very relation to the other, in respect of symmetry, which the crystalline and optical phenomena have. Or, again, instead of supposing two different degrees of elasticity in different directions, we may suppose three such different degrees in directions at right angles to each other. This kind of generalization was tolerably obvious to a practised mathematician.

But what shall call into play all these elasticities at once, and produce waves governed by each of them? And what shall explain the different polarization of the rays which these separate waves carry with them? These were difficult questions, to the solution of which mathematical calculation had hitherto been unable to offer any aid.

It was here that the conception of transverse vibrations came in, like a beam of sunlight, to disclose the possibility of a mechanical connexion of all these facts. If transverse vibrations, travelling through a uniform medium, come to a medium not uniform, but constituted so that the elasticity shall be different in different directions, in the manner we have described, what will be the course and condition of the waves in the second medium? Will the effects of such waves agree with the phenomena of doubly-refracted light in biaxal crystals? Here was a problem, striking to the mathematician for its generality and difficulty, and of deep interest to the physical philosopher, because the fate of a great theory depended upon its solution.

The solution, obtained by great mathematical skill, was laid before the French Institute by Fresnel in November, 1821, and was carried {103} further in two Memoirs presented in 1822. Its import is very curious. The undulations which, coming from a distant centre, fall upon such a medium as we have described, are, it appears from the principles of mechanics, propagated in a manner quite different from anything which had been anticipated. The "surface of the waves" (that is, the surface which would bound undulations diverging from a point), is a very complex, yet symmetrical curve surface; which, in the case of uniaxal crystals, resolves itself

into a sphere and a spheroid; but which, in general, forms a continuous double envelope of the central point to which it belongs, intersecting itself and returning into itself. The directions of the rays are determined by this curve surface in biaxal crystals, as in uniaxal crystals they are determined by the sphere and the spheroid; and the result is, that in biaxal crystals, _both_ rays suffer _extraordinary_ refraction according to determinate laws. And the positions of the planes of polarization of the two rays follow from the same investigation; the plane of polarization in every case being supposed to be that which is perpendicular to the transverse vibrations. Now it appeared that the polarization of the two rays, as determined by Fresnel's theory, would be in directions, not indeed exactly accordant with the law deduced by M. Biot from experiment, but deviating so little from those directions, that there could be small doubt that the empirical formula was wrong, and the theoretical one right.

The theory was further confirmed by an experiment showing that, in a biaxal crystal (topaz), neither of the rays was refracted according to the ordinary law, though it had hitherto been supposed that one of them was so; a natural inaccuracy, since the error was small.[85\9] Thus this beautiful theory corrected, while it explained, the best of the observations which had previously been made; and offered itself to mathematicians with an almost irresistible power of conviction. The explanation of laws so strange and diverse as those of double refraction and polarization, by the same general and symmetrical theory, could not result from anything but the truth of the theory.

[Note 85\9: _An. Ch._ xxviii. p. 264.]

"Long," says Fresnel,[86\9] "before I had conceived this theory, I had convinced myself by a pure contemplation of the facts, that it was not possible to discover the true explanation of double refraction, without explaining, at the same time, the phenomena of polarization, which always goes along with it; and accordingly, it was after having found {104} what mode of vibration constituted polarization, that I caught sight of the mechanical causes of double refraction."

[Note 86\9: _Sur la Double Réf., Mém. Inst._ 1826, p. 174.]

Having thus got possession of the principle of the mechanism of polarization, Fresnel proceeded to apply it to the other cases of polarized light, with a rapidity and sagacity which reminds us of the spirit in which Newton traced out the consequences of the principle of universal gravitation. In the execution of his task, indeed, Fresnel was forced upon several precarious assumptions, which make, even yet, a wide difference between the theory of gravitation and that of light. But the mode in which these were confirmed by experiment, compels us to admire the happy apparent boldness of the calculator.

The subject of _polarization by reflection_ was one of those which seemed most untractable; but, by means of various artifices and conjectures, it was broken up and subdued. Fresnel began with the simplest case, the reflection of light polarized in the plane of reflection; which he solved by means of the laws of collision of elastic bodies. He then took the reflection of light polarized perpendicularly to this plane; and here, adding to the general mechanical principles a hypothetical assumption, that the

communication of the resolved motion parallel to the refracting surface, takes place according to the laws of elastic bodies, he obtains his formula. These results were capable of comparison with experiment; and the comparison, when made by M. Arago, confirmed the formulæ. They accounted, too, for Sir D. Brewster's law concerning the polarizing angle (see Chap. vi.); and this could not but be looked upon as a striking evidence of their having some real foundation. Another artifice which MM. Fresnel and Arago employed, in order to trace the effect of reflection upon common light, was to use a ray polarized in a plane making half a right angle with the plane of reflection; for the quantities of the oppositely[87\9] polarized light in such an incident ray are equal, as they are in common light; but the relative quantities of the oppositely polarized light in the reflected ray are indicated by the new plane of polarization; and thus these relative quantities become known for the case of common light. The results thus obtained were also confirmed by facts; and in this manner, all that was doubtful in the process of Fresnel's reasoning, seemed to be authorized by its application to real cases. {105}

[Note 87\9: It will be recollected all along, that _oppositely_ polarized rays are those which are polarized in two planes _perpendicular_ to each other. See above, chap. vi.]

These investigations were published[88\9] in 1821. In succeeding years, Fresnel undertook to extend the application of his formulæ to a case in which they ceased to have a meaning, or, in the language of mathematicians, became _imaginary_; namely, to the case of internal reflection at the surface of a transparent body. It may seem strange to those who are not mathematicians, but it is undoubtedly true, that in many cases in which the solution of a problem directs impossible arithmetical or algebraical operations to be performed, these directions may be so interpreted as to point out a true solution of the question. Such an interpretation Fresnel attempted[89\9] in the case of which we now speak; and the result at which he arrived was, that the reflection of light through a rhomb of glass of a certain form (since called _Fresnel's rhomb_, would produce a polarization of a kind altogether different from those which his theory had previously considered, namely, that kind which we have spoken of as _circular polarization_. The complete confirmation of this curious and unexpected result by trial, is another of the extraordinary triumphs which have distinguished the history of the theory at every step since the commencement of Fresnel's labors.

[Note 88\9: _An. Chim._ t. xvii.]

[Note 89\9: _Bullet. des Sc._ Feb. 1823.]

But anything further which has been done in this way, may be treated of more properly in relating the verification of the theory. And we have still to speak of the most numerous and varied class of facts to which rival theories of light were applied, and of the establishment of the undulatory doctrine in reference to that department; I mean the phenomena of depolarized, or rather, as I have already said, _di_polarized light.

_Sect._ 5.--_Explanation of Dipolarization by the Undulatory Theory._

WHEN Arago, in 1811, had discovered the colors produced by polarized light passing through certain crystals,[90\9] it was natural that attempts should be made to reduce them to theory. M. Biot, animated by the success of Malus in detecting the laws of double refraction, and Young, knowing the resources of his own theory, were the first persons to enter upon this undertaking. M. Biot's theory, though in the end displaced by its rival, is well worth notice in the history of the subject. It was what he called the doctrine of _moveable polarization_. He conceived that when the molecules of light pass through {106} thin crystalline plates, the plane of polarization undergoes an oscillation which carries it backwards and forwards through a certain angle, namely, twice the angle contained between the original plane of polarization and the principal section of the crystal. The intervals which this oscillation occupies are lengths of the path of the ray, very minute, and different for different colors, like Newton's fits of easy transmission; on which model, indeed, the new theory was evidently framed.[91\9] The colors produced in the phenomena of dipolarization really do depend, in a periodical manner, on the length of the path of the light through the crystal, and a theory such as M. Biot's was capable of being modified, and was modified, so as to include the leading features of the facts as then known; but many of its conditions being founded on special circumstances in the experiments, and not on the real conditions of nature, there were in it several incongruities, as well as the general defect of its being an arbitrary and unconnected hypothesis.

[Note 90\9: See chap. ix.]

[Note 91\9: See MM. Arago and Biot's Memoirs, _Mém. Inst._ for 1811; the whole volume for 1812 is a Memoir of M. Biot's (published 1814); also _Mém. Inst._ for 1817; M. Biot's Mem. read in 1818, published in 1819 and for 1818.]

Young's mode of accounting for the brilliant phenomena of dipolarization appeared in the _Quarterly Review_ for 1814. After noticing the discoveries of MM. Arago, Brewster, and Biot, he adds, "We have no doubt that the surprise of these gentlemen will be as great as our own satisfaction in finding that they are perfectly reducible, like other causes of recurrent colors, to the general laws of the interference of light which have been established in this country;" giving a reference to his former statements. The results are then explained by the interference of the ordinary and extraordinary ray. But, as M. Arago properly observes, in his account of this matter,[92\9] "It must, however, be added that Dr. Young had not explained either in what circumstances the interference of the rays can take place, nor why we see no colors unless the crystallized plates are exposed to light previously polarized." The explanation of these circumstances depends on the laws of interference of polarized light which MM. Arago and Fresnel established in 1816. They then proved, by direct experiment, that when polarized light was treated so as to bring into view the most marked phenomena of interference, namely, the bands of shadows; pencils of light which have a common origin, and which are polarized in the parallel planes, interfere completely, while those which are {107} polarized in _opposite_ (that is, perpendicular,) planes do not interfere at all.[93\9] Taking these principles into the account, Fresnel explained very completely, by means of the interference of

undulations, all the circumstances of colors produced by crystallized plates; showing the necessity of the _polarization_ in the first instance; the _dipolarizing_ effect of the crystal; and the office of the _analysing plate_, by which certain portions of each of the two rays in the crystal are made to interfere and produce color. This he did, as he says,[94\9] without being aware, till Arago told him, that Young had, to some extent, anticipated him.

[Note 92\9: _Enc. Brit._ Supp. art. _Polarization._]

[Note 93\9: _Ann. Chim._ tom. x.]

[Note 94\9: Ib. tom. xvii. p. 402.]

When we look at the history of the emission-theory of light, we see exactly what we may consider as the natural course of things in the career of a false theory. Such a theory may, to a certain extent, explain the phenomena which it was at first contrived to meet; but every new class of facts requires a new supposition,--an addition to the machinery; and as observation goes on, these incoherent appendages accumulate, till they overwhelm and upset the original frame-work. Such was the history of the hypothesis of solid epicycles; such has been the history of the hypothesis of the material emission of light. In its simple form, it explained reflection and refraction; but the colors of thin plates added to it the hypothesis of fits of easy transmission and reflection; the phenomena of diffraction further invested the particles with complex hypothetical laws of attraction and repulsion; polarization gave them sides; double refraction subjected them to peculiar forces emanating from the axes of crystals; finally, dipolarization loaded them with the complex and unconnected contrivance of moveable polarization; and even when all this had been assumed, additional mechanism was wanting. There is here no unexpected success, no happy coincidence, no convergence of principles from remote quarters; the philosopher builds the machine, but its parts do not fit; they hold together only while he presses them: this is not the character of truth.

In the undulatory theory, on the other hand, all tends to unity and simplicity. We explain reflection and refraction by undulations; when we come to thin plates, the requisite "fits" are already involved in our fundamental hypothesis, for they are the length of an undulation; the phenomena of diffraction also require such intervals; and the intervals thus required agree exactly with the others in magnitude, {108} so that no new property is needed. Polarization for a moment checks us; but not long; for the direction of our vibrations is hitherto arbitrary;--we allow polarization to decide it. Having done this for the sake of polarization, we find that it also answers an entirely different purpose, that of giving the law of double refraction. Truth may give rise to such a coincidence; falsehood cannot. But the phenomena become more numerous, more various, more strange; no matter: the Theory is equal to them all. It makes not a single new physical hypothesis; but out of its original stock of principles it educes the counterpart of all that observation shows. It accounts for, explains, simplifies, the most entangled cases; corrects known laws and facts; predicts and discloses unknown ones; becomes the guide of its former teacher, Observation; and, enlightened by mechanical conceptions, acquires an insight which pierces through

shape and color to force and cause.

We thus reach the philosophical _moral_ of this history, so important in reference to our purpose; and here we shall close the account of the discovery and promulgation of the undulatory theory. Any further steps in its development and extension, may with propriety be noticed in the ensuing chapters, respecting its reception and verification.

[2nd Ed.] [In the _Philosophy of the Inductive Sciences_, B. xi. ch. iii. Sect. 11, I have spoken of the _Consilience of Inductions_ as one of the characters of scientific truth. We have several striking instances of such consilience in the history of the undulatory theory. The phenomena of fringes of shadows and colored bands in crystals _jump together_ in the Theory of Vibrations. The phenomena of polarization and double refraction _jump together_ in the Theory of Crystalline Vibrations. The phenomena of polarization and of the interference of polarized rays _jump together_ in the Theory of Transverse Vibrations.

The proof of what is above said of the undulatory theory is contained in the previous history. This theory has "accounted for, explained, and simplified the most entangled cases;" as the cases of fringes of shadows; shadows of gratings; colored bands in biaxal crystals, and in quartz. There are no optical phenomena more entangled than these. It has "corrected experimental laws," as in the case of M. Biot's law of the direction of polarization in biaxal crystals. It has done this, "without making any new physical hypothesis;" for the transverse direction of vibrations, the different optical elasticities of crystals in different directions, and (if it be adopted) the hypothesis of finite {109} intervals of the particles (see chap. x. and hereafter, chap. xiii.), are only limitations of what was indefinite in the earlier form of the hypothesis. And so far as the properties of visible radiant light are concerned, I do not think it at all too much to say, as M. Schwerd has said, that "the undulation theory accounts for the phenomena as completely as the theory of gravitation does for the facts of the solar system."

This we might say, even if some facts were not yet fully explained; for there were till very lately, if there are not still, such unexplained facts with regard to the theory of gravitation, presented to us by the solar system. With regard to the undulatory theory, these exceptions are, I think, disappearing quite as rapidly and as completely as in the case of gravitation. It is to be observed that no presumption against the theory can with any show of reason be collected from the cases in which classes of phenomena remain unexplained, the theory having never been applied to them by any mathematician capable of tracing its results correctly. The history of the theory of gravitation may show us abundantly how necessary it is to bear in mind this caution; and the results of the undulatory theory cannot be traced without great mathematical skill and great labor, any more than those of gravitation.

This remark applies to such cases as that of the _transverse fringes of grooved surfaces_. The general phenomena of these cases are perfectly explained by the theory. But there is an interruption in the light in an oblique direction, which has not yet been explained; but looking at what has been done in other cases, it is impossible

to doubt that this phenomenon depends upon the results of certain integrations, and would be explained if these were rightly performed.

The phenomena of _crystallized surfaces_, and especially their effects upon the plane of polarization, were examined by Sir D. Brewster, and laws of the phenomena made out by him with his usual skill and sagacity. For a time these were unexplained by the theory. But recently Mr. Mac Cullagh has traced the consequences of the theory in this case,[95\9] and obtained a law which represents with much exactness, Sir D. Brewster's observation.

[Note 95\9: Prof. Lloyd's _Report, Brit. Assoc._ 1834, p. 374.]

The phenomena which Sir D. Brewster, in 1837, called a _new property of light_, (certain appearances of the spectrum when the pupil of the eye is half covered with a thin glass or crystal,) have been explained by Mr. Airy in the _Phil. Trans._ for 1840.

Mr. Airy's explanation of the phenomena termed by Sir D. {110} Brewster a _new property of light_, is completed in the _Philosophical Magazine_ for November, 1846. It is there shown that a dependence of the breadth of the bands upon the aperture of the pupil, which had been supposed to result from the theory, and which does not appear in the experiment, did really result from certain limited conditions of the hypothesis, which conditions do not belong to the experiment; and that when the problem is solved without those limitations, the discrepance of theory and observation vanishes; so that, as Mr. Airy says, "this very remarkable experiment, which long appeared inexplicable, seems destined to give one of the strongest confirmations to the Undulatory Theory."

I may remark also that there is no force in the objection which has been urged against the admirers of the undulatory theory, that by the fulness of their assent to it, they discourage further researches which may contradict or confirm it. We must, in this point of view also, look at the course of the theory of gravitation and its results. The acceptance of that theory did not prevent mathematicians and observers from attending to the apparent exceptions, but on the contrary, stimulated them to calculate and to observe with additional zeal, and still does so. The acceleration of the Moon, the mutual disturbances of Jupiter and Saturn, the motions of Jupiter's Satellites, the effect of the Earth's oblateness on the Moon's motion, the motions of the Moon about her own centre, and many other phenomena, were studied with the greater attention, _because_ the general theory was deemed so convincing: and the same cause makes the remaining exceptions objects of intense interest to astronomers and mathematicians. The mathematicians and optical experimenters who accept the undulatory theory, will of course follow out their conviction in the same manner. Accordingly, this has been done and is still doing, as in Mr. Airy's mathematical investigation of the effect of an annular aperture; Mr. Earnshaw's, of the effect of a triangular aperture; Mr. Talbot's explanation of the effect of interposing a film of mica between a part of the pupil and the pure spectrum, so nearly approaching to the phenomena which have been spoken of as a new Polarity of Light; besides other labors of eminent mathematicians, elsewhere mentioned in these pages.

The phenomena of the _absorption_ of light have no especial bearing upon the undulatory theory. There is not much difficulty in explaining the _possibility_ of absorption upon the theory. When the light is absorbed, it ceases to belong to the theory. {111}

For, as I have said, the theory professes only to explain the phenomena of _radiant visible_ light. We know very well that light has other bearings and properties. It produces chemical effects. The optical polarity of crystals is connected with the chemical polarity of their constitution. The natural colors of bodies, too, are connected with their chemical constitution. Light is also connected with heat. The undulatory theory does not undertake to explain these properties and their connexion. If it did, it would be a Theory of Heat and of Chemical Composition, as well as a Theory of Light.

Dr. Faraday's recent experiments have shown that the magnetic polarity is directly connected with that optical polarity by which the plane of polarization is affected. When the lines of magnetic force pass through certain transparent bodies, they communicate to them a certain kind of circular polarizing power; yet different from the circular polarizing power of quartz, and certain fluids mentioned in chapter ix.

Perhaps I may be allowed to refer to this discovery as a further illustration of the views I have offered in the _Philosophy of the Inductive Sciences_ respecting the _Connexion of Co-existent Polarities_. (B. v. Chap. ii.)]

## CHAPTER XII.

SEQUEL TO THE EPOCH OF YOUNG AND FRESNEL. RECEPTION OF THE UNDULATORY THEORY.

WHEN Young, in 1800, published his assertion of the Principle of Interferences, as the true theory of optical phenomena, the condition of England was not very favorable to a fair appreciation of the value of the new opinion. The men of science were strongly pre-occupied in favor of the doctrine of emission, not only from a national interest in Newton's glory, and a natural reverence for his authority, but also from deference towards the geometers of France, who were looked up to as our masters in the application of mathematics to physics, and who were understood to be Newtonians in this as in other subjects. A general tendency to an atomic philosophy, which had begun to appear from the time of Newton, operated powerfully; and {112} the hypothesis of emission was so easily conceived, that, when recommended by high authority, it easily became popular; while the hypothesis of luminiferous undulations, unavoidably difficult to comprehend, even by the aid of steady thought, was neglected, and all but forgotten.

Yet the reception which Young's opinions met with was more harsh than he might have expected, even taking into account all these considerations. But there was in England no visible body of men, fitted by their knowledge and character to pronounce judgment on such a question, or to give the proper impulse and bias to public opinion. The Royal Society, for instance, had not, for a long time, by custom or institution,

possessed or aimed at such functions. The writers of "Reviews" alone, self-constituted and secret tribunals, claimed this kind of authority. Among these publications, by far the most distinguished about this period was the _Edinburgh Review_; and, including among its contributors men of eminent science and great talents, employing also a robust and poignant style of writing (often certainly in a very unfair manner), it naturally exercised great influence. On abstruse doctrines, intelligible to few persons, more than on other subjects, the opinions and feelings expressed in a Review must be those of the individual reviewer. The criticism on some of Young's early papers on optics was written by Mr. (afterwards Lord) Brougham, who, as we have seen, had experimented on diffraction, following the Newtonian view, that of inflexion. Mr. Brougham was perhaps at this time young enough[96\9] to be somewhat intoxicated with the appearance of judicial authority in matters of science, which his office of anonymous reviewer gave him: and even in middle-life, he was sometimes considered to be prone to indulge himself in severe and sarcastic expressions. In January, 1803, was published[97\9] his critique on Dr. Young's Bakerian Lecture, _On the Theory of Light and Colors_, in which lecture the doctrine of undulations and the law of interferences was maintained. This critique was an uninterrupted strain of blame and rebuke. "This paper," the reviewer said, "contains nothing which deserves the name either of experiment or discovery." He charged the writer with "dangerous relaxations of the principles of physical logic." "We wish," he cried, "to recall philosophers to the strict and severe methods of investigation," describing them as those pointed out by Bacon, Newton, and the like. Finally, Dr. Young's speculations {113} were spoken of as a hypothesis, which is a mere work of fancy; and the critic added, "we cannot conclude our review without entreating the attention of the Royal Society, which has admitted of late so many hasty and unsubstantial papers into its _Transactions_;" which habit he urged them to reform. The same aversion to the undulatory theory appears soon after in another article by the same reviewer, on the subject of Wollaston's measures of the refraction of Iceland spar; he says, "We are much disappointed to find that so acute and ingenious an experimentalist should have adopted the wild optical theory of vibrations." The reviewer showed ignorance as well as prejudice in the course of his remarks; and Young drew up an answer, which was ably written, but being published separately had little circulation. We can hardly doubt that these Edinburgh reviews had their effect in confirming the general disposition to reject the undulatory theory.

[Note 96\9: His age was twenty-four.]

[Note 97\9: _Edin. Review_, vol. i. p. 450.]

We may add, however, that Young's mode of presenting his opinions was not the most likely to win them favor; for his mathematical reasonings placed them out of the reach of popular readers, while the want of symmetry and system in his symbolical calculations, deprived them of attractiveness for the mathematician. He himself gave a very just criticism of his own style of writing, in speaking on another of his works:[98\9] "The mathematical reasoning, for want of mathematical symbols, was not understood, even by tolerable mathematicians. From a dislike of the affectation of algebraical formality which he had observed in some foreign authors, he was led into

something like an affectation of simplicity, which was equally inconvenient to a scientific reader."

[Note 98\9: See _Life of Young_, p. 54.]

Young appears to have been aware of his own deficiency in the power of drawing public favor, or even notice, to his discoveries. In 1802, Davy writes to a friend, "Have you seen the theory of my colleague, Dr. Young, on the undulations of an ethereal medium as the cause of light? It is not likely to be a popular hypothesis, after what has been said by Newton concerning it. He would be very much flattered if you could offer any observations upon it, _whether for or against it_." Young naturally felt confident in his power of refuting objections, and wanted only the opportunity of a public combat.

Dr. Brewster, who was, at this period, enriching optical knowledge with so vast a train of new phenomena and laws, shared the general aversion to the undulatory theory, which, indeed, he hardly overcame {114} thirty years later. Dr. Wollaston was a person whose character led him to look long at the laws of phenomena, before he attempted to determine their causes; and it does not appear that he had decided the claims of the rival theories in his own mind. Herschel (I now speak of the son) had at first the general mathematical prejudice in favor of the emission doctrine. Even when he had himself studied and extended the laws of dipolarized phenomena, he translated them into the language of the theory of moveable polarization. In 1819, he refers to, and corrects, this theory; and says, it is now "relieved from every difficulty, and entitled to rank with the fits of easy transmission and reflection as a general and simple physical law;" a just judgment, but one which now conveys less of praise than he then intended. At a later period, he remarked that we cannot be certain that if the theory of emission had been as much cultivated as that of undulation, it might not have been as successful; an opinion which was certainly untenable after the fair trial of the two theories in the case of diffraction, and extravagant after Fresnel's beautiful explanation of double refraction and polarization. Even in 1827, in a _Treatise on Light_, published in the _Encyclopædia Metropolitana_, he gives a section to the calculations of the Newtonian theory; and appears to consider the rivalry of the theories as still subsisting. But yet he there speaks with a proper appreciation of the advantages of the new doctrine. After tracing the prelude to it, he says, "But the unpursued speculations of Newton, and the opinions of Hooke, however distinct, must not be put in competition, and, indeed, ought scarcely to be mentioned, with the elegant, simple, and comprehensive theory of Young,--a theory which, if not founded in nature, is certainly one of the happiest fictions that the genius of man ever invented to grasp together natural phenomena, which, at their first discovery, seemed in irreconcileable opposition to it. It is, in fact, in all its applications and details, one succession of _felicities_; insomuch, that we may almost be induced to say, if it be not true, it deserves to be so."

In France, Young's theory was little noticed or known, except perhaps by M. Arago, till it was revived by Fresnel. And though Fresnel's assertion of the undulatory theory was not so rudely received as Young's had been, it met with no small opposition from the older mathematicians, and made its way slowly to the notice and comprehension

of men of science. M. Arago would perhaps have at once adopted the conception of transverse vibrations, when it was suggested by his fellow-laborer, Fresnel, if it had not been that he was a member of the {115} Institute, and had to bear the brunt of the war, in the frequent discussions on the undulatory theory; to which theory Laplace, and other leading members, were so vehemently opposed, that they would not even listen with toleration to the arguments in its favor. I do not know how far influences of this kind might operate in producing the delays which took place in the publication of Fresnel's papers. We have seen that he arrived at the conception of transverse vibrations in 1816, as the true key to the understanding of polarization. In 1817 and 1818, in a memoir read to the Institute, he analysed and explained the perplexing phenomena of quartz, which he ascribed to a _circular polarization_. This memoir had not been printed, nor any extract from it inserted in the scientific journals, in 1822, when he confirmed his views by further experiments.[99\9] His remarkable memoir, which solved the extraordinary and capital problem of the connexion of double refraction and crystallization, though written in 1821, was not published till 1827. He appears by this time to have sought other channels of publication. In 1822, he gave,[100\9] in the _Annales de Chimie et de Physique_, an explanation of refraction on the principles of the undulatory theory; alleging, as the reason for doing so, that the theory was still little known. And in succeeding years there appeared in the same work, his theory of reflection. His memoir on this subject (_Mémoire sur la Loi des Modifications que la Réflexion imprime à la Lumière Polarisée_,) was read to the Academy of Sciences in **1823. But the original paper was mislaid, and, for a time, supposed to be lost; it has since been recovered among the papers of M. Fourier, and printed in the eleventh volume of the Memoirs of the Academy.[101\9] Some of the speculations to which he refers, as communicated to the Academy, have never yet appeared.[102\9]

[Note 99\9: Hersch. _Light_, p. 539.]

[Note 100\9: _Ann. de Chim._ 1822, tom. xxi. p. 235.]

[Note 101\9: Lloyd. _Report on Optics_, p. 363. (Fourth Rep. of Brit. Ass.)]

[Note 102\9: Ib. p. 316, _note._]

Still Fresnel's labors were, from the first, duly appreciated by some of the most eminent of his countrymen. His _Memoir on Diffraction_ was, as we have seen, crowned in 1819: and, in 1822, a Report upon his _Memoir on Double Refraction_ was drawn up by a commission consisting of MM. Ampère, Fourier, and Arago. In this report[103\9] Fresnel's theory is spoken of as confirmed by the most delicate tests. The reporters add, respecting his "theoretical ideas on the particular kind of undulations which, according to him, constitute light," that "it would be impossible for them to pronounce at present a decided {116} judgment," but that "they have not thought it right to delay any longer making known a work of which the difficulty is attested by the fruitless efforts of the most skilful philosophers, and in which are exhibited in the same brilliant degree, the talent for experiment and the spirit of invention."

[Note 103\9: _Ann. Chim._ tom. xx. p. 343.]

In the meantime, however, a controversy between the theory of undulations and the theory of moveable polarization which M. Biot had proposed with a view of accounting for the colors produced by dipolarizing crystals, had occurred among the French men of science. It is clear that in some main features the two theories coincide; the intervals of interference in the one theory being represented by the intervals of the oscillations in the other. But these intervals in M. Biot's explanations were arbitrary hypotheses, suggested by these very facts themselves; in Fresnel's theory, they were essential parts of the general scheme. M. Biot, indeed, does not appear to have been averse from a coalition; for he allowed[104\9] to Fresnel that "the theory of undulations took the phenomena at a higher point and carried them further." And M. Biot could hardly have dissented from M. Arago's account of the matter, that Fresnel's views "_linked together_"[105\9] the oscillations of moveable polarization. But Fresnel, whose hypothesis was all of one piece, could give up no part of it, although he allowed the usefulness of M. Biot's formulæ. Yet M. Biot's speculations fell in better with the views of the leading mathematicians of Paris. We may consider as evidence of the favor with which they were looked upon, the large space they occupy in the volumes of the Academy for 1811, 1812, 1817, and 1818. In 1812, the entire volume is filled with a memoir of M. Biot's on the subject of moveable polarization. This doctrine also had some advantage in coming early before the world in a didactic form, in his _Traité de Physique_, which was published in 1816, and was the most complete treatise on general physics which had appeared up to that time. In this and others of this author's writings, he expresses facts so entirely in the terms of his own hypothesis, that it is difficult to separate the two. In the sequel M. Arago was the most prominent of M. Biot's opponents; and in his report upon Fresnel's memoir on the colors of crystalline plates, he exposed the weaknesses of the theory of moveable polarization with some severity. The details of this controversy need not occupy us; but we may observe that this may be considered as the last struggle {117} in favor of the theory of emission among mathematicians of eminence. After this crisis of the war, the theory of moveable polarization lost its ground; and the explanations of the undulatory theory, and the calculations belonging to it, being published in the _Annales de Chimie et de Physique_, of which M. Arago was one of the conductors, soon diffused it over Europe.

[Note 104\9: _Ann. Chim._ tom. xvii. p. 251.]

[Note 105\9: "Nouait".]

It was probably in consequence of the delays to which we have referred, in the publication of Fresnel's memoirs, that as late as December, 1826, the Imperial Academy at St Petersburg proposed, as one of their prize-questions for the two following years, this,--"To deliver the optical system of waves from all the objections which have (as it appears) with justice been urged against it, and to apply it to the polarization and double refraction of light." In the programme to this announcement, Fresnel's researches on the subject are not alluded to, though his memoir on diffraction is noticed; they were, therefore, probably not known to the Russian Academy.

Young was always looked upon as a person of marvellous variety of attainments and extent of knowledge; but during his life he hardly held that elevated place among great discoverers which posterity will probably assign him. In 1802, he was constituted Foreign Secretary of the Royal Society, an office which he held during life; in 1827 he was elected one of the eight Foreign Members of the Institute of France; perhaps the greatest honor which men of science usually receive. The fortune of his life in some other respects was of a mingled complexion. His profession of a physician occupied, sufficiently to fetter, without rewarding him; while he was Lecturer at the Royal Institution, he was, in his lectures, too profound to be popular; and his office of Superintendent of the _Nautical Almanac_ subjected him to much minute labor, and many petulant attacks of pamphleteers. On the other hand, he had a leading part in the discovery of the long-sought key to the Egyptian hieroglyphics; and thus the age which was marked by two great discoveries, one in science and one in literature, owed them both in a great measure to him. Dr. Young died in 1829, when he had scarcely completed his fifty-sixth year. Fresnel was snatched from science still more prematurely, dying, in 1827, at the early age of thirty-nine.

We need not say that both these great philosophers possessed, in an eminent degree, the leading characteristics of the discoverer's mind, perfect clearness of view, rich fertility of invention, and intense love of knowledge. We cannot read without great interest a letter of {118} Fresnel to Young,[106\9] in November, 1824: "For a long time that sensibility, or that vanity, which people call love of glory, is much blunted in me. I labor much less to catch the suffrages of the public, than to obtain an inward approval which has always been the sweetest reward of my efforts. Without doubt I have often wanted the spur of vanity to excite me to pursue my researches in moments of disgust and discouragement. But all the compliments which I have received from MM. Arago, De Laplace, or Biot, never gave me so much pleasure as the discovery of a theoretical truth, or the confirmation of a calculation by experiment."

[Note 106\9: I was able to give this, and some other extracts, from the then unedited correspondence of Young and Fresnel, by the kindness of (the Dean of Ely) Professor Peacock, of Trinity College, Cambridge, whose Life of Dr. Young has since been published.]

Though Young and Fresnel were in years the contemporaries of many who are now alive, we must consider ourselves as standing towards them in the relation of posterity. The Epoch of Induction in Optics is past; we have now to trace the Verification and Application of the true theory.

## CHAPTER XIII.

CONFIRMATION AND EXTENSION OF THE UNDULATORY THEORY.

AFTER the undulatory theory had been developed in all its main features, by its great authors, Young and Fresnel, although it bore marks of truth that could hardly be fallacious, there was still here, as in the case of other great theories, a period in which difficulties were to be removed, objections answered, men's minds familiarized

to the new conceptions thus presented to them; and in which, also, it might reasonably be expected that the theory would be extended to facts not at first included in its domain. This period is, indeed, that in which we are living; and we might, perhaps with propriety, avoid the task of speaking of our living contemporaries. But it would be unjust to the theory not to notice some of the remarkable events, characteristic of such a period, which have already occurred; and this may be done very simply. {119}

In the case of this great theory, as in that of gravitation, by far the most remarkable of these confirmatory researches were conducted by the authors of the discovery, especially Fresnel. And in looking at what he conceived and executed for this purpose, we are, it appears to me, strongly reminded of Newton, by the wonderful inventiveness and sagacity with which he devised experiments, and applied to them mathematical reasonings.

1. _Double Refraction of Compressed Glass._--One of these confirmatory experiments was the production of double refraction by the compression of glass. Fresnel observes,[107\9] that though Sir D. Brewster had shown that glass under compression produced colors resembling those which are given by doubly-refracting crystals, "very skilful physicists had not considered those experiments as a sufficient proof of the bifurcation of the light." In the hypothesis of moveable polarization, it is added, there is no apparent connexion between these phenomena of coloration and double refraction; but on Young's theory, that the colors arise from two rays which have traversed the crystal with different velocities, it appears almost unavoidable to admit also a difference of path in the two rays.

[Note 107\9: _Ann. de Chim._ 1822, tom. xx. p. 377.]

"Though," he says, "I had long since adopted this opinion, it did not appear to me so completely demonstrated, that it was right to neglect an experimental verification of it;" and therefore, in 1819, he proceeded to satisfy himself of the fact, by the phenomena of diffraction. The trial left no doubt on the subject; but he still thought it would be interesting actually to produce two images in glass by compression; and by a highly-ingenious combination, calculated to exaggerate the effect of the double refraction, which is very feeble, even when the compression is most intense, he obtained two distinct images. This evidence of the dependence of dipolarizing structure upon a doubly-refracting state of particles, thus excogitated out of the general theory, and verified by trial, may well be considered, as he says, "as a new occasion of proving the infallibility of the principle of interferences."

2. _Circular Polarization._--Fresnel then turned his attention to another set of experiments, related to this indeed, but by a tie so recondite, that nothing less than his clearness and acuteness of view could have detected any connexion. The optical properties of quartz had been perceived to be peculiar, from the period of the discovery {120} of dipolarized colors by MM. Arago and Biot. At the end of the Notice just quoted, Fresnel says,[108\9] "As soon as my occupations permit me, I propose to employ a pile of prisms similar to that which I have described, in order to study the double refraction of the rays which traverse crystals of quartz in the

direction of the axis." He then ventures, without hesitation, to describe beforehand what the phenomena will be. In the _Bulletin des Sciences_[109\9] for December, 1822, it is stated that experiment had confirmed what he had thus announced.

[Note 108\9: _Ann. de Chim._ 1822, tom. xx. p. 382.]

[Note 109\9: Ib. _Ann. de Chim._ 1822, tom. xx. p. 191.]

The phenomena are those which have since been spoken of as _circular polarization_; and the term first occurs in this notice.[110\9] They are very remarkable, both by their resemblances to, and their differences from, the phenomena of _plane-polarized_ light. And the manner in which Fresnel was led to this anticipation of the facts is still more remarkable than the facts themselves. Having ascertained by observation that two differently-polarized rays, totally reflected at the internal surface of glass, suffer different _retardations_ of their undulations, he applied the formulæ which he had obtained for the polarizing effect of reflection to this case. But in this case the formulæ expressed an impossibility; yet as algebraical formulæ, even in such cases, have often some meaning, "I interpreted," he says,[111\9] "in the manner which appeared to me most natural and most probable, what the analysis indicated by this imaginary form;" and by such an interpretation he collected the law of the difference of undulation of the two rays. He was thus able to predict that by two internal reflections in a _rhomb_, or parallelopiped of glass, of a certain form and position, a polarized ray would acquire a circular undulation of its particles; and this constitution of the ray, it appeared, by reasoning further, would show itself by its possessing peculiar properties, partly the same as those of polarized light, and partly different. This extraordinary anticipation was exactly confirmed; and thus the apparently bold and strange guess of the author was fully justified, or at least assented to, even by the most cautious philosophers. "As I cannot appreciate the mathematical evidence for the nature of circular polarization," says Prof. Airy,[112\9] "I shall mention the experimental evidence on which I receive it." The conception has since been universally adopted.

[Note 110\9: Ib. p. 194.]

[Note 111\9: _Bullet. des Sc._ 1823, p. 33.]

[Note 112\9: _Camb. Trans._ vol. iv. p. 81, 1831.]

But Fresnel, having thus obtained circularly-polarized rays, saw {121} that he could account for the phenomena of quartz, already observed by M. Arago, as we have noticed in Chap. ix., by supposing two circularly-polarized rays to pass, with different velocities, along the axis. The curious succession of colors, following each other in right-handed or left-handed circular order, of which we have already spoken, might thus be hypothetically explained.

But was this hypothesis of two circularly-polarized rays, travelling along the axis of such crystals, to be received, merely because it accounted for the phenomena? Fresnel's ingenuity again enabled him to avoid such a defect in theorizing. If there were two such rays, they might be visibly separated[113\9] by the same artifice, of a pile of prisms properly achromatized, which he had used for compressed glass. The

result was, that he did obtain a visible separation of the rays; and this result has since been confirmed by others, for instance. Professor Airy.[114\9] The rays were found to be in all respects identical with the circularly-polarized rays produced by the internal reflections in Fresnel's rhomb. This kind of double refraction gave a hypothetical explanation of the laws which M. Biot had obtained for the phenomena of this class; for example,[115\9] the rule, that the deviation of the plane of polarization of the emergent ray is inversely as the square of the length of an undulation for each kind of rays. And thus the phenomena produced by light passing along the axis of quartz were reduced into complete conformity with the theory.

[Note 113\9: _Bull. des Sc._ 1822, p. 193.]

[Note 114\9: _Cambridge Trans._ iv. p. 80.]

[Note 115\9: _Bull. des Sc._ 1822, p. 197.]

[2nd Ed.] [I believe, however, Fresnel did not deduce the phenomenon from the mathematical formula, without the previous suggestion of experiment. He _observed_ appearances which implied a difference of retardation in the two differently-polarized rays at total reflection; as Sir D. Brewster observed in reflection of metals phenomena having a like character. The general fact being observed, Fresnel used the theory to discover the law of this retardation, and to determine a construction in which, one ray being a quarter of an undulation retarded more than the other, circular polarization would be produced. And this anticipation was verified by the construction of his _rhomb_.

As a still more curious verification of this law, another of Fresnel's experiments may be mentioned. He found the proper angles for a circularly-polarizing glass rhomb on the supposition that there were {122} _four_ internal reflections instead of two; two of the four taking place when the surface of the glass was dry, and two when it was wet. The rhomb was made; and when all the points of reflection were dry, the light was not circularly polarized; when two points were wet, the light was circularly polarized; and when all four were wet, it was not circularly polarized.]

3. _Elliptical Polarization in Quartz._--We now come to one of the few additions to Fresnel's theory which have been shown to be necessary. He had accounted fully for the colors produced by the rays which travel _along the axis_ of quartz crystals; and thus, for the colors and changes of the central spot which is produced when polarized light passes through a transverse plate of such crystals. But this central spot is surrounded by rings of colors. How is the theory to be extended to these?

This extension has been successfully made by Professor Airy.[116\9] His hypothesis is, that as rays passing along the axis of a quartz crystal are circularly polarized, rays which are oblique to the axis are elliptically polarized, the amount of ellipticity depending, in some unknown manner, upon the obliquity; and that each ray is separated by double refraction into two rays polarized elliptically; the one right-handed, the other left-handed. By means of these suppositions, he not only was enabled to account for the simple phenomena of single plates of quartz; but for many most complex and intricate appearances which arise from the superposition of two plates, and which at first sight might appear to defy all attempts to reduce them to

law and symmetry; such as spirals, curves approaching to a square form, curves broken in four places. "I can hardly imagine," he says,[117\9] very naturally, "that any other supposition would represent the phenomena to such extreme accuracy. I am not so much struck with the accounting for the continued dilatation of circles, and the general representation of the forms of spirals, as with the explanations of the minute deviations from symmetry; as when circles become almost square, and crosses are inclined to the plane of polarization. And I believe that any one who shall follow my investigation, and imitate my experiments, will be surprised at their perfect agreement."

[Note 116\9: _Camb. Trans._, iv. p. 83, &c.]

[Note 117\9: _Camb. Trans._, iv. p. 122.]

4. _Differential Equations of Elliptical Polarization._--Although circular and elliptical polarization can be clearly conceived, and their existence, it would seem, irresistibly established by the phenomena, it {123} is extremely difficult to conceive any arrangement of the particles of bodies by which such motions can mechanically be produced; and this difficulty is the greater, because some fluids and some gases impress a circular polarization upon light; in which cases we cannot imagine any definite arrangement of the particles, such as might form the mechanism requisite for the purpose. Accordingly, it does not appear that any one has been able to suggest even a plausible hypothesis on that subject. Yet, even here, something has been done. Professor Mac Cullagh, of Dublin, has discovered that by slightly modifying the _analytical expressions_ resulting from the common case of the propagation of light, we may obtain other expressions which would give rise to such motions as produce circular and elliptical polarization. And though we cannot as yet assign the mechanical interpretation of the language of analysis thus generalized, this generalization brings together and explains by one common numerical supposition, two distinct classes of facts;--a circumstance which, in all cases, entitles an hypothesis to a very favorable consideration.

Mr. Mac Cullagh's assumption consists in adding to the two equations of motion which are expressed by means of second differentials, two other terms involving third differentials in a simple and symmetrical manner. In doing this, he introduces a coefficient, of which the magnitude determines both the amount of rotation of the polarization of a ray passing along the axis, as observed and measured by Biot, and the ellipticity of the polarization of a ray which is oblique to the axis, according to Mr. Airy's theory, of which ellipticity that philosopher also had obtained certain measures. The agreement between the two sets of measures[118\9] thus brought into connexion is such as very strikingly to confirm Mr. Mac Cullagh's hypothesis. It appears probable, too, that the confirmation of this hypothesis involves, although in an obscure and oracular form, a confirmation of the undulatory theory, which is the starting-point of this curious speculation.

[Note 118\9: _Royal I. A. Trans._ 1836.]

5. _Elliptical Polarization of Metals._--The effect of metals upon the light which they reflect, was known from the first to be different from that which transparent

bodies produce. Sir David Brewster, who has recently examined this subject very fully,[119\9] has described the modification thus produced, as _elliptic polarization_. In employing this term, "he seems to have been led," it has been observed,[120\9] "by a {124} desire to avoid as much as possible all reference to theory. The laws which he has obtained, however, belong to elliptically-polarized light in the sense in which the term was introduced by Fresnel." And the identity of the light produced by metallic reflection with the elliptically-polarized light of the wave-theory, is placed beyond all doubt, by an observation of Professor Airy, that the rings of uniaxal crystals, produced by Fresnel's elliptically-polarized light, are exactly the same as those produced by Brewster's metallic light.

[Note 119\9: _Phil. Trans._ 1830.]

[Note 120\9: Lloyd, _Report on Optics_, p. 372. (Brit. Assoc.)]

6. _Newton's Rings by Polarized Light._--Other modifications of the phenomena of thin plates by the use of polarized light, supplied other striking confirmations of the theory. These were in one case the more remarkable, since the result was foreseen by means of a rigorous application of the conception of the vibratory motion of light, and confirmed by experiment. Professor Airy, of Cambridge, was led by his reasonings to see, that if Newton's rings are produced between a lens and a plate of metal, by polarized light, then, up to the polarizing angle, the central spot will be black, and instantly beyond this, it will be white. In a note,[121\9] in which he announced this, he says, "This I anticipated from Fresnel's expressions; it is confirmatory of them, and defies emission." He also predicted that when the rings were produced between two substances of very different refractive powers, the centre would twice pass from black to white and from white to black, by increasing the angle; which anticipation was fulfilled by using a diamond for the higher refraction.[122\9]

[Note 121\9: Addressed to myself, dated May 28, 1831. I ought, however, to notice, that this experiment had been made by M. Arago, fifteen years earlier, and published: though not then recollected by Mr. Airy.]

[Note 122\9: _Camb. Trans._ vol. ii. p. 409.]

7. _Conical Refraction._--In the same manner. Professor Hamilton of Dublin pointed out that according to the Fresnelian doctrine of double refraction, there is a certain direction of a crystal in which a single ray of light will be refracted so as to form a _conical pencil_. For the direction of the refracted ray is determined by a plane which touches the wave surface, the rule being that the ray must pass from the centre of the surface to the point of contact; and though in general this contact gives a single point only, it so happens, from the peculiar inflected form of the wave surface, which has what is called _a cusp_, that in one particular position, the plane can touch the surface in an entire circle. Thus the general rule which assigns the path of {125} the refracted ray, would, in this case, guide it from the centre of the surface to every point in the circumference of the circle, and thus make it a cone. This very curious and unexpected result, which Professor Hamilton thus obtained from the theory, his friend Professor Lloyd verified as an experimental fact. We may notice, also, that Professor Lloyd found the light of the conical pencil to be polarized

according to a law of an unusual kind; but one which was easily seen to be in complete accordance with the theory.

8. _Fringes of Shadows._--The phenomena of the _fringes of shadows_ of small holes and groups of holes, which had been the subject of experiment by Fraunhofer, were at a later period carefully observed in a vast variety of cases by M. Schwerd of Spires, and published in a separate work,[123\9] _Beugungs-erscheinungen_ (Phenomena of Inflection), 1836. In this Treatise, the author has with great industry and skill calculated the integrals which, as we have seen, are requisite in order to trace the consequences of the theory; and the accordance which he finds between these and the varied and brilliant results of observation is throughout exact. "I shall," says he, in the preface,[124\9] "prove by the present Treatise, that all inflection-phenomena, through openings of any form, size, and arrangement, are not only explained by the undulation-theory, but that they can be represented by analytical expressions, determining the intensity of the light in any point whatever." And he justly adds, that the undulation-theory accounts for the phenomena of light, as completely as the theory of gravitation does for the facts of the solar system.

[Note 123\9: _Die Beugungs-erscheinungen, aus dem Fundamental-gesetz der Undulations-Theorie analytisch entwickelt und in Bildern dargestellt_, von F. M. Schwerd. Mannheim, 1835.]

[Note 124\9: Dated Speyer, Aug. 1835.]

9. _Objections to the Theory._--We have hitherto mentioned only cases in which the undulatory theory was either entirely successful in explaining the facts, or at least hypothetically consistent with them and with itself. But other objections were started, and some difficulties were long considered as very embarrassing. Objections were made to the theory by some English experimenters, as Mr. Potter, Mr. Barton, and others. These appeared in scientific journals, and were afterwards answered in similar publications. The objections depended partly on the measure of the _intensity_ of light in the different points of the phenomena (a datum which it is very difficult to obtain with accuracy {126} by experiment), and partly on misconceptions of the theory; and I believe there are none of them which would now be insisted on.

We may mention, also, another difficulty, which it was the habit of the opponents of the theory to urge as a reproach against it, long after it had been satisfactorily explained: I mean the _half-undulation_ which Young and Fresnel had found it necessary, in some cases, to assume as gained or lost by one of the rays. Though they and their followers could not analyse the mechanism of reflection with sufficient exactness to trace out all the circumstances, it was not difficult to see, upon Fresnel's principles, that reflection from the interior and exterior surface of glass must be of opposite kinds, which might be expressed by supposing one of these rays to lose half an undulation. And thus there came into view a justification of the step which had originally been taken upon empirical grounds alone.

10. _Dispersion, on the Undulatory Theory._--A difficulty of another kind occasioned a more serious and protracted embarrassment to the cultivators of this theory. This was the apparent impossibility of accounting, on the theory, for the

prismatic dispersion of color. For it had been shown by Newton that the amount of refraction is different for every color; and the amount of refraction depends on the velocity with which light is propagated. Yet the theory suggested no reason why the velocity should be different for different colors: for, by mathematical calculation, vibrations of all degrees of rapidity (in which alone colors differ) are propagated with the same speed. Nor does analogy lead us to expect this variety. There is no such difference between quick and slow waves of air. The sounds of the deepest and the highest bells of a peal are heard at any distance in the same order. Here, therefore, the theory was at fault.

But this defect was far from being a fatal one. For though the theory did not explain, it did not contradict, dispersion. The suppositions on which the calculations had been conducted, and the analogy of sound, were obviously in no small degree precarious. The velocity of propagation might differ for different rates of undulation, in virtue of many causes which would not affect the general theoretical results.

Many such hypothetical causes were suggested by various eminent mathematicians, as solutions of this conspicuous difficulty. But without dwelling upon these conjectures, it may suffice to notice that hypothesis upon which the attention of mathematicians was soon concentrated. This was the _hypothesis of finite intervals_ between the {127} particles of the ether. The length of one of those undulations which produce light, is a very small quantity, its mean value being 1⁄50,000th of an inch; but in the previous investigations of the consequences of the theory, it had been assumed that the distance from each other, of the particles of the ether, which, by their attractions or repulsions, caused the undulations to be propagated, is indefinitely less than this small quantity;--so that its amount might be neglected in the cases in which the length of the undulation was one of the quantities which determined the result. But this assumption was made arbitrarily, as a step of simplification, and because it was imagined that, in this way, a nearer approach was made to the case of a continuous fluid ether, which the supposition of distinct particles imperfectly represented. It was still free for mathematicians to proceed upon the opposite assumption, of particles of which the distances were finite, either as a mathematical basis of calculation, or as a physical hypothesis; and it remained to be seen if, when this was done, the velocity of light would still be the same for different lengths of undulation, that is, for different colors. M. Cauchy, calculating, upon the most general principles, the motion of such a collection of particles as would form an elastic medium, obtained results which included the new extension of the previous hypothesis. Professor Powell, of Oxford, applied himself to reduce to calculation, and to compare with experiment, the result of these researches. And it appeared that, on M. Cauchy's principles, a variation in the velocity of light is produced by a variation in the length of the wave, provided that the interval between the molecules of the ether bears a sensible ratio to the length of an undulation.[125\9] Professor Powell obtained also, from the general expressions, a formula expressing the relation between the refractive index of a ray, and the length of a wave, or the color of light.[126\9] It then became his task to ascertain whether this relation obtained experimentally; and he found a very close agreement between the numbers which resulted from the formula and those observed by Fraunhofer, for

ten different kinds of media, namely, certain glasses and fluids.[127\9] To these he afterwards added ten other cases of crystals observed by M. Rudberg.[128\9] Mr. Kelland, of Cambridge, also calculated, in a manner somewhat different, the results of the same hypothesis of finite intervals;[129\9] and, obtaining {128} formulæ not exactly the same as Professor Powell, found also an agreement between these and Fraunhofer's observations.

[Note 125\9: _Phil. Mag._ vol. vi. p. 266.]

[Note 126\9: Ib. vol. vii. 1835, p. 266.]

[Note 127\9: _Phil. Trans._ 1835, p. 249.]

[Note 128\9: Ib. 1836, p. 17.]

[Note 129\9: _Camb. Trans._ vol. vi. p. 153.]

It may be observed, that the refractive indices observed and employed in these comparisons, were not those determined by the color of the ray, which is not capable of exact identification, but those more accurate measures which Fraunhofer was enabled to make, in consequence of having detected in the spectrum the black lines which he called B, C, D, E, F, G, H. The agreement between the theoretical formulæ and the observed numbers is remarkable, throughout all the series of comparisons of which we have spoken. Yet we must at present hesitate to pronounce upon the hypothesis of finite intervals, as proved by these calculations; for though this hypothesis has given results agreeing so closely with experiment, it is not yet clear that other hypotheses may not produce an equal agreement. By the nature of the case, there must be a certain gradation and continuity in the succession of colors in the spectrum, and hence, any supposition which will account for the general fact of the whole dispersion, may possibly account for the amount of the intermediate dispersions, because these must be interpolations between the extremes. The result of this hypothetical calculation, however, shows very satisfactorily that there is not, in the fact of dispersion, anything which is at all formidable to the undulatory theory.

11. _Conclusion._--There are several other of the more recondite points of the theory which may be considered as, at present, too undecided to allow us to speak historically of the discussions which they have occasioned.[130\9] For example, it was conceived, for some time, that the vibrations of polarized light are perpendicular to the plane of polarization. But this assumption was not an essential part of the theory; and all the phenomena would equally allow us to suppose the vibrations to be in the polarization plane; the main requisite being, that light polarized in planes at right angles to each other, should also have the vibrations at right angles. Accordingly, for some time, this point was left undecided by Young and Fresnel, and, more recently, some mathematicians have come to the opinion that ether vibrates in the plane of polarization. The theory of transverse vibrations is equally stable, whichever supposition may be finally confirmed.

[Note 130\9: For on account of these, see Professor Lloyd's _Report on Physical Optics_. (Brit. Assoc. Report, 1834.)]

We may speak, in the same manner, of the suppositions which, from {129} the time of Young and Fresnel, the cultivators of this theory have been led to make respecting the mechanical constitution of the ether, and the forces by which transverse vibrations are produced. It was natural that various difficulties should arise upon such points, for transverse vibrations had not previously been made the subject of mechanical calculation, and the forces which occasion them must act in a different manner from those which were previously contemplated. Still, we may venture to say, without entering into these discussions, that it has appeared, from all the mathematical reasonings which have been pursued, that there is not, in the conception of transverse vibrations, anything inconsistent either with the principles of mechanics, or with the best general views which we can form, of the forces by which the universe is held together.

I willingly speak as briefly as the nature of my undertaking allows, of those points of the undulatory theory which are still under deliberation among mathematicians. With respect to these, an intimate acquaintance with mathematics and physics is necessary to enable any one to understand the steps which are made from day to day; and still higher philosophical qualifications would be requisite in order to pronounce a judgment upon them. I shall, therefore, conclude this survey by remarking the highly promising condition of this great department of science, in respect to the character of its cultivators. Nothing less than profound thought and great mathematical skill can enable any one to deal with this theory, in any way likely to promote the interests of science. But there appears, in the horizon of the scientific world, a considerable class of young mathematicians, who are already bringing to these investigations the requisite talents and zeal; and who, having acquired their knowledge of the theory since the time when its acceptation was doubtful, possess, without effort, that singleness and decision of view as to its fundamental doctrines, which it is difficult for those to attain whose minds have had to go through the hesitation, struggle, and balance of the epoch of the establishment of the theory. In the hands of this new generation, it is reasonable to suppose the Analytical Mechanics of light will be improved as much as the Analytical Mechanics of the solar system was by the successors of Newton. We have already had to notice many of this younger race of undulationists. For besides MM. Cauchy, Poisson, and Ampère, M. Lamé has been more recently following these researches in France.[131\9] In {130} Belgium, M. Quetelet has given great attention to them; and, in our own country, Sir William Hamilton, and Professor Lloyd, of Dublin, have been followed by Mr. Mac Cullagh. Professor Powell, of Oxford, has continued his researches with unremitting industry; and, at Cambridge, Professor Airy, who did much for the establishment and diffusion of the theory before he was removed to the post of Astronomer Royal, at Greenwich, has had the satisfaction to see his labors continued by others, even to the most recent time; for Mr. Kelland,[132\9] whom we have already mentioned, and Mr. Archibald Smith,[133\9] the two persons who, in 1834 and 1836, received the highest mathematical honors which that university can bestow, have both of them published investigations respecting the undulatory theory.

[Note 131\9: Prof. Lloyd's _Report_, p. 392.]

[Note 132\9: _On the Dispersion of Light, as explained by the Hypothesis of Finite Intervals._ Camb. Trans. vol. vi. p. 153.]

[Note 133\9: _Investigation of the Equation to Fresnel's Wave Surface_, ib. p. 85. See also, in the same volume, _Mathematical Considerations on the Problem of the Rainbow_, showing it to belong to Physical Optics, by R. Potter, Esq., of Queen's College.]

We may be permitted to add, as a reflection obviously suggested by these facts, that the cause of the progress of science is incalculably benefited by the existence of a body of men, trained and stimulated to the study of the higher mathematics, such as exist in the British universities, who are thus prepared, when an abstruse and sublime theory comes before the world with all the characters of truth, to appreciate its evidence, to take steady hold of its principles, to pursue its calculations, and thus to convert into a portion of the permanent treasure and inheritance of the civilized world, discoveries which might otherwise expire with the great geniuses who produced them, and be lost for ages, as, in former times, great scientific discoveries have sometimes been.

The reader who is acquainted with the history of recent optical discovery, will see that we have omitted much which has justly excited admiration; as, for example, the phenomena produced by glass under heat or pressure, noticed by MM. Lobeck, and Biot, and Brewster, and many most curious properties of particular minerals. We have omitted, too, all notice of the phenomena and laws of the absorption of light, which hitherto stand unconnected with the theory. But in this we have not materially deviated from our main design; for our end, in what we have done, has been to trace the advances of Optics {131} towards perfection as a theory; and this task we have now nearly executed as far as our abilities allow.

We have been desirous of showing that the _type_ of this progress, in the histories of the two great sciences, Physical Astronomy and Physical Optics, is the same. In both we have many _Laws of Phenomena_ detected and accumulated by acute and inventive men; we have _Preludial_ guesses which touch the true theory, but which remain for a time imperfect, undeveloped, unconfirmed: finally we have the _Epoch_ when this true theory, clearly apprehended by great philosophical geniuses, is recommended by its fully explaining what it was first meant to explain, and confirmed by its explaining what it was not meant to explain. We have then its _Progress_ struggling for a little while with adverse prepossessions and difficulties; finally overcoming all these, and moving onwards, while its triumphal procession is joined by all the younger and more vigorous men of science.

It would, perhaps, be too fanciful to attempt to establish a parallelism between the prominent persons who figure in these two histories. If we were to do this, we must consider Huyghens and Hooke as standing in the place of Copernicus, since, like him, they announced the true theory, but left it to a future age to give it development and mechanical confirmation; Malus and Brewster, grouping them together, correspond to Tycho Brahe and Kepler, laborious in accumulating observations, inventive and happy in discovering laws of phenomena; and Young and Fresnel combined, make up

the Newton of optical science.

[2nd Ed.] [In the _Report on Physical Optics_, (_Brit. Ass. Reports_, 1834,) by Prof. Lloyd, the progress of the mathematical theory after Fresnel's labors is stated more distinctly than I have stated it, to the following effect. Ampère, in 1828, proved Fresnel's mathematical results directly, which Fresnel had only proved indirectly, and derived from his proof Fresnel's beautiful geometrical construction. Prof. Mac Cullagh not long after gave a concise demonstration of the same theorem, and of the other principal points of Fresnel's theory. He represents the elastic force by means of an ellipsoid whose axes are inversely proportional to those of Fresnel's generating ellipsoid, and deduces Fresnel's construction geometrically. In the third Supplement to his _Essay on the Theory of Systems of Rays_ (_Trans. R. I. Acad._ vol. xvii.), Sir W. Hamilton has presented that portion of Fresnel's theory which relates to the fundamental problem of the determination of the velocity and polarization of a plane wave, in a very elegant and analytical form. This he does by means of what he calls the {132} _characteristic function_ of the optical system to which the problem belongs. From this function is deduced the _surface of wave-slowness_ of the medium; and by means of this surface, the direction of the rays refracted into the medium. From this construction also Sir W. Hamilton was led to the anticipation of _conical refraction_, mentioned above.

The investigations of MM. Cauchy and Lamé refer to the laws by which the particles of the ether act upon each other and upon the particles of other bodies;--a field of speculation which appears to me not yet ripe for the final operations of the analyst.

Among the mathematicians who have supplied defects in Fresnel's reasoning on this subject, I may mention Mr. Tovey, who treated it in several papers in the _Philosophical Magazine_ (1837-40). Mr. Tovey's early death must be deemed a loss to mathematical science.

Besides investigating the motion of symmetrical systems of particles which may be supposed to correspond to biaxal crystals, Mr. Tovey considered the case of unsymmetrical systems, and found that the undulations propagated would, in the general case, be elliptical; and that in a particular case, circular undulations would take place, such as are propagated along the axis of quartz. It appears to me, however, that he has not given a definite meaning to those limitations of his general hypothesis which conduct him to this result. Perhaps if the hypothetical conditions of this result were traced into detail, they would be found to reside in a _screw-like_ arrangement of the elementary particles, in some degree such as crystals of quartz themselves exhibit in their forms, when they have plagihedral faces at both ends.

Such crystals of quartz are, some like a right-handed and some like a left-handed screw; and, as Sir John Herschel discovered, the circular polarization is right-handed or left-handed according as the plagihedral form is so. In Mr. Tovey's hypothetical investigation it does not appear upon what part of the hypothesis this difference of right and left-handed depends. The definition of this part of the hypothesis is a very desirable step.

When crystals of Quartz are right-handed at one end, they are right-handed at the other end: but there is a different kind of plagihedral form, which occurs in some other crystals, for instance, in Apatite: in these the plagihedral faces are right-handed at the one extremity and left-handed at the other. For the sake of distinction, we may call the former _homologous_ plagihedral faces, since, at both ends, they have the same name; and the latter _heterologous_ plagihedral faces. {133}

The homologous plagihedral faces of Quartz crystals are accompanied by homologous circular polarization of the same name. I do not know that heterologous circular polarization has been observed in any crystal, but it has been discovered by Dr. Faraday to occur in glass, &c., when subjected to powerful magnetic action.

Perhaps it was presumptuous in me to attempt to draw such comparisons, especially with regard to living persons, as I have done in the preceding pages of this Book. Having published this passage, however, I shall not now suppress it. But I may observe that the immense number and variety of the beautiful optical discoveries which we owe to Sir David Brewster makes the comparison in his case a very imperfect representation of his triumphs over nature; and that, besides his place in the history of the Theory of Optics, he must hold a most eminent position in the history of Optical Crystallography, whenever the discovery of a True Optical Theory of Crystals supplies us with the _Epoch_ to which his labors in this field form so rich a _Prelude_. I cordially assent to the expression employed by Mr. Airy in the _Phil. Trans._ for 1840, in which he speaks of Sir David Brewster as "the Father of Modern Experimental Optics."]

{{135}} BOOK X.

_SECONDARY MECHANICAL SCIENCES._ (CONTINUED.)

HISTORY OF THERMOTICS AND ATMOLOGY.

Et primum faciunt ignem se vortere in auras Aëris; hinc imbrem gigni terramque creari Ex imbri; retroque a terrâ cuncta revorti, Humorem primum, post aëra deinde calorem; Nec cessare hæc inter se mutare, meare, De cœlo ad terram de terrâ ad sidera mundi. LUCRETIUS, i. 783.

Water, and Air, and Fire, alternate run Their endless circle, multiform, yet one. For, moulded by the fervor's latent beams, Solids flow loose, and fluids flash to steams, And elemental flame, with secret force, Pursues through earth, air, sky, its stated course.

{{137}} INTRODUCTION.

_Of Thermotics and Atmology._

I EMPLOY the term _Thermotics_ to include all the doctrines respecting Heat, which have hitherto been established on proper scientific grounds. Our survey of the history of this branch of science must be more rapid and less detailed than it has been in those subjects of which we have hitherto treated: for our knowledge is, in this case, more vague and uncertain than in the others, and has made less progress towards a general and certain theory. Still, the narrative is too important and too

instructive to be passed over.

The distinction of Formal Thermotics and Physical Thermotics,--of the discovery of the mere Laws of Phenomena, and the discovery of their causes,--is applicable here, as in other departments of our knowledge. But we cannot exhibit, in any prominent manner, the latter division of the science now before us; since no general theory of heat has yet been propounded, which affords the means of calculating the circumstances of the phenomena of conduction, radiation, expansion, and change of solid, liquid, and gaseous form. Still, on each of these subjects there have been proposed, and extensively assented to, certain general views, each of which explains its appropriate class of phenomena; and, in some cases, these principles have been clothed in precise and mathematical conditions, and thus made bases of calculation.

These principles, thus possessing a generality of a limited kind, connecting several observed laws of phenomena, but yet not connecting all the observed classes of facts which relate to heat, will require our separate attention. They may be described as the Doctrine of Conduction, the Doctrine of Radiation, the Doctrine of Specific Heat, and the Doctrine of Latent Heat; and these, and similar doctrines respecting heat, make up the science which we may call _Thermotics proper_.

But besides these collections of principles which regard heat by itself, the relations of heat and moisture give rise to another and important collection of laws and principles, which I shall treat of in connexion with Thermotics, and shall term _Atmology_, borrowing {138} the term from the Greek word (ἄτμος,) which signifies _vapor_. The _Atmosphere_ was so named by the Greeks, as being a sphere of vapor; and, undoubtedly, the most general and important of the phenomena which take place in the air, by which the earth is surrounded, are those in which water, of one _consistence_ or other (ice, water, or steam,) is concerned. The knowledge which relates to what takes place in the atmosphere has been called _Meteorology_, in its collective form: but such knowledge is, in fact, composed of parts of many different sciences. And it is useful for our purpose to consider separately those portions of Meteorology which have reference to the laws of aqueous vapor, and these we may include under the term Atmology.

The instruments which have been invented for the purpose of measuring the moisture of the air, that is, the quantity of vapor which exists in it, have been termed _Hygrometers_; and the doctrines on which these instruments depend, and to which they lead, have been called _Hygrometry_; but this term has not been used in quite so extensive a sense as that which we intend to affix to _Atmology_.

In treating of Thermotics, we shall first describe the earlier progress of men's views concerning Conduction, Radiation, and the like, and shall then speak of the more recent corrections and extensions, by which they have been brought nearer to theoretical generality.

{{139}} THERMOTICS PROPER.

## CHAPTER I.

## THE DOCTRINES OF CONDUCTION AND RADIATION.

### _Section_ 1.--_Introduction of the Doctrine of Conduction._

BY _conduction_ is meant the propagation of heat from one part to another of a continuous body; or from one body to another in contact with it; as when one end of a poker stuck in the fire heats the other end, or when this end heats the hand which takes hold of it. By _radiation_ is meant the diffusion of heat from the surface of a body to points not in contact. It is clear in both these cases, that, in proportion as the hot portion is hotter, it produces a greater effect in warming the cooler portion; that is, it _communicates more Heat_ to it, if _Heat_ be the abstract conception of which this effect is the measure. The simplest rule which can be proposed is, that the heat thus communicated in a given instant is proportional to the excess of the heat of the hot body over that of the contiguous bodies; there are no obvious phenomena which contradict the supposition that this is the true law; and it was thence assumed by Newton as the true law for radiation and by other writers for conduction. This assumption was confirmed approximately, and afterwards corrected, for the case of Radiation; in its application to Conduction, it has been made the basis of calculation up to the present time. We may observe that this statement takes for granted that we have attained to a measure of heat (or of _temperature_, as heat thus measured is termed), corresponding to the law thus assumed; and, in fact, as we shall have occasion to explain in speaking of the _measures_ of sensible qualities, {140} the thermometrical scale of heat according to the expansion of liquids (which is the measure of temperature here adopted), was constructed with a reference to Newton's law of radiation of heat; and thus the law is necessarily consistent with the scale.

In any case in which the parts of a body are unequally hot, the temperature will vary _continuously_ in passing from one part of the body to another; thus, a long bar of iron, of which one end is kept red hot, will exhibit a gradual diminution of temperature at successive points, proceeding to the other end. The law of temperature of the parts of such a bar might be expressed by the ordinates of a _curve_ which should run alongside the bar. And, in order to trace mathematically the consequences of the assumed law, some of those processes would be necessary, by which mathematicians are enabled to deal with the properties of curves; as the method of infinitesimals, or the differential calculus; and the truth or falsehood of the law would be determined, according to the usual rules of inductive science, by a comparison of results so deduced from the principle, with the observed phenomena.

It was easily perceived that this comparison was the task which physical inquirers had to perform; but the execution of it was delayed for some time; partly, perhaps, because the mathematical process presented some difficulties. Even in a case so simple as that above mentioned, of a linear bar with a stationary temperature at one end, _partial differentials_ entered; for there were three variable quantities, the time, as well as the place of each point and its temperature. And at first, another scruple occurred to M. Biot when, about 1804, he undertook this problem.[1\10] "A difficulty," says Laplace,[2\10] in 1809, "presents itself, which has not yet been solved. The quantities of heat received and communicated in an instant (by any point of the bar) must be infinitely small quantities of the same order as the excess of the

heat of a slice of the body over that of the contiguous slice; therefore the _excess_ of the heat received by any slice over the heat communicated, is an infinitely small quantity of the second order; and the accumulation in a finite time (which depends on this excess) cannot be finite." I conceive that this difficulty arises entirely from an arbitrary and unnecessary assumption concerning the relation of the infinitesimal parts of the body. Laplace resolved the difficulty by further reasoning founded upon the same assumption which occasioned {141} it; but Fourier, who was the most distinguished of the cultivators of this mathematical doctrine of conduction, follows a course of reasoning in which the difficulty does not present itself. Indeed it is stated by Laplace, in the Memoir above quoted,[3\10] that Fourier had already obtained the true fundamental equations by views of his own.

[Note 1\10: Biot, _Traité de Phys._ iv. p. 669.]

[Note 2\10: Laplace, _Mém. Inst._ for 1809, p. 332.]

[Note 3\10: Laplace, _Mém. Inst._ for 1809, p. 538.]

The remaining part of the history of the doctrine of conduction is principally the history of Fourier's labors. Attention having been drawn to the subject, as we have mentioned, the French Institute, in January, 1810, proposed, as their prize question, "To give the mathematical theory of the laws of the propagation of heat, and to compare this theory with exact observations." Fourier's Memoir (the sequel of one delivered in 1807,) was sent in September, 1811; and the prize (3000 francs) adjudged to it in 1812. In consequence of the political confusion which prevailed in France, or of other causes, these important Memoirs were not published by the Academy till 1824; but extracts had been printed in the _Bulletin des Sciences_ in 1808, and in the _Annales de Chimie_ in 1816; and Poisson and M. Cauchy had consulted the manuscript itself.

It is not my purpose to give, in this place,[4\10] an account of the analytical processes by which Fourier obtained his results. The skill displayed in these Memoirs is such as to make them an object of just admiration to mathematicians; but they consist entirely of deductions from the fundamental principle which I have noticed,--that the quantity of heat conducted from a hotter to a colder point is proportional to the excess of heat, modified by the _conductivity_, or conducting power of each substance. The equations which flow from this principle assume nearly the same forms as those which occur in the most general problems of hydrodynamics. Besides Fourier's solution, Laplace, Poisson, and M. Cauchy have also exercised their great analytical skill in the management of these formulæ. We shall briefly speak of the comparison of the results of these reasonings with experiment, and notice some other consequences to which they lead. But before we can do this, we must pay some attention to the subject of radiation. {142}

[Note 4\10: I have given an account of Fourier's mathematical results in the _Reports of the British Association_ for 1835.]

_Sect._ 2.--_Introduction of the Doctrine of Radiation._

A HOT body, as a mass of incandescent iron, emits heat, as we perceive by our senses when we approach it; and by this emission of heat the hot body cools down. The first step in our systematic knowledge of the subject was made in the _Principia_. "It was in the destiny of that great work," says Fourier, "to exhibit, or at least to indicate, the causes of the principal phenomena of the universe." Newton assumed, as we have already said, that the rate at which a body cools, that is, parts with its heat to surrounding bodies, is proportional to its heat; and on this assumption he rested the verification of his scale of temperatures. It is an easy deduction from this law, that if times of cooling be taken in arithmetical progression, the heat will decrease in geometrical progression. Kraft, and after him Richman, tried to verify this law by direct experiments on the cooling of vessels of warm water; and from these experiments, which have since been repeated by others, it appears that for differences of temperature which do not exceed 50 degrees (boiling water being 100), this geometrical progression represents, with tolerable (but not with complete) accuracy, the process of cooling.

This principle of radiation, like that of conduction, required to be followed out by mathematical reasoning. But it required also to be corrected in the first place, for it was easily seen that the rate of cooling depended, not on the absolute temperature of the body, but on the excess of its temperature above the surrounding objects to which it communicated its heat in cooling. And philosophers were naturally led to endeavor to explain or illustrate this process by some physical notions. Lambert in 1765 published[5\10] an _Essay on the Force of Heat_, in which he assimilates the communication of heat to the flow of a fluid out of one vessel into another by an excess of pressure; and mathematically deduces the laws of the process on this ground. But some additional facts suggested a different view of the subject. It was found that heat is propagated by radiation according to straight lines, like light; and that it is, as light is, capable of being reflected by mirrors, and thus brought to a focus of intenser action. In this manner the radiative effect of a body could be more precisely traced. A fact, however, came under notice, which, at first sight, appeared to {143} offer some difficulty. It appeared that cold was reflected no less than heat. A mass of ice, when its effect was concentrated on a thermometer by a system of mirrors, made the thermometer fall, just as a vessel of hot water placed in a similar situation made it rise. Was cold, then, to be supposed a real substance, no less than heat?

[Note 5\10: _Act. Helvet._ tom. ii. p. 172.]

The solution of this and similar difficulties was given by Pierre Prevost, professor at Geneva, whose theory of radiant heat was proposed about 1790. According to this theory, heat, or _caloric_, is constantly radiating from every point of the surface of all bodies in straight lines; and it radiates the more copiously, the greater is the quantity of heat which the body contains. Hence a constant exchange of heat is going on among neighboring bodies; and a body grows hotter or colder, according as it receives more caloric than it emits, or the contrary. And thus a body is cooled by rectilinear rays from a cold body, because along these paths it sends rays of heat in greater abundance than those which return the same way. This _theory of exchanges_

is simple and satisfactory, and was soon generally adopted; but we must consider it rather as the simplest mode of expressing the dependence of the communication of heat on the excess of temperature, than as a proposition of which the physical truth is clearly established.

A number of curious researches on the effect of the different kinds of surface of the heating and of the heated body, were made by Leslie and others. On these I shall not dwell; only observing that the relative amount of this radiative and receptive energy may be expressed by numbers, for each kind of surface; and that we shall have occasion to speak of it under the term _exterior conductivity_; it is thus distinguished from _interior conductivity_, which is the relative rate at which heat is conducted in the interior of bodies.[6\10]

[Note 6\10: The term employed by Fourier, _conductibility_ or _conducibility_, suggests expressions altogether absurd, as if the bodies could be called _conductible_, or _conducible_, with respect to heat: I have therefore ventured upon a slight alteration of the word, and have used the abstract term which analogy would suggest, if we suppose bodies to be _conductive_ in this respect.]

_Sect._ 3.--_Verifications of the Doctrines of Conduction and Radiation._

THE interior and exterior conductivity of bodies are numbers, which enter as elements, or _coefficients_, into the mathematical calculations founded on the doctrines of conduction and radiation. These {144} coefficients are to be determined for each case by appropriate experiments: when the experimenters had obtained these data, as well as the mathematical solutions of the problems, they could test the truth of their fundamental principles by a comparison of the theoretical and actual results in properly-selected cases. This was done for the law of conduction in the simple cases of metallic bars heated at one end, by M. Biot,[7\10] and the accordance with experiment was sufficiently close. In the more complex cases of conduction which Fourier considered, it was less easy to devise a satisfactory mode of comparison. But some rather curious relations which he demonstrated to exist among the temperatures at different points of an _armille_, or ring, afforded a good criterion of the value of the calculations, and confirmed their correctness.[8\10]

[Note 7\10: _Tr. de Phys._ iv. 671.]

[Note 8\10: _Mém. Inst._ 1819, p. 192, published 1824.]

We may therefore presume these doctrines of radiation and conduction to be sufficiently established; and we may consider their application to any remarkable case to be a portion of the history of science. We proceed to some such applications.

_Sect._ 4.--_The Geological and Cosmological Application of Thermotics._

BY far the most important case to which conclusions from these doctrines have been applied, is that of the globe of the earth, and of those laws of climate to which the modifications of temperature give rise; and in this way we are led to inferences concerning other parts of the universe. If we had any means of observing these terrestrial and cosmical phenomena to a sufficient extent, they would be valuable facts on which we might erect our theories; and they would thus form part, not of the

corollaries, but of the foundations of our doctrine of heat. In such a case, the laws of the propagation of heat, as discovered from experiments on smaller bodies, would serve to explain these phenomena of the universe, just as the laws of motion explain the celestial movements. But since we are almost entirely without any definite indications of the condition of the other bodies in the solar system as to heat; and since, even with regard to the earth, we know only the temperature of the parts at or very near the surface, our knowledge of the part which heat plays in the earth and the heavens must be in a great measure, not a generalization of observed facts, but a deduction from theoretical principles. Still, such knowledge, whether obtained {145} from observation or from theory, must possess great interest and importance. The doctrines of this kind which we have to notice refer principally to the effect of the sun's heat on the earth, the laws of climate,--the thermotical condition of the interior of the earth,--and that of the planetary spaces.

1. _Effect of Solar Heat on the Earth._--That the sun's heat passes into the interior of the earth in a variable manner, depending upon the succession of days and nights, summers and winters, is an obvious consequence of our first notions on this subject. The mode in which it proceeds into the interior, after descending below the surface, remained to be gathered, either from the phenomena, or from reasoning. Both methods were employed.[9\10] Saussure endeavored to trace its course by digging, in 1785, and thus found that at the depth of about thirty-one feet, the annual variation of temperature is about 1⁄12th what it is at the surface. Leslie adopted a better method, sinking the bulbs of thermometers deep in the earth, while their stems appeared above the surface. In 1813, '16, and '17, he observed thus the temperatures at the depths of one, two, four, and eight feet, at Abbotshall, in Fifeshire. The results showed that the extreme annual oscillations of the temperature diminish as we descend. At the depth of one foot, the yearly range of oscillation was twenty-five degrees (Fahrenheit); at two feet it was twenty degrees; at four feet it was fifteen degrees; at eight feet it was only nine degrees and a half. And the time at which the heat was greatest was later and later in proceeding to the lower points. At one foot, the maximum and minimum were three weeks after the solstice of summer and of winter; at two feet, they were four or five weeks; at four feet, they were two months; and at eight feet, three months. The mean temperature of all the thermometers was nearly the same. Similar results were obtained by Ott at Zurich in 1762, and by Herrenschneider at Strasburg in 1821, '2, '3.[10\10]

[Note 9\10: Leslie, art. _Climate_, Supp. _Enc. Brit._ 179.]

[Note 10\10: Pouillet, _Météorol._ t. ii. p. 643.]

These results had already been explained by Fourier's theory of conduction. He had shown[11\10] that when the surface of a sphere is affected by a periodical heat, certain alternations of heat travel uniformly into the interior, but that the extent of the alternation diminishes in geometrical progression in this descent. This conclusion applies to the effect of days and years on the temperature of the earth, and shows that such facts as those observed by Leslie are both exemplifications of {146} the general circumstances of the earth, and are perfectly in accordance with the principles on which Fourier's theory rests.

[Note 11\10: _Mém. Inst._ for 1821 (published 1826), p. 162.]

2. _Climate._--The term _climate_, which means _inclination_, was applied by the ancients to denote that inclination of the axis of the terrestrial sphere from which result the inequalities of days in different latitudes. This inequality is obviously connected also with a difference of thermotical condition. Places near the poles are colder, on the whole, than places near the equator. It was a natural object of curiosity to determine the law of this variation.

Such a determination, however, involves many difficulties, and the settlement of several preliminary points. How is the temperature of any place to be estimated? and if we reply, by its _mean_ temperature, how are we to learn this mean? The answers to such questions require very multiplied observations, exact instruments, and judicious generalizations; and cannot be given here. But certain first approximations may be obtained without much difficulty; for instance, the mean temperature of any place may be taken to be the temperature of deep springs, which is probably identical with the temperature of the soil below the reach of the annual oscillations. Proceeding on such facts, Mayer found that the mean temperature of any place was nearly proportional to the square of the cosine of the latitude. This, as a law of phenomena, has since been found to require considerable correction; and it appears that the mean temperature does not depend on the latitude alone, but on the distribution of land and water, and on other causes. M. de Humboldt has expressed these deviations[12\10] by his map of _isothermal lines_, and Sir D. Brewster has endeavored to reduce them to a law by assuming two _poles of maximum cold_.

[Note 12\10: British Assoc. 1833. Prof. Forbes's _Report on Meteorology_, p. 215.]

The expression which Fourier finds[13\10] for the distribution of heat in a homogeneous sphere, is not immediately comparable with Mayer's empirical formula, being obtained on a certain hypothesis, namely, that the equator is kept constantly at a fixed temperature. But there is still a general agreement; for, according to the theory, there is a diminution of heat in proceeding from the equator to the poles in such a case; the heat is propagated from the equator and the neighboring parts, and radiates out from the poles into the surrounding space. And thus, in the case of the earth, the solar heat enters in the tropical {147} parts, and constantly flows towards the polar regions, by which it is emitted into the planetary spaces.

[Note 13\10: Fourier. _Mém. Inst._ tom. v. p. 173.]

Climate is affected by many thermotic influences, besides the conduction and radiation of the solid mass of the earth. The atmosphere, for example, produces upon terrestrial temperatures effects which it is easy to see are very great; but these it is not yet in the power of calculation to appreciate;[14\10] and it is clear that they depend upon other properties of air besides its power to transmit heat. We must therefore dismiss them, at least for the present.

[Note 14\10: _Mém. Inst._ tom. vii. p. 584]

3. _Temperature of the Interior of the Earth._--The question of the temperature of the interior of the earth has excited great interest, in consequence of its bearing on

other branches of knowledge. The various facts which have been supposed to indicate the fluidity of the central parts of the terrestrial globe, belong, in general, to geological science; but so far as they require the light of thermotical calculations in order to be rightly reasoned upon, they properly come under our notice here.

The principal problem of this kind which has been treated of is this:--If in the globe of the earth there be a certain original heat, resulting from its earlier condition, and independent of the action of the sun, to what results will this give rise? and how far do the observed temperatures of points below the surface lead us to such a supposition? It has, for instance, been asserted, that in many parts of the world the temperature, as observed in mines and other excavations, increases in descending, at the rate of one degree (centesimal) in about forty yards. What inference does this justify?

The answer to this question was given by Fourier and by Laplace. The former mathematician had already considered the problem of the cooling of a large sphere, in his Memoirs of 1807, 1809, and 1811. These, however, lay unpublished in the archives of the Institute for many years. But in 1820, when the accumulation of observations which indicated an increase of the temperature of the earth as we descend, had drawn observation to the subject, Fourier gave, in the Bulletin of the Philomathic Society,[15\10] a summary of his results, as far as they bore on this point. His conclusion was, that such an increase of temperature in proceeding towards the centre of the earth, can arise from nothing but the remains of a primitive heat;--that the heat which the sun's action would communicate, would, in its final and {148} permanent state, be uniform in the same vertical line, as soon as we get beyond the influence of the superficial oscillations of which we have spoken;--and that, before the distribution of temperature reaches this limit, it will decrease, not increase, in descending. It appeared also, by the calculation, that this remaining existence of the primitive heat in the interior of the earth's mass, was quite consistent with the absence of all perceptible traces of it at the surface; and that the same state of things which produces an increase of one degree of heat in descending forty yards, does not make the surface a quarter of a degree hotter than it would otherwise be. Fourier was led also to some conclusions, though necessarily very vague ones, respecting the time which the earth must have taken to cool from a supposed original state of incandescence to its present condition, which time it appeared must have been very great; and respecting the extent of the future cooling of the surface, which it was shown must be insensible. Everything tended to prove that, within the period which the history of the human race embraces, no discoverable change of temperature had taken place from the progress of this central cooling. Laplace further calculated the effect[16\10] which any contraction of the globe of the earth by cooling would produce on the length of the day. He had already shown, by astronomical reasoning, that the day had not become shorter by 1⁄200th of a second, since the time of Hipparchus; and thus his inferences agreed with those of Fourier. As far as regards the smallness of the perceptible effect due to the past changes of the earth's temperature, there can be no doubt that all the curious conclusions just stated are deduced in a manner quite satisfactory, from the fact of a general increase of heat in descending below the surface of the earth; and thus our principles of

speculative science have a bearing upon the history of the past changes of the universe, and give us information concerning the state of things in portions of time otherwise quite out of our reach.

[Note 15\10: _Bullet. des Sc._ 1820, p. 58.]

[Note 16\10: _Conn. des Tems_, 1823.]

4. _Heat of the Planetary Spaces._--In the same manner, this portion of science is appealed to for information concerning parts of space which are utterly inaccessible to observation. The doctrine of heat leads to conclusions concerning the temperatures of the spaces which surround the earth, and in which the planets of the solar system revolve. In his Memoir, published in 1827,[17\10] Fourier states that he conceives it to follow from his principles, that these planetary spaces {149} are not absolutely cold, but have a "proper heat" independent of the sun and of the planets. If there were not such a heat, the cold of the polar regions would be much more intense than it is, and the alternations of cold and warmth, arising from the influence of the sun, would be far more extreme and sudden than we find them. As the cause of this heat in the planetary spaces, he assigns the radiation of the innumerable stars which are scattered through the universe.

[Note 17\10: _Mém. Inst._ tom. vii. p. 580.]

Fourier says,[18\10] "We conclude from these various remarks, and principally from the mathematical examination of the question," that this is so. I am not aware that the mathematical calculation which bears peculiarly upon this point has anywhere been published. But it is worth notice, that Svanberg has been led[19\10] to the opinion of the same temperature in these spaces which Fourier had adopted (50 centigrade below zero), by an entirely different course of reasoning, founded on the relation of the atmosphere to heat.

[Note 18\10: _Mém. Inst._ tom. vii. p. 581.]

[Note 19\10: Berzel. _Jahres Bericht_, xi. p. 50.]

In speaking of this subject, I have been led to notice incomplete and perhaps doubtful applications of the mathematical doctrine of conduction and radiation. But this may at least serve to show that Thermotics is a science, which, like Mechanics, is to be established by experiments on masses capable of manipulation, but which, like that, has for its most important office the solution of geological and cosmological problems. I now return to the further progress of our thermotical knowledge.

_Sect._ 5.--_Correction of Newton's Law of Cooling._

IN speaking of the establishment of Newton's assumption, that the temperature communicated is proportional to the excess of temperature, we stated that it was approximately verified, and afterwards corrected (chap. i., sect. 1.)**. This correction was the result of the researches of MM. Dulong and Petit in 1817, and the researches by which they were led to the true law, are an admirable example both of laborious experiment and sagacious induction. They experimented through a very great range of temperature (as high as two hundred and forty degrees centigrade), which was

necessary because the inaccuracy of Newton's law becomes considerable only at high temperatures. They removed the effect of the surrounding medium, by making their experiments in a vacuum. They selected with great {150} judgment the conditions of their experiments and comparisons, making one quantity vary while the others remained constant. In this manner they found, that _the quickness of cooling for a constant excess of temperature, increases in geometrical progression, when the temperature of the surrounding space increases in arithmetical progression_; whereas, according to the Newtonian law, this quickness would not have varied at all. Again, this variation being left out of the account, it appeared that _the quickness of cooling, so far as it depends on the excess of temperature of the hot body, increases as the terms of a geometrical progression diminished by a constant number, when the temperature of the hot body increases in arithmetical progression_. These two laws, with the coefficients requisite for their application to particular substances, fully determine the conditions of cooling in a vacuum.

Starting from this determination, MM. Dulong and Petit proceeded to ascertain the effect of the medium, in which the hot body is placed, upon its rate of cooling; for this effect became a _residual phenomenon_,[20\10] when the cooling in the vacuum was taken away. We shall not here follow this train of research; but we may briefly state, that they were led to such laws as this;--that the rapidity of cooling due to any gaseous medium in which the body is placed, is the same, so long as the excess of the body's temperature is the same, although the temperature itself vary;--that the cooling power of a gas varies with the elasticity, according to a determined law; and other similar rules.

[Note 20\10: See _Phil. Ind. Sciences_, B. xiii. c. 7, Sect. iv.]

In reference to the process of their induction, it is worthy of notice, that they founded their reasonings upon Prevost's law of exchanges; and that, in this way, the second of their laws above stated, respecting the quickness of cooling, was a mathematical consequence of the first. It may be observed also, that their temperatures are measured by means of the air-thermometer, and that if they were estimated on another scale, the remarkable simplicity and symmetry of their results would disappear. This is a strong argument for believing such a measure of temperature to have a natural prerogative of simplicity. This belief is confirmed by other considerations; but these, depending on the laws of _expansion_ by heat, cannot be here referred to; and we must proceed to finish our survey of the mathematical theory of heat, as founded on the phenomena of radiation and conduction, which alone have as yet been traced up to general principles.

We may observe, before we quit this subject, that this correction of {151} Newton's law will materially affect the mathematical calculations on the subject, which were made to depend on that law both by Fourier, Laplace, and Poisson. Probably, however, the general features of the results will be the same as on the old supposition. M. Libri, an Italian mathematician, has undertaken one of the problems of this kind, that of the armil, with Dulong and Petit's law for his basis, in a Memoir read to the Institute of France in 1825, and since published at Florence.[21\10]

[Note 21\10: _Mém. de Math. et de Phys._ 1829.]

_Sect._ 6.--_Other Laws of Phenomena with respect to Radiation._

THE laws of radiation as depending upon the surface of radiating bodies, and as affecting screens of various kinds interposed between the hot body and the thermometer, were examined by several inquirers. I shall not attempt to give an account of the latter course of research, and of the different laws which luminous and non-luminous heat have been found to follow in reference to bodies, whether transparent or opaque, which intercept them. But there are two or three laws of the phenomena, depending upon the effects of the surfaces of bodies, which are important.

1. In the first place, the powers of bodies to _emit_ and to _absorb_ heat, as far as depends upon their surface, appear to be in the same proportion. If we blacken the surface of a canister of hot water, it radiates heat more copiously; and in the same measure, it is more readily heated by radiation.

2. In the next place, as the radiative power increases, the power of reflection diminishes, and the contrary. A bright metal vessel reflects much heat; on this very account it does not emit much; and hence a hot fluid which such a vessel contains, remains hot longer than it does in an unpolished case.

3. The heat is emitted from every point of the surface of a hot body in all directions; but by no means in all directions with equal intensity. The intensity of the heating ray is as the sine of the angle which it makes with the surface.

The last law is entirely, the two former in a great measure, due to the researches of Leslie, whose _Experimental Inquiry into the Nature and Propagation of Heat_, published in 1804, contains a great number of curious and striking results and speculations. The laws now just {152} stated bear, in a very important manner, upon the formation of the theory; and we must now proceed to consider what appears to have been done in this respect; taking into account, it must still be borne in mind, only the phenomena of conduction and radiation.

_Sect._ 7.--_Fourier's Theory of Radiant Heat._

THE above laws of phenomena being established, it was natural that philosophers should seek to acquire some conception of the physical action by which they might account, both for these laws, and for the general fundamental facts of Thermotics; as, for instance, the fact that all bodies placed in an inclosed space assume, in time, the temperature of the inclosure. Fourier's explanation of this class of phenomena must be considered as happy and successful; for he has shown that the supposition to which we are led by the most simple and general of the facts, will explain, moreover, the less obvious laws. It is an obvious and general fact, that bodies which are included in the space tend to acquire the same temperature. And this identity of temperature of neighboring bodies requires an hypothesis, which, it is found, also accounts for Leslie's law of the sine, in radiation.

This hypothesis is, that the radiation takes place, not from the surface alone of the hot body, but from all particles situated within a certain small depth of the surface. It

is easy to see[22\10] that, on this supposition, a ray emitted obliquely from an internal particle, will be less intense than one sent forth perpendicular to the surface, because the former will be intercepted in a greater degree, having a greater length of path within the body; and Fourier shows, that whatever be the law of this intercepting power, the result will be, that the radiative intensity is as the sine of the angle made by the ray with the surface.

[Note 22\10: _Mém. Inst._ t. v. 1821, p. 204.]

But this law is, as I have said, likewise necessary, in order that neighboring bodies may tend to assume the same temperature: for instance, in order that a small particle placed within a spherical shell, should finally assume the temperature of the shell. If the law of the sines did not obtain, the final temperature of such a particle would depend upon its place in the inclosure;[23\10] and within a shell of ice we should have, at certain points, the temperature of boiling water and of melting iron.

[Note 23\10: _An. Chim._ iv. 1817, p. 129.]

This proposition may at first appear strange and unlikely; but it may {153} be shown to be a necessary consequence of the assumed principle, by very simple reasoning, which I shall give in a general form in a Note.[24\10]

[Note 24\10: The following reasoning may show the connexion of the law of the sines in radiant heat with the general principle of ultimate identity of neighboring temperatures. The equilibrium and identity of temperature between an including shell and an included body, cannot obtain upon the whole, except it obtain between each pair of parts of the two surfaces of the body and of the shell; that is, any part of the one surface, in its exchanges with any part of the other surface, must give and receive the same quantity of heat. Now the quantity exchanged, so far as it depends on the receiving surface, will, by geometry, be proportional to the sine of the obliquity of that surface: and as, in the exchanges, each may be considered as receiving, the quantity transferred must be proportional to the sines of the two obliquities; that is, to that of the giving as well as of the receiving surface.

Nor is this conclusion disturbed by the consideration, that all the rays of heat which fall upon a surface are not absorbed, some being reflected according to the nature of the surface. For, by the other above-mentioned laws of phenomena, we know that, in the same measure in which the surface loses the power of admitting, it loses the power of emitting, heat; and the superficial parts gain, by absorbing their own radiation, as much as they lose by not absorbing the incident heat; so that the result of the preceding reasoning remains unaltered.]

This reasoning is capable of being presented in a manner quite satisfactory, by the use of mathematical symbols, and proves that Leslie's law of the sines is rigorously and mathematically true on Fourier's hypothesis. And thus Fourier's theory of _molecular extra-radiation_ acquires great consistency.

_Sect._ 8.--_Discovery of the Polarization of Heat._

THE laws of which the discovery is stated in the preceding Sections of this Chapter, and the explanations given of them by the theories of conduction and

radiation, all tended to make the conception of a material heat, or _caloric_, communicated by an actual flow and emission, familiar to men's minds; and, till lately, had led the greater part of thermotical philosophers to entertain such a view, as the most probable opinion concerning the nature of heat. But some steps have recently been made in thermotics, which appear to be likely to overturn this belief, and to make the doctrine of emission as untenable with regard to heat, as it had been found to be with regard to light. I speak of the discovery of the polarization of heat. It being ascertained that rays of heat are polarized in the same manner as rays of {154} light, we cannot retain the doctrine that heat radiates by the emanation of material particles, without supposing those particles of caloric to have poles; an hypothesis which probably no one would embrace; for, besides that the ill fortune which attended that hypothesis in the case of light must deter speculators from it, the intimate connexion of heat and light would hardly allow us to suppose polarization in the two cases to be produced by two different kinds of machinery.

But, without here tracing further the influence which the polarization of heat must exercise upon the formation of our theories of heat, we must briefly notice this important discovery, as a law of phenomena.

The analogies and connexions between light and heat are so strong, that when the polarization of light had been discovered, men were naturally led to endeavor to ascertain whether heat possessed any corresponding property. But partly from the difficulty of obtaining any considerable effect of heat separated from light, and partly from the want of a thermometrical apparatus sufficiently delicate, these attempts led, for some time, to no decisive result. M. Berard took up the subject in 1813. He used Malus's apparatus, and conceived that he found heat to be polarized by reflection at the surface of glass, in the same manner as light, and with the same circumstances.[25\10] But when Professor Powell, of Oxford, a few years later (1830), repeated these experiments with a similar apparatus, he found[26\10] that though the heat which is conveyed along with light is, of course, polarizable, "simple radiant heat," as he terms it, did not offer the smallest difference in the two rectangular azimuths of the second glass, and thus showed no trace of polarization.

[Note 25\10: _Ann. Chim._ March, 1813.]

[Note 26\10: _Edin. Journ. of Science_, 1830, vol. ii. p. 303.]

Thus, with the old thermometers, the point remained doubtful. But soon after this time, MM. Melloni and Nobili invented an apparatus, depending on certain galvanic laws, of which we shall have to speak hereafter, which they called a _thermomultiplier_; and which was much more sensitive to changes of temperature than any previously-known instrument. Yet even with this instrument, M. Melloni failed; and did not, at first, detect any perceptible polarization of heat by the tourmaline;[27\10] nor did M. Nobili,[28\10] in repeating M. Berard's experiment. But in this experiment the attempt was made to polarize heat by reflection from glass, as light is polarized: and the quantity {155} reflected is so small that the inevitable errors might completely disguise the whole difference in the two opposite positions. When Prof. Forbes, of Edinburgh, (in 1834,) employed mica in the like

experiments, he found a very decided polarizing effect; first, when the heat was transmitted through several films of mica at a certain angle, and afterwards, when it was reflected from them. In this case, he found that with non-luminous heat, and even with the heat of water below the boiling point, the difference of the heating power in the two positions of opposite polarity (parallel and _crossed_) was manifest. He also detected by careful experiments,[29\10] the polarizing effect of tourmaline. This important discovery was soon confirmed by M. Melloni. Doubts were suggested whether the different effect in the opposite positions might not be due to other circumstances; but Professor Forbes easily showed that these suppositions were inadmissible; and the property of a difference of _sides_, which at first seemed so strange when ascribed to the rays of light, also belongs, it seems to be proved, to the rays of heat. Professor Forbes also found, by interposing a plate of mica to intercept the ray of heat in an intermediate point, an effect was produced in certain positions of the mica analogous to what was called _depolarization_ in the case of light; namely, a partial destruction of the differences which polarization establishes.

[Note 27\10: _Ann. de Chimie_, vol. lv.]

[Note 28\10: _Bibliothèque Universelle_.]

[Note 29\10: _Ed. R. S. Transactions_, vol. xiv.; and _Phil. Mag._ 1835, vol. v. p. 209. Ib. vol. vii. p. 349.]

Before this discovery, M. Melloni had already proved by experiment that heat is _refracted_ by transparent substances as light is. In the case of light, the _depolarizing_ effect was afterwards found to be really, as we have seen, a _dipolarizing_ effect, the ray being divided into two rays by _double refraction_. We are naturally much tempted to put the same interpretation upon the dipolarizing effect in the case of heat; but perhaps the assertion of the analogy between light and heat to this extent is as yet insecure.

It is the more necessary to be cautious in our attempt to identify the laws of light and heat, inasmuch as along with all the resemblances of the two agents, there are very important differences. The power of transmitting light, _the diaphaneity_ of bodies, is very distinct from their power of transmitting heat, which has been called _diathermancy_ by M. Melloni. Thus both a plate of alum and a plate of rock-salt transmit nearly the whole light; but while the first stops nearly the whole heat, the second stops very little of it; and a plate of opake {156} quartz, nearly impenetrable by light, allows a large portion of the heat to pass. By passing the rays through various media, the heat may be, as it were, _sifted_ from the light which accompanies it.

[2nd Ed.] [The diathermancy of bodies is distinct from their diaphaneity, in so far that the same bodies do not exercise the same powers of selection and suppression of certain rays on heat and on light; but it appears to be proved by the investigations of modern thermotical philosophers (MM. De la Roche, Powell, Melloni, and Forbes), that there is a close analogy between the absorption of certain colors by transparent bodies, and the absorption of certain kinds of heat by diathermanous bodies. Dark sources of heat emit rays which are analogous to blue and violet rays of light; and

highly luminous sources emit rays which are analogous to red rays. And by measuring the angle of total reflection for heat of different kinds, it has been shown that the former kind of calorific rays are really less refrangible than the latter.[30\10]

[Note 30\10: See Prof. Forbes's _Third Series of Researches on Heat_, _Edinb. R.S. Trans._ vol. xiv.]

M. Melloni has assumed this analogy as so completely established, that he has proposed for this part of thermotics the name _Thermochroology_ (Qu. _Chromothermotics_?); and along with this term, many others derived from the Greek, and founded on the same analogy. If it should appear, in the work which he proposes to publish on this subject, that the doctrines which he has to state cannot easily be made intelligible without the use of the terms he suggests, his nomenclature will obtain currency; but so large a mass of etymological innovations is in general to be avoided in scientific works.

M. Melloni's discovery of the extraordinary power of _rock-salt_ to transmit heat, and Professor Forbes's discovery of the extraordinary power of _mica_ to polarize and depolarize heat, have supplied thermotical inquirers with two new and most valuable instruments.[31\10]]

[Note 31\10: For an account of many thermotical researches, which I have been obliged to pass unnoticed here, see two Reports by Prof. Powell on the present state of our knowledge respecting Radiant Heat, in the _Reports of the British Association_ for 1832 and 1840.]

Moreover, besides the laws of conduction and radiation, many other laws of the phenomena of heat have been discovered by philosophers; and these must be taken into account in judging any theory of heat. To these other laws we must now turn our attention. {157}

## CHAPTER II.

THE LAWS OF CHANGES OCCASIONED BY HEAT.

_Sect._ 1.--_Expansion by Heat.--The Law of Dalton and Gay-Lussac for Gases._

ALMOST all bodies expand by heat; solids, as metals, in a small degree; fluids, as water, oil, alcohol, mercury, in a greater degree. This was one of the facts first examined by those who studied the nature of heat, because this property was used for the measure of heat. In the _Philosophy of the Inductive Sciences_, Book iv., Chap. iv., I have stated that secondary qualities, such as Heat, must be measured by their effects: and in Sect. 4 of that Chapter I have given an account of the successive attempts which have been made to obtain measures of heat. I have there also spoken of the results which were obtained by comparing the rate at which the expansion of different substances went on, under the same degrees of heat; or as it was called, the different _thermometrical march_ of each substance. Mercury appears to be the liquid which is most uniform in its thermometrical march; and it has been taken as the most common material of our thermometers; but the expansion of mercury is not proportional to the heat. De Luc was led, by his experiments, to conclude "that the

dilatations of mercury follow an accelerated march for equal augmentations of heat." Dalton conjectured that water and mercury both expand as the square of the _real temperature_ from the point of greatest contraction: the real temperature being measured so as to lead to such a result. But none of the rules thus laid down for the expansion of solids and fluids appear to have led, as yet, to any certain general laws.

With regard to gases, thermotical inquirers have been more successful. Gases expand by heat; and their expansion is governed by a law which applies alike to all degrees of heat, and to all gaseous fluids. The law is this: that _for equal increments of temperature they expand by the same fraction of their own bulk_; which fraction is _three-eights_ {158} in proceeding from freezing to boiling water. This law was discovered by Dalton and M. Gay-Lussac independently of each other;[32\10] and is usually called by both their names, _the law of Dalton and Gay-Lussac_. The latter says,[33\10] "The experiments which I have described, and which have been made with great care, prove incontestably that oxygen, hydrogen, azotic acid, nitrous acid, ammoniacal acid, muriatic acid, sulphurous acid, carbonic acid, gases, expand equally by equal increments of heat." "Therefore," he adds with a proper inductive generalization, "the result does not depend upon physical properties, and I collect that _all gases expand equally by heat_." He then extends this to vapors, as ether. This must be one of the most important foundation-stones of any sound theory of heat.

[Note 32\10: _Manch. Mem._ vol. v. 1802; and _Ann. Chim._ xliii. p. 137.]

[Note 33\10: Ib. p. 272.]

[2nd Ed.] Yet MM. Magnus and Regnault conceive that they have overthrown this law of Dalton and Gay-Lussac, and shown that the different gases do not expand alike for the same increment of heat. Magnus found the ratio to be for atmospheric air, 1•366; for hydrogen, 1•365; for carbonic acid, 1•369; for sulphurous-acid gas, 1•385. But these differences are not greater than the differences obtained for the same substances by different observers; and as this law is referred to in Laplace's hypothesis, hereafter to be discussed, I do not treat the law as disproved.

Yet that the rate of expansion of gas in certain circumstances is different for different substances, must be deemed very probable, after Dr. Faraday's recent investigations _On the Liquefaction and Solidification of Bodies generally existing as Gases_,[34\10] by which it appears that the elasticity of vapors _in contact with their fluids_ increases at different rates in different substances. "That the force," he says, "of vapor increases in a geometrical ratio for equal increments of heat is true for all bodies, but the ratio is not the same for all. . . . For an increase of pressure from two to six atmospheres, the following number of degrees require to be added to the bodies named:--water 69°, sulphureous acid 63°, cyanogen 64°•5, ammonia 60°, arseniuretted hydrogen 54°, sulphuretted hydrogen 56°•5, muriatic acid 43°, carbonic acid 32°•5, nitrous oxide 30°."]

[Note 34\10: _Phil. Trans._ 1845, Pt. 1.]

We have already seen that the opinion that the air-thermometer is a true measure of heat, is strongly countenanced by the symmetry which, by using it, we introduce

into the laws of radiation. If we {159} accept the law of Dalton and Gay-Lussac, it follows that this result is independent of any peculiar properties in the air employed; and thus this measure has an additional character of generality and simplicity which make it still more probable that it is the true standard. This opinion is further supported by the attempts to include such facts in a theory; but before we can treat of such theories, we must speak of some other doctrines which have been introduced.

_Sect._ 2.--_Specific Heat.--Change of Consistence._

IN the attempts to obtain measures of heat, it was found that bodies had different capacities for heat; for the same quantity of heat, however measured, would raise, in different degrees, the temperature of different substances. The notion of different capacities for heat was thus introduced, and each body was thus assumed to have a specific _capacity for heat_, according to the quantity of heat which it required to raise it through a given scale of heat.[35\10] The term "capacity for heat" was introduced by Dr. Irvine, a pupil of Dr. Black. For this term, Wilcke, the Swedish physicist, substituted "specific heat;" in analogy with "specific gravity."

[Note 35\10: See Crawford, _On Heat_, for the History of Specific Heat.]

It was found, also, that the capacity of the same substance was different in the same substance at different temperatures. It appears from experiments of MM. Dulong and Petit, that, in general, the capacity of liquids and solids increases as we ascend in the scale of temperature.

But one of the most important thermotic facts is, that by the sudden contraction of any mass, its temperature is increased. This is peculiarly observable in gases, as, for example, common air. The amount of the increase of temperature by sudden condensation, or of the cold produced by sudden rarefaction, is an important datum, determining the velocity of sound, as we have already seen, and affecting many points of meteorology. The coefficient which enters the calculation in the former case depends on the ratio of two specific heats of air under different conditions; one belonging to it when, varying in density, the pressure is constant by which the air is contained; the other, when, varying in density, it is contained in a constant space.

A leading fact, also, with regard to the operation of heat on bodies {160} is, that it changes their _form_, as it is often called, that is, their condition as solid, liquid, or air. Since the term "form" is employed in too many and various senses to be immediately understood when it is intended to convey this peculiar meaning, I shall use, instead of it, the term _consistence_, and shall hope to be excused, even when I apply this word to gases, though I must acknowledge such phraseology to be unusual. Thus there is a change of consistence when solids become liquid, or liquids gaseous; and the laws of such changes must be fundamental facts of our thermotical theories. We are still in the dark as to many of the laws which belong to this change; but one of them, of great importance, has been discovered, and to that we must now proceed.

_Sect._ 3.--_The Doctrine of Latent Heat._

The Doctrine of Latent Heat refers to such changes of consistence as we have just spoken of. It is to this effect; that during the conversion of solids into liquids, or of

liquids into vapors, there is communicated to the body heat which is not indicated by the thermometer. The heat is absorbed, or becomes _latent_; and, on the other hand, on the condensation of the vapor to a liquid, or the liquid to a solid consistency, this heat is again given out and becomes sensible. Thus a pound of ice requires twenty times as long a time, in a warm room, to raise its temperature seven degrees, as a pound of ice-cold water does. A kettle placed on a fire, in four minutes had its temperature raised to the boiling point, 212°: and this temperature continued stationary for twenty minutes, when the whole was boiled away. Dr. Black inferred from these facts that a large quantity of heat is absorbed by the ice in becoming water, and by the water in becoming steam. He reckoned from the above experiments, that ice, in melting, absorbs as much heat as would raise ice-cold water through 140° of temperature: and that water, in evaporating, absorbs as much heat as would raise it through 940°.

That snow requires a great quantity of heat to melt it; that water requires a great quantity of heat to convert it into steam; and that this heat is not indicated by a rise in the thermometer, are facts which it is not difficult to observe; but to separate these from all extraneous conditions, to group the cases together, and to seize upon the general law by which they are connected, was an effort of inductive insight, which has been considered, and deservedly, as one of the most striking {161} events in the modern history of physics. Of this step the principal merit appears to belong to Black.

[2nd Ed.] [In the first edition I had mentioned the names of De Luc and of Wilcke, in connexion with the discovery of Latent Heat, along with the name of Black. De Luc had observed, in 1755, that ice, in melting, did not rise above the freezing-point of temperature till the whole was melted. De Luc has been charged with plagiarizing Black's discovery, but, I think, without any just ground. In his _Idées sur la Météorologique_ (1787), he spoke of Dr. Black as "the first who had attempted the determinations of the quantities of latent heat." And when Mr. Watt pointed out to him that from this expression it might be supposed that Black had not discovered the fact itself, he acquiesced, and redressed the equivocal expression in an Appendix to the volume.[36\10]

[Note 36\10: See his _Letter_ to the Editors of the _Edinburgh Review_, No. xii. p. 502, of the _Review_.]

Black never published his own account of the doctrine of Latent Heat: but he delivered it every year after 1760 in his Lectures. In 1770, a surreptitious publication of his Lectures was made by a London bookseller, and this gave a view of the leading points of Dr. Black's doctrine. In 1772, Wilcke, of Stockholm, read a paper to the Royal Society of that city, in which the absorption of heat by melting ice is described; and in the same year, De Luc of Geneva published his _Recherches sur les Modifications de l'Atmosphère_, which has been alleged to contain the doctrine of latent heat, and which the author asserts to have been written in ignorance of what Black had done. At a later period, De Luc, adopting, in part. Black's expression, gave the name of _latent fire_ to the heat absorbed.[37\10]

[Note 37\10: See _Ed. Rev._ No. vi. p. 20.]

It appears that Cavendish determined the amount of heat produced by condensing steam, and by thawing snow, as early as 1765. He had perhaps already heard something of Black's investigations, but did not accept his term "latent heat".**[38\10]]

[Note 38\10: See Mr. V. Harcourt's _Address_ to the Brit. Assoc. in 1839, and the _Appendix_.]

The consequences of Black's principle are very important, for upon it is founded the whole doctrine of evaporation; besides which, the principle of latent heat has other applications. But the relations of aqueous vapor to air are so important, and have been so long a {162} subject of speculation, that we may with advantage dwell a little upon them. The part of science in which this is done may be called, as we have said, Atmology; and to that division of Thermotics the following chapters belong.

{{163}} ATMOLOGY.

## CHAPTER III.

THE RELATION OF VAPOR AND AIR.

_Sect._ 1.--_The Boylean Law of the Air's Elasticity._

IN the Sixth Book (Chap. iv. Sect. 1.) we have already seen how the conception on the laws of fluid equilibrium was, by Pascal and others, extended to air, as well as water. But though air presses and is pressed as water presses and is pressed, pressure produces upon air an effect which it does not, in any obvious degree, produce upon water. Air which is pressed is also _compressed_, or made to occupy a smaller space; and is consequently also made more dense, or _condensed_; and on the other hand, when the pressure upon a portion of air is diminished, the air expands or is rarefied. These broad facts are evident. They are expressed in a general way by saying that air is an _elastic_ fluid, yielding in a certain degree to pressure, and recovering its previous dimensions when the pressure is removed.

But when men had reached this point, the questions obviously offered themselves, in what degree and according to what law air yields to pressure; when it is compressed, what relation does the density bear to the pressure? The use which had been made of tubes containing columns of mercury, by which the pressure of portions of air was varied and measured, suggested obvious modes of devising experiments by which this question might be answered. Such experiments accordingly were made by Boyle about 1650; and the result at which he arrived was, that when air is thus compressed, the density is as the pressure. Thus if the pressure of the atmosphere in its common state be equivalent to 30 inches of mercury, as shown by the barometer; if air included in a tube be pressed by 30 additional inches of {164} mercury, its density will be doubled, the air being compressed into one half the space. If the pressure be increased threefold, the density is also trebled; and so on. The same law was soon afterwards (in 1676) proved experimentally by Mariotte. And this law of the air's elasticity, that the density is as the pressure, is sometimes called the _Boylean Law_, and sometimes the _Law of Boyle and Mariotte_.

Air retains its aerial character permanently; but there are other aerial substances which appear as such, and then disappear or change into some other condition. Such are termed _vapors_. And the discovery of their true relation to air was the result of a long course of researches and speculations.

[2nd Ed.] [It was found by M. Cagniard de la Tour (in 1823), that at a certain temperature, a liquid, under sufficient pressure, becomes clear transparent vapor or gas, having the same bulk as the liquid. This condition Dr. Faraday calls the _Cagniard de la Tour_ state, (the _Tourian_ state?) It was also discovered by Dr. Faraday that carbonic-acid gas, and many other gases, which were long conceived to be permanently elastic, are really reducible to a liquid state by pressure.[39\10] And in 1835, M. Thilorier found the means of reducing liquid carbonic acid to a solid form, by means of the cold produced in evaporation. More recently Dr. Faraday has added several substances usually gaseous to the list of those which could previously be shown in the liquid state, and has reduced others, including ammonia, nitrous oxide, and sulphuretted hydrogen, to a solid consistency.[40\10] After these discoveries, we may, I think, reasonably doubt whether all bodies are not capable of existing in the three _consistencies_ of solid, liquid, and air.

[Note 39\10: _Phil. Trans._ 1823.]

[Note 40\10: Ib. Pt. 1. 1845.]

We may note that the law of Boyle and Mariotte is not exactly true near the limit at which the air passes to the liquid state in such cases as that just spoken of. The diminution of bulk is then more rapid than the increase of pressure.

The transition of fluids from a liquid to an airy consistence appears to be accompanied by other curious phenomena. See Prof. Forbes's papers on the _Color of Steam under certain circumstances_, and on the _Colors of the Atmosphere_, in the _Edin. Trans._ vol. xiv.] {165}

_Sect._ 2.--_Prelude to Dalton's Doctrine of Evaporation._

VISIBLE clouds, smoke, distillation, gave the notion of Vapor; vapor was at first conceived to be identical with air, as by Bacon.[41\10] It was easily collected, that by heat, water might be converted into vapor. It was thought that air was thus produced, in the instrument called the _æolipile_, in which a powerful blast is caused by a boiling fluid; but Wolfe showed that the fluid was not converted into air, by using camphorated spirit of wine, and condensing the vapor after it had been formed. We need not enumerate the doctrines (if very vague hypotheses may be so termed) of Descartes, Dechales, Borelli.[42\10] The latter accounted for the rising of vapor by supposing it a mixture of fire and water; and thus, fire being much lighter than air, the mixture also was light. Boyle endeavored to show that vapors do not permanently float _in vacuo_. He compared the mixture of vapor with air to that of salt with water. He found that the pressure of the atmosphere affected the heat of boiling water; a very important fact. Boyle proved this by means of the air-pump; and he and his friends were much surprised to find that when air was removed, water only just warm boiled violently. Huyghens mentions an experiment of the same kind made by Papin about 1673.

[Note 41\10: Bacon's _Hist. Nat._ Cent. i. p. 27.]

[Note 42\10: They may be seen in Fischer, _Geschichte der Physik_, vol. ii. p. 175.]

The ascent of vapor was explained in various ways in succession, according to the changes which physical science underwent. It was a problem distinctly treated of, at a period when hydrostatics had accounted for many phenomena; and attempts were naturally made to reduce this fact to hydrostatical principles. An obvious hypothesis, which brought it under the dominion of these principles, was, to suppose that the water, when converted into vapor, was divided into small hollow globules;--thin pellicles including air or heat. Halley gave such an explanation of evaporation; Leibnitz calculated the dimensions of these little bubbles; Derham managed (as he supposed) to examine them with a magnifying glass: Wolfe also examined and calculated on the same subject. It is curious to see so much confidence in so lame a theory; for if water became hollow globules in order to rise as vapor, we require, in order to explain the formation of these globules, new laws of nature, which are not even hinted at by {166} the supporters of the doctrine, though they must be far more complex than the hydrostatical law by which a hollow sphere floats.

Newton's opinion was hardly more satisfactory; he[43\10] explained evaporation by the repulsive power of heat; the parts of vapors, according to him, being small, are easily affected by this force, and thus become lighter than the atmosphere.

[Note 43\10: _Opticks_, Qu. 31.]

Muschenbroek still adhered to the theory of globules, as the explanation of evaporation; but he was manifestly discontented with it; and reasonably apprehended that the pressure of the air would destroy the frail texture of these bubbles. He called to his aid a rotation of the globules (which Descartes also had assumed); and, not satisfied with this, threw himself on electrical action as a reserve. Electricity, indeed, was now in favor, as hydrostatics had been before; and was naturally called in, in all cases of difficulty. Desaguliers, also, uses this agent to account for the ascent of vapor, introducing it into a kind of sexual system of clouds; according to him, the male fire (heat) does a part, and the female fire (electricity) performs the rest. These are speculations of small merit and no value.

In the mean time, Chemistry made great progress in the estimation of philosophers, and had its turn in the explanation of the important facts of evaporation. Bouillet, who, in 1742, placed the particles of water in the interstices of those of air, may be considered as approaching to the chemical theory. In 1748, the Academy of Sciences of Bourdeaux proposed the ascent of vapors as the subject of a prize; which was adjudged in a manner very impartial as to the choice of a theory; for it was divided between Kratzenstein, who advocated the bubbles, (the coat of which he determined to be 1⁄50,000th of an inch thick,) and Hamberger, who maintained the truth to be the adhesion of particles of water to those of air and fire. The latter doctrine had become much more distinct in the author's mind when seven years afterwards (1750) he published his _Elementa Physices_. He then gave the explanation of evaporation in a phrase which has since been adopted,--the _solution of water in air_; which he conceived to be of the same kind as other chemical

solutions.

This theory of solution was further advocated and developed by Le Roi;[44\10] and in his hands assumed a form which has been extensively adopted up to our times, and has, in many instances, tinged the language commonly used. He conceived that air, like other solvents, {167} might be _saturated_; and that when the water was beyond the amount required for saturation, it appeared in a visible form. The saturating quantity was held to depend mainly on warmth and wind.

[Note 44\10: _Ac. R. Sc._ Paris, 1750.]

This theory was by no means devoid of merit; for it brought together many of the phenomena, and explained a number of the experiments which Le Roi made. It explained the facts of the transparency of vapor, (for perfect solutions are transparent,) the precipitation of water by cooling, the disappearance of the visible moisture by warming it again, the increased evaporation by rain and wind; and other observed phenomena. So far, therefore, the introduction of the notion of the chemical solution of water in air was apparently very successful. But its defects are of a very fatal kind; for it does not at all apply to the facts which take place when air is excluded.

In Sweden, in the mean time,[45\10] the subject had been pursued in a different, and in a more correct manner. Wallerius Ericsen had, by various experiments, established the important fact, that water evaporates in a _vacuum_. His experiments are clear and satisfactory; and he inferred from them the falsity of the common explanation of evaporation by the solution of water in _air_. His conclusions are drawn in a very intelligent manner. He considers the question whether water can be changed into air, and whether the atmosphere is, in consequence, a mere collection of vapors; and on good reasons, decides in the negative, and concludes the existence of permanently-elastic air different from vapor. He judges, also, that there are two causes concerned, one acting to produce the first ascent of vapors, the other to support them afterwards. The first, which acts in a vacuum, he conceives to be the mutual repulsion of the particles; and since this force is independent of the presence of other substances, this seems to be a sound induction. When the vapors have once ascended into the air, it may readily be granted that they are carried higher, and driven from side to side by the currents of the atmosphere. Wallerius conceives that the vapor will rise till it gets into air of the same density as itself, and being then in equilibrium, will drift to and fro.

[Note 45\10: Fischer, _Gesch. Phys._ vol. v. p. 63.]

The two rival theories of evaporation, that of _chemical solution_ and that of _independent vapor_, were, in various forms, advocated by the next generation of philosophers. De Saussure may be considered as the leader on one side, and De Luc on the other. The former maintained the solution theory, with some modifications of his own. De {168} Luc denied all solution, and held vapor to be a combination of the particles of water with fire, by which they became lighter than air. According to him, there is always fire enough present to produce this combination, so that evaporation goes on at all temperatures.

This mode of considering independent vapor as a combination of fire with water, led the attention of those who adopted that opinion to the thermometrical changes which take place when vapor is formed and condensed. These changes are important, and their laws curious. The laws belong to the induction of latent heat, of which we have just spoken; but a knowledge of them is not absolutely necessary in order to enable us to understand the manner in which steam exists in air.

De Luc's views led him[46\10] also to the consideration of the effect of pressure on vapor. He explains the fact that pressure will condense vapor, by supposing that it brings the particles within the distance at which the repulsion arising from fire ceases. In this way, he also explains the fact, that though external pressure does thus condense steam, the mixture of a body of air, by which the pressure is equally increased, will not produce the same effect; and therefore, vapors can exist in the atmosphere. They make no fixed proportion of it; but at the same temperature we have the same pressure arising _from them_, whether they are in air or not. As the heat increases, vapor becomes capable of supporting a greater and greater pressure, and at the boiling heat, it can support the pressure of the atmosphere.

[Note 46\10: Fischer, vol. vii. p. 453. _Nouvelles Idées sur la Météorologie_, 1787.]

De Luc also marked very precisely (as Wallerius had done) the difference between vapor and air; the former being capable of change of _consistence_ by cold or pressure, the latter not so. Pictet, in 1786, made a hygrometrical experiment, which appeared to him to confirm De Luc's views; and De Luc, in 1792, published a concluding essay on the subject in the _Philosophical Transactions_. Pictet's _Essay on Fire_, in 1791, also demonstrated that "all the train of hygrometrical phenomena takes place just as well, indeed rather quicker, in a vacuum than in air, provided the same quantity of moisture is present." This essay, and De Luc's paper, gave the death-blow to the theory of the solution of water in air.

Yet this theory did not fall without an obstinate struggle. It was taken up by the new school of French chemists, and connected with their views of heat. Indeed, it long appears as the prevalent opinion. {169} Girtanner,[47\10] in his _Grounds of the Antiphlogistic Theory_, may be considered as one of the principal expounders of this view of the matter. Hube, of Warsaw, was, however, the strongest of the defenders of the theory of solution, and published upon it repeatedly about 1790. Yet he appears to have been somewhat embarrassed with the increase of the air's elasticity by vapor. Parrot, in 1801, proposed another theory, maintaining that De Luc had by no means successfully attacked that of solution, but only De Saussure's superfluous additions to it.

[Note 47\10: Fischer, vol. vii. 473.]

It is difficult to see what prevented the general reception of the doctrine of independent vapor; since it explained all the facts very simply, and the agency of air was shown over and over again to be unnecessary. Yet, even now, the solution of water in air is hardly exploded. M. Gay Lussac,[48\10] in 1800, talks of the quantity of water "held in solution" by the air; which, he says, varies according to its temperature and density by a law which has not yet been discovered. And Professor Robison, in

the article "Steam," in the _Encyclopædia Britannica_ (published about 1800), says,[49\10] "Many philosophers imagine that spontaneous evaporation, at low temperatures, is produced in this way (by elasticity alone). But we cannot be of this opinion; and must still think that this kind of evaporation is produced by the dissolving power of the air." He then gives some reasons for his opinion. "When moist air is suddenly rarefied, there is always a precipitation of water. But by this new doctrine the very contrary should happen, because the tendency of water to appear in the elastic form is promoted by removing the external pressure." Another main difficulty in the way of the doctrine of the mere mixture of vapor and air was supposed to be this; that if they were so mixed, the heavier fluid would take the lower part, and the lighter the higher part, of the space which they occupied.

[Note 48\10: _Ann. Chim._ tom. xliii.]

[Note 49\10: Robison's _Works_, ii. 37.]

The former of these arguments was repelled by the consideration that in the rarefaction of air, its specific heat is changed, and thus its temperature reduced below the constituent temperature of the vapor which it contains. The latter argument is answered by a reference to Dalton's law of the mixture of gases. We must consider the establishment of this doctrine in a new section, as the most material step to the true notion of evaporation. {170}

_Sect._ 3.--_Dalton's Doctrine of Evaporation._

A PORTION of that which appears to be the true notion of evaporation was known, with greater or less distinctness, to several of the physical philosophers of whom we have spoken. They were aware that the vapor which exists in air, in an invisible state, may be condensed into water by cold: and they had noticed that, in any state of the atmosphere, there is a certain temperature lower than that of the atmosphere, to which, if we depress bodies, water forms upon them in fine drops like dew; this temperature is thence called the _dew-point_. The vapor of water which exists anywhere may be reduced below the degree of heat which is necessary to constitute it vapor, and thus it ceases to be vapor. Hence this temperature is also called the _constituent temperature_. This was generally known to the meteorological speculators of the last century, although, in England, attention was principally called to it by Dr. Wells's _Essay on Dew_, in 1814. This doctrine readily explains how the cold produced by rarefaction of air, descending below the constituent temperature of the contained vapor, may precipitate a dew; and thus, as we have said, refutes one obvious objection to the theory of independent vapor.

The other difficulty was first fully removed by Mr. Dalton. When his attention was drawn to the subject of vapor, he saw insurmountable objections to the doctrine of a chemical union of water and air. In fact, this doctrine was a mere nominal explanation; for, on closer examination, no chemical analogies supported it. After some reflection, and in the sequel of other generalizations concerning gases, he was led to the persuasion, that when air and steam are mixed together, each follows its separate laws of equilibrium, the particles of each being elastic with regard to those of their own kind only: so that steam may be conceived as flowing among the

particles of air[50\10] "like a stream of water among pebbles;" and the resistance which air offers to evaporation arises, not from its weight, but from the inertia of its particles.

[Note 50\10: _Manchester Memoirs_, vol. v. p. 581.]

It will be found that the theory of independent vapor, understood with these conditions, will include all the facts of the case;--gradual evaporation in air; sudden evaporation in a vacuum; the increase of {171} the air's elasticity by vapor; condensation by its various causes; and other phenomena.

But Mr. Dalton also made experiments to prove his fundamental principle, that if two different gases communicate, they will diffuse themselves through each other;[51\10]--slowly, if the opening of communication be small. He observes also, that all the gases had equal solvent powers for vapor, which could hardly have happened, had chemical affinity been concerned. Nor does the density of the air make any difference.

[Note 51\10: _New System of Chemical Philosophy_, vol. i. p. 151.]

Taking all these circumstances into the account, Mr. Dalton abandoned the idea of solution. "In the autumn of 1801," he says, "I hit upon an idea which seemed to be exactly calculated to explain the phenomena of vapor: it gave rise to a great variety of experiments," which ended in fixing it in his mind as a true idea. "But," he adds, "the theory was almost universally misunderstood, and consequently reprobated."

Mr. Dalton answers various objections. Berthollet had urged that we can hardly conceive the particles of an elastic substance added to those of another, without increasing its elasticity. To this Mr. Dalton replies by adducing the instance of magnets, which repel each other, but do not repel other bodies. One of the most curious and ingenious objections is that of M. Gough, who argues, that if each gas is elastic with regard to itself alone, we should hear, produced by one stroke, four sounds; namely, _first_, the sound through aqueous vapor; _second_, the sound through azotic gas; _third_, the sound through oxygen gas; _fourth_, the sound through carbonic acid. Mr. Dalton's answer is, that the difference of time at which these sounds would come is very small; and that, in fact, we do hear, sounds double and treble.

In his _New System of Chemical Philosophy_, Mr. Dalton considers the objections of his opponents with singular candor and impartiality. He there appears disposed to abandon that part of the theory which negatives the mutual repulsion of the particles of the two gases, and to attribute their diffusion through one another to the different size of the particles, which would, he thinks,[52\10] produce the same effect.

[Note 52\10: _New System_, vol. i. p. 188.]

In selecting, as of permanent importance, the really valuable part of this theory, we must endeavor to leave out all that is doubtful or unproved. I believe it will be found that in all theories hitherto {172} promulgated, all assertions respecting the properties of the particles of bodies, their sizes, distances, attractions, and the like, are insecure and superfluous. Passing over, then, such hypotheses, the inductions

which remain are these;--that two gases which are in communication will, by the elasticity of each, diffuse themselves in one another, quickly or slowly; and--that the quantity of steam contained in a certain space of air is the same, whatever be the air, whatever be its density, and even if there be a vacuum. These propositions may be included together by saying, that one gas is _mechanically mixed_ with another; and we cannot but assent to what Mr. Dalton says of the latter fact,--"this is certainly the touchstone of the mechanical and chemical theories." This _doctrine of the mechanical mixture of gases_ appears to supply answers to all the difficulties opposed to it by Berthollet and others, as Mr. Dalton has shown;[53\10] and we may, therefore, accept it as well established.

[Note 53\10: _New System_, vol. i. p. 160, &c.]

This doctrine, along with the _principle of the constituent temperature of steam_, is applicable to a large series of meteorological and other consequences. But before considering the applications of theory to natural phenomena, which have been made, it will be proper to speak of researches which were carried on, in a great measure, in consequence of the use of steam in the arts: I mean the laws which connect its elastic force with its constituent temperature.

_Sect._ 4.--_Determination of the Laws of the Elastic Force of Steam._

THE expansion of aqueous vapor at different temperatures is governed, like that of all other vapors, by the law of Dalton and Gay-Lussac, already mentioned; and from this, its elasticity, when its expansion is resisted, will be known by the law of Boyle and Mariotte; namely, by the rule that the pressure of airy fluids is as the condensation. But it is to be observed, that this process of calculation goes on the supposition that the steam is cut off from contact with water, so that no more steam can be generated; a case quite different from the common one, in which the steam is more abundant as the heat is greater. The examination of the force of vapor, when it is in contact with water, must be briefly noticed.

During the period of which we have been speaking, the progress of the investigation of the laws of aqueous vapor was much accelerated {173} by the growing importance of the steam-engine, in which those laws operated in a practical form. James Watts, the main improver of that machine, was thus a great contributor to speculative knowledge, as well as to practical power. Many of his improvements depended on the laws which regulate the quantity of heat which goes to the formation or condensation of steam; and the observations which led to these improvements enter into the induction of latent heat. Measurements of the force of steam, at all temperatures, were made with the same view. Watts's attention had been drawn to the steam-engine in 1759, by Robison, the former being then an instrument-maker, and the latter a student at the University of Glasgow.[54\10] In 1761 or 1762, he tried some experiments on the force of steam in a Papin's Digester;[55\10] and formed a sort of working model of a steam-engine, feeling already his vocation to develope the powers of that invention. His knowledge was at that time principally derived from Desaguliers and Belidor, but his own experiments added to it rapidly. In 1764 and 1765, he made a more systematical course of

experiments, directed to ascertain the force of steam. He tried this force, however, only at temperatures above the boiling-point; and inferred it at lower degrees from the supposed continuity of the law thus obtained. His friend Robison, also, was soon after led, by reading the account of some experiments of Lord Charles Cavendish, and some others of Mr. Nairne, to examine the same subject. He made out a table of the correspondence of the elasticity and the temperature of vapor, from thirty-two to two hundred and eighty degrees of Fahrenheit's thermometer.[56\10] The thing here to be remarked, is the establishment of a law of the pressure of steam, down to the freezing-point of water. Ziegler of Basle, in 1769, and Achard of Berlin, in 1782, made similar experiments. The latter examined also the elasticity of the vapor of alcohol. Betancourt, in 1792, published his Memoir on the expansive force of vapors; and his tables were for some time considered the most exact. {174} Prony, in his _Architecture Hydraulique_ (1796), established a mathematical formula,[57\10] on the experiments of Betancourt, who began his researches in the belief that he was first in the field, although he afterwards found that he had been anticipated by Ziegler. Gren compared the experiments of Betancourt and De Luc with his own. He ascertained an important fact, that when water _boils_, the elasticity of the steam is equal to that of the atmosphere. Schmidt at Giessen endeavored to improve the apparatus used by Betancourt; and Biker, of Rotterdam, in 1800, made new trials for the same purpose.

[Note 54\10: Robison's _Works_, vol. ii. p. 113.]

[Note 55\10: Denis Papin, who made many of Boyle's experiments for him, had discovered that if the vapor be prevented from rising, the water becomes hotter than the usual boiling-point; and had hence invented the instrument called _Papin's Digester_. It is described in his book, _La manière d'amolir les os et de faire cuire toutes sorts de viandes en fort peu de temps et à peu de frais_. Paris, 1682.]

[Note 56\10: These were afterwards published in the _Encyclopædia Britannica_; in the article "Steam," written by Robison.]

[Note 57\10: _Architecture Hydraulique_, Seconde Partie, p. 163.]

In 1801, Mr. Dalton communicated to the Philosophical Society of Manchester his investigations on this subject; observing truly, that though the forces at high temperatures are most important when steam is considered as a mechanical agent, the progress of philosophy is more immediately interested in accurate observations on the force at low temperatures. He also found that his elasticities for equidistant temperatures resembled a _geometrical progression_, but with a ratio constantly diminishing. Dr. Ure, in 1818, published in the _Philosophical Transactions_ of London, experiments of the same kind, valuable from the high temperatures at which they were made, and for the simplicity of his apparatus. The law which he thus obtained approached, like Dalton's, to a _geometrical progression_. Dr. Ure says, that a formula proposed by M. Biot gives an error of near nine inches out of seventy-five, at a temperature of 266 degrees. This is very conceivable, for if the formula be wrong at all, the geometrical progress rapidly inflames the error in the higher portions of the scale. The elasticity of steam, at high temperatures, has also been experimentally

examined by Mr. Southern, of Soho, and Mr. Sharpe, of Manchester. Mr. Dalton has attempted to deduce certain general laws from Mr. Sharpe's experiments; and other persons have offered other rules, as those which govern the force of steam with reference to the temperature: but no rule appears yet to have assumed the character of an established scientific truth. Yet the law of the expansive force of steam is not only required in order that the steam-engine may be employed with safety and to the best advantage; but must also be an important point in every consistent thermotical theory.

[2nd Ed.] [To the experiments on steam made by private physicists, are to be added the experiments made on a grand scale by order of the governments of France and of America, with a view to {175} legislation on the subject of steam-engines. The French experiments were made in 1823, under the direction of a commission consisting of some of the most distinguished members of the Academy of Sciences; namely, MM. de Prony, Arago, Girard, and Dulong. The American experiments were placed in the hands of a committee of the Franklin Institute of the State of Pennsylvania, consisting of Prof. Bache and others, in 1830. The French experiments went as high as 435° of Fahrenheit's thermometer, corresponding to a pressure of 60 feet of mercury, or 24 atmospheres. The American experiments were made up to a temperature of 346°, which corresponded to 274 inches of mercury, more than 9 atmospheres. The extensive range of these experiments affords great advantages for determining the law of the expansive force. The French Academy found that their experiments indicated an increase of the elastic force according to the _fifth_ power of a binominal 1 + _mt_, where _t_ is the temperature. The American Institute were led to a _sixth_ power of a like binominal. Other experimenters have expressed their results, not by powers of the temperature, but by geometrical ratios. Dr. Dalton had supposed that the expansion of mercury being as the square of the true temperature above its freezing-point, the expansive force of steam increases in geometrical ratio for equal increments of temperature. And the author of the article _Steam_ in the Seventh Edition of the _Encyclopædia Britannica_ (Mr. J. S. Russell), has found that the experiments are best satisfied by supposing mercury, as well as steam, to expand in a geometrical ratio for equal increments of the true temperature.

It appears by such calculation, that while dry gas increases in the ratio of 8 to 11, by an increase of temperature from freezing to boiling water; steam in contact with water, by the same increase of temperature above boiling water, has its expansive force increased in the proportion of 1 to 12. By an equal increase of temperature, mercury expands in about the ratio of 8 to 9.

Recently, MM. Magnus of Berlin, Holzmann and Regnault, have made series of observations on the relation between temperature and elasticity of steam.[58\10]

[Note 58\10: See Taylor's _Scientific Memoirs_, Aug. 1845, vol. iv. part xiv., and _Ann. de Chimie_.]

Prof. Magnus measured his temperatures by an air-thermometer; a process which, I stated in the first edition, seemed to afford the best promise of simplifying the law of expansion. His result is, that the {176} elasticity proceeds in a geometric series when

the temperature proceeds in an arithmetical series nearly; the differences of temperature for equal augmentations of the ratio of elasticity being somewhat greater for the higher temperatures.

The forces of the vapors of other liquids in contact with their liquids, determined by Dr. Faraday, as mentioned in Chap. ii. Sect. 1, are analogous to the elasticity of steam here spoken of.]

_Sect._ 5.--_Consequences of the Doctrine of Evaporation.--Explanation of Rain, Dew, and Clouds._

THE discoveries concerning the relations of heat and moisture which were made during the last century, were principally suggested by meteorological inquiries, and were applied to meteorology as fast as they rose. Still there remains, on many points of this subject, so much doubt and obscurity, that we cannot suppose the doctrines to have assumed their final form; and therefore we are not here called upon to trace their progress and connexion. The principles of atmology are pretty well understood; but the difficulty of observing the conditions under which they produce their effects in the atmosphere is so great, that the precise theory of most meteorological phenomena is still to be determined.

We have already considered the answers given to the question: According to what rules does transparent aqueous vapor resume its form of visible water? This question includes, not only the problems of Rain and Dew, but also of Clouds; for clouds are not vapor, but water, vapor being always invisible. An opinion which attracted much notice in its time, was that of Hutton, who, in 1784, endeavored to prove that if two masses of air saturated with transparent vapor at different temperatures are mixed together, the precipitation of water in the form either of cloud or of drops will take place. The reason he assigned for the opinion was this: that the temperature of the mixture is a mean between the two temperatures, but that the force of the vapor in the mixture, which is the mean of the forces of the two component vapors, will be greater than that which corresponds to the mean temperature, since the force increases faster than the temperature;[59\10] and hence some part of the vapor will be precipitated. This doctrine, it will be seen, speaks of vapor as "saturating" air, and is {177} therefore, in this form, inconsistent with Dalton's principle; but it is not difficult to modify the expression so as to retain the essential part of the explanation.

[Note 59\10: _Edin. Trans._ vol. 1. p. 42.]

_Dew._--The principle of a "constituent temperature" of steam, and the explanation of the "dew-point," were known, as we have said (chap. iii. sect. 3,) to the meteorologists of the last century; but we perceive how incomplete their knowledge was, by the very gradual manner in which the consequences of this principle were traced out. We have already noticed, as one of the books which most drew attention to the true doctrine, in this country at least, Dr. Wells's _Essay on Dew_, published in 1814. In this work the author gives an account of the progress of his opinions;[60\10] "I was led," he says, "in the autumn of 1784, by the event of a rude experiment, to think it probable that the formation of dew is attended with the production of cold." This was confirmed by the experiments of others. But some years after, "upon

considering the subject more closely, I began to suspect that Mr. Wilson, Mr. Six, and myself, had all committed an error in regarding the cold which accompanies the dew, as an _effect_ of the formation of the dew." He now considered it rather as the _cause_: and soon found that he was able to account for the circumstances of this formation, many of them curious and paradoxical, by supposing the bodies on which dew is deposited, to be cooled down, by radiation into the clear night-sky, to the proper temperature. The same principle will obviously explain the formation of mists over streams and lakes when the air is cooler than the water; which was put forward by Davy, even in 1810, as a new doctrine, or at least not familiar.

[Note 60\10: _Essay on Dew_, p. 1.]

_Hygrometers._--According as air has more or less of vapor in comparison with that which its temperature and pressure enable it to contain, it is more or less humid; and an instrument which measures the degrees of such a gradation is a _hygrometer_. The hygrometers which were at first invented, were those which measured the moisture by its effect in producing expansion or contraction in certain organic substances; thus De Saussure devised a hair-hygrometer, De Luc a whalebone-hygrometer, and Dalton used a piece of whipcord. All these contrivances were variable in the amount of their indications under the same circumstances; and, moreover, it was not easy to know the physical meaning of the degree indicated. The dew-point, or constituent temperature of the vapor which exists in the air, is, on {178} the other hand, both constant and definite. The determination of this point, as a datum for the moisture of the atmosphere, was employed by Le Roi, and by Dalton (1802), the condensation being obtained by cold water:[61\10] and finally, Mr. Daniell (1812) constructed an instrument, where the condensing temperature was produced by evaporation of ether, in a very convenient manner. This invention (_Daniell's Hygrometer_) enables us to determine the quantity of vapor which exists in a given mass of the atmosphere at any time of observation.

[Note 61\10: Daniell, _Met. Ess._ p. 142. _Manch. Mem._ vol. v. p. 581.]

[2nd Ed.] [As a happy application of the Atmological Laws which have been discovered, I may mention the completion of the theory and use of the _Wet-bulb Hygrometer_; an instrument in which, from the depression of temperature produced by wetting the bulb of a thermometer, we infer the further depression which would produce dew. Of this instrument the history is thus summed up by Prof. Forbes:--"Hutton invented the method; Leslie revived and extended it, giving probably the earliest, though an imperfect theory; Gay-Lussac, by his excellent experiments and reasoning from them, completed the theory, so far as perfectly dry air is concerned; Ivory extended the theory; which was reduced to practice by Auguste and Bohnenberger, who determined the constant with accuracy. English observers have done little more than confirm the conclusions of our industrious Germanic neighbors; nevertheless the experiments of Apjohn and Prinsep must ever be considered as conclusively settling the value of the coefficient near the one extremity of the scale, as those of Kæmtz have done for the other."[62\10]

[Note 62\10: _Second Report on Meteorology_, p. 101.]

Prof. Forbes's two Reports _On the Recent Progress and Present State of Meteorology_ given among the _Reports of the British Association_ for 1832 and 1840, contain a complete and luminous account of recent researches on this subject. It may perhaps be asked why I have not given Meteorology a place among the Inductive Sciences; but if the reader refers to these accounts, or any other adequate view of the subject, he will see that Meteorology is not a single Inductive Science, but the application of several sciences to the explanation of terrestrial and atmospheric phenomena. Of the sciences so applied, Thermotics and Atmology are the principal ones. But others also come into play; as Optics, in the explanation of Rainbows, Halos, {179} Parhelia, Coronæ, Glories, and the like; Electricity, in the explanation of Thunder and Lightning, Hail, Aurora Borealis; to which others might be added.]

_Clouds._--When vapor becomes visible by being cooled below its constituent temperature, it forms itself into a very fine watery powder, the diameter of the particles of which this powder consists being very small: they are estimated by various writers, from 1⁄100,000th to 1⁄20,000th of an inch.[63\10] Such particles, even if solid, would descend very slowly; and very slight causes would suffice for their suspension, without recurring to the hypothesis of vesicles, of which we have already spoken. Indeed that hypothesis will not explain the fact, except we suppose these vesicles filled with a rarer air than that of the atmosphere; and, accordingly, though this hypothesis is still maintained by some,[64\10] it is asserted as a fact of observation, proved by optical or other phenomena, and not deduced from the suspension of clouds. Yet the latter result is still variously explained by different philosophers: thus, M. Gay-Lussac[65\10] accounts for it by upward currents of air, and Fresnel explains it by the heat and rarefaction of air in the interior of the cloud.

[Note 63\10: Kæmtz, _Met._ i. 393.]

[Note 64\10: Ib. i. 393. Robison, ii. 13.]

[Note 65\10: _Ann. Chim._ xxv. 1822.]

_Classification of Clouds._--A classification of clouds can then only be consistent and intelligible when it rests upon their atmological conditions. Such a system was proposed by Mr. Luke Howard, in 1802-3. His primary modifications are, _Cirrus_, _Cumulus_, and _Stratus_, which the Germans have translated by terms equivalent in English to _feather-cloud_, _heap-cloud_, and _layer-cloud_. The cumulus increases by accumulations on its top, and floats in the air with a horizontal base; the stratus grows from below, and spreads along the earth; the cirrus consists of fibres in the higher regions of the atmosphere, which grow every way. Between the simple modifications are intermediate ones, _cirro-cumulus_ and _cirro-stratus_; and, again, compound ones, the _cumulo-stratus_ and the _nimbus_, or _rain-cloud_. These distinctions have been generally accepted all over Europe: and have rendered a description of all the processes which go on in the atmosphere far more definite and clear than it could be made before their use.

I omit a mass of facts and opinions, supposed laws of phenomena and assigned causes, which abound in meteorology more than in any other science. The slightest consideration will show us what a great {180} amount of labor, of persevering and

combined observation, the progress of this branch of knowledge requires. I do not even speak of the condition of the more elevated parts of the atmosphere. The diminution of temperature as we ascend, one of the most marked of atmospheric facts, has been variously explained by different writers. Thus Dalton[66\10] (1808) refers it to a principle "that each atom of air, in the same perpendicular column, is possessed of the same degree of heat," which principle he conceives to be entirely empirical in this case. Fourier says[67\10] (1817), "This phenomenon results from several causes: one of the principal is the progressive extinction of the rays of heat in the successive strata of the atmosphere."

[Note 66\10: _New Syst. of Chem._ vol. i. p. 125.]

[Note 67\10: _Ann. Chim._ vi. 285.]

Leaving, therefore, the application of thermotical and atmological principles in particular cases, let us consider for a moment the general views to which they have led philosophers.

## CHAPTER IV.

PHYSICAL THEORIES OF HEAT.

WHEN we look at the condition of that branch of knowledge which, according to the phraseology already employed, we must call _Physical Thermotics_, in opposition to Formal Thermotics, which gives us detached laws of phenomena, we find the prospect very different from that which was presented to us by physical astronomy, optics, and acoustics. In these sciences, the maintainers of a distinct and comprehensive theory have professed at least to show that it explains and includes the principal laws of phenomena of various kinds; in Thermotics, we have only attempts to explain a part of the facts. We have here no example of an hypothesis which, assumed in order to explain one class of phenomena, has been found also to account exactly for another; as when central forces led to the precession of the equinoxes, or when the explanation of polarization explained also double refraction; or when the pressure of the atmosphere, as measured by the barometer, gave the true velocity of sound. Such coincidences, or _consiliences_, as I have elsewhere called them, are the test of truth; and thermotical theories cannot yet exhibit credentials of this kind. {181}

On looking back at our view of this science, it will be seen that it may be distinguished into two parts; the Doctrines of Conduction and Radiation, which we call Thermotics proper; and the Doctrines respecting the relation of Heat, Airs, and Moisture, which we have termed Atmology. These two subjects differ in their bearing on our hypothetical views.

_Thermotical Theories._--The phenomena of radiant heat, like those of radiant light, obviously admit of general explanation in two different ways;--by the emission of material particles, or by the propagation of undulations. Both these opinions have found supporters. Probably most persons, in adopting Prevost's theory of exchanges, conceive the radiation of heat to be the radiation of matter. The undulation

hypothesis, on the other hand, appears to be suggested by the production of heat by friction, and was accordingly maintained by Rumford and others. Leslie[68\10] appears, in a great part of his _Inquiry_, to be a supporter of some undulatory doctrine, but it is extremely difficult to make out what his undulating medium is; or rather, his opinions wavered during his progress. In page 31, he asks, "What is this calorific and frigorific fluid? and after keeping the reader in suspense for a moment, he replies, "Quod petis hic est. It is merely the ambient AIR." But at page 150, he again asks the question, and, at page 188, he answers, "It is the same subtile matter that, according to its different modes of existence, constitutes either heat or light." A person thus vacillating between two opinions, one of which is palpably false, and the other laden with exceeding difficulties which he does not even attempt to remove, had little right to protest against[69\10] "the sportive freaks of some intangible _aura_;" to rank all other hypotheses than his own with the "occult qualities of the schools;" and to class the "prejudices" of his opponents with the tenets of those who maintained the _fuga vacui_ in opposition to Torricelli. It is worth while noticing this kind of rhetoric, in order to observe, that it may be used just as easily on the wrong side as on the right.

[Note 68\10: _An Experimental Inquiry into the Nature and Propagation of Heat_, 1804.]

[Note 69\10: Ib. p. 47.]

Till recently, the theory of material heat, and of its propagation by emission, was probably the one most in favor with those who had studied mathematical thermotics. As we have said, the laws of {182} conduction, in their ultimate analytical form, were almost identical with the laws of motion of fluids. Fourier's principle also, that the radiation of heat takes place from points below the surface, and is intercepted by the superficial particles, appears to favor the notion of material emission.

Accordingly, some of the most eminent modern French mathematicians have accepted and extended the hypothesis of a material caloric. In addition to Fourier's doctrine of molecular extra-radiation, Laplace and Poisson have maintained the hypothesis of _molecular intra-radiation_, as the mode in which conduction takes place; that is, they say that the particles of bodies are to be considered as _discrete_, or as points separated from each other, and acting on each other at a distance; and the conduction of heat from one part to another, is performed by radiation between all neighboring particles. They hold that, without this hypothesis, the differential equations expressing the conditions of conduction cannot be made homogeneous: but this assertion rests, I conceive, on an error, as Fourier has shown, by dispensing with the hypothesis. The necessity of the hypothesis of discrete molecular action in bodies, is maintained in all cases by M. Poisson; and he has asserted Laplace's theory of capillary attraction to be defective on this ground, as Laplace asserted Fourier's reasoning respecting heat to be so. In reality, however, this hypothesis of discrete molecules cannot be maintained as a physical truth; for the law of molecular action, which is assumed in the reasoning, after answering its purpose in the progress of calculation, vanishes in the result; the conclusion is the same, whatever law of the intervals of the molecules be assumed. The definite integral, which expresses the

whole action, no more proves that this action is actually made of the differential parts by means of which it was found, than the processes of finding the weight of a body by integration, prove it to be made up of differential weights. And therefore, even if we were to adopt the emission theory of heat, we are by no means bound to take along with it the hypothesis of discrete molecules.

But the recent discovery of the refraction, polarization, and depolarization of heat, has quite altered the theoretical aspect of the subject, and, almost at a single blow, ruined the emission theory. Since heat is reflected and refracted like light, analogy would lead us to conclude that the mechanism of the processes is the same in the two cases. And when we add to these properties the property of polarization, it is scarcely possible to believe otherwise than that heat consists in {183} transverse vibrations; for no wise philosopher would attempt an explanation by ascribing poles to the emitted particles, after the experience which Optics affords, of the utter failure of such machinery.

But here the question occurs, If heat consists in vibrations, whence arises the extraordinary identity of the laws of its propagation with the laws of the flow of matter? How is it that, in conducted heat, this vibration creeps slowly from one part of the body to another, the part first heated remaining hottest; instead of leaving its first place and travelling rapidly to another, as the vibrations of sound and light do? The answer to these questions has been put in a very distinct and plausible form by that distinguished philosopher, M. Ampère, who published a _Note on Heat and Light considered as the results of Vibratory Motion_,[70\10] in 1834 and 1835; and though this answer is an hypothesis, it at least shows that there is no fatal force in the difficulty.

[Note 70\10: _Bibliothèque Universelle de Genève_, vol. xlix. p. 225. _Ann. Chim._ tom. lvii. p. 434.]

M. Ampère's hypothesis is this; that bodies consist of solid molecules, which may be considered as arranged at intervals in a very rare ether; and that the vibrations of the molecules, causing vibrations of the ether and caused by them, constitute heat. On these suppositions, we should have the phenomena of conduction explained; for if the molecules at one end of a bar be hot, and therefore in a state of vibration, while the others are at rest, the vibrating molecules propagate vibrations in the ether, but these vibrations do not produce heat, except in proportion as they put the quiescent molecules of the bar in vibration; and the ether being very rare compared with the molecules, it is only by the repeated impulses of many successive vibrations that the nearest quiescent molecules are made to vibrate; after which they combine in communicating the vibration to the more remote molecules. "We then find necessarily," M. Ampère adds, "the same equations as those found by Fourier for the distribution of heat, setting out from the same hypothesis, that the temperature or heat transmitted is proportional to the difference of the temperatures."

Since the undulatory hypothesis of heat can thus answer all obvious objections, we may consider it as upon its trial, to be confirmed or modified by future discoveries; and especially by an enlarged knowledge of the laws of the polarization of heat.

[2nd Ed.] [Since the first edition was written, the analogies between light and heat have been further extended, as I have already stated. It {184} has been discovered by MM. Biot and Melloni that quartz impresses a circular polarization upon heat; and by Prof. Forbes that mica, of a certain thickness, produces phenomena such as would be produced by the impression of circular polarization of the supposed transversal vibrations of radiant heat; and further, a rhomb of rock-salt, of the shape of the glass rhomb which verified Fresnel's extraordinary anticipation of the circular polarization of light, verified the expectation, founded upon other analogies, of the polarization of heat. By passing polarized heat through various thicknesses of mica, Prof. Forbes has attempted to calculate the length of an undulation for heat.

These analogies cannot fail to produce a strong disposition to believe that light and heat, essences so closely connected that they can hardly be separated, and thus shown to have so many curious properties in common, are propagated by the same machinery; and thus we are led to an Undulatory Theory of Heat.

Yet such a Theory has not yet by any means received full confirmation. It depends upon the analogy and the connexion of the Theory of Light, and would have little weight if those were removed. For the separation of the rays in double refraction, and the phenomena of periodical intensity, the two classes of facts out of which the Undulatory Theory of Optics principally grew, have neither of them been detected in thermotical experiments. Prof. Forbes has assumed alternations of heat for increasing thicknesses of mica, but in his experiments we find only one _maximum_. The occurrence of alternate maxima and minima under the like circumstances would exhibit visible waves of heat, as the fringes of shadows do of light, and would thus add much to the evidence of the theory.

Even if I conceived the Undulatory Theory of Heat to be now established, I should not venture, as yet, to describe its establishment as an event in the history of the Inductive Sciences. It is only at an interval of time after such events have taken place that their history and character can be fully understood, so as to suggest lessons in the Philosophy of Science.]

_Atmological Theories._--Hypotheses of the relations of heat and air almost necessarily involve a reference to the forces by which the composition of bodies is produced, and thus cannot properly be treated of, till we have surveyed the condition of chemical knowledge. But we may say a few words on one such hypothesis; I mean the hypothesis on the subject of the atmological laws of heat, proposed by Laplace, in the twelfth Book of the _Mécanique Céléste_, and published in 1823. {185} It will be recollected that the main laws of phenomena for which we have to account, by means of such an hypothesis, are the following:--

(1.) The law of Boyle and Mariotte, that the elasticity of an air varies as its density. See Chap. iii., Sect. 1 of this Book.

(2.) The Law of Gay-Lussac and Dalton, that all airs expand equally by heat. See Chap. ii. Sect. 1.

(3.) The production of heat by sudden compression. See Chap. ii. Sect. 2.

(4.) Dalton's principle of the mechanical mixture of airs. See Chap. iii. Sect. 3.

(5.) The Law of expansion of solids and fluids by heat. See Chap. ii. Sect. 1.

(6.) Changes of consistence by heat, and the doctrine of latent heat. See Chap. ii. Sect. 3.

(7.) The Law of the expansive force of steam. See Chap. iii. Sect. 4.

Besides these, there are laws of which it is doubtful whether they are or are not included in the preceding, as the low temperature of the air in the higher parts of the atmosphere. (See Chap. iii. Sect. 5.)

Laplace's hypothesis[71\10] is this:--that bodies consist of particles, each of which gathers round it, by its attraction, a quantity of caloric: that the particles of the bodies attract each other, besides attracting the caloric, and that the particles of the caloric repel each other.

[Note 71\10: _Méc. Cél._ t. v. p. 89.]

In gases, the particles of the bodies are so far removed, that their mutual attraction is insensible, and the matter tends to expand by the mutual repulsion of the caloric. He conceives this caloric to be constantly radiating among the particles; the density of this internal radiation is the _temperature_, and he proves that, on this supposition, the elasticity of the air will be as the density, and as this temperature. Hence follow the three first rules above stated. The same suppositions lead to Dalton's principle of mixtures (4), though without involving his mode of conception; for Laplace says that whatever the mutual action of two gases be, the whole pressure will be equal to the sum of the separate pressures.[72\10] Expansion (5), and the changes of consistence (6), are explained by supposing[73\10] that in solids, the mutual attraction of the particles of the body is the greatest force; in liquids, the attraction of the particles for the caloric; in airs, the repulsion of {186} the caloric. But the doctrine of latent heat again modifies[74\10] the hypothesis, and makes it necessary to include latent heat in the calculation; yet there is not, as we might suppose there would be if the theory were the true one, any confirmation of the hypothesis resulting from the new class of laws thus referred to. Nor does it appear that the hypothesis accounts for the relation between the elasticity and the temperature of steam.

[Note 72\10: Ib. p. 110.]

[Note 73\10: Ib. p. 92.]

[Note 74\10: _Méc. Cél._ t. v. p. 93.]

It will be observed that Laplace's hypothesis goes entirely upon the materiality of heat, and is inconsistent with any vibratory theory; for, as Ampère remarks, "It is clear that if we admit heat to consist in vibrations, it is a contradiction to attribute to heat (or caloric) a repulsive force of the particles which would be a cause of vibration."

An unfavorable judgment of Laplace's Theory of Gases is suggested by looking for that which, in speaking of Optics, was mentioned as the great characteristic of a true theory; namely, that the hypotheses, which were assumed in order to account for one class of facts, are found to explain another class of a different nature:--the consilience of inductions. Thus, in thermotics, the law of an intensity of radiation proportional to the sine of the angle of the ray with the surface, which is founded on direct experiments of radiation, is found to be necessary in order to explain the tendency of neighboring bodies to equality of temperature; and this leads to the higher generalization, that heat is radiant from points below the surface. But in the doctrine of the relation of heat to gases, as delivered by Laplace, there is none of this unexpected confirmation; and though he explains some of the leading laws, his assumptions bear a large proportion to the laws explained. Thus, from the assumption that the repulsion of gases arises from the mutual repulsion of the particles of caloric, he finds that the pressure in any gas is as the square of the density and of the quantity of caloric;[75\10] and from the assumption that the temperature is the internal radiation, he finds that this temperature is as the density and the square of the caloric.[76\10] Hence he obtains the law of Boyle and Mariotte, and that of Dalton and Gay-Lussac. But this view of the subject requires other assumptions when we come to latent heat; and accordingly, he introduces, to express the latent heat, a new quantity.[77\10] Yet this quantity produces no effect on his calculations, nor does he apply his reasoning to any problem in which latent heat is concerned. {187}

[Note 75\10: P = 2 π H K ρ^2_c_^2 (1) p. 107.]

[Note 76\10: _q_' Π (_a_) = ρ_c_^2 (2) p. 108.]

[Note 77\10: The quantity _i_, p. 113.]

Without, then, deciding upon this theory, we may venture to say that it is wanting in all the prominent and striking characteristics which we have found in those great theories which we look upon as clearly and indisputably established.

_Conclusion._--We may observe, moreover, that heat has other bearings and effects, which, as soon as they have been analysed into numerical laws of phenomena, must be attended to in the formation of thermotical theories. Chemistry will probably supply many such; those which occur to us, we must examine hereafter. But we may mention as examples of such, MM. De la Rive and Marcet's law, that the specific heat of all gases is the same;[78\10] and MM. Dulong and Petit's law, that single atoms of all simple bodies have the same capacity for heat.[79\10] Though we have not yet said anything of the relation of different gases, or explained the meaning of _atoms_ in the chemical sense, it will easily be conceived that these are very general and important propositions.

[Note 78\10: _Ann. Chim._ xxxv. (1827.)]

[Note 79\10: Ib. x. 397.]

Thus the science of Thermotics, imperfect as it is, forms a highly-instructive part of our survey; and is one of the cardinal points on which the doors of those chambers of

physical knowledge must turn which hitherto have remained closed. For, on the one hand, this science is related by strong analogies and dependencies to the most complete portions of our knowledge, our mechanical doctrines and optical theories; and on the other, it is connected with properties and laws of a nature altogether different,--those of chemistry; properties and laws depending upon a new system of notions and relations, among which clear and substantial general principles are far more difficult to lay hold of and with which the future progress of human knowledge appears to be far more concerned. To these notions and relations we must now proceed; but we shall find an intermediate stage, in certain subjects which I shall call the _Mechanico-chemical_ Sciences; viz., those which have to do with Magnetism, Electricity, and Galvanism.

{{189}} BOOK XI.

# _THE MECHANICO-CHEMICAL SCIENCES._

HISTORY OF ELECTRICITY.

PARVA metu primo: mox sese extollit in auras, Ingrediturque solo, et caput inter nubila condit. _Æn._ iv. 176.

A timid breath at first, a transient touch, How soon it swells from little into much! Runs o'er the ground, and springs into the air, And fills the tempest's gloom, the lightning's glare; While denser darkness than the central storm Conceals the secrets of its inward form.

{{191}} INTRODUCTION.

_Of the Mechanico-Chemical Sciences._

UNDER the title of Mechanico-Chemical Sciences, I include the laws of Magnetism, Electricity, Galvanism, and the other classes of phenomena closely related to these, as Thermo-electricity. This group of subjects forms a curious and interesting portion of our physical knowledge; and not the least of the circumstances which give them their interest, is that double bearing upon mechanical and chemical principles, which their name is intended to imply. Indeed, at first sight they appear to be purely Mechanical Sciences; the attractions and repulsions, the pressure and motion, which occur in these cases, are referrible to mechanical conceptions and laws, as completely as the weight or fall of terrestrial bodies, or the motion of the moon and planets. And if the phenomena of magnetism and electricity had directed us only to such laws, the corresponding sciences must have been arranged as branches of mechanics. But we find that, on the other side, these phenomena have laws and bearings of a kind altogether different. Magnetism is associated with Electricity by its mechanical analogies; and, more recently, has been discovered to be still more closely connected with it by physical influence; electric is identified with galvanic agency; but in galvanism, decomposition, or some action of that kind, universally appears; and these appearances lead to very general laws. Now composition and decomposition are the subjects of Chemistry; and thus we find that we are insensibly but irresistibly led into the domain of that science. The highest

generalizations to which we can look, in advancing from the elementary facts of electricity and galvanism, must involve chemical notions; we must therefore, in laying out the platform of these sciences, make provision for that convergence of mechanical and chemical theory, which they are to exhibit as we ascend.

We must begin, however, with stating the mechanical phenomena of these sciences, and the reduction of such phenomena to laws. In this point of view, the phenomena of which we have to speak are those in which bodies exhibit attractions and repulsions, peculiarly determined by their nature and circumstances; as the magnet, and a {192} piece of amber when rubbed. Such results are altogether different from the universal attraction which, according to Newton's discovery, prevails among all particles of matter, and to which cosmical phenomena are owing. But yet the difference of these special attractions, and of cosmical attraction, was at first so far from being recognized, that the only way in which men could be led to conceive or assent to an action of one body upon another at a distance, in cosmical cases, was by likening it to magnetic attraction, as we have seen in the history of Physical Astronomy. And we shall, in the first part of our account, not dwell much upon the peculiar conditions under which bodies are magnetic or electric, since these conditions are not readily reducible to mechanical laws; but, taking the magnetic or electric character for granted, we shall trace its effects.

The habit of considering magnetic action as the type or general case of attractive and repulsive agency, explains the early writers having spoken of Electricity as a kind of Magnetism. Thus Gilbert, in his book _De Magnete_ (1600), has a chapter,[1\11] _De coitione Magniticâ, primumque de Succini attractione, sive verius corporum ad Succinum applicatione_. The manner in which he speaks, shows us how mysterious the fact of attraction then appeared; so that, as he says, "the magnet and amber were called in aid by philosophers as illustrations, when our sense is in the dark in abstruse inquiries, and when our reason can go no further. Gilbert speaks of these phenomena like a genuine inductive philosopher, reproving[2\11] those who before him had "stuffed the booksellers' shops by copying from one another extravagant stories concerning the attraction of magnets and amber, without giving any reason from experiment." He himself makes some important steps in the subject. He distinguishes magnetic from _electric_ forces,[3\11] and is the inventor of the latter name, derived from ἤλεκτρον, _electron_, amber. He observes rightly, that the electric force attracts all light bodies, while the magnetic force attracts iron only; and he devises a satisfactory apparatus by which this is shown. He gives[4\11] a considerable list of bodies which possess the electric property; "Not only amber and agate attract small bodies, as some think, but diamond, sapphire, carbuncle, opal, amethyst, Bristol gem, beryli, crystal, glass, glass of antimony, spar of various kinds, sulphur, mastic, sealing-wax," and other substances which he mentions. Even his speculations on the general laws of these phenomena, though vague and erroneous, as {193} at that period was unavoidable, do him no discredit when compared with the doctrines of his successors a century and a half afterwards. But such speculations belong to a succeeding part of this history.

[Note 1\11: Lib. ii. cap. 2.]

[Note 2\11: _De Magnete_, p. 48.]

[Note 3\11: Ib. p. 52.]

[Note 4\11: Ib. p. 48.]

In treating of these Sciences, I will speak of Electricity in the first place; although it is thus separated by the interposition of Magnetism from the succeeding subjects (Galvanism, &c.) with which its alliance seems, at first sight, the closest, and although some general notions of the laws of magnets were obtained at an earlier period than a knowledge of the corresponding relations of electric phenomena: for the theory of electric attraction and repulsion is somewhat more simple than of magnetic; was, in fact, the first obtained; and was of use in suggesting and confirming the generalization of magnetic laws.

## CHAPTER 1.

DISCOVERY OF LAWS OF ELECTRIC PHENOMENA.

WE have already seen what was the state of this branch of knowledge at the beginning of the seventeenth century; and the advances made by Gilbert. We must now notice the additions which it subsequently received, and especially those which led to the discovery of general laws, and the establishment of the theory; events of this kind being those of which we have more peculiarly to trace the conditions and causes. Among the facts which we have thus especially to attend to, are the electric attractions of small bodies by amber and other substances when rubbed. Boyle, who repeated and extended the experiments of Gilbert, does not appear to have arrived at any new general notions; but Otto Guericke of Magdeburg, about the same time, made a very material step, by discovering that there was an electric force of repulsion as well as of attraction. He found that when a globe of sulphur had attracted a feather, it afterwards repelled it, till the feather had been in contact with some other body. This, when verified under a due generality of circumstances, forms a capital fact in our present subject. Hawkesbee, who wrote in 1709 (_Physico-Mechanical Experiments_) also observed various of the effects of attraction and repulsion upon threads hanging loosely. But the person who appears to have first fully seized the general law of these facts, is {194} Dufay, whose experiments appear in the Memoirs of the French Academy, in 1733, 1734, and 1737.[5\11] "I discovered," he says, "a very simple principle, which accounts for a great part of the irregularities, and, if I may use the term, the caprices that seem to accompany most of the experiments in electricity. This principle is, that electric bodies attract all those that are not so, and repel them as soon as they are become electric by the vicinity or contact of the electric body. . . . Upon applying this principle to various experiments of electricity, any one will be surprised at the number of obscure and puzzling facts which it clears up." By the help of this principle, he endeavors to explain several of Hawkesbee's experiments.

[Note 5\11: Priestley's _History of Electricity_, p. 45, and the Memoirs quoted.]

A little anterior to Dufay's experiments were those of Grey, who, in 1729, discovered the properties of _conductors_. He found that the attraction and repulsion which appear in electric bodies are exhibited also by other bodies in contact with the electric. In this manner he found that an ivory ball, connected with a glass tube by a stick, a wire, or a packthread, attracted and repelled a feather, as the glass itself would have done. He was then led to try to extend this communication to considerable distances, first by ascending to an upper window and hanging down his ball, and, afterwards, by carrying the string horizontally supported on loops. As his success was complete in the former case, he was perplexed by failure in the latter; but when he supported the string by loops of silk instead of hempen cords, he found it again become a conductor of electricity. This he ascribed at first to the smaller thickness of the silk, which did not carry off so much of the electric virtue; but from this explanation he was again driven, by finding that wires of brass still thinner than the silk destroyed the effect. Thus Grey perceived that the efficacy of the support depended on its being silk, and he soon found other substances which answered the same purpose. The difference, in fact, depended on the supporting substance being electric, and therefore not itself a conductor; for it soon appeared from such experiments, and especially[6\11] from those made by Dufay, that substances might be divided into _electrics per se_, and _non-electrics_, or _conductors_. These terms were introduced by Desaguliers,[7\11] and gave a permanent currency to the results of the labors of Grey and others.

[Note 6\11: _Mém. Acad. Par._ 1734.]

[Note 7\11: Priestley, p. 66.]

Another very important discovery belonging to this period is, that {195} of the two kinds of electricity. This also was made by Dufay. "Chance," says he, "has thrown in my way another principle more universal and remarkable than the preceding one, and which casts a new light upon the subject of electricity. The principle is, that there are two distinct kinds of electricity, very different from one another; one of which I call _vitreous_, the other _resinous_, electricity. The first is that of glass, gems, hair, wool, &c.; the second is that of amber, gum-lac, silk, &c. The characteristic of these two electricities is, that they repel themselves and attract each other." This discovery does not, however, appear to have drawn so much attention as it deserved. It was published in 1735; (in the Memoirs of the Academy _for_ 1733;) and yet in 1747, Franklin and his friends at Philadelphia, who had been supplied with electrical apparatus and information by persons in England well acquainted with the then present state of the subject, imagined that they were making observations unknown to European science, when they were led to assert two conditions of bodies, which were in fact the opposite electricities of Dufay, though the American experimenters referred them to a single element, of which electrized bodies might have either excess or defect. "Hence," Franklin says, "have arisen some new terms among us: we say B," who receives a spark from glass, "and bodies in like circumstances, is electrized _positively_; A," who communicates his electricity to glass, "_negatively_; or rather B is electrized _plus_, A _minus_." Dr. (afterwards Sir William) Watson had, about the same time, arrived at the same conclusions, which he expresses by saying

that the electricity of A was _more rare_, and that of B _more dense_, than it naturally would have been.[8\11] But that which gave the main importance to this doctrine was its application to some remarkable experiments, of which we must now speak.

[Note 8\11: Priestley, p. 115.]

Electric action is accompanied, in many cases, by light and a crackling sound. Otto Guericke[9\11] observes that his sulphur-globe, when rubbed in a dark place, gave faint flashes, such as take place when sugar is crushed. And shortly after, a light was observed at the surface of the mercury in the barometer, when shaken, which was explained at first by Bernoulli, on the then prevalent Cartesian principles; but, afterwards, more truly by Hawkesbee, as an electrical phenomenon. Wall, in 1708, found sparks produced by rubbing amber, and Hawkesbee observed the light and the _snapping_, as he calls it, under various modifications. But the electric spark from a living body, which, as {196} Priestley says,[10\11] "makes a principal part of the diversion of gentlemen and ladies who come to see experiments in electricity," was first observed by Dufay and the Abbé Nollet. Nollet says[11\11] he "shall never forget the surprise which the first electric spark ever drawn from the human body excited, both in M. Dufay and in himself." The drawing of a spark from the human body was practised in various forms, one of which was familiarly known as the "electrical kiss." Other exhibitions of electrical light were the electrical star, electrical rain, and the like.

[Note 9\11: _Experimenta Magdeburgica_, 1672, lib. iv. cap. 15.]

[Note 10\11: P. p. 47.]

[Note 11\11: Priestley, p. 47. Nollet, _Leçons de Physique_, vol. vi. p. 408.]

As electricians determined more exactly the conditions of electrical action, they succeeded in rendering more intense those sudden actions which the spark accompanies, and thus produced the electric _shock_. This was especially done in the _Leyden phial_. This apparatus received its name, while the discovery of its property was attributed to Cunæus, a native of Leyden, who, in 1746, handling a vessel containing water in communication with the electrical machine, and happening thus to bring the inside and the outside into connexion, received a sudden shock in his arms and breast. It appears, however,[12\11] that a shock had been received under nearly the same circumstances in 1746, by Von Kleist, a German prelate, at Camin, in Pomerania. The strangeness of this occurrence, and the suddenness of the blow, much exaggerated the estimate which men formed of its force. Muschenbroek, after taking one shock, declared he would not take a second for the kingdom of France; though Boze, with a more magnanimous spirit, wished[13\11] that he might die by such a stroke, and have the circumstances of the experiment recorded in the Memoirs of the Academy. But we may easily imagine what a new fame and interest this discovery gave to the subject of electricity. It was repeated in all parts of the world, with various modifications: and the shock was passed through a line of several persons holding hands; Nollet, in the presence of the king of France, sent it through a circle of 180 men of the guards, and along a line of men and wires of 900 toises;[14\11] and experiments of the same kind were made in England, principally

under the direction of Watson, on a scale so large as to excite the admiration of Muschenbroek; who says, in a letter to Watson, "Magnificentissimis tuis experimentis superasti conatus omnium." The result was, that the transmission of electricity through a length of 12,000 feet was, to sense, instantaneous. {197}

[Note 12\11: Fischer, v. 490.]

[Note 13\11: Fischer, p. 84.]

[Note 14\11: Ibid. v. 512.]

The essential circumstances of the electric shock were gradually unravelled. Watson found that it did not increase in proportion either to the contents of the phial or the size of the globe by which the electricity was excited; that the outside coating of the glass (which, in the first form of the experiment, was only a film of water), and its contents, might be varied in different ways. To Franklin is due the merit of clearly pointing out most of the circumstances on which the efficacy of the Leyden phial depends. He showed, in 1747,[15\11] that the inside of the bottle is electrized positively, the outside negatively; and that the shock is produced by the restoration of the equilibrium, when the outside and inside are brought into communication suddenly. But in order to complete this discovery, it remained to be shown that the electric matter was collected entirely at the surface of the glass, and that the opposite electricities on the two opposite sides of the glass were accumulated by their mutual attraction. Monnier the younger discovered that the electricity which bodies can receive, depends upon their surface rather than their mass, and Franklin[16\11] soon found that "the whole force of the bottle, and power of giving a shock, is in the glass itself." This they proved by decanting the water out of an electrized into another bottle, when it appeared that the second bottle did not become electric, but the first remained so. Thus it was found "that the non-electrics, in contact with the glass, served only to unite the force of the several parts."

[Note 15\11: _Letters_, p. 13.]

[Note 16\11: _Letters_, iv. Sect. 16.]

So far as the effect of the coating of the Leyden phial is concerned, this was satisfactory and complete: but Franklin was not equally successful in tracing the action of the electric matter upon itself, in virtue of which it is accumulated in the phial; indeed, he appears to have ascribed the effect to some property of the glass. The mode of describing this action varied, accordingly as two electric _fluids_ were supposed (with Dufay,) or one, which was the view taken by Franklin. On this latter supposition the parts of the electric fluid repel each other, and the excess in one surface of the glass expels the fluid from the other surface. This kind of action, however, came into much clearer view in the experiments of Canton, Wilcke, and Æpinus. It was principally manifested in the attractions and repulsions which objects exert when they are in the neighborhood of electrized bodies; or in the _electrical atmosphere_, using the phraseology of the time. At present we say that bodies are electrized _by induction_, when they are {198} thus made electric by the electric attraction and repulsion of other bodies. Canton's experiments were communicated to the Royal Society in 1753, and show that the electricity on each body acts upon the

electricity of another body, at a distance, with a repulsive energy. Wilcke, in like manner, showed that parts of non-electrics, plunged in electric atmospheres, acquire an electricity opposite to that of such atmospheres. And Æpinus devised a method of examining the nature of the electricity at any part of the surface of a body, by means of which he ascertained its distribution, and found that it agreed with such a law of self-repulsion. His attempt to give mathematical precision to this induction was one of the most important steps towards electrical theory, and must be spoken of shortly, in that point of view. But in the mean time we may observe, that this doctrine was applied to the explanation of the Leyden jar; and the explanation was confirmed by charging a plate of air, and obtaining a shock from it, in a manner which the theory pointed out.

Before we proceed to the history of the theory, we must mention some other of the laws of phenomena which were noticed, and which theory was expected to explain. Among the most celebrated of these, were the effect of sharp points in conductors, and the phenomena of electricity in the atmosphere. The former of these circumstances was one of the first which Franklin observed as remarkable. It was found that the points of needles and the like throw off and draw off the electric virtue; thus a bodkin, directed towards an electrized ball, at six or eight inches' distance, destroyed its electric action. The latter subject, involving the consideration of thunder and lightning, and of many other meteorological phenomena, excited great interest. The comparison of the electric spark to lightning had very early been made; but it was only when the discharge had been rendered more powerful in the Leyden jar, that the comparison of the effects became very plausible. Franklin, about 1750, had offered a few somewhat vague conjectures[17\11] respecting the existence of electricity in the clouds; but it was not till Wilcke and Æpinus had obtained clear notions of the effect of electric matter at a distance, that the real condition of the clouds could be well understood. In 1752, however,[18\11] D'Alibard, and other French philosophers, were desirous of verifying Franklin's conjecture of the analogy of thunder and electricity. This they did by erecting a pointed iron rod, forty feet high, {199} at Marli: the rod was found capable of giving out electrical sparks when a thunder-cloud passed over the place. This was repeated in various parts of Europe, and Franklin suggested that a communication with the clouds might be formed by means of a kite. By these, and similar means, the electricity of the atmosphere was studied by Canton in England, Mazeas in France, Beccaria in Italy, and others elsewhere. These essays soon led to a fatal accident, the death of Richman at Petersburg, while he was, on Aug. 6th, 1753, observing the electricity collected from an approaching thunder-cloud, by means of a rod which he called an electrical gnomon: a globe of blue fire was seen to leap from the rod to the head of the unfortunate professor, who was thus struck dead.

[Note 17\11: Letter v.]

[Note 18\11: Franklin, p. 107.]

[2nd Ed.] [As an important application of the doctrines of electricity, I may mention the contrivances employed to protect ships from the effects of lightning. The use of conductors in such cases is attended with peculiar difficulties. In 1780 the French

began to turn their attention to this subject, and Le Roi was sent to Brest and the various sea-ports of France for that purpose. Chains temporarily applied in the rigging had been previously suggested, but he endeavored to place, he says, such conductors in ships as might be fixed and durable. He devised certain long linked rods, which led from a point in the mast-head along a part of the rigging, or in divided stages along the masts, and were fixed to plates of metal in the ship's sides communicating with the sea. But these were either unable to stand the working of the rigging, or otherwise inconvenient, and were finally abandoned.[19\11]

[Note 19\11: See Le Roi's Memoir in the _Hist. Acad. Sc._ for 1790.]

The conductor commonly used in the English Navy, till recently, consisted of a flexible copper chain, tied, when occasion required, to the mast-head, and reaching down into the sea; a contrivance recommended by Dr. Watson in 1762. But notwithstanding this precaution, the shipping suffered greatly from the effects of lightning.

Mr. Snow Harris (now Sir William Snow Harris), whose electrical labors are noticed above, proposed to the Admiralty, in 1820, a plan which combined the conditions of ship-conductors, so desirable, yet so difficult to secure:--namely, that they should be permanently fixed, and sufficiently large, and yet should in no way interfere with the motion of the rigging, or with the sliding masts. The method which he proposed was to make the masts themselves conductors of electricity, {200} by incorporating with them, in a peculiar way, two laminæ of sheet-copper, uniting these with the metallic masses in the hull by other laminæ, and giving the whole a free communication with the sea. This method was tried experimentally, both on models and to a large extent in the navy itself; and a Commission appointed to examine the result reported themselves highly satisfied with Mr. Harris's plan, and strongly recommended that it should be fully carried out in the Navy.[20\11]]

[Note 20\11: See Mr. Snow Harris's paper in _Phil. Mag._ March, 1841.]

It is not here necessary to trace the study of atmospheric electricity any further: and we must now endeavor to see how these phenomena and laws of phenomena which we have related, were worked up into consistent theories; for though many experimental observations and measures were made after this time, they were guided by the theory, and may be considered as having rather discharged the office of confirming than of suggesting it.

We may observe also that we have now described the period of most extensive activity and interest in electrical researches. These naturally occurred while the general notions and laws of the phenomena were becoming, and were not yet become, fixed and clear. At such a period, a large and popular circle of spectators and amateurs feel themselves nearly upon a level, in the value of their trials and speculations, with more profound thinkers: at a later period, when the subject is become a science, that is, a study in which all must be left far behind who do not come to it with disciplined, informed, and logical minds, the cultivators are far more few, and the shout of applause less tumultuous and less loud. We may add, too, that the experiments, which are the most striking to the senses, lose much of their

impressiveness with their novelty. Electricity, to be now studied rightly, must be reasoned upon mathematically; how slowly such a mode of study makes its way, we shall see in the progress of the theory, which we must now proceed to narrate.

[2nd Ed.] [A new mode of producing electricity has excited much notice lately. In October, 1840, one of the workmen in attendance upon a boiler belonging to the Newcastle and Durham Railway, reported that the boiler was full of fire; the fact being, that when he placed his hand near it an electrical spark was given out. This drew the attention of Mr. Armstrong and Mr. Pattinson, who made the circumstance publicly known.[21\11] Mr. Armstrong pursued the investigation {201} with great zeal, and after various conjectures was able to announce[22\11] that the electricity was excited at the point where the steam is subject to friction in its emission. He found too that he could produce a like effect by the emission of condensed air. Following out his views, he was able to construct, for the Polytechnic Institution in London, a "Hydro-electric Machine," of greater power than any electrical machine previously made. Dr. Faraday took up the investigation as the subject of the Eighteenth Series of his _Researches_, sent to the Royal Society, Jan. 26, 1842; and in this he illustrated, with his usual command of copious and luminous experiments, a like view;--that the electricity is produced by the friction of the particles of the water carried along by the **steam. And thus this is a new manifestation of that electricity, which, to distinguish it from voltaic electricity, is sometimes called _Friction Electricity_ or _Machine Electricity_. Dr. Faraday has, however, in the course of this investigation, brought to light several new electrical relations of bodies.]

[Note 21\11: _Phil. Mag._ Oct 1840.]

[Note 22\11: _Phil. Mag._ Jan. 1848, dated Dec. 9, 1841.]

## CHAPTER II.

THE PROGRESS OF ELECTRICAL THEORY.

THE cause of electrical phenomena, and the mode of its operation, were naturally at first spoken of in an indistinct and wavering manner. It was called the electric _fire_, the electric _fluid_; its effects were attributed to _virtues_, _effluvia_, _atmospheres_. When men's mechanical ideas became somewhat more distinct, the motions and tendencies to motion were ascribed to _currents_, in the same manner as the cosmical motions had been in the Cartesian system. This doctrine of currents was maintained by Nollet, who ascribed all the phenomena of electrized bodies to the contemporaneous afflux and efflux of electrical matter. It was an important step towards sound theory, to get rid of this notion of moving fluids, and to consider attraction and repulsion as statical forces; and this appears to have been done by others about the same time. Dufay[23\11] considered that he had proved the existence of two electricities, the vitreous and the resinous, and conceived each {202} of these to be a fluid which repelled its own parts and attracted those of the other: this is, in fact, the outline of the theory which recently has been considered as the best established; but from various causes it was not at once, or at least not generally adopted. The hypothesis of the excess and defect of a single fluid is capable

of being so treated as to give the same results with the hypothesis of two opposite fluids and happened to obtain the preference for some time. We have already seen that this hypothesis, according to which electric phenomena arose from the excess and defect of a generally diffused fluid, suggested itself to Watson and Franklin about 1747. Watson found that when an electric body was excited, the electricity was not created, but collected; and Franklin held, that when the Leyden jar was charged, the quantity of electricity was unaltered, though its distribution was changed. Symmer[24\11] maintained the existence of two fluids; and Cigna supplied the main defect which belonged to this tenet in the way in which Dufay held it, by showing that the two opposite electricities were usually produced at the same time. Still the apparent simplicity of the hypothesis of one fluid procured it many supporters. It was that which Franklin adopted, in his explanation of the Leyden experiment; and though after the first conception of an electrical charge as a disturbance of equilibrium, there was nothing in the development or details of Franklin's views which deserved to win for them any peculiar authority, his reputation, and his skill as a writer, gave a considerable influence to his opinions. Indeed, for a time he was considered, over a large part of Europe, as the creator of the science, and the terms[25\11] _Franklinism_, _Franklinist_, _Franklinian system_, occur in almost every page of continental publications on the subject. Yet the electrical phenomena to the knowledge of which Franklin added least, those of induction, were those by which the progress of the theory was most promoted. These, as we have already said, were at first explained by the hypothesis of electrical atmospheres. Lord Mahon wrote a treatise, in which this hypothesis was mathematically treated; yet the hypothesis was very untenable, for it would not account for the most obvious cases of induction, such as the Leyden jar, except the atmosphere was supposed to penetrate glass.

[Note 23\11: _Ac. Par._ 1733, p. 467]

[Note 24\11: _Phil. Trans._ 1759.]

[Note 25\11: Priestley, p. 160.]

The phenomena of electricity by induction, when fairly considered by a person of clear notions of the relations of space and force, were seen to accommodate themselves very generally to the conception {203} introduced by Dufay;[26\11] of two electricities each repelling itself and attracting the other. If we suppose that there is only one fluid, which repels itself and attracts all other matter, we obtain, in many cases, the same general results as if we suppose two fluids; thus, if an electrized body, overcharged with the single fluid, act upon a ball, it drives the electric fluid in the ball to the further side by its repulsion, and then attracts the ball by attracting the matter of the ball more than it repels the fluid which is upon the ball. If we suppose two fluids, the positively electrized body draws the negative fluid to the nearer side of the ball, repels the positive fluid to the opposite side, and attracts the ball on the whole, because the attracted fluid is nearer than that which is repelled. The verification of either of these hypotheses, and the determination of their details, depended necessarily upon experiment and calculation. It was under the hypothesis of a single fluid that this trial was first properly made. Æpinus of Petersburg published, in 1759, his _Tentamen Theoriæ Electricitatis et Magnetismi_;

in which he traces mathematically the consequences of the hypothesis of an electric fluid, attracting all other matter, but repelling itself; the law of force of this repulsion and attraction he did not pretend to assign precisely, confining himself to the supposition that the mutual force of the particles increases as the distance decreases. But it was found, that in order to make this theory tenable, an additional supposition was required, namely, that the particles of bodies repel each other as much as they attract the electric fluid.[27\11] For if two bodies, A and B, be in their natural electrical condition, they neither attract nor repel each other. Now, in this case, the fluid in A attracts the matter in B and repels the fluid in B with equal energy, and thus no tendency to motion results from the fluid in A; and if we further suppose that the _matter_ in A attracts the fluid in B and _repels the matter_ in B with equal energy, we have the resulting mutual inactivity of the two bodies explained; but without the latter supposition, there would be a mutual attraction: or we may put the truth more simply thus; two negatively electrized bodies repel each other; if negative electrization were merely the abstraction of the fluid which is the repulsive element, this result could not follow except there were a repulsion in the bodies themselves, independent of the fluid. And thus Æpinus found himself compelled to assume this mutual repulsion of material particles; he had, in fact, the {204} alternative of this supposition, or that of two fluids, to choose between, for the mathematical results of both hypotheses are the same. Wilcke, a Swede, who had at first asserted and worked out the Æpinian theory in its original form, afterwards inclined to the opinion of Symmer; and Coulomb, when, at a later period, he confirmed the theory by his experiments and determined the law of force, did not hesitate to prefer[28\11] the theory of two fluids, "because," he says, "it appears to me contradictory to admit at the same time, in the particles of bodies, an attractive force in the inverse ratio of the squares of the distances, which is demonstrated by universal gravitation, and a repulsive force in the same inverse ratio of the squares of the distances; a force which would necessarily be infinitely great relatively to the action of gravitation." We may add, that by forcing us upon this doctrine of the universal repulsion of matter, the theory of a single fluid seems quite to lose that superiority in the way of simplicity which had originally been its principal recommendation.

[Note 26\11: _Mém. A. P._ 1733, p. 467.]

[Note 27\11: Robison, vol. iv. p. 18.]

[Note 28\11: _Mém. Ac. P._ 1788, p. 671.]

The mathematical results of the supposition of Æpinus, which are, as Coulomb observes,[29\11] the same as of that of the two fluids, were traced by the author himself in the work referred to, and shown to agree, in a great number of cases, with the observed facts of electrical induction, attraction, and repulsion. Apparently this work did not make its way very rapidly through Europe; for in 1771, Henry Cavendish stated[30\11] the same hypothesis in a paper read before the Royal Society; which he prefaces by saying, "Since I first wrote the following paper, I find that this way of accounting for the phenomena of electricity is not new. Æpinus, in his _Tentamen Theoriæ Electricitatis et Magnetismi_, has made use of the same, or nearly the same hypothesis that I have; and the conclusions he draws from it agree nearly with mine

as far as he goes."

[Note 29\11: _Ac. P._ 1788, p. 672.]

[Note 30\11: _Phil. Trans._ 1771, vol. lxi.]

The confirmation of the theory was, of course, to be found in the agreement of its results with experiment; and in particular, in the facts of electrical induction, attraction, and repulsion, which suggested the theory. Æpinus showed that such a confirmation appeared in a number of the most obvious cases; and to these, Cavendish added others, which, though not obvious, were of such a nature that the calculations, in general difficult or impossible, could in these instances be easily performed; as, for example, cases in which there are plates or globes at the two extremities of a long wire. In all these cases of {205} electrical action the theory was justified. But in order to give it full confirmation, it was to be considered whether any other facts, not immediately assumed in the foundation of the theory, were explained by it; a circumstance which, as we have seen, gave the final stamp of truth to the theories of astronomy and optics. Now we appear to have such confirmation, in the effect of points, and in the phenomena of the electrical discharge. The theory of neither of these was fully understood by Cavendish, but he made an approach to the true view of them. If one part of a conducting body be a sphere of small radius, the electric fluid upon the surface of this sphere will, it appears by calculation, be more dense, and tend to escape more energetically, in proportion as the radius of the sphere is smaller; and, therefore, if we consider a point as part of the surface of a sphere of imperceptible radius, it follows from the theory that the effort of the fluid to escape at that place will be enormous; so that it may easily be supposed to overcome the resisting causes. And the discharge may be explained in nearly the same manner; for when a conductor is brought nearer and nearer to an electrized body, the opposite electricity is more and more accumulated by attraction on the side next to the electrized body; its tension becomes greater by the increase of its quantity and the diminution of the distance, and at last it is too strong to be contained, and leaps out in the form of a spark.

The light, sound, and mechanical effects produced by the electric discharge, made the electric _fluid_ to be not merely considered as a mathematical hypothesis, useful for reducing phenomena to formulæ (as for a long time the magnetic fluid was), but caused it to be at once and universally accepted as a physical reality, of which we learn the existence by the common use of the senses, and of which measures and calculations are only wanted to teach us the laws.

The applications of the theory of electricity which I have principally considered above, are those which belong to conductors, in which the electric fluid is perfectly moveable, and can take that distribution which the forces require. In non-conducting or electric bodies, the conditions to which the fluid is subject are less easy to determine; but by supposing that the fluid moves with great difficulty among the particles of such bodies,--that nevertheless it may be dislodged and accumulated in parts of the surface of such bodies, by friction and other modes of excitement; and that the earth is an inexhaustible reservoir of electric matter,--the principal facts of

excitation and the like receive a tolerably satisfactory explanation. {206}

The theory of Æpinus, however, still required to have the law of action of the particles of the fluid determined. If we were to call to mind how momentous an event in physical astronomy was the determination of the law of the cosmical forces, the inverse square of the distance, and were to suppose the importance and difficulty of the analogous step in this case to be of the same kind, this would be to mistake the condition of science at that time. The leading idea, the conception of the possibility of explaining natural phenomena by means of the action of forces, on rigorously mechanical principles, had already been promulgated by Newton, and was, from the first, seen to be peculiarly applicable to electrical phenomena; so that the very material step of clearly proposing the problem, often more important than the solution of it, had already been made. Moreover the confirmation of the truth of the assumed cause in the astronomical case depended on taking the right law; but the electrical theory could be confirmed, in a general manner at least, without this restriction. Still it was an important discovery that the law of the inverse square prevailed in these as well as in cosmical attractions.

It was impossible not to conjecture beforehand that it would be so. Cavendish had professed in his calculations not to take the exponent of the inverse power, on which the force depended, to be strictly 2, but to leave it indeterminate between 1 and 3; but in his applications of his results, he obviously inclines to the assumption that it is 2. Experimenters tried to establish this in various ways. Robison,[31\11] in 1769, had already proved that the law of force is very nearly or exactly the inverse square; and Meyer[32\11] had discovered, but not published, the same result. The clear and satisfactory establishment of this truth is due to Coulomb, and was one of the first steps in his important series of researches on this subject. In his first paper[33\11] in the _Memoirs_ of the Academy for 1785, he proves this law for small globes; in his second Memoir he shows it to be true for globes one and two feet in diameter. His invention of the _torsion-balance_, which measures very small forces with great certainty and exactness, enabled him to set this question at rest for ever.

[Note 31\11: _Works_, iv. p. 68.]

[Note 32\11: _Biog. Univ._ art. _Coulumb_, by Biot.]

[Note 33\11: _Mém. A. P._ 1785, pp. 569, 578.]

The law of force being determined for the particles of the electric fluid, it now came to be the business of the experimenter and the {207} mathematician to compare the results of the theory in detail with those of experimental measures. Coulomb undertook both portions of the task. He examined the electricity of portions of bodies by means of a little disk (his _tangent plane_) which he applied to them and then removed, and which thus acted as a sort of electric _taster_. His numerical results (the intensity being still measured by the torsion-balance) are the fundamental facts of the theory of the electrical fluid. Without entering into detail, we may observe that he found the electricity to be entirely collected at the surface of conductors (which Beccaria had before shown to be the case), and that he examined and recorded the electric intensity at the surface of globes, cylinders, and other

conducting bodies, placed within each other's influence in various ways.

The mathematical calculation of the distribution of two fluids, all the particles of which attract and repel each other according to the above law, was a problem of no ordinary difficulty; as may easily be imagined, when it is recollected that the attraction and repulsion determine the distribution, and the distribution reciprocally determines the attraction and repulsion. The problem was of the same nature as that of the figure of the earth; and its rigorous solution was beyond the powers of the analysis of Coulomb's time. He obtained, however, approximate solutions with much ingenuity; for instance, in a case in which it was obvious that the electric fluid would be most accumulated at and near the equator of a certain sphere, he calculated the action of the sphere on two suppositions: first, that the fluid was all collected precisely at the equator; and next, that it was uniformly diffused over the surface; and he then assumed the actual case to be intermediate between these two. By such artifices he was able to show that the results of his experiments and of his calculations gave an agreement sufficiently near to entitle him to consider the theory as established on a solid basis.

Thus, at this period, mathematics was behind experiment; and a problem was proposed, in which theoretical numerical results were wanted for comparison with observation, but could not be accurately obtained; as was the case in astronomy also, till the time of the approximate solution of the Problem of Three Bodies, and the consequent formation of the Tables of the Moon and Planets on the theory of universal gravitation. After some time, electrical theory was relieved from this reproach, mainly in consequence of the progress which astronomy had occasioned in pure mathematics. About 1801, {208} there appeared in the _Bulletin des Sciences_,[34\11] an exact solution of the problem of the distribution of electric fluid on a spheroid, obtained by M. Biot, by the application of the peculiar methods which Laplace had invented for the problem of the figure of the planets. And in 1811, M. Poisson applied Laplace's artifices to the case of two spheres acting upon one another in contact, a case to which many of Coulomb's experiments were referrible; and the agreement of the results of theory and observation, thus extricated from Coulomb's numbers, obtained above forty years previously, was very striking and convincing.[35\11] It followed also from Poisson's calculations, that when two electrized spheres are brought near each other, the accumulation of the opposite electricities on their nearest points increases without limit as the spheres approach to contact; so that before the contact takes place, the external resistance will be overcome, and a _spark_ will pass.

[Note 34\11: No. li.]

[Note 35\11: _Mém. A. P._ 1811.]

Though the relations of non-conductors to electricity, and various other circumstances, leave many facts imperfectly explained by the theory, yet we may venture to say that, as a theory which gives the laws of the phenomena, and which determines the distribution of those elementary forces, on the surface of electrized bodies, from which elementary forces (whether arising from the presence of a fluid or

not,) the total effects result, the doctrine of Dufay and Coulomb, as developed in the analysis of Poisson, is securely and permanently established. This part of the subject has been called _statical electricity_. In the establishment of the theory of this branch of science, we must, I conceive, allow to Dufay more merit than is generally ascribed to him; since he saw clearly, and enunciated in a manner which showed that he duly appreciated their capital character, the two chief principles,--the conditions of electrical attraction and repulsion, and the apparent existence of two kinds of electricity. His views of attraction are, indeed, partly expressed in terms of the Cartesian hypothesis of vortices, then prevalent in France; but, at the time when he wrote, these forms of speech indicated scarcely anything besides the power of attraction. Franklin's real merit as a discoverer was, that he was one of the first who distinctly conceived the electrical _charge_ as a derangement of equilibrium. The great fame which, in his day, he enjoyed, arose from the clearness and spirit with which he narrated his discoveries; from his dealing with electricity in the imposing form of thunder and lightning; and partly, perhaps, from his character as an {209} American and a politician; for he was already, in 1736, engaged in public affairs as clerk to the General Assembly of Pennsylvania, though it was not till a later period of his life that his admirers had the occasion of saying of him Eripuit cœlis fulmen sceptrumque tyrannis; Born to control all lawless force, all fierce and baleful sway, The thunder's bolt, the tyrant's rod, alike he wrenched away.

Æpinus and Coulomb were two of the most eminent physical philosophers of the last century, and labored in the way peculiarly required by that generation; whose office it was to examine the results, in particular subjects, of the general conception of attraction and repulsion, as introduced by Newton. The reasonings of the Newtonian period had, in some measure, anticipated all possible theories resembling the electrical doctrine of Æpinus and Coulomb; and, on that account, this doctrine could not be introduced and confirmed in a sudden and striking manner, so as to make a great epoch. Accordingly, Dufay, Symmer, Watson, Franklin, Æpinus and Coulomb, have all a share in the process of induction. With reference to these founders of the theory of electricity, Poisson holds the same place which Laplace holds with reference to Newton.

The reception of the Coulombian theory (so we most call it, for the Æpinian theory implies one fluid only,) has hitherto not been so general as might have been reasonably expected from its very beautiful accordance with the facts which it contemplates. This has partly been owing to the extreme abstruseness of the mathematical reasoning which it employs, and which put it out of the reach of most experimenters and writers of works of general circulation. The theory of Æpinus was explained by Robison in the _Encyclopædia Britannica_; the analysis of Poisson has recently been presented to the public in the _Encyclopædia Metropolitana_, but is of a kind not easily mastered even by most mathematicians. On these accounts probably it is, that in English compilations of science, we find, even to this day, the two theories of one and of two fluids stated as if they were nearly on a par in respect of their experimental evidence. Still we may say that the Coulombian theory is probably assented to by all who have examined it, at least as giving the laws of phenomena; and I have not heard of any denial of it from such a quarter, or of any attempt to show

it to be erroneous by detailed and measured experiments. Mr. Snow Harris {210} has recently[36\11] described some important experiments and measures; but his apparatus was of such a kind that the comparison of the results with the Coulombian theory was not easy; and indeed the mathematical problems which Mr. Harris's combinations offered, require another Poisson for their solution. Still the more obvious results are such as agree with the theory, even in the cases in which their author considered them to be inexplicable. For example, he found that by doubling the quantity of electricity of a conductor, it attracted a body with four times the force; but the body not being insulated, would have its electricity also doubled by induction, and thus the fact was what the theory required.

[Note 36\11: _Phil. Trans._ 1834, p. 2.]

Though it is thus highly probable that the Coulombian theory of electricity (or the Æpinian, which is mathematically equivalent) will stand as a true representation of the law of the elementary actions, we must yet allow that it has not received that complete evidence, by means of experiments and calculations added to those of its founders, which the precedents of other permanent sciences have led us to look for. The experiments of Coulomb, which he used in the establishment of the theory, were not very numerous, and they were limited to a peculiar form of bodies, namely spheres. In order to form the proper _sequel_ to the promulgation of this theory, to give a full _confirmation_, and to ensure its general _reception_, we ought to have experiments more numerous and more varied (such as those of Mr. Harris are) shown to agree in all respects with results calculated from the theory. This would, as we have said, be a task of labor and difficulty; but the person who shall execute it will deserve to be considered as one of the real founders of the true doctrine of electricity. To show that the coincidence between theory and observation, which has already been proved for spherical conductors, obtains also for bodies of other forms, will be a step in electricity analogous to what was done in astronomy, when it was shown that the law of gravitation applied to comets as well as to planets.

But although we consider the views of Æpinus or Coulomb in a very high degree probable as a _formal theory_, the question is very different when we come to examine them as a _physical theory_;--that is, when we inquire whether there really is a material electric fluid or fluids.

_Question of One or Two Fluids._--In the first place as to the question whether the fluids are one or two;--Coulomb's introduction of {211} the hypothesis of two fluids has been spoken of as a reform of the theory of Æpinus; it would probably have been more safe to have called his labors an advance in the calculation, and in the comparison of hypothesis with experiment, than to have used language which implied that the question, between the rival hypotheses of one or two fluids, could be treated as settled. For, in reality, if we assume, as Æpinus does, the mutual repulsion of all the particles of matter, in addition to the repulsion of the particles of the electric fluid for one another and their attraction for the particles of matter, the one fluid of Æpinus will give exactly the same results as the two fluids of Coulomb. The mathematical formulæ of Coulomb and of Poisson express the conditions of the one case as well as of the other; the interpretation only being somewhat different. The

place of the forces of the resinous fluid is supplied by the excess of the forces ascribed to the matter above the forces of the fluid, in the parts where the electric fluid is deficient.

The obvious argument against this hypothesis is, that we ascribe to the particles of matter a mutual repulsion, in addition to the mutual attraction of universal gravitation, and that this appears incongruous. Accordingly, Æpinus says, that when he was first driven to this proposition it horrified him.[37\11] But we may answer it in this way very satisfactorily:--If we suppose the mutual repulsion of matter to be somewhat less than the mutual attraction of matter and electric fluid, it will follow, as a consequence of the hypothesis, that besides all obvious electrical action, the particles of matter would attract each other with forces varying inversely as the square of the distance. Thus gravitation itself becomes an electrical phenomenon, arising from the residual excess of attraction over repulsion; and the fact which is urged against the hypothesis becomes a confirmation of it. By this consideration the prerogative of simplicity passes over to the side of the hypothesis of one fluid; and the rival view appears to lose at least all its superiority.

[Note 37\11: Neque diffiteor cum ipsa se mihi offerret . . . . me ad ipsam quodammodo exhorruisse. _Tentamen Theor. Elect._ p. 39.]

Very recently, M. Mosotti[38\11] has calculated the results of the Æpinian theory in a far more complete manner than had previously been performed; using Laplace's coefficients, as Poisson had done for the {212} Coulombian theory. He finds that, from the supposition of a fluid and of particles of matter exercising such forces as that theory assumes (with the very allowable additional supposition that the particles are small compared with their distances), it follows that the particles would exert a force, repulsive at the smallest distances, a little further on vanishing, afterwards attractive, and at all sensible distances attracting in proportion to the inverse square of the distance. Thus there would be a position of stable equilibrium for the particles at a very small distance from each other, which may be, M. Mosotti suggests, that equilibrium on which their physical structure depends. According to this view, the resistance of bodies to compression and to extension, as well as the phenomena of statical electricity and the mutual gravitation of matter, are accounted for by the same hypothesis of a single fluid or ether. A theory which offers a prospect of such a generalization is worth attention; but a very clear and comprehensive view of the doctrines of several sciences is requisite to prepare us to estimate its value and probable success.

[Note 38\11: _Sur les Forces qui régissent la Constitution Intérieure des Corps._ Turin. 1836.]

_Question of the Material Reality of the Electric Fluid._--At first sight the beautiful accordance of the experiments with calculations founded upon the attractions and repulsions of the two hypothetical fluids, persuade us that the hypotheses must be the real state of things. But we have already learned that we must not trust to such evidence too readily. It is a curious instance of the mutual influence of the histories of two provinces of science, but I think it will be allowed to be just, to say that the

discovery of the polarization of heat has done much to shake the theory of the electric fluids as a physical reality. For the doctrine of a material caloric appeared to be proved (from the laws of conduction and radiation) by the same kind of mathematical evidence (the agreement of laws respecting the elementary actions with those of fluids), which we have for the doctrine of material electricity. Yet we now seem to see that heat cannot be matter, since its rays have _sides_, in a manner in which a stream of particles of matter cannot have sides without inadmissible hypotheses. We see, then, that it will not be contrary to precedent, if our electrical theory, representing with perfect accuracy the _laws_ of the actions, in all their forms, simple and complex, should yet be fallacious as a view of the _cause_ of the actions.

Any true view of electricity must include, or at least be consistent with, the other classes of the phenomena, as well as this statical electrical action; such as the conditions of excitation and retention of {213} electricity; to which we may add, the connexion of electricity with magnetism and with chemistry;--a vast field, as yet dimly seen. Now, even with regard to the simplest of these questions, the cause of the retention of electricity at the surface of bodies, it appears to be impossible to maintain Coulomb's opinion, that this is effected by the resistance of air to the passage of electricity. The other questions are such as Coulomb did not attempt to touch; they refer, indeed, principally to laws not suspected at his time. How wide and profound a theory must be which deals worthily with these, we shall obtain some indications in the succeeding part of our history.

But it may be said on the other side, that we have the evidence of our senses for the reality of an electric fluid;--we see it in the spark; we hear it in the explosion; we feel it in the shock; and it produces the effects of mechanical violence, piercing and tearing the bodies through which it passes. And those who are disposed to assert a real fluid on such grounds, may appear to be justified in doing so, by one of Newton's "Rules of Philosophizing," in which he directs the philosopher to assume, in his theories, "causes which are true." The usual interpretation of a "vera causa," has been, that it implies causes which, independently of theoretical calculations, are known to exist by their mechanical effects; as gravity was familiarly known to exist on the earth, before it was extended to the heavens. The electric fluid might seem to be such a _vera causa_.

To this I should venture to reply, that this reasoning shows how delusive the Newtonian rule, so interpreted, may be. For a moment's consideration will satisfy us that none of the circumstances, above adduced, can really prove material currents, rather than vibrations, or other modes of agency. The spark and shock are quite insufficient to supply such a proof. Sound is vibrations,--light is vibrations; vibrations may affect our nerves, and may rend a body, as when glasses are broken by sounds. Therefore all these supposed indications of the reality of the electric fluid are utterly fallacious. In truth, this mode of applying Newton's rule consists in elevating our first rude and unscientific impressions into a supremacy over the results of calculation, generalization, and systematic induction.[**39\11] {214}

[Note **39\11: On the subject of this Newtonian Rule of Philosophizing, see further _Phil. Ind. Sc._ B. xii. c. 13. I have given an account of the history and evidence of the

Theory of Electricity in the _Reports of the British Association_ for 1835. I may seem there to have spoken more favorably of the Theory as a Physical Theory than I have done here. This difference is principally due to a consideration of the present aspect of the Theory of Heat.]

Thus our conclusion with regard to this subject is, that if we wish to form a stable physical theory of electricity, we must take into account not only the laws of statical electricity, which we have been chiefly considering, but the laws of other kinds of agency, different from the electric, yet connected with it. For the electricity of which we have hitherto spoken, and which is commonly excited by friction, is identical with galvanic action, which is a result of chemical combinations, and belongs to chemical philosophy. The connexion of these different kinds of electricity with one another leads us into a new domain; but we must, in the first place, consider their mechanical laws. We now proceed to another branch of the same subject, Magnetism.

{{215}} BOOK XII.

# _MECHANICO-CHEMICAL SCIENCES._

## (CONTINUED.)

HISTORY OF MAGNETISM.

EFFICE, ut interea fera munera militiaï Per maria ac terras omneis sopita quiescant. Nam tu sola potes tranquilla pace juvare Mortales; quoniam belli fera munera Mavors Armipotens regit, in gremium qui sæpe tuum se Rejicit, æterno devictus vulnere amoris; Atque ita suspiciens tereti cervice reposta, Pascit amore avidos inhians in te, Dea, visus, Eque tuo pendet resupini spiritus ore. Hunc tu, Diva, tuo recubantem corpore sancto Circumfusa super, suaves ex ore loquelas Funde, petens placidam Romanis, incluta, pacem. LUCRET. i. 31.

O charming Goddess, whose mysterious sway The unseen hosts of earth and sky obey; To whom, though cold and hard to all besides, The Iron God by strong affection glides. Flings himself eager to thy close embrace, And bends his head to gaze upon thy face; Do thou, what time thy fondling arms are thrown Around his form, and he is all thy own, Do thou, thy Rome to save, thy power to prove, Beg him to grant a boon for thy dear love; Beg him no more in battle-fields to deal. Or crush the nations with his mailed heel. But, touched and softened by a worthy flame, Quit sword and spear, and seek a better fame. Bid him to make all war and slaughter cease, And ply his genuine task in arts of peace; And by thee guided o'er the trackless surge, Bear wealth and joy to ocean's farthest verge.

{{217}} CHAPTER I.

DISCOVERY OF LAWS OF MAGNETIC PHENOMENA.

THE history of Magnetism is in a great degree similar to that of Electricity, and many of the same persons were employed in the two trains of research. The general fact, that the magnet attracts iron, was nearly all that was known to the ancients, and

is frequently mentioned and referred to; for instance, by Pliny, who wonders and declaims concerning it, in his usual exaggerated style.[1\12] The writers of the Stationary Period, in this subject as in others, employed themselves in collecting and adorning a number of extravagant tales, which the slightest reference to experiment would have disproved; as, for example, that a magnet, when it has lost its virtue, has it restored by goat's blood. Gilbert, whose work _De Magnete_ we have already mentioned, speaks with becoming indignation and pity of this bookish folly, and repeatedly asserts the paramount value of experiments. He himself, no doubt, acted up to his own precepts; for his work contains all the fundamental facts of the science, so fully examined indeed, that even at this day we have little to add to them. Thus, in his first Book, the subjects of the third, fourth, and fifth Chapters are,--that the magnet has poles,--that we may call these poles the north and the south pole,--that in two magnets the north pole of each attracts the south pole and repels the north pole of the other. This is, indeed, the cardinal fact on which our generalizations rest; and the reader will perceive at once its resemblance to the leading phenomena of statical electricity.

[Note 1\12: _Hist. Nat._ lib. xxxvi. c. 25.]

But the doctrines of magnetism, like those of heat, have an additional claim on our notice from the manner in which they are exemplified in the globe of the earth. The subject of _terrestrial magnetism_ forms a very important addition to the general facts of magnetic attraction and repulsion. The property of the magnet by which it directs its poles exactly or nearly north and south, when once discovered, was of immense importance to the mariner. It does not {218} appear easy to trace with certainty the period of this discovery. Passing over certain legends of the Chinese, as at any rate not bearing upon the progress of European science,[2\12] the earliest notice of this property appears to be contained in the Poem of Guyot de Provence, who describes the needle as being magnetized, and then placed in or on a straw, (floating on water, as I presume:) Puis se torne la pointe toute Contre l'estoile sans doute; that is, it turns towards the pole-star. This account would make the knowledge of this property in Europe anterior to 1200. It was afterwards found[3\12] that the needle does not point exactly towards the north. Gilbert was aware of this deviation, which he calls the _variation_, and also, that it is different in different places.[4\12] He maintained on theoretical principles also,[5\12] that at the same place the variation is constant; probably in his time there were not any recorded observations by which the truth of this assertion could be tested; it was afterwards found to be false. The alteration of the variation in proceeding from one place to another was, it will be recollected, one of the circumstances which most alarmed the companions of Columbus in 1492. Gilbert says,[6\12] "Other learned men have, in long navigations, observed the differences of magnetic variations, as Thomas Hariot, Robert Hues, Edward Wright, Abraham Kendall, all Englishmen: others have invented magnetic instruments and convenient modes of observation, such as are requisite for those who take long voyages, as William Borough in his Book concerning the variation of the compass, William Barlo in his supplement, William Norman in his _New Attractive_. This is that Robert Norman (a good seaman and an ingenious artificer,) who first discovered the _dip_ of magnetic iron." This important discovery was made[7\12] in

1576. From the time when the difference of the variation of the compass in different places became known, it was important to mariners to register the variation in all parts of the world. Halley was appointed to the command of a ship in the Royal Navy by the Government of William and Mary, with orders "to seek by observation the discovery of the rule for the variation of the compass." He published Magnetic Charts, which {219} have been since corrected and improved by various persons. The most recent are those of Mr. Yates in 1817, and of M. Hansteen. The dip, as well as the variation, was found to be different in different places. M. Humboldt, in the course of his travels, collected many such observations. And both the observations of variation and of dip seemed to indicate that the earth, as to its effect on the magnetic needle, may, approximately at least, be considered as a magnet, the poles of which are not far removed from the earth's poles of rotation. Thus we have a _magnetic equator_, in which the needle has no dip, and which does not deviate far from the earth's equator; although, from the best observations, it appears to be by no means a regular circle. And the phenomena, both of the dip and of the variation, in high northern latitudes, appear to indicate the existence of a pole below the surface of the earth to the north of Hudson's Bay. In his second remarkable expedition into those regions, Captain Ross is supposed to have reached the place of this pole; the dipping-needle there pointing vertically downwards, and the variation-compass turning towards this point in the adjacent regions. We shall hereafter have to consider the more complete and connected views which have been taken of terrestrial magnetism.

[Note 2\12: _Enc. Met._ art. _Magnetism_, p. 736.]

[Note 3\12: Before 1269. _Enc. Met._ p. 737.]

[Note 4\12: _De Magnete_, lib. iv. c. 1.]

[Note 5\12: c. 3.]

[Note 6\12: Lib. i. c. 1.]

[Note 7\12: _Enc. Met._ p. 738.]

In 1633, Gellibrand discovered that the variation is not constant, as Gilbert imagined, but that at London it had diminished from eleven degrees east in 1580, to four degrees in 1633. Since that time the variation has become more and more westerly; it is now about twenty-five degrees west, and the needle is supposed to have begun to travel eastward again.

The next important fact which appeared with respect to terrestrial magnetism was, that the position of the needle is subject to a small _diurnal_ variation: this was discovered in 1722, by Graham, a philosophical instrument-maker, of London. The daily variation was established by one thousand observations of Graham, and confirmed by four thousand more made by Canton, and is now considered to be out of dispute. It appeared also, by Canton's researches, that the diurnal variation undergoes an annual inequality, being nearly a quarter of a degree in June and July, and only half that quantity in December and January.

Having thus noticed the principal facts which belong to terrestrial magnetism, we must return to the consideration of those phenomena which gradually led to a consistent magnetic theory. Gilbert observed that both smelted iron and hammered iron have the magnetic virtue, {220} though in a weaker degree than the magnet itself,[8\12] and he asserted distinctly that the magnet is merely an ore of iron, (lib. i. c. 16, Quod magnes et vena ferri idem sunt.) He also noted the increased energy which magnets acquire by being _armed_; that is, fitted with a cap of polished iron at each pole.[9\12] But we do not find till a later period any notice of the distinction which exists between the magnetical properties of soft iron and of hard steel;--the latter being susceptible of being formed into _artificial magnets_, with permanent poles; while soft iron is only _passively magnetic_, receiving a temporary polarity from the action of a magnet near it, but losing this property when the magnet is removed. About the middle of the last century, various methods were devised of making artificial magnets, which exceeded in power all magnetic bodies previously known.

[Note 8\12: Lib. i. c. 9-13.]

[Note 9\12: Lib. ii. c. 17.]

The remaining experimental researches had so close an historical connexion with the theory, that they will be best considered along with it, and to that, therefore, we now proceed.

## CHAPTER II.

PROGRESS OF MAGNETIC THEORY.

THEORY OF MAGNETIC ACTION.--The assumption of a fluid, as a mode of explaining the phenomena, was far less obvious in magnetic than in electric cases, yet it was soon arrived at. After the usual philosophy of the middle ages, the "forms" of Aquinas, the "efflux" of Cusanus, the "vapors" of Costæus, and the like, which are recorded by Gilbert,[10\12] we have his own theory, which he also expresses by ascribing the effects to a "formal efficiency;"--a "_form_ of primary globes; the proper entity and existence of their homogeneous parts, which we may call a primary and radical and astral _form_;"--of which forms there is one in the sun, one in the moon, one in the earth, the latter being the magnetic virtue.

[Note 10\12: Gilb. lib. ii. c. 3, 4]

Without attempting to analyse the precise import of these expressions, we may proceed to Descartes's explanation of magnetic phenomena. The mode in which he presents this subject[11\12] is, perhaps, the {221} most persuasive of his physical attempts. If a magnet be placed among iron filings, these arrange themselves in curved lines, which proceed from one pole of the magnet to the other. It was not difficult to conceive these to be the traces of currents of ethereal matter which circulate through the magnet, and which are thus rendered sensible even to the eye. When phenomena could not be explained by means of one vortex, several were introduced. Three Memoirs on Magnetism, written on such principles, had the prize

adjudged[12\12] by the French Academy of Sciences in 1746.

[Note 11\12: _Prin. Phil._ pars c. iv. 146.]

[Note 12\12: Coulomb, 1789, p. 482.]

But the Cartesian philosophy gradually declined; and it was not difficult to show that the _magnetic curves_, as well as other phenomena, would, in fact, result from the attraction and repulsion of two poles. The analogy of magnetism with electricity was so strong and clear, that similar theories were naturally proposed for the two sets of facts; the distinction of bodies into conductors and electrics in the one case, corresponding to the distinction of soft and hard steel, in their relations to magnetism. Æpinus published a theory of magnetism and electricity at the same time (1759); and the former theory, like the latter, explained the phenomena of the opposite poles as results of the excess and defect of a magnetic "fluid," which was dislodged and accumulated in the ends of the body, by the repulsion of its own particles, and by the attraction of iron or steel, as in the case of induced electricity. The Æpinian theory of magnetism, as of electricity, was recast by Coulomb, and presented in a new shape, with two fluids instead of one. But before this theory was reduced to calculation, it was obviously desirable, in the first place, to determine the law of force.

In magnetic, as in electric action, the determination of the law of attraction of the particles was attended at first with some difficulty, because the action which a finite magnet exerts is a compound result of the attractions and repulsions of many points. Newton had imagined the attractive force of magnetism to be inversely as the cube of the distance; but Mayer in 1760, and Lambert a few years later, asserted the law to be, in this as in other forces, the inverse square. Coulomb has the merit of having first clearly confirmed this law, by the use of his torsion-balance.[13\12] He established, at the same time, other very important facts, for instance, "that the directive magnetic force, which the earth exerts upon a needle, is a constant quantity, parallel {222} to the magnetic meridian, and passing through the same point of the needle whatever be its position." This was the more important, because it was necessary, in the first place, to allow for the effect of the terrestrial force, before the mutual action of the magnets could be extricated from the phenomena.[14\12] Coulomb then proceeded to correct the theory of magnetism.

[Note 13\12: _Mem. A. P._ 1784, 2d Mem. p. 593.]

[Note 14\12: p. 603.]

Coulomb's reform of the Æpinian theory, in the case of magnetism, as in that of electricity, substituted two fluids (an _austral_ and a _boreal_ fluid,) for the single fluid; and in this way removed the necessity under which Æpinus found himself, of supposing all the particles of iron and steel and other magnetic bodies to have a peculiar repulsion for each other, exactly equal to their attraction for the magnetic fluid. But in the case of magnetism, another modification was necessary. It was impossible to suppose here, as in the electrical phenomena, that one of the fluids was accumulated on one extremity of a body, and the other fluid on the other extremity; for though this might appear, at first sight, to be the case in a magnetic needle, it was

found that when the needle was cut into two halves, the half in which the austral fluid had seemed to predominate, acquired immediately a boreal pole opposite to its austral pole, and a similar effect followed in the other half. The same is true, into however many parts the magnetic body be cut. The way in which Coulomb modified the theory so as to reconcile it with such facts, is simple and satisfactory. He supposes[15\12] the magnetic body to be made up of "molecules or integral parts," or, as they were afterwards called by M. Poisson, "magnetic elements." In each of these elements, (which are extremely minute,) the fluids can be separated, so that each element has an austral and a boreal pole; but the austral pole of an element which is adjacent to the boreal pole of the next, neutralizes, or nearly neutralizes, its effect; so that the sensible magnetism appears only towards the extremities of the body, as it would do if the fluids could permeate the body freely. We shall have exactly the same result, as to sensible magnetic force, on the one supposition and on the other, as Coulomb showed.[16\12]

[Note 15\12: _Mem. A. P._ 1789, p. 488.]

[Note 16\12: _Mem. A. P._ p. 492.]

The theory, thus freed from manifest incongruities, was to be reduced to calculation, and compared with experiment; this was done in Coulomb's Seventh Memoir.[17\12] The difficulties of calculation in this, as in the electric problem, could not be entirely surmounted by the analysis of Coulomb; but by various artifices, he obtained theoretically the {223} relative amount of magnetism at several points of a needle,[18\12] and the proposition that the directive force of the earth on similar needles saturated with magnetism, was as the cube of their dimensions; conclusions which agreed with experiment.

[Note 17\12: _A. P._ 1789.]

[Note 18\12: p. 485.]

The agreement thus obtained was sufficient to give a great probability to the theory; but an improvement of the methods of calculation and a repetition of experiments, was, in this as in other cases, desirable, as a confirmation of the labors of the original theorist. These requisites, in the course of time, were supplied. The researches of Laplace and Legendre on the figure of the earth had (as we have already stated,) introduced some very peculiar analytical artifices, applicable to the attractions of spheroids; and these methods were employed by M. Biot in 1811, to show that on an elliptical spheroid, the thickness of the fluid in the direction of the radius would be as the distance from the centre.[19\12] But the subject was taken up in a more complete manner in 1824 by M. Poisson, who obtained general expressions for the attractions or repulsions of a body of any form whatever, magnetized by influence, upon a given point; and in the case of spherical bodies was able completely to solve the equations which determine these forces.[20\12]

[Note 19\12: _Bull. des Sc._ No. li.]

[Note 20\12: _A. P._ for 1821 and 2, published 1826.]

Previously to these theoretical investigations, Mr. Barlow had made a series of experiments on the effect of an iron sphere upon a compass needle; and had obtained empirical formulæ for the amount of the deviation of the needle, according to its dependence upon the position and magnitude of the sphere. He afterwards deduced the same formulæ from a theory which was, in fact, identical with that of Coulomb, but which he considered as different, in that it supposed the magnetic fluids to be entirely collected at the surface of the sphere. He had indeed found, by experiment, that the surface was the only part in which there was any sensible magnetism; and that a thin shell of iron would produce the same effect as a solid ball of the same diameter.

But this was, in fact, a most complete verification of Coulomb's theory. For though that theory did not suppose the magnetism to be collected solely at the surface, as Mr. Barlow found it, it followed from the theory, that the _sensible_ magnetic intensity assumed the same distribution (namely, a surface distribution,) as if the fluids could permeate the whole body, instead of the "magnetic elements" only. Coulomb, indeed, had not expressly noticed the result, that the sensible {224} magnetism would be confined to the surface of bodies; but he had found that, in a long needle, the magnetic fluid might be supposed to be concentrated very near the extremities, just as it is in a long electric body. The theoretical confirmation of this rule among the other consequences of the theory,--that the sensible magnetism would be collected at the surface,--was one of the results of Poisson's analysis. For it appeared that if the sum of the electric elements of the body was equal to the whole body, there would be no difference between the action of a solid sphere and very thin shell.

We may, then, consider the Coulombian theory to be fully established and verified, as a representation of the laws of magnetical phenomena. We may add, as a remarkable and valuable example of an ulterior step in the course of sciences, the application of the laws of the distribution of magnetism to the purposes of navigation. It had been found that the mass of iron which exists in a ship produces a deviation in the direction of the compass-needle, which was termed "local attraction," and which rendered the compass an erroneous guide. Mr. Barlow proposed to correct this by a plate of iron placed near the compass; the plate being of comparatively small mass, but, in consequence of its expanded form, and its proximity to the needle, of equivalent effect to the disturbing cause.

[2nd Ed.] [This proposed arrangement was not successful, because as the ship turns into different positions, it may be considered as revolving round a vertical axis; and as this does not coincide with the magnetic axis, the relative magnetic position of the disturbing parts of the ship, and of the correcting plate, will be altered, so that they will not continue to counteract each other. In high magnetic latitudes the correcting plate was used with success.

But when iron ships became common, a correction of the effect of the iron upon the ship's compass in the general case became necessary. Mr. Airy devised the means of making this correction. By placing a magnet and a mass of iron in certain positions relative to the compass, the effect of the rest of the iron in the ship is completely

counteracted in all positions.[21\12]]

[Note 21\12: See _Phil. Trans._ 1836.]

But we have still to trace the progress of the theory of terrestrial magnetism.

_Theory of Terrestrial Magnetism._--Gilbert had begun a plausible course of speculation on this point. "We must reject," he says,[22\12] "in {225} the first place, that vulgar opinion of recent writers concerning magnetic mountains, or a certain magnetic rock, or an imaginary pole at a certain distance from the pole of the earth." For, he adds, "we learn by experience, that there is no such fixed pole or term in the earth for the variation." Gilbert describes the whole earth as a magnetic globe, and attributes the variation to the irregular form of its protuberances, the solid parts only being magnetic. It was not easy to confirm or refute this opinion, but other hypotheses were tried by various writers; for instance, Halley had imagined, from the forms of the lines of equal variation, that there must be four magnetic poles; but Euler[23\12] showed that the "Halleian lines" would, for the most part, result from the supposition of two magnetic poles, and assigned their position so as to represent pretty well the known state of the variation all over the world in 1744. But the variation was not the only phenomenon which required to be taken into account; the dip at different places, and also the intensity of the force, were to be considered. We have already mentioned M. de Humboldt's collection of observations of the dip. These were examined by M. Biot, with the view of reducing them to the action of two poles in the supposed terrestrial magnetic axis. Having, at first, made the distance of these poles from the centre of the earth indefinite, he found that his formulæ agreed more and more nearly with the observations, as the poles were brought nearer; and that fact and theory coincided tolerably well when both poles were at the centre. In 1809,[24\12] Krafft simplified this result, by showing that, on this supposition, the tangent of the dip was twice the tangent of the latitude of the place as measured from the magnetic equator. But M. Hansteen, who has devoted to the subject of terrestrial magnetism a great amount of labor and skill, has shown that, taking together all the observations which we possess, we are compelled to suppose four magnetic poles; two near the north pole, and two near the south pole, of the terrestrial globe; and that these poles, no two of which are exactly opposite each other, are all in motion, with different velocities, some moving to the east and some to the west. This curious collection of facts awaits the hand of future theorists, when the ripeness of time shall invite them to the task.

[Note 22\12: Lib. iv. c. 1. _De Variatione._]

[Note 23\12: _Ac. Berlin_, 1757.]

[Note 24\12: _Enc. Met._ p. 742.]

[2nd Ed.] [I had thus written in the first edition. The theorist who was needed to reduce this accumulation of facts to their laws, {226} had already laid his powerful hand upon them; namely, M. Gauss, a mathematician not inferior to any of the great men who completed the theory of gravitation. And institutions had been established for extending the collection of the facts pertaining to it, on a scale which elevates Magnetism into a companionship with Astronomy. M. Hansteen's _Magnetismus der

Erde_ was published in 1819. His conclusions respecting the position of the four magnetic "poles" excited so much interest in his own country, that the Norwegian _Storthing_, or parliament, by a unanimous vote, provided funds for a magnetic expedition which he was to conduct along the north of Europe and Asia; and this they did at the very time when they refused to make a grant to the king for building a palace at Christiania. The expedition was made in 1828-30, and verified Hansteen's anticipations as to the existence of a region of magnetic convergence in Siberia, which he considered as indicating a "pole" to the north of that country. M. Erman also travelled round the earth at the same time, making magnetic observations.

About the same time another magnetical phenomenon attracted attention. Besides the general motion of the magnetic poles, and the diurnal movements of the needle, it was found that small and irregular disturbances take place in its position, which M. de Humboldt termed _magnetic storms_. And that which excited a strong interest on this subject was the discovery that these magnetic storms, seen only by philosophers who watch the needle with microscopic exactness, rage simultaneously over large tracts of the surface of our globe. This was detected about 1825 by a comparison of the observations of M. Arago at Paris with simultaneous observations of M. Kupffer at Kasan in Russia, distant more than 47 degrees of longitude.

At the instance of M. de Humboldt, the Imperial Academy of Russia adopted with zeal the prosecution of this inquiry, and formed a chain of magnetic stations across the whole of the Russian empire. Magnetic observations were established at Petersburg and at Kasan, and corresponding observations were made at Moscow, at Nicolaieff in the Crimea, and Barnaoul and Nertchinsk in Siberia, at Sitka in Russian America, and even at Pekin. To these magnetic stations the Russian government afterwards added, Catharineburg in Russia Proper, Helsingfors in Finland, Teflis in Georgia. A comparison of the results obtained at four of these stations made by MM. de Humboldt and Dove, in the year 1830, showed that the magnetic disturbances were simultaneous, and were for the most parallel in their progress. {227}

Important steps in the prosecution of this subject were soon after made by M. Gauss, the great mathematician of Göttingen. He contrived instruments and modes of observation far more perfect than any before employed, and organized a system of comparative observations throughout Europe. In 1835, stations for this purpose were established at Altona, Augsburg, Berlin, Breda, Breslau, Copenhagen, Dublin, Freiberg, Göttingen, Greenwich, Hanover, Leipsic, Marburg, Milan, Munich, Petersburg, Stockholm, and Upsala. At these places, six times in the year, observations were taken simultaneously, at intervals of five minutes for 24 hours. The _Results of the Magnetic Association_ (Resultaten des Magnetischen Vereins) were published by MM. Gauss and Weber, beginning in 1836.

British physicists did not at first take any leading part in these plans. But in 1836, Baron Humboldt, who by his long labors and important discoveries in this subject might be considered as peculiarly entitled to urge its claims, addressed a letter to the Duke of Sussex, then President of the Royal Society, asking for the co-operation of this country in so large and hopeful a scheme for the promotion of science. The Royal Society willingly entertained this appeal; and the progress of the cause was still

further promoted when it was zealously taken up by the British Association for the Advancement of Science, assembled at Newcastle in 1838. The Association there expressed its strong interest in the German system of magnetic observations; and at the instigation of this body, and of the Royal Society, four complete magnetical observatories were established by the British government, at Toronto, St. Helena, the Cape of Good Hope, and Van Diemen's Land. The munificence of the Directors of the East India Company founded and furnished an equal number at Simla (in the Himalayah), Madras, Bombay, and Sincapore. Sir Thomas Brisbane added another at his own expense at Kelso, in Scotland. Besides this, the government sent out a naval expedition to make discoveries (magnetic among others), in the Antarctic regions, under the command of Sir James Ross. Other states lent their assistance also, and founded or reorganized their magnetic observatories. Besides those already mentioned, one was established by the French government at Algiers; one by the Belgian, at Brussels; two by Austria, at Prague and Milan; one by Prussia, at Breslau; one by Bavaria, at Munich; one by Spain, at Cadiz; there are two in the United States, at Philadelphia and Cambridge; one at Cairo, founded by the Pasha of Egypt; and in India, one at Trevandrum, established by the Rajah of Travancore; and one by {228} the King of Oude, at Lucknow. At all these distant stations the same plan was followed out, by observations strictly simultaneous, made according to the same methods, with the same instrumental means. Such a scheme, combining world-wide extent with the singleness of action of an individual mind, is hitherto without parallel.

At first, the British stations were established for three years only; but it was thought advisable to extend this period three years longer, to end in 1845. And when the termination of that period arrived, a discussion was held among the magneticians themselves, whether it was better to continue the observations still, or to examine and compare the vast mass of observations already collected, so as to see to what results and improvements of methods they pointed. This question was argued at the meeting of the British Association at Cambridge in that year; and the conference ended in the magneticians requesting to have the observations continued, at some of the observatories for an indefinite period, at others, till the year 1848. In the mean time the Antarctic expedition had brought back a rich store of observations, fitted to disclose the magnetic condition of those regions which it had explored. These were _discussed_, and their results exhibited, in the _Philosophical Transactions_ for 1843, by Col. Sabine, who had himself at various periods, made magnetic observations in the Arctic regions, and in several remote parts of the globe, and had always been a zealous laborer in this fruitful field. The general mass of the observations was placed under the management of Professor Lloyd, of Dublin, who has enriched the science of magnetism with several valuable instruments and methods, and who, along with Col. Sabine, made a magnetic survey of the British Isles in 1835 and 1836.

I do not dwell upon magnetic surveys of various countries made by many excellent observers; as MM. Quetelet, Forbes, Fox, Bache and others.

The facts observed at each station were, the _intensity_ of the magnetic force; the _declination_ of the needle from the meridian, sometimes called the _variation_; and its _inclination_ to the horizon, _the dip_;--or at least, some elements equivalent to

these. The values of these elements at any given time, if known, can be expressed by charts of the earth's surface, on which are drawn the _isodynamic_, _isogonal_, and _isoclinal_ curves. The second of these kinds of charts contain the "Halleian lines" spoken of in a previous page. Moreover the magnetic elements at each place are to be observed in such a {229} manner as to determine both their _periodical_ variations (the changes which occur in the period of a day, and of a year), the _secular_ changes, as the gradual increase or diminution of the declination at the same place for many years; and the _irregular_ fluctuations which, as we have said, are simultaneous over a large part, or the whole, of the earth's surface.

When these Facts have been ascertained over the whole extent of the earth's surface, we shall still have to inquire what is the Cause of the changes in the forces which these phenomena disclose. But as a basis for all speculation on that subject, we must know the law of the phenomena, and of the forces which immediately produce them. I have already said that Euler tried to account for the Halleian lines by means of _two_ magnetic "poles," but that M. Hansteen conceived it necessary to assume _four_. But an entirely new light has been thrown upon this subject by the beautiful investigations of Gauss, in his _Theory of Terrestrial Magnetism_, published in 1839. He remarks that the term "poles," as used by his predecessors, involves an assumption arbitrary, and, as it is now found, false; namely, that certain definite points, two, four, or more, acting according to the laws of ordinary magnetical poles, will explain the phenomena. He starts from a more comprehensive assumption, that magnetism is distributed throughout the mass of the earth in an unknown manner. On this assumption he obtains a function _V_, by the differentials of which the elements of the magnetic force at any point will be expressed. This function _V_ is well known in physical astronomy, and is obtained by summing all the elements of magnetic force in each particle, each multiplied by the reciprocal of its distance; or as we may express it, by taking the sum of each element and its proximity jointly. Hence it has been proposed[25\12] to term this function the "_integral proximity_" of the attracting mass.[26\12] By using the most refined {230} mathematical artifices for deducing the values of _V_ and its differentials in converging series, he is able to derive the coefficients of these series from the observed magnetic elements at certain places, and hence, to calculate them for all places. The comparison of the calculation with the observed results is, of course, the test of the truth of the theory.

[Note 25\12: _Quart. Rev._ No. 131, p. 283.]

[Note 26\12: The function V is of constant occurrence in investigations respecting attractions. It is introduced by Laplace in his investigations respecting the attractions of spheroids, _Méc. Cél._ Livr. III. Art. 4. Mr. Green and Professor Mac Cullagh have proposed to term this function the _Potential_ of the system; but this term (though suggested, I suppose, by analogy with the substantive _Exponential_), does not appear convenient in its form. On the other hand, the term _Integral Proximity_ does not indicate that which gives the function its peculiar claim to distinction; namely, that its differentials express the power or attraction of the system. Perhaps _Integral Potentiality_, or _Integral Attractivity_, would be a term combining the recommendations of both the others.]

The degree of convergence of the series depends upon the unknown distribution of magnetism within the earth. "If we could venture to assume," says M. Gauss, "that the members have a sensible influence only as far as the fourth order, complete observations from eight points would be sufficient, theoretically considered, for the determination of the coefficients." And under certain limitations, making this assumption, as the best we can do at present, M. Gauss obtains from eight places, 24 coefficients (each supplying three elements), and hence calculates the magnetic elements (intensity, variation and dip) at 91 places in all parts of the earth. He finds his calculations approach the observed values with a degree of exactness which appears to be quite convincing as to the general truth of his results; especially taking into account how entirely unlimited is his original hypothesis.

It is one of the most curious results of this investigation that according to the most simple meaning which we can give to the term "pole" the earth has only _two_ magnetic poles; that is, two points where the direction of the magnetic force is vertical. And thus the _isogonal curves_ may be looked upon as _deformations_ of the curves deduced by Euler from the supposition of two poles, the deformation arising from this, that the earth does not contain a single definite magnet, but irregularly diffused magnetical elements, which still have collectively a distinct resemblance to a single magnet. And instead of Hansteen's Siberian pole, we have a Siberian region in which the needles converge; but if the apparent convergence be pursued it nowhere comes to a point; and the like is the case in the Antarctic region. When the 24 Gaussian elements at any time are known the magnetic condition of the globe is known, just as the mechanical condition of the solar system is known, when we know the elements of the orbits of the satellites and planets and the mass of each. And the comparison of this magnetic condition of the globe at distant periods of time cannot fail to supply materials for future researches and speculations with regard to the agencies by which the condition of the earth is determined. The condition of which we here speak must necessarily be its _mechanico-chemical_ condition, being expressed, as it will be, in terms of the mechanico-chemical sciences. The {231} investigations I have been describing belong to the mechanical side of the subject: but when philosophers have to consider the causes of the secular changes which are found to occur in this mechanical condition, they cannot fail to be driven to electrical, that is, chemical agencies and laws.

I can only allude to Gauss's investigations respecting the _Absolute Measure_ of the Earth's Magnetic Force. To determine the ratio of the magnetic force of the earth to that of a known magnet, Poisson proposed to observe the time of vibration of a second magnet. The method of Gauss, now universally adopted, consists in observing the position of equilibrium of the second magnet when deflected by the first.

The manner in which the business of magnetic observation has been taken up by the governments of our time makes this by far the greatest scientific undertaking which the world has ever seen. The result will be that we shall obtain in a few years a knowledge of the magnetic constitution of the earth which otherwise it might have required centuries to accumulate. The secular magnetic changes must still require a long time to reduce to their laws of phenomena, except observation be anticipated or

assisted by some happy discovery as to the cause of these changes. But besides the special gain to magnetic science by this great plan of joint action among the nations of the earth, there is thereby a beginning made in the recognition and execution of the duty of forwarding science in general by national exertions. For at most of the magnetic observatories, meteorological observations are also carried on; and such observations, being far more extensive, systematic, and permanent than those which have usually been made, can hardly fail to produce important additions to science. But at any rate they do for science that which nations can do, and individuals cannot; and they seek for scientific truths in a manner suitable to the respect now professed for science and to the progress which its methods have made. Nor are we to overlook the effect of such observations as means of training men in the pursuit of science. "There is amongst us," says one of the magnetic observers, "a growing recognition of the importance, both for science and for practical life, of forming exact observers of nature. Hitherto astronomy alone has afforded a very partial opportunity for the formation of fine observers, of which few could avail themselves. Experience has shown that magnetic observations may serve as excellent training schools in this respect."[27\12]] {232}

[Note 27\12: _Letter_ of W. Weber. _Brit. Assoc. Rep._ 1845, p. 17.]

The various other circumstances which terrestrial magnetism exhibits,--the diurnal and annual changes of the position of the compass-needle;--the larger secular change which affects it in the course of years;--the difference of intensity at different places, and other facts, have naturally occupied philosophers with the attempt to determine, both the laws of the phenomena and their causes. But these attempts necessarily depend, not upon laws of statical magnetism, such as they have been explained above; but upon the laws by which the production and intensity of magnetism in different cases are regulated;--laws which belong to a different province, and are related to a different set of principles. Thus, for example, we have not attempted to explain the discovery of the laws by which heat influences magnetism; and therefore we cannot now give an account of those theories of the facts relating to terrestrial magnetism, which depend upon the influence of temperature. The conditions of excitation of magnetism are best studied by comparing this force with other cases where the same effects are produced by very different apparent agencies; such as galvanic and thermo-electricity. To the history of these we shall presently proceed.

_Conclusion._--The hypothesis of magnetic fluids, as physical realities, was never widely or strongly embraced, as that of electric fluids was. For though the hypothesis accounted, to a remarkable degree of exactness, for large classes of the phenomena, the presence of a material fluid was not indicated by facts of a different kind, such as the spark, the discharge from points, the shock, and its mechanical effects. Thus the belief of a peculiar magnetic fluid or fluids was not forced upon men's minds; and the doctrine above stated was probably entertained by most of its adherents, chiefly as a means of expressing the laws of phenomena in their elementary form.

One other observation occurs here. We have seen that the supposition of a fluid moveable from one part of bodies to another, and capable of accumulation in different parts of the surface, appeared at first to be as distinctly authorized by

magnetic as by electric phenomena; and yet that it afterwards appeared, by calculation, that this must be considered as a derivative result; no real transfer of fluid taking place except within the limits of the insensible particles of the body. Without attempting to found a formula of philosophizing on this circumstance, we may observe, that this occurrence, like the disproof of heat as a material fluid, shows the possibility of an hypothesis which shall very exactly satisfy many phenomena, and yet be incomplete: it {233} shows, too, the necessity of bringing facts of all kinds to bear on the hypothesis; thus, in this case it was requisite to take into account the facts of junction and separation of magnetic bodies, as well as their attractions and repulsions.

If we have seen reason to doubt the doctrine of electric fluids as physical realities, we cannot help pronouncing upon the magnetic fluids as having still more insecure claims to a material existence, even on the grounds just stated. But we may add considerations still more decisive; for at a further stage of discovery, as we shall see, magnetic and electric action were found to be connected in the closest manner, so as to lead to the persuasion of their being different effects of one common cause. After those discoveries, no philosopher would dream of assuming electric fluids and magnetic fluids as two distinct material agents. Yet even now the nature of the dependence of magnetism upon any other cause is extremely difficult to conceive. But till we have noticed some of the discoveries to which we have alluded, we cannot even speculate about that dependence. We now, therefore, proceed to sketch the history of these discoveries.

{{235}} BOOK XIII.

# _MECHANICO-CHEMICAL SCIENCES._

# (CONTINUED.)

HISTORY OF GALVANISM, OR VOLTAIC ELECTRICITY.

Percusssæ gelido trepidant sub pectore fibræ, Et nova desuetis subrepens vita medullis Miscetur morti: tunc omnis palpitat artus Tenduntur nervi; nec se tellure cadaver Paullatim per membra levat; terrâque repulsum est Erectumque simul. LUCAN. vi. 752.

The form which lay before inert and dead, Sudden a piercing thrill of change o'erspread; Returning life gleams in the stony face, The fibres quiver and the sinews brace, Move the stiff limbs;--nor did the body rise With tempered strength which genial life supplies, But upright starting, its full stature held, As though the earth the supine corse repelled.

{{237}} CHAPTER I.

DISCOVERY OF VOLTAIC ELECTRICITY.

WE have given the name of _mechanico-chemical_ to the class of sciences now under our consideration; for these sciences are concerned with cases in which

mechanical effects, that is, attractions and repulsions, are produced; while the conditions under which these effects occur, depend, as we shall hereafter see, on chemical relations. In that branch of these sciences which we have just treated of, Magnetism, the mechanical phenomena were obvious, but their connexion with chemical causes was by no means apparent, and, indeed, has not yet come under our notice.

The subject to which we now proceed, Galvanism, belongs to the same group, but, at first sight, exhibits only the other, the chemical, portion of the features of the class; for the connexion of galvanic phenomena with chemical action was soon made out, but the mechanical effects which accompany them were not examined till the examination was required by a new train of discovery. It is to be observed, that I do not include in the class of mechanical effects the convulsive motions in the limbs of animals which are occasioned by galvanic action; for these movements are produced, not by attraction and repulsion, but by muscular irritability; and though they indicate the existence of a peculiar agency, cannot be used to measure its intensity and law.

The various examples of the class of agents which we here consider,--magnetism, electricity, galvanism, electro-magnetism, thermo-electricity,--differ from each other principally in the circumstances by which they are called into action; and these differences are in reality of a chemical nature, and will have to be considered when we come to treat of the inductive steps by which the general principles of chemical theory are established. In the present part of our task, therefore, we must take for granted the chemical conditions on which the excitation of these various kinds of action depends, and trace the history of the discovery of their mechanical laws only. This rule will much abridge the account we have here to give of the progress of discovery in the provinces to which I have just referred. {238}

The first step in this career of discovery was that made by Galvani, Professor of Anatomy at Bologna. In 1790, electricity, as an experimental science, was nearly stationary. The impulse given to its progress by the splendid phenomena of the Leyden phial had almost died away; Coulomb was employed in systematizing the theory of the electric fluid, as shown by its statical effects; but in all the other parts of the subject, no great principle or new result had for some time been detected. The first announcement of Galvani's discovery in 1791 excited great notice, for it was given forth as a manifestation of electricity under a new and remarkable character; namely, as residing in the muscles of animals.[1\13] The limbs of a dissected frog were observed to move, when touched with pieces of two different metals; the agent which produced these motions was conceived to be identified with electricity, and was termed _animal electricity_; and Galvani's experiments were repeated, with various modifications, in all parts of Europe, exciting much curiosity, and giving rise to many speculations.

[Note 1\13: _De Viribus Electricis in Motu Musculari_, Comm. Bonon. t. vii. 1792.]

It is our business to determine the character of each great discovery which appears in the progress of science. Men are fond of repeating that such discoveries are most commonly the result of accident; and we have seen reason to reject this

opinion, since that preparation of thought by which the accident produces discovery is the most important of the conditions on which the successful event depends. Such accidents are like a spark which discharges a gun already loaded and pointed. In the case of Galvani, indeed, the discovery may, with more propriety than usual, be said to have been casual; but in the form in which it was first noted, it exhibited no important novelty. His frog was lying on a table near the conductor of an electrical machine, and the convulsions appeared only when a spark was taken from the machine. If Galvani had been as good a physicist as he was an anatomist, he would probably have seen that the movements so occasioned proved only that the muscles or nerves, or the two together, formed a very sensitive indicator of electrical action. It was when he produced such motions by contact of metals alone, that he obtained an important and fundamental fact in science.

The analysis of this fact into its real and essential conditions was the work of Alexander Volta, another Italian professor. Volta, indeed, possessed that knowledge of the subject of electricity which made a hint like that of Galvani the basis of a new science. Galvani appears {239} never to have acquired much general knowledge of electricity: Volta, on the other hand, had labored at this branch of knowledge from the age of eighteen, through a period of nearly thirty years; and had invented an _electrophorus_ and an _electrical condenser_, which showed great experimental skill. When he turned his attention to the experiments made by Galvani, he observed that the author of them had been far more surprised than he needed to be, at those results in which an electrical spark was produced; and that it was only in the cases in which no such apparatus was employed, that the observations could justly be considered as indicating a new law, or a new kind of electricity.[2\13] He soon satisfied himself[3\13] (about 1794) that the essential conditions of this kind of action depended on the metals; that it is brought into play most decidedly when two different metals touch each other, and are connected by any moist body;--and that the parts of animals which had been used discharged the office both of such moist bodies, and of very sensitive electrometers. The _animal_ electricity of Galvani might, he observed, be with more propriety called _metallic_ electricity.

[Note 2\13: _Phil. Trans._ 1793, p. 21.]

[Note 3\13: See Fischer, viii. 625.]

The recognition of this agency as a peculiar kind of _electricity_, arose in part perhaps, at first, from the confusion made by Galvani between the cases in which his electrical machine was, and those in which it was not employed. But the identity was confirmed by its being found that the known difference of electrical conductors and non-conductors regulated the conduction of the new influence. The more exact determination of the new facts to those of electricity was a succeeding step of the progress of the subject.

The term "animal electricity" has been superseded by others, of which _galvanism_ is perhaps the most familiar. I think it will appear from what has been said, that Volta's office in this discovery is of a much higher and more philosophical kind than that of Galvani; and it would, on this account, be more fitting to employ the term

_voltaic electricity_; which, indeed, is very commonly used, especially by our most recent and comprehensive writers.

Volta more fully still established his claim as the main originator of this science by his next step. When some of those who repeated the experiments of Galvani had expressed a wish that there was some method of multiplying the effect of _this_ electricity, such as the Leyden phial supplies for common electricity, they probably thought their wishes far from a realization. But the _voltaic pile_, which Volta {240} described in the _Philosophical Transactions_ for 1800, completely satisfies this aspiration; and was, in fact, a more important step in the history of electricity than the Leyden jar had been. It has since undergone various modifications, of which the most important was that introduced by **Cruickshanks, who[4\13] substituted a trough for a pile. But in all cases the principle of the instrument was the same;--a continued repetition of the triple combination of two metals and a fluid in contact, so as to form a circuit which returns into itself.

[Note 4\13: Fischer, viii. p. 683.]

Such an instrument is capable of causing effects of great intensity; as seen both in the production of light and heat, and in chemical changes. But the discovery with which we are here concerned, is not the details and consequences of the effects, (which belong to chemistry,) but the analysis of the conditions under which such effects take place; and this we may consider as completed by Volta at the epoch of which we speak.

## CHAPTER II.

RECEPTION AND CONFIRMATION OF THE DISCOVERY OF VOLTAIC ELECTRICITY.

GALVANI'S experiments excited a great interest all over Europe, in consequence partly of a circumstance which, as we have seen, was unessential, the muscular contractions and various sensations which they occasioned. Galvani himself had not only considered the animal element of the circuit as the origin of the electricity, but had framed a theory,[5\13] in which he compared the muscles to charged jars, and the nerves to the discharging wires; and a controversy was, for some time, carried on, in Italy, between the adherents of Galvani and those of Volta.[6\13]

[Note 5\13: Ib. viii. 613.]

[Note 6\13: Ib. viii. 619.]

The galvanic experiments, and especially those which appeared to have a physiological bearing, were verified and extended by a number of the most active philosophers of Europe, and especially William von Humboldt. A commission of the Institute of France, appointed in 1797, repeated many of the known experiments, but does not seem to have decided any disputed points. The researches of this {241} commission referred rather to the discoveries of Galvani than to those of Volta: the latter were, indeed, hardly known in France till the conquest of Italy by Bonaparte, in 1801. France was, at the period of these discoveries, separated from all other

countries by war, and especially from England,[7\13] where Volta's Memoirs were published.

[Note 7\13: _Biog. Univ._, art. _Volta_, (by Biot.)]

The political revolutions of Italy affected, in very different manners, the two discoverers of whom we speak. Galvani refused to take an oath of allegiance to the Cisalpine republic, which the French conqueror established; he was consequently stripped of all his offices; and deprived, by the calamities of the times, of most of his relations, he sank into poverty, melancholy, and debility. At last his scientific reputation induced the republican rulers to decree his restoration to his professorial chair; but his claims were recognised too late, and he died without profiting by this intended favor, in 1798.

Volta, on the other hand, was called to Paris by Bonaparte as a man of science, and invested with honors, emoluments, and titles. The conqueror himself, indeed, was strongly interested by this train of research.[8\13] He himself founded valuable prizes, expressly with a view to promote its prosecution. At this period, there was something in this subject peculiarly attractive to his Italian mind; for the first glimpses of discoveries of great promise have always excited an enthusiastic activity of speculation in the philosophers of Italy, though generally accompanied with a want of precise thought. It is narrated[9\13] of Bonaparte, that after seeing the decomposition of the salts by means of the voltaic pile, he turned to Corvisart, his physician, and said, "Here, doctor, is the image of life; the vertebral column is the pile, the liver is the negative, the bladder the positive, pole." The importance of voltaic researches is not less than it was estimated by Bonaparte; but the results to which it was to lead were of a kind altogether different from those which thus suggested themselves to his mind. The connexion of mechanical and chemical action was the first great point to be dealt with; and for this purpose the laws of the mechanical action of voltaic electricity were to be studied.

[Note 8\13: Becquerel, _Traité d'Electr._ t. i. p. 107.]

[Note 9\13: Ib. t. i. p. 108.]

It will readily be supposed that the voltaic researches, thus begun, opened a number of interesting topics of examination and discussion. These, however, it does not belong to our place to dwell upon at present; since they formed parts of the theory of the subject, which {242} was not completed till light had been thrown upon it from other quarters. The identity of galvanism with electricity, for instance, was at first, as we have intimated, rather conjectured than proved. It was denied by Dr. Fowler, in 1793; was supposed to be confirmed by Dr. Wells two years later; but was, still later, questioned by Davy. The nature of the operation of the pile was variously conceived. Volta himself had obtained a view of it which succeeding researches confirmed, when he asserted,[10\13] in 1800, that it resembled an electric battery feebly charged and constantly renewing its charge. In pursuance of this view, the common electrical action was, at a later period (for instance by Ampère, in 1820), called _electrical tension_, while the voltaic action was called the _electrical current_, or _electromotive action_. The different effects produced, by increasing the size and

the number of the plates in the voltaic trough, were also very remarkable. The power of producing heat was found to depend on the size of the plates; the power of producing chemical changes, on the other hand, was augmented by the number of plates of which the battery consisted. The former effect was referred to the increased _quantity_, the latter to the _intensity_, of the electric fluid. We mention these distinctions at present, rather for the purpose of explaining the language in which the results of the succeeding investigations are narrated, than with the intention of representing the hypotheses and measures which they imply, as clearly established, at the period of which we speak. For that purpose new discoveries were requisite, which we have soon to relate.

[Note 10\13: _Phil. Trans._ p. 403.]

## CHAPTER III.

DISCOVERY OF THE LAWS OF THE MUTUAL ATTRACTION AND REPULSION OF VOLTAIC CURRENTS.--AMPÈRE.

IN order to show the place of voltaic electricity among the mechanico-chemical sciences, we must speak of its mechanical laws as separate from the laws of electro-magnetic action; although, in fact, it was only in consequence of the forces which conducting voltaic wires exert upon magnets, that those forces were detected which they exert upon each {243} other. This latter discovery was made by M. Ampère; and the extraordinary rapidity and sagacity with which he caught the suggestion of such forces, from the electro-magnetic experiments of M. Oersted, (of which we shall speak in the next chapter,) well entitle him to be considered as a great and independent discoverer. As he truly says,[11\13] "it by no means followed, that because a conducting wire exerted a force on a magnet, two conducting wires must exert a force on each other; for two pieces of soft iron, both of which affect a magnet, do not affect each other." But immediately on the promulgation of Oersted's experiments, in 1820, Ampère leapt forwards to a general theory of the facts, of which theory the mutual attraction and repulsion of conducting voltaic wires was a fundamental supposition. The supposition was immediately verified by direct trial; and the laws of this attraction and repulsion were soon determined, with great experimental ingenuity, and a very remarkable command of the resources of analysis. But the experimental and analytical investigation of the mutual action of voltaic or electrical currents, was so mixed up with the examination of the laws of electro-magnetism, which had given occasion to the investigation, that we must not treat the two provinces of research as separate. The mention in this place, premature as it might appear, of the labors of Ampère, arises inevitably from his being the author of a beautiful and comprehensive generalization, which not only included the phenomena exhibited by the new combinations of Oersted, but also disclosed forces which existed in arrangements already familiar, although they had never been detected till the theory pointed out how they were to be looked for.

[Note 11\13: _Théorie des Phénom. Electrodynamiques_, p. 113.]

## CHAPTER IV.

DISCOVERY OF ELECTRO-MAGNETIC ACTION.--OERSTED.

THE impulse which the discovery of galvanism, in 1791, and that of the voltaic pile, in 1800, had given to the study of electricity as a mechanical science, had nearly died away in 1820. It was in that year that M. Oersted, of Copenhagen, announced that the conducting {244} wire of a voltaic circuit, acts upon a magnetic needle; and thus recalled into activity that endeavor to connect magnetism with electricity, which, though apparently on many accounts so hopeful, had hitherto been attended with no success. Oersted found that the needle has a tendency to place itself _at right angles_ to the wire;--a kind of action altogether different from any which had been suspected.

This observation was of vast importance; and the analysis of its conditions and consequences employed the best philosophers in Europe immediately on its promulgation. It is impossible, without great injustice, to refuse great merit to Oersted as the author of the discovery. We have already said that men appear generally inclined to believe remarkable discoveries to be accidental, and the discovery of Oersted has been spoken of as a casual insulated experiment.[12\13] Yet Oersted had been looking for such an _accident_ probably more carefully and perseveringly than any other person in Europe. In 1807, he had published[13\13] a work, in which he professed that his purpose was "to ascertain whether electricity, in its most latent state, had any effect on the magnet." And he, as I know from his own declaration, considered his discovery as the natural sequel and confirmation of his early researches; as, indeed, it fell in readily and immediately with speculations on these subjects then very prevalent in Germany. It was an accident like that by which a man guesses a riddle on which his mind has long been employed.

[Note 12\13: See _Schelling ueber Faraday's Entdeckung_, p. 27.]

[Note 13\13: Ampère, p. 69.]

Besides the confirmation of Oersted's observations by many experimenters, great additions were made to his facts: of these, one of the most important was due to Ampère. Since the earth is in fact magnetic, the voltaic wire ought to be affected by terrestrial magnetism alone, and ought to tend to assume a position depending on the position of the compass-needle. At first, the attempts to produce this effect failed, but soon, with a more delicate apparatus, the result was found to agree with the anticipation.

It is impossible here to dwell on any of the subsequent researches, except so far as they are essential to our great object, the progress towards a general theory of the subject. I proceed, therefore, immediately to the attempts made towards this object. {245}

## CHAPTER V.

DISCOVERY OF THE LAWS OF ELECTRO-MAGNETIC ACTION.

ON attempting to analyse the electro-magnetic phenomena observed by Oersted and others into their simplest forms, they appeared, at least at first sight, to be different from any mechanical actions which had yet been observed. It seemed as if the conducting wire exerted on the pole of the magnet a force which was not attractive or repulsive, but _transverse_;--not tending to draw the point acted on nearer, or to push it further off, in the line which reached from the acting point, but urging it to move at right angles to this line. The forces appeared to be such as Kepler had dreamt of in the infancy of mechanical conceptions; rather than such as those of which Newton had established the existence in the solar system, and such as he, and all his successors, had supposed to be the only kinds of force which exist in nature. The north pole of the needle moved as if it were impelled by a vortex revolving round the wire in one direction, while the south pole seemed to be driven by an opposite vortex. The case seemed novel, and almost paradoxical.

It was soon established by experiments, made in a great variety of forms, that the mechanical action was really of this transverse kind. And a curious result was obtained, which a little while before would have been considered as altogether incredible;--that this force would cause a constant and rapid revolution of either of the bodies about the other;--of the conducting wire about the magnet, or of the magnet about the conducting wire. This was effected by Mr. Faraday in 1821.

The laws which regulated the intensity of this force, with reference to the distance and position of the bodies, now naturally came to be examined. MM. Biot and Savart in France, and Mr. Barlow in England, instituted such measures; and satisfied themselves that the elementary force followed the law of magnitude of all known elementary forces, in being inversely as the square of the distance; although, in its direction, it was so entirely different from other forces. But the investigation of the _laws of phenomena_ of the subject was too closely connected with the choice of a mechanical theory, to be established {246} previously and independently, as had been done in astronomy. The experiments gave complex results, and the analysis of these into their elementary actions was almost an indispensable step in order to disentangle their laws. We must, therefore, state the progress of this analysis.

## CHAPTER VI.

THEORY OF ELECTRODYNAMICAL ACTION.

AMPÈRE'S THEORY.--Nothing can show in a more striking manner the advanced condition of physical speculation in 1820, than the reduction of the strange and complex phenomena of electromagnetism to a simple and general theory as soon as they were published. Instead of a gradual establishment of laws of phenomena, and of theories more and more perfect, occupying ages, as in the case of astronomy, or generations, as in the instances of magnetism and electricity, a few months sufficed for the whole process of generalization; and the experiments made at Copenhagen were announced at Paris and London, almost at the same time with the skilful analysis and comprehensive inductions of Ampère.

Yet we should err if we should suppose, from the celerity with which the task was executed, that it was an easy one. There were required in the author of such a theory, not only those clear conceptions of the relations of space and force, which are the first conditions of all sound theory, and a full possession of the experiments; but also a masterly command of the mathematical arms by which alone the victory could be gained, and a sagacious selection of proper experiments which might decide the fate of the proposed hypothesis.

It is true, that the nature of the requisite hypothesis was not difficult to see in a certain vague and limited way. The conducting-wire and the magnetic needle had a tendency to arrange themselves at right angles to one another. This might be represented by supposing the wire to be made up of transverse magnetic needles, or by supposing the needle to be made up of transverse conducting-wires; for it was easy to conceive forces which should bring corresponding elements, either magnetic or voltaic, into parallel positions; and then the {247} general phenomena above stated would be accounted for. And the choice between the two modes of conception, appeared at first sight a matter of indifference. The majority of philosophers at first adopted, or at least employed, the former method, as Oersted in Germany, Berzelius in Sweden, Wollaston in England.

Ampère adopted the other view, according to which the magnet is made up of conducting-wires in a transverse position. But he did for his hypothesis what no one did or could do for the other: he showed that it was the only one which would account, without additional and arbitrary suppositions, for the facts of _continued_ motion in electromagnetic cases. And he further elevated his theory to a higher rank of generality, by showing that it explained,--not only the action of a conducting-wire upon a magnet, but also two other classes of facts, already spoken of in this history,--the action of magnets upon each other,--and the action of conducting-wires upon each other.

The deduction of such particular cases from the theory, required, as may easily be imagined, some complex calculations: but the deduction being satisfactory, it will be seen that Ampère's theory conformed to that description which we have repeatedly had to point out as the usual character of a true and stable theory; namely, that besides accounting for the class of phenomena which suggested it, it supplies an unforeseen explanation of other known facts. For the mutual action of magnets, which was supposed to be already reduced to a satisfactory theoretical form by Coulomb, was not contemplated by Ampère in the formation of his hypothesis; and the mutual action of voltaic currents, though tried only in consequence of the suggestion of the theory, was clearly a fact distinct from electromagnetic action; yet all these facts flowed alike from the theory. And thus Ampère brought into view a class of forces for which the term "electromagnetic" was too limited, and which he designated[14\13] by the appropriate term _electrodynamic_; distinguishing them by this expression, as the forces of an electric _current_, from the _statical_ effects of electricity which we had formerly to treat of. This term has passed into common use among scientific writers, and remains the record and stamp of the success of the Amperian induction.

[Note 14\13: _Ann. de Chim._, tom. xx. p. 60 (1822).]

The first promulgation of Ampère's views was by a communication to the French Academy of Sciences, September the 18th, 1820; Oersted's discoveries having reached Paris only in the preceding July. {248} At almost every meeting of the Academy during the remainder of that year and the beginning of the following one, he had new developements or new confirmations of his theory to announce. The most hypothetical part of his theory,--the proposition that magnets might be considered in their effects as identical with spiral voltaic wires,--he asserted from the very first. The mutual attraction and repulsion of voltaic wires,--the laws of this action,--the deduction of the observed facts from it by calculation,--the determination, by new experiments, of the constant quantities which entered into his formulæ,--followed in rapid succession. The theory must be briefly stated. It had already been seen that parallel voltaic currents attracted each other; when, instead of being parallel, they were situate in any directions, they still exerted attractive and repulsive forces depending on the distance, and on the directions of each element of both currents. Add to this doctrine the hypothetical constitution of magnets, namely, that a voltaic current runs round the axis of each particle, and we have the means of calculating a vast variety of results which may be compared with experiment. But the laws of the elementary forces required further fixation. What _functions_ are the forces of the distance and the directions of the elements?

To extract from experiment an answer to this inquiry was far from easy, for the elementary forces were mathematically connected with the observed facts, by a double mathematical integration;--a long, and, while the constant coefficients remained undefined, hardly a possible operation. Ampère made some trials in this way, but his happier genius suggested to him a better path. It occurred to him, that if his integrals, without being specially found, could be shown to vanish upon the whole, under certain conditions of the problem, this circumstance would correspond to arrangements of his apparatus in which a state of equilibrium was preserved, however the form of some of the parts might be changed. He found two such cases, which were of great importance to the theory. The first of these cases proved that the force exerted by any element of the voltaic wire might be resolved into other forces by a theorem resembling the well-known proposition of the parallelogram of forces. This was proved by showing that the action of a straight wire is the same with that of another wire which joins the same extremities, but is bent and contorted in any way whatever. But it still remained necessary to determine two fundamental quantities; one which expressed the _power_ of the distance according to which the force varied; the other, the {249} degree in which the force is affected by the _obliquity_ of the elements. One of the general causes of equilibrium, of which we have spoken, gave a relation between these two quantities;[15\13] and as the power was naturally, and, as it afterwards appeared, rightly conjectured to be the inverse square, the other quantity also was determined; and the general problem of electrodynamical action was fully solved.

[Note 15\13: Communication to the Acad. Sc., June 10, 1822. See Ampère, _Recueil_, p. 292.]

If Ampère had not been an accomplished analyst, he would not have been able to discover the condition on which the nullity of the integral in this case depended.[16\13] And throughout his labors, we find reason to admire, both his mathematical skill, and his steadiness of thought; although these excellences are by no means accompanied throughout with corresponding clearness and elegance of exposition in his writings.

[Note 16\13: _Recueil_, p. 314.]

_Reception of Ampère's Theory._--Clear mathematical conceptions, and some familiarity with mathematical operations, were needed by readers also, in order to appreciate the evidence of the theory; and, therefore, we need not feel any surprise if it was, on its publication and establishment, hailed with far less enthusiasm than so remarkable a triumph of generalizing power might appear to deserve. For some time, indeed, the greater portion of the public were naturally held in suspense by the opposing weight of rival names. The Amperian theory did not make its way without contention and competition. The electro-magnetic experiments, from their first appearance, gave a clear promise of some new and wide generalization; and held out a prize of honor and fame to him who should be first in giving the right interpretation of the riddle. In France, the emulation for such reputation is perhaps more vigilant and anxious than it is elsewhere; and we see, on this as on other occasions, the scientific host of Paris springing upon a new subject with an impetuosity which, in a short time, runs into controversies for priority or for victory. In this case, M. Biot, as well as Ampère, endeavored to reduce the electro-magnetic phenomena to general laws. The discussion between him and Ampère turned on some points which are curious. M. Biot was disposed to consider as an elementary action, the force which an element of a voltaic wire exerts upon a magnetic particle, and which is, as we have seen, at right angles to their mutual distance; and he conceived that {250} the equal reaction which necessarily accompanies this action acts oppositely to the action, not in the same line, but in a parallel line, at the other extremity of the distance; thus forming a primitive _couple_, to use a technical expression borrowed from mechanics. To this Ampère objected,[17\13] that the _direct_ opposition of all elementary action and reaction was a universal and necessary mechanical law. He showed too that such a couple as had been assumed, would follow as a _derivative_ result from his theory. And in comparing his own theory with that in which the voltaic wire is assimilated to a collection of transverse magnets, he was also able to prove that no such assemblage of forces acting to and from fixed points, as the forces of magnets do act, could produce a continued motion like that discovered by Faraday. This, indeed, was only the well-known demonstration of the impossibility of a perpetual motion. If, instead of a collection of magnets, the adverse theorists had spoken of a magnetic _current_, they might probably interpret their expressions so as to explain the facts; that is, if they considered every element of such a current as a magnet, and consequently, every point of it as being a north and a south point at the same instant. But to introduce such a conception of a magnetic current was to abandon all the laws of magnetic action hitherto established; and consequently to lose all that gave the hypothesis its value. The idea of an electric current, on the other hand, was so far from being a new and hazardous assumption, that it had already been forced upon

philosophers from the time of Volta; and in this current, the relation of _preceding_ and _succeeding_, which necessarily existed between the extremities of any element, introduced that relative polarity on which the success of the explanations of the facts depended. And thus in this controversy, the theory of Ampère has a great and undeniable superiority over the rival hypotheses.

[Note 17\13: Ampère, _Théorie_, p. 154.]

## CHAPTER VII.

CONSEQUENCES OF THE ELECTRODYNAMIC THEORY.

IT is not necessary to state the various applications which were soon made of the electro-magnetic discoveries. But we may notice one {251} of the most important,--the _Galvanometer_, an instrument which, by enabling the philosopher to detect and to measure extremely minute electrodynamic actions, gave an impulse to the subject similar to that which it received from the invention of the Leyden Phial, or the Voltaic Pile. The strength of the voltaic current was measured, in this instrument, by the deflection produced in a compass-needle; and its sensibility was multiplied by making the wire pass repeatedly above and below the needle. Schweigger, of Halle, was one of the first devisers of this apparatus.

The substitution of electro-magnets, that is, of spiral tubes composed of voltaic wires, for common magnets, gave rise to a variety of curious apparatus and speculations, some of which I shall hereafter mention.

[2nd Ed.] [When a voltaic apparatus is in action, there may be conceived to be a current of electricity running through its various elements, as stated in the text. The force of this current in various parts of the circuit has been made the subject of mathematical investigation by M. Ohm.[18\13] The problem is in every respect similar to that of the flow of heat through a body, and taken generally, leads to complex calculations of the same kind. But Dr. Ohm, by limiting the problem in the first place by conditions which the usual nature and form of voltaic apparatus suggest, has been able to give great simplicity to his reasonings. These conditions are, the linear form of the conductors (wires) and the steadiness of the electric state. For this part of the problem Dr. Ohm's reasonings are as simple and as demonstrative as the elementary propositions of Mechanics. The formulæ for the electric force of a voltaic current to which he is led have been experimentally verified by others, especially Fechner,[19\13] Gauss,[20\13] Lenz, Jacobi, Poggendorf, and Pouillet.

[Note 18\13: _Die Galvanische Kette Mathematisch bearbeitet von Dr. G. S. Ohm_, Berlin, 1827.]

[Note 19\13: _**Mass-bestimmungen über die Galvanische Kette._ Leipzig, 1831.]

[Note 20\13: _Results of the Magnetic Association._]

Among ourselves, Mr. Wheatstone has confirmed and applied the views of M. Ohm, in a Memoir[21\13] _On New Instruments and Processes for determining the Constants of a Voltaic Circuit_. He there remarks that the clear ideas of electromotive

forces and resistances, substituted by Ohm for the vague notions of quantity and intensity which have long been prevalent, give satisfactory explanations of the most important difficulties, and express the laws of a vast number of phenomena {252} in formulæ of remarkable simplicity and generality. In this Memoir, Professor Wheatstone describes an instrument which he terms _Rheostat_, because it brings to a common standard the voltaic currents which are compared by it. He generalizes the language of the subject by employing the term _rheomotor_ for any apparatus which originates an electric current (whether voltaic or thermoelectric, &c.) and _rheometer_ for any instrument to measure the force of such a current. It appears that the idea of constructing an instrument of the nature of the Rheostat had occurred also to Prof. Jacobi, of St Petersburg.]

[Note 21\13: _Phil. Trans._ 1843. Pt. 11.]

The galvanometer led to the discovery of another class of cases in which the electrodynamical action was called into play, namely, those in which a circuit, composed of two metals only, became electro-magnetic by _heating_ one part of it. This discovery of _thermo-electricity_ was made by Professor Seebeck of Berlin, in 1822, and prosecuted by various persons; especially by Prof. Cumming[22\13] of Cambridge, who, early in 1823, extended the examination of this property to most of the metals, and determined their thermo-electric order. But as these investigations exhibited no new mechanical effects of electromotive forces, they do not now further concern us; and we pass on, at present, to a case in which such forces act in a manner different from any of those already described.

[Note 22\13: _Camb. Trans._ vol. ii. p. 62. _On the Development of Electro-Magnetism by Heat._]

DISCOVERY OF DIAMAGNETISM.

[2nd Ed.] [By the discoveries just related, a cylindrical spiral of wire through which an electric current is passing is identified with a magnet; and the effect of such a spiral is increased by placing in it a core of soft iron. By the use of such a combination under the influence of a voltaic battery, magnets are constructed far more powerful than those which depend upon the permanent magnetism of iron. The electro-magnet employed by Dr. Faraday in some of his experiments would sustain a hundred-weight at either end.

By the use of such magnets Dr. Faraday discovered that, besides iron, nickel and cobalt, which possess magnetism in a high degree, many bodies are magnetic in a slight degree. And he made the further very important discovery, that of those substances which are not magnetic, many, perhaps all, possess an opposite property, in virtue of which he terms them _diamagnetic_. The opposition is of this {253} kind;--that magnetic bodies in the form of bars or needles, if free to move, arrange themselves in the _axial_ line joining the poles; diamagnetic bodies under the same circumstances arrange themselves in an _equatorial_ position, perpendicular to the axial line. And this tendency he conceives to be the result of one more general; that whereas magnetic bodies are attracted to the poles of a magnet, diamagnetic bodies are repelled from the poles. The list of diamagnetic bodies includes all kinds of

substances; not only metals, as antimony, bismuth, gold, silver, lead, tin, zinc, but many crystals, glass, phosphorus, sulphur, sugar, gum, wood, ivory; and even flesh and fruit.

It appears that M. le Bailli had shown, in 1829, that both bismuth and antimony and bismuth repelled the magnetic needle; and as Dr. Faraday remarks, it is astonishing that such an experiment should have remained so long without further results. M. Becquerel in 1827 observed, and quoted Coulomb as having also observed, that a needle of wood under certain conditions pointed across the magnetic curves; and also stated that he had found a needle of wood place itself parallel to the wires of a galvanometer. This he referred to a magnetism transverse to the length. But he does not refer the phenomena to elementary repulsive action, nor show that they are common to an immense class of bodies, nor distinguish this diamagnetic from the magnetic class, as Faraday has taught us to do.

I do not dwell upon the peculiar phenomena of copper which, in the same series of researches, are traced by Dr. Faraday to the combined effect of its diamagnetic character, and the electric currents excited in it by the electro-magnet; nor to the optical phenomena manifested by certain transparent diamagnetic substances under electric action; as already stated in Book ix.[23\13]]

[Note 23\13: See the _Twentieth Series of Experimental Researches in Electricity_, read to the Royal Society, Dec. 18, 1845.]

## CHAPTER VIII.

DISCOVERY OF THE LAWS OF MAGNETO-ELECTRIC INDUCTION.--FARADAY.

IT was clearly established by Ampère, as we have seen, that magnetic action is a peculiar form of electromotive actions, and that, in {254} this kind of agency, action and reaction are equal and opposite. It appeared to follow almost irresistibly from these considerations, that magnetism might be made to produce electricity, as electricity could be made to imitate all the effects of magnetism. Yet for a long time the attempts to obtain such a result were fruitless. Faraday, in 1825, endeavored to make the conducting-wire of the voltaic circuit excite electricity in a neighboring wire by induction, as the conductor charged with common electricity would have done, but he obtained no such effect. If this attempt had succeeded, the magnet, which, for all such purposes, is an assemblage of voltaic circuits, might also have been made to excite electricity. About the same time, an experiment was made in France by M. Arago, which really involved the effect thus sought; though this effect was not extricated from the complex phenomenon, till Faraday began his splendid career of discovery on this subject in 1832. Arago's observation was, that the rapid revolution of a conducting-plate in the neighborhood of a magnet, gave rise to a force acting on the magnet. In England, Messrs. Barlow and Christie, Herschel and Babbage, repeated and tried to analyse this experiment; but referring the forces only to conditions of space and time, and overlooking the real cause, the electrical currents produced by the motion, these philosophers were altogether unsuccessful in their labors. In 1831, Faraday again sought for electro-dynamical induction, and after

some futile trials, at last found it in a form different from that in which he had looked for it. It was then seen, that at the precise time of making or breaking the contact which closed the galvanic circuit, a momentary effect was induced in a neighboring wire, but disappeared instantly.[24\13] Once in possession of this fact, Mr. Faraday ran rapidly up the ladder of discovery, to the general point of view.--Instead of suddenly making or breaking the contact of the inducing circuit, a similar effect was produced by removing the inducible wire nearer to or further from the circuit;[25\13]--the effects were increased by the proximity of soft iron;[26\13]--when the soft iron was affected by an ordinary magnet instead of the voltaic wire, the same effect still recurred;[27\13]--and thus it appeared, that by making and breaking magnetic contact, a momentary electric current was produced. It was produced also by moving the magnet;[28\13]--or by moving the wire with reference to the magnet.[29\13] Finally, it was found that the earth might supply the place of a magnet {255} in this as in other experiments;[30\13] and the mere motion of a wire, under proper circumstances, produced in it, it appeared, a momentary electric current.[31\13] These facts were curiously confirmed by the results in special cases. They explained Arago's experiments: for the momentary effect became permanent by the revolution of the plate. And without using the magnet, a revolving plate became an electrical machine;[32\13]--a revolving globe exhibited electro-magnetic action,[33\13] the circuit being complete in the globe itself without the addition of any wire;--and a mere motion of the wire of a galvanometer produced an electro-dynamic effect upon its needle.[34\13]

[Note 24\13: _Phil. Trans._ 1832, p. 127, First Series, Art. 10.]

[Note 25\13: Art. 18.]

[Note 26\13: Art. 28.]

[Note 27\13: Art. 37.]

[Note 28\13: Art. 39.]

[Note 29\13: Art. 53.]

[Note 30\13: Second Series, _Phil. Trans._ p. 163.]

[Note 31\13: Art. 141.]

[Note 32\13: Art. 150.]

[Note 33\13: Art. 164.]

[Note 34\13: Art. 171.]

But the question occurs, What is the general law which determines the direction of electric currents thus produced by the joint effects of motion and magnetism? Nothing but a peculiar steadiness and clearness in his conceptions of space, could have enabled Mr. Faraday to detect the law of this phenomenon. For the question required that he should determine the mutual relations in space which connect the magnetic poles, the position of the wire, the direction of the wire's motion, and the electrical current produced in it. This was no easy problem; indeed, the mere relation

of the magnetic to the electric forces, the one set being perpendicular to the other, is of itself sufficient to perplex the mind; as we have seen in the history of the electrodynamical discoveries. But Mr. Faraday appears to have seized at once the law of the phenomena. "The relation," he says,[35\13] "which holds between the magnetic pole, the moving wire or metal, and the direction of the current evolved, is very simple (so it seemed to him) although rather difficult to express." He represents it by referring position and motion to the "magnetic curves," which go from a magnetic pole to the opposite pole. The current in the wire sets one way or the other, according to the direction in which the motion of the wire cuts these curves. And thus he was enabled, at the end of his Second Series of _Researches_ (December, 1831), to give, in general terms, the law of nature to which may be referred the extraordinary number of new and curious experiments which he has stated;[36\13]--namely, that if a wire move so as to cut a magnetic curve, a power is called into action which tends to urge a magnetic current through the wire; and that if a mass move so that its parts do not move in the same direction across the magnetic curves, {256} and with the same angular velocity, electrical currents are called into play in the mass.

[Note 35\13: First Series, Art. 114.]

[Note 36\13: Art. 256-264.]

This rule, thus simple from its generality, though inevitably complex in every special case, may be looked upon as supplying the first demand of philosophy, _the law of the phenomena_; and accordingly Dr. Faraday has, in all his subsequent researches on magneto-electric induction, applied this law to his experiments; and has thereby unravelled an immense amount of apparent inconsistency and confusion, for those who have followed him in his mode of conceiving the subject.

But yet other philosophers have regarded these phenomena in other points of view, and have stated the laws of the phenomena in a manner different from Faraday's, although for the most part equivalent to his. And these attempts to express, in the most simple and general form, the law of the phenomena of magneto-electrical induction, have naturally been combined with the expression of other laws of electrical and magnetical phenomena. Further, these endeavors to connect and generalize the Facts have naturally been clothed in the garb of various Theories:--the _laws of phenomena_ have been expressed in terms of the supposed _causes of the phenomena_; as fluids, attractions and repulsions, particles with currents running through them or round them, physical lines of force, and the like. Such views, and the conflict of them, are the natural and hopeful prognostics of a theory which shall harmonize their discords and include all that each contains of Truth. The fermentation at present is perhaps too great to allow us to see clearly the truth which lies at the bottom. But a few of the leading points of recent discussions on these subjects will be noticed in the Additions to this volume.

## CHAPTER IX.

TRANSITION TO CHEMICAL SCIENCE.

THE preceding train of generalization may justly appear extensive, and of itself well worthy of admiration. Yet we are to consider all that has there been established as only one-half of the science to which it belongs,--one limb of the colossal form of Chemistry. We {257} have ascertained, we will suppose, the laws of Electric Polarity; but we have then to ask, What is the relation of this Polarity to Chemical Composition? This was the great problem which, constantly present to the minds of electro-chemical inquirers, drew them on, with the promise of some deep and comprehensive insight into the mechanism of nature. Long tasks of research, though only subsidiary to this, were cheerfully undertaken. Thus Faraday[37\13] describes himself as compelled to set about satisfying himself of the identity of common, animal, and voltaic electricity, as "the decision of a doubtful point which interfered with the extension of his views, and destroyed the strictness of reasoning." Having established this identity, he proceeded with his grand undertaking of electro-chemical research.

[Note 37\13: Dec. 1832. _Researches_, 266.]

The connexion of electrical currents with chemical action, though kept out of sight in the account we have hitherto given, was never forgotten by the experimenters; for, in fact, the modes in which electrical currents were excited, were chemical actions;--the action of acids and metals on each other in the voltaic trough, or in some other form. The dependence of the electrical effect on these chemical actions, and still more, the chemical actions produced by the agency of the poles of the circuit, had been carefully studied; and we must now relate with what success.

But in what terms shall we present this narration? We have spoken of chemical actions,--but what kind of actions are these? _Decomposition_; the _resolution_ of compounds into their ingredients; the separation of _acids_ from _bases_; the reduction of bodies to _simple elements_. These names open to us a new drama; they are words which belong to a different set of relations of things, a different train of scientific inductions, a different system of generalizations, from any with which we have hitherto been concerned. We must learn to understand these phrases, before we can advance in our history of human knowledge.

And how are we to learn the meaning of this collection of words? In what other language shall it be explained? In what terms shall we define these new expressions? To this we are compelled to reply, that we cannot translate these terms into any ordinary language;--that we cannot define them in any terms already familiar to us. Here, as in all other branches of knowledge, the meaning of words is to be sought in the progress of thought; the history of science is our {258} dictionary; the steps of scientific induction are our definitions. It is only by going back through the successful researches of men respecting the composition and elements of bodies, that we can learn in what sense such terms must be understood, so as to convey real knowledge. In order that they may have a meaning for us, we must inquire what meaning they had in the minds of the authors of our discoveries.

And thus we cannot advance a step, till we have brought up our history of Chemistry to the level of our history of Electricity;--till we have studied the progress

of the analytical, as well as the mechanical sciences. We are compelled to pause and look backwards here; just as happened in the history of astronomy, when we arrived at the brink of the great mechanical inductions of Newton, and found that we must trace the history of Mechanics, before we could proceed to mechanical Astronomy. The terms "force, attraction, inertia, momentum," sent us back into preceding centuries then, just as the terms "composition" and "element" send us back now.

Nor is it to a small extent that we have thus to double back upon our past advance. Next to Astronomy, Chemistry is one of the most ancient of sciences;--the field of the earliest attempts of man to command and understand nature. It has held men for centuries by a kind of fascination; and innumerable and endless are the various labors, the failures and successes, the speculations and conclusions, the strange pretences and fantastical dreams, of those who have pursued it. To exhibit all these, or give any account of them, would be impossible; and for our design, it would not be pertinent. To extract from the mass that which is to our purpose, is difficult; but the attempt must be made. We must endeavor to analyse the history of Chemistry, so far as it has tended towards the establishment of general principles. We shall thus obtain a sight of generalizations of a new kind, and shall prepare ourselves for others of a higher order.

{{259}} BOOK XIV.

## _THE ANALYTICAL SCIENCE._

HISTORY OF CHEMISTRY.

. . . . . Soon had his crew Opened into the hill a spacious wound, And digged out ribs of gold . . . . Anon out of the earth a fabric huge Rose like an exhalation, with the sound Of dulcet symphonies and voices sweet, Built like a temple. MILTON. _Paradise Lost_, i.

{{261}} CHAPTER I.

IMPROVEMENT OF THE NOTION OF CHEMICAL ANALYSIS, AND RECOGNITION OF IT AS THE SPAGIRIC ART.

THE doctrine of "the four elements" is one of the oldest monuments of man's speculative nature; goes back, perhaps, to times anterior to Greek philosophy; and as the doctrine of Aristotle and Galen, reigned for fifteen hundred years over the Gentile, Christian, and Mohammedan world. In medicine, taught as the doctrine of the four "elementary qualities," of which the human body and all other substances are compounded, it had a very powerful and extensive influence upon medical practice. But this doctrine never led to any attempt actually to analyse bodies into their supposed elements: for composition was inferred from the resemblance of the qualities, not from the separate exhibition of the ingredients; the supposed analysis was, in short, a decomposition of the body into adjectives, not into substances.

This doctrine, therefore, may be considered as a negative state, antecedent to the very beginning of chemistry; and some progress beyond this mere negation was made, as soon as men began to endeavor to compound and decompound substances

by the use of fire or mixture, however erroneous might be the opinions and expectations which they combined with their attempts. Alchemy is a step in chemistry, so far as it implies the recognition of the work of the cupel and the retort, as the produce of analysis and synthesis. How perplexed and perverted were the forms in which this recognition was clothed,--how mixed up with mythical follies and extravagancies, we have already seen; and the share which Alchemy had in the formation of any sounder knowledge, is not such as to justify any further notice of that pursuit.

The result of the attempts to analyse bodies by heat, mixture, and the like processes, was the doctrine that the first principles of things are _three_, not four; namely, _salt_, _sulphur_, and _mercury_; and that, of these three, all things are compounded. In reality, the doctrine, as thus stated, contained no truth which was of any value; for, though the chemist could extract from most bodies portions which he called salt, {262} and sulphur, and mercury, these names were given, rather to save the hypothesis, than because the substances were really those usually so called: and thus the supposed analyses proved nothing, as Boyle justly urged against them.[1\14]

[Note 1\14: Shaw's Boyle. _Skeptical Chymist_, pp. 312, 313. &c.]

The only real advance in chemical theory, therefore, which we can ascribe to the school of _the three principles_, as compared with those who held the ancient dogma of the four elements, is, the acknowledgment of the changes produced by the chemist's operations, as being changes which were to be accounted for by the union and separation of substantial elements, or, as they were sometimes called, of _hypostatical principles_. The workmen of this school acquired, no doubt, a considerable acquaintance with the results of the kinds of processes which they pursued; they applied their knowledge to the preparation of new medicines; and some of them, as Paracelsus and Van Helmont, attained, in this way, to great fame and distinction: but their merits, as regards theoretical chemistry, consist only in a truer conception of the problem, and of the mode of attempting its solution, than their predecessors had entertained.

This step is well marked by a word which, about the time of which we speak, was introduced to denote the chemist's employment. It was called the _Spagiric art_, (often misspelt _Spagyric_,) from two Greek words, (σπάω, ἀγείρω,) which mean to _separate_ parts, and to _unite_ them. These two processes, or in more modern language, _analysis_ and _synthesis_, constitute the whole business of the chemist. We are not making a fanciful arrangement, therefore, when we mark the recognition of this object as a step in the progress of chemistry. I now proceed to consider the manner in which the conditions of this analysis and synthesis were further developed.

## CHAPTER II.

DOCTRINE OF ACID AND ALKALI.--SYLVIUS.

AMONG the results of mixture observed by chemists, were many instances in which two ingredients, each in itself pungent or destructive, being put together,

became mild and inoperative; each {263} counteracting and neutralizing the activity of the other. The notion of such opposition and neutrality is applicable to a very wide range of chemical processes. The person who appears first to have steadily seized and generally applied this notion is Francis de la Boé Sylvius; who was born in 1614, and practised medicine at Amsterdam, with a success and reputation which gave great currency to his opinions on that art.[2\14] His chemical theories were propounded as subordinate to his medical doctrines; and from being thus presented under a most important practical aspect, excited far more attention than mere theoretical opinions on the composition of bodies could have done. Sylvius is spoken of by historians of science, as the founder of the _iatro-chemical_ sect among physicians; that is, the sect which considers the disorders in the human frame as the effects of chemical relations of the fluids, and applies to them modes of cure founded upon this doctrine. We have here to speak, not of his physiological, but of his chemical views.

[Note 2\14: Sprengel. _Geschichte der Arzneykunde_, vol. iv. Thomson's _History of Chemistry_ in the corresponding part is translated from Sprengel.]

The distinction of _acid_ and _alkaline_ bodies (_acidum_, _lixivum_) was familiar before the time of Sylvius; but he framed a system, by considering them both as eminently acrid and yet opposite, and by applying this notion to the human frame. Thus[3\14] the lymph contains an acid, the bile an alkaline salt. These two opposite acrid substances, when they are brought together, _neutralize_ each other (_infringunt_), and are changed into an intermediate and milder substance.

[Note 3\14: _De Methodo Medendi_, Amst. 1679. Lib. ii. cap. 28, sects. 8 and 53.]

The progress of this doctrine, as a physiological one, is an important part of the history of medical science in the seventeenth century; but with that we are not here concerned. But as a chemical doctrine, this notion of the opposition of acid and alkali, and of its very general applicability, struck deep root, and has not been eradicated up to our own time. Boyle, indeed, whose disposition led him to suspect all generalities, expressed doubts with regard to this view;[4\14] and argued that the supposition of acid and alkaline parts in all bodies was precarious, their offices arbitrary, and the notion of them unsettled. Indeed it was not difficult to show, that there was no one certain criterion to which all supposed acids conformed. Yet the general conception of such a combination as that of acid and alkali was supposed to {264} be, served so well to express many chemical facts, that it kept its ground. It is found, for instance, in Lemery's _Chemistry_, which was one of those in most general use before the introduction of the phlogistic theory. In this work (which was translated into English by Keill, in 1698) we find alkalies defined by their effervescing with acids.[5\14] They were distinguished as the _mineral_ alkali (soda), the _vegetable_ alkali (potassa), and the _volatile_ alkali (ammonia). Again, in Macquer's _Chemistry_, which was long the text-book in Europe during the reign of phlogiston, we find acids and alkalies, and their union, in which they rob each other of their characteristic properties, and form neutral salts, stated among the leading principles of the science.[6\14]

[Note 4\14: Shaw's _Boyle_, iii. p. 432.]

[Note 5\14: Lemery, p. 25.]

[Note 6\14: Macquer, p. 19.]

In truth, the mutual relation of acids to alkalies was the most essential part of the knowledge which chemists possessed concerning them. The importance of this relation arose from its being the first distinct form in which the notion of chemical attraction or affinity appeared. For the acrid or caustic character of acids and alkalies is, in fact, a tendency to alter the bodies they touch, and thus to alter themselves; and the neutral character of the compounds **is the absence of any such proclivity to change. Acids and alkalies have a strong disposition to unite. They combine, often with vehemence, and produce neutral salts; they exhibit, in short, a prominent example of the chemical attraction, or affinity, by which two ingredients are formed into a compound. The relation of _acid_ and _base_ in a salt is, to this day, one of the main grounds of all theoretical reasonings.

The more distinct development of the notion of such chemical attraction, gradually made its way among the chemists of the latter part of the seventeenth and the beginning of the eighteenth century, as we may see in the writings of Boyle, Newton, and their followers. Beecher speaks of this attraction as a _magnetism_; but I do not know that any writer in particular, can be pointed out as the person who firmly established the general notion of _chemical attraction_.

But this idea of chemical attraction became both more clear and more extensively applicable, when it assumed the form of the doctrine of _elective_ attractions, in which shape we must now speak of it. {265}

## CHAPTER III.

DOCTRINE OF ELECTIVE ATTRACTIONS. GEOFFROY. BERGMAN.

THOUGH the chemical combinations of bodies had already been referred to attraction, in a vague and general manner, it was impossible to explain the changes that take place, without supposing the attraction to be greater or less, according to the nature of the body. Yet it was some time before the necessity of such a supposition was clearly seen. In the history of the French Academy for 1718 (published 1719), the writer of the introductory notice (probably Fontenelle) says, "That a body which is united to another, for example, a solvent which has penetrated a metal, should quit it to go and unite itself with another which we present to it, is a thing of which the possibility had never been guessed by the most subtle philosophers, and of which the explanation even now is not easy." The doctrine had, in fact, been stated by Stahl, but the assertion just quoted shows, at least, that it was not familiar. The principle, however, is very clearly stated[7\14] in a memoir in the same volume, by Geoffroy, a French physician of great talents and varied knowledge, "We observe in chemistry," he says, "certain relations amongst different bodies, which cause them to unite. These relations have their _degrees_ and their _laws_. We observe their different degrees in this;--that among different matters jumbled together, which have a certain disposition to unite, we find that one of these

substances always unites constantly with a certain other, preferably to all the rest." He then states that those which unite by preference, have "plus de rapport," or, according to a phrase afterwards used, more _affinity_. "And I have satisfied myself," he adds, "that we may deduce, from these observations, the following proposition, which is very extensively true, though I cannot enunciate it as universal, not having been able to examine all the possible combinations, to assure myself that I should find no exception." The proposition which he states in this admirable spirit of philosophical caution, is this: "In all cases where two substances, {266} which have any disposition to combine, are united; if there approaches them a third, which has more affinity with one of the two, this one unites with the third and lets go the other." He then states these affinities in the form of a Table; placing a substance at the head of each column, and other substances in succession below it, according to the order of their affinities for the substance which stands at the head. He allows that the separation is not always complete (an imperfection which he ascribes to the glutinosity of fluids and other causes), but, with such exceptions, he defends very resolutely and successfully his Table, and the notions which it implies.

[Note 7\14: _Mém. Acad. Par._ 1718, p. 202.]

The value of such a tabulation was immense at the time, and is even still very great; it enabled the chemist to trace beforehand the results of any operation; since, when the ingredients were given, he could see which were the strongest of the affinities brought into play, and, consequently, what compounds would be formed. Geoffroy himself gave several good examples of this use of his table. It was speedily adopted into works on chemistry. For instance, Macquer[8\14] places it at the end of his book; "taking it," as he says, "to be of great use at the end of an elementary tract, as it collects into one point of view, the most essential and fundamental doctrines which are dispersed through the work."

[Note 8\14: Pref., p. 13.]

The doctrine of _Elective Attraction_, as thus promulgated, contained so large a mass of truth, that it was never seriously shaken, though it required further development and correction. In particular the celebrated work of Torbern Bergman, professor at Upsala, _On Elective Attractions_, published in 1775, introduced into it material improvements. Bergman observed, that not only the order of attractions, but the sum of those attractions which had to form the new compounds, must be taken account of, in order to judge of the result. Thus,[9\14] if we have a combination of two elements, _P_, _s_, (potassa and vitriolic acid), and another combination, _L_, _m_, (lime and muriatic acid,) though _s_ has a greater affinity for _P_ than for _L_, yet the sum of the attractions of _P_ to _m_, and of _L_ to _s_, is greater than that of the original compounds, and therefore if the two combinations are brought together, the new compounds, _P_, _m_, and _L_, _s_, are formed.

[Note 9\14: _Elect. Attract._, p. 19.]

The Table of Elective Attractions, modified by Bergman in pursuance of these views, and corrected according to the advanced knowledge of the time, became still more important than before. The next step {267} was to take into account the

quantities of the elements which combined; but this leads us into a new train of investigation, which was, indeed, a natural sequel to the researches of Geoffroy and Bergman.

In 1803, however, a chemist of great eminence, Berthollet, published a work (_Essai de Statique Chimique_), the tendency of which appeared to be to throw the subject back into the condition in which it had been before Geoffroy. For Berthollet maintained that the rules of chemical combination were not definite, and dependent on the nature of the substances alone, but indefinite, depending on the quantity present, and other circumstances. Proust answered him, and as Berzelius says,[10\14] "Berthollet defended himself with an acuteness which makes the reader hesitate in his judgment; but the great mass of facts finally decided the point in favor of Proust." Before, however, we trace the result of these researches, we must consider Chemistry as extending her inquiries to combustion as well as mixture, to airs as well as fluids and solids, and to weight as well as quality. These three steps we shall now briefly treat of.

[Note 10\14: _Chem._ t. iii. p. 23.]

## CHAPTER IV.

DOCTRINE OF ACIDIFICATION AND COMBUSTION.--PHLOGISTIC THEORY.

PUBLICATION _of the Theory by Beccher and Stahl._--It will be recollected that we are tracing the history of the _progress_ only of Chemistry, not of its errors;--that we are concerned with doctrines only so far as they are true, and have remained part of the received system of chemical truths. The Phlogistic Theory was deposed and succeeded by the Theory of Oxygen. But this circumstance must not lead us to overlook the really sound and permanent part of the opinions which the founders of the phlogistic theory taught. They brought together, as processes of the same kind, a number of changes which at first appeared to have nothing in common; as acidification, combustion, respiration. Now this classification is true; and its importance remains undiminished, whatever are the explanations which we adopt of the processes themselves.

The two chemists to whom are to be ascribed the merit of this step, and the establishment of the _phlogistic theory_ which they connected {268} with it, are John Joachim Beccher and George Ernest Stahl; the former of whom was professor at Mentz, and physician to the Elector of Bavaria (born 1625, died 1682); the latter was professor at Halle, and afterwards royal physician at Berlin (born 1660, died 1734). These two men, who thus contributed to a common purpose, were very different from each other. The first was a frank and ardent enthusiast in the pursuit of chemistry, who speaks of himself and his employments with a communicativeness and affection both amusing and engaging. The other was a teacher of great talents and influence, but accused of haughtiness and moroseness; a character which is well borne out by the manner in which, in his writings, he anticipates an unfavorable reception, and defies it. But it is right to add to this that he speaks of Beccher, his predecessor, with an ungrudging acknowledgment of obligations to him, and a vehement assertion of

his merit as the founder of the true system, which give a strong impression of Stahl's justice and magnanimity.

Beccher's opinions were at first promulgated rather as a correction than a refutation of the doctrine of the three principles, salt, sulphur, and mercury. The main peculiarity of his views consists in the offices which he ascribes to his _sulphur_, these being such as afterwards induced Stahl to give the name of _Phlogiston_ to this element. Beccher had the sagacity to see that the reduction of metals to an earthy form (_calx_), and the formation of sulphuric acid from sulphur, are operations connected by a general analogy, as being alike processes of combustion. Hence the metal was supposed to consist of an earth, and of something which, in the process of combustion, was separated from it; and, in like manner, sulphur was supposed to consist of the sulphuric acid, which remained after its combustion, and of the combustible part or true sulphur, which flew off in the burning. Beccher insists very distinctly upon this difference between his element sulphur and the "sulphur" of his Paracelsian predecessors.

It must be considered as indicating great knowledge and talent in Stahl, that he perceived so clearly what part of the views of Beccher was of general truth and permanent value. Though he[11\14] everywhere gives to Beccher the credit of the theoretical opinions which he promulgates, ("Beccheriana sunt quæ profero,") it seems certain that he had the merit, not only of proving them more completely, and applying them more widely than his forerunner, but also of conceiving them {269} with a distinctness which Beccher did not attain. In 1697, appeared Stahl's _Zymotechnia Fundamentalis_ (the Doctrine of Fermentation), "simulque _experimentum novum_ sulphur verum arte producendi." In this work (besides other tenets which the author considered as very important), the opinion published by Beccher was now maintained in a very distinct form;--namely, that the process of forming sulphur from sulphuric acid, and of restoring the metals from their calces, are analogous, and consist alike in the addition of some combustible element, which Stahl termed _phlogiston_ (φλογίστον, _combustible_). The experiment most insisted on in the work now spoken of,[12\14] was the formation of sulphur from sulphate of potass (or of soda) by fusing the salt with an alkali, and throwing in coals to supply phlogiston. This is the "experimentum novum." Though Stahl published an account of this process, he seems to have regretted his openness. "He denies not," he says, "that he should peradventure have dissembled this experiment as the true foundation of the Beccherian assertion concerning the nature of sulphur, if he had not been provoked by the pretending arrogance of some of his contemporaries."

[Note 11\14: **Stahl, _Præf. ad Specim. Becch._ 1703.]

[Note 12\14: P. 117.]

From this time, Stahl's confidence in his theory may be traced becoming more and more settled in his succeeding publications. It is hardly necessary to observe here, that the explanations which his theory gives are easily transformed into those which the more recent theory supplies. According to modern views, the addition of oxygen takes place in the formation of acids and of calces, and in combustion, instead of the

subtraction of phlogiston. The coal which Stahl supposed to supply the combustible in his experiment, does in fact absorb the liberated oxygen. In like manner, when an acid corrodes a metal, and, according to existing theory, combines with and oxidates it, Stahl supposed that the phlogiston separated from the metal and combined with the acid. That the explanations of the phlogistic theory are so generally capable of being translated into the oxygen theory, merely by inverting the supposed transfer of the combustible element, shows us how important a step towards the modern doctrines the phlogistic theory really was.

The question, whether these processes were in fact addition or subtraction, was decided by the balance, and belongs to a succeeding period of the science. But we may observe, that both Beccher and Stahl were aware of the increase of weight which metals undergo in {270} calcination; although the time had not yet arrived in which this fact was to be made one of the bases of the theory.

It has been said,[13\14] that in the adoption of the phlogistic theory, that is, in supposing the above-mentioned processes to be addition rather than subtraction, "of two possible roads the wrong was chosen, as if to prove the perversity of the human mind." But we must not forget how natural it was to suppose that some part of a body was _destroyed_ or _removed_ by combustion; and we may observe, that the merit of Beccher and Stahl did not consist in the selection of one road or two, but in advancing so far as to reach this point of separation. That, having done this, they went a little further on the wrong line, was an error which detracted little from the merit or value of the progress really made. It would be easy to show, from the writings of phlogistic chemists, what important and extensive truths their theory enabled them to express simply and clearly.

[Note 13\14: Herschel's _Introd. to Nat. Phil._ p. 300.]

That an enthusiastic temper is favorable to the production of great discoveries in science, is a rule which suffers no exception in the character of Beccher. In his preface[14\14] addressed "to the benevolent reader" of his _Physica Subterranea_, he speaks of the chemists as a strange class of mortals, impelled by an almost insane impulse to seek their pleasure among smoke and vapor, soot and flame, poisons and poverty. "Yet among all these evils," he says, "I seem to myself to live so sweetly, that, may I die if I would change places with the Persian king." He is, indeed, well worthy of admiration, as one of the first who pursued the labors of the furnace and the laboratory, without the bribe of golden hopes. "My kingdom," he says, "is not of this world. I trust that I have got hold of my pitcher by the right handle,--the true method of treating this study. For the _Pseudochymists_ seek gold; but the _true philosophers_, science, which is more precious than any gold."

[Note 14\14: Frankfort, 1681.]

The _Physica Subterranea_ made no converts. Stahl, in his indignant manner, says,[15\14] "No one will wonder that it never yet obtained a physician or a chemist as a disciple, still less as an advocate." And again, "This work obtained very little reputation or estimation, or, to speak ingenuously, as far as I know, none whatever." In 1671, Beccher published a supplement to his work, in which he showed how metal

might be extracted from mud and sand. He offered to execute {271} this at Vienna; but found that people there cared nothing about such novelties. He was then induced, by Baron D'Isola, to go to Holland for similar purposes. After various delays and quarrels, he was obliged to leave Holland for fear of his creditors; and then, I suppose, came to Great Britain, where he examined the Scottish and Cornish mines. He is said to have died in London in 1682.

[Note 15\14: Præf. _Phys. Sub._ 1703.]

Stahl's publications appear to have excited more notice, and led to controversy on the "so-called sulphur." The success of the experiment had been doubted, which, as he remarks, it was foolish to make a matter of discussion, when any one might decide the point by experiment; and finally, it had been questioned whether the substance obtained by this process were pure sulphur. The originality of his doctrine was also questioned, which, as he says, could not with any justice be impugned. He published in defence and development of his opinion at various intervals, as the _Specimen Beccherianum_ in 1703, the _Documentum Theoriæ Beecherianæ_, a Dissertation _De Anatomia Sulphuris Artificialis_; and finally, _Casual Thoughts on the so-called Sulphur_, in 1718, in which he gave (in German) both a historical and a systematic view of his opinions on the nature of salts and of his Phlogiston.

_Reception and Application of the Theory._--The theory that the formation of sulphuric acid, and the restoration of metals from their calces, are analogous processes, and consist in the addition of _phlogiston_, was soon widely received; and the Phlogistic School was thus established. From Berlin, its original seat, it was diffused into all parts of Europe. The general reception of the theory may be traced, not only in the use of the term "phlogiston," and of the explanations which it implies; but in the adoption of a nomenclature founded on those explanations, which, though not very extensive, is sufficient evidence of the prevalence of the theory. Thus when Priestley, in 1774, discovered oxygen, and when Scheele, a little later, discovered chlorine, these gases were termed _dephlogisticated air_, and _dephlogisticated marine acid_; while azotic acid gas, having no disposition to combustion, was supposed to be saturated with phlogiston, and was called _phlogisticated air_.

This phraseology kept its ground, till it was expelled by the antiphlogistic, or oxygen theory. For instance. Cavendish's papers on the chemistry of the airs are expressed in terms of it, although his researches led him to the confines of the new theory. We must now give an account of such researches, and of the consequent revolution in the science. {272}

## CHAPTER V.

CHEMISTRY OF GASES.--BLACK. CAVENDISH.

THE study of the properties of aëriform substances, or Pneumatic Chemistry, as it was called, occupied the chemists of the eighteenth century, and was the main occasion of the great advances which the science made at that period. The most material general truths which came into view in the course of these researches, were,

that gases were to be numbered among the constituent elements of solid and fluid bodies; and that, in these, as in all other cases of composition, the compound was equal to the sum of its elements. The latter proposition, indeed, cannot be looked upon as a discovery, for it had been frequently acknowledged, though little applied; in fact, it could not be referred to with any advantage, till the aëriform elements, as well as others, were taken into the account. As soon as this was done, it produced a revolution in chemistry.

[2nd Ed.] [Though the view of the mode in which gaseous elements become fixed in bodies and determine their properties, had great additional light thrown upon it by Dr. Black's discoveries, as we shall see, the notion that solid bodies involve such gaseous elements was not new at that period. Mr. Vernon Harcourt has shown[16\14] that Newton and Boyle admitted into their speculations airs of various kinds, capable of fixation in bodies. I have, in the succeeding chapter (chap. vi.), spoken of the views of Rey, Hooke, and Mayow, connected with the function of airs in chemistry, and forming a prelude to the Oxygen Theory.]

[Note 16\14: _Phil. Mag._ 1846.]

Notwithstanding these preludes, the credit of the first great step in pneumatic chemistry is, with justice, assigned to Dr. Black, afterwards professor at Edinburgh, but a young man of the age of twenty-four at the time when he made his discovery.[17\14] He found that the difference between caustic lime and common limestone arose from this, that the latter substance consists of the former, combined with a certain air, which, being thus fixed in the solid body, he called _fixed air_ (carbonic {273} acid gas). He found, too, that magnesia, caustic potash, and caustic soda, would combine with the same air, with similar results. This discovery consisted, of course, in a new interpretation of observed changes. Alkalies appeared to be made caustic by contact with quicklime: at first Black imagined that they underwent this change by acquiring igneous matter from the quicklime; but when he perceived that the lime gained, not lost, in magnitude as it became mild, he rightly supposed that the alkalies were rendered caustic by imparting their air to the lime. This discovery was announced in Black's inaugural dissertation, pronounced in 1755, on the occasion of his taking his degree of Doctor in the University of Edinburgh.

[Note 17\14: Thomson's _Hist. Chem._ i. 317.]

The chemistry of airs was pursued by other experimenters. The Honorable Henry Cavendish, about 1765, invented an apparatus, in which aërial fluids are confined by water, so that they can be managed and examined. This hydro-pneumatic apparatus, or as it is sometimes called, _the pneumatic trough_, from that time was one of the most indispensable parts of the chemist's apparatus. Cavendish,[18\14] in 1766, showed the identity of the properties of fixed air derived from various sources; and pointed out the peculiar qualities of _inflammable air_ (afterwards called hydrogen gas), which, being nine times lighter than common air, soon attracted general notice by its employment for raising balloons. The promise of discovery which this subject now offered, attracted the confident and busy mind of Priestley, whose _Experiments and Observations on different kinds of Air_ appeared in 1744-79. In these volumes, he

describes an extraordinary number of trials of various kinds; the results of which were, the discovery of new kinds of air, namely, _phlogisticated air_ (azotic gas), _nitrous air_ (nitrous gas), and _dephlogisticated air_ (oxygen gas).

[Note 18\14: _Phil. Trans._ 1766.]

But the discovery of new substances, though valuable in supplying chemistry with materials, was not so important as discoveries respecting their modes of composition. Among such discoveries, that of Cavendish, published in the _Philosophical Transactions_ for 1784, and disclosing the composition of water by the union of two gases, oxygen and hydrogen, must be considered as holding a most distinguished place. He states,[19\14] that his "experiments were made principally with a view to find out the cause of the diminution which common air is well known to suffer, by all the various ways in which it is phlogisticated." And, after describing various unsuccessful attempts, he finds {274} that when inflammable air is used in this phlogistication (or burning), the diminution of the common air is accompanied by the formation of a dew in the apparatus.[20\14] And thus he infers[21\14] that "almost all the inflammable air, and one-fifth of the common air, are turned into pure water."

[Note 19\14: _Phil. Trans._ 1784, p. 119.]

[Note 20\14: _Phil. Trans._ 1784, p. 128.]

[Note 21\14: Ib. p. 129.]

Lavoisier, to whose researches this result was, as we shall soon see, very important, was employed in a similar attempt at the same time (1783), and had already succeeded,[22\14] when he learned from Dr. Blagden, who was present at the experiment, that Cavendish had made the discovery a few months sooner. Monge had, about the same time, made the same experiments, and communicated the result to Lavoisier and Laplace immediately afterwards. The synthesis was soon confirmed by a corresponding analysis. Indeed the discovery undoubtedly lay in the direct path of chemical research at the time. It was of great consequence in the view it gave of experiments in composition; for the small quantity of water produced in many such processes, had been quite overlooked; though, as it now appeared, this water offered the key to the whole interpretation of the change.

[Note 22\14: _A. P._ 1781, p. 472]

Though some objections to Mr. Cavendish's view were offered by Kirwan,[23\14] on the whole they were generally received with assent and admiration. But the bearing of these discoveries upon the new theory of Lavoisier, who rejected phlogiston, was so close, that we cannot further trace the history of the subject without proceeding immediately to that theory.

[Note 23\14: _P. T._ 1784, p. 154.]

[2nd Ed.] [I have elsewhere stated,[24\14]--with reference to recent attempts to deprive Cavendish of the credit of his discovery of the composition of water, and to transfer it to Watt,--that Watt not only did not anticipate, but did not fully appreciate the discovery of Cavendish and Lavoisier; and I have expressed my concurrence with

Mr. Vernon Harcourt's views, when he says,[25\14] that "Cavendish pared off from the current hypotheses their theory of combustion, and their affinities of imponderable for ponderable matter, as complicating chemical with physical considerations; and he then corrected and adjusted them with admirable skill to the actual phenomena, not binding the facts to the theory, but adapting the theory to the facts."

[Note 24\14: _Philosophy_, b. vi. c. 4.]

[Note 25\14: _Address to the British Association_, 1839.]

I conceive that the discussion which the subject has recently received, has left no doubt on the mind of any one who has perused the {275} documents, that Cavendish is justly entitled to the honor of this discovery, which in his own time was never contested. The publication of his Journals of Experiments[26\14] shows that he succeeded in establishing the point in question in July, 1781. His experiments are referred to in an abstract of a paper of Priestley's, made by Dr. Maty, the secretary of the Royal Society, in June, 1783. In June, 1783, also, Dr. Blagden communicated the result of Cavendish's experiments to Lavoisier, at Paris. Watt's letter, containing his hypothesis that "water is composed of dephlogisticated air and phlogiston deprived of part of their latent or elementary heat; and that phlogisticated or pure air is composed of water deprived of its phlogiston and united to elementary heat and light," was not read till Nov. 1783; and even if it could have suggested such an experiment as Cavendish's (which does not appear likely), is proved, by the dates, to have had no share in doing so.

[Note 26\14: _Appendix_ to Mr. V. Harcourt's _Address_.]

Mr. Cavendish's experiment was suggested by an experiment in which Warltire, a lecturer on chemistry at Birmingham, exploded a mixture of hydrogen and common air in a close vessel, in order to determine whether heat were ponderable.]

## CHAPTER VI.

EPOCH OF THE THEORY OF OXYGEN.--LAVOISIER.

_Sect._ 1.--_Prelude to the Theory.--Its Publication._

WE arrive now at a great epoch in the history of Chemistry. Few revolutions in science have immediately excited so much general notice as the introduction of the theory of oxygen. The simplicity and symmetry of the modes of combination which it assumed; and, above all, the construction and universal adoption of a nomenclature which applied to all substances, and which seemed to reveal their inmost constitution by their name, naturally gave it an almost irresistible sway over men's minds. We must, however, dispassionately trace the course of its introduction. {276}

Antoine Laurent Lavoisier, an accomplished French chemist, had pursued, with zeal and skill, researches such as those of Black, Cavendish, and Priestley, which we have described above. In 1774, he showed that, in the calcination of metals in air, the metal acquires as much weight as the air loses. It might appear that this discovery at

once overturned the view which supposed the metal to be phlogiston _added_ to the calx. Lavoisier's contemporaries were, however, far from allowing this; a greater mass of argument was needed to bring them to this conclusion. Convincing proofs of the new opinion were, however, rapidly supplied. Thus, when Priestley had discovered dephlogisticated air, in 1774, Lavoisier showed, in 1776, that fixed air consisted of charcoal and the dephlogisticated or pure air; for the mercurial calx which, heated by itself, gives out pure air, gives out, when heated with charcoal, fixed air,[27\14] which has, therefore, since been called _carbonic acid gas_.

[Note 27\14: _Mém. Ac. Par._ 1775.]

Again, Lavoisier showed that the atmospheric air consists of pure or vital air, and of an _unvital_ air, which he thence called _azot_. The vital air he found to be the agent in combustion, acidification, calcination, respiration; all of these processes were analogous: all consisted in a decomposition of the atmospheric air, and a fixation of the pure or vital portion of it.

But he thus arrived at the conclusion, that this pure air was added, in all the cases in which, according to the received theory, _phlogiston_ was subtracted, and _vice versâ_. He gave the name[28\14] of _oxygen_ (_principe oxygène_) to "the substance which thus unites itself with metals to form their calces, and with combustible substances to form acids."

[Note 28\14: _Mém. Ac. Par._ 1781, p. 448.]

A new theory was thus produced, which would account for all the facts which the old one would explain, and had besides the evidence of the balance in its favor. But there still remained some apparent objections to be removed. In the action of dilute acids on metals, inflammable air was produced. Whence came this element? The discovery of the decomposition of water sufficiently answered this question, and converted the objection into an argument on the side of the theory: and thus the decomposition of water was, in fact, one of the most critical events for the fortune of the Lavoisierian doctrine, and one which, more than any other, decided chemists in its favor. In succeeding years, Lavoisier showed the consistency of his theory with {277} all that was discovered concerning the composition of alcohol, oil, animal and vegetable substances, and many other bodies.

It is not necessary for us to consider any further the evidence for this theory, but we must record a few circumstances respecting its earlier history. Rey, a French physician, had in 1630, published a book, in which he inquires into the grounds of the increase of the weight of metals by calcination.[29\14] He says, "To this question, then, supported on the grounds already mentioned, I answer, and maintain with confidence, that the increase of weight arises from the air, which is condensed, rendered heavy and adhesive, by the heat of the furnace." Hooke and Mayow had entertained the opinion that the air contains a "nitrous spirit," which is the supporter of combustion. But Lavoisier disclaimed the charge of having derived anything from these sources; nor is it difficult to understand how the received generalizations of the phlogistic theory had thrown all such narrower explanations into obscurity. The merit of Lavoisier consisted in his combining the generality of Stahl with the verified

conjectures of Rey and Mayow.

[Note 29\14: Thomson, _Hist. Chem._ ii. 95.]

No one could have a better claim, by his early enthusiasm for science, his extensive knowledge, and his zealous labors, to hope that a great discovery might fall to his share, than Lavoisier. His father,[30\14] a man of considerable fortune, had allowed him to make science his only profession; and the zealous philosopher collected about him a number of the most active physical inquirers of his time, who met and experimented at his house one day in the week. In this school, the new chemistry was gradually formed. A few years after the publication of Priestley's first experiments, Lavoisier was struck with the presentiment of the theory which he was afterwards to produce. In 1772, he deposited[31\14] with the secretary of the Academy, a note which contained the germ of his future doctrines. "At that time," he says, in explaining this step, "there was a kind of rivalry between France and England in science, which gave importance to new experiments, and which sometimes was the cause that the writers of the one or other of the nations disputed the discovery with the real author." In 1777, the editor of the Memoirs of the Academy speaks of his theory as overturning that of Stahl; but the general acceptance of the new opinion did not take place till later. {278}

[Note 30\14: _Biogr. Univ._ (Cuvier.)]

[Note 31\14: Thomson, ii. 99.]

_Sect._ 2.--_Reception and Confirmation of the Theory of Oxygen._

THE Oxygen Theory made its way with extraordinary rapidity among the best philosophers.[32\14] In 1785, that is, soon after Cavendish's synthesis of water had removed some of the most formidable objections to it, Berthollet, already an eminent chemist, declared himself a convert. Indeed it was so soon generally adopted in France, that Fourcroy promulgated its doctrines under the name of "La Chimie Française," a title which Lavoisier did not altogether relish. The extraordinary eloquence and success of Fourcroy as a lecturer at the Jardin des Plantes, had no small share in the diffusion of the oxygen theory; and the name of "the apostle of the new chemistry" which was at first given him in ridicule, was justly held by him to be a glorious distinction.[33\14]

[Note 32\14: Thomson, ii. 130.]

[Note 33\14: Cuvier, _Eloges_, i. p. 20.]

Guyton de Morveau, who had at first been a strenuous advocate of the phlogistic theory, was invited to Paris, and brought over to the opinions of Lavoisier; and soon joined in the formation of the nomenclature founded upon the theory. This step, of which we shall shortly speak, fixed the new doctrine, and diffused it further. Delametherie alone defended the phlogistic theory with vigor, and indeed with violence. He was the editor of the _Journal de Physique_, and to evade the influence which this gave him, the antiphlogistians[34\14] established, as the vehicle of their opinions, another periodical, the _Annales de Chimie_.

[Note 34\14: Thomson, ii. 133.]

In England, indeed, their success was not so immediate. Cavendish,[35\14] in his Memoir of 1784, speaks of the question between the two opinions as doubtful. "There are," he says, "several Memoirs of M. Lavoisier, in which he entirely discards phlogiston; and as not only the foregoing experiments, but most other phenomena of nature, seem explicable as well, or nearly as well, upon this as upon the commonly believed principle of phlogiston," Cavendish proceeds to explain his experiments according to the new views, expressing no decided preference, however, for either system. But Kirwan, another English chemist, contested the point much more resolutely. His theory identified inflammable air, or hydrogen, with phlogiston; and in this view, he wrote a work which was intended as a confutation of {279} the essential part of the oxygen theory. It is a strong proof of the steadiness and clearness with which the advocates of the new system possessed their principles, that they immediately translated this work, adding, at the end of each chapter, a refutation of the phlogistic doctrines which it contained. Lavoisier, Berthollet, De Morveau, Fourcroy, and Monge, were the authors of this curious specimen of scientific polemics. It is also remarkable evidence of the candor of Kirwan, that notwithstanding the prominent part he had taken in the controversy, he allowed himself at last to be convinced. After a struggle of ten years, he wrote[36\14] to Berthollet in 1796, "I lay down my arms, and abandon the cause of phlogiston." Black followed the same course. Priestley alone, of all the chemists of great name, would never assent to the new doctrines, though his own discoveries had contributed so much to their establishment. "He saw," says Cuvier,[37\14] "without flinching, the most skilful defenders of the ancient theory go over to the enemy in succession; and when Kirwan had, almost the last of all, abjured phlogiston, Priestley remained alone on the field of battle, and threw out a new challenge, in a memoir addressed to the principal French chemists." It happened, curiously enough, that the challenge was accepted, and the arguments answered by M. Adet, who was at that time (1798,) the French ambassador to the United States, in which country Priestley's work was published. Even in Germany, the birth-place and home of the phlogistic theory, the struggle was not long protracted. There was, indeed, a controversy, the older philosophers being, as usual, the defenders of the established doctrines; but in 1792, Klaproth repeated, before the Academy of Berlin, all the fundamental experiments; and "the result was a full conviction on the part of Klaproth and the Academy, that the Lavoisierian theory was the true one."[38\14] Upon the whole, the introduction of the Lavoisierian theory in the scientific world, when compared with the great revolution of opinion to which it comes nearest in importance, the introduction of the Newtonian theory, shows, by the rapidity and temper with which it took place, a great improvement, both in the means of arriving at truth, and in the spirit with which they were used.

[Note 35\14: _Phil. Trans._ 1784, p. 150.]

[Note 36\14: Pref. to Fourcroy's _Chemistry_, xiv.]

[Note 37\14: Cuvier, _Eloge de Priestley_, p. 208.]

[Note 38\14: Thomson, vol. ii. p. 136.]

Some English writers[39\14] have expressed an opinion that there was {280} little that was original in the new doctrines. But if they were so obvious, what are we to say of eminent chemists, as Black and Cavendish, who hesitated when they were presented, or Kirwan and Priestley, who rejected them? This at least shows that it required some peculiar insight to see the evidence of these truths. To say that most of the materials of Lavoisier's theory existed before him, is only to say that his great merit was, that which must always be the great merit of a new theory, his generalization. The effect which the publication of his doctrines produced, shows us that he was the first person who, possessing clearly the idea of quantitative composition, applied it steadily to a great range of well-ascertained facts. This is, as we have often had to observe, precisely the universal description of an inductive discoverer. It has been objected, in like manner, to the originality of Newton's discoveries, that they were contained in those of Kepler. They were so, but they needed a Newton to find them there. The originality of the theory of oxygen is proved by the conflict, short as it was, which accompanied its promulgation; its importance is shown by the changes which it soon occasioned in every part of the science.

[Note 39\14: Brande, _Hist. Diss._ in _Enc, Brit._ p. 182. Lunn, _Chem._ in _Enc. Met._ p. 596.]

Thus Lavoisier, far more fortunate than most of those who had, in earlier ages, produced revolutions in science, saw his theory accepted by all the most eminent men of his time, and established over a great part of Europe within a few years from its first promulgation. In the common course of events, it might have been expected that the later years of his life would have been spent amid the admiration and reverence which naturally wait upon the patriarch of a new system of acknowledged truths. But the times in which he lived allowed no such euthanasia to eminence of any kind. The democracy which overthrew the ancient political institutions of France, and swept away the nobles of the land, was not, as might have been expected, enthusiastic in its admiration of a great revolution in science, and forward to offer its homage to the genuine nobility of a great discoverer. Lavoisier was thrown into prison on some wretched charge of having, in the discharge of a public office which he had held, adulterated certain tobacco; but in reality, for the purpose of confiscating his property.[40\14] In his imprisonment, his philosophy was his resource; and he employed himself in the preparation of his papers for printing. When he was brought before the revolutionary tribunal, he begged for a respite of a few days, in order to complete some researches, the results of which {281} were, he said, important to the good of humanity. The brutish idiot, whom the state of the country at that time had placed in the judgment-seat, told him that the republic wanted no sçavans. He was dragged to the guillotine, May the 8th, 1794, and beheaded, in the fifty-second year of his age; a melancholy proof that, in periods of political ferocity, innocence and merit, private virtues and public services, amiable manners and the love of friends, literary fame and exalted genius, are all as nothing to protect their possessor from the last extremes of violence and wrong, inflicted under judicial forms.

[Note 40\14: _Biog. Univ._ (Cuvier.)]

_Sect._ 3.--_Nomenclature of the Oxygen Theory._

AS we have already said, a powerful instrument in establishing and diffusing the new chemical theory, was a Systematic Nomenclature founded upon it, and applicable to all chemical compounds, which was soon constructed and published by the authors of the theory. Such a nomenclature made its way into general use the more easily, in that the want of such a system had already been severely felt; the names in common use being fantastical, arbitrary, and multiplied beyond measure. The number of known substances had become so great, that a list of names with no regulative principle, founded on accident, caprice, and error, was too cumbrous and inconvenient to be tolerated. Even before the currency which Lavoisier's theory obtained, these evils had led to attempts towards a more convenient set of names. Bergman and Black had constructed such lists; and Guyton de Morveau, a clever and accomplished lawyer of Dijon, had formed a system of nomenclature in 1782, before he had become a convert to Lavoisier's theory, in which task he had been exhorted and encouraged by Bergman and Macquer. In this system,[41\14] we do not find most of the characters of the method which was afterwards adopted. But a few years later, Lavoisier, De Morveau, Berthollet and Fourcroy, associated themselves for the purpose of producing a nomenclature which should correspond to the new theoretical views. This appeared in 1787, and soon made its way into general use. The main features of this system are, a selection of the simplest radical words, by which substances are designated, and a systematic distribution of terminations, to express their relations. Thus, sulphur, combined with oxygen in two different proportions, forms two acids, the {282} sulphur_ous_ and the sulphur_ic_; and these acids form, with earthy or alkaline bases, sulph_ides_ and sulph_ates_; while sulphur directly combined with another element, forms a sulph_uret_. The term _oxyd_ (now usually written _oxide_) expressed a lower degree of combination with oxygen than the acids. The _Méthode de Nomenclature Chimique_ was published in 1787; and in 1789, Lavoisier published a treatise on chemistry in order further to explain this method. In the preface to this volume, he apologizes for the great amount of the changes, and pleads the authority of Bergman, who had exhorted De Morveau "to spare no improper names; those who are learned will always be learned, and those who are ignorant will thus learn sooner." To this maxim they so far conformed, that their system offers few anomalies; and though the progress of discovery, and the consequent changes of theoretical opinion, which have since gone on, appear now to require a further change of nomenclature, it is no small evidence of the skill with which this scheme was arranged, that for half a century it was universally used, and felt to be far more useful and effective than any nomenclature in any science had ever been before.

[Note 41\14: _Journal de Physique_, 1782, p. 370.]

## CHAPTER VII.

APPLICATION AND CORRECTION OF THE OXYGEN THEORY.

SINCE a chemical theory, as far as it is true, must enable us to obtain a true view of the intimate composition of all bodies whatever, it will readily be supposed that the new chemistry led to an immense number of analyses and researches of various kinds. These it is not necessary to dwell upon; nor will I even mention the names of any of the intelligent and diligent men who have labored in this field. Perhaps one of the most striking of such analyses was Davy's decomposition of the earths and alkalies into metallic bases and oxygen, in 1807 and 1808; thus extending still further that analogy between the earths and the calces of the metals, which had had so large a share in the formation of chemical theories. This discovery, however, both in the means by which it was made, and in the views to which it led, bears upon subjects hereafter to be treated of.

The Lavoisierian theory also, wide as was the range of truth which it embraced, required some limitation and correction. I do not now {283} speak of some erroneous opinions entertained by the author of the theory; as, for instance, that the heat produced in combustion, and even in respiration, arose from the conversion of oxygen gas to a solid consistence, according to the doctrine of latent heat. Such opinions not being necessarily connected with the general idea of the theory, need not here be considered. But the leading generalization of Lavoisier, that acidification was _always_ combination with oxygen, was found untenable. The point on which the contest on this subject took place was the constitution of the _oxymuriatic_ and _muriatic_ acids;--as they had been termed by Berthollet, from the belief that muriatic acid contained oxygen, and oxymuriatic a still larger dose of oxygen. In opposition to this, a new doctrine was put forward in 1809 by Gay-Lussac and Thenard in France, and by Davy in England;--namely, that oxymuriatic acid was a simple substance, which they termed _chlorine_, and that muriatic acid was a combination of chlorine with hydrogen, which therefore was called _hydrochloric acid_. It may be observed, that the point in dispute in the controversy on this subject was nearly the same which had been debated in the course of the establishment of the oxygen theory; namely, whether in the formation of muriatic acid from chlorine, oxygen is subtracted, or hydrogen added, and the water concealed.

In the course of this dispute, it was allowed on both sides, that the combination of dry muriatic acid and ammonia afforded an _experimentum crucis_; since, if water was produced from these elements, oxygen must have existed in the acid. Davy being at Edinburgh in 1812, this experiment was made in the presence of several eminent philosophers; and the result was found to be, that though a slight dew appeared in the vessel, there was not more than might be ascribed to unavoidable imperfection in the process, and certainly not so much as the old theory of muriatic acid required. The new theory, after this period, obtained a clear superiority in the minds of philosophical chemists, and was further supported by new analogies.[42\14]

[Note 42\14: Paris, _Life of Davy_, i. 337.]

For, the existence of one _hydracid_ being thus established, it was found that other substances gave similar combinations; and thus chemists obtained the _hydriodic_, _hydrofluoric_, and _hydrobromic_ acids. These acids, it is to be observed, form salts with bases, in the same manner as the oxygen acids do. The analogy of the muriatic

and fluoric compounds was first clearly urged by a philosopher who was {284} not peculiarly engaged in chemical research, but who was often distinguished by his rapid and happy generalizations, M. Ampère. He supported this analogy by many ingenious and original arguments, in letters written to Davy, while that chemist was engaged in his researches on fluor spar, as Davy himself declares.[43\14]

[Note 43\14: Paris, _Life of Davy_, i. 370.]

Still further changes have been proposed, in that classification of elementary substances to which the oxygen theory led. It has been held by Berzelius and others, that other elements, as, for example, sulphur, form _salts_ with the alkaline and earthy metals, rather than sulphurets. The character of these _sulpho-salts_, however, is still questioned among chemists; and therefore it does not become us to speak as if their place in history were settled. Of course, it will easily be understood that, in the same manner in which the oxygen theory introduced its own proper nomenclature, the overthrow or material transformation of the theory would require a change in the nomenclature; or rather, the anomalies which tended to disturb the theory, would, as they were detected, make the theoretical terms be felt as inappropriate, and would suggest the necessity of a reformation in that respect. But the discussion of this point belongs to a step of the science which is to come before us hereafter.

It may be observed, that in approaching the limits of this part of our subject, as we are now doing, the doctrine of the combination of _acids_ and _bases_, of which we formerly traced the rise and progress, is still assumed as a fundamental relation by which other relations are tested. This remark connects the stage of chemistry now under our notice with its earliest steps. But in order to point out the chemical bearing of the next subjects of our narrative, we may further observe, that _metals_, _earths_, _salts_, are spoken of as known _classes_ of substances; and in like manner the newly-discovered elements, which form the last trophies of chemistry, have been distributed into such classes according to their analogies; thus _potassium_, _sodium_, _barium_, have been asserted to be metals; _iodine_, _bromine_, _fluorine_, have been arranged as analogical to _chlorine_. Yet there is something vague and indefinite in the boundaries of such classifications and analogies; and it is precisely where this vagueness falls, that the science is still obscure or doubtful. We are led, therefore, to see the dependence of Chemistry upon Classification; and it is to Sciences of Classification which we shall next proceed; as soon as we have noticed the most general views {285} which have been given of chemical relations, namely, the views of the electro-chemists.

But before we do this, we must look back upon a law which obtains in the combination of elements, and which we have hitherto not stated; although it appears, more than any other, to reveal to us the intimate constitution of bodies, and to offer a basis for future generalizations. I speak of the _Atomic Theory_, as it is usually termed; or, as we might rather call it, the Doctrine of Definite, Reciprocal, and Multiple Proportions.

## CHAPTER VIII.

## THEORY OF DEFINITE, RECIPROCAL, AND MULTIPLE PROPORTIONS.

### _Sect._ 1.--_Prelude to the Atomic Theory, and its Publication by Dalton._

THE general laws of chemical combination announced by Mr. Dalton are truths of the highest importance in the science, and are now nowhere contested; but the view of matter as constituted of _atoms_, which he has employed in conveying those laws, and in expressing his opinion of their cause, is neither so important nor so certain. In the place which I here assign to his discovery, as one of the great events of the history of chemistry, I speak only of the _law of phenomena_, the rules which govern the quantities in which elements combine.

This law may be considered as consisting of three parts, according to the above description of it;--that elements combine in _definite_ proportions;--that these determining proportions operate _reciprocally_;--and that when, between the same elements, several combining proportions occur, they are related as _multiples_.

That elements combine in certain definite proportions of quantity, and in no other, was implied, as soon as it was supposed that chemical compounds had any definite properties. Those who first attempted to establish regular formulæ[44\14] for the constitution of salts, minerals, and {286} other compounds, assumed, as the basis of this process, that the elements in different specimens had the same proportion. Wenzel, in 1777, published his _Lehre von der Verwandschaft der Körper_; or, _Doctrine of the Affinities of Bodies_; in which he gave many good and accurate analyses. His work, it is said, never grew into general notice. Berthollet, as we have already stated, maintained that chemical compounds were not definite; but this controversy took place at a later period. It ended in the establishment of the doctrine, that there is, for each combination, only one proportion of the elements, or at most only two or three.

[Note 44\14: Thomson, _Hist. Chem._ vol. ii. p. 279.]

Not only did Wenzel, by his very attempt, presume the first law of chemical composition, the definiteness of the proportions, but he was also led, by his results, to the second rule, that they are reciprocal. For he found that when two _neutral_ salts decompose each other, the resulting salts are also neutral. The neutral character of the salts shows that they are definite compounds; and when the two elements of the one salt, _P_ and _s_, are presented to those of the other, _B_ and _n_, if _P_ be in such quantity as to combine definitely with _n_, _B_ will also combine definitely with _s_.[45\14]

[Note 45\14: I am told that Wenzel (whose book I have not seen), though he adduces many cases in which double decomposition gives neutral salts, does not express the proposition in a general form, nor use letters in expressing it.]

Views similar to those of Wenzel were also published by Jeremiah Benjamin Richter[46\14] in 1792, in his _Anfangsgründe der Stöchyometrie, oder Messkunst Chymischer Elemente_, (_Principles of the Measure of Chemical Elements_) in which he took the law, just stated, of reciprocal proportions, as the basis of his researches, and determined the numerical quantities of the common bases and acids which would

saturate each other. It is clear that, by these steps, the two first of our three rules may be considered as fully developed. The change of general views which was at this time going on, probably prevented chemists from feeling so much interest as they might have done otherwise, in these details; the French and English chemists, in particular, were fully employed with their own researches and controversies.

[Note 46\14: Thomson, _Hist. Chem._ vol. ii. p. 283.]

Thus the rules which had already been published by Wenzel and Richter had attracted so little notice, that we can hardly consider Mr. Dalton as having been anticipated by those writers, when, in 1803, he began to communicate his views on the chemical constitution of {287} bodies; these views being such as to include both these two rules in their most general form, and further, the rule, at that time still more new to chemists, of _multiple_ proportions. He conceived bodies as composed of atoms of their constituent elements, grouped, either one and one, or one and two, or one and three, and so on. Thus, if _C_ represent an atom of carbon and _O_ one of oxygen, _O C_ will be an atom of _carbonic oxide_, and _O C O_ an atom of _carbonic acid_; and hence it follows, that while both these bodies have a definite quantity of oxygen to a given quantity of carbon, in the latter substance this quantity is _double_ of what it is in the former.

The consideration of bodies as consisting of compound atoms, each of these being composed of elementary atoms, naturally led to this law of multiple proportions. In this mode of viewing bodies, Mr. Dalton had been preceded (unknown to himself) by Mr. Higgins, who, in 1789, published[47\14] his _Comparative View of the Phlogistic and Antiphlogistic Theories_. He there says,[48\14] "That in volatile vitriolic acid, a single ultimate particle of sulphur is united only to a single particle of dephlogisticated air; and that in perfect vitriolic acid, every single particle of sulphur is united to two of dephlogisticated air, being the quantity necessary to saturation;" and he reasons in the same manner concerning the constitution of water, and the compounds of nitrogen and oxygen. These observations of Higgins were, however, made casually, and not followed out, and cannot affect Dalton's claim to original merit.

[Note 47\14: Turner's _Chem._ p. 217.]

[Note 48\14: P. 36 and 37.]

Mr. Dalton's generalization was first suggested[49\14] during his examination of olefiant gas and carburetted hydrogen gas; and was asserted generally, on the strength of a few facts, being, as it were, irresistibly recommended by the clearness and simplicity which the notion possessed. Mr. Dalton himself represented the compound atoms of bodies by symbols, which professed to exhibit the arrangement of the elementary atoms in space as well as their numerical proportion; and he attached great importance to this part of his scheme. It is clear, however, that this part of his doctrine is not essential to that numerical comparison of the law with facts, on which its establishment rests. These hypothetical configurations of atoms have no value till they are confirmed by corresponding facts, such as the optical or crystalline properties of bodies may perhaps one day furnish. {288}

[Note 49\14: Thomson, vol. ii. p. 291.]

_Sect._ 2.--_Reception and Confirmation of the Atomic Theory._

IN order to give a sketch of the progress of the Atomic Theory into general reception, we cannot do better than borrow our information mainly from Dr. Thomson, who was one of the earliest converts and most effective promulgators of the doctrine. Mr. Dalton, at the time when he conceived his theory, was a teacher of mathematics at Manchester, in circumstances which might have been considered narrow, if he himself had been less simple in his manner of life, and less moderate in his worldly views. His experiments were generally made with apparatus of which the simplicity and cheapness corresponded to the rest of his habits. In 1804, he was already in possession of his atomic theory, and explained it to Dr. Thomson, who visited him at that time. It was made known to the chemical world in Dr. Thomson's _Chemistry_, in 1807; and in Dalton's own _System of Chemistry_ (1808) the leading ideas of it were very briefly stated. Dr. Wollaston's memoir, "on superacid and subacid salts," which appeared in the _Philosophical Transactions_ for 1808, did much to secure this theory a place in the estimation of chemists. Here the author states, that he had observed, in various salts, the quantities of acid combined with the base in the neutral and in the superacid salts to be as one to two: and he says that, thinking it likely this law might obtain generally in such compounds, it was his design to have pursued this subject, with the hope of discovering the cause to which so regular a relation may be ascribed. But he adds, that this appears to be superfluous after the publication of Dalton's theory by Dr. Thomson, since all such facts are but special cases of the general law. We cannot but remark here, that the scrupulous timidity of Wollaston was probably the only impediment to his anticipating Dalton in the publication of the rule of multiple proportions; and the forwardness to generalize, which belongs to the character of the latter, justly secured him, in this instance, the name of the discoverer of this law. The rest of the English chemists soon followed Wollaston and Thomson, though Davy for some time resisted. They objected, indeed, to Dalton's assumption of atoms, and, to avoid this hypothetical step, Wollaston used the phrase _chemical equivalents_, and Davy the word _proportions_, for the numbers which expressed Dalton's atomic weights. We may, however, venture to say that the term "atom" is the most convenient, and it need not be understood as claiming our assent to the hypothesis of indivisible molecules. {289}

As Wollaston and Dalton were thus arriving independently at the same result in England, other chemists, in other countries, were, unknown to each other, travelling towards the same point.

In 1807, Berzelius,[50\14] intending to publish a system of chemistry, went through several works little read, and among others the treatises of Richter. He was astonished, he tells us, at the light which was there thrown upon composition and decomposition, and which had never been turned to profit. He was led to a long train of experimental research, and, when he received information of Dalton's ideas concerning multiple proportions, he found, in his own collection of analyses, a full confirmation of this theory.

[Note 50\14: Berz. _Chem._ B. iii. p. 27.]

Some of the Germans, indeed, appear discontented with the partition of reputation which has taken place with respect to the Theory of Definite Proportions. One[51\14] of them says, "Dalton has only done this;--he has wrapt up the good Richter (whom he knew; compare Schweigger, T, older series, vol. x., p. 381;) in a ragged suit, patched together of atoms; and now poor Richter comes back to his own country in such a garb, like Ulysses, and is not recognized." It is to be recollected, however, that Richter says nothing of multiple proportions.

[Note 51\14: Marx. _Gesch. der Cryst._ p. 202.]

The general doctrine of the atomic theory is now firmly established over the whole of the chemical world. There remain still several controverted points, as, for instance, whether the atomic weights of all elements are exact multiples of the atomic weight of hydrogen. Dr. Prout advanced several instances in which this appeared to be true, and Dr. Thomson has asserted the law to be of universal application. But, on the other hand, Berzelius and Dr. Turner declare that this hypothesis is at variance with the results of the best analyses. Such controverted points do not belong to our history, which treats only of the progress of scientific truths already recognized by all competent judges.

Though Dalton's discovery was soon generally employed, and universally spoken of with admiration, it did not bring to him anything but barren praise, and he continued in the humble employment of which we have spoken, when his fame had filled Europe, and his name become a household word in the laboratory. After some years he was appointed a corresponding member of the Institute of France; which may be considered as a European recognition of the importance {290} of what he had done; and, in 1826, two medals for the encouragement of science having been placed at the disposal of the Royal Society by the King of England, one of them was assigned to Dalton, "for his development of the atomic theory." In 1833, at the meeting of the British Association for the Advancement of Science, which was held in Cambridge, it was announced that the King had bestowed upon him a pension of 150_l._; at the preceding meeting at Oxford, that university had conferred upon him the degree of Doctor of Laws, a step the more remarkable, since he belonged to the sect of Quakers. At all the meetings of the British Association he has been present, and has always been surrounded by the reverence and admiration of all who feel any sympathy with the progress of science. May he long remain among us thus to remind us of the vast advance which Chemistry owes to him!

[2nd Ed.] [Soon after I wrote these expressions of hope, the period of Dalton's sojourn among us terminated. He died on the 27th of July, 1844, aged 78.

His fellow-townsmen, the inhabitants of Manchester, who had so long taken a pride in his residence among them, soon after his death came to a determination to perpetuate his memory by establishing in his honor a Professor of Chemistry at Manchester.]

_Sect._ 3.--_The Theory of Volumes.--Gay-Lussac._

THE atomic theory, at the very epoch of its introduction into France, received a modification in virtue of a curious discovery then made. Soon after the publication of Dalton's system, Gay-Lussac and Humboldt found a rule for the combination of substances, which includes that of Dalton as far as it goes, but extends to combinations of gases only. This law is the _theory of volumes_; namely, that gases unite together _by volume_ in very simple and definite proportions. Thus water is composed exactly of 100 measures of oxygen and 200 measures of hydrogen. And since these simple ratios 1 and 1, 1 and 2, 1 and 3, alone prevail in such combinations, it may easily be shown that laws like Dalton's law of multiple proportions, must obtain in such cases as he considered.

[2nd Ed.] [M. Schröder, of Mannheim, has endeavored to extend to solids a law in some degree resembling Gay-Lussac's law of the volumes of gases. According to him, the volumes of the chemical equivalents {291} of simple substances and their compounds are as whole numbers.[52\14] MM. Kopp, Playfair, and Joule have labored in the same field.]

[Note 52\14: _Die molecular-volume der Chemischen Verbindungen in festen und flüssingen Zustande_, 1843.]

I cannot now attempt to trace other bearings and developments of this remarkable discovery. I hasten on to the last generalization of chemistry; which presents to us chemical forces under a new aspect, and brings us back to the point from which we departed in commencing the history of this science.

## CHAPTER IX.

EPOCH OF DAVY AND FARADAY.

_Sect._ 1.--_Promulgation of the Electro-chemical Theory by Davy._

THE reader will recollect that the History of Chemistry, though highly important and instructive in itself, has been an interruption of the History of Electro-dynamic Research:--a necessary interruption, however; for till we became acquainted with Chemistry in general, we could not follow the course of Electro-chemistry: we could not estimate its vast yet philosophical theories, nor even express its simplest facts. We have now to endeavor to show what has thus been done, and by what steps;--to give a fitting view of the Epoch of Davy and Faraday.

This is, doubtless, a task of difficulty and delicacy. We cannot execute it at all, except we suppose that the great truths, of which the discovery marks this epoch, have already assumed their definite and permanent form. For we do not learn the just value and right place of imperfect attempts and partial advances in science, except by seeing to what they lead. We judge properly of our trials and guesses only when we have gained our point and guessed rightly. We might personify philosophical theories, and might represent them to ourselves as figures, all pressing eagerly onwards in the same {292} direction, whom we have to pursue: and it is only in proportion as we ourselves overtake those figures in the race, and pass beyond them, that we are enabled to look back upon their faces; to discern their real aspects, and to catch the

true character of their countenances. Except, therefore, I were of opinion that the great truths which Davy brought into sight have been firmly established and clearly developed by Faraday, I could not pretend to give the history of this striking portion of science. But I trust, by the view I have to offer of these beautiful trains of research and their result, to justify the assumption on which I thus proceed.

I must, however, state, as a further appeal to the reader's indulgence, that, even if the great principles of electro-chemistry have now been brought out in their due form and extent, the discovery is but a very few years, I might rather say a few months, old, and that this novelty adds materially to the difficulty of estimating previous attempts from the point of view to which we are thus led. It is only slowly and by degrees that the mind becomes sufficiently imbued with those new truths, of which the office is, to change the face of a science. We have to consider familiar appearances under a new aspect; to refer old facts to new principles; and it is not till after some time, that the struggle and hesitation which this employment occasions, subsides into a tranquil equilibrium. In the newly acquired provinces of man's intellectual empire, the din and confusion of conquest pass only gradually into quiet and security. We have seen, in the history of all capital discoveries, how hardly they have made their way, even among the most intelligent and candid philosophers of the antecedent schools: we must, therefore, not expect that the metamorphosis of the theoretical views of chemistry which is now going on, will be effected without some trouble and delay.

I shall endeavor to diminish the difficulties of my undertaking, by presenting the earlier investigations in the department of which I have now to speak, as much as possible according to the most deliberate view taken of them by the great discoverers themselves, Davy and Faraday; since these philosophers are they who have taught us the true import of such investigations.

There is a further difficulty in my task, to which I might refer;--the difficulty of speaking, without error and without offence, of men now alive, or who were lately members of social circles which exist still around us. But the scientific history in which such persons play a part, is so important to my purpose, that I do not hesitate to incur {293} the responsibility which the narration involves; and I have endeavored earnestly, and I hope not in vain, to speak as if I were removed by centuries from the personages of my story.

The phenomena observed in the Voltaic apparatus were naturally the subject of many speculations as to their cause, and thus gave rise to "Theories of the Pile." Among these phenomena there was one class which led to most important results: it was discovered by Nicholson and Carlisle, in 1800, that water was _decomposed_ by the pile of Volta; that is, it was found that when the wires of the pile were placed with their ends near each other in the fluid, a stream of bubbles of air arose from each wire, and these airs were found on examination to be oxygen and hydrogen: which, as we have had to narrate, had already been found to be the constituents of water. This was, as Davy says,[53\14] the true origin of all that has been done in electro-chemical science. It was found that other substances also suffered a like decomposition under the same circumstances. Certain metallic solutions were decomposed, and an alkali

was separated on the negative plates of the apparatus. Cruickshank, in pursuing these experiments, added to them many important new results; such as the decomposition of muriates of magnesia, soda, and ammonia by the pile; and the general observation that the alkaline matter always appeared at the _negative_, and the acid at the _positive_, pole.

[Note 53\14: _Phil. Trans._ 1826, p. 386.]

Such was the state of the subject when one who was destined to do so much for its advance, first contributed his labors to it. Humphry Davy was a young man who had been apprenticed to a surgeon at Penzance, and having shown an ardent love and a strong aptitude for chemical research, was, in 1798, made the superintendent of a "Pneumatic Institution," established at Bristol by Dr. Beddoes, for the purpose of discovering medical powers of factitious airs.[54\14] But his main attention was soon drawn to galvanism; and when, in consequence of the reputation he had acquired, he was, in 1801, appointed lecturer at the Royal Institution in London (then recently established), he was soon put in possession of a galvanic apparatus of great power; and with this he was not long in obtaining the most striking results.

[Note 54\14: Paris, _Life of Davy_, i. 58.]

His first paper on the subject[55\14] is sent from Bristol, in September, 1800; and describes experiments, in which he had found that the decompositions observed by Nicholson and Carlisle go on, although the {294} water, or other substance in which the two wires are plunged, be separated into two portions, provided these portions are connected by muscular or other fibres. This use of muscular fibres was, probably, a remnant of the original disposition, or accident, by which galvanism had been connected with physiology, as much as with chemistry. Davy, however, soon went on towards the conclusion, that the phenomena were altogether chemical in their nature. He had already conjectured,[56\14] in 1802, that all decompositions might be _polar_; that is, that in all cases of chemical decomposition, the elements might be related to each other as electrically _positive_ and _negative_; a thought which it was the peculiar glory of his school to confirm and place in a distinct light. At this period such a view was far from obvious; and it was contended by many, on the contrary, that the elements which the voltaic apparatus brought to view, were not liberated from combinations, but generated. In 1806, Davy attempted the solution of this question; he showed that the ingredients which had been supposed to be produced by electricity, were due to impurities in the water, or to the decomposition of the vessel; and thus removed all preliminary difficulties. And then he says,[57\14] "referring to my experiments of 1800, 1801, and 1802, and to a number of new facts, which showed that inflammable substances and oxygen, alkalies and acids, and oxidable and noble metals, were in electrical relations of positive and negative, I drew the conclusion, _that the combinations and decompositions by electricity were referrible to the law of electrical attractions and repulsions_," and advanced the hypothesis, "_that chemical and electrical attractions were produced by the same cause, acting in the one case on particles, in the other on masses; . . . and that the same property, under different modifications, was the cause of all the phenomena exhibited by different voltaic combinations._"

[Note 55\14: Nicholson's _Journal_, 4to. iv. 275.]

[Note 56\14: _Phil. Trans._ 1826.]

[Note 57\14: Ib. 1826, p. 389.]

Although this is the enunciation, in tolerably precise terms, of the great discovery of his epoch, it was, at the period of which we speak, conjectured rather than proved; and we shall find that neither Davy nor his followers, for a considerable period, apprehended it with that distinctness which makes a discovery complete. But in a very short time afterwards, Davy drew great additional notice to his researches by effecting, in pursuance, as it appeared, of his theoretical views, the decomposition of potassa into a metallic base and oxygen. This was, as he truly said, in the memorandum written in his journal at the {295} instant, "a capital experiment." This discovery was soon followed by that of the decomposition of soda; and shortly after, of other bodies of the same kind; and the interest and activity of the whole chemical world were turned to the subject in an intense degree.

At this period, there might be noticed three great branches of speculation on this subject; _the theory of the pile_, _the theory of electrical decomposition_, and **_the theory of the identity of chemical and electrical forces_; which last doctrine, however, was found to include the other two, as might have been anticipated from the time of its first suggestion.

It will not be necessary to say much on the theories of the voltaic pile, as separate from other parts of the subject. The _contact-theory_, which ascribed the action to the contact of different metals, was maintained by Volta himself; but gradually disappeared, as it was proved (by Wollaston[58\14] especially,) that the effect of the pile was inseparably connected with oxidation or other chemical changes. The theories of electro-chemical decomposition were numerous, and especially after the promulgation of Davy's _Memoir_ in 1806; and, whatever might be the defects under which these speculations for a long time labored, the subject was powerfully urged on in the direction in which truth lay, by Davy's discoveries and views. That there remained something still to be done, in order to give full evidence and consistency to the theory, appears from this;--that some of the most important parts of Davy's results struck his followers as extraordinary paradoxes;--for instance, the fact that the decomposed elements are transferred from one part of the circuit to another, in a form which escapes the cognizance of our senses, through intervening substances for which they have a strong affinity. It was found afterwards that the circumstance which appeared to make the process so wonderful, was, in fact, the condition of its going on at all. Davy's expressions often seem to indicate the most exact notions: for instance, he says, "It is very natural to suppose that the repellent and attractive energies are communicated from one particle to another of the same kind, so as to establish a conducting _chain_ in the fluid; and that the locomotion takes place in consequence;"[59\14] and yet at other times he speaks of the element as _attracted_ and _repelled_ by the metallic surfaces which form the _poles_;--a different, and, as it appeared afterwards, an untenable view. Mr. Faraday, who supplied what was wanting, justly notices this vagueness. {296} He says,[60\14] that though, in Davy's

celebrated Memoir of 1806, the points established are of the utmost value, the mode of action by which the effects take place is stated very generally; so generally, indeed, that probably a dozen precise schemes of electro-chemical action might be drawn up, differing essentially from each other, yet all agreeing with the statement there given." And at a period a little later, being reproached by Davy's brother with injustice in this expression, he substantiated his assertion by an enumeration of twelve such schemes which had been published.

[Note 58\14: _Phil. Trans._ 1801, p. 427.]

[Note 59\14: Paris, i. 154.]

[Note 60\14: _Researches_, 482.]

But yet we cannot look upon this Memoir of 1806, otherwise than as a great event, perhaps the most important event of the epoch now under review. And as such it was recognized at once all over Europe. In particular, it received the distinguished honor of being crowned by the Institute of France, although that country and England were then engaged in fierce hostility. Buonaparte had proposed a prize of sixty thousand francs "to the person who by his experiments and discoveries should advance the knowledge of electricity and galvanism, as much as Franklin and Volta did;" and "of three thousand francs for the best experiment which should be made in the course of each year on the galvanic fluid;" the latter prize was, by the First Class of the Institute, awarded to Davy.

From this period he rose rapidly to honors and distinctions, and reached a height of scientific fame as great as has ever fallen to the lot of a discoverer in so short a time. I shall not, however, dwell on such circumstances, but confine myself to the progress of my subject.

_Sect._ 2.--_Establishment of the Electro-chemical Theory by Faraday._

THE defects of Davy's theoretical views will be seen most clearly by explaining what Faraday added to them. Michael Faraday was in every way fitted and led to become Davy's successor in his great career of discovery. In 1812, being then a bookseller's apprentice, he attended the lectures of Davy, which at that period excited the highest admiration.[61\14] "My desire to escape from trade," Mr. Faraday says, "which I thought vicious and selfish, and to enter into the service of science, which I imagined made its pursuers amiable and liberal, induced me at last to take the bold and simple step of writing to Sir H. Davy." He was favorably received, and, in the next year, became {297} Davy's assistant at the Institution; and afterwards his successor. The Institution which produced such researches as those of these two men, may well be considered as a great school of exact and philosophical chemistry. Mr. Faraday, from the beginning of his course of inquiry, appears to have had the consciousness that he was engaged on a great connected work. His _Experimental Researches_, which appeared in a series of Memoirs in the _Philosophical Transactions_, are divided into short paragraphs, numbered into a continued order from 1 up to 1160, at the time at which I write;[62\14] and destined, probably, to extend much further. These paragraphs are connected by a very rigorous method of investigation and reasoning which runs through the whole body of them. Yet this

unity of purpose was not at first obvious. His first two Memoirs were upon subjects which we have already treated of (B. xiii. c. 5 and c. 8), Voltaic Induction, and the evolution of Electricity from Magnetism. His "Third Series" has also been already referred to. Its object was, as a preparatory step towards further investigation, to show the identity of voltaic and animal electricity with that of the electrical machine; and as machine electricity differs from other kinds in being successively in a state of tension and explosion, instead of a continued current, Mr. Faraday succeeded in identifying it with them, by causing the electrical discharge to pass through a bad conductor into a discharging-train of vast extent; nothing less, indeed, than the whole fabric of the metallic gas-pipes and water-pipes of London. In this Memoir[63\14] it is easy to see already traces of the general theoretical views at which he had arrived; but these are not expressly stated till his "Fifth Series;" his intermediate Fourth Series being occupied by another subsidiary labor on the conditions of conduction. At length, however, in the Fifth Series, which was read to the Royal Society in June, 1833, he approaches the theory of electro-chemical decomposition. Most preceding theorists, and Davy amongst the number, had referred this result to _attractive powers_ residing in the _poles_ of the apparatus; and had even pretended to compare the intensity of this attraction at different distances from the poles. By a number of singularly beautiful and skilful experiments, Mr. Faraday shows that the phenomena can with no propriety be {298} ascribed to the attraction of the poles.[64\14] "As the substances evolved in cases of electro-chemical decomposition may be made to appear against air,[65\14] which, according to common language, is not a conductor, nor is decomposed; or against water,[66\14] which is a conductor, and can be decomposed; as well as against the metal poles, which are excellent conductors, but undecomposable; there appears but little reason to consider this phenomenon generally as due to the attraction or attractive powers of the latter, when used in the ordinary way, since similar attractions can hardly be imagined in the former instances."

[Note 61\14: Paris, ii. 3.]

[Note 62\14: December, 1835. (At present, when I am revising the second edition, September, 1846, Dr. Faraday has recently published the "Twenty-first Series" of his _Researches_ ending with paragraph 2453.)]

[Note 63\14: _Phil. Trans._ 1833.]

[Note 64\14: _Researches_, Art. 497]

[Note 65\14: _Researches_, Arts. 465, 469.]

[Note 66\14: 495.]

Faraday's opinion, and, indeed, the only way of expressing the results of his experiments, was, that the chemical elements, in obedience to the direction of the voltaic currents established in the decomposing substance, were evolved, or, as he prefers to say, _ejected_ at its extremities.[67\14] He afterwards states that the influence which is present in the electric current may be described[68\14] as _an axis of power, having_ [at each point] _contrary forces exactly equal in amount in contrary directions_.

[Note 67\14: 493.]

[Note 68\14: 517.]

Having arrived at this point, Faraday rightly wished to reject the term _poles_, and other words which could hardly be used without suggesting doctrines now proved to be erroneous. He considered, in the case of bodies electrically decomposed, or, as he termed them, _electrolytes_, the elements as travelling in two opposite directions; which, with reference to the direction of terrestrial magnetism, might be considered as naturally east and west; and he conceived elements as, in this way, arriving at the doors or outlets at which they finally made their separate appearance. The doors he called _electrodes_, and, separately, the _anode_ and the _cathode_;[69\14] and the elements which thus travel he termed the _anïon_ and the _catïon_ (or _cathïon_).[70\14] By means of this nomenclature he was able to express his general results with much more distinctness and facility.

[Note 69\14: 663.]

[Note 70\14: The analogy of the Greek derivation requires _catïon_; but to make the relation to _cathode_ obvious to the English reader, and to avoid a violation of the habits of English pronunciation, I should prefer _cathïon_.]

But this general view of the electrolytical process required to be pursued further, in order to explain the nature of the action. The identity of electrical and chemical forces, which had been hazarded as {299} a conjecture by Davy, and adopted as the basis of chemistry by Berzelius, could only be established by exact measures and rigorous proofs. Faraday had, in his proof of the identity of voltaic and electric agency, attempted also to devise such a measure as should give him a comparison of their quantity; and in this way he proved that[71\14] a voltaic group of two small wires of platinum and zinc, placed near each other, and immersed in dilute acid for three seconds, yields as much electricity as the electrical battery, charged by ten turns of a large machine; and this was established both by its momentary electro-magnetic effect, and by the amount of its chemical action.[72\14]

[Note 71\14: _Researches_, Art. 371.]

[Note 72\14: 537.]

It was in his "Seventh Series," that he finally established a principle of definite measurement of the amount of electrolytical action, and described an instrument which he termed[73\14] a _volta-electrometer_. In this instrument the amount of action was measured by the quantity of water decomposed: and it was necessary, in order to give validity to the mensuration, to show (as Faraday did show) that neither the size of the electrodes, nor the intensity of the current, nor the strength of the acid solution which acted on the plates of the pile, disturbed the accuracy of this measure. He proved, by experiments upon a great variety of substances, of the most different kinds, that the electro-chemical action is definite in amount according to the measurement of the new instrument.[74\14] He had already, at an earlier period,[75\14] asserted, that _the chemical power of a current of electricity is in direct proportion to the absolute quantity of electricity which passes_; but the

volta-electrometer enabled him to fix with more precision the meaning of this general proposition, as well as to place it beyond doubt.

[Note 73\14: 739.]

[Note 74\14: Arts. 758, 814.]

[Note 75\14: 377.]

The vast importance of this step in chemistry soon came into view. By the use of the volta-electrometer, Faraday obtained, for each elementary substance, a number which represented the relative amount of its decomposition, and which might properly[76\14] be called its "electro-chemical equivalent." And the question naturally occurs, whether these numbers bore any relation to any previously established chemical measures. The answer is remarkable. _They were no other than the atomic weights of the Daltonian theory_, which formed the climax of the previous ascent of chemistry; and thus here, as everywhere in {300} the progress of science, the generalizations of one generation are absorbed in the wider generalizations of the next.

[Note 76\14: 792.]

But in order to reach securely this wider generalization, Faraday combined the two branches of the subject which we have already noticed;--the _theory of electrical decomposition_ with the _theory of the pile_. For his researches on the origin of activity of the voltaic circuit (his Eighth Series), led him to see more clearly than any one before him, what, as we have said, the most sagacious of preceding philosophers had maintained, that the current in the pile was due to the mutual chemical action of its elements. He was led to consider the processes which go on in the _exciting-cell_ and in the decomposing place as of the same kind, but opposite in direction. The chemical _composition_ of the fluid with the zinc, in the common apparatus, produces, when the circuit is completed, a current of electric influence in the wire; and this current, if it pass through an electrolyte, manifests itself by _decomposition_, overcoming the chemical affinity which there resists it. An electrolyte cannot conduct without being decomposed. The forces at the point of composition and the point of decomposition are of the same kind, and are opposed to each other by means of the conducting-wire; the wire may properly be spoken of[77\14] as _conducting chemical affinity_: it allows two forces of the same kind to oppose one another;[78\14] electricity is only another mode of the exertion of chemical forces;[79\14] and we might express all the circumstances of the voltaic pile without using any other term than chemical affinity, though that of electricity may be very convenient.[80\14] Bodies are held together by a definite power, which, when it ceases to discharge that office, may be thrown into the condition of an electric current.[81\14]

[Note 77\14: Researches Art. 918.]

[Note 78\14: 910.]

[Note 79\14: 915.]

[Note 80\14: 917.]

[Note 81\14: 855.]

Thus the great principle of the identity of electrical and chemical action was completely established. It was, as Faraday with great candor says,[82\14] a confirmation of the general views put forth by Davy, in 1806, and might be expressed in his terms, that "chemical and electrical attractions are produced by the same cause;" but it is easy to see that neither was the full import of these expressions understood nor were the quantities to which they refer conceived as measurable quantities, nor was the assertion anything but a sagacious conjecture, till Faraday gave the interpretation, measure, and proof, of which we have spoken. The evidence of the incompleteness of the views of his predecessor we have already adduced, in speaking of his vague and {301} inconsistent theoretical account of decomposition. The confirmation of Davy's discoveries by Faraday is of the nature of Newton's confirmation of the views of Borelli and Hooke respecting gravity, or like Young's confirmation of the undulatory theory of Huyghens.

[Note 82\14: 965.]

We must not omit to repeat here the moral which we wish to draw from all great discoveries, that they depend upon the combination of _exact facts_ with _clear ideas_. The former of these conditions is easily illustrated in the case of Davy and Faraday, both admirable and delicate experimenters. Davy's rapidity and resource in experimenting were extraordinary,[83\14] and extreme elegance and ingenuity distinguish almost every process of Faraday. He had published, in 1829, a work on _Chemical Manipulation_, in which directions are given for performing in the neatest manner all chemical processes. Manipulation, as he there truly says, is to the chemist like the external senses to the mind;[84\14] and without the supply of fit materials which such senses only can give, the mind can acquire no real knowledge.

[Note 83\14: Paris, i. 145.]

[Note 84\14: _Pref._ p. ii.]

But still the operations of the mind as well as the information of the senses, ideas as well as facts, are requisite for the attainment of any knowledge; and all great steps in science require a peculiar distinctness and vividness of thought in the discoverer. This it is difficult to exemplify in any better way than by the discoveries themselves. Both Davy and Faraday possessed this vividness of mind; and it was a consequence of this endowment, that Davy's **lectures upon chemistry, and Faraday's upon almost any subject of physical philosophy, were of the most brilliant and captivating character. In discovering the nature of voltaic action, the essential intellectual requisite was to have a distinct conception of that which Faraday expressed by the remarkable phrase,[85\14] "_an axis of power having equal and opposite forces_;" and the distinctness of this idea in Faraday's mind shines forth in every part of his writings. Thus he says, the force which determines the decomposition of a body is _in_ the body, not in the poles.[86\14] But for the most part he can of course only convey this fundamental idea by illustrations. Thus[87\14] he represents the voltaic circuit by a double circle, studded with the elements of the circuit, and shows how the _anïons_ travel round it in one direction, and the _cathïons_ in the opposite. He

considers[88\14] the powers at the two places of action as balancing against each other through the medium of the conductors, in a manner {302} analogous to that in which mechanical forces are balanced against each other by the intervention of the lever. It is impossible to him[89\14] to resist the idea, that the voltaic current must be preceded by a state of tension in its interrupted condition, which is relieved when the circuit is completed. He appears to possess the idea of this kind of force with the same eminent distinctness with which Archimedes in the ancient, and Stevinus in the modern history of science, possessed the idea of pressure, and were thus able to found the science of mechanics.[90\14] And when he cannot obtain these distinct modes of conception, he is dissatisfied, and conscious of defect. Thus in the relation between magnetism and electricity,[91\14] "there appears to be a link in the chain of effects, a wheel in the physical mechanism of the action, as yet unrecognized." All this variety of expression shows how deeply seated is the thought. This conception of Chemical Affinity as a peculiar influence of force, which, acting in opposite directions, combines and resolves bodies;--which may be liberated and thrown into the form of a voltaic current, and thus be transferred to remote points, and applied in various ways; is essential to the understanding, as it was to the making, of these discoveries.

[Note 85\14: Art. 517.]

[Note 86\14: Art. 661.]

[Note 87\14: **963.]

[Note 88\14: 917.]

[Note 89\14: Art. 950.]

[Note 90\14: 990.]

[Note 91\14: 1114.]

By those to whom this conception has been conveyed, I venture to trust that I shall be held to have given a faithful account of this important event in the history of science. We may, before we quit the subject, notice one or two of the remarkable subordinate features of Faraday's discoveries.

_Sect._ 3.--_Consequences of Faraday's Discoveries._

FARADAY'S volta-electrometer, in conjunction with the method he had already employed, as we have seen, for the comparison of voltaic and common electricity, enabled him to measure the actual quantity of electricity which is exhibited, in given cases, in the form of chemical affinity. His results appeared in numbers of that enormous amount which so often comes before us in the expression of natural laws. One grain of water[92\14] will require for its decomposition as much electricity as would make a powerful flash of lightning. By further calculation, he finds this quantity to be not less than 800,000 charges of his Leyden battery;[93\14] and this is, by his theory of the identity of the combining with the decomposing force, the quantity of electricity {303} which is naturally associated with the elements of the grain of water, endowing them with their mutual affinity.

[Note 92\14: **853.]

[Note 93\14: 861.]

Many of the subordinate facts and laws which were brought to light by these researches, clearly point to generalizations, not included in that which we have had to consider, and not yet discovered: such laws do not properly belong to our main plan, which is to make our way _up to_ the generalizations. But there is one which so evidently promises to have an important bearing on future chemical theories, that I will briefly mention it. The class of bodies which are capable of electrical decomposition is limited by a very remarkable law: they are such binary compounds only as consist of _single_ proportionals of their elementary principles. It does not belong to us here to speculate on the possible import of this curious law; which, if not fully established, Faraday has rendered, at least, highly probable:[94\14] but it is impossible not to see how closely it connects the Atomic with the Electro-chemical Theory; and in the connexion of these two great members of Chemistry, is involved the prospect of its reaching wider generalizations, and principles more profound than we have yet caught sight of.

[Note 94\14: Art. 697.]

As another example of this connexion, I will, finally, notice that Faraday has employed his discoveries in order to decide, in some doubtful cases, what is the true chemical equivalent;[95\14] "I have such conviction," he says, "that the power which governs electro-decomposition and ordinary chemical attractions is the same; and such confidence in the overruling influence of those natural laws which render the former definite, as to feel no hesitation in believing that the latter must submit to them too. Such being the case, I can have no doubt that, assuming hydrogen as 1, and dismissing small fractions for the simplicity of expression, the equivalent number or atomic weight of oxygen is 8, of chlorine 36, of bromine 78·4, of lead 103·5, of tin 59, &c.; notwithstanding that a very high authority doubles several of these numbers."

[Note 95\14: 851.]

_Sect._ 4.--_Reception of the Electro-chemical Theory._

THE epoch of establishment of the electro-chemical theory, like other great scientific epochs, must have its sequel, the period of its reception and confirmation, application and extension. In that period we {304} are living, and it must be the task of future historians to trace its course.

We may, however, say a word on the reception which the theory met with, in the forms which it assumed, anterior to the labors of Faraday. Even before the great discovery of Davy, Grotthuss, in 1805, had written upon the theory of electro-chemical decomposition; but he and, as we have seen, Davy, and afterwards other writers, as Riffault and Chompré, in 1807, referred the effects to the poles.[96\14] But the most important attempt to appropriate and employ the generalization which these discoveries suggested, was that of Berzelius; who adopted at once the view of the identity, or at least the universal connexion, of electrical relations with chemical affinity. He considered,[97\14] that in all chemical

combinations the elements may be considered as electro-positive and electro-negative; and made this opposition the basis of his chemical doctrines; in which he was followed by a large body of the chemists of Germany. He held too that the heat and light, evolved during cases of powerful combination, are the consequence of the electric discharge which is at that moment taking place: a conjecture which Faraday at first spoke of with praise.[98\14] But at a later period he more sagely says,[99\14] that the flame which is produced in such cases exhibits but a small portion of the electric power which really acts. "These therefore may not, cannot, be taken as evidences of the nature of the action; but are merely incidental results, incomparably small in relation to the forces concerned, and supplying no information of the way in which the particles are active on each other, or in which their forces are finally arranged." And comparing the evidence which he himself had given of the principle on which Berzelius's speculations rested, with the speculations themselves, Faraday justly conceived, that he had transferred the doctrine from the domain of what he calls _doubtful knowledge_, to that of inductive certainty.

[Note 96\14: Faraday (_Researches_, Art. 481, 492).]

[Note 97\14: _Ann. Chim._ lxxxvi. 146, for 1813.]

[Note 98\14: _Researches_, Art. 870]

[Note 99\14: 960.]

Now that we are arrived at the starting-place, from which this well-proved truth, the identity of electric and chemical forces, must make its future advances, it would be trifling to dwell longer on the details of the diffusion of that doubtful knowledge which preceded this more certain science. Our history of chemistry is, therefore, here at an end. I have, as far as I could, executed my task; which was, to mark all the {305} great steps of its advance, from the most unconnected facts and the most imperfect speculations, to the highest generalization at which chemical philosophers have yet arrived.

Yet it will appear to our purpose to say a few words on the connexion of this science with those of which we are next to treat; and that I now proceed to do.

## CHAPTER X.

TRANSITION FROM THE CHEMICAL TO THE CLASSIFICATORY SCIENCES.

IT is the object and the boast of chemistry to acquire a knowledge of bodies which is more exact and constant than any knowledge borrowed from their sensible qualities can be; since it penetrates into their intimate constitution, and discloses to us the invariable laws of their composition. But yet it will be seen, on a little reflection, that such knowledge could not have any existence, if we were not also attentive to their sensible qualities.

The whole fabric of chemistry rests, even at the present day, upon the opposition of acids and bases: an acid was certainly at first known by its sensible qualities, and how otherwise, even now, do we perceive its quality? It was a great discovery of

modern times that earths and alkalies have for their bases metals: but what are _metals_? or how, except from lustre, hardness, weight, and the like, do we recognize a body as a metal? And how, except by such characters, even before its analysis, was it known to be an earth or an alkali? We must suppose some classification established, before we can make any advance by experiment or observation.

It is easy to see that all attempts to avoid this difficulty by referring to processes and analogies, as well as to substances, bring us back to the same point in a circle of fallacies. If we say that an acid and alkali are known by combining with each other, we still must ask, What is the criterion that they have _combined_? If we say that the distinctive qualities of metals and earths are, that metals become earths by oxidation, we must still inquire how we recognize the process of _oxidation_? We have seen how important a part combustion plays in the history of chemical speculation; and we may usefully form such classes of {306} bodies as _combustibles_ and _supporters of combustion_. But even _combustion_ is not capable of being infallibly known, for it passes by insensible shades into oxidation. We can find no basis for our reasonings, which does not assume a classification of obvious facts and qualities.

But any classification of substances on such grounds, appears, at first sight, to involve us in vagueness, ambiguity, and contradiction. Do we really take the sensible qualities of an acid as the criterion of its being an acid?--for instance, its sourness? Prussic acid, arsenious acid, are not sour. "I remember," says Dr. Paris,[100\14] "a chemist having been exposed to much ridicule from speaking of a _sweet_ acid,--why not?" When Davy had discovered potassium, it was disputed whether it was a metal; for though its lustre and texture are metallic, it is so light as to swim on water. And if potassium be allowed to be a metal, is silicium one, a body which wants the metallic lustre, and is a non-conductor of electricity? It is clear that, at least, the _obvious_ application of a classification by physical characters, is attended with endless perplexity.

[Note 100\14: _Life of Davy_, i. 263.]

But since we cannot even begin our researches without assuming a classification, and since the forms of such a classification which first occur, end in apparent confusion, it is clear that we must look to our philosophy for a solution of this difficulty; and must avoid the embarrassments and contradictions of casual and unreflective classification, by obtaining a consistent and philosophical arrangement. We must employ external characters and analogies in a connected and systematic manner; we must have _Classificatory Sciences_, and these must have a bearing even on Chemistry.

Accordingly, the most philosophical chemists now proceed upon this principle. "The method which I have followed," says M. Thenard, in his _Traité de Chimie_, published in 1824, "is, to unite in one group all analogous bodies; and the advantage of this method, which is that employed by naturalists, is very great, especially in the study of the metals and their compounds."[101\14] In this, as in all good systems of chemistry, which have appeared since the establishment of the phlogistic theory, combustion, and the analogous processes, are one great element in the arrangement,

while the difference of metallic and non-metallic, is another element. Thus Thenard, in the first place, speaks of Oxygen; in the next place, of the Non-metallic Combustibles, as Hydrogen, Carbon, Sulphur, Chlorine; and in the next place, of Metals. But the Metals are again divided into six Sections, with reference, {307} principally, to their facility of combination with oxygen. Thus, the First Section is the Metals of the Earths; the Second, the Metals of the Alkalies; the Third, the Easily Oxidable Metals, as Iron; the Fourth, Metals Less Oxidable, as Copper and Lead; the Fifth Section contains only Mercury and Osmium; and the Sixth, what were at an earlier period termed the _Noble_ Metals, Gold, Silver, Platinum, and others.

[Note 101\14: Pref., p. viii.]

How such principles are to be applied, so as to produce a definite and consistent arrangement, will be explained in speaking of the philosophy of the Classificatory Sciences; but there are one or two peculiarities in the classes of bodies thus recognized by modern chemistry, which it may be useful to notice.

1. The distinction of Metallic and Non-metallic is still employed, as of fundamental importance. The discovery of new metals is so much connected with the inquiries concerning chemical elements, that we may notice the general progress of such discoveries. _Gold_, _Silver_, _Iron_, _Copper_, _Quicksilver_, _Lead_, _Tin_, were known from the earliest antiquity. In the beginning of the sixteenth century, mine-directors, like George Agricola, had advanced so far in practical metallurgy, that they had discovered the means of extracting three additional metals, _Zinc_, _Bismuth_, _Antimony_. After this, there was no new metal discovered for a century, and then such discoveries were made by the theoretical chemists, a race of men who had not existed before Beccher and Stahl. Thus _Arsenic_ and _Cobalt_ were made known by Brandt, in the middle of the eighteenth century, and we have a long list of similar discoveries belonging to the same period; _Nickel_, _Manganese_, and _Tungsten_, which were detected by Cronstedt, Gahn, and Scheele, and Delhuyart, respectively; metals of a very different kind, _Tellurium_ and _Molybdenum_, which were brought to light by Müller, Scheele, Bergman, and Hielm; _Platinum_, which was known as early as 1741, but with the ore of which, in 1802 and 1803, the English chemists, Wollaston and Tennant, found that no less than four other new metals (_Palladium_, _Rhodium_, _Iridium_ and _Osmium_) were associated. Finally, (omitting some other new metals,) we have another period of discovery, opened in 1807, by Davy's discovery of _Potassium_, and including the resolution of all, or almost all, the alkalies and earths into metallic bases.

[2nd Ed.] [The next few years made some, at least some conjectural, additions to the list of simple substances, detected by a more minute scrutiny of known substances. _Thorium_ was discovered by Berzelius in 1828; and _Vanadium_ by Professor Sefström in 1830. A {308} metal named _Cerium_, was discovered in 1803, by Hisinger and Berzelius, in a rare Swedish mineral known by the name of Cerit. Mosander more recently has found combined with Cerium, other new metals which he has called _Lanthanium_, _Didymium_, _Erbium_, and _Terbium_: M. Klaus has found a new metal, _Ruthenium_, in the ore of Platinum; and Rose has discovered in Tantalite two other new metals, which he has announced under the names of

_Pelopium_ and _Niobium_. Svanberg is said to have discovered a new earth in Eudialyt, which is supposed to have, like the rest, a new radical. If these last discoveries be confirmed, the number of simple substances will be raised to _sixty-two_.]

2. Attempts have been made to indicate the classification of chemical substances by some peculiarity in the Name; and the Metals, for example, have been designated generally by names in _um_, like the Latin names of the ancient metals, _aurum_, _ferrum_. This artifice is a convenient nomenclature for the purpose of marking a recognized difference; and it would be worth the while of chemists to agree to make it universal, by writing molybden_um_ and platin_um_; which is sometimes done, but not always.

3. I am not now to attempt to determine how far this class,--Metals,--extends; but where the analogies of the class cease to hold there the nomenclature must also change. Thus, some chemists, as Dr. Thomson, have conceived that the base of Silica is more analogous to Carbon and Boron, which form acids with oxygen, than it is to the metals: and he has accordingly associated this base with these substances, and has given it the same termination, _Silicon_. But on the validity of this analogy chemists appear not to be generally agreed.

4. There is another class of bodies which have attracted much notice among modern chemists, and which have also been assimilated to each other in the form of their names; the English writers calling them _Chlorine_, _Fluorine_, _Iodine_, _Bromine_, while the French use the terms _Chlore_, _Phtore_, _Iode_, _Brome_. We have already noticed the establishment of the doctrine--that muriatic acid is formed of a base, chlorine, and of hydrogen,--as a great reform in the oxygen theory; with regard to which rival claims were advanced by Davy, and by MM. Gay-Lussac and Thenard in 1800. Iodine, a remarkable body which, from a dark powder, is converted into a violet-colored gas by the application of heat, was also, in 1813, the subject of a similar rivalry between the same English and French chemists. Bromine {309} was only discovered as late as 1826; and Fluorine, or _Phtore_, as, from its destructive nature, it has been proposed to term it, has not been obtained as a separate substance, and is inferred to exist by analogy only. The analogies of these bodies (Chlore, Phtore, &c.) are very peculiar; for instance, by combination with metals they form salts; by combination with hydrogen they form very strong acids; and all, at the common temperature of the atmosphere, operate on other bodies in the most energetic manner. Berzelius[102\14] proposes to call them _halogenous_ bodies, or _halogenes_.

[Note 102\14: _Chem._ i. 262.]

5. The number of Elementary Substances which are at present presented in our treatises of chemistry[103\14] is _fifty-three_, [or rather, as we have said above, _sixty-two_.] It is naturally often asked what evidence we have, that all these are _elementary_, and what evidence that they are _all_ the elementary bodies;--how we know that new elements may not hereafter be discovered, or these supposed simple bodies resolved into simpler still? To these questions we can only answer, by

referring to the history of chemistry;--by pointing out what chemists have understood by analysis, according to the preceding narrative. They have considered, as the analysis of a substance, that elementary constitution of it which gives the only intelligible explanation of the results of chemical manipulation, and which is proved to be complete as to quantity, by the balance, since the whole can only be equal to all its parts. It is impossible to maintain that new substances may not hereafter be discovered; for they may lurk, even in familiar substances, in doses so minute that they have not yet been missed amid the inevitable slight inaccuracies of all analysis; in the way in which iodine and bromine remained so long undetected in sea-water; and new minerals, or old ones not yet sufficiently examined, can hardly fail to add something to our list. As to the possibility of a further analysis of our supposed simple bodies, we may venture to say that, in regard to such supposed simple bodies as compose a numerous and well-characterized class, no such step can be made, except through some great change in chemical theory, which gives us a new view of all the general relations which chemistry has yet discovered. The proper evidence of the reality of any supposed new analysis is, that it is more consistent with the known analogies of chemistry, to suppose the process analytical than synthetical. Thus, as has already been said, chemists admit the existence of fluorine, from the analogy of chlorine; and Davy, when it was found {310} that ammonia formed an amalgam with mercury, was tempted to assign to it a metallic basis. But then he again hesitates,[104\14] and doubts whether the analogies of our knowledge are not better preserved by supposing that ammonia, as a compound of hydrogen and another principle, is "a type of the composition of the metals."

[Note 103\14: Turner, p. 971.]

[Note 104\14: _Elem. Chem. Phil._ 1812, p. 481.]

Our history, which is the history of what we know, has little to do with such conjectures. There are, however, some not unimportant principles which bear upon them, and which, as they are usually employed, belong to the science which next comes under our review, Mineralogy.

{{311}} BOOK XV.

## _THE ANALYTICO-CLASSIFICATORY SCIENCE._

HISTORY OF MINERALOGY.

Κρύσταλλον φαέθοντα διαυγέα λάζεο χερσὶ, Λᾶαν ἀπόῤῥοιαν περιφεγγέος ἀμβρότου αἴγλης, Αἰθέρι δ' ἀθανάτων μέγα τέρπεται ἄφθιτον ἦτορ. Τόν κ' εἴπερ μετὰ χειρὰς ἔχων, περὶ νηὸν ἵκηαι, Οὔτις τοι μακάρων ἀρνήσεται εὐχωλῆσι. ORPHEUS. _Lithica._

Now, if the bold but pious thought be thine, To reach our spacious temple's inner shrine, Take in thy reverent hands the crystal stone, Where heavenly light in earthy shroud is shown:-- Where, moulded into measured form, with rays Complex yet clear, the eternal Ether plays; This if thou firmly hold and rightly use, Not long the gods thy ardent wish refuse.

{{313}} INTRODUCTION.

_Sect._ 1.--_Of the Classificatory Sciences._

THE horizon of the sciences spreads wider and wider before us, as we advance in our task of taking a survey of the vast domain. We have seen that the existence of Chemistry as a science which declares the ingredients and essential constitution of all kinds of bodies, implies the existence of another corresponding science, which shall divide bodies into kinds, and point out steadily and precisely what bodies they are which we have analysed. But a science thus dividing and defining bodies, is but one member of an order of sciences, different from those which we have hitherto described; namely, of the _classificatory sciences_. Such sciences there must be, not only having reference to the bodies with which chemistry deals, but also to all things respecting which we aspire to obtain any general knowledge, as, for instance, plants and animals. Indeed it will be found, that it is with regard to these latter objects, to organized beings, that the process of scientific classification has been most successfully exercised; while with regard to inorganic substances, the formation of a satisfactory system of arrangement has been found extremely difficult; nor has the necessity of such a system been recognised by chemists so distinctly and constantly as it ought to be. The best exemplification of these branches of knowledge, of which we now have to speak, will, therefore, be found in the organic world, in Botany and Zoology; but we will, in the first place, take a brief view of the science which classifies inorganic bodies, and of which Mineralogy is hitherto the very imperfect representative.

The principles and rules of the Classificatory Sciences, as well as of those of the other orders of sciences, must be fully explained when we come to treat of the Philosophy of the Sciences; and cannot be introduced here, where we have to do with history only. But I may observe very briefly, that with the process of _classing_, is joined the process of _naming_;--that names imply classification;--and that even the rudest and earliest application of language presupposes a distribution of objects according to their kinds;--but that such a spontaneous {314} and unsystematic distribution cannot, in the cases we now have to consider, answer the purposes of exact and general knowledge. Our classification of objects must be made consistent and systematic, in order to be scientific; we must discover marks and characters, properties and conditions, which are constant in their occurrence and relations; we must form our classes, we must impose our names, according to such marks. We can thus, and thus alone, arrive at that precise, certain, and systematic knowledge, which we seek; that is, at science. The object, then, of the classificatory sciences is to obtain FIXED CHARACTERS of the kinds of things; and the criterion of the fitness of names is, that THEY MAKE GENERAL PROPOSITIONS POSSIBLE.

I proceed to review the progress of certain sciences on these principles, and first, though briefly, the science of Mineralogy.

_Sect._ 2.--_Mineralogy as the Analytico-classificatory Science._

MINERALOGY, as it has hitherto been cultivated, is, as I have already said, an imperfect representative of the department of human knowledge to which it belongs.

The attempts at the science have generally been made by collecting various kinds of information respecting mineral bodies; but the science which we require is a complete and consistent classified system of all inorganic bodies. For chemistry proceeds upon the principle that the constitution of a body invariably determines its properties; and, consequently, its kind: but we cannot apply this principle, except we can speak with precision of the _kind_ of a body, as well as of its composition. We cannot attach any sense to the assertion, that "soda or baryta has a metal for its base," except we know what _a metal_ is, or at least what properties it implies. It may not be, indeed it is not, possible, to define the kinds of bodies by words only; but the classification must proceed by some constant and generally applicable process; and the knowledge which has reference to the classification will be precise as far as this process is precise, and vague as far as this is vague.

There must be, then, as a necessary supplement to Chemistry, a Science of those properties of bodies by which we divide them into _kinds_. Mineralogy is the branch of knowledge which has discharged the office of such a science, so far as it has been discharged; and, indeed, Mineralogy has been gradually approaching to a clear consciousness of her real place, and of her whole task; I shall give the history of some of the advances which have thus been made. They are, principally, {315} the establishment and use of External Characters, especially of _Crystalline Form_, as a fixed character of definite substances; and the attempts to bring into view the connexion of Chemical Constitution and External Properties, made in the shape of mineralogical _Systems_; both those in which _chemical methods of arrangement_ are adopted, and those which profess to classify by the _natural-history method_.

{{316}} CRYSTALLOGRAPHY.

## CHAPTER I.

PRELUDE TO THE EPOCH OF DE LISLE AND HAÜY.

OF all the physical properties of bodies, there is none so fixed, and in every way so remarkable, as this;--that the same chemical compound always assumes, with the utmost precision, the same geometrical form. This identity, however, is not immediately obvious; it is often obscured by various mixtures and imperfections in the substance; and even when it is complete, it is not immediately recognized by a common eye, since it consists, not in the equality of the sides or faces of the figures, but in the equality of their angles. Hence it is not surprising that the constancy of form was not detected by the early observers. Pliny says,[1\15] "Why crystal is generated in a hexagonal form, it is difficult to assign a reason; and the more so, since, while its faces are smoother than any art can make them, the pyramidal points are _not all of the same kind_." The quartz crystals of the Alps, to which he refers, are, in some specimens, very regular, while in others, one side of the pyramid becomes much the largest; yet the angles remain constantly the same. But when the whole shape varied so much, the angles also seemed to vary. Thus Conrad Gessner, a very learned naturalist, who, in 1564, published at Zurich his work, _De rerum Fossilium, Lapidum et Gemmarum maxime, Figuris_, says,[2\15] "One crystal differs

from another in its angles, and consequently in its figure." And Cæsalpinus, who, as we shall find, did so much in establishing fixed characters in botany, was led by some of his general views to disbelieve the fixity of the form of crystals. In his work _De Metallicis_, published at Nuremberg in 1602, he says,[3\15] "To ascribe to inanimate bodies a definite form, does not appear consentaneous to reason; for it is the office of organization to produce a definite form;" {317} an opinion very natural in one who had been immersed in the study of the general analogies of the forms of plants. But though this is excusable in Cæsalpinus, the rejection of this definiteness of form a hundred years later, when its existence had been proved, and its laws developed by numerous observers, cannot be ascribed to anything but strong prejudice; yet this was the course taken by no less a person than Buffon. "The form of crystallization," says he,[4\15] "is _not a constant character_, but is more equivocal and more variable than any other of the characters by which minerals are to be distinguished." And accordingly, he makes no use of this most important feature in his history of minerals. This strange perverseness may perhaps be ascribed to the dislike which Buffon is said to have entertained for Linnæus, who had made crystalline form a leading character of minerals.

[Note 1\15: _Nat. Hist._ xxvii. 2.]

[Note 2\15: p. 25.]

[Note 3\15: p. 97.]

[Note 4\15: _Hist. des Min._ p. 343.]

It is not necessary to mark all the minute steps by which mineralogists were gradually led to see clearly the nature and laws of the fixity of crystalline forms. These forms were at first noticed in that substance which is peculiarly called rock-crystal or quartz; and afterwards in various stones and gems, in salts obtained from various solutions, and in snow. But those who observed the remarkable regular figures which these substances assume, were at first impelled onwards in their speculations by the natural tendency of the human mind to generalize and guess, rather than to examine and measure. They attempted to snatch at once the general laws of geometrical regularity of these occurrences, or to connect them with some doctrine concerning formative causes. Thus Kepler,[5\15] in his _Harmonics of the World_, asserts a "_formatrix facultas_, which has its seat in the entrails of the earth, and, after the manner of a pregnant woman, expresses the five regular geometrical solids in the forms of gems." But Philosophers, in the course of time, came to build more upon observation, and less upon abstract reasonings. Nicolas Steno, a Dane, published, in 1669, a dissertation _De Solido intra Solidum Naturaliter contento_, in which he says,[6\15] that though the sides of the hexagonal crystal may vary, _the angles are not changed_. And Dominic Gulielmini, in a _Dissertation on Salts_, published in 1707, says,[7\15] in a true inductive spirit, "Nature does not employ all figures, but only certain ones of those which are possible; and of these, the determination is not to be fetched from the brain, or proved _à priori_, but obtained by experiments and observations." And {318} he speaks[8\15] with entire decision on this subject: "Nevertheless since there is here a principle of crystallization, the

inclination of the planes and of the angles is always constant." He even anticipates, very nearly, the views of later crystallographers as to the mode in which crystals are formed from elementary molecules. From this time, many persons labored and speculated on this subject; as Cappeller, whose _Prodromus Crystallographiæ_ appeared at Lucern in 1723; Bourguet, who published _Lettres Philosophiques sur la Formation de Sels et de Cristaux_, at Amsterdam, in 1792; and Henckel, the "Physicus" of the Elector of Saxony, whose _Pyritologia_ came forth in 1725. In this last work we have an example of the description of the various forms of special classes of minerals, (iron pyrites, copper pyrites, and arsenic pyrites;) and an example of the enthusiasm which this apparently dry and laborious study can excite: "Neither tongue nor stone," he exclaims,[9\15] "can express the satisfaction which I received on setting eyes upon this sinter covered with galena; and thus it constantly happens, that one must have more pleasure in what seems worthless rubbish, than in the purest and most precious ores, if we know aught of minerals."

[Note 5\15: Linz. 1619, p. 161.]

[Note 6\15: p. 69.]

[Note 7\15: p. 19.]

[Note 8\15: p. 83.]

[Note 9\15: p. 343.]

Still, however, Henckel[10\15] disclaims the intention of arranging minerals according to their mathematical forms; and this, which may be considered as the first decided step in the formation of crystallographic mineralogy, appears to have been first attempted by Linnæus. In this attempt, however, he was by no means happy; nor does he himself appear to have been satisfied. He begins his preface by saying, "Lithology is not what I plume myself upon." (_Lithologia mihi cristas non eriget_.) Though his sagacity, as a natural historian, led him to see that crystalline form was one of the most definite, and therefore most important, characters of minerals, he failed in profiting by this thought, because, in applying it, he did not employ the light of geometry, but was regulated by what appeared to him resemblances, arbitrarily selected, and often delusive.[11\15] Thus he derived the form of pyrites from that of vitriol;[12\15] and brought together alum and diamond on account of their common octohedral form. But he had the great merit of animating to this study one to whom, more perhaps than to any other person, it owes its subsequent progress; I mean Romé de Lisle. "Instructed," this writer says, in his preface to his _Essais de Crystallographie_, "by the works of the celebrated Von Linnée, how {319} greatly the study of the angular form of crystals might become interesting, and fitted to extend the sphere of our mineralogical knowledge, I have followed them in all their metamorphoses with the most scrupulous attention." The views of Linnæus, as to the importance of this character, had indeed been adopted by several others; as John Hill, the King's gardener at Kew, who, in 1777, published his _Spathogenesia_; and Grignon, who, in 1775, says, "These crystallizations may give the means of finding a new theory of the generation of crystalline gems."

[Note 10\15: p. 167.]

[Note 11\15: Marx. _Gesch._ p. 97.]

[Note 12\15: _Syst. Nat._ vi. p. 220.]

The circumstance which threw so much difficulty in the way of those who tried to follow out his thought was, that in consequence of the apparent irregularity of crystals, arising from the extension or contraction of particular sides of the figure, each kind of substance may really appear under many different forms, connected with each other by certain geometrical relations. These may be conceived by considering a certain fundamental form to be cut into new forms in particular ways. Thus if we take a cube, and cut off all the eight corners, till the original faces disappear, we make it an octohedron; and if we stop short of this, we have a figure of fourteen faces, which has been called a _cubo-octohedron_. The first person who appears distinctly to have conceived this _truncation_ of angles and edges, and to have introduced the word, is Démeste;[13\15] although Wallerius[14\15] had already said, in speaking of the various crystalline forms of calcspar, "I conceive it would be better not to attend to all differences, lest we be overwhelmed by the number." And Werner, in his celebrated work _On the External Characters of Minerals_,[15\15] had formally spoken of _truncation_, _acuation_, and _acumination_, or replacement by a plane, an edge, a point respectively, (_abstumpfung_, _zuschärfung_, _zuspitzung_,) as ways in which the forms of crystals are modified and often disguised. He applied this process in particular to show the connexion of the various forms which are related to the cube. But still the extension of the process to the whole range of minerals and other crystalline bodies, was due to Romé de Lisle. {320}

[Note 13\15: _Lettres_, 1779, i. 48.]

[Note 14\15: _Systema Mineralogicum_, 1772-5, i. 143.]

[Note 15\15: Leipzig, 1774.]

## CHAPTER II.

EPOCH OF ROMÉ DE LISLE AND HAÜY.--ESTABLISHMENT OF THE FIXITY OF CRYSTALLINE ANGLES, AND THE SIMPLICITY OF THE LAWS OF DERIVATION.

WE have already seen that, before 1780, several mineralogists had recognized the constancy of the angles of crystals, and had seen (as Démeste and Werner,) that the forms were subject to modifications of a definite kind. But neither of these two thoughts was so apprehended and so developed, as to supersede the occasion for a discoverer who should put forward these principles as what they really were, the materials of a new and complete science. The merit of this step belongs jointly to Romé de Lisle and to Haüy. The former of these two men had already, in 1772, published an _Essai de Crystallographie_, in which he had described a number of crystals. But in this work his views are still rude and vague; he does not establish any connected sequence of transitions in each kind of substance, and lays little or no stress on the angles. But in 1783, his ideas[16\15] had reached a maturity which, by comparison, excites our admiration. In this he asserts, in the most distinct manner, the _invariability_ of the angles of crystals of each kind, under all the changes of

relative dimension which the faces may undergo;[17\15] and he points out that this invariability applies only to the _primitive forms_, from each of which many secondary forms are derived by various changes.[18\15] Thus we cannot deny him the merit of having taken steady hold on both the handles of this discovery, though something still remained for another to do. Romé pursues his general ideas into detail with great labor and skill. He gives drawings of more than five hundred regular forms (in his first work he had inserted only one hundred and ten; Linnæus only knew forty); and assigns them to their proper substances; for instance, thirty to calcspar, and sixteen to felspar. He also invented and used a goniometer. We cannot doubt that he would have been {321} looked upon as a great discoverer, if his fame had not been dimmed by the more brilliant success of his contemporary Haüy.

[Note 16\15: _Cristallographie, ou Description de Formes propres à tous les Corps du Règne Minéral._ 3 vols. and 1 vol. of plates.]

[Note 17\15: p. 68.]

[Note 18\15: p. 73.]

Réné-Just Haüy is rightly looked upon as the founder of the modern school of crystallography; for all those who have, since him, pursued the study with success, have taken his views for their basis. Besides publishing a system of crystallography and of mineralogy, far more complete than any which had yet appeared, the peculiar steps in the advance which belong to him are, the discovery of the importance of _cleavage_, and the consequent expression of the laws of derivation of secondary from primary forms, by means of the _decrements_ of the successive layers of _integrant molecules_.

The latter of these discoveries had already been, in some measure, anticipated by Bergman, who had, in 1773, conceived a hexagonal prism to be built up by the juxtaposition of solid rhombs on the planes of a rhombic nucleus.[19\15] It is not clear[20\15] whether Haüy was acquainted with Bergman's Memoir, at the time when the cleavage of a hexagonal prism of calcspar, accidentally obtained, led him to the same conception of its structure. But however this might be, he had the indisputable credit of following out this conception with all the vigor of originality, and with the most laborious and persevering earnestness; indeed he made it the business of his life. The hypothesis of a solid, built up of small solids, had this peculiar advantage in reference to crystallography; it rendered a reason of this curious fact;--that a certain series of forms occur in crystals of the same kind, while other forms, apparently intermediate between those which actually occur, are rigorously excluded. The doctrine of decrements explained this; for by placing a number of regularly-decreasing rows of equal solids, as, for instance, of bricks, upon one another, we might form a regular equal-sided triangle, as the gable of a house; and if the breadth of the gable were one hundred bricks, the height of the triangle might be one hundred, or fifty, or twenty-five; but it would be found that if the height were an intermediate number, as fifty-seven, or forty-three, the edge of the wall would become irregular; and such irregularity is assumed to be inadmissible in the regular structure of crystals. Thus this mode of conceiving crystals allows of certain definite

secondary forms, and no others.

[Note 19\15: _De Formis Crystallorum._ Nov. Act. Reg. Soc. Sc. Ups. 1773.]

[Note 20\15: _Traité de Minér._ 1822, i. 15.]

The mathematical deduction of the dimensions and proportions {322} of these secondary forms;--the invention of a notation to express them;--the examination of the whole mineral kingdom in accordance with these views;--the production of a work[21\15] in which they are explained with singular clearness and vivacity;--are services by which Haüy richly earned the admiration which has been bestowed upon him. The wonderful copiousness and variety of the forms and laws to which he was led, thoroughly exercised and nourished the spirit of deduction and calculation which his discoveries excited in him. The reader may form some conception of the extent of his labors, by being told--that the mere geometrical propositions which he found it necessary to premise to his special descriptions, occupy a volume and a half of his work;--that his diagrams are nearly a thousand in number;--that in one single substance (calcspar) he has described forty-seven varieties of form;--and that he has described one kind of crystal (called by him _fer sulfuré parallélique_) which has one hundred and thirty-four faces.

[Note 21\15: _Traité de Minéralogie_, 1801, 5 vols.]

In the course of a long life, he examined, with considerable care, all the forms he could procure of all kinds of mineral; and the interpretation which he gave of the laws of those forms was, in many cases, fixed, by means of a name applied to the mineral in which the form occurred; thus, he introduced such names as _équiaxe_, _métastatique_, _unibinaire_, _perihexahèdre_, _bisalterne_, and others. It is not now desirable to apply separate names to the different forms of the same mineral species, but these terms answered the purpose, at the time, of making the subjects of study more definite. A symbolical notation is the more convenient mode of designating such forms, and such a notation Haüy invented; but the symbols devised by him had many inconveniences, and have since been superseded by the systems of other crystallographers.

Another of Haüy's leading merits was, as we have already intimated, to have shown, more clearly than his predecessors had done, that the crystalline angles of substances are a criterion of the substances; and that this is peculiarly true of the _angles of cleavage_;--that is, the angles of those edges which are obtained by cleaving a crystal in two different directions;--a mode of division which the structure of many kinds of crystals allowed him to execute in the most complete manner. As an instance of the employment of this criterion, I may mention his separation of the sulphates of baryta and strontia, which had {323} previously been confounded. Among crystals which in the collections were ranked together as "heavy spar," and which were so perfect as to admit of accurate measurement, he found that those which were brought from Sicily, and those of Derbyshire, differed in their cleavage angle by three degrees and a half. "I could not suppose," he says,[22\15] "that this difference was the effect of any law of decrement; for it would have been necessary to suppose so rapid and complex a law, that such an hypothesis might have been justly

regarded as an abuse of the theory." He was, therefore, in great perplexity. But a little while previous to this, Klaproth had discovered that there is an earth which, though in many respects it resembles baryta, is different from it in other respects; and this earth, from the place where it was found (in Scotland), had been named _Strontia_. The French chemists had ascertained that the two earths had, in some cases, been mixed or confounded; and Vauquelin, on examining the Sicilian crystals, found that their base was strontia, and not, as in the Derbyshire ones, baryta. The riddle was now read; all the crystals with the larger angle belong to the one, all those with the smaller, to the other, of these two sulphates; and crystallometry was clearly recognized as an authorized test of the difference of substances which nearly resemble each other.

[Note 22\15: _Traité_, ii. 320.]

Enough has been said, probably, to enable the reader to judge how much each of the two persons, now under review, contributed to crystallography. It would be unwise to compare such contributions to science with the great discoveries of astronomy and chemistry; and we have seen how nearly the predecessors of Romé and Haüy had reached the point of knowledge on which these two crystallographers took their stand. But yet it is impossible not to allow, that in these discoveries, which thus gave form and substance to the science of crystallography, we have a manifestation of no common sagacity and skill. Here, as in other discoveries, were required ideas and facts;--clearness of geometrical conception which could deal with most complex relations of form; a minute and extensive acquaintance with actual crystals; and the talent and habit of referring these facts to the general ideas. Haüy, in particular, was happily endowed for his task. Without being a great mathematician, he was sufficiently a geometer to solve all the problems which his undertaking demanded; and though the mathematical reasoning might have been made more compendious {324} by one who was more at home in mathematical generalization, probably this could hardly have been done without making the subject less accessible and less attractive to persons moderately disciplined in mathematics. In all his reasonings upon particular cases, Haüy is acute and clear; while his general views appear to be suggested rather by a lively fancy than by a sage inductive spirit: and though he thus misses the character of a great philosopher, the vivacity of style, and felicity and happiness of illustration, which grace his book, and which agree well with the character of an Abbé of the old French monarchy, had a great and useful influence on the progress of the subject.

Unfortunately Romé de Lisle and Haüy were not only rivals, but in some measure enemies. The former might naturally feel some vexation at finding himself, in his later years (he died in 1790), thrown into shade by his more brilliant successor. In reference to Haüy's use of cleavage, he speaks[23\15] of "innovators in crystallography, who may properly be called _crystalloclasts_." Yet he adopted, in great measure, the same views of the formation of crystals by laminæ,[24\15] which Haüy illustrated by the destructive process at which he thus sneers. His sensitiveness was kept alive by the conduct of the Academy of Sciences, which took no notice of him and his labors;[25\15] probably because it was led by Buffon, who disliked

Linnæus, and might dislike Romé as his follower; and who, as we have seen, despised crystallography. Haüy revenged himself by rarely mentioning Romé in his works, though it was manifest that his obligations to him were immense; and by recording his errors while he corrected them. More fortunate than his rival, Haüy was, from the first, received with favor and applause. His lectures at Paris were eagerly listened to by persons from all quarters of the world. His views were, in this manner, speedily diffused; and the subject was soon pursued, in various ways, by mathematicians and mineralogists in every country of Europe.

[Note 23\15: Pref. p. xxvii.]

[Note 24\15: T. ii. p. 21.]

[Note 25\15: Marx. _Gesch. d. Cryst._ 130.]

## CHAPTER III.

RECEPTION AND CORRECTIONS OF THE HAUÏAN CRYSTALLOGRAPHY.

I HAVE not hitherto noticed the imperfections of the crystallographic views and methods of Haüy, because my business in the last section {325} was to mark the permanent additions he made to the science. His system did, however, require completion and rectification in various points; and in speaking of the crystallographers of the subsequent time, who may all be considered as the cultivators of the Hauïan doctrines, we must also consider what they did in correcting them.

The three main points in which this improvement was needed were;--a better determination of the crystalline forms of the special substances;--a more general and less arbitrary method of considering crystalline forms according to their symmetry; and a detection of more general conditions by which the crystalline angle is regulated. The first of these processes may be considered as the natural sequel of the Hauïan epoch: the other two must be treated as separate steps of discovery.

When it appeared that the angle of natural or of cleavage faces could be used to determine the differences of minerals, it became important to measure this angle with accuracy. Haüy's measurements were found very inaccurate by many succeeding crystallographers: Mohs says[26\15] that they are so generally inaccurate, that no confidence can be placed in them. This was said, of course, according to the more rigorous notions of accuracy to which the establishment of Haüy's system led. Among the persons who principally labored in ascertaining, with precision, the crystalline angles of minerals, were several Englishmen, especially Wollaston, Phillips, and Brooke. Wollaston, by the invention of his Reflecting Goniometer, placed an entirely new degree of accuracy within the reach of the crystallographer; the angle of two faces being, in this instrument, measured by means of the reflected images of bright objects seen in them, so that the measure is the more accurate the more minute the faces are. In the use of this instrument, no one was more laborious and successful than William Phillips, whose power of apprehending the most complex forms with steadiness and clearness, led Wollaston

to say that he had "a geometrical sense." Phillips published a Treatise on Mineralogy, containing a great collection of such determinations; and Mr. Brooke, a crystallographer of the same exact and careful school, has also published several works of the same kind. The precise measurement of crystalline angles must be the familiar employment of all who study crystallography; and, therefore, any further enumeration of those {326} who have added in this way to the stock of knowledge, would be superfluous.

[Note 26\15: Marx. p. 153.]

Nor need I dwell long on those who added to the knowledge which Haüy left, of derived forms. The most remarkable work of this kind was that of Count Bournon, who published a work on a single mineral (calcspar) in three quarto volumes.[27\15] He has here given representations of seven hundred forms of crystals, of which, however, only fifty-six are essentially different. From this example the reader may judge what a length of time, and what a number of observers and calculators, were requisite to exhaust the subject.

[Note 27\15: _Traité complet de la Chaux Carbonatée et d'Aragonite_, par M. le Comte de Bournon. London, 1808.]

If the calculations, thus occasioned, had been conducted upon the basis of Haüy's system, without any further generalization, they would have belonged to that process, the natural sequel of inductive discoveries, which we call _deduction_; and would have needed only a very brief notice here. But some additional steps were made in the upward road to scientific truth, and of these we must now give an account.

## CHAPTER IV.

ESTABLISHMENT OF THE DISTINCTION OF SYSTEMS OF CRYSTALLIZATION.--WEISS AND MOHS.

IN Haüy's views, as generally happens in new systems, however true, there was involved something that was arbitrary, something that was false or doubtful, something that was unnecessarily limited. The principal points of this kind were;--his having made the laws of crystalline derivation depend so much upon cleavage;--his having assumed an atomic constitution of bodies as an essential part of his system; and his having taken a set of primary forms, which, being selected by no general view, were partly superfluous, and partly defective.

How far evidence, such as has been referred to by various philosophers, has proved, or can prove, that bodies are constituted of indivisible atoms, will be more fully examined in the work which treats of the Philosophy of this subject. There can be little doubt that the {327} portion of Haüy's doctrine which most riveted popular attention and applause, was his dissection of crystals, in a manner which was supposed to lead actually to their ultimate material elements. Yet it is clear, that since the solids given by cleavage are, in many cases, such as cannot make up a solid space, the primary conception of a necessary geometrical identity between the results of division and the elements of composition, which is the sole foundation of

the supposition that crystallography points out the actual elements, disappears on being scrutinized: and when Haüy, pressed by this difficulty, as in the case of fluor-spar, put his integrant octohedral molecules together, touching by the edges only, his method became an empty geometrical diagram, with no physical meaning.

The real fact, divested of the hypothesis which was contained in the fiction of decrements, was, that when the relation of the derivative to the primary faces is expressed by means of numerical indices, these numbers are integers, and generally very small ones; and this was the form which the law gradually assumed, as the method of derivation was made more general and simple by Weiss and others.

"When, in 1809, I published my Dissertation," says Weiss,[28\15] "I shared the common opinion as to the necessity of the assumption and the reality of the existence of a primitive form, at least in a sense not very different from the usual sense of the expression. While I sought," he adds, referring to certain doctrines of general philosophy which he and others entertained, "a _dynamical_ ground for this, instead of the untenable atomistic view, I found that, out of my primitive forms, there was gradually unfolded to my hands, that which really governs them, and is not affected by their casual fluctuations, the fundamental relations of those Dimensions according to which a multiplicity of internal oppositions, necessarily and mutually interdependent, are developed in the mass, each having its own polarity; so that the crystalline character is co-extensive with these polarities."

[Note 28\15: _Mem. Acad. Berl._ 1816, p. 307.]

The "Dimensions" of which Weiss here speaks, are the _Axes of Symmetry_ of the crystal; that is, those lines in reference to which, every face is accompanied by other faces, having like positions and properties. Thus a rhomb, or more properly a _rhombohedron_,[29\15] of {328} calcspar may be placed with one of its obtuse corners uppermost, so that all the three faces which meet there are equally inclined to the vertical line. In this position, every derivative face, which is obtained by any modification of the faces or edges of the rhombohedron, implies either three or six such derivative faces; for no one of the three upper faces of the rhombohedron has any character or property different from the other two; and, therefore, there is no reason for the existence of a derivative from one of these primitive faces, which does not equally hold for the other primitive faces. Hence the derivative forms will, in all cases, contain none but faces connected by this kind of correspondence. The axis thus made vertical will be an Axis of Symmetry, and the crystal will consist of three divisions, ranged round this axis, and exactly resembling each other. According to Weiss's nomenclature, such a crystal is "three-and-three-membered."

[Note 29\15: I use this name for the solid figure, since _rhomb_ has always been used for a plane figure.]

But this is only one of the kinds of symmetry which crystalline forms may exhibit. They may have _three axes_ of complete and _equal_ symmetry at right angles to each other, as the cube and the regular octohedron;--or, _two axes_ of equal symmetry, perpendicular to each other and to a _third axis_, which is not affected with the same symmetry with which they are; such a figure is a square pyramid;--or they may have

_three_ rectangular _axes_, all of _unequal_ symmetry, the modifications referring to each axis separately from the other two.

These are essential and necessary distinctions of crystalline form; and the introduction of a classification of forms founded on such relations, or, as they were called, _Systems of Crystallization_, was a great improvement upon the divisions of the earlier crystallographers, for those divisions were separated according to certain arbitrarily-assumed primary forms. Thus Romé de Lisle's fundamental forms were, the tetrahedron, the cube, the octohedron, the rhombic prism, the rhombic octohedron, the dodecahedron with triangular faces: Haüy's primary forms are the cube, the rhombohedron, the oblique rhombic prism, the right rhombic prism, the rhombic dodecahedron, the regular octohedron, tetrahedron, and six-sided prism, and the bipyramidal dodecahedron. This division, as I have already said, errs both by excess and defect, for some of these primary forms might be made derivatives from others; and no solid reason could be assigned why they were not. Thus the cube may be derived from the tetrahedron, by truncating the edges; and the rhombic dodecahedron again from the cube, by truncating its edges; while the square pyramid could not be legitimately identified with the derivative of any of these forms; for if we were to {329} derive it from the rhombic prism, why should the acute angles always suffer decrements corresponding in a certain way to those of the obtuse angles, as they must do in order to give rise to a square pyramid?

The introduction of the method of reference to Systems of Crystallization has been a subject of controversy, some ascribing this valuable step to Weiss, and some to Mohs.[30\15] It appears, I think, on the whole, that Weiss first published works in which the method is employed; but that Mohs, by applying it to all the known species of minerals, has had the merit of making it the basis of real crystallography. Weiss, in 1809, published a Dissertation _On the mode of investigating the principal geometrical character of crystalline forms_, in which he says,[31\15] "No part, line, or quantity, is so important as the axis; no consideration is more essential or of a higher order than the relation of a crystalline plane to the axis;" and again, "An axis is any line governing the figure, about which all parts are similarly disposed, and with reference to which they correspond mutually." This he soon followed out by examination of some difficult cases, as Felspar and Epidote. In the Memoirs of the Berlin Academy,[32\15] for 1814-15, he published _An Exhibition of the natural Divisions of Systems of Crystallization_. In this Memoir, his divisions are as follows:--The _regular_ system, the _four-membered_, the _two-and-two-membered_, the _three-and-three-membered_, and some others of inferior degrees of symmetry. These divisions are by Mohs (_Outlines of Mineralogy_, 1822), termed the _tessular_, _pyramidal_, _prismatic_, and _rhombohedral_ systems respectively. Hausmann, in his _Investigations concerning the Forms of Inanimate Nature_,[33\15] makes a nearly corresponding arrangement;--the _isometric_, _monodimetric_, _trimetric_, and _monotrimetic_; and one or other of these sets of terms have been adopted by most succeeding writers.

[Note 30\15: _Edin. Phil. Trans._ 1823, vols. xv. and xvi.]

[Note 31\15: pp. 16, 42.]

[Note 32\15: Ibid.]

[Note 33\15: Göttingen, 1821.]

In order to make the distinctions more apparent, I have purposely omitted to speak of the systems which arise when the _prismatic_ system loses some part of its symmetry;--when it has only half or a quarter its complete number of faces;--or, according to Mohs's phraseology, when it is _hemihedral_ or _tetartohedral_. Such systems are represented by the singly-oblique or doubly-oblique prism; they are termed by Weiss _two-and-one-membered_, and _one-and-one-membered_; by other writers, _Monoklinometric_, and _Triklinometric_ Systems. There are also other {330} peculiarities of Symmetry, such, for instance, as that of the _plagihedral_ faces of quartz, and other minerals.

The introduction of an arrangement of crystalline forms into systems, according to their degree of symmetry, was a step which was rather founded on a distinct and comprehensive perception of mathematical relations, than on an acquaintance with experimental facts, beyond what earlier mineralogists had possessed. This arrangement was, however, remarkably confirmed by some of the properties of minerals which attracted notice about the time now spoken of, as we shall see in the next chapter.

## CHAPTER V.

RECEPTION AND CONFIRMATION OF THE DISTINCTION OF SYSTEMS OF CRYSTALLIZATION.

DIFFUSION OF THE DISTINCTION OF SYSTEMS.--The distinction of systems of crystallization was so far founded on obviously true views, that it was speedily adopted by most mineralogists. I need not dwell on the steps by which this took place. Mr. Haidinger's translation of Mohs was a principal occasion of its introduction in England. As an indication of dates, bearing on this subject, perhaps I may be allowed to notice, that there appeared in the _Philosophical Transactions for_ 1825, _A General Method of Calculating the **Angles of Crystals_, which I had written, and in which I referred only to Haüy's views; but that in 1826,[34\15] I published a Memoir _On the Classification of Crystalline Combinations_, founded on the methods of Weiss and Mohs, especially the latter; with which I had in the mean time become acquainted, and which appeared to me to contain their own evidence and recommendation. General methods, such as was attempted in the Memoir just quoted, are part of that process in the history of sciences, by which, when the principles are once established, the mathematical operation of deducing their consequences is made more and more general and symmetrical: which we have seen already exemplified in the history of celestial mechanics after the time of Newton. It does not enter into our plan, to dwell upon the various steps in this way {331} made by Levy, Naumann, Grassmann, Kupffer, Hessel, and by Professor Miller among ourselves. I may notice that one great improvement was, the method introduced by Monteiro and Levy, of determining the laws of derivation of forces by means of the _parallelisms of edges_; which was afterwards extended so that faces were

considered as belonging to _zones_. Nor need I attempt to enumerate (what indeed it would be difficult to describe in words) the various methods of _notation_ by which it has been proposed to represent the faces of crystals, and to facilitate the calculations which have reference to them.

[Note 34\15: _Camb. Trans._ vol. ii. p. 391.]

[2nd Ed.] [My Memoir of 1825 depended on the views of Haüy in so far as that I started from his "primitive forms;" but being a general method of expressing all forms by co-ordinates, it was very little governed by these views. The mode of representing crystalline forms which I proposed seemed to contain its own evidence of being more true to nature than Haüy's theory of decrements, inasmuch as my method expressed the faces at much lower numbers. I determine a face by means of the dimensions of the primary form _divided_ by certain numbers; Haüy had expressed the face virtually by the same dimensions _multiplied_ by numbers. In cases where my notation gives such numbers as (3, 4, 1), (1, 3, 7), (5, 1, 19), his method involves the higher numbers (4, 3, 12), (21, 7, 3), (19, 95, 5). My method however has, I believe, little value as a method of "_calculating_ the angles of crystals."

M. Neumann, of Königsberg, introduced a very convenient and elegant mode of representing the position of faces of crystals by corresponding points on the surface of a circumscribing sphere. He gave (in 1823) the laws of the derivation of crystalline faces, expressed geometrically by the intersection of zones, (_Beiträge zur Krystallonomie_.) The same method of indicating the position of faces of crystals was afterwards, together with the notation, re-invented by M. Grassmann, (_Zur Krystallonomie und Geometrischen Combinationslehre_, 1829.) Aiding himself by the suggestions of these writers, and partly adopting my method, Prof. Miller has produced a work on Crystallography remarkable for mathematical elegance and symmetry; and has given expressions really useful for calculating the angles of crystalline faces, (_A Treatise on Crystallography_. Cambridge, 1839.)]

_Confirmation of the Distinction of Systems by the Optical Properties of Minerals.--Brewster._--I must not omit to notice the striking confirmation which the distinction of systems of crystallization received from optical discoveries, especially those of Sir D. Brewster. Of the {332} history of this very rich and beautiful department of science, we have already given some account, in speaking of Optics. The first facts which were noticed, those relating to double refraction, belonged exclusively to crystals of the rhombohedral system. The splendid phenomena of the rings and lemniscates produced by dipolarizing crystals, were afterwards discovered; and these were, in 1817, classified by Sir David Brewster, according to the crystalline forms to which they belong. This classification, on comparison with the distinction of Systems of Crystallization, resolved itself into a necessary relation of mathematical symmetry: all crystals of the pyramidal and rhombohedral systems, which from their geometrical character have a single axis of symmetry, are also optically uniaxal, and produce by dipolarization circular rings; while the prismatic system, which has no such single axis, but three unequal axes of symmetry, is optically biaxal, gives lemniscates by dipolarized light, and according to Fresnel's theory, has three rectangular axes of unequal elasticity.

[2nd Ed.] [I have placed Sir David Brewster's arrangement of crystalline forms in this chapter, as an event belonging to the _confirmation_ of the distinctions of forms introduced by Weiss and Mohs; because that arrangement was established, not on crystallographical, but on optical grounds. But Sir David Brewster's optical discovery was a much greater step in science than the systems of the two German crystallographers; and even in respect to the crystallographical principle, Sir D. Brewster had an independent share in the discovery. He divided crystalline forms into three classes, enumerating the Hauïan "primitive forms" which belonged to each; and as he found some exceptions to this classification, (such as idocrase, &c.,) he ventured to pronounce that in those substances the received primitive forms were probably erroneous; a judgment which was soon confirmed by a closer crystallographical scrutiny. He also showed his perception of the mineralogical importance of his discovery by publishing it, not only in the _Phil. Trans._ (1818), but also in the _Transactions of the Wernerian Society of Natural History_. In a second paper inserted in this later series, read in 1820, he further notices Mohs's System of Crystallography, which had then recently appeared, and points out its agreement with his own.

Another reason why I do not make his great optical discovery a cardinal point in the history of crystallography is, that as a crystallographical system it is incomplete. Although we are thus led to distinguish the _tessular_ and the _prismatic_ systems (using Mohs's terms) {333} from the _rhombohedral_ and the _square prismatic_, we are not led to distinguish the latter two from each other; inasmuch as they have no optical difference of character. But this distinction is quite essential in crystallography; for these two systems have faces formed by laws as different as those of the other two systems.

Moreover, Weiss and Mohs not only divided crystalline forms into certain classes, but showed that by doing this, the derivation of all the existing forms from the fundamental ones assumed a new aspect of simplicity and generality; and this was the essential part of what they did.

On the other hand, I do not think it is too much to say as I have elsewhere said[35\15] that "Sir D. Brewster's optical experiments must have led to a classification of crystals into the above systems, or something nearly equivalent, even if crystals had not been so arranged by attention to their forms."]

[Note 35\15: _Philosophy of the Inductive Sciences_, B. viii. C. iii. Art. 3.]

Many other most curious trains of research have confirmed the general truth, that the degree and kind of geometrical symmetry corresponds exactly with the symmetry of the optical properties. As an instance of this, eminently striking for its singularity, we may notice the discovery of Sir John Herschel, that the _plagihedral_ crystallization of quartz, by which it exhibits faces _twisted_ to the right or the left, is accompanied by right-handed or left-handed circular polarization respectively. No one acquainted with the subject can now doubt, that the correspondence of geometrical and optical symmetry is of the most complete and fundamental kind.

[2nd Ed.] [Our knowledge with respect to the positions of the optical axes of the oblique prismatic crystals is still imperfect. It appears to be ascertained that, in singly oblique crystals, one of the axes of optical elasticity coincides with the rectangular crystallographic axis. In doubly oblique crystals, one of the axes of optical elasticity is, in many cases, coincident with the axis of a principal zone. I believe no more determinate laws have been discovered.]

Thus the highest generalization at which mathematical crystallographers have yet arrived, may be considered as fully established; and the science of Crystallography, in the condition in which these place it, is fit to be employed as one of the members of Mineralogy, and thus to fill its appropriate place and office. {334}

## CHAPTER VI.

CORRECTION OF THE LAW OF THE SAME ANGLE FOR THE SAME SUBSTANCE.

DISCOVERY OF ISOMORPHISM. MITSCHERLICH.--The discovery of which we now have to speak may appear at first sight too large to be included in the history of crystallography, and may seem to belong rather to chemistry. But it is to be recollected that crystallography, from the time of its first assuming importance in the hands of Haüy, founded its claim to notice entirely upon its connexion with chemistry; crystalline forms were properties of _something_; but _what_ that something was, and how it might be modified without becoming something else, no crystallographer could venture to decide, without the aid of chemical analysis. Haüy had assumed, as the general result of his researches, that the same chemical elements, combined in the same proportions, would always exhibit the same crystalline form; and reciprocally, that the same form and angles (except in the obvious case of the tessular system, in which the angles are determined by its _being_ the tessular system,) implied the same chemical constitution. But this dogma could only be considered as an approximate conjecture; for there were many glaring and unexplained exceptions to it. The explanation of several of these was beautifully described by the discovery that there are various elements which are _isomorphous_ to each other; that is, such that one may take the place of another without altering the crystalline form; and thus the chemical composition may be much changed, while the crystallographic character is undisturbed.

This truth had been caught sight of, probably as a guess only, by Fuchs as early as 1815. In speaking of a mineral which had been called Gehlenite, he says, "I hold the oxide of iron, not for an essential component part of this genus, but only as a _vicarious_ element, replacing so much lime. We shall find it necessary to consider the results of several analyses of mineral bodies in this point of view, if we wish, on the one hand, to bring them into agreement with the doctrine of chemical proportions, and on the other, to avoid unnecessarily splitting up genera." In a lecture _On the Mutual Influence of_ {335} _Chemistry and Mineralogy_,[36\15] he again draws attention to his term _vicarious_ (_vicarirende_), which undoubtedly expresses the nature of the general law afterwards established by Mitscherlich in 1822.

[Note 36\15: Munich, 1820.]

But Fuchs's conjectural expression was only a prelude to Mitscherlich's experimental discovery of isomorphism. Till many careful analyses had given substance and signification to this conception of vicarious elements, it was of small value. Perhaps no one was more capable than Berzelius of turning to the best advantage any ideas which were current in the chemical world; yet we find him,[37\15] in 1820, dwelling upon a certain vague view of these cases,--that "oxides which contain equal doses of oxygen must have their general properties common;" without tracing it to any definite conclusions. But his scholar, Mitscherlich, gave this proposition a real crystallographical import. Thus he found that the carbonates of lime (calcspar,) of magnesia, of protoxide of iron, and of protoxide of manganese, agree in many respects of form, while the homologous angles vary through one or two degrees only; so again the carbonates of baryta, strontia, lead, and lime (arragonite), agree nearly; the different kinds of felspar vary only by the substitution of one alkali for another; the phosphates are almost identical with the arseniates of several bases. These, and similar results, were expressed by saying that, in such cases, the bases, lime, protoxide of iron, and the rest, are _isomorphous_; or in the latter instance, that the arsenic and phosphoric acids are isomorphous.

[Note 37\15: _Essay on the Theory of Chemical Proportions_, p. 122.]

Since, in some of these cases, the substitution of one element of the isomorphous group for another does alter the angle, though slightly, it has since been proposed to call such groups _plesiomorphous_.

This discovery of isomorphism was of great importance, and excited much attention among the chemists of Europe. The history of its reception, however, belongs, in part, to the classification of minerals; for its effect was immediately to metamorphose the existing chemical systems of arrangement. But even those crystallographers and chemists who cared little for general systems of classification, received a powerful impulse by the expectation, which was now excited, of discovering definite laws connecting chemical constitution with crystalline form. Such investigations were soon carried on with great activity. Thus, at a recent period, Abich analysed a number of tessular minerals, spinelle, pleonaste, gahnite, franklinite, and chromic iron oxide; and {336} seems to have had some success in **giving a common type to their chemical formulæ, as there is a common type in their crystallization.

[2nd Ed.] [It will be seen by the above account that Prof. Mitscherlich's merit in the great discovery of Isomorphism is not at all narrowed by the previous conjectures of M. Fuchs. I am informed, moreover, that M. Fuchs afterwards (in Schweigger's _Journal_) retracted the opinions he had put forward on this subject.]

_Dimorphism._--My business is, to point out the connected truths which have been obtained by philosophers, rather than insulated difficulties which still stand out to perplex them. I need not, therefore, dwell on the curious cases of _dimorphism_; cases in which the same definite chemical compound of the same elements appears to have two different forms; thus the carbonate of lime has two forms, _calcspar_ and

_arragonite_, which belong to different systems of crystallization. Such facts may puzzle us; but they hardly interfere with any received general truths, because we have as yet no truths of very high order respecting the connexion of chemical constitution and crystalline form. Dimorphism does not interfere with isomorphism; the two classes of facts stand at the same stage of inductive generalization, and we wait for some higher truth which shall include both, and rise above them.

[2nd Ed.] [For additions to our knowledge of the Dimorphism of Bodies, see Professor Johnstone's valuable _Report_ on that subject in the _Reports of the British Association_ for 1837. Substances have also been found which are _trimorphous_. We owe to Professor Mitscherlich the discovery of dimorphism, as well as of isomorphism: and to him also we owe the greater part of the knowledge to which these discoveries have led.]

## CHAPTER VII.

ATTEMPTS TO ESTABLISH THE FIXITY OF OTHER PHYSICAL PROPERTIES.--WERNER.

THE reflections from which it appeared, (at the end of the last Book,) that in order to obtain general knowledge respecting bodies, we must give scientific fixity to our appreciation of their properties, applies to their other properties as well as to their crystalline {337} form. And though none of the other properties have yet been referred to standards so definite as that which geometry supplies for crystals, a system has been introduced which makes their measures far more constant and precise than they are to a common undisciplined sense.

The author of this system was Abraham Gottlob Werner, who had been educated in the institutions which the Elector of Saxony had established at the mines of Freiberg. Of an exact and methodical intellect, and of great acuteness of the senses, Werner was well fitted for the task of giving fixity to the appreciation of outward impressions; and this he attempted in his _Dissertation on the external Characters of Fossils_, which was published at Leipzig in 1774. Of the precision of his estimation of such characters, we may judge from the following story, told by his biographer Frisch.[38\15] One of his companions had received a quantity of pieces of amber, and was relating to Werner, then very young, that he had found in the lot one piece from which he could extract no signs of electricity. Werner requested to be allowed to put his hand in the bag which contained these pieces, and immediately drew out the unelectrical piece. It was yellow chalcedony, which is distinguishable from amber by its weight and coldness.

[Note 38\15: _Werner's Leben_, p. 26.]

The principal external characters which were subjected by Werner to a systematic examination were color, lustre, hardness, and specific gravity. His subdivisions of the first character (_Color_), were very numerous; yet it cannot be doubted that if we recollect them by the eye, and not by their names, they are definite and valuable characters, and especially the metallic colors. Breithaupt, merely by the aid of this

character, distinguished two new compounds among the small grains found along with the grains of platinum, and usually confounded with them. The kinds of _Lustre_, namely, _glassy_, _fatty_, _adamantine_, _metallic_, are, when used in the same manner, equally valuable. _Specific Gravity_ obviously admits of a numerical measure; and the _Hardness_ of a mineral was pretty exactly defined by the substances which it would scratch, and by which it was capable of being scratched.

Werner soon acquired a reputation as a mineralogist, which drew persons from every part of Europe to Freiberg in order to hear his lectures; and thus diffused very widely his mode of employing external characters. It was, indeed, impossible to attend so closely to {338} these characters as the Wernerian method required, without finding that they were more distinctive than might at first sight be imagined; and the analogy which this mode of studying Mineralogy established between that and other branches of Natural History, recommended the method to those in whom a general inclination to such studies was excited. Thus Professor Jameson of Edinburgh, who had been one of the pupils of Werner at Freiberg, not only published works in which he promulgated the mineralogical doctrines of his master, but established in Edinburgh a "Wernerian Society," having for its object the general cultivation of Natural History.

Werner's standards and nomenclature of external characters were somewhat modified by Mohs, who, with the same kinds of talents and views, succeeded him at Freiberg. Mohs reduced hardness to numerical measure by selecting ten known minerals, each harder than the other in order, from _talc_ to _corundum_ and _diamond_, and by making the place which these minerals occupy in the list, the numerical measure of the hardness of those which are compared with them. The result of the application of this fixed measurement and nomenclature of external characters will appear in the History of Classification, to which we now proceed.

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## CHAPTER VIII.

ATTEMPTS AT THE CLASSIFICATION OF MINERALS.

_Sect._ 1.--_Proper object of Classification._

THE fixity of the crystalline and other physical properties of minerals is turned to account by being made the means of classifying such objects. To use the language of Aristotle,[39\15] Classification is the _architectonic_ science, to which Crystallography and the Doctrine of External Characters are subordinate and ministerial, as the art of the bricklayer and carpenter are to that of the architect. But classification itself is useful only as subservient to an ulterior science, which shall furnish us with knowledge concerning things so classified. To classify is to divide and to name; and the value of the Divisions which we thus make, and of the names which we give them, is this;--that they render exact knowledge and general propositions possible. Now the knowledge which we principally seek concerning minerals is a knowledge of their chemical composition; the general propositions to which we hope

to be led are such as assert relations between their intimate constitution and their external attributes. Thus our Mineralogical Classification must always have an eye turned towards Chemistry. We cannot get rid of the fundamental conviction, that the elementary composition of bodies, since it fixes their essence, must determine their properties. Hence all mineralogical arrangements, whether they profess it or not, must be, in effect, chemical; they must have it for their object to bring into view a set of relations, which, whatever else they may be, are at least chemical relations. We may begin with the outside, but it is only in order to reach the inner {340} structure. We may classify without reference to chemistry; but if we do so, it is only that we may assert chemical propositions with reference to our classification.

[Note 39\15: _Eth. Nicom._ i. 2.]

But, as we have already attempted to show, we not only may, but we _must_ classify, by other than chemical characters, in order to be able to make our classification the basis of chemical knowledge. In order to assert chemical truths concerning bodies, we must have the bodies known by some tests not chemical. The chemist cannot assert that Arragonite does or does not contain Strontia, except the mineralogist can tell him whether any given specimen is or is not _Arragonite_. If chemistry be called upon to supply the _definitions_ as well as the _doctrines_ of mineralogy, the science can only consist of identical propositions.

Yet chemistry has been much employed in mineralogical classifications, and, it is generally believed, with advantage to the science: How is this consistent with what has been said?

To this the answer is, that when this _has_ been done with advantage, the authority of external characters, as well as of chemical constitution, has really been brought into play. We have two sets of properties to compare, chemical and physical; to exhibit the connexion of these is the object of scientific mineralogy. And though this connexion would be most distinctly asserted, if we could keep the two sets of properties distinct, yet it may be brought into view in a great degree, by classifications in which both are referred to as guides. Since the governing principle of the attempts at classification is the conviction that the chemical constitution and the physical properties have a definite relation to each other, we appear entitled to use both kinds of evidence, in proportion as we can best obtain each; and then the general consistency and convenience of our system will be the security for its containing substantial knowledge, though this be not presented in a rigorously logical or systematic form.

Such _mixed systems_ of classification, resting partly on chemical and partly on physical characters, naturally appeared as the earliest attempts in this way, before the two members of the subject had been clearly separated in men's minds; and these systems, therefore, we must first give an account of.

_Sect._ 2.--_Mixed Systems of Classification._

_Early Systems._--The first attempts at classifying minerals went upon the ground of those differences of general aspect which had been {341} recognized in the formation of common language; as _earths_, _stones_, _metals_. But such

arrangements were manifestly vague and confused; and when chemistry had advanced to power and honor, her aid was naturally called in to introduce a better order. "Hiarne and Bromell were, as far as I know," says[40\15] Cronstedt, "the first who founded any mineral system upon chemical principles; to them we owe the three known divisions of the most simple mineral bodies; viz., the _calcarei_, _vitrescentes_, and _apyri_." But Cronstedt's own _Essay towards a System of Mineralogy_, published in Swedish in 1758, had perhaps more influence than any other, upon succeeding systems. In this, the distinction of earths and stones, and also of vitrescent and non-vitrescent earths (_apyri_), is rejected. The earths are classed as _calcareous_, _siliceous_, _argillaceous_, and the like. Again, calcareous earth is pure (_calc spar_), or united with acid of vitriol (_gypsum_), or united with the muriatic add (_sal ammoniac_), and the like. It is easy to see that this is the method, which, in its general principle, has been continued to our own time. In such methods, it is supposed that we can recognize the substance by its general appearance, and on this assumption, its place in the system conveys to us chemical knowledge concerning it.

[Note 40\15: _Mineralogy_, Pref. p. viii.]

But as the other branches of Natural History, and especially Botany, assumed a systematic form, many mineralogists became dissatisfied with this casual and superficial mode of taking account of external characters; they became convinced, that in Mineralogy as in other sciences, classification must have its system and its rules. The views which Werner ascribes to his teacher, Pabst van Ohain,[41\15] show the rise of those opinions which led through Werner to Mohs: "He was of opinion that a natural mineral system must be constructed by chemical determinations, and external characters at the same time (_methodus mixta_); but that along with this, mineralogists ought also to construct and employ what he called an _artificial system_, which might serve us as a guide (_loco indicis_) how to introduce newly-discovered fossils into the system, and how to find easily and quickly those already known and introduced." Such an artificial system, containing not the grounds of classification, but marks for recognition, was afterwards attempted by Mohs, and termed by him the _Characteristic_ of his system.

[Note 41\15: Frisch. _Werner's Leben_, p. 15.]

_Werner's System._--But, in the mean time, Werner's classification had an extensive reign, and this was still a mixed system. Werner himself, indeed, never published a system of mineralogy. "We might {342} almost imagine," Cuvier says,[42\15] "that when he had produced his nomenclature of external characters, he was affrighted with his own creation; and that the reason of his writing so little after his first essay, was to avoid the shackles which he had imposed upon others." His system was, indeed, made known both in and out of Germany, by his pupils; but in consequence of Werner's unwillingness to give it on his own authority, it assumed, in its published forms, the appearance of an extorted secret imperfectly told. A _Notice of the Mineralogical Cabinet of Mine-Director Pabst von Ohain_, was, in 1792, published by Karsten and Hoffman, under Werner's direction; and conveyed by example, his views of mineralogical arrangement; and[43\15] in 1816 his _Doctrine of Classification_ was surreptitiously copied from his manuscript, and published in a

German Journal, termed _The Hesperus_. But it was only in 1817, after his death, that there appeared _Werner's Last Mineral System_, edited from his papers by Breithaupt and Köhler: and by this time, as we shall soon see, other systems were coming forwards on the stage.

[Note 42\15: Cuv. _El._ ii. 314.]

[Note 43\15: Frisch. p. 52.]

A very slight notice of Werner's arrangement will suffice to show that it was, as we have termed it, a Mixed System. He makes four great Classes of fossils, _Earthy_, _Saline_, _Combustible_, _Metallic_: the earthy fossils are in eight Genera--Diamond, Zircon, Silica, Alumina, Talc, Lime, Baryta, Hallites. It is clear that these genera are in the main chemical, for chemistry alone can definitely distinguish the different Earths which characterize them. Yet the Wernerian arrangement supposed the distinctions to be practically made by reference to those external characters which the teacher himself could employ with such surpassing skill. And though it cannot be doubted, that the chemical views which prevailed around him had a latent influence on his classification in some cases, he resolutely refused to bend his system to the authority of chemistry. Thus,[44\15] when he was blamed for having, in opposition to the chemists, placed diamond among the earthy fossils, he persisted in declaring that, mineralogically considered, it was a stone, and could not be treated as anything else.

[Note 44\15: Frisch. p. 62.]

This was an indication to that tendency, which, under his successor, led to a complete separation of the two grounds of classification. But before we proceed to this, we must notice what was doing at this period in other parts of Europe.

_Haüy's System._--Though Werner, on his own principles, ought to {343} have been the first person to see the immense value of the most marked of external characters, crystalline form, he did not, in fact, attach much importance to it. Perhaps he was in some measure fascinated by a fondness for those characters which he had himself systematized, and the study of which did not direct him to look for geometrical relations. However this may be, the glory of giving to Crystallography its just importance in Mineralogy is due to France: and the Treatise of Haüy, published in 1801, is the basis of the best succeeding works of mineralogy. In this work, the arrangement is professedly chemical; and the classification thus established is employed as the means of enunciating crystallographic and other properties. "The principal object of this Treatise," says the author,[45\15] "is the exposition and development of a method founded on certain principles, which may serve as a frame-work for all the knowledge which Mineralogy can supply, aided by the different sciences which can join hands with her and march on the same line.**" It is worthy of notice, as characteristic of this period of Mixed Systems, that the classification of Haüy, though founded on principles so different from the Wernerian ones, deviates little from it in the general character of the divisions. Thus, the first Order of the first Class of Haüy is _Acidiferous Earthy Substances_; the first genus is _Lime_; the species are, _Carbonate of Lime_, _Phosphate of Lime_, _Fluate of Lime_, _Sulphate of Lime_, and so on.

[Note 45\15: Disc. Prél. p. xvii.]

_Other Systems._--Such mixed methods were introduced also into this country, and have prevailed, we may say, up to the present time. The _Mineralogy_ of William Phillips, which was published in 1824, and which was an extraordinary treasure of crystallographic facts, was arranged by such a mixed system; that is, by a system professedly chemical; but, inasmuch as a rigid chemical system is impossible, and the assumption of such a one leads into glaring absurdities, the system was, in this and other attempts of the same kind, corrected by the most arbitrary and lax application of other considerations.

It is a curious example of the difference of national intellectual character, that the manifest inconsistencies of the prevalent systems, which led in Germany, as we shall see, to bold and sweeping attempts at reform, produced in England a sort of contemptuous despair with regard to systems in general;--a belief that no system could be consistent or useful;--and a persuasion that the only valuable knowledge is the accumulation of particular facts. This is not the place to {344} explain how erroneous and unphilosophical such an opinion is. But we may notice that while such a temper prevails among us, our place in this science can never be found in advance of that position which we are now considering as exemplified in the period of Werner and Haüy. So long as we entertain such views respecting the objects of Mineralogy, we can have no share in the fortunes of the succeeding period of its history, to which I now proceed.

## CHAPTER IX.

ATTEMPTS AT THE REFORM OF MINERALOGICAL SYSTEMS.--SEPARATION OF THE CHEMICAL AND NATURAL HISTORY METHODS.

_Sect._ 1.--_Natural History System of Mohs._

THE chemical principle of classification, if pursued at random, as in the cases just spoken of leads to results at which a philosophical spirit revolts; it separates widely substances which are not distinguishable; joins together bodies the most dissimilar; and in hardly any instance does it bring any truth into view. The vices of classifications like that of Haüy could not long be concealed; but even before time had exposed the weakness of his system, Haüy himself had pointed out, clearly and without reserve,[46\15] that a chemical system is only one side of the subject, and supposes, as its counterpart, a science of external characters. In the mean time, the Wernerians were becoming more and more in love with the form which they had given to such a science. Indeed, the expertness which Werner and his scholars acquired in the use of external characters, justified some partiality for them. It is related of him,[47\15] that, by looking at a piece of iron-ore, and poising it in his hand, he was able to tell, almost precisely, the proportion of pure metal which it contained. And in the last year of his life,[48\15] he had marked out, as the employment of the ensuing winter, the study of the system of Berzelius, with a view to find out the laws of combination as disclosed by external characters. In the same spirit, his pupil {345} Breithaupt[49\15] attempted to discover the ingredients of

minerals by their peculiarities of crystallization. The persuasion that there must be _some_ connexion between composition and properties, transformed itself, in their minds, into a belief that they could seize the nature of the connexion by a sort of instinct.

[Note 46\15: See his Disc. Prél.]

[Note 47\15: Frisch. _Werner's Leben_, p. 78.]

[Note 48\15: Frisch. 3.]

[Note 49\15: _Dresdn. Auswahl_, vol. ii. p. 97.]

This opinion of the independency of the science of external characters, and of its sufficiency for its own object, at last assumed its complete form in the bold attempt to construct a system which should borrow nothing from chemistry. This attempt was made by Frederick Mohs, who had been the pupil of Werner, and was afterwards his successor in the school of Freiberg; and who, by the acute and methodical character of his intellect, and by his intimate knowledge of minerals, was worthy of his predecessor. Rejecting altogether all divisions of which the import was chemical, Mohs turned for guidance, or at least for the light of analogy, to botany. His object was to construct a _Natural System_ of mineralogy. What the conditions and advantages of a natural system of any province of nature are, we must delay to explain till we have before us, in botany, a more luminous example of such a scheme. But further; in mineralogy, as in botany, besides the Natural System, by which we _form_ our classes, it is necessary to have an _Artificial System_ by which we _recognize_ them;--a principle which, we have seen, had already taken root in the school of Freiberg. Such an artificial system Mohs produced in his _Characteristic of the Mineral Kingdom_, which was published at Dresden in 1820; and which, though extending only to a few pages, excited a strong interest in Germany, where men's minds were prepared to interpret the full import of such a work. Some of the traits of such a "Characteristic" had, indeed, been previously drawn by others; as for example, by Haüy, who notices that each of his Classes has peculiar characters. For instance, his First Class (acidiferous substances,) alone possesses these combinations of properties; "division into a regular octohedron, without being able to scratch glass; specific gravity above 3·5, without being able to scratch glass." The extension of such characters into a scheme which should exhaust the whole mineral kingdom, was the undertaking of Mohs.

Such a collection of marks of classes, implied a classification previously established, and accordingly, Mohs had created his own mineral system. His aim was to construct it, as we shall hereafter see that other natural systems are constructed, by taking into account _all_ the {346} resemblances and differences of the objects classified. It is obvious that to execute such a work, implied a most intimate and universal acquaintance with minerals;--a power of combining in one vivid survey the whole mineral kingdom. To illustrate the spirit in which Professor Mohs performed his task, I hope I may be allowed to refer to my own intercourse with him. At an early period of my mineralogical studies, when the very conception of a Natural System was new to me, he, with great kindliness of temper, allowed me habitually to propose

to him the scruples which arose in my mind, before I could admit principles which appeared to me then so vague and indefinite; and answered my objections with great patience and most instructive clearness. Among other difficulties, I one day propounded to him this;--"You have published a Treatise on Mineralogy, in which you have described _all_ the important properties of all known minerals. On your principles, then, it ought to be possible, merely by knowing the descriptions in your book, and without seeing any minerals, to construct a natural system; and this natural system ought to turn out identical with that which you have produced, by so careful an examination of the minerals themselves." He pondered a moment, and then he answered, "It is true; but what an enormous _imagination_ (_einbildungskraft_, _power of inward imagining_), a man must have for such a work!" Vividness of conception of sensible properties, and the steady intuition (_anschauung_) of objects, were deemed by him, and by the Wernerian school in general, to be the most essential conditions of complete knowledge.

It is not necessary to describe Mohs's system in detail; it may sufficiently indicate its form to state that the following substances, such as I before gave as examples of other arrangements, calcspar, gypsum, fluor spar, apatite, heavy spar, are by Mohs termed respectively, _Rhombohedral Lime Haloide_, _Gyps Haloide_, _Octohedral Fluor Haloide_, _Rhombohedral Fluor Haloide_, _Prismatic Hal Baryte_. These substances are thus referred to the _Orders_ Haloide, and Baryte; to _Genera_ Lime Haloide, Fluor Haloide, Hal Baryte; and the _Species_ is an additional particularization.

Mohs not only aimed at framing such a system, but was also ambitious of giving to all minerals _Names_ which should accord with the system. This design was too bold to succeed. It is true, that a new nomenclature was much needed in mineralogy: it is true, too, that it was reasonable to expect, from an improved classification, an improved nomenclature, such as had been so happily obtained in botany by the {347} reform of Linnæus. But besides the defects of Mohs's system, he had not prepared his verbal novelties with the temperance and skill of the great botanical reformer. He called upon mineralogists to change the name of almost every mineral with which they were acquainted; and the proposed appellations were mostly of a cumbrous form, as the above example may serve to show. Such names could have obtained general currency, only after a general and complete acceptance of the system; and the system did not possess, in a sufficient degree, that evidence which alone could gain it a home in the belief of philosophers,--the coincidence of its results with those of Chemistry. But before I speak finally of the fortunes of the Natural-history System, I will say something of the other attempt which was made about the same time to introduce a Reform into Mineralogy from the opposite extremity of the science.

_Sect._ 2.--_Chemical System of Berzelius and others._

IF the students of external characters were satisfied of the independence of their method, the chemical analysts were naturally no less confident of the legitimate supremacy of their principles: and when the beginning of the present century had been distinguished by the establishment of the theory of definite proportions, and by discoveries which pointed to the electro-chemical theory, it could not appear

presumption to suppose, that the classification of bodies, so far as it depended on chemistry, might be presented in a form more complete and scientific than at any previous time.

The attempt to do this was made by the great Swedish chemist Jacob Berzelius. In 1816, he published his _Essay to establish a purely Scientific System of Mineralogy, by means of the Application of the Electro-chemical Theory and the Chemical Doctrine of Definite Proportions_. It is manifest that, for minerals which are constituted by the law of Definite Proportions, this constitution must be a most essential part of their character. The electro-chemical theory was called in aid, in addition to the composition, because, distinguishing the elements of all compounds as electro-positive and electro-negative, and giving to every element a place in a series, and a place defined by the degree of these relations, it seemed to afford a rigorous and complete principle of arrangement. Accordingly, Berzelius, in his First System, arranged minerals according to their electro-positive element, and the elements according to their electro-positive rank; {348} and supposed that he had thus removed all that was arbitrary and vague in the previous chemical systems of mineralogy.

Though the attempt appeared so well justified by the state of chemical science, and was so plausible in its principle, it was not long before events showed that there was some fallacy in these specious appearances. In 1820, Mitscherlich discovered Isomorphism: by that discovery it appeared that bodies containing very different electro-positive elements could not be distinguished from each other; it was impossible, therefore, to put them in distant portions of the classification;--and thus the first system of Berzelius crumbled to pieces.

But Berzelius did not so easily resign his project. With the most unhesitating confession of his first failure, but with undaunted courage, he again girded himself to the task of rebuilding his edifice. Defeated at the electro-positive position, he now resolved to make a stand at the electro-negative element. In 1824, he published in the Transactions of the Swedish Academy, a Memoir _On the Alterations in the Chemical Mineral System, which necessarily follow from the Property exhibited by Isomorphous Bodies, of replacing each other in given Proportions_. The alteration was, in fact, an inversion of the system, with an attempt still to preserve the electro-chemical principle of arrangement. Thus, instead of arranging metallic minerals according to the _metal_, under iron, copper, &c., all the _sulphurets_ were classed together, all the _oxides_ together, all the _sulphates_ together, and so in other respects. That such an order was a great improvement on the preceding one, cannot be doubted; but we shall see, I think, that as a strict scientific system it was not successful. The discovery of isomorphism, however, naturally led to such attempts. Thus Gmelin also, in 1825, published a mineral system,[50\15] which, like that of Berzelius, founded its leading distinctions on the electro-negative, or, as it was sometimes termed, the _formative_ element of bodies; and, besides this, took account of the _numbers_ of atoms or proportions which appear in the composition of the body; distinguishing, for instance, Silicates, as simple silicates, double silicates, and so on, to _quintuple_ silicate (_Pechstein_) and _sextuple_ silicate (_Perlstein_). In like

manner, Nordenskiöld devised a system resting on the same bases, taking into account also the crystalline form. In 1824, Beudant published his _Traité Elémentaire de Minéralogie_, in which he professes to found his arrangement on the electro-negative element, and on Ampère's circular {349} arrangement of elementary substances. Such schemes exhibit rather a play of the mere logical faculty, exercising itself on assumed principles, than any attempt at the real interpretation of nature. Other such pure chemical systems may have been published, but it is not necessary to accumulate instances. I proceed to consider their result.

[Note 50\15: _Zeitsch. der Min._ 1825, p. 435.]

_Sect._ 3.--_Failure of the Attempts at Systematic Reform._

IT may appear presumptuous to speak of the failure of those whom, like Berzelius and Mohs, we acknowledge as our masters, at a period when, probably, they and some of their admirers still hold them to have succeeded in their attempt to construct a consistent system. But I conceive that my office as an historian requires me to exhibit the fortunes of this science in the most distinct form of which they admit, and that I cannot evade the duty of attempting to seize the true aspect of recent occurrences in the world of science. Hence I venture to speak of the failure of both the attempts at framing a pure scientific system of mineralogy,--that founded on the chemical, and that founded on the natural-history principle; because it is clear that they have not obtained that which alone we could, according to the views here presented, consider as success,--a coincidence of each with the other. A Chemical System of arrangement, which should bring together, in all cases, the substances which come nearest each other in external properties;--a Natural-history System, which should be found to arrange bodies in complete accordance with their chemical constitution:--if such systems existed, they might, with justice, claim to have succeeded. Their agreement would be their verification. The interior and exterior system are the type and the antitype, and their entire correspondence would establish the mode of interpretation beyond doubt. But nothing less than this will satisfy the requisitions of science. And when, therefore, the chemical and the natural-history system, though evidently, as I conceive, tending towards each other, are still far from coming together, it is impossible to allow that either method has been successful in regard to its proper object.

But we may, I think, point out the fallacy of the principles, as well as the imperfection of the results, of both of those methods. With regard to that of Berzelius, indeed, the history of the subject obviously betrays its unsoundness. The electro-positive principle was, in a very short time after its adoption, proved and acknowledged to be utterly untenable: what security have we that the electro-negative element is {350} more trustworthy? Was not the necessity of an entire change of system, a proof that the ground, whatever that was, on which the electro-chemical principle was adopted, was an unfounded assumption? And, in fact, do we not find that the same argument which was allowed to be fatal to the First System of Berzelius, applies in exactly the same manner against the Second? If the electro-positive elements be often isomorphous, are not the electro-negative elements sometimes isomorphous also? for instance, the arsenic and phosphoric acids. But to

go further, what _is_ the ground on which the electro-chemical arrangement is adopted? Granted that the electrical relations of bodies are important; but how do we come to know that these relations have anything to do with mineralogy? How does it appear that on them, principally, depend those external properties which mineralogy must study? How does it appear that because sulphur is the electro-negative part of one body, and an acid the electro-negative part of another, these two elements similarly affect the compounds? How does it appear that there is any analogy whatever in their functions? We allow that the composition must, in _some way_, determine the classified place of the mineral,--but why in _this_ way?

I do not dwell on the remark which Berzelius himself[51\15] makes on Nordenskiöld's system;--that it assumes a perfect knowledge of the composition in every case; although, considering the usual discrepancies of analyses of minerals, this objection must make all pure chemical systems useless. But I may observe, that mineralogists have not yet determined what characters are sufficiently affixed to determine a species of minerals. We have seen that the ancient notion of the composition of a species, has been unsettled by the discovery of isomorphism. The tenet of the constancy of the angle is rendered doubtful by cases of plesiomorphism. The optical properties, which are so closely connected with the crystalline, are still so imperfectly known, that they are subject to changes which appear capricious and arbitrary. Both the chemical and the optical mineralogists have constantly, of late, found occasion to separate species which had been united, and to bring together those which had been divided. Everything shows that, in this science, we have our classification still to begin. The detection of that fixity of characters, on which a right establishment of species must rest, is not yet complete, great as the progress is which we have made, by acquiring a knowledge of the laws of crystallization and of {351} definite chemical constitution. Our ignorance may surprise us; but it may diminish our surprise to recollect, that the knowledge which we seek is that of the laws of the physical constitution of all bodies whatever; for to us, as mineralogists, all chemical compounds are minerals.

[Note 51\15: _Jahres Bericht._ viii. 188.]

The defect of the principle of the natural-history classifiers may be thus stated:--in studying the external characters of bodies, they take for granted that they can, without any other light, discover the relative value and importance of those characters. The grouping of Species into a Genus, of Genera into an Order, according to the method of this school, proceeds by no definite rules, but by a latent talent of appreciation,--a sort of classifying instinct. But this course cannot reasonably be expected to lead to scientific truth; for it can hardly be hoped, by any one who looks at the general course of science, that we shall discover the relation between external characters and chemical composition, otherwise than by tracing their association in cases where both are known. It is urged that in other classificatory sciences, in botany, for example, we obtain a natural classification from external characters without having recourse to any other source of knowledge. But this is not true in the sense here meant. In framing a natural system of botany, we have constantly before our eyes the principles of physiology; and we estimate the value of the characters of a

plant by their bearing on its functions,--by their place in its organization. In an unorganic body, the chemical constitution is the law of its being; and we shall never succeed in framing a science of such bodies but by studiously directing our efforts to the interpretation of that law.

On these grounds, then, I conceive, that the bold attempts of Mohs and of Berzelius to give new forms to mineralogy, cannot be deemed successful in the manner in which their authors aspired to succeed. Neither of them can be marked as a permanent reformation of the science. I shall not inquire how far they have been accepted by men of science, for I conceive that their greatest effect has been to point out improvements which might be made in mineralogy without going the whole length either of the _pure_ chemical, or of the _pure_ natural-history system.

_Sect._ 4.--_Return to Mixed Systems with Improvements._

IN spite of the efforts of the purists, mineralogists returned to mixed systems of classification; but these systems are much better than they were before such efforts were made. {352}

The Second System of Berzelius, though not tenable in its rigorous form, approaches far nearer than any previous system to a complete character, bringing together like substances in a large portion of its extent. The System of Mohs also, whether or not unconsciously swayed by chemical doctrines, forms orders which have a community of chemical character; thus, the minerals of the order _Haloide_ are salts of oxides, and those of the order _Pyrites_ are sulphurets of metals. Thus the two methods appear to be converging to a common centre; and though we are unable to follow either of them to this point of union, we may learn from both in what direction we are to look for it. If we regard the best of the pure systems hitherto devised as indications of the nature of that system, perfect both as a chemical and as a natural-history system, to which a more complete condition of mineralogical knowledge may lead us, we may obtain, even at present, a tolerably good approximation to a complete classification; and such a one, if we recollect that it must be imperfect, and is to be held as provisional only, may be of no small value and use to us.

The best of the mixed systems produced by this compromise again comes from Freiberg, and was published by Professor Naumann in 1828. Most of his orders have both a chemical character and great external resemblances. Thus his _Haloides_, divided into _Unmetallic_ and _Metallic_, and these again into _Hydrous_ and _Anhydrous_, give good natural groups. The most difficult minerals to arrange in all systems are the siliceous ones. These M. Naumann calls _Silicides_, and subdivides them into _Metallic_, _Unmetallic_, and _Amphoteric_ or mixed; and again, into _Hydrous_ and _Anhydrous_. Such a system is at least a good basis for future researches; and this is, as we have said, all that we can at present hope for. And when we recollect that the natural-history principle of classification has begun, as we have already seen, to make its appearance in our treatises of chemistry, we cannot doubt that some progress is making towards the object which I have pointed out. But we know not yet how far we are from the end. The combination of chemical,

crystallographical, physical and optical properties into some lofty generalization, is probably a triumph reserved for future and distant years.

_Conclusion._--The history of Mineralogy, both in its successes and by its failures, teaches us this lesson;--that in the sciences of classification, the establishment of the fixity of characters, and the discovery of such characters as are fixed, are steps of the first importance in the progress of these sciences. The recollection of this maxim may aid us in {353} shaping our course through the history of other sciences of this kind; in which, from the extent of the subject, and the mass of literature belonging to it, we might at first almost despair of casting the history into distinct epochs and periods. To the most prominent of such sciences, Botany, I now proceed.

{{355}} BOOK XVI.

# _CLASSIFICATORY SCIENCES._

HISTORY OF SYSTEMATIC BOTANY AND ZOOLOGY.

. . . . . Vatem aspicies quæ rupe sub altâ Fata canit, foliisque notas et nomina mandat. Quæcunque in foliis descripsit carmina virgo Digerit in numerum atque antro seclusa relinquit Illa manent immorta locis neque ab ordine cedunt. VIRGIL. _Æn._ iii. 443.

Behold the Sibyl!--Her who weaves a long, A tangled, full, yet sweetly flowing song. Wondrous her skill; for leaf on leaf she frames Unerring symbols and enduring names; And as her nicely measured line she binds, For leaf on leaf a fitting place she finds; Their place once found, no more the leaves depart, But fixed rest:--such is her magic art.

{{357}} INTRODUCTION.

WE now arrive at that study which offers the most copious and complete example of the sciences of classification, I mean Botany. And in this case, we have before us a branch of knowledge of which we may say, more properly than of any of the sciences which we have reviewed since Astronomy, that it has been constantly advancing, more or less rapidly, from the infancy of the human race to the present day. One of the reasons of this resemblance in the fortunes of two studies so widely dissimilar, is to be found in a simplicity of principle which they have in common; the ideas of Likeness and Difference, on which the knowledge of plants depends, are, like the ideas of Space and Time, which are the foundation of astronomy, readily apprehended with clearness and precision, even without any peculiar culture of the intellect. But another reason why, in the history of Botany, as in that of Astronomy, the progress of knowledge forms an unbroken line from the earliest times, is precisely the great difference of the kind of knowledge which has been attained in the two cases. In Astronomy, the discovery of general truths began at an early period of civilization; in Botany, it has hardly yet begun; and thus, in each of these departments of study, the lore of the ancient is homogeneous with that of the modern times, though in the one case it is science, in the other, the absence of science, which pervades all ages. The resemblance of the form of their history arises from the

diversity of their materials.

I shall not here dwell further upon this subject, but proceed to trace rapidly the progress of _Systematic Botany_, as the classificatory science is usually denominated, when it is requisite to distinguish between that and Physiological Botany. My own imperfect acquaintance with this study admonishes me not to venture into its details, further than my purpose absolutely requires. I trust that, by taking my views principally from writers who are generally allowed to possess the best insight into the science, I may be able to draw the larger features of its history with tolerable correctness; and if I succeed in this, I shall attain an object of great importance in my general scheme. {358}

## CHAPTER I.

IMAGINARY KNOWLEDGE OF PLANTS.

THE apprehension of such differences and resemblances as those by which we group together and discriminate the various kinds of plants and animals, and the appropriation of words to mark and convey the resulting notions, must be presupposed, as essential to the very beginning of human knowledge. In whatever manner we imagine man to be placed on the earth by his Creator, these processes must be conceived to be, as our Scriptures represent them, contemporaneous with the first exertion of reason, and the first use of speech. If we were to indulge ourselves in framing a hypothetical account of the origin of language, we should probably assume as the first-formed words, those which depend on the visible likeness or unlikeness of objects; and should arrange as of subsequent formation, those terms which imply, in the mind, acts of wider combination and higher abstraction. At any rate, it is certain that the names of the kinds of vegetables and animals are very abundant even in the most uncivilized stages of man's career. Thus we are informed[1\16] that the inhabitants of New Zealand have a distinct name of every tree and plant in their island, of which there are six or seven hundred or more different kinds. In the accounts of the rudest tribes, in the earliest legends, poetry, and literature of nations, pines and oaks, roses and violets, the olive and the vine, and the thousand other productions of the earth, have a place, and are spoken of in a manner which assumes, that in such kinds of natural objects, permanent and infallible distinctions had been observed and universally recognized.

[Note 1\16: Yate's _New Zealand_, p. 238.]

For a long period, it was not suspected that any ambiguity or confusion could arise from the use of such terms; and when such inconveniences did occur, (as even in early times they did,) men were far from divining that the proper remedy was the construction of a science of classification. The loose and insecure terms of the language of common life retained their place in botany, long after their {359} defects were severely felt: for instance, the vague and unscientific distinction of vegetables into _trees_, _shrubs_, and _herbs_, kept its ground till the time of Linnæus.

While it was thus imagined that the identification of a plant, by means of its name, might properly be trusted to the common uncultured faculties of the mind, and to what we may call the instinct of language, all the attention and study which were bestowed on such objects, were naturally employed in learning and thinking upon such circumstances respecting them as were supplied by any of the common channels through which knowledge and opinion flow into men's minds.

The reader need hardly be reminded that in the earlier periods of man's mental culture, he acquires those opinions on which he loves to dwell, not by the exercise of observation subordinate to reason; but, far more, by his fancy and his emotions, his love of the marvellous, his hopes and fears. It cannot surprise us, therefore, that the earliest lore concerning plants which we discover in the records of the past, consists of mythological legends, marvellous relations, and extraordinary medicinal qualities. To the lively fancy of the Greeks, the Narcissus, which bends its head over the stream, was originally a youth who in such an attitude became enamored of his own beauty: the hyacinth,[2\16] on whose petals the notes of grief were traced (A I, A I), recorded the sorrow of Apollo for the death of his favorite Hyacinthus: the beautiful lotus of India,[3\16] which floats with its splendid flower on the surface of the water, is the chosen seat of the goddess Lackshmi, the daughter of Ocean.[4\16] In Egypt, too,[5\16] Osiris swam on a lotus-leaf and Harpocrates was cradled in one. The lotus-eaters of Homer lost immediately their love of home. Every one knows how easy it would be to accumulate such tales of wonder or religion.

[Note 2\16: Lilium martagon. Ipse suos gemitus foliis inscribit et A I, A I, Flos habet inscriptum funestaque litera ducta est.--OVID.]

[Note 3\16: Nelumbium speciosum.]

[Note 4\16: Sprengel, _Geschichte der Botanik_, i. 27.]

[Note 5\16: Ib. i. 28.]

Those who attended to the effects of plants, might discover in them some medicinal properties, and might easily imagine more; and when the love of the marvellous was added to the hope of health, it is easy to believe that men would be very credulous. We need not dwell upon the examples of this. In Pliny's Introduction to that book of his {360} Natural History which treats of the medicinal virtues of plants, he says,[6\16] "Antiquity was so much struck with the properties of herbs, that it affirmed things incredible. Xanthus, the historian, says, that a man killed by a dragon, will be restored to life by an herb which he calls _balin_; and that Thylo, when killed by a dragon, was recovered by the same plant. Democritus asserted, and Theophrastus believed, that there was an herb, at the touch of which, the wedge which the woodman had driven into a tree would leap out again. Though we cannot credit these stories, most persons believe that almost anything might be effected by means of herbs, if their virtues were fully known." How far from a reasonable estimate of the reality of such virtues were the persons who entertained this belief we may judge from the many superstitious observances which they associated with the gathering and using of medicinal plants. Theophrastus speaks of these;[7\16] "The drug-sellers and the rhizotomists (root-cutters) tell us," he says, "some things which

may be true, but other things which are merely solemn quackery;[8\16] thus they direct us to gather some plants, standing from the wind, and with our bodies anointed; some by night, some by day, some before the sun falls on them. So far there may be something in their rules. But others are too fantastical and far fetched. It is, perhaps, not absurd to use a prayer in plucking a plant; but they go further than this. We are to draw a sword three times round the mandragora, and to cut it looking to the west: again, to dance round it, and to use obscene language, as they say those who sow cumin should utter blasphemies. Again, we are to draw a line round the black hellebore, standing to the east and praying; and to avoid an eagle either on the right or on the left; for, say they, 'if an eagle be near, the cutter will die in a year.'"

[Note 6\16: Lib. xxv. 5.]

[Note 7\16: _De Plantis_, ix. 9.]

[Note 8\16: Ἐπιτραγῳδοῦντες.]

This extract may serve to show the extent to which these imaginations were prevalent, and the manner in which they were looked upon by Theophrastus, our first great botanical author. And we may now consider that we have given sufficient attention to these fables and superstitions, which have no place in the history of the progress of real knowledge, except to show the strange chaos of wild fancies and legends out of which it had to emerge. We proceed to trace the history of the knowledge of plants. {361}

## CHAPTER II.

UNSYSTEMATIC KNOWLEDGE OF PLANTS.

A STEP was made towards the formation of the Science of Plants, although undoubtedly a slight one, as soon as men began to collect information concerning them and their properties, from a love and reverence for knowledge, independent of the passion for the marvellous and the impulse of practical utility. This step was very early made. The "wisdom" of Solomon, and the admiration which was bestowed upon it, prove, even at that period, such a working of the speculative faculty: and we are told, that among other evidences of his being "wiser than all men," "he spake of trees, from the cedar-tree that is in Lebanon even unto the hyssop that springeth out of the wall."[9\16] The father of history, Herodotus, shows us that a taste for natural history had, in his time, found a place in the minds of the Greeks. In speaking of the luxuriant vegetation of the Babylonian plain,[10\16] he is so far from desiring to astonish merely, that he says, "the blades of wheat and barley are full four fingers wide; but as to the size of the trees which grow from millet and sesame, though I could mention it, I will not; knowing well that those who have not been in that country will hardly believe what I have said already." He then proceeds to describe some remarkable circumstances respecting the fertilization of the date-palms in Assyria.

[Note 9\16: 1 Kings iv. 33.]

[Note 10\16: Herod. i. 193.]

This curious and active spirit of the Greeks led rapidly, as we have seen in other instances, to attempts at collecting and systematizing knowledge on almost every subject: and in this, as in almost every other department, Aristotle may be fixed upon, as the representative of the highest stage of knowledge and system which they ever attained. The vegetable kingdom, like every other province of nature, was one of the fields of the labors of this universal philosopher. But though his other works on natural history have come down to us, and are a most valuable monument of the state of such knowledge in his time, his Treatise on Plants is lost. The book _De Plantis_ {362} which appears with his name, is an imposture of the middle ages, full of errors and absurdities.[11\16]

[Note 11\16: Mirbel, _Botanique_, ii. 505.]

His disciple, friend, and successor, Theophrastus of Eresos, is, as we have said already, the first great writer on botany whose works we possess; and, as may be said in most cases of the first great writer, he offers to us a richer store of genuine knowledge and good sense than all his successors. But we find in him that the Greeks of his time, who aspired, as we have said, to collect and _systematize_ a body of information on every subject, failed in one half of their object, as far as related to the vegetable world. Their attempts at a systematic distribution of plants were altogether futile. Although Aristotle's divisions of the animal kingdom are, even at this day, looked upon with admiration by the best naturalists, the arrangements and comparisons of plants which were contrived by Theophrastus and his successors, have not left the slightest trace in the modern form of the science; and, therefore, according to our plan, are of no importance in our history. And thus we can treat all the miscellaneous information concerning vegetables which was accumulated by the whole of this school of writers, in no other way than as something antecedent to the first progress towards systematic knowledge.

The information thus collected by the unsystematic writers is of various kinds; and relates to the economical and medicinal uses of plants, their habits, mode of cultivation, and many other circumstances: it frequently includes some description; but this is always extremely imperfect, because the essential conditions of description had not been discovered. Of works composed of materials so heterogeneous, it can be of little use to produce specimens; but I may quote a few words from Theophrastus, which may serve to connect him with the future history of the science, as bearing upon one of the many problems respecting the identification of ancient and modern plants. It has been made a question whether the following description does not refer to the potato.[12\16] He is speaking of the differences of roots: "Some roots," he says, "are still different from those which have been described; as that of the _arachidna_[13\16] plant: for this bears fruit underground as well as above: the fleshy part sends one thick root deep into the ground, but the others, which bear the fruit, are more slender {363} and higher up, and ramified. It loves a sandy soil, and has no leaf whatever."

[Note 12\16: Theoph. i. 11.]

[Note 13\16: Most probably the _Arachnis hypogæa_, or ground-nut.]

The books of Aristotle and Theophrastus soon took the place of the Book of Nature in the attention of the degenerate philosophers who succeeded them. A story is told by Strabo[14\16] concerning the fate of the works of these great naturalists. In the case of the wars and changes which occurred among the successors of Alexander, the heirs of Theophrastus tried to secure to themselves his books, and those of his master, by burying them in the ground. There the manuscripts suffered much from damp and worms; till Apollonicon, a book-collector of those days, purchased them, and attempted, in his own way, to supply what time had obliterated. When Sylla marched the Roman troops into Athens, he took possession of the library of Apollonicon; and the works which it contained were soon circulated among the learned of Rome and Alexandria, who were thus enabled to _Aristotelize_[15\16] on botany as on other subjects.

[Note 14\16: Strabo, lib. xiii. c. i. § 54.]

[Note 15\16: Ἀριστοτλίζειν.]

The library collected by the Attalic kings of Pergamus, and the Alexandrian Museum, founded and supported by the Ptolemies of Egypt, rather fostered the commentatorial spirit than promoted the increase of any real knowledge of nature. The Romans, in this as in other subjects, were practical, not speculative. They had, in the times of their national vigor, several writers on agriculture, who were highly esteemed; but no author, till we come to Pliny, who dwells on the mere knowledge of plants. And even in Pliny, it is easy to perceive that we have before us a writer who extracted his information principally from books. This remarkable man,[16\16] in the middle of a public and active life, of campaigns and voyages, contrived to accumulate, by reading and study, an extraordinary store of knowledge of all kinds. So unwilling was he to have his reading and note-making interrupted, that, even before day-break in winter, and from his litter as he travelled, he was wont to dictate to his amanuensis, who was obliged to preserve his hand from the numbness which the cold occasioned, by the use of gloves.[17\16]

[Note 16\16: Sprengel, i. 163.]

[Note 17\16: Plin. Jun. Epist. 3, 5.]

It has been ingeniously observed, that we may find traces in the botanical part of his Natural History, of the errors which this hurried and broken habit of study produced; and that he appears frequently to have had books read to him and to have heard them amiss.[18\16] Thus, {364} among several other instances, Theophrastus having said that the plane-tree is in Italy rare,[19\16] Pliny, misled by the similarity of the Greek word (_spanian_, rare), says that the tree occurs in Italy and Spain.[20\16] His work has, with great propriety, been called the Encyclopædia of Antiquity; and, in truth, there are few portions of the learning of the times to which it does not refer. Of the thirty-seven Books of which it consists, no less than sixteen (from the twelfth to the twenty-seventh) relate to plants. The information which is collected in these books, is of the most miscellaneous kind; and the author admits, with little distinction, truth and error, useful knowledge and absurd fables. The declamatory style, and the comprehensive and lofty tone of thought which we have already spoken

of as characteristic of the Roman writers, are peculiarly observable in him. The manner of his death is well known: it was occasioned by the eruption of Vesuvius, A.D. 79, to which, in his curiosity, he ventured so near as to be suffocated.

[Note 18\16: Sprengel, i. 163.]

[Note 19\16: Theoph. iv. 7. Ἔν μὲν γὰρ τῷ Ἀδρίᾳ πλάτανον οὐ φασὶν εἶναι πλῆν περὶ το Διομήδους ἱερόν, _σπανίαν_ δὲ καὶ ἐν Ἰταλίᾳ πάσῃ]

[Note 20\16: Plin. Nat. Hist. xii. 3. Et alias (platanos) fuisse in Italia, ac nominatim _Hispania_, apud auctores invenitur.]

Pliny's work acquired an almost unlimited authority, as one of the standards of botanical knowledge, in the middle ages; but even more than his, that of his contemporary, Pedanius Dioscorides, of Anazarbus in Cilicia. This work, written in Greek, is held by the best judges[21\16] to offer no evidence that the author observed for himself. Yet he says expressly in his Preface, that his love of natural history, and his military life, have led him into many countries, in which he has had opportunity to become acquainted with the nature of herbs and trees.[22\16] He speaks of six hundred plants, but often indicates only their names and properties, giving no description by which they can be identified. The main cause of his great reputation in subsequent times was, that he says much of the medicinal virtues of vegetables.

[Note 21\16: Mirbel, 510.]

[Note 22\16: Sprengel, i. 136.]

We come now to the ages of darkness and lethargy, when the habit of original thought seems to die away, as the talent of original observation had done before. Commentators and mystics succeed to the philosophical naturalists of better times. And though a new race, altogether distinct in blood and character from the Greek, appropriates to itself the stores of Grecian learning, this movement does not, as might be expected, break the chains of literary slavery. The Arabs {365} bring, to the cultivation of the science of the Greeks, their own oriental habit of submission, their oriental love of wonder; and thus, while they swell the herd of commentators and mystics, they produce no philosopher.

Yet the Arabs discharged an important function in the history of human knowledge,[23\16] by preserving, and transmitting to more enlightened times, the intellectual treasures of antiquity. The unhappy dissensions which took place in the Christian church had scattered these treasures over the East, at a period much antecedent to the rise of the Saracen power. In the fifth century, the adherents of Nestorius, bishop of Constantinople, were declared heretical by the Council of Ephesus (A.D. 431), and driven into exile. In this manner, many of the most learned and ingenious men of the Christian world were removed to the Euphrates, where they formed the _Chaldean_ church, erected the celebrated Nestorian school of Edessa, and gave rise to many offsets from this in various regions. Already, in the fifth century, Hibas, Cumas, and Probus, translated the writings of Aristotle into Syriac. But the learned Nestorians paid an especial attention to the art of medicine, and were the most zealous students of the works of the Greek physicians. At Djondisabor, in

Khusistan, they became an ostensible medical school, who distributed academical honors as the result of public disputations. The califs of Bagdad heard of the fame and the wisdom of the doctors of Djondisabor, summoned some of them to Bagdad, and took measures for the foundation of a school of learning in that city. The value of the skill, the learning, and the virtues of the Nestorians, was so strongly felt, that they were allowed by the Mohammedans the free exercise of the Christian religion, and intrusted with the conduct of the studies of those of the Moslemin, whose education was most cared for. The affinity of the Syriac and Arabic languages made the task of instruction more easy. The Nestorians translated the works of the ancients out of the former into the latter language: hence there are still found Arabic manuscripts of Dioscorides, with Syriac words in the margin. Pliny and Aristotle likewise assumed an Arabic dress; and were, as well as Dioscorides, the foundation of instruction in all the Arabian academies; of which a great number were established throughout the Saracen empire, from Bokhara in the remotest east, to Marocco and Cordova in the west. After some time, the Mohammedans themselves began to translate and {366} extract from their Syriac sources; and at length to write works of their own. And thus arose vast libraries, such as that of Cordova, which contained 250,000 volumes.

[Note 23\16: Sprengel, i. 203.]

The Nestorians are stated[24\16] to have first established among the Arabs those collections of medicinal substances (_Apothecæ_), from which our term _Apothecary_ is taken; and to have written books (_Dispensatoria_) containing systematic instructions for the employment of these medicaments; a word which long continued to be implied in the same sense, and which we also retain, though in a modified application (_Dispensary_).

[Note 24\16: Sprengel, i. 205.]

The directors of these collections were supposed to be intimately acquainted with plants; and yet, in truth, the knowledge of plants owed but little to them; for the Arabic Dioscorides was the source and standard of their knowledge. The flourishing commerce of the Arabians, their numerous and distant journeys, made them, no doubt, practically acquainted with the productions of lands unknown to the Greeks and Romans. Their Nestorian teachers had established Christianity even as far as China and Malabar; and their travellers mention[25\16] the camphor of Sumatra, the aloe-wood of Socotra near Java, the tea of China. But they never learned the art of converting their practical into speculative knowledge. They treat of plants only in so far as their use in medicine is concerned,[26\16] and followed Dioscorides in the description, and even in the order of the plants, except when they arrange them according to the Arabic alphabet. With little clearness of view, they often mistake what they read:[27\16] thus when Dioscorides says that _ligusticon_ grows on the _Apennine_, a mountain not far from the _Alps_; Avicenna, misled by a resemblance of the Arabic letters, quotes him as saying that the plant grows on _Akabis_, a mountain near _Egypt_.

[Note 25\16: Sprengel, i. 206.]

[Note 26\16: Ib. i. 207.]

[Note 27\16: Ib. i. 211.]

It is of little use to enumerate such writers. One of the most noted of them was Mesuë, physician of the Calif of Kahirah. His work, which was translated into Latin at a later period, was entitled, _On Simple Medicines_; a title which was common to many medical treatises, from the time of Galen in the second century. Indeed, of this opposition of _simple_ and _compound_ medicines, we still have traces in our language: {367} He would ope his leathern scrip, And show me _simples_ of a thousand names, Telling their strange and vigorous faculties. MILTON, _Comus_.

Where the subject of our history is so entirely at a stand, it is unprofitable to dwell on a list of names. The Arabians, small as their science was, were able to instruct the Christians. Their writings were translated by learned Europeans, for instance Michael Scot, and Constantine of Africa, a Carthaginian who had lived forty years among the Saracens[28\16] and who died A.D. 1087. Among his works, is a Treatise, _De Gradibus_, which contains the Arabian medicinal lore. In the thirteenth century occur Encyclopædias, as that of Albertus Magnus, and of Vincent of Beauvais; but these contain no natural history except traditions and fables. Even the ancient writers were altogether perverted and disfigured. The Dioscorides of the middle ages varied materially from ours.[29\16] Monks, merchants, and adventurers travelled far, but knowledge was little increased. Simon of Genoa,[30\16] a writer on plants in the fourteenth century, boasts that he perambulated the East in order to collect plants. "Yet in his _Clavis Sanationis_," says a modern botanical writer,[31\16] "we discover no trace of an acquaintance with nature. He merely compares the Greek, Arabic, and Latin names of plants, and gives their medicinal effect after his predecessors:"--so little true is it, that the use of the senses alone necessarily leads to real knowledge.

[Note 28\16: Sprengel, i. 230.]

[Note 29\16: Ib. i. 239.]

[Note 30\16: Ib. i. 241.]

[Note 31\16: Ib. ib.]

Though the growing activity of thought in Europe, and the revived acquaintance with the authors of Greece in their genuine form, were gradually dispelling the intellectual clouds of the middle ages, yet during the fifteenth century, botany makes no approach to a scientific form. The greater part of the literature of this subject consisted of Herbals, all of which were formed on the same plan, and appeared under titles such as _Hortus_, or _Ortus Sanitatis_. There are, for example, three[32\16] such German Herbals, with woodcuts, which date about 1490. But an important peculiarity in these works is that they contain some indigenous species placed side by side with the old ones. In 1516, _The Grete Herbal_ was published in England, also with woodcuts. It contains an account of more than four hundred vegetables, and their {368} products; of which one hundred and fifty are English, and are no way distinguished from the exotics by the mode in which they are inserted in the work.

[Note 32\16: Augsburg, 1488. Mainz, 1491. Lubec, 1492.]

We shall see, in the next chapter, that when the intellect of Europe began really to apply itself to the observation of nature, the progress towards genuine science soon began to be visible, in this as in other subjects; but before this tendency could operate freely, the history of botany was destined to show, in another instance, how much more grateful to man, even when roused to intelligence and activity, is the study of tradition than the study of nature. When the scholars of Europe had become acquainted with the genuine works of the ancients in the original languages, the pleasure and admiration which they felt, led them to the most zealous endeavors to illustrate and apply what they read. They fell into the error of supposing that the plants described by Theophrastus, Dioscorides, Pliny, must be those which grew in their own fields. And thus Ruellius,[33\16] a French physician, who only travelled in the environs of Paris and Picardy, imagined that he found there the plants of Italy and Greece. The originators of genuine botany in Germany, Brunfels and Tragus (Bock), committed the same mistake; and hence arose the misapplication of classical names to many genera. The labors of many other learned men took the same direction, of treating the ancient writers as if they alone were the sources of knowledge and truth.

[Note 33\16: _De Natura Stirpium_, 1536.]

But the philosophical spirit of Europe was already too vigorous to allow this superstitious erudition to exercise a lasting sway. Leonicenus, who taught at Ferrara till he was almost a hundred years old, and died in 1524,[34\16] disputed, with great freedom, the authority of the Arabian writers, and even of Pliny. He saw, and showed by many examples, how little Pliny himself knew of nature, and how many errors he had made or transmitted. The same independence of thought with regard to other ancient writers, was manifested by other scholars. Yet the power of ancient authority melted away but gradually. Thus Antonius Brassavola, who established on the banks of the Po the first botanical garden of modern times, published in 1536, his _Examen omnium Simplicium Medicamentorum_; and, as Cuvier says,[35\16] though he studied plants in nature, his book (written in the {369} Platonic form of dialogue), has still the character of a commentary on the ancients.

[Note 34\16: Sprengel, i. 252.]

[Note 35\16: _Hist. des Sc. Nat._ partie ii. 169.]

The Germans appear to have been the first to liberate themselves from this thraldom, and to publish works founded mainly on actual observation. The first of the botanists who had this great merit is Otho Brunfels of Mentz, whose work, _Herbarum Vivæ Icones_, appeared in 1530. It consists of two volumes in folio, with wood-cuts; and in 1532, a German edition was published. The plants which it contains are given without any arrangement, and thus he belongs to the period of unsystematic knowledge. Yet the progress towards the formation of a system manifested itself so immediately in the series of German botanists to which he belongs, that we might with almost equal propriety transfer him to the history of that progress; to which we now proceed.

## CHAPTER III.

FORMATION OF A SYSTEM OF ARRANGEMENT OF PLANTS.

_Sect._ 1.--_Prelude to the Epoch of Cæsalpinus._

THE arrangement of plants in the earliest works was either arbitrary, or according to their use, or some other extraneous circumstance, as in Pliny. This and the division of vegetables by Dioscorides into _aromatic_, _alimentary_, _medicinal_, _vinous_, is, as will be easily seen, a merely casual distribution. The Arabian writers, and those of the middle ages, showed still more clearly their insensibility to the nature of system, by adopting an alphabetical arrangement; which was employed also in the Herbals of the sixteenth century. Brunfels, as we have said, adopted no principle of order; nor did his successor, Fuchs. Yet the latter writer urged his countrymen to put aside their Arabian and barbarous Latin doctors, and to observe the vegetable kingdom for themselves; and he himself set the example of doing this, examined plants with zeal and accuracy, and made above fifteen hundred drawings of them.[36\16] {370}

[Note 36\16: His _Historia Stirpium_ was published at Basil in 1542.]

The difficulty of representing plants in any useful way by means of drawings, is greater, perhaps, than it at first appears. So long as no distinction was made of the importance of different organs of the plant, a picture representing merely the obvious general appearance and larger parts, was of comparatively small value. Hence we are not to wonder at the slighting manner in which Pliny speaks of such records. "Those who gave such pictures of plants," he says, "Crateuas, Dionysius, Metrodorus, have shown nothing clearly, except the difficulty of their undertaking. A picture may be mistaken, and is changed and disfigured by copyists; and, without these imperfections, it is not enough to represent the plant in one state, since it has four different aspects in the four seasons of the year."

The diffusion of the habit of exact drawing, especially among the countrymen of Albert Durer and Lucas Cranach, and the invention of wood-cuts and copper-plates, remedied some of these defects. Moreover, the conviction gradually arose in men's minds that the structure of the flower and the fruit are the most important circumstances in fixing the identity of the plant. Theophrastus speaks with precision of the organs which he describes, but these are principally the leaves, roots, and stems. Fuchs uses the term _apices_ for the anthers, and _gluma_ for the blossom of grasses, thus showing that he had noticed these parts as generally present.

In the next writer whom we have to mention, we find some traces of a perception of the real resemblances of plants beginning to appear. It is impossible to explain the progress of such views without assuming in the reader some acquaintance with plants; but a very few words may suffice to convey the requisite notions. Even in plants which most commonly come in our way, we may perceive instances of the resemblances of which we speak. Thus, Mint, Marjoram, Basil, Sage, Lavender, Thyme, Dead-nettle, and many other plants, have a tubular flower, of which the mouth is divided into two lips; hence they are formed into a family, and termed _Labiatæ_. Again, the Stock, the Wall-flower, the Mustard, the Cress, the Lady-smock, the Shepherd's purse, have, among other similarities, their blossoms with four petals arranged crosswise; these are all of the order _Cruciferæ_. Other

flowers, apparently more complex, still resemble each other, as Daisy. Marigold, Aster, and Chamomile; these belong to the order _Compositæ_. And though the members of each such family may differ widely in their larger parts, their stems and leaves, the close study of nature leads the botanist irresistibly to consider their resemblances as {371} occupying a far more important place than their differences. It is the general establishment of this conviction and its consequences which we have now to follow.

The first writer in whom we find the traces of an arrangement depending upon these natural resemblances, is Hieronymus Tragus, (Jerom Bock,) a laborious German botanist, who, in 1551, published a herbal. In this work, several of the species included in those natural families to which we have alluded,[37\16] as for instance the Labiatæ, the Cruciferæ, the Compositæ, are for the most part brought together; and thus, although with many mistakes as to such connexions, a new principle of order is introduced into the subject.

[Note 37\16: Sprengel, i. 270.]

In pursuing the development of such principles of natural order, it is necessary to recollect that the principles lead to an assemblage of divisions and groups, successively subordinate, the lower to the higher, like the brigades, regiments, and companies of an army, or the provinces, towns, and parishes of a kingdom. Species are included in Genera, Genera in Families or Orders, and orders in Classes. The perception that there is some connexion among the species of plants, was the first essential step; the detection of different marks and characters which should give, on the one hand, limited groups, on the other, comprehensive divisions, were other highly important parts of this advance. To point out every successive movement in this progress would be a task of extreme difficulty, but we may note, as the most prominent portions of it, the establishment of the groups which immediately include Species, that is, _the formation of Genera_; and the invention of a method which should distribute into consistent and distinct divisions the whole vegetable kingdom, that is, _the construction of a System_.

To the second of these two steps we have no difficulty in assigning its proper author. It belongs to Cæsalpinus, and marks the first great epoch of this science. It is less easy to state to what botanist is due the establishment of Genera; yet we may justly assign the greater part of the merit of this invention, as is usually done, to Conrad Gessner of Zurich. This eminent naturalist, after publishing his great work on animals, died[38\16] of the plague in 1565, at the age of forty-nine, while he was preparing to publish a History of Plants, a sequel to his History of Animals. The fate of the work thus left {372} unfinished was remarkable. It fell into the hands of his pupil, Gaspard Wolf, who was to have published it, but wanting leisure for the office, sold it to Joachim Camerarius, a physician and botanist of Nuremberg, who made use of the engravings prepared by Gessner, in an Epitome which he published in 1586. The text of Gessner's work, after passing through various hands, was published in 1754 under the title of _Gessneri Opera Botanica per duo Sæcula desiderata, &c._, but is very incomplete.

[Note 38\16: Cuvier, _Leçons sur l'Hist. des Sciences Naturelles_, partie ii. p. 193.]

The imperfect state in which Gessner left his botanical labors, makes it necessary to seek the evidence of his peculiar views in scattered passages of his correspondence and other works. One of his great merits was, that he saw the peculiar importance of the flower and fruit as affording the characters by which the affinities of plants were to be detected; and that he urged this view upon his contemporaries. His plates present to us, by the side of each plant, its flower and its fruit, carefully engraved. And in his communications with his botanical correspondents, he repeatedly insists on these parts. Thus[39\16] in 1565 he writes to Zuinger concerning some foreign plants which the latter possessed: "Tell me if your plants have fruit and flower, as well as stalk and leaves, for those are of much the greater consequence. By these three marks,--flower, fruit, and seed,--I find that Saxifraga and Consolida Regalis are related to Aconite." These characters, derived from the _fructification_ (as the assemblage of flower and fruit is called), are the means by which genera are established, and hence, by the best botanists, Gessner is declared to be the inventor of genera.[40\16] {373}

[Note 39\16: _Epistolæ_, fol. 113 a; see also fol. 65 b.]

[Note 40\16: Haller, _Biblio Botanica_, i. 284. Methodi Botanicæ rationem primus pervidit;--dari nempe et genera quæ plures species comprehenderent et classes quæ multa genera. Varias etiam classes naturales expressit. Characterem in flore inque semine posuit, &c.--_Rauwolfio Socio Epist._ Wolf, p. 39.

Linnæus, _Genera Plantarum_, Pref. xiii. "A fructificatione plantas distinguere in genera, infinitæ sapientiæ placuisse, detexit posterior ætas, et quidem primus, sæculi sui ornamentum, Conradus Gessnerus, uti patet ex Epistolis ejus postremis, et Tabulis per Carmerarium editis."

Cuvier says (_Hist. des Sc. Nat._ 2^e p^e, p. 193), after speaking to the same effect, "Il fit voir encore que toutes les plantes qui ont des fleurs et des fruits semblables se ressemblent par leurs propriétés, et que quand on rapproche ces plantes on obtient ainsi une classification naturelle." I do not know if he here refers to any particular passages of Gessner's work.]

The labors of Gessner in botany, both on account of the unfinished state in which he left the application of his principles, and on account of the absence of any principles manifestly applicable to the whole extent of the vegetable kingdom, can only be considered as a prelude to the epoch in which those defects were supplied. To that epoch we now proceed.

_Sect._ 2.--_Epoch of Cæsalpinus.--Formation of a System of Arrangement._

IF any one were disposed to question whether Natural History truly belongs to the domain of Inductive Science;--whether it is to be prosecuted by the same methods, and requires the same endowments of mind as those which lead to the successful cultivation of the Physical Sciences,--the circumstances under which Botany has made its advance appear fitted to remove such doubts. The first decided step in this study was merely the construction of a classification of its subjects. We shall, I trust,

be able to show that such a classification includes, in reality, the establishment of one general principle, and leads to more. But without here dwelling on this point, it is worth notice that the person to whom we owe this classification, Andreas Cæsalpinus of Arezzo, was one of the most philosophical men of his time, profoundly skilled in the Aristotelian lore which was then esteemed, yet gifted with courage and sagacity which enabled him to weigh the value of the Peripatetic doctrines, to reject what seemed error, and to look onwards to a better philosophy. "How are we to understand," he inquires, "that we must proceed from universals to particulars (as Aristotle directs), when particulars are better known?"[41\16] Yet he treats the Master with deference, and, as has been observed,[42\16] we see in his great botanical work deep traces of the best features of the Aristotelian school, logic and method; and, indeed, in this work he frequently refers to his _Quæstiones Peripateticæ_. His book, entitled _De Plantis libri_ xvi. appeared at Florence in 1583. The aspect under which his task presented itself to his mind appears to me to possess so much interest, that I will transcribe a few of his reflections. After speaking of the splendid multiplicity of the productions of nature, and the confusion which has hitherto prevailed among writers on plants, {374} the growing treasures of the botanical world; he adds,[43\16] "In this immense multitude of plants, I see that want which is most felt in any other unordered crowd: if such an assemblage be not arranged into brigades like an army, all must be tumult and fluctuation. And this accordingly happens in the treatment of plants: for the mind is overwhelmed by the confused accumulation of things, and thus arise endless mistake and angry altercation." He then states his general view, which, as we shall see, was adopted by his successors. "_Since all science consists in the collection of similar, and the distinction of dissimilar things_, and since the consequence of this is a distribution into genera and species, which are to be natural classes governed by real differences, I have attempted to execute this task in the whole range of plants;--ut si quid pro ingenii mei tenuitate in hujusmodi studio profecerim, ad communem utilitatem proferam." We see here how clearly he claims for himself the credit of being the first to execute this task of arrangement.

[Note 41\16: _Quæstiones Peripateticæ_, (1569,) lib. i. quæst. i.]

[Note 42\16: Cuvier, p. 198.]

[Note 43\16: Dedicatio, a 2.]

After certain preparatory speculations, he says,[44\16] "Let us now endeavor to mark the kinds of plants by essential circumstances in the fructification." He then observes, "In the constitution of organs three things are mainly important--the number, the position, the figure." And he then proceeds to exemplify this: "Some have under one flower, ONE _seed_, as _Amygdala_, or ONE seed-_receptacle_, as _Rosa_; or TWO _seeds_, as _Ferularia_, or TWO seed-_receptacles_, as _Nasturtium_; or three, as the _Tithymalum_ kind have THREE _seeds_, the _Bulbaceæ_ THREE _receptacles_; or four, as _Marrubium_, FOUR _seeds_, _Siler_ FOUR _receptacles_; or more, as _Cicoraceæ_, and _Acanaceæ_ have MORE _seeds_, _Pinus_, MORE _receptacles_."

[Note 44\16: Lib. i. c. 13, 14.]

It will be observed that we have here ten classes made out by means of number alone, added to the consideration of whether the seed is alone in its covering, as in a cherry, or contained in a receptacle with several others, as in a berry, pod, or capsule. Several of these divisions are, however, further subdivided according to other circumstances, and especially according as the vital part of the seed, which he called the heart (_cor_[45\16]), is situated in the upper or lower part of the seed. As our object here is only to indicate the principle of the method of Cæsalpinus, I need not further dwell on the details, and still less on the defects by which it is disfigured, as, for instance, the retention of the old distinction of Trees, Shrubs, and Herbs. {375}

[Note 45\16: _Corculum_, of Linnæus.]

To some persons it may appear that this arbitrary distribution of the vegetable kingdom, according to the number of parts of a particular kind, cannot deserve to be spoken of as a great discovery. And if, indeed, the distribution had been arbitrary, this would have been true; the real merit of this and of every other system is, that while it is artificial in its form, it is natural in its results. The plants which are associated by the arrangement of Cæsalpinus, are those which have the closest resemblances in the most essential points. Thus, as Linnæus says, though the first in attempting to form natural orders, he observed as many as the most successful of later writers. Thus his _Legumina_[46\16] correspond to the natural order _Leguminosæ_; his _genus Ferulaceum_[47\16] to the _Umbellatæ_; his _Bulbaceæ_[48\16] to _Liliaceæ_; his _Anthemides_[49\16] to the _Compositæ_; in like manner, the _Borragineæ_ are brought together,[50\16] and the _Labiatæ_. That such assemblages are produced by the application of his principles, is a sufficient evidence that they have their foundation in the general laws of the vegetable world. If this had not been the case, the mere application of number or figure alone as a standard of arrangement, would have produced only intolerable anomalies. If, for instance, Cæsalpinus had arranged plants by the number of flowers on the same stalk, he would have separated individuals of the same species; if he had distributed them according to the number of leaflets which compose the leaves, he would have had to place far asunder different species of the same genus. Or, as he himself says,[51\16] "If we make one genus of those which have a round root, as Rapum, Aristolochia, Cyclaminus, Aton, we shall separate from this genus those which most agree with it, as Napum and Raphanum, which resemble Rapum, and the long Aristolochia, which resembles the round; while we shall join the most remote kinds, for the nature of Cyclaminus and Rapum is altogether diverse in all other respects. Or if we attend to the differences of stalk, so as to make one genus of those which have a naked stalk, as the Junci, Cæpe, Aphacæ, along with Cicoraceæ, Violæ, we shall still connect the most unlike things, and disjoin the closest affinities. And if we note the differences of leaves, or even flowers, we fall into the same difficulty; for many plants very different in kind have leaves very similar, as Polygonum and Hypericum, Ernea and Sesamois, Apium and Ranunculus; and plants of the same genus have sometimes very different {376} leaves, as the several species of Ranunculus and of Lactuca. Nor

will color or shape of the flowers help us better; for what has Vitis in common with Œnanthe, except the resemblance of the flower?" He then goes on to say, that if we seek a too close coincidence of all the characters we shall have no Species; and thus shows us that he had clearly before his view the difficulty, which he had to attack, and which it is his glory to have overcome, that of constructing Natural Orders.

[Note 46\16: Lib. vi.]

[Note 47\16: Lib. vii.]

[Note 48\16: Lib. x.]

[Note 49\16: Lib. xii.]

[Note 50\16: Lib. xi.]

[Note 51\16: Lib. i. cap. xii. p. 25.]

But as the principles of Cæsalpinus are justified, on the one hand, by their leading to _Natural Orders_, they are recommended on the other by their producing a _System_ which applies through the whole extent of the vegetable kingdom. The parts from which he takes his characters must occur in all flowering-plants, for all such plants have seeds. And these seeds, if not very numerous for each flower, will be of a certain definite number and orderly distribution. And thus every plant will fall into one part or other of the same system.

It is not difficult to point out, in this induction of Cæsalpinus, the two elements which we have so often declared must occur in all inductive processes; the exact acquaintance with _facts_, and the general and applicable _ideas_ by which these facts are brought together. Cæsalpinus was no mere dealer in intellectual relations or learned traditions, but a laborious and persevering collector of plants and of botanical knowledge. "For many years," he says in his Dedication, "I have been pursuing my researches in various regions, habitually visiting the places in which grew the various kinds of herbs, shrubs, and trees; I have been assisted by the labors of many friends, and by gardens established for the public benefit, and containing foreign plants collected from the most remote regions." He here refers to the first garden directed to the public study of Botany, which was that of Pisa,[52\16] instituted in 1543, by order of the Grand Duke Cosmo the First. The management of it was confided first to Lucas Ghini, and afterwards to Cæsalpinus. He had collected also a herbarium of dried plants, which he calls the rudiment of his work. "Tibi enim," he says, in his dedication to Francis Medici, Grand Duke of Etruria, "apud quem extat ejus rudimentum ex plantis libro agglutinatis a me compositum." And, throughout, he speaks with the most familiar and vivid acquaintance of the various vegetables which he describes.

[Note 52\16: Cuv. 187.]

But Cæsalpinus also possessed fixed and general views concerning the relation and functions of the parts of plants, and ideas of symmetry {377} and system; without which, as we see in other botanists of his and succeeding times, the mere accumulation of a knowledge of details does not lead to any advance in science. We

have already mentioned his reference to general philosophical principles, both of the Peripatetics and of his own. The first twelve chapters of his work are employed in explaining the general structure of plants, and especially that point to which he justly attaches so much importance, the results of the different situation of the _cor_ or _corculum_ of the seed. He shows[53\16] that if we take the root, or stem, or leaves, or blossom, as our guide in classification, we shall separate plants obviously alike, and approximate those which have merely superficial resemblances. And thus we see that he had in his mind ideas of fixed resemblance and symmetrical distribution, which he sedulously endeavored to apply to plants; while his acquaintance with the vegetable kingdom enabled him to see in what manner these ideas were not, and in what manner they were, really applicable.

[Note 53\16: Lib. i. cap. xii.]

The great merit and originality of Cæsalpinus have been generally allowed, by the best of the more modern writers on Botany. Linnæus calls him one of the founders of the science; "Primus verus systematicus;"[54\16] and, as if not satisfied with the expression of his admiration in prose, hangs a poetical garland on the tomb of his hero. The following distich concludes his remarks on this writer: Quisquis hic extiterit primos concedet honores Cæsalpine tibi; primaque serta dabit: and similar language of praise has been applied to him by the best botanists up to Cuvier,[55\16] who justly terms his book "a work of genius."

[Note 54\16: _Philosoph. Bot._ p. 19.]

[Note 55\16: Cuv. _Hist._ 193.]

Perhaps the great advance made in this science by Cæsalpinus, is most strongly shown by this; that no one appeared, to follow the path which he had opened to system and symmetry, for nearly a century. Moreover, when the progress of this branch of knowledge was resumed, his next successor, Morison, did not choose to acknowledge that he had borrowed so much from so old a writer; and thus, hardly mentions his name, although he takes advantage of his labors, and even transcribes his words without acknowledgement, as I shall show. The pause between the great invention of Cæsalpinus, and its natural sequel, the developement and improvement of his method, is so marked, that I {378} will, in order to avoid too great an interruption of chronological order, record some of its circumstances in a separate section.

_Sect._ 3.--_Stationary Interval._

THE method of Cæsalpinus was not, at first, generally adopted. It had, indeed, some disadvantages. Employed in drawing the boundary-lines of the larger divisions of the vegetable kingdom, he had omitted those smaller groups, Genera, which were both most obvious to common botanists, and most convenient in the description and comparison of plants. He had also neglected to give the Synonyms of other authors for the plants spoken of by him; an appendage to botanical descriptions, which the increase of botanical information and botanical books had now rendered indispensable. And thus it happened, that a work, which must always be considered as forming a great epoch in the science to which it refers, was probably little read,

and in a short time could be treated as if it were quite forgotten.

In the mean time, the science was gradually improved in its details. Clusius, or Charles de l'Ecluse, first taught botanists to describe well. "Before him," says Mirbel,[56\16] "the descriptions were diffuse, obscure, indistinct; or else concise, incomplete, vague. Clusius introduced exactitude, precision, neatness, elegance, method: he says nothing superfluous; he omits nothing necessary." He travelled over great part of Europe, and published various works on the more rare of the plants which he had seen. Among such plants, we may note now one well known, the potato; which he describes as being commonly used in Italy in 1586;[57\16] thus throwing doubt, at least, on the opinion which ascribes the first introduction of it into Europe to Sir Walter Raleigh, on his return from Virginia, about the same period. As serving to illustrate, both this point, and the descriptive style of Clusius, I quote, in a note, his description of the flower of this plant.[58\16] {379}

[Note 56\16: _Physiol. Veg._ p. 525.]

[Note 57\16: Clusius. _Exotic_. iv. c. 52, p. lxxix.]

[Note 58\16: "Papas Peruanorum. Arachidna, Theoph. forte. Flores elegantes, uncialis amplitudinis aut majores, angulosi, singulari folio constantes, sed ita complicato ut quinque folia discreta videantur, coloris exterius ex purpura candicantis, interius purpurascentis, radiis quinque herbaceis ex umbilico stellæ instar prodeuntibus, et totidem staminibus flavis in umbonem coeuntibus."

He says that the Italians do not know whence they had the plant, and that they call it _Taratouffli_. The name _Potato_ was, in England, previously applied to the Sweet Potato (_Convolvulus batatas_), which was the _common_ Potato, in distinction to the _Virginian _Potato, at the time of Gerard's Herbal. (1597?) Gerard's figures of both plants are copied from those of Clusius.

It may be seen by the description of Arachidna, already quoted from Theophrastus, (above,) that there is little plausibility in Clusius's conjecture of the plant being known to the ancients. I need not inform the botanist that this opinion is untenable.]

The addition of exotic species to the number of known plants was indeed going on rapidly during the interval which we are now considering. Francis Hernandez, a Spaniard, who visited America towards the end of the sixteenth century, collected and described many plants of that country, some of which were afterwards published by Recchi.[59\16] Barnabas Cobo, who went as a missionary to America in 1596, also described plants.[60\16] The Dutch, among other exertions which they made in their struggle with the tyranny of Spain, sent out an expedition which, for a time, conquered the Brazils; and among other fruits of this conquest, they published an account of the natural history of the country.[61\16] To avoid interrupting the connexion of such labors, I will here carry them on a little further in the order of time. Paul Herman, of Halle, in Saxony, went to the Cape of Good Hope and to Ceylon; and on his return, astonished the botanists of Europe by the vast quantity of remarkable plants which he introduced to their knowledge.[62\16] Rheede, the Dutch governor of Malabar, ordered descriptions and drawings to be made of many curious species, which were published in a large work in twelve folio volumes.[63\16] Rumphe,

another Dutch consul at Amboyna,[64\16] labored with zeal and success upon the plants of the Moluccas. Some species which occur in Madagascar figured in a description of that island composed by the French Commandant Flacourt.[65\16] Shortly afterwards, Engelbert Kæmpfer,[66\16] a Westphalian of great acquirements and undaunted courage, visited Persia, Arabia Felix, the Mogul Empire, Ceylon, Bengal, Sumatra, Java, Siam, Japan; Wheler travelled in Greece and Asia Minor; and Sherard, the English consul, published an account of the plants of the neighborhood of Smyrna. {380}

[Note 59\16: _Nova Plantarum Regni Mexicana Historia_, Rom. 1651, fol.]

[Note 60\16: Sprengel, _Gesch. der Botanik_, ii. 62.]

[Note 61\16: _Historia Naturalis Brasiliæ_, L. B. 1648, fol. (Piso and Maregraf).]

[Note 62\16: _Museum Zeylanicum_, L. B. 1726.]

[Note 63\16: _Hortus Malabaricus_, 1670-1703.]

[Note 64\16: _Herbarium Amboinense_, Amsterdam, 1741-51, fol.]

[Note 65\16: _Histoire de la grande Isle Madagascar_, Paris, 1661.]

[Note 66\16: _Amœnitates Exoticæ_, Lemgov. 1712. 4to.]

At the same time, the New World excited also the curiosity of botanists. Hans Sloane collected the plants of Jamaica; John Banister those of Virginia; William Vernon, also an Englishman, and David Kriege, a Saxon, those of Maryland; two Frenchmen, Surian and Father Plumier, those of Saint Domingo.

We may add that public botanical gardens were about this time established all over Europe. We have already noticed the institution of that of Pisa in 1543; the second was that of Padua in 1545; the next, that of Florence in 1556; the fourth, that of Bologna, 1568; that of Rome, in the Vatican, dates also from 1568.

The first transalpine garden of this kind arose at Leyden in 1577; that of Leipzig in 1580. Henry the Fourth of France established one at Montpellier in 1597. Several others were instituted in Germany; but that of Paris did not begin to exist till 1626; that of Upsal, afterwards so celebrated, took its rise in 1657, that of Amsterdam in 1684. Morison, whom we shall soon have to mention, calls himself, in 1680, the first Director of the Botanical Garden at Oxford.

[2nd Ed.] [To what is above said of Botanical Gardens and Botanical Writers, between the times of Cæsalpinus and Morison, I may add a few circumstances. The first academical garden in France was that at Montpellier, which was established by Peter Richier de Belleval, at the end of the sixteenth century. About the same period, rare flowers were cultivated at Paris, and pictures of them made, in order to supply the embroiderers of the court-robes with new patterns. Thus figures of the most beautiful flowers in the garden of Peter Robins were published by the court-embroiderer Peter Vallet, in 1608, under the title of _Le Jardin du Roi Henry IV_. But Robins' works were of great service to botany; and his garden assisted the studies of Renealmus (Paul Reneaulme), whose _Specimen Historiæ Plantarum_

(Paris, 1611), is highly spoken of by the best botanists. Recently, Mr. Robert Brown has named after him a new genus of _Irideæ_ (RENEALMIA); adding, "Dixi in memoriam PAULI RENEALMI, botanici sui ævi accuratissimi, atque staminum primi scrutatoris; qui non modo eorum numerum et situm, sed etiam filamentorum proportionem passim descripsit, et characterem tetradynamicum siliquosarum perspexit." (_Prodromus Floræ Novæ Hollandiæ_, p. 448.)

The oldest Botanical Garden in England is that at Hampton Court, founded by Queen Elizabeth, and much enriched by Charles II. and William III. (Sprengel, _Gesch. d. Bot._ vol. ii. p. 96.)]

In the mean time, although there appeared no new system which {381} commanded the attention of the botanical world, the feeling of the importance of the affinities of plants became continually more strong and distinct.

Lobel, who was botanist to James the First, and who published his _Stirpium Adversaria Nova_ in 1571, brings together the natural families of plants more distinctly than his predecessors, and even distinguishes (as Cuvier states,[67\16]) monocotyledonous from dicotyledonous plants; one of the most comprehensive division-lines of botany, of which succeeding times discovered the value more completely. Fabius Columna,[68\16] in 1616, gave figures of the fructification of plants on copper, as Gessner had before done on wood. But the elder Bauhin (John), notwithstanding all that Cæsalpinus had done, retrograded, in a work published in 1619, into the less precise and scientific distinctions of--trees with nuts; with berries; with acorns; with pods; creeping plants, gourds, &c.: and no clear progress towards a system was anywhere visible among the authors of this period.

[Note 67\16: Cuv. _Leçons, &c._ 198.]

[Note 68\16: Ib. 206.]

While this continued to be the case, and while the materials, thus destitute of order, went on accumulating, it was inevitable that the evils which Cæsalpinus had endeavored to remedy, should become more and more grievous. "The nomenclature of the subject[69\16] was in such disorder, it was so impossible to determine with certainty the plants spoken of by preceding writers, that thirty or forty different botanists had given to the same plant almost as many different names. Bauhin called by one appellation, a species which Lobel or Matheoli designated by another. There was an actual chaos, a universal confusion, in which it was impossible for men to find their way." We can the better understand such a state of things, from having, in our own time, seen another classificatory science, Mineralogy, in the very condition thus described. For such a state of confusion there is no remedy but the establishment of a true system of classification; which by its real foundation renders a reason for the place of each species; and which, by the fixity of its classes, affords a basis for a standard nomenclature, as finally took place in Botany. But before such a remedy is obtained, men naturally try to alleviate the evil by tabulating the synonyms of different writers, as far as they are able to do so. The task of constructing such a _Synonymy_ of botany at the period of which we speak, was undertaken by Gaspard Bauhin, the brother of John, but nineteen years younger. This work, the _Pinax

Theatri Botanici_, was printed {382} at Basil in 1623. It was a useful undertaking at the time; but the want of any genuine order in the _Pinax_ itself, rendered it impossible that it should be of great permanent utility.

[Note 69\16: Ib. 212.]

After this period, the progress of almost all the sciences became languid for a while; and one reason of this interruption was, the wars and troubles which prevailed over almost the whole of Europe. The quarrels of Charles the First and his parliament, the civil wars and the usurpation, in England; in France, the war of the League, the stormy reign of Henry the Fourth, the civil wars of the minority of Louis the Thirteenth, the war against the Protestants and the war of the Fronde in the minority of Louis the Fourteenth; the bloody and destructive Thirty Years' War in Germany; the war of Spain with the United Provinces and with Portugal;--all these dire agitations left men neither leisure nor disposition to direct their best thoughts to the promotion of science. The baser spirits were brutalized; the better were occupied by high practical aims and struggles of their moral nature. Amid such storms, the intellectual powers of man could not work with their due calmness, nor his intellectual objects shine with their proper lustre.

At length a period of greater tranquillity gleamed forth, and the sciences soon expanded in the sunshine. Botany was not inert amid this activity, and rapidly advanced in a new direction, that of physiology; but before we speak of this portion of our subject, we must complete what we have to say of it as a classificatory science.

_Sect._ 4.--_Sequel to the Epoch of Cæsalpinus. Further Formation and Adoption of Systematic Arrangement._

SOON after the period of which we now speak, that of the restoration of the Stuarts to the throne of England, systematic arrangements of plants appeared in great numbers; and in a manner such as to show that the minds of botanists had gradually been ripening for this improvement, through the influence of preceding writers, and the growing acquaintance with plants. The person whose name is usually placed first on this list, Robert Morison, appears to me to be much less meritorious than many of those who published very shortly after him; but I will give him the precedence in my narrative. He was a Scotchman, who was wounded fighting on the royalist side in the civil wars of England. On the triumph of the republicans, he withdrew to France, when he became director of the garden of Gaston, Duke of Orléans at Blois; and there he came under the notice of our Charles {383} the Second; who, on his restoration, summoned Morison to England, where he became Superintendent of the Royal Gardens, and also of the Botanic Garden at Oxford. In 1669, he published _Remarks on the Mistakes of the two Bauhins_, in which he proves that many plants in the _Pinax_ are erroneously placed, and shows considerable talent for appreciating natural families and genera. His great systematic work appeared from the University press at Oxford in 1680. It contains a system, but a system, Cuvier says,[70\16] which approaches rather to a natural method than to a rigorous distribution, like that of his predecessor Cæsalpinus, or that of his successor Ray. Thus the herbaceous plants are divided into _climbers_, _leguminous_,

_siliquose_, _unicapsalar_, _bicapsular_, _tricapsular_, _quadricapsular_, _quinquecapsular_; this division being combined with characters derived from the number of petals. But along with these numerical elements, are introduced others of a loose and heterogeneous kind, for instance, the classification of herbs as _lactescent_ and _emollient_. It is not unreasonable to say, that such a scheme shows no talent for constructing a complete system; and that the most distinct part of it, that dependent on the fruit, was probably borrowed from Cæsalpinus. That this is so, we have, I think, strong proof; for though Morison nowhere, I believe, mentions Cæsalpinus, except in one place in a loose enumeration of botanical writers,[71\16] he must have made considerable use of his work. For he has introduced into his own preface a passage copied literally[72\16] from the dedication of Cæsalpinus; which passage we have already quoted (p. 374,) beginning, "Since all science consists in the collection of similar, and the distinction of dissimilar things." And that the mention of the original is not omitted by accident, appears from this; that Morison appropriates also the conclusion of the passage, which has a personal reference, "_Conatus sum id præstare in universa plantarum historia, ut si quid pro ingenii mei tenuitate in hujusmodi studio profecerim, ad communem utilitatem proferrem._" That Morison, thus, at so long an interval after the publication of the work of Cæsalpinus, borrowed from him without acknowledgement, and adopted his system so as to mutilate it, proves that he had neither the temper nor the talent of a discoverer; and justifies us withholding from him the credit which belongs to those, who, in his time, resumed the great undertaking of constructing a vegetable system.

[Note 70\16: Cuv. _Leçons_, &c. p. 486.]

[Note 71\16: Pref. p. i.]

[Note 72\16: Ib. p. ii.]

Among those whose efforts in this way had the greatest and earliest {384} influence, was undoubtedly our countryman, John Ray, who was Fellow of Trinity College, Cambridge, at the same time with Isaac Newton. But though Cuvier states[73\16] that Ray was the model of the systematists during the whole of the eighteenth century, the Germans claim a part of his merit for one of their countrymen, Joachim Jung, of Lubeck, professor at Hamburg.[74\16] Concerning the principles of this botanist, little was known during his life. But a manuscript of his book was communicated[75\16] to Ray in 1660, and from this time forwards, says Sprengel, there might be noticed in the writings of Englishmen, those better and clearer views to which Jung's principles gave birth. Five years after the death of Jung, his _Doxoscopia Physica_ was published, in 1662; and in 1678, his _Isagoge Phytoscopica_. But neither of these works was ever much read; and even Linnæus, whom few things escaped which concerned botany, had, in 1771, seen none of Jung's works.

[Note 73\16: _Leçons Hist. Sc._ p. 487.]

[Note 74\16: Sprengel, ii. 27.]

[Note 75\16: Ray acknowledges this in his _Index Plant. Agri Cantab._ p. 87, and quotes from it the definition of _caulis_.]

I here pass over Jung's improvements of botanical language, and speak only of those which he is asserted to have suggested in the arrangement of plants. He examines, says Sprengel,[76\16] the value of characters of species, which, he holds, must not be taken from the thorns, nor from color, taste, smell, medicinal effects, time and place of blossoming. He shows, in numerous examples, what plants must be separated, though called by a common name, and what most be united, though their names are several.

[Note 76\16: Sprengel, ii. 29.]

I do not see in this much that interferes with the originality of Ray's method,[77\16] of which, in consequence of the importance ascribed to it by Cuvier, as we have already seen, I shall give an account, following that great naturalist.[78\16] I confine myself to the ordinary plants, and omit the more obscure vegetables, as mushrooms, mosses, ferns, and the like.

[Note 77\16: _Methodus Plantarum Nova_, 1682. _Historia Plantarum_, 1686.]

[Note 78\16: Cuv. _Leçons Hist. Sc._ 488.]

Such plants are _composite_ or _simple_. The _composite_ flowers are those which contain many florets in the same _calyx_.[79\16] These are subdivided according as they are composed altogether of complete florets, {385} or of half florets, or of a centre of complete florets, surrounded by a circumference or ray of demi-florets. Such are the divisions of the _corymbiferæ_, or _compositæ_.

[Note 79\16: _Involucrum_, in modern terminology.]

In the _simple_ flowers, the seeds are _naked_, or in a _pericarp_. Those with _naked_ seeds are arranged according to the number of the seeds, which may be one, two, three, four, or more. If there is only one, no subdivision is requisite: if there are two, Ray makes a subdivision, according as the flower has five petals, or a continuous corolla. Here we come to several natural families. Thus, the flowers with two seeds and five petals are the _Umbelliferous_ plants; the monopetalous flowers with two seeds are the _Stellatæ_. He founds the division of four-seeded flowers on the circumstance of the leaves being opposite, or alternate; and thus again, we have the natural families of _Asperifoliæ_, as _Echium_, &c., which have the leaves alternate, and the _Verticillatæ_, as _Salvia_, in which the leaves are opposite. When the flower has more than four seeds, he makes no subdivision.

So much for simple flowers with naked seeds. In those where the seeds are surrounded by a _pericarp_, or fruit, this fruit is large, soft, and fleshy, and the plants are _pomiferous_; or it is small and juicy, and the fruit is a berry, as a Gooseberry.

If the fruit is not juicy, but _dry_, it is multiple or simple. If it be simple, we have the _leguminose_ plants. If it be multiple, the form of the flower is to be attended to. The flower may be _monopetalous_, or _tetrapetalous_, or _pentapetalous_, or with still _more_ divisions. The monopetalous may be _regular_ or _irregular_; so may the tetrapetalous. The regular tetrapetalous flowers are, for example, the _Cruciferæ_, as Stock and Cauliflower; the irregular, are the _papilionaceous_ plants, Peas, Beans, and Vetches; and thus we again come to natural families. The remaining plants are

divided in the same way, into those with _imperfect_, and those with _perfect_, flowers. Those with _imperfect_ flowers are the _Grasses_, the _Rushes_ (_Junci_), and the like; among those with _perfect_ flowers, are the _Palmaceæ_, and the _Liliaceæ_.

We see that the division of plants is complete as a system; all flowers must belong to one or other of the divisions. Fully to explain the characters and further subdivisions of these families, would be to write a treatise on botany; but it is easily seen that they exhaust the subject as far as they go.

Thus Ray constructed his system partly on the fruit and partly on the flower; or more properly, according to the expression of Linnæus, {386} comparing his earlier with his later system, he began by being a _fructicist_, and ended by being a _corollist_.[80\16]

[Note 80\16: Ray was a most industrious herbalizer, and I cannot understand on what ground Mirbel asserts (_Physiol. Veg._, tom. ii. p. 531,) that he was better acquainted with books than with plants.]

As we have said, a number of systems of arrangement of plants were published about this time, some founded on the fruit, some on the corolla, some on the calyx, and these employed in various ways. Rivinus[81\16] (whose real name was Bachman,) classified by the flower alone; instead of combining it with the fruit, as Ray had done.[82\16] He had the further merit of being the first who rejected the old division, of _woody_ and _herbaceous_ plants; a division which, though at variance with any system founded upon the structure of the plants was employed even by Tournefort, and only finally expelled by Linnæus.

[Note 81\16: Cuv. _Leçons_, 491.]

[Note 82\16: _Historia Generalis ad rem Herbariam_, 1690.]

It would throw little light upon the history of botany, especially for our purpose, to dwell on the peculiarities of these transitory systems. Linnæus,[83\16] after his manner, has given a classification of them. Rivinus, as we have just seen, was a _corollist_, according to the regularity and number of the petals; Hermann was a _fructicist_. Christopher Knaut[84\16] adopted the system of Ray, but inverted the order of its parts; Christian Knaut did nearly the same with regard to that of Rivinus, taking number before regularity in the flower.[85\16]

[Note 83\16: _Philos. Bot._ p. 21.]

[Note 84\16: _Enumeratio Plantarum_, &c., 1687.]

[Note 85\16: Linn.]

Of the systems which prevailed previous to that of Linnæus, Tournefort's was by far the most generally accepted. Joseph Pitton de Tournefort was of a noble family in Provence, and was appointed professor at the Jardin du Roi in 1683. His well-known travels in the Levant are interesting on other subjects, as well as botany. His _Institutio Rei Herbariæ_, published in 1700, contains his method, which is that of a _corollist_. He is guided by the regularity or irregularity of the flowers, by their form,

and by the situation of the receptacle of the seeds below the calyx, or within it. Thus his classes are--those in which the flowers are _campaniform_, or bell-shaped; those in which they are _infundibuliform_, or funnel-shaped, as Tobacco; then the irregular flowers, as the _Personatæ_, which resemble an ancient mask; the _Labiatæ_, with their two lips; the _Cruciform_; the _Rosaceæ_, with flowers like a rose; the _Umbelliferæ_; the _Caryophylleæ_, as the {387} Pink; the _Liliaceæ_, with six petals, as the Tulip, Narcissus, Hyacinth, Lily; the _Papilionaceæ_, which are leguminous plants, the flower of which resembles a butterfly, as Peas and Beans; and finally, the _Anomalous_, as Violet, Nasturtium, and others.

Though this system was found to be attractive, as depending, in an evident way, on the most conspicuous part of the plant, the flower, it is easy to see that it was much less definite than systems like that of Rivinus, Hermann, and Ray, which were governed by number. But Tournefort succeeded in giving to the characters of genera a degree of rigor never before attained, and abstracted them in a separate form. We have already seen that the reception of botanical Systems has depended much on their arrangement into Genera.

Tournefort's success was also much promoted by the author inserting in his work a figure of a flower and fruit belonging to each genus; and the figures, drawn by Aubriet, were of great merit. The study of botany was thus rendered easy, for it could be learned by turning over the leaves of a book. In spite of various defects, these advantages gave this writer an ascendancy which lasted, from 1700, when his book appeared, for more than half a century. For though Linnæus began to publish in 1735, his method and his nomenclature were not generally adopted till 1760.

## CHAPTER IV.

THE REFORM OF LINNÆUS.

_Sect._ 1.--_Introduction of the Reform._

ALTHOUGH, perhaps, no man of science ever exercised a greater sway than Linnæus, or had more enthusiastic admirers, the most intelligent botanists always speak of him, not as a great discoverer, but as a judicious and strenuous _Reformer_. Indeed, in his own lists of botanical writers, he places himself among the "Reformatores;" and it is apparent that this is the nature of his real claim to admiration; for the doctrine of the sexes of plants, even if he had been the first to establish it, was a point of botanical physiology, a province of the {388} science which no one would select as the peculiar field of Linnæus's glory; and the formation of a system of arrangement on the basis of this doctrine, though attended with many advantages, was not an improvement of any higher order than those introduced by Ray and Tournefort. But as a Reformer of the state of Natural History in his time, Linnæus was admirable for his skill, and unparalleled in his success. And we have already seen, in the instance of the reform of mineralogy, as attempted by Mohs and Berzelius, that men of great talents and knowledge may fail in such an undertaking.

It is, however, only by means of the knowledge which he displays, and of the beauty and convenience of the improvements which he proposes, that any one can acquire such an influence as to procure his suggestions to be adopted. And even if original circumstances of birth or position could invest any one with peculiar prerogatives and powers in the republic of science, Karl Linné began his career with no such advantages. His father was a poor curate in Smaland, a province of Sweden; his boyhood was spent in poverty and privation; it was with great difficulty that, at the age of twenty-one, he contrived to subsist at the University of Upsal, whither a strong passion for natural history had urged him. Here, however, he was so far fortunate, that Olaus Rudbeck, the professor of botany, committed to him the care of the Botanic Garden.[86\16] The perusal of the works of Vaillant and Patrick Blair suggested to him the idea of an arrangement of plants, formed upon the sexual organs, the stamens and pistils; and of such an arrangement he published a sketch in 1731, at the age of twenty-four.

[Note 86\16: Sprengel, ii. 232.]

But we must go forwards a few years in his life, to come to the period to which his most important works belong. University and family quarrels induced him to travel; and, after various changes of scene, he was settled in Holland, as the curator of the splendid botanical garden of George Clifford, an opulent banker. Here it was[87\16] that he laid the foundation of his future greatness. In the two years of his residence at Harlecamp, he published nine works. The first, the _Systema Naturæ_, which contained a comprehensive sketch of the whole domain of Natural History, excited general astonishment, by the acuteness of the observations, the happy talent of combination, and the clearness of the systematic views. Such a work could not fail to procure considerable respect for its author. His _Hortus Cliffortiana_ {389} and _Musa Cliffortiana_ added to this impression. The weight which he had thus acquired, he proceeded to use for the improvement of botany. His _Fundamenta Botanica_ and _Bibliotheca Botanica_ appeared in 1736; his _Critica Botanica_ and _Genera Plantarum_ in 1737; his _Classes Plantarum_ in 1738; his _Species Plantarum_ was not published till 1753; and all these works appeared in many successive editions, materially modified.

[Note 87\16: Ibid. 234.]

This circulation of his works showed that his labors were producing their effect. His reputation grew; and he was soon enabled to exert a personal, as well as a literary, influence, on students of natural history. He became Botanist Royal, President of the Academy of Sciences at Stockholm, and Professor in the University of Upsal; and this office he held for thirty-six years with unrivalled credit; exercising, by means of his lectures, his constant publications, and his conversation, an extraordinary power over a multitude of zealous naturalists, belonging to every part of the world.

In order to understand more clearly the nature and effect of the reforms introduced by Linnæus into botany, I shall consider them under the four following heads;--_Terminology_, _Nomenclature_, _Artificial System_, and _Natural System_.

_Sect._ 2.--_Linnæan Reform of Botanical Terminology._

IT must be recollected that I designate as _Terminology_, the system of _terms_ employed in the _description_ of objects of natural history; while by _Nomenclature_, I mean the collection of the _names_ of _species_. The reform of the descriptive part of botany was one of the tasks first attempted by Linnæus; and his terminology was the instrument by which his other improvements were effected.

Though most readers, probably, entertain, at first, a persuasion that a writer ought to content himself with the use of common words in their common sense, and feel a repugnance to technical terms and arbitrary rules of phraseology, as pedantic and troublesome; it is soon found, by the student of any branch of science that, without technical terms and fixed rules, there can be no certain or progressive knowledge. The loose and infantine grasp of common language cannot hold objects steadily enough for scientific examination, or lift them from one stage of generalization to another. They must be secured by the rigid mechanism of a scientific phraseology. This necessity had been felt in all the sciences, from the earliest periods of their progress. But the {390} conviction had never been acted upon so as to produce a distinct and adequate descriptive botanical language. Jung, indeed,[88\16] had already attempted to give rules and precepts which should answer this purpose; but it was not till the _Fundamenta Botanica_ appeared, that the science could be said to possess a fixed and complete terminology.

[Note 88\16: _Isagoge Phytoscopica_, 1679.]

To give an account of such a terminology, is, in fact, to give a description of a dictionary and grammar, and is therefore what cannot here be done in detail. Linnæus's work contains about a thousand terms of which the meaning and application are distinctly explained; and rules are given, by which, in the use of such terms, the botanist may avoid all obscurity, ambiguity, unnecessary prolixity and complexity, and even inelegance and barbarism. Of course the greater part of the words which Linnæus thus recognized had previously existed in botanical writers; and many of them had been defined with technical precision. Thus Jung[89\16] had already explained what was a _composite_, what a _pinnate_ leaf; what kind of a bunch of flowers is a _spike_, a _panicle_, an _umbel_, a _corymb_, respectively. Linnæus extended such distinctions, retaining complete clearness in their separation. Thus, with him, composite leaves are further distinguished as _digitate_, _pinnate_, _bipinnate_, _pedate_, and so on; pinnate leaves are _abruptly_ so, or _with an odd_ one, or _with a tendril_; they are pinnate _oppositely_, _alternately_, _interruptedly_, _articulately_, _decursively_. Again, the _inflorescence_, as the mode of assemblage of the flowers is called, may be a _tuft_ (fasciculus), a _head_ (capitulum), a _cluster_ (racemus), a _bunch_ (thyrsus), a _panicle_, a _spike_, a _catkin_ (amentum), a _corymb_, an _umbel_, a _cyme_, a _whorl_ (verticillus). And the rules which he gives, though often apparently arbitrary and needless, are found, in practice, to be of great service by their fixity and connexion. By the good fortune of having had a teacher with so much delicacy of taste as Linnæus, in a situation of so much influence, Botany possesses a descriptive language which will long stand as a model for all other subjects.

[Note 89\16: Sprengel, ii. 28.]

It may, perhaps, appear to some persons, that such a terminology as we have here described must be enormously cumbrous; and that, since the terms are arbitrarily invested with their meaning, the invention of them requires no knowledge of nature. With respect to the former doubt, we may observe, that technical description is, in reality, the only description which is clearly intelligible; but that technical language cannot be understood without being learnt as any other {391} language is learnt; that is, the reader must connect the terms immediately with his own sensations and notions, and not mediately, through a verbal explanation; he must not have to guess their meaning, or to discover it by a separate act of interpretation into more familiar language as often as they occur. The language of botany must be the botanist's most familiar tongue. When the student has thus learnt to _think_ in botanical language, it is no idle distinction to tell him that a _bunch_ of grapes is not a _cluster_; that is, a _thyrsus_ not a _raceme_. And the terminology of botany is then felt to be a useful implement, not an oppressive burden. It is only the schoolboy that complains of the irksomeness of his grammar and vocabulary. The accomplished student possesses them without effort or inconvenience.

As to the other question, whether the construction of such a botanical grammar and vocabulary implies an extensive and accurate acquaintance with the facts of nature, no one can doubt who is familiar with any descriptive science. It is true, that a person might construct an arbitrary scheme of distinctions and appellations, with no attention to natural objects; and this is what shallow and self-confident persons often set about doing, in some branch of knowledge with which they are imperfectly acquainted. But the slightest attempt to use such a phraseology leads to confusion; and any continued use of it leads to its demolition. Like a garment which does not fit us, if we attempt to work in it we tear it in pieces.

The formation of a good descriptive language is, in fact, an inductive process of the same kind as those which we have already noticed in the progress of natural history. It requires the _discovery of fixed characters_, which discovery is to be marked and fixed, like other inductive steps, by appropriate _technical terms_. The characters must be so far fixed, that the things which they connect must have a more permanent and real association than the things which they leave unconnected. If one bunch of grapes were really a racemus, and another a thyrsus, according to the definition of these terms, this part of the Linnæan language would lose its value; because it would no longer enable us to assert a general proposition with respect to one kind of plants.

_Sect._ 3.--_Linnæan Reform of Botanical Nomenclature._

IN the ancient writers each recognized kind of plants had a distinct name. The establishment of Genera led to the practice of designating {392} Species by the name of the genus, with the addition of a "phrase" to distinguish the species. These phrases, (expressed in Latin in the ablative case,) were such as not only to mark, but to describe the species, and were intended to contain such features of the plant as were sufficient to distinguish it from others of the same genus. But in this way the designation of a plant often became a long and inconvenient assemblage of words.

Thus different kinds of Rose were described as, Rosa campestris, spinis carens, biflora (_Rosa alpina_.) Rosa aculeata, foliis odoratis subtus rubiginosis (_R. eglanteria_.) Rosa carolina fragrans, foliis medio tenus serratis (_R. carolina_.) Rosa sylvestris vulgaris, flore odorato incarnato (_R. canina_.) And several others. The prolixity of these appellations, their variety in every different author, the insufficiency and confusion of the distinctions which they contained, were felt as extreme inconveniences. The attempt of Bauhin to remedy this evil by a Synonymy, had, as we have seen, failed at the time, for want of any directing principle; and was become still more defective by the lapse of years and the accumulation of fresh knowledge and new books. Haller had proposed to distinguish the species of each genus by the numbers 1, 2, 3, and so on; but botanists found that their memory could not deal with such arbitrary abstractions. The need of some better nomenclature was severely felt.

The remedy which Linnæus finally introduced was the use of _trivial_ names; that is, the designation of each species by the name of the genus along with a _single_ conventional word, imposed without any general rule. Such names are added above in parentheses, to the specimens of the names previously in use. But though this remedy was found to be complete and satisfactory, and is now universally adopted in every branch of natural history, it was not one of the reforms which Linnæus at first proposed. Perhaps he did not at first see its full value; or, if he did, we may suppose that it required more self-confidence than he possessed, to set himself to introduce and establish ten thousand new names in the botanical world. Accordingly, the first attempts of Linnæus at the improvement of the nomenclature of botany were, the proposal of fixed and careful rules for the generic name, and for the descriptive phrase. Thus, in his _Critica Botanica_, he gives many precepts concerning the selection of the names of {393} genera, intended to secure convenience or elegance. For instance, that they are to be single words;[90\16] he substitutes _atropa_ for _bella donna_, and _leontodon_ for _dens leonis_; that they are not to depend upon the name of another genus,[91\16] as _acriviola_, _agrimonoides_; that they are not[92\16] to be "sesquipedalia;" and, says he, any word is sesquipedalian to me, which has more than twelve letters, as _kalophyllodendron_, for which he substitutes _calophyllon_. Though some of these rules may seem pedantic, there is no doubt that, taken altogether, they tend exceedingly, like the labors of purists in other languages, to exclude extravagance, caprice, and barbarism in botanical speech.

[Note 90\16: _Phil. Bot._ 224.]

[Note 91\16: Ib. 228, 229.]

[Note 92\16: Ib. 252.]

The precepts which he gives for the matter of the "descriptive phrase," or, as it is termed in the language of the Aristotelian logicians, the "differentia," are, for the most part, results of the general rule, that the most fixed characters which can be found are to be used; this rule being interpreted according to all the knowledge of plants which had then been acquired. The language of the rules was, of course, to be regulated by the terminology, of which we have already spoken.

Thus, in the _Critica Botanica_, the name of a plant is considered as consisting of a generic word and a specific phrase; and these are, he says,[93\16] the right and left hands of the plant. But he then speaks of another kind of name; the _trivial_ name, which is opposed to the scientific. Such names were, he says,[94\16] those of his predecessors, and especially of the most ancient of them. Hitherto[95\16] no rules had been given for their use. He manifestly, at this period, has small regard for them. "Yet," he says, "trivial names may, perhaps, be used on this account,--that the _differentia_ often turns out too long to be convenient in common use, and may require change as new species are discovered. However," he continues, "in this work we set such names aside altogether, and attend only to the _differentiæ_."

[Note 93\16: Ib. 266.]

[Note 94\16: Ib. 261.]

[Note 95\16: Ib. 260.]

Even in the _Species Plantarum_, the work which gave general currency to these trivial names, he does not seem to have yet dared to propose so great a novelty. They only stand in the margin of the work. "I have placed them there," he says in his Preface, "that, without circumlocution, we may call every herb by a single name; I have done this without selection, which would require more time. And I beseech all sane botanists to avoid most religiously ever {394} proposing a trivial name without a sufficient specific distinction, lest the science should fall into its former barbarism."

It cannot be doubted, that the general reception of these trivial names of Linnæus, as the current language among botanists, was due, in a very great degree, to the knowledge, care, and skill with which his characters, both of genera and of species, were constructed. The rigorous rules of selection and expression which are proposed in the _Fundamenta Botanica_ and _Critica Botanica_, he himself conformed to; and this scrupulosity was employed upon the results of immense labor. "In order that I might make myself acquainted with the species of plants," he says, in the preface to his work upon them, "I have explored the Alps of Lapland, the whole of Sweden, a part of Norway, Denmark, Germany, Belgium, England, France: I have examined the Botanical Gardens of Paris, Oxford, Chelsea, Harlecamp, Leyden, Utrecht, Amsterdam, Upsal, and others: I have turned over the Herbals of Burser, Hermann, Clifford, Burmann, Oldenland, Gronovius, Royer, Sloane, Sherard, Bobart, Miller, Tournefort, Vaillant, Jussieu, **Surian, Beck, Brown, &c.: my dear disciples have gone to distant lands, and sent me plants from thence; Kerlen to Canada, Hasselquist to Egypt, Asbech to China, Toren to Surat, Solander to England, Alstrœmer to Southern Europe, Martin to Spitzbergen, Pontin to Malabar, Kœhler to Italy, Forskähl to the East, Lœfling to Spain, Montin to Lapland: my botanical friends have sent me many seeds and dried plants from various countries: Lagerström many from the East Indies; Gronovius most of the Virginian; Gmelin all the Siberian; Burmann those of the Cape." And in consistency with this habit of immense collection of materials, is his maxim,[96\16] that "a person is a better botanist in proportion as he knows more species." It will easily be seen that this maxim, like Newton's declaration that discovery requires patient thought alone, refers only to the exertions of which the

man of genius is conscious; and leaves out of sight his peculiar endowments, which he does not see because they are part of his power of vision. With the taste for symmetry which dictated the _Critica Botanica_, and the talent for classification which appears in the _Genera Plantarum_, and the _Systema Naturæ_, a person must undoubtedly rise to higher steps of classificatory knowledge and skill, as he became acquainted with a greater number of facts.

[Note 96\16: _Phil. Bot._ 259.]

The acknowledged superiority of Linnæus in the knowledge of the {395} matter of his science, induced other persons to defer to him in what concerned its form; especially when his precepts were, for the most part, recommended strongly both by convenience and elegance. The trivial names of the _Species Plantarum_ were generally received; and though some of the details may have been altered, the immense advantage of the scheme ensures its permanence.

_Sect._ 4.--_Linnæus's Artificial System._

WE have already seen, that, from the time of Cæsalpinus, botanists had been endeavoring to frame a systematic arrangement of plants. All such arrangements were necessarily both artificial and natural: they were _artificial_, inasmuch as they depended upon assumed principles, the number, form, and position of certain parts, by the application of which the whole vegetable kingdom was imperatively subdivided; they were _natural_, inasmuch as the justification of this division was, that it brought together those plants which were naturally related. No system of arrangement, for instance, would have been tolerated which, in a great proportion of cases, separated into distant parts of the plan the different species of the same genus. As far as the main body of the genera, at least, all systems are natural.

But beginning from this line, we may construct our systems with two opposite purposes, according as we endeavor to carry our assumed principle of division rigorously and consistently through the system, or as we wish to associate natural families of a wider kind than genera. The former propensity leads to an artificial, the latter to a natural method. Each is a _System of Plants_; but in the first, the emphasis is thrown on the former word of the title, in the other, on the latter.

The strongest recommendation of an artificial system, (besides its approaching to a natural method,) is, that it shall be capable of easy use; for which purpose, the facts on which it depends must be apparent in their relations, and universal in their occurrence. The system of Linnæus, founded upon the number, position, and other circumstances of the stamina and pistils, the reproductive organs of the plants, possessed this merit in an eminent degree, as far as these characters are concerned; that is, as far as the classes and orders. In its further subdivision into genera, its superiority was mainly due to the exact observation and description, which we have already had to notice as talents which Linnæus peculiarly possessed.

The Linnæan system of plants was more definite than that of {396} Tournefort, which was governed by the corolla; for number is more definite than irregular form. It was more readily employed than any of those which depend on the fruit, for the flower is a more obvious object, and more easily examined. Still, it can hardly be

doubted, that the circumstance which gave the main currency to the system of Linnæus was its physiological signification: it was the _Sexual System_. The relation of the parts to which it directed the attention, interested both the philosophical faculty and the imagination. And when, soon after the system had become familiar in our own country, the poet of _The Botanic Garden_ peopled the bell of every flower with "Nymphs" and "Swains," his imagery was felt to be by no means forced and far-fetched.

The history of the doctrine of the sexes of plants, as a point of physiology, does not belong to this place; and the Linnæan system of classification need not be longer dwelt upon for our present purpose. I will only explain a little further what has been said, that it is, up to a certain point, a natural system. Several of Linnæus's classes are, in a great measure, natural associations, kept together in violation of his own artificial rules. Thus the class _Diadelphia_, in which, by the system, the filaments of the stamina should be bound together in two parcels, does, in fact, contain many genera which are_ monadelphous_, the filaments of the stamina all cohering so as to form one bundle only; as in _Genista_, _Spartium_, _Anthyllis_, _Lupinus_, &c. And why is this violation of rule? Precisely because these genera all belong to the natural tribe of Papilionaceous plants, which the author of the system could not prevail upon himself to tear asunder. Yet in other cases Linnæus was true to his system, to the injury of natural alliances, as he was, for instance, in another portion of this very tribe of _Papilionaceæ_; for there are plants which undoubtedly belong to the tribe, but which have ten separate stamens; and these he placed in the order _Decandria_. Upon the whole, however, he inclines rather to admit transgression of art than of nature.

The reason of this inclination was, that he rightly considered an artificial method as instrumental to the investigation of a natural one; and to this part of his views we now proceed.

_Sect._ 5.--_Linnæus's Views on a Natural Method._

THE admirers of Linnæus, the English especially, were for some time in the habit of putting his Sexual System in opposition to the Natural Method, which about the same time was attempted in France. And {397} as they often appear to have imagined that the ultimate object of botanical methods was to know the name of plants, they naturally preferred the Swedish method, which is excellent as a _finder_. No person, however, who wishes to know botany as a science, that is, as a body of general truths, can be content with making names his ultimate object. Such a person will be constantly and irresistibly led on to attempt to catch sight of the natural arrangement of plants, even before he discovers, as he will discover by pursuing such a course of study, that the knowledge of the natural arrangement is the knowledge of the essential construction and vital mechanism of plants. He will consider an artificial method as a means of arriving at a natural method. Accordingly, however much some of his followers may have overlooked this, it is what Linnæus himself always held and taught. And though what he executed with regard to this object was but little,[97\16] the distinct manner in which he presented the relations of an artificial and natural method, may justly be looked upon as one of the great improvements which he

introduced into the study of his science.

[Note 97\16: The natural orders which he proposed are a bare enumeration of genera, and have not been generally followed.]

Thus in the _Classes Plantarum_ (1747), he speaks of the difficulty of the task of discovering the natural orders, and of the attempts made by others. "Yet," he adds, "I too have labored at this, have done something, have much still to do, and shall labor at the object as long as I live." He afterwards proposed sixty-seven orders, as the fragments of a natural method, always professing their imperfection.[98\16] And in others of his works[99\16] he lays down some antitheses on the subject after his manner. "The natural orders teach us the nature of plants; the artificial orders enable us to recognize plants. The natural orders, without a key, do not constitute a Method; the Method ought to be available without a master."

[Note 98\16: _Phil. Bot._ p. 80.]

[Note 99\16: _Genera Plantarum_, 1764. See _Prælect. in Ord. Nat._ p. xlviii.]

That extreme difficulty must attend the formation of a Natural Method, may be seen from the very indefinite nature of the Aphorisms upon this subject which Linnæus has delivered, and which the best botanists of succeeding times have assented to. Such are these;--the Natural Orders must be formed by attention, not to one or two, but to _all_ the parts of plants;--the same organs are of great importance in regulating the divisions of one part of the system, and {398} of small importance in another part;[100\16]--the Character does not constitute the Genus, but the Genus the Character;--the Character is necessary, not to make the Genus, but to recognize it. The vagueness of these maxims is easily seen; the rule of attending to all the parts, implies, that we are to estimate their relative importance, either by physiological considerations (and these again lead to arbitrary rules, as, for instance, the superiority of the function of nutrition to that of reproduction), or by a sort of latent naturalist instinct, which Linnæus in some passages seems to recognize. "The Habit of a plant," he says,[101\16] "must be secretly consulted. A practised botanist will distinguish, at the first glance, the plants of different quarters of the globe, and yet will be at a loss to tell by what mark he detects them. There is, I know not what look,--sinister, dry, obscure in African plants; superb and elevated, in the Asiatic; smooth and cheerful, in the American; stunted and indurated, in the Alpine."

[Note 100\16: _Phil. Bot._ p. 172.]

[Note 101\16: Ib. p. 171.]

Again, the rule that the same parts are of very different value in different Orders, not only leaves us in want of rules or reasons which may enable us to compare the marks of different Orders, but destroys the systematic completeness of the natural arrangement. If some of the Orders be regulated by the flower and others by the fruit, we may have plants, of which the flower would place them in one Order, and the fruit in another. The answer to this difficulty is the maxim already stated;--that no Character _makes_ the Order; and that if a Character do not enable us to recognize the Order, it does not answer its purpose, and ought to be changed for another.

This doctrine, that the Character is to be employed as a servant and not as a master, was a stumbling-block in the way of those disciples who looked only for dogmatical and universal rules. One of Linnæus's pupils, Paul Dietrich Giseke, has given us a very lively account of his own perplexity on having this view propounded to him, and of the way in which he struggled with it. He had complained of the want of intelligible grounds, in the collection of natural orders given by Linnæus. Linnæus[102\16] wrote in answer, "You ask me for the characters of the Natural Orders: I confess I cannot give them." Such a reply naturally increased Giseke's difficulties. But afterwards, in 1771, he had the good fortune to spend some time at Upsal; and he narrates a conversation which he held with the great {399} teacher on this subject, and which I think may serve to show the nature of the difficulty;--one by no means easily removed, and by the general reader, not even readily comprehended with distinctness. Giseke began by conceiving that an Order _must_ have that attribute from which its name is derived;--that the _Umbellatæ_ must have their flower disposed in an umbel. The "mighty master" smiled,[103\16] and told him not to look at names, but at nature. "But" (said the pupil) "what is the use of the name, if it does not mean what it professes to mean?" "It is of small import" (replied Linnæus) "_what_ you _call_ the Order, if you take a proper series of plants and give it some name, which is clearly understood to apply to the plants which you have associated. In such cases as you refer to, I followed the logical rule, of borrowing a name _a potiori_, from the principal member. Can you" (he added) "give me the character of any single Order?" _Giseke._ "Surely, the character of the _Umbellatæ_ is, that they have an umbel?" _Linnæus._ "Good; but there are plants which have an umbel, and are not of the _Umbellatæ_." _G._ "I remember. We must therefore add, that they have two naked seeds." _L._ "Then, _Echinophora_, which has only one seed, and _Eryngium_, which has not an umbel, will not be _Umbellatæ_; and yet they are of the Order." _G._ "I would place _Eryngium_ among the _Aggregatæ_. _L._ "No; both are beyond dispute _Umbellatæ_. _Eryngium_ has an involucrum, five stamina, two pistils, &c. Try again for your Character." _G._ "I would transfer such plants to the end of the Order, and make them form the transition to the next Order. _Eryngium_ would connect the _Umbellatæ_ with the _Aggregatæ_." _L._ "Ah! my good friend, the _Transition_ from Order to Order is one thing; the _Character_ of an Order is another. The Transitions I could indicate; but a Character of a Natural Order is impossible. I will not give my reasons for the distribution of Natural Orders which I have published. You or some other person, after twenty or after fifty years, will discover them, and see I was in the right."

[Note 102\16: _Linnæi Prælectiones_, Pref. p. xv.]

[Note 103\16: "Subrisit ὁ πανυ."]

I have given a portion of this curious conversation in order to show that the attempt to establish Natural Orders leads to convictions which are out of the domain of the systematic grounds on which they profess to proceed. I believe the real state of the case to be that the systematist, in such instances, is guided by an unformed and undeveloped apprehension of physiological functions. The ideas of the form, {400} number, and figure of parts are, in some measure, overshadowed and superseded by

the rising perception of organic and vital relations; and the philosopher who aims at a Natural Method, while he is endeavoring merely to explore the apartment in which he had placed himself, that of Arrangement, is led beyond it, to a point where another light begins, though dimly, to be seen; he is brought within the influence of the ideas of Organization and Life.

The sciences which depend on these ideas will be the subject of our consideration hereafter. But what has been said may perhaps serve to explain the acknowledged and inevitable imperfection of the unphysiological Linnæan attempts towards a natural method. "Artificial Glasses are," Linnæus says, "a substitute for Natural, till Natural are detected." But we have not yet a Natural Method. "Nor," he says, in the conversation above cited, "can we have a Natural Method; for a Natural Method implies Natural Classes and Orders; and these Orders must have Characters." "And they," he adds in another place,[104\16] "who, though they cannot obtain a complete Natural Method, arrange plants according to the fragments of such a method, to the rejection of the Artificial, seem to me like persons who pull down a convenient vaulted room, and set about building another, though they cannot turn the vault which is to cover it."

[Note 104\16: _Gen. Plant. in Prælect._ p. xii.]

How far these considerations deterred other persons from turning their main attention to a natural method, we shall shortly see; but in the mean time, we must complete the history of the Linnæan Reform.

_Sect._ 6.--_Reception and Diffusion of the Linnæan Reform._

WE have already seen that Linnæus received, from his own country, honors and emoluments which mark his reputation as established, as early as 1740; and by his publications, his lectures, and his personal communications, he soon drew round him many disciples, whom he impressed strongly with his own doctrines and methods. It would seem that the sciences of classification tend, at least in modern times more than other sciences, to collect about the chair of the teacher a large body of zealous and obedient pupils; Linnæus and Werner were by far the most powerful heads of schools of any men who appeared in the course of the last century. Perhaps one reason of this is, that in these sciences, consisting of such an enormous multitude of species, of descriptive {401} particulars, and of previous classifications, the learner is dependent upon the teacher more completely, and for a longer time than in other subjects of speculation: he cannot so soon or so easily cast off the aid and influence of the master, to pursue reasonings and hypotheses of his own. Whatever the cause may be, the fact is, that the reputation and authority of Linnæus, in the latter part of his life, were immense. He enjoyed also royal favor, for the King and Queen of Sweden were both fond of natural history. In 1753, Linnæus received from the hand of his sovereign the knighthood of the Polar Star, an honor which had never before been conferred for literary merit; and in 1756, was raised to the rank of Swedish nobility by the title of Von Linné; and this distinction was confirmed by the Diet in 1762. He lived, honored and courted, to the age of seventy-one; and in 1778 was buried in the cathedral of Upsal, with many testimonials of public respect and veneration.

De Candolle[105\16] assigns, as the causes of the successes of the Linnæan system,--the specific names,--the characteristic phrase,--the fixation of descriptive language,--the distinction of varieties and species,--the extension of the method to all the kingdoms of nature,--and the practice of introducing into it the species most recently discovered. This last course Linnæus constantly pursued; thus making his works the most valuable for matter, as they were the most convenient in form. The general diffusion of his methods over Europe may be dated, perhaps, a few years after 1760, when the tenth and the succeeding editions of the _Systema Naturæ_ were in circulation, professing to include every species of organized beings. But his pupils and correspondents effected no less than his books, in giving currency to his system. In Germany,[106\16] it was defended by Ludwig, Gesner, Fabricius. But Haller, whose reputation in physiology was as great as that of Linnæus in methodology, rejected it as too merely artificial. In France, it did not make any rapid or extensive progress: the best French botanists were at this time occupied with the solution of the great problem of the construction of a Natural Method. And though the rhetorician Rousseau charmed, we may suppose, with the elegant precision of the _Philosophia Botanica_, declared it to be the most philosophical work he had ever read in his life, Buffon and Andanson, describers and philosophers of a more ambitious school, felt a repugnance to the rigorous rules, and limited, but finished, undertakings of the Swedish naturalist. To resist his {402} criticism and his influence, they armed themselves with dislike and contempt.

[Note 105\16: _Théor. Elém._ p. 40.]

[Note 106\16: Sprengel, ii. 244.]

In England the Linnæan system was very favorably received:--perhaps the more favorably, for being a strictly artificial system. For the indefinite and unfinished form which almost inevitably clings to a natural method, appears to be peculiarly distasteful to our countrymen. It might seem as if the suspense and craving which comes with knowledge confessedly incomplete were so disagreeable to them, that they were willing to avoid it, at any rate whatever; either by rejecting system altogether, or by accepting a dogmatical system without reserve. The former has been their course in recent times with regard to Mineralogy; the latter was their proceeding with respect to the Linnæan Botany. It is in this country alone, I believe, that _Wernerian_ and _Linnæan_ Societies have been instituted. Such appellations somewhat remind us of the Aristotelian and Platonic schools of ancient Greece. In the same spirit it was, that the Artificial System was at one time here considered, not as subsidiary and preparatory to the Natural Orders, but as opposed to them. This was much as if the disposition of an army in a review should be considered as inconsistent with another arrangement of it in a battle.

When Linnæus visited England in 1736, Sloane, then the patron of natural history in this country, is said to have given him a cool reception, such as was perhaps most natural from an old man to a young innovator; and Dillenius, the Professor at Oxford, did not accept the sexual system. But as Pulteney, the historian of English Botany, says, when his works became known, "the simplicity of the classical characters, the uniformity of the generic notes, all confined to the parts of the fructification, and the

precision which marked the specific distinctions, merits so new, soon commanded the assent of the unprejudiced."

Perhaps the progress of the introduction of the Linnæan System into England will be best understood from the statement of T. Martyn, who was Professor of Botany in the University of Cambridge, from 1761 to 1825. "About the year 1750," he says,[107\16] "I was a pupil of the school of our great countryman Ray; but the rich vein of knowledge, the profoundness and precision, which I remarked everywhere in the _Philosophia Botanica_, (published in 1751,) withdrew me from my first master, and I became a decided convert to that system of botany which has since been generally received. In 1753, the _Species_ {403} _Plantarum_, which first introduced the specific names, made me a Linnæan completely." In 1763, he introduced the system in his lectures at Cambridge, and these were the first Linnæan lectures in England. Stillingfleet had already, in 1757, and Lee, in 1760, called the attention of English readers to Linnæus. Sir J. Hill, (the king's gardener at Kew,) in his _Flora Britannica_, published in 1760, had employed the classes and generic characters, but not the nomenclature; but the latter was adopted by Hudson, in 1762, in the _Flora Anglica_.

[Note 107\16: Pref. to _Language of Botany_, 3rd edit. 1807.]

Two young Swedes, pupils of Linnæus, Dryander and Solander, settled in England, and were in intimate intercourse with the most active naturalists, especially with Sir Joseph Banks, of whom the former was librarian, and the latter a fellow-traveller in Cook's celebrated voyage. James Edward Smith was also one of the most zealous disciples of the Linnæan school; and, after the death of Linnæus, purchased his Herbariums and Collections. It is related,[108\16] as a curious proof of the high estimation in which Linnæus was held, that when the Swedish government heard of this bargain, they tried, though too late, to prevent these monuments of their countryman's labor and glory being carried from his native land, and even went so far as to send a frigate in pursuit of the ship which conveyed them to England. Smith had, however, the triumph of bringing them home in safety. On his death they were purchased by the Linnæan Society. Such relics serve, as will easily be imagined, not only to warm the reverence of his admirers, but to illustrate his writings: and since they have been in this country, they have been the object of the pilgrimage of many a botanist, from every part of Europe.

[Note 108\16: Trapp's _Transl. of Stower's Life of Linnæus_, p. 314.]

I have purposely confined myself to the history of the Linnæan system in the cases in which it is most easily applicable, omitting all consideration of more obscure and disputed kinds of vegetables, as ferns, mosses, fungi, lichens, sea-weeds, and the like. The nature and progress of a classificatory science, which it is our main purpose to bring into view, will best be understood by attending, in the first place, to the cases in which such a science has been pursued with the most decided success; and the advances which have been made in the knowledge of the more obscure vegetables, are, in fact, advances in artificial classification, only in as far as they are advances in natural classification, and in physiology.

To these subjects we now proceed. {404}

## CHAPTER V.

PROGRESS TOWARDS A NATURAL SYSTEM OF BOTANY.

WE have already said, that the formation of a Natural System of classification must result from a comparison of _all_ the resemblances and differences of the things classed; but that, in acting upon this maxim, the naturalist is necessarily either guided by an obscure and instinctive feeling, which is, in fact, an undeveloped recognition of physiological relations, or else acknowledges physiology for his guide, though he is obliged to assume arbitrary rules in order to interpret its indications. Thus all Natural Classification of organized beings, either begins or soon ends in Physiology; and can never advance far without the aid of that science. Still, the progress of the Natural Method in botany went to such a length before it was grounded entirely on the anatomy of plants, that it will be proper, and I hope instructive, to attempt a sketch of it here.

As I have already had occasion to remark, the earlier systems of plants were natural; and they only ceased to be so, when it appeared that the problem of constructing a _system_ admitted of a very useful solution, while the problem of devising a _natural system_ remained insoluble. But many botanists did not so easily renounce the highest object of their science. In France, especially, a succession of extraordinary men labored at it with no inconsiderable success: and they were seconded by worthy fellow-laborers in Germany and elsewhere.

The precept of taking into account all the parts of plants according to their importance, may be applied according to arbitrary rules. We may, for instance, assume that the fruit is the most important part; or we may make a long list of parts, and look for agreement in the greatest possible number of these, in order to construct our natural orders. The former course was followed by Gærtner;[109\16] the latter by Adanson. Gærtner's principles, deduced from the dissection of more than a thousand kinds of fruits,[110\16] exercised, in the sequel, a great and {405} permanent influence on the formation of natural classes. Adanson's attempt, bold and ingenious, belonged, both in time and character, to a somewhat earlier stage of the subject.[111\16] Enthusiastic and laborious beyond belief but self-confident, and contemptuous of the labors of others, Michael Adanson had collected, during five years spent in Senegal, an enormous mass of knowledge and materials; and had formed plans for the systems which he conceived himself thus empowered to reach, far beyond the strength and the lot of man.[112\16] In his _Families of Plants_, however, all agree that his labors were of real value to the science. The method which he followed is thus described by his eloquent and philosophical eulogist.[113\16]

[Note 109\16: _De Fructibus et Seminibus Plantarum_. Stuttg. 1788-1791.]

[Note 110\16: Sprengel, ii. 290.]

[Note 111\16: _Familles des Plantes_, 1763.]

[Note 112\16: Cuvier's _Eloge_.]

[Note 113\16: Cuv. _Eloges_, tom. i. p. 282.]

Considering each organ by itself, he formed, by pursuing its various modifications, a system of division, in which he arranged all known species according to that organ alone. Doing the same for another organ, and another, and so for many, he constructed a collection of systems of arrangement, each artificial,--each founded upon one assumed organ. The species which come together in all these systems are, of all, naturally the nearest to each other; those which are separated in a few of the systems, but contiguous in the greatest number, are naturally near to each other, though less near than the former; those which are separated in a greater number, are further removed from each other in nature; and they are the more removed, the fewer are the systems in which they are associated.

Thus, by this method, we obtain the means of estimating precisely the degree of natural affinity of all the species which our systems include, independent of a physiological knowledge of the influence of the organs. But the method has, Cuvier adds, the inconvenience of presupposing another kind of knowledge, which, though it belongs only to descriptive natural history, is no less difficult to obtain;--the knowledge, namely, of all species, and of all the organs of each. A single one neglected, may lead to relations the most false; and Adanson himself, in spite of the immense number of his observations, exemplifies this in some instances.

We may add, that in the division of the structure into organs, and in the estimation of the gradations of these in each artificial system, there is still room for arbitrary assumption.

In the mean time, the two Jussieus had presented to the world a "Natural Method," which produced a stronger impression than the {406} "Universal Method" of Adanson. The first author of the system was Bernard de Jussieu, who applied it in the arrangement of the garden of the Trianon, in 1759, though he never published upon it. His nephew, Antoine Laurent de Jussieu, in his _Treatise of the Arrangement of the Trianon_,[114\16] gave an account of the principles and orders of his uncle, which he adopted when he succeeded him; and, at a later period, published his _Genera Plantarum secundum Ordines Naturales disposita_; a work, says Cuvier, which perhaps forms as important an epoch in the sciences of observation, as the _Chimie_ of Lavoisier does in the sciences of experiment. The object of the Jussieus was to obtain a system which should be governed by the natural affinities of the plants, while, at the same time, the characters by which the orders were ostensibly determined, should be as clear, simple, and precise, as those of the best artificial system. The main points in these characters were the number of the cotyledons, and the structure of the seed: and subordinate to this, the insertion of the stamina, which they distinguished as _epigynous_, _perigynous_, and _hypogynous_, according as they were inserted over, about, or under, the germen. And the classes which were formed by the Jussieus, though they have since been modified by succeeding writers, have been so far retained by the most profound botanists, notwithstanding all the new care and new light which have been bestowed upon the subject, as to show that what was done at first, was a real and important step in the solution of the problem.

[Note 114\16: _Mém. Ac. P._ 1774.]

The merit of the formation of this natural method of plants must be divided between the two Jussieus. It has been common to speak of the nephew, Antoine Laurent, as only the publisher of his uncle's work.[115\16] But this appears, from a recent statement,[116\16] to be highly unjust. Bernard left nothing in writing but the catalogues of the garden of the Trianon, which he had arranged according to his own views; but these catalogues consist merely of a series of names without explanation or reason added. The nephew, in 1773, undertook and executed for himself the examination of a natural family, the _Ranunculaceæ_; and he was wont to relate (as his son informs us) that it {407} was this employment which first opened his eyes and rendered him a botanist. In the memoir which he wrote, he explained fully the relative importance of the characters of plants, and the subordination of some to others;--an essential consideration, which Adanson's scheme had failed to take account of. The uncle died in 1777; and his nephew, in speaking of him, compares his arrangement to the _Ordines Naturales_ of Linnæus: "Both these authors," he says, "have satisfied themselves with giving a catalogue of genera which approach each other in different points, without explaining the motives which induced them to place one order before another, or to arrange a genus under a certain order. These two arrangements may be conceived as problems which their authors have left for botanists to solve. Linnæus published his; that of M. de Jussieu is only known by the manuscript catalogues of the garden of the Trianon."

[Note 115\16: _Prodromus Floræ Penins. Ind. Orient._ Wight and Walker-Arnott, Introd. p. xxxv.]

[Note 116\16: By Adrien de Jussieu, son of Antoine Laurent, in the _Annales des Sc. Nat._, Nov. 1834.]

It was not till the younger Jussieu had employed himself for nineteen years upon botany, that he published, in 1789, his _Genera Plantarum_; and by this time he had so entirely formed his scheme in his head, that he began the impression without having written the book, and the manuscript was never more than two pages in advance of the printer's type.

When this work appeared, it was not received with any enthusiasm; indeed, at that time, the revolution of states absorbed the thoughts of all Europe, and left men little leisure to attend to the revolutions of science. The author himself was drawn into the vortex of public affairs, and for some years forgot his book. The method made its way slowly and with difficulty: it was a long time before it was comprehended and adopted in France, although the botanists of that country had, a little while before, been so eager in pursuit of a natural system. In England and Germany, which had readily received the Linnæan method, its progress was still more tardy.

There is only one point, on which it appears necessary further to dwell. A main and fundamental distinction in all natural systems, is that of the Monocotyledonous and Dicotyledonous plants; that is, plants which unfold themselves from an embryo with two little leaves, or with one leaf only. This distinction produces its effects in the systems which are regulated by numbers; for the flowers and fruit of the

monocotyledons are generally referrible to some law in which the number _three_ prevails; a type which rarely occurs in dicotyledons, these affecting most commonly an arrangement founded on the number _five_. But it appears, when we attempt to rise towards a natural {408} method, that this division according to the cotyledons is of a higher order than the other divisions according to number; and corresponds to a distinction in the general structure and organization of the plant. The apprehension of the due rank of this distinction has gradually grown clearer. Cuvier[117\16] conceives that he finds such a division clearly marked in Lobel, in 1581, and employed by Ray as the basis of his classification a century later. This difference has had its due place assigned it in more recent systems of arrangement; but it is only later still that its full import has been distinctly brought into view. Desfontaines discovered[118\16] that the ligneous fibre is developed in an opposite manner in vegetables with one and with two cotyledons;--towards the inside in the former case, and towards the outside in the latter; and hence these two great classes have been since termed _endogenous_ and _exogenous_.

[Note 117\16: _Hist. Sc. Nat._ ii. 197.]

[Note 118\16: _Hist. Sc. Nat._ i. pp. 196, 290.]

Thus this division, according to the cotyledons, appears to have the stamp of reality put upon it, by acquiring a physiological meaning. Yet we are not allowed to forget, even at this elevated point of generalization, that _no one_ character can be imperative in a natural method. Lamarck, who employed his great talents on botany, before he devoted himself exclusively to other branches of natural history, published his views concerning methods, systems,[119\16] and characters. His main principle is, that no single part of a plant, however essential, can be an absolute rule for classification; and hence he blames the Jussieuian method, as giving this inadmissible authority to the cotyledons. Roscoe[120\16] further urges that some plants, as _Orchis morio_, and _Limodorum verecundum_, have no visible cotyledons. Yet De Candolle, who labored along with Lamarck, in the new edition of the _Flore Française_, has, as we have already intimated, been led, by the most careful application of the wisest principles, to a system of Natural Orders, of which Jussieu's may be looked upon as the basis; and we shall find the greatest botanists, up to the most recent period, recognizing, and employing themselves in improving, Jussieu's Natural Families; so that in the progress of this part of our knowledge, vague and perplexing as it is, we have no exception to our general aphorism, that no real acquisition in science is ever discarded. {409}

[Note 119\16: Sprengel, ii. 296; and, there quoted, _Flore Française_, t. i. 3, 1778. _Mém. Ac. P._ 1785. _Journ. Hist. Nat._ t. i. For Lamarck's _Méthode Analytique_, see Dumeril, _Sc. Nat._ i. Art. 390.]

[Note 120\16: Roscoe, _Linn. Tr._ vol. xi. _Cuscuta_ also has no cotyledons.]

The reception of the system of Jussieu in this country was not so ready and cordial as that of Linnæus. As we have already noticed, the two systems were looked upon as rivals. Thus Roscoe, in 1810,[121\16] endeavored to show that Jussieu's system was not more natural than the Linnæan, and was inferior as an artificial system: but he

argues his points as if Jussieu's characters were the grounds of his distribution; which, as we have said, is to mistake the construction of a natural system. In 1803, Salisbury[122\16] had already assailed the machinery of the system, maintaining that there are no cases of perigynous stamens, as Jussieu assumes; but this he urges with great expressions of respect for the author of the method. And the more profound botanists of England soon showed that they could appreciate and extend the natural method. Robert Brown, who had accompanied Captain Flinders to New Holland in 1801, and who, after examining that country, brought home, in 1805, nearly four thousand species of plants, was the most distinguished example of this. In his preface to the _Prodromus Floræ Novæ Hollandiæ_, he says, that he found himself under the necessity of employing the natural method, as the only way of avoiding serious error, when he had to deal with so many new genera as occur in New Holland; and that he has, therefore, followed the method of Jussieu; the greater part of whose orders are truly natural, "although their arrangement in classes, as is," he says, "conceded by their author, no less candid than learned, is often artificial, and, as appears to me, rests on doubtful grounds."

[Note 121\16: _Linn. Tr._ vol. xi. p. 50.]

[Note 122\16: Ibid. vol. viii.]

From what has already been said, the reader will, I trust, see what an extensive and exact knowledge of the vegetable world, and what comprehensive views of affinity, must be requisite in a person who has to modify the natural system so as to make it suited to receive and arrange a great number of new plants, extremely different from the genera on which the arrangement was first formed, as the New Holland genera for the most part were. He will also see how impossible it must be to convey by extract or description any notion of the nature of these modifications: it is enough to say, that they have excited the applause of botanists wherever the science is studied, and that they have induced M. de Humboldt and his fellow-laborers, themselves botanists of the first rank, to dedicate one of their works to him in terms of the strongest admiration.[123\16] Mr. Brown has also published {410} special disquisitions on parts of the Natural System; as on Jussieu's _Proteaceæ_;[124\16] on the _Asclepiadeæ_, a natural family of plants which must be separated from Jussieu's _Apocyneæ_;[125\16] and other similar labors.

[Note 123\16: Roberto Brown, Britanniarum gloriæ atque ornamento, totam Botanices scientiam ingenio mirifico complectenti. &c.]

[Note 124\16: _Linn. Tr._ vol. x. 1809.]

[Note 125\16: _Mem. of Wernerian N. H. Soc._ vol. i. 1809.]

We have, I think, been led, by our survey of the history of Botany, to this point;--that a Natural Method directs us to the study of Physiology, as the only means by which we can reach the object. This conviction, which in botany comes at the end of a long series of attempts at classification, offers itself at once in the natural history of animals, where the physiological signification of the resemblances and differences is so much more obvious. I shall not, therefore, consider any of these branches of natural history in detail as examples of mere classification. They will come before us,

if at all, more properly when we consider the classifications which depend on the functions of organs, and on the corresponding modifications which they necessarily undergo; that is, when we trace the results of Physiology. But before we proceed to sketch the history of that part of our knowledge, there are a few points in the progress of Zoology, understood as a mere classificatory science, which appear to me sufficiently instructive to make it worth our while to dwell upon them.

[2nd Ed.] [Mr. Lindley's recent work, _The Vegetable Kingdom_ (1846), may be looked upon as containing the best view of the recent history of Systematic Botany. In the Introduction to this work, Mr. Lindley has given an account of various recent works on the subject; as Agardh's _Classes Plantarum_ (1826); Perleb's _Lehrbuch der Naturgeschichte der Pflanzenreich_ (1826); Dumortier's _Florula Belgica_ (1827); Bartling's _Ordines Naturales Plantarum_ (1830); Hess's _Uebersicht der Phanerogenischen Natürlichen Pflanzenfamilien_ (1832); Schulz's _Natürliches System des Pflanzenreich's_ (1832); Horaninow's _Primæ Lineæ Systematis Naturæ_ (1834); Fries's _Corpus Florarum provincialium Sueciæ_ (1835); Martins's _Conspectus Regni Vegetablis secundum Characteres Morphologicos_ (1835); Sir Edward F. Bromhead's System, as published in the _Edinburgh Journal_ and other Journals (1836-1840); Endlicher's _Genera Plantarum secundum Ordines Naturales disposita_ (1836-1840); Perleb's _Clavis Classicum Ordinum et Familiarum_ (1838); Adolphe Brongniart's _Enumération des Genres de Plantes_ (1843); Meisner's _Plantarum vascularium Genera secundum Ordines Naturales digesta_ (1843); Horaninow's _Tetractys Naturæ, seu Systema quinquemembre omnium Naturalium_ {411} (1843); Adrien de Jussieu's _**Cours Elémentaire d'Histoire Naturelle. Botanique_ (1844).

Mr. Lindley, in this as in all his works, urges strongly the superior value of natural as compared with artificial systems; his principles being, I think, nearly such as I have attempted to establish in the _Philosophy of the Sciences_, Book viii., Chapter ii. He states that the leading idea which has been kept in view in the compilation of his work is this maxim of Fries: "Singula sphæra (sectio) _ideam quandam_ exponit, indeque ejus character notione simplici optime exprimitur;" and he is hence led to think that the true characters of all natural assemblages are extremely simple.

One of the leading features in Mr. Lindley's system is that he has thrown the Natural Orders into groups subordinate to the higher divisions of Classes and Sub-classes. He had already attempted this, in imitation of Agardh and Bartling, in his _Nixus Plantarum_ (1838). The groups of Natural Orders were there called _Nixus_ (tendencies); and they were denoted by names ending in _ales_; but these groups were further subordinated to _Cohorts_. Thus the first member of the arrangement was Class 1. EXOGENÆ. Sub-class 1. POLYPETALÆ. Cohort 1. ALBUMINOSÆ. _Nixus_ 1. _Ranales_. Natural Orders included in this _Nixus_, Ranunculaceæ, Saraceniceæ, Papaveraceæ, &c. In the _Vegetable Kingdom_, the groups of Natural Orders are termed _Alliances_. In this work, the Sub-classes of the EXOGENS are four: I. DICLINOUS; II. HYPOGYNOUS; III. PERIGYNOUS; IV. EPIGYNOUS; and the Alliances are subordinated to these without the intervention of _Cohorts_.

Mr. Lindley has also, in this as in other works, given English names for the Natural Orders. Thus for _Nymphaceæ_, _Ranunculaceæ_, _Tamaricaceæ_, _Zygophyllaceæ_, _Eleatrinaceæ_, he substitutes Water-Lilies, Crowfoots, Tamarisks, Bean-Capers, and Water-Peppers; for _Malvaceæ_, _Aurantiaceæ_, _Gentianaceæ_, _Primulaceæ_, _Urtiaceæ_, _Euphorbiaceæ_, he employs Mallow-worts, Citron-worts, Gentian-worts, Prim-worts, Nettle-worts, Spurge-worts; and the terms Orchids, Hippurids, Amaryllids, Irids, Typhads, Arads, Cucurbits, are taken as English equivalents for _Orchidaceæ_, _Haloragaceæ_, _Amaryllidaceæ_, _Iridaceæ_, _Typhaceæ_, _Araceæ_, _Cucurbitaceæ_. All persons who wish success to the study of botany in England must rejoice to see it tend to assume this idiomatic shape.] {412}

## CHAPTER VI.

THE PROGRESS OF SYSTEMATIC ZOOLOGY.

THE history of Systematic Botany, as we have presented it, may be considered as a sufficient type of the general order of progression in the sciences of classification. It has appeared, in the survey which we have had to give, that this science, no less than those which we first considered, has been formed by a series of inductive processes, and has, in its history, Epochs at which, by such processes, decided advances were made. The important step in such cases is, the seizing upon some artificial mark which conforms to natural resemblances;--some basis of arrangement and nomenclature by means of which true propositions of considerable generality can be enunciated. The advance of other classificatory sciences, as well as botany, must consist of such steps; and their course, like that of botany, must (if we attend only to the real additions made to knowledge,) be gradual and progressive, from the earliest times to the present.

To exemplify this continued and constant progression in the whole range of Zoology, would require vast knowledge and great labor; and is, perhaps, the less necessary, after we have dwelt so long on the history of Botany, considered in the same point of view. But there are a few observations respecting Zoology in general which we are led to make in consequence of statements recently promulgated; for these statements seem to represent the history of Zoology as having followed a course very different from that which we have just ascribed to the classificatory sciences in general. It is held by some naturalists, that not only the formation of a systematic classification in Zoology dates as far back as Aristotle; but that his classification is, in many respects, superior to some of the most admired and recent attempts of modern times.

If this were really the case, it would show that at least the idea of a Systematic Classification had been formed and developed long previous to the period to which we have assigned such a step; and it would be difficult to reconcile such an early maturity of Zoology with the conviction, which we have had impressed upon us by the other {413} parts of our history, that not only labor but time, not only one man of genius but several, and those succeeding each other, are requisite to the formation of any considerable science.

But, in reality, the statements to which we refer, respecting the scientific character of Aristotle's Zoological system, are altogether without foundation; and this science confirms the lessons taught us by all the others. The misstatements respecting Aristotle's doctrines are on this account so important, and are so curious in themselves, that I must dwell upon them a little.

Aristotle's nine Books _On Animals_ are a work enumerating the differences of animals in almost all conceivable respects;--in the organs of sense, of motion, of nutrition, the interior anatomy, the exterior covering, the manner of life, growth, generation, and many other circumstances. These differences are very philosophically estimated. "The corresponding parts of animals," he says,[126\16] "besides the differences of quality and circumstance, differ in being more or fewer, greater or smaller, and, speaking generally, in excess and defect. Thus some animals have crustaceous coverings, others hard shells; some have long beaks, some short; some have many wings, some have few; Some again have parts which others want, as crests and spurs." He then makes the following important remark: "Some animals have parts which correspond to those of others, not as being the same in species, nor by excess and defect, but by _analogy_; thus a claw is analogous to a thorn, and a nail to a hoof, and a hand to the nipper of a lobster, and a feather to a scale; for what a feather is in a bird, that is a scale in a fish."

[Note 126\16: Lib. i. c. i.]

It will not, however, be necessary, in order to understand Aristotle for our present purpose, that we should discuss his notion of Analogy. He proceeds to state his object,[127\16] which is, as we have said, to describe the differences of animals in their structure and habits. He then observes, that for structure, we may take Man for our type,[128\16] as being best known to us; and the remainder of the first Book is occupied with a description of man's body, beginning from the head, and proceeding to the extremities.

[Note 127\16: Lib. i. c. ii.]

[Note 128\16: c. iii.]

In the next Book, (from which are taken the principal passages in which his modern commentators detect his system,) he proceeds to compare the differences of parts in different animals, according to the order which he had observed in man. In the first chapter he speaks {414} of the head and neck of animals; in the second, of the parts analogous to arms and hands; in the third, of the breast and paps, and so on; and thus he comes, in the seventh chapter, to the legs, feet, and toes: and in the eleventh, to the teeth, and so to other parts.

The construction of a classification consists in the selection of certain parts, as those which shall eminently and peculiarly determine the place of each species in our arrangement. It is clear, therefore, that such an enumeration of differences as we have described, supposing it complete, contains the materials of all possible classifications. But we can with no more propriety say that the author of such an enumeration of differences is the author of any classification which can be made by means of them, than we can say that a man who writes down the whole alphabet

writes down the solution of a given riddle or the answer to a particular question.

Yet it is on no other ground than this enumeration, so far as I can discover, that Aristotle's "System" has been so decidedly spoken of,[129\16] and exhibited in the most formal tabular shape. The authors of this _Systema Aristotelicum_, have selected, I presume, the following passages from the work _On Animals_, as they might have selected any other; and by arranging them according to a subordination unknown to Aristotle himself have made for him a scheme which undoubtedly bears a great resemblance to the most complete systems of modern times.

[Note 129\16: _Linnæan Transactions_, vol. xvi. p. 24.]

Book I., chap. v.--"Some animals are viviparous, some oviparous, some vermiparous. The viviparous are such as man, and the horse, and all those animals which have hair; and of aquatic animals, the whale kind, as the dolphin and cartilaginous fishes."

Book II., chap. vii.--"Of quadrupeds which have blood and are viviparous, some are (as to their extremities,) many-cloven, as the hands and feet of man. For some are many-toed, as the lion, the dog, the panther; some are bifid, and have hoofs instead of nails, as the sheep, the goat, the elephant, the hippopotamus; and some have undivided feet, as the solid-hoofed animals, the horse and ass. The swine kind share both characters."

Chap. ii.--"Animals have also great differences in the teeth, both when compared with each other and with man. For all quadrupeds which have blood and are viviparous, have teeth. And in the first place, some are ambidental,[130\16] (having teeth in both jaws;) and some {415} are not so, wanting the front teeth in the upper jaw. Some have neither front teeth nor horns, as the camel; some have tusks,[131\16] as the boar, some have not. Some have serrated[132\16] teeth, as the lion, the panther, the dog; some have the teeth unvaried,[133\16] as the horse and the ox; for the animals which vary their cutting-teeth have all serrated teeth. No animal has both tusks and horns; nor has any animal with serrated teeth either of those weapons. The greater part have the front teeth cutting, and those within broad."

[Note 130\16: Ἀμφόδοντα.]

[Note 131\16: Χαυλιόδοντα.]

[Note 132\16: Καρχαρόδοντα.]

[Note 133\16: Ἀνεπάλλακτα.]

These passages undoubtedly contain most of the differences on which the asserted Aristotelian classification rests; but the classification is formed by using the characters drawn from the teeth, in order to subdivide those taken from the feet; whereas in Aristotle these two sets of characters stand side by side, along with dozens of others; any selection of which, employed according to any arbitrary method of subordination, might with equal justice be called Aristotle's system.

Why, for instance, in order to form subdivisions of animals, should we not go on with Aristotle's continuation of the second of the above quoted passages, instead of

capriciously leaping to the third? "Of these some have horns, some have none . . . Some have a fetlock-joint,[134\16] some have none . . . Of those which have horns, some have them solid throughout, as the stag; others, for the most part, hollow . . . Some cast their horns, some do not." If it be replied, that we could not, by means of such characters, form a tenable zoological system; we again ask by what right we assume Aristotle to have made or attempted a systematic arrangement, when what he has written, taken in its natural order, does not admit of being construed into a system.

[Note 134\16: Ἀστράγαλον.]

Again, what is the object of any classification? This, at least, among others. To enable the person who uses it to study and describe more conveniently the objects thus classified. If, therefore, Aristotle had formed or adopted any system of arrangement, we should see it in the order of the subjects in his work. Accordingly, so far as he has a system, he professes to make this use of it. At the beginning of the fifth Book, where he is proceeding to treat of the different modes of generation of animals, he says, "As we formerly made a Division of animals according to their kinds, we must now, in the same manner, give a general survey of their History (θεωρίαν). Except, indeed, that in the former case we made our commencement by a description {416} of man, but in the present instance we must speak of him last, because he requires most study. We must begin then with those animals which have shells; we must go on to those which have softer coverings, as crustacea, soft animals, and insects; after these, fishes, both viviparous and oviparous; then birds; then land animals, both viviparous and oviparous."

It is clear from this passage that Aristotle had certain wide and indefinite views of classification, which though not very exact, are still highly creditable to him; but it is equally clear that he was quite unconscious of the classification that has been ascribed to him. If he had adopted that or any other system, this was precisely the place in which he must have referred to and employed it.

The honor due to the stupendous accumulation of zoological knowledge which Aristotle's works contain, cannot be tarnished by our denying him the credit of a system which he never dreamt of and which, from the nature of the progress of science, could not possibly be constructed at that period. But, in reality, we may exchange the mistaken claims which we have been contesting for a better, because a truer praise. Aristotle does show, as far as could be done at his time, a perception of the need of groups, and of names of groups, in the study of the animal kingdom; and thus may justly be held up as the great figure in the Prelude to the Formation of Systems which took place in more advanced scientific times.

This appears, in some measure, from the passage last quoted. For not only is there, in that, a clear recognition of the value and object of a method in natural history; but the general arrangement of the animal kingdom there proposed has considerable scientific merit, and is, for the time, very philosophical. But there are passages in his work in which he shows a wish to carry the principle of arrangement more into detail. Thus, in the first Book, before proceeding to his survey of the differences of

animals,[135\16] after speaking of such classes as Quadrupeds. Birds, Fishes, Cetaceous, Testaceous, Crustaceous Animals, Mollusks, Insects, he says, (chap. vii.) "Animals cannot be divided into large genera, in which one kind includes many kinds. For some kinds are unique, and have no difference of species, as _man_. Some have such kinds, but have no names for them. Thus all quadrupeds which have not wings, have blood. But of these, some are viviparous, some oviparous. Those which are {417} viviparous have not all hair; those which are oviparous have scales." We have here a manifestly intentional subordination of characters: and a kind of regret that we have not names for the classes here indicated; such, for instance, as viviparous quadrupeds having hair. But he follows the subject into further detail. "Of the class of viviparous quadrupeds," he continues, "there are many genera,[136\16] but these again are without names, except specific names, such as _man_, _lion_, _stag_, _horse_, _dog_, and the like. Yet there is a genus of animals that have names, as the horse, the ass, the _oreus_, the _ginnus_, the _innus_, and the animal which in Syria is called _heminus_ (mule); for these are called _mules_, from their resemblance only; not being mules, for they breed of their own kind. Wherefore," he adds, that is, because we do not possess recognized genera and generic names of this kind, "we must take the species separately, and study the nature of each."

[Note 135\16: Γένη.]

[Note 136\16: **Εἴδη.]

These passages afford us sufficient ground for placing Aristotle at the head of those naturalists to whom the first views of the necessity of a zoological system are due. It was, however, very long before any worthy successor appeared, for no additional step was made till modern times. When Natural History again came to be studied in Nature, the business of Classification, as we have seen, forced itself upon men's attention, and was pursued with interest in animals, as in plants. The steps of its advance were similar in the two cases;--by successive naturalists, various systems of artificial marks were selected with a view to precision and convenience;--and these artificial systems assumed the existence of certain natural groups, and of a natural system to which they gradually tended. But there was this difference between botany and zoology:--the reference to physiological principles, which, as we have remarked, influenced the natural systems of vegetables in a latent and obscure manner, botanists being guided by its light, but hardly aware that they were so, affected the study of systematic zoology more directly and evidently. For men can neither overlook the general physiological features of animals, nor avoid being swayed by them in their judgments of the affinities of different species. Thus the classifications of zoology tended more and more to a union with comparative anatomy, as the science was more and more improved.[137\16] But comparative anatomy belongs to the subject of the next Book; and anything it may be proper to say respecting its influence upon zoological arrangements, will properly find a place there. {418}

[Note 137\16: Cuvier, _Leç. d'Anat. Comp._ vol. i. p. 17.]

It will appear, and indeed it hardly requires to be proved, that those steps in systematic zoology which are due to the light thrown upon the subject by physiology,

are the result of a long series of labors by various naturalists, and have been, like other advances in science, led to and produced by the general progress of such knowledge. We can hardly expect that the classificatory sciences can undergo any material improvement which is not of this kind. Very recently, however, some authors have attempted to introduce into these sciences certain principles which do not, at first sight, appear as a continuation and extension of the previous researches of comparative anatomists. I speak, in particular, of the doctrines of a _Circular Progression_ in the series of affinity; of a _Quinary Division_ of such circular groups; and of a relation of _Analogy_ between the members of such groups, entirely distinct from the relation of _Affinity_.

The doctrine of Circular Progression has been propounded principally by Mr. Macleay; although, as he has shown,[138\16] there are suggestions of the same kind to be found in other writers. So far as this view negatives the doctrine of a mere linear progression in nature, which would place each genus in contact only with the preceding and succeeding ones, and so far as it requires us to attend to more varied and ramified resemblances, there can be no doubt that it is supported by the result of all the attempts to form natural systems. But whether that assemblage of circles of arrangement which is now offered to naturalists, be the true and only way of exhibiting the natural relations of organized bodies, is a much more difficult question, and one which I shall not here attempt to examine; although it will be found, I think, that those analogies of science which we have had to study, would not fail to throw some light upon such an inquiry. The prevalence of an invariable numerical law in the divisions of natural groups, (as the number _five_ is asserted to prevail by Mr. Macleay, the number _ten_ by Fries, and other numbers by other writers), would be a curious fact, if established; but it is easy to see that nothing short of the most consummate knowledge of natural history, joined with extreme clearness of view and calmness of judgment, could enable any one to pronounce on the attempts which have been made to establish such a principle. But the doctrine of a relation of _Analogy_ distinct from Affinity, in the manner which has recently been taught, seems to be obviously at variance with that gradual approximation of the classificatory to the {419} physiological sciences, which has appeared to us to be the general tendency of real knowledge. It seems difficult to understand how a reference to such relations as those which are offered as examples of analogy[139\16] can be otherwise than a retrograde step in science.

[Note 138\16: _Linn. Trans._ vol. xvi. p. 9.]

[Note 139\16: For example, the goatsucker has an _affinity_ with the swallow; but it has an _analogy_ with the bat, because both fly at the same hour of the day, and feed in the same manner.--Swainson, _Geography and Classification of Animals_, p. 129.]

Without, however, now dwelling upon these points, I will treat a little more in detail of one of the branches of Zoology.

[2nd Ed.] [For the more recent progress of Systematic Zoology, see in the _Reports_ of the British Association, in 1834, Mr. L. Jenyns's _Report an the Recent

Progress and Present State of Zoology_, and in 1844, Mr. Strickland's _Report on the Recent Progress and Present State of Ornithology_. In these Reports, the questions of the Circular Arrangement, the Quinary System, and the relation of Analogy and Affinity are discussed.]

## CHAPTER VII.

THE PROGRESS OF ICHTHYOLOGY.

IF it had been already observed and admitted that sciences of the same kind follow, and must follow, the same course in the order of their development, it would be unnecessary to give a history of any special branch of Systematic Zoology; since botany has already afforded us a sufficient example of the progress of the classificatory sciences. But we may be excused for introducing a sketch of the advance of one department of zoology, since we are led to the attempt by the peculiar advantage we possess in having a complete history of the subject written with great care, and brought up to the present time, by a naturalist of unequalled talents and knowledge. I speak of Cuvier's _Historical View of Ichthyology_, which forms the first chapter of his great work on that part of natural history. The place and office in the progress of this science, which is assigned to each person by Cuvier, will probably not be lightly contested. It will, therefore, be no small confirmation of the justice of the views on which the {420} distribution of the events in the history of botany was founded, if Cuvier's representation of the history of ichthyology offers to us obviously a distribution almost identical.

We shall find that this is so;--that we have, in zoology as in botany, a period of unsystematic knowledge; a period of misapplied erudition; an epoch of the discovery of fixed characters; a period in which many systems were put forward; a struggle of an artificial and a natural method; and a gradual tendency of the natural method to a manifestly physiological character. A few references to Cuvier's history will enable us to illustrate these and other analogies.

_Period of Unsystematic Knowledge._--It would be easy to collect a number of the fabulous stories of early times, which formed a portion of the imaginary knowledge of men concerning animals as well as plants. But passing over these, we come to a long period and a great collection of writers, who, in various ways, and with various degrees of merit, contributed to augment the knowledge which existed concerning fish, while as yet there was hardly ever any attempt at a classification of that province of the animal kingdom. Among these writers, Aristotle is by far the most important. Indeed he carried on his zoological researches under advantages which rarely fall to the lot of the naturalist; if it be true, as Athenæus and Pliny state,[140\16] that Alexander gave him sums which amounted to nine hundred talents, to enable him to collect materials for his history of animals, and put at his disposal several thousands of men to be employed in hunting, fishing, and procuring information for him. The works of his on Natural History which remain to us are, nine Books _Of the History of Animals_; four, _On the Parts of Animals_; five, _On the Generation of Animals_; one, _On the Going of Animals_; one, _Of the Sensations, and the Organs of them_; one,

_On Sleeping and Waking_; one, _On the Motion of Animals_; one, _On the Length and Shortness of Life_; one, _On Youth and Old Age_; one, _On Life and Death_; one, _On Respiration_. The knowledge of the external and internal conformation of animals, their habits, instincts, and uses, which Aristotle displays in these works, is spoken of as something wonderful even to the naturalists of our own time. And he may be taken as a sufficient representative of the whole of the period of which we speak; for he is, says Cuvier,[141\16] not only the first, but the only one of the ancients who has treated of the natural history of fishes (the province to which {421} we now confine ourselves,) in a scientific point of view, and in a way which shows genius.

[Note 140\16: Cuv. _Hist. Nat. des Poissons_, i. 13.]

[Note 141\16: Cuv. p. 18.]

We may pass over, therefore, the other ancient authors from whose writings Cuvier, with great learning and sagacity, has levied contributions to the history of ichthyology; as Theophrastus, Ovid, Pliny, Oppian, Athenæus, Ælian, Ausonius, Galen. We may, too, leave unnoticed the compilers of the middle ages, who did little but abstract and disfigure the portions of natural history which they found in the ancients. Ichthyological, like other knowledge, was scarcely sought except in books, and on that very account was not understood when it was found.

_Period of Erudition._--Better times at length came, and men began to observe nature for themselves. The three great authors who are held to be the founders of modern ichthyology, appeared in the middle of the sixteenth century; these were Bélon, Rondelet, and Salviani, who all published about 1555. All the three, very different from the compilers who filled the interval from Aristotle to them, themselves saw and examined the fishes which they describe, and have given faithful representations of them. But, resembling in that respect the founders of modern botany, Briassavola, Ruellius, Tragus, and others, they resembled them in this also, that they attempted to make their own observations a commentary upon the ancient writers. Faithful to the spirit of their time, they are far more careful to make out the names which each fish bore in the ancient world, and to bring together scraps of their history from the authors in whom these names occur, than to describe them in a lucid manner; so that without their figures, says Cuvier, it would be almost as difficult to discover their species as those of the ancients.

The difficulty of describing and naming species so that they can be recognized, is little appreciated at first, although it is in reality the main-spring of the progress of the sciences of classification. Aristotle never dreamt that the nomenclature which was in use in his time could ever become obscure;[142\16] hence he has taken no precaution to enable his readers to recognize the species of which he speaks; and in him and in other ancient authors, it requires much labor and great felicity of divination to determine what the names mean. The perception of this difficulty among modern naturalists led to systems, and to nomenclature founded upon system; but these did not come into {422} being immediately at the time of which we speak; nor till the evil had grown to a more inconvenient magnitude.

[Note 142\16: Cuvier, p. 17.]

_Period of Accumulation of Materials. Exotic Collections._--The fishes of Europe were for some time the principal objects of study; but those of distant regions soon came into notice.[143\16] In the seventeenth century the Dutch conquered Brazil, and George Margrave, employed by them, described the natural productions of the country, and especially the fishes. Bontius, in like manner, described some of those of Batavia. Thus these writers correspond to Romphius and Rheede in the history of botany. Many others might be mentioned; but we must hasten to the formation of systems, which is our main object of attention.

[Note 143\16: Cuv. p. 43.]

_Epoch of the Fixation of Characters. Ray and Willoughby._--In botany, as we have seen, though Ray was one of the first who invented a connected system, he was preceded at a considerable interval by Cæsalpinus, who had given a genuine solution of the same problem. It is not difficult to assign reasons why a sound classification should be discovered for plants at an earlier period than for fishes. The vastly greater number of the known species, and the facilities which belong to the study of vegetables, give the botanist a great advantage; and there are numerical relations of a most definite kind (for instance, the number of parts of the seed-vessel employed by Cæsalpinus as one of the bases of his system), which are tolerably obvious in plants, but which are not easily discovered in animals. And thus we find that in ichthyology, Ray, with his pupil and friend Willoughby, appears as the first founder of a tenable system.[144\16]

[Note 144\16: Francisci Willoughbeii, Armigeri, _de Historia Piscium_, libri iv. jussu et sumptibus Societatis Regiæ Londinensis editi, &c. Totum opus recognovit, coaptavit, supplevit, librum etiam primum et secundum adjecit Joh. Raius. Oxford, 1668.]

The first great division in this system is into _cartilaginous_ and _bony_ fishes; a primary division, which had been recognized by Aristotle, and is retained by Cuvier in his latest labors. The subdivisions are determined by the general form of the fish (as long or flat), by the teeth, the presence or absence of ventral fins, the number of dorsal fins, and the nature of the spines of the fins, as soft or prickly. Most of these characters have preserved their importance in later systems; especially the last, which, under the terms _malacopterygian_ and _acanthopterygian_, holds a place in the best recent arrangements. {423}

That this system was a true first approximation to a solution of the problem, appears to be allowed by naturalists. Although, says Cuvier,[145\16] there are in it no genera well defined and well limited, still in many places the species are brought together very naturally, and in such a way that a few words of explanation would suffice to form, from the groups thus presented to us, several of the genera which have since been received. Even in botany, as we have seen, genera were hardly maintained with any degree of precision, till the binary nomenclature of Linnæus made this division a matter of such immense convenience.

[Note 145\16: Cuvier, p. 57.]

The amount of this convenience, the value of a brief and sure nomenclature, had not yet been duly estimated. The work of Willoughby forms an epoch,[146\16] and a happy epoch, in the history of ichthyology; for the science, once systematized, could distinguish the new from the old, arrange methodically, describe clearly. Yet, because Willoughby had no nomenclature of his own, and no fixed names for his genera, his immediate influence was not great. I will not attempt to trace this influence in succeeding authors, but proceed to the next important step in the progress of system.

[Note 146\16: p. 58.]

_Improvement of the System. Artedi._--Peter Artedi was a countryman and intimate friend of Linnæus; and rendered to ichthyology nearly the same services which Linnæus rendered to botany. In his _Philosophia Ichthyologica_, he analysed[147\16] all the interior and exterior parts of animals; he created a precise terminology for the different forms of which these parts are susceptible; he laid down rules for the nomenclature of genera and species; besides his improvements of the subdivisions of the class. It is impossible not to be struck with the close resemblance between these steps, and those which are due to the _Fundamenta Botanica_. The latter work appeared in 1736, the former was published by Linnæus, after the death of the author, in 1738; but Linnæus had already, as early as 1735, made use of Artedi's manuscripts in the ichthyological part of his _Systema Naturæ_. We cannot doubt that the two young naturalists (they were nearly of the same age), must have had a great influence upon each other's views and labors; and it would be difficult now to ascertain what portion of the peculiar merits of the Linnæan reform was derived from Artedi. But we may remark that, in ichthyology at least, Artedi appears to have been a naturalist of more original views and profounder philosophy than his friend and editor, who afterwards himself took up the subject. {424} The reforms of Linnæus, in all parts of natural history, appear as if they were mainly dictated by a love of elegance, symmetry, clearness, and definiteness; but the improvement of the ichthyological system by Artedi seems to have been a step in the progress to a natural arrangement. His genera,[148\16] which are forty-five in number, are so well constituted, that they have almost all been preserved; and the subdivisions which the constantly-increasing number of species has compelled his successors to introduce, have very rarely been such that they have led to the transposition of his genera.

[Note 147\16: p. 20.]

[Note 148\16: Cuvier, p. 71.]

In its bases, however, Artedi's was an artificial system. His characters were positive and decisive, founded in general upon the number of rays of the membrane of the gills, of which he was the first to mark the importance;--upon the relative position of the fins, upon their number, upon the part of the mouth where the teeth are found, upon the conformation of the scales. Yet, in some cases, he has recourse to the interior anatomy.

Linnæus himself at first did not venture to deviate from the footsteps of a friend, who, in this science, had been his master. But in 1758, in the tenth edition of the _Systema Naturæ_, he chose to depend upon himself and devised a new

ichthyological method. He divided some genera, united others, gave to the species trivial names and characteristic phrases, and added many species to those of Artedi. Yet his innovations are for the most part disapproved of by Cuvier; as his transferring the _chondropterygian_ fishes of Artedi to the class of reptiles, under the title of _Amphybia nantes_; and his rejecting the distinction of acanthopterygian and malacopterygian, which, as we have seen, had prevailed from the time of Willoughby, and introducing in its stead a distribution founded on the presence or absence of the ventral fins, and on their situation with regard to the pectoral fins. "Nothing," says Cuvier, "more breaks the true connexions of genera than these orders of _apodes_, _jugulares_, _thoracici_, and _abdominales_."

Thus Linnæus, though acknowledging the value and importance of natural orders, was not happy in his attempts to construct a system which should lead to them. In his detection of good characters for an artificial system he was more fortunate. He was always attentive to number, as a character; and he had the very great merit[149\16] of introducing into the classification the number of rays of the fins of each species. This mark is one of great importance and use. And this, as well as {425} other branches of natural history, derived incalculable advantages from the more general merits of the illustrious Swede;[150\16]--the precision of the characters, the convenience of a well-settled terminology, the facility afforded by the binary nomenclature. These recommendations gave him a pre-eminence which was acknowledged by almost all the naturalists of his time, and displayed by the almost universal adoption of his nomenclature, in zoology, as well as in botany; and by the almost exclusive employment of his distributions of classes, however imperfect and artificial they might be.

[Note 149\16: p. 74.]

[Note 150\16: Cuvier, p. 85.]

And even[151\16] if Linnæus had had no other merit than the impulse he gave to the pursuit of natural science, this alone would suffice to immortalize his name. In rendering natural history easy, or at least in making it appear so, he diffused a general taste for it. The great took it up with interest; the young, full of ardor, rushed forwards in all directions, with the sole intention of completing his system. The civilized world was eager to build the edifice which Linnæus had planned.

[Note 151\16: Ib. p. 88.]

This spirit, among other results, produced voyages of natural historical research, sent forth by nations and sovereigns. George the Third of England had the honor of setting the example in this noble career, by sending out the expeditions of Byron, Wallis, and Carteret, in 1765. These were followed by those of Bougainville, Cook, Forster, and others. Russia also scattered several scientific expeditions through her vast dominions; and pupils of Linnæus sought the icy shores of Greenland and Iceland, in order to apply his nomenclature to the productions of those climes. But we need not attempt to convey any idea of the vast stores of natural historical treasures which were thus collected from every part of the globe.

I shall not endeavor to follow Cuvier in giving an account of the great works of natural history to which this accumulation of materials gave rise; such as the magnificent work of Bloch on Fishes, which appeared in 1782-1785; nor need I attempt, by his assistance, to characterize or place in their due position the several systems of classification proposed about this time. But in the course of these various essays, the distinction of the artificial and natural methods of classification came more clearly into view than before; and this is a point so important to the philosophy of the subject, that we must devote a few words to it. {426}

_Separation of the Artificial and Natural Methods in Ichthyology._--It has already been said that all so-called _artificial methods_ of classification must be natural, at least as to the narrowest members of the system; thus the artificial Linnæan method is natural as to species, and even as to genera. And on the other hand, all proposed natural methods, so long as they remain unmodified, are artificial as to their characteristic marks. Thus a Natural Method is an attempt to provide positive and distinct _characters_ for the _wider_ as well as for the narrower _natural groups_. These considerations are applicable to zoology as well as to botany. But the question, how we know natural groups before we find marks for them, was, in botany, as we have seen, susceptible only of vague and obscure answers:--the mind forms them, it was said, by taking the aggregate of all the characters; or by establishing a subordination of characters. And each of these answers had its difficulty, of which the solution appeared to be, that in attempting to form natural orders we are really guided by a latent undeveloped estimate of physiological relations. Now this principle, which was so dimly seen in the study of vegetables, shines out with much greater clearness when we come to the study of animals, in which the physiological relations of the parts are so manifest that they cannot be overlooked, and have so strong an attraction for our curiosity that we cannot help having our judgments influenced by them. Hence the superiority of natural systems in zoology would probably be far more generally allowed than in botany; and no arrangement of animals which, in a large number of instances, violated strong and clear natural affinities, would be tolerated because it answered the purpose of enabling us easily to find the name and place of the animal in the artificial system. Every system of zoological arrangement may be supposed to aspire to be a natural system. But according to the various habits of the minds of systematizers, this object was pursued more or less steadily and successfully; and these differences came more and more into view with the increase of knowledge and the multiplication of attempts.

Bloch, whose ichthyological labors have been mentioned, followed in his great work the method of Linnæus. But towards the end of his life he had prepared a general system, founded upon one single numerical principle;--the number of fins; just as the sexual system of Linnæus is founded upon the number of stamina; and he made his subdivisions according to the position of the ventral and pectoral fins; the same character which Linnæus had employed for his primary {427} division. He could not have done better, says Cuvier,[152\16] if his object had been to turn into ridicule all artificial methods, and to show to what absurd combinations they may lead.

[Note 152\16: p. 108.]

Cuvier himself who always pursued natural systems with a singularly wise and sagacious consistency, attempted to improve the ichthyological arrangements which had been proposed before him. In his _Règne Animal_, published in 1817, he attempts the problem of arranging this class; and the views suggested to him, both by his successes and his failures, are so instructive and philosophical, that I cannot illustrate the subject better than by citing some of them.

"The class of fishes," he says,[153\16] "is, of all, that which offers the greatest difficulties, when we wish to subdivide it into orders, according to fixed and obvious characters. After many trials, I have determined on the following distribution, which in some instances is wanting in precision, but which possesses the advantage of keeping the natural families entire.

[Note 153\16: _Règne Animal_, vol. ii. p. 110.]

"Fish form two distinct series;--that of _chondropterygians_ or _cartilaginous fish_, and that of _fish_ properly so called.

"The _first_ of these series has for its character, that the palatine bones replace, in it, the bones of the upper jaw: moreover the whole of its structure has evident analogies, which we shall explain.

"It divides itself into three ORDERS: "The CYCLOSTOMES, in which the jaws are soldered (_soudées_) into an immovable ring, and the bronchiæ are open in numerous holes.

"The SELACIANS, which have the bronchiæ like the preceding, but not the jaws.

"The STURONIANS, in which the bronchiæ are open as usual by a slit furnished with an operculum.

"The second series, or that of _ordinary fishes_, offers me, in the first place, a primary division, into those of which the maxillary bone and the palatine arch are dovetailed (_engrenés_) to the skull. Of these I make an order of PECTOGNATHS, divided into two families; the _gymnodonts_ and the _scleroderms_.

"After these I have the fishes with complete jaws, but with bronchiæ which, instead of having the form of combs, as in all the others, have the form of a series of little tufts (_houppes_). Of these I again form an order, which I call LOPHOBRANCHS, which only includes one family. {428}

"There then remains an innumerable quantity of fishes, to which we can no longer apply any characters except those of the exterior organs of motion. After long examination, I have found that the least bad of these characters is, after all, that employed by Ray and Artedi, taken from the nature of the first rays of the dorsal and of the anal fin. Thus ordinary fishes are divided into MALACOPTERYGIANS, of which all the rays are soft, except sometimes the first of the dorsal fin or the pectorals;--and ACANTHOPTERYGIANS, which have always the first portion of the dorsal, or of the first dorsal when there are two, supported by spinous rays, and in which the anal has also some such rays, and the ventrals, at least, each one.

"The former may be subdivided without inconvenience, according to their ventral fins, which are sometimes situate behind the abdomen, sometimes adherent to the apparatus of the shoulder, or, finally, are sometimes wanting altogether.

"We thus arrive at the three orders of ABDOMINAL MALACOPTERYGIANS, of SUBBRACHIANS, and of APODES; each of which includes some natural families which we shall explain: the first, especially, is very numerous.

"But this basis of division is absolutely impracticable with the Acanthopterygians; and the problem of establishing among these any other subdivision than that of the natural families has hitherto remained for me insoluble. Fortunately several of these families offer characters almost as precise as those which we could give to true orders.

"In truth, we cannot assign to the families of fishes, ranks as marked, as for example, to those of mammifers. Thus the Chondropterygians on the one hand hold to reptiles by the organs of the senses, and by those of generation in some; and they are related to mollusks and worms by the imperfection of the skeleton in others.

"As to Ordinary Fishes, if any part of the organization is found more developed in some than in others, there does not result from this any pre-eminence sufficiently marked, or of sufficient influence upon their whole system, to oblige us to consult it in the methodical arrangement.

"We shall place them, therefore, nearly in the order in which we have just explained their characters."

I have extracted the whole of this passage, because, though it is too technical to be understood in detail by the general reader, those who have followed with any interest the history of the attempts at a natural classification in any department in nature, will see here a fine example of the problems which such attempts propose, of the {429} difficulties which it may present, and of the reasonings, labors, cautions, and varied resources, by means of which its solution is sought, when a great philosophical naturalist girds himself to the task. We see here, most instructively, how different the endeavor to frame such a natural system, is from the procedure of an artificial system, which carries imperatively through the whole of a class of organized beings, a system of marks either arbitrary, or conformable to natural affinities in a partial degree. And we have not often the advantage of having the reasons for a systematic arrangement so clearly and fully indicated, as is done here, and in the descriptions of the separate orders.

This arrangement Cuvier adhered to in all its main points, both in the second edition of the _Règne Animal_, published in 1821, and in his _Histoire Naturelle des Poissons_, of which the first volume was published in 1828, but which unfortunately was not completed at the time of his death. It may be supposed, therefore, to be in accordance with those views of zoological philosophy, which it was the business of his life to form and to apply; and in a work like the present, where, upon so large a question of natural history, we must be directed in a great measure by the analogy of the history of science, and by the judgments which seem most to have the character of wisdom, we appear to be justified in taking Cuvier's ichthyological system as the

nearest approach which has yet been made to a natural method in that department.

The true natural method is only one: artificial methods, and even good ones, there may be many, as we have seen in botany; and each of these may have its advantages for some particular use. On some methods of this kind, on which naturalists themselves have hardly yet had time to form a stable and distinct opinion, it is not our office to decide. But judging, as I have already said, from the general analogy of the natural sciences, I find it difficult to conceive that the ichthyological method of M. Agassiz, recently propounded with an especial reference to fossil fishes, can be otherwise than an artificial method. It is founded entirely on one part of the animal, its scaly covering, and even on a single scale. It does not conform to that which almost all systematic ichthyologists hitherto have considered as a permanent natural distinction of a high order; the distinction of bony and cartilaginous fishes; for it is stated that each order contains examples of both.[154\16] I do not know what general anatomical or physiological {430} truths it brings into view; but they ought to be very important and striking ones, to entitle them to supersede those which led Cuvier to his system. To this I may add, that the new ichthyological classification does not seem to form, as we should expect that any great advance towards a natural system would form, a connected sequel to the past history of ichthyology;--a step to which anterior discoveries and improvements have led, and in which they are retained.

[Note 154\16: Dr. Buckland's _Bridgewater Treatise_, p. 270.]

But notwithstanding these considerations, the method of M. Agassiz has probably very great advantages for his purpose; for in the case of fossil fish, the parts which are the basis of his system often remain, when even the skeleton is gone. And we may here again refer to a principle of the classificatory sciences which we cannot make too prominent;--all arrangements and nomenclatures are good, which enable us to assert general propositions. Tried by this test, we cannot fail to set a high value on the arrangement of M. Agassiz; for propositions of the most striking generality respecting fossil remains of fish, of which geologists before had never dreamt, are enunciated by means of his groups and names. Thus only the two first orders, the _Placoïdians_ and _Ganoïdians_, existed before the commencement of the cretaceous formation: the third and fourth orders, the _Ctenoïdians_ and _Cycloïdians_, which contain three-fourths of the eight thousand known species of living Fishes, appear for the first time in the cretaceous formation: and other geological relations of these orders, no less remarkable, have been ascertained by M. Agassiz.

But we have now, I trust, pursued these sciences of classification sufficiently far; and it is time for us to enter upon that higher domain of Physiology to which, as we have said. Zoology so irresistibly directs us.

[2nd Ed.] [I have retained the remarks which I ventured at first to make on the System of M. Agassiz; but I believe the opinion of the most philosophical ichthyologists to be that Cuvier's System was too exclusively based on the internal skeleton, as Agassiz's was on the external skeleton. In some degree both systems have been superseded, while all that was true in each has been retained. Mr. Owen, in his _Lectures on Vertebrata_ (1846), takes Cuvierian characters from the

endo-skeleton, Agassizian ones from the exo-skeleton, Linnæan ones from the ventral fins, Müllerian ones from the air-bladder, and combines them by the light of his own researches, with the view of forming a system more truly natural than any preceding one.

As I have said above, naturalists, in their progress towards a Natural {431} System, are guided by physiological relations, latently in Botany, but conspicuously in Zoology. From the epoch of Cuvier's _Règne Animal_, the progress of Systematic Zoology is inseparably dependent on the progress of Comparative Anatomy. Hence I have placed Cuvier's Classification of animal forms in the next Book, which treats of Physiology.]

{{433}} BOOK XVII.

# _ORGANICAL SCIENCES._

HISTORY OF PHYSIOLOGY AND COMPARATIVE ANATOMY.

Fearful and wondrous is the skill which moulds Our body's vital plan, And from the first dim hidden germ unfolds The perfect limbs of man. Who, who can pierce the secret? tell us how Something is drawn from naught, Life from the inert mass? Who, Lord! but thou, Whose hand the whole has wrought? Of this corporeal substance, still to be, Thine eye a survey took; And all my members, yet unformed by thee, Were written in thy book. PSALM cxxxix. 13-16.

{{435}} INTRODUCTION.

_Of the Organical Sciences_

THOUGH the general notion of _life_ is acknowledged by the most profound philosophers to be dim and mysterious, even up to the present time; and must, in the early stages of human speculation, have been still more obscure and confused; it was sufficient, even then, to give interest and connexion to men's observations upon their own bodies and those of other animals. It was seen, that in living things, certain peculiar processes were constantly repeated, as those of breathing and of taking food, for example; and that a certain conformation of the parts of the animal was subservient to these processes; and thus were gradually formed the notions of _Function_ and of _Organization_. And the sciences of which these notions formed the basis are clearly distinguishable from all those which we have hitherto considered. We conceive an _organized_ body to be one in which the parts are there for the sake of the whole, in a manner different from any mechanical or chemical connexion; we conceive a _function_ to be not merely a process of change, but of change connected with the general vital process. When mechanical or chemical processes occur in the living body, they are instrumental to, and directed by, the peculiar powers of life. The sciences which thus consider organization and vital functions may be termed _organical_ sciences.

When men began to speculate concerning such subjects, the general mode of apprehending the process in the cases of some functions, appeared to be almost obvious; thus it was conceived that the growth of animals arose from their frame

appropriating to itself a part of the substance of the food through the various passages of the body. Under the influence of such general conceptions, speculative men were naturally led to endeavor to obtain more clear and definite views of the course of each of such processes, and of the mode in which the separate parts contributed to it. Along with the observation of the living person, the more searching examination which could be carried on in the dead body, and the comparison of various kinds of animals, soon showed that this pursuit was rich in knowledge and in interest. {436} Moreover, besides the interest which the mere speculative faculty gave to this study, the Art of Healing added to it a great practical value; and the effects of diseases and of medicines supplied new materials and new motives for the reasonings of the philosopher.

In this manner anatomy or physiology may be considered as a science which began to be cultivated in the earliest periods of civilization. Like most other ancient sciences, its career has been one of perpetual though variable progress; and as in others, so in this, each step has implied those which had been previously made, and cannot be understood aright except we understand them. Moreover, the steps of this advance have been very many and diverse; the cultivators of anatomy have in all ages been numerous and laborious; the subject is one of vast extent and complexity; almost every generation had added something to the current knowledge of its details; and the general speculations of physiologists have been subtle, bold, and learned. It must, therefore, be difficult or impossible for a person who has not studied the science with professional diligence and professional advantages, to form just judgments of the value of the discoveries of various ages and persons, and to arrange them in their due relation to each other. To this we may add, that though all the discoveries which have been made with respect to particular functions or organizations are understood to be subordinate to one general science, the Philosophy of Life, yet the principles and doctrines of this science nowhere exist in a shape generally received and assented to among physiologists; and thus we have not, in this science, the advantage which in some others we have possessed;--of discerning the true direction of its first movements, by knowing the point to which they ultimately tend;--of running on beyond the earlier discoveries, and thus looking them in the face, and reading their true features. With these disadvantages, all that we can have to say respecting the history of Physiology must need great indulgence on the part of the reader.

Yet here, as in other cases, we may, by guiding our views by those of the greatest and most philosophical men who have made the subject their study, hope to avoid material errors. Nor can we well evade making the attempt. To obtain some simple and consistent view of the progress of physiological science, is in the highest degree important to the completion of our views of the progress of physical science. For the physiological or organical sciences form a class to which the classes already treated of, the mechanical, chemical, and classificatory sciences, are subordinate and auxiliary. Again, another {437} circumstance which makes physiology an important part of our survey of human knowledge is, that we have here a science which is concerned, indeed, about material combinations, but in which we are led almost beyond the borders of the material world, into the region of sensation and perception,

thought and will. Such a contemplation may offer some suggestions which may prepare us for the transition from physical to metaphysical speculations.

In the survey which we must, for such purposes, take of the progress of physiology, it is by no means necessary that we should exhaust the subject, and attempt to give the history of every branch of the knowledge of the phenomena and laws of living creatures. It will be sufficient, if we follow a few of the lines of such researches, which may be considered as examples of the whole. We see that life is accompanied and sustained by many processes, which at first offer themselves to our notice as separate functions, however they may afterwards be found to be connected and identified; such are feeling, digestion, respiration, the action of the heart and pulse, generation, perception, voluntary motion. The analysis of any one of these functions may be pursued separately. And since in this, as in all genuine sciences, our knowledge becomes real and scientific, only in so far as it is verified in particular facts, and thus established in general propositions, such an original separation of the subjects of research is requisite to a true representation of the growth of real knowledge. The loose hypotheses and systems, concerning the connexion of different vital faculties and the general nature of living things, which have often been promulgated, must be excluded from this part of our plan. We do not deny all value and merit to such speculations; but they cannot be admitted in the earlier stages of the history of physiology, treated of as an inductive science. If the doctrine so propounded have a solid and permanent truth, they will again come before us when we have travelled through the range of more limited truths, and are prepared to ascend with security and certainty into the higher region of general physiological principles. If they cannot be arrived at by such a road, they are then, however plausible and pleasing, no portion of that real and progressive science with which alone our history is concerned.

We proceed, therefore, to trace the establishment of some of the more limited but certain doctrines of physiology. {438}

## CHAPTER I.

DISCOVERY OF THE ORGANS OF VOLUNTARY MOTION.

_Sect._ 1.--_Knowledge of Galen and his Predecessors._

IN the earliest conceptions which men entertained of their power of moving their own members, they probably had no thought of any mechanism or organization by which this was effected. The foot and the hand, no less than the head, were seen to be endowed with life; and this pervading life seemed sufficiently to explain the power of motion in each part of the frame, without its being held necessary to seek out a special seat of the will, or instruments by which its impulses were made effective. But the slightest inspection of dissected animals showed that their limbs were formed of a curious and complex collection of cordage, and communications of various kinds, running along and connecting the bones of the skeleton. These cords and communications we now distinguish as muscles, nerves, veins, arteries, &c.; and among these, we assign to the muscles the office of moving the parts to which they

are attached, as cords move the parts of a machine. Though this action of the muscles on the bones may now appear very obvious, it was, probably, not at first discerned. It is observed that Homer, who describes the wounds which are inflicted in his battles with so much apparent anatomical precision, nowhere employs the word _muscle_. And even Hippocrates of Cos, the most celebrated physician of antiquity, is held to have had no distinct conception of such an organ.[1\17] He always employs the word _flesh_ when he means _muscle_, and the first explanation of the latter word (μῦς) occurs in a spurious work ascribed to him. For nerves, sinews, ligaments,[2\17] he used indiscriminately the same terms; (τόνος or νεῦρον;) and of these nerves (νεῦρα) he asserts that they contract the limbs. Nor do we find much more distinctness on this subject even in Aristotle, a generation or two later. "The origin of the νεῦρα," he says,[3\17] "is from the heart; they connect {439} the bones, and surround the joints." It is clear that he means here the muscles, and therefore it is with injustice that he has been accused of the gross error of deriving the nerves from the heart. And he is held to have really had the merit[4\17] of discovering the nerves of sensation, which he calls the "canals of the brain" (πόροι τοῦ ἐγκεφάλου); but the analysis of the mechanism of motion is left by him almost untouched. Perhaps his want of sound mechanical notions, and his constant straining after verbal generalities, and systematic classifications of the widest kind, supply the true account of his thus missing the solution of one of the simplest problems of Anatomy.

[Note 1\17: Sprengel, _Geschichte der Arzneikunde_, i. 382.]

[Note 2\17: Sprengel, _Gesch. Arz._ i. 385.]

[Note 3\17: _Hist. Anim._ iii. 5.]

[Note 4\17: Ib. i. 456.]

In this, however, as in other subjects, his immediate predecessors were far from remedying the deficiencies of his doctrines. Those who professed to study physiology and medicine were, for the most part, studious only to frame some general system of abstract principles, which might give an appearance of connexion and profundity to their tenets. In this manner the successors of Hippocrates became a medical school, of great note in its day, designated as the _Dogmatic_ school;[5\17] in opposition to which arose an _Empiric_ sect, who professed to deduce their modes of cure, not from theoretical dogmas, but from experience. These rival parties prevailed principally in Asia Minor and Egypt, during the time of Alexander's successors,--a period rich in names, but poor in discoveries; and we find no clear evidence of any decided advance in anatomy, such as we are here attempting to trace.

[Note 5\17: Sprengel, _Gesch. Arz._ i. 583.]

The victories of Lucullus and Pompeius, in Greece and Asia, made the Romans acquainted with the Greek philosophy; and the consequence soon was, that shoals of philosophers, rhetoricians, poets, and physicians[6\17] streamed from Greece, Asia Minor, and Egypt, to Rome and Italy, to traffic their knowledge and their arts for Roman wealth. Among these, was one person whose name makes a great figure in the history of medicine, Asclepiades of Prusa in Bithynia. This man appears to have been a quack, with the usual endowments of his class;--boldness, singularity, a

contemptuous rejection of all previously esteemed opinions, a new classification of diseases, a new list of medicines, and the assertion of some wonderful cures. He would not, on such accounts, deserve a place in the history of science, but that he became the founder of a new school, the _Methodic_, which professed to hold itself separate both from the Dogmatics and the Empirics. {440}

[Note 6\17: Sprengel, _Gesch. Arz._ ii. 5.]

I have noticed these schools of medicine, because, though I am not able to state distinctly their respective merits in the cultivation of anatomy, a great progress in that science was undoubtedly made during their domination, of which the praise must, I conceive, be in some way divided among them. The amount of this progress we are able to estimate, when we come to the works of Galen, who flourished under the Antonines, and died about A.D. 203. The following passage from his works will show that this progress in knowledge was not made without the usual condition of laborious and careful experiment, while it implies the curious fact of such experiment being conducted by means of family tradition and instruction, so as to give rise to a _caste_ of dissectors. In the opening of his Second Book _On Anatomical Manipulations_, he speaks thus of his predecessors: "I do not blame the ancients, who did not write books on anatomical manipulation; though I praise Marinus, who did. For it was superfluous for them to compose such records for themselves or others, while they were, from their childhood, exercised by their parents in dissecting, just as familiarly as in writing and reading; so that there was no more fear of their forgetting their anatomy, than of forgetting their alphabet. But when grown men, as well as children, were taught, this thorough discipline fell off; and, the art being carried out of the family of the Asclepiads, and declining by repeated transmission, books became necessary for the student."

That the general structure of the animal frame, as composed of bones and muscles, was known with great accuracy before the time of Galen, is manifest from the nature of the mistakes and deficiencies of his predecessors which he finds it necessary to notice. Thus he observes, that some anatomists have made one muscle into two, from its having two heads;--that they have overlooked some of the muscles in the face of an ape, in consequence of not skinning the animal with their own hands;--and the like. Such remarks imply that the current knowledge of this kind was tolerably complete. Galen's own views of the general mechanical structure of an animal are very clear and sound. The skeleton, he observes, discharges[7\17] the office of the pole of a tent, or the walls of a house. With respect to the action of the muscles, his views were anatomically and mechanically correct; in some instances, he showed what this action was, by severing the muscle.[8\17] He himself added considerably to the existing knowledge of {441} this subject; and his discoveries and descriptions, even of very minute parts of the muscular system, are spoken of with praise by modern anatomists.[9\17]

[Note 7\17: _De Anatom. Administ._ i. 2.]

[Note 8\17: Sprengel, ii. 157.]

[Note 9\17: Sprengel, ii. 150.]

We may consider, therefore, that the doctrine of the muscular system, as a collection of cords and sheets, by the contraction of which the parts of the body are moved and supported, was firmly established, and completely followed into detail, by Galen and his predecessors. But there is another class of organs connected with voluntary motion, the nerves, and we must for a moment trace the opinions which prevailed respecting these. Aristotle, as we have said, noticed some of the nerves of sensation. But Herophilus, who lived in Egypt in the time of the first Ptolemy, distinguished nerves as the organs of the will,[10\17] and Rufus, who lived in the time of Trajan,[11\17] divides the nerves into sensitive and motive, and derives them all from the brain. But this did not imply that men had yet distinguished the nerves from the muscles. Even Galen maintained that every muscle consists of a bundle of nerves and sinews.[12\17] But the important points, the necessity of the nerve, and the origination of all this apparatus of motion from the brain, he insists upon with great clearness and force. Thus he proved the necessity experimentally, by cutting through some of the bundles of nerves,[13\17] and thus preventing the corresponding motions. And it is, he says,[14\17] allowed by all, both physicians and philosophers, that where the origin of the nerve is, there the seat of the soul (ἡγημονικὸν τῆς ψυχῆς) must be: now this, he adds, is in the brain, and not in the heart.

[Note 10\17: Ib. i. 534.]

[Note 11\17: Ib. ii. 67.]

[Note 12\17: Ibid. ii. 152. Galen, _De Motu Musc._, p. 553.]

[Note 13\17: Ib. 157.]

[Note 14\17: _De Hippocr. et Plat. Dog._ viii. 1.]

Thus the general construction and arrangement of the organization by which voluntary motion is effected, was well made out at the time of Galen, and is found distinctly delivered in his works. We cannot, perhaps, justly ascribe any large portion of the general discovery to him: indeed, the conception of the mechanism of the skeleton and muscles was probably so gradually unfolded in the minds of anatomical students, that it would be difficult, even if we knew the labors of each person, to select one, as peculiarly the author of the discovery. But it is clear that all those who did materially contribute to the establishment of this doctrine, must have possessed the qualifications which we find in Galen for such a task; namely, clear mechanical views of what the {442} tensions of collections of strings could do, and an exact practical acquaintance with the muscular cordage which exists in the animal frame;--in short, in this as in other instances of real advance in science, there must have been clear ideas and real facts, unity of thought and extent of observation, brought into contact.

_Sect._ 2.--_Recognition of Final Causes in Physiology. Galen._

THERE is one idea which the researches of the physiologist and the anatomist so constantly force upon him, that he cannot help assuming it as one of the guides of his speculations; I mean, the idea of a _purpose_, or, as it is called in Aristotelian phrase, a _final cause_, in the arrangements of the animal frame. It is impossible to doubt

that the motive nerves run along the limbs, _in order that_ they may convey to the muscles the impulses of the will; and that the muscles are attached to the bones, _in order that_ they may move and support them. This conviction prevails so steadily among anatomists, that even when the use of any part is altogether unknown, it is still taken for granted that it has some use. The developement of this conviction,--of a purpose in the parts of animals,--of a function to which each portion of the organization is subservient,--contributed greatly to the progress of physiology; for it constantly urged men forwards in their researches respecting each organ, till some definite view of its purpose was obtained. The assumption of hypothetical final causes in Physics may have been, as Bacon asserts it to have been, prejudicial to science; but the assumption of unknown final causes in Physiology, has given rise to the science. The two branches of speculation, Physics and Physiology, were equally led, by every new phenomenon, to ask their question, "Why?" But, in the former case, "why" meant "through what cause?" in the latter, "for what end?" And though it may be possible to introduce into physiology the doctrine of efficient causes, such a step can never obliterate the obligations which the science owes to the pervading conception of a purpose contained in all organization.

This conception makes its appearance very early. Indeed, without any special study of our structure, the thought, that we are fearfully and wonderfully made, forces itself upon men, with a mysterious impressiveness, as a suggestion of our Maker. In this bearing, the thought is developed to a considerable extent in the well-known passage in Xenophon's _Conversations of Socrates_. Nor did it ever lose its hold on sober-minded and instructed men. The Epicureans, indeed, {443} held that the eye was not made for seeing, nor the ear for hearing; and Asclepiades, whom we have already mentioned as an impudent pretender, adopted this wild dogma.[15\17] Such assertions required no labor. "It is easy," says Galen,[16\17] "for people like Asclepiades, when they come to any difficulty, to say that Nature has worked to no purpose." The great anatomist himself pursues his subject in a very different temper. In a well-known passage, he breaks out into an enthusiastic scorn of the folly of the atheistical notions.[17\17] "Try," he says, "if you can imagine a shoe made with half the skill which appears in the skin of the foot." Some one had spoken of a structure of the human body which he would have preferred to that which it now has. "See," Galen exclaims, after pointing out the absurdity of the imaginary scheme, "see what brutishness there is in this wish. But if I were to spend more words on such cattle, reasonable men might blame me for desecrating my work, which I regard as a religious hymn in honor of the Creator."

[Note 15\17: Sprengel, ii. 15.]

[Note 16\17: _De Usu Part._ v. 5, (on the kidneys.)]

[Note 17\17: _De Usu Part._ iii. 10.]

Galen was from the first highly esteemed as an anatomist. He was originally of Pergamus; and after receiving the instructions of many medical and philosophical professors, and especially of those of Alexandria, which was then the metropolis of the learned and scientific world, he came to Rome, where his reputation was soon so

great as to excite the envy and hatred of the Roman physicians. The emperors Marcus Aurelius and Lucius Verus would have retained him near them; but he preferred pursuing his travels, directed principally by curiosity. When he died, he left behind him numerous works, all of them of great value for the light they throw on the history of anatomy and medicine; and these were for a long period the storehouse of all the most important anatomical knowledge which the world possessed. In the time of intellectual barrenness and servility, among the Arabians and the Europeans of the dark ages, the writings of Galen had almost unquestioned authority;[18\17] and it was only by an uncommon effort of independent thinking that Abdollatif ventured to assert, that even Galen's assertions must give way to the evidence of the senses. In more modern times, when Vesalius, in the sixteenth century, accused Galen of mistakes, he drew upon himself the hostility of the whole body of physicians. Yet the mistakes were such as might have {444} been pointed out and confessed[19\17] without acrimony, if, in times of revolution, mildness and moderation were possible; but an impatience of the superstition of tradition on the part of the innovators, and an alarm of the subversion of all recognized truths on the part of the established teachers, inflame and pervert all such discussions. Vesalius's main charge against Galen is, that his dissections were performed upon animals, and not upon the human body. Galen himself speaks of the dissection of apes as a very familiar employment, and states that he killed them by drowning. The natural difficulties which, in various ages, have prevented the unlimited prosecution of human dissection, operated strongly among the ancients, and it would have been difficult, under such circumstances, to proceed more judiciously than Galen did.

[Note 18\17: Sprengel, ii. 359.]

[Note 19\17: Cuv. _Leçons sur l'Hist. des Sc. Nat._ p. 25.]

I shall now proceed to the history of the discovery of another and less obvious function, the circulation of the blood, which belongs to modern times.

## CHAPTER II.

DISCOVERY OF THE CIRCULATION OF THE BLOOD.

_Sect._ 1.--_Prelude to the Discovery._

THE blood-vessels, the veins and arteries, are as evident and peculiar in their appearance as the muscles; but their function is by no means so obvious. Hippocrates[20\17] did not discriminate Veins and Arteries; both are called by the same name (φλέβες) and the word from which artery comes (ἀρτηρίη) means, in his works, the windpipe. Aristotle, scanty as was his knowledge of the vessels of the body, has yet the merit of having traced the origin of all the veins to the heart. He expressly contradicts those of his predecessors who had derived the veins from the head;[21\17] and refers to dissection for the proof. If the book _On the Breath_ be genuine (which is doubted), Aristotle was aware of the distinction between veins and arteries. "Every artery," {445} it is there asserted, "is accompanied by a vein; the former are filled only with breath or air."[22\17] But whether or no this passage be

Aristotle's, he held opinions equally erroneous; as, that the windpipe conveys air into the heart.[23\17] Galen[24\17] was far from having views respecting the blood-vessels, as sound as those which he entertained concerning the muscles. He held the liver to be the origin of the veins, and the heart of the arteries. He was, however, acquainted with their junctions, or _anastomoses_. But we find no material advance in the knowledge of this subject, till we overleap the blank of the middle ages, and reach the dawn of modern science.

[Note 20\17: Sprengel, i. 383.]

[Note 21\17: _Hist. Animal._ iii. 3.]

[Note 22\17: _De Spiritu_, v. 1078.]

[Note 23\17: Spr. i. 501.]

[Note 24\17: Ib. ii. 152.]

The father of modern anatomy is held to be Mondino,[25\17] who dissected and taught at Bologna in 1315. Some writers have traced in him the rudiments of the doctrine of the circulation of the blood; for he says that the heart transmits blood to the lungs. But it is allowed, that he afterwards destroys the merit of his remark, by repeating the old assertion that the left ventricle ought to contain spirit or air, which it generates from the blood.

[Note 25\17: _Encyc. Brit._ 692. Anatomy.]

Anatomy was cultivated with great diligence and talent in Italy by Achillini, Carpa, and Messa, and in France by Sylvius and Stephanus (Dubois and Etienne). Yet still these empty assumptions respecting the heart and blood-vessels kept their ground. Vesalius, a native of Brussels, has been termed the founder of human anatomy, and his great work _De Humani Corporis Fabricâ_ is, even yet, a splendid monument of art, as well as science. It is said that his figures were designed by Titian; and if this be not exactly true, says Cuvier,[26\17] they must, at least, be from the pencil of one of the most distinguished pupils of the great painter; for to this day, though we have more finished drawings, we have no designs that are more artist-like. Fallopius, who succeeded Vesalius at Padua, made some additions to the researches of his predecessor; but in his treatise _De Principio Venarum_, it is clearly seen[27\17] that the circulation of the blood was unknown to him. Eustachius also, whom Cuvier groups with Vesalius and Fallopius, as the three great founders of modern anatomy, wrote a treatise on the vein _azygos_[28\17] which is a little treatise on comparative anatomy; but the discovery of the functions of the veins came from a different quarter. {446}

[Note 26\17: _Leçons sur l'Hist. des Sc. Nat._ p. 21.]

[Note 27\17: Cuv. _Sc. Nat._ p. 32.]

[Note 28\17: Ib. p. 34.]

The unfortunate Servetus, who was burnt at Geneva as a heretic in 1553, is the first person who speaks distinctly of the small circulation, or that which carries the

blood from the heart to the lungs, and back again to the heart. His work entitled _Christianismi Restitutio_ was also burnt; and only two copies are known to have escaped the flames. It is in this work that he asserts the doctrine in question, as a collateral argument or illustration of his subject. "The communication between the right and left ventricle of the heart, is made," he says, "not as is commonly believed, through the partition of the heart, but by a remarkable artifice (_magno artificio_) the blood is carried from the right ventricle by a long circuit through the lungs; is elaborated by the lungs, made yellow, and transfused from the _vena arteriosa_ into the _arteria venosa_." This truth is, however, mixed with various of the traditional fancies concerning the "_vital spirit_, which has its origin in the left ventricle." It may be doubted, also, how far Servetus formed his opinion upon conjecture, and on a hypothetical view of the formation of this vital spirit. And we may, perhaps, more justly ascribe the real establishment of the pulmonary circulation as an inductive truth, to Realdus Columbus, a pupil and successor of Vesalius at Padua, who published a work _De Re Anatomicâ_ in 1559, in which he claims this discovery as his own.[29\17]

[Note 29\17: _Encyc. Brit._]

Andrew Cæsalpinus, who has already come under our notice as one of the fathers of modern inductive science, both by his metaphysical and his physical speculations, described the pulmonary circulation still more completely in his _Quæstiones Peripateticæ_, and even seemed to be on the eve of discovering the great circulation; for he remarked the swelling of veins below ligatures, and inferred from it a refluent motion of blood in these vessels.[30\17] But another discovery of structure was needed, to prepare the way for this discovery of function; and this was made by Fabricius of Acquapendente, who succeeded in the grand list of great professors at Padua, and taught there for fifty years.[31\17] Sylvius had discovered the existence of the valves of the veins; but Fabricius remarked that they are all turned towards the heart. Combining this disposition with that of the valves of the heart, and with the absence of valves in the arteries, he might have come to the conclusion[32\17] that the blood moves in a different direction in the arteries and in the veins, and might thus have discovered the circulation: but this glory was reserved for William Harvey: so true {447} is it, observes Cuvier, that we are often on the brink of a discovery without suspecting that we are so;--so true is it, we may add, that a certain succession of time and of persons is generally necessary to familiarize men with one thought, before they can advance to that which is the next in order.

[Note 30\17: Ib.]

[Note 31\17: Cuv. p. 44.]

[Note 32\17: p. 45.]

_Sect._ 2.--_The Discovery of the Circulation made by Harvey._

WILLIAM HARVEY was born in 1578, at Folkestone in Kent.[33\17] He first studied at Cambridge: he afterwards went to Padua, where the celebrity of Fabricius of Acquapendente attracted from all parts those who wished to be instructed in anatomy and physiology. In this city, excited by the discovery of the valves of the veins, which

his master had recently made, and reflecting on the direction of the valves which are at the entrance of the veins into the heart, and at the exit of the arteries from it, he conceived the idea of making experiments, in order to determine what is the course of the blood in its vessels. He found that when he tied up veins in various animals, they swelled below the ligature, or in the part furthest from the heart; while arteries, with a like ligature, swelled on the side next the heart. Combining these facts with the direction of the valves, he came to the conclusion that the blood is impelled, by the left side of the heart, in the arteries to the extremities, and thence returns by the veins into the right side of the heart. He showed, too, how this was confirmed by the phenomena of the pulse, and by the results of opening the vessels. He proved, also, that the circulation of the lungs is a continuation of the larger circulation; and thus the whole doctrine of the double circulation was established.

[Note 33\17: Cuv. p. 51.]

Harvey's experiments had been made in 1616 and 1618; it is commonly said that he first promulgated his opinion in 1619; but the manuscript of the lectures, delivered by him as lecturer to the College of Physicians, is extant in the British Museum, and, containing the propositions on which the doctrine is founded, refers them to April, 1616. It was not till 1628 that he published, at Frankfort, his _Exercitatio Anatomica de Motu Cordis et Sanguinis_; but he there observes that he had for above nine years confirmed and illustrated his opinion in his lectures, by arguments grounded upon ocular demonstrations. {448}

_Sect._ 3.--_Reception of the Discovery._

WITHOUT dwelling long upon the circumstances of the general reception of this doctrine, we may observe that it was, for the most part, readily accepted by his countrymen, but that abroad it had to encounter considerable opposition. Although, as we have seen, his predecessors had approached so near to the discovery, men's minds were by no means as yet prepared to receive it. Several physicians denied the truth of the opinion, among whom the most eminent was Riolan, professor at the Collège de France. Other writers, as usually happens in the case of great discoveries, asserted that the doctrine was ancient, and even that it was known to Hippocrates. Harvey defended his opinion with spirit and temper; yet he appears to have retained a lively recollection of the disagreeable nature of the struggles in which he was thus involved. At a later period of his life, Ent,[34\17] one of his admirers, who visited him, and urged him to publish the researches on generation, on which he had long been engaged, gives this account of the manner in which he received the proposal: "And would you then advise me, (smilingly replies the doctor,) to quit the tranquillity of this haven, wherein I now calmly spend my days, and again commit myself to the unfaithful ocean? You are not ignorant how great troubles my lucubrations, formerly published, have raised. Better it is, certainly, at some time, to endeavor to grow wise at home in private, than by the hasty divulgation of such things to the knowledge whereof you have attained with vast labor, to stir up tempests that may deprive you of your leisure and quiet for the future."

[Note 34\17: Epist. Dedic. to _Anatom. Exercit._]

His merits were, however, soon generally recognized. He was[35\17] made physician to James the First, and afterwards to Charles the First, and attended that unfortunate monarch in the civil war. He had the permission of the parliament to accompany the king on his leaving London; but this did not protect him from having his house plundered in his absence, not only of its furniture, but, which he felt more, of the records of his experiments. In 1652, his brethren of the College of Physicians placed a marble bust of him in their hall, with an inscription recording his discoveries; and two years later, he was nominated to the office of President of the College, which however he {449} declined in consequence of his age and infirmities. His doctrine soon acquired popular currency; it was, for instance, taken by Descartes[36\17] as the basis of his physiology in his work _On Man_; and Harvey had the pleasure, which is often denied to discoverers, of seeing his discovery generally adopted during his lifetime.

[Note 35\17: _Biog. Brit._]

[Note 36\17: Cuv. 53.]

_Sect._ 4.--_Bearing of the Discovery on the Progress of Physiology._

IN considering the intellectual processes by which Harvey's discoveries were made, it is impossible not to notice, that the recognition of a creative purpose, which, as we have said, appears in all sound physiological reasonings, prevails eminently here. "I remember," says Boyle, "that when I asked our famous Harvey what were the things that induced him to think of a circulation of the blood, he answered me, that when he took notice that the valves in the veins of so many parts of the body were so placed, that they gave a free passage to the blood towards the heart, but opposed the passage of the venal blood the contrary way; he was incited to imagine that so provident a cause as Nature had not placed so many valves without design; and no design seemed more probable than that the blood should be sent through the arteries, and return through the veins, whose valves did not oppose its course that way."

We may notice further, that this discovery implied the usual conditions, distinct general notions, careful observation of many facts, and the mental act of bringing together these elements of truth. Harvey must have possessed clear views of the motions and pressures of a fluid circulating in ramifying tubes, to enable him to see how the position of valves, the pulsation of the heart, the effects of ligatures, of bleeding, and of other circumstances, ought to manifest themselves in order to confirm his view. That he referred to a multiplied and varied experience for the evidence that it was so confirmed, we have already said. Like all the best philosophers of his time, he insists rigidly upon the necessity of such experience. "In every science," he says,[37\17] "be it what it will, a diligent observation is requisite, and sense itself must be frequently consulted. We must not rely upon other men's experience, but our own, without which no man is a proper disciple of any part of natural knowledge." And by publishing his experiments, he trusts, he adds, that he has enabled his reader "to be an equitable {450} umpire between Aristotle and Galen;" or rather, he might have said, to see how, in the promotion of science, sense

and reason, observation and invention, have a mutual need of each other.

[Note 37\17: _Generation of Animals_, Pref.]

We may observe further, that though Harvey's glory, in the case now before us, rested upon his having proved the reality of certain mechanical movements and actions in the blood, this discovery, and all other physiological truths, necessarily involved the assumption of some peculiar agency belonging to living things, different both from mechanical agency, and from chemical; and in short, something _vital_, and not physical merely. For when it was seen that the pulsation of the heart, its _systole_ and _diastole_, caused the circulation of the blood, it might still be asked, what force caused this constantly-recurring contraction and expansion. And again, circulation is closely connected with respiration; the blood is, by the circulation, carried to the lungs, and is there, according to the expression of Columbus and Harvey, mixed with air. But by what mechanism does this _mixture_ take place, and what is the real nature of it? And when succeeding researches had enabled physiologists to give an answer to this question, as far as chemical relations go, and to say, that the change consists in the abstraction of the carbon from the blood by means of the oxygen of the atmosphere; they were still only led to ask further, how this chemical change was effected, and how such a change of the blood fitted it for its uses. Every function of which we explain the course, the mechanism, or the chemistry, is connected with other functions,--is subservient to them, and they to it; and all together are parts of the general vital system of the animal, ministering to its life, but deriving their activity from the life. Life is not a collection of forces, or polarities, or affinities, such as any of the physical or chemical sciences contemplate; it has powers of its own, which often supersede those subordinate relations; and in the cases where men have traced such agents in the animal frame, they have always seen, and usually acknowledged, that these agents were ministerial to some higher agency, more difficult to trace than these, but more truly the cause of the phenomena.

The discovery of the mechanical and chemical conditions of the vital functions, as a step in physiology, may be compared to the discovery of the laws of phenomena in the heavens by Kepler and his predecessors, while the discovery of the force by which they were produced was still reserved in mystery for Newton to bring to light. The subordinate relation of the facts, their **dependence on space and time, their reduction to order and cycle, had been fully performed; but the {451} reference of them to distinct ideas of causation, their interpretation as the results of mechanical force, was omitted or attempted in vain. The very notion of such Force, and of the manner in which motions were determined by it, was in the highest degree vague and vacillating; and a century was requisite, as we have seen, to give to the notion that clearness and fixity which made the Mechanics of the Heavens a possible science. In like manner, the notion of Life, and of Vital Forces, is still too obscure to be steadily held. We cannot connect it distinctly with severe inductions from facts. We can trace the motions of the animal fluids as Kepler traced the motions of the planets; but when we seek to render a reason for these motions, like him, we recur to terms of a wide and profound, but mysterious import; to Virtues, Influences, undefined Powers. Yet we are not on this account to despair. The very instance to which I am referring

shows us how rich is the promise of the future. Why, says Cuvier,[38\17] may not Natural History one day have its Newton? The idea of the vital forces may gradually become so clear and definite as to be available in science; and future generations may include, in their physiology, propositions elevated as far above the circulation of the blood, as the doctrine of universal gravitation goes beyond the explanation of the heavenly motions by epicycles.

[Note 38\17: _Ossem. Foss._ Introd.]

If, by what has been said, I have exemplified sufficiently the nature of those steps in physiology, which, like the discovery of the Circulation, give an explanation of the process of some of the animal functions, it is not necessary for me to dwell longer on the subject; for to write a history, or even a sketch of the history of Physiology, would suit neither my powers nor my purpose. Some further analysis of the general views which have been promulgated by the most eminent physiologists, may perhaps be attempted in treating of the Philosophy of Inductive Science; but the estimation of the value of recent speculations and investigations must be left to those who have made this vast subject the study of their lives. A few brief notices may, however, be here introduced. {452}

## CHAPTER III.

DISCOVERY OF THE MOTION OF THE CHYLE, AND CONSEQUENT SPECULATIONS.

_Sect._ 1.--_The Discovery of the Motion of the Chyle._

IT may have been observed in the previous course of this History of the Sciences, that the discoveries in each science have a peculiar physiognomy: something of a common type may be traced in the progress of each of the theories belonging to the same department of knowledge. We may notice something of this common form in the various branches of physiological speculation. In most, or all of them, we have, as we have noticed the case to be with respect to the circulation of the blood, clear and certain discoveries of mechanical and chemical processes, succeeded by speculations far more obscure, doubtful, and vague, respecting the relation of these changes to the laws of life. This feature in the history of physiology may be further instanced, (it shall be done very briefly), in one or two other cases. And we may observe, that the lesson which we are to collect from this narrative, is by no means that we are to confine ourselves to the positive discovery, and reject all the less clear and certain speculations. To do this, would be to lose most of the chances of ulterior progress; for though it may be, that our conceptions of the nature of organic life are not yet sufficiently precise and steady to become the guides to positive inductive truths, still the only way in which these peculiar physiological ideas can be made more distinct and precise, and thus brought more nearly into a scientific form, is by this struggle with our ignorance or imperfect knowledge. This is the lesson we have learnt from the history of physical astronomy and other sciences. We must strive to refer facts which are known and understood, to higher principles, of which we cannot doubt the existence, and of which, in some degree, we can see the place; however dim and

shadowy may be the glimpses we have hitherto been able to obtain of their forms. We may often fail in such attempts, but without the attempt we can never succeed. {453}

That the food is received into the stomach, there undergoes a change of its consistence, and is then propelled along the intestines, are obvious facts in the animal economy. But a discovery made in the course of the seventeenth century brought into clearer light the sequel of this series of processes, and its connexion with other functions. In the year 1622, Asellius or Aselli[39\17] discovered certain minute vessels, termed _lacteals_, which absorb a white liquid (the _chyle_) from the bowels, and pour it into the blood. These vessels had, in fact, been discovered by Eristratus, in the ancient world,[40\17] in the time of Ptolemy; but Aselli was the first modern who attended to them. He described them in a treatise entitled _De Venis Lacteis, cum figuris elegantissimis_, printed at Milan in 1627, the year after the death of the author. The work is remarkable as the first which exhibits _colored_ anatomical figures; the arteries and veins are represented in red, the lacteals in black.

[Note 39\17: Mayo, _Physiology_, p. 156.]

[Note 40\17: Cuv. _Hist. Sc._ p. 50.]

Eustachius,[41\17] at an earlier period, had described (in the horse) the thoracic duct by which the chyle is poured into the subclavian vein, on the right side of the neck. But this description did not excite so much notice as to prevent its being forgotten, and rediscovered in 1550, after the knowledge of the circulation of the blood had given more importance to such a discovery. Up to this time,[42\17] it had been supposed that the lacteals carried the chyle to the liver, and that the blood was manufactured there. This opinion had prevailed in all the works of the ancients and moderns; its falsity was discovered by Pecquet, a French physician, and published in 1651, in his _New Anatomical Experiments_; in which are discovered a receptacle of the chyle, unknown till then, and the vessel which conveys it to the subclavian vein. Pecquet himself and other anatomists, soon connected this discovery with the doctrine, then recently promulgated, of the circulation of the blood. In 1665, these vessels, and the _lymphatics_ which are connected with them, were further illustrated by Ruysch in his exhibition of their valves. (_Dilucidatio valvularum in vasis lymphaticis et lacteis_.)

[Note 41\17: Cuv. _Hist._ p. 34.]

[Note 42\17: Ib. p. 365.]

_Sect._ 2.--_The Consequent Speculations. Hypotheses of Digestion._

THUS it was shown that aliments taken into the stomach are, by its action, made to produce _chyme_; from the chyme, gradually changed {454} in its progress through the intestines, _chyle_ is absorbed by the lacteals; and this, poured into the blood by the thoracic duct, repairs the waste and nourishes the growth of the animal. But by what powers is the food made to undergo these transformations? Can we explain them on mechanical or on chemical principles? Here we come to a part of physiology less certain than the discovery of vessels, or of the motion of fluids. We have a number of opinions on this subject, but no universally acknowledged truth. We have a

collection of _Hypotheses of Digestion_ and _Nutrition_.

I shall confine myself to the former class; and without dwelling long upon these, I shall mention some of them. The philosophers of the Academy _del Cimento_, and several others, having experimented on the stomach of gallinaceous birds, and observed the astonishing force with which it breaks and grinds substances, were led to consider the digestion which takes place in the stomach as a kind of _trituration_.[43\17] Other writers thought it was more properly described as _fermentation_; others again spoke of it as a _putrefaction_. Varignon gave a merely physical account of the first part of the process, maintaining that the division of the aliments was the effect of the disengagement of the air introduced into the stomach, and dilated by the heat of the body. The opinion that digestion is a _solution_ of the food by the gastric juice has been more extensively entertained.

[Note 43\17: Bourdon, _Physiol. Comp._ p. 514.]

Spallanzani and others made many experiments on this subject. Yet it is denied by the best physiologists, that the changes of digestion can be adequately represented as chemical changes only. The nerves of the stomach (the _pneumo-gastric_) are said to be essential to digestion. Dr. Wilson Philip has asserted that the influence of these nerves, when they are destroyed, may be replaced by a galvanic current.[44\17] This might give rise to a supposition that digestion depends on galvanism. Yet we cannot doubt that all these hypotheses,--mechanical, physical, chemical, galvanic--are altogether insufficient. "The stomach must have," as Dr. Prout says,[45\17] "the power of {455} organizing and vitalizing the different elementary substances. It is impossible to imagine that this organizing agency of the stomach can be chemical. This agency is _vital_, and its nature completely unknown."

[Note 44\17: Müller (_Manual of Physiology_, B. iii. Sect. 1, Chap. iii.) speaks of Dr. Wilson Philip's assertion that the nerves of the stomach being cut, and a galvanic current kept up in them, digestion is still accomplished. He states that he and other physiologists have repeated such experiments on an extensive scale, and have found no effect of this kind.]

[Note 45\17: _Bridgewater Tr._ p. 493.]

## CHAPTER IV.

EXAMINATION OF THE PROCESS OF REPRODUCTION IN ANIMALS AND PLANTS, AND CONSEQUENT SPECULATIONS.

_Sect._ 1.--_The Examination of the Process of Reproduction in Animals._

IT would not, perhaps, be necessary to give any more examples of what has hitherto been the general process of investigations on each branch of physiology; or to illustrate further the combination which such researches present, of certain with uncertain knowledge;--of solid discoveries of organs and processes, succeeded by indefinite and doubtful speculation concerning vital forces. But the reproduction of organized beings is not only a subject of so much interest as to require some notice, but also offers to us laws and principles which include both the vegetable and the

animal kingdom; and which, therefore, are requisite to render intelligible the most general views to which we can attain, respecting the world of organization.

The facts and laws of reproduction were first studied in detail in animals. The subject appears to have attracted the attention of some of the philosophers of antiquity in an extraordinary degree: and indeed we may easily imagine that they hoped, by following this path, if any, to solve the mystery of creation. Aristotle appears to have pursued it with peculiar complacency; and his great work _On animals_ contains[46\17] an extraordinary collection of curious observations relative to this subject. He had learnt the modes of reproduction of most of the animals with which he was acquainted; and his work is still, as a writer of our own times has said,[47\17] "original after so many copies, and young after two thousand years." His observations referred principally to the external circumstances of generation: the anatomical examination was {456} left to his successors. Without dwelling on the intermediate labors, we come to modern times, and find that this examination owes its greatest advance to those who had the greatest share in the discovery of the circulation of the blood;--Fabricius of Acquapendente, and Harvey. The former[48\17] published a valuable work on the Egg and the Chick. In this are given, for the first time, figures representing the developement of the chick, from its almost imperceptible beginning, to the moment when it breaks the shell. Harvey pursued the researches of his teacher. Charles[49\17] the First had supplied him with the means of making the experiments which his purpose required, by sacrificing a great number of the deer in Windsor Park in the state of gestation: but his principal researches were those respecting the egg, in which he followed out the views of Fabricius. In the troubles which succeeded the death of the unfortunate Charles the house of Harvey was pillaged; and he lost the whole of the labors he had bestowed on the generation of insects. His work, _Exercitationes de Generatione Animalium_, was published at London in 1651; it is more detailed and perfect than that of Fabricius; but the author was prevented by the unsettled condition of the country from getting figures engraved to accompany his descriptions.

[Note 46\17: Bourdon, p. 161.]

[Note 47\17: Ib. p. 101.]

[Note 48\17: Cuv. _Hist. Sc. Nat._ p. 46.]

[Note 49\17: Ib. p. 53.]

Many succeeding anatomists pursued the examination of the series of changes in generation, and of the organs which are concerned in them, especially Malpighi, who employed the microscope in this investigation, and whose work on the Chick was published in 1673. It is impossible to give here any general view of the result of these laborious series of researches: but we may observe, that they led to an extremely minute and exact survey of all the parts of the fœtus, its envelopes and appendages, and, of course, to a designation of these by appropriate names. These names afterwards served to mark the attempts which were made to carry the analogy of animal generation into the vegetable kingdom.

There is one generalization of Harvey which deserves notice.[50\17] He was led by his researches to the conclusion, that all living things may be properly said to come from eggs: "Omne vivum ex ovo." Thus not only do oviparous animals produce by means of eggs, but in those which are viviparous, the process of generation begins with the developement of a small vesicle, which comes from the ovary, and which exists before the embryo: and thus viviparous or suckling-beasts, {457} notwithstanding their name, are born from eggs, as well as birds, fishes, and reptiles.[51\17] This principle also excludes that supposed production of organized beings without parents (of worms in corrupted matter, for instance,) which was formerly called _spontaneous generation_; and the best physiologists of modern times agree in denying the reality of such a mode of generation.[52\17]

[Note 50\17: Exerc. lxiii.]

[Note 51\17: Bourdon, p. 221.]

[Note 52\17: Ib. p. 49.]

_Sect._ 2.--_The Examination of the Process of Reproduction in Vegetables._

THE extension of the analogies of animal generation to the vegetable world was far from obvious. This extension was however made;--with reference to the embryo plant, principally by the microscopic observers, Nehemiah Grew, Marcello Malpighi, and Antony Leeuwenhoek;--with respect to the existence of the sexes, by Linnæus and his predecessors.

The microscopic labors of Grew and Malpighi were patronized by the Royal Society of London in its earliest youth. Grew's book, _The Anatomy of Plants_, was ordered to be printed in 1670. It contains plates representing extremely well the process of germination in various seeds, and the author's observations exhibit a very clear conception of the relation and analogies of different portions of the seed. On the day on which the copy of this work was laid before the Society, a communication from Malpighi of Bologna, _Anatomes Plantarum Idea_, stated his researches, and promised figures which should illustrate them. Both authors afterwards went on with a long train of valuable observations, which they published at various times, and which contain much that has since become a permanent portion of the science.

Both Grew and Malpighi were, as we have remarked, led to apply to vegetable generation many terms which imply an analogy with the generation of animals. Thus, Grew terms the innermost coat of the seed, the _secundine_; speaks of the _navel-fibres_, &c. Many more such terms have been added by other writers. And, as has been observed by a modern physiologist,[53\17] the resemblance is striking. Both in the vegetable seed and in the fertilized animal egg, we have an _embryo_, _chalazæ_, a _placenta_, an _umbilical cord_, a _cicatricula_, an _amnios_, _membranes_, _nourishing vessels_. The _cotyledons_ of the seed are the equivalent of the _vitellus_ of birds, or of the _umbilical vesicle_ of **suckling-beasts: {458} the _albumen_ or _perisperm_ of the grain is analogous to the _white of the egg_ of birds, or the _allantoid_ of viviparous animals.

[Note 53\17: Ib. p. 384.]

_Sexes of Plants._--The attribution of sexes to plants, is a notion which was very early adopted; but only gradually unfolded into distinctness and generality.[54\17] The ancients were acquainted with the fecundation of vegetables. Empedocles, Aristotle, Theophrastus, Pliny, and some of the poets, make mention of it; but their notions were very incomplete, and the conception was again lost in the general shipwreck of human knowledge. A Latin poem, composed in the fifteenth century by Jovianus Pontanus, the preceptor of Alphonso, King of Naples, is the first modern work in which mention is made of the sex of plants. Pontanus sings the loves of two date-palms, which grew at the distance of fifteen leagues from each other: the male at Brundusium, the female at Otranto. The distance did not prevent the female from becoming fruitful, as soon as the palms had raised their heads above the surrounding trees, so that nothing intervened directly between them, or, to speak with the poet, so that they were able to see each other.

[Note 54\17: Mirbel, _El._ ii. 538.]

Zaluzian, a botanist who lived at the end of the fifteenth century, says that the greater part of the species of plants are _androgynes_, that is, have the properties of the male and of the female united in the same plant; but that some species have the two sexes in separate individuals; and he adduces a passage of Pliny relative to the fecundation of the date-palm. John Bauhin, in the middle of the seventeenth century, cites the expressions of Zaluzian; and forty years later, a professor of Tübingen, Rudolph Jacob Camerarius, pointed out clearly the organs of generation, and proved by experiments on the mulberry, on maize, and on the plant called Mercury (_mercurialis_), that when by any means the action of the stamina upon the pistils is intercepted, the seeds are barren. Camerarius, therefore, a philosopher in other respects of little note, has the honor assigned him of being the author of the discovery of the sexes of plants in modern times.[55\17]

[Note 55\17: Mirbel, ii. 539.]

The merit of this discovery will, perhaps, appear more considerable when it is recollected that it was rejected at first by very eminent botanists. Thus Tournefort, misled by insufficient experiments, maintained that the stamina are excretory organs; and Reaumur, at the beginning of the eighteenth century, inclined to the same doctrine. {459} Upon this, Geoffroy, an apothecary at Paris, scrutinized afresh the sexual organs; he examined the various forms of the pollen, already observed by Grew and Malpighi; he pointed out the excretory canal, which descends through the style, and the _micropyle_, or minute orifice in the coats of the ovule, which is opposite to the extremity of this canal; though he committed some mistakes with regard to the nature of the pollen. Soon afterwards, Sebastian Vaillant, the pupil of Tournefort, but the corrector of his error on this subject, explained in his public lectures the phenomenon of the fecundation of plants, described the explosion of the anthers, and showed that the _florets_ of composite flowers, though formed on the type of an _androgynous_ flower, are sometimes male, sometimes female, and sometimes neuter.

But though the sexes of plants had thus been noticed, the subject drew far more attention when Linnæus made the sexual parts the basis of his classification. Camerarius and Burkard had already entertained such a thought, but it was Linnæus who carried into effect, and thus made the notion of the sexes of vegetables almost as familiar to us as that of the sexes of animals.

_Sect._ 3.--_The Consequent Speculations.--Hypotheses of Generation._

THE views of the processes of generation, and of their analogies throughout the whole of the organic world, which were thus established and diffused, form an important and substantial part of our physiological knowledge. That a number of curious but doubtful hypotheses should be put forward, for the purpose of giving further significance and connexion to these discoveries, was to be expected. We must content ourselves with speaking of these very briefly. We have such hypotheses in the earliest antiquity of Greece; for as we have already said, the speculations of cosmogony were the source of the Greek philosophy; and the laws of generation appeared to offer the best promise of knowledge respecting the mystery of creation. Hippocrates explained the production of a new animal by the _mixture of seed_ of the parents; and the offspring was male or female as the seminal principle of the father or of the mother was the more powerful. According to Aristotle, the mother supplied the _matter_, and the father the _form_. Harvey's doctrine was, that the ovary of the female is fertilized by a _seminal contagion_ produced by the seed of the male. But an opinion which obtained far more general reception was, that {460} the _embryo pre-existed_ in the mother, before any union of the sexes.[56\17] It is easy to see that this doctrine is accompanied with great difficulties;[57\17] for if the mother, at the beginning of life, contain in her the embryos of all her future children; these embryos again must contain the children which they are capable of producing; and so on indefinitely; and thus each female of each species contains in herself the germs of infinite future generations. The perplexity which is involved in this notion of an endless series of creatures, thus encased one within another, has naturally driven inquirers to attempt other suppositions. The microscopic researches of Leeuwenhoek and others led them to the belief that there are certain animalcules contained in the seed of the male, which are the main agents in the work of reproduction. This system ascribes almost everything to the male, as the one last mentioned does to the female. Finally, we have the system of Buffon;--the famous hypothesis of _organic molecules_. That philosopher asserted that he found, by the aid of the microscope, all nature full of moving globules, which he conceived to be, not animals as Leeuwenhoek imagined, but bodies capable of producing, by their combination, either animals or vegetables, in short, all organized bodies. These globules he called _organic molecules_.[58\17] And if we inquire how these organic molecules, proceeding from all parts of the two parents, unite into a whole, as perfect as either of the progenitors, Buffon answers, that this is the effect of the _interior mould_; that is, of a system of internal laws and tendencies which determine the form of the result as an external mould determines the shape of the cast.

[Note 56\17: Bourdon, p. 204.]

[Note 57\17: Ib. p. 209.]

[Note 58\17: Ib. p. 219.]

An admirer of Buffon, who has well shown the untenable character of this system, has urged, as a kind of apology for the promulgation of the hypothesis,[59\17] that at the period when its author wrote, he could not present his facts with any hope of being attended to, if he did not connect them by some common tie, some dominant idea which might gratify the mind; and that, acting under this necessity, he did well to substitute for the extant theories, already superannuated and confessedly imperfect, conjectures more original and more probable. Without dissenting from this view, we may observe, that Buffon's theory, like those which preceded it, is excusable, and even deserving of admiration, so far as it groups the facts consistently; because in doing this, it exhibits the necessity, which the physiological speculator ought to feel, of aspiring to definite and solid general principles; and that thus, though {461} the theory may not be established as true, it may be useful by bringing into view the real nature and application of such principles.

[Note 59\17: Ib. p. 221.]

It is, therefore, according to our views, unphilosophical to derive despair, instead of hope, from the imperfect success of Buffon and his predecessors. Yet this is what is done by the writer to whom we refer. "For me," says he,[60\17] "I vow that, after having long meditated on the system of Buffon,--a system so remarkable, so ingenious, so well matured, so wonderfully connected in all its parts, at first sight so probable;--I confess that, after this long study, and the researches which it requires, I have conceived in consequence, a distrust of myself a skepticism, a disdain of hypothetical systems, a decided predilection and exclusive taste for pure and rational observation, in short, a disheartening, which I had never felt before."

[Note 60\17: Bourdon, p. 274.]

The best remedy of such feelings is to be found in the history of science. Kepler, when he had been driven to reject the solid epicycles of the ancients, or a person who had admired Kepler as M. Bourdon admires Buffon, but who saw that his magnetic virtue was an untenable fiction, might, in the same manner, have thrown up all hope of a sound theory of the causes of the celestial motions. But astronomers were too wise and too fortunate to yield to such despondency. The predecessors of Newton substituted a solid science of Mechanics for the vague notions of Kepler; and the time soon came when Newton himself reduced the motions of the heavens to a Law as distinctly conceived as the Motions had been before.

## CHAPTER V.

EXAMINATION OF THE NERVOUS SYSTEM, AND CONSEQUENT SPECULATIONS.

_Sect._ 1.--_The Examination of the Nervous System._

IT is hardly necessary to illustrate by further examples the manner in which anatomical observation has produced conjectural and hypothetical attempts to connect structure and action with some {462} higher principle, of a more peculiarly physiological kind. But it may still be instructive to notice a case in which the

principle, which is thus brought into view, is far more completely elevated above the domain of matter and mechanism than in those we have yet considered;--a case where we have not only Irritation, but Sensation;--not only Life, but Consciousness and Will. A part of science in which suggestions present themselves, brings us, in a very striking manner, to the passage from the physical to the hyperphysical sciences.

We have seen already (chap. i.) that Galen and his predecessors had satisfied themselves that the nerves are the channels of perception; a doctrine which had been distinctly taught by Herophilus[61\17] in the Alexandrian school. Herophilus, however, still combined, under the common name of Nerves, the Tendons; though he distinguished such Nerves from those which arise from the brain and the spinal marrow, and which are subservient to the will. In Galen's time this subject had been prosecuted more into detail. That anatomist has left a Treatise expressly upon _The Anatomy of the Nerves_; in which he describes the successive _Pairs_ of Nerves: thus, the First Pair are the visual nerves: and we see, in the language which Galen uses, the evidence of the care and interest with which he had himself examined them. "These nerves," he says, "are not resolved into many fibres, like all the other nerves, when they reach the organs to which they belong; but spread out in a different and very remarkable manner, which it is not easy to describe or to believe, without actually seeing it." He then gives a description of the retina. In like manner he describes the Second Pair, which is distributed to the muscles of the eyes; the Third and Fourth Pairs, which go to the tongue and palate; and so on to the Seventh Pair. This division into Seven Pairs was established by Marinus,[62\17] but Vesalius found it to be incomplete. The examination which is the basis of the anatomical enumeration of the Nerves at present recognized was that of Willis. His book, entitled _Cerebri Anatome, cui accessit Nervorum descriptio et usus_, appeared at London in 1664. He made important additions to the knowledge of this subject.[63\17] Thus he is the first who describes in a distinct manner what has been called the _Nervous Centre_,[64\17] the pyramidal eminences which, according to more recent anatomists, are the communication of the brain with the spinal marrow: and of which the _Decussation_, described by Santorini, affords the explanation of the action of a part {463} of the brain upon the nerves of the opposite side. Willis proved also that the _Rete Mirabile_, the remarkable net-work of arteries at the base of the brain, observed by the ancients in ruminating animals, does not exist in man. He described the different Pairs of Nerves with more care than his predecessors; and his mode of numbering them is employed up to the present time. He calls the Olfactory Nerves the First Pair; previously to him, these were not reckoned a Pair: and thus the optic nerves were, as we have seen, called the first. He added the Sixth and the Ninth Pairs, which the anatomists who preceded him did not reckon. Willis also examined carefully the different _Ganglions_, or knots which occur upon the nerves. He traced them wherever they were to be found, and he gave a general figure of what Cuvier calls the _nervous skeleton_, very superior to that of Vesalius, which was coarse and inexact. Willis also made various efforts to show the connexion of the parts of the brain. In the earlier periods of anatomy, the brain had been examined by slicing it, so as to obtain a section. Varolius endeavored to unravel it, and was followed by Willis. Vicq d'Azyr, in modern times, has carried the method of section to greater perfection

than had before been given it;[65\17] as Vieussens and Gall have done with respect to the method of Varolius and Willis. Recently Professor Chaussier[66\17] makes three kinds of Nerves:--the _Encephalic_, which proceed from the head, and are twelve on each side;--the _Rachidian_, which proceed from the spinal marrow, and are thirty on each side;--and _Compound Nerves_, among which is the _Great Sympathetic_ Nerve.

[Note 61\17: Spr. i. 534.]

[Note 62\17: _Dic. Sc. Med._ xxxv. 467.]

[Note 63\17: Cuv. _Sc. Nat._ p. 385.]

[Note 64\17: Ibid.]

[Note 65\17: Cuv. p. 40.]

[Note 66\17: _Dict. Sc. Nat._ xxxv. 467.]

One of the most important steps ever made in our knowledge of the nerves is, the distinction which Bichat is supposed to have established, of a _ganglionic system_, and a _cerebral system_. And we may add, to the discoveries in nervous anatomy, the remarkable one, made in our own time, that the two offices--of conducting the motive impressions from the central seat of the will to the muscles, and of propagating sensations from the surface of the body and the external organs of sense to the sentient mind--reside in two distinct portions of the nervous substance:--a discovery which has been declared[67\17] to be "doubtless the most important accession to physiological (anatomical) knowledge since the time of Harvey." This doctrine was first published and taught by Sir Charles Bell: after an interval of some {464} years, it was more distinctly delivered in the publications of Mr. John Shaw, Sir C. Bell's pupil. Soon afterwards it was further confirmed, and some part of the evidence corrected, by Mr. Mayo, another pupil of Sir C. Bell, and by M. Majendie.[68\17]

[Note 67\17: Dr. Charles Henry's _Report of Brit. Assoc._ iii. p. 62.]

[Note 68\17: As authority for the expressions which I have now used in the text, I will mention Müller's _Manual of Physiology_ (4th edition, 1844). In Book iii. Section 2, Chap. i., "On the Nerves of Sensation and Motion," Müller says, "Charles Bell was the first who had the ingenious thought that the posterior roots of the nerves of the spine--those which are furnished with a ganglion--govern sensation only; that the anterior roots are appointed for motion; and that the primitive fibres of these roots, after being united in a single nervous cord, are mingled together in order to supply the wants of the skin and muscles. He developed this idea in a little work (_An Idea of a new Anatomy of the Brain_, London, 1811), which was not intended to travel beyond the circle of his friends." Müller goes on to say, that eleven years later, Majendie prosecuted the same theory. But Mr. Alexander Shaw, in 1839, published _A Narrative of the Discoveries of Sir Charles Bell in the Nervous System_, in which it appears that Sir Charles Bell had further expounded his views in his lectures to his pupils (p. 89), and that one of these, Mr. John Shaw, had in various publications, in 1821 and 1822, further insisted upon the same views; especially in a Memoir _On Partial Paralysis_ (p. 75). MM. Mayo and Majendie both published Memoirs in August, 1822; and these and subsequent works confirmed the doctrine of Bell. Mr.

Alexander Shaw states (p. 97), that a mistake of Sir Charles Bell's, in an experiment which he had made to prove his doctrine, was discovered through the joint labors of M. Majendie and Mr. Mayo.]

_Sect._ 2.--_The Consequent Speculations. Hypotheses respecting Life, Sensation, and Volition._

I SHALL not attempt to explain the details of these anatomical investigations; and I shall speak very briefly of the speculations which have been suggested by the obvious subservience of the nerves to life, sensation, and volition. Some general inferences from their distribution were sufficiently obvious; as, that the seat of sensation and volition is in the brain. Galen begins his work, _On the Anatomy of the Nerves_, thus: "That none of the members of the animal either exercises voluntary motion, or receives sensation, and that if the nerve be cut, the part immediately becomes inert and insensible, is acknowledged by all physicians. But that the origin of the nerves is partly from the brain, and partly from the spinal marrow, I proceed to explain." And in his work _On the Doctrines of Plato and Hippocrates_, he proves at {465} great length[69\17] that the brain is the origin of sensation and motion, refuting the opinions of earlier days, as that of Chrysippus,[70\17] who placed the _hegemonic_ or master-principle of the soul, in the heart. But though Galen thought that the rational soul resides in the brain, he was disposed to agree with the poets and philosophers, according to whom the heart is the seat of courage and anger, and the liver the seat of love.[71\17] The faculties of the soul were by succeeding physiologists confined to the brain; but the disposition still showed itself, to attribute to them distinct localities. Thus Willis[72\17] places the imagination in the _corpus callosum_, the memory in the folds of the _hemispheres_, the perception in the _corpus striatum_. In more recent times, a system founded upon a similar view has been further developed by Gall and his followers. The germ of Gall's system may be considered as contained in that of Willis; for Gall represents the hemispheres as the folds of a great membrane which is capable of being unwrapped and spread out, and places the different faculties of man in the different regions of this membrane. The chasm which intervenes between matter and motion on the one side, and thought and feeling on the other, is brought into view by all such systems; but none of the hypotheses which they involve can effectually bridge it over.

[Note 69\17: Lib. vii.]

[Note 70\17: Lib. iii. c. 1.]

[Note 71\17: Lib. vi. c. 8.]

[Note 72\17: Cuv. _Sc. Nat._ p. 384.]

The same observation may be made respecting the attempts to explain the manner in which the nerves operate as the instruments of sensation and volition. Perhaps a real step was made by Glisson,[73\17] professor of medicine in the University of Cambridge, who distinguished in the fibres of the muscles of motion a peculiar property, different from any merely mechanical or physical action. His work _On the Nature of the Energetic Substance, or on the Life of Nature and of its Three First Faculties, The Perceptive, Appetitive, and Motive_, which was published in 1672, is

rather metaphysical than physiological. But the principles which he establishes in this treatise he applies more specially to physiology in a treatise _On the Stomach and Intestines_ (Amsterdam, 1677). In this he ascribes to the fibres of the animal body a peculiar power which he calls _Irritability_. He divides _irritation_ into natural, vital, and animal; and he points out, though briefly, the gradual differences of irritability in different organs. "It is hardly comprehensible," says Sprengel,[74\17] "how this {466} lucid and excellent notion of the Cambridge teacher was not accepted with greater alacrity, and further unfolded by his contemporaries." It has, however, since been universally adopted.

[Note 73\17: Cuv. _Sc. Nat._ p. 434.]

[Note 74\17: Spr. iv. 47.]

But though the discrimination of muscular irritability as a peculiar power might be a useful step in physiological research, the explanations hitherto offered, of the way in which the nerves operate on this irritability, and discharge their other offices, present only a series of hypotheses. Glisson[75\17] assumed the existence of certain vital spirits, which, according to him, are a mild, sweet fluid, resembling the spirituous part of white of egg, and residing in the nerves.--This hypothesis, of a very subtle humor or spirit existing in the nerves, was indeed very early taken up.[76\17] This nervous spirit had been compared to air by Erasistratus, Asclepiades, Galen, and others. The chemical tendencies of the seventeenth century led to its being described as acid, sulphureous or nitrous. At the end of that century, the hypothesis of an _ether_ attracted much notice as a means of accounting for many phenomena; and this ether was identified with the nervous fluid. Newton himself inclines to this view, in the remarkable Queries which are annexed to his _Opticks_. After ascribing many physical effects to his ether, he adds (Query 23), "Is not vision performed chiefly by the vibrations of this medium, excited in the bottom of the eye by the rays of light, and propagated through the solid, pellucid, and uniform capillamenta of the nerves into the place of sensation?" And (Query 24), "Is not animal motion performed by the vibrations of this medium, excited in the brain by the power of the will, and propagated from thence through the capillamenta of the nerves into the muscles for contracting and dilating them?" And an opinion approaching this has been adopted by some of the greatest of modern physiologists; as Haller, who says,[77\17] that, though it is more easy to find what this nervous spirit is not than what it is, he conceives that, while it must be far too fine to be perceived by the sense, it must yet be more gross than fire, magnetism, or electricity; so that it may be contained in vessels, and confined by boundaries. And Cuvier speaks to the same effect:[78\17] "There is a great probability that it is by an imponderable fluid that the nerve acts on the fibre, and that this nervous fluid is drawn from the blood, and secreted by the medullary matter."

[Note 75\17: Spr. iv. 38.]

[Note 76\17: Haller, _Physiol._ iv. 365.]

[Note 77\17: _Physiol._ iv. 381, lib. x. sect. viii. § 15.]

[Note 78\17: _Règne Animal_, Introd. p. 30.]

Without presuming to dissent from such authorities on a point of {467} anatomical probability, we may venture to observe, that these hypotheses do not tend at all to elucidate the physiological principle which is here involved; for this principle cannot be mechanical, chemical, or physical, and therefore cannot be better understood by embodying it in a fluid; the difficulty we have in conceiving what the moving force _is_, is not got rid of by explaining the machinery by which it is merely _transferred_. In tracing the phenomena of sensation and volition to their cause, it is clear that we must call in some peculiar and hyperphysical principle. The hypothesis of a fluid is not made more satisfactory by attenuating the fluid; it becomes subtle, spirituous, ethereal, imponderable, to no purpose; it must cease to be a fluid, before its motions can become sensation and volition. This, indeed, is acknowledged by most physiologists; and strongly stated by Cuvier.[79\17] "The impression of external objects upon the ME, the production of a sensation, of an image, is a mystery impenetrable for our thoughts." And in several places, by the use of this peculiar phrase, "_the me_," (_le moi_) for the sentient and volent faculty, he marks, with peculiar appropriateness and force, that phraseology borrowed from the world of matter will, in this subject, no longer answer our purpose. We have here to go from Nouns to Pronouns, from Things to Persons. We pass from the Body to the Soul, from Physics to Metaphysics. We are come to the borders of material philosophy; the next step is into the domain of Thought and Mind. Here, therefore, we begin to feel that we have reached the boundaries of our present subject. The examination of that which lies beyond them must be reserved for a philosophy of another kind, and for the labors of the future; if we are ever enabled to make the attempt to extend into that loftier and wider scene, the principles which we gather on the ground we are now laboriously treading.

[Note 79\17: _Règne Animal_, Introd. p. 47.]

Such speculations as I have quoted respecting the nervous fluid, proceeding from some of the greatest philosophers who ever lived, prove only that hitherto the endeavor to comprehend the mystery of perception and will, of life and thought, have been fruitless and vain. Many anatomical truths have been discovered, but, so far as our survey has yet gone, no genuine physiological principle. All the trains of physiological research which we have followed have begun in exact examination of organization and function, and have ended in wide conjectures and arbitrary hypotheses. The stream of knowledge in all such cases is {468} clear and lively at its outset; but, instead of reaching the great ocean of the general truths of science, it is gradually spread abroad among sands and deserts till its course can be traced no longer.

Hitherto, therefore, we must consider that we have had to tell the story of the _failures_ of physiological speculation. But of late there have come into view and use among physiologists certain principles which may be considered as peculiar to organized subjects; and of which the introduction forms a real advance in organical science. Though these have hitherto been very imperfectly developed, we must endeavor to exhibit, in some measure, their history and bearing.

[2nd Ed.] [In order to show that I am not unaware how imperfect the sketch given in this work is, as a History of Physiology, I may refer to the further discussions on these subjects contained in the _Philosophy of the Inductive Sciences_, Book ix. I have there (Chap. ii.) noticed the successive _Biological Hypotheses_ of the Mystical, the Iatrochemical, and Iatromathematical Schools, the Vital-Fluid School, and the Psychical School. I have (Chaps. iii., iv., v.) examined several of the attempts which have been made to analyze the Idea of Life, to classify Vital Functions, and to form Ideas of Separate Vital Forces. I have considered in particular, the attempts to form a distinct conception of Assimilation and Secretion, of Generation, and of Voluntary Motion; and I have (Chap. vi.) further discussed the Idea of Final Causes as employed in Biology.]

## CHAPTER VI.

INTRODUCTION OF THE PRINCIPLE OF DEVELOPED AND METAMORPHOSED SYMMETRY.

_Sect._ 1.--_Vegetable Morphology. Göthe. De Candolle._

BEFORE we proceed to consider the progress of principles which belong to animal and human life, such as have just been pointed at, we must look round for such doctrines, if any such there be, as apply alike to all organized beings, conscious or unconscious, fixed or locomotive;--to the laws which regulate vegetable as well as animal forms and functions. Though we are very far from being able to present a {469} clear and connected code of such laws, we may refer to one law, at least, which appears to be of genuine authority and validity; and which is worthy our attention as an example of a properly organical or physiological principle, distinct from all mechanical, chemical, or other physical forces; and such as cannot even be conceived to be resolvable into those. I speak of the tendency which produces such results as have been brought together in recent speculations upon _Morphology_.

It may perhaps be regarded as indicating how peculiar are the principles of organic life, and how far removed from any mere mechanical action, that the leading idea in these speculations was first strongly and effectively apprehended, not by a laborious experimenter and reasoner, but by a man of singularly brilliant and creative fancy; not by a mathematician or chemist, but by a poet. And we may add further, that this poet had already shown himself incapable of rightly apprehending the relation of physical facts to their principles; and had, in trying his powers on such subjects, exhibited a signal instance of the ineffectual and perverse operation of the method of philosophizing to which the constitution of his mind led him. The person of whom we speak, is John Wolfgang Göthe, who is held, by the unanimous voice of Europe, to have been one of the greatest poets of our own, or of any time, and whose _Doctrine of Colors_ we have already had to describe, in the History of Optics, as an entire failure. Yet his views on the laws which connect the forms of plants into one simple system, have been generally accepted and followed up. We might almost be led to think that this writer's poetical endowments had contributed to this scientific discovery;--the love of beauty of form, by fixing the attention upon the symmetry of

plants; and the creative habit of thought, by making constant developement of a familiar process.[80\17] {470}

[Note 80\17: We may quote some of the poet's own verses as an illustration of his feelings on this subject. They are addressed to a lady.

Dich verwirret, geliebte, die tausendfältige mischung Dieses blumengewühls über dem garten umher; Viele namen hörest du an, und immer verdränget, Mit barbarischem klang, einer den andern im ohr. Alle gestalten sind **ähnlich und keine gleichet der andern; Und so deutet das chor auf ein geheimes gesetz, Auf ein heiliges räthsel. O! könnte ich dich, liebliche freundinn, Ueberliefern so gleich glücklich das lösende wort.

Thou, my love, art perplext with the endless seeming confusion Of the luxuriant wealth which in the garden is spread; Name upon name thou hearest, and in thy dissatisfied hearing, With a barbarian noise one drives another along. All the forms resemble, yet none is the same as another; Thus the whole of the throng points at a deep hidden law. Points at a sacred riddle. Oh! could I to thee, my beloved friend, Whisper the fortunate word by which the riddle is read!]

But though we cannot but remark the peculiarity of our being indebted to a poet for the discovery of a scientific principle, we must not forget that he himself held, that in making this step, he had been guided, not by his invention, but by observation. He repelled, with extreme repugnance, the notion that he had substituted fancy for fact, or imposed ideal laws on actual things. While he was earnestly pursuing his morphological speculations, he attempted to impress them upon Schiller. "I expounded to him, in as lively a manner as possible, the metamorphosis of plants, drawing on paper, with many characteristic strokes, a symbolic plant before his eyes. He heard me," Göthe says,[81\17] "with much interest and distinct comprehension; but when I had done, he shook his head, and said, 'That is not Experience; that is an Idea:' I stopt with some degree of irritation; for the point which separated us was marked most luminously by this expression." And in the same work he relates his botanical studies and his habit of observation, from which it is easily seen that no common amount of knowledge and notice of details, were involved in the course of thought which led him to the principle of the Metamorphosis of Plants.

[Note 81\17:_ Zur Morphologie_, p. 24.]

Before I state the history of this principle, I may be allowed to endeavor to communicate to the reader, to whom this subject is new, some conception of the principle itself. This will not be difficult, if he will imagine to himself a flower, for instance, a common wild-rose, or the blossom of an apple-tree, as consisting of a series of parts disposed in _whorls_, placed one over another on an _axis_. The lowest whorl is the calyx with its five sepals; above this is the corolla with its five petals; above this are a multitude of stamens, which may be considered as separate whorls of five each, often repeated; above these is a whorl composed of the ovaries, or what become the seed-vessels in the fruit, which are five united together in the apple, but indefinite in number and separate in the rose. Now the morphological view is {471} this;--that the members of each of these whorls are in their nature identical, and the

same as if they were whorls of ordinary leaves, brought together by the shortening their common axis, and modified in form by the successive elaboration of their nutriment. Further, according to this view, a whorl of leaves itself is to be considered as identical with several detached leaves dispersed spirally along the axis, and brought together because the axis is shortened. Thus all the parts of a plant are, or at least represent, the successive metamorphoses of the same elementary member. The root-leaves thus pass into the common leaves;--these into bracteæ;--these into the sepals;--these into the petals;--these into the stamens with their anthers;--these into the ovaries with their styles and stigmas;--these ultimately become the fruit; and thus we are finally led to the seed of a new plant.

Moreover the same notion of metamorphosis may be applied to explain the existence of flowers which are not symmetrical like those we have just referred to, but which have an irregular corolla or calyx. The papilionaceous flower of the pea tribe, which is so markedly irregular, may be deduced by easy gradations from the regular flower, (through the _mimoseæ_,) by expanding one petal, joining one or two others, and modifying the form of the intermediate ones.

Without attempting to go into detail respecting the proofs of that identity of all the different organs, and all the different forms of plants, which is thus asserted, we may observe, that it rests on such grounds as these;--the transformations which the parts of flowers undergo by accidents of nutriment or exposure. Such changes, considered as monstrosities where they are very remarkable, show the tendencies and possibilities belonging to the organization in which they occur. For instance, the single wild-rose, by culture, transforms many of its numerous stamens into petals, and thus acquires the deeply folded flower of the double garden-rose. We cannot doubt of the reality of this change, for we often see stamens in which it is incomplete. In other cases we find petals becoming leaves, and a branch growing out of the centre of the flower. Some pear-trees, when in blossom, are remarkable for their tendencies to such monstrosities.[82\17] Again, we find that flowers which are usually irregular, occasionally become regular, and conversely. The common snap-dragon (_Linaria vulgaris_) affords a curious instance of this.[83\17] The usual form of this plant is "personate," the corolla being divided into two lobes, which differ in form, and {472} together present somewhat the appearance of an animal's face; and the upper portion of the corolla is prolonged backwards into a tube-like "spur." No flower can be more irregular; but there is a singular variety of this plants termed _Peloria_, in which the corolla is strictly symmetrical, consisting of a conical tube, narrowed in front, elongated behind into five equal spurs, and containing five stamens of equal length, instead of the two unequal pairs of the didynamous Linaria. These and the like appearances show that there is in nature a capacity for, and tendency to, such changes as the doctrine of metamorphosis asserts.

[Note 82\17: Lindley, _Nat. Syst._ p. 84.]

[Note 83\17: Henslow, _Principles of Botany_, p. 116.]

Göthe's _Metamorphosis of Plants_ was published 1790: and his system was the result of his own independent course of thoughts. The view which it involved was not,

however, absolutely new, though it had never before been unfolded in so distinct and persuasive a manner. Linnæus considered the leaves, calyx, corolla, stamens, each as evolved in succession from the other; and spoke of it as _prolepsis_ or _anticipation_,[84\17] when the leaves changed accidentally into bracteæ, these into a calyx, this into a corolla, the corolla into stamens, or these into the pistil. And Caspar Wolf apprehended in a more general manner the same principle. "In the whole plant," says he,[85\17] "we see nothing but leaves and stalk;" and in order to prove what is the situation of the leaves in all their later forms, he adduces the cotyledons as the first leaves.

[Note 84\17: Sprengel, _Bot._ ii. 302. _Amœn. Acad._ vi. 324, 365.]

[Note 85\17: _Nov. Con. Ac. Petrop._ xii. 403, xiii. 478.]

Göthe was led to his system on this subject by his general views of nature. He saw, he says,[86\17] that a whole life of talent and labor was requisite to enable any one to arrange the infinitely copious organic forms of a single kingdom of nature. "Yet I felt," he adds, "that for me there must be another way, analogous to the rest of my habits. The appearance of the changes, round and round, of organic creatures had taken strong hold on my mind. Imagination and Nature appeared to me to vie with each other which could go on most boldly yet most consistently." His observation of nature, directed by such a thought, led him to the doctrine of the metamorphosis.

[Note 86\17: _Zur Morph._ i. 30.]

In a later republication of his work (_Zur Morphologie_, 1817,) he gives a very agreeable account of the various circumstances which affected the reception and progress of his doctrine. Willdenow[87\17] quoted {473} him thus:--"The life of plants is, as Mr. Göthe very prettily says, an expansion and contraction, and these alternations make the various periods of life." "This '_prettily_,'" says Göthe, "I can be well content with, but the '_egregie_,' of Usteri is much more pretty and obliging." Usteri had used this term respecting Göthe in an edition of Jussieu.

[Note 87\17: _Zur Morph._ i. 121.]

The application of the notion of metamorphosis to the explanation of double and monstrous flowers had been made previously by Jussieu. Göthe's merit was, to have referred to it the _regular_ formation of the flower. And as Sprengel justly says,[88\17] his view had so profound a meaning, made so strong an appeal by its simplicity, and was so fruitful in the most valuable consequences, that it was not to be wondered at if it occasioned further examination of the subject; although many persons pretend to slight it. The task of confirming and verifying the doctrine by a general application of it to all cases,--a labor so important and necessary after the promulgation of any great principle,--Göthe himself did not execute. At first he collected specimens and made drawings with some such view,[89\17] but he was interrupted and diverted to other matters. "And now," says he, in his later publication, "when I look back on this undertaking, it is easy to see that the object which I had before my eyes was, for me, in my position, with my habits and mode of thinking, unattainable. For it was no less than this: that I was to take that which I had stated in general, and presented to the conception, to the mental intuition, in words;

and that I should, in a particularly visible, orderly, and gradual manner, present it to the eye; so as to show to the outward sense that out of the germ of this idea might grow a tree of physiology fit to overshadow the world."

[Note 88\17: _Gesch. Botan._ ii. 304.]

[Note 89\17: _Zur Morph._ i. **129.]

Voigt, professor at Jena, was one of the first who adopted Göthe's view into an elementary work, which he did in 1808. Other botanists labored in the direction which had thus been pointed out. Of those who have thus contributed to the establishment and developement of the metamorphic doctrine. Professor De Candolle, of Geneva, is perhaps the most important. His Theory of Developement rests upon two main principles, _abortion_ and _adhesion_. By considering some parts as degenerated or absent through the abortion of the buds which might have formed them, and other parts as adhering together, he holds that all plants may be reduced to perfect symmetry: and the actual and constant occurrence of such incidents is shown beyond {474} all doubt. And thus the snap-dragon, of which we have spoken above, is derived from the Peloria, which is the normal condition of the flower, by the abortion of one stamen, and the degeneration of two others. Such examples are too numerous to need to be dwelt on.

_Sect._ 2.--_Application of Vegetable Morphology._

THE doctrine, being thus fully established, has been applied to solve different problems in botany; for instance, to explain the structure of flowers which appear at first sight to deviate widely from the usual forms of the vegetable world. We have an instance of such an application in Mr. Robert Brown's explanation of the real structure of various plants which had been entirely misunderstood: as, for example, the genus _Euphorbia_. In this plant he showed that what had been held to be a jointed filament, was a pedicel with a filament above it, the intermediate corolla having evanesced. In _Orchideæ_ (the orchis tribe), he showed that the peculiar structure of the plant arose from its having six stamens (two sets of three each), of which five are usually abortive. In _Coniferæ_ (the cone-bearing trees), it was made to appear that the seed was naked, while the accompanying appendage, corresponding to a seed-vessel, assumed all forms, from a complete leaf to a mere scale. In like manner it was proved that the _pappus_, or down of _composite_ plants (as thistles), is a transformed calyx.

Along with this successful application of a profound principle, it was natural that other botanists should make similar attempts. Thus Mr. Lindley was led to take a view[90\17] of the structure of _Reseda_ (mignonette) different from that usually entertained; which, when published, attracted a good deal of attention, and gained some converts among the botanists of Germany and France. But in 1833, Mr. Lindley says, with great candor, "Lately, Professor Henslow has satisfactorily proved, in part by the aid of a monstrosity in the common _Mignonette_, in part by a severe application of morphological rules, that my hypothesis must necessarily be false." Such an agreement of different botanists respecting the consequences of morphological rules, proves the reality and universality of the rules.

[Note 90\17: Lindley, _Brit. Assoc. Report_, iii. 50.]

We find, therefore, that a principle which we may call the _Principle of Developed and Metamorphosed Symmetry_, is firmly established {475} and recognized, and familiarly and successfully applied by botanists. And it will be apparent, on reflection, that though _symmetry_ is a notion which applies to inorganic as well as to organic things, and is, in fact, a conception of certain relations of space and position, such _developement_ and _metamorphosis_ as are here spoken of, are ideas entirely different from any of those to which the physical sciences have led us in our previous survey; and are, in short, genuine _organical_ or _physiological_ ideas;--real elements of the philosophy of _life_.

We must, however imperfectly, endeavor to trace the application of this idea in the other great department of the world of life; we must follow the history of Animal Morphology.

## CHAPTER VII.

PROGRESS OF ANIMAL MORPHOLOGY.

_Sect._ 1.--_Rise of Comparative Anatomy._

THE most general and constant relations of the form of the organs, both in plants and animals, are the most natural grounds of classification. Hence the first scientific classifications of animals are the first steps in animal morphology. At first, a _zoology_ was constructed by arranging animals, as plants were at first arranged, according to their external parts. But in the course of the researches of the anatomists of the seventeenth century, it was seen that the internal structure of animals offered resemblances and transitions of a far more coherent and philosophical kind, and the Science of _Comparative Anatomy_ rose into favor and importance. Among the main cultivators of this science[91\17] at the period just mentioned, we find Francis Redi, of Arezzo; Guichard-Joseph Duvernay, who was for sixty years Professor of Anatomy at the Jardin du Roi at Paris, and during this lapse of time had for his pupils almost all the greatest anatomists of the greater part of the eighteenth century; Nehemiah Grew, secretary to the Royal Society of London, whose _Anatomy of Plants_ we have already noticed.

[Note 91\17: Cuv. _Leçons sur l'Hist. des Sc. Nat._ 414, 420.]

But Comparative Anatomy, which had been cultivated with ardor {476} to the end of the seventeenth century, was, in some measure, neglected during the first two-thirds of the eighteenth. The progress of botany was, Cuvier sagaciously suggests,[92\17] one cause of this; for that science had made its advances by confining itself to external characters, and rejecting anatomy; and though Linnæus acknowledged the dependence of zoology upon anatomy[93\17] so far as to make the number of teeth his characters, even this was felt, in his method, as a bold step. But his influence was soon opposed by that of Buffon, Daubenton, and Pallas; who again brought into view the importance of comparative anatomy in Zoology; at the same time that Haller proved how much might be learnt from it in Physiology. John Hunter

in England, the two Monros in Scotland, Camper in Holland, and Vicq d'Azyr in France, were the first to follow the path thus pointed out. Camper threw the glance of genius on a host of interesting objects, but almost all that he produced was a number of sketches; Vicq d'Azyr, more assiduous, was stopt in the midst of a most brilliant career by a premature death.

[Note 92\17: Cuv. _Hist. Sc. Nat._ i. 301.]

[Note 93\17: Ib.]

Such is Cuvier's outline of the earlier history of comparative anatomy. We shall not go into detail upon this subject; but we may observe that such studies had fixed in the minds of naturalists the conviction of the possibility and the propriety of considering large divisions of the animal kingdom as modifications of one common _type_. Belon, as early as 1555, had placed the skeleton of a man and a bird side by side, and shown the correspondence of parts. So far as the case of vertebrated animals extends, this correspondence is generally allowed; although it required some ingenuity to detect its details in some cases; for instance, to see the analogy of parts between the head of a man and a fish.

In tracing these less obvious correspondencies, some curious steps have been made in recent times. And here we must, I conceive, again ascribe no small merit to the same remarkable man who, as we have already had to point out, gave so great an impulse to vegetable morphology. Göthe, whose talent and disposition for speculating on all parts of nature were truly admirable, was excited to the study of anatomy by his propinquity to the Duke of Weimar's cabinet of natural history. In 1786, he published a little essay, the object of which was to show that in man, as well as in beasts, the upper jaw contains an intermaxillary bone, although the sutures are obliterated. After 1790,[94\17] animated and impelled by the same passion for natural {477} observation and for general views, which had produced his Metamorphosis of Plants, he pursued his speculations on these subjects eagerly and successfully. And in 1795, he published a _Sketch of a Universal Introduction into Comparative Anatomy, beginning with Osteology_; in which he attempts to establish an "osteological type," to which skeletons of all animals may be referred. I do not pretend that Göthe's anatomical works have had any influence on the progress of the science comparable with that which has been exercised by the labors of professional anatomists; but the ingenuity and value of the views which they contained was acknowledged by the best authorities; and the clearer introduction and application of the principle of developed and metamorphosed symmetry may be dated from about this time. Göthe declares that, at an early period of these speculations, he was convinced[95\17] that the bony head of beasts is to be derived from six vertebræ. In 1807, Oken published a "Program" _On the Signification of the Bones of the Skull_, in which he maintained that these bones are equivalent to four vertebræ); and Meckel, in his _Comparative Anatomy_, in 1811, also resolved the skull into vertebræ. But Spix, in his elaborate work _Cephalogenesis_, in 1815, reduced the vertebræ of the head to three. "Oken," he says,[96\17] "published opinions merely theoretical, and consequently contrary to those maintained in this work, which are drawn from observation." This resolution of the head into vertebræ is assented to by many of the

best physiologists, as explaining the distribution of the nerves, and other phenomena. Spix further extended the application of the vertebral theory to the heads of all classes of vertebrate animals; and Bojanus published a Memoir expressly on the vertebral structure of the skulls of fishes in Oken's _Isis_ for 1818. Geoffroy Saint-Hilaire presented a lithographic plate to the French Academy in February 1824, entitled _Composition de la Tête **osseuse chez l'Homme et les Animaux_, and developed his views of the vertebral composition of the skull in two Memoirs published in the _Annales des Sciences Naturelles_ for 1824. We cannot fail to recognize here the attempt to apply to the skeleton of animals the principle which leads botanists to consider all the parts of a flower as transformations of the same organs. How far the application of the principle, as here proposed, is just, I must leave philosophical physiologists to decide.

[Note 94\17: _Zur Morphologie_, i. 234.]

[Note 95\17: _Zur Morphologie_, 250.]

[Note 96\17: Spix, _Cephalogenesis_.]

By these and similar researches, it is held by the best physiologists {478} that the skull of all vertebrate animals is pretty well reduced to a uniform structure, and the laws of its variations nearly determined.[97\17]

[Note 97\17: Cuv. _Hist. Sc. Nat._ iii. 442.]

The vertebrate animals being thus reduced to a single type, the question arises how far this can be done with regard to other animals, and how many such types there are. And here we come to one of the important services which Cuvier rendered to natural history.

_Sect._ 2.--_Distinction of the General Types of the Forms of Animals.--Cuvier._

ANIMALS were divided by Lamarck into vertebrate and invertebrate; and the general analogies of all vertebrate animals are easily made manifest. But with regard to other animals, the point is far from clear. Cuvier was the first to give a really philosophical view of the animal world in reference to the plan on which each animal is constructed. There are,[98\17] he says, four such plans;--four forms on which animals appear to have been modelled; and of which the ulterior divisions, with whatever titles naturalists have decorated them, are only very slight modifications, founded on the development or addition of some parts which do not produce any essential change in the plan.

[Note 98\17: _Règne Animal_, p. 57.]

These four great branches of the animal world are the _vertebrata_, _mollusca_, _articulata_, _radiata_; and the differences of these are so important that a slight explanation of them may be permitted.

The _vertebrata_ are those animals which (as man and other sucklers, birds, fishes, lizards, frogs, serpents) have a backbone and a skull with lateral appendages, within which the viscera are included, and to which the muscles are attached.

The _mollusca_, or soft animals, have no bony skeleton; the muscles are attached to the skin, which often includes stony plates called _shells_; such molluscs are shell-fish; others are cuttle-fish, and many pulpy sea-animals.

The _articulata_ consist of _crustacea_ (lobsters, &c.), _insects_, _spiders_, and _annulose worms_, which consist of a head and a number of successive annular portions of the body _jointed_ together (to the interior of which the muscles are attached), whence the name.

Finally, the _radiata_ include the animals known under the name of _zoophytes_. In the preceding three branches the organs of motion and of sense were distributed symmetrically on the two sides of an axis, {479} so that the animal has a right and a left side. In the radiata the similar members radiate from the axis in a circular manner, like the petals of a regular flower.

The whole value of such a classification cannot be understood without explaining its use in enabling us to give general descriptions, and general laws of the animal functions of the classes which it includes; but in the present part of our work our business is to exhibit it as an exemplification of the reduction of animals to laws of Symmetry. The bipartite Symmetry of the form of vertebrate and articulate animals is obvious; and the reduction of the various forms of such animals to a common type has been effected, by attention to their anatomy, in a manner which has satisfied those who have best studied the subject. The molluscs, especially those in which the head disappears, as oysters, or those which are rolled into a spiral, as snails, have a less obvious Symmetry, but here also we can apply certain general types. And the Symmetry of the radiated zoophytes is of a nature quite different from all the rest, and approaching, as we have suggested, to the kind of Symmetry found in plants. Some naturalists have doubted whether[99\17] these zoophytes are not referrible to two types (_acrita_ or polypes, and true _radiata_,) rather than to one.

[Note 99\17: _Brit. Assoc. Rep._ iv. 227.]

This fourfold division was introduced by Cuvier.[100\17] Before him, naturalists followed Linnæus, and divided non-vertebrate animals into two classes, insects and worms. "I began," says Cuvier, "to attack this view of the subject, and offered another division, in a Memoir read at the Society of Natural History of Paris, the 21st of Floreal, in the year III. of the Republic (May 10, 1795,) printed in the _Décade Philosophique_: in this, I mark the characters and the limits of molluscs, insects, worms, echinoderms, and zoophytes. I distinguish the red-blooded worms or annelides, in a Memoir read to the Institute, the 11th Nivose, year X. (December 31, 1801.) I afterwards distributed these different classes into three branches, each co-ordinate to the branch formed by the vertebrate animals, in a Memoir read to the Institute in July, 1812, printed in the _Annales du Muséum d'Histoire Naturelle_, tom. xix." His great systematic work, the _Règne Animal_, founded on this distribution, was published in 1817; and since that time the division has been commonly accepted among naturalists.

[Note 100\17: _Règne A._ 61.]

[2nd Ed.] [The question of the Classification of Animals is discussed in the first of Prof. Owen's _Lectures on the Invertebrate_ {480} _Animals_ (1843). Mr. Owen observes that the arrangement of animals into _Vertebrate_ and _Invertebrate_ which prevailed before Cuvier, was necessarily bad, inasmuch as no _negative_ character in Zoology gives true natural groups. Hence the establishment of the _sub-kingdoms_, _Mollusca_, _Articulata_, _Radiata_, as co-ordinate with _Vertebrata_, according to the arrangement of the nervous system, was a most important advance. But Mr. Owen has seen reason to separate the _Radiata_ of Cuvier into two divisions; the _Nematoneura_, in which the nervous system can be traced in a filamentary form (including _Echinoderma_, _Ciliobrachiata_, _Cœlelmintha_, _Rotifera_,) and the _Acrita_ or lowest division of the animal kingdom, including _Acalepha_, _Nudibrachiata_, _Sterelmintha_, _Polygastria_.]

_Sect._ 3.--_Attempts to establish the Identity of the Types of Animal Forms._

SUPPOSING this great step in Zoology, of which we have given an account,--the reduction of all animals to four types or plans,--to be quite secure, we are then led to ask whether any further advance is possible;--whether several of these types can be referred to one common form by any wider effort of generalization. On this question there has been a considerable difference of opinion. Geoffroy Saint-Hilaire,[101\17] who had previously endeavored to show that all vertebrate animals were constructed so exactly upon the same plan as to preserve the strictest analogy of parts in respect to their osteology, thought to extend this unity of plan by demonstrating, that the hard parts of crustaceans and insects are still only modifications of the skeleton of higher animals, and that therefore the type of vertebrata must be made to include them also:--the segments of the articulata are held to be strictly analogous to the vertebras of the higher animals, and thus the former live _within_ their vertebral column in the same manner as the latter live _without_ it. Attempts have even been made to reduce molluscous and vertebrate animals to a community of type, as we shall see shortly.

[Note 101\17: Mr. Jenyns, _Brit. Assoc. Rep._ iv. 150.]

Another application of the principle, according to which creatures the most different are developments of the same original type, may be discerned[102\17] in the doctrine, that the embryo of the higher forms of animal life passes by gradations through those forms which are {481} permanent in inferior animals. Thus, according to this view, the human fœtus assumes successively the plan of the zoophyte, the worm, the fish, the turtle, the bird, the beast. But it has been well observed, that "in these analogies we look in vain for the precision which can alone support the inference that has been deduced;"[103\17] and that at each step, the higher embryo and the lower animal which it is supposed to resemble, differ in having each different organs suited to their respective destinations.

[Note 102\17: Dr. Clark, _Report_, Ib. iv. 113.]

[Note 103\17: Dr. Clark, p. 114.]

Cuvier[104\17] never assented to this view, nor to the attempts to refer the different divisions of his system to a common type. "He could not admit," says his

biographer, "that the lungs or gills of the vertebrates are in the same connexion as the branchiæ of molluscs and crustaceans, which in the one are situated at the base of the feet, or fixed on the feet themselves, and in the other often on the back or about the arms. He did not admit the analogy between the skeleton of the vertebrates and the skin of the articulates; he could not believe that the tænia and the sepia were constructed on the same plan; that there was a similarity of composition between the bird and the echinus, the whale and the snail; in spite of the skill with which some persons sought gradually to efface their discrepancies."

[Note 104\17: Laurillard, _Elog. de Cuvier_, p. 66.]

Whether it may be possible to establish, among the four great divisions of the "Animal Kingdom," some analogies of a higher order than those which prevail within each division, I do not pretend to conjecture. If this can be done, it is clear that it must be by comparing the types of these divisions under their most general forms: and thus Cuvier's arrangement, so far as it is itself rightly founded on the unity of composition of each branch, is the surest step to the discovery of a unity pervading and uniting these branches. But those who generalize surely, and those who generalize rapidly, may travel in the same direction, they soon separate so widely, that they appear to move from each other. The partisans of a universal "unity of composition" of animals, accused Cuvier of being too inert in following the progress of physiological and zoological science. Borrowing their illustration from the political parties of the times, they asserted that he belonged to the science of _resistance_, not to the science of the _movement_. Such a charge was highly honorable to him; for no one acquainted with the history of zoology can doubt that he had a great share in the impulse by which the "movement" was occasioned; or that he {482} himself made a large advance with it; and it was because he was so poised by the vast mass of his knowledge, so temperate in his love of doubtful generalizations, that he was not swept on in the wilder part of the stream. To such a charge, moderate reformers, who appreciate the value of the good which exists, though they try to make it better, and who know the knowledge, thoughtfulness, and caution, which are needful in such a task, are naturally exposed. For us, who can only decide on such a subject by the general analogies of the history of science, it may suffice to say, that it appears doubtful whether the fundamental conceptions of affinity, analogy, transition, and developement, have yet been fixed in the minds of physiologists with sufficient firmness and clearness, or unfolded with sufficient consistency and generality, to make it likely that any great additional step of this kind can for some time be made.

We have here considered the doctrine of the identity of the seemingly various types of animal structure, as an attempt to extend the correspondencies which were the basis of Cuvier's division of the animal kingdom. But this doctrine has been put forward in another point of view, as the antithesis to the doctrine of final causes. This question is so important a one, that we cannot help attempting to give some view of its state and bearings.

## CHAPTER VIII.

## THE DOCTRINE OF FINAL CAUSES IN PHYSIOLOGY.

### _Sect._ 1.--_Assertion of the Principle of Unity of Plan._

WE have repeatedly seen, in the course of our historical view of Physiology, that those who have studied the structure of animals and plants, have had a conviction forced upon them, that the organs are constructed and combined in subservience to the life and functions of the whole. The parts have a _purpose_, as well as a _law_;--we can trace Final Causes, as well as Laws of Causation. This principle is peculiar to physiology; and it might naturally be expected that, in the progress of the science, it would come under special consideration. This accordingly has happened; and the principle has been drawn {483} into a prominent position by the struggle of two antagonistic schools of physiologists. On the one hand, it has been maintained that this doctrine of final causes is altogether unphilosophical, and requires to be replaced by a more comprehensive and profound principle: on the other hand, it is asserted that the doctrine is not only true, but that, in our own time, it has been fixed and developed so as to become the instrument of some of the most important discoveries which have been made. Of the views of these two schools we must endeavor to give some account.

The disciples of the former of the two schools express their tenets by the phrases _unity_ of _plan_, _unity_ of _composition_; and the more detailed developement of these doctrines has been termed the _Theory of Analogies_, by Geoffroy Saint-Hilaire, who claims this theory as his own creation. According to this theory, the structure and functions of animals are to be studied by the guidance of their analogy only; our attention is to be turned, not to the fitness of the organization for any end of life or action, but to its resemblance to other organizations by which it is gradually derived from the original type.

According to the rival view of this subject, we must not assume, and cannot establish, that the plan of all animals is the same, or their composition similar. The existence of a single and universal system of analogies in the construction of all animals is entirely unproved, and therefore cannot be made our guide in the study of their properties. On the other hand, the plan of the animal, the purpose of its organization in the support of its life, the necessity of the functions to its existence, are truths which are irresistibly apparent, and which may therefore be safely taken as the bases of our reasonings. This view has been put forward as the doctrine of the _conditions of existence_: it may also be described as the principle of _a purpose in organization_; the structure being considered as having the function for its end. We must say a few words on each of these views.

It had been pointed out by Cuvier, as we have seen in the last chapter, that the animal kingdom may be divided into four great branches; in each of which the _plan_ of the animal is different, namely, _vertebrata_, _articulata_, _mollusca_, _radiata_. Now the question naturally occurs, is there really no resemblance of construction in these different classes? It was maintained by some, that there is such a resemblance. In 1820,[105\17] M. Audouin, a young naturalist of Paris, {484} endeavored to fill up the chasm which separates insects from other animals; and by examining carefully

the portions which compose the solid frame-work of insects, and following them through their various transformations in different classes, he conceived that he found relations of position and function, and often of number and form, which might be compared with the relations of the parts of the skeleton in vertebrate animals. He thought that the first segment of an insect, the head,[106\17] represents one of the three vertebræ which, according to Spix and others, compose the vertebrate head: the second segment of the insects, (the _prothorax_ of Audouin,) is, according to M. Geoffroy, the second vertebra of the head of the vertebrata, and so on. Upon this speculation Cuvier[107\17] does not give any decided opinion; observing only, that even if false, it leads to active thought and useful research.

[Note 105\17: Cuv. _Hist. Sc. Nat._ iii. 422.]

[Note 106\17: Ib. 437.]

[Note 107\17: Cuv. _Hist. Sc. Nat._ iii. 441.]

But when an attempt was further made to identify the plan of another branch of the animal world, the mollusca, with that of the vertebrata, the radical opposition between such views and those of Cuvier, broke out into an animated controversy.

Two French anatomists, MM. Laurencet and Meyranx, presented to the Academy of Sciences, in 1830, a Memoir containing their views on the organization of molluscous animals; and on the sepia or cuttle-fish in particular, as one of the most complete examples of such animals. These creatures, indeed, though thus placed in the same division with shell-fish of the most defective organization and obscure structure, are far from being scantily organized. They have a brain,[108\17] often eyes, and these, in the animals of this class, (_cephalopoda_) are more complicated than in any vertebrates;[109\17] they have sometimes ears, salivary glands, multiple stomachs, a considerable liver, a bile, a complete double circulation, provided with auricles and ventricles; in short, their vital activity is vigorous, and their senses are distinct.

[Note 108\17: Geoffroy Saint-Hilaire denies this. _Principes de Phil. Zoologique discutés en_ 1830, p. 68.]

[Note 109\17: Geoffroy Saint-Hilaire, _Principes de Phil. Zoologique discutés en_ 1830, p. 55.]

But still, though this organization, in the abundance and diversity of its parts, approaches that of vertebrate animals, it had not been considered as composed in the same manner, or arranged in the same order, Cuvier had always maintained that the plan of molluscs is not a continuation of the plan of vertebrates. {485}

MM. Laurencet and Meyranx, on the contrary, conceived that the sepia might be reduced to the type of a vertebrate creature, by considering the back-bone of the latter bent double backwards, so as to bring the root of the tail to the nape of the neck; the parts thus brought into contact being supposed to coalesce. By this mode of conception, these anatomists held that the viscera were placed in the same connexion as in the vertebrate type, and the functions exercised in an analogous manner.

To decide on the reality of the analogy thus asserted, clearly belonged to the jurisdiction of the most eminent anatomists and physiologists. The Memoir was committed to Geoffroy Saint-Hilaire and Latreille, two eminent zoologists, in order to be reported on. Their report was extremely favorable; and went almost to the length of adopting the views of the authors.

Cuvier expressed some dissatisfaction with this report on its being read;[110\17] and a short time afterwards,[111\17] represented Geoffroy Saint-Hilaire as having asserted that the new views of Laurencet and Meyranx refuted completely the notion of the great interval which exists between molluscous and vertebrate animals. Geoffroy protested against such an interpretation of his expressions; but it soon appeared, by the controversial character which the discussions on this and several other subjects assumed, that a real opposition of opinions was in action.

[Note 110\17: _Princ. de Phil. Zool. discutés en_ 1830, p. 36.]

[Note 111\17: p. 50.]

Without attempting to explain the exact views of Geoffroy, (we may, perhaps, venture to say that they are hardly yet generally understood with sufficient distinctness to justify the mere historian of science in attempting such an explanation,) their general tendency may be sufficiently collected from what has been said; and from the phrases in which his views are conveyed.[112\17] _The principle of connexions, the elective affinities of organic elements, the equilibrization of organs_;--such are the designations of the leading doctrines which are unfolded in the preliminary discourse of his _Anatomical Philosophy_. Elective affinities of organic elements are the forces by which the vital structures and varied forms of living things are produced; and the principles of connexion and equilibrium of these forces in the various parts of the organization prescribe limits and conditions to the variety and developement of such forms.

[Note 112\17: _Phil. Zool._ 15.]

The character and tendency of this philosophy will be, I think, {486} much more clear, if we consider what it excludes and denies. It rejects altogether all conception of a plan and purpose in the organs of animals, as a principle which has determined their forms, or can be of use in directing our reasonings. "I take care," says Geoffroy, "not to ascribe to God any intention."[113\17] And when Cuvier speaks of the combination of organs in such order that they may be in consistence with the part which the animal _has to play_ in nature; his rival rejoins,[114\17] I "know nothing of animals which _have to play_ a part in nature." Such a notion is, he holds, unphilosophical and dangerous. It is an abuse of final causes which makes the cause to be engendered by the effect. And to illustrate still further his own view, he says, "I have read concerning fishes, that because they live in a medium which resists more than air, their motive forces are calculated so as to give them the power of progression under those circumstances. By this mode of reasoning, you would say of a man who makes use of crutches, that he was originally destined to the misfortune of having a leg paralysed or amputated."

[Note 113\17: "Je me garde de prêter à Dieu aucune intention." _Phil. Zool._ 10.]

[Note 114\17: "Je ne connais point d'animal qui DOIVE jouer un rôle dans la nature." p. 65.]

How far this doctrine of unity in the plan in animals, is admissible or probable in physiology when kept within proper limits, that is, when not put in opposition to the doctrine of a purpose involved in the plan of animals, I do not pretend even to conjecture. The question is one which appears to be at present deeply occupying the minds of the most learned and profound physiologists; and such persons alone, adding to their knowledge and zeal, judicial sagacity and impartiality, can tell us what is the general tendency of the best researches on this subject.[115\17] But when the anatomist expresses such opinions, and defends them by such illustrations as those which I have just quoted,[116\17] we perceive that he quits the entrenchments of his superior science, in which he might {487} have remained unassailable so long as the question was a professional one; and the discussion is open to those who possess no peculiar knowledge of anatomy. We shall, therefore, venture to say a few words upon it.

[Note 115\17: So far as this doctrine is generally accepted among the best physiologists, we cannot doubt the propriety of Meckel's remark, (_Comparative Anatomy_, 1821, Pref. p. xi.) that it cannot be truly asserted either to be new, or to be peculiarly due to Geoffroy Saint-Hilaire.]

[Note 116\17: It is hardly worth while answering such illustrations, but I may remark, that the one quoted above, irrelevant and unbecoming as it is, tells altogether against its author. The fact that the wooden leg is of the same length as the other, proves, and would satisfy the most incredulous man, that it was _intended_ for walking.]

_Sect._ 2.--_Estimate of the Doctrine of Unity of Plan._

IT has been so often repeated, and so generally allowed in modern times, that Final Causes ought not to be made our guides in natural philosophy, that a prejudice has been established against the introduction of any views to which this designation can be applied, into physical speculations. Yet, in fact, the assumption of an end or purpose in the structure of organized beings, appears to be an intellectual habit which no efforts can cast off. It has prevailed from the earliest to the latest ages of zoological research; appears to be fastened upon us alike by our ignorance and our knowledge; and has been formally accepted by so many great anatomists, that we cannot feel any scruple in believing the rejection of it to be the superstition of a false philosophy, and a result of the exaggeration of other principles which are supposed capable of superseding its use. And the doctrine of unity of plan of all animals, and the other principles associated with this doctrine, so far as they exclude the conviction of an intelligible scheme and a discoverable end, in the organization of animals, appear to be utterly erroneous. I will offer a few reasons for an opinion which may appear presumptuous in a writer who has only a general knowledge of the subject.

1. In the first place, it appears to me that the argumentation on the case in question, the Sepia, does by no means turn out to the advantage of the new

hypothesis. The arguments in support of the hypothetical view of the structure of this mollusc were, that by this view the relative position of the parts was explained, and confirmations which had appeared altogether anomalous, were reduced to rule; for example, the beak, which had been supposed to be in a position the reverse of all other beaks, was shown, by the assumed posture, to have its upper mandible longer than the lower, and thus to be regularly placed. "But," says Cuvier,[117\17] "supposing the posture, in order that the side on which the funnel of the sepia is folded should be the back of the animal, considered as similar to a vertebrate, the brain with {488} regard to the beak, and the œsophagus with regard to the liver, should have positions corresponding to those in vertebrates; but the positions of these organs are exactly contrary to the hypothesis. How, then, can you say," he asks, "that the cephalopods and vertebrates have _identity of composition_, _unity of composition_, without using words in a sense entirely different from their common meaning?"

[Note 117\17: _G. S. H. Phil. Zool._ p. 70.]

This argument appears to be exactly of the kind on which the value of the hypothesis must depend.[118\17] It is, therefore, interesting to see the reply made to it by the theorist. It is this: "I admit the facts here stated, but I deny that they lead to the notion of a different sort of animal composition. Molluscous animals had been placed too high in the zoological scale; but if they are only the embryos of its lower stages, if they are only beings in which far fewer organs come into play, it does not follow that the organs are destitute of the relations which the power of successive generations may demand. The organ A will be in an unusual relation with the organ C, if B has not been produced;--if a stoppage of the developement has fallen upon this latter organ, and has thus prevented its production. And thus," he says, "we see how we may have different arrangements, and divers constructions as they appear to the eye."

[Note 118\17: I do not dwell on other arguments which were employed. It was given as a circumstance suggesting the supposed posture of the type, that in this way the back was colored, and the belly was white. On this Cuvier observes (_Phil. Zool._ pp. 93, 68), "I must say, that I do not know any naturalist so ignorant as to suppose that the back is determined by its dark color, or even by its position when the animal is in motion; they all know that the badger has a black belly and a white back; that an infinity of other animals, especially among insects, are in the same case; and that many fishes swim on their side, or with their belly upwards."]

It seems to me that such a concession as this entirely destroys the theory which it attempts to defend; for what arrangement does the principle of unity of composition _exclude_, if it admits unusual, that is, various arrangements of some organs, accompanied by the total absence of others? Or how does this differ from Cuvier's mode of stating the conclusion, except in the introduction of certain arbitrary hypotheses of developement and stoppage? "I reduce the facts," Cuvier says, "to their true expression, by saying that Cephalopods have several organs which are common to them and vertebrates, and which discharge the same offices; but that these organs are in them differently distributed, and often constructed in a different manner;

{489} and they are accompanied by several other organs which vertebrates have not; while these on the other hand have several which are wanting in cephalopods."

We shall see afterwards the general principles which Cuvier himself considered as the best guides in these reasonings. But I will first add a few words on the disposition of the school now under consideration, to reject all assumption of an end.

2. That the parts of the bodies of animals are made in order to discharge their respective offices, is a conviction which we cannot believe to be otherwise than an irremovable principle of the philosophy of organization, when we see the manner in which it has constantly forced itself upon the minds of zoologists and anatomists in all ages; not only as an inference, but as a guide whose indications they could not help following. I have already noticed expressions of this conviction in some of the principal persons who occur in the history of physiology, as Galen and Harvey. I might add many more, but I will content myself with adducing a contemporary of Geoffroy's whose testimony is the more remarkable, because he obviously shares with his countryman in the common prejudice against the use of final causes. "I consider," he says, in speaking of the provisions for the reproduction of animals,[119\17] "with the great Bacon, the philosophy of final causes as sterile; but I have elsewhere acknowledged that it was very difficult for the most cautious man never to have recourse to them in his explanations." After the survey which we have had to take of the history of physiology, we cannot but see that the assumption of final causes in this branch of science is so far from being sterile, that it has had a large share in every discovery which is included in the existing mass of real knowledge. The use of every organ has been discovered by starting from the assumption that it must have _some_ use. The doctrine of the circulation of the blood was, as we have seen, clearly and professedly due to the persuasion of a purpose in the circulatory apparatus. The study of comparative anatomy is the study of the adaption of animal structures to their purposes. And we shall soon have to show that this conception of final causes has, in our own times, been so far from barren, that it has, in the hands of Cuvier and others, enabled us to become intimately acquainted with vast departments of zoology to which we have no other mode of access. It has placed before us in a complete state, {490} animals, of which, for thousands of years, only a few fragments have existed, and which differ widely from all existing animals; and it has given birth, or at least has given the greatest part of its importance and interest, to a science which forms one of the brightest parts of the modern progress of knowledge. It is, therefore, very far from being a vague and empty assertion, when we say that final causes are a real and indestructible element in zoological philosophy; and that the exclusion of them, as attempted by the school of which we speak, is a fundamental and most mischievous error.

[Note 119\17: Cabanis, _Rapports du Physique et du Morale de l'Homme_, i. **299.]

3. Thus, though the physiologist may persuade himself that he ought not to refer to final causes, we find that, practically, he cannot help doing this; and that the event shows that his practical habit is right and well-founded. But he may still cling to the speculative difficulties and doubts in which such subjects may be involved by _à

priori_ considerations. He may say, as Saint-Hilaire does say,[120\17] "I ascribe no intention to God, for I mistrust the feeble powers of my reason. I observe facts merely, and go no further. I only pretend to the character of the historian of _what is_." "I cannot make Nature an intelligent being who does nothing in vain, who acts by the shortest mode, who does all for the best."

[Note 120\17: _Phil. Zool._ p. 10.]

I am not going to enter at any length into this subject, which, thus considered, is metaphysical and theological, rather than physiological. If any one maintain, as some have maintained, that no manifestation of means apparently used for ends in nature, can prove the existence of design in the Author of nature, this is not the place to refute such an opinion in its general form. But I think it may be worth while to show, that even those who incline to such an opinion, still cannot resist the necessity which compels men to assume, in organized beings, the existence of an end.

Among the philosophers who have referred our conviction of the being of God to our moral nature, and have denied the possibility of demonstration on mere physical grounds, Kant is perhaps the most eminent. Yet he has asserted the reality of such a principle of physiology as we are now maintaining in the most emphatic manner. Indeed, this assumption of an end makes his very definition of an organized being. "An organized product of nature is that in which all the parts are mutually ends and means."[121\17] And this, he says, is a universal and necessary maxim. He adds, "It is well known that the {491} anatomizers of plants and animals, in order to investigate their structure, and to obtain an insight into the grounds why and to what end such parts, why such a situation and connexion of the parts, and exactly such an internal form, come before them, assume, as indispensably necessary, this maxim, that in such a creature nothing is _in vain_, and proceed upon it in the same way in which in general natural philosophy we proceed upon the principle that _nothing happens by chance_. In fact, they can as little free themselves from this _teleological_ principle as from the general physical one; for as, on omitting the latter, no experience would be possible, so on omitting the former principle, no clue could exist for the observation of a kind of natural objects which can be considered teleologically under the conception of natural ends."

[Note 121\17: _Urtheilskraft_, p. 296.]

Even if the reader should not follow the reasoning of this celebrated philosopher, he will still have no difficulty in seeing that he asserts, in the most distinct manner, that which is denied by the author whom we have before quoted, the propriety and necessity of assuming the existence of an end as our guide in the study of animal organization.

4. It appears to me, therefore, that whether we judge from the arguments, the results, the practice of physiologists, their speculative opinions, or those of the philosophers of a wider field, we are led to the same conviction, that in the organized world we may and must adopt the belief that organization exists for its purpose, and that the apprehension of the purpose may guide us in seeing the meaning of the organization. And I now proceed to show how this principle has been brought into

additional clearness and use by Cuvier.

In doing this, I may, perhaps, be allowed to make a reflection of a kind somewhat different from the preceding remarks, though suggested by them. In another work,[122\17] I endeavored to show that those who have been discoverers in science have generally had minds, the disposition of which was to believe in an intelligent Maker of the universe; and that the scientific speculations which produced an opposite tendency, were generally those which, though they might deal familiarly with known physical truths, and conjecture boldly with regard to the unknown, did not add to the number of solid generalizations. In order to judge whether this remark is distinctly applicable in the case now considered, I should have to estimate Cuvier in comparison with other physiologists of his time, which I do not presume to do. But I may {492} observe, that he is allowed by all to have established, on an indestructible basis, many of the most important generalizations which zoology now contains; and the principal defect which his critics have pointed out, has been, that he did not generalize still more widely and boldly. It appears, therefore, that he cannot but be placed among the great discoverers in the studies which he pursued; and this being the case, those who look with pleasure on the tendency of the thoughts of the greatest men to an Intelligence far higher than their own, most be gratified to find that he was an example of this tendency; and that the acknowledgement of a creative purpose, as well as a creative power, not only entered into his belief but made an indispensable and prominent part of his philosophy.

[Note 122\17: _Bridgewater Treatise_, B. iii. c. vii. and viii. On Inductive Habits of Thought, and on Deductive Habits of Thought.]

_Sect._ 3.--_Establishment and Application of the Principle of the Conditions of Existence of Animals.--Cuvier._

WE have now to describe more in detail the doctrine which Cuvier maintained in opposition to such opinions as we have been speaking of; and which, in his way of applying it, we look upon as a material advance in physiological knowledge, and therefore give to it a distinct place in our history. "Zoology has," he says,[123\17] in the outset of his _Règne Animal_, "a principle of reasoning which is peculiar to it, and which it employs with advantage on many occasions: this is the principle of _the Conditions of Existence_, vulgarly the principle of _Final Causes_. As nothing can exist if it do not combine all the conditions which render its existence possible, the different parts of each being must be co-ordinated in such a manner as to render the total being possible, not only in itself, but in its relations to those which surround it; and the analysis of these conditions often leads to general laws, as clearly demonstrated as those which result from calculation or from experience."

[Note 123\17: _Règne An._ p. 6.]

This is the enunciation of his leading principle in general terms. To our ascribing it to him, some may object on the ground of its being self-evident in its nature,[124\17] and having been very anciently applied. But to this we reply, that the principle must be considered as a real discovery in the hands of him who first shows how to make it an instrument of other discoveries. It is true, in other cases as well as in this, that

some vague apprehension, of true general principles, such as _à_ {493} _priori_ considerations can supply, has long preceded the knowledge of them as real and verified laws. In such a way it was seen, before Newton, that the motions of the planets must result from attraction; and so, before Dufay and Franklin, it was held that electrical actions must result from a fluid. Cuvier's merit consisted, not in seeing that an animal cannot exist without combining all the conditions of its existence; but in perceiving that this truth may be taken as a guide in our researches concerning animals;--that the mode of their existence may be collected from one part of their structure, and then applied to interpret or detect another part. He went on the supposition not only that animal forms have _some_ plan, _some_ purpose, but that they have an intelligible plan, a discoverable purpose. He proceeded in his investigations like the decipherer of a manuscript, who makes out his alphabet from one part of the context, and then applies it to read the rest. The proof that his principle was something very different from an identical proposition, is to be found in the fact, that it enabled him to understand and arrange the structures of animals with unprecedented clearness and completeness of order; and to restore the forms of the extinct animals which are found in the rocks of the earth, in a manner which has been universally assented to as irresistibly convincing. These results cannot flow from a trifling or barren principle; and they show us that if we are disposed to form such a judgment of Cuvier's doctrine, it must be because we do not fully apprehend its import.

[Note 124\17: Swainson. _Study of Nat. Hist._ p. 85.]

To illustrate this, we need only quote the statement which he makes, and the uses to which he applies it. Thus in the Introduction to his great work on _Fossil Remains_ he says, "Every organized being forms an entire system of its own, all the parts of which mutually correspond, and concur to produce a certain definite purpose by reciprocal reaction, or by combining to the same end. Hence none of these separate parts can change their forms without a corresponding change in the other parts of the same animal; and consequently each of these parts, taken separately, indicates all the other parts to which it has belonged. Thus, if the viscera of an animal are so organized as only to be fitted for the digestion of recent flesh, it is also requisite that the jaws should be so constructed as to fit them for devouring prey; the claws must be constructed for seizing it and tearing it to pieces; the teeth for cutting and dividing its flesh; the entire system of the limbs or organs of motion for pursuing and overtaking it; and the organs of sense for discovering it at a distance. Nature must also have endowed the brain of the animal with instincts sufficient for concealing itself and for laying plans to {494} catch its necessary victims."[125\17] By such considerations he has been able to reconstruct the whole of many animals of which parts only were given;--a positive result, which shows both the reality and the value of the truth on which he wrought.

[Note 125\17: _Theory of the Earth_, p. 90.]

Another great example, equally showing the immense importance of this principle in Cuvier's hands, is the reform which, by means of it, he introduced into the classification of animals. Here again we may quote the view he himself has

given[126\17] of the character of his own improvements. In studying the physiology of the natural classes of vertebrate animals, he found, he says, "in the respective quantity of their respiration, the reason of the quantity of their motion, and consequently of the kind of locomotion. This, again, furnishes the reason for the forms of their skeletons and muscles; and the energy of their senses, and the force of their digestion, are in a necessary proportion to the same quantity. Thus a division which had till then been established, like that of vegetables, only upon observation, was found to rest upon causes appreciable, and applicable to other cases." Accordingly, he applied this view to invertebrates;--examined the modifications which take place in their organs of circulation, respiration, and sensation; and having calculated the necessary results of these modifications, he deduced from it a new division of those animals, in which they are arranged according to their true relations.

[Note 126\17: _Hist. Sc. Nat._ i. 293.]

Such have been some of the results of the principle of the Conditions of Existence, as applied by its great assertor.

It is clear, indeed, that such a principle could acquire its practical value only in the hands of a person intimately acquainted with anatomical details, with the functions of the organs, and with their variety in different animals. It is only by means of such nutriment that the embryo truth could be developed into a vast tree of science. But it is not the less clear, that Cuvier's immense knowledge and great powers of thought led to their results, only by being employed under the guidance of this master-principle: and, therefore, we may justly consider it as the distinctive feature of his speculations, and follow it with a gratified eye, as the thread of gold which runs through, connects, and enriches his zoological researches:--gives them a deeper interest and a higher value than can belong to any view of the organical sciences, in which the very essence of organization is kept out of sight. {495}

The real philosopher, who knows that all the kinds of truth are intimately connected, and that all the best hopes and encouragements which are granted to our nature must be consistent with truth, will be satisfied and confirmed, rather than surprised and disturbed, thus to find the Natural Sciences leading him to the borders of a higher region. To him it will appear natural and reasonable, that after journeying so long among the beautiful and orderly laws by which the universe is governed, we find ourselves at last approaching to a Source of order and law, and intellectual beauty:--that, after venturing into the region of life and feeling and will, we are led to believe the Fountain of life and will not to be itself unintelligent and dead, but to be a living Mind, a Power which aims as well as acts. To us this doctrine appears like the natural cadence of the tones to which we have so long been listening; and without such a final strain our ears would have been left craving and unsatisfied. We have been lingering long amid the harmonies of law and symmetry, constancy and development; and these notes, though their music was sweet and deep, must too often have sounded to the ear of our moral nature, as vague and unmeaning melodies, floating in the air around us, but conveying no definite thought, moulded into no intelligible announcement. But one passage which we have again and again

caught by snatches, though sometimes interrupted and lost, at last swells in our ears full, clear, and decided; and the religious "Hymn in honor of the Creator," to which Galen so gladly lent his voice, and in which the best physiologists of succeeding times have ever joined, is filled into a richer and deeper harmony by the greatest philosophers of these later days, and will roll on hereafter the "perpetual song" of the temple of science.

{{497}} BOOK XVIII.

# _THE PALÆTIOLOGICAL SCIENCES._

HISTORY OF GEOLOGY.

Di quibus imperium est animarum, Umbræque silentes, Et Chaos, et Phlegethon, loca nocte silentia late, Sit mihi fas audita loqui; sit, numine vestro Pandere res alta terrâ et caligine mersas. VIRGIL. _Æn._ vi. 264.

Ye Mighty Ones, who sway the Souls that go Amid the marvels of the world below! Ye, silent Shades, who sit and hear around! Chaos! and Streams that burn beneath the ground! All, all forgive, if by your converse stirred, My lips shall utter what my ears have heard; If I shall speak of things of doubtful birth, Deep sunk in darkness, as deep sunk in earth.

{{499}} INTRODUCTION.

_Of the Palætiological Sciences._

WE now approach the last Class of Sciences which enter into the design of the present work; and of these, Geology is the representative, whose history we shall therefore briefly follow. By the Class of Sciences to which I have referred it, I mean to point out those researches in which the object is, to ascend from the present state of things to a more ancient condition, from which the present is derived by intelligible causes.

The sciences which treat of causes have sometimes been termed _ætiological_, from αἰτία, _a cause_: but this term would not sufficiently describe the speculations of which we now speak; since it might include sciences which treat of Permanent Causality, like Mechanics, as well as inquiries concerning Progressive Causation. The investigations which I now wish to group together, deal, not only with the possible, but with the actual past; and a portion of that science on which we are about to enter, Geology, has properly been termed _Palæontology_, since it treats of beings which formerly existed.[1\18] Hence, combining these two notions,[2\18] _Palætiology_ appears to be a term not inappropriate, to describe those speculations which thus refer to actual past events, and attempt to explain them by laws of causation.

[Note 1\18: Πάλαι, ὄντα]

[Note 2\18: Πάλαι, αἰτία]

Such speculations are not confined to the world of inert matter; we have examples of them in inquiries concerning the monuments of the art and labor of distant ages; in

examinations into the origin and early progress of states and cities, customs and languages; as well as in researches concerning the causes and formations of mountains and rocks, the imbedding of fossils in strata, and their elevation from the bottom of the ocean. All these speculations are connected by this bond,--that they endeavor to ascend to a past state of things, by the aid of the evidence of the present. In asserting, with Cuvier, that {500} "The geologist is an antiquary of a new order," we do not mark a fanciful and superficial resemblance of employment merely, but a real and philosophical connexion of the principles of investigation. The organic fossils which occur in the rock, and the medals which we find in the ruins of ancient cities, are to be studied in a similar spirit and for a similar purpose. Indeed, it is not always easy to know where the task of the geologist ends, and that of the antiquary begins. The study of ancient geography may involve us in the examination of the causes by which the forms of coasts and plains are changed; the ancient mound or scarped rock may force upon us the problem, whether its form is the work of nature or of man; the ruined temple may exhibit the traces of time in its changed level, and sea-worn columns; and thus the antiquarian of the earth may be brought into the very middle of the domain belonging to the antiquarian of art.

Such a union of these different kinds of archæological investigations has, in fact, repeatedly occurred. The changes which have taken place in the temple of Jupiter Serapis, near Puzzuoli, are of the sort which have just been described; and this is only one example of a large class of objects;--the monuments of art converted into records of natural events. And on a wider scale, we find Cuvier, in his inquiries into geological changes, bringing together historical and physical evidence. Dr. Prichard, in his _Researches into the Physical History of Man_, has shown that to execute such a design as his, we must combine the knowledge of the physiological laws of nature with the traditions of history and the philosophical comparison of languages. And even if we refuse to admit, as part of the business of geology, inquiries concerning the origin and physical history of the present population of the globe; still the geologist is compelled to take an interest in such inquiries, in order to understand matters which rigorously belong to his proper domain; for the ascertained history of the present state of things offers the best means of throwing light upon the causes of _past_ changes. Mr. Lyell quotes Dr. Prichard's book more frequently than any geological work of the same extent.

Again, we may notice another common circumstance in the studies which we are grouping together as palætiological, diverse as they are in their subjects. In all of them we have the same kind of manifestations of a number of successive changes, each springing out of a preceding state; and in all, the phenomena at each step become more and more complicated, by involving the results of all that has preceded, modified by supervening agencies. The general aspect of all these {501} trains of change is similar, and offers the same features for description. The relics and ruins of the earlier states are preserved, mutilated and dead, in the products of later times. The analogical figures by which we are tempted to express this relation are philosophically true. It is more than a mere fanciful description, to say that in languages, customs, forms of Society, political institutions, we see a number of formations super-imposed upon one another, each of which is, for the most part, an

assemblage of fragments and results of the preceding condition. Though our comparison might be bold, it would be just, if we were to assert, that the English language is a conglomerate of Latin words, bound together in a Saxon cement; the fragments of the Latin being partly portions introduced directly from the parent quarry, with all their sharp edges, and partly pebbles of the same material, obscured and shaped by long rolling in a Norman or some other channel. Thus the study of palætiology in the materials of the earth, is only a type of similar studies with respect to all the elements, which, in the history of the earth's inhabitants, have been constantly undergoing a series of connected changes.

But, wide as is the view which such considerations give us of the class of sciences to which geology belongs, they extend still further. "The science of the changes which have taken place in the organic kingdoms of nature," (such is the description which has been given of Geology,[3\18]) may, by following another set of connexions, be extended beyond "the modifications of the surface of our own planet." For we cannot doubt that some resemblance of a closer or looser kind, has obtained between the changes and causes of change, on other bodies of the universe, and on our own. The appearances of something of the kind of volcanic action on the surface of the moon, are not to be mistaken. And the inquiries concerning the origin of our planet and of our solar system, inquiries to which Geology irresistibly impels her students, direct us to ask what information the rest of the universe can supply, bearing upon this subject. It has been thought by some, that we can trace systems, more or less like our solar system, in the process of formation; the nebulous matter, which is at first expansive and attenuated, condensing gradually into suns and planets. Whether this _Nebular Hypothesis_ be tenable or not, I shall not here inquire; but the discussion of such a question would be closely connected with {502} geology, both in its interests and in its methods. If men are ever able to frame a science of the past changes by which the universe has been brought into its present condition, this science will be properly described as _Cosmical Palætiology_.

[Note 3\18: Lyell, _Principles of Geology_, p. 1.]

These palætiological sciences might properly be called _historical_, if that term were sufficiently precise: for they are all of the nature of history, being concerned with the succession of events: and the part of history which deals with the past causes of events, is, in fact, a moral palætiology. But the phrase _Natural History_ has so accustomed us to a use of the word _history_ in which we have nothing to do with time, that, if we were to employ the word _historical_ to describe the palætiological sciences, it would be in constant danger of being misunderstood. The fact is, as Mohs has said, that Natural History, when systematically treated, rigorously excludes all that is _historical_; for it classes objects by their permanent and universal properties, and has nothing to do with the narration of particular and casual facts. And this is an inconsistency which we shall not attempt to rectify.

All palætiological sciences, since they undertake to refer changes to their causes, assume a certain classification of the phenomena which change brings forth, and a knowledge of the operation of the causes of change. These phenomena, these causes, are very different, in the branches of knowledge which I have thus classed together.

The natural features of the earth's surface, the works of art, the institutions of society, the forms of language, taken together, are undoubtedly a very wide collection of subjects of speculation; and the kinds of causation which apply to them are no less varied. Of the causes of change in the inorganic and organic world,--the peculiar principles of Geology--we shall hereafter have to speak. As these must be studied by the geologist, so, in like manner, the tendencies, instincts, faculties, principles, which direct man to architecture and sculpture, to civil government, to rational and grammatical speech, and which have determined the circumstances of his progress in these paths, must be in a great degree known to the Palætiologist of Art, of Society, and of Language, respectively, in order that he may speculate soundly upon his peculiar subject. With these matters we shall not here meddle, confining ourselves, in our exemplification of the conditions and progress of such sciences, to the case of Geology.

The journey of survey which we have attempted to perform over the field of human knowledge, although carefully directed according to the paths and divisions of the physical sciences, has already {503} conducted us to the boundaries of physical science, and gives us a glimpse of the region beyond. In following the history of Life, we found ourselves led to notice the perceptive and active faculties of man; it appeared that there was a ready passage from physiology to psychology, from physics to metaphysics. In the class of sciences now under notice, we are, at a different point, carried from the world of matter to the world of thought and feeling,--from things to men. For, as we have already said, the science of the causes of change includes the productions of Man as well as of Nature. The history of the earth, and the history of the earth's inhabitants, as collected from phenomena, are governed by the same principles. Thus the portions of knowledge which seek to travel back towards the origin, whether of inert things or of the works of man, resemble each other. Both of them treat of events as connected by the thread of time and causation. In both we endeavor to learn accurately what the present is, and hence what the past has been. Both are _historical_ sciences in the same sense.

It must be recollected that I am now speaking of history as ætiological;--as it investigates causes, and as it does this in a scientific, that is, in a rigorous and systematic, manner. And I may observe here, though I cannot now dwell on the subject, that all ætiological sciences will consist of three portions; the Description of the facts and phenomena;--the general Theory of the causes of change appropriate to the case;--and the Application of the theory to the facts. Thus, taking Geology for our example, we must have, first _Descriptive_ or _Phenomenal_ Geology; next, the exposition of the general principles by which such phenomena can be produced, which we may term _Geological Dynamics_; and, lastly, doctrines hence derived, as to what have been the causes of the existing state of things, which we may call _Physical Geology_.

These three branches of geology may be found frequently or constantly combined in the works of writers on the subject, and it may not always be easy to discriminate exactly what belongs to each subject.[4\18] But the analogy of this science with others, its present {504} condition and future fortunes, will derive great illustration

from such a distribution of its history; and in this point of view, therefore, we shall briefly treat of it; dividing the history of Geological Dynamics, for the sake of convenience, into two Chapters, one referring to inorganic, and one to organic, phenomena.

[Note 4\18: The Wernerians, in distinguishing their study from _Geology_, and designating it as _Geognosy_, the _knowledge_ of the earth, appear to have intended to select Descriptive Geology for their peculiar field. In like manner, the original aim of the Geological Society of London, which was formed (1807) "with a view to record and multiply observations," recognized the possibility of a Descriptive Geology separate from the other portions of the science.]

{{505}} DESCRIPTIVE GEOLOGY.

## CHAPTER I.

PRELUDE TO SYSTEMATIC DESCRIPTIVE GEOLOGY.

_Sect._ 1.--_Ancient Notices of Geological Facts._

THE recent history of Geology, as to its most important points, is bound up with what is doing at present from day to day; and that portion of the history of the science which belongs to the past, has been amply treated by other writers.[5\18] I shall, therefore, pass rapidly over the series of events of which this history consists; and shall only attempt to mention what may seem to illustrate and confirm my own view of its state and principles.

[Note 5\18: As MM. Lyell, Fitton, Conybeare, in our own country.]

Agreeably to the order already pointed out, I shall notice, in the first place, Phenomenal Geology, or the description of the facts, as distinct from the inquiry into their causes. It is manifest that such a merely descriptive kind of knowledge may exist; and it probably will not be contested, that such knowledge ought to be collected, before we attempt to frame theories concerning the causes of the phenomena. But it must be observed, that we are here speaking of the formation of a _science_; and that it is not a collection of miscellaneous, unconnected, unarranged knowledge that can be considered as constituting science; but a methodical, coherent, and, as far as possible, complete body of facts, exhibiting fully the condition of the earth as regards those circumstances which are the subject matter of geological speculation. Such a Descriptive Geology is a pre-requisite to Physical Geology, just as Phenomenal Astronomy necessarily preceded Physical Astronomy, or as Classificatory Botany is a necessary accompaniment to Botanical Physiology. We may observe also that Descriptive Geology, such as we now speak of, is one of the classificatory sciences, like {506} Mineralogy or Botany: and will be found to exhibit some of the features of that class of sciences.

Since, then, our History of Descriptive Geology is to include only systematic and scientific descriptions of the earth or portions of it, we pass over, at once, all the casual and insulated statements of facts, though they may be geological facts, which occur in early writers; such, for instance, as the remark of Herodotus,[6\18] that

there are shells in the mountains of Egypt; or the general statements which Ovid puts in the mouth of Pythagoras:[7\18] Vidi ego quod fuerat solidissima tellus, Esse fretum; vidi factas ex æquore terras, Et procul a pelago conchæ jacuere marinæ.

[Note 6\18: ii. 12.]

[Note 7\18: Met. xv. 262.]

We may remark here already how generally there are mingled with descriptive notices of such geological facts, speculations concerning their causes. Herodotus refers to the circumstance just quoted, for the purpose of showing that Egypt was formerly a gulf of the sea; and the passage of the Roman poet is part of a series of exemplifications which he gives of the philosophical tenet, that nothing perishes but everything changes. It will be only by constant attention that we shall be able to keep our provinces of geology distinct.

_Sect._ 2.--_Early Descriptions and Collections of Fossils._

IF we look, as we have proposed to do, for systematic and exact knowledge of geological facts, we find nothing which we can properly adduce till we come to modern times. But when facts such as those already mentioned, (that sea-shells and other marine objects are found imbedded in rocks,) and other circumstances in the structure of the Earth, had attracted considerable attention, the exact examination, collection, and record of these circumstances began to be attempted. Among such steps in Descriptive Geology, we may notice descriptions and pictures of fossils, descriptions of veins and mines, collections of organic and inorganic fossils, maps of the mineral structure of countries, and finally, the discoveries concerning the superposition of strata, the constancy of their organic contents, their correspondence in different countries, and such great general relations of the materials and features of the earth as have been discovered up to the present time. {507} Without attempting to assign to every important advance its author, I shall briefly exemplify each of the modes of contributing to descriptive geology which I have just enumerated.

The study of organic fossils was first pursued with connexion and system in Italy. The hills which on each side skirt the mountain-range of the Apennines are singularly rich in remains of marine animals. When these remarkable objects drew the attention of thoughtful men, controversies soon arose whether they really were the remains of living creatures, or the productions of some capricious or mysterious power by which the forms of such creatures were mimicked; and again, if the shells were really the spoils of the sea, whether they had been carried to the hills by the deluge of which the Scripture speaks, or whether they indicated revolutions of the earth of a different kind. The earlier works which contain the descriptions of the phenomena have, in almost all instances, by far the greater part of their pages occupied with these speculations; indeed, the facts could not be studied without leading to such inferences, and would not have been collected but for the interest which such reasonings possessed. As one of the first persons who applied a sound and vigorous intellect to these subjects, we may notice the celebrated painter Leonardo da Vinci, whom we have already had to refer to as one of the founders of the modern

mechanical sciences. He strenuously asserts the contents of the rocks to be real shells, and maintains the reality of the changes of the domain of land and sea which these spoils of the ocean imply. "You will tell me," he says, "that nature and the influence of the stars have formed these shelly forms in the mountains; then show me a place in the mountains where the stars at the present day make shelly forms of different ages, and of different species in the same place. And how, with that, will you explain the gravel which is hardened in stages at different heights in the mountains?" He then mentions several other particulars respecting these evidences that the existing mountains were formerly in the bed of the sea. Leonardo died in 1519. At present we refer to geological essays like his, only so far as they are descriptive. Going onwards with this view, we may notice Fracastoro, who wrote concerning the petrifactions which were brought to light in the mountains of Verona, when, in 1517, they were excavated for the purpose of repairing the city. Little was done in the way of collection of facts for some time after this. In 1669, Steno, a Dane resident in Italy, put forth his treatise, _De Solido intra Solidum naturaliter contento_; and the {508} following year, Augustino Scilla, a Sicilian painter, published a Latin epistle, _De Corporibus Marinis Lapidescentibus_, illustrated by good engravings of fossil-shells, teeth, and corals.[8\18] After another interval of speculative controversy, we come to Antonio Vallisneri, whose letters, _De' Corpi Marini che su' Monti si trovano_, appeared at Venice in 1721. In these letters he describes the fossils of Monte Bolca, and attempts to trace the extent of the marine deposits of Italy,[9\18] and to distinguish the most important of the fossils. Similar descriptions and figures were published with reference to our own country at a later period. In 1766, Brander's _Fossilia Hantoniensia_, or Hampshire Fossils, appeared; containing excellent figures of fossil shells from a part of the south coast of England; and similar works came forth in other parts of Europe.

[Note 8\18: Augustine Scilla's original drawings of fossil shells, teeth, and corals, from which the engravings mentioned in the text were executed, as well as the natural objects from which the drawings were made, were bought by Woodward, and are now in the Woodwardian Museum at Cambridge.]

[Note 9\18: p. 20.]

However exact might be the descriptions and figures thus produced, they could not give such complete information as the objects themselves, collected and permanently preserved in museums. Vallisneri says,[10\18] that having begun to collect fossils for the purpose of forming a grotto, he selected the best, and preserved them "as a noble diversion for the more curious." The museum of Calceolarius at Verona contained a celebrated collection of such remains. A copious description of it appeared in 1622. Such collections had been made from an earlier period, and catalogues of them published. Thus Gessner's work, _De Rerum Fossilium, Lapidum et Gemmarum Figuris_ (1565), contains a catalogue of the cabinet of petrifactions collected by John Kentman; many catalogues of the same kind appeared in the seventeenth century.[11\18] Lhwyd's _**Lythophylacii Britannici Iconographia_, published at Oxford in 1669, and exhibiting a very ample catalogue of English Fossils contained in the Ashmolean Museum, may be noticed as one of these.

[Note 10\18: p. 1.]

[Note 11\18: Parkinson, _Organic Remains_, vol. i. p. 20.]

One of the most remarkable occurrences in the progress of descriptive geology in England, was the formation of a geological museum by William Woodward as early as 1695. This collection, formed with great labor, systematically arranged, and carefully catalogued, he bequeathed to the University of Cambridge; founding and endowing {509} at the same time a professorship of the study of geology. The Woodwardian Museum still subsists, a monument of the sagacity with which its author so early saw the importance of such a collection.

Collections and descriptions of fossils, including in the term specimens of minerals of all kinds, as well as organic remains, were frequently made, and especially in places where mining was cultivated; but under such circumstances, they scarcely tended at all to that general and complete knowledge of the earth of which we are now tracing the progress.

In more modern times, collections may be said to be the most important books of the geologist, at least next to the strata themselves. The identifications and arrangements of our best geologists, the immense studies of fossil anatomy by Cuvier and others, have been conducted mainly by means of collections of specimens. They are more important in this study than in botany, because specimens which contain important geological information are both more rare and more permanent. Plants, though each individual is perishable, perpetuate and diffuse their kind; while the organic impression on a stone, if lost, may never occur in a second instance; but, on the other hand, if it be preserved in the museum, the individual is almost as permanent in this case, as the species in the other.

I shall proceed to notice another mode in which such information was conveyed.

_Sect._ 3.--_First Construction of Geological Maps._

DR. LISTER, a learned physician, sent to the Royal Society, in 1683, a proposal for maps of soils or minerals; in which he suggested that in the map of England, for example, each soil and its boundaries might be distinguished by color, or in some other way. Such a mode of expressing and connecting our knowledge of the materials of the earth was, perhaps, obvious, when the mass of knowledge became considerable. In 1720, Fontenelle, in his observations on a paper of De Reaumur's, which contained an account of a deposit of fossil-shells in Touraine, says, that in order to reason on such cases, "we must have a kind of geographical charts, constructed according to the collection of shells found in the earth." But he justly adds, "What a quantity of observations, and what time would it not require to form such maps!"

The execution of such projects required, not merely great labor, but {510} several steps in generalization and classification, before it could take place. Still such attempts were made. In 1743, was published, _A new Philosophico-chorographical Chart of East Kent, invented and delineated_ by Christopher Packe, M.D.; in which, however, the main object is rather to express the course of the valleys than the

materials of the country. Guettard formed the project of a mineralogical map of France, and Monnet carried this scheme into effect in 1780,[12\18] "by order of the king." In these maps, however, the country is not considered as divided into soils, still less strata; but each part is marked with its predominant mineral only. The spirit of generalization which constitutes the main value of such a work is wanting.

[Note 12\18: _Atlas et Description Minéralogique de la France, entrepris par ordre du Roi_, par MM. Guettard et Monnet, Paris, 1780, pp. 212, with 31 maps.]

Geological maps belong strictly to Descriptive Geology; they are free from those wide and doubtful speculations which form so large a portion of the earlier geological books. Yet even geological maps cannot be usefully or consistently constructed without considerable steps of classification and generalization. When, in our own time, geologists were become weary of controversies respecting theory, they applied themselves with extraordinary zeal to the construction of stratigraphical maps of various countries; flattering themselves that in this way they were merely recording incontestable facts and differences. Nor do I mean to intimate that their facts were doubtful, or their distinctions arbitrary. But still they were facts interpreted, associated, and represented, by means of the classifications and general laws which earlier geologists had established; and thus even Descriptive Geology has been brought into existence as a science by the formation of systems and the discovery of principles. At this we cannot be surprized, when we recollect the many steps which the formation of Classificatory Botany required. We must now notice some of the discoveries which tended to the formation of Systematic Descriptive Geology. {511}

## CHAPTER II.

FORMATION OF SYSTEMATIC DESCRIPTIVE GEOLOGY.

_Sect._ 1.--_Discovery of the Order and Stratification of the Materials of the Earth._

THAT the substances of which the earth is framed are not scattered and mixed at random, but possess identity and continuity to a considerable extent, Lister was aware, when he proposed his map. But there is, in his suggestions, nothing relating to stratification; nor any order of position, still less of time, assigned to these materials. Woodward, however, appears to have been fully aware of the general law of stratification. On collecting information from all parts, "the result was," he says, "that in time I was abundantly assured that the circumstances of these things in remoter countries were much the same with those of ours here: that the stone, and other terrestrial matter in France, Flanders, Holland, Spain, Italy, Germany, Denmark, and Sweden, was distinguished into _strata or layers_, as it is in England; that these strata were divided by parallel fissures; that there were enclosed in the stone and all the other denser kinds of terrestrial matter, great numbers of the shells, and other productions of the sea, in the same manner as in that of this island."[13\18] So remarkable a truth, thus collected from a copious collection of particulars by a patient induction, was an important step in the science.

[Note 13\18: _Natural History of the Earth_, 1723.]

These general facts now began to be commonly recognized, and followed into detail. **Stukeley the antiquary[14\18] (1724), remarked an important feature in the strata of England, that their _escarpments_, or steepest sides, are turned towards the west and north-west; and Strachey[15\18] (1719), gave a stratigraphical description of certain coal-mines near Bath.[16\18] Michell, appointed Woodwardian Professor at Cambridge {512} in 1762, described this stratified structure of the earth far more distinctly than his predecessors, and pointed out, as the consequence of it, that "the same kinds of earths, stones, and minerals, will appear at the surface of the earth in long parallel slips, parallel to the long ridges of mountains; and so, in fact, we find them."[17\18]

[Note 14\18: _Itinerarium Curiosum_, 1724.]

[Note 15\18: _Phil. Trans._ 1719, and _Observations on Strata, &c._ 1729.]

[Note 16\18: Fitton, _Annals of Philosophy_, N. S. vol. i. and ii. (1832, '3), p. 157.]

[Note 17\18: _Phil. Trans._ 1760.]

Michell (as appeared by papers of his which were examined after his death) had made himself acquainted with the series of English strata which thus occur from Cambridge to York;--that is, from the chalk to the coal. These relations of position required that geological maps, to complete the information they conveyed, should be accompanied by geological _Sections_, or imaginary representations of the order and mode of superpositions, as well as of the superficial extent of the strata, as in more recent times has usually been done. The strata, as we travel from the higher to the lower, come from under each other into view; and this _out-cropping_, _basseting_, or by whatever other term it is described, is an important feature in their description.

It was further noticed that these relations of position were combined with other important facts, which irresistibly suggested the notion of a relation in time. This, indeed, was implied in all theories of the earth; but observations of the facts most require our notice. Steno is asserted by Humboldt[18\18] to be the first who (in 1669) distinguished between rocks anterior to the existence of plants and animals upon the globe, containing therefore no organic remains; and rocks super-imposed on these, and full of such remains; "turbidi maris sedimenta sibi invicem imposita".

[Note 18\18: _Essai **Géognostique_.]

Rouelle is stated, by his pupil Desmarest, to have made some additional and important observations. "He saw," it is said, "that the shells which occur in rocks were not the same in all countries; that certain species occur together, while others do not occur in the same beds; that there is a constant order in the arrangement of these shells, certain species lying in distinct bands."[19\18]

[Note 19\18: _Encycl. Méthod. Geogr. Phys._ tom. i. p. 416, as quoted by Fitton as above, p. 159.]

Such divisions as these required to be marked by technical names. A distinction was made of _l'ancienne terre_ and _la nouvelle terre_, to which Rouelle added a _travaille intermédiaire_. Rouelle died in 1770, having been known by lectures, not by

books. Lehman, in 1756, claims for himself the credit of being the first to observe and describe correctly the structure of stratified countries; being ignorant, {513} probably, of the labors of Strachey in England. He divided mountains into three classes;[20\18] _primitive_, which were formed with the world;--those which resulted from a partial destruction of the primitive rocks;--and a third class resulting from local or universal deluges. In 1759, also, Arduine,[21\18] in his Memoirs on the mountains of Padua, Vicenza, and Verona, deduced, from original observations, the distinction of rocks into _primary_, _secondary_, and _tertiary_.

[Note 20\18: Lyell, i. 70.]

[Note 21\18: Ib. 72.]

The relations of position and fossils were, from this period, inseparably connected with opinions concerning succession in time. Odoardi remarked,[22\18] that the strata of the **Sub-Apennine hills are _unconformable_ to those of the Apennine, (as Strachey had observed, that the strata above the coal were unconformable to the coal;[23\18]) and his work contained a clear argument respecting the different ages of these two classes of hills. Fuchsel was, in 1762, aware of the distinctness of strata of different ages in Germany. Pallas and Saussure were guided by general views of the same kind in observing the countries which they visited: but, perhaps, the general circulation of such notions was most due to Werner.

[Note 22\18: Ib. 74.]

[Note 23\18: Fitton, p. 157.]

_Sect._ 2.--_Systematic form given to Descriptive Geology.--Werner._

WERNER expressed the general relations of the strata of the earth by means of classifications which, so far as general applicability is concerned, are extremely imperfect and arbitrary; he promulgated a theory which almost entirely neglected all the facts previously discovered respecting the grouping of fossils,--which was founded upon observations made in a very limited district of Germany,--and which was contradicted even by the facts of this district. Yet the acuteness of his discrimination in the subjects which he studied, the generality of the tenets he asserted, and the charm which he threw about his speculations, gave to Geology, or, as he termed it, _Geognosy_, a popularity and reputation which it had never before possessed. His system had asserted certain universal formations, which followed each other in a constant order;--granite the lowest,--then mica-slate and clay-slate;--upon these _primitive_ rocks, generally highly inclined, rest other _transition_ strata;--upon these, lie _secondary_ ones, which being more nearly horizontal, are called _flötz_ or flat. The term _formation_, {514} which we have thus introduced, indicating groups which, by evidence of all kinds,--of their materials, their position, and their organic contents,--are judged to belong to the same period, implies no small amount of theory: yet this term, from this time forth, is to be looked upon as a term of classification solely, so far as classification can be separately attended to.

Werner's distinctions of strata were for the most part drawn from mineralogical constitution. Doubtless, he could not fail to perceive the great importance of organic

fossils. "I was witness," says M. de Humboldt, one of his most philosophical followers, "of the lively satisfaction which he felt when, in 1792, M. de **Schlottheim, one of the most distinguished geologists of the school of Freiberg, began to make the relations of fossils to strata the principal object of his studies." But Werner and the disciples of his school, even the most enlightened of them, never employed the characters derived from organic remains with the same boldness and perseverance as those who had from the first considered them as the leading phenomena: thus M. de Humboldt expresses doubts which perhaps many other geologists do not feel when, in 1823, he says, "Are we justified in concluding that all formations are characterized by particular species? that the fossil-shells of the chalk, the muschelkalk, the Jura limestone, and the Alpine limestone, are all different? I think this would be pushing the induction much too far."[24\18] In Prof. Jamieson's _Geognosy_, which may be taken as a representation of the Wernerian doctrines, organic fossils are in no instance referred to as characters of formations or strata. After the curious and important evidence, contained in organic fossils, which had been brought into view by the labors of Italian, English, and German writers, the promulgation of a system of Descriptive Geology, in which all this evidence was neglected, cannot be considered otherwise than as a retrograde step in science.

[Note 24\18: _Gissement des Roches_, p. 41.]

Werner maintained the aqueous deposition of all strata above the primitive rocks; even of those _trap_ rocks, to which, from their resemblance to lava and other phenomena, Raspe, Arduino, and others, had already assigned a volcanic origin. The fierce and long controversy between the _Vulcanists_ and _Neptunists_, which this dogma excited, does not belong to this part of our history; but the discovery of veins of granite penetrating the superincumbent slate, to which the controversy led, was an important event in descriptive geology. Hutton, the {515} author of the theory of igneous causation which was in this country opposed to that of Werner, sought and found this phenomenon in the Grampian hills, in 1785. This supposed verification of his system "filled him with delight, and called forth such marks of joy and exultation, that the guides who accompanied him were persuaded, says his biographer,[25\18] that he must have discovered a vein of silver or gold."[26\18]

[Note 25\18: Playfair's _Works_, vol. iv. p. 75.]

[Note 26\18: Lyell, i. 90.]

Desmarest's examination of Auvergne (1768) showed that there was there an instance of a country which could not even be described without terms implying that the basalt, which covered so large a portion of it, had flowed from the craters of extinct volcanoes. His map of Auvergne was an excellent example of a survey of such a country, thus exhibiting features quite different from those of common stratified countries.[27\18]

[Note 27\18: Lyell, i. 86.]

The facts connected with metalliferous veins were also objects of Werner's attention. A knowledge of such facts is valuable to the geologist as well as to the miner, although even yet much difficulty attends all attempts to theorize concerning

them. The facts of this nature have been collected in great abundance in all mining districts; and form a prominent part of the descriptive geology of such districts; as, for example, the Hartz, and Cornwall.

Without further pursuing the history of the knowledge of the inorganic phenomena of the earth, I turn to a still richer department of geology, which is concerned with organic fossils.

_Sect._ 3.--_Application of Organic Remains as a Geological Character.--Smith._

ROUELLE and Odoardi had perceived, as we have seen, that fossils were grouped in bands: but from this general observation to the execution of a survey of a large kingdom, founded upon this principle, would have been a vast stride, even if the author of it had been aware of the doctrines thus asserted by these writers. In fact, however, William Smith executed such a survey of England, with no other guide or help than his own sagacity and perseverance. In his employments as a civil engineer, he noticed the remarkable continuity and constant order of the strata in the neighborhood of Bath, as discriminated by their fossils; and about the year 1793, he[28\18] drew up a Tabular View of the {516} strata of that district, which contained the germ of his subsequent discoveries. Finding in the north of England the same strata and associations of strata with which he had become acquainted in the west, he was led to name them and to represent them by means of maps, according to their occurrence over the whole face of England. These maps appeared[29\18] in 1815; and a work by the same author, entitled _The English Strata identified by Organic Remains_, came forth later. But the views on which this identification of strata rests, belong to a considerably earlier date; and had not only been acted upon, but freely imparted in conversation many years before.

[Note 28\18: Fitton, p. 148.]

[Note 29\18: Brit. Assoc. 1832. Conybeare, p. 373.]

In the meantime the study of fossils was pursued with zeal in various countries. Lamarck and Defrance employed themselves in determining the fossil shells of the neighborhood of Paris;[30\18] and the interest inspired by this subject was strongly nourished and stimulated by the memorable work of Cuvier and Brongniart, _On the Environs of Paris_, published in 1811, and by Cuvier's subsequent researches on the subjects thus brought under notice. For now, not only the distinction, succession, and arrangement, but many other relations among fossil strata, irresistibly arrested the attention of the philosopher. Brongniart[31\18] showed that very striking resemblances occurred in their fossil remains, between certain strata of Europe and of North America; and proved that a rock may be so much disguised, that the identity of the stratum can only be recognized by geological characters.[32\18]

[Note 30\18: Humboldt, _Giss. d. R._ p. 35.]

[Note 31\18: _Hist. Nat. des Crustacés Fossiles_, pp. 57, 62.]

[Note 32\18: Humboldt, _Giss. d. R._ p. 45.]

The Italian geologists had found in their hills, for the most part, the same species of shells which existed in their seas; but the German and English writers, as Gesner,[33\18] Raspe,[34\18] and Brander,[35\18] had perceived that the fossil-shells were either of unknown species, or of such as lived in distant latitudes. To decide that the animals and plants, of which we find the remains in a fossil state, were of species now extinct, obviously required an exact and extensive knowledge of natural history. And if this were so, to assign the relations of the past to the existing tribes of beings, and the peculiarities of their vital processes and habits, were tasks which could not be performed without the most consummate physiological skill and talent. Such tasks, however, have been the familiar employments of geologists, and naturalists incited and {517} appealed to by geologists, ever since Cuvier published his examination of the fossil inhabitants of the Paris basin. Without attempting a history of such labors, I may notice a few circumstances connected with them.

[Note 33\18: Lyell, i. 70.]

[Note 34\18: Ib. 74.]

[Note 35\18: Ib. 76.]

_Sect._ 4.--_Advances in Palæontology.--Cuvier._

SO long as the organic fossils which were found in the strata of the earth were the remains of marine animals, it was very difficult for geologists to be assured that the animals were such as did not exist in any part or clime of the existing ocean. But when large land and river animals were discovered, different from any known species, the persuasion that they were of extinct races was forced upon the naturalist. Yet this opinion was not taken up slightly, nor acquiesced in without many struggles.

Bones supposed to belong to fossil elephants, were some of the first with regard to which this conclusion was established. Such remains occur in vast numbers in the soil and gravel of almost every part of the world; especially in Siberia, where they are called the bones of the _mammoth_. They had been noticed by the ancients, as we learn from Pliny;[36\18] and had been ascribed to human giants, to elephants imported by the Romans, and to many other origins. But in 1796, Cuvier had examined these opinions with a more profound knowledge than his predecessors; and he thus stated the result of his researches.[37\18] "With regard to what have been called the fossil remains of elephants, from Tentzelius to Pallas, I believe that I am in the condition to prove, that they belong to animals which were very clearly different in species from our existing elephants, although they resembled them sufficiently to be considered as belonging to the same genera." He had founded this conclusion principally on the structure of the teeth, which he found to differ in the Asiatic and African elephant; while, in the fossil animal, it was different from both. But he also reasoned in part on the form of the skull, of which the best-known example had been described in the _Philosophical Transactions_ as early as 1737.[38\18] "As soon," says Cuvier, at a later period, "as I became acquainted with Messerschmidt's drawing, and joined to the differences which it presented, those which I had myself observed in the inferior jaw and the {518} molar teeth, I no longer doubted that the fossil elephants

were of a species different from the Indian elephant. This idea, which I announced to the Institute in the month of January, 1796, opened to me views entirely new respecting the theory of the earth; and determined me to devote myself to the long researches and to the assiduous labors which have now occupied me for twenty-five years."[39\18]

[Note 36\18: _Hist. Nat._ lib. xxxvi. 18.]

[Note 37\18: _Mém. Inst. Math. et Phys._ tom. ii. p. 4.]

[Note 38\18: Described by Breyne from a specimen found in Siberia by Messerschmidt in 1722. _Phil. Trans._ xl. 446.]

[Note 39\18: _Ossemens Fossiles_, second edit. i. 178.]

We have here, then, the starting-point of those researches concerning extinct animals, which, ever since that time, have attracted so large a share of notice from geologists and from the world. Cuvier could hardly have anticipated the vast storehouse of materials which lay under his feet, ready to supply him occupation of the most intense interest in the career on which he had thus entered. The examination of the strata on which Paris stands, and of which its buildings consist, supplied him with animals, not only different from existing ones, but some of them of great size and curious peculiarities. A careful examination of the remains which these strata contain was undertaken soon after the period we have referred to. In 1802, Defrance had collected several hundreds of undescribed species of shells; and Lamarck[40\18] began a series of Memoirs upon them; remodelling the whole of Conchology, in order that they might be included in its classifications. And two years afterwards (1804) appears the first of Cuvier's grand series of Memoirs containing the restoration of the vertebrate animals of these strata. In this vast natural museum, and in contributions from other parts of the globe, he discovered the most extraordinary creatures:--the Palæotherium,[41\18] which is intermediate between the horse and the pig; the Anoplotherium, which stands nearest to the rhinoceros and the tapir; the Megalonix and Megatherium, animals of the sloth tribe, but of the size of the ox and the rhinoceros. The Memoirs which contained these and many other discoveries, set the naturalists to work in every part of Europe.

[Note 40\18: _Annales du Muséum d'Hist. Nat._ tom. i. p. 308, and the following volumes.]

[Note 41\18: Daubuisson, ii. 411.]

Another very curious class of animals was brought to light principally by the geologists of England; animals of which the bones, found in the _lias_ stratum, were at first supposed to be those of crocodiles. But in 1816,[42\18] Sir Everard Home says, "In truth, on a consideration of this skeleton, we cannot but be inclined to believe, that among the animals destroyed by the catastrophes of remote antiquity, there had {519} been some at least that differ so entirely in their structure from any which now exist as to make it impossible to arrange their fossil remains with any known class of animals." The animal thus referred to, being clearly intermediate between fishes and lizards, was named by Mr. König, _Ichthyosaurus_; and its

structure and constitution were more precisely determined by Mr. Conybeare in 1821, when he had occasion to compare with it another extinct animal of which he and Mr. de la Beche had collected the remains. This animal, still more nearly approaching the lizard tribe, was by Mr. Conybeare called _Plesiosaurus_.[43\18] Of each of these two genera several species were afterwards found.

[Note 42\18: _Phil. Trans._ 1816, p. 20.]

[Note 43\18: _Geol. Trans._ vol. v.]

Before this time, the differences of the races of animals and plants belonging to the past and the present periods of the earth's history, had become a leading subject of speculation among geological naturalists. The science produced by this study of the natural history of former states of the earth has been termed _Palæontology_; and there is no branch of human knowledge more fitted to stir men's wonder, or to excite them to the widest physiological speculations. But in the present part of our history this science requires our notice, only so far as it aims at the restoration of the types of ancient animals, on clear and undoubted principles of comparative anatomy. To show how extensive and how conclusive is the science when thus directed, we need only refer to Cuvier's _Ossemens Fossiles_;[44\18] a work of vast labor and profound knowledge, which has opened wide the doors of this part of geology. I do not here attempt even to mention the labors of the many other eminent contributors to Palæontology; as Brocchi, Des Hayes, Sowerby, Goldfuss, Agassiz, who have employed themselves on animals, and Schlottheim, Brongniart, Hutton, Lindley, on plants.

[Note 44\18: The first edition appeared in 1812, consisting principally of the Memoirs to which reference has already been made.]

[2nd Ed.] [Among the many valuable contributions to Palæontology in more recent times, I may especially mention Mr. Owen's _Reports on British Fossil Reptiles_, _on British Fossil Mammalia_, and _on the Extinct Animals of Australia_, with descriptions of certain Fossils indicative of large Marsupial Pachydermata: and M. Agassiz's _Report on the Fossil Fishes of the Devonian System_, his _Synoptical Table of British Fossil Fishes_, and his _Report on the Fishes of the London Clay_. All these are contained in the volumes produced by the British Association from 1839 to 1845. {520}

A new and most important instrument of palæontological investigation has been put in the geologist's hand by Prof. Owen's discovery, that the internal structure of teeth, as disclosed by the microscope, is a means of determining the kind of the animal. He has carried into every part of the animal kingdom an examination founded upon this discovery, and has published the results of this in his _Odontography_. As an example of the application of this character of animals, I may mention that a tooth brought from Riga by Sir R. Murchison was in this way ascertained by Mr. Owen to belong to a fish of the genus _Dendrodus_. (_Geology of Russia_, i. 67.)]

When it had thus been established, that the strata of the earth are characterized by innumerable remains of the organized beings which formerly inhabited it, and that anatomical and physiological considerations must be carefully and skilfully applied in

order rightly to interpret these characters, the geologist and the palæontologist obviously had, brought before them, many very wide and striking questions. Of these we may give some instances; but, in the first place, we may add a few words concerning those eminent philosophers to whom the science owed the basis on which succeeding speculations were to be built.

_Sect._ 5.--_Intellectual Characters of the Founders of Systematic Descriptive Geology._

IT would be in accordance with the course we have pursued in treating of other subjects, that we should attempt to point out in the founders of the science now under consideration, those intellectual qualities and habits to which we ascribe their success. The very recent date of the generalizations of geology, which has hardly allowed us time to distinguish the calm expression of the opinion of the wisest judges, might, in this instance, relieve us from such a duty; but since our plan appears to suggest it, we will, at least, endeavor to mark the characters of the founders of geology, by a few of their prominent lines.

The three persons who must be looked upon as the main authors of geological classification are, Werner, Smith, and Cuvier. These three men were of very different mental constitution; and it will, perhaps, not be difficult to compare them, in reference to those qualities which we have all along represented as the main features of the discoverer's genius, clearness of ideas, the possession of numerous facts, and the power of bringing these two elements into contact. {521}

In the German, considering him as a geologist, the ideal element predominated. That Werner's powers of external discrimination were extremely acute, we have seen in speaking of him as a mineralogist; and his talent and tendency for classifying were, in his mineralogical studies, fully fed by an abundant store of observation; but when he came to apply this methodizing power to geology, the love of system, so fostered, appears to have been too strong for the collection of facts he had to deal with. As we have already said, he promulgated, as representing the world, a scheme collected from a province, and even too hastily gathered from that narrow field. Yet his intense spirit of method in some measure compensated for other deficiencies, and enabled him to give the character of a science to what had been before a collection of miscellaneous phenomena. The ardor of system-making produced a sort of fusion, which, however superficial, served to bind together the mass of incoherent and mixed materials, and thus to form, though by strange and anomalous means, a structure of no small strength and durability, like the ancient vitrified structures which we find in some of our mountain regions.

Of a very different temper and character was William Smith. No literary cultivation of his youth awoke in him the speculative love of symmetry and system; but a singular clearness and precision of the classifying power, which he possessed as a native talent, was exercised and developed by exactly those geological facts among which his philosophical task lay. Some of the advances which he made, had, as we have seen, been at least entered upon by others who preceded him: but of all this he was ignorant; and, perhaps, went on more steadily and eagerly to work out his own ideas,

from the persuasion that they were entirely his own. At a later period of his life, he himself published an account of the views which had animated him in his earlier progress. In this account[45\18] he dates his attempts to discriminate and connect strata from the year 1790, at which time he was twenty years old. In 1792, he "had considered how he could best represent the order of superposition--continuity of course--and general eastern declination of the strata." Soon after, doubts which had arisen were removed by the "discovery of a mode of identifying the strata by the organized fossils respectively imbedded therein." And "thus stored with ideas," as he expresses himself, he began to communicate them to his friends. In all this, we see great vividness {522} of thought and activity of mind, unfolding itself exactly in proportion to the facts with which it had to deal. We are reminded of that cyclopean architecture in which each stone, as it occurs, is, with wonderful ingenuity, and with the least possible alteration of its form, shaped so as to fit its place in a solid and lasting edifice.

[Note 45\18: _Phil. Mag._ 1833, vol. i. p. 38.]

Different yet again was the character (as a geological discoverer) of the great naturalist of the beginning of the nineteenth century. In that part of his labors of which we have now to speak, Cuvier's dominant ideas were rather physiological than geological. In his views of past physical changes, he did not seek to include any ranges of facts which lay much beyond the narrow field of the Paris basin. But his sagacity in applying his own great principle of the Conditions of Existence, gave him a peculiar and unparalleled power in interpreting the most imperfect fossil records of extinct anatomy. In the constitution of his mind, all philosophical endowments were so admirably developed and disciplined, that it was difficult to say, whether more of his power was due to genius or to culture. The talent of classifying which he exercised in geology, was the result of the most complete knowledge and skill in zoology; while his views concerning the revolutions which had taken place in the organic and inorganic world, were in no small degree aided by an extraordinary command of historical and other literature. His guiding ideas had been formed, his facts had been studied, by the assistance of all the sciences which could be made to bear upon them. In his geological labors we seem to see some beautiful temple, not only firm and fair in itself, but decorated with sculpture and painting, and rich in all that art and labor, memory and imagination, can contribute to its beauty.

[2nd Ed.] [Sir Charles Lyell (B. i. c. iv.) has quoted with approval what I have elsewhere said, that the advancement of three of the main divisions of geology in the beginning of the present century was promoted principally by the three great nations of Europe,--the German, the English, and the French:--Mineralogical Geology by the German school of Werner:--Secondary Geology by Smith and his English successors;--Tertiary Geology by Cuvier and his fellow-laborers in France.] {523}

## CHAPTER III.

SEQUEL TO THE FORMATION OF SYSTEMATIC DESCRIPTIVE GEOLOGY.

_Sect._ 1.--_Reception and Diffusion of Systematic Geology._

IF our nearness to the time of the discoveries to which we have just referred, embarrasses us in speaking of their authors, it makes it still more difficult to narrate the reception with which these discoveries met. Yet here we may notice a few facts which may not be without their interest.

The impression which Werner made upon his hearers was very strong; and, as we have already said, disciples were gathered to his school from every country, and then went forward into all parts of the world, animated by the views which they had caught from him. We may say of him, as has been so wisely said of a philosopher of a very different kind,[46\18] "He owed his influence to various causes; at the head of which may be placed that genius for system, which, though it cramps the growth of knowledge, perhaps finally atones for that mischief by the zeal and activity which it rouses among followers and opponents, who discover truth by accident, when in pursuit of weapons for their warfare." The list of Werner's pupils for a considerable period included most of the principal geologists of Europe; Freisleben, Mohs, Esmark, d'Andrada, Raumer, Engelhart, Charpentier, Brocchi. Alexander von Humboldt and Leopold von Buch went forth from his school to observe America and Siberia, the Isles of the Atlantic, and the coast of Norway. Professor Jameson established at Edinburgh a Wernerian Society; and his lecture-room became a second centre of Wernerian doctrines, whence proceeded many zealous geological observers; among these we may mention as one of the most distinguished, M. Ami Boué, though, like several others, he soon cast away the peculiar opinions of the Wernerian school. The classifications of this school were, however, diffused over the civilized world with {524} extraordinary success; and were looked upon with great respect, till the study of organic fossils threw them into the shade.

[Note 46\18: Mackintosh _on Hobbes_, Dissert. p. 177.]

Smith, on the other hand, long pursued his own thoughts without aid and without sympathy. About 1799 he became acquainted with a few gentlemen (Dr. Anderson, Mr. Richardson, Mr. Townsend, and Mr. Davies), who had already given some attention to organic fossils, and who were astonished to find his knowledge so much more exact and extensive than their own. From this time he conceived the intention of publishing his discoveries; but the want of literary leisure and habits long prevented him. His knowledge was orally communicated without reserve to many persons; and thus gradually and insensibly became part of the public stock. When this diffusion of his views had gone on for some time, his friends began to complain that the author of them was deprived of his well-merited share of fame. His delay in publication made it difficult to remedy this wrong; for soon after he published his Geological Map of England, another appeared, founded upon separate observations; and though, perhaps, not quite independent of his, yet in many respects much more detailed and correct. Thus, though his general ideas obtained universal currency, he did not assume his due prominence as a geologist. In 1818, a generous attempt was made to direct a proper degree of public gratitude to him, in an article in the _Edinburgh Review_, the production of Dr. Fitton, a distinguished English geologist. And when the eminent philosopher, Wollaston, had bequeathed to the Geological Society of London a fund from which a gold medal was to be awarded to geological

services, the first of such medals was, in 1831, "given to Mr. William Smith, in consideration of his being a great original discoverer in English geology; and especially for his having been the first in this country to discover and to teach the identification of strata, and to determine their succession by means of their imbedded fossils."

Cuvier's discoveries, on the other hand, both from the high philosophic fame of their author, and from their intrinsic importance, arrested at once the attention of scientific Europe; and, notwithstanding the undoubted priority of Smith's labors, for a long time were looked upon as the starting-point of our knowledge of organic fossils. And, in reality, although Cuvier's memoirs derived the greatest part of their value from his zoological conclusions, they reflected back no small portion of interest on the classifications of strata which were involved in his inferences. And the views which he presented gave to geology an attractive and striking character, and a connexion with {525} large physiological as well as physical principles, which added incomparably to its dignity and charm.

In tracing the reception and diffusion of doctrines such as those of Smith and Cuvier, we ought not to omit to notice more especially the formation and history of the Geological Society of London, just mentioned. It was established in 1807, with a view to multiply and record observations, and patiently to await the result of some future period; that is, its founders resolved to apply themselves to Descriptive Geology, thinking the time not come for that theoretical geology which had then long fired the controversial ardor of Neptunists and Plutonists. The first volume of the Transactions of this society was published in 1811. The greater part of the contents of this volume[47\18] savor of the notions of the Wernerian school; and there are papers on some of the districts in England most rich in fossils, which Mr. Conybeare says, well exhibit the low state of secondary geology at that period. But a paper by Mr. Parkinson refers to the discoveries both of Smith and of Cuvier; and in the next volume, Mr. Webster gives an account of the Isle of Wight, following the admirable model of Cuvier and Brongniart's account of the Paris basin. "If we compare this memoir of Mr. Webster with the preceding one of Dr. Berger (also of the Isle of Wight), they at once show themselves to belong to two very distinct eras of science; and it is difficult to believe that the interval which elapsed between their respective publication was only three or four years."[48\18]

[Note 47\18: Conybeare, _Report. Brit. Assoc._ p. 372.]

[Note 48\18: Conybeare, _Report_, p. 372.]

Among the events belonging to the diffusion of sound geological views in this country, we may notice the publication of a little volume entitled, _The Geology of England and Wales_, by Mr. Conybeare and Mr. Phillips, in 1821; an event far more important than, from the modest form and character of the work, it might at first sight appear. By describing in detail the geological structure and circumstances of England (at least as far downwards as the coal), it enabled a very wide class of readers to understand and verify the classifications which geology had then very recently established; while the extensive knowledge and philosophical spirit of Mr.

Conybeare rendered it, under the guise of a topographical enumeration, in reality a profound and instructive scientific treatise. The vast impulse which it gave to the study of sound descriptive geology was felt and acknowledged in other countries, as well as in Britain. {526}

Since that period, Descriptive Geology in England has constantly advanced. The advance has been due mainly to the labors of the members of the Geological Society; on whose merits as cultivators of their science, none but those who are themselves masters of the subject, have a right to dwell. Yet some parts of the scientific character of these men may be appreciated by the general speculator; for they have shown that there are no talents and no endowments which may not find their fitting employment in this science. Besides that they have united laborious research and comprehensive views, acuteness and learning, zeal and knowledge; the philosophical eloquence with which they have conducted their discussions has had a most beneficial influence on the tone of their speculations; and their researches in the field, which have carried them into every country and every class of society, have given them that prompt and liberal spirit, and that open and cordial bearing, which results from intercourse with the world on a large and unfettered scale. It is not too much to say, that in our time, Practical Geology has been one of the best schools of philosophical and general culture of mind.

_Sect._ 2.--_Application of Systematic Geology. Geological Surveys and Maps._

SUCH surveys as that which Conybeare and Phillips's book presented with respect to England, were not only a means of disseminating the knowledge implied in the classifications of such a work, but they were also an essential part of the Application and Extension of the principles established by the founders of Systematic Geology. As soon as the truth of such a system was generally acknowledged, the persuasion of the propriety of geological surveys and maps of each country could not but impress itself on men's minds.

When the earlier writers, as Lister and Fontenelle, spoke of mineralogical and fossilological maps, they could hardly be said to know the meaning of the terms which they thus used. But when subsequent classifications had shown how such a suggestion might be carried into effect, and to what important consequences it might lead, the task was undertaken in various countries in a vigorous and consistent manner. In England, besides Smith's map, another, drawn up by Mr. Greenough, was published by the Geological Society in 1819; and, being founded on very numerous observations of the author and his friends, made with great labor and cost, was not only an important {527} correction and confirmation of Smith's labors, but a valuable storehouse and standard of what had then been done in English geology. Leopold von Buch had constructed a geological map of a large portion of Germany, about the same period; but, aware of the difficulty of the task he had thus attempted, he still forbore to publish it. At a later period, and as materials accumulated, more detailed maps of parts of Germany were produced by Hoffmann and others. The French government entrusted to a distinguished Professor of the School of Mines (M. Brochant de Villiers), the task of constructing a map of France on the model of Mr. Greenough's; associating with him two younger persons, selected for their energy and talents, MM.

Beaumont and Dufrénoy. We shall have occasion hereafter to speak of the execution of this survey. By various persons, geological maps of almost every country and province of Europe, and of many parts of Asia and America, have been published. I need not enumerate these, but I may refer to the account given of them by Mr. Conybeare, in the _Reports of the British Association for_ 1832, p. 384. These various essays may be considered as contributions, though hitherto undoubtedly very imperfect ones, to that at which Descriptive Geology ought to aim, and which is requisite as a foundation for sound theory;--a complete geological survey of the whole earth. But we must say a few words respecting the language in which such a survey must be written.

As we have already said, that condition which made such maps and the accompanying descriptions possible, was that the strata and their contents had previously undergone classification and arrangement at the hands of the fathers of geology. Classification, in this as in other cases, implied names which should give to the classes distinctness and permanence; and when the series of strata belonging to one country were referred to in the description of another, in which they appeared, as was usually the case, under an aspect at least somewhat different, the supposed identification required a peculiar study of each case; and thus Geology had arrived at the point, which we have before had to notice as one of the stages of the progress of Classificatory Botany, at which a technical _nomenclature_ and a well-understood _synonymy_ were essential parts of the science.

_Sect._ 3.--_Geological Nomenclature._

BY Nomenclature we mean a _system_ of names; and hence we can {528} not speak of a Geological Nomenclature till we come to Werner and Smith. The earlier mineralogists had employed names, often artificial and arbitrary, for special minerals, but no technical and constant names for strata. The elements of Werner's names for the members of his geological series were words in use among miners, as _Gneiss_, _Grauwacke_, _Thonschiefer_, _Rothe todte liegende_, _Zechstein_; or arbitrary names of the mineralogists, as Syenite, Serpentine, Porphyry, Granite. But the more technical part of his phraseology was taken from that which is the worst kind of name, arbitrary numeration. Thus he had his _first_ sandstone formation, _second_ sandstone, _third_ sandstone; _first_ flötz limestone, _second_ flötz limestone, _third_ flötz limestone. Such names are, beyond all others, liable to mistake in their application, and likely to be expelled by the progress of knowledge; and accordingly, though the Wernerian names for rocks mineralogically distinguished, have still some currency, his sandstones and limestones, after creating endless confusion while his authority had any sway, have utterly disappeared from good geological works.

The nomenclature of Smith was founded upon English provincial terms of very barbarous aspect, as _Cornbrash_, _Lias_, _Gault_, _Clunch Clay_, _Coral Rag_. Yet these terms were widely diffused when his classification was generally accepted; they kept their place, precisely because they had no systematic signification; and many of them are at present part of the geological language of the whole civilized world.

Another kind of names which has been very prevalent among geologists are those borrowed from places. Thus the Wernerians spoke of Alpine Limestone and Jura Limestone; the English, of Kimmeridge Clay and Oxford Clay, Purbeck Marble, and Portland Rock. These names, referring to the stratum of a known locality as a type, were good, as far as an identity with that type had been traced; but when this had been incompletely done, they were liable to great ambiguity. If the Alps or the Jura contain several formations of limestone, such terms as we have noticed, borrowed from those mountains, cease to be necessarily definite, and may give rise to much confusion.

Descriptive names, although they might be supposed to be the best, have, in fact, rarely been fortunate. The reason of this is obvious;--the mark which has been selected for description may easily fail to be essential; and the obvious connexions of natural facts may overleap the arbitrary definition. As we have already stated in the history of botany, the establishment of descriptive marks of real classes presupposes the important but difficult step, of the discovery of such marks. {529} Hence those descriptive names only have been really useful in geology which had been used without any scrupulous regard to the appropriateness of the description. The _Green Sand_ may be white, brown, or red; the _Mountain Limestone_ may occur only in valleys; the _Oolite_ may have no roe-like structure; and yet these may be excellent geological names, if they be applied to formations geologically identical with those which the phrases originally designated. The signification may assist the memory, but must not be allowed to subjugate the faculty of natural classification.

The terms which have been formed by geologists in recent times have been drawn from sources similar to those of the older ones, and will have their fortune determined by the same conditions. Thus Mr. Lyell has given to the divisions of the tertiary strata the appellations _Pleiocene_, _Meiocene_, _Eocene_, accordingly as they contain a _majority_ of recent species of shells, a _minority_ of such species, or a small proportion of living species, which may be looked upon as indicating the _dawn_ of the existing state of the animate creation. But in this case, he wisely treats his distinctions, not as definitions, but as the marks of natural groups. "The plurality of species indicated by the name _pleiocene_ must not," he says,[49\18] "be understood to imply an absolute majority of recent fossil shells in all cases, but a comparative preponderance wherever the pleiocene are contrasted with strata of the period immediately preceding."

[Note 49\18: _Geol._ iii. 392.]

Mr. Lyell might have added, that no precise percentage of recent species, nor any numerical criterion whatever, can be allowed to overbear the closer natural relations of strata, proved by evidence of a superior kind, if such can be found. And this would be the proper answer to the objection made by De la Beche to these names; namely, that it may happen that the _meiocene_ rocks of one country may be of the same date as the _pleiocene_ of another; the same formation having in one place a majority, in another a minority, of existing species. We are not to run into this incongruity, for we are not so to apply the names. The formation which has been called pleiocene, must continue to be so called, even where the majority of recent species fails; and all rocks

that agree with that in date, without further reference to the numerical relations of their fossils, must also share in the name.

To invent good names for these large divisions of the series of strata is indeed extremely difficult. The term _Oolite_ is an instance in which {530} a descriptive word has become permanent in a case of this kind; and, in imitation of it, _Pœcilite_ (from ποικίλος, various,**) has been proposed by Mr. Conybeare[50\18] as a name for the group of strata inferior to the oolites, of which the _Variegated_ Sandstone (Bunter Sandstein, Grès Bigarré,) is a conspicuous member. For the series of formations which lies immediately over the rocks in which no organic remains are found, the term _Transition_ was long used, but with extreme ambiguity and vagueness. When this series, or rather the upper part of it, was well examined in South Wales, where it consists of many well-marked members, and may be probably taken as a type for a large portion of the rest of the world, it became necessary to give to the group thus explored a name not necessarily leading to assumption or controversy. Mr. Murchison selected the term _Silurian_, borrowed from the former inhabitants of the country in which his types were found; and this is a term excellent in many respects; but one which will probably not quite supersede "Transition," because, in other places, transition rocks occur which correspond to none of the members of the Silurian region.

[Note 50\18: _Report_, p. 379.]

Though new names are inevitable accompaniments of new views of classification, and though, therefore, the geological discoverer must be allowed a right to coin them, this is a privilege which, for the sake of his own credit, and the circulation of his tokens, he must exercise with great temperance and judgment. M. Brongniart may be taken as an example of the neglect of this caution. Acting upon the principle, in itself a sound one, that inconveniences arise from geological terms which have a mineralogical signification, he has given an entirely new list of names of the members of the geological series. Thus the primitive unstratified rocks are _terrains agalysiens_; the transition semi-compact are _hemilysiens_; the sedimentary strata are _yzemiens_; the diluvial deposits are _clysmiens_; and these divisions are subdivided by designations equally novel; thus of the "terrains yzemiens," members are--the terrains _clastiques_, _tritoniens_, _protéïques_, _palæotheriens_, _epilymniques_, _thalassiques_.[51\18] Such a nomenclature appears to labor under great inconveniences, since the terms are descriptive in their derivation, yet are not generally intelligible, and refer to theoretical views yet have not the recommendation of systematic connexion. {531}

[Note 51\18: Brongniart, _Tableau des Terrains_, 1829.]

_Sect._ 4.--_Geological Synonymy, or Determination of Geological Equivalents._

IT will easily be supposed that with so many different sources of names as we have mentioned, the same stratum may be called by different designations; and thus a synonymy may be necessary for geology; as it was for botany in the time of Bauhin, when the same plants had been spoken of by so many different appellations in different authors. But in reality, the synonymy of geology is a still more important

part of the subject than the analogy of botany would lead us to suppose. For in plants, the species are really fixed, and easily known when seen; and the ambiguity is only in the imperfect communication or confused ideas of the observers. But in geology, the identity of a stratum or formation in different places, though not an arbitrary, may be a very doubtful matter, even to him who has seen and examined. To assign its right character and place to a stratum in a new country, is, in a great degree, to establish the whole geological history of the country. To assume that the same names may rightly be applied to the strata of different countries, is to take for granted, not indeed the Wernerian dogma of universal formations, but a considerable degree of generality and uniformity in the known formations. And how far this generality and uniformity prevail, observation alone can teach. The search for geological synonyms in different countries brings before us two questions;--first, _are_ there such synonyms? and only in the second place, and as far as they occur, _what_ are they?

In fact, it is found that although formations which must be considered as geologically identical (because otherwise no classification is possible,) do extend over large regions, and pass from country to country, their identity includes certain modifications; and the determination of the identity and of the modifications are inseparably involved with each other, and almost necessarily entangled with theoretical considerations. And in two countries, in which we find this modified coincidence, instead of saying that the strata are identical, and that their designations are synonyms, we may, with more propriety, consider them as two corresponding series; of which the members of the one may be treated as the _Representatives_ or _Equivalents_ of the members of the other.

This doctrine of Representatives or Equivalents supposes that the geological phenomena in the two countries have been the results of {532} similar series of events, which have, in some measure, coincided in time and order; and thus, as we have said, refers us to a theory. But yet, considered merely as a step in classification, the comparison of the geological series of strata in different countries is, in the highest degree, important and interesting. Indeed in the same manner in which the separation of Classificatory from Chemical Mineralogy is necessary for the completion of mineralogical science, the comparative Classification of the strata of different countries according to their resemblances and differences alone, is requisite as a basis for a Theory of their causes. But, as will easily be imagined from its nature, this part of descriptive geology deals with the most difficult and the most elevated problems; and requires a rare union of laborious observation with a comprehensive spirit of philosophical classification.

In order to give instances of this process (for of the vast labor and great talents which have been thus employed in England, France, and Germany, it is only instances that we can give,) I may refer to the geological survey of France, which was executed, as we have already stated, by order of the government. In this undertaking it was intended to obtain a knowledge of the whole mineral structure of France; but no small portion of this knowledge was brought into view, when a synonymy had been established between the Secondary Rocks of France and the corresponding members of the English and German series, which had been so well studied as to have become

classical points of standard reference. For the purpose of doing this, the principal directors of the survey, MM. Brochant de Villiers, De Beaumont, and Dufrénoy, came to England in 1822, and following the steps of the best English geologists, in a few months made themselves acquainted with the English series. They then returned to France, and, starting from the chalk of Paris in various directions, travelled on the lines which carried them over the edges of the strata which emerge from beneath the chalk, identifying, as they could, the strata with their foreign analogues. They thus recognized almost all of the principal beds of the oolitic series of England.[52\18] At the same time they found differences as well as resemblances. Thus the Portland and Kimmeridge beds of France were found to contain in abundance a certain shell, the _gryphæa virgula_, which had not before been much remarked in those beds in England. With regard to the synonyms in Germany, on the other hand, a difference of opinion {538} arose between M. Elie de Beaumont and M. Voltz,[53\18] the former considering the _Grès de Vosges_ as the equivalent of the _Rothe todte liegende_, which occurs beneath the Zechstein, while M. Voltz held that it was the lower portion of the Red or _Variegated Sandstone_ which rests on the Zechstein.

[Note 52\18: De la Beche, _Manual_, 305.]

[Note 53\18: De la Beche, _Manual_, 381.]

In the same manner, from the first promulgation of the Wernerian system, attempts were made to identify the English with the German members of the geological alphabet; but it was long before this alphabet was rightly read. Thus the English geologists who first tried to apply the Wernerian series to this country, conceived the Old and New Red Sandstone of England to be the same with the Old and New Red Sandstone of Werner; whereas Werner's Old Red, the Rothe todte liegende, is above the coal, while the English Old Red is below it. This mistake led to a further erroneous identification of our Mountain Limestone with Werner's First Flötz Limestone; and caused an almost inextricable confusion, which, even at a recent period, has perplexed the views of German geologists respecting this country. Again, the Lias of England was, at first, supposed to be the equivalent of the Muschelkalk of Germany. But the error of this identification was brought into view by examinations and discussions in which MM. Œyenhausen and Dechen took the lead; and at a later period, Professor Sedgwick, by a laborious examination of the strata of England, was enabled to show the true relation of this part of the geology of the two countries. According to him, the New Red Sandstone of England, considered as one great complex formation, may be divided into seven members, composed of sandstones, limestones, and marls; five of which represent respectively the _Rothe todte liegende_; the _Kupfer schiefer_; the _Zechstein_, (with the _Rauchwacké_, _Asche_, and _Stinkstein_ of the Thuringenwald;) the _Bunter sandstein_; and the _Keuper_: while the _Muschelkalk_, which lies between the two last members of the German list, has not yet been discovered in our geological series. "Such a coincidence," he observes,[54\18] "in the subdivisions of two distant mechanical deposits, even upon the supposition of their being strictly contemporaneous, is truly astonishing. It has not been assumed hypothetically, but is the fair result of the facts which are recorded in this paper."

[Note 54\18: _Geol. Trans._ Second Series, iii. 121.]

As an example in which the study of geological equivalents becomes still more difficult, we may notice the attempts to refer the strata of {534} the Alps to those of the north-west of Europe. The dark-colored marbles and schists resembling mica slate[55\18] were, during the prevalence of the Wernerian theory, referred, as was natural, to the transition class. The striking physical characters of this mountain region, and its long-standing celebrity as a subject of mineralogical examination, made a complete subversion of the received opinion respecting its place in the geological series, an event of great importance in the history of the science. Yet this was what occurred when Dr. Buckland, in 1820, threw his piercing glance upon this district. He immediately pointed out that these masses, by their fossils, approach to the Oolitic Series of this country. From this view it followed, that the geological equivalents of that series were to be found among rocks in which the mineralogical characters were altogether different, and that the loose limestones of England represent some of the highly-compact and crystalline marbles of Italy and Greece. This view was confirmed by subsequent investigations; and the correspondence was traced, not only in the general body of the formations, but in the occurrence of the Red Marl at its bottom, and the Green Sand and Chalk at its top.

[Note 55\18: De la Beche, _Manual_, 313.]

The talents and the knowledge which such tasks require are of no ordinary kind; nor, even with a consummate acquaintance with the well-ascertained formations, can the place of problematical strata be decided without immense labor. Thus the examination and delineation of hundreds of shells by the most skilful conchologists, has been thought necessary in order to determine whether the calcareous beds of Maestricht and of Gosau are or are not intermediate, as to their organic contents, between the chalk and the tertiary formations. And scarcely any point of geological classification can be settled without a similar union of the accomplished naturalist with the laborious geological collector.

It follows from the views already presented, of this part of geology, that no attempt to apply to distant countries the names by which the well-known European strata have been described, can be of any value, if not accompanied by a corresponding attempt to show how far the European series is really applicable. This must be borne in mind in estimating the import of the geological accounts which have been given of various parts of Asia, Africa, and America. For instance, when the carboniferous group and the new red sandstone are stated to {535} be found in India, we require to be assured that these formations are, in some way, the equivalents of their synonyms in countries better explored. Till this is done, the results of observation in such places would be better conveyed by a nomenclature implying only those facts of resemblance, difference, and order, which have been ascertained in the country so described. We know that serious errors were incurred by the attempts made to identify the Tertiary strata of other countries with those first studied in the Paris basin. Fancied points of resemblance, Mr. Lyell observes, were magnified into undue importance, and essential differences in mineral character and organic contents were slurred over.

[2nd Ed.] [The extension of geological surveys, the construction of geological maps, and the determination of the geological equivalents which replace each other in various countries, have been carried on in continuation of the labors mentioned above, with enlarged activity, range, and means. It is estimated that one-third of the land of each hemisphere has been geologically explored; and that thus Descriptive Geology has now been prosecuted so far, that it is not likely that even the extension of it to the whole globe would give any material novelty of aspect to Theoretical Geology. The recent literature of the subject is so voluminous that it is impossible for me to give any account of it here; very imperfectly acquainted, as I am, even with the English portion, and still more, with what has been produced in other countries.

While I admire the energetic and enlightened labors by which the philosophers of France, Belgium, Germany, Italy, Russia, and America, have promoted scientific geology, I may be allowed to rejoice to see in the very phraseology of the subject, the evidence that English geologists have not failed to contribute their share to the latest advances in the science. The following order of strata proceeding upwards is now, I think, recognized throughout Europe. The _Silurian_; the _Devonian_ (Old Red Sandstone;) the _Carboniferous_; the _Permian_, (Lower part of the new Red Sandstone series;) the _Trias_, (Upper three members of the New Red Sandstone series;) the _Lias_; the _Oolite_, (in which are reckoned by M. D'Orbigny the Etages _Bathonien_, _Oxonien_, _Kimmeridgien_, and _Portlandien_;) the _Neocomien_, (Lower Green Sand,) the Chalk; and above these, Tertiary and Supra-Tertiary beds. Of these, the Silurian, described by Sir R. Murchison from its types in South Wales, has been traced by European Geologists through the Ardennes, Servia, Turkey, the shores of the Gulf of Finland, the valley {536} of the Mississippi, the west coast of North America, and the mountains of South America. Again, the labors of Prof. Sedgwick and Sir R. Murchison, in 1836, '7, and '8, aided by the sagacity of Mr. Lonsdale, led to their placing certain rocks of Devon and Cornwall as a formation intermediate between the Silurian and Carboniferous Series; and the Devonian System thus established has been accepted by geologists in general, and has been traced, not only in various parts of Europe, but in Australia and Tasmania, and in the neighborhood of the Alleganies.

Above the Carboniferous Series, Sir R. Murchison and his fellow laborers, M. de Verneuil and Count Keyserling, have found in Russia a well-developed series of rocks occupying the ancient kingdom of Permia, which they have hence called the _Permian formation_; and this term also has found general acceptance. The next group, the Keuper, Muschelkalk, and Bunter Sandstein of Germany, has been termed _Trias_ by the continental geologists. The _Neocomien_ is called from Neuchatel, where it is largely developed. Below all these rocks come, in England, the _Cambrian_ on which Prof. Sedgwick has expended so many years of valuable labor. The comparison of the Protozoic and Hypozoic rocks of different countries is probably still incomplete.

The geologists of North America have made great progress in decyphering and describing the structure of their own country; and they have wisely gone, in a great measure, upon the plan which I have commended at the end of the third Chapter;--they have compared the rocks of their own country with each other, and

given to the different beds and formations names borrowed from their own localities. This course will facilitate rather than impede the redaction of their classification to its synonyms and equivalents in the old world.

Of course it is not to be expected nor desired that books belonging to Descriptive Geology shall exclude the other two branches of the subject, Geological Dynamics and Physical Geology. On the contrary, among the most valuable contributions to both these departments have been speculations appended to descriptive works. And this is naturally and rightly more and more the case as the description embraces a wider field. The noble work _On the Geology of Russia and the Urals_, by Sir Roderick Murchison and his companions, is a great example of this, as of other merits in a geological book. The author introduces into his pages the various portions of geological dynamics of which I shall have to speak afterwards; and thus endeavors to make out the {537} physical history of the region, the boundaries of its raised sea bottoms, the shores of the great continent on which the mammoths lived, the period when the gold ore was formed, and when the watershed of the Ural chain was elevated.]

## CHAPTER IV.

ATTEMPTS TO DISCOVER GENERAL LAWS IN GEOLOGY.

_Sect._ 1.--_General Geological Phenomena._

BESIDES thus noticing such features in the rocks of each country as were necessary to the identification of the strata, geologists have had many other phenomena of the earth's surface and materials presented to their notice; and these they have, to a certain extent, attempted to generalize, so as to obtain on this subject what we have elsewhere termed the Laws of Phenomena, which are the best materials for physical theory. Without dwelling long upon these, we may briefly note some of the most obvious. Thus it has been observed that mountain ranges often consist of a ridge of subjacent rock, on which lie, on each side, strata sloping from the ridge. Such a ridge is an _Anticlinal Line_, a _Mineralogical Axis_. The sloping strata present their _Escarpements_, or steep edges, to this axis. Again, in mining countries, the _Veins_ which contain the ore are usually a system of _parallel_ and nearly vertical partitions in the rock; and these are, in very many cases, intersected by another system of veins parallel to each other and nearly _perpendicular_ to the former. Rocky regions are often intersected by _Faults_, or fissures interrupting the strata, in which the rock on one side the fissure appears to have been at first continuous with that on the other, and shoved aside or up or down after the fracture. Again, besides these larger fractures, rocks have _Joints_,--separations, or tendencies to separate in some directions rather than in others; and a _slaty Cleavage_, in which the parallel subdivisions may be carried on, so as to produce laminæ of indefinite thinness. As an example of those laws of phenomena of which we have spoken, we may instance the general law asserted by Prof. {538} Sedgwick (not, however, as free from exception), that in one particular class of rocks the slaty Cleavage _never_ coincides with the Direction of the strata.

The phenomena of metalliferous veins may be referred to, as another large class of facts which demand the notice of the geologist. It would be difficult to point out briefly any general laws which prevail in such cases; but in order to show the curious and complex nature of the facts, it may be sufficient to refer to the description of the metallic veins of Cornwall by Mr. Carne;[56\18] in which the author maintains that their various contents, and the manner in which they cut across, and _stop_, or _shift_, each other, leads naturally to the assumption of veins of no less than six or eight different ages in one kind of rock.

[Note 56\18: _Transactions of the Geol. Soc. of Cornwall_, vol. ii.]

Again, as important characters belonging to the physical history of the earth, and therefore to geology, we may notice all the general laws which refer to its temperature;--both the laws of climate, as determined by the _isothermal lines_, which Humboldt has drawn, by the aid of very numerous observations made in all parts of the world; and also those still more curious facts, of the increase of temperature which takes place as we descend in the solid mass. The latter circumstance, after being for a while rejected as a fable, or explained away as an accident, is now generally acknowledged to be the true state of things in many distant parts of the globe, and probably in all.

Again, to turn to cases of another kind: some writers have endeavored to state in a general manner laws according to which the members of the geological series succeed each other; and to reduce apparent anomalies to order of a wider kind. Among those who have written with such views, we may notice Alexander von Humboldt, always, and in all sciences, foremost in the race of generalization. In his attempt to extend the doctrine of geological equivalents from the rocks of Europe[57\18] to those of the Andes, he has marked by appropriate terms the general modes of geological succession. "I have insisted," he says[58\18] "principally upon the phenomena of _alternation_, _oscillation_, and _local suppression_, and on those presented by the _passages_ of formations from one to another, by the effect of an _interior developement_."

[Note 57\18: _Gissement des Roches dans les deux Hemisphères_, 1823.]

[Note 58\18: Pref. p. vi.]

The phenomena of alternation to which M. de Humboldt here refers are, in fact, very curious: as exhibiting a mode in which the transitions from one formation to another may become gradual and insensible, {539} instead of sudden and abrupt. Thus the coal measures in the south of England are above the mountain limestone; and the distinction of the formations is of the most marked kind. But as we advance northward into the coal-field of Yorkshire and Durham, the subjacent limestone begins to be subdivided by thick masses of sandstone and carbonaceous strata, and passes into a complex deposit, not distinguishable from the overlying coal measures; and in this manner the transition from the limestone to the coal is made by alternation. Thus, to use another expression of M. de Humboldt's in ascending from the limestone, the coal, before we quit the subjacent stratum, _preludes_ to its fuller exhibition in the superior beds.

Again, as to another point: geologists have gone on up to the present time endeavoring to discover general laws and facts, with regard to the position of mountain and mineral masses upon the surface of the earth. Thus M. Von Buch, in his physical description of the Canaries, has given a masterly description of the lines of volcanic action and volcanic products, all over the globe. And, more recently, M. Elie de Beaumont has offered some generalizations of a still wider kind. In this new doctrine, those mountain ranges, even in distant parts of the world, which are of the same age, according to the classifications already spoken of, are asserted to be parallel[59\18] to each other, while those ranges which are of different ages lie in different directions. This very wide and striking proposition may be considered as being at present upon its trial among the geologists of Europe.[60\18]

[Note 59\18: We may observe that the notion of parallelism, when applied to lines drawn on _remote_ portions of a globular surface, requires to be interpreted in so arbitrary a manner, that we can hardly imagine it to express a physical law.]

[Note 60\18: Mr. Lyell, in the sixth edition of his _Principles_, B. i. c. xii., has combated the hypothesis of M. Elie de Beaumont, stated in the text. He has argued both against the catastrophic character of the elevation of mountain chains, and the parallelism of the contemporaneous ridges. It is evident that the former doctrine may be true, though the latter be shown to be false.]

Among the organic phenomena, also, which have been the subject of geological study, general laws of a very wide and comprehensive kind have been suggested, and in a greater or less degree confirmed by adequate assemblages of facts. Thus M. Adolphe Brongniart has not only, in his _Fossil Flora_, represented and skilfully restored a vast number of the plants of the ancient world; but he has also, in the _Prodromus_ of the work, presented various important and striking views of the general character of the vegetation of former periods, as {540} insular or continental, tropical or temperate. And M. Agassiz, by the examination of an incredible number of specimens and collections of fossil fish, has been led to results which, expressed in terms of his own ichthyological classification, form remarkable general laws. Thus, according to him,[61\18] when we go below the lias, we lose all traces of two of the four orders under which he comprehends all known kinds of fish; namely, the _Cycloïdean_ and the _Ctenoïdean_; while the other two orders, the _Ganoïdean_ and _Placoïdean_, rare in our days, suddenly appear in great numbers, together with large sauroid and carnivorous fishes. Cuvier, in constructing his great work on ichthyology, transferred to M. Agassiz the whole subject of fossil fishes, thus showing how highly he esteemed his talents as a naturalist. And M. Agassiz has shown himself worthy of his great predecessor in geological natural history, not only by his acuteness and activity, but by the comprehensive character of his zoological philosophy, and by the courage with which he has addressed himself to the vast labors which lie before him. In his _Report on the Fossil Fish discovered in England_, published in 1835, he briefly sketches some of the large questions which his researches have suggested; and then adds,[62\18] "Such is the meagre outline of a history of the highest interest, full of curious episodes, but most difficult to relate. To unfold the details which it contains will be the business of my life."

[Note 61\18: Greenough, _Address to Geol. Soc._ 1835, p. 19.]

[Note 62\18: _Brit. Assoc. Report_, p. 72.]

[2nd Ed.] [In proceeding downwards through the series of formations into which geologists have distributed the rocks of the earth, one class of organic forms after another is found to disappear. In the Tertiary Period we find all the classes of the present world: Mammals, Birds, Reptiles, Fishes, Crustaceans, Mollusks, Zoophytes. In the Secondary Period, from the Chalk down to the New Red Sandstone, Mammals are not found, with the minute exception of the marsupial _amphitherium_ and _phascolotherium_ in the Stonesfield slate. In the Carboniferous and Devonian period we have no large Reptiles, with, again, a minute amount of exception. In the lower part of the Silurian rocks, Fishes vanish, and we have no animal forms but Mollusks, Crustaceans and Zoophytes.

The Carboniferous, Devonian and Silurian formations, thus containing the oldest forms of life, have been termed _palæozoic_. The boundaries of the life-bearing series have not yet been determined; but the series in which vertebrated animals do not appear has been {541} provisionally termed _protozoic_, and the lower Silurian rocks may probably be looked upon as its upper members. Below this, geologists place a _hypozoic_ or _azoic_ series of rocks.

Geologists differ as to the question whether these changes in the inhabitants of the globe were made by determinate steps or by insensible gradations. M. Agassiz has been led to the conviction that the organized population of the globe was renewed in the interval of each principal member of its formations.[63\18] Mr. Lyell, on the other hand, conceives that the change in the collection of organized beings was gradual, and has proposed on this subject an hypothesis which I shall hereafter consider.]

[Note 63\18: _Brit. Assoc. Report_ 1842, p. 83.]

_Sect._ 2.--_Transition to Geological Dynamics._

WHILE we have been giving this account of the objects with which Descriptive Geology is occupied, it must have been felt how difficult it is, in contemplating such facts, to confine ourselves to description and classification. Conjectures and reasonings respecting the causes of the phenomena force themselves upon us at every step; and even influence our classification and nomenclature. Our Descriptive Geology impels us to endeavor to construct a Physical Geology. This close connexion of the two branches of the subject by no means invalidates the necessity of distinguishing them: as in Botany, although the formation of a Natural System necessarily brings us to physiological relations, we still distinguish Systematic from Physiological Botany.

Supposing, however, our Descriptive Geology to be completed, as far as can be done without considering closely the causes by which the strata have been produced, we have now to enter upon the other province of the science, which treats of those causes, and of which we have already spoken, as _Physical Geology_. But before we can treat this department of speculation in a manner suitable to the conditions of science, and to the analogy of other parts of our knowledge, a certain intermediate

and preparatory science must be formed, of which we shall now consider the origin and progress.

{{542}} GEOLOGICAL DYNAMICS.

## CHAPTER V.

INORGANIC GEOLOGICAL DYNAMICS.

_Sect._ 1.--_Necessity and Object of a Science of Geological Dynamics._

WHEN the structure and arrangement which men observed in the materials of the earth instigated them to speculate concerning the past changes and revolutions by which such results had been produced, they at first supposed themselves sufficiently able to judge what would be the effects of any of the obvious agents of change, as water or volcanic fire. It did not at once occur to them to suspect, that their common and extemporaneous judgment on such points was far from sufficient for sound knowledge;--they did not foresee that they must create a special science, whose object should be to estimate the general laws and effects of assumed causes, before they could pronounce whether such causes had actually produced the particular facts which their survey of the earth had disclosed to them.

Yet the analogy of the progress of knowledge on other subjects points out very clearly the necessity of such a science. When phenomenal astronomy had arrived at a high point of completeness, by the labors of ages, and especially by the discovery of Kepler's laws, astronomers were vehemently desirous of knowing the causes of these motions; and sanguine men, such as Kepler, readily conjectured that the motions were the effects of certain virtues and influences, by which the heavenly bodies acted upon each other. But it did not at first occur to him and his fellow-speculators, that they had not ascertained what motions the influences of one body upon another could produce: and that, therefore, they were not prepared to judge whether such causes as they spoke of, did really regulate the motions of the planets. Yet such was found to be the necessary course of sound inference. Men needed a science of motion, in order to arrive at a science of the {543} heavenly motions: they could not advance in the study of the Mechanics of the heavens, till they had learned the Mechanics of terrestrial bodies. And thus they were, in such speculations, at a stand for nearly a century, from the time of Kepler to the time of Newton, while the science of Mechanics was formed by Galileo and his successors. Till that task was executed, all the attempts to assign the causes of cosmical phenomena were fanciful guesses and vague assertions; after that was done, they became demonstrations. The science of _Dynamics_ enabled philosophers to pass securely and completely from _Phenomenal Astronomy_ to _Physical Astronomy_.

In like manner, in order that we may advance from Phenomenal Geology to Physical Geology, we need a science of _Geological Dynamics_;--that is, a science which shall investigate and determine the laws and consequences of the known causes of changes such as those which Geology considers:--and which shall do this, not in an occasional, imperfect, and unconnected manner, but by systematic,

complete, and conclusive methods;--shall, in short, be a Science, and not a promiscuous assemblage of desultory essays.

The necessity of such a study, as a distinct branch of geology, is perhaps hardly yet formally recognized, although the researches which belong to it have, of late years, assumed a much more methodical and scientific character than they before possessed. Mr. Lyell's work (_Principles of Geology_), in particular, has eminently contributed to place Geological Dynamics in its proper prominent position. Of the four books of his Treatise, the second and third are upon this division of the subject; the second book treating of aqueous and igneous causes of change, and the third, of changes in the organic world.

There is no difficulty in separating this auxiliary geological science from theoretical Geology itself, in which we apply our principles to the explanation of the actual facts of the earth's surface. The former, if perfected, would be a demonstrative science dealing with general cases; the latter is an ætiological view having reference to special facts; the one attempts to determine what always must be under given conditions; the other is satisfied with knowing what is and has been, and why it has been; the first study has a strong resemblance to Mechanics, the other to philosophical Archæology.

Since this portion of science is still so new, it is scarcely possible to give any historical account of its progress, or any complete survey of its shape and component parts. I can only attempt a few notices, {544} which may enable us in some measure to judge to what point this division of our subject is tending.

We may remark, in this as in former cases, that since we have here to consider the formation and progress of a _science_, we must treat as unimportant preludes to its history, the detached and casual observations of the effects of causes of change which we find in older writers. It is only when we come to systematic collections of information, such as may afford the means of drawing general conclusions; or to rigorous deductions from known laws of nature;--that we can recognise the separate existence of geological dynamics, as a path of scientific research.

The following may perhaps suffice, for the present, as a sketch of the subjects of which this science treats:--the aqueous causes of change, or those in which water adds to, takes from, or transfers, the materials of the land:--the igneous causes; volcanoes, and, closely connected with them, earthquakes, and the forces by which they are produced;--the calculations which determine, on physical principles, the effects of assumed mechanical causes acting upon large portions of the crust of the earth;--the effect of the forces, whatever they be, which produce the crystalline texture of rocks, their fissile structure, and the separation of materials, of which we see the results in metalliferous veins. Again, the estimation of the results of changes of temperature in the earth, whether operating by pressure, expansion, or in any other way;--the effects of assumed changes in the superficial condition, extent, and elevation, of terrestrial continents upon the climates of the earth;--the effect of assumed cosmical changes upon the temperature of this planet;--and researches of the same nature as these.

These researches are concerned with the causes of change in the inorganic world; but the subject requires no less that we should investigate the causes which may modify the forms and conditions of organic things; and in the large sense in which we have to use the phrase, we may include researches on such subjects also as parts of Geological Dynamics; although, in truth, this department of physiology has been cultivated, as it well deserves to be, independently of its bearing upon geological theories. The great problem which offers itself here, in reference to Geology, is, to examine the value of any hypotheses by which it may be attempted to explain the succession of different races of animals and plants in different strata; and though it may be difficult, in this inquiry, to arrive at any positive result, we {545} may at least be able to show the improbability of some conjectures which have been propounded.

I shall now give a very brief account of some of the attempts made in these various departments of this province of our knowledge; and in the present chapter, of Inorganic Changes.

_Sect._ 2.--_Aqueous Causes of Change._

THE controversies to which the various theories of geologists gave rise, proceeding in various ways upon the effects of the existing causes of change, led men to observe, with some attention and perseverance, the actual operation of such causes. In this way, the known effect of the Rhine, in filling up the Lake of Geneva at its upper extremity, was referred to by De Luc, Kirwan, and others, in their dispute with the Huttonians; and attempts were even made to calculate how distant the period was, when this alluvial deposit first began. Other modern observers have attended to similar facts in the natural history of rivers and seas. But the subject may be considered as having first assumed its proper form, when taken up by Mr. Von Hoff; of whose _History of the Natural Changes of the Earth's surface which are proved by Tradition_, the first part, treating of aqueous changes, appeared in 1822. This work was occasioned by a Prize Question of the Royal Society of Göttingen, promulgated in 1818; in which these changes were proposed as the subject of inquiry, with a special reference to geology. Although Von Hoff does not attempt to establish any general inductions upon the facts which his book contains, the collection of such a body of facts gave almost a new aspect to the subject, by showing that changes in the relative extent of land and water were going on at every time, and almost at every place; and that mutability and fluctuation in the form of the solid parts of the earth, which had been supposed by most persons to be a rare exception to the common course of events, was, in fact, the universal rule. But it was Mr. Lyell's _Principles of Geology, being an attempt to explain the former Changes of the Earth's Surface by the Causes now in action_ (of which the first volume was published in 1830), which disclosed the full effect of such researches on geology; and which attempted to present such assemblages of special facts, as examples of general laws. Thus this work may, as we have said, be looked upon as the beginning of Geological Dynamics, at least among us. Such generalizations and applications as it contains give the most lively {546} interest to a thousand observations respecting rivers and floods, mountains and morasses, which otherwise appear without aim or meaning; and thus this department of science cannot fail to be constantly augmented by

contributions from every side. At the same time it is clear, that these contributions, voluminous as they must become, must, from time to time, be resolved into laws of greater and greater generality; and that thus alone the progress of this, as of all other sciences, can be furthered.

I need not attempt any detailed enumeration of the modes of aqueous action which are here to be considered. Some are destructive, as when the rivers erode the channels in which they flow; or when the waves, by their perpetual assault, shatter the shores, and carry the ruins of them into the abyss of the ocean. Some operations of the water, on the other hand, add to the land; as when _deltas_ are formed at the mouths of rivers or when calcareous springs form deposits of _travertin_. Even when bound in icy fetters, water is by no mean deprived of its active power; the _glacier_ carries into the valley masses of its native mountain, and often, becoming ice-bergs, float with a lading of such materials far into the seas of the temperate zone. It is indisputable that vast beds of worn down fragments of the existing land are now forming into strata at the bottom of the ocean; and that many other effects are constantly produced by existing aqueous causes, which resemble some, at least, of the facts which geology has to explain.

[2nd Ed.] [The effects of glaciers above mentioned are obvious; but the mechanism of these bodies,--the mechanical cause of their motions,--was an unsolved problem till within a very few years. That they slide as rigid masses;--that they advance by the expansion of their mass;--that they advance as a collection of rigid fragments; were doctrines which were held by eminent physicists; though a very slight attention to the subject shows these opinions to be untenable. In Professor James Forbes's theory on the subject (published in his _Travels through the Alps_, 1843,) we find a solution of the problem, so simple, and yet so exact, as to produce the most entire conviction. In this theory, the ice of a glacier is, on a great scale, supposed to be a plastic or viscous mass, though small portions of it are sensibly rigid. It advances down the slope of the valley in which it lies as a plastic mass would do, accommodating itself to the varying shape and size of its bed, and showing by its crevasses its mixed character between fluid and rigid. It shows this character still more curiously by a _ribboned_ {547} _structure_ on a small scale, which is common in the solid ice of the glacier. The planes of these _ribbons_ are, for the most part, at right angles to the crevasses, near the sides of the glacier, while, near its central line, they _dip_ towards the upper part of the glacier. This structure appears to arise from the difference of velocities of contiguous moving filaments of the icy mass, as the crevasses themselves arise from the tension of larger portions. Mr. Forbes has, in successive publications, removed the objections which have been urged against this theory. In the last of them, a Memoir in the _Phil. Trans._, 1846, (_Illustrations of the Viscous Theory of Glacier Motion_,) he very naturally expresses astonishment at the opposition which has been made to the theory on the ground of the rigidity of small pieces of ice. He has himself shown that the ice of glaciers has a plastic flexibility, by marking forty-five points in a transverse straight line upon the Mer de Glace, and observing them for several days. The straight line in that time not only became oblique to the side, but also became visibly curved.

Both Mr. Forbes and other philosophers have made it in the highest degree probable that glaciers have existed in many places in which they now exist no longer, and have exercised great powers in transporting large blocks of rock, furrowing and polishing the rocks along which they slide, and leaving lines and masses of detritus or _moraine_ which they had carried along with them or pushed before them. It cannot be doubted that extinct glaciers have produced some of the effects which the geologist has to endeavor to explain. But this part of the machinery of nature has been worked by some theorists into an exaggerated form, in which it cannot, as I conceive, have any place in an account of Geological Dynamics which aims at being permanent.

The great problem of the diffusion of drift and erratic blocks from their parent rocks to great distances, has driven geologists to the consideration of other hypothetical machinery by which the effects may be accounted for: especially the great _northern drift_ and _boulders_,--the rocks from the Scandinavian chain which cover the north of Europe on a vast area, having a length of 2000 and breadth of from 400 to 800 miles. The diffusion of these blocks has been accounted for by supposing them to be imbedded in icebergs, detached from the shore, and floated into oceanic spaces, where they have grounded and been deposited by the melting of the ice. And this mode of action may to some extent be safely admitted into geological speculation. For it is a matter of fact, that our navigators in arctic and antarctic regions have {548} repeatedly seen icebergs and icefloes sailing along laden with such materials.

The above explanation of the phenomena of drift supposes the land on which the travelled materials are found to have been the bottom of a sea where they were deposited. But it does not, even granting the conditions, account for some of the facts observed;--that the drift and the boulders are deposited in "trainées" or streaks, which, in direction, diverge from the parent rock;--and that the boulders are of smaller and smaller size, as they are found more remote from that centre. These phenomena rather suggest the notion of currents of water as the cause of the distribution of the materials into their present situations. And though the supposition that the whole area occupied by drift and boulders was a sea-bottom when they were scattered over it much reduces the amount of violence which it is necessary to assume in order to distribute the loose masses, yet still the work appears to be beyond the possible effect of ordinary marine currents, or any movements which would be occasioned by a slow and gradual rising of the centre of distribution.

It has been suggested that a _sudden_ rise of the centre of distribution would cause a motion in the surrounding ocean sufficient to produce such an effect: and in confirmation of this reference has been made to Mr. Scott Russell's investigations with respect to waves, already referred to. (Book VIII.) The wave in this case would be the _wave of translation_, in which the motion of the water is as great at the bottom as at the top; and it has hence been asserted that by paroxysmal elevations of 100 or 200 feet, a current of 25 or 30 miles an hour might be accounted for. But I think it has not been sufficiently noted that at each point this "current" is transient: it lasts only while the wave is passing over the point, and therefore it would only either

carry a single mass the whole way with its own velocity, or move through a short distance a series of masses over which it successively passed. It does not appear, therefore, that we have here a complete account of the transport of a collection of materials, in which each part is transferred through great distances:--except, indeed, we were to suppose a numerous succession of paroxysmal elevations. Such a _battery_ might, by successive shocks, transmitting their force through the water, diffuse the fragments of the central mass over any area, however wide.

The fact that the erratic blocks are found to rest on the lower drift, is well explained by supposing the latter to have been spread on the {549} sea bottom while rock-bearing ice-masses floated on the surface till they deposited their lading.

Sir R. Murchison has pointed out another operation of ice in producing mounds of rocky masses; namely, the effects of rivers and lakes, in climates where, as in Russia, the waters carry rocky fragments entangled in the winter ice, and leave them in heaps at the highest level which the waters attain.

The extent to which the effects of glaciers, now vanished, are apparent in many places, especially in Switzerland and in England, and other phenomena of the like tendency, have led some of the most eminent geologists to the conviction that, interior to the period of our present temperature, there was a _Glacial Period_, at which the temperature of Europe was lower than it now is.]

Although the study of the common operations of water may give the geologist such an acquaintance with the laws of his subject as may much aid his judgment respecting the extent to which such effects may proceed, a long course of observation and thought must be requisite before such operations can be analysed into their fundamental principles, and become the subjects of calculation, or of rigorous reasoning in any manner which is as precise and certain as calculation. Various portions of Hydraulics have an important bearing upon these subjects, including some researches which have been pursued with no small labor by engineers and mathematicians; as the effects of currents and waves, the laws of tides and of rivers, and many similar problems. In truth, however, such subjects have not hitherto been treated by mathematicians with much success; and probably several generations must elapse before this portion of geological dynamics can become an exact science.

_Sect._ 3.--_Igneous Causes of Change.--Motions of the Earth's Surface._

THE effects of volcanoes have long been noted as important and striking features in the physical history of our globe; and the probability of their connexion with many geological phenomena, had not escaped notice at an early period. But it was not till more recent times, that the full import of these phenomena was apprehended. The person who first looked at such operations with that commanding general view which showed their extensive connexion with physical geology, was Alexander von Humboldt, who explored the volcanic phenomena {550} of the New World, from 1799 to 1804. He remarked[64\18] the linear distribution of volcanic domes, considering them as vents placed along the edge of vast fissures communicating with reservoirs of igneous matter, and extending across whole continents. He observed, also, the frequent sympathy of volcanic and terremotive action in remote districts of

the earth's surface, thus showing how deeply seated must be the cause of these convulsions. These views strongly excited and influenced the speculations of geologists; and since then, phenomena of this kind have been collected into a general view as parts of a natural-historical science. Von Hoff, in the second volume of the work already mentioned, was one of the first who did this; "At least," he himself says,[65\18] (1824,) "it was not known to him that any one before him had endeavored to combine so large a mass of facts with the general ideas of the natural philosopher, so as to form a whole." Other attempts were, however, soon made. In 1825, M. von Ungern-Sternberg published his book _On the Nature and Origin of Volcanoes_,[66\18] in which, he says, his object is, to give an empirical representation of these phenomena. In the same year, Mr. Poulett Scrope published a work in which he described the known facts of volcanic action; not, however, confining himself to description; his purpose being, as his title states, to consider "the probable causes of their phenomena, the laws which determine their march, the disposition of their products, and their connexion with the present state and past history of the globe; leading to the establishment of a new theory of the earth." And in 1826, Dr. Daubeny, of Oxford, produced _A Description of Active_ and Extinct _Volcanoes_, including in the latter phrase the volcanic rocks of central France, of the Rhine, of northern and central Italy, and many other countries. Indeed, the near connexion between the volcanic effects now going on, and those by which the basaltic rocks of Auvergne and many other places had been produced, was, by this time, no longer doubted by any; and therefore the line which here separates the study of existing causes from that of past effects may seem to melt away. But yet it is manifest that the assumption of an identity of scale and mechanism between volcanoes now active, and the igneous catastrophes of which the products have {551} survived great revolutions on the earth's surface, is hypothetical; and all which depends on this assumption belongs to theoretical geology.

[Note 64\18: Humboldt, _Relation Historique_; and his other works.]

[Note 65\18: Vol. ii. Prop. 5.]

[Note 66\18: _Werden und Seyn des Vulkanischen Gebirges_. Carlsruhe, 1825.]

Confining ourselves, then, to volcanic effects, which have been produced, certainly or probably, since the earth's surface assumed its present form, we have still an ample exhibition of powerful causes of change, in the streams of lava and other materials emitted in eruptions; and still more in the earthquakes which, as men easily satisfied themselves, are produced by the same causes as the eruptions of volcanic fire.

Mr. Lyell's work was important in this as in other portions of this subject. He extended the conceptions previously entertained of the effects which such causes may produce, not only by showing how great these operations are historically known to have been, and how constantly they are going on, if we take into our survey the whole surface of the earth; but still more, by urging the consequences which would follow in a long course of time from the constant repetition of operations in themselves of no extraordinary amount. A lava-stream many miles long and wide, and

several yards deep, a subsidence or elevation of a portion of the earth's surface of a few feet, are by no means extraordinary facts. Let these operations, said Mr. Lyell, be repeated thousands of times; and we have results of the same order with the changes which geology discloses.

The most mitigated earthquakes have, however, a character of violence. But it has been thought by many philosophers that there is evidence of a change of level of the land in cases where none of these violent operations are going on. The most celebrated of these cases is Sweden; the whole of the land from Gottenburg to the north of the Gulf of Bothnia has been supposed in the act of rising, slowly and insensibly, from the surrounding waters. The opinion of such a change of level has long been the belief of the inhabitants; and was maintained by Celsius in the beginning of the eighteenth century. It has since been conceived to be confirmed by various observations of marks cut on the face of the rock; beds of shells, such as now live in the neighboring seas, raised to a considerable height; and other indications. Some of these proofs appear doubtful; but Mr. Lyell, after examining the facts upon the spot in 1834, says, "In regard to the proposition that the land, in certain parts of Sweden, is gradually rising, I have no hesitation in assenting to it, after my visit to the districts above alluded to."[67\18] If this conclusion be generally accepted by {552} geologists, we have here a daily example of the operation of some powerful agent which belongs to geological dynamics; and which, for the purposes of the geological theorist, does the work of the earthquake upon a very large scale, without assuming its terrors.

[Note 67\18: _Phil. Trans._ 1835, p. 32.]

[2nd Ed.] [Examples of changes of level of large districts occurring at periods when the country has been agitated by earthquakes are well ascertained, as the rising of the coast of Chili in 1822, and the subsidence of the district of Cutch, in the delta of the Indus, in 1819. (Lyell, B. II. c. xv.) But the cases of more slow and tranquil movement seem also to be established. The gradual secular rise of the shore of the Baltic, mentioned in the text, has been confirmed by subsequent investigation. It appears that the rate of elevation increases from Stockholm, where it is only a few inches in a century, to the North Cape, where it is several feet. It appears also that several other regions are in a like state of secular change. The coast of Greenland is sinking. (Lyell, B. II. c. xviii.) And the existence of "raised beaches" along various coasts is now generally accepted among geologists. Such beaches, anciently forming the margin of the sea, but now far above it, exist in many places; for instance, along a great part of the Scotch coast; and among the raised beaches of that country we ought probably, with Mr. Darwin, to include the "parallel roads" of Glenroy, the subject, in former days, of so much controversy among geologists and antiquaries.

Connected with the secular rise and fall of large portions of the earth's surface, another agency which plays an important part in Geological dynamics has been the subject of some bold yet singularly persuasive speculations by Mr. Darwin. I speak of the formation of Coral, and Coral Reefs. He says that the coral-building animal works only at small and definite distances below the surface. How then are we to account for the vast number of coral islands, rings, and reefs, which are scattered over the

Pacific and Indian Oceans! Can we suppose that there are so many mountains, craters, and ridges, all exactly within a few feet of the same height through this vast portion of the globe's surface? This is incredible. How then are we to explain the facts? Mr. Darwin replies, that if we suppose the land to subside slowly beneath the sea, and at the same time suppose the coralline zoophytes to go on building, so that their structure constantly rises nearly to the surface of the water, we shall have the facts explained. A submerged island will produce a ring; a long coast, a barrier reef; and so on. Mr. Darwin also notes other phenomena, as {553} elevated beds of coral, which, occurring in other places, indicate a recent rising of the land; and on such grounds as these he divides the surface of those parts of the ocean into regions of elevation and of depression.

The labors of coralline zoophytes, as thus observed, form masses of coral, such as are found fossilized in the strata of the earth. But our knowledge of the laws of life which have probably affected the distribution of marine remains in strata, has received other very striking accessions by the labors of Prof. Edward Forbes in observing the marine animals of the Ægean Sea. He found that, even in their living state, the mollusks and zoophytes are already distributed into strata. Dividing the depth into eight regions, from 2 to 230 fathoms, he found that each region had its peculiar inhabitants, which disappeared speedily either in ascending or in descending. The zero of animal life appeared to occur at about 300 fathoms. This curious result bears in various ways upon geology. Mr. Forbes himself has given an example of the mode in which it may be applied, by determining the depth at which the submarine eruption took place which produced the volcanic isle of Neokaimeni in 1707. By an examination of the fossils embedded in the pumice, he showed that it came from the fourth region.[68\18]

[Note 68\18: _British Assoc. Reports_, 1843, p. 177.]

To the modes in which organized beings operate in producing the materials of the earth, we must add those pointed out by the extraordinary microscopic discoveries of Professor Ehrenberg. It appears that whole beds of earthy matter consist of the cases of certain infusoria, the remains of these creatures being accumulated in numbers which it confounds our thoughts to contemplate.]

Speculations concerning the _causes_ of volcanoes and earthquakes, and of the rising and sinking of land, are a highly important portion of this science, at least as far as the calculation of the possible results of definite causes is concerned. But the various hypotheses which have been propounded on this subject can hardly be considered as sufficiently matured for such calculation. A mass of matter in a state of igneous fusion, extending to the centre of the earth, even if we make such an hypothesis, requires some additional cause to produce eruption. The supposition that this fire may be produced by intense chemical action between combining elements, requires further, not only some agency to bring together such elements, but some reason why {554} they should be originally separate. And if any other causes have been suggested, as electricity or magnetism, this has been done so vaguely as to elude all possibility of rigorous deduction from the hypothesis. The doctrine of a Central Heat, however, has occupied so considerable a place in theoretical geology,

that it ought undoubtedly to form an article in geological dynamics.

_Sect._ 4.--_The Doctrine of Central Heat._

THE early geological theorists who, like Leibnitz and Buffon, assumed that the earth was originally a mass in a state of igneous fusion, naturally went on to deduce from this hypothesis, that the crust consolidated and cooled before the interior, and that there might still remain a central heat, capable of producing many important effects. But it is in more recent times that we have measures of such effects, and calculations which we can compare with measures. It was found, as we have said, that in descending below the surface of the earth, the temperature of its materials increased. Now it followed from Fourier's mathematical investigations of the distribution of heat in the earth, that if there be no primitive heat (_chaleur d'origine_), the temperature, when we descend below the crust, will be constant in each vertical line. Hence an observed increase of temperature in descending, appeared to point out a central heat resulting from some cause now no longer in action.

The doctrine of a central heat has usually been combined with the supposition of a central igneous fluidity; for the heat in the neighborhood of the centre must be very intense, according to any law of its increase in descending which is consistent with known principles. But to this central fluidity it has been objected that such a fluid must be in constant circulation by the cooling of its exterior. Mr. Daniell found this to be the case in all fused metals. It has also been objected that there must be, in such a central fluid, _tides_ produced by the moon and sun; but this inference would require several additional suppositions and calculations to give it a precise form.

Again, the supposition of a central heat of the earth, considered as the effect of a more ancient state of its mass, appeared to indicate that its cooling must still be going on. But if this were so, the earth might contract, as most bodies do when they cool; and this contraction might lead to mechanical results, as the shortening of the day. Laplace satisfied himself, by reference to ancient astronomical records, that no such {555} alteration in the length of the day had taken place, even to the amount of one two-hundredth of a second; and thus, there was here no confirmation of the hypothesis of a primitive heat of the earth.

Though we find no evidence of the secular contraction of the earth in the observations with which astronomy deals, there are some geological facts which at first appear to point to the reality of a refrigeration within geological periods; as the existence of the remains of plants and shells of tropical climates, in the strata of countries which are now near to or within the frigid zones. These facts, however, have given rise to theories of the changes of climate, which we must consider separately.

But we may notice, as connected with the doctrine of central heat, the manner in which this hypothesis has been applied to explain volcanic and geological phenomena. It does not enter into my plan, to consider explanations in which this central heat is supposed to give rise to an expansive force,[69\18] without any distinct reference to known physical laws. But we may notice; as more likely to

become useful materials of the science now before us, such speculations as those of Mr. Babbage; in which he combines the doctrine of central heat with other physical laws;[70\18] as, that solid rocks _expand_ by being heated, but that clay contracts; that different rocks and strata _conduct_ heat differently; that the earth _radiates_ heat differently, or at different parts of its surface, according as it is covered with forests, with mountains, with deserts, or with water. These principles, applied to large masses, such as those which constitute the crust of the earth, might give rise to changes as great as any which geology discloses. For example: when the bed of a sea is covered by a thick deposit of new matter worn from the shores, the strata below the bed, being protected by a bad conductor of heat, will be heated, and, being heated, maybe expanded; or, as Sir J. Herschel has observed, may produce explosion by the conversion of their moisture into steam. Such speculations, when founded on real data and sound calculations, may hereafter be of material use in geology.

[Note 69\18: Scrope _On Volcanoes_, p. 192.]

[Note 70\18: _On the Temple of Serapis_, 1834. See also _Journal of the Royal Inst._ vol. ii., quoted in Conyb. and Ph. p. xv. Lyell, B. ii. c. xix. p. 383, (4th ed.) on Expansion of Stone.]

The doctrine of central heat and fluidity has been rejected by some eminent philosophers. Mr. Lyell's reasons for this rejection belong {556} rather to Theoretical Geology; but I may here notice M. **Poisson's opinion. He does not assent to the conclusion of Fourier, that once the temperature increases in descending, there must be some primitive central heat. On the contrary, he considers that such an increase may arise from this;--that the earth, at some former period, passed (by the motion of the solar system in the universe,) **through a portion of space which was warmer than the space in which it now revolves (by reason, it may be, of the heat of other stars to which it was then nearer). He supposes that, since such a period, the surface has cooled down by the influence of the surrounding circumstances; while the interior, for a certain unknown depth, retains the trace of the former elevation of temperature. But this assumption is not likely to expel the belief is the terrestrial origin of the subterraneous heat. For the supposition of such an inequality in the temperature of the different regions in which the solar system is placed at different times, is altogether arbitrary; and, if pushed to the amount to which it must be carried, in order to account for the phenomenon, is highly improbable.[71\18] The doctrine of central heat, on the other hand, (which need not be conceived as implying the _universal_ fluidity of the mass,) is not only naturally suggested by the subterraneous increase of temperatures, but explains the spheroidal figure of the earth; and falls in with almost any theory which can be devised, of volcanoes, earthquakes, and great geological changes.

[Note 71\18: For this hypothesis would make it necessary to suppose that the earth has, at some former period, derived from some other star or stars more heat than she now derives from the sun. But this would imply, as highly probable, that at some period some other star or stars must have produced also a _mechanical_ effect upon the solar system, greater than the effect of the sun. Now such a past operation of forces, fitted to obliterate all order and symmetry, is quite inconsistent with the

simple, regular, and symmetrical relation which the whole solar system, as far as Uranus, bears to the present central body.]

_Sect._ 5.--_Problems respecting Elevations and Crystalline Forces._

OTHER problems respecting the forces by which great masses of the earth's crust have been displaced, have also been solved by various mathematicians. It has been maintained by Von Buch that there occur, in various places, _craters of elevation_; that is, mountain-masses resembling the craters of volcanoes, but really produced by an expansive force from below, bursting an aperture through horizontal strata, {557} and elevating them in a conical form. Against this doctrine, as exemplified in the most noted instances, strong arguments have been adduced by other geologists. Yet the protrusion of fused rock by subterraneous forces upon a large scale is not denied: and how far the examples of such operations may, in any cases, be termed craters of elevation, must be considered as a question not yet decided. On the supposition of the truth of Von Buch's doctrine, M. de Beaumont has calculated the relations of position, the fissures, &c., which would arise. And Mr. Hopkins,[72\18] of Cambridge, has investigated in a much more general manner, upon mechanical principles, the laws of the elevations, fissures, faults, veins, and other phenomena which would result from an elevatory force, acting simultaneously at every point beneath extensive portions of the crust of the earth. An application of mathematical reasoning to the illustration of the phenomena of veins had before been made in Germany by Schmidt and Zimmerman.[73\18] The conclusion which Mr. Hopkins has obtained, respecting the two sets of fissures, at right angles to each other, which would in general be produced by such forces as he supposes, may suggest interesting points of examination respecting the geological phenomena of fissured districts.

[Note 72\18: _Trans. Camb. Phil. Soc._ vol. vi. 1836.]

[Note 73\18: _Phil. Mag._ July, 1836, p. 2.]

[2nd Ed.] [The theory of craters of elevation probably errs rather by making the elevation of a point into a particular class of volcanic agency, than by giving volcanic agency too great a power of elevation.

A mature consideration of the subject will make us hesitate to ascribe much value to the labors of those writers who have applied mathematical reasoning to geological questions. Such reasoning, when it is carried to the extent which requires symbolical processes, has always been, I conceive, a source, not of knowledge, but of error, and confusion; for in such applications the real questions are slurred over in the hypothetical assumptions of the mathematician, while the calculation misleads its followers by a false aspect of demonstration. All symbolical reasonings concerning the fissures of a semi-rigid mass produced by elevatory or other forces, appear to me to have turned out valueless. At the same time it cannot be too strongly borne in mind, that mathematical and mechanical habits of thought are requisite to all clear thinking on such subjects.]

Other forces, still more secure in their nature and laws, have played a very important part in the formation of the earth's crust. I speak of the forces by which the crystalline, slaty, and jointed structure of {558} mineral masses has been produced.

These forces are probably identical, on the one hand, with the cohesive forces from which rocks derive their solidity and their physical properties; while, on the other hand, they are closely connected with the forces of chemical attraction. No attempts, of any lucid and hopeful kind, have yet been made to bring such forces under definite mechanical conceptions: and perhaps mineralogy, to which science, as the point of junction of chemistry and crystallography, such attempts would belong, is hardly yet ripe for such speculations. But when we look at the universal prevalence of crystalline forms and cleavages, at the extent of the phenomena of slaty cleavage, and at the _segregation_ of special minerals into veins and nodules, which has taken place in some unknown manner, we cannot doubt that the forces of which we now speak have acted very widely and energetically. Any elucidation of their nature would be an important step in Geological Dynamics.

[2nd Ed.] [A point of Geological Dynamics of great importance is, the change which rocks undergo in structure after they are deposited, either by the action of subterraneous heat, or by the influence of crystalline or other corpuscular forces. By such agencies, sedimentary rocks may be converted into crystalline, the traces of organic fossils may be obliterated, a slaty cleavage may be produced, and other like effects. The possibility of such changes was urged by Dr. Hutton in his Theory; and Sir James Hall's very instructive and striking experiments were made for the purpose of illustrating this theory. In these experiments, powdered chalk was, by the application of heat under pressure, converted into crystalline calcspar. Afterwards Dr. McCulloch's labors had an important influence in satisfying geologists of the reality of corresponding changes in nature. Dr. McCulloch, by his very lively and copious descriptions of volcanic regions, by his representations of them, by his classification of igneous rocks, and his comprehensive views of the phenomena which they exhibit, probably was the means of converting many geologists from the Wernerian opinions.

Rocks which have undergone changes since they were deposited are termed by Mr. Lyell _metamorphic_. The great extent of metamorphic rock changed by heat is now uncontested. The internal changes which are produced by the crystalline forces of mountain masses have been the subjects of important and comprehensive speculations by Professor Sedgwick.] {559}

_Sect._ 6.--_Theories of Changes of Climate._

AS we have already stated, Geology offers to us strong evidence that the climate of the ancient periods of the earth's history was hotter than that which now exists in the same countries. This, and other circumstances, have led geologists to the investigation of the effects of any hypothetical causes of such changes of condition in respect of heat.

The love of the contemplation of geometrical symmetry, as well as other reasons, suggested the hypothesis that the earth's axis had originally no obliquity, but was perpendicular to the equator. Such a construction of the world had been thought of before the time of Milton,[74\18] as what might be supposed to have existed when man was expelled from Paradise; and Burnet, in his _Sacred Theory of the Earth_

(1690), adopted this notion of the paradisiacal condition of the globe: The spring Perpetual smiled on earth with verdant flowers, Equal in days and nights.

[Note 74\18: Some said he bade his angels turn askance The poles of earth twice ten degrees and more From the sun's axle, &c.--_Paradise Lost_, x. 214.]

In modern times, too, some persons have been disposed to adopt this hypothesis, because they have conceived that the present polar distribution of light is inconsistent with the production of the fossil plants which are found in those regions,[75\18] even if we could, in some other way, account for the change of temperature. But this alteration in the axes of a revolution could not take place without a subversion of the equilibrium of the surface, such as does not appear to have occurred; and the change has of late been generally declared impossible by physical astronomers.

[Note 75\18: Lyell, i. 155. Lindley, _Fossil Flora_.]

The effects of other astronomical changes have been calculated by Sir John Herschel. He has examined, for instance, the thermotical consequences of the diminution of the eccentricity of the earth's orbit, which has been going on for ages beyond the records of history. He finds[76\18] that, on this account, the annual effect of solar radiation would increase as we go back to remoter periods of the past; but (probably at least) not in a degree sufficient to account for the apparent past {560} changes of climate. He finds, however, that though the effect of this change on the mean temperature of the year may be small, the effect on the extreme temperature of the seasons will be much more considerable; "so as to produce alternately, in the same latitude of either hemisphere, a perpetual spring, or the extreme vicissitudes of a burning summer and a rigorous winter."[77\18]

[Note 76\18: _Geol. Trans._ vol. iii. p. 295.]

[Note 77\18: _Geol. Trans._ vol. iii. p. 298.]

Mr. Lyell has traced the consequences of another hypothesis on this subject, which appears at first sight to promise no very striking results, but which yet is found, upon examination, to involve adequate causes of very great changes: I refer to the supposed various distribution of land and water at different periods of the earth's history. If the land were all gathered into the neighborhood of the poles, it would become the seat of constant ice and snow, and would thus very greatly reduce the temperature of the whole surface of the globe. If, on the other hand, the polar regions were principally water, while the tropics were occupied with a belt of land, there would be no part of the earth's surface on which the frost could fasten a firm hold, while the torrid zone would act like a furnace to heat the whole. And, supposing a cycle of terrestrial changes in which these conditions should succeed each other, the winter and summer of this "great year" might differ much more than the elevated temperature which we are led to ascribe to former periods of the globe, can be judged to have differed from the present state of things.

The ingenuity and plausibility of this theory cannot be doubted: and perhaps its results may hereafter be found not quite out of the reach of calculation. Some

progress has already been made in calculating the movement of heat into, through, and out of the earth; but when we add to this the effects of the currents of the ocean and the atmosphere, the problem, thus involving so many thermotical and atmological laws, operating under complex conditions, is undoubtedly one of extreme difficulty. Still, it is something, in this as in all cases, to have the problem even stated; and none of the elements of the solution appears to be of such a nature that we need allow ourselves to yield to despair, respecting the possibility of dealing with it in a useful manner, as our knowledge becomes more complete and definite. {561}

## CHAPTER VI.

PROGRESS OF THE GEOLOGICAL DYNAMICS OF ORGANIZED BEINGS.

_Sect._ 1.--_Objects of this Science._

PERHAPS in extending the term _Geological Dynamics_ to the causes of changes in organized beings, I shall be thought to be employing a forced and inconvenient phraseology. But it will be found that, in order to treat geology in a truly scientific manner, we must bring together all the classes of speculations concerning known causes of change; and the Organic Dynamics of Geology, or of Geography, if the reader prefers the word, appears not an inappropriate phrase for one part of this body of researches.

As has already been said, the species of plants and animals which are found embedded in the strata of the earth, are not only different from those which now live in the same regions, but, for the most part, different from any now existing on the face of the earth. The remains which we discover imply a past state of things different from that which now prevails; they imply also that the whole organic creation has been renewed, and that this renewal has taken place several times. Such extraordinary general facts have naturally put in activity very bold speculations.

But it has already been said, we cannot speculate upon such facts in the past history of the globe, without taking a large survey of its present condition. Does the present animal and vegetable population differ from the past, in the same way in which the products of one region of the existing earth differ from those of another? Can the creation and diffusion of the fossil species be explained in the same manner as the creation and diffusion of the creatures among which we live? And these questions lead us onwards another step, to ask,--What _are_ the laws by which the plants and animals of different parts of the earth differ? What was the manner in which they were originally diffused?--Thus we have to include, as portions of our subject, {562} the _Geography of Plants_, and _of Animals_, and the _History of their change and diffusion_; intending by the latter subject, of course, _palætiological_ history,--the examination of the causes of what has occurred, and the inference of past events, from what we know of causes.

It is unnecessary for me to give at any length a statement of the problems which are included in these branches of science, or of the progress which has been made in them; since Mr. Lyell, in his _Principles of Geology_, has treated these subjects in a

very able manner, and in the same point of view in which I am thus led to consider them. I will only briefly refer to some points, availing myself of his labors and his ideas.

_Sect._ 2.--_Geography of Plants and Animals._

WITH regard both to plants and animals, it appears,[78\18] that besides such differences in the products of different regions as we may naturally suppose to be occasioned by climate and other external causes; an examination of the whole organic population of the globe leads us to consider the earth as divided into _provinces_, each province being occupied by its own group of species, and these groups not being mixed or interfused among each other to any great extent. And thus, as the earth is occupied by various nations of men, each appearing at first sight to be of a different stock, so each other tribe of living things is scattered over the ground in a similar manner, and distributed into its separate _nations_ in distant countries. The places where species are thus peculiarly found, are, in the case of plants, called their _stations_. Yet each species in its own region loves and selects some peculiar conditions of shade or exposure, soil or moisture: its place, defined by the general description of such conditions, is called its _habitation_.

[Note 78\18: Lyell, _Principles_, B. iii. c. v.]

Not only each species thus placed in its own province, has its position further fixed by its own habits, but more general groups and assemblages are found to be determined in their situation by more general conditions. Thus it is the character of the _flora_ of a collection of islands, scattered through a wide ocean in a tropical and humid climate, to contain an immense preponderance of tree-ferns. In the same way, the situation and depth at which certain genera of shells are found have been tabulated[79\18] by Mr. Broderip. Such general inferences, if {563} they can be securely made, are of extreme interest in their bearing on geological speculations.

[Note 79\18: Greenough, _Add._ 1835, p. 20.]

The means by which plants and animals are now diffused from one place to another, have been well described by Mr. Lyell.[80\18] And he has considered also, with due attention, the manner in which they become imbedded in mineral deposits of various kinds.[81\18] He has thus followed the history of organized bodies, from the germ to the tomb, and thence to the cabinet of the geologist.

[Note 80\18: Lyell, B. iii. c. v. vi. vii.]

[Note 81\18: B. iii. c. xiii. xiv. xv. xvi.]

But, besides the fortunes of individual plants and animals, there is another class of questions, of great interest, but of great difficulty;--the fortunes of each species. In what manner do species which were not, begin to be? as geology teaches us that they many times have done; and, as even our own reasonings convince us they must have done, at least in the case of the species among which we live.

We here obviously place before us, as a subject of research, the Creation of Living Things;--a subject shrouded in mystery, and not to be approached without reverence.

But though we may conceive, that, on this subject, we are not to seek our belief from science alone, we shall find, it is asserted, within the limits of allowable and unavoidable speculation, many curious and important problems which may well employ our physiological skill. For example, we may ask:--how we are to recognize the species which were originally created distinct?--whether the population of the earth at one geological epoch could pass to the form which it has at a succeeding period, by the agency of natural causes alone?--and if not, what other account we can give of the succession which we find to have taken place?

The most remarkable point in the attempts to answer these and the like questions, is the controversy between the advocates and the opponents of the doctrine of the _transmutation of species_. This question is, even from its mere physiological import, one of great interest; and the interest is much enhanced by our geological researches, which again bring the question before us in a striking form, and on a gigantic scale. We shall, therefore, briefly state the point at issue.

_Sect._ 3.--_Question of the Transmutation of Species._

WE see that animals and plants may, by the influence of breeding, and of external agents operating upon their constitution, be greatly {564} modified, so as to give rise to varieties and races different from what before existed. How different, for instance, is one kind and breed of dog from another! The question, then, is, whether organized beings can, by the mere working of natural causes, pass from the type of one species to that of another? whether the wolf may, by domestication, become the dog? whether the ourang-outang may, by the power of external circumstances, be brought within the circle of the human species? And the dilemma in which we are placed is this;--that if species are not thus interchangeable, we must suppose the fluctuations of which each species is capable, and which are apparently indefinite, to be bounded by rigorous limits; whereas, if we allow such a _transmutation of species_, we abandon that belief in the adaptation of the structure of every creature to its destined mode of being, which not only most persons would give up with repugnance, but which, as we have seen, has constantly and irresistibly impressed itself on the minds of the best naturalists, as the true view of the order of the world.

But the study of Geology opens to us the spectacle of many groups of species which have, in the course of the earth's history, succeeded each other at vast intervals of time; one set of animals and plants disappearing, as it would seem, from the face of our planet, and others, which did not before exist, becoming the only occupants of the globe. And the dilemma then presents itself to us anew:--either we must accept the doctrine of the transmutation of species, and must suppose that the organized species of one geological epoch were transmuted into those of another by some long-continued agency of natural causes; or else, we must believe in many successive acts of creation and extinction of species, out of the common course of nature; acts which, therefore, we may properly call miraculous.

This latter dilemma, however, is a question concerning the facts which have happened in the history of the world; the deliberation respecting it belongs to physical geology itself, and not to that subsidiary science which we are now

describing, and which is concerned only with such causes as we know to be in constant and orderly action.

The former question, of the limited or unlimited extent of the modifications of animals and plants, has received full and careful consideration from eminent physiologists; and in their opinions we find, I think, an indisputable preponderance to that decision which rejects the transmutation of species, and which accepts the former side of the dilemma; namely, that the changes of which each species is {565} susceptible, though difficult to define in words, are limited in fact. It is extremely interesting and satisfactory thus to receive an answer in which we can confide, to inquiries seemingly so wide and bold as those which this subject involves. I refer to Mr. Lyell, Dr. Prichard, Mr. Lawrence, and others, for the history of the discussion, and for the grounds of the decision; and I shall quote very briefly the main points and conclusions to which the inquiry has led.[82\18]

[Note 82\18: Lyell, B. iii. c. iv.]

It may be considered, then, as determined by the over-balance of physiological authority, that there is a capacity in all species to accommodate themselves, to a certain extent, to a change of external circumstances; this extent varying greatly according to the species. There may thus arise changes of appearance or structure, and some of these changes are transmissible to the offspring: but the mutations thus superinduced are governed by constant laws, and confined within certain limits. Indefinite divergence from the original type is not possible; and the extreme limit of possible variation may usually be reached in a brief period of time: in short, _species have a real existence in nature_, and a transmutation from one to another does not exist.

Thus, for example, Cuvier remarks, that notwithstanding all the differences of size, appearance, and habits, which we find in the dogs of various races and countries, and though we have (in the Egyptian mummies) skeletons of this animal as it existed three thousand years ago, the relation of the bones to each other remains essentially the same; and, with all the varieties of their shape[83\18] and size, there are characters which resist all the influences both of external nature, of human intercourse, and of time.

[Note 83\18: _Ossem. Foss._ Disc. Prél. p. 61.]

_Sect._ 4.--_Hypothesis of Progressive Tendencies._

WITHIN certain limits, however, as we have said, external circumstances produce changes in the forms of organized beings. The causes of change, and the laws and limits of their effects, as they obtain in the existing state of the organic creation, are in the highest degree interesting. And, as has been already intimated, the knowledge thus obtained, has been applied with a view to explain the origin of the existing population of the world, and the succession of its past conditions. But those who have attempted such an explanation, have found it necessary to assume certain additional laws, in order to enable themselves to {566} deduce, from the tenet of the transmutability of the species of organized beings, such a state of things as we see about us, and such a succession of states as is evidenced by geological researches.

And here, again, we are brought to questions of which we must seek the answers from the most profound physiologists. Now referring, as before, to those which appear to be the best authorities, it is found that these additional positive laws are still more inadmissible than the primary assumption of indefinite capacity of change. For example, in order to account, on this hypothesis, for the seeming adaptation of the endowments of animals to their wants, it is held that the endowments are the result of the wants; that the swiftness of the antelope, the claws and teeth of the lion, the trunk of the elephant, the long neck of the giraffe have been produced by a certain plastic character in the constitution of animals, operated upon, for a long course of ages, by the attempts which these animals made to attain objects which their previous organization did not place within their reach. In this way, it is maintained that the most striking attributes of animals, those which apparently imply most clearly the providing skill of their Creator, have been brought forth by the long-repeated efforts of the creatures to attain the object of their desire; thus animals with the highest endowments have been gradually developed from ancestral forms of the most limited organization; thus fish, bird, and beast, have grown from _small gelatinous bodies_, "petits corps gelatineux," possessing some obscure principle of life, and the capacity of development; and thus man himself with all his intellectual and moral, as well as physical privileges, has been derived from some creature of the ape or baboon tribe, urged by a constant tendency to improve, or at least to alter his condition.

As we have said, in order to arrive even hypothetically at this result, it is necessary to assume besides a mere capacity for change, other positive and active principles, some of which we may notice. Thus, we must have as the direct productions of nature on this hypothesis, certain monads or rough draughts, the primary _rudiments_ of plants and animals. We must have, in these, a constant _tendency to progressive improvement_, to the attainment of higher powers and faculties than they possess; which tendency is again perpetually modified and controlled by the _force of external circumstances_. And in order to account for the simultaneous existence of animals in every stage of this imaginary progress, we must suppose that nature is compelled to be _constantly_ producing those elementary beings, from which all animals are successively developed. {567}

I need not stay to point out how extremely arbitrary every part of this scheme is; and how complex its machinery would be, even if it did account for the facts. It may be sufficient to observe, as others have done,[84\18] that the capacity of change, and of being influenced by external circumstances, such as we really find it in nature, and therefore such as in science we must represent it, is a tendency, not to improve, but to deteriorate. When species are modified by external causes, they usually degenerate, and do not advance. And there is no instance of a species acquiring an entirely new sense, faculty, or organ, in addition to, or in the place of, what it had before.

[Note 84\18: Lyell, B. III. c. iv.]

Not only, then, is the doctrine of the transmutation of species in itself disproved by the best physiological reasonings, but the additional assumptions which are requisite,

to enable its advocates to apply it to the explanation of the geological and other phenomena of the earth, are altogether gratuitous and fantastical.

Such is the judgment to which we are led by the examination of the discussions which have taken place on this subject. Yet in certain speculations, occasioned by the discovery of the _Sivatherium_, a new fossil animal from the Sub-Himalaya mountains of India, M. Geoffroy Saint-Hilaire speaks of the belief in the immutability of species as a conviction which is fading away from men's minds. He speaks too of the termination of the age of Cuvier, "la clôture du siècle de Cuvier," and of the commencement of a better zoological philosophy.[85\18] But though he expresses himself with great animation, I do not perceive that he adduces, in support of his peculiar opinions, any arguments in addition to those which he urged during the lifetime of Cuvier. And the reader[86\18] may recollect that the consideration of that controversy led us to very different anticipations from his, respecting the probable future progress of physiology. The discovery of the Sivatherium supplies no particle of proof to the hypothesis, that the existing species of animals are descended from extinct creatures which are specifically distinct: and we cannot act more wisely than in listening to the advice of that eminent naturalist, M. de Blainville.[87\18] "Against this hypothesis, which, up to the present time, I regard as purely gratuitous, and likely to turn geologists out of the sound and excellent road in which they now are, I willingly raise my voice, with the most absolute conviction of being in the right." {568}

[Note 85\18: _Compte Rendu de l'Acad. des Sc._ 1837, No. 3, p. 81.]

[Note 86\18: See B. XVII. c. vii.]

[Note 87\18: _Compte Rendu_, 1837, No. 5, p. 168.]

[2nd Ed.] [The hypothesis of the progressive developement of species has been urged recently, in connexion with the physiological tenet of Tiedemann and De Serres, noticed in B. XVII. c. vii. sect. 3;--namely, that the embryo of the higher forms of animals passes by gradations through those forms which are permanent in inferior animals. Assuming this tenet as exact, it has been maintained that the higher animals which are found in the more recent strata may have been produced by an ulterior development of the lower forms in the embryo state; the circumstances being such as to favor such a developement. But all the best physiologists agree in declaring that such an extraordinary developement of the embryo is inconsistent with physiological possibility. Even if the progression of the embryo in time have a general correspondence with the order of animal forms as more or less perfectly organized (which is true in an extremely incomplete and inexact degree), this correspondence must be considered, not as any indication of causality, but as one of those marks of universal analogy and symmetry which are stamped upon every part of the creation.

Mr. Lyell[88\18] notices this doctrine of Tiedemann and De Serres; and observes, that though nature presents us with cases of animal forms degraded by incomplete developement, she offers none of forms exalted by extraordinary developement. Mr. Lyell's own hypothesis of the introduction of new species upon the earth, not having any physiological basis, hardly belongs to this chapter.]

[Note 88\18: _Principles_, B. III. c. iv.]

_Sect._ 5.--_Question of Creation as related to Science._

BUT since we reject the production of new species by means of external influence, do we then, it may be asked, accept the other side of the dilemma which we have stated; and admit a series of creations of species, by some power beyond that which we trace in the ordinary course of nature?

To this question, the history and analogy of science, I conceive, teach us to reply as follows:--All palætiological sciences, all speculations which attempt to ascend from the present to the remote past, by the chain of causation, do also, by an inevitable consequence, urge us to look for the beginning of the state of things which we thus contemplate; but in none of these cases have men been able, by the aid of science, to arrive at a beginning which is homogeneous with the {569} known course of events. The first origin of language, of civilization, of law and government, cannot be clearly made out by reasoning and research; just as little, we may expect, will a knowledge of the origin of the existing and extinct species of plants and animals, be the result of physiological and geological investigation.

But, though philosophers have never yet demonstrated, and perhaps never will be able to demonstrate, what was that primitive state of things in the social and material worlds, from which the progressive state took its first departure; they can still, in all the lines of research to which we have referred, go very far back;--determine many of the remote circumstances of the past sequence of events;--ascend to a point which, from our position at least, seems to be near the origin;--and exclude many suppositions respecting the origin itself. Whether, by the light of reason alone, men will ever be able to do more than this, it is difficult to say. It is, I think, no irrational opinion, even on grounds of philosophical analogy alone, that in all those sciences which look back and seek a beginning of things, we may be unable to arrive at a consistent and definite belief, without having recourse to other grounds of truth, as well as to historical research and scientific reasoning. When our thoughts would apprehend steadily the creation of things, we find that we are obliged to summon up other ideas than those which regulate the pursuit of scientific truths;--to call in other powers than those to which we refer natural events: it cannot, then, be considered as very surprizing, if, in this part of our inquiry, we are compelled to look for other than the ordinary evidence of science.

Geology, forming one of the palætiological class of sciences, which trace back the history of the earth and its inhabitants on philosophical grounds, is thus associated with a number of other kinds of research, which are concerned about language, law, art, and consequently about the internal faculties of man, his thoughts, his social habits, his conception of right, his love of beauty. Geology being thus brought into the atmosphere of moral and mental speculations, it may be expected that her investigations of the probable past will share an influence common to them; and that she will not be allowed to point to an origin of her own, a merely physical beginning of things; but that, as she approaches towards such a goal, she will be led to see that it is the origin of many trains of events, the point of convergence of many lines. It may

be, that instead of being allowed to travel up to this focus of being, we are only able to estimate its place and nature, and {570} to form of it such a judgment as this;--that it is not only the source of mere vegetable and animal life, but also of rational and social life, language and arts, law and order; in short, of all the progressive tendencies by which the highest principles of the intellectual and moral world have been and are developed, as well as of the succession of organic forms, which we find scattered, dead or living, over the earth.

This reflection concerning the natural scientific view of creation, it will be observed, has not been sought for, from a wish to arrive at such conclusions; but it has flowed spontaneously from the manner in which we have had to introduce geology into our classification of the sciences; and this classification was framed from an unbiassed consideration of the general analogies and guiding ideas of the various portions of our knowledge. Such remarks as we have made may on this account be considered more worthy of attention.

But such a train of thought must be pursued with caution. Although it may not be possible to arrive at a right conviction respecting the origin of the world, without having recourse to other than physical considerations, and to other than geological evidence: yet extraneous considerations, and extraneous evidence, respecting the nature of the beginning of things, must never be allowed to influence our physics or our geology. Our geological dynamics, like our astronomical dynamics, may be inadequate to carry us back to an origin of that state of things, of which it explains the progress: but this deficiency must be supplied, not by adding supernatural to natural geological dynamics, but by accepting, in their proper place, the views supplied by a portion of knowledge of a different character and order. If we include in our Theology the speculations to which we have recourse for this purpose, we must exclude from them our Geology. The two sciences may conspire, not by having any part in common: but because, though widely diverse in their lines, both point to a mysterious and invisible origin of the world.

All that which claims our assent on those higher grounds of which theology takes cognizance, must claim such assent as is consistent with those grounds; that is, it must require belief in respect of all that bears upon the highest relations of our being, those on which depend our duties and our hopes. Doctrines of this kind may and must be conveyed and maintained, by means of information concerning the past history of man, and his social and material, as well as moral and spiritual fortunes. He who believes that a Providence has {571} ruled the affairs of mankind, will also believe that a Providence has governed the material world. But any language in which the narrative of this government of the material world can be conveyed, must necessarily be very imperfect and inappropriate; being expressed in terms of those ideas which have been selected by men, in order to describe appearances and relations of created things as they affect one another. In all cases, therefore, where we have to attempt to interpret such a narrative, we must feel that we are extremely liable to err; and most of all, when our interpretation refers to those material objects and operations which are most foreign to the main purpose of a history of providence. If we have to consider a communication containing a view of such a government of the world,

imparted to us, as we may suppose, in order to point out the right direction for our feelings of trust, and reverence, and hope, towards the Governor of the world, we may expect that we shall be in no danger of collecting from our authority erroneous notions with regard to the power, and wisdom, and goodness of His government; or with respect to our own place, duties, and prospects, and the history of our race so far as our duties and prospects are concerned. But that we shall rightly understand the detail of all events in the history of man, or of the skies, or of the earth, which are narrated for the purpose of thus giving a right direction to our minds, is by no means equally certain; and I do not think it would be too much to say, that an immunity from perplexity and error, in such matters, is, on general grounds, very improbable. It cannot then surprise us to find, that parts of such narrations which seem to refer to occurrences like those of which astronomers and geologists have attempted to determine the laws, have given rise to many interpretations, all inconsistent with one another, and most of them at variance with the best established principles of astronomy and geology.

It may be urged, that all truths must be consistent with all other truths, and that therefore the results of true geology or astronomy cannot be irreconcileable with the statements of true theology. And this universal consistency of truth with itself must be assented to; but it by no means follows that we must be able to obtain a full insight into the nature and manner of such a consistency. Such an insight would only be possible if we could obtain a clear view of that central body of truth, the source of the principles which appear in the separate lines of speculation. To expect that we should see clearly how the providential government of the world is consistent with the unvarying laws {572} by which its motions and developements are regulated, is to expect to understand thoroughly the laws of motion, of developement, and of providence; it is to expect that we may ascend from geology and astronomy to the creative and legislative centre, from which proceeded earth and stars; and then descend again into the moral and spiritual world, because its source and centre are the same as those of the material creation. It is to say that reason, whether finite or infinite, must be consistent with itself; and that, therefore, the finite must be able to comprehend the infinite, to travel from any one province of the moral and material universe to any other, to trace their bearing, and to connect their boundaries.

One of the advantages of the study of the history and nature of science in which we are now engaged is, that it warns us of the hopeless and presumptuous character of such attempts to understand the government of the world by the aid of science, without throwing any discredit upon the reality of our knowledge;--that while it shows how solid and certain each science is, so long as it refers its own facts to its own ideas, it confines each science within its own limits, and condemns it as empty and helpless, when it pronounces upon those subjects which are extraneous to it. The error of persons who should seek a geological narrative in theological records, would be rather in the search itself than in their interpretation of what they might find; and in like manner the error of those who would conclude against a supernatural beginning, or a providential direction of the world, upon geological or physiological reasonings, would be, that they had expected those sciences alone to place the origin or the government of the world in its proper light.

Though these observations apply generally to all the palætiological sciences, they may be permitted here, because they have an especial bearing upon some of the difficulties which have embarrassed the progress of geological speculation; and though such difficulties are, I trust, nearly gone by, it is important for us to see them in their true bearing.

From what has been said, it follows that geology and astronomy are, of themselves, incapable of giving us any distinct and satisfactory account of the origin of the universe, or of its parts. We need not wonder, then, at any particular instance of this incapacity; as, for example, that of which we have been speaking, the impossibility of accounting by any natural means for the production of all the successive tribes of plants and animals which have peopled the world in the {573} various stages of its progress, as geology teaches us. That they were, like our own animal and vegetable contemporaries, profoundly adapted to the condition in which they were placed, we have ample reason to believe; but when we inquire whence they came into this our world, geology is silent. The mystery of creation is not within the range of her legitimate territory; she says nothing, but she points upwards.

_Sect._ 6.--_The Hypothesis of the regular Creation and Extinction of Species._

1. _Creation of Species._--We have already seen, how untenable, as a physiological doctrine, is the principle of the transmutability and progressive tendency of species; and therefore, when we come to apply to theoretical geology the principles of the present chapter, this portion of the subject will easily be disposed of. I hardly know whether I can state that there is any other principle which has been applied to the solution of the geological problem, and which, therefore, as a general truth, ought to be considered here. Mr. Lyell, indeed, has spoken[89\18] of an hypothesis that "the successive creation of species may constitute a regular part of the economy of nature:" but he has nowhere, I think, so described this process as to make it appear in what department of science we are to place the hypothesis. Are these new species created by the production, at long intervals, of an offspring different in species from the parents? Or are the species so created produced without parents? Are they gradually evolved from some embryo substance? or do they suddenly start from the ground, as in the creation of the poet? . . . . . Perfect forms Limbed and full-grown: out of the ground up rose As from his lair, the wild beast where he wons In forest wild, in thicket, brake, or den; . . . The grassy clods now calved; now half appeared The tawny lion, pawing to get free His hinder parts; then springs as broke from bounds, And rampant shakes his brinded mane; &c. &c. _Paradise Lost_, B. vii.

[Note 89\18: B. III. c. xi. p. 234.]

Some selection of one of these forms of the hypothesis, rather than the others, with evidence for the selection, is requisite to entitle us to {574} place it among the known causes of change which in this chapter we are considering. The bare conviction that a creation of species has taken place, whether once or many times, so long as it is unconnected with our organical sciences, is a tenet of Natural Theology rather than of Physical Philosophy.

[2nd Ed.] [Mr. Lyell has explained his theory[90\18] by supposing man to people a great desert, introducing into it living plants and animals: and he has traced, in a very interesting manner, the results of such a hypothesis on the distribution of vegetable and animal species. But he supposes the agents who do this, before they import species into particular localities, to study attentively the climate and other physical conditions of each spot, and to use various precautions. It is on account of the notion of design thus introduced that I have, above, described this opinion as rather a tenet of Natural Theology than of Physical Philosophy.

[Note 90\18: B. III. c. viii. p. 166.]

Mr. Edward Forbes has published some highly interesting speculations on the distribution of existing species of animals and plants. It appears that the manner in which animal and vegetable forms are now diffused requires us to assume centres from which the diffusion took place by no means limited by the present divisions of continents and islands. The changes of land and water which have thus occurred since the existing species were placed on the earth must have been very extensive, and perhaps reach into the glacial period of which I have spoken above.[91\18]

[Note 91\18: See, in _Memoirs of the Geological Survey of Great Britain_, vol. i. p. 336, Professor Forbes's Memoir "On the Connection between the Distribution of the existing Fauna and Flora of the British Isles, and the Geological Changes which have affected their area, especially during the epoch of the Northern Drift."]

According to Mr. Forbes's views, for which he has offered a great body of very striking and converging reasons, the present vegetable and animal population of the British Isles is to be accounted for by the following series of events. The marine deposits of the _meiocine_ formation were elevated into a great Atlantic continent, yet separate from what is now America, and having its western shore where now the great semi-circular belt of gulf-weed ranges from the 15th to the 45th parallel of latitude. This continent then became stocked with life, and of its vegetable population, the flora of the west of Ireland, which has many points in common with the flora of Spain and the {575} Atlantic islands (the _Asturian_ flora), is the record. The region between Spain and Ireland, and the rest of this meiocene continent, was destroyed by some geological movement, but there were left traces of the connexion which still remain. Eastwards of the flora just mentioned, there is a flora common to Devon and Cornwall, to the south-east part of Ireland, the Channel Isles, and the adjacent provinces of France;--a flora passing to a southern character; and having its course marked by the remains of a great rocky barrier, the destruction of which probably took place anterior to the formation of the narrower part of the channel. Eastward from this _Devon_ or _Norman_ flora, again, we have the _Kentish_ flora, which is an extension of the flora of North-western France, insulated by the breach which formed the straits of Dover. Then came the _Glacial period_, when the east of England and the north of Europe were submerged, the northern drift was distributed, and England was reduced to a chain of islands or ridges, formed by the mountains of Wales, Cumberland, and Scotland, which were connected with the land of Scandinavia. This was the period of glaciers, of the dispersion of boulders, of the grooving and scratching of rocks as they are now found. The climate being then much

colder than it now is, the flora, even down to the water's edge, consisted of what are now Alpine plants; and this _Alpine_ flora is common to Scandinavia and to our mountain-summits. And these plants kept their places, when, by the elevation of the land, the whole of the present German Ocean became a continent connecting Britain with central Europe. For the increased elevation of their stations counterbalanced the diminished cold of the succeeding period. Along the dry bed of the German Sea, thus elevated, the principal part of the existing flora of England, the _Germanic_ flora, migrated. A large portion of our existing animal population also came over through the same region; and along with those, came hyenas, tigers, rhinoceros, aurochs, elk, wolves, beavers, which are extinct in Britain, and other animals which are extinct altogether, as the primigenian elephant or mammoth. But then, again, the German Ocean and the Irish Channel were scooped out; and the climate again changed. In our islands, so detached, many of the larger beasts perished, and their bones were covered up in peat-mosses and caves, where we find them. This distinguished naturalist has further shown that the population of the sea lends itself to the same view. Mr. Forbes says that the writings of Mr. Smith, of Jordan-hill, "On the last Changes in the relative Levels of the Land and Sea in the British Islands," published in the _Memoirs of the_ {576} _Wernerian Society for_ 1837-8, must be esteemed the foundation of a critical investigation of this subject in Britain.]

2. _Extinction of Species._--With regard to the extinction of species Mr. Lyell has propounded a doctrine which is deserving of great attention here. Brocchi, when he had satisfied himself, by examination of the Sub-Apennines, that about half the species which had lived at the period of their deposition, had since become extinct, suggested as a possible cause for this occurrence, that the vital energies of a species, like that of an individual, might gradually decay in the progress of time and of generations, till at last the prolific power might fail, and the species wither away. Such a property would be conceivable as a physiological fact; for we see something of the kind in fruit-trees propagated by cuttings: after some time, the stock appears to wear out, and loses its peculiar qualities. But we have no sufficient evidence that this is the case in generations of creatures continued by the reproductive powers. Mr. Lyell conceives, that, without admitting any inherent constitutional tendency to deteriorate, the misfortunes to which plants and animals are exposed by the change of the physical circumstances of the earth, by the alteration of land and water, and by the changes of climate, must very frequently occasion the loss of several species. We have historical evidence of the extinction of one conspicuous species, the Dodo, a bird of large size and singular form, which inhabited the Isle of France when that island was first discovered, and which now no longer exists. Several other species of animals and plants seem to be in the course of vanishing from the face of the earth, even under our own observation. And taking into account the greater changes of the surface of the globe which geology compels us to assume, we may imagine many or all the existing species of living things to be extirpated. If, for instance, that reduction of the climate of the earth which appears, from geological evidence, to have taken place already, be supposed to go on much further, the advancing snow and cold of the polar regions may destroy the greater part of our plants and animals, and drive the remainder, or those of them which possess the requisite faculties of migration and

accommodation, to seek an asylum near the equator. And if we suppose the temperature of the earth to be still further reduced, this zone of now-existing life, having no further place of refuge, will perish, and the whole earth will be tenanted, if at all, by a new creation. Other causes might produce the same effect as a change of climate; and, without supposing such causes to affect the whole globe, it is easy to {577} imagine circumstances such as might entirely disturb the equilibrium which the powers of diffusion of different species have produced;--might give to some the opportunity of invading and conquering the domain of others; and in the end, the means of entirely suppressing them, and establishing themselves in their place.

That this extirpation of certain species, which, as we have seen, happens in a few cases under common circumstances, might happen upon a greater scale, if the range of external changes were to be much enlarged, cannot be doubted. The extent, therefore, to which natural causes may account for the extinction of species, will depend upon the amount of change which we suppose in the physical conditions of the earth. It must be a task of extreme difficulty to estimate the effect upon the organic world, even if the physical circumstances were given. To determine the physical condition to which a given state of the earth would give rise, I have already noted as another very difficult problem. Yet these two problems must be solved, in order to enable us to judge of the sufficiency of any hypothesis of the extinction of species; and in the mean time, for the mode in which new species come into the places of those which are extinguished, we have (as we have seen) no hypothesis which physiology can, for a moment, sanction.

_Sect._ 7.--_The Imbedding of Organic Remains._

THERE is still one portion of the Dynamics of Geology, a branch of great and manifest importance, which I have to notice, but upon which I need only speak very briefly. The mode in which the spoils of existing plants and animals are imbedded in the deposits now forming, is a subject which has naturally attracted the attention of geologists. During the controversy which took place in Italy respecting the fossils of the Sub-Apennine hills, Vitaliano Donati,[92\18] in 1750, undertook an examination of the Adriatic, and found that deposits containing shells and corals, extremely resembling the strata of the hills, were there in the act of formation. But without dwelling on other observations of like kind, I may state that Mr. Lyell has treated this subject, and all the topics connected with it, in a very full and satisfactory manner. He has explained,[93\18] by an excellent collection of illustrative facts, how deposits of various substance and contents are formed; how plants and animals become fossil in peat, in blown sand, in volcanic matter, in {578} alluvial soil, in caves, and in the beds of lakes and seas. This exposition is of the most instructive character, as a means of obtaining right conclusions concerning the causes of geological phenomena. Indeed, in many cases, the similarity of past effects with operations now going on, is so complete, that they may be considered as identical; and the discussion of such cases belongs, at the same time, to Geological Dynamics and to Physical Geology; just as the problem of the fall of meteorolites may be considered as belonging alike to mechanics and to physical astronomy. The growth of modern peat-mosses, for example, fully explains the formation of the most ancient: objects are buried in the

same manner in the ejections of active and of extinct volcanoes; within the limits of history, many estuaries have been filled up; and in the deposits which have occupied these places, are strata containing shells,[94\18] as in the older formations.

[Note 92\18: Lyell, B. I. c. iii. p. 67. (4th ed.)]

[Note 93\18: B. III. c. xiii. xiv. xv. xvi. xvii.]

[Note 94\18: Lyell, B. III. c. xvii. p. 286. See also his Address to the Geological Society in 1837, for an account of the Researches of Mr. Stokes and of Professor Göppert, on the lapidification of vegetables.]

{{579}} PHYSICAL GEOLOGY.

## CHAPTER VII.

PROGRESS OF PHYSICAL GEOLOGY.

_Sect._ 1.--_Object and Distinctions of Physical Geology._

BEING, in consequence of the steps which we have attempted to describe, in possession of two sciences, one of which traces the laws of action of known causes, and the other describes the phenomena which the earth's surface presents, we are now prepared to examine how far the attempts to refer the facts to their causes have been successful: we are ready to enter upon the consideration of Theoretical or _Physical_ Geology, as, by analogy with Physical Astronomy, we may term this branch of speculation.

The distinction of this from other portions of our knowledge is sufficiently evident. In former times, Geology was always associated with Mineralogy, and sometimes confounded with it; but the mistake of such an arrangement must be clear, from what has been said. Geology is connected with Mineralogy, only so far as the latter science classifies a large portion of the objects which Geology employs as evidence of its statements. To confound the two is the same error as it would be to treat philosophical history as identical with the knowledge of medals. Geology procures evidence of her conclusions wherever she can; from minerals or from seas; from inorganic or from organic bodies; from the ground or from the skies. The geologist's business is to learn the past history of the earth; and he is no more limited to one or a few kinds of documents, as his sources of information, than is the historian of man, in the execution of a similar task.

Physical Geology, of which I now speak, may not be always easily separable from Descriptive Geology: in fact, they have generally been combined, for few have been content to describe, without attempting in some measure to explain. Indeed, if they had done so, it is {580} probable that their labors would have been far less zealous, and their expositions far less impressive. We by no means regret, therefore, the mixture of these two kinds of knowledge, which has so often occurred; but still, it is our business to separate them. The works of astronomers before the rise of sound physical astronomy, were full of theories, but these were advantageous, not prejudicial, to the progress of the science.

Geological theories have been abundant and various; but yet our history of them must be brief. For our object is, as must be borne in mind, to exhibit these, only so far as they are steps discoverably tending to the _true_ theory of the earth: and in most of them we do not trace this character. Or rather, the portions of the labors of geologists which do merit this praise, belong to the two preceding divisions of the subject, and have been treated of there.

The history of Physical Geology, considered as the advance towards a science as real and stable as those which we have already treated of (and this is the form in which we ought to trace it), hitherto consists of few steps. We hardly know whether the progress is begun. The history of Physical Astronomy almost commences with Newton, and few persons will venture to assert that the Newton of Geology has yet appeared.

Still, some examination of the attempts which have been made is requisite, in order to explain and justify the view which the analogy of scientific history leads us to take, of the state of the subject. Though far from intending to give even a sketch of all past geological speculations, I must notice some of the forms such speculations have at different times assumed.

_Sect._ 2.--_Of Fanciful Geological Opinions._

REAL and permanent geological knowledge, like all other physical knowledge, can be obtained only by inductions of classification and law from many clearly seen phenomena. The labor of the most active, the talent of the most intelligent, are requisite for such a purpose. But far less than this is sufficient to put in busy operation the inventive and capricious fancy. A few appearances hastily seen, and arbitrarily interpreted, are enough to give rise to a wondrous tale of the past, full of strange events and supernatural agencies. The mythology and early poetry of nations afford sufficient evidence of man's love of the wonderful, and of his inventive powers, in early stages of intellectual development. The scientific faculty, on the other hand, {581} and especially that part of it which is requisite for the induction of laws from facts, emerges slowly and with difficulty from the crowd of adverse influences, even under the most favorable circumstances. We have seen that in the ancient world, the Greeks alone showed themselves to possess this talent; and what they thus attained to, amounted only to a few sound doctrines in astronomy, and one or two extremely imperfect truths in mechanics, optics, and music, which their successors were unable to retain. No other nation, till we come to the dawn of a better day in modern Europe, made any positive step at all in sound physical speculation. Empty dreams or useless exhibitions of ingenuity, formed the whole of their essays at such knowledge.

It must, therefore, independently of positive evidence, be considered as extremely improbable, that any of these nations should, at an early period, have arrived, by observation and induction, at wide general truths, such as the philosophers of modern times have only satisfied themselves of by long and patient labor and thought. If resemblances should be discovered between the assertions of ancient writers and the discoveries of modern science, the probability in all cases, the certainty in most, is that these are accidental coincidences;--that the ancient opinion

is no anticipation of the modern discovery, but is one guess among many, not a whit the more valuable because its expression agrees with a truth. The author of the guess could not intend the truth, because his mind was not prepared to comprehend it. Those of the ancients who spoke of the _harmony_ which binds all things together, could not mean the Newtonian gravitation, because they had never been led to conceive an attractive force, governed by definite mathematical laws in its quantity and operation.

In agreement with these views, we must, I conceive, estimate the opinions which we find among the ancients, respecting the changes which the earth's surface has undergone. These opinions, when they are at all of a general kind, are arbitrary fictions of the fancy, showing man's love of generality indeed, but indulging it without that expense of labor and thought which alone can render it legitimate.

We might, therefore, pass by all the traditions and speculations of Oriental, Egyptian, and Greek cosmogony, as extraneous to our subject. But since these have recently been spoken of, as conclusions collected, however vaguely, from observed facts,[95\18] we may make a remark or two upon them. {582}

[Note 95\18: Lyell, B. i. c. ii. p. 8. (4th ed.)]

The notion of a series of creations and destructions of worlds, which appears in the sacred volume of the Hindoos, which formed part of the traditionary lore of Egypt, and which was afterwards adopted into the poetry and philosophy of Greece, must be considered as a mythological, not a physical, doctrine. When this doctrine was dwelt upon, men's thoughts were directed, not to the terrestrial facts which it seemed to explain, but to the attributes of the deities which it illustrated. The conception of a Supreme power, impelling and guiding the progress of events, which is permanent among all perpetual change, and regular among all seeming chance, was readily entertained by contemplative and enthusiastic minds; and when natural phenomena were referred to this doctrine, it was rather for the purpose of fastening its impressiveness upon the senses, than in the way of giving to it authority and support. Hence we perceive that in the exposition of this doctrine, an attempt was always made to fill and elevate the mind with the notions of marvellous events, and of infinite times, in which vast cycles of order recurred. The "great year," in which all celestial phenomena come round, offered itself as capable of being calculated; and a similar great year was readily assumed for terrestrial and human events. Hence there were to be brought round by great cycles, not only deluges and conflagrations which were to destroy and renovate the earth, but also the series of historical occurrences. Not only the sea and land were to recommence their alternations, but there was to be another Argo, which should carry warriors on the first sea-foray,[96\18] and another succession of heroic wars. Looking at the passages of ancient authors which refer to terrestrial changes in this view, we shall see that they are addressed almost entirely to the love of the marvellous and the infinite, and cannot with propriety be taken as indications of a spirit of physical philosophy. For example, if we turn to the celebrated passage in Ovid,[97\18] where Pythagoras is represented as asserting that land becomes sea, and sea land, and many other changes which geologists have verified, we find that these observations are associated with many fables, as being matter of

exactly the same kind;--the fountain of Ammon which was cold by day and warm by night;[98\18]--the waters of Salmacis which effeminate men;--the Clitorian spring which makes them loathe wine;--the Simplegades islands which were once moveable;--the Tritonian lake which covered men's bodies with feathers;--and many similar marvels. And the general purport of {583} the whole is, to countenance the doctrine of the metempsychosis, and the Pythagorean injunction of not eating animal food. It is clear, I think, that facts so introduced must be considered as having been contemplated rather in the spirit of poetry than of science.

[Note 96\18: Virg. _Eclog._ 4.]

[Note 97\18: _Met._ Lib. xv.]

[Note 98\18: V. 309, &c.]

We must estimate in the same manner, the very remarkable passage brought to light by M. Elie de Beaumont,[99\18] from the Arabian writer, Kazwiri; in which we have a representation of the same spot of ground, as being, at successive intervals of five hundred years, a city, a sea, a desert, and again a city. This invention is adduced, I conceive, rather to feed the appetite of wonder, than to fix it upon any reality: as the title of his book, _The Marvels of Nature_ obviously intimates.

[Note 99\18: _Ann. des Sc. Nat._ xxv. 380.]

The speculations of Aristotle, concerning the exchanges of land and sea which take place in long periods, are not formed in exactly the same spirit, but they are hardly more substantial; and seem to be quite as arbitrary, since they are not confirmed by any examples and proofs. After stating,[100\18] that the same spots of the earth are not always land and always water, he gives the reason. "The principle and cause of this is," he says, "that the inner parts of the earth, like the bodies of plants and animals, have their ages of vigor and of decline; but in plants and animals all the parts are in vigor, and all grow old, at once: in the earth different parts arrive at maturity at different times by the operation of cold and heat: they grow and decay on account of the sun and the revolution of the stars, and thus the parts of the earth acquire different power, so that for a certain time they remain moist, and then become dry and old: and then other places are revivified, and become partially watery." We are, I conceive, doing no injustice to such speculations by classing them among _fanciful_ geological opinions.

[Note 100\18: _Meteorol._ i. 14.]

We must also, I conceive, range in the same division another class of writers of much more modern times;--I mean those who have trained their geology by interpretations of Scripture. I have already endeavored to show that such an attempt is a perversion of the purpose of a divine communication, and cannot lead to any physical truth. I do not here speak of geological speculations in which the Mosaic account of the deluge has been referred to; for whatever errors may have been committed on that subject, it would be as absurd to disregard the most ancient historical record, in attempting to trace back the history of the earth, as it would be, gratuitously to reject any other {584} source of information. But the interpretations

of the account of the creation have gone further beyond the limits of sound philosophy: and when we look at the arbitrary and fantastical inventions by which a few phrases of the writings of Moses have been moulded into complete systems, we cannot doubt that these interpretations belong to the present Section.

I shall not attempt to criticize, nor even to enumerate, these Scriptural Geologies,--_Sacred Theories of the Earth_, as Burnet termed his. Ray, Woodward, Whiston, and many other persons to whom science has considerable obligations, were involved, by the speculative habits of their times, in these essays; and they have been resumed by persons of considerable talent and some knowledge, on various occasions up to the present day; but the more geology has been studied on its own proper evidence, the more have geologists seen the unprofitable character of such labors.

I proceed now to the next step in the progress of Theoretical Geology.

_Sect._ 3.--_Of Premature Geological Theories._

WHILE we were giving our account of Descriptive Geology, the attentive reader would perceive that we did, in fact, state several steps in the advance towards general knowledge; but when, in those cases, the theoretical aspect of such discoveries softened into an appearance of mere classification, the occurrence was assigned to the history of Descriptive rather than of Theoretical Geology. Of such a kind was the establishment, by a long and vehement controversy, of the fact, that the impressions in rocks are really the traces of ancient living things; such, again, were the division of rocks into Primitive, Secondary, Tertiary; the ascertainment of the orderly succession of organic remains: the consequent fixation of a standard series of formations and strata; the establishment of the igneous nature of trap rocks; and the like. These are geological truths which are assumed and implied in the very language which geology uses; thus showing how in this, as in all other sciences, the succeeding steps involve the preceding. But in the history of geological theory, we have to consider the wider attempts to combine the facts, and to assign them to their causes.

The close of the last century produced two antagonist theories of this kind, which long maintained a fierce and doubtful struggle;--that of Werner and that of Hutton: the one termed _Neptunian_, from its {585} ascribing the phenomena of the earth's surface mainly to aqueous agency; the other _Plutonian_ or _Vulcanian_, because it employed the force of subterraneous fire as its principal machinery. The circumstance which is most worthy of notice in these remarkable essays is, the endeavor to give, by means of such materials as the authors possessed, a complete and simple account of all the facts of the earth's history. The Saxon professor, proceeding on the examination of a small district in Germany, maintained the existence of a chaotic fluid, from which a series of universal formations had been precipitated, the position of the strata being broken up by the falling in of subterraneous cavities, in the intervals between these depositions. The Scotch philosopher, who had observed in England and Scotland, thought himself justified in declaring that the existing causes were sufficient to spread new strata on the bottom of the ocean, and that they are consolidated, elevated, and fractured by volcanic heat, so as to give rise to new continents.

It will hardly be now denied that all that is to remain as permanent science in each of these systems must be proved by the examination of many cases and limited by many conditions and circumstances. Theories so wide and simple, were consistent only with a comparatively scanty collection of facts, and belong to the early stage of geological knowledge. In the progress of the science, the "theory" of each part of the earth must come out of the examination of that part, combined with all that is well established, concerning all the rest; and a general theory must result from the comparison of all such partial theoretical views. Any attempt to snatch it before its time must fail; and therefore we may venture at present to designate general theories, like those of Hutton and Werner, as _premature_.

This, indeed, is the sentiment of most of the good geologists of the present day. The time for such general systems, and for the fierce wars to which the opposition of such generalities gives rise, is probably now past for ever; and geology will not again witness such a controversy as that of the Wernerian and Huttonian schools. . . . . As when two black clouds With heaven's artillery fraught, come rattling on Over the Caspian: then stand front to front, Hovering a space, till winds the signal blow To join their dark encounter in mid-air. So frowned the mighty combatants, that hell Grew darker at their frown; so matched they stood: For never but once more was either like To meet so great a foe. {586}

The main points really affecting the progress of sound theoretical geology, will find a place in one of the two next Sections.

[2nd Ed.] [I think I do no injustice to Dr. Hutton in describing his theory of the earth as _premature_. Prof. Playfair's elegant work, _Illustrations of the Huttonian Theory_ (1802,) so justly admired, contains many doctrines which the more mature geology of modern times rejects; such as the igneous origin of chalk-flints, siliceous pudding stone, and the like; the universal formation of river-beds by the rivers themselves; and other points. With regard to this last-mentioned question, I think all who have read Deluc's _Geologie_ (1810) will deem his refutation of Playfair complete.

But though Hutton's theory was premature, as well as Werner's, the former had a far greater value as an important step on the road to truth. Many of its boldest hypotheses and generalizations have become a part of the general creed of geologists; and its publication is perhaps the greatest event which has yet occurred in the progress of Physical Geology.]

## CHAPTER VIII.

THE TWO ANTAGONIST DOCTRINES OF GEOLOGY.

_Sect._ 1.--_Of the Doctrine of Geological Catastrophes._

THAT great changes, of a kind and intensity quite different from the common course of events, and which may therefore properly be called _catastrophes_, have taken place upon the earth's surface, was an opinion which appeared to be forced upon men by obvious facts. Rejecting, as a mere play of fancy, the notions of the

destruction of the earth by cataclysms or conflagrations, of which we have already spoken, we find that the first really scientific examination of the materials of the earth, that of the Sub-Apennine hills, led men to draw this inference. Leonardo da Vinci, whom we have already noticed for his early and strenuous assertion of the real marine origin of fossil impressions of shells, also maintained that the bottom of the sea had become the top of the mountain; yet his mode of explaining this may perhaps be claimed by the modern advocates of uniform causes as more allied to their {587} opinion, than to the doctrine of catastrophes.[101\18] But Steno, in 1669, approached nearer to this doctrine; for he asserted that Tuscany must have changed its face at intervals, so as to acquire six different configurations, by the successive breaking down of the older strata into inclined positions, and the horizontal deposit of new ones upon them. Strabo, indeed, at an earlier period had recourse to earthquakes, to explain the occurrence of shells in mountains; and Hooke published the same opinion later. But the Italian geologists prosecuted their researches under the advantage of having, close at hand, large collections of conspicuous and consistent phenomena. Lazzaro Moro, in 1740, attempted to apply the theory of earthquakes to the Italian strata; but both he and his expositor, Cirillo Generelli, inclined rather to reduce the violence of these operations within the ordinary course of nature,[102\18] and thus leant to the doctrine of uniformity, of which we have afterwards to speak. Moro was encouraged in this line of speculation by the extraordinary occurrence, as it was deemed by most persons, of the rise of a new volcanic island from a deep part of the Mediterranean, near Santorino, in 1707.[103\18] But in other countries, as the geological facts were studied, the doctrine of catastrophes appeared to gain ground. Thus in England, where, through a large part of the country, the coal-measures are extremely inclined and contorted, and covered over by more horizontal fragmentary beds, the opinion that some violent catastrophe had occurred to dislocate them, before the superincumbent strata were deposited, was strongly held. It was conceived that a period of violent and destructive action must have succeeded to one of repose; and that, for a time, some unusual and paroxysmal forces must have been employed in elevating and breaking the pre-existing strata, and wearing their fragments into smooth pebbles, before nature subsided into a new age of tranquillity and vitality. In like manner Cuvier, from the alternations of fresh-water and salt-water species in the strata of Paris, collected the opinion of a series of great revolutions, in which "the thread of induction was broken." Deluc and others, to whom we owe the first steps in geological dynamics, attempted carefully to distinguish between causes now in action, and those which have ceased to act; in which latter class they reckoned the causes which have {588} elevated the existing continents. This distinction was assented to by many succeeding geologists. The forces which have raised into the clouds the vast chains of the Pyrenees, the Alps, the Andes, must have been, it was deemed, something very different from any agencies now operating.

[Note 101\18: "Here is a part of the earth which has become more light, and which rises, while the opposite part approaches nearer to the centre, and what was the bottom of the sea is become the top of the mountain."--Venturi's _Léonardo da Vinci_.]

[Note 102\18: Lyell, i. 3. p. 64. (4th ed.)]

[Note 103\18: Ib. p. 60.]

This opinion was further confirmed by the appearance of a complete change in the forms of animal and vegetable life, in passing from one formation to another. The species of which the remains occurred, were entirely different, it was said, in two successive epochs: a new creation appears to have intervened; and it was readily believed that a transition, so entirely out of the common course of the world, might be accompanied by paroxysms of mechanical energy. Such views prevail extensively among geologists up to the present time: for instance, in the comprehensive theoretical generalizations of Elie de Beaumont and others, respecting mountain-chains, it is supposed that, at certain vast intervals, systems of mountains, which may be recognized by the parallelism of course of their inclined beds, have been disturbed and elevated, lifting up with them the aqueous strata which had been deposited among them in the intervening periods of tranquillity, and which are recognized and identified by means of their organic remains: and according to the adherents of this hypothesis, these sudden elevations of mountain-chains have been followed, again and again, by mighty waves, desolating whole regions of the earth.

The peculiar bearing of such opinions upon the progress of physical geology will be better understood by attending to the _doctrine of uniformity_, which is opposed to them, and with the consideration of which we shall close our survey of this science, the last branch of our present task.

_Sect._ 2.--_Of the Doctrine of Geological Uniformity._

THE opinion that the history of the earth had involved a serious of catastrophes, confirmed by the two great classes of facts, the symptoms of mechanical violence on a very large scale, and of complete changes in the living things by which the earth had been tenanted, took strong hold of the geologists of England, France, and Germany. Hutton, though he denied that there was evidence of a beginning of the present state of things, and referred many processes in the formation of strata to existing causes, did not assert that the elevatory forces which raise continents from the bottom of the ocean, were of the same order, {589} as well as of the same kind, with the volcanoes and earthquakes which now shake the surface. His doctrine of uniformity was founded rather on the supposed analogy of other lines of speculation, than on the examination of the amount of changes now going on. "The Author of nature," it was said, "has not permitted in His works any symptom of infancy or of old age, or any sign by which we may estimate either their future or their past duration:" and the example of the planetary system was referred to in illustration of this.[104\18] And a general persuasion that the champions of this theory were not disposed to accept the usual opinions on the subject of creation, was allowed, perhaps very unjustly, to weigh strongly against them in the public opinion.

[Note 104\18: Lyell, i. 4, p. 94.]

While the rest of Europe had a decided bias towards the doctrine of geological catastrophes, the phenomena of Italy, which, as we have seen, had already tended to soften the rigor of that doctrine, in the progress of speculation from Steno to Generelli, were destined to mitigate it still more, by converting to the belief of

uniformity transalpine geologists who had been bred up in the catastrophist creed. This effect was, indeed, gradual. For a time the distinction of the _recent_ and the _tertiary_ period was held to be marked and strong. Brocchi asserted that a large portion of the Sub-Apennine fossil shells belonged to a living species of the Mediterranean Sea: but the geologists of the rest of Europe turned an incredulous ear to this Italian tenet; and the persuasion of the distinction of the tertiary and the recent period was deeply impressed on most geologists by the memorable labors of Cuvier and Brongniart on the Paris basin. Still, as other tertiary deposits were examined, it was found that they could by no means be considered as contemporaneous, but that they formed a chain of posts, advancing nearer and nearer to the recent period. Above the strata of the basins of London and Paris,[105\18] lie the newer strata of Touraine, of Bourdeaux, of the valley of the Bormida and the Superga near Turin, and of the basin of Vienna, explored by M. Constant Prevost. Newer and higher still than these, are found the Sub-Apennine formations of Northern Italy, and probably of the same period, the English "crag" of Norfolk and Suffolk. And most of these marine formations are associated with volcanic products and fresh-water deposits, so as to imply apparently a long train of alternations of corresponding processes. It may easily be supposed that, when the subject had assumed this form, the boundary of the present and past condition of the earth {590} was in some measure obscured. But it was not long before a very able attempt was made to obliterate it altogether. In 1828, Mr. Lyell set out on a geological tour through France and Italy.[106\18] He had already conceived the idea of classing the tertiary groups by reference to the number of recent species which were found in a fossil state. But as he passed from the north to the south of Italy, he found, by communication with the best fossil conchologists, Borelli at Turin, Guidotti at Parma, Costa at Naples, that the number of extinct species decreased; so that the last-mentioned naturalist, from an examination of the fossil shells of Otranto and Calabria, and of the neighboring seas, was of opinion that few of the tertiary shells were of extinct species. To complete the series of proof, Mr. Lyell himself explored the strata of Ischia, and found, 2000 feet above the level of the sea, shells, which were all pronounced to be of species now inhabiting the Mediterranean; and soon after, he made collections of a similar description on the flanks of Etna, in the Val di Noto, and in other places.

[Note 105\18: Lyell, 1st ed. vol. iii. p. 61.]

[Note 106\18: 1st ed. vol. iii. Pref.]

The impression produced by these researches is described by himself.[107\18] "In the course of my tour I had been frequently led to reflect on the precept of Descartes, that a philosopher should once in his life doubt everything he had been taught; but I still retained so much faith in my early geological creed as to feel the most lively surprize on visiting Sortino, Pentalica, Syracuse, and other parts of the Val di Noto, at beholding a limestone of enormous thickness, filled with recent shells, or sometimes with mere casts of shells, resting on marl in which shells of Mediterranean species were imbedded in a high state of preservation. All idea of [necessarily] attaching a high antiquity to a regularly-stratified limestone, in which the casts and impressions

of shells alone were visible, vanished at once from my mind. At the same time, I was struck with the identity of the associated igneous rocks of the Val di Noto with well-known varieties of 'trap' in Scotland and other parts of Europe; varieties which I had also seen entering largely into the structure of Etna.

[Note 107\18: Lyell, 1st ed. Pref. x.]

"I occasionally amused myself," Mr. Lyell adds, "with speculating on the different rate of progress which geology might have made, had it been first cultivated with success at Catania, where the phenomena above alluded to, and the great elevation of the modern tertiary beds in the Val di Noto, and the changes produced in the historical era by the Calabrian earthquakes, would have been familiarly known." {591}

Before Mr. Lyell entered upon his journey, he had put into the hands of the printer the first volume of his "Principles of Geology, being an attempt to explain the former Changes of the Earth's Surface _by reference to the Causes now in Operation_." And after viewing such phenomena as we have spoken of, he, no doubt, judged that the doctrine of catastrophes of a kind entirely different from the existing course of events, would never have been generally received, if geologists had at first formed their opinions upon the Sicilian strata. The boundary separating the present from the anterior state of things crumbled away; the difference of fossil and recent species had disappeared, and, at the same time, the changes of position which marine strata had undergone, although not inferior to those of earlier geological periods, might be ascribed, it was thought, to the same kind of earthquakes as those which still agitate that region. Both the supposed proofs of catastrophic transition, the organical and the mechanical changes, failed at the same time; the one by the removal of the fact, the other by the exhibition of the cause. The powers of earthquakes, even such as they now exist, were, it was supposed, if allowed to operate for an illimitable time, adequate to produce all the mechanical effects which the strata of all ages display. And it was declared that all evidence of a beginning of the present state of the earth, or of any material alteration in the energy of the forces by which it has been modified at various epochs, was entirely wanting.

Other circumstances in the progress of geology tended the same way. Thus, in cases where there had appeared in one country a sudden and violent transition from one stratum to the next, it was found, that by tracing the formations into other countries, the chasm between them was filled up by intermediate strata; so that the passage became as gradual and gentle as any other step in the series. For example, though the conglomerates, which in some parts of England overlie the coal-measures, appear to have been produced by a complete discontinuity in the series of changes; yet in the coal-fields of Yorkshire, Durham, and Cumberland, the transition is smoothed down in such a way that the two formations pass into each other. A similar passage is observed in Central-Germany, and in Thuringia is so complete, that the coal-measures have sometimes been considered as subordinate to the _todtliegendes_.[108\18]

[Note 108\18: De la Beche, p. 414, _Manual_.]

Upon such evidence and such arguments, the doctrine of {592} catastrophes was rejected with some contempt and ridicule; and it was maintained, that the operation of the causes of geological change may properly and philosophically be held to have been uniform through all ages and periods. On this opinion, and the grounds on which it he been urged, we shall make a few concluding remarks.

It must be granted at once, to the advocates of this geological uniformity, that we are not arbitrarily to assume the existence of catastrophes. The degree of uniformity and continuity with which terremotive forces have acted, must be collected, not from any gratuitous hypothesis, but from the facts of the case. We must suppose the causes which have produced geological phenomena, to have been as similar to existing causes, and as dissimilar, as the effects teach us. We are to avoid all bias in favor of powers deviating in kind and degree from those which act at present; a bias which, Mr. Lyell asserts, has extensively prevailed among geologists.

But when Mr. Lyell goes further, and considers it a merit in a course of geological speculation that it _rejects_ any difference between the intensity of existing and of past causes, we conceive that he errs no less than those whom he censures. "An _earnest and patient endeavor to reconcile_ the former indication of change,"[109\18] with _any_ restricted class of causes,--a habit which he enjoins,--is not, we may suggest, the temper in which science ought to be pursued. The effects must themselves teach us the nature and intensity of the causes which have operated; and we are in danger of error, if we seek for slow and shun violent agencies further than the facts naturally direct us, no less than if we were parsimonious of time and prodigal of violence. _Time_, inexhaustible and ever accumulating his efficacy, can undoubtedly do much for the theorist in geology; but _Force_, whose limits we cannot measure, and whose nature we cannot fathom, is also a power never to be slighted: and to call in the one to protect us from the other, is equally presumptuous, to whichever of the two our superstition leans. To invoke Time, with ten thousand earthquakes, to overturn and set on edge a mountain-chain, should the phenomena indicate the change to have been sudden and not successive, would be ill excused by pleading the obligation of first appealing to known causes.[110\18] {593}

[Note 109\18: Lyell, B. iv. c. i. p. 328, 4th ed.]

[Note 110\18: [2nd Ed.] [I have, in the text, quoted the fourth edition of Mr. Lyell's _Principles_, in which he recommends "an earnest and patient endeavor to reconcile the former indications of change with the evidence of gradual mutation now in progress." In the sixth edition, in that which is, I presume, the corresponding passage, although it is transferred from the fourth to the first Book (B. i. c. xiii. p. 325) he recommends, instead, "an earnest and patient inquiry how far geological appearances are reconcileable with the effect of changes now in progress." But while Mr. Lyell has thus softened the advocate's character in his language in this passage, the transposition which I have noticed appears to me to have an opposite tendency. For in the former edition, the causes now in action were first described in the second and third Books, and the great problem of Geology, stated in the first Book, was attempted to be solved in the fourth. But by incorporating this fourth Book with the first, and thus prefixing to the study of existing causes arguments against the belief

of their geological insufficiency, there is an appearance as if the author wished his reader to be prepared by a previous pleading against the doctrine of catastrophes, before he went to the study of existing causes. The Doctrines of Catastrophes and of Uniformity, and the other leading questions of the Palætiological Sciences, are further discussed in the _Philosophy of the Inductive Sciences_, Book x.]]

In truth, we know causes only by their effects; and in order to learn the nature of the causes which modify the earth, we must study them through all ages of their action, and not select arbitrarily the period in which we live as the standard for all other epochs. The forces which have produced the Alps and Andes are known to us by experience, no less than the forces which have raised Etna to its present height; for we learn their amount in both cases by their results. Why, then, do we make a merit of using the latter case as a measure for the former? Or how can we know the true scale of such force, except by comprehending in our view all the facts which we can bring together?

In reality when we speak of the _uniformity_ of nature, are we not obliged to use the term in a very large sense, in order to make the doctrine at all tenable? It includes catastrophes and convulsions of a very extensive and intense kind; what is the limit to the violence which we must allow to these changes? In order to enable ourselves to represent geological causes as operating with uniform energy through all time, we must measure our time by long cycles, in which repose and violence alternate; how long may we extend this cycle of change, the repetition of which we express by the word _uniformity_?

And why must we suppose that all our experience, geological as well as historical, includes more than _one_ such cycle? Why must we insist upon it, that man has been long enough an observer to obtain the _average_ of forces which are changing through immeasurable time? {594}

The analogy of other sciences has been referred to, as sanctioning this attempt to refer the whole train of facts to known causes. To have done this, it has been said, is the glory of Astronomy: she seeks no hidden virtues, but explains all by the force of gravitation, which we witness operating at every moment. But let us ask, whether it would really have been a merit in the founders of Physical Astronomy, to assume that the celestial revolutions resulted from any selected class of known causes? When Newton first attempted to explain the motions of the moon by the force of gravity, and failed because the measures to which he referred were erroneous, would it have been philosophical in him, to insist that the difference which he found ought to be overlooked, since otherwise we should be compelled to go to causes other than those which we usually witness in action? Or was there any praise due to those who assumed the celestial forces to be the same with gravity, rather than to those who assimilated them with any other known force, as magnetism, till the calculation of the laws and amount of these forces, from the celestial phenomena, had clearly sanctioned such an identification? We are not to select a conclusion now well proved, to persuade ourselves that it would have been wise to assume it anterior to proof, and to attempt to philosophize in the method thus recommended.

Again, the analogy of Astronomy has been referred to, as confirming the assumption of perpetual uniformity. The analysis of the heavenly motions, it has been said, supplies no trace of a beginning, no promise of an end. But here, also, this analogy is erroneously applied. Astronomy, as the science of cyclical motions, has nothing in common with Geology. But look at Astronomy where she has an analogy with Geology; consider our knowledge of the heavens as a palætiological science;--as the study of a past condition, from which the present is derived by causes acting in time. Is there then no evidence of a beginning, or of a progress? What is the import of the Nebular Hypothesis? A luminous matter is condensing, solid bodies are forming, are arranging themselves into systems of cyclical motion; in short, we have exactly what we are told, on this analogy, we ought not to have;--the beginning of a world. I will not, to justify this argument, maintain the truth of the nebular hypothesis; but if geologists wish to borrow maxims of philosophizing from astronomy, such speculations as have led to that hypothesis must be their model.

Or, let them look at any of the other provinces of palætiological speculation; at the history of states, of civilization, of languages. We {595} may assume some _resemblance_ or connexion between the principles which determined the progress of government, or of society, or of literature, in the earliest ages, and those which now operate; but who has speculated successfully, assuming an _identity_ of such causes? Where do we now find a language in the process of formation, unfolding itself in inflexions, terminations, changes of vowels by grammatical relations, such as characterize the oldest known languages? Where do we see a nation, by its natural faculties, inventing writing, or the arts of life, as we find them in the most ancient civilized nations? We may assume hypothetically, that man's faculties develop themselves in these ways; but we see no such effects produced by these faculties, in our own time, and now in progress, without the influence of foreigners.

Is it not clear, in all these cases, that history does not exhibit a series of cycles, the aggregate of which may be represented as a uniform state, without indication of origin or termination? Does it not rather seem evident that, in reality, the whole course of the world, from the earliest to the present times, is but one cycle, yet unfinished;--offering, indeed, no clear evidence of the mode of its beginning; but still less entitling us to consider it as a repetition or series of repetitions of what had gone before?

Thus we find, in the analogy of the sciences, no confirmation of the doctrine of uniformity, as it has been maintained in Geology. Yet we discern, in this analogy, no ground for resigning our hope, that future researches, both in Geology and in other palætiological sciences, may throw much additional light on the question of the uniform or catastrophic progress of things, and on the earliest history of the earth and of man. But when we see how wide and complex is the range of speculation to which our analogy has referred us, we may well be disposed to pause in our review of science;--to survey from our present position the ground that we have passed over;--and thus to collect, so far as we may, guidance and encouragement to enable us to advance in the track which lies before us.

Before we quit the subject now under consideration, we may, however, observe, that what the analogy of science really teaches us, as the most promising means of promoting this science, is the strenuous cultivation of the two subordinate sciences, Geological Knowledge of Facts, and Geological Dynamics. These are the two provinces of knowledge--corresponding to Phenomenal Astronomy, and Mathematical Mechanics--which may lead on to the epoch of the Newton of {596} geology. We may, indeed, readily believe that we have much to do in both these departments. While so large a portion of the globe is geologically unexplored;--while all the general views which are to extend our classifications satisfactorily from one hemisphere to another, from one zone to another, are still unformed; while the organic fossils of the tropics are almost unknown, and their general relation to the existing state of things has not even been conjectured;--how can we expect to speculate rightly and securely, respecting the history of the whole of our globe? And if Geological Classification and Description are thus imperfect, the knowledge of Geological Causes is still more so. As we have seen, the necessity and the method of constructing a science of such causes, are only just beginning to be perceived. Here, then, is the point where the labors of geologists may be usefully applied; and not in premature attempts to decide the widest and abstrusest questions which the human mind can propose to itself.

It has been stated,[111\18] that when the Geological Society of London was formed, their professed object was to multiply and record observations, and patiently to await the result at some future time; and their favorite maxim was, it is added, that the time was not yet come for a General System of Geology. This was a wise and philosophical temper, and a due appreciation of their position. And even now, their task is not yet finished; their mission is not yet accomplished. They have still much to do, in the way of collecting Facts; and in entering upon the exact estimation of Causes, they have only just thrown open the door of a vast Labyrinth, which it may employ many generations to traverse, but which they must needs explore, before they can penetrate to the Oracular Chamber of Truth.

[Note 111\18: Lyell, B. i. c. iv. p. 103.]

I REJOICE, on many accounts, to find myself arriving at the termination of the task which I have attempted. One reason why I am glad to close my history is, that in it I have been compelled, especially in the latter part of my labors, to speak as a judge respecting eminent philosophers whom I reverence as my Teachers in those very sciences on which I have had to pronounce a judgment;--if, indeed, even the appellation of Pupil be not too presumptuous. But I doubt not that such men are as full of candor and tolerance, as they are of knowledge and thought. And if they deem, as I did, that such a history of {597} science ought to be attempted, they will know that it was not only the historian's privilege, but his duty, to estimate the import and amount of the advances which he had to narrate; and if they judge, as I trust they will, that the attempt has been made with full integrity of intention and no want of labor, they will look upon the inevitable imperfections of the execution of my work with indulgence and hope.

There is another source of satisfaction in arriving at this point of my labors. If, after our long wandering through the region of physical science, we were left with

minds unsatisfied and unraised, to ask, "Whether this be all?"--our employment might well be deemed weary and idle. If it appeared that all the vast labor and intense thought which has passed under our review had produced nothing but a barren Knowledge of the external world, or a few Arts ministering merely to our gratification; or if it seemed that the methods of arriving at truth, so successfully applied in these cases, aid us not when we come to the higher aims and prospects of our being;--this History might well be estimated as no less melancholy and unprofitable than those which narrate the wars of states and the wiles of statesmen. But such, I trust, is not the impression which our survey has tended to produce. At various points, the researches which we have followed out, have offered to lead us from matter to mind, from the external to the internal world; and it was not because the thread of investigation snapped in our hands, but rather because we were resolved to confine ourselves, for the present, to the material sciences, that we did not proceed onwards to subjects of a closer interest. It will appear, also, I trust, that the most perfect method of obtaining speculative truth,--that of which I have had to relate the result,--is by no means confined to the least worthy subjects; but that the Methods of learning what is really true, though they must assume different aspects in cases where a mere contemplation of external objects is concerned, and where our own internal world of thought, feeling, and will, supplies the matter of our speculations, have yet a unity and harmony throughout all the possible employments of our minds. To be able to trace such connexions as this, is the proper sequel, and would be the high reward, of the labor which has been bestowed on the present work. And if a persuasion of the reality of such connexions, and a preparation for studying them, have been conveyed to the reader's mind while he has been accompanying me through our long survey, his time may not have been employed on {598} these pages in vain. However vague and hesitating and obscure may be such a persuasion, it belongs, I doubt not, to the dawning of a better Philosophy, which it may be my lot, perhaps, to develop more fully hereafter, if permitted by that Superior Power to whom all sound philosophy directs our thoughts.

{{599}} ADDITIONS TO THE THIRD EDITION.

# BOOK VIII.

ACOUSTICS.

## CHAPTER III.

SOUND.

_The Velocity of Sound in Water._

THE Science of which the history is narrated in this Book has for its objects, the minute Vibrations of the parts of bodies such as those by which Sounds are produced, and the properties of Sounds. The Vibrations of bodies are the result of a certain tension of their structure which we term _Elasticity_. The Elasticity determines the rate of Vibration: the rate of Vibration determines the audible note: the Elasticity

determines also the velocity with which the vibration travels through the substance. These points of the subject, Elasticity, Rate of Vibration, Velocity of Propagation, Audible Note, are connected in each substance, and are different in different substances.

In the history of this Science, considered as tending to a satisfactory general theory, the Problems which have obviously offered themselves were, to explain the properties of Sounds by the relations of their constituent vibrations; and to explain the existence of vibrations by the elasticity of the substances in which they occurred: as in Optics, philosophers have explained the phenomenon of light and colors by the Undulatory Theory, and are still engaged in explaining the requisite modulations by means of the elasticity of the Ether. But the _Undulatory Theory of Sound_ was seen to be true at an early period of the Science: and the explanation, in a general way at least, of all kinds of such undulations by means of the elasticity of the vibrating substances has been performed by a series of mathematicians of whom I have given an account in this Book. Hence the points of the subject already mentioned (Elasticity, Vibrations and their Propagations, {600} and Note), have a known material dependence, and each may be employed in determining the other: for instance, the Note may be employed in determining the velocity of sound and the elasticity of the vibrating substance.

Chladni,[1\B] and the Webers,[2\B] had made valuable experimental inquiries on such subjects. But more complete investigations of this kind have been conducted with care and skill by M. Wertheim.[3\B] For instance, he has determined the velocity with which sound travels in water, by making an organ-pipe to sound by the passage of water through it. This is a matter of some difficulty; for the mouthpiece of an organ-pipe, if it be not properly and carefully constructed, produces sounds of its own, which are not the genuine musical note of the pipe. And though the note depends mainly upon the length of the pipe, it depends also, in a small degree, on the breadth of the pipe and the size of the mouthpiece.

[Note 1\B: _Traité d'Acoustique_, 1809.]

[Note 2\B: _Wellenlehre_, 1852.]

[Note 3\B: _Mémoires de Physique Mécanique_. Paris, 1848.]

If the pipe were a mere line, the time of a vibration would be the time in which a vibration travels from one end of the pipe to the other; and thus the note for a given length (which is determined by the time of vibration), is connected with the velocity of vibration. He thus found that the velocity of a vibration along the pipe in sea-water is 1157 _mètres_ per second.

But M. Wertheim conceived that he had previously shown, by general mathematical reasoning, that the velocity with which sound travels in an unlimited expanse of any substance, is to the velocity with which it travels along a pipe or linear strip of the same substance as the square root of 3 to the square root of 2. Hence the velocity of sound in sea-water would be 1454 _mètres_ a second. The velocity of sound in air is 332 _mètres_.

M. Wertheim also employed the vibrations of rods of steel and other metals in order to determine their _modulus of elasticity_--that is, the quantity which determines for each substance, the extent to which, in virtue of its elasticity, it is compressed and expanded by given pressures or tensions. For this purpose he caused the rod to vibrate near to a tuning-fork of given pitch, so that both the rod and the tuning-fork by their vibrations traced undulating curves on a revolving disk. The curves traced by the two could be compared so as to give their relative rate, and thus to determine the elasticity of the substance.

{{601}} BOOK IX.

PHYSICAL OPTICS.

_Photography._

I HAVE, at the end of Chapter xi., stated that the theory of which I have endeavored to sketch the history professes to explain only the phenomena of radiant visible light; and that though we know that light has other properties--for instance, that it produces chemical effects--these are not contemplated as included within the domain of the theory. The chemical effects of light cannot as yet be included in exact and general truths, such as those which constitute the undulatory theory of radiant visible light. But though the present age has not yet attained to a _Science_ of the chemistry of Light, it has been enriched with a most exquisite _Art_, which involves the principles of such a science, and may hereafter be made the instrument of bringing them into the view of the philosopher. I speak of the Art of _Photography_, in which chemistry has discovered the means of producing surfaces almost as sensitive to the modifications of light as the most sensitive of organic textures, the retina of the eye: and has given permanence to images which in the eye are only momentary impressions. Hereafter, when the laws shall have been theoretically established, which connect the chemical constitution of bodies with the action of light upon them, the prominent names in the Prelude to such an Epoch must be those who by their insight, invention, and perseverance, discovered and carried to their present marvellous perfection the processes of photographic Art:--Niepce and Daguerre in France, and our own accomplished countryman, Mr. Fox Talbot.

_Fluorescence._

As already remarked, it is not within the province of the undulatory theory to explain the phenomena of the absorption of light which take place in various ways when the light is transmitted through various {602} mediums. I have, at the end of Chapter iii., given the reasons which prevent my assenting to the assertion of a special analysis of light by absorption. In the same manner, with regard to other effects produced by media upon light, it is sufficient for the defence of the theory that it should be consistent with the possibility of the laws of phenomena which are observed, not that it should explain those laws; for they belong, apparently, to another province of philosophy.

Some of the optical properties of bodies which have recently attracted notice appear to be of this kind. It was noticed by Sir John Herschel,[4\B] that a certain liquid, sulphate of quinine, which is under common circumstances colorless, exhibits

in certain aspects and under certain incidences of light, a beautiful celestial blue color. It appeared that this color proceeded from the surface on which the light first fell; and color thus produced Sir J. Herschel called _epipolic_ colors, and spoke of the light as _epipolized_. Sir David Brewster had previously noted effects of color in transparent bodies which he ascribed to internal dispersion:[5\B] and he conceived that the colors observed by Sir J. Herschel were of the same class. Professor Stokes[6\B] of Cambridge applied himself to the examination of these phenomena, and was led to the conviction that they arise from a power which certain bodies possess, of changing the color, and with it, the refrangibility of the rays of light which fall upon them: and he traced this property in various substances, into various remarkable consequences. As this change of refrangibility always makes the rays _less_ refrangible, it was proposed to call it a _degradation_ of the light; or again, _dependent emission_, because the light is emitted in the manner of self-luminous bodies, but only in dependence upon the active rays, and so long as the body is under their influence. In this respect it differs from _phosphorescence_, in which light is emitted without such dependence. The phenomenon occurs in a conspicuous and beautiful manner in certain kinds of fluor spar: and the term _fluorescence_, suggested by Professor Stokes, has the advantage of inserting no hypothesis, and will probably be found the most generally acceptable.[7\B]

[Note 4\B: _Phil. Trans._ 1845.]

[Note 5\B: _Edinb. Trans._ 1833.]

[Note 6\B: _Phil. Trans._ 1852 and 1854.]

[Note 7\B: _Phil. Trans._ 1852.]

It may be remarked that Professor Stokes rejects altogether the doctrine that light of definite refrangibility may still be compound, and maybe analysed by absorption. He says, "I have not overlooked the remarkable effect of absorbing media in causing apparent changes {603} of color in a pure spectrum; but this I believe to be a subjective phenomenon depending upon contrast."

## CHAPTER XIII.

UNDULATORY THEORY.

_Direction of the Transverse Vibrations in Polarization._

IN the conclusion of Chapter xiii. I have stated that there is a point in the undulatory theory which was regarded as left undecided by Young and Fresnel, and on which the two different opinions have been maintained by different mathematicians; namely, whether the vibrations of polarized light are perpendicular to the plane of polarization or in that plane. Professor Stokes of Cambridge has attempted to solve this question in a manner which is, theoretically, exceedingly ingenious, though it is difficult to make the requisite experiments in a decisive manner. The method may be briefly described.

If polarized light be _diffracted_ (see Chap. xi. sect. 2), each ray will be bent from its position, but will still be polarized. The original ray and the diffracted ray, thus forming a broken line, may be supposed to be connected at the angle by a universal joint (called a _Hooke's Joint_), such that when the original ray turns about its axis, the diffracted ray also turns about its axis; as in the case of the long handle of a telescope and the screw which is turned by it. Now if the motion of the original ray round its axis be uniform, the motion of the diffracted ray round its axis is not uniform: and hence if, in a series of cases, the planes of polarization of the original ray differ by equal angles, in the diffracted ray the planes of polarization will differ by unequal angles. Then if vibrations be perpendicular to the plane of polarization, the planes of polarization in the diffracted rays will be crowded together in the neighborhood of the plane in which the diffraction takes place, and will be more rarely distributed in the neighborhood of the plane perpendicular to this, in which is the diffracting thread or groove.

On making the experiment, Prof. Stokes conceived that he found, in his experiments, such a crowding of the planes of diffracted polarization towards the plane of diffraction; and thus he held that the {604} hypothesis that the transverse vibrations which constitute polarization are perpendicularly transverse to the plane of polarization was confirmed.[8\B]

[Note 8\B: _Camb. Trans._, vol. ix. part i. 1849.]

But Mr. Holtzmann,[9\B] who, assenting to the reasoning, has made the experiment in a somewhat different manner, has obtained an opposite result; so that the point may be regarded as still doubtful.

[Note 9\B: _Phil. Mag._, Feb. 1857.]

_Final Disproof of the Emission Theory._

As I have stated in the History, we cannot properly say that there ever was an Emission Theory of Light which was the _rival_ of the Undulatory Theory: for while the undulatory theory provided explanations of new classes of phenomena as fast as they arose, and exhibited a _consilience_ of theories in these explanations, the hypothesis of emitted particles required new machinery for every new set of facts, and soon ceased to be capable even of expressing the facts. The simple cases of the ordinary reflexion and refraction of light were explained by Newton on the supposition that the transmission of light is the motion of particles: and though his explanation includes a somewhat harsh assumption (that a refracting surface exercises an attractive force through a _fixed finite_ space), the authority of his great name gave it a sort of permanent notoriety, and made it to be regarded as a standard point of comparison between a supposed "Emission Theory" and the undulation theory. And the way in which the theories were to be tested in this case was obvious: in the Newtonian theory, the velocity of light is increased by the refracting medium; in the undulatory theory, it is diminished. On the former hypothesis the velocity of light in air and in water is as 3 to 4; in the latter, as 4 to 3.

But the immense velocity of light made it appear impossible to measure it, within the limits of any finite space which we can occupy with refracting matter. The

velocity of light is known from astronomical phenomena;--from the eclipses of Jupiter's satellites, by which it appears that light occupies 8 minutes in coming from the sun to the earth; and from the aberration of light, by which its velocity is shown to be 10,000 times the velocity of the earth in its orbit. Is it, then, possible to make apparent so small a difference as that between its passing through a few yards of air and of water?

Mr. Wheatstone, in 1831, invented a machine by which this could {605} be done. His object was to determine the velocity of the electric shock. His apparatus consisted in a small mirror, turning with great velocity about an axis which is in its own plane, like a coin spinning on its edge. The velocity of spinning may be made so great, that an object reflected shall change its place perceptibly after an almost inconceivably small fraction of a second. The application of this contrivance to measure the velocity of light, was, at the suggestion of Arago, who had seen the times of the rival theories of light, undertaken by younger men at Paris, his eyesight not allowing him to prosecute such a task himself. It was necessary that the mirrors should turn more than 1000 times in a second, in order that the two images, produced, one by light coming through air, and the other by light coming through an equal length of water, should have places perceptibly different. The mechanical difficulties of the experiment consisted in keeping up this great velocity by the machinery without destroying the machinery, and in transmitting the light without too much enfeebling it. These difficulties were overcome in 1850, by M. Fizeau and M. Léon Foucault separately: and the result was, that the velocity of light was found to be less in water than in air. And thus the Newtonian explanation of refraction, the last remnant of the Emission Theory, was proved to be false.

{{606}} BOOK X.

THERMOTICS.--ATMOLOGY.

## CHAPTER III.

THE RELATION OF VAPOUR AND AIR.

_Sect._ 4.--_Force of Steam._

THE experiments on the elastic force of steam made by the French Academy are fitted in an especial manner to decide the question between rival formulæ, in consequence of the great amount of force to which they extend; namely, 60 feet of mercury, or 24 atmospheres: for formulæ which give results almost indistinguishable in the lower part of the scale diverge widely at those elevated points. Mr. Waterston[10\B] has reduced both these and other experiments to a rule in the following manner:--He takes the zero of gaseous tension, determined by other experimenters (Rudberg, Magnus, and Regnault,) to be 461° below the zero of Fahrenheit, or 274° below the zero of the centigrade scale: and temperatures reckoned from this zero he calls "G temperatures." The square root of the G temperatures is the element to which the elastic force is referred (for certain theoretical reasons), and it is found that the density of steam is as the _sixth power_

of this element. The agreement of this rule with the special results is strikingly close. A like rule was found by him to apply generally to many other gases in contact with their liquids.

[Note 10\B: _Phil. Trans._ 1852.]

But M. Regnault has recently investigated the subject in the most complete and ample manner, and has obtained results somewhat different.[11\B] He is led to the conclusion that no formula proceeding by {607} a power of the temperature can represent the experiments. He also finds that the rule of Dalton (that as the temperatures increase in arithmetical progression, the elastic force increases in geometric progression) deviates from the observations, especially at high temperatures. Dalton's rule would be expressed by saying that the variable part of the elastic force is as _a^t_, where _t_ is the temperature. This failing, M. Regnault makes trial of a formula suggested by M. Biot, consisting of a sum of two terms, one of which is as _a^t_, and the other is _b^t_: and in this way satisfies the experiments very closely. But this can only be considered as a formula of interpolation, and has no theoretical basis. M. Roche had proposed a formula in which the force is as _a^z_, _z_ depending upon the temperature by an equation[12\B] to which he had been led by theoretical considerations. This agrees better with observation than any other formula which includes only the same number of coefficients.

[Note 11\B: _Mém. de l'Institut_, vol. xxi. (1847). M. Regnault's Memoir occupies 767 pages.]

[Note 12\B: The equation _z_ = _t_/(1 + _mt_).]

Among the experimental thermotical laws referred to by M. Regnault are, the Law of Watt,[13\B] that "the quantity of heat which is required to convert a pint of water at a temperature of zero into steam, is the same whatever be the pressure." Also, the Law of Southern, that "the latent heat of vaporization, that is the heat absorbed in the passage from the liquid to the gaseous consistence, is constant for all purposes: and that we obtain the total heat in adding to the constant latent heat the number which represents the latent heat of steam." Southern found the latent heat of the steam of water to be represented by about 950 degrees of Fahrenheit.[14\B]

[Note 13\B: See Robison's _Mechanical Philosophy_, vol. ii. p. 8.]

[Note 14\B: Ib. p. 160.]

_Sect._ 5.--_Temperature of the Atmosphere._

I MAY notice, as important additions to our knowledge on this subject, the results of four balloon ascents made in 1852,[15\B] by the Committee of the Meteorological Observatory established at Kew by the British Association for the Advancement of Science. In these ascents the observers mounted to more than 13,000, 18,000, and 19,000 feet, and in the last to 22,370; by which ascent the temperature fell from 49 degrees to nearly 10 degrees below zero; and the dew-point fell from 37° to 12°. Perhaps the most marked result of these observations is the {608} following:--The temperature of the air decreases uniformly as we ascend above the earth's surface; but this decrease does not go on continuously. At a certain elevation, varying on

different days, the decrease is arrested: and for a depth of two or three thousand feet of air, the temperature decreases little, or even increases in ascending. Above this, the diminution again takes place at nearly the same rate as in the lower regions. This intermediate region of undecreasing temperature extended in the various ascents, from about altitude 4000 to 6000 feet, 6500 to 10,000, 2000 to 4500, and 4000 to 8000. This interruption in the decrease of temperature is accompanied by a large and abrupt fall in the temperature of the dew-point, or by an actual condensation of vapor. Thus, this region is the _region of the clouds_, and the increase of heat appears to arise from the latent heat liberated when aqueous vapor is formed into clouds.

[Note 15\B: _Phil. Trans._ 1853.]

## CHAPTER IV.

THEORIES OF HEAT.

_The Dynamical Theory of Heat._

THAT the transmission of _radiant_ Heat takes place by means of the vibrations of a medium, as the transmission of Sound certainly does, and the transmission of Light most probably, is a theory which, as I have endeavored to explain, has strong arguments and analogies in its favor. But that Heat itself, in its essence and quantity, is Motion is a hypothesis of quite another kind. This hypothesis has been recently asserted and maintained with great ability. The doctrine thus asserted is, that Motion may be converted into Heat, and Heat into Motion; that Heat and Motion may produce each other, as we see in the rarefaction and condensation of air, in steam-engines, and the like: and that in all such cases the Motion produced and the Heat expended exactly measure each other. The foundation of this theory is conceived to have been laid by Mr. Joule of Manchester, in 1844: and it has since been prosecuted by him and by Professor Thomson of Glasgow, by experimental investigations of various kinds. It is difficult to make these experiments so as to be quite satisfactory; for it is {609} difficult to measure _all_ the heat gained or lost in any of the changes here contemplated. That friction, agitation of fluids, condensation of gases, conversion of gases into fluids and liquids into solids, produce heat, is undoubted: and that the quantity of such heat may be measured by the mechanical force which produces it, or which it produces, is a generalization which will very likely be found a fertile source of new propositions, and probably of important consequences.

As an example of the conclusions which Professor Thomson draws from this doctrine of the mutual conversion of motion and heat, I may mention his speculations concerning the cause which produces and sustains the heat of the sun.[16\B] He conceives that the support of the solar heat must be meteoric matter which is perpetually falling towards the globe of the sun, and has its motion converted into heat. He inclines to think that the meteors containing the stores of energy for future Sun-light must be principally within the earth's orbit; and that we actually see them there as the "Zodiacal Light," an illuminated shower, or rather tornado, of stones. The

inner parts of this tornado are always getting caught in the Sun's atmosphere, and drawn to his mass by gravitation.

[Note 16\B: On the Mechanical Energies of the Solar System. _Edinb. Trans._ vol. XXI. part i. (1854), p. 67.]

{{610}} BOOK XI.

ELECTRICITY.

GENERAL REMARKS.

ELECTRICITY in the form in which it was originally studied--Franklinic, frictional, or statical electricity--has been so completely identified with electricity in its more comprehensive form--Voltaic, chemical, or dynamical electricity--that any additions we might have to make to the history of the earlier form of the subject are included in the later science.

There are, however, several subjects which may still be regarded rather as branches of Electricity than of the Cognate Sciences. Such are, for instance, Atmospheric Electricity, with all that belongs to Thunderstorms and Lightning Conductors. The observation of Atmospheric Electricity has been prosecuted with great zeal at various meteorological observatories; and especially at the Observatory established by the British Association at Kew. The Aurora Borealis, again, is plainly an electrical phenomenon; but probably belonging rather to dynamical than to statical electricity. For it strongly affects the magnetic needle, and its position has reference to the direction of magnetism; but it has not been observed to affect the electroscope. The general features of this phenomenon have been described by M. de Humboldt, and more recently by M. de Bravais; and theories of the mode of its production have been propounded by MM. Biot, De la Rive, Kaemtz, and others.

Again, there are several fishes which have the power of giving an electrical shock:--the torpedo, the gymnotus, and the silurus. The agency of these creatures has been identified with electricity in the most general sense. The peculiar energy of the animal has been made to produce the effects which are produced by an electrical discharge or a voltaic current:--not only to destroy life in small animals, but to {611} deflect a magnet, to make a magnet, to decompose water, and to produce a spark.

_Dr. Faraday's Views of Statical Electric Induction._

According to the theories of electricity of Æpinus and Coulomb, which in this Book of our History are regarded as constituting a main part of the progress of this portion of science, the particles of the electric fluid or fluids exert forces, attractive and repulsive, upon each other in straight lines at a distance, in the same way in which, in the Newtonian theory of the universe, the particles of matter are conceived as exerting attractive forces upon each other. An electrized body presented a conducting body of any form, determines a new arrangement of the electric fluids in the conductor, attracting the like fluid to its own side, and repelling the opposite fluid to the opposite side. This is Electrical _Induction_. And as, by the theory, the attraction is greater at the smaller distances, the distribution of the fluid upon the conductor in virtue of this Induction will not be symmetrical, but will be governed by

laws which it will require a complex and difficult calculation to determine--as we have seen was the case in the investigations of Coulomb, Poisson, and others.

Instead of this action at a distance. Dr. Faraday has been led to conceive Electrical Induction to be the result of an action taking place between the electrized body and the conductor through lines of contiguous particles in the mass of the intermediate body, which he calls the _Dielectric_. And the irregularities of the distribution of the electricity in these cases of Induction, and indeed the existence of an action in points protected from direct action by the protuberant sides of the conductor, are the causes, I conceive, which lead him to the conclusion that Induction takes place in _curved lines_[17\B] of such contiguous particles.

[Note 17\B: _Researches_, 1165, &c.]

With reference to this, I may remark that, as I have said, the distribution of electricity on a conductor in the presence of an electrized body is so complex a mathematical problem that I do not conceive any merely popular way of regarding the result can entitle us to say, that the distribution which we find cannot be explained by the Coulombian theory, and must force us upon the assumption of an action in curved lines:--which is, indeed, itself a theory, and so vague a one {612} that it requires to be made much more precise before we can say what consequences it does or does not lead to. Professor W. Thomson has arrived at a mathematical proof that the effect of induction on the view of Coulomb and of Faraday must, under certain conditions, be necessarily and universally the same.

With regard to the influence of different _Dielectrics_ upon Induction, the inquiry appears to be of the highest importance; and may certainly necessitate some addition to the theory.

{{613}} BOOK XII.

MAGNETISM.

_Recent Progress of Terrestrial Magnetism._

IN Chapter II., I have noticed the history of Terrestrial Magnetism; Hansteen's map published in 1819; the discovery of "magnetic storms" about 1825; the chain of associated magnetic observations, suggested by M. de Humboldt, and promoted by the British Association and the Royal Society; the demand for the continuation of these till 1848; the magnetic observations made in several voyages; the magnetic surveys of various countries. And I have spoken also of Gauss's theory of Terrestrial Magnetism, and his directions and requirements concerning the observations to be made. I may add a few words with regard to the more recent progress of the subject.

The magnetic observations made over large portions of the Earth's surface by various persons, and on the Ocean by British officers, have been transmitted to Woolwich, where they have been employed by General Sabine in constructing magnetic maps of the Earth for the year 1840.[18\B] Following the course of inquiry described in the part of the history referred to, these maps exhibit the declination, inclination, and intensity of the magnetic force at every point of the earth's surface. The curves which mark equal amounts of each of these three elements (the _lines of

equal declination_, _inclination_, and _force_:--the _isogonal_, the _isoclinal_, and the _isodynamic_ lines,) are, in their general form, complex and irregular; and it has been made a matter of question (the facts being agreed upon) whether it be more proper to say that they indicate four poles, as Halley and as Hansteen said, or only two poles, as Gauss asserts. The matter appears to become more clear if we draw magnetic _meridians_; that is, lines obtained by following the directions, or pointings, of the magnetic needle to the north or to {614} the south, till we arrive at the points of convergence of all their directions; for there are only two such poles, one in the Arctic and one in the Antarctic region. But in consequence of the irregularity of the magnetic constitution of the earth, if we follow the inclination of the magnetic force round the earth on any parallel of latitude, we find that it has two _maxima_ and two _minima_, as if there were four magnetic poles. The isodynamic map is a new presentation of the facts of this subject; the first having been constructed by Colonel Sabine in 1837.

[Note 18\B: These maps are published in Mr. Keith Johnstone's _Physical Atlas_.]

I have stated also that the magnetic elements at each place are to be observed in such a manner as to bring into view both their _periodical_, their _secular_, and their _irregular_ or _occasional_ changes. The observations made at Toronto in Canada, and at Hobart Town in Van Diemen's Land, two stations at equal distances from the two poles of the earth, and also at St Helena, a station within the tropics, have been discussed by General Sabine with great care, and with an amount of labor approaching to that employed upon reductions of astronomical observations. And the results have been curious and unexpected.

The declination was first examined.[19\B] This magnetical element is, as we have already seen (p. 232), liable both to a diurnal and to an annual inequality; and also to irregular perturbations which have been termed magnetic storms. Now it was found that all these inequalities went on increasing gradually and steadily from 1843 to 1848, so as to become, at the end of that time, above twice as large as they were at the beginning of it. A new periodical change in all these elements appeared to be clearly established by this examination. M. Lamont, of Munich, had already remarked indications of a decennial period in the diurnal variation of the declination of the needle. The duration of the period from minimum to maximum being about five years, and therefore the whole period about ten years. The same conclusion was found to follow still more decidedly from the observations of the dip and intensity.

[Note 19\B: _Phil. Trans._ 1852 and 1856.]

This period of ten years had no familiar meaning in astronomy; and if none such had been found for it, its occurrence as a magnetic period must have been regarded, as General Sabine says,[20\B] in the light of a fragmentary fact. But it happened about this time that the scientific world was made aware of the existence of a like period in a {615} phenomenon which no one would have guessed to be connected with terrestrial magnetism, namely, the spots in the Sun. M. Schwabe, of Dessau, had observed the Sun's disk with immense perseverance for 24 years:--often examining it more than 300 days in the year; and had found that the spots had, as to their quantity

and frequency, a periodical character. The years of maximum are 1828, 1838, 1848, in which there were respectively 225,[21\B] 282, 330 groups of spots. The minimum years, 1833, 1843, had only 33 and 34 such groups. This curious fact[22\B] was first made public by M. de Humboldt, in the third volume of his _Kosmos_ (1850). The coincidence of the periods and epochs of these two classes of facts was pointed out by General Sabine in a Memoir presented to the Royal Society in March, 1852.

[Note 20\B: _Phil. Trans._ 1856, p. 382.]

[Note 21\B: In 1837 there were 333.]

[Note 22\B: The observations up to 1844 were published in Poggendorf's _Annalen_.]

Of course it was natural to suppose, even before this discovery, that the diurnal and annual inequalities of the magnetic element at each place depend upon the action of the sun, in some way or other.

Dr. Faraday had endeavored to point out how the effect of the solar heat upon the atmosphere would, according to the known relations of heat and magnetism, explain many of the phenomena. But this new feature of the phenomena, their quinquennial increase and decrease, makes us doubt whether such an explanation can really be the true one.

Of the _secular_ changes in the magnetic elements, not much more is known than was known some years ago. These changes go on, but their laws are imperfectly known, and their causes not even conjectured. M. Hansteen, in a recent memoir,[23\B] says that the decrease of the inclination goes on progressively diminishing. With us this rate of decrease appears to be at present nearly uniform. We cannot help conjecturing that the sun, which has so plain a connexion with the diurnal, annual, and occasional movements of the needle, must also have some connexion with its secular movements.

[Note 23\B: See K. Johnstone's _Physical Atlas_.]

In 1840 the observations made at various places had to a great extent enabled Gauss, in connexion with W. Weber, to apply his Theory to the actual condition of the Earth;[24\B] and he calculated the Declination, Inclination, and Intensity at above 100 places, and found {616} the agreement, as he says, far beyond his hopes. They show, he says, that the Theory comes near to the Truth.

[Note 24\B: _Atlas des Erdmagnetismus nach den Elementen der Theorie Entworfen_. See Preface.]

_Correction of Ship's Compasses._

The magnetic needle had become of importance when it was found that it always pointed to the North. Since that time the history of magnetism has had its events reflected in the history of navigation. The change of the declination arising from a change of place terrified the companions of Columbus. The determination of the laws of this change was the object of the voyage of Halley; and has been pursued with the utmost energy in the Arctic and Antarctic regions by navigators up to the present

time. Probably the dependence of the magnetic declination upon place is now known well enough for the purposes of navigation. But a new source of difficulty has in the meantime come into view; the effect of the iron in the ship upon the Compass. And this has gone on increasing as guns, cables, stays, knees, have been made of iron; then steam-engines with funnels, wheels, and screws, have been added; and finally the whole ship has been made of iron. How can the compass be trusted in such cases?

I have already said in the history that Mr. Barlow proposed to correct the error of the compass by placing near to the compass an iron plate, which from its proximity to the compass might counterbalance magnetically the whole effect of the ship's iron upon the compass. This correction was not effectual, because the magnetic forces of the plate and of the ship do not change their direction and value according to the same law, with the change of position. I have further stated that Mr. Airy devised other means of correcting the error. I may add a few words on the subject; for the subject has been further examined by Mr. Airy[25\B] and by others.

[Note 25\B: _Phil. Trans._ 1856.]

It appears, by mathematical reasoning, that the magnetic effect of the iron in a ship may be regarded as producing two kinds of deviation which are added together;--a "polar-magnet deviation," which changes from positive to negative as the direction of the ship's keel, in a horizontal revolution, passes from semicircle to semicircle; and a "quadrantal deviation," which changes from positive to negative as the keel turns from quadrant to quadrant. The latter deviation may be remedied completely by a mass of unmagnetized iron placed on a level {617} with the compass, either in the athwartship line or in the fore-and-aft line, according to circumstances. "The polar-magnet-deviation" may be corrected at _any given place_ by a magnet or magnets, but the magnets thus applied at one place will not always correct the deviation in another magnetic latitude. For it appears that this deviation arises partly from a magnetism inherent in the materials of the ship, not changing with the change of magnetic position, and partly from the effect of terrestrial magnetism upon the ship's iron. But the errors arising from both sources may be remedied by adjusting, at a new locality, the positions of the corrective magnets.

The inherent magnetism of the ship, of which I have spoken, may be much affected by the position in which the ship was built; and may change from time to time; for instance, by the effect of the battering of the waves, and other causes. Hence it is called by Mr. Airy "sub-permanent magnetism."

Another method of correcting the errors of a ship's compass has been proposed, and is used to some extent; namely, by _swinging_ the ship round (in harbor) to all points of azimuth, and thus constructing a _Table of Compass Errors_ for that particular ship. But to this method it is objected that the Table loses its value in a new magnetic latitude much more than the correction by magnets does; besides the inconveniences of steering a ship by a Table.

{{618}} BOOK XIII.

VOLTAIC ELECTRICITY.

# CHAPTER VII.

MAGNETO-ELECTRIC INDUCTION.

FARADAY'S discovery that, in combinations like those in which a voltaic current was known to produce motion, motion would produce a voltaic current, naturally excited great attention among the scientific men of Europe. The general nature of his discovery was communicated by letter[26\B] to M. Hachette at Paris, in December, 1831; and experiments having the like results were forthwith made by MM. Becquerel and Ampère at Paris, and MM. Nobili and Antinori at Florence.

[Note 26\B: _Ann. de Chimie_, vol. xlviii. (1831), p. 402.]

It was natural also that in a case in which the relations of space which determine the results are so complicated, different philosophers should look at them in different ways. There had been, from the first discovery by Oersted of the effect of a voltaic current upon a magnet, two rival methods of regarding the facts. Electric and magnetic lines exert an effort to place themselves transverse to each other (see chapter iv. of this Book), and (as I have already said) two ways offered themselves of simplifying this general truth:--to suppose an electric current made up of transverse magnetic lines; or to suppose magnetic lines made up of transverse electric currents. On either of these assumptions, the result was expressed by saying that _like_ currents or lines (electric or magnetic) tend to place themselves parallel; which is a law more generally intelligible than the law of transverse position. Faraday had adopted the former view; had taken the lines of magnetic force for the fundamental lines of his system, and defined the direction of the magneto-electric current of induction by the relation {619} of the motion to these lines. Ampère, on the other hand, supposed the magnet to be made up of transverse electric currents (chap. vi.); and had deduced all the facts of electro-dynamical action, with great felicity, from this conception. The question naturally arose, in what manner, on this view, were the new facts of magneto-electric induction by motion to be explained, or even expressed?

Various philosophers attempted to answer this question. Perhaps the form in which the answer has obtained most general acceptance is that in which it was put by Lenz, who discoursed on the subject to the Academy of St. Petersburg in 1833.[27\B] His general rule is to this effect: when a wire moves in the neighborhood of an electric current or a magnet, a current takes place in it, such as, existing independently, would have produced a motion opposite to the actual motion. Thus two parallel _forward_ currents move towards each other:--hence if a current move towards a parallel wire, it produces in it a _backward_ current. A moveable wire conducting a current _downwards_ will move round the north pole of a magnet in the direction N., W., S., E.:--hence if, when the wire have in it no current, we move it in the direction N., W., S., E., we produce in the wire an _upward_ current. And thus, as M. de la Rive remarks,[28\B] in cases in which the mutual action of two currents produces a limited motion, as attraction or repulsion, or a deviation right or left, the corresponding magneto-electric induction produces an instantaneous current only; but when the electrodynamic action produces a continued motion, the corresponding motion

produces, by induction, a continued current.

[Note 27\B: _Acad. Petrop._ Nov. 29, 1833. _Pogg. Ann._ vol. xxxi. p. 483.]

[Note 28\B: _Traité de l'Electricité_, vol. i. p. 441 (1854).]

Looking at this mode of stating the law, it is impossible not to regard this effect as a sort of reaction; and accordingly, this view was at once taken of it. Professor Ritchie said, in 1833, "The law is founded on the universal principle that action and reaction are equal." Thus, if voltaic electricity induce magnetism under certain arrangements, magnetism will, by similar arrangements, react on a conductor and induce voltaic electricity.[29\B]

[Note 29\B: On the Reduction of Mr. Faraday's discoveries in Magneto-electric Induction to a General Law. _Trans._ of R. S. in _Phil. Mag._ N.S. vol. iii. 37, and vol. iv. p. 11. In the second edition of this history I used the like expressions.]

There are still other ways of looking at this matter. I have elsewhere pointed out that where polar properties co-exist, they are {620} generally found to be connected,[30\B] and have illustrated this law in the case of electrical, magnetical, and chemical polarities. If we regard motion backwards and forwards, to the right and the left, and the like, as _polar_ relations, we see that magneto-electric induction gives us a new manifestation of connected polarities.

[Note 30\B: _Phil. Ind. Sc._ B. v. c. ii.]

_Diamagnetic Polarity_.

But the manifestation of co-existent polarities which are brought into view in this most curious department of nature is not yet exhausted by those which we have described. I have already spoken (chap. **vii.) of Dr. Faraday's discovery that there are diamagnetic as well as magnetic bodies; bodies which are repelled by the pole of a magnet, as well as bodies which are attracted. Here is a new opposition of properties. What is the exact definition of this opposition in connexion with other polarities? To this, at present, different philosophers give different answers. Some say that diamagnetism is completely the opposite of ordinary magnetism, or, as Dr. Faraday has termed it for the sake of distinction, of _paramagnetism_. They say that as a north pole of a magnet gives to the neighboring extremity of a piece of soft iron a south pole, so it gives to the neighboring extremity of a piece of bismuth a north pole, and that the bismuth becomes for a time an inverted magnet; and hence, arranges itself across the line of magnetised force, instead of along it. Dr. Faraday himself at first adopted this view;[31\B] but he now conceives that the bismuth is not made polar, but is simply repelled by the magnet; and that the transverse position which it assumes, arises merely from its elongated form, each end trying to recede as far as possible from the repulsive pole of the magnet.

[Note 31\B: Faraday's _Researches_, Art. 2429, 2430.]

Several philosophers of great eminence, however, who have examined the subject with great care, adhere to Dr. Faraday's first view of the nature of Diamagnetism--as W. Weber,[32\B] Plücker, and Mr. Tyndall among ourselves. If we translate this view

into the language of Ampère's theory, it comes to this:--that as currents are induced in iron and magnetics parallel to those existing in the inducing magnet or battery wire; so in bismuth, heavy glass, and other diamagnetic bodies, the currents induced are in the contrary {621} directions:--these hypothetical currents being in non-conducting diamagnetic, as in magnetic bodies, not in the mass, but round the particles of the matter.

[Note 32\B: Poggendorf's _Ann. Jou._ 1848.]

_Magneto-optic Effects and Magnecrystallic Polarity._

Not even yet have we terminated the enumeration of the co-existent polarities which in this province of nature have been brought into view. Light has polar properties; the very term _polarization_ is the record of the discovery of these. The forces which determine the crystalline forms of bodies are of a polar nature: crystalline forms, when complete, may be defined as those forms which have a certain degree of symmetry in reference to opposite poles. Now has this optical and crystalline polarity any relation to the electrical polarity of which we have been speaking?

However much we might be disposed beforehand to conjecture that there is some relation between these two groups of polar properties, yet in this as in the other parts of this history of discoveries respecting polarities, no conjecture hits the nature of the relation, such as experiment showed it to be. In November, 1846, Faraday announced the discovery of what he then called "the action of magnets on light." But this action was manifested, not on light directly, but on light passing through certain kinds of glass.[33\B] When this glass, subjected to the action of the powerful magnets which he used, transmitted a ray of light parallel to the line of magnetic force, an effect was produced upon the light. But of what nature was this effect? When light was ordinary light, no change in its condition was discoverable. But if the light were light polarized in any plane, the plane of polarization was turned round through a certain angle while the ray passed through the glass:--a greater angle, in proportion as the magnetic force was greater, and the thickness of the glass greater.

[Note 33\B: Silicated borate of lead. See _Researches_, § 2151, &c. Also flint glass, rock salt, water (2215).]

A power in some respects of this kind, namely, a power to rotate the plane of polarization of a ray passing through them, is possessed by some bodies in their natural state; for instance, quartz crystals, and oil of turpentine. But yet, as Dr. Faraday remarks,[34\B] there is a great difference in the two cases. When polarized rays pass through oil of turpentine, in whatever direction they pass, they all of them have their {622} plane of polarization rotated in the same direction; that is, all to the right or all to the left; but when a ray passes through the heavy glass, the power of rotation exists only in a plane perpendicular to the magnetic line, and its direction as right or left-handed is reversed by reversing the magnetic polarity.

[Note 34\B: _Researches_, Art. 2231.]

In this case, we have optical properties, which do not depend on crystalline form, affected by the magnetic force. But it has also been found that crystalline form, which is so fertile a source of optical properties, affords indications of magnetic forces. In 1847, M. Plücker,[35\B] of the University of Bonn, using a powerful magnetic apparatus, similar to Faraday's, found that crystals in general are magnetic, in this sense, that the axes of crystalline form tend to assume a certain position with reference to the magnetic lines of force. The possession of one optic axis or of two is one of the broad distinctions of the different crystalline forms: and using this distinction, M. Plücker found that a crystal having a single optic axis tends to place itself with this axis transverse to the magnetic line of force, as if its optic axis were repelled by each magnetic pole; and crystals with two axes act as if each of these axes were repelled by the magnetic poles. This force is independent of the magnetic or diamagnetic character of the crystal; and is a directive, more properly than an attractive or repulsive force.

[Note 35\B: Taylor's _Scientific Memoirs_, vol. v.]

Soon afterwards (in 1848) Faraday also discovered[36\B] an effect of magnetism depending on crystalline form, which at first sight appeared to be different from the effects observed by M. Plücker. He found that a crystal of bismuth, of which the form is nearly a cube, but more truly a rhombohedron with one diagonal a little longer than the others, tends to place itself with this diagonal in the direction of the lines of magnetic force. At first he conceived[37\B] the properties thus detected to be different from those observed by M. Plücker; since in this case the force of a crystalline axis is axial, whereas in those, it was equatorial. But a further consideration of the subject, led him[38\B] to a conviction that these forces must be fundamentally identical: for it was easy to conceive a combination of bismuth crystals which would behave in the magnetic field as a crystal of calcspar does; or a combination of calcspar crystals which would behave as a crystal of bismuth does.

[Note 36\B: _Researches_, Art. 2454, &c.]

[Note 37\B: Art. 2469.]

[Note 38\B: Art 2593, 2601.]

And thus we have fresh examples to show that the Connexion of coexistent Polarities is a thought deeply seated in the minds of the {623} profoundest and most sagacious philosophers, and perpetually verified and illustrated, by unforeseen discoveries in unguessed forms, through the labors of the most skilful experimenters.

_Magneto-electric Machines._

The discovery that a voltaic wire moved in presence of a magnet, has a current generated in it, was employed as the ground of the construction of machines to produce electrical effects. In Saxton's machine two coils of wire including a core of soft iron revolved opposite to the ends of a horseshoe magnet, and thus, as the two coils came opposite to the N. and S. and to the S. and N. poles of the magnet, currents were generated alternately in the wires in opposite directions. But by arranging the connexions of the ends of the wires, the successive currents might be

made to pass in corresponding directions. The alternations or successions of currents in such machines are governed by a contrivance which alternately interrupts and permits the action; this contrivance has been called a _rheotome_. Clarke gave a new form to a machine of the same nature as Saxton's. But the like effect may be produced by using an electro-magnet instead of a common magnet. When this is done, a current is produced which by induction produces a current in another wire, and the action is alternately excited and interrupted. When the inducing current is interrupted, a momentary current _in an opposite direction_ is produced in the induced wire; and when this current stops, it produces in the inducing wire a current _in the original direction_, which may be adjusted so as to reinforce the resumed action of the original current. This was pointed out by M. De la Rive in 1843.[39\B] Machines have been constructed on such principles by him and others. Of such machines the most powerful hitherto known is that constructed by M. Ruhmkorff. The effects of this instrument are exceedingly energetic.

[Note 39\B: _Traité de l'Elect._ i. 391.]

_Applications of Electrodynamic Discoveries._

The great series of discoveries of which I have had to speak have been applied in many important ways to the uses of life. The _Electric Telegraph_ is one of the most remarkable of these. By wires extended to the most distant places, the electric current is transmitted {624} thither in an imperceptible time; and by means of well-devised systems of operation, is made to convey from man to man words, which are now most emphatically "winged words." In the most civilised states such wires now form a net-work across the land, which is familiar to our thoughts as the highway is to our feet; and wide seas have such pathways of human thought buried deep in their waves from shore to shore. Again, by using the chemical effects of electrodynamic action, of which we shall have to speak in the next Book, a new means has been obtained of copying, with an exactness unattainable before, any forms which art or nature has produced, and of covering them with a surface of metal. The _Electrotype Process_ is now one of the great powers which manufacturing art employs.

But these discoveries have also been employed in explaining natural phenomena, the causes of which had before been altogether inscrutable. This is the case with regard to the diurnal variation of the magnetic needle; a fact which as to its existence is universal in all places, and which yet is so curiously diverse in its course at different places. Dr. Faraday has shown that some of the most remarkable of these diversities, and probably all, seem to be accounted for by the different magnetic effects of air at different temperatures: although, as I have already said, **(Book xii.) the discovery of a decennial period in the diurnal changes of magnetic declination shows that any explanation of those changes which refers them to causes existing in the atmosphere must be very incomplete.[40\B]

[Note 40\B: _Researches_, Art. 2892.]

{{625}} BOOK XIV.

CHEMISTRY.

# CHAPTER IX.

THE ELECTRO-CHEMICAL THEORY.

AMONG the consequences of the Electro-chemical Theory, must be ranged the various improvements which have been made in the voltaic battery. Daniel introduced between the two metals a partition permeable by chemical action, but such as to allow of two different acid solutions being in contact with the two metals. Mr. Grove's battery, in which the partition is of porous porcelain, and the metals are platinum and amalgamated zinc, is one of the most powerful hitherto known. Another has been constructed by Dr. Callan, in which the negative or conducting plate is a cylinder of cast iron, and the positive element a cylinder of amalgamated zinc placed in a porous cell. This also has great energy.

_The Number of Elementary Substances._

There have not been, I believe, any well-established additions to the list of the simple substances recognized by chemists. Indeed the tendency at present appears to be rather to deny the separate elementary character of some already announced as such substances. Pelopium and Niobium were, as I have said, two of the new metals. But Naumann, in his _Elemente der Mineralogie_ (4th ed. 1855), says, in a foot note (page 25): "_Pelopium_ is happily again got rid of; for Pelopic Acid and Niobic Acid possess the same Radical. _Donarium_ had a still shorter existence."

In the same way, when Hermann imagined that he had discovered a new simple metallic substance in the mineral Samarskite from Miask, the discovery was disproved by H. Rose (_Pogg. Ann._ B. 73, s. 449). {626}

In general the insulation of the new simple substances, the metallic bases of the earths, and the like,--their separation from their combinations, and the exhibition of them in a metallic form--has been a difficult chemical process, and has rarely been executed on any considerable scale. But in the case of _Aluminium_, the basis of the earth Alumina, the process of its extraction has recently been so much facilitated, that the metal can be produced in abundance. This being the case, it will probably soon be applied to special economical uses, for which it is fitted by possessing special properties.

{{626}} BOOK XV.

MINERALOGY.

BY the kindness of W. H. Miller, Esq., Professor of Mineralogy in the University of Cambridge, I am able to add to this part the following notices of books and memoirs.

1. _Crystallography._

_Elemente der Krystallographie, nebst einer tabellarischen Uebersicht der Mineralien nach der Krystallformen_, von Gustav Rose. 2. Auflage. Berlin, 1838. The crystallographic method here adopted is, for the most part, that of Weiss. The method of this work has been followed in

_A System of Crystallography, with its Applications to Mineralogy_. By John Joseph Griffin. Glasgow, 1841. Mr. Griffin has, however, modified the notation of Rose. He has constructed a series of models of crystalline forms.

Frankenheim's _System der Krystalle_. 1842. This work adopts nearly the Mohsian systems of crystallization. It contains Tables of the chemical constitution, inclinations of the axis, and magnitude of the axes of all the crystals of which a description was to be found, including those formed in the laboratory, as well as those usually called minerals; 713 in all.

Fr. Aug. Quenstedt, _Methode der Krystallographie_, 1840, employs a fanciful method of representing a crystal by projecting upon one face of the crystal all the other faces. This invention appears to be more curious than useful.

Dr. Karl Naumann, who is spoken of in Chap. ix. of this Book, as the author of the best of the Mixed Systems of Classification, published also _Grundriss der Krystallographie_, Leipzig, 1826. In this and other works he modifies the notation of Mohs in a very advantageous manner. {628}

Professor Dana, in his _System of Mineralogy_, New Haven (U.S.), 1837, follows Naumann for the most part, both in crystallography and in mineral classification. In the latter part of the subject, he has made the attempt, which in all cases is a source of confusion and of failure, to introduce a whole system of new names of the members of his classification.

The geometry of crystallography has been investigated in a very original manner by M. Bravais, in papers published in the Journal of the Ecole Polytechnique, entitled _Mémoires sur les Systèmes formés par des Points_. 1850. _Etudes Crystallographiques_. 1851.

Hermann Kopp (_Einleitung in die Krystallographie_, Braunschweig. 1849) has given the description and measurement of the angles of a large number of laboratory crystals.

Rammelsberg (_Krystallographische Chemie_, Berlin, 1855) has collected an account of the systems, simple forms and angles of all the laboratory crystals of which he could obtain descriptions.

Schabus of Vienna (_Bestimmung der Krystallgestalten **in Chemischen Laboratorien erzeugten Producte_, Wien, 1855; a successful Prize Essay) has given a description, accompanied by measurements, of 90 crystalline species from his own observations.

To these attempts made in other countries to simplify and improve crystallography, I may add a remarkable Essay very recently made here by Mr. Brooke, and suggested to him by his exact and familiar knowledge of Mineralogy. It is to this effect. All the crystalline forms of any given mineral species are derived from the _primitive form_ of that species; and the degree of symmetry, and the _parameters_, of this form determine the angles of all derivative forms. But how is this primitive form selected and its parameters determined? The selection of the kind of the primitive form depends upon the _degree of symmetry_ which appears in all the derivative forms;

according to which they belong to the _rhombohedral_, _prismatic_, _square pyramidal_, or some other _system_: and this determination is commonly clear. But the parameters, or the angles, of the primitive form, are commonly determined by the _cleavage_ of the mineral. Is this a sufficient and necessary ground of such determination? May not a simplification be effected, in some cases, by taking some other parameters? by taking a primitive form which belongs to the proper system, but which has some other angles than those given by cleavage? Mr. Brooke has tried whether, for instance, crystals of the rhombohedral system may not be referred with advantage to primitive rhombohedrons which have, in all {629} the species, nearly the same angles. The advantage to be obtained by such a change would be the simplification of the laws of derivation in the derivative forms: and therefore we have to ask, whether the indices of derivation are smaller numbers in this way or with the hitherto accepted fundamental angles. It appears to me, from the examples given, that the advantage of simplicity in the indices is on the side of the old system: but whether this be so or not, it was a great benefit to crystallography to have the two methods compared. Mr. Brooke's Essay is a Memoir presented to the Royal Society in 1856.

2. _Optical Properties of Minerals._

The _Handbuch der Optik_, von F. W. G. Radicke, Berlin, 1839, contains a chapter on the optical properties of crystals. The author's chief authority is Sir D. Brewster, as might be expected.

M. Haidinger has devoted much attention to experiments on the _pleochroism_ of minerals. He has invented an instrument which makes the dichroism of minerals more evident by exhibiting the two colors side by side.

The pleochroism of minerals, and especially the remarkable clouds that in the cases of Iolite, Andalusite, Augite, Epidote, and Axinite, border the positions of either optical axis, have been most successfully imitated by M. de Senarmont by means of artificial crystallizations. (_Ann. de Chim._ 3 _Ser._ xli. p. 319.)

M. Pasteur has found that Racemic Acid consists of two different acids, having the same density and composition. The salts of these acids, with bases of Ammonia and of Potassa, are hemihedral, the hemihedral faces which occur in the one being wanting in the other. The acids of these different crystals have circular polarization of opposite kinds. (_Ann. de Chim._ 3 _Ser._ xxviii. 56, 99.) This discovery was marked by the assignation of the Rumford Medal to M. Pasteur in 1856.

M. Marbach has discovered that crystals of chlorate of soda, which apparently belongs to the cubic or tessular system, exhibit hemihedral faces of a peculiar character; and that the crystals have circular polarization of opposite kinds in accordance with the differences of the plagihedral faces. (_Poggendorf's Annalen_, xci. 482.)

M. Seybolt of Vienna has found a means of detecting plagihedral faces in quartz crystals which do not reveal them externally. (_Akad. d. Wissenschaft zu Wien_, B. xv. s. 59.) {630}

3. _Classification of Minerals._

In the _Philosophy of the Inductive Sciences_, B. VIII. C. iii., I have treated of the Application of the Natural-history Method of Classification to Mineralogy, and have spoken of the Systems of this kind which have been proposed. I have there especially discussed the system proposed in the treatise of M. Necker, _Le Règne Minéral ramené aux Méthodes d'Histoire Naturelle_ (Paris, 1835). More recently have been published M. Beudant's _Cours élémentaire d'Histoire Naturelle, Minéralogie_ (Paris, 1841); and M. A. Dufresnoy's _Traité de Minéralogie_ (Paris, 1845). Both these works are so far governed by mere chemical views that they lapse into the inconveniences and defects which are avoided in the best systems of German mineralogists.

The last mineral system of Berzelius has been developed by M. Rammelsberg (Nürnberg, 1847). It is in principle such as we have described it in the history.

M. **Nordenskiöld's system (3rd Ed. 1849,) has been criticised by G. Rose, who observes that it removes the defects of the system of Berzelius only in part. He himself proposes what he calls a "Krystallo-Chemisches System," in which the crystalline form determines the genus and the chemical composition the species. His classes are-- 1. Simple Substances. 2. Combinations of Sulphur, Selenium, Titanium, Arsenic, Antimony. 3. Chlorides, Fluorides, Bromides, Iodides. 4. Combinations with Oxygen.

We have already said that for us, all chemical compounds are _minerals_, in so far that they are included in our classifications. The propriety of this mode of dealing with the subject is confirmed by our finding that there is really no tenable distinction between native minerals and the products of the laboratory. A great number of eminent chemists have been employed in producing, by artificial means, crystals which had before been known only as native products.

{{631}} BOOK XVI.

CLASSIFICATORY SCIENCES.

BOTANY.

FOR the purpose of giving to my reader some indication of the present tendency of Botanical Science, I conceive that I cannot do better than direct his attention to the reflections, procedure, and reasonings which have been suggested by the most recent extensions of man's knowledge of the vegetable world. And as a specimen of these, I may take the labors of Dr. Joseph Hooker, on the Flora of the Antarctic Regions,[41\B] and especially of New Zealand. Dr. Hooker was the Botanist to an expedition commanded by Sir James Ross, sent out mainly for the purpose of investigating the phenomena of Terrestrial Magnetism near the South Pole; but directed also to the improvement of Natural History. The extension of botanical descriptions and classifications to a large mass of new objects necessarily suggests wider views of the value of classes (genera, species, &c.,) and the conclusions to be drawn from their constancy or inconstancy. A few of Dr. Hooker's remarks may show the nature of the views taken under such circumstances.

[Note 41\B: _The Botany of the Antarctic Voyage of H. M. Discovery Ships Erebus and Terror, in the years_ 1839-40. Published 1847. _Flora Novæ Zelandiæ_. 1853.]

I may notice, in the first place, (since this work is intended for general rather than for scientific readers,) Dr. Hooker's testimony to the value of a technical descriptive language for a classificatory science--a Terminology, as it is called. He says, "It is impossible to write Botanical descriptions which a person ignorant of Botany can understand, although it is supposed by many unacquainted with science that this can and should be done." And hence, he says, the state of botanical science demands Latin descriptions of the plants; and this is a lesson which he especially urges upon the Colonists who study the indigenous plants. {632}

Dr. Hooker's remarks on the limits of species, their dispersion and variation, are striking and instructive. He is of opinion that species vary more, and are more widely diffused, than is usually supposed. Hence he conceives that the number of species has been needlessly and erroneously multiplied, by distinguishing the specimens which occur in different places, and vary in unessential features. He says that though, according to the lowest estimate of compilers, 100,000 is the commonly received number of known plants, he thinks that half that number is much nearer the truth. "This," he says, "may be well conceived, when it is notorious that nineteen species have been made of the Common Potatoe, and many more of _Solanum nigrum_ alone. _Pteris aquilina_ has given rise to numerous book species; _Vernonia cinerea_ of India to fifteen at least. . . . . Many more plants are common to most countries than is supposed; I have found 60 New Zealand flowering plants and 9 Ferns to be European ones, besides inhabiting numerous intermediate countries. . . . . So long ago as 1814, Mr. Brown drew attention to the importance of such considerations, and gave a list of 150 European plants common to Australia."

As an example of the extent to which unessential differences may go, he says (p. xvii.,) "The few remaining native Cedars of Lebanon may be abnormal states of the tree which was once spread over the whole of the Lebanon; for there are now growing in England varieties of it which have no existence in a wild state. Some of them closely resemble the Cedars of Atlas and of the Himalayas (_Deodar_;) and the absence of any valid botanical differences tends to prove that all, though generally supposed to be different species, are one."

Still the great majority of the species of plants in those Southern regions are peculiar. "There are upwards of 100 genera, subgenera, or other well marked groups of plants, entirely or nearly confined to New Zealand, Australia, and extra-tropical South America. They are represented by one or more species in two or more of those countries, and thus effect a botanical relationship or affinity between them all which every botanist appreciates."

In reference to the History of Botany, I have received corrections and remarks from Dr. Hooker, with which I am allowed to enrich my pages.

"P. 359. Note ^3. ~= Note 3\16~ _Nelumbium speciosum_, the Lotus of India. The _Nelumbium_ does not float, but raises both leaf and flower several feet above the water: the _Nymphæa Lotus_ has floating leaves. Both enter largely into the

symbolism of the Hindoos, and are often confounded. {633}

"P. 362. Note ^5. ~= Note 13\16~ For _Arachnis_ read _Arachis_. The _Arachidna_ of Theophrastus cannot, however, be the _Arachis_ or ground-nut.

"Pp. 388 and 394. For _Harlecamp_ read _Hartecamp_.

"P. 394. For _Kerlen_ read _Kalm_.

"P. 394. For _Asbech_ read _Osbeck_.

"P. 386. _John Ray_. Ray was further the author of the present Natural System in its most comprehensive sense. He first divided plants into Flowerless and Flowering; and the latter into Monocotyledonous and Dicotyledonous:--'Floriferas dividemus in DICOTYLEDONES, quarum semina sata binis foliis, seminalibus dictis, quæ cotyledonorum usum præstant, e terra exeunt, vel in binos saltem lobos dividuntur, quamvis eos supra terram foliorum specie non efferant; et MONOCOTYLEDONES, quæ nec folia bina seminalia efferunt nec lobos binos condunt. Hæc divisio ad arbores etiam extendi potest; siquidem Palmæ et congeneres hoc respectu eodem modo a reliquis arboribus differunt quo Monocotyledones a reliquis herbis.'

"P. 408. _Endogenous and Exogenous Growth._ The exact course of the wood fibres which traverse the stems of both Monocotyledonous and Dicotyledonous plants has been only lately discovered. In the Monocotyledons, those fibres are collected in bundles, which follow a very peculiar course:--from the base of each leaf they may be followed downwards and inwards, towards the axis of the trunk, when they form an arch with the convexity to the centre; and curving outwards again reach the circumference, where they are lost amongst the previously deposited fibres. The intrusion of the bases of these bundles amongst those already deposited, causes the circumference of the stem to be harder than the centre; and as all these arcs have a short course (their chords being nearly equal), the trunk does not increase in girth, and grows at the apex only. The wood-bundles are here definite. In the Dicotyledonous trunks, the layers of wood run in parallel courses from the base to the top of the trunk, each externally to that last formed, and the trunk increases both in height and girth; the wood-bundles are here indefinite.

"With regard to the Cotyledons, though it is often difficult to distinguish a Monocotyledonous Embryo from a Dicotyledonous, they may always be discriminated when germinating. The Cotyledons, when two or more, and primordial leaves (when no Cotyledons are visible) of a Monocotyledon, are alternate; those of a Dicotyledon are opposite.

"A further physiological distinction between Monocotyledons and {634} Dicotyledons is observed in germination, when the Dicotyledonous radicle elongates and forms the root of the young plant; the Monocotyledonous radicle does not elongate, but pushes out rootlets from itself at once. Hence the not very good terms, _exorhizal_ for Dicotyledonous, and _endorhizal_ for Monocotyledonous.

"The highest physiological generalization in the vegetable kingdom is between _Phænogama_ and _Cryptogama_. In the former, fertilization is effected by a pollen-tube touching the nucleus of an ovule; in Cryptogams, the same process is

effected by the contact of a sperm-cell, usually ciliated (_antherozoid_), upon another kind of cell called a germ-cell. In Phænogams, further, the organs of fructification are all modified leaves; those of Cryptogams are not homologous." (J. D. H.)

ZOOLOGY.

I have exemplified the considerations which govern zoological classification by quoting the reflexions which Cuvier gives us, as having led him to his own classification of Fishes. Since the varieties of Quadrupeds, or _Mammals_ (omitting whales, &c.), are more familiar to the common reader than those of Fishes, I may notice some of the steps in their classification; the more so as some curious questions have recently arisen thereupon.

Linnæus first divides Mammals into two groups, as they have Claws, or Hoofs (_unguiculata_, _ungulata_.) But he then again divides them into six orders (omitting whales, &c.), according to their number of _incisor_, _laniary_, and _molar_ teeth; namely:-- _Primates_. (Man, Monkey, &c.) _Bruta_. (Rhinoceros, Elephant, &c.) _Feræ_. (Dog, Cat, Bear, Mole, &c.) _Glires_. (Mouse, Squirrel, Hare, &c.) _Pecora_. (Camel, Giraffe, Stag, Goat, Sheep, Ox, &c.) _Belluæ_. (Horse, Hippopotamus, Tapir, Sow, &c.)

In the place of these, Cuvier, as I have stated in the _Philosophy_ (_On the Language of Sciences_, Aphorism xvi.), introduced the following orders: _Bimanes_, _Quadrumanes_, _Carnassiers_, _Rongeurs_, _Edentés_, _Pachyderms_, _Ruminans_. Of these, the _Carnassiers_ correspond to the _Feræ_ of Linnæus; the _Rongeurs_ to his _Glires_; the _Edentés_ are a new order, taking the Sloths, Ant-eaters, &c., from the _Bruta_ of Linnæus, the Megatherium from extinct animals, and the Ornithorhynchus, &c., from the new animals of Australia; the _Ruminans_ agree with the {635} _Pecora_; the _Pachyderms_ include some of the _Bruta_ and the _Belluæ_, comprehending also extinct animals, as _Anoplotherium_ and _Palæotherium_.

But the two orders of Hoofed Animals, the Pachyderms and the Ruminants, form a group which is held by Mr. Owen to admit of a better separation, on the ground of a character already pointed out by Cuvier; namely, as to whether they are _two-toed_ or _three-toed_. According to this view, the Horse is connected with the Tapir, the Palæotherium, and the Rhinoceros, not only by his teeth, but by his feet, for he has really three digits. And Cuvier notices that in the two-toed or even-toed Pachyderms, the astragalus bone has its face divided into two equal parts by a ridge; while in the uneven-toed pachyderms it has a narrow cuboid face. Mr. Owen has adopted this division of Pachyderms and Ruminants, giving the names _artiodactyla_ and _perissodactyla_ to the two groups; the former including the Ox, Hog, Peccary, Hippopotamus, &c.; the latter comprehending the Horse, Tapir, Rhinoceros, Hyrax, &c. And thus the Ruminants take their place as a subordinate group of the great natural even-toed Division of the Hoofed Section of Mammals; and the Horse is widely separated from them, inasmuch as he belongs to the odd-toed division.[42\B]

[Note 42\B: Owen, _Odontography_.]

As we have seen, these modern classifications are so constructed as to include extinct as well as living species of animals; and indeed the species which have been

discovered in a fossil state have tended to fill up the gaps in the series of zoological forms which had marred the systems of modern zoologists. This has been the case with the division of which we are speaking.

Mr. Owen had established two genera of extinct Herbivorous Animals, on the strength of fossil remains brought from South America:--_Toxodon_, and _Nesodon_. In a recent communication to the Royal Society[43\B] he has considered the bearing of these genera upon the divisions of odd-toed and even-toed animals. He had already been led to the opinion that the three sections, _Proboscidea_, _Perissodactyla_, and _Artiodactyla_, formed a natural division of Ungulata; and he is now led to think that this division implies another group, "a distinct division of the _Ungulata_, of equal value, if not with the _Perissodactyla_ and _Artiodactyla_ at least with the _Proboscidea_. This group he proposes to call _Toxodonta_.

[Note 43\B: _Phil. Trans._, 1853.]

{{636}} BOOK XVII.

PHYSIOLOGY AND COMPARATIVE ANATOMY.

VEGETABLE MORPHOLOGY.

_Morphology in Linnæus._

I HAVE stated that Linnæus had some views on this subject. Dr. Hooker conceives these views to be more complete and correct than is generally allowed, though unhappily clothed in metaphorical language and mixed with speculative matter. By his permission I insert some remarks which I have received from him.

The fundamental passage on this subject is in the _Systema Naturæ_; in the Introduction to which work the following passage occurs:--

"Prolepsis (Anticipation) exhibits the mystery of the metamorphosis of plants, by which the herb, which is the _larva_ or imperfect condition, is changed into the declared fructification: for the plant is capable of producing either a leafy herb or a fructification. . . . . .

"When a tree produces a flower, nature anticipates the produce of five years where these come out all at once; forming of the bud-leaves of the next year _bracts_; of those of the following year, the _calyx_; of the following, the _corolla_; of the next, the _stamina_; of the subsequent, the _pistils_, filled with the granulated marrow of the seed, the terminus of the life of a vegetable."

Dr. Hooker says, "I derive my idea of his having a better knowledge of the subject than most Botanists admit, not only from the Prolepsis, but from his paper called _Reformatio Botanices_ (_Amœn. Acad._ vol. vi.); a remarkable work, in respect of his candor in speaking of his predecessors' labors, and the sagacity he shows in indicating researches to be undertaken or completed. Amongst the latter is V. 'Prolepsis plantarum, ulterius extendenda per earum metamorphoses.' The last word occurs rarely in his _Prolepsis_; but when it does it seems to me that he uses it as indicating a normal change and not an accidental one. {637}

"In the _Prolepsis_ the speculative matter, which Linnæus himself carefully distinguishes as such, must be separated from the rest, and this may I think be done in most of the sections. He starts with explaining clearly and well the origin and position of buds, and their constant presence, whether developed or not, in the axil of the leaf: adding abundance of acute observations and experiments to prove his statements. The leaf he declares to be the first effort of the plant in spring: he proceeds to show, successively, that bracts, calyx, corolla, stamens, and pistil are each of them metamorphosed leaves, in every case giving MANY EXAMPLES, both from monsters and from characters presented by those organs in their normal condition.

"The (to me) obscure and critical part of the _Prolepsis_ was that relating to the change of the style of _Carduus_ into two leaves. Mr. Brown has explained this. He says it was a puzzle to him, till he went to Upsala and consulted Fries and Wahlenberg, who informed him that such monstrous _Cardui_ grew in the neighborhood, and procured him some. Considering how minute and masked the organs of _Compositæ_ are, it shows no little skill in Linnæus, and a very clear view of the whole matter, to have traced the metamorphosis of all their floral organs into leaves, except their stamens, of which he says, 'Sexti anni folia e staminibus me non in compositis vidisse fateor, sed illorum loco folia pistillacea, quæ in compositis aut plenis sunt frequentissima.' I must say that nothing could well be clearer to my mind than the full and accurate appreciation which Linnæus shows of the whole series of phenomena, and their _rationale_. He over and over again asserts that these organs are leaves, every one of them,--I do not understand him to say that the prolepsis is an accidental change of leaves into bracts, of bracts into calyx, and so forth. Even were the language more obscure, much might be inferred from the wide range and accuracy of the observations he details so scientifically. It is inconceivable that a man should have traced the sequence of the phenomena under so many varied aspects, and shown such skill, knowledge, ingenuity, and accuracy in his methods of observing and describing, and yet missed the _rationale_ of the whole. Eliminate the speculative parts and there is not a single error of observation or judgment; whilst his history of the developement of buds, leaves, and floral organs, and of various other obscure matters of equal interest and importance, are of a very high order of merit, are, in fact, for the time profound.

"There is nothing in all this that detracts from the merit of Goethe's {638} re-discovery. With Goethe it was, I think, a deductive process,--with Linnæus an inductive. Analyse Linnæus's observations and method, and I think it will prove a good example of inductive reasoning.

"P. 473. Perhaps Professor Auguste St Hilaire of Montpellier should share with De Candolle the honor of contributing largely to establish the metamorphic doctrine;--their labors were cotemporaneous.

"P. **474. Linnæus pointed out that the pappus was calyx: 'Et _pappum_ gigni ex quarti anni foliis, in jam nominatis Carduis.'--_Prol. Plant._ 338." (_J. D. H._)

## CHAPTER VII.

ANIMAL MORPHOLOGY.

THE subject of Animal Morphology has recently been expanded into a form strikingly comprehensive and systematic by Mr. Owen; and supplied by him with a copious and carefully-chosen language; which in his hands facilitates vastly the comparison and appreciation of the previous labors of physiologists, and opens the way to new truths and philosophical generalizations. Though the steps which have been made had been prepared by previous anatomists, I will borrow my view of them mainly from him; with the less scruple, inasmuch as he has brought into full view the labors of his predecessors.

I have stated in the History that the skeletons of all vertebrate animals are conceived to be reducible to a single Type, and the skull reducible to a series of vertebræ. But inasmuch as this reduction includes not only a detailed correspondence of the bones of man with those of beasts, but also with those of birds, fishes, and reptiles, it may easily be conceived that the similarities and connexions are of a various and often remote kind. The views of such relations, held by previous Comparative Anatomists, have led to the designations of the bones of animals which have been employed in anatomical descriptions; and these designations having been framed and adopted by anatomists looking at the subject from different sides, and having different views of analogies and relations, have been very various and unstable; besides being often of cumbrous length and inconvenient form.

The corresponding parts in different animals are called _homologues_, {639} a term first applied to anatomy by the philosophers of Germany; and this term Mr. Owen adopts, to the exclusion of terms more loosely denoting identity or similarity. And the Homology of the various bones of vertebrates having been in a great degree determined by the labors of previous anatomists, Mr. Owen has proposed names for each of the bones: the condition of such names being, that the homologues in all vertebrates shall be called by the same name, and that these names shall be founded upon the terms and phrases in which the great anatomists of the 16th, 17th, and 18th centuries expressed the results of their researches respecting the human skeleton. These names, thus selected, so far as concerned the bones of the Head of Fishes, one of the most difficult cases of this Special Homology, he published in a Table,[44\B] in which they were compared, in parallel columns, with the names or phrases used for the like purpose by Cuvier, Agassiz, Geoffroy, Hallman, Sœmmering, Meckel, and Wagner. As an example of the considerations by which this selection of names was determined, I may quote what he says with regard to one of these bones of the skull.

[Note 44\B: _Lectures on Vertebrates_. 1846, p. 158. And _On the Archetype and Homologies of the Vertebrate Skeleton_. 1848, p. 172.]

"With regard to the 'squamosal' (_squamosum_. Lat. pars squamosa ossis temporis.--Sœmmering), it might be asked why the term 'temporal' might not be retained for this bone. I reply, because that term has long been, and is now universally, understood in human anatomy to signify a peculiarly anthropotomical coalesced congeries of bones, which includes the 'squamosal' together with the 'petrosal,' the 'tympanic,' the 'mastoid,' and the 'stylohyal.' It seems preferable,

therefore, to restrict the signification of the term 'temporal' to the whole (in Man) of which the 'squamosal' is a part. To this part Cuvier has unfortunately applied the term 'temporal' in one class, and 'jugal' in another; and he has also transferred the term 'temporal' to a third equally distinct bone in fishes; while to increase the confusion M. Agassiz has shifted the name to a fourth different bone in the skull of fishes. Whatever, therefore, may be the value assigned to the arguments which will be presently set forth, as to the special homologies of the 'pars squamosa ossis temporis,' I have felt compelled to express the conclusion by a definite term, and in the present instance, have selected that which recalls the best accepted anthropomorphical designation of the part; although 'squamosal' must be understood and applied in an arbitrary sense; and not as descriptive of a scale-like {640} form; which in reference to the bone so called, is rather its exceptional than normal figure in the vertebrate series."

The principles which Mr. Owen here adopts in the selection of names for the parts of the skeleton are wise and temperate. They agree with the aphorisms concerning the language of science which I published in the _Philosophy of the Inductive Sciences_; and Mr. Owen does me the great honor of quoting with approval some of those Aphorisms. I may perhaps take the liberty of remarking that the system of terms which he has constructed, may, according to our principles, be called rather a _Terminology_ **than a _Nomenclature_: that is, they are analogous more nearly to the _terms_ by which botanists describe the parts and organs of plants, than to the _names_ by which they denote genera and species. As we have seen in the History, plants as well as animals are subject to morphological laws; and the names which are given to organs in consequence of those laws are a part of the Terminology of the science. Nor is this distinction between Terminology and Nomenclature without its use; for the rules of prudence and propriety in the selection of words in the two cases are different. The Nomenclature of genera and species may be arbitrary and casual, as is the case to a great extent in Botany and in Zoology, especially of fossil remains; names being given, for instance, simply as marks of honor to individuals. But in a Terminology, such a mode of derivation is not admissible: some significant analogy or idea must be adopted, at least as the origin of the name, though not necessarily true in all its applications, as we have seen in the case of the "squamosal" just quoted. This difference in the rules respecting two classes of scientific words is stated in the _Aphorisms_ xiii. and xiv. _concerning the Language of Science_.

Such a Terminology of the bones of the skeletons of all vertebrates as Mr. Owen has thus propounded, cannot be otherwise than an immense acquisition to science, and a means of ascending from what we know already to wider truths and new morphological doctrines.

With regard to one of these doctrines, the resolution of the human head into vertebræ, Mr. Owen now regards it as a great truth, and replies to the objections of Cuvier and M. Agassiz, in detail.[45\B] He gives a Table in which the Bones of the Head are resolved into four vertebræ, which he terms the Occipital, Parietal, Frontal, and Nasal Vertebra, respectively. These four vertebræ agree in general with what Oken called the Ear-vertebra, the Jaw-vertebra, the Eye-vertebra, and {641} the

Nose-vertebra, in his work _On the Signification of the Bones of the Skull_, published in 1807: and in various degrees, with similar views promulgated by Spix (1815), Bojanus (1818), Geoffroy (1824), Carus **(1828). And I believe that these views, bold and fanciful as they at first appeared, have now been accepted by most of the principal physiologists of our time.

[Note 45\B: _Archetype and Homologies of the Vertebrate Skeleton_. 1848, p. 141.]

But another aspect of this generalization has been propounded among physiologists; and has, like the others, been extended, systematized, and provided with a convenient language by Mr. Owen. Since animal skeletons are thus made up of vertebræ and their parts are to be understood as developements of the parts of vertebræ, Geoffroy (1822), Carus (1828), Müller (1834), Cuvier (1836), had employed certain terms while speaking of such developements; Mr. Owen in the _Geological Transactions_ in 1838, while discussing the osteology of certain fossil Saurians, used terms of this kind, which are more systematic than those of his predecessors, and to which he has given currency by the quantity of valuable knowledge and thought which he has embodied in them.

According to his Terminology,[46\B] a vertebra, in its typical completeness, consists of a central part or _centrum_; at the back of this, two plates (the _neural apophyses_) and a third outward projecting piece (the _neural spine_), which three, with the centrum, form a canal for the spinal marrow; at the front of the centrum two other plates (the _hæmal apophyses_) and a projecting piece, forming a canal for a vascular trunk. Further lateral elements (_pleuro-apophyses_) and other projections, are in a certain sense dependent on these principal bones; besides which the vertebra may support _diverging appendages_. These parts of the vertebra are fixed together, so that a vertebra is by some anatomists described as a single bone; but the parts now mentioned are usually developed from distinct and independent centres, and are therefore called by Mr. Owen "autogenous" elements.

[Note 46\B: _Archetype and Homologies of the Vertebrate Skeleton_. 1848, p. 81.]

The _General_ Homology of the vertebral skeleton is the reference of all the parts of a skeleton to their true types in a series of vertebræ: and thus, as _special_ homology refers all the parts of skeletons to a given type of skeleton, say that of Man, _general_ homology refers all the parts of every skeleton, say that of Man, to the parts of a series of Vertebræ. And thus as Oken propounded his views of the Head as a resolution of the Problem of _the Signification of the Bones of the Head_, {642} so have we in like manner, for the purposes of General Homology, to solve the Problem of _the Signification of Limbs_. The whole of the animal being a string of vertebræ, what are arms and legs, hands and paws, claws and fingers, wings and fins, and the like? This inquiry Mr. Owen has pursued as a necessary part of his inquiries. In giving a public lecture upon the subject in 1849,[47\B] he conceived that the phrase which I have just employed would not be clearly apprehended by an English Audience, and entitled his Discourse "On the _Nature_ of Limbs:" and in this discourse he explained the modifications by which the various kinds of limbs are derived from their rudiments in an archetypal skeleton, that is, a mere series of vertebræ without head,

arms, legs, wings, or fins.

[Note 47\B: _On the Nature of Limbs_, a discourse delivered at a Meeting of the Royal Institution, 1849.]

_Final Causes_

It has been mentioned in the History that in the discussions which took place concerning the Unity of Plan of animal structure, this principle was in some measure put in opposition to the principle of Final Causes: Morphology was opposed to Teleology. It is natural to ask whether the recent study of Morphology has affected this antithesis.

If there be advocates of Final Causes in Physiology who would push their doctrines so far as to assert that every feature and every relation in the structure of animals have a purpose discoverable by man, such reasoners are liable to be perpetually thwarted and embarrassed by the progress of anatomical knowledge; for this progress often shows that an arrangement which had been explained and admired with reference to some purpose, exists also in cases where the purpose disappears; and again, that what had been noted as a special teleological arrangement is the result of a general morphological law. Thus to take an example given by Mr. Owen: that the ossification of the head originates in several centres, and thus in its early stages admits of compression, has been pointed out as a provision to facilitate the birth of viviparous animals; but our view of this provision is disturbed, when we find that the same mode of the formation of the bony framework takes place in animals which are born from an egg. And the number of points from which ossification begins, depends in a wider sense on the general homology of the animal frame, according to which each part is composed of a certain number of autogenous vertebral elements. In this {643} way, the admission of a new view as to Unity of Plan will almost necessarily displace or modify some of the old views respecting Final Causes.

But though the view of Final Causes is displaced, it is not obliterated; and especially if the advocate of Purpose is also ready to admit visible correspondences which have not a discoverable object, as well as contrivances which have. And in truth, how is it possible for the student of anatomy to shut his eyes to either of these two evident aspects of nature? The arm and hand of man are made for taking and holding, the wing of the sparrow is made for flying; and each is adapted to its end with subtle and manifest contrivance. There is plainly Design. But the arm of man and the wing of the sparrow correspond to each other in the most exact manner, bone for bone. Where is the Use or the Purpose of this correspondence? If it be said that there may be a purpose though we do not see it, that is granted. But Final Causes _for us_ are contrivances of which _we see_ the end; and nothing is added to the evidence of Design by the perception of a unity of plan which in no way tends to promote the design.

It may be said that the design appears in the modification of the plan in special ways for special purposes;--that the vertebral plan of an animal being given, the fore limbs are modified in Man and in Sparrow, as the nature and life of each require. And this is truly said; and is indeed the truth which we are endeavoring to bring into

view:--that there are in such speculations, two elements; one given, the other to be worked out from our examination of the case; the _datum_ and the _problem_; the homology and the teleology.

Mr. Owen, who has done so much for the former of these portions of our knowledge, has also been constantly at the same time contributing to the other. While he has been aiding our advances towards the Unity of Nature, he has been ever alive to the perception of an Intelligence which pervades Nature. While his morphological doctrines have moved the point of view from which he sees Design, they have never obscured his view of it, but, on the contrary, have led him to present it to his readers in new and striking aspects. Thus he has pointed out the final purposes in the different centres of ossification of the long bones of the limbs of mammals, and shown how and why they differ in this respect from reptiles (_Archetype_, p. 104). And in this way he has been able to point out the insufficiency of the rule laid down both by Geoffroy St. Hilaire and Cuvier, for ascertaining the true number of bones in each species. {644}

Final Causes, or Evidences of Design, appear, as we have said, not merely as contrivances for evident purposes, but as modifications of a given general Plan for special given ends. If the general Plan be discovered after the contrivance has been noticed, the discovery may at first seem to obscure our perception of Purpose; but it will soon be found that it merely transfers us to a higher point of view. The adaptation of the Means to the End remains, though the Means are parts of a more general scheme than we were aware of. No generalization of the Means can or ought permanently to shake our conviction of the End; because we must needs suppose that the Intelligence which contemplates the End is an intelligence which can see at a glance along a vista of Means, however long and complex. And on the other hand, no special contrivance, however clear be its arrangement, can be unconnected with the general correspondences and harmonies by which all parts of nature are pervaded and bound together. And thus no luminous teleological point can be extinguished by homology; nor, on the other hand, can it be detached from the general expanse of homological light.

The reference to Final Causes is sometimes spoken of as unphilosophical, in consequence of Francis Bacon's comparison of Final Causes in Physics to Vestal Virgins devoted to God, and barren. I have repeatedly shown that, in Physiology, almost all the great discoveries which have been made, have been made by the assumption of a purpose in animal structures. With reference to Bacon's simile, I have elsewhere said that if he had had occasion to develope its bearings, full of latent meaning as his similes so often are, he would probably have said that to those Final Causes barrenness was no reproach, seeing they ought to be not the Mothers but the Daughters of our Natural Sciences; and that they were barren, not by imperfection of their nature, but in order that they might be kept pure and undefiled, and so fit ministers in the temple of God. I might add that in Physiology, if they are not Mothers, they are admirable Nurses; skilful and sagacious in perceiving the signs of pregnancy, and helpful in bringing the Infant Truth into the light of day.

There is another aspect of the doctrine of the Archetypal Unity of Composition of Animals, by which it points to an Intelligence from which the frame of nature proceeds; namely this:--that the Archetype of the Animal Structure being of the nature of an _Idea_, implies a mind in which this Idea existed; and that thus Homology itself points the way to the Divine Mind. But while we acknowledge the full {645} value of this view of theological bearing of physiology, we may venture to say that it is a view quite different from that which is described by speaking of "Final Causes," and one much more difficult to present in a lucid manner to ordinary minds.

{{646}} BOOK XVIII.

GEOLOGY.

WITH regard to Geology, as a Palætiological Science, I do not know that any new light of an important kind has been thrown upon the general doctrines of the science. Surveys and examinations of special phenomena and special districts have been carried on with activity and intelligence; and the animals of which the remains people the strata, have been reconstructed by the skill and knowledge of zoologists:--of such reconstructions we have, for instance, a fine assemblage in the publications of the Palæontological Society. But the great questions of the manner of the creation and succession of animal and vegetable species upon the earth remain, I think, at the point at which they were when I published the last edition of the History.

I may notice the views propounded by some chemists of certain bearings of Mineralogy upon Geology. As we have, in mineral masses, organic remains of former organized beings, so have we crystalline remains of former crystals; namely, what are commonly called _pseudomorphoses_--the shape of one crystal in the substance of another. M. G. Bischoff[48\B] considers the study of pseudomorphs as important in geology, and as frequently the only means of tracing processes which have taken place and are still going on in the mineral kingdom.

[Note 48\B: _Chemical and Physical Geology_.]

I may notice also Professor Breithaupt's researches on the order of succession of different minerals, by observing the mode in which they occur and the order in which different crystals have been deposited, promise to be of great use in following out the geological changes which the crust of the globe has undergone. (_Die Paragenesis der Mineralien_. Freiberg. 1849.)

In conjunction with these may be taken M. de Senarmont's experiments on the formation of minerals in veins; and besides Bischoff's {647} _Chemical Geology_, Sartorius von Walterhausen's Observations on the occurrence of minerals in Amygdaloid.

As a recent example of speculations concerning Botanical Palætiology, I may give Dr. Hooker's views of the probable history of the Flora of the Pacific.

In speculating upon this question, Dr. Hooker is led to the discussion of geological doctrines concerning the former continuity of tracts of land which are now separate, the elevation of low lands into mountain ranges in the course of ages, and the like. We have already seen, in the speculations of the late lamented Edward Forbes, (see

Book xviii. chap. vi. of this History,) an example of a hypothesis propounded to account for the existing Flora of England: a hypothesis, namely, of a former Connexion of the West of the British Isles with Portugal, of the Alps of Scotland with those of Scandinavia, and of the plains of East Anglia with those of Holland. In like manner Dr. Hooker says (p. xxi.) that he was led to speculate on the possibility of the plants of the Southern Ocean being the remains of a Flora that had once spread over a larger and more continuous tract of land than now exists in the ocean; and that the peculiar Antarctic genera and species may be the vestiges of a Flora characterized by the predominance of plants which are now scattered throughout the Southern islands. He conceives this hypothesis to be greatly supported by the observations and reasonings of Mr. Darwin, tending to show that such risings and sinkings are in active progress over large portions of the continents and islands of the Southern hemisphere: and by the speculations of Sir C. Lyell respecting the influence of climate on the migrations of plants and animals, and the influence of geological changes upon climate.

In Zoology I may notice (following Mr. Owen)[49\B] recent discoveries of the remains of the animals which come nearest to man in their structure. At the time of Cuvier's death, in 1832, no evidence had been obtained of fossil Quadrumana; and he supposed that these, as well as Bimana, were of very recent introduction. Soon after, in the oldest (eocene) tertiary deposits of Suffolk, remains were found proving the existence of a monkey of the genus Macacus. In the Himalayan tertiaries were found petrified bones of a Semnopithecus; in Brazil, remains of an extinct platyrhine monkey of great size; and lastly, in the middle tertiary series of the South of France, was discovered a fragment of the jaw of the long-armed ape (_Hylobates_). But no fossil human {648} remains have been discovered in the regularly deposited layers of any divisions (not even the pleiocene) of the tertiary series; and thus we have evidence that the placing of man on the earth was the last and peculiar act of Creation.

[Note 49\B: _Brit. Asso._ 1854, p. 112.]

THE END.

Transcriber's Notes

Whewell's book was originally published in 3 volumes in London in 1837. A second edition appeared in 1847, and a third in 1857. A 2-volume version of the 3rd edition was published in New York in 1858, reprinted 1875. This Project Gutenberg text, combining both volumes in sequence, was derived from the 1875 version, relying upon resources kindly provided by the Internet Archive.

Three items have been added to the Contents of the First Volume; they are marked off by ~ ~, as are any other additions to the text.

Printed page numbers have been transcribed in { }; pages without a printed number have been indicated by {{ }}. Where words were hyphenated across pages, the number has been placed before the word.

Fractions have been transcribed as numerator/denominator, occasionally using parentheses to disambiguate. The original sometimes has numerator over a line with denominator below, at other times numerator hyphen denominator. Superscripted characters are marked by a ^ before the character.

Footnotes in the original text were numbered by chapter; here they have been numbered by Book (the number of which is given after a \, for the two appendices to the 3rd edition A has been used for volume 1, B for volume 2). They are placed after the paragraph in which they occur, and are transcribed [Note m\n: ...]. Footnote anchors are transcribed [m\n]. All other square brackets are in the original text.

One difficult item is the use of numbers within a ring as names of asteroids; here the numbers are in ( ).

Corrections to the text have been marked with **. They are listed below, and were usually confirmed by reference to English printings of the text. Inconsistencies, especially with respect to accents and formatting, are numerous and have in general not been adjusted, though Greek quotations have been checked against other versions where available. Nor have Whewell's unbalanced quotation marks been modernised. The English versions have been used to restore Whewell's "gesperrt" emphases in some Greek passages.

Location 1875 Text Correction Vol. 1 p. 25 Cruikshanks Cruickshank p. 30 19 65 p. 30 : ; p. 33 (thrice) 184 182 p. 36 184 182 p. 71 Arisotelians Aristotelians p. 75 " p. 79 σερματικοὶ σπερματικοὶ p. 101 " note 1\2 6 7 p. 175 ecliptical elliptical note 1\4 iv. vi. note 75\4 Summæ Summa note 10\5 iii. iv. p. 271 (twice) Mastlin Mæstlin p. 282 _Dialogo "_Dialogo p. 284 semil semel p. 287 endeaver endeavor note 7\6 1. i. note 8\6 Dial. i. p. 40. p. 141. note 9\6 _Speculutionum _Speculationum p. 325 Gualtier Gualter p. 341 and 342 Marsenne Mersenne p. 374 of p. 377 prependicularity perpendicularity p. 403 " note 30\7 Cosmotheros Cosmotheoros p. 415 _casual_ _causal_ p. 416 ) p. 419 ] p. 431 _a_ a note 69\7 1453 1753 note 84\7 Ast. Ass. p. 463 Philosphical Philosophical p. 471 ] p. 564 prevalance prevalence Vol. 2 p. 50 Ὑφιφάνη Ὑψιφάνη ἄρισπον ἄριστον οὔδιον εὔδιον p. 84 ] p. 85 viii. vii. p. 115 1853 1823 p. 149 , . p. 162 Footnote number missing in text p. 201 stream steam p. 213 and note 39\11 same number as the preceding note p. 240 Cruikshanks Cruickshank note 18\13 Mass-bestimmengen Mass-bestimmungen p. 264 in is note 11\14 _Stahl Stahl p. 295 the _the note 78\14 the entire text of this note is missing p. 301 lecture lectures note 87\14 96. 963. note 92\14 153 853 p. 330 Angels Angles p. 336 given giving p. 343 " p. 394 Surien Surian p. 411 _Couérs Elmentaire_ _Cours Elémentaire_ note 136\16 Εἴδην Εἴδη p. 450 dependance dependence p. 457 sucking-beasts suckling-beasts note 80\17 ähnlich ähnlich note 89\17 229 129 p. 477 osseuze osseuse note 119\17 229 299* p. 508 Lythophylaccii Lythophylacii p. 511 Stukely Stukeley note 18\18 Géognastique Géognostique p. 513 Sabapennine Sub-Apennine p. 514 Schlotheim Schlottheim p. 530 , ( ,) p. 556 Poissons Poisson's p. 620 iv. vii. p. 624 [ ( p. 628 in (not italicised in text) p. 630 Nordenskiold's Nordenskiöld's p. 638 390 474 p. 640 then than p. 641 1828 (1828)

* This is the page number given in the English edition. In the only version of the text referred to that I have found, the quotation is in a footnote on page 352 of the second edition (1805); the note was not in the first edition.